LOCAL STUDY OF SCHEMES AND THEIR MORPHISMS (EGA IV)

CUMULATIVE ENGLISH TRANSLATION — §§1–21

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Contents

Summary 5
1. Relative finiteness conditions. Constructible sets in preschemes 6
   1.1 Quasi-compact morphisms 7
   1.2 Quasi-separated morphisms 8
   1.3 Morphisms locally of finite type 10
   1.4 Morphisms locally of finite presentation 11
   1.5 Morphisms of finite type 14
   1.6 Morphisms of finite presentation 15
   1.7 Improvements of earlier results 16
   1.8 Morphisms of finite presentation and constructible sets 17
   1.9 Pro-constructible sets and ind-constructible sets 20
   1.10 Application to open morphisms 25
2. Base change and flatness 26
   2.1 Flat Modules on preschemes 26
   2.2 Faithfully flat Modules on preschemes 29
   2.3 Topological properties of flat morphisms 32
   2.4 Universally open morphisms and flat morphisms 36
   2.5 Permanence of properties of Modules by faithfully flat descent 38
   2.6 Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent 42
   2.7 Permanence of various properties of morphisms by faithfully flat descent 44
   2.8 Preschemes over a regular base of dimension 1; closure of a closed sub-prescheme of the generic fibre 47
3. Associated prime cycles and primary decompositions 49
   3.1 Prime cycles associated to a Module 49
   3.2 Irredundant decompositions 52
   3.3 Relations with flatness 54
   3.4 Properties of the sheaves $\mathscr{F}/t\mathscr{F}$ 56
4. Change of base field for algebraic preschemes 59
   4.1 If X is an algebraic prescheme over a field k, and K is an extension of k, the prescheme $X \otimes_k K$ and its underlying topological space 59
   4.2 Algebraic preschemes over a field, and extension of the base field 62
   4.3 Reminders on tensor products of fields 62
   4.4 Irreducible preschemes and connected preschemes over an algebraically closed field 63
   4.5 Geometrically irreducible and geometrically connected preschemes 64
   4.6 Geometrically reduced preschemes 69
   4.7 Multiplicities in the primary decomposition on an algebraic prescheme 74
   4.8 Field of definition 78
   4.9 Field of definition of a part of a prescheme 81

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5. Dimension and depth of locally Noetherian preschemes 82
5.1 Dimension of preschemes 82
5.2 Dimension of an algebraic prescheme 85
5.3 Dimension of the support of a Module and Hilbert polynomial 86
5.4 Dimension of the image of a morphism 87
5.5 Dimension formula for a morphism of finite type 88
5.6 Dimension formula and universally catenary rings 90
5.7 Depth and property (S_k) 94
5.8 Regular preschemes and property (R_k). Serre's normality criterion 97
5.9 Z-pure and Z-closed modules 98
5.10 Property (S_2) and Z-closure 102
5.11 Coherence criteria for the Modules $\mathscr{H}^0_{X/Z}(\mathscr{F})$ 107
5.12 Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA 110
5.13 Properties of permanence by passage to the inductive limit 114
6. Flat morphisms of locally Noetherian preschemes 116
6.1 Flatness and dimension 116
6.2 Flatness and projective dimension 118
6.3 Flatness and depth 119
6.4 Flatness and property (S_k) 121
6.5 Flatness and property (R_k) 122
6.6 Properties of transitivity 123
6.7 Application to base changes in algebraic preschemes 124
6.8 Regular, normal, reduced, smooth morphisms 126
6.9 The theorem of generic flatness 128
6.10 Dimension and depth of a Module normally flat along a closed sub-prescheme 129
6.11 Criteria for the sets $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ or $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ to be open 132
6.12 Nagata's criteria for $\mathrm{Reg}(\mathrm{X})$ to be open 136
6.13 Criteria for $\mathrm{Nor}(\mathrm{X})$ to be open 139
6.14 Base change and integral closure 141
6.15 Geometrically unibranch preschemes 146
7. Formal properties of morphisms; Noetherian approximation 149
7.1 Formal equidimensionality and formally catenary rings 149
7.2 Strictly equidimensional rings and strictly formally catenary rings 153
7.3 Formal fibres of Noetherian local rings 156
7.4 Permanence of properties of formal fibres 160
7.5 A criterion for $\boldsymbol{P}$-morphisms 163
7.6 Applications: I. Japanese local rings 165
7.7 Applications: II. Universally Japanese rings 168
7.8 Excellent rings 169
7.9 Excellent rings and resolution of singularities 172
8. Projective limits of preschemes 175
8.1 Introduction 175
8.2 Projective limits of preschemes 177
8.3 Constructible subsets of a projective limit of preschemes 180
8.4 Criteria of irreducibility and connectedness for projective limits of preschemes 184
8.5 Modules of finite presentation on a projective limit of preschemes 185
8.6 Sub-preschemes of finite presentation of a projective limit of preschemes 189
8.7 Criteria for a projective limit of preschemes to be a reduced (resp. integral) prescheme 190
8.8 Preschemes of finite presentation over a projective limit of preschemes 191
8.9 First applications to the elimination of Noetherian hypotheses 194
8.10 Properties of permanence of morphisms by passage to the projective limit 195
8.11 Application to quasi-finite morphisms 198

8.12 New proof and generalisation of the "Main Theorem" of Zariski 199
8.13 Translation in terms of pro-objects 203
8.14 Characterisation of a prescheme locally of finite presentation over another, in terms of the functor it represents 205
9. Constructible properties 206
9.1 The finite extension principle 206
9.2 Constructible and ind-constructible properties 208
9.3 Constructible properties of morphisms of algebraic preschemes 211
9.4 Constructibility of certain properties of modules 212
9.5 Constructibility of topological properties 215
9.6 Constructibility of certain properties of morphisms 217
9.7 Constructibility of properties of geometric separability, geometric irreducibility, and geometric connectedness 221
9.8 Primary decomposition in the neighbourhood of a generic fibre 225
9.9 Constructibility of local properties of fibres 228
10. Jacobson preschemes 232
10.1 Very dense subsets of a topological space 232
10.2 Quasi-homeomorphisms 234
10.3 Jacobson spaces 236
10.4 Jacobson preschemes and Jacobson rings 237
10.5 Noetherian Jacobson preschemes 239
10.6 Dimension in Jacobson preschemes 241
10.7 Examples and counter-examples 243
10.8 Rectified depth 243
10.9 Maximal spectra and ultra-preschemes 245
10.10 Algebraic spaces of Serre 247
11. Topological properties of finitely presented flat morphisms. Flatness criteria 248
11.1 Flatness loci (Noetherian case) 248
11.2 Flatness of a projective limit of preschemes 250
11.3 Application to the elimination of Noetherian hypotheses 259
11.4 Descent of flatness by arbitrary morphisms: the case of an Artinian base 265
11.5 Descent of flatness by arbitrary morphisms: the general case 270
11.6 Descent of flatness by arbitrary morphisms: the case of a unibranch base 273
11.7 Counter-examples 274
11.8 A valuative criterion for flatness 276
11.9 Separating and universally separating families of homomorphisms of sheaves of modules 277
11.10 Schematically dominant families of morphisms and families of schematically dense sub-preschemes 284
12. Study of fibres of finitely presented flat morphisms 286
12.1 Introduction 286
12.2 Local properties of fibres of a flat morphism locally of finite presentation 286
12.3 Local and global properties of fibres of a proper, flat morphism of finite presentation 290
12.4 Local cohomological properties of fibres of a flat morphism, locally of finite presentation 294
13. Equidimensional morphisms 296
13.1 Chevalley's semi-continuity theorem 297
13.2 Equidimensional morphisms: case of dominant morphisms of irreducible preschemes 299
13.3 Equidimensional morphisms: general case 302
14. Universally open morphisms 305
14.1 Open morphisms 305
14.2 Open morphisms and the dimension formula 306
14.3 Universally open morphisms 308

14.4 Chevalley's criterion for universally open morphisms 312
14.5 Universally open morphisms and quasi-sections 317
15. Study of fibres of a universally open morphism 321
15.1 Multiplicities of fibres of a universally open morphism 321
15.2 Flatness of universally open morphisms with geometrically reduced fibres 323
15.3 Applications: criteria for reduction and irreducibility 325
15.4 Complements on Cohen-Macaulay morphisms 326
15.5 Separable rank of fibres of a quasi-finite and universally open morphism. Application to connected components of fibres of a proper morphism 327
15.6 Connected components of fibres along a section 331
15.7 Appendix: Valuative criteria for local properness 336
16. Differential invariants. Differentially smooth morphisms 339
16.1 Normal invariants of an immersion 340
16.2 Functorial properties of the normal invariants of an immersion 343
16.3 Fundamental differential invariants of morphisms of preschemes 347
16.4 Functorial properties of differential invariants 348
16.5 Relative tangent sheaves and bundles; derivations. 356
16.6 Sheaf of p-differentials and exterior differentials. 361
16.7 The $\mathscr{P}^n_{X/S}(\mathscr{F})$. 363
16.8 Differential operators. 365
16.9 Regular and quasi-regular immersions 370
16.10 Differentially smooth morphisms. 374
16.11 Differential operators on a differentially smooth S-prescheme 375
16.12 Case of characteristic zero: Jacobian criterion for differentially smooth morphisms. 377
17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms. 377
17.1 Formally smooth morphisms, formally unramified morphisms, formally étale morphisms. 378
17.2 General properties of differentials 380
17.3 Smooth morphisms, unramified morphisms, étale morphisms 381
17.4 Characterizations of unramified morphisms. 383
18. Supplement on étale morphisms. Henselian local rings and strictly local rings 383
18.1 A remarkable equivalence of categories. 383
18.2 Coverings. 384
18.3 Finite algebras and étale algebras. 387
18.4 Local structure of unramified morphisms and of étale morphisms. 389
18.5 Henselian local rings. 393
18.6 Henselization. 400
18.7 Henselization and excellent rings. 405
18.8 Strictly local rings and strict Henselization. 406
18.9 Formal fibres of Noetherian Henselian rings. 410
18.10 Étale preschemes over a geometrically unibranch or normal prescheme 415
18.11 Application to Noetherian local algebras complete over a field 423
18.12 Applications of étale localisation to quasi-finite morphisms 430
19. Regular immersions and normal flatness 433
19.1 Properties of regular immersions 433
19.2 Transversally regular immersions 437
19.3 Relative complete intersections (flat case) 440
19.4 Application: criteria of regularity and smoothness for blown-up preschemes 442
19.5 Criteria for M-regularity 447
19.6 Regular sequences relative to a filtered quotient module 450
19.7 Hironaka's normal flatness criterion 452
19.8 Properties of passage to the projective limit 458
19.9 $\mathscr{F}$-regular sequences and depth 459
20. Meromorphic functions and pseudo-morphisms 461

20.0 Introduction 461
20.1 Meromorphic functions 461
20.2 Pseudo-morphisms and pseudo-functions 465
21. Divisors 465
21.1 Divisors on a ringed space 465
21.2 Divisors and invertible fractionary Ideals 467
21.3 Linear equivalence of divisors 470
21.4 Inverse images of divisors 472
21.5 Direct images of divisors 473
21.6 1-codimensional cycle associated to a divisor 475
21.7 Interpretation of positive 1-codimensional cycles in terms of sub-preschemes 480
21.8 Divisors and normalization 482
21.9 Divisors on preschemes of dimension 1 484
21.10 Inverse images and direct images of 1-codimensional cycles 488
21.11 Factoriality of regular local rings 497
21.12 The Van der Waerden purity theorem for the ramification locus of a birational morphism 498
21.13 Parafactorial pairs. Parafactorial local rings 504
21.14 The Ramanujam-Samuel theorem 510
21.15 Relative divisors 513
References 514

Summary

IV-1 | 222

§1. Relative finiteness conditions. Constructible sets in preschemes.
§2. Base change and flatness.
§3. Associated prime cycles and primary decomposition.
§4. Change of base field for algebraic preschemes.
§5. Dimension, depth, and regularity for locally Noetherian preschemes.
§6. Flat morphisms of locally Noetherian preschemes.
§7. Relations between a local Noetherian ring and its completion. Excellent rings.
§8. Projective limits of preschemes.
§9. Constructible properties.
§10. Jacobson preschemes.
§11. Topological properties of finitely presented flat morphisms. Flatness criteria.
§12. Study of fibres of finitely presented flat morphisms.
§13. Equidimensional morphisms.
§14. Universally open morphisms.
§15. Study of fibres of a universally open morphism.
§16. Differential invariants. Differentially smooth morphisms.
§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.
§18. Supplement on étale morphisms. Henselian local rings and strictly local rings.
§19. Regular immersions and normal flatness.
§20. Meromorphic functions and pseudo-morphisms
§21. Divisors.

The subjects discussed in the chapter call for the following remarks. IV-1 | 223

(a) The common property of all the subjects discussed is that they all related to *local* properties of preschemes or morphisms, i.e. considered at a point, or the points of a fibre, or on a (non-specified) neighbourhood of a point or of a fibre. These properties are generally of a *topological*, *differential*, or *dimensional* nature (i.e. bringing the ideas of *dimension* and *depth* into play), and are linked to the properties of the *local rings* at the points considered. One type of problem is the relating, for a given morphisms $f : X \to Y$ and point $x \in X$, of the properties of X at x with those of Y at $y = f(x)$ and those of the fibre $X_y = f^{-1}(y)$ at x. Another is the determining of the topological nature (for example, the constructibility, or the fact of being open or closed) of the set of points $x \in X$ at which X has a certain property, or for which the fibre $X_{f(x)}$ passing through x has a certain property at x. Similarly, we are interested in the topological nature of the set of points $y \in Y$ such that X has a certain property at all the points of the fibre X_y, or those such that this fibre itself has a certain property.

(b) The most important idea for the following chapters is that of *flat morphisms of finite presentation*, as well as the particular cases of *smooth morphisms* and *étale morphisms*. Their detailed study (as well as that of connected questions) really starts in §11.

(c) Sections §§1–10 can be considered as being preliminary in nature, and as developing three types of techniques, used, not only in the other sections of the chapter, but also, of course, in the follow chapters:

- (c1) Sections §§1–4 are envisaged as treating the diverse aspects of the idea of *change of base*, above all in relation with the conditions of *finiteness* or *flatness*; we there initiate the technique of *descent*, with its most elementary aspects (the questions of "effectiveness" linked to this technique will be studied in Chapter V).
- (c2) Sections §§5–7 are focused on what we may call *Noetherian* techniques, since the preschemes considered are always locally Noetherian, whereas, on the contrary, there is generally no finiteness condition imposed on the *morphisms*; this is essentially due to the fact that the ideas of dimension and depth are hardly manageable except in the case of Noetherian local rings. Recall that §7 constitutes a "delicate" theory of Noetherian local rings, not much used in what follows in the chapter.
- (c3) Sections §§8–10 describe, amongst other things, the means of *eliminating the Noetherian hypotheses* on the preschemes considered, by substituting such hypotheses for suitable ones of *finiteness* ("finite presentation") on the *morphisms* considered: the advantage of this substitution is that the latter such hypotheses (those of finiteness on the morphisms) are *stable under base change*, which is not the case for the Noetherian hypotheses on the preschemes. The technique permitting this substitution relies, in some part, on the use of the idea of the *projective limit* of preschemes, thanks to which we can reduce a question to the same question with *Noetherian* hypotheses; on the other hand, it relies on the systematic use of *constructible sets*, which have the double interest of being preserved under taking inverse images (of arbitrary morphisms) and by direct images (of morphisms of finite presentation), and having manageable topological properties in locally Noetherian preschemes. The same techniques often even allow to restrict to the case of more specific Noetherian rings, for example the **Z**-*algebras of finite type*, and it is here that the properties of "excellent" rings (studied in §7) intervene in a decisive manner. Independently of the question of elimination of Noetherian hypotheses, the techniques of §§8–10, elementary in nature, find constant use in nearly all applications. **IV-1** | 224

§1. Relative finiteness conditions. Constructible sets in preschemes

In this section, we will resume the exposé of "finiteness conditions" for a morphism of preschemes $f : X \to Y$ given in (**I**, 6.3 and 6.6). There are essentially two notions of "finiteness" of a *global* nature on X, that of *quasi-compact* morphism (defined in (**I**, 6.6.1)) and that of a *quasi-separated* morphism; on the other hand, there are two notions of "finiteness" of a *local* nature on X, that of a morphism *locally of finite type* (defined in (**I**, 6.6.2)) and that of a morphism *locally of finite presentation*. By combining these local notions with the preceding global notions, we obtain the notion of a morphism *of finite type* (defined in (**I**, 6.3.1)) and of a morphism *of finite presentation*. For the convenience of the reader, we will give again in this section the properties stated in (**I**, 6.3 and 6.6), referring to their labels in Chapter I for their proofs.

In n^os 1.8 and 1.9, we complete, in the context of preschemes, and making use of the previous notions of finiteness, the results on constructible sets given in ($\mathbf{0}_{\text{III}}$, §9).

1.1. Quasi-compact morphisms

Definition (1.1.1). — We say that a morphism of preschemes $f : X \to Y$ is *quasi-compact* if the continuous map f from the topological space X to the topological space Y is quasi-compact (**0**, 9.1.1), in other words, if the inverse image $f^{-1}(U)$ of every quasi-compact open subset U of Y is quasi-compact (cf. (**I**, 6.6.1)).

If $\mathfrak{B}$ is a basis for the topology of Y consisting of affine open sets, then for f to be quasi-compact, it is necessary and sufficient that for all $V \in \mathfrak{B}$, $f^{-1}(V)$ is a *finite union of affine open sets*. For example, if Y is affine and X is quasi-compact, *every* morphism $f : X \to Y$ is quasi-compact (**I**, 6.6.1).

If $f : X \to Y$ is a quasi-compact morphism, then it is clear that for every open subset V of Y, the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \to V$. Conversely, if (U_α) is an open cover of Y and $f : X \to Y$ is a morphism such that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ are quasi-compact, then f is quasi-compact. As a result, if $f : X \to Y$ is an S-morphism of S-preschemes, and if there **IV-1** | 225
exists an open cover (S_λ) of S such that the restrictions $g^{-1}(S_\lambda) \to h^{-1}(S_\lambda)$ of f (where g and h are the structure morphisms) are quasi-compact, then f is quasi-compact.

Proposition (1.1.2). —

label=(i) *An immersion $X \to Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally Noetherian or if the underlying space of X is Noetherian.*

lbbel=(ii) *The composition of two quasi-compact morphisms is quasi-compact.*

lcbel=(iii) *If $f : X \to Y$ is an S-morphism quasi-compact, the same is true of $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every extension $g : S' \to S$ of the base prescheme.*

ldbel=(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms quasi-compacts,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is quasi-compact.

lebel=(v) *If the composition $g \circ f$ of two morphisms $f : X \to Y$, $g : Y \to Z$ is quasi-compact, and if g is separated, or the underlying space of X is locally Noetherian, f is quasi-compact.*

lfbel=(vi) *For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be so.*

For the proof, see (**I**, 6.6.4). We note that the assertion (vi) is also a consequence of the following more general proposition:

Proposition (1.1.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If $g \circ f$ is quasi-compact and f is surjective, then g is quasi-compact.*

Proof. Indeed, if V is a quasi-compact open subset of Z, $f^{-1}(g^{-1}(V))$ is quasi-compact by hypothesis, and we have $g^{-1}(V) = f(f^{-1}(g^{-1}(V)))$, since f is surjective; so $g^{-1}(V)$ is quasi-compact. □

Corollary (1.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; suppose g is quasi-compact and surjective (resp. surjective). Then, for f to be quasi-compact (resp. dominant), it suffices that f' be so.*

Proof. If $g' : X' \to X$ is the canonical projection, g' is a surjective morphism (**I**, 3.5.2), and we have $f \circ g' = g \circ f'$; if f' is quasi-compact (resp. dominant), the same is true of $g \circ f'$ since g is quasi-compact (1.1.2) (resp. surjective); as $f \circ g'$ is quasi-compact (resp. dominant) and g' is surjective, we deduce that f is quasi-compact (1.1.3) (resp. dominant). □

To abbreviate, we will call *maximal point* of a prescheme X every generic point of an irreducible component of X; if $X = \operatorname{Spec}(A)$ is affine, the maximal points of X are the *minimal prime ideals* of A. **IV-1** | 226

Proposition (1.1.5). — *Let $f : X \to Y$ be a quasi-compact morphism. The following conditions are equivalent:*

label=a) *f is dominant;*

lbbel=b) *for every maximal point y of Y, we have $f^{-1}(y) \neq 0$;*

lcbel=c) *for every maximal point y of Y, $f^{-1}(y)$ contains a maximal point of X.*

The equivalence of *a)* and *c)* was proved in (**I**, 6.6.5). It is clear that *c)* implies *b)*; on the other hand, if X' is an irreducible component of X meeting $f^{-1}(y)$, $f(X')$ is irreducible and contains y, so it is contained in the irreducible component $Y' = \overline{\{y\}}$ of Y (**0**, 2.1.6); if x is the generic point of X', we therefore necessarily have $f(x) = y$ (**0**, 2.1.5); so *b)* implies *c)*.

Proposition (1.1.6). — *Let $f' : X' \to Y$, $f'' : X'' \to Y$ be two morphisms, X the prescheme which is the direct sum $X' \coprod X''$, $f : X \to Y$ the morphism equal to f' in X' and to f'' in X''. For f to be quasi-compact, it is necessary and sufficient that f' and f'' be so.*

This follows immediately from the definition.

1.2. Quasi-separated morphisms

Definition (1.2.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *quasi-separated* (or that X is *quasi-separated over* Y) if the diagonal morphism $\Delta_f = (1_X, 1_X)_Y : X \to X \times_Y X$ is quasi-compact. We say that a prescheme X is *quasi-separated* if it is quasi-separated over $\mathrm{Spec}(\mathbf{Z})$.

By definition, every separated morphism is quasi-separated, Δ_f being a closed immersion (**I**, 5.4.1), (i)); a *scheme* X is a quasi-separated prescheme, being separated over $\mathrm{Spec}(\mathbf{Z})$ (**I**, 5.4.1).

Proposition (1.2.2). —

label=(i) *Every monomorphism of preschemes $f : X \to Y$ (in particular every immersion) is quasi-separated.*
lbbel=(ii) *If $f : X \to Y$ and $g : Y \to Z$ are two quasi-separated morphisms, $g \circ f : X \to Z$ is quasi-separated.*
lcbel=(iii) *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism quasi-separated. Then, for every base change $g : S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is quasi-separated.*
ldbel=(iv) *If $f : X \to Y$, $f' : X' \to Y'$ are two quasi-separated S-morphisms, then*

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'$$

is quasi-separated.
lebel=(v) *If $f : X \to Y$, $g : Y \to Z$ are two morphisms such that $g \circ f$ is quasi-separated, then f is quasi-separated.*
lfbel=(vi) *If $f : X \to Y$ is a quasi-separated morphism, $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is quasi-separated.*

Proof. By virtue of (**I**, 5.5.12), it suffices to prove (i), (ii) and (iii).

The assertion (i) is trivial, every monomorphism being separated (**I**, 5.5.1). To prove (iii), we reduce immediately to the case where $Y = S$ (**I**, 3.3.11) and the assertion follows from the fact that $\Delta_{f_{(S')}} = (\Delta_f)_{(S')}$ (**I**, 5.3.4) and from (1.1.2), (iii). To prove (ii), consider the projections p and q of $X \times_Y X$ over X; if $i = (p, q)_Z$, we know that the diagram

$$\begin{tikzcd} X \times_Y X \arrow[r, "i"] & X \times_Z X \\ Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

is the same as that of i (1.1.2), (iii)); if furthermore f is quasi-separated, Δ_f is quasi-compact, so it is the same for $i \circ \Delta_f$ (1.1.2), (ii)) which is equal to $\Delta_{g \circ f}$. □ **IV-1** | 227

Corollary (1.2.3). —

label=(i) *Let X be a quasi-separated prescheme. Then every morphism $f : X \to Y$ is quasi-separated.*
lbbel=(ii) *Let Y be a quasi-separated prescheme. For a morphism $f : X \to Y$ to be quasi-separated, it is necessary and sufficient that the prescheme X be quasi-separated.*

This follows immediately from (1.2.2), (ii) and (v) applied to X, Y and $\mathrm{Spec}(\mathbf{Z})$.

Proposition (1.2.4). — *Let $f : X \to Y$ be a morphism, $g : Y \to Z$ a morphism. If $g \circ f$ is quasi-compact, then so is f.*

Proof. We know that f is the morphism composed $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{\mathrm{pr}_2} Y$ (**I**, 5.3.13); on the other hand, pr_2 is identified with $(g \circ f) \times_Z 1_Y$ (**I**, 3.3.4) and since $g \circ f$ is quasi-compact, so is pr_2 (1.1.2), (iv)). Finally, we have the commutative diagram

(1.2.4.1)
$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_Z Y \arrow[d, "f \times 1_Y"] \\ Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

which identifies $X \times_Y Y$-preschemes Y and $X \times_Z Y$ (**I**, 5.3.7). As by hypothesis Δ_g is a quasi-compact morphism, so is Γ_f (1.1.2), (iii)); the conclusion then follows from (1.1.2), (ii)). □

Proposition (1.2.5). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If g is quasi-compact and surjective and f' quasi-separated, then f is quasi-separated.*

Proof. We have indeed $(X' \times_{Y'} X') = (X \times_Y X)_{(Y')}$ and $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4); as the projection $X' \times_{Y'} X' \to X \times_Y X$ is quasi-compact (1.1.2), (iii)) and surjective (**I**, 3.5.2), we can, by virtue of (**I**, 3.3.11), apply (1.1.4), which shows that since $\Delta_{f'}$ is a quasi-compact morphism, the same is true of Δ_f. □

Proposition (1.2.6). — *Let $f : X \to Y$ be a morphism, (U_α) a covering of Y by open subsets such that the induced preschemes U_α are quasi-separated. For f to be quasi-separated, it is necessary and sufficient that each of the induced preschemes on the open subsets $f^{-1}(U_\alpha)$ be quasi-separated.*

Proof. The inverse image of $X \times_Y X$ of U_α is $X_\alpha \times_{U_\alpha} X_\alpha$, writing $X_\alpha = f^{-1}(U_\alpha)$, and the restriction $X_\alpha \to X_\alpha \times_{U_\alpha} X_\alpha$ of Δ_f is none other than Δ_{f_α}, denoting by f_α the restriction $X_\alpha \to U_\alpha$ of f; by virtue of the definition (1.2.1) and the local character of the notion of quasi-compact morphism (1.1.1), for f to be quasi-separated, it is necessary and sufficient that each of the morphisms f_α be so. But since by hypothesis the morphism $U_\alpha \to \mathrm{Spec}(\mathbf{Z})$ is quasi-separated, it amounts to saying that f_α is quasi-separated or that the composition $X_\alpha \xrightarrow{f_\alpha} U_\alpha \to \mathrm{Spec}(\mathbf{Z})$ is (1.2.2), (ii) and (v)), hence the proposition. □

To verify that a morphism is quasi-separated, we are therefore reduced to giving criteria for a prescheme to be quasi-separated: **IV-1** | 228

Proposition (1.2.7). — *Let X be a prescheme, (U_α) a covering of X formed by quasi-compact open subsets. The following properties are equivalent:*

label=a) *X is quasi-separated.*
lbbel=b) *For every quasi-compact open subset U of X, the canonical immersion $U \to X$ is quasi-compact (in other words, U is retrocompact in X).*
b') *The intersection of two quasi-compact open subsets of X is quasi-compact.*
lcbel=c) *For every pair of indices (α, β), the intersection $U_\alpha \cap U_\beta$ is quasi-compact.*

Proof. The properties *b)* and *b')* are trivially equivalent by definition of a quasi-compact morphism. As a quasi-compact open subset is a finite union of affine open subsets, for two quasi-compact open subsets U, V of X, $U \times_{\mathbf{Z}} V$ is a quasi-compact open subset of $X \times_{\mathbf{Z}} X$ (**I**, 3.2.7), whose inverse image under Δ_X is $U \cap V$, so *a)* implies *b')*. It is trivial that *b')* implies *c)*; finally, if *c)* is satisfied, the $U_\alpha \times_{\mathbf{Z}} U_\beta$ form a covering of $X \times_{\mathbf{Z}} X$ by quasi-compact open subsets, and we know that for the morphism $\Delta_X : X \to X \times_{\mathbf{Z}} X$ to be quasi-compact, it suffices that the inverse images $U_\alpha \cap U_\beta$ under Δ_X be quasi-compact (1.1.1); so *c)* implies *a)*. □

Corollary (1.2.8). — *Every prescheme X whose underlying space is locally Noetherian (for example a locally Noetherian prescheme) is quasi-separated; every morphism $f : X \to Y$ is then quasi-separated.*

Proposition (1.2.9). — *Let X', X'' be two preschemes, $X = X' \coprod X''$ their direct sum, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism which coincides with f' in X' and with f'' in X''. For f to be quasi-separated, it is necessary and sufficient that f' and f'' be so.*

Proof. Indeed, $X \times_Y X$ is the direct sum of the four preschemes $X' \times_Y X'$, $X' \times_Y X''$, $X'' \times_Y X'$ and $X'' \times_Y X''$, and Δ_f is the morphism which coincides with $\Delta_{f'}$ in X' and with $\Delta_{f''}$ in X''; the proposition then follows immediately from the definitions. □

1.3. Morphisms locally of finite type

(1.3.1). Recall that if A is a ring, a (commutative) A-algebra B is said to be *of finite type* if it is generated by a finite number of elements of B, or, what amounts to the same thing, if it is isomorphic to a quotient of a polynomial algebra $A[T_1, \ldots, T_n]$ by an ideal of this algebra. It is clear that for every (commutative) A-algebra A', $B \otimes_A A'$ is then an A'-algebra of finite type. If B is an A-algebra of finite type and C a B-algebra of finite type, C is an A-algebra of finite type, for if C is a quotient of $B[T_1, \ldots, T_n]$ and B a quotient of $A[S_1, \ldots, S_m]$, $B[T_1, \ldots, T_n]$ is isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{a}$, where $\mathfrak{a}$ is the kernel of the surjection $A[S_1, \ldots, S_m, T_1, \ldots, T_n] \to B[T_1, \ldots, T_n]$; hence C is also a quotient of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$, and so it is indeed an A-algebra of finite type.

Definition (1.3.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite type at x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and the ring $A(U)$ is an $A(V)$-algebra of finite type. We say that f is *locally of finite type* if it is of finite type at every point of X.

IV-1 | 229

Proposition (1.3.3). — *If Y is a locally Noetherian prescheme and $f : X \to Y$ a morphism locally of finite type, then X is locally Noetherian.*

For the proof, see (**I**, 6.3.7).

Proposition (1.3.4). —

label=(i) *Every local isomorphism is locally of finite type.*
lbbel=(ii) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite type, it is the same for $g \circ f$.*
lcbel=(iii) *If $f : X \to Y$ is an S-morphism locally of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite type for every extension $S' \to S$ of the base prescheme.*
ldbel=(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite type, $f \times_S g$ is locally of finite type.*
lebel=(v) *If the composition $g \circ f$ of two morphisms is locally of finite type, f is locally of finite type.*
lfbel=(vi) *If a morphism f is locally of finite type, the same is true of f_{red}.*

For the proof, see (**I**, 6.6.6).

Corollary (1.3.5). — *Let $f : X \to Y$ be a morphism locally of finite type. For every morphism $Y' \to Y$ such that Y' is locally Noetherian, $X \times_Y Y'$ is locally Noetherian.*

Proof. Indeed, $f_{(Y')} : X \times_Y Y' \to Y'$ is locally of finite type by (1.3.4), (iii)), and it suffices to apply (1.3.3). □

Proposition (1.3.6). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ to be locally of finite type, it is necessary and sufficient that B be an A-algebra of finite type.*

For the proof, see (**I**, 6.3.3), taking into account that f is quasi-compact and (**I**, 6.6.3).

Proposition (1.3.7). — *For a morphism locally of finite type $f : X \to Y$ to be surjective, it is necessary and sufficient that, for every algebraically closed field Ω, the map $X(\Omega) \to Y(\Omega)$ corresponding to f (**I**, 3.4.1) be surjective.*

For the proof, see (**I**, 6.3.10).

(1.3.8). Let A be a ring. We say that an A-algebra B is *essentially of finite type* if B is A-isomorphic to an A-algebra of the form $S^{-1}C$, where C is an A-algebra *of finite type* and S a multiplicative subset of C.

Proposition (1.3.9). —

label=(i) *If B is an A-algebra essentially of finite type and C a B-algebra essentially of finite type, then C is an A-algebra essentially of finite type.*

lbbel=(ii) *Let B be an A-algebra essentially of finite type, A' an A-algebra; then $B' = B \otimes_A A'$ is an A'-algebra essentially of finite type.*

Proof. label=(i) Let $B = S'^{-1}B'$ and $C = T'^{-1}C'$, where B' (resp. C') is an A-algebra (resp. B-algebra) of finite type and S' (resp. T') a multiplicative subset of B' (resp. C'); then C' is of the form $S'^{-1}C''$, where C'' is also a B'-algebra of finite type, so C is also a ring of fractions of C''; as C'' is an A-algebra of finite type, this proves our assertion.

lbbel=(ii) If B is a ring of fractions of an A-algebra of finite type C, B' is a ring of fractions of the A'-algebra of finite type $C \otimes_A A'$, hence (ii).

□

(1.3.10). If B is a *local* A-algebra essentially of finite type, B is of the form $C_{\mathfrak{q}}$, where C is an A-algebra of finite type and $\mathfrak{q}$ a prime ideal of C (Bourbaki, *Alg. comm.*, chap. II, § 5, prop. 11). Let $\mathfrak{p}$ be the inverse image in A of the ideal $\mathfrak{q}$; if we set $S = A - \mathfrak{p}$, $C_{\mathfrak{q}}$ is also a local ring over a prime ideal of $S^{-1}C$; as $S^{-1}C$ is an algebra of finite type over $A_{\mathfrak{p}} = S^{-1}A$, we see that B is also an $A_{\mathfrak{p}}$-algebra *essentially of finite type*, and furthermore the homomorphism $A_{\mathfrak{p}} \to B$ is *local*.

Proposition (1.3.11). — *If B is a local A-algebra essentially of finite type, B is A-isomorphic to a quotient of an A-algebra of the form $C_{\mathfrak{q}}$, where C is an algebra of polynomials $A[T_1, \ldots, T_n]$ and $\mathfrak{q}$ a prime ideal of C.*

Proof. Indeed, by definition, B is isomorphic to $C'_{\mathfrak{q}'}$, where C' is an A-algebra of finite type and $\mathfrak{q}'$ a prime ideal of C'; but $C' = C/\mathfrak{b}$, where $C = A[T_1, \ldots, T_n]$ and $\mathfrak{b}$ is an ideal of C; so $\mathfrak{q}' = \mathfrak{q}/\mathfrak{b}$, where $\mathfrak{q}$ is a prime ideal of C, and $C'_{\mathfrak{q}'}$ is isomorphic to $C_{\mathfrak{q}}/\mathfrak{b}C_{\mathfrak{q}}$. □

1.4. Morphisms locally of finite presentation

(1.4.1). Given a ring A, we say that an A-algebra B (commutative) is *of finite presentation* if it is isomorphic to a quotient $A[T_1, \ldots, T_n]/\mathfrak{a}$ of a polynomial algebra over A by an ideal *of finite type* $\mathfrak{a}$ of $A[T_1, \ldots, T_n]$. For every (commutative) A-algebra A', $B \otimes_A A'$ is then an A'-algebra of finite presentation, being isomorphic to the quotient of $A'[T_1, \ldots, T_n]$ by the image of the ideal $\mathfrak{a} \otimes_A A'$ in this ring, which is evidently an $A'[T_1, \ldots, T_n]$-module of finite type. If B is an A-algebra of finite presentation and C a B-algebra of finite presentation, then C is an A-algebra of finite presentation. Indeed, let $B = A[S_1, \ldots, S_m]/\mathfrak{a}$ and $C = B[T_1, \ldots, T_n]/\mathfrak{b}$, where $\mathfrak{a}$ (resp. $\mathfrak{b}$) is an ideal of finite type of $A[S_1, \ldots, S_m]$ (resp. $B[T_1, \ldots, T_n]$); the ring $B[T_1, \ldots, T_n]$ is isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{a}'$, where $\mathfrak{a}'$ is the canonical image of $\mathfrak{a} \otimes_A A[T_1, \ldots, T_n]$, and is therefore an ideal of finite type of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$; the ideal $\mathfrak{b}$ of $B[T_1, \ldots, T_n]$ is of the form $\mathfrak{b}'/\mathfrak{a}'$, where $\mathfrak{b}'$ is an ideal of finite type of $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$; as $\mathfrak{a}'$ and $\mathfrak{b}'/\mathfrak{a}'$ are modules of finite type over $A[S_1, \ldots, S_m, T_1, \ldots, T_n]$, so is $\mathfrak{b}'$, and C, being isomorphic to $A[S_1, \ldots, S_m, T_1, \ldots, T_n]/\mathfrak{b}'$, is therefore an A-algebra of finite presentation. We note finally that if A is *Noetherian*, then every A-algebra of finite type is an algebra of finite presentation. IV-1 | 326

Definition (1.4.2). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X, $y = f(x)$. We say that f is *of finite presentation at x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $\mathscr{O}(U)$ is an $\mathscr{O}(V)$-algebra of finite presentation. We say that f is *locally of finite presentation* if it is of finite presentation at every point of X.

If Y is locally Noetherian, it amounts to the same to say that f is locally of finite type or locally of finite presentation.

Proposition (1.4.3). — (i) *Every local isomorphism is locally of finite presentation.*

(ii) *If two morphisms $f : X \to Y$, $g : Y \to Z$ are locally of finite presentation, so is $g \circ f$.*

IV-1 | 327

(iii) If $f : X \to Y$ is an S-morphism locally of finite presentation, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is locally of finite presentation for every base extension $S' \to S$ of the base prescheme.

(iv) If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms locally of finite presentation, $f \times_S g$ is locally of finite presentation.

(v) If the composition $g \circ f$ of two morphisms is locally of finite presentation and if g is locally of finite type, then f is locally of finite presentation.

Proof. Assertion (i) is trivial. To prove (iii), we may limit to the case where $S = Y$ (**I**, 3.3.11); let x' be a point of $X_{(S')}$, s' and x its projections on S' and X respectively, s the common projection of s' and x on S. By hypothesis, there exists an affine open neighbourhood V (resp. U) of s (resp. x) such that the image of U is contained in V and such that $\mathscr{O}(U)$ is an $\mathscr{O}(V)$-algebra of finite presentation; let V' be an affine open neighbourhood of s' whose image in S is contained in V; then $U \times_V V'$ is an affine open neighbourhood of x' whose image in S' is contained in V' and whose ring $\mathscr{O}(U) \otimes_{\mathscr{O}(V)} \mathscr{O}(V')$ is an $\mathscr{O}(V')$-algebra of finite presentation (1.4.1).

To prove (ii), consider a point $x \in X$, and let $y = f(x)$, $z = g(f(x))$. There is by hypothesis an affine open neighbourhood $W = \operatorname{Spec}(C)$ (resp. $V = \operatorname{Spec}(B)$) of z (resp. y) such that $g(V) \subset W$ and such that B is a C-algebra of finite presentation. On the other hand, there is an affine open neighbourhood $V' = \operatorname{Spec}(B')$ (resp. $U = \operatorname{Spec}(A)$) of y (resp. x) such that $f(U) \subset V'$ and such that A is a B'-algebra of finite presentation. Let $s \in B$ be such that the open set $D(s) = \operatorname{Spec}(B_s)$ in V is a neighbourhood of y contained in V', and let $U' = f^{-1}(D(s)) \subset U$; we have $U' = \operatorname{Spec}(A \otimes_{B'} B_s)$, and $A \otimes_{B'} B_s$ is a B_s-algebra of finite presentation (1.4.1); on the other hand $B_s = B[T]/(1 - sT)$ is a B-algebra of finite presentation by definition; hence by (1.4.1), $A \otimes_{B'} B_s$ is a C-algebra of finite presentation, which proves (ii).

We know that (iv) follows from (i), (ii), and (iii) (**I**, 3.5.1). To prove (v), consider the commutative diagram

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] & X \times_Z Y \arrow[d, "f \times 1"] \\ Y \arrow[r, "\Delta_g"] & Y \times_Z Y \end{tikzcd}$$

which identifies X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z Y$ (**I**, 5.3.7), Δ_g being the diagonal morphism. If we know that Δ_g is locally of finite presentation, it will follow by (iii) that so is Γ_f. On the other hand, we have the factorisation of f as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ (**I**, 5.3.13), where the projection p_2 is equal to $(g \circ f) \times_Z 1_Y$; as $g \circ f$ is assumed locally of finite presentation, so is p_2 by (iv), and it will follow finally from (ii) that f is also locally of finite presentation. We thus see that we are reduced to proving the following. □

Corollary (1.4.3.1). — *Let $g : Y \to Z$ be a morphism locally of finite type; then the diagonal morphism $\Delta_g : Y \to Y \times_Z Y$ is locally of finite presentation.*

IV-1 | 328

Proof. We may reduce to the case where $Z = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, B being an A-algebra of finite type; the morphism Δ_g then corresponds to the *augmentation homomorphism* $\delta : B \otimes_A B \to B$ such that $\delta(x \otimes y) = xy$. In view of the definition (1.4.2), it suffices to show that the kernel $\mathfrak{J}$ of δ is an ideal *of finite type*, and this follows from (**0**, 20.4.4). □

Proposition (1.4.4). — *Let A be a ring, B an A-algebra, B' a polynomial algebra $A[T_1, \ldots, T_n]$, $u : B' \to B$ a surjective homomorphism of A-algebras. For B to be an A-algebra of finite presentation, it is necessary and sufficient that the kernel $\mathfrak{J}$ of u be an ideal of finite type of B'.*

Proof. The condition is sufficient by definition (1.4.1). To see that it is necessary, we remark that the morphism $g : \operatorname{Spec}(B') \to \operatorname{Spec}(A)$ is locally of finite type; if B is an A-algebra of finite presentation, the morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(B')$ corresponding to u is such that $g \circ f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is locally of finite presentation, and so it follows from (1.4.3), (v) that f is also locally of finite presentation. Now, to show that the ideal $\mathfrak{J}$ is of finite type, it suffices to prove that $\widetilde{\mathfrak{J}}$ is a $\widetilde{B'}$-Module of finite type (**II**, 6.1.4.1). Returning to the definition (1.4.2), we are finally reduced to proving the following. □

Lemma (1.4.4.1). — *Let $\mathfrak{a}$ be an ideal of a ring C, s an element of C, such that $C_s/\mathfrak{a}_s$ is a C-algebra of finite presentation; then $\mathfrak{a}_s$ is an ideal of finite type in the ring C_s.*

Proof. By hypothesis, there exists a C-algebra of polynomials $C' = C[T_1, \ldots, T_n]$ and a surjective C-homomorphism $v : C' \to C_s/\mathfrak{a}_s$ whose kernel $\mathfrak{b}$ is of finite type. Let $p : C_s \to C_s/\mathfrak{a}_s$ be the canonical homomorphism; for each i, there is a $t_i \in C$ such that $p(t_i/s^k) = v(T_i)$ (we may assume that the exponent k is the same for all indices i). Consider then the C_s-algebra $D = C_s[T_1, \ldots, T_n]$, and the C_s-algebra homomorphism $w : D \to C_s$ such that $w(T_i) = t_i/s^k$; w is evidently surjective, and so is $v' = p \circ w : D \to C_s/\mathfrak{a}_s$. On the other hand, every polynomial P of D can be written as P/s^m, where $P \in C' = C[T_1, \ldots, T_n]$, and we have $v'(P/s^m) = (1/s^m)v(P)$; as $1/s$ is invertible in C_s, the relation $v'(P/s^m) = 0$ in $C_s/\mathfrak{a}_s$ is equivalent to $v(P) = 0$, and hence the kernel $\mathfrak{b}'$ of v' is generated by the image of $\mathfrak{b}$ in the ring D, and *a fortiori* is *of finite type* in D. But $\mathfrak{b}' = v'^{-1}(0) = w^{-1}(p^{-1}(0)) = w^{-1}(\mathfrak{a}_s)$, and as w is surjective, $\mathfrak{a}_s = w(\mathfrak{b}')$, which proves that $\mathfrak{a}_s$ is an ideal of finite type of C_s. □

Corollary (1.4.5). — *Let X, Y be two preschemes, $j : X \to Y$ an immersion, U an open subset of Y such that $j(X)$ is closed in U, $\mathscr{J}$ the quasi-coherent ideal of $\mathscr{O}_U$ that defines the closed subprescheme of Y associated to j. For j to be locally of finite presentation, it is necessary and sufficient that $\mathscr{J}$ be an $\mathscr{O}_U$-Module of finite type.*

Proposition (1.4.6). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be locally of finite presentation, it is necessary and sufficient that B be an A-algebra of finite presentation.*

Proof. The condition being trivially sufficient, it remains to prove that it is necessary. If f is locally of finite presentation, it already follows from (1.3.6) that B is an A-algebra of finite type, so there exists **IV-1** | 329
a surjective homomorphism $u : B' = A[T_1, \ldots, T_n] \to B$ of A-algebras; it follows from (1.4.3), (v) that the immersion $j : \operatorname{Spec}(B) \to \operatorname{Spec}(B')$ is locally of finite presentation; hence by (1.4.5), if $\mathfrak{J}$ is the kernel of u, the $\widetilde{B'}$-Module $\widetilde{\mathfrak{J}}$ is of finite type, and hence by (**II**, 6.1.4.1) $\mathfrak{J}$ is an ideal of finite type. □

Proposition (1.4.7). — *Let $\rho : A \to B$ be a homomorphism of rings making B an A-algebra of finite presentation. For B to be an A-module of finite presentation, it is necessary and sufficient that it be an A-module of finite type.*

We will use the following.

Lemma (1.4.7.1). — *If B is an A-algebra of finite presentation, there exists a surjective A-homomorphism of algebras $u : B' \to B$, where B' is an A-algebra of finite presentation and a free A-module.*

Proof. Indeed, there is a finite system $(a_i)_{1 \leqslant i \leqslant m}$ of generators of the A-algebra B such that each a_i satisfies a relation $F_i(a_i) = 0$, where $F_i \in A[X]$ is a monic polynomial of degree > 0; the A-algebra quotient $B'_i = A[X]/(F_i)$ is therefore free of finite rank over A; let c_i be the class of X in B'_i. There exists an A-homomorphism of algebras $u_i : B'_i \to B$ such that $u_i(c_i) = a_i$; it then suffices to take for B' the tensor product $B'_1 \otimes_A B'_2 \otimes \ldots \otimes_A B'_m$ and to consider the homomorphism $u_1 \otimes u_2 \otimes \ldots \otimes u_m : B' \to B$, which is surjective by construction. Furthermore, if $B'' = A[T_1, \ldots, T_m]$ and if b'_i denotes the canonical image of c_i in B', the A-homomorphism of algebras $v : B'' \to B'$ such that $v(T_i) = b'_i$ $(1 \leqslant i \leqslant m)$ is surjective, and its kernel $\mathfrak{b}'$ is the ideal of B'' generated by the $F_i(T_i)$, and therefore of finite type; this proves that B' is an A-algebra of finite presentation. □

Proof of (1.4.7). This lemma being proved, let us keep the same notation, and let $w = v \circ u : B'' \to B$. Let $\mathfrak{b}$ (resp. $\mathfrak{a}$) be the kernel of w (resp. u); we have $\mathfrak{a} = v(\mathfrak{b})$ since v is surjective. Now by (1.4.4), if B is an A-algebra of finite presentation, $\mathfrak{b}$ is an ideal of finite type of B'', so $\mathfrak{a}$ is an A-module of finite type since B is an A-algebra of finite presentation. As B' is a free A-module, B is an A-module of finite presentation. Conversely, if B' is an A-module of finite presentation, $\mathfrak{a}$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. I, § 2, nº 8, lemme 9) and *a fortiori* an ideal of finite type of B'; hence, B is by definition a B'-algebra of finite presentation, and as B' is an A-algebra of finite presentation, B is an A-algebra of finite presentation. □

1.5. Morphisms of finite type

Proposition (1.5.1). — *Let $f : X \to Y$ be a morphism, (U_α) a covering of Y by open affines. The following conditions are equivalent:*

a) *f is locally of finite type and quasi-compact.*
b) *For all α, $f^{-1}(U_\alpha)$ is a finite union of affine open sets $V_{\alpha i}$ such that each ring $\mathscr{O}(V_{\alpha i})$ is an $\mathscr{O}(U_\alpha)$-algebra of finite type.*
c) *For every affine open subset U of Y, $f^{-1}(U)$ is a finite union of affine open sets V_j such that the rings $\mathscr{O}(V_j)$ are $\mathscr{O}(U)$-algebras of finite type.*

Proof. For the proof, see (**I**, 6.3.2) and (**I**, 6.6.3). □

Definition (1.5.2). — We say that a morphism f is *of finite type* if it satisfies the equivalent conditions of the proposition (1.5.1).

IV-1 | 330

Proposition (1.5.3). — *Let $f : X \to Y$ be a morphism of finite type; if Y is Noetherian, so is X.*

Proof. For the proof, see (**I**, 6.3.7). □

Proposition (1.5.4). —

(i) *Let $j : X \to Y$ be a quasi-compact immersion. If j is a closed immersion, or if the underlying space of X is Noetherian, or if the space underlying Y is locally Noetherian, then j is of finite type.*
(ii) *The composition of two morphisms of finite type is of finite type.*
(iii) *If $f : X \to Y$ is an S-morphism of finite type, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite type for every base extension $S' \to S$ of the base prescheme.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms of finite type, $f \times_S g$ is of finite type.*
(v) *If the composition $g \circ f$ of two morphisms is of finite type and if g is quasi-separated or if X is Noetherian, f is of finite type.*
(vi) *If a morphism f is of finite type, so is f_{red}.*

Proof. This follows immediately (except for the case (v) where X is Noetherian, proved in (**I**, 6.3.6)) from the definitions and from (1.1.2), (1.2.4), and (1.3.4). □

Corollary (1.5.5). — *Let $f : X \to Y$ be a morphism of finite type. For every morphism $Y' \to Y$ such that Y' is Noetherian, $X \times_Y Y'$ is Noetherian.*

Proof. This follows from (1.5.4), (iii) and from (1.5.3). □

Corollary (1.5.6). — *Let X be a prescheme of finite type over a locally Noetherian prescheme S. Then every S-morphism $f : X \to Y$ is of finite type.*

Proof. For the proof, see (**I**, 6.3.9). □

Proposition (1.5.7). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ to be of finite type, it is necessary and sufficient that φ make B an A-algebra of finite type.*

Proof. For the proof, see (**I**, 6.3.3). □

1.6. Morphisms of finite presentation

Definition (1.6.1). — Let X, Y be two preschemes. We say that a morphism $f : X \to Y$ is *of finite presentation* if it satisfies the following three conditions:

(i) f is locally of finite presentation;

(ii) f is quasi-compact (which, when (i) is satisfied, is equivalent to saying that f is of finite type (1.5.2));

(iii) f is quasi-separated.

We then also say that X is *of finite presentation over* Y, or a Y*-prescheme of finite presentation*.

Condition (iii) is automatically satisfied if f is separated, or if X is locally Noetherian (1.2.8); when Y is locally Noetherian, it amounts to the same to say that f is *of finite presentation* or that f is *of finite type* (the latter condition implying that X is locally Noetherian (1.3.3)).

IV-1 | 331

Proposition (1.6.2). — (i) *Every quasi-compact immersion that is locally of finite presentation (in particular every quasi-compact open immersion) is of finite presentation.*

(ii) *The composition of two morphisms of finite presentation is of finite presentation.*

(iii) *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism of finite presentation. Then, for every morphism $S' \to S$, the morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ is of finite presentation.*

(iv) *Let $f : X \to Y$, $f' : X' \to Y'$ be two S-morphisms of finite presentation; then $f \times_S f' : X \times_S X' \to Y \times_S Y'$ is of finite presentation.*

(v) *If the composition $g \circ f$ of two morphisms is of finite presentation and if g is quasi-separated and locally of finite presentation, then f is of finite presentation.*

Proof. This follows immediately from (1.1.2), (1.2.2), (1.2.4), and (1.4.3). □

It follows in particular from (1.6.2), (iii) that if f is a morphism of finite presentation and U an open subset of Y, the restriction $f^{-1}(U) \to U$ of f is a morphism of finite presentation. Conversely, let (U_α) be a covering of Y by open affines, and suppose that the restrictions $f^{-1}(U_\alpha) \to U_\alpha$ of f are morphisms of finite presentation; then f is evidently locally of finite presentation, quasi-compact by virtue of (1.1.1), and quasi-separated by virtue of (1.2.6). We can therefore say that the property for a morphism $f : X \to Y$ of being of finite presentation is *local on* Y.

If X is a quasi-separated prescheme, every morphism $f : X \to Y$ is quasi-separated (1.2.3). Hence, if f is quasi-compact and locally of finite presentation, f is of finite presentation.

In particular:

Corollary (1.6.3). — *Let $\varphi : A \to B$ be a homomorphism of rings. For the corresponding morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ to be of finite presentation, it is necessary and sufficient that φ make B an A-algebra of finite presentation.*

Proof. The necessity of the condition follows from (1.4.6). □

Remark (1.6.4). — In the definition (1.6.1), condition (iii) is not a consequence of the two others. Let for example Y be an affine scheme whose underlying space is not Noetherian, and let U be an open subset that is not quasi-compact in Y. Let X be the prescheme obtained by gluing two preschemes Y_1, Y_2 isomorphic to Y along the open subsets U_1, U_2 corresponding to U (**0**, 4.1.7), so that X is the union of two open subsets isomorphic respectively to Y_1 and Y_2 (and that we identify with them), with $Y_1 \cap Y_2 = U$. Furthermore, there is a morphism $f : X \to Y$ that coincides in Y_i with the isomorphism $Y_i \to Y$ ($i = 1, 2$). It is clear that f satisfies condition (i) of (1.6.1), being a local isomorphism; it also satisfies (ii), the inverse image of an open quasi-compact subset of Y being the union of its images in Y_1 and Y_2; but as $Y_1 \cap Y_2$ is not quasi-separated (1.2.7).

Proposition (1.6.5). — *Let X', X'' be two preschemes, $X = X' \amalg X''$ their sum, $f' : X' \to Y$, $f'' : X'' \to Y$ two morphisms, $f : X \to Y$ the morphism that coincides with f' in X' and with f'' in X''. For f to be of finite presentation, it is necessary and sufficient that f' and f'' be so.*

Proof. It suffices to show that for f to possess one of the three properties of the definition (1.6.1), it is necessary and sufficient that f' and f'' possess it; this is evident for the property of being locally of finite presentation, which is local on X; for the property of being quasi-compact, this was seen in (1.1.6), and for the property of being quasi-separated, in (1.2.9). □ IV-1 | 332

1.7. Improvements of earlier results We will give in this number a list of propositions proved in the preceding chapters, whose statements can be improved with the aid of the new finiteness conditions introduced above.

(1.7.1). In the statements of (**I**, 6.4.2), (**I**, 6.4.3), and (**I**, 6.4.9), we can, instead of assuming that X and Y are of finite type over the field K, assume only that they are *locally of finite type* over K; for (**I**, 6.4.2), it suffices to observe that every point of a prescheme X that is closed in every open affine of X is closed in X, and a scheme affine locally of finite type over K is *ipso facto* of finite type over K (1.5.1). For (**I**, 6.4.9), we use (1.3.7) and (1.5.4), (v).

(1.7.2). In (**I**, 6.5.1), (ii), we can, instead of assuming that S is locally Noetherian and Y of finite type over S, assume only that Y is *locally of finite presentation* over S, as one shows immediately by applying the definition (1.4.2). Similarly in (**I**, 6.5.4), (ii) and (**I**, 6.5.5), (ii) it suffices to assume that f is an S-morphism, Y being *locally of finite presentation* over S.

(1.7.3). In (**I**, 7.1.11), for the second assertion, instead of assuming S locally Noetherian and Y of finite type over S, we can assume only Y locally of finite presentation over S (the proof depending on (**I**, 6.5.1), (ii)); similarly in (**I**, 7.1.12) to (**I**, 7.1.15).

(1.7.4). In (**I**, 9.2.2), we can replace the three hypotheses (each implying the others) by the single hypothesis that f is *quasi-compact and quasi-separated*, by virtue of (1.2.6) and (1.2.7). We note that in (**I**, 9.2.1), the hypothesis means exactly that f is quasi-compact and quasi-separated.

(1.7.5). In (**I**, 9.3.1), (**I**, 9.3.2), and (**I**, 9.3.3), we can replace the two hypotheses on X by the (weaker) hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using these two properties (by virtue of (1.2.7)).

(1.7.6). In (**I**, 9.3.4) and (**I**, 9.3.5), it suffices to assume X quasi-compact, $\mathscr{F}$ and $\mathscr{G}$ quasi-coherent and of finite type.

(1.7.7). In (**I**, 9.4.7), we can weaken hypothesis b) by assuming only X *quasi-separated*, since it suffices that the canonical immersion $U \to X$ be quasi-compact (1.2.7). We can make the same modification in (**I**, 9.4.8), (**I**, 9.4.9), and (**I**, 9.4.10).

(1.7.8). In (**I**, 9.5.1), the conditions indicated in parentheses in the statement can be replaced by the condition that f be quasi-compact and quasi-separated.

(1.7.9). In (**I**, 9.6.5), we can replace the hypotheses on X by the weaker hypothesis that X is *quasi-* **IV-1** | 333
compact and quasi-separated, as the proof shows, which uses only (**I**, 9.4.7). In (**I**, 9.6.6), the hypothesis "X is a quasi-compact scheme" may be replaced by "X is a prescheme quasi-compact and quasi-separated".

(1.7.10). In (**II**, 1.7.15), we can suppose only that Y is quasi-compact and quasi-separated, the proof using only (**I**, 9.6.5).

(1.7.11). In the second assertion of (**II**, 3.4.5), we can substitute for the two hypotheses on Y the weaker hypothesis that Y is quasi-compact and quasi-separated. Similarly in (**II**, 3.4.8).

(1.7.12). In (**II**, 3.8.5), we can similarly suppose only that Y is quasi-compact and quasi-separated, this hypothesis being covered by (**I**, 9.4.7).

(1.7.13). In (**II**, 4.4.1), (**II**, 4.4.6), and (**II**, 4.4.7), it suffices to assume that Y is *quasi-compact and quasi-separated*, in view of (1.2.3) in the proof of (**II**, 4.4.7). Similarly, in (**II**, 4.4.10.1), it suffices to assume that Z is quasi-compact and quasi-separated.

(1.7.14). In (**II**, 4.5.2) and (**II**, 4.5.5), it suffices to assume X *quasi-compact and quasi-separated*, this hypothesis being covered by (**I**, 9.3.1) and (**I**, 9.3.2).

(1.7.15). In (**II**, 4.6.8), it suffices still to assume X *quasi-compact and quasi-separated*, this hypothesis being covered by (**I**, 9.4.7).

(1.7.16). In (**II**, 5.1.2), it suffices to assume that X is quasi-compact and quasi-separated (the hypothesis being covered by (**II**, 4.5.2) and (**II**, 4.5.5)). Similarly in (**II**, 5.1.9), where we use (**I**, 9.6.5).

(1.7.17). In (**II**, 5.2.1), it suffices always to assume X quasi-compact and quasi-separated (the hypothesis being covered by (**II**, 4.5.2)).

(1.7.18). In (**II**, 5.2.2), one must assume f *quasi-compact* and X *quasi-separated*; the current proof is in fact insufficient (with the hypotheses made), for from the fact that for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ we have $\mathrm{R}^1 f_*(\mathscr{F}) = 0$, it does not follow, necessarily, for an open affine U of Y, that the same property holds for the $(\mathscr{O}_X|f^{-1}(U))$-Modules quasi-coherent, if one does not know that such a Module is the restriction of a quasi-coherent $\mathscr{O}_X$-Module (a remark applying to conditions b) and c′)); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (**I**, 9.4.7), modified above (1.7.7).

(1.7.19). In (**II**, 5.3.2), it suffices to assume Y *quasi-compact and quasi-separated* (using (**II**, 4.4.6) and (**II**, 4.4.7)). Similarly in (**II**, 5.5.3), (ii).

(1.7.20). In (**III**, 1.2.2), (**III**, 1.2.3), and (**III**, 1.2.4), we can replace the two hypotheses on X by the weaker hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using only (**I**, 9.3.3).

(1.7.21). In (**III**, 1.4.10) to (**III**, 1.4.14), we can replace "separated" by "quasi-separated", this hypothesis serving only to be able to apply (**I**, 9.2.2) (see above (1.7.4)). Similarly, in (**III**, 1.4.15), it suffices to assume that the morphism u is *quasi-separated and quasi-compact*.

(1.7.22). In (**III**, 2.1.3), we can again replace the hypotheses by the weaker hypothesis that X is a prescheme quasi-compact and quasi-separated, the reasoning being the same as in (**III**, 1.2.2).

1.8. Morphisms of finite presentation and constructible sets **IV-1 | 334**

(1.8.1). Let X be a prescheme; we know that every quasi-compact open subset of X is a finite union of affine open subsets, and conversely. For an open subset U of X to be *retrocompact* in X (**0**, 9.1.1), it is necessary and sufficient that for every *open affine* subset V of X, $U \cap V$ be quasi-compact. When X is *quasi-separated* (1.2.1), every quasi-compact open subset of X is retrocompact in X (1.2.7), so every locally constructible part of X is retrocompact in X (**0**, 9.1.13); if *in addition* X is *quasi-compact*, there is identity between constructible parts and locally constructible parts of X (**0**, 9.1.12), and between constructible open parts and quasi-compact open parts (**0**, 9.1.5).

Proposition (1.8.2). — *Let X, Y be two preschemes, $f : X \to Y$ a morphism. For every constructible (resp. locally constructible) part Z of Y, $f^{-1}(Z)$ is a constructible (resp. locally constructible) part of X.*

Proof. Suppose Z locally constructible; for every $x \in X$, there will be an affine open neighbourhood V of $f(x)$ such that $Z \cap V$ is constructible in V; if the proposition is proved for constructible parts, $f^{-1}(Z) \cap f^{-1}(V)$ will be constructible in $f^{-1}(V)$, so $f^{-1}(Z)$ will be locally constructible. It therefore suffices to consider the case where Z is constructible, and taking into account (**0**, 9.1.3), we are reduced to the case where Z is open and *retrocompact* in Y, in other words such that the canonical injection $Z \to Y$ is a quasi-compact morphism; it then suffices to see that $f^{-1}(Z)$ is *retrocompact in* X, in other words such that the canonical injection $f^{-1}(Z) \to X$ is a quasi-compact morphism; but this follows from (1.1.2), (iii)), since $f^{-1}(Z) = Z \times_Y X$ (**I**, 4.4.1). □

Lemma (1.8.3). — *Let X be a prescheme quasi-compact and quasi-separated, Z a constructible part of X. There exists then an affine scheme X' and a morphism of finite presentation $f : X' \to X$ such that $f(X') = Z$.*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets X_i ($i \in \mathrm{I}$), and since X is quasi-separated, it follows from (1.2.7) that each of the canonical immersions $g_i : X_i \to X$ is of finite presentation; one concludes that if $f_i : X'_i \to X_i$ is a morphism of finite presentation, the same is true of $g_i \circ f_i : X'_i \to X$ (1.6.2), and if X' is the prescheme sum of the X'_i, the morphism $h : X' \to X$ which coincides with $g_i \circ f_i$ in each X'_i is also of finite presentation (1.6.5). Since $Z \cap X_i$ is constructible in X_i (**0**, 9.1.8), we see that we are reduced to proving the lemma when X_i is *affine* of ring A. Since Z is then a finite union of sets of the form $U \cap \complement V$, where U and V are quasi-compact open subsets **IV-1 | 335**
(**0**, 9.1.3), one sees, by considering yet another suitable sum of preschemes, that one can reduce to the case where $Z = U \cap \complement V$; since moreover U is a finite union of affine open subsets, one can even restrict ourselves to the case where U is *affine*. Since V is quasi-compact, it is a finite union of open

subsets of the form X_{f_i}, where $f_i \in A$; let $\mathfrak{J}$ be the ideal of A generated by the f_i, and let Z' be the closed affine sub-scheme of X that it defines (and which is by construction of finite presentation over X); one has by definition $V = \complement Z'$, so $U \cap \complement V = U \cap Z'$. Consider the affine scheme $X' = Z' \times_X U$ and let $f : X' \to X$ be the structural morphism; we just saw that the canonical immersion $Z' \to X$ is of finite presentation, and it is the same for the open immersion $U \to X$, which is quasi-compact since U and X are affine; one concludes therefore from (1.6.2), (iv)) that f is of finite presentation, and one has $f(X') = Z' \cap U$ (**I**, 3.4.8), which completes the proof. □

Theorem (1.8.4). — (Chevalley). — *Let $f : X \to Y$ be a morphism of finite presentation. For every locally constructible part Z of X, $f(Z)$ is locally constructible in Y.*

Proof. Let $y \in Y$ and V an open affine neighbourhood of y; since f is quasi-compact and quasi-separated, the same is true of its restriction $f^{-1}(V) \to V$, so $f^{-1}(V)$ is a prescheme quasi-compact and quasi-separated (1.2.3); since $Z \cap f^{-1}(V)$ is constructible in $f^{-1}(V)$ (1.8.1), we see that one can restrict ourselves to the case where Y is affine and Z constructible; X is then quasi-compact and quasi-separated, so (1.8.3) there exists a morphism of finite presentation $g : X' \to X$ such that $Z = g(X')$; since $f \circ g$ is of finite presentation, we see that we are reduced to the case where $Z = X$; in other words, it will suffice to prove the □

Lemma (1.8.4.1). — *Let Y be an affine scheme, $f : X \to Y$ a morphism quasi-compact and locally of finite presentation; then $f(X)$ is a constructible part of Y.*

Proof. Since X is quasi-compact, it is a finite union of affine open subsets; one can therefore restrict ourselves to the case where $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, B being an A-algebra of finite type (1.4.6). Or, one has the □

Lemma (1.8.4.2). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in \mathrm{L}}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; if B is an A_0-algebra of finite presentation, there exist a λ and an A_λ-algebra of finite presentation B_λ such that B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$.*

Proof. By hypothesis, B is isomorphic to an algebra of the form $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]/\mathfrak{J}$, where the T_i are indeterminates and $\mathfrak{J}$ an ideal of finite type; let (F_j) $(1 \leqslant j \leqslant m)$ be a system of generators of $\mathfrak{J}$. Let $\varphi_\lambda : A_\lambda \to A$ be the canonical homomorphism. Since L is filtering, there exists λ such that each of the coefficients of each of the polynomials F_j is the image by φ_λ of an element of A_λ. In other words, there exists a system of polynomials $(\mathrm{F}_{j\lambda})_{1 \leqslant j \leqslant m}$ of $A_\lambda[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ such that $\mathrm{F}_j = \varphi_\lambda(\mathrm{F}_{j\lambda})$ for $1 \leqslant j \leqslant m$. If $\mathfrak{J}_\lambda$ is the ideal of $A_\lambda[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ generated by the $\mathrm{F}_{j\lambda}$, $\mathfrak{J}$ is the image of $\mathfrak{J}_\lambda \otimes_{A_\lambda} A$ in $A[\mathrm{T}_1, \ldots, \mathrm{T}_n] = A_\lambda[\mathrm{T}_1, \ldots, \mathrm{T}_n] \otimes_{A_\lambda} A$; the ring $B_\lambda = A_\lambda[\mathrm{T}_1, \ldots, \mathrm{T}_n]/\mathfrak{J}_\lambda$ answers to the question. □

One will apply this lemma here by considering A as inductive limit of its sub-$\mathbf{Z}$-algebras *of finite type*. One sees that there exist such a sub-$\mathbf{Z}$-algebra A_0 of A, and an A_0-algebra of finite presentation B_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$; let us set $X_0 = \operatorname{Spec}(B_0)$, $Y_0 = \operatorname{Spec}(A_0)$, so that $X = X_0 \times_{Y_0} Y$, f being the projection $X \to Y$; let $f_0 : X_0 \to Y_0$, $g_0 : Y \to Y_0$ be the structural morphisms; it follows from (**I**, 3.4.8) that $f(X) = g_0^{-1}(f_0(X_0))$; taking into account (1.8.2), it therefore suffices to show that $f_0(X_0)$ is constructible. In other words, one is finally reduced to proving the

IV-1 | 336

Corollary (1.8.5). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a morphism of finite type. For every constructible part Z of X, $f(Z)$ is a constructible part of Y.*

Proof. One reduces as above to proving that $f(X)$ is constructible. Let us apply the criterion (**0**, 9.2.3): one must prove that for every irreducible closed part T of Y, $T \cap f(X) = f(f^{-1}(T))$ is rare in T or contains a non-empty open subset of T; taking T as a sub-prescheme of Y with underlying space T (**I**, 5.2.1) and considering its inverse image by f, one sees finally (taking into account (1.5.4), (iii)) that one is reduced to proving that if one supposes Y *irreducible and f dominant*, $f(X)$ contains a non-empty open subset of Y. One can moreover suppose Y affine; then X is a finite union of affine open subsets V_i, and since $f(X)$ is dense in Y, the same is true of at least one of the $f(V_i)$. One can therefore also suppose X affine; if (X_j) is the (finite) family of irreducible components of X, one sees as above that at least one of the $f(X_j)$ is dense in Y; one can therefore suppose X irreducible. Finally, by replacing f by f_{red} (1.5.4), (vi)) one can suppose X and Y *integral*. Then

$X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, where A is an integral Noetherian ring, B an integral A-algebra of finite type containing A (**I**, 1.2.7). It suffices to show that there exists $g \in A$ such that, for every $y \in \mathrm{D}(g)$ (that is to say such that $g \notin \mathfrak{i}_y$) the ideal $\mathfrak{i}_y$ is the intersection of A and a prime ideal of B, for this will show that $\mathrm{D}(g) \subset f(X)$. Finally it suffices to prove the □

Lemma (1.8.5.1). — *Let A be an integral ring, B an integral A-algebra of finite type. There exists $g \in A$ such that every homomorphism of A into an algebraically closed field Ω, non-zero on g, extends to a homomorphism of B into Ω.*

Proof. This is a classical result of commutative algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 3 of th. 1). □

Corollary (1.8.6). — *Let Y be an irreducible prescheme, η its generic point, $f : X \to Y$ a morphism locally of finite type. If $f^{-1}(\eta)$ is not empty, there exists a neighbourhood U open of η in Y such that $f^{-1}(y)$ is non-empty for every $y \in U$. If $\mathscr{F}$ is an $\mathscr{O}_X$-Module quasi-coherent of finite type, and if $\mathscr{F}_\eta = \mathscr{F} \otimes_{\mathscr{O}_Y} k(\eta)$ is not zero, then there exists a neighbourhood U' of η in Y such that $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is not zero for every $y \in U'$.*

Proof. If p_y is the canonical projection of the fibre $f^{-1}(y) = X \times_Y \mathrm{Spec}(k(y))$ in X, one has $\mathscr{F}_y = p_y^*(\mathscr{F})$, so $\mathrm{Supp}(\mathscr{F}_y) = p_y^{-1}(\mathrm{Supp}(\mathscr{F})) = \mathrm{Supp}(\mathscr{F}) \cap f^{-1}(y)$ by virtue of (**I**, 9.1.13.1) and (**I**, 3.6.1), since $\mathscr{F}$ is of finite type; moreover $\mathrm{Supp}(\mathscr{F})$ is closed in X (**0**, 5.2.2), and if Z is a closed sub-prescheme of X having as underlying space $\mathrm{Supp}(\mathscr{F})$ (**I**, 5.2.1), and j the canonical immersion $Z \to X$, $f \circ j$ is locally of finite type (1.3.4), (vi)); this proves that the first assertion implies the second. To prove the first assertion, let us first note that one can suppose X and Y reduced (1.3.4), (vi)), in other words Y *integral*. Let ξ be a point of $f^{-1}(\eta)$; one can replace X and Y by affine open neighbourhoods of ξ and η respectively, and thus suppose that $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, B being an A-algebra of finite type. Moreover, if Z is the reduced closed sub-prescheme of X having $\{\overline{\xi}\}$ as underlying set (**I**, 5.2.1), one can, as above, replace X by Z, in other words suppose B *integral* (**I**, 5.1.4). Since the morphism f is then dominant, the corresponding homomorphism $\varphi : A \to B$ is injective (**I**, 1.2.7); the corollary therefore follows finally from the lemma (1.8.5.1). □

Proposition (1.8.7). — *Let S be a prescheme, $g : X \to S$, $h : Y \to S$ two morphisms of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \mathrm{Spec}(k(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set of $s \in S$ such that f_s is surjective (resp. radical) is locally constructible.*

Proof. Let E be the set of $s \in S$ such that f_s is surjective; the set $S - E$ is equal to $h(Y - f(X))$; now f is of finite presentation (1.6.2), (v)) so $f(X)$ is locally constructible in Y (1.8.4); since h is of finite presentation, $h(Y - f(X))$ is locally constructible in S, and therefore also E.

To show that the set E' of $s \in S$ such that f_s is radical is locally constructible, we will use the □

Lemma (1.8.7.1). — *Let U, V be two preschemes; for a morphism $f : U \to V$ to be radical, it is necessary and sufficient that the diagonal morphism $\Delta_f : U \to U \times_V U$ is surjective. Consequently, every radical morphism is separated.*

Proof. Indeed, Δ_f being an immersion (**I**, 5.3.9), is a morphism locally of finite type (1.3.4); to say that Δ_f is surjective therefore means that for every algebraically closed field K, the corresponding map $U(K) \to (U \times_V U)(K) = U(K) \times_{V(K)} U(K)$ is surjective (1.3.7) and (**I**, 3.4.2.1); but by definition of a fibre product of sets, this means that the map $U(K) \to V(K)$ corresponding to f is injective, and this is equivalent to saying that f is radical (**I**, 3.5.5).

This being so, Δ_{f_s} is deduced from $\Delta_f : X \to X \times_Y X$ by the base change $\mathrm{Spec}(k(s)) \to S$ (**I**, 5.3.4). Since f is of finite presentation, it is the same for the structural morphism $X \times_Y X \to Y$ (1.6.2), (iv)), so also for Δ_f (1.6.2), (v)); it therefore suffices to apply the first part of the proposition to Δ_f, using the lemma (1.8.7.1). □

1.9. Pro-constructible sets and ind-constructible sets

Lemma (1.9.1). — *Let $(A_\alpha)_{\alpha\in \mathrm{I}}$ be an inductive system of rings with index set* I *filtering to the right, and let $A = \varinjlim A_\alpha$. For $A = 0$, it is necessary and sufficient that there exists an index γ such that $A_\gamma = 0$ (and one then has $A_\beta = 0$ for every $\beta \geqslant \gamma$).*

Proof. Indeed, for every $\alpha \in \mathrm{I}$, the canonical homomorphism $\varphi_\alpha : A_\alpha \to A$ sends the unit element to the unit element; to say that $A = 0$ means that $\varphi_\alpha(1) = 0$, or again $\varphi_\alpha(1) = \varphi_\alpha(0)$; one knows that this is equivalent to saying that there exists an index $\gamma \geqslant \alpha$ such that the homomorphism $\varphi_{\gamma\alpha} : A_\alpha \to A_\gamma$ is such that $\varphi_{\gamma\alpha}(1) = \varphi_{\gamma\alpha}(0)$, and this last relation is equivalent to $A_\gamma = 0$, whence the lemma. □

Proposition (1.9.2). — *Let B be a ring, $(A_\alpha)_{\alpha\in\mathrm{I}}$ an inductive system of B-algebras with index set* I *filtering to the right, and let $A = \varinjlim A_\alpha$; set $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(A)$, $X_\alpha = \mathrm{Spec}(A_\alpha)$, and let $u : X \to Y$, $u_\alpha : X_\alpha \to Y$ be the morphisms corresponding to the structural homomorphisms $B \to A$, $B \to A_\alpha$. Then:*

(1.9.2.i). *For $X = \varnothing$, it is necessary and sufficient that there exists $\gamma \in \mathrm{I}$ such that $X_\gamma = \varnothing$, and one then has $X_\alpha = \varnothing$ for $\alpha \geqslant \gamma$.*

(1.9.2.ii). *We have*

(1.9.2.1)
$$u(X) = \bigcap_{\alpha\in\mathrm{I}} u_\alpha(X_\alpha).$$

The assertion (i) is nothing other than the translation of (1.9.1). To prove (ii), note that, since u factors as $X \xrightarrow{u_\alpha} X_\alpha \to Y$, the first member is contained in the second. Conversely, let $y \in Y - u(X)$; one has $u^{-1}(y) = \varnothing$, and $u^{-1}(y)$ is the space underlying the $k(y)$-prescheme $\mathrm{Spec}(A \otimes_B k(y))$; since in the category of B-modules the functor $\varinjlim$ commutes with the tensor product, $A \otimes_B k(y)$ is the inductive limit of the system of rings $A_\alpha \otimes_B k(y)$; the lemma (1.9.1) shows that there exists α such that $A_\alpha \otimes_B k(y) = 0$, that is to say $u_\alpha^{-1}(y) = \varnothing$, so $y \notin u_\alpha(X_\alpha)$.

Proposition (1.9.3). — *Let X be a prescheme quasi-compact and quasi-separated, $(X_\alpha)_{\alpha\in\mathrm{I}}$ an open cover of X, and for every $\alpha \in \mathrm{I}$, let $E_\alpha = E \cap X_\alpha$. The following conditions are equivalent:*

a) There exists a quasi-compact morphism $f : X' \to X$ such that $E = f(X')$.
a′) For every α, there exists a quasi-compact morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $E_\alpha = f_\alpha(X'_\alpha)$.
a″) For every α, there exist an affine scheme X'_α and a morphism $f_\alpha : X'_\alpha \to X_\alpha$ such that $f_\alpha(X'_\alpha) = E_\alpha$.
b) E is intersection of constructible parts of X.
b′) $F = X - E$ is union of constructible parts of X.

Proof. It is clear that *b*) and *b′*) are equivalent. The condition *a*) implies *a′*) by taking for X'_α the prescheme induced on the open subset $f^{-1}(X_\alpha)$ of X' and for f_α the restriction of f. To see that *a′*) implies *a″*), it suffices to remark that X'_α is a union of affine open subsets $X''_{\alpha\lambda}$ ($\lambda \in \mathrm{L}$); let X''_α be the prescheme *sum* of the $X''_{\alpha\lambda}$ ($\lambda \in \mathrm{L}$), which is an affine scheme; if $g_\alpha : X''_\alpha \to X'_\alpha$ is the morphism coinciding with the identity in each $X''_{\alpha\lambda}$, it is clear that if one sets $f'_\alpha = f_\alpha \circ g_\alpha$, one has $E_\alpha = f'_\alpha(X''_\alpha)$ since g_α is surjective. To show that *a″*) implies *a*), note that there is a finite part J of I such that the X_α for $\alpha \in \mathrm{J}$ cover X; let X' be the prescheme sum of the X'_α for $\alpha \in \mathrm{J}$, and let $f : X' \to X$ be the morphism coinciding with $j_\alpha \circ f_\alpha$ for every $\alpha \in \mathrm{J}$, where $j_\alpha : X_\alpha \to X$ is the canonical injection. Since E is the union of the $E \cap X_\alpha$ for $\alpha \in \mathrm{J}$, one has $E = f(X')$ and it remains to see that f is a quasi-compact morphism; but the hypothesis that X is quasi-separated implies that j_α is quasi-compact (1.2.7), so it is the same for $j_\alpha \circ f_\alpha$ (1.1.1) and (1.1.2) and finally of f (1.1.6).

It remains therefore to prove the equivalence of *a″*) and *b*). Let us prove first that *a″*) implies *b*): **IV-1** | 338
let us consider a finite part J of I such that the X_α for $\alpha \in \mathrm{J}$ cover X; it will suffice to show that, for every $\alpha \in \mathrm{J}$, E_α is intersection of constructible parts $\mathrm{F}_{\alpha\lambda}$ of X_α ($\lambda \in \mathrm{L}_\alpha$); indeed, every $\mathrm{F}_{\alpha\lambda}$ is also a constructible part of X by virtue of (**0**, 9.1.8), (ii)), since the hypothesis that X is quasi-separated implies that every X_α is retrocompact in X (1.2.7); E being union of the E_α for $\alpha \in \mathrm{J}$, is intersection of the finite unions $\bigcup_{\alpha\in\mathrm{J}} \mathrm{F}_{\alpha,\lambda(\alpha)}$, for all choices of $\lambda(\alpha) \in \mathrm{L}_\alpha$; these finite unions being constructible parts of X, this establishes our assertion. One can therefore restrict ourselves to the case where **IV-1** | 339
$X = \mathrm{Spec}(A)$ and where $E = f(X')$, where $X' = \mathrm{Spec}(A')$ is affine, $f : X' \to X$ a morphism corresponding to a homomorphism of rings $A \to A'$. Let us note now that A' is inductive limit of

the family (A'_λ) of its sub-A-algebras *of finite type*, ordered by inclusion; if $X'_\lambda = \mathrm{Spec}(A'_\lambda)$, f factors as $X' \to X'_\lambda \xrightarrow{f_\lambda} X$ and it follows from (1.9.2), (ii)) that $f(X') = \bigcap_\lambda f_\lambda(X'_\lambda)$; if one shows that $f_\lambda(X'_\lambda)$ is intersection of constructible parts of X, it will be the same for $f(X')$. One can therefore restrict ourselves to the case where A' is an A-algebra *of finite type*. Or, one has the following lemma: □

Lemma (1.9.3.1). — *Let A be a ring, A' an A-algebra of finite type. Then A' is A-isomorphic to a filtering inductive limit $\varinjlim A''_\mu$ where the A''_μ are A-algebras of finite presentation (1.4.1).*

Proof. Indeed, one can write $A' = B/\mathfrak{J}$, with $B = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ and $\mathfrak{J}$ an ideal of B. But $\mathfrak{J}$ is inductive limit of the ideals *of finite type* $\mathfrak{J}'_\mu$ of B, contained in $\mathfrak{J}$, ordered by inclusion; since in the category of B-modules the functor $\varinjlim$ is exact, one concludes that $A' = \varinjlim B/\mathfrak{J}'_\mu$ up to A-isomorphism. □

The same reasoning as above then permits us to reduce to the case where A' is an A-algebra of finite presentation; but then $f(X')$ is constructible in X by virtue of the theorem of Chevalley (1.8.5), which proves *b*).

Let us prove finally that *b*) implies *a''*). If E is intersection of a family $(G_\lambda)_{\lambda \in \mathrm{L}}$ of constructible parts of X, each E_α is intersection of the $G_\lambda \cap X_\alpha$, which are constructible in X_α (**0**, 9.1.8), (i)), and one is reduced to the case where $X = \mathrm{Spec}(A)$ is affine.

One knows then (1.8.3) that for every $\lambda \in \mathrm{L}$ there exists a morphism $f_\lambda : X'_\lambda \to X$, where $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ is affine, such that $f_\lambda(X'_\lambda) = G_\lambda$. It therefore suffices to prove the following lemma:

Lemma (1.9.3.2). — *Let $(A'_\lambda)_{\lambda \in \mathrm{L}}$ be a family of A-algebras, $X = \mathrm{Spec}(A)$, $X'_\lambda = \mathrm{Spec}(A'_\lambda)$, and let $f_\lambda : X'_\lambda \to X$ be the structural morphism. For every finite part J of L, set $A'_\mathrm{J} = \bigotimes_{\lambda \in \mathrm{J}} A'_\lambda$ (tensor product of A-algebras), $X'_\mathrm{J} = \mathrm{Spec}(A'_\mathrm{J})$, and let $f_\mathrm{J} : X'_\mathrm{J} \to X$ be the structural morphism. One then has $f_\mathrm{J}(X'_\mathrm{J}) = \bigcap_{\lambda \in \mathrm{J}} f_\lambda(X'_\lambda)$. If $A' = \varinjlim A'_\mathrm{J}$, J running through the set of finite parts of L, $X' = \mathrm{Spec}(A')$ and $f : X' \to X$ is the structural morphism, one has $f(X') = \bigcap_{\lambda \in \mathrm{L}} f_\lambda(X'_\lambda)$.*

Proof. The first assertion follows from (**I**, 3.4.7); the second follows from (1.9.2), (ii)), which gives the relation $f(X') = \bigcap_\mathrm{J} f_\mathrm{J}(X'_\mathrm{J})$. □

Definition (1.9.4). — Let X be a topological space. We say that a part E of X is *pro-constructible* (resp. *ind-constructible*) in X if, for every $x \in X$, there exists an open neighbourhood U of x in X such that $E \cap U$ is intersection (resp. union) of locally constructible parts of U.

Taking into account (1.2.7) and (**0**, 9.1.8) and (**0**, 9.1.12), the equivalent conditions of the proposition (1.9.3) are therefore expressed by saying that E is *pro-constructible* in X, the locally constructible parts of X being identical to the constructible parts of X under the hypotheses of (1.9.3).

IV-1 | 340

Proposition (1.9.5). — *Let X be a prescheme.*

(i) *For a part E of X to be pro-constructible, it is necessary and sufficient that $X - E$ be ind-constructible.*
(ii) *Every finite union and every finite intersection of pro-constructible parts of X are pro-constructible. Every finite intersection and every finite union of ind-constructible parts of X are ind-constructible.*
(iii) *Every intersection (resp. union) of locally constructible parts of X is pro-constructible (resp. ind-constructible). Conversely, if X is quasi-compact and quasi-separated, every pro-constructible (resp. ind-constructible) part of X is intersection (resp. union) of constructible parts of X.*
(iv) *Let (U_α) be an open cover of X. For a part E of X to be pro-constructible (resp. ind-constructible) in X, it is necessary and sufficient that for every α, $E \cap U_\alpha$ be pro-constructible (resp. ind-constructible) in U_α.*
(v) *Every pro-constructible part E of X is retrocompact in X. For a locally closed part E of X to be retrocompact, it is necessary and sufficient that it be pro-constructible in X.*
(vi) *Let $f : X' \to X$ be a morphism of preschemes; for every pro-constructible (resp. ind-constructible) part E of X, $f^{-1}(E)$ is pro-constructible (resp. ind-constructible) in X'.*
(vii) *Let $f : X' \to X$ be a quasi-compact morphism; for every pro-constructible part E' of X', $f(E')$ is pro-constructible in X; in particular $f(X')$ is pro-constructible in X.*
(viii) *Let $f : X' \to X$ be a morphism locally of finite presentation; for every ind-constructible part E' of X', $f(E')$ is ind-constructible in X; in particular $f(X')$ is ind-constructible in X.*

(ix) *Suppose that X is quasi-compact and quasi-separated; then, for a part E of X to be pro-constructible, it is necessary and sufficient that there exist an affine scheme X' and a (necessarily quasi-compact) morphism $f : X' \to X$ such that $E = f(X')$.*

Proof. Properties (i), (ii), (iv) and the first assertion of (iii) follow from the definition (1.9.4) and are valid for any topological space, using the mutual distributivity of intersection and union and the fact that if T is locally constructible in X, $T \cap U$ is locally constructible in U for every open subset U of X. If X is quasi-compact and quasi-separated and E is pro-constructible in X, there is a finite open cover (U_i) of X formed by affine open subsets such that $E \cap U_i$ is intersection of constructible parts $\mathrm{M}_{i\lambda}$ of U_i ($\lambda \in \mathrm{L}_i$); the $\mathrm{M}_{i\lambda}$ are also constructible parts of X by virtue of (**0**, 9.1.8), (ii)), since the hypothesis that X is quasi-separated implies that every X_α is retrocompact in X (1.2.7); E being union of the E_α for $\alpha \in \mathrm{J}$, is intersection of the finite unions $\bigcup \mathrm{M}_{i,\lambda(i)}$ (where $\lambda(i) \in \mathrm{L}_i$ for every i), which are constructible parts of X; this proves the second assertion of (iii). The assertion (ix) follows from (iii) and (1.9.3). To prove the first assertion of (v), one can limit ourselves to the case where X is affine, and prove then that E is quasi-compact (**0**, 9.1.1); but since X is then quasi-separated, there exists by virtue of (ix) a quasi-compact morphism $f : X' \to X$ such that $E = f(X')$; since X' is quasi-compact, the same is true of E by a continuous map.

To prove (vi), one can restrict ourselves, by virtue of (iv), to the case where X is affine; the conclusion follows then from (iii) and (1.8.2).

To prove (vii) as well, one can restrict ourselves to the case where X is affine; then X' is a finite union of affine open subsets X'_i and $f(E')$ is the union of the $f(E' \cap X'_i)$, so one can also suppose that X' is affine, by virtue of (ii); but then $E' = g(X'')$, where $g : X'' \to X'$ is a quasi-compact morphism, by virtue of (ix), and one has $f(E') = f(g(X''))$, so $f \circ g$ is quasi-compact, and therefore $f(E')$ is pro-constructible.

One deduces from (vii) the proof of the second assertion of (v): indeed, let E be a locally closed and retrocompact part of X, and consider the sub-prescheme of X having E as underlying space (**I**, 5.2.1); the canonical injection $j : E \to X$ is a quasi-compact morphism, and it follows from (vii) that $E = j(E)$ is pro-constructible in X.

Finally, to prove (viii), one can still suppose X affine; moreover, if (X'_α) is an open affine cover of X' such that $E' \cap X'_\alpha$ is union of constructible parts of X'_α ((iii) and (iv)), $f(E')$ is union of the $f(E' \cap X'_\alpha)$, and by virtue of the theorem of Chevalley (1.8.4), each $f(E' \cap X'_\alpha)$ is union of constructible parts, whence the conclusion. □

(1.9.6). Examples (1.9.6). — Every finite part of a prescheme X is pro-constructible: indeed, it suffices to consider the parts $\{x\}$ reduced to a single point (1.9.5), (ii)) and $\{x\}$ is the image of $\mathrm{Spec}(k(x))$ by the canonical morphism $\mathrm{Spec}(k(x)) \to X$, which is quasi-compact; whence the conclusion by (1.9.5), (vii)). On the other hand, a part $\{x\}$ reduced to a point is not necessarily ind-constructible; for example, let A be a Noetherian integral ring having an infinity of maximal ideals, and let x be the generic point of $X = \mathrm{Spec}(A)$; if $\{x\}$ were ind-constructible in X, it would be constructible by virtue of (1.9.5), (iii)) and therefore would contain a non-empty open subset of X (**0**, 9.2.2); but this contradicts the hypothesis that A has an infinity of maximal ideals, by virtue of the Artin–Tate theorem (**0**, 16.3.3).

Every closed part of a prescheme X is pro-constructible, by (1.9.5), (v)). Every open part of X is therefore ind-constructible, but an open part U of X is pro-constructible only if it is *retrocompact*, by virtue again of (1.9.5), (v)).

Finally, if X is a *Noetherian* prescheme, therefore quasi-separated (1.2.8), it follows from (1.9.5), (iii)) and (**0**, 9.1.7) that the ind-constructible parts of X are the *unions of locally closed parts* of X.

Theorem (1.9.7). — *Let X be a quasi-compact prescheme, E an ind-constructible part of X, $(\mathrm{F}_\lambda)_{\lambda \in \mathrm{L}}$ a family of pro-constructible parts of X such that $\bigcap_{\lambda \in \mathrm{L}} \mathrm{F}_\lambda \subset E$; then there exists a finite part* J *of* L *such that $\bigcap_{\lambda \in \mathrm{J}} \mathrm{F}_\lambda \subset E$.*

Proof. Note that the sets $F = X - E$ and $F \cap \mathrm{F}_\lambda$ are pro-constructible, so one is reduced to the □

Corollary (1.9.8). — *Let X be a quasi-compact prescheme, $(\mathrm{F}_\lambda)_{\lambda \in \mathrm{L}}$ a family of pro-constructible parts of X whose intersection is empty; then there exists a finite subfamily $(\mathrm{F}_\lambda)_{\lambda \in \mathrm{J}}$ whose intersection is empty.*

Proof. Covering X by a finite number of affine open subsets, and using (1.9.5), (iv)), one is reduced to the case where X is affine; by virtue of (1.9.5), (ix)), one has $\mathrm{F}_\lambda = f_\lambda(X'_\lambda)$, where $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ **IV-1** | 341 is affine, and f_λ is a morphism $X'_\lambda \to X$; then, with the notations of (1.9.3.2), on the one hand, by hypothesis $f(X') = \varnothing$, hence $X' = \varnothing$; but this implies by (1.9.2), (i)) that there exists a finite part J of L such that $X'_\mathrm{J} = \varnothing$, hence $f_\mathrm{J}(X'_\mathrm{J}) = \varnothing$; but $f_\mathrm{J}(X'_\mathrm{J}) = \bigcap_{\lambda\in\mathrm{J}} \mathrm{F}_\lambda = \varnothing$. □

Corollary (1.9.9). — *Let X be a quasi-compact prescheme, F a pro-constructible part of X, $(\mathrm{E}_\lambda)_{\lambda\in\mathrm{L}}$ a family of ind-constructible parts of X such that $\mathrm{F} \subset \bigcup_{\lambda\in\mathrm{L}} \mathrm{E}_\lambda$; then there exists a finite part J of L such that $\mathrm{F} \subset \bigcup_{\lambda\in\mathrm{J}} \mathrm{E}_\lambda$. In particular, from every covering of X by ind-constructible parts, one can extract a finite covering of X.*

Proof. This follows from (1.9.7) by passing to complements. □

Proposition (1.9.10). — *Let X be a prescheme. For a part E of X to be ind-constructible, it is necessary that, for every $x \in X$, the intersection $E \cap \overline{\{x\}}$ be a neighbourhood of x in $\overline{\{x\}}$. If X is locally Noetherian, this condition is sufficient.*

Proof. Suppose E ind-constructible, and let Y be the intersection of $\overline{\{x\}}$ and an open affine subset of X containing x; then there is thus a sub-prescheme of X having Y for underlying space (**I**, 5.2.1), and if $j : Y \to X$ is the canonical injection, $E \cap Y = j^{-1}(E)$ is ind-constructible in Y (1.9.5), (vi)). As Y is quasi-compact and separated, $E \cap Y$ is therefore a union of constructible parts of Y (1.9.5), (iii)); consequently, there are two open subsets U, V in Y such that $x \in U \cap \complement V \subset E \cap Y$ (**0**, 9.1.3). But as x is the generic point of the irreducible space Y, V is necessarily empty and we have $U \subset E \cap Y$.

Suppose now X locally Noetherian, and let E be a part of X satisfying the condition of the statement. By virtue of the definition (1.9.4), one can restrict to the case where X is *Noetherian*. Then, for every $x \in E$, $E \cap \overline{\{x\}}$ contains a non-empty open part of the form $U \cap \overline{\{x\}}$, where U is open in X; this shows that E is a union of locally closed parts of X, and hence is ind-constructible (1.9.6). □

Proposition (1.9.11). — *Let X be a prescheme, E a part of X. The following properties are equivalent:*

a) *E is locally constructible.*
b) *E is ind-constructible and pro-constructible.*
c) *E and $X - E$ are pro-constructible.*
d) *E and $X - E$ are ind-constructible.*

Proof. It suffices evidently to prove that *d*) implies *a*), and one can restrict to the case where X is affine. Then (1.9.5), (iii)), E (resp. $X - E$) is a union of a family (E_λ) (resp. (E'_μ)) of constructible parts of X; as the E_λ and the E'_μ form a covering of X, it follows from (1.9.9) that there are finitely many indices λ_i, μ_j such that the E_{λ_i} and the E'_{μ_j} form a covering of X; this implies that E is a union of the E_{λ_i}, and hence is constructible. □

Corollary (1.9.12). — *Let $f : X \to Y$ be a surjective morphism, either quasi-compact, or locally of finite presentation. For a part E of Y to be locally constructible (resp. pro-constructible, resp. ind-constructible), it is necessary and sufficient that $f^{-1}(E)$ be so in X.*

IV-1 | 342

Proof. We know that the condition is necessary (1.8.2) and (1.9.5), (vi))); to show that it is sufficient, one is reduced to the case where X is affine, by virtue of (1.9.5), (iv)); in addition, by virtue of (1.9.11), one can restrict to the case where $f^{-1}(E)$ is pro-constructible, *or* the case where $f^{-1}(E)$ is ind-constructible. If f is surjective and quasi-compact, and $f^{-1}(E)$ is pro-constructible, it follows from (1.9.5), (vii)) that $E = f(f^{-1}(E))$ is pro-constructible. If f is surjective and locally of finite presentation, and $f^{-1}(E)$ is ind-constructible, $E = f(f^{-1}(E))$ is ind-constructible by (1.9.5), (viii)), which completes the proof. □

(1.9.13). Let X be a prescheme, $\mathscr{T}$ its topology; it follows from (1.9.5), (i) and (ii)) that the ind-constructible (resp. pro-constructible) parts of X are the open parts (resp. closed parts) for a topology on X, called the *constructible topology* and which we will denote by $\mathscr{T}^{\mathrm{cons}}$; we will denote by X^{cons} the set X equipped with the topology $\mathscr{T}^{\mathrm{cons}}$.

Proposition (1.9.14). — *Let X be a prescheme, $\mathscr{T}$ its topology, $\mathscr{T}^{\mathrm{cons}}$ the constructible topology on X.*

(i) *The topology $\mathscr{T}^{\mathrm{cons}}$ is finer than $\mathscr{T}$.*
(ii) *The locally constructible parts of X are identical to the parts that are both open and closed in the space X^{cons}.*
(iii) *For every morphism $f : X \to Y$, the underlying map of X^{cons} to Y^{cons} is continuous; we denote it f^{cons}.*
(iv) *If the morphism $f : X \to Y$ is quasi-compact, f^{cons} is a closed map; in particular, if f is quasi-compact and bijective, f^{cons} is a homeomorphism.*
(v) *If the morphism $f : X \to Y$ is locally of finite presentation, f^{cons} is an open map; in particular, if f is bijective and locally of finite presentation, f^{cons} is a homeomorphism.*
(vi) *For every open subset U of X, the topology induced by $\mathscr{T}^{\mathrm{cons}}$ on U is identical to the topology of U^{cons}.*

Proof. Indeed, (i) follows from the fact that every open set for $\mathscr{T}$ is open for $\mathscr{T}^{\mathrm{cons}}$ (1.9.6), and (ii) is a restatement of (1.9.11); (iii), (iv) and (v) are restatements respectively of (1.9.5), (vi), (vii) and (viii). Finally, to prove (vi), it suffices to remark that the canonical injection $j : U \to X$ is locally of finite presentation (1.4.3), hence $j^{\mathrm{cons}} : U^{\mathrm{cons}} \to X^{\mathrm{cons}}$ is a continuous open map. □

Proposition (1.9.15). — *Let X be a prescheme.*

(i) *For X^{cons} to be quasi-compact, it is necessary and sufficient that X be quasi-compact.*
(ii) *For X^{cons} to be separated, it is necessary and sufficient that X be quasi-separated, and then X^{cons} is locally compact and totally disconnected.*
(iii) *For X^{cons} to be compact, it is necessary and sufficient that X be quasi-compact and quasi-separated.*
(iv) *Every point of X admits, for the topology $\mathscr{T}^{\mathrm{cons}}$, an open and compact neighbourhood.*
(v) *For a morphism $f : X \to Y$ to be quasi-compact, it is necessary and sufficient that the map $f^{\mathrm{cons}} : X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ be proper.*

IV-1 | 343

Proof. (i) As $\mathscr{T}^{\mathrm{cons}}$ is finer than $\mathscr{T}$, it is clear that if X^{cons} is quasi-compact, so is X; the converse follows from (1.9.9).

(ii) Suppose X quasi-separated, and let us show that X^{cons} is totally disconnected: indeed, if x, y are two distinct points of X and if for example $x \notin \overline{\{y\}}$, there exists an open affine subset U of X not containing y; as X is quasi-separated, U is retrocompact in X (1.2.7), hence U and $X - U$ are locally constructible in X, and therefore open in X^{cons} by virtue of (1.9.11), whence our assertion. As for every open affine subset U of X, the topology induced by $\mathscr{T}^{\mathrm{cons}}$ is that of U^{cons} by virtue of (1.9.14), (vi)), it follows from the above that X^{cons} is locally compact, since U is open in X^{cons} and U^{cons} is compact. It remains to show that if X^{cons} is separated, then X is quasi-separated; consider indeed an open affine subset U of X; the topology induced on U by $\mathscr{T}^{\mathrm{cons}}$ is that of U^{cons} (1.9.14), (vi)), hence U is compact for this induced topology, since U^{cons} is compact by the first part of the reasoning. If V is a second open affine subset in X, $U \cap V$ is therefore the intersection of two compact parts of the separated space X^{cons}; as the topology induced by $\mathscr{T}^{\mathrm{cons}}$ on $U \cap V$ is that of $(U \cap V)^{\mathrm{cons}}$ (1.9.14), (vi)) and the identity $(U \cap V)^{\mathrm{cons}} \to U \cap V$ is continuous, one concludes that $U \cap V$ is quasi-compact for the topology induced by $\mathscr{T}$; X is therefore quasi-separated by virtue of (1.2.7).

(iii) is an immediate corollary of (i) and (ii).

(iv) For every $x \in X$, an open affine neighbourhood U of x for $\mathscr{T}$ is also a neighbourhood of x for $\mathscr{T}^{\mathrm{cons}}$ and it is compact by virtue of (iii) and (1.9.14), (vi)), the topology induced on U by $\mathscr{T}^{\mathrm{cons}}$ being identical to that of U^{cons}.

(v) Suppose f quasi-compact; then we already know (1.9.14), (iv)) that f^{cons} is a closed map. Let on the other hand y be a point of Y, and let us set

$$Z = f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y));$$

Z is quasi-compact (1.1.2), (iii)) and as the canonical morphism $\rho : Z \to X$ is injective, it follows from (i) and from the fact that the map ρ^{cons} is continuous, that the topology induced by that of X^{cons} on $f^{-1}(y)$ makes $f^{-1}(y)$ a quasi-compact space. This proves that f^{cons} is a proper map (Bourbaki, *Top. gén.*, chap. I$^{\text{er}}$, 3$^{\text{e}}$ éd., § 10, n$^{\text{o}}$ 2, th. 1). Conversely, suppose the continuous map f^{cons} proper, and let

V be an open quasi-compact subset of Y; if $h : V \to Y$ is the canonical injection, $h^{\text{cons}} : V^{\text{cons}} \to Y^{\text{cons}}$ is continuous and injective and V^{cons} is quasi-compact by (i), hence the topology induced on V by that of Y^{cons} makes V a quasi-compact space. The hypothesis that f^{cons} is proper implies that the topology induced on $f^{-1}(V)$ by that of X^{cons} makes $f^{-1}(V)$ a quasi-compact space (*loc. cit.*, prop. 6), hence $f^{-1}(V)$ is also a quasi-compact subspace of X, which shows that the morphism f is quasi-compact. □

(1.9.16). We will show later how one can, for every prescheme X, equip the space X^{cons} with a sheaf of rings which is in fact a prescheme whose local rings are all fields, identical to the residue fields at the points of X. Such preschemes arise naturally in the translation, into the language of schemes, of the constructions of Néron [**?**] in his theory of the reduction of abelian varieties.

IV-1 | 344

1.10. Application to open morphisms

Theorem (1.10.1). — *Let X be a prescheme, E an ind-constructible part of X (1.9.4), x a point of X. For x to be interior to E, it is necessary and sufficient that every generisation* (**0**, 2.1.2) *x' of x belong to E, in other words* (**I**, 2.4.2) *that* $\operatorname{Spec}(\mathscr{O}_{X,x}) \subset E$.

Proof. The condition is evidently necessary, every neighbourhood of x containing the generisations of x. To show that it is sufficient, one can evidently restrict to the case where $X = \operatorname{Spec}(A)$ is affine, x being a prime ideal $\mathfrak{p}$ of A. We know then (1.9.5), (ix)) that there exists an affine scheme $X' = \operatorname{Spec}(A')$ and a morphism $f : X' \to X$ such that $f(X') = X - E$. Let $Y = \operatorname{Spec}(\mathscr{O}_{X,x})$ and $Y' = X' \times_X Y$; the hypothesis means (by virtue of (**I**, 3.6.5)) that $Y' = \varnothing$, in other words $A' \otimes_A A_{\mathfrak{p}} = 0$. As $A_{\mathfrak{p}} = \varinjlim A_t$ (**0**, 1.4.5), where t runs through $A - \mathfrak{p}$, and the functor $\varinjlim$ commutes with the tensor product, by (1.9.1), there exists $t \in A - \mathfrak{p}$ such that $A'_t = 0$, and $\mathrm{D}(t)$ is then an open neighbourhood of x contained in E. □

Corollary (1.10.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, and let $y \in Y$. For y to be interior to $f(X)$, it is necessary and sufficient that every generisation y' of y belong to $f(X)$.*

Proof. We know indeed (1.9.5), (viii)) that $f(X)$ is ind-constructible in Y and it suffices to apply (1.10.1). □

We will say that a continuous map $f : X \to Y$ is *open at a point $x \in X$* if the image by f of *every* neighbourhood of x in X is a neighbourhood of $f(x)$ in Y.

Proposition (1.10.3). — *Let $f : X \to Y$ be a morphism, x a point of X, $y = f(x)$. Consider the following conditions:*

a) *f is open at the point x.*
b) *For every generisation y' of y, there exists a generisation x' of x such that $f(x') = y'$.*
b′) *The image by f of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *is* $\operatorname{Spec}(\mathscr{O}_{Y,y})$.
c) *For every irreducible closed part Y' of Y containing y, there exists a component of $X' = f^{-1}(Y')$ containing x and dominating Y'.*

Then we have the implications a) ⇒ b) ⇔ b′) ⇔ c). *If in addition f is locally of finite presentation, the four conditions are equivalent.*

IV-1 | 345

Proof. By definition of a generisation, the image by f of $\operatorname{Spec}(\mathscr{O}_{X,x})$ is always contained in $\operatorname{Spec}(\mathscr{O}_{Y,y})$, so b) and b′) are equivalent. It is immediate that c) implies b), since if y' is a generisation of y, $Y' = \overline{\{y'\}}$ is an irreducible closed part of Y containing y; if x' is the generic point of an irreducible component of $f^{-1}(Y')$ containing x and dominating Y', we have $f(x') = y'$ (**0**, 2.1.5) and x' is a generisation of x. Conversely, b) implies c): indeed, let y' be the generic point of Y' and let x' be a generisation of x such that $f(x') = y'$; let Z' be an irreducible component of $f^{-1}(Y')$ containing x' (and hence x); as $f(x')$ is already dense in Y', so *a fortiori* is $f(Z')$.

To show that a) implies b′), one can evidently restrict to the case where $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, x being a prime ideal $\mathfrak{q}$ of B, so that $\mathscr{O}_{X,x} = B_{\mathfrak{q}} = \varinjlim B_t$, where t runs through $B - \mathfrak{q}$. One has, by (1.9.2), $f(\mathrm{Spec}(\mathscr{O}_{X,x})) = \bigcap_t f(\mathrm{Spec}(B_t)) = \bigcap_t f(\mathrm{D}(t))$, and by hypothesis, y is interior to $f(\mathrm{D}(t))$, hence $f(\mathrm{D}(t))$ contains $\mathrm{Spec}(\mathscr{O}_{Y,y})$; whence $\mathrm{Spec}(\mathscr{O}_{Y,y}) \subset f(\mathrm{Spec}(\mathscr{O}_{X,x}))$, which establishes our assertion. Finally, when f is locally of finite presentation, b) implies a) by virtue of (1.10.2) applied to the restriction of f to an arbitrary open neighbourhood U of x. □ **IV-1 | 346**

Corollary (1.10.4). — *Let $f : X \to Y$ be a morphism. Consider the following conditions:*

a) *f is open.*
b) *For every $x \in X$ and every generisation y' of $y = f(x)$, there exists a generisation x' of x such that $f(x') = y'$.*
b′) *For every $x \in X$, the image by f of $\mathrm{Spec}(\mathscr{O}_{X,x})$ is $\mathrm{Spec}(\mathscr{O}_{Y,f(x)})$.*
c) *For every irreducible closed part Y' of Y, every component of $f^{-1}(Y')$ dominates Y'.*

Then we have the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) $\Leftrightarrow$ c). *If in addition f is locally of finite presentation, the four conditions are equivalent.*

Proof. To say that f is open means that f is open at every point $x \in X$, hence the implications a) $\Rightarrow$ b) $\Leftrightarrow$ b′) follow from the analogous implications in (1.10.3), as well as the implication b) $\Rightarrow$ a) when f is locally of finite presentation. The implication c) $\Rightarrow$ b) also follows from the analogous implication in (1.10.3); let us finally prove that *b)* implies *c)*. Indeed, let x' be the generic point of an irreducible component of $f^{-1}(Y')$, and let us show that $y' = f(x')$ is the generic point of Y'. Let y'' be a generisation of y'; by hypothesis there exists a generisation x'' of x' such that $f(x'') = y''$, and as $x'' \in f^{-1}(Y')$, we have necessarily $x'' = x'$, hence $y'' = y'$, which completes the proof. □

§2. BASE CHANGE AND FLATNESS

This section (unlike § 6) only exceptionally makes use of Noetherian techniques. The n[os] 1 and 2 are essentially nothing more than translations of elementary properties of flatness in commutative Algebra (cf. Bourbaki, *Alg. comm.*, chap. I[er]), and are included here for the convenience of references. The following numbers are primarily devoted to properties of "descent" by flat or faithfully flat morphisms: if $g : Y' \to Y$ is such a morphism, we want to be able to assert that a part of Y, or an $\mathscr{O}_Y$-Module, or a morphism $X \to Y$, has a certain property, when we know that its *inverse image* by g possesses this property. We limit ourselves here to properties that do not make use of the general technique of "descent", which will be developed in Chapter V. **IV-2 | 5**

2.1. Flat Modules on preschemes

(2.1.1). Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module; recall (**0**, 6.7.1) that $\mathscr{F}$ is said to be *f-flat* (or *Y-flat*) *at a point* $x \in X$ if $\mathscr{F}_x$ is an $\mathscr{O}_{f(x)}$-module flat, *f-flat* (or *Y-flat*) if it is f-flat at every point $x \in X$, and finally that the morphism f is said to be *flat at the point* $x \in X$ (resp. *flat*) if $\mathscr{O}_X$ is f-flat at the point x (resp. f-flat). When $f = 1_X$, we will simply say that an $\mathscr{O}_X$-Module $\mathscr{F}$ is *flat at the point* x (resp. *flat*) if it is X-flat at this point (resp. at every point $x \in X$), that is to say if $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module (resp. if this is the case for every $x \in X$). Recall that we have shown (**III**, 1.4.15.1) the following property:

Proposition (2.1.2). — *Let A, B be two rings, $\varphi : A \to B$ a ring homomorphism, $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, $f : X \to Y$ the corresponding morphism. For $\mathscr{F} = \widetilde{M}$ to be f-flat, it is necessary and sufficient that M be a flat A-module.*

Proposition (2.1.3). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

*label=*a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules is exact.*
a′) *Condition* a) *is satisfied for all canonical morphisms $g : \mathrm{Spec}(\mathscr{O}_y) \to Y$ where y ranges over Y.*
*lbbel=*b) *$\mathscr{F}$ is f-flat.*

Proof. The questions being local on X and Y, we can restrict to the case where $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(A)$, $\mathscr{F} = \widetilde{M}$, where M is an A-module. It is clear that *a)* implies *a′)*; condition *a′)* implies that for every $x \in X$, the functor $N \mapsto M_{\mathfrak{n}} \otimes_{B_{\mathfrak{n}}} N$ is exact in the category of $B_{\mathfrak{n}}$-modules, $\mathfrak{n}$ being the ideal $\mathfrak{j}_{f(x)}$ of B; this means that $M_\mathfrak{n}$ is a flat $B_\mathfrak{n}$-module, and it follows from (**0**, 6.3.3) and from (2.1.2) that $\mathscr{F}$ is f-flat. Finally, to see that *b)* implies *a)*, we can also restrict to the case where $Y' = \operatorname{Spec}(A')$ is affine and $\mathscr{G}' = \widetilde{N'}$, where N' is an A'-module; the conclusion then follows from (2.1.2) and from the definition of flatness, since $(M \otimes_A N')^\sim = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$. □

Proposition (2.1.4). — *Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, $y = g(y')$, $y' = f'(x')$, $x = g'(x')$. If $\mathscr{F}$ is f-flat at the point x, $\mathscr{F}'$ is f'-flat at the point x'; in particular if $\mathscr{F}$ is f-flat, f' is flat, f is flat.* IV-2 | 6

Proof. It suffices to prove the first assertion; applying three times (**I**, 3.6.5), as well as (**I**, 2.4.4), we can reduce to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$, $X = \operatorname{Spec}(\mathscr{O}_x)$, $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$, $\mathscr{F} = \widetilde{M}$, where M is an A-module. The hypothesis and (2.1.2) imply that $\mathscr{F}$ is f-flat, and the conclusion then also follows from (2.1.2) and from the definition of flatness. □

Proposition (2.1.5). — *Consider a commutative diagram of morphisms of preschemes*

$$\begin{tikzcd} X \arrow[d, "t"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ Y & Y' \arrow[l, "g"] \\ & Z \arrow[u, "h"] \end{tikzcd}$$

where $X' = X \times_Y Y'$ and $f' = f_{(Y')}$. Let x' be a point of X', and set $x = g'(x')$, $y' = f'(x')$, $y = f(x) = g(y')$, $z = h(y')$. Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module f-flat at the point x (resp. f-flat), and $\mathscr{G}'$ a quasi-coherent $\mathscr{O}_{Y'}$-Module h-flat at the point y' (resp. h-flat); then $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is a quasi-coherent $\mathscr{O}_{X'}$-Module $(h \circ f')$-flat at the point x' (resp. $(h \circ f')$-flat).

Proof. As in (2.1.4), we reduce to the case where $X = \operatorname{Spec}(\mathscr{O}_x)$, $Y = \operatorname{Spec}(\mathscr{O}_y)$, $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$ and $Z = \operatorname{Spec}(\mathscr{O}_z)$, and it suffices then to prove that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is $(h \circ f')$-flat. Taking into account (2.1.2), the proposition follows from Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, § 2, n$^{\text{o}}$ 7, prop. 8. □ IV-2 | 7

Corollary (2.1.6). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module. If $\mathscr{F}$ is f-flat at the point $x \in X$ and if g is flat at the point $f(x)$, $\mathscr{F}$ is $(g \circ f)$-flat at the point x. In particular, if f and g are flat morphisms, so is $g \circ f$.*

Proof. This is the special case of (2.1.5) with $Y' = Y$, $\mathscr{G}' = \mathscr{O}_Y$. □

Corollary (2.1.7). — *If $f : X \to X'$, $g : Y \to Y'$ are two flat S-morphisms, $f \times_S g : X \times_S Y \to X' \times_S Y'$ is a flat morphism.*

Proof. This follows from (2.1.4) and (2.1.6) (cf. (**I**, 3.5.1)). □

Proposition (2.1.8). — *Let $f : X \to Y$ be a morphism of preschemes,*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules, such that $\mathscr{F}''$ is Y-flat.

label=(i) For every morphism $g : Y' \to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the sequence $0 \to \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F}'' \otimes_{\mathscr{O}_Y} \mathscr{G}' \to 0$ of $\mathscr{O}_{X'}$-Modules (where $X' = X \times_Y Y'$) is exact.

lbbel=(ii) For $\mathscr{F}$ to be Y-flat, it is necessary and sufficient that $\mathscr{F}'$ be so.

Proof. We can evidently suppose X, Y, Y' affine; the conclusion then follows from (2.1.2) and from (**0**, 6.1.2). □

Corollary (2.1.9). — *Let $\mathscr{L}^\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules, i an index such that if $d^i : \mathscr{L}^i \to \mathscr{L}^{i+1}$ is the derivation operator, $\mathscr{B}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Im}(d^i)$ and $\mathscr{Z}^{i+1}(\mathscr{L}^\bullet) = \operatorname{Coker}(d^i)$ are Y-flat. Then, with the notations of (2.1.8), the canonical homomorphism*

$$\mathscr{H}^i(\mathscr{L}^\bullet) \otimes_{\mathscr{O}_Y} \mathscr{G}' \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{G}')$$

is bijective.

Proof. As the tensor product is right exact, we have

$$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}' = \operatorname{Coker}(d^i\otimes 1) = \mathscr{Z}^{i+1}(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')$$

and $\mathscr{Z}^i(\mathscr{L}^\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}' = \mathscr{Z}^i(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')$. Moreover, in the exact sequence

$$0\longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet)\longrightarrow \mathscr{L}^{i+1}\longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet)\longrightarrow 0$$

$\mathscr{Z}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, so it follows from (2.1.8), (i) that the exact sequence

$$0\longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}'\longrightarrow \mathscr{L}^{i+1}\otimes_{\mathscr{O}_Y}\mathscr{G}'\longrightarrow \mathscr{Z}^{i+1}(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')\longrightarrow 0$$

whence $\mathscr{B}^{i+1}(\mathscr{L}^\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}' = \operatorname{Im}(d^i\otimes 1) = \mathscr{B}^{i+1}(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')$. Then, as in the exact sequence

$$0\longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet)\longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet)\longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet)\longrightarrow 0$$

$\mathscr{B}^{i+1}(\mathscr{L}^\bullet)$ is Y-flat, it follows from (2.1.8), (i) and from what precedes that the sequence

$$0\longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}'\longrightarrow \mathscr{Z}^i(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')\longrightarrow \mathscr{B}^{i+1}(\mathscr{L}^\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}')\longrightarrow 0$$

is exact, which proves the corollary. □

Corollary (2.1.10). — *Let $f : X\to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module* **IV-2 | 8** *which is Y-flat, $\mathscr{L}_\bullet = (\mathscr{L}_i)$ a left resolution of $\mathscr{F}$ formed of quasi-coherent $\mathscr{O}_X$-Modules which are Y-flat. Then, for every morphism $g : Y'\to Y$ and every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{G}'$, the complex $\mathscr{L}_\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}' = (\mathscr{L}_i\otimes_{\mathscr{O}_Y}\mathscr{G}')_{i\geqslant 0}$ is a left resolution of $\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{G}'$. Moreover, if $\mathscr{Z}_i(\mathscr{L}_\bullet) = \operatorname{Ker}(\mathscr{L}_i\to\mathscr{L}_{i-1})$, the $\mathscr{Z}_i(\mathscr{L}_\bullet)$ are Y-flat, and we have $\mathscr{Z}_i(\mathscr{L}_\bullet)\otimes_{\mathscr{O}_Y}\mathscr{G}' = \mathscr{Z}_i(\mathscr{L}_\bullet\otimes_{\mathscr{O}_Y}\mathscr{G}') = \operatorname{Ker}(\mathscr{L}_i\otimes_{\mathscr{O}_Y}\mathscr{G}'\to\mathscr{L}_{i-1}\otimes_{\mathscr{O}_Y}\mathscr{G}')$. Setting $\mathscr{R}_i = \operatorname{Im}(\mathscr{L}_{i+1}\to\mathscr{L}_i) = \mathscr{Z}_i(\mathscr{L}_\bullet)$; we then have the exact sequences*

$$0\leftarrow\mathscr{F}\leftarrow\mathscr{L}_0\leftarrow\mathscr{R}_0\leftarrow 0$$

$$0\leftarrow\mathscr{R}_i\leftarrow\mathscr{L}_{i+1}\leftarrow\mathscr{R}_{i+1}\leftarrow 0\qquad (i\geqslant 0)$$

and since $\mathscr{F}$ and $\mathscr{L}_i$ are Y-flat, we deduce from (2.1.8), (ii) by induction that all the $\mathscr{R}_i$ are also Y-flat; using (2.1.8), (i), we thus have the exact sequences

$$0\leftarrow\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow\mathscr{L}_0\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow\mathscr{R}_0\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow 0$$

$$0\leftarrow\mathscr{R}_i\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow\mathscr{L}_{i+1}\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow\mathscr{R}_{i+1}\otimes_{\mathscr{O}_Y}\mathscr{G}'\leftarrow 0\qquad (i\geqslant 0)$$

which prove the corollary.

Proposition (2.1.11). — *Let $f : X\to Y$ be a flat morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_Y$-Module of finite presentation. If $\mathscr{J}$ is the Ideal of $\mathscr{O}_Y$ annihilator of $\mathscr{F}$, $f^*(\mathscr{J})$ is the Ideal of $\mathscr{O}_X$ annihilator of $f^*(\mathscr{F})$.*

Proof. We have indeed by definition an exact sequence (**0**, 5.3.7)

$$0\longrightarrow\mathscr{J}\longrightarrow\mathscr{O}_Y\longrightarrow\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F},\mathscr{F})$$

whence, since f is flat, an exact sequence

$$0\longrightarrow f^*(\mathscr{J})\longrightarrow\mathscr{O}_X\longrightarrow f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F},\mathscr{F}))$$

and since $\mathscr{F}$ is an $\mathscr{O}_Y$-Module of finite presentation, $f^*(\mathscr{H}om_{\mathscr{O}_Y}(\mathscr{F},\mathscr{F}))$ canonically identifies with $\mathscr{H}om_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{F}))$ (**0**, 6.7.6), whence the conclusion. □

Proposition (2.1.12). — *Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation, x a point of X. The following conditions are equivalent:*

label=a) *$\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module.*
lbbel=b) *There exists an open neighbourhood U of x such that $\mathscr{F}|U$ is a locally free $(\mathscr{O}_X|U)$-Module.*

Proof. Indeed, $\mathscr{F}_x$ is a module of finite presentation over $\mathscr{O}_x$ a local ring; it amounts to the same thing to say that $\mathscr{F}_x$ is a flat $\mathscr{O}_x$-module or a free $\mathscr{O}_x$-module (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 5); whence the conclusion, taking into account (**0**, 5.2.7). We note that the proposition is valid for any *ringed space with local rings*. □

Proposition (2.1.13). — *Let $f : X\to Y$ be a morphism of preschemes. If f is flat at a point $x\in X$ and if the ring $\mathscr{O}_x$ is reduced (resp. integral and integrally closed), then the ring $\mathscr{O}_{f(x)}$ is reduced (resp. integral and* **IV-2 | 9** *integrally closed). If f is faithfully flat and if X is reduced (resp. normal), then Y is reduced (resp. normal).*

Proof. Set $\mathscr{O}_{f(x)} = A$, $\mathscr{O}_x = B$. If B is a flat A-module, it is also a faithfully flat A-module (**0**, 6.6.2), so A identifies with a subring of B; if B is reduced, so is A. Suppose now B integral and integrally closed, then $A \subset B$ is integral; let L be the field of fractions of A and let $K \subset L$ be its field of fractions; denote by K the field of fractions. The hypothesis implies that $B \cap K = A$ (Bourbaki, *Alg. comm.*, chap. I, § 3, nº 5, prop. 10). If then $t \in K$ is integral over A, it is also integral over B, so belongs to B by hypothesis, and thus $t \in A$, which proves that A is integrally closed. □

Proposition (2.1.14). — *Let $f : X \to Y$ be a faithfully flat morphism. If X is locally integral and if the underlying space is locally Noetherian, then Y is locally integral.*

Proof. Indeed, every $y \in Y$ is of the form $f(x)$ for some $x \in X$ and by hypothesis $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_x$ (**0**, 6.6.1); as $\mathscr{O}_x$ is integral, so is $\mathscr{O}_y$ and this proves the proposition (**I**, 5.1.4). □

2.2. Faithfully flat Modules on preschemes

Proposition (2.2.1). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

label=a) *For every base change $g : Y' \to Y$, if we set $X' = X \times_Y Y'$, the functor $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$, from the category of quasi-coherent $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules, is exact and faithful.*

a′) *Condition* a) *is satisfied for all canonical morphisms $g : \mathrm{Spec}(\mathscr{O}_y) \to Y$, where y ranges over Y.*

a″) *Condition* a) *is satisfied for all canonical immersions $Y' \to Y$, where Y' ranges over the set of affine open sub-preschemes of Y.*

lbbel=b) *$\mathscr{F}$ is Y-flat and, for every $y \in Y$, if we denote by X_y the fibre $f^{-1}(y)$, the $\mathscr{O}_{X_y}$-Module $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$ is $\neq 0$.*

Proof. It is clear that *a)* implies *a′)* and *a″)*; condition *a′)* implies first that $\mathscr{F}$ is Y-flat (2.1.3); on the other hand it implies that for every $y \in Y$, the functor $N \mapsto \mathscr{F} \otimes_{\mathscr{O}_y} \widetilde{N}$ is faithful in the category of $\mathscr{O}_y$-modules; taking in particular $N = k(y)$, we obtain the second assertion of *b)*. To show that *b)* implies *a)*, we can restrict to the case where Y is affine. Likewise, to prove that *a″)* implies *a)*, the question being local on Y. In other words, we are reduced to proving the following more precise proposition: □

Proposition (2.2.2). — *Let $Y = \mathrm{Spec}(A)$ be an affine scheme, $f : X \to Y$ a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. Condition* a) *of* (2.2.1) *is equivalent to each of the following:* **IV-2** | 10

b′) *$\mathscr{F}$ is Y-flat and, for every closed point y of Y, we have $\mathscr{F}_y \neq 0$.*

c) *The functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ from the category of quasi-coherent $\mathscr{O}_Y$-Modules to that of quasi-coherent $\mathscr{O}_X$-Modules is exact and faithful.*

Proof. If *b′)* is satisfied, there is at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x \neq 0$; let $U = \mathrm{Spec}(B)$, where M is a B-module. Then *b′)* implies (since M is an A-module flat (2.1.2)) that $M \otimes_A A_y$ is an A_y-module faithfully flat (**0**, 6.4.5). The relation $(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}) \otimes_{\mathscr{O}_y} \mathscr{G}_y = 0$ for a closed point y of Y implies $(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}) \otimes_{\mathscr{O}_y} \mathscr{G}_y = 0$, so $\mathscr{G}_y = 0$; but if we have $\mathscr{G}_y = 0$ for every closed point y of Y, we conclude $\mathscr{G} = 0$, for if $\mathscr{G} = \widetilde{N}$, the annihilator of N is not contained in any maximal ideal of A, so A is all of it. The relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$ implies therefore $\mathscr{G} = 0$, in other words the functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is faithful; we know moreover that this functor is exact (2.1.3), which shows that *b′)* implies *c)*.

Finally, to see that *c)* implies *a)*, we can restrict to the case where Y' is also affine; as $g : Y' \to X$ is a flat morphism, so is it the same, by the more (**II**, 1.5.5); de plus, the functor $\mathscr{H}' \mapsto g'_*(\mathscr{H}')$ is then exact in the category of quasi-coherent $\mathscr{O}_{X'}$-Modules (**I**, 1.6.4), and if $g'_*(\mathscr{H}') = \mathscr{H}$, on a $\mathscr{H}' = \widetilde{\mathscr{F}}$, so the preceding functor is also faithful; to see that $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ is exact and faithful, it suffices therefore to see that the functor $\mathscr{G}' \mapsto g'_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}') = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G}')$; the fact that g is affine implies that we have a canonical isomorphism

(2.2.2.1) $$\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(g^*(\mathscr{F}) \otimes_{\mathscr{O}_{X'}} f'^*(\mathscr{G}')).$$

Indeed, we know (**II**, 1.5.2) that we have a canonical isomorphism

$$f^*(g_*(\mathscr{G}')) \xrightarrow{\sim} g'_*(f'^*(\mathscr{G}')),$$

and on the other hand a canonical homomorphism $\mathscr{F} \to g'_*(g'^*(\mathscr{F}))$ (**0**, 4.4.3.2); composing the homomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} f^*(g_*(\mathscr{G}')) \to g'_*(g'^*(\mathscr{F})) \otimes_{\mathscr{O}_X} g'_*(f'^*(\mathscr{G}'))$ with the canonical homomorphism (**0**, 4.2.2.1), we deduce the homomorphism (2.2.2.1); the verification that it is an isomorphism is immediate by reducing to the case where X is affine. This being so, the functor $\mathscr{G}' \mapsto g_*(\mathscr{G}')$ is exact and faithful and by hypothesis it is the same of $\mathscr{M} \mapsto \mathscr{M}' \otimes_{\mathscr{O}_Y} \mathscr{F}$ (from the category of $\mathscr{O}_Y$-Modules to that of quasi-coherent $\mathscr{O}_X$-Modules); the composition is exact and faithful, which completes the proof of (2.2.1) and (2.2.2). □

Corollary (2.2.3). — *Let $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ be two affine schemes, $f : X \to Y$ a morphism, $\mathscr{F} = \widetilde{M}$ a quasi-coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to satisfy the equivalent conditions of* (2.2.1) (*or* (2.2.2)), *it is necessary and sufficient that the A-module M be faithfully flat.*

Proof. Indeed, condition *c)* of (2.2.2) means that the functor $N \mapsto M \otimes_A N$ from the category of A-modules to that of B-modules is exact and faithful, and the conclusion follows from (**0**, 6.4.1). □

Definition (2.2.4). — When the equivalent conditions of (2.2.1) are satisfied, we say that the quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is *faithfully flat relative to f* (or to Y).

We note that this notion is local over Y, but *not over X*; in particular we can have $\mathscr{F}_x = 0$ for certain $x \in X$, in other words $\operatorname{Supp}(\mathscr{F})$ is not necessarily equal to X. However, it follows immediately from the criterion (2.2.1), *b)* that for every $y \in Y$, there exists at least one $x \in f^{-1}(y)$ such that $(\mathscr{F}_y)_x = \mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \neq 0$, and *a fortiori* $\mathscr{F}_x \neq 0$; in other terms: **IV-2 | 11**

Corollary (2.2.5). — *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module, faithfully flat relative to f, we have $f(\operatorname{Supp}(\mathscr{F})) = Y$, and a fortiori f is a surjective morphism.*

This result admits a partial converse:

Corollary (2.2.6). — *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module of finite type. For $\mathscr{F}$ to be faithfully flat relative to f, it is necessary and sufficient that $\mathscr{F}$ be f-flat and that $f(\operatorname{Supp}(\mathscr{F})) = Y$.*

Proof. Indeed (**I**, 9.1.13) and (2.3.6.1) we have $\operatorname{Supp}(\mathscr{F}_y) = f^{-1}(y) \cap \operatorname{Supp}(\mathscr{F})$, so in this case the criterion (2.2.1), *b)* is nothing else than the condition of the corollary. In particular, the $\mathscr{O}_X$-Module $\mathscr{O}_X$ is faithfully flat relative to f if and only if f is flat and f is surjective, that is to say if and only if the *morphism f is faithfully flat* (**0**, 6.7.8). □

Let us spell out the properties intervening in definition (2.2.4):

Proposition (2.2.7). — *Let $f : X \to Y$ be a morphism of preschemes, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to f. Then, for a sequence $\mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$ of quasi-coherent $\mathscr{O}_Y$-Modules to be exact, it is necessary and sufficient that the corresponding sequence $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ be so. In particular, for a homomorphism $u : \mathscr{G} \to \mathscr{G}'$ of quasi-coherent $\mathscr{O}_Y$-Modules to be injective* (*resp. surjective, bijective, zero*), *it is necessary and sufficient that $1_{\mathscr{F}} \otimes f^*(u) : \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} \to \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be so. For a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$ to be zero, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G} = 0$. For an $\mathscr{O}_Y$-Module quasi-coherent $\mathscr{G}$, the map $\mathscr{G}' \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ from the set of quasi-coherent sub-$\mathscr{O}_Y$-Modules of $\mathscr{G}$ to the set of sub-$\mathscr{O}_X$-Modules quasi-coherent of $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is injective.*

Proof. To prove the last assertion, that is to say that for two quasi-coherent sub-$\mathscr{O}_Y$-Modules $\mathscr{G}'$, $\mathscr{G}''$ of $\mathscr{G}$, the relation $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}''$ implies $\mathscr{G}' = \mathscr{G}''$, we can reduce to the case where $\mathscr{G}' \subset \mathscr{G}''$ (by replacing $\mathscr{G}''$ by $\mathscr{G}' + \mathscr{G}''$), and it then suffices to apply the second assertion of the statement to the injection $u : \mathscr{G}' \to \mathscr{G}''$. □

Corollary (2.2.8). — *Let $f : X \to Y$ be a faithfully flat morphism, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-Module. The canonical map*

(2.2.8.1) $$\Gamma(Y, \mathscr{G}) \longrightarrow \Gamma(X, f^*(\mathscr{G}))$$

is injective.

Proof. Indeed, $\Gamma(Y, \mathscr{G})$ canonically identifies with $\operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{O}_Y, \mathscr{G})$ (**0**, 5.1.1) and similarly $\Gamma(X, f^*(\mathscr{G}))$ with $\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, f^*(\mathscr{G}))$. By virtue of (2.2.1) and (2.2.4), the hypothesis implies that the functor $\mathscr{G} \mapsto f^*(\mathscr{G})$ is exact and faithful in the category of $\mathscr{O}_Y$-Modules quasi-coherent, and thus a homomorphism $u : \mathscr{O}_Y \to \mathscr{G}$ is zero if and only if $f^*(u) : f^*(\mathscr{O}_Y) = \mathscr{O}_X \to f^*(\mathscr{G})$ is zero. □

Remark (2.2.9). — The assertions of (2.2.7) and (2.2.8) are still true when the $\mathscr{O}_Y$-Modules $\mathscr{G}', \mathscr{G}, \mathscr{G}''$ appearing there are arbitrary (not necessarily quasi-coherent). Indeed, for every $y \in Y$, there exists $x \in f^{-1}(y)$ such that $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module faithfully flat, and thus the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ is **IV-2** | 12
faithful; since moreover for *every* $x \in f^{-1}(y)$ the functor $\mathscr{G}_y \mapsto \mathscr{G}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ is exact, we immediately deduce the conclusion.

Proposition (2.2.10). — *Let $f : X \to Y$, $g : Y' \to Y$, $h : Y' \to Z$ be three morphisms of preschemes, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y, $\mathscr{G}'$ a quasi-coherent $\mathscr{O}_{Y'}$-Module. For $\mathscr{G}'$ to be flat* (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ be a flat* (*resp. faithfully flat*) *$\mathscr{O}_{X'}$-Module relative to Z.*

Proof. We know already that if $\mathscr{G}'$ is Z-flat, so is $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}'$ (2.1.5). Consider a base change $Z'' \to Z$ and set $X'' = X' \times_Z Z''$; if $\mathscr{G}'$ is faithfully flat relative to Z, the functor

(2.2.10.1) $$\mathscr{H}'' \longmapsto \mathscr{H}'' \otimes_Z \mathscr{G}' \longrightarrow (\mathscr{H}'' \otimes_Z \mathscr{G}') \otimes_{\mathscr{O}_Y} \mathscr{F} = \mathscr{H}'' \otimes_Z (\mathscr{G}' \otimes_{\mathscr{O}_Y} \mathscr{F})$$

from the category of $\mathscr{O}_{Z''}$-Modules to that of $\mathscr{O}_{X''}$-Modules is composed of two exact and faithful functors, so this functor is also exact and faithful. Conversely, if ce fonсteur $\mathscr{H}'' \mapsto \mathscr{H}'' \otimes_Z \mathscr{G}'$ is exact (resp. exact and faithful), since the functor $\mathscr{M} \mapsto \mathscr{M}' \otimes_{\mathscr{O}_Y} \mathscr{F}$ (from the category of $\mathscr{O}_{Y'}$-Modules to that of quasi-coherent $\mathscr{O}_{X'}$-Modules) is exact and faithful by hypothesis. □

Corollary (2.2.11). —

label=(i) Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to Y, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$ is faithfully flat relative to Y'.

lbbel=(ii) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module faithfully flat relative to Y. For g to be a faithfully flat morphism, it is necessary and sufficient that $\mathscr{F}$ be faithfully flat relative to Z.

lcbel=(iii) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Y$-Module. Suppose the morphism f is faithfully flat. For $\mathscr{G}$ to be flat (*resp. faithfully flat*) *relative to Z, it is necessary and sufficient that $f^*(\mathscr{G})$ be flat* (*resp. faithfully flat*) *relative to Z.*

ldbel=(iv) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, x a point of X. Suppose f is flat at the point x. For $g \circ f$ to be flat at the point x, it is necessary and sufficient that g be flat at the point $f(x)$.

Proof. To prove (i), we apply (2.2.10) replacing Z par Y' and $\mathscr{G}'$ by $\mathscr{O}_{Y'}$. To prove (ii), we apply (2.2.10) taking $Y' \to Y$ the identity and replacing $\mathscr{G}'$ by $\mathscr{O}_Y$. To prove (iii), we apply (2.2.10) taking $Y' \to Y$ the identity, and replacing $\mathscr{F}$ par $\mathscr{O}_X$ and $\mathscr{G}'$ by $\mathscr{G}$. Finally (iv) is deduced from (ii) by replacing X by $\operatorname{Spec}(\mathscr{O}_x)$, $\mathscr{F}$ by $\mathscr{O}_X$, Y by $\operatorname{Spec}(\mathscr{O}_{f(x)})$ and Z by $\operatorname{Spec}(\mathscr{O}_{g(f(x))})$, and taking into account (**0**, 6.6.2). □

Corollary (2.2.12). — *Let Y be an affine scheme, $f : X \to Y$ a quasi-compact morphism, $\mathscr{F}$ a quasi-coherent* **IV-2** | 13
$\mathscr{O}_X$-Module. If $\mathscr{F}$ is faithfully flat relative to f, there exists an affine scheme X' and a faithfully flat morphism $g : X' \to X$ such that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$.

Proof. Indeed, X is quasi-compact, so a finite union of open affine sets X_i; it suffices to take for X' the affine scheme *direct sum* of the preschemes induced on the open sets X_i and for $g : X' \to X$ the canonical morphism; it is clear that g is faithfully flat, and the hypothesis implies that $g^*(\mathscr{F})$ is faithfully flat relative to $f \circ g$, by virtue of (2.2.11), (iii). □

Corollary (2.2.13). —

label=(i) Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If f is a faithfully flat morphism, so is f'.

lbbel=(ii) If $f : X \to X'$, $g : Y \to Y'$ are two flat S-morphisms,

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is faithfully flat.

lcbel=(iii) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms such that f is faithfully flat. For g to be a flat (resp. faithfully flat) morphism, it is necessary and sufficient that $g \circ f$ be a flat (resp. faithfully flat) morphism.*

Proposition (2.2.14). — *Let $f : X \to Y$ be a faithfully flat morphism and quasi-compact. If X is locally Noetherian, so is Y.*

Proof. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine; as f is quasi-compact, it follows from (2.2.12) that we can also restrict to the case where $X = \operatorname{Spec}(B)$ is affine. Then B is a Noetherian ring by hypothesis, and a faithfully flat A-module (2.2.3); so A is Noetherian (**0**, 6.5.2). □

Proposition (2.2.15). — *Let $f : X \to Y$ be a faithfully flat morphism. If $\mathfrak{S}(X)$ and $\mathfrak{S}(Y)$ denote respectively the sets of sub-preschemes of X and of Y, the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{S}(Y)$ to $\mathfrak{S}(X)$ is injective.*

Proof. Since f is surjective, for a sub-prescheme Z of Y, $f(f^{-1}(Z)) = Z$. On the other hand, if U is an open subset of Y containing Z and in which Z is closed, $f^{-1}(Z)$ is closed in $f^{-1}(U)$, and the restriction $f^{-1}(U) \to U$ of f is a faithfully flat morphism. We can therefore restrict to considering only the *closed* sub-preschemes of Y. Now, if Z is a closed sub-prescheme of Y corresponding to a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_Y$, we know that $f^{-1}(Z)$ corresponds to the quasi-coherent Ideal $f^*(\mathscr{J})\mathscr{O}_X$ of $\mathscr{O}_X$ (**I**, 4.4.5), and since f is flat, $f^*(\mathscr{J})\mathscr{O}_X$ identifies with $f^*(\mathscr{J})$. But the map $\mathscr{J} \mapsto f^*(\mathscr{J})$ from the set of quasi-coherent Ideals of $\mathscr{O}_X$ to the set of quasi-coherent Ideals of $\mathscr{O}_X$ is injective (2.2.7), whence the conclusion. □

Corollary (2.2.16). — *Let X, Y be two S-preschemes; if $S' \to S$ is a faithfully flat morphism, the map $f \mapsto f_{(S')}$ from $\operatorname{Hom}_S(X, Y)$ to $\operatorname{Hom}_{S'}(X_{(S')}, Y_{(S')})$ is injective.*

Proof. We have $X_{(S')} \times_{S'} Y_{(S')} = (X \times_S Y)_{(S')}$ (**I**, 3.3.10), so the morphism projection $X_{(S')} \times_{S'} Y_{(S')} \to X \times_S Y$ is faithfully flat (2.2.13). The elements of $\operatorname{Hom}_S(X, Y)$ correspond bijectively to sub-preschemes of $X \times_S Y$ that are *graphs* of $f \in \operatorname{Hom}_S(X, Y)$, and if Z_f is the graph of f, we have $Z_{f_{(S')}} = g^{-1}(Z_f)$ (**I**, 5.3.12). It suffices therefore to apply (2.2.15). □

Proposition (2.2.17). — *Let A be a ring, B an A-algebra such that B is a faithfully flat A-module and of finite presentation. Then the structural homomorphism $\varphi : A \to B$ is an isomorphism of the A-module A onto a direct summand of the A-module B. If A is a local ring, B is a free A-module and there exists a basis of this module containing the identity element of B.*

Proof. By virtue of Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, prop. 12, it suffices to prove the proposition when A is a local ring; we know then (*loc. cit.*, n° 2, cor. 2 of prop. 5) that B is an A-module free of finite type, and the conclusion follows from *loc. cit.*, prop. 5. □ **IV-2 | 14**

2.3. Topological properties of flat morphisms

Lemma (2.3.1). — *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, $g : Y' \to Y$ a flat morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the canonical homomorphism*

$$g^*(f_*(\mathscr{F})) \longrightarrow f'_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}) \tag{2.3.1.1}$$

is bijective.

Proof. This is a special case of (**III**, 1.4.15) (improved in (1.7.21)). □

Proposition (2.3.2). — *Let S be a prescheme, $f : X \to Y$ an S-morphism quasi-compact and quasi-separated, Z the sub-prescheme which is the closed image of X by f ((**I**, 9.5.3) and (1.7.8)), and let $j : Z \to Y$ be the injection canonical, so that $f = j \circ g$, where $g : X \to Z$ is a morphism* (loc. cit.). *Let $h : S' \to S$ be a morphism, and set $f' = f_{(S')} : X_{(S')} \to Y_{(S')}$; then $j' = j_{(S')} : Z_{(S')} \to Y_{(S')}$ identifies with the canonical injection of the sub-prescheme of $Y_{(S')}$, closed image of $X_{(S')}$ by f'.*

Proof. Since the morphism $Y_{(S')} \to Y$ is flat (2.1.4), we can restrict to the case where $S = Y$ (**I**, 3.3.11); if $f = (\psi, \theta)$, the canonical injection of Z is the sub-prescheme of Y defined by the Ideal $\mathscr{J}$ of $\mathscr{O}_S$ kernel of $\theta : \mathscr{O}_S \to f_*(\mathscr{O}_X)$. As h is a flat morphism, the Ideal $\mathscr{J}\mathscr{O}_{S'}$ identifies with the kernel of $h^*(\theta) : h^*(\mathscr{O}_S) \to h^*(f_*(\mathscr{O}_X))$ (**0**, 6.3.1)); as f is quasi-compact and quasi-separated, the canonical homomorphism (2.3.1.1) $h^*(f_*(\mathcal{O}_X)) \to f'_*(\mathcal{O}_{X'})$ is an isomorphism, the conclusion follows from (2.3.1) and from (**I**, 9.5.2), since f' is quasi-compact (1.1.2 and 1.2.2). □

(2.3.3). We will say that a morphism $f : X \to Y$ is *quasi-flat* if there exists a quasi-coherent $\mathcal{O}_X$-Module $\mathscr{F}$ *of finite type* which is f-flat and whose support is equal to X. We will say that f is *quasi-faithfully flat* if it is quasi-flat and surjective. Every morphism flat (resp. faithfully flat) is quasi-flat (resp. quasi-faithfully flat), for then $\mathscr{F} = \mathcal{O}_X$ satisfies the preceding conditions.

It follows immediately from (2.1.4) and from (**I**, 9.1.13), that if f is quasi-flat, then, for every morphism $Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is quasi-flat. Similarly (taking into account (**I**, 3.5.2)), if f is quasi-faithfully flat, so is $f_{(Y')}$.

Proposition (2.3.4). — *Let $f : X \to Y$ be a quasi-flat morphism* (2.3.3)*. Then f possesses the following properties* (*which are equivalent by virtue of (1.10.4)*):

label=(i) *For every $x \in X$ and every generization y' of $y = f(x)$, there exists a generization x' of x such that $f(x') = y'$.*

lbbel=(ii) *For every $x \in X$, the image by f of* $\operatorname{Spec}(\mathcal{O}_{X,x})$ *is* $\operatorname{Spec}(\mathcal{O}_{Y,y})$. **IV-2** | 15

(iii) *For every closed irreducible part Y' of Y, every irreducible component of $f^{-1}(Y')$ dominates Y'.*

Proof. It suffices, for example, to prove (ii). By hypothesis, there is an $\mathcal{O}_X$-Module quasi-coherent of finite type $\mathscr{F}$, which is f-flat and such that $\operatorname{Supp}(\mathscr{F}) = X$. For every $x \in X$, $\mathscr{F}_x$ is then an $\mathcal{O}_x$-module of finite type, nonzero, and moreover $\mathscr{F}_x$ is an $\mathcal{O}_y$-module *flat*, for the homomorphism $\rho : \mathcal{O}_y \to \mathcal{O}_x$. As the latter is local and $\mathscr{F}_x \neq 0$, the Nakayama lemma shows that $\mathscr{F}_x \otimes_{\mathcal{O}_y} k(y) \neq 0$, so $\mathscr{F}_x$ is a faithfully flat $\mathcal{O}_y$-module (**0**, 6.4.1). It follows that for every prime ideal $\mathfrak{q}$ of $\mathcal{O}_y$, there exists a prime ideal $\mathfrak{p}$ of $\mathcal{O}_x$ such that $\mathfrak{q} = \rho^{-1}(\mathfrak{p})$ (**0**, 6.5.1), which proves (ii). □

Corollary (2.3.5). — *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of* (2.3.4) (*in particular a quasi-flat or open morphism (1.10.4)*).

label=(i) *Let Z, Z' be two closed irreducible subsets of Y such that $Z \subset Z'$, and let T be an irreducible component of $f^{-1}(Z)$; then there exists an irreducible component T' of $f^{-1}(Z')$ containing T* (*and dominating Z'*).

lbbel=(ii) *For every irreducible component T of X, $\overline{f(\mathrm{T})}$ is an irreducible component of Y.*

lcbel=(iii) *Suppose Y irreducible, and, if y is its generic point, suppose that $f^{-1}(y)$ is irreducible. Then X is irreducible.*

Proof. (i) It suffices to apply (2.3.4), (i) by taking for x the generic point of T ($y = f(x)$ is then the generic point of Z) and for y' the generic point of Z'.

(ii) It is clear that $\overline{f(\mathrm{T})} = Z$ is irreducible, and in virtue of (i), Z cannot be strictly contained in a closed irreducible part of Y.

(iii) By virtue of (ii), every irreducible component of X dominates Y, and if y is the generic point of Y, its inverse image $f^{-1}(y)$ meets each of these sets $f^{-1}(y_i)$ (2.3.4); the conclusion then follows from (**0**, 2.1.8). □

Proposition (2.3.6). — *Let Y be a prescheme whose set of irreducible components is locally finite* (*which is the case if the underlying space is locally Noetherian (cf.* (**I**, 6.1.9))).

label=(i) *Every closed subset W of Y stable by generization* (**0**, 2.1.2) *is open. In particular, every connected component of Y is open.*

lbbel=(ii) *Let $f : X \to Y$ be a closed morphism satisfying in addition the equivalent conditions (i), (ii), (iii) of* (2.3.4) (*which will be the case if f is quasi-flat or open*)*. Then the image by f of every connected component C of X is a connected component of Y.*

lcbel=(iii) *Let $f : X \to Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of* (2.3.4)*, and such that moreover for every $y \in Y$, the set $f^{-1}(y)$ is finite* (*which will be the case if f is quasi-finite or radicial*)*. Then the set of irreducible components of X is locally finite.*

Proof. (i) If $y \in W$, the generic points η_i ($1 \leqslant i \leqslant m$) of the irreducible components Y_i of Y containing y belong to W by hypothesis, so these components themselves are contained in W; since there is by hypothesis a neighbourhood U of y such that U is a union of the $U \cap Y_i$ (**0**, 2.1.6), we have $U \subset W$, **IV-2** | 16 so W is open. Since for every $y \in Y$, $\overline{\{y\}}$ is connected, a connected component of Y is stable by generization, so the second assertion follows immediately from the first.

(ii) Since C is closed in X, $f(C)$ is closed in Y by hypothesis. Moreover, since C is stable by generization, the hypothesis on f implies that $f(C)$ is stable by generization; so, by virtue of (i), $f(C)$ is open and closed, and since it is connected, it is a connected component of Y.

(iii) Let $x \in X$; by hypothesis, there is an open neighbourhood U such that $y = f(x)$ meets only a finite number of irreducible components Y_i of Y ($1 \leqslant i \leqslant n$); let y_i be the generic point of Y_i. For every irreducible component Z of X meeting $f^{-1}(U)$, the generic point z of Z is necessarily in one of the sets $f^{-1}(y_i)$ (2.3.4). Since each of these sets is finite by hypothesis, this proves our assertion. □

Proposition (2.3.7). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a quasi-flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$.*

label=(i) If f is quasi-compact and dominant, so is f'.

lbbel=(ii) If every irreducible component of X dominates an irreducible component of Y, then every irreducible component of X' dominates an irreducible component of Y'.

Proof. Denote by $g' : X' \to X$ the canonical projection, which is a quasi-flat morphism (2.3.3).

(i) We know already (1.1.2) that f' is quasi-compact; moreover, if y' is a maximal point of Y', $y = g(y')$ is a maximal point of Y, as it follows from (2.3.4), (ii). By hypothesis (1.1.5), there exists $x \in X$ such that $f(x) = y$; so (**I**, 3.4.7), there exists $x' \in X'$ such that $f'(x') = y'$.

(ii) Let x' be a maximal point of X', and let $x = g'(x')$; it follows from (2.3.4), (ii) that x is a maximal point of X, and by hypothesis $y = f(x)$ is a maximal point of Y. Set $y' = f'(x')$, so that $g(y') = y$; it remains to show that y' is a maximal point of Y'. By virtue of (**I**, 5.1.7) and (2.3.3), we can restrict to the case where $\mathscr{O}_x$ and $\mathscr{O}_y$ are *fields*, so $\mathscr{O}_x$ and $\mathscr{O}_y$ are reduced, and by virtue of (**I**, 3.6.5) applied twice and of (**I**, 2.4.4), we can restrict to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$ and $Y' = \operatorname{Spec}(\mathscr{O}_{y'})$. Then f is a morphism *flat* since $\mathscr{O}_y$ is a field (2.1.2), and it is of the same of f' (2.1.4); so it follows from (2.3.4), (ii) that $y' = f'(x')$ is a maximal point of Y'. □

Corollary (2.3.8). — (Zariski). *Let A, B be two Noetherian local rings, $\mathfrak{m}$, $\mathfrak{n}$ their maximal ideals respectively, $\varphi : A \to B$ a local homomorphism. Suppose the following hypotheses satisfied:*

1° *B is an A-algebra essentially of finite type (1.3.6).*

2° *The $\mathfrak{m}$-adic completion $\widehat{A}$ of A for the $\mathfrak{m}$-adic topology is integral.*

3° *φ is injective.*

Then the $\mathfrak{m}$-adic topology of A is induced by the $\mathfrak{n}$-adic topology of B.

Proof. Set $B' = B \otimes_A \widehat{A}$; by hypothesis 1°, B' is of the form $S^{-1}(C \otimes_A \widehat{A})$, where C is an A-algebra **IV-2** | 17
of finite type and S is a multiplicative subset of C. As A identifies with a subring of $\widehat{A}$ (**0**, 7.3.5), A is integral by hypothesis 2°. Hypothesis 3° implies that there exists a prime ideal $\mathfrak{q}$ of B inducing the prime ideal $\mathfrak{q}$ of A of the homomorphism local $A \to B/\mathfrak{q}$ injective. We can therefore restrict to proving the conclusion of (2.3.8) under the added hypothesis that B is a local ring which is *integral*. We apply (2.3.7), (ii) to $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(\widehat{A})$ and $X' = \operatorname{Spec}(B')$; since the morphism $Y' \to Y$ is flat and X dominates the integral scheme Y (integral by hypothesis), every irreducible component of X' dominates Y'. If y, x, y' are the closed points of Y, X and Y' respectively, and designating by $\mathfrak{r}$ the ideal maximal of $\mathscr{O}_{x'}$, the intersection of ideals $\mathfrak{r}^k \cap \widehat{A}$ is null (**0**, 7.3.5); since $\widehat{A}$ is *complete*, this implies (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8) that the topology of $\widehat{A}$ is *induced* by the $\mathfrak{r}$-preadic topology of $\mathscr{O}_{x'}$; *a fortiori* it is the same for the topology of A (**0**, 7.3.5). Moreover, on a $\mathfrak{m}^k \cap A \subset \mathfrak{r}^k \cap A$, so $\mathfrak{m}^k \subset \mathfrak{r}^k \cap A$; but since $\mathfrak{m}^k \supset \mathfrak{r}^k \cap A$, these two topologies are *identical*. □

Remark (2.3.9). — We will see further on (7.8.3, (vii)) that for the Noetherian local rings A most commonly used in Algebraic Geometry, the hypothesis that A is integral and integrally closed implies that it has the same of $\widehat{A}$. This is why, in Algebraic Geometry over a base field, one generally states (2.3.8) under the hypothesis that A is *integral and integrally closed*.

Theorem (2.3.10). — *Let $f : X \to Y$ be a quasi-flat morphism (2.3.3). Then, for every pro-constructible part (1.9.4) Z of Y, we have $\overline{f^{-1}(Z)} = f^{-1}(\overline{Z})$.*

Proof. Since f is continuous, we have $\overline{f^{-1}(Z)} \subset f^{-1}(\overline{Z})$ and it amounts to proving that for every $x \in X$ such that $f(x) \in \overline{Z}$, x is adherent to $f^{-1}(Z)$; it is clear that the question is local on Y, so we can suppose Y affine. By virtue of the hypothesis, there exists an affine scheme Y' and a morphism $g : Y' \to Y$ such that $g(Y') = Z$ (1.9.5, (ix)). Let Y_1 be the closed image of Y' by g (**I**, 9.5.3), and let X_1 be the closed sub-prescheme $f^{-1}(Y_1)$ of X; if $f_1 : X_1 \to Y_1$ is the restriction of f to X_1, we know that f_1 is quasi-flat (2.3.3); we can therefore replace X, Y by X_1, Y_1 respectively, in other words suppose that g is *dominant*. Set then $X' = X \times_Y Y'$, and let f' and g' be the projections of X' into Y' and X respectively, so that we have the commutative diagram

$$\begin{tikzcd} X \arrow[d, "t"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ Y & Y' \arrow[l, "g"] \end{tikzcd}$$

Since f is quasi-flat, g quasi-compact and dominant, it follows from (2.3.7) (with the roles of f and g interchanged) that g' is a dominant morphism, which proves the theorem. ☐

Corollary (2.3.11). — *Let f be a quasi-flat and quasi-compact morphism, F a closed part of X such that $F = f^{-1}(f(F))$; then $F = f^{-1}(\overline{f(F)})$.*

Proof. Let Y' be the reduced sub-prescheme of X having F as underlying space, and let X_1 be its closed image, so that $Z = f(F)$ is pro-constructible in Y (1.9.5, (vii)), and the corollary follows from the fact that F is closed. **IV-2** | 18

We can further write the result of (2.3.11) in the form $F = f^{-1}(f(X) \cap \overline{f(F)})$, in other words, the closed subsets of the subspace $f(X)$ of Y are the parts $Z \subset f(X)$ such that $f^{-1}(Z)$ is closed (resp. *open*) in X; this also means that *the topology induced on Y by f on $f(X)$ is the quotient of that of X by the equivalence relation defined by f*. In particular: ☐

Corollary (2.3.12). — *Let $f : X \to Y$ be a quasi-faithfully flat morphism* (2.3.3) *and quasi-compact. Then the topology of Y is the quotient of that of X by the equivalence relation defined by f, in other words, for $Z \subset Y$ to be open* (*resp. closed*) *in Y, it is necessary and sufficient that $f^{-1}(Z)$ be open* (*resp. closed*) *in X.*

Proof. Indeed, we then have $f(X) = Y$. ☐

Corollary (2.3.13). — *Let X, Y be two S-preschemes, $f : X \to Y$ an S-morphism faithfully flat and quasi-compact. For Y to be separated over S, it is necessary and sufficient that the canonical immersion $j : X \times_Y X \to X \times_S X$* (**I**, 5.3.10) *be* closed.

Proof. Let us note for this that we have the commutative diagram (**I**, 5.3.5)

$$\begin{tikzcd} X \times_Y X \arrow[r, "j"] \arrow[d] & X \times_S X \arrow[d, "f \times_S f"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

identifying $X \times_Y X$ with the product of the $(Y \times_S Y)$-preschemes Y and $X \times_S X$. Since f is surjective, so is $f \times_S f$, so the diagonal $\Delta_Y(Y)$ has for inverse image by $f \times_S f$ the image $j(X \times_Y X)$ (**I**, 3.4.8). Since $f \times_S f$ is faithfully flat and quasi-compact (1.1.2 and 2.2.13), it suffices to apply (2.3.12) to this morphism. ☐

Corollary (2.3.14). — *Let $f : X \to Y$ be a quasi-faithfully flat morphism, and quasi-compact, and let Z be a part of Y. For Z to be a locally closed part pro-constructible of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a locally closed part pro-constructible of X.*

Proof. We already know (1.9.12) that, for Z to be a pro-constructible part of Y, it is necessary and sufficient that $f^{-1}(Z)$ be a pro-constructible part of X. The condition is evidently necessary. To prove that it is sufficient, consider the closed reduced sub-prescheme Y_1 of Y having $\overline{Z}$ as underlying space and let X_1 be its inverse image, so that $f^{-1}(\overline{Z}) = \overline{f^{-1}(Z)}$ by virtue of (2.3.10). As $f^{-1}(Z)$ is locally closed in X, it is open in $\overline{f^{-1}(Z)}$, so in X_1; now, the morphism $f_1 : X_1 \to Y_1$ deduced from f

by restriction is quasi-faithfully flat and quasi-compact (1.1.2 and 2.3.3); we deduce from (2.3.12) that Z is open in $\overline{Z}$, which shows that Z is locally closed in Y. □

Remark (2.3.15). — It does not suffice, in (2.3.12), to suppose only that f is *faithfully flat*. For example, take for Y a reduced irreducible algebraic curve (**II**, 7.4.2) of generic point y, for X the prescheme direct sum $\coprod \mathrm{Spec}(\mathscr{O}_\eta)$, where η ranges over Y (**I**, 3.1), for f the canonical morphism, which is faithfully flat (**I**, 2.4.2); if η denotes the generic point of Y, $Z = \{\eta\}$ is not open in Y (**II**, 7.4.3), but $f^{-1}(Z) = \mathrm{Spec}(\mathscr{O}_\eta)$ is open in X, since $\mathscr{O}_\eta$ is a field, and thus $f^{-1}(Z)$ is open in X. **IV-2** | 19

2.4. Universally open morphisms and flat morphisms

(2.4.1). We have already defined (**II**, 5.4.9) the notion of *universally closed* morphism; in the same manner, we give the following definitions:

Definition (2.4.2). — We say that a morphism of preschemes $f : X \to Y$ is *universally open* (resp. *universally bicontinuous*, resp. a *universal homeomorphism*) if, for every morphism $g : Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is open (resp. a homeomorphism onto its image, resp. a homeomorphism onto Y').

We will see further on (14.3.2) that when Y is locally Noetherian, the definition of universally open morphism given here is equivalent to the definition (**III**, 4.3.9) for morphisms of finite type; the reader can verify that we do not use the latter definition before § 14.

Proposition (2.4.3). —

label=(i) *An immersion* (*resp. an open immersion, resp. a closed immersion*) *is universally bicontinuous* (*resp. universally open, resp. universally closed*).

lbbel=(ii) *The composition of two universally open morphisms* (*resp. universally closed, resp. universal homeomorphisms*) *is also so.*

lcbel=(iii) *If $f : X \to Y$ is an S-morphism universally open* (*resp. universally closed, resp. a universal homeomorphism*), *so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every base change $S' \to S$.*

ldbel=(iv) *If $f : X \to X'$, $g : Y \to Y'$ are two S-morphisms universally open* (*resp. universally closed, resp. universal homeomorphisms*), *so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*

lebel=(v) *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms such that f is surjective; if $g \circ f$ is universally open* (*resp. universally closed, resp. a universal homeomorphism*), *so is g.*

lfbel=(vi) *For f to be a universally open morphism* (*resp. universally closed, resp. a universal homeomorphism*), *it is necessary and sufficient that f_{red} be so.*

lgbel=(vii) *Let (U_α) be an open covering of Y. For $f : X \to Y$ to be universally open* (*resp. universally closed, resp. a universal homeomorphism*), *it is necessary and sufficient that for every α, its restriction $f^{-1}(U_\alpha) \to U_\alpha$ be so.*

Proof. Assertion (i) follows from (**I**, 4.3.2). Assertion (ii) also follows immediately from the definitions, and it is the same of (iii) by reducing to the case where $Y = S$, $Y' = S'$, which one can do by (**I**, 3.3.11); we know that (iv) follows from (ii) and (iii) (**I**, 3.5.1). To prove (v), let us note that for every morphism $Z' \to Z$, $f_{(Z)} : X_{(Z)} \to Y_{(Z)}$ is surjective (**I**, 3.5.2); on the other hand, to note that a morphism g is open (resp. closed, resp. a homeomorphism onto its image, resp. a bijective homeomorphism), it is the same of g, one can then restrict to the case where $g \circ f$ is open (resp. closed), the fact that g is then open (resp. closed) follows from Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., § 5, n^o 1, prop. 1; for the two other cases, we can restrict to assuming that $g(f(X)) = g(Y) = Z$, in other words that $g \circ f$ is a homeomorphism of X onto Z; since f is surjective, g is necessarily bijective, and since g is a continuous open map by what precedes, g is indeed a homeomorphism of Y onto Z.

To prove (vi), let us note that to say that a morphism g is open (resp. closed, resp. a bijective homeomorphism) is equivalent to saying that g_{red} has the same property. On the other hand (**I**, 5.1.8), for every morphism $Y'' \to Y$, on a $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y''_{\mathrm{red}})_{\mathrm{red}} = (X \times_Y Y'')_{\mathrm{red}}$, so the preceding remark shows that if f_{red} is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), so is f. The converse is proved in the same manner, noting here that for every morphism $Y'' \to Y_{\mathrm{red}}$, we have $(X_{\mathrm{red}} \times_{Y_{\mathrm{red}}} Y'')_{\mathrm{red}} = (X \times_Y Y'')_{\mathrm{red}}$ (**I**, 5.1.8). **IV-2** | 20

Finally, the necessity of condition (vii) follows immediately from (iii). Conversely, suppose the condition satisfied, and let $g : Y' \to Y$ be a morphism; then the $g^{-1}(U_\alpha) = U'_\alpha$ form an open covering of Y' and if one notes f_α the restriction $f^{-1}(U_\alpha) \to U_\alpha$ of f, and f' the morphism $f_{(Y')}$, the restriction $f'^{-1}(U'_\alpha) \to U'_\alpha$ is nothing other than $(f_\alpha)_{(U'_\alpha)}$. We can therefore restrict to proving that f is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism onto Y'), which is immediate. □

Proposition (2.4.4). — *A universally bicontinuous morphism $f : X \to Y$ is radicial* (*and hence separated (1.7.7.1)*).

Proof. Indeed, f is universally injective by hypothesis (**I**, 3.5.11). □

Proposition (2.4.5). —

label=(i) *A morphism $f : X \to Y$ which is entire, surjective, and radicial is a universal homeomorphism.*

lbbel=(ii) *Conversely, suppose Y locally Noetherian. Then, if a morphism of finite type $f : X \to Y$ is a universal homeomorphism, it is surjective, finite and radicial.*

Proof. (i) It suffices to observe that the three properties of f are conserved by base change (**I**, 3.5.2), (**I**, 3.5.7) and (**II**, 6.1.5), and since an entire morphism is closed (**II**, 6.1.10), it is clear that f is a homeomorphism of X onto Y.

(ii) Since f is of finite type and universally closed by hypothesis and since it is separated by (2.4.4), and for every $y \in Y$, $f^{-1}(y)$ is reduced to a single element; so it is clear that f is surjective, and it is radicial since it is universally injective (**I**, 3.5.11). Since f is of finite type and universally closed, it is proper by hypothesis (**II**, 5.4.4), and for every $y \in Y$, $f^{-1}(y)$ is finite; so (**III**, 4.4.2) f is finite, and it is radicial since it is universally injective (**I**, 3.5.11). □

Theorem (2.4.6). — *Let $f : X \to Y$ be a quasi-flat morphism (2.3.3) and locally of finite presentation. Then f is universally open. In particular, a flat morphism locally of finite presentation is universally open.*

Proof. We know that for every base change $Y' \to Y$, $f_{(Y')}$ is quasi-flat (2.3.3) and locally of finite presentation (1.4.3, (iii)). It suffices therefore to prove that f is an open morphism. But this follows from the criterion (1.10.4) for morphisms locally of finite presentation, conditions *b*), *b′*), and *c*) of (1.10.4) being nothing other than conditions (i), (ii), (iii) of (2.3.4). □ **IV-2 | 21**

Corollary (2.4.7). — *For every prescheme Y, the structural morphism $\mathbf{V}^n_Y = Y \otimes_{\mathbf{Z}} \operatorname{Spec}(\mathbf{Z}[T_1, \ldots, T_n])$, also denoted $Y[T_1, \ldots, T_n]$) is universally open.*

Proof. Indeed, for $Y = \operatorname{Spec}(A)$, $Y[T_1, \ldots, T_n] = \operatorname{Spec}(A[T_1, \ldots, T_n])$, and $A[T_1, \ldots, T_n]$ is a free A-algebra of finite presentation. □

Remark (2.4.8). —

label=(i) We note that a morphism $f : X \to Y$ faithfully flat and quasi-compact is *not necessarily open*, even when X and Y are Noetherian. Take for example for Y a reduced irreducible algebraic curve (**II**, 7.4.2) of generic point y, and let X be the prescheme direct sum $Y \coprod \operatorname{Spec}(k(y))$, f being the canonical morphism; it is clear that f is flat and surjective, hence quasi-compact, but the image by f of the open part $\operatorname{Spec}(k(y))$ of X is the set $\{y\}$, which is not open in Y (**II**, 7.4.3).

lbbel=(ii) For every prescheme X, the canonical morphism $f : X_{\text{red}} \to X$ is a closed immersion and a *universal homeomorphism* (2.4.5), (vi); but when X is locally Noetherian, f is not *flat* unless X is reduced, so $f = 1_X$ (2.2.17).

Proposition (2.4.9). — *Let Y be a discrete prescheme. Then every morphism $f : X \to Y$ is universally open.*

Proof. The question being local on Y (2.4.3), (vii), we can restrict to the case where the underlying space of Y is reduced to a point; replacing f by f_{red} (2.4.3), (vi), we can moreover suppose that Y is the spectrum of a *field* k; on the other hand, for every base change $Y' \to Y$, the inverse images of affine open sets of X cover X', so we can suppose $X = \operatorname{Spec}(B)$ affine, B being a *k-algebra*. It suffices then to prove that for every k-algebra A', if we set $B' = B \otimes_k A'$, the image of every open subset of the form $U = D(t)$ ($t \in B'$) is open in $\operatorname{Spec}(A')$. Now, B is the inductive limit of the filtered increasing family (B_α) of its sub-k-algebras *of finite type* (since the functor $\varinjlim$ commuting with the tensor product), B' is the inductive limit of the A'-algebras $B_\alpha \otimes_k A' = B'_\alpha$; there exists α such that t is the image in B' of an element $t_\alpha \in B'_\alpha$, so $D(t)$ is the inverse image by the canonical morphism $u_\alpha : \mathrm{Spec}(B') \to \mathrm{Spec}(B'_\alpha)$ of the open $U_\alpha = D(t_\alpha)$ (**I**, 1.2.2). But since k is a field, B_α is a k-algebra of finite presentation and a k-module flat, hence B'_α is an A'-algebra of finite presentation and a flat A'-module; we thus conclude from (2.4.6) that the image of U_α by $f'_\alpha : \mathrm{Spec}(B'_\alpha) \to \mathrm{Spec}(A')$ is *open* in $\mathrm{Spec}(A')$. It therefore remains to see that $f'(U) = f'_\alpha(U_\alpha)$ and it remains to see that for every point $y' \in f'_\alpha(U_\alpha)$, the intersection $V = U \cap f'^{-1}(y')$ is not empty. Now $V = u_\alpha^{-1}(V_\alpha)$, where $V_\alpha = U_\alpha \cap f'^{-1}_\alpha(y')$; in other words, V_α is a non-empty open subset of the prescheme $\mathrm{Spec}(B'_\alpha \otimes_{A'} k(y')) = \mathrm{Spec}(B_\alpha \otimes_k k(y'))$ and V is its inverse image by the morphism $v : \mathrm{Spec}(B \otimes_k k(y')) \to \mathrm{Spec}(B_\alpha \otimes_k k(y'))$. Since B_α is a sub-algebra of B and k is a field, the homomorphism $B_\alpha \otimes_k k(y') \to B \otimes_k k(y')$ is *injective* by flatness, so the morphism v is dominant (**I**, 1.2.7), which shows that V is not empty. □

IV-2 | 22

Corollary (2.4.10). — *Let k be a field, X, Y two k-preschemes; then the morphism projection $X \times_k Y \to X$ is universally open. In particular, for every extension K of k, the morphism projection $X_{(K)} \to X$ is universally open.*

Proof. It suffices to apply (2.4.9) to the structural morphism $Y \to \mathrm{Spec}(k)$. □

Remark (2.4.11). — If $f : X \to Y$ is an *open* morphism, we know that, for every part E of Y, we have $f^{-1}(\overline{E}) = \overline{f^{-1}(E)}$ (Bourbaki, *Top. gén.*, chap. I, 3e éd., § 5, no 4, prop. 7). This remark applies for example when f is a morphism *flat* locally of finite presentation (2.4.6), or a morphism projection $X \times_k Y \to X$ where X, Y are preschemes over a field k (2.4.10), and generalises then (2.3.10).

2.5. Permanence of properties of Modules by faithfully flat descent

Proposition (2.5.1). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, $g : Y' \to Y$ a morphism faithfully flat, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$. For $\mathscr{F}$ to be flat (resp. faithfully flat) relative to f, it is necessary and sufficient that $\mathscr{F}'$ be flat (resp. faithfully flat) relative to f'.*

Proof. It suffices to apply (2.2.10) by replacing X, Y′, Z, $\mathscr{F}$, $\mathscr{G}'$ by Y′, X, Y, $\mathscr{O}_{Y'}$, $\mathscr{F}$ respectively; we then conclude that for $\mathscr{F}$ to be flat (resp. faithfully flat) relative to f, it is necessary and sufficient that $\mathscr{F}'$ be flat (resp. faithfully flat) relative to $g \circ f'$. But if $g' : X' \to X$ is the canonical projection, g' is faithfully flat (2.2.13) and on a $g \circ f' = f \circ g'$; therefore for $\mathscr{F}'$ to be flat (resp. faithfully flat) relative to $f \circ g'$, it is necessary and sufficient that $\mathscr{F}$ be so relative to f (2.2.11), (iii)). □

Proposition (2.5.2). — *Let $f : X' \to X$ be a morphism faithfully flat and quasi-compact, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, $\mathscr{F}' = f^*(\mathscr{F})$. Consider, for a quasi-coherent Module, the property of being:*

label=(i) of finite type;
lbbel=(ii) of finite presentation;
lcbel=(iii) locally free of finite type;
ldbel=(iv) locally free of rank n.

Then, if **P** *denotes one of the preceding properties, for $\mathscr{F}$ to possess the property* **P**, *it is necessary and it suffices that $\mathscr{F}'$ possess it.*

IV-2 | 22

Proof. For a quasi-coherent $\mathscr{O}_X$-Module to be locally free of finite type, it is necessary and sufficient that it be flat over X and of finite presentation (Bourbaki, *Alg. comm.*, chap. II, § 5, no. 2, cor. 2 du th. 1, taking into account (2.1.2)); as $\mathscr{F}$ is flat over X if and only if $\mathscr{F}'$ is flat over X' by virtue of (2.5.1) (applied by taking for f the identity), we see that to prove the proposition in case (iii), it suffices to have done so in cases (i) and (ii); it is the same to prove (iv), since $f^*(\mathscr{O}_X^n) = \mathscr{O}_{X'}^n$, so if $\mathscr{F}$ and $\mathscr{F}'$ are locally free of finite type and $x = f(x')$, the rank of $\mathscr{F}$ at x is equal to that of $\mathscr{F}'$ at x', and our assertion follows from the fact that f is surjective. To treat cases (i) and (ii),

we note that the question is local over X, and one cannot therefore suppose X affine; we know (2.2.12) that there is an affine scheme X'' and a faithfully flat morphism $g : X'' \to X'$ that is a local isomorphism; it therefore amounts to the same to say that $f^*(\mathscr{F})$ possesses the property **P**, or that $g^*(f^*(\mathscr{F}))$ possesses it. One is thus reduced to the case where $X' = \mathrm{Spec}(A')$, $X' = \mathrm{Spec}(A')$; taking into account (2.2.3) and (**II**, 6.1.4.1), it therefore suffices to prove the □ **IV-2** | 23

Lemma (2.5.3). — *Let* A *be a ring,* A′ *a faithfully flat* A*-algebra, and* M *an* A*-module,* $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$*. For* M *to be of finite type (resp. of finite presentation), it is necessary and it suffices that* M′ *be so.*

Proof. For the proof, see Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, § 3, no. 6, prop. 11. □

Remark (2.5.4). — The assertions of (2.5.2) for the properties (i) and (ii) are still valid if one only supposes f *quasi-faithfully flat* (2.3.3) and quasi-compact. One is indeed reduced (Bourbaki, *loc. cit.*) to proving the

Lemma (2.5.4.1). — *Let* $\rho : \mathrm{A} \to \mathrm{A}'$ *be a homomorphism of rings, such that the morphism* $f : \mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ *is surjective; suppose that there exists an* A′*-module* N′ *of finite type, which is* A*-flat and has support* $\mathrm{Spec}(\mathrm{A}')$*. Then, if* $u : \mathrm{P} \to \mathrm{Q}$ *is a homomorphism of* A*-modules such that* $u \otimes 1 : \mathrm{P} \otimes_{\mathrm{A}} \mathrm{A}' \to \mathrm{Q} \otimes_{\mathrm{A}} \mathrm{A}'$ *is surjective,* u *is surjective.*

Proof. Indeed, we deduce that the homomorphism $u \otimes 1_{\mathrm{N}'} : (\mathrm{P}\otimes_{\mathrm{A}'})\otimes_{\mathrm{A}'}\mathrm{N}' \to (\mathrm{Q}\otimes'_{\mathrm{A}}) \otimes_{\mathrm{A}'} \mathrm{N}'$ corresponding is surjective, and it can be written $u_{\mathfrak{p}} \otimes 1 : \mathrm{P}_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{N}'_{\mathfrak{q}} \to \mathrm{Q}_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{N}'_{\mathfrak{q}}$. By hypothesis on a $\mathrm{N}'_{\mathfrak{q}} \neq 0$ and $\mathrm{N}'_{\mathfrak{q}}$ is an $\mathrm{A}_{\mathfrak{p}}$-module flat (**0**, 6.3.1); by virtue of the lemma of Nakayama, on a $\mathfrak{q}\mathrm{N}'_{\mathfrak{q}} \neq \mathrm{N}'_{\mathfrak{q}}$, and *a fortiori* $\mathfrak{p}\mathrm{N}'_{\mathfrak{q}} \neq \mathrm{N}'_{\mathfrak{q}}$, therefore $\mathrm{N}'_{\mathfrak{q}}$ is an $\mathrm{A}_{\mathfrak{p}}$-module faithfully flat (**0**, 6.4.1). It follows that $u_{\mathfrak{p}}$ is surjective (**0**, 6.4.1), and this being true for all $\mathfrak{p} \in \mathrm{Spec}(\mathrm{A})$, since f is surjective, we finally conclude that u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, th. 1). □

Proposition (2.5.5). — *Let* $f : X \to Y$ *be a morphism,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module quasi-coherent of finite type and* f*-flat,* $\mathscr{E}$ *an* $\mathscr{O}_Y$*-Module quasi-coherent of finite type; for all* $y \in Y$*, set* $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$*. For a point* $x \in X$ *to be a maximal point of* $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$*, it is necessary and it suffices that* $y = f(x)$ *be a maximal point of* $\mathrm{Supp}(\mathscr{E})$*, and that* x *be a maximal point of* $\mathrm{Supp}(\mathscr{F}_y)$ *in* $f^{-1}(y)$*. If this is the case, we have*

(2.5.5.1).

$$\mathrm{length}((\mathscr{E} \otimes_Y \mathscr{F})_x) = \mathrm{length}(\mathscr{E}_y) \cdot \mathrm{length}((\mathscr{F}_y)_x)$$

Proof. It is clear that $f(\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})) \subset \mathrm{Supp}(\mathscr{E})$ (**0**, 5.2.2); the image by f of every irreducible component of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$ is therefore contained in a single irreducible component of $\mathrm{Supp}(\mathscr{E})$. We can therefore restrict to the case where $\mathrm{Supp}(\mathscr{F}) = X$. Indeed (**Err$_{\mathbf{III}}$**, 30), since $\mathscr{F}$ is of finite type, there exists a closed sub-prescheme X' of X having $\mathrm{Supp}(\mathscr{F})$ for underlying space and an $\mathscr{O}_{X'}$-Module quasi-coherent of finite type $\mathscr{G}$ such that, if $j : X' \to X$ is the canonical injection, we have $\mathscr{F} = j_*(\mathscr{G})$. Setting $f' = f \circ j$, it is clear that $\mathscr{G}$ is f'-flat and that we have $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F}) = \mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{G})$.

Suppose therefore $\mathrm{Supp}(\mathscr{F}) = X$; if Z is an irreducible component of $\mathrm{Supp}(\mathscr{E})$, on a $f^{-1}(\mathrm{Z}) \subset \mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$ (**I**, 9.1.13), and it follows from (2.3.4) that every irreducible component of $f^{-1}(\mathrm{Z})$ dominates Z; in other words, if x is the generic point of such an irreducible component of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$ contained in $f^{-1}(\mathrm{Z})$, $y = f(x)$ is the generic point of Z; moreover (**0**, 2.1.8), x is the generic point of an irreducible component of $f^{-1}(y) = \mathrm{Supp}(\mathscr{F}_y)$ (**I**, 9.1.13), and conversely every generic point of one of these components is the generic point of an irreducible component of $f^{-1}(\mathrm{Z})$.

It remains to prove (2.5.5.1); we have $(\mathscr{E} \otimes_Y \mathscr{F})_x = \mathscr{E}_y \otimes_{\mathscr{O}_y} \mathscr{F}_x$ (**I**, 9.1.12) and $(\mathscr{F}_y)_x = \mathscr{F}_x / \mathfrak{m}_y \mathscr{F}_x$; **IV-2** | 24
we are therefore reduced to proving the □

Lemma (2.5.5.2). — *Let* A, B *be two local rings,* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* $\mathfrak{m}$ *the maximal ideal of* A*. Let* M *be an* A*-module,* N *a* B*-module which is a faithfully flat* A*-module and such that* $\mathrm{N}/\mathfrak{m}\mathrm{N}$ *is a* B*-module of finite length; then we have*

(2.5.5.3).

$$\mathrm{length}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \mathrm{length}_{\mathrm{A}}(\mathrm{M}) \cdot \mathrm{length}_{\mathrm{B}}(\mathrm{N}/\mathfrak{m}\mathrm{N}).$$

Proof. If M is of infinite length, it is likewise of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$, since for every strictly increasing chain of n sub-modules M_i of M ($1 \leqslant i \leqslant n$), the $\mathrm{M}_i \otimes_{\mathrm{A}} \mathrm{N}$ are identified with sub-B-modules of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ that are pairwise distinct; as $\mathrm{N} \neq \mathfrak{m}\mathrm{N}$ since N is a faithfully flat A-module, formula (2.5.5.3) is true in this case. Suppose therefore M of finite length. If $\mathrm{M} = 0$ the two members of (2.5.5.3) are zero, so we can suppose $\mathrm{M} \neq 0$. The $\mathfrak{m}^k\mathrm{M}$ are sub-modules of M, therefore of finite length, and if $\mathfrak{m}^k\mathrm{M} \neq 0$, it follows from the lemma of Nakayama that $\mathfrak{m}^{k+1}\mathrm{M} \neq \mathfrak{m}^k\mathrm{M}$; therefore there is necessarily an integer r such that $\mathfrak{m}^r\mathrm{M} = 0$. The $\mathfrak{m}^k\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ are identified with sub-B-modules of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$,

$(\mathfrak{m}^k \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N})/(\mathfrak{m}^{k+1}\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N})$ being isomorphic to $(\mathfrak{m}^k\mathrm{M}/\mathfrak{m}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{m}} (\mathrm{N}/\mathfrak{m}\mathrm{N})$ $(0 \leqslant k \leqslant r-1)$; the length of this last, considered as a B-module, is therefore the product of $\mathrm{length}_{\mathrm{B}}(\mathrm{N}/\mathfrak{m}\mathrm{N})$ and the rank of the $(\mathrm{A}/\mathfrak{m})$-vector space $\mathfrak{m}^k\mathrm{M}/\mathfrak{m}^{k+1}\mathrm{M}$, a rank which is also the length of the A-module $\mathfrak{m}^k\mathrm{M}/\mathfrak{m}^{k+1}\mathrm{M}$. By addition (for $0 \leqslant k \leqslant r-1$), we immediately deduce formula (2.5.5.3). □

Remark (2.5.4$_{\mathbf{bis}}$). — We note that when N is an A-module *of finite type*, to say that N is a faithfully flat A-module is equivalent to saying that $\mathrm{N} \neq 0$ and that N is a flat A-module; indeed the lemma of Nakayama then shows that $\mathfrak{m}\mathrm{N} \neq \mathrm{N}$.

Lemma (2.5.6). — *Let* B *be a not necessarily commutative ring,* V, W *two* B*-modules to the left isomorphic,* C *the* C*-module isomorphic to* C_d*, and let* $\mathrm{M} = \mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$*, equipped with its canonical structure of* C*-module to the right. Then* M *is a* C*-module, and moreover, for all* $u \in \mathrm{M}$*, the following conditions are equivalent:*

label=a) $\{u\}$ *is a basis of the* C*-module* M.
lbbel=b) u *is an isomorphism of* V *onto* W.

Proof. If u is an isomorphism of V onto W, the map $v \mapsto u \circ v$ from C to M is evidently a bijection, therefore *b)* implies *a)*. Conversely, suppose that $\{u\}$ is a basis of the C-module M. By hypothesis u' is an isomorphism of V onto W, and $\{u'\}$ is likewise a basis of the C-module M; by hypothesis there is an invertible element w of C (that is to say, an automorphism of V) such that $u = u' \circ w$, which implies that u is an isomorphism of V onto W. □

Corollary (2.5.7). — *The hypotheses on* B, V *and* W *being those of* (2.5.6)*, suppose moreover that one of the following conditions is satisfied:*

label=(i) V *and* W *are Noetherian* B*-modules;*
lbbel=(ii) V *and* W *are modules of finite presentation over a commutative subring of* B.

Then conditions a) *and* b) *of* (2.5.6) *are also equivalent to the following:*

label=a$'$) u *is a generator of the* C*-module* M.
b$'$) u *is an epimorphism of* V *onto* W.

Proof. One knows indeed that an epimorphism of an A-module E onto itself is *bijective* in the two following cases: 1° E is a Noetherian A-module (Bourbaki, *Alg.*, chap. VIII, § 2, no. 2, lemme 3); 2° A is commutative and E is an A-module of finite presentation (8.9.3)[1]; therefore *b)* and *b$'$)* are equivalent. On the other hand, if u generates M □

IV-2 | 25

Proposition (2.5.8). — *Let* A *be a commutative semi-local ring,* B *an* A*-algebra* (*not necessarily commutative*)*,* V *and* W *two* B*-modules. Let* A' *be a commutative* A*-algebra which is a faithfully flat* A*-module; set* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{V}' = \mathrm{V} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{W}' = \mathrm{W} \otimes_{\mathrm{A}} \mathrm{A}'$*, so that* B' *is an* A'*-algebra,* V' *and* W' *are* B'*-modules. Suppose moreover that one of the following conditions is satisfied:*

label=(i) A *and* A' *are Noetherian,* V *and* W *are* A*-modules of finite type.*
lbbel=(ii) B *is an* A*-module of finite type,* V *is a* B*-module projective of finite type and an* A*-module of finite presentation.*

Then, if V' *and* W' *are* B'*-modules isomorphic,* V *and* W *are* B*-modules isomorphic.*

Proof. We note that in case (ii), W, being A'-isomorphic to V', is an A'-module of finite type, whence it follows that W is an A-module of finite type (Bourbaki, *Alg.*, chap. I$^{\mathrm{er}}$, § 3, no. 6, prop. 11); therefore in all cases V and W are A-modules of finite type. Moreover:

(2.5.8.1). *Under one of the hypotheses* (i), (ii), $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ *is an* A*-module of finite type.*

This is evident in case (i), since $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V}, \mathrm{W})$ is then an A-module of finite type, and $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ is a sub-A-module of $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V}, \mathrm{W})$. In case (ii), V is a direct factor of a free B-module B_0^n, therefore $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$ is a direct factor of $\mathrm{Hom}_{\mathrm{B}}(\mathrm{B}_0^n, \mathrm{W}) = \mathrm{W}^n$, and as W is an A-module of finite type, it is likewise of $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W})$. Set

$$\mathrm{C} = \mathrm{End}_{\mathrm{B}}(\mathrm{V}), \qquad \mathrm{M} = \mathrm{Hom}_{\mathrm{B}}(\mathrm{V}, \mathrm{W}).$$

[1] The reader may verify that (2.5.7) and (2.5.8) are not used before § 9.

which are A-modules of finite type in cases (i) and (ii). We know that under one of the conditions (i), (ii), the canonical homomorphism

(2.5.8.2).

$$\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}'\longrightarrow\mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}')$$

is *bijective* (Bourbaki, *Alg. comm.*, chap. II, § 2, no. 10, prop. 11). As A' is a flat A-module, $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}'$ is identified canonically with a sub-A'-module of $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}'$. The image of this sub-module by the homo- morphism (2.5.8.2) is contained in $\mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}')$, since if $u\in\mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})$ and $a'\in\mathrm{A}'$, the image of $u\otimes a'$ by (2.5.8.2) is the homomorphism $u':V'\to W'$ such that $u'(x\otimes 1)=u(x)\otimes a'$; for all $b\in\mathrm{B}$, we then have $u'((b\otimes 1)(x\otimes 1))=u(bx)\otimes a'=bu(x)\otimes a'=(b\otimes 1)(u(x)\otimes a')$, whence our assertion. This being so:

(2.5.8.3). *Under one of the hypotheses* (i), (ii), *the homomorphism*

(2.5.8.4).

$$\mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}'\longrightarrow\mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}')$$

is bijective.

For all $b\in\mathrm{B}$, denote $h(b)$ (resp. $h'(b)$) the homothety $x\mapsto bx$ in V (resp. W), which is an A-endomorphism. Let $(b_\alpha)_{\alpha\in\mathrm{I}}$ be a system of generators of the A-algebra B; the map

$$u\longmapsto(h'(b_\alpha)\circ u-u\circ h(b_\alpha))_{\alpha\in\mathrm{I}}$$

from $\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})$ to $(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}$ is A-linear, and by definition, its *kernel* is none other than $\mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})$; in other words, we have an exact sequence

$$0\longrightarrow\mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})\longrightarrow\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\longrightarrow(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}.$$

The same reasoning applies by replacing A, B, V, W by A', B', V', W'; moreover, we have a diagram

(2.5.8.5).

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_{\mathrm{B}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}' \arrow[r, "r"] \arrow[d] & \mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}' \arrow[r] \arrow[d, "s"] & (\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}\otimes_{\mathrm{A}}\mathrm{A}' \arrow[d, "t"] \\
0 \arrow[r] & \mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}') \arrow[r] & \mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}') \arrow[r] & (\mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}'))^{\mathrm{I}}
\end{tikzcd}$$

where r is the homomorphism (2.5.8.4), s the homomorphism (2.5.8.2) and t the composite homomorphism

$$(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W}))^{\mathrm{I}}\otimes_{\mathrm{A}}\mathrm{A}'\xrightarrow{w}(\mathrm{Hom}_{\mathrm{A}}(\mathrm{V},\mathrm{W})\otimes_{\mathrm{A}}\mathrm{A}')^{\mathrm{I}}\xrightarrow{s^{\mathrm{I}}}(\mathrm{Hom}_{\mathrm{A}'}(\mathrm{V}',\mathrm{W}'))^{\mathrm{I}}$$

w being the canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3ᵉ éd., § 3, no. 7). One verifies immediately that the diagram (2.5.8.5) is commutative, and as A' is a flat A-module, its rows are exact. Finally, we have seen
that s is an isomorphism, and it is likewise of s^{I}; in case (ii), we can take I finite, and we know then **IV-2** | 26
that w is bijective (Bourbaki, *loc. cit.*, prop. 7); in case (i), we note that if B' (resp. B'') is the image of B by h (resp. A' dans $\mathrm{End}_{\mathrm{A}}(\mathrm{V})$ (resp. $\mathrm{End}_{\mathrm{A}}(\mathrm{W})$)), B' and B'' are A-modules of finite type, so we can still take I finite. Thus, in all cases, t is bijective, and one concludes therefore that it is the same of r.

(2.5.8.6). It follows therefore from (2.5.8.4) that if one sets

$$\mathrm{C}'=\mathrm{C}\otimes_{\mathrm{A}}\mathrm{A}',\qquad\mathrm{M}'=\mathrm{M}\otimes_{\mathrm{A}}\mathrm{A}'$$

we have canonical bijections

$$\mathrm{C}'\xrightarrow{\sim}\mathrm{End}_{\mathrm{B}'}(\mathrm{V}'),\qquad\mathrm{M}'\xrightarrow{\sim}\mathrm{Hom}_{\mathrm{B}'}(\mathrm{V}',\mathrm{W}')$$

the first being an isomorphism of A'-algebras, the second constituting with the first a di-isomorphism of C'-modules to the right.

(2.5.8.7). *Reduction to the case where* $V = B_s$. The hypothesis that V' and W' are B'-modules isomorphic implies that C'_d and M' are C'-modules to the right *isomorphic* (2.5.6). Let us show that to prove (2.5.8), it suffices to find an element $u \in M$ which is a *generator* of M as a C-module to the right. Indeed, $u' = u \otimes 1$ will be a generator of M' in its capacity as C'-module to the right; or, in case (i), V' and W' are A'-modules of finite type, therefore Noetherian since A' is Noetherian; in case (ii), the torsion modules of V' are of finite presentation. One can therefore apply (2.5.7) to A', B', V', W' and to conclude that u' is bijective (**0**, 6.4.1), that is to say the conclusion of (2.5.8). But as A' is faithfully flat over A, this implies that u is a B'-isomorphism of V' onto W'. But as A' is faithfully flat over A, and one concludes that u' is a B'-isomorphism is reduced (by changing the notations) to proving (2.5.8) in the particular case where $V = B_s$, and it suffices to prove the existence of a *generator* of the B-module W. We note that B is an A-module *of finite type*.

(2.5.8.8). *Reduction to the case where* A *and* A' *are fields*, B an *Artinian simple algebra over the centre* A.

Suppose now that B is a *finite product of fields* k_i ($1 \leqslant i \leqslant n$), A' is composed of direct factors of k_i's ($1 \leqslant i \leqslant n$), each of the A'_i being cancelled by the k_j of index $j \neq i$; the hypothesis that A' is a faithfully flat A-module implies that the A'_i are $\neq 0$; thus there exists for each an extension A''_i of A'_i which is a field, and A'', composed of direct factors of A''_i, is a faithfully flat A-module and a quotient of A'. If one considers then the ring $B'' = B \otimes_A A''$, $W'' = W \otimes_A A''$ and a B''-module isomorphic to B''_s; one can then restrict ourselves to showing that (2.5.8) by replacing A' by A'', that is to say one can also suppose that A' is a *product of fields*.

Let Z be the centre of B, which is a finite product of fields and an A-module of finite type; we note that B and W are Z-modules of finite type, projective as Z-modules; moreover, on a $B' = B \otimes_Z Z'$ and $W' = W \otimes_Z Z'$ by setting $Z' = Z \otimes_A A'$, and Z' is an A-module faithfully flat. Thus one can reduce showing (2.5.8) by replacing A' by Z'; one can then restrict ourselves to proving (2.5.8) in the case where A and A' are fields and B a simple algebra over its centre A.

Let now A and A' be fields and suppose for the rest of the argument that B is a finite product of simple k_i-algebras ($1 \leqslant i \leqslant n$), k_i being an extension of k_i and being cancelled by the k_j of index $j \neq i$; moreover, the reasoning made above shows that one can suppose that A' is a product of fields k'_i ($1 \leqslant i \leqslant n$), k'_i being an extension of k_i and being cancelled by the k_j of index $j \neq i$; in this case, the $B_i \otimes_{k_i} k'_i$ and $W_i \otimes_{k_i} k'_i$ are $(B_i \otimes_{k_i} k'_i)$-modules isomorphic; it therefore suffices to prove (2.5.8) when $n = 1$, that is to say in the case where A and A' are fields and B a *simple algebra over its centre* A.

(2.5.8.9). *End of the proof.* One knows then (Bourbaki, *Alg.*, chap. VIII, § 5, prop. 6 and 8) that every B-module is a direct sum of modules isomorphic to a minimal ideal of B, and that two B-modules of finite rank over A are therefore isomorphic if and only if they have the same *rank* over A. By hypothesis, on a $[W' : A'] = [B'_s : A']$. But on a $[W' : A'] = [W : A]$ and $[B'_s : A'] = [B_s : A]$; therefore $[W : A] = [B_s : A]$, which concludes the proof.

□

IV-2 | 27

2.6. Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent

Proposition (2.6.1). — *Let* $f : X \to Y$ *be an* S*-morphism of* S*-preschemes,* $g : S' \to S$ *a surjective morphism,* $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \to Y'$. *Consider, for a morphism, the property of being:*

label=(i) *surjective;*
lbbel=(ii) *injective;*
lcbel=(iii) *with finite fibres (as sets);*
ldbel=(iv) *bijective;*
lebel=(v) *radicial.*

Then, if **P** *denotes one of the preceding properties and if* f' *possesses the property* **P**, *the same is true of* f.

Proof. As the projection morphism $Y' \to Y$ is itself surjective (**I**, 3.5.2), one can, by virtue of (**I**, 3.3.11), restrict to the case where $Y = S$, $Y' = S'$. For all $y \in Y$ (resp. $y' \in Y'$) denote by X_y (resp. $X'_{y'}$) the fibre prescheme $f^{-1}(y)$ (resp. $f'^{-1}(y)$) (**I**, 3.6.2); we know that for $y' \in Y'$, $y = g(y')$, we have an isomorphism $X'_{y'} \xrightarrow{\sim} X_y \otimes_{k(y)} k(y')$ (**I**, 3.6.4); as the morphism $\operatorname{Spec}(k(y')) \to \operatorname{Spec}(k(y))$ is surjective, it is the same of the projection $X'_{y'} \to X_y$ (**I**, 3.5.2); therefore if $X'_{y'}$ is non-empty (resp. has at most one point, resp. is a finite set), the same is true of X_y; as $g : Y' \to Y$ is surjective, this proves the proposition in cases (i) and (iii), and (iv) follows from (i) and (ii); finally, to prove (v), it suffices to show that if f' is universally injective (**I**, 3.5.11), the same is true of f; or, let $Y_1 \to Y$ be any morphism; set $X_1 = X \times_Y Y_1$ and $f_1 = f_{(Y_1)}$. On the other hand, set $Y'_1 = Y_1 \times_Y Y'$, $X'_1 = X' \times_{Y'} Y'_1 = X_1 \times_{Y_1} Y'_1$, $f'_1 = f'_{(Y'_1)} = (f_1)_{(Y'_1)} : X'_1 \to Y'_1$; as $g' = g_{(Y_1)} : Y'_1 \to Y'$ is surjective (**I**, 3.5.2) and f' is universally injective, f'_1 is injective, and it follows from (ii) that f_1 is injective, whence our assertion. □

Proposition (2.6.2). — *With the notations being those of (2.6.1), suppose the morphism $g : \mathrm{S}' \to \mathrm{S}$ faithfully flat and quasi-compact. Consider, for a morphism, the property of being:*

label=(i) open;

lbbel=(ii) closed;

lcbel=(iii) quasi-compact and universally bicontinuous;

ldbel=(iv) a universal homeomorphism.

Then, if **P** *denotes one of the preceding properties and if f' possesses the property* **P**, *the same is true of f.*

IV-2 | 28

Proof. As the morphism $Y' \to Y$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), one can, by virtue of (**I**, 3.3.11), restrict to the case where $Y = \mathrm{S}$, $Y' = \mathrm{S}'$. If g' is the projection $X' \to X$, we know that we have, for every subset M of X, $g^{-1}(f(\mathrm{M})) = f'(g'^{-1}(\mathrm{M}))$ (**I**, 3.4.8). If f' is an open morphism (resp. closed) then, for every open subset (resp. closed subset) M of X, $f'(g'^{-1}(\mathrm{M}))$ is open (resp. closed) in Y', and as g is faithfully flat and quasi-compact, we conclude that $f(\mathrm{M})$ is open (resp. closed) in Y by virtue of (2.3.12). This proves the proposition in cases (i) and (ii). Let us prove it in case (iii) (since this implies in case (iv), taking into account (2.6.1), (iv)). By virtue of (2.6.1), (ii) f is injective, and it remains to prove that f, considered as a map from X to $f(X)$, is a quasi-compact and *open* map. As f' is quasi-compact, it is the same of f (1.1.4). It therefore suffices to show that for every closed subset Z of X, $\mathrm{Z} = f^{-1}(\overline{f(\mathrm{Z})})$; as g' is surjective, this relation is equivalent to $g'^{-1}(\mathrm{Z}) = g'^{-1}(f^{-1}(\overline{f(\mathrm{Z})}))$ or equivalently $g'^{-1}(\mathrm{Z}) = f'^{-1}(g^{-1}(\overline{f(\mathrm{Z})}))$. Or, as f is quasi-compact, it is the same of its composite with the canonical injection $Z \to X$ (Z being here the sub-prescheme reduced of X having Z as underlying space); applying (2.3.10) to the part $f(Z)$ of Y (image of the morphism $f|Z$), we get $g^{-1}(\overline{f(\mathrm{Z})}) = \overline{g^{-1}(f(\mathrm{Z}))}$; the formula to prove is therefore equivalent to the following: $Z' = f'^{-1}(f'(Z'))$, where we have set $Z' = g'^{-1}(\mathrm{Z})$; but this formula results from the hypothesis that f' is a homeomorphism onto the sub-space image of X' over $f'(X')$. □

Remark (2.6.3). — In cases (i) and (ii), the conclusions of (2.6.2) are still valid when one only supposes g *quasi-faithfully flat* (2.3.3) and *quasi-compact*; indeed, by virtue of (2.1.4), (**I**, 3.5.2) and (**I**, 9.1.13.1), one can reduce to the case where $Y = \mathrm{S}$, $Y' = \mathrm{S}'$; the conclusion results then from (2.3.12). In cases (iii) and (iv), the conclusions remain valid when one only supposes g *quasi-faithfully flat*, provided one already supposes f quasi-compact; indeed, we then only use (2.3.10) and the fact that g is surjective. Finally, if g is faithfully flat and locally of finite presentation, or if g is surjective and S discrete, the conclusion of (2.6.2) is valid when **P** is the property:

(iii *bis*) to be a homeomorphism onto the sub-space image;

this results indeed from the proof given in (2.6.2) and Remark (2.4.11).

Corollary (2.6.4). — *With the notations being those of (2.6.1), suppose the morphism $g : \mathrm{S}' \to \mathrm{S}$ faithfully flat and quasi-compact. Consider, for a morphism, the property of being:*

label=(i) universally open;

lbbel=(ii) universally closed;

lcbel=(iii) quasi-compact and universally bicontinuous;

ldbel=(iv) a universal homeomorphism;

lebel=(v) quasi-compact;

lfbel=(vi) quasi-compact and dominant.

Then, if **P** *denotes one of the preceding properties, for f to possess the property* **P**, *it is necessary and it suffices that f' possess it.*

Proof. Properties (v) and (vi) are only given for the record, being consequences of (1.1.4), (1.1.6) and (2.3.7). As far as the others are concerned, the condition is necessary by virtue of (2.4.3). Conversely, suppose for example that f' is universally open, and let $Y_1 \to Y$ be any morphism; set $X_1 = X \times_Y Y_1$ and $f_1 = f_{(Y_1)}$. On the other hand, set $Y'_1 = Y_1 \times_Y Y'$, $X'_1 = X' \times_{Y'} Y'_1 = X_1 \times_{Y_1} Y'_1$, $f'_1 = f'_{(Y'_1)} = f_{1(Y'_1)} : X'_1 \to Y'_1$; as $g' = g_{(Y_1)} : Y'_1 \to Y'$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), and f' is universally open, f'_1 is open and it follows from (2.6.2) that f_1 is open; therefore f is universally open. The same reasoning applies for the other cases.

We note again here that one can replace "faithfully flat" by "quasi-faithfully flat" and, when g is moreover locally of finite presentation, or g is surjective and S discrete, one can replace the property (iii) by:

(iii *bis*) universally bicontinuous.

□

IV-2 | 29

2.7. Permanence of various properties of morphisms by faithfully flat descent

Proposition (2.7.1). — *Let $f : X \to Y$ be an* S*-morphism of* S*-preschemes, $g : S' \to S$ a faithfully flat and quasi-compact morphism, $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \to Y'$. Consider, for a morphism, the property of being:*

label=(i) separated;
lbbel=(ii) quasi-separated;
lcbel=(iii) locally of finite type;
ldbel=(iv) locally of finite presentation;
lebel=(v) of finite type;
lfbel=(vi) of finite presentation;
lgbel=(vii) proper;
lhbel=(viii) an isomorphism;
libel=(ix) a monomorphism;
ljbel=(x) an open immersion;
lkbel=(xi) a quasi-compact immersion;
llbel=(xii) a closed immersion;
lmbel=(xiii) affine;
lnbel=(xiv) quasi-affine;
lobel=(xv) finite;
lpbel=(xvi) quasi-finite;
lqbel=(xvii) integral.

Then, if **P** *denotes one of the preceding properties, for f to possess the property* **P***, it is necessary and it suffices that f' possess it.*

Proof. It has been proved in chapters I^er, II and in chapter IV, § 1, that if f possesses one of the properties **P** above, the same is true of f' (without hypothesis on the morphism $g : S' \to S$). It therefore remains to prove the converse; as the projection $Y' \to Y$ is

a faithfully flat and quasi-compact morphism (2.2.13) and (1.1.2), one can restrict to the case where $S = Y$, $S' = Y'$ by virtue of (**I**, 3.3.11). IV-2 | 30

(i) To say that f is separated means that the diagonal morphism $\Delta_f : X \to X \times_Y X$ is *closed*; as $\Delta_{f'} = (\Delta_f)_{(Y')}$ (**I**, 5.3.4), if $\Delta_{f'}$ is closed, it is the same of Δ_f by virtue of (2.6.2), therefore f is separated.

(ii) has already been proved under weaker hypotheses (1.2.5).

(iii) and (iv): The question is evidently local over X and Y, and, taking into account (2.2.12), it suffices therefore to prove the

Lemma (2.7.1.1). — *Let* A *be a ring,* B *an* A*-algebra,* A′ *an* A*-algebra which is a faithfully flat* A*-module,* $B' = B \otimes_A A'$. *For* B *to be an* A′*-algebra of finite type* (*resp. of finite presentation*)*, it is necessary and it suffices that* B′ *be an* A′*-algebra of finite type* (*resp. of finite presentation*).

We already know that the condition is necessary without hypothesis on A' (1.3.4), (1.3.6), (1.4.3) and also $\mathrm{B}' = \varinjlim(\mathrm{B}_\alpha \otimes_\mathrm{A} \mathrm{A}')$, the tensor product commuting with inductive limits; if (x'_i) is a finite system of generators of the A'-algebra B', there exists an index α such that all the x'_i belong to the sub-algebra $\mathrm{B}_\alpha \otimes_\mathrm{A} \mathrm{A}'$ of B', whence $\mathrm{B}' = \mathrm{B}_\alpha \otimes_\mathrm{A} \mathrm{A}'$, and as A' is faithfully flat, $\mathrm{B} = \mathrm{B}_\alpha \otimes_\mathrm{A} \mathrm{A}'$.

Suppose now that B' is an A'-algebra of finite presentation; we already know that B is an A'-algebra of finite type, therefore there exists an A-algebra of polynomials $\mathrm{C} = \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_m]$ and an A-homomorphism surjective of algebras $\mathrm{C} \to \mathrm{B}$; let $\mathfrak{J}$ be the kernel of this homomorphism, so that we have an exact sequence $0 \to \mathfrak{J} \to \mathrm{C} \to \mathrm{B} \to 0$, and therefore also an exact sequence $0 \to \mathfrak{J}' \to \mathrm{C}' \to \mathrm{B}' \to 0$ (since A' is A-flat), by setting $\mathrm{C}' = \mathrm{C} \otimes_\mathrm{A} \mathrm{A}' = \mathrm{A}'[\mathrm{T}_1, \dots, \mathrm{T}_m]$, and $\mathfrak{J}' = \mathfrak{J} \otimes_\mathrm{A} \mathrm{A}'$ (identified to an ideal of C'). As B' is an A'-algebra of finite presentation, $\mathfrak{J}'$ is a C'-module of finite type (1.4.4); but on a $\mathfrak{J}' = \mathfrak{J} \otimes_\mathrm{C} \mathrm{C}'$ and C' is a C-module faithfully flat (2.2.13) and (2.2.3); one knows then that the hypothesis that $\mathfrak{J}'$ is a C'-module of finite type implies that $\mathfrak{J}$ is a C-module of finite type (Bourbaki, *Alg. comm.*, chap. I$^{\text{er}}$, § 3, no. 6, prop. 11), therefore B is an A-algebra of finite presentation.

(v) results from (iii) and from (2.6.2), (v) by virtue of (1.5.2).

(vi) results similarly from (iv), (v) and (ii) by virtue of (1.6.1).

(vii) results from (i), (v) and (2.6.4), (ii) (**II**, 5.4.1).

(viii) We note first that as f' is an isomorphism, it is a homeomorphism universal, therefore the same is true of f (2.6.4); one concludes already that f is quasi-compact and separated (2.4.4). Set $f = (\psi, \theta)$, where ψ is therefore a homeomorphism; it is a question of proving that $\theta : \mathscr{O}_Y \to f_*(\mathscr{O}_X)$ is an isomorphism of $\mathscr{O}_Y$-Modules. Or, if one sets $f' = (\psi', \theta')$, the homomorphism $\theta' : \mathscr{O}_{Y'} \to f'_*(\mathscr{O}_{X'})$ is composed of the canonical homomorphism $g^*(f_*(\mathscr{O}_X)) \to f'_*(\mathscr{O}_{X'})$ and of $g^*(\theta)$ (2.3.2); but the first of these two homo- morphisms is bijective by virtue of the hypothesis on g (2.3.1), therefore if θ' is bijective, the same is true of $g^*(\theta)$, and as g is faithfully flat, θ is bijective (2.2.7), which proves (viii).

(ix) The proposition results from (viii), from (**I**, 5.3.4) and from (**I**, 5.3.8) which reduces monomorphisms to isomorphisms.

(x) If f' is an open immersion, $f'(X')$ is open in Y', and we have $f'(X') = g^{-1}(f(X))$ (**I**, 3.4.8); it follows from (2.3.12) that $f(X)$ is open. One can then replace Y (resp. Y') by the sub-prescheme induced on the open set $f(X)$ (resp. $f'(X')$); taking into account (1.1.2) and (2.2.13); then f' becomes an isomorphism, therefore it is the same of f by (viii), and this establishes (x).

(xi) If f' is a quasi-compact immersion, f' is a quasi-compact morphism and quasi-separated **IV-2 | 31** (1.2.2), therefore the same is true of f by (ii) and (2.6.2), (v). Let Z be the closed sub-prescheme of Y, closed image of X by f (1.7.8), and set $f = j \circ g$ with $j' = j_{(Y')}$, $g' = g_{(Y')}$, and we know that j' identifies with the canonical injection $Z' \to Y'$ of the sub-prescheme Z' of Y', closed image of X' by f' (2.3.2). The hypothesis on f' means then that g' is an open immersion ouverte (**I**, 9.5.10), therefore the same is true of g by (x), and this shows that f is an immersion.

(xii) To say that f (resp. f') is a closed immersion means that f (resp. f') is a quasi-compact immersion and a closed morphism; we see therefore that (xii) results from (xi) and from (2.6.2), (ii).

(xiii) and (xiv) Suppose f' affine (resp. quasi-affine); note then that f' is quasi- compact and quasi-separated (**II**, 5.1.1), therefore the same is true of f by (ii) and (2.6.2), (v). Set $\mathscr{A} = f_*(\mathscr{O}_X)$, $\mathscr{A}' = f'_*(\mathscr{O}_{X'})$; by virtue of (2.3.1), the canonical homomorphism of $\mathscr{O}_{Y'}$-Algebras $g^*(\mathscr{A}) \to \mathscr{A}'$ is bijective; consequently, if $h : Z = \operatorname{Spec}(\mathscr{A}) \to Y$ is the structural morphism, the structural morphism $h' : Z' = \operatorname{Spec}(\mathscr{A}') \to Y'$ identifies to $h_{(Y')}$ (**II**, 1.5.2). Let then $u : X \to Z$ (resp. $u' : X' \to Z'$) be the Y-morphism (resp. Y'-morphism) canonical corresponding to the identical homomorphism of $\mathscr{A}$ (resp. $\mathscr{A}'$) (**II**, 1.2.7); as on the commutative diagram

$$\begin{tikzcd}
X \arrow[d, "u"'] & X' \arrow[l] \arrow[d, "u'"] \\
Z \arrow[d, "h"'] & Z' \arrow[l] \arrow[d, "h'"] \\
Y & Y' \arrow[l, "g"']
\end{tikzcd}$$

and $h' \circ u' = f'$, it follows from (**II**, 1.2.7) that $u' = u_{(Z')}$. Moreover, g' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). The hypothesis on f' signifies that u' is an isomorphism (resp. an open immersion) (**II**, 5.1.6); it follows then from (viii) (resp. (x)) that u is an isomorphism (resp. an open immersion), whence (xiii) (resp. (xiv)).

(xv) If f' is finite, it is affine, therefore f is also affine by (xiii); moreover, with the notations of the proof of (xiii), $\mathscr{A}'$ is an $\mathscr{O}_{Y'}$-Module of finite type, and $\mathscr{A}'$ is isomorphic to $g^*(\mathscr{A})$; it follows from (2.5.2) that $\mathscr{A}$ is an $\mathscr{O}_Y$-Module of finite type, therefore f is a finite morphism. **IV-2** | 32

(xvi) To say that f is quasi-finite means that f is a morphism of finite type and that for all $y \in Y$, $f^{-1}(y)$ is finite (**II**, 6.2.2) and **I**, (6.4.4); the conclusion results therefore from (v) and from (xv).

(xvii) One argues as in (xv) that f is affine. One can restrict to the case where $Y = \operatorname{Spec}(A)$, $Y' = \operatorname{Spec}(A')$, and then $X = \operatorname{Spec}(B)$, $X' = \operatorname{Spec}(B')$, where $B' = B \otimes_A A'$; B is equal to the inductive limit of its sub-A-algebras of finite type B_α, therefore $B' = \varinjlim B'_\alpha$ where $B'_\alpha = B_\alpha \otimes_A A'$, and B'_α is an A'-algebra of finite type. But by hypothesis B' is integral over A', therefore B'_α is an A'-*module* of finite type, and B_α is therefore also an A-module of finite type (2.5.2). □

Corollary (2.7.2). — *With the hypotheses and notations being those of (2.7.1), suppose f quasi- compact; let $\mathscr{L}$ be an $\mathscr{O}_X$-Module invertible, $\mathscr{L}' = \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$ its inverse image. For $\mathscr{L}$ to be ample* (*resp. very ample*) *for f, it is necessary and it suffices that $\mathscr{L}'$ be ample* (*resp. very ample*) *for f'.*

Proof. The condition is necessary without hypothesis on $g : S' \to S$ (**II**, 4.4.10) and (4.6.13); to see that it is sufficient, one can, as in (2.7.1) limit to the case where $S = Y$, $S' = Y'$. The hypothesis on $\mathscr{L}'$ implies that f' is quasi-compact and separated (**II**, 4.6.1), therefore it is the same of f by ((2.6.2, (v)) and (2.7.1), (i)). Setting $\mathscr{E} = f_*(\mathscr{L})$, $\mathscr{E}' = f'_*(\mathscr{L}')$; it follows from (2.3.1) that the canonical homomorphism $u : g^*(\mathscr{E}) \to \mathscr{E}'$ is bijective. If $\mathscr{L}'$ is very ample for f, the canonical homomorphism $\sigma' : f'^*(\mathscr{E}') \to \mathscr{L}'$ is surjective, and the morphism $r' = r_{\mathscr{L}',\sigma'} : X' \to \mathbf{P}(\mathscr{E}')$ is an immersion (**II**, 4.4.4), *b*)) necessarily quasi-compact (1.1.2), (v)). The fact that $u : g^*(\mathscr{E}) \to \mathscr{E}'$ is bijective implies that if $h : \mathbf{P}(\mathscr{E}) \to Y$, $h' : \mathbf{P}(\mathscr{E}') \to Y'$ are the structural morphisms, h' identifies to $h_{Y'}$ (**II**, 4.1.3); on the other hand, by designating by g' the projection $X' \to X$, g' is faithfully flat (2.2.13), and on a $f \circ g' = g \circ f'$, and one verifies easily that the homomorphism $g^*(\sigma) : g^*(f^*(f_*(\mathscr{L}))) \to g^*(\mathscr{L})$ is identical to the composite homomorphism

$$f'^*(g^*(f_*(\mathscr{L}))) \xrightarrow{f'^*(u)} f'^*(f'_*(\mathscr{L}')) \xrightarrow{\sigma'} \mathscr{L}'$$

(for example by reducing to the case where Y and Y' are affine). As σ' is surjective and $f'^*(u)$ bijective, it is the same of σ' therefore the same of σ (2.2.7). One concludes therefore that the morphism $r = r_{\mathscr{L},\sigma} : X \to \mathbf{P}(\mathscr{E})$ is defined everywhere (**II**, 3.7.4); moreover, if l' is the projection $P' \to P$, and if g'' is the projection r' identifies to $r_{(P')}$ (**II**, 4.2.10) and g'' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). On concludes therefore from (2.7.1), (xi)) that r is an immersion, and therefore $\mathscr{L}$ is *very ample* (**II**, 4.4.4), *b*)).

Suppose now that $\mathscr{L}'$ is ample for f'; to prove that $\mathscr{L}$ is ample for f, one can restrict to the case where Y is affine (**II**, 4.6.4), and by virtue of (2.2.12) and of (**II**, 4.6.13), one can also suppose that Y' is affine. Then X and X' are quasi-compact schemes, and to prove that $\mathscr{L}$ is f-ample, one can apply the criterion of (**II**, 4.6.8), *c*)). Let therefore $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent of finite type; if $\sigma : f^*(f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n})) \to \mathscr{F} \otimes \mathscr{L}^{\otimes n}$ is the canonical homomorphism, we see as above that $g^*(\sigma)$ is the composite homomorphism

$$f'^*(g^*(f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n}))) \xrightarrow{f'^*(u)} f'^*(f'_*(\mathscr{F}' \otimes \mathscr{L}'^{\otimes n})) \xrightarrow{\sigma'} \mathscr{F}' \otimes \mathscr{L}'^{\otimes n}$$

by setting $\mathscr{F}' = g^*(\mathscr{F})$, taking into account (**0**, 4.3.3.1) and designating by u the canonical homomorphism $g^*(f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n})) \to f'_*(\mathscr{F}' \otimes \mathscr{L}'^{\otimes n})$. Or, we know that u is bijective for all n (2.3.1); on the other hand, as $\mathscr{F}'$ is quasi-compact and of finite type, the hypothesis that $\mathscr{L}'$ is ample for f' implies the existence of an n_0 such that for $n \geqslant n_0$, σ' is surjective; we see therefore that $g^*(\sigma)$ is surjective for $n \geqslant n_0$, and as g' is faithfully flat, σ is surjective for these values of n (2.2.7), which completes the proof. □

IV-2 | 33

Remarks (2.7.3). — label=(i) It follows from (2.6.1), (2.6.4) and (2.5.4.1) that the conclusions of (2.7.1) are still valid in cases (i), (iii), (v), (vii) and (xvi) when one only supposes g quasi-compact and *quasi-faithfully flat* (2.3.3); on has already remarked that (2.7.1) is valid in case (ii) assuming only g surjective and quasi-compact.

lbbel=(ii) With the hypotheses and notations of (2.7.1), one can assert that f is *proper* and f' is *projective* without f being a *quasi-projectif*. Indeed, Hironaka [?] gave an example of a proper non-projective morphism $f : X \to Y$, where X and Y are two algebraic schemes regular (**0**, 4.1.4) over the same field k, Y being projective over k; moreover Y is the union of two affine open sets Y_i ($i = 1, 2$) such that $f_i : X \times_Y Y_i \to Y_i$ is projective for $i = 1, 2$. Let then $Y' = Y_1 \amalg Y_2$ the prescheme coproduct; it is clear that the canonical morphism $g : Y' \to Y$ coinciding with the canonical injections from Y_1 and Y_2, is faithfully flat, and it is quasi-compact by virtue of (**I**, 5.5.10); however, $f' : X \times_Y Y' \to Y'$ (coinciding with f_i in each of the Y_i) is projective (**II**, 5.5.6), it is nonetheless not true of f itself. There exists therefore an $\mathscr{O}_X$-Module invertible $\mathscr{L}'$ which is f'-ample but which *is not of the form $g^*(\mathscr{L})$ for an $\mathscr{O}_X$-Module invertible $\mathscr{L}$*, by virtue of (2.7.2).

lcbel=(iii) Under the hypotheses of (2.7.1), one can assert that f' is a *local* isomorphism without f being a local immersion. Indeed, let k be a field, $\bar{k}$ an algebraic closure of k, K a separable extension of finite degree of k, distinct from k; then the structural morphism $f : X \to Y$, where $X = \mathrm{Spec}(\mathrm{K})$ and $Y = \mathrm{Spec}(k)$, is not a local immersion, but if one takes $Y' = \mathrm{Spec}(\bar{k})$, the morphism $Y' \to Y$ is faithfully flat and quasi-compact, and f' is a local isomorphism, since $X' = X \times_Y Y'$ is a coproduct of a finite number of schemes isomorphic to Y'.

2.8. Preschemes over a regular base of dimension 1; closure of a closed sub-prescheme of the generic fibre

Proposition (2.8.1). — *Let Y be a prescheme locally Noetherian, regular, irreducible, and of dimension* 1*, with generic point η, $f : X \to Y$ a morphism, $\mathrm{X}_\eta = f^{-1}(\eta)$ the fibre at the point generic, $i : \mathrm{X}_\eta \to X$ the canonical morphism. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent, $\mathscr{F}_\eta = i^*(\mathscr{F})$, $\mathscr{G}'$ an $\mathscr{O}_{\mathrm{X}_\eta}$-Module quotient of $\mathscr{F}_\eta$, and let $\overline{\mathscr{G}'}$ be the $\mathscr{O}_X$-Module, image of $\mathscr{F}$ by the composite homomorphism* (**0**, 4.4.3.2) **IV-2 | 34**

$$\mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} i_*(i^*(\mathscr{F})) \longrightarrow i_*(\mathscr{G}').$$

Then $\overline{\mathscr{G}'}$ is an $\mathscr{O}_X$-Module quotient of $\mathscr{F}$, quasi-coherent and f-flat, such that $i^*(\overline{\mathscr{G}'}) = \mathscr{G}'$, and it is the unique $\mathscr{O}_X$-Module quotient of $\mathscr{F}$ having these properties.

Proof. As i_* is quasi-compact and quasi-separated (1.1.2) and (1.2.2), it follows from (1.7.4) that for all $\mathscr{O}_{\mathrm{X}_\eta}$-Module quasi-coherent $\mathscr{H}'$, $i_*(\mathscr{H}')$ is an $\mathscr{O}_X$-Module quasi-coherent; moreover, for every open subset U of X, we have $(i_*(\mathscr{H}'))|U = (i|U \cap \mathrm{X}_\eta)_*(\mathscr{H}'|U \cap \mathrm{X}_\eta)$ by definition (**0**, 3.4.1). If one proves the proposition when X and Y are affine, it follows by gluing in the general case, in view of the assertion of uniqueness valid in the affine case. In other words, one is reduced to proving the

Lemma (2.8.1.1). — *Let* A *be a Noetherian regular ring* (**0**, 17.3.6)*, integral of dimension* 1, K *its field of fractions,* M *an* A*-module,* N' *a* K*-module quotient of* $\mathrm{M}_{(\mathrm{K})} = \mathrm{M} \otimes_\mathrm{A} \mathrm{K}$ *by a sub-*K*-module* P', $\overline{\mathrm{N}}'$ *the image of* M *by the composite homomorphism* $\mathrm{M} \to \mathrm{M}_{(\mathrm{K})} \to \mathrm{N}'$*. Then* $\overline{\mathrm{N}}'$ *is a flat* A*-module, and is the unique quotient module* N *of* M *which is a flat* A*-module and such that the kernel of the surjective homomorphism* $\mathrm{M}_{(\mathrm{K})} \to \mathrm{N}_{(\mathrm{K})}$ *is equal to* P'*.*

Proof. As for every maximal ideal $\mathfrak{m}$ of A, $\mathrm{A}_\mathfrak{m}$ is a local regular ring of dimension 1, and therefore a discrete valuation ring, it amounts to the same to say that an A-module N is flat or that it is *without torsion* (**0**, 6.3.4). As N' is a K-vector space, therefore without torsion, $\overline{\mathrm{N}'}$ is the same as $\overline{\mathrm{N}'}$, a sub-module of N'; moreover, it is immediately verified that $\overline{\mathrm{N}'}_{(\mathrm{K})}$ is identified with N'. Conversely, if N is a flat A-module quotient of M, the fact that N is an A-module flat implies that the canonical homomorphism $\mathrm{N} \to \mathrm{N}_{(\mathrm{K})} = \mathrm{N} \otimes_\mathrm{A} \mathrm{K}$ is *injective*. As $\mathrm{N}_{(\mathrm{K})}$ is identified with N', the conclusion results from the commutativity of the diagram

$$\begin{tikzcd} M \arrow[r] & N \\ M_{(K)} \arrow[u] \arrow[r] & N_{(K)} \arrow[u] \end{tikzcd}$$

□

□

Corollary (2.8.2). — *Under the conditions of (2.8.1), for $\mathscr{F}$ to be f-flat, it is necessary and it suffices that the canonical homomorphism $\mathscr{F} \to i_*(\mathscr{F}_\eta) = i_*(i^*(\mathscr{F}))$ be injective.*

(2.8.3). The formation of the $\mathscr{O}_X$-Module $\overline{\mathscr{G}'}$ is *functorial* in $\mathscr{F}$ and $\mathscr{G}'$: to be precise, if $\mathscr{F}_1$, $\mathscr{F}_2$ are two $\mathscr{O}_X$-Modules quasi-coherent, $u : \mathscr{F}_1 \to \mathscr{F}_2$ an $\mathscr{O}_X$-homomorphism, $\mathscr{G}'_i$ an $\mathscr{O}_{X_\eta}$-Module quotient of $(\mathscr{F}_i)_\eta$ $(i = 1, 2)$ and $v : \mathscr{G}'_1 \to \mathscr{G}'_2$ a homomorphism making the diagram

$$\begin{tikzcd} (\mathscr{F}_1)_\eta \arrow[r, "i^*(u)"] & (\mathscr{F}_2)_\eta \\ \mathscr{G}'_1 \arrow[u] \arrow[r, "v"'] & \mathscr{G}'_2 \arrow[u] \end{tikzcd}$$

commutative, then there is a unique homomorphism $w : \overline{\mathscr{G}'_1} \to \overline{\mathscr{G}'_2}$ making the diagram

$$\begin{tikzcd} \mathscr{F}_1 \arrow[r, "u"] & \mathscr{F}_2 \\ \overline{\mathscr{G}'_1} \arrow[u] \arrow[r, "w"'] & \overline{\mathscr{G}'_2} \arrow[u] \end{tikzcd}$$

commutative (determined uniquely, when it exists, by this property),
and this is a homomorphism determined uniquely (when it exists) by this property. **IV-2** | 35

Proposition (2.8.4). — *The hypotheses on Y being those of (2.8.1), let X_1, X_2 be two Y-preschemes, $\mathscr{F}_i$ an $\mathscr{O}_{X_i}$-Module quasi-coherent, $\mathscr{G}'_i$ an $\mathscr{O}_{(X_i)_\eta}$-Module quotient of $(\mathscr{F}_i)_\eta$ $(i = 1, 2)$. Then we have*

(2.8.4.1).

$$\overline{\mathscr{G}'_1 \otimes_{\mathscr{O}_Y} \mathscr{G}'_2} = \overline{\mathscr{G}'_1} \otimes_{k(\eta)} \overline{\mathscr{G}'_2}.$$

Proof. Indeed, setting $X = X_1 \times_Y X_2$; the first member of (2.8.3.1) is an $\mathscr{O}_X$-Module quasi-coherent which is Y-flat (**0**, 6.2.1), whose inverse image in X_η is $\mathscr{G}'_1 \otimes_{k(\eta)} \mathscr{G}'_2$ (**I**, 9.1.5) and which is a quotient of $\mathscr{F}_1 \otimes_{\mathscr{O}_Y} \mathscr{F}_2$; the conclusion results therefore from the property of uniqueness of (2.8.1). □

Proposition (2.8.5). — *The hypotheses on X and Y being those of (2.8.1), let Z' be a closed sub-prescheme of X_η. There exists then a unique closed sub-prescheme $\overline{Z'}$ of X which is Y-flat and such that $i^{-1}(\overline{Z'}) = Z'$.*

Proof. If $\mathscr{J}'$ is the quasi-coherent Ideal of $\mathscr{O}_{X_\eta}$ defining Z', it suffices to apply (2.8.1) to the case where $\mathscr{F} = \mathscr{O}_X$ and $\mathscr{G}' = \mathscr{O}_{X_\eta}/\mathscr{J}'$; if $\overline{\mathscr{G}'} = \mathscr{O}_X/\mathscr{J}$, we have indeed $\mathscr{J}' = i^*(\mathscr{J})\mathscr{O}_{X_\eta}$, therefore $i^{-1}(\overline{Z'}) = Z'$ (**I**, 4.4.5). □

We note that the prescheme $\overline{Z'}$ is the *closed image* of Z' by the composite morphism $Z' \to X_\eta \xrightarrow{i} X$, where the first arrow is the canonical injection (**I**, 9.5.3); its underlying space is the closure *in* X of Z' (**I**, 9.5.4), which justifies the adopted notation. We say moreover that $\overline{Z'}$ is the sub-prescheme of X *closure* of Z'.

Corollary (2.8.6). — *Let X_1, X_2 be two Y-preschemes, Z'_i a closed sub-prescheme of $(X_i)_\eta$ $(i = 1, 2)$. Then we have*

(2.8.6.1).

$$\overline{Z'_1 \times_Y Z'_2} = \overline{Z'_1} \times_{k(\eta)} \overline{Z'_2}.$$

Proof. This results from (2.8.4) and (2.8.5). □

§3. Associated prime cycles and primary decompositions

In this paragraph, we mainly give the translation of results on modules presented in Bourbaki, *Alg. comm.*, chap. IV, which we follow very closely. The notions that follow seem to be of interest only in the case of *locally Noetherian* preschemes. **IV-2** | 36

3.1. Prime cycles associated to a Module

Definition (3.1.1). — Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. We say that a point $x \in X$ is *associated* to $\mathscr{F}$ if the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$ is associated to the $\mathscr{O}_x$-Module $\mathscr{F}_x$ (in other words, if $\mathfrak{m}_x$ is the annihilator of an element of $\mathscr{F}_x$). We denote by $\mathrm{Ass}(\mathscr{F})$ the set of $x \in X$ associated to $\mathscr{F}$. We say that an irreducible closed subset Z of X is a *prime cycle associated* to $\mathscr{F}$ if its generic point is associated to $\mathscr{F}$. When $\mathscr{F} = \mathscr{O}_X$, we also say that the points (resp. *prime cycles*) associated to $\mathscr{F}$ are associated to the *prescheme* X.

We say that a prime cycle associated to $\mathscr{F}$ (resp. X) is *immersed* if it is contained in another prime cycle associated to $\mathscr{F}$ (resp. X) (in other words, if it is not *maximal* in the set of these cycles).

If X is locally Noetherian, the *irreducible components* of X are none other than the *maximal* (or *non-immersed*) prime cycles associated to X.

It is clear that if $x \in \mathrm{Ass}(\mathscr{F})$, then $\mathscr{F}_x \neq 0$, in other words

(3.1.1.1) $$\mathrm{Ass}(\mathscr{F}) \subset \mathrm{Supp}(\mathscr{F}).$$

If $x \in X$ is associated to $\mathscr{F}$, it is evidently associated to $\mathscr{F}|U$ for every open neighbourhood U of x, and conversely, if it is associated to $\mathscr{F}|U$ for one of these neighbourhoods, it is associated to $\mathscr{F}$.

We note finally that for a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the existence of immersed associated prime cycles is a *local* question, for if y and z are two points of $\mathrm{Ass}(\mathscr{F})$ such that $y \in \overline{\{z\}}$, every neighbourhood of y contains z.

Proposition (3.1.2). — *Let A be a Noetherian ring, M an A-module, $X = \mathrm{Spec}(A)$, $\mathscr{F} = \widetilde{M}$; for a point $x \in X$ to be associated to $\mathscr{F}$, it is necessary and sufficient that the prime ideal $\mathfrak{j}_x$ of A be associated to the module M (in other words, be the annihilator of an element $f \in M$).*

Proof. This follows from the definition (3.1.1) and from Bourbaki, *loc. cit.*, § 1, no 2, cor. of prop. 5, applied to $S = A - \mathfrak{j}_x$. ☐

Proposition (3.1.3). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, x a point of X, Z the closed reduced sub-prescheme of X having $\overline{\{x\}}$ as underlying space* (**I**, 5.2.1). *The following conditions are equivalent:*

label=a) $x \in \mathrm{Ass}(\mathscr{F})$. **IV-2** | 37

lbbel=b) *There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is equal to* $\mathrm{Supp}(\mathscr{O}_U \cdot f)$.

b′) *There exist an open neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{F})$ such that $U \cap Z$ is an irreducible component of* $\mathrm{Supp}(\mathscr{O}_U \cdot f)$.

lcbel=c) *There exist an open neighbourhood U of x and a sub-Module of $\mathscr{F}|U$ isomorphic to $\mathscr{O}_Z|U$ ($\mathscr{O}_Z$ being identified with a quotient of $\mathscr{O}_X$).*

c′) *There exist an open neighbourhood U of x and a coherent sub-Module $\mathscr{G}$ of $\mathscr{F}|U$ such that $U \cap Z$ is an irreducible component of* $\mathrm{Supp}(\mathscr{G})$.

Proof. It is clear that *c)* implies *b)*, for it suffices to take for f the element of $\Gamma(U, \mathscr{F})$ which corresponds to the unit section of $\mathscr{O}_Z|U$. As $U \cap Z$ is irreducible (**0**, 2.1.6), *b)* implies *b′)*, and *b′)* implies *c′)* since $\mathscr{O}_X$ is coherent (**0**, 5.3.4). To see that *c′)* implies *a)*, we can reduce to the case where $U = X = \mathrm{Spec}(A)$ is affine, A being Noetherian, and where $\mathscr{F} = \widetilde{M}$, where M is an A-module, and $\mathscr{G} = \widetilde{N}$, where N is a sub-module of M, of finite type. We then know that the minimal elements of $\mathrm{Supp}(\mathscr{G})$ are the maximal points of $V(\mathrm{Ann}(N))$ (**0**, 1.7.4), and these are also the minimal elements of $\mathrm{Ass}(N)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 3, cor. 1 of prop. 7); as $\mathrm{Ass}(N) \subset \mathrm{Ass}(M) = \mathrm{Ass}(\mathscr{F})$, we see that *c′)* implies *a)*. Finally, *a)* implies *c)*, by taking X affine again, $\mathscr{F} = \widetilde{M}$, and Z defined by the ideal fA (with the notations of (3.1.2)). ☐

Corollary (3.1.4). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Then the maximal prime cycles associated to $\mathscr{F}$ are the irreducible components of* $\operatorname{Supp}(\mathscr{F})$, *and the generic points of these components are the points $x \in X$ such that $\mathscr{F}_x$ is an $\mathscr{O}_x$-Module of finite length and $\neq 0$.*

Proof. Indeed, if x is the generic point of one of the irreducible components Z of $\operatorname{Supp}(\mathscr{F})$, it follows from the equivalence of *a)* and *c′)* in (3.1.3) that x belongs to $\operatorname{Ass}(\mathscr{F})$, and Z is a prime cycle associated to $\mathscr{F}$, necessarily maximal by virtue of (3.1.1.1); finally, the last assertion follows trivially from (3.1.1.1); indeed, the last assertion follows from a local result from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 2 of prop. 7. □

Corollary (3.1.5). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. For $\mathscr{F} = 0$, it is necessary and sufficient that $\operatorname{Ass}(\mathscr{F}) = \varnothing$.*

Proof. The question being local, we reduce to the case where X is affine, and the conclusion follows immediately from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, cor. 1 of prop. 2. □

Proposition (3.1.6). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent; then $\operatorname{Ass}(\mathscr{F})$ is* locally finite *(that is to say, every point of X admits a neighbourhood whose intersection with $\operatorname{Ass}(\mathscr{F})$ is finite).*

Proof. It suffices to consider the case where X is affine, hence Noetherian, and then the proposition follows from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of th. 2. □

Proposition (3.1.7). — *Let X be a prescheme.*

label=(i) *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules. Then $\operatorname{Ass}(\mathscr{F}') \subset \operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{F}') \cup \operatorname{Ass}(\mathscr{F}'')$.*
lbbel=(ii) *Let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-Module, $(\mathscr{F}_\alpha)$ a family of quasi-coherent sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$ such that $\mathscr{F}$ is the union of the $\mathscr{F}_\alpha$. Then $\operatorname{Ass}(\mathscr{F}) = \bigcup_\alpha \operatorname{Ass}(\mathscr{F}_\alpha)$.*
lcbel=(iii) *For every family $(\mathscr{F}_\alpha)$ of quasi-coherent $\mathscr{O}_X$-Modules, we have $\operatorname{Ass}(\bigoplus_\alpha \mathscr{F}_\alpha) = \bigcup_\alpha \operatorname{Ass}(\mathscr{F}_\alpha)$.*

Proof. We immediately reduce to the corresponding propositions for modules (Bourbaki, *loc. cit.*, § 1, no 1, formula (1), prop. 3 and cor. 1 of prop. 3). □

IV-2 | 38

Proposition (3.1.8). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, U an open subset of X, $\mathscr{J}$ a coherent Ideal of $\mathscr{O}_X$ defining a closed sub-prescheme of X having $X - U$ as underlying space. The following conditions are equivalent:*

*label=*a) $\operatorname{Ass}(\mathscr{F}) \subset U$.
*lbbel=*b) *For every affine open subset V, every section of $\mathscr{F}$ over V, whose restriction to $V \cap U$ is zero, is zero.*
*lcbel=*c) *The canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{F})$ is injective.*

Proof. The question being local, we can suppose $X = \operatorname{Spec}(A)$ affine, A being a Noetherian ring, $\mathscr{F} = \widetilde{M}$, where M is an A-module, and $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A. The homomorphism $M \to \operatorname{Hom}_A(\mathfrak{J}, M)$ sends $m \in M$ to the homomorphism $x \mapsto xm$ of $\mathfrak{J}$ into M; to say that it is not injective means that there exists $m \neq 0$ in M such that $\mathfrak{J}m = 0$.

Let us first show that *c)* implies *a)*: indeed, if $\operatorname{Ass}(\mathscr{F})$ met $X - U$, there would be a prime ideal $\mathfrak{p} \in \operatorname{Ass}(M)$ containing $\mathfrak{J}$, hence an element $m \neq 0$ of M such that $\mathfrak{J}m = 0$ for a $m \neq 0$ in M; indeed, if l had $\mathfrak{J}m = 0$ for all $a \in \mathfrak{J}$ not in $\mathfrak{q}$, for every prime ideal $\mathfrak{q} \neq \mathfrak{J}$ there exists $a \in \mathfrak{J}$ not in $\mathfrak{q}$, hence the relation $am = 0$ implies that the canonical image of m in $M_\mathfrak{q}$ is 0, so otherwise said the canonical image of m in $M_\mathfrak{q}$ is 0 for all $\mathfrak{q} \notin \mathfrak{J}$. In the second place, *b)* implies *c)*: indeed, if l had $\mathfrak{J}m = 0$ for a $m \neq 0$ in M, the section of $\mathscr{F}$ over U whose restriction to U is zero would be a section of $\mathscr{F}$ whose restriction to U is zero. Finally, *a)* implies *b)*. To see this, note that the canonical homomorphism $M \to \prod_{\mathfrak{p} \in \operatorname{Ass}(M)} M_\mathfrak{p}$ is *injective*: indeed, if N is the kernel of this homomorphism, we have $\operatorname{Ass}(N) \subset \operatorname{Ass}(M)$; if there existed $\mathfrak{p} \in \operatorname{Ass}(N)$, there would be an element $n \in N$ with $\mathfrak{p}$ as annihilator; but by definition of N, there exists $s \notin \mathfrak{p}$ such that $sn = 0$, which is absurd; hence $\operatorname{Ass}(N) = \varnothing$, whence $N = 0$. Now, condition *a)* implies that if $m \in M$ is such that the canonical image of m in $M_\mathfrak{p}$ is zero for all $\mathfrak{p} \in \operatorname{Ass}(M)$, then $m = 0$. □

Corollary (3.1.9). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, f a section of $\mathscr{O}_X$ over X. For f to be $\mathscr{F}$-regular* (**0**, 15.2.2), *it is necessary and sufficient that* $\mathrm{Ass}(\mathscr{F}) \subset X_f$.

Proof. Indeed, it is immediate that to say that f is $\mathscr{F}$-regular means that the canonical homomorphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(f\mathscr{O}_X, \mathscr{F})$ is injective, and it suffices to apply (3.1.8) to the Ideal $\mathscr{J} = f\mathscr{O}_X$. □

Proposition (3.1.10). — *Let X, Y be locally Noetherian preschemes, $f : X \to Y$ a whole morphism. Then, for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we have $f(\mathrm{Ass}(\mathscr{F})) = \mathrm{Ass}(f_*(\mathscr{F}))$.*

Proof. The question being local on Y and the morphism f being affine, we are immediately reduced to the case where Y is affine, in other words: □

IV-2 | 39

Lemma (3.1.10.1). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a ring homomorphism making B a whole A-algebra, M a B-module. Then the prime ideals $\mathfrak{p} \in \mathrm{Ass}(M_{[\rho]})$ are the inverse images by ρ of the prime ideals $\mathfrak{q} \in \mathrm{Ass}(M)$.*

Proof. Indeed, if $\mathfrak{q} \in \mathrm{Ass}(M)$, $\mathfrak{q}$ is the annihilator in B of an element $x \in M$, hence $\rho^{-1}(\mathfrak{q})$ is the annihilator in A of x. Conversely, suppose $\mathfrak{p} \in \mathrm{Ass}(M_{[\rho]})$, so that $\mathfrak{p}$ is the image by ρ of the annihilator $\mathfrak{b}$ in B of an element $x \in M$; it follows from the prime theorem of Cohen–Seidenberg that there exists a prime ideal $\mathfrak{q}$ of B containing $\mathfrak{b}$ and whose inverse image is $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, cor. 2 of th. 1); by considering a minimal prime ideal among those contained in $\mathfrak{q}$ and containing $\mathfrak{b}$, we can evidently suppose that $\mathfrak{q}$ is itself one of these minimal primes. But as $B.x \subset M$ is isomorphic to $B/\mathfrak{b}$, we know that one then has $\mathfrak{q} \in \mathrm{Ass}(B/\mathfrak{b}) \subset \mathrm{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 2). □

Corollary (3.1.11). — *Under the hypotheses of (3.1.10), for $\mathscr{F}$ to be without associated immersed prime cycle, it suffices that the same be true of $f_*(\mathscr{F})$.*

Proof. Suppose indeed that $f_*(\mathscr{F})$ has no immersed associated prime cycle. Note that if A is a whole algebra over a field k, all prime ideals of A are maximal (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1); it follows from (**I**, 6.2.2) that the fibres of f are *discrete* spaces; if x, x' are two distinct points of $\mathrm{Ass}(\mathscr{F})$, none of the two can be adherent to the other if $f(x) = f(x')$; and if $f(x) \neq f(x')$, (3.1.10) and the hypothesis imply that neither of the two points $f(x)$, $f(x')$ can be adherent to the other, hence the same is true of x and x'. □

Remark (3.1.12). — Under the hypotheses of (3.1.10), it may happen on the contrary that $\mathscr{F}$ is without associated immersed prime cycle, but not $f_*(\mathscr{F})$. Take for example $Y = \mathrm{Spec}(k[\mathrm{T}])$ where k is a field (an "affine line"), and X the *sum* of $X_1 = Y$ and $X_2 = \mathrm{Spec}(k)$, the morphism $X_2 \to Y$ corresponding to the canonical homomorphism $k[\mathrm{T}] \to k[\mathrm{T}]/\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal (T). It is clear that the morphism $f : X \to Y$ is finite; if one takes $\mathscr{F} = \mathscr{O}_X$, $\mathscr{F}$ is without associated immersed prime cycle, but $f_*(\mathscr{F}) = \widetilde{M}$, where M is the $k[\mathrm{T}]$-module direct sum of $k[\mathrm{T}]$ and k, hence $\mathrm{Ass}(M)$ consists of the generic point (0) of Y and of the point $\mathfrak{m}$.

Proposition (3.1.13). — *Let X be a prescheme locally Noetherian, U an open subset of X, $i : U \to X$ the canonical injection. For every quasi-coherent $\mathscr{O}_U$-Module $\mathscr{F}$, we have $\mathrm{Ass}(i_*(\mathscr{F})) = \mathrm{Ass}(\mathscr{F})$.*

Proof. Recall that $i_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-Module (**I**, 2.2) and (**I**, 7.4); as $i_*(\mathscr{F})|U = \mathscr{F}$, on has $(\mathrm{Ass}(i_*(\mathscr{F}))) \cap U = \mathrm{Ass}(\mathscr{F})$, and it remains therefore to prove that $\mathrm{Ass}(i_*(\mathscr{F})) \subset U$. But by (3.1.8), this relation means that for every affine open subset V of X, every section of $i_*(\mathscr{F})$ over V which is zero on $U \cap V$, is zero, a condition which is trivially satisfied since $\Gamma(V, i_*(\mathscr{F})) = \Gamma(U \cap V, \mathscr{F}) = \Gamma(U \cap V, i_*(\mathscr{F}))$. □

IV-2 | 40

3.2. Irredundant decompositions

Proposition (3.2.1). — *Let X be a prescheme locally Noetherian, U an open dense subset of X. The following conditions are equivalent:*

label=a) *X is reduced.*
lbbel=b) *The sub-prescheme induced on U is reduced and X is without associated immersed prime cycle.*
lcbel=c) *X is without associated immersed prime cycle and for every generic point x of an irreducible component of X, we have* $\operatorname{length}(\mathscr{O}_x) = 1$.

The prime cycles associated to X are then identical to the irreducible components of X.

Proof. It is clear that if X is reduced then so is the sub-prescheme induced on U. In addition, the existence of associated immersed prime cycles being a local question, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, A being Noetherian. If A is reduced, we know that the minimal prime ideals of A form a reduced primary decomposition of (0) (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, prop. 10) and these are all the elements of $\operatorname{Ass}(A)$, hence there are no immersed prime ideals associated to A, which shows that *a)* implies *b)*. It is immediate that *b)* implies *c)*, for a generic point x of an irreducible component belongs to U, hence $\mathscr{O}_x$ is a field. Finally, *c)* implies *a)*: it suffices indeed to note that if $\mathscr{N}$ is the Nilradical of $\mathscr{O}_X$, which is a coherent Ideal, $\operatorname{Supp}(\mathscr{N})$ cannot contain any of the generic points of the irreducible components of X by hypothesis; if $\operatorname{Supp}(\mathscr{N})$ were non-empty and if x was one of its maximal points, the criterion (3.1.3), *c′*)) would show that $x \in \operatorname{Ass}(\mathscr{O}_X)$, and $\overline{\{x\}}$ would therefore be an associated immersed prime cycle of X, contradicting the hypothesis; hence $\mathscr{N} = 0$. □

Definition (3.2.2). — Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. We say that $\mathscr{F}$ is *reduced* if it satisfies the following two conditions: 1° $\mathscr{F}$ is without associated immersed prime cycle; 2° for every maximal point x of $\operatorname{Supp}(\mathscr{F})$, we have $\operatorname{length}(\mathscr{F}_x) = 1$.

Condition 1° means that the prime cycles associated to $\mathscr{F}$ are the irreducible components of $\operatorname{Supp}(\mathscr{F})$ (3.1.4), and condition 2° means that for every generic point x of such a component, we have $\operatorname{length}(\mathscr{F}_x) = 1$.

For an affine scheme X, this definition gives in particular the notion of a *reduced module* over a Noetherian ring A; a finitely generated A-module M is said to be *reduced* if it has no immersed associated prime ideals and if, for all $\mathfrak{p} \in \operatorname{Ass}(M)$, $\operatorname{length}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we say that $\mathscr{F}$ is *reduced at a point* $x \in X$ if $\mathscr{F}_x$ is a reduced $\mathscr{O}_x$-module; this means that x does not belong to any associated immersed prime cycle of $\mathscr{F}$ and that $\operatorname{length}(\mathscr{F}_z) = 1$ *for all maximal points z of* $\operatorname{Supp}(\mathscr{F})$ *such that* $x \in \overline{\{z\}}$. It is clear that if $\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module reduced, it is reduced at every point of X; conversely, if $\mathscr{F}$ is *reduced at a point* x, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is an $\mathscr{O}_X$-Module *reduced*: it suffices indeed to take U such that $\mathscr{F}|U$ does not meet any associated immersed prime cycle of $\mathscr{F}$ (such a neighbourhood exists since these cycles form a locally finite set of closed subsets of X). To say that $\mathscr{O}_X$ is reduced at a point x is equivalent to saying that X is *reduced* at the point x.

We say that the locally Noetherian prescheme X is *irredundant* if $\mathscr{O}_X$ is irredundant (which implies that X is *irreducible*); for X to be *integral*, it is necessary and sufficient that the $\mathscr{O}_X$-Module $\mathscr{O}_X$ be integral ((**I**, 2.1.8) and (3.2.1)). If $\mathscr{O}_X$ is integral at a point x, that is to say if the ring $\mathscr{O}_x$ is integral, we say that X is *integral at the point* x. We say that a closed sub-prescheme Y of X is *primary* in X if the Ideal $\mathscr{J}$ of $\mathscr{O}_X$ which defines Y is primary in $\mathscr{O}_X$. **IV-2** | 41

Proposition (3.2.3). — *Let X be a prescheme locally Noetherian, U an open subset of X, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent such that $U \cap \operatorname{Supp}(\mathscr{F})$ is dense in $\operatorname{Supp}(\mathscr{F})$. The following conditions are equivalent:*

label=a) *$\mathscr{F}$ is reduced.*
lbbel=b) *$\mathscr{F}|U$ is reduced, and $\mathscr{F}$ is without associated immersed prime cycle.*
lcbel=c) *There exists a closed reduced sub-prescheme X' of X and a coherent $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ without torsion and of rank 1 on every irreducible component of X', such that if $j : X' \to X$ is the canonical injection, we have $j_*(\mathscr{F}') = \mathscr{F}$.*

Furthermore, when this is so, the sub-prescheme X' is defined by the Ideal $\mathscr{J}$ of $\mathscr{O}_X$ annihilator of $\mathscr{F}$.

Proof. To see that *c)* implies *a)*, we can, by virtue of (3.1.3), limit ourselves to the case where $X' = X$ is integral, of generic point x, and one can in addition suppose $X = \operatorname{Spec}(A)$ affine and $\mathscr{F} = \widetilde{M}$, where A is integral and M is an A-module without torsion of rank 1; the annihilator of every element of M being then reduced to 0, we have $\operatorname{Ass}(\mathscr{F}) = \{x\}$ and $\mathscr{F}_x$ is isomorphic to $\mathscr{O}_x$, the field of fractions of A, hence the conditions of the definition (3.2.2) are satisfied. As the existence of associated immersed prime cycles is a local question, it is clear that if $\mathscr{F}$ has no such cycles and if $U \cap \operatorname{Supp}(\mathscr{F})$ is dense in $\operatorname{Supp}(\mathscr{F})$, we have $\operatorname{Ass}(\mathscr{F}) = \operatorname{Ass}(\mathscr{F}|U)$, hence *a)* and *b)* are equivalent. If *a)* is satisfied, take for X' the closed reduced sub-prescheme of X whose $\operatorname{Supp}(\mathscr{F})$ is the underlying space (**I**, 5.2.1), and let $\mathscr{F}' = j^*(\mathscr{F})$; a point x of $\operatorname{Ass}(\mathscr{F})$ is necessarily a maximal point of X' and as $\operatorname{length}(\mathscr{F}_x) = 1$, $\mathscr{F}'_x$ is isomorphic to $\mathscr{F}_x$, hence to the residue field $\boldsymbol{k}(x) = \mathscr{O}_{X',x}$, which proves that $\mathscr{F}'$ is without torsion and of rank 1 (**I**, 7.4.6) and (**I**, 7.4.2). Finally, the last assertion is trivial, since for all $y \in X'$, the annihilator of the $\mathscr{O}_{X',y}$-Module $\mathscr{F}'_y$ is zero. □

Definition (3.2.4). — Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. We say that $\mathscr{F}$ is *irredundant* if $\operatorname{Ass}(\mathscr{F})$ is reduced to a single element x; if $\mathscr{F}$ is of finite type, we say that $\mathscr{F}$ is *integral* if in addition $\mathscr{F}$ is reduced (in other words if $\operatorname{length}(\mathscr{F}_x) = 1$). We say that a quasi-coherent sub-$\mathscr{O}_X$-Module $\mathscr{G}$ of a quasi-coherent $\mathscr{O}_X$-Module coherent $\mathscr{F}$ is *primary* in $\mathscr{F}$ if $\mathscr{F}/\mathscr{G}$ is irredundant.

For an affine scheme X, this definition gives in particular the notion of *integral module* over a Noetherian ring A; a finitely generated A-module M is said to be *integral* if it is of finite type, if $\operatorname{Ass}(M)$ is reduced to a single prime ideal $\mathfrak{p}$) and if in addition $\operatorname{length}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we say that $\mathscr{F}$ is *integral at a point* $x \in X$ if $\mathscr{F}_x$ is an integral $\mathscr{O}_x$-module: this means that *x belongs to at most a single associated prime cycle of $\mathscr{F}$ and that* $\operatorname{length}(\mathscr{F}_z) = 1$ *at its generic point z*. It is clear that if $\mathscr{F}$ is a coherent integral $\mathscr{O}_X$-Module, it is integral at every point of X; conversely, if $\mathscr{F}$ is *integral at a point* x, there exists an open neighbourhood U of x such that $\mathscr{F}|U$ is an $\mathscr{O}_U$-Module *integral*: it suffices indeed to take U such that $\mathscr{F}|U$ is an $\mathscr{O}_X$-Module reduced.

IV-2 | 42

Definition (3.2.5). — Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. An *irredundant decomposition* of $\mathscr{F}$ is called a family $(\mathscr{F}_\alpha)_{\alpha \in I}$ of $\mathscr{O}_X$-Module quotients of $\mathscr{F}$ such that the $\mathscr{F}_\alpha$ are irredundant, that the family $(\operatorname{Supp}(\mathscr{F}_\alpha))$ is locally finite, and that the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is injective. We say that such a decomposition is *reduced* if the sets $\operatorname{Ass}(\mathscr{F}_\alpha)$ are pairwise distinct, and if there does not exist a proper subset $J \subsetneq I$ such that the subfamily $(\mathscr{F}_\alpha)_{\alpha \in J}$ is also an irredundant decomposition of $\mathscr{F}$.

If $(\mathscr{F}_\alpha)_{\alpha \in I}$ is an irredundant decomposition (resp. irredundant reduced decomposition) of $\mathscr{F}$, and if one sets $\mathscr{F}_\alpha = \mathscr{F}/\mathscr{G}_\alpha$, we say that the family $(\mathscr{G}_\alpha)_{\alpha \in I}$ of sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$ is a *primary decomposition* (resp. *irredundant reduced primary decomposition*) of $\mathscr{F}$; one observes that the hypothesis of injectivity of the canonical homomorphism $\mathscr{F} \to \bigoplus_{\alpha \in I} \mathscr{F}_\alpha$ is equivalent to the condition $\bigcap_{\alpha \in I} \mathscr{G}_\alpha = 0$.

If $(\mathscr{F}_\alpha)$ is an irredundant decomposition of $\mathscr{F}$, to say that it is reduced amounts to saying that IV-2 | 42
the sets $\operatorname{Ass}(\mathscr{F}_\alpha)$ are pairwise distinct and contained in $\operatorname{Ass}(\mathscr{F})$; if $\operatorname{Ass}(\mathscr{F}_\alpha) = \{x_\alpha\}$ for all $\alpha \in I$, $\alpha \mapsto x_\alpha$ is then a bijection of I onto $\operatorname{Ass}(\mathscr{F})$: these properties are local in nature and follow from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 4.

Proposition (3.2.6). — *Let X be a prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Then there exists a reduced irredundant decomposition $(\mathscr{F}^{(x)})_{x \in \operatorname{Ass}(\mathscr{F})}$, formed of coherent $\mathscr{O}_X$-Modules, such that for every $x \in \operatorname{Ass}(\mathscr{F})$, we have $\operatorname{Ass}(\mathscr{F}^{(x)}) = \{x\}$. For every $x \in \operatorname{Ass}(\mathscr{F})$ such that $\overline{\{x\}}$ is not immersed, $\mathscr{F}^{(x)}$ is uniquely determined as the image of the canonical homomorphism $\mathscr{F} \to i_*(i^*(\mathscr{F}))$, where i is the canonical morphism* $\operatorname{Spec}(\boldsymbol{k}(x)) \to X$.

Proof. For every $x \in \operatorname{Ass}(\mathscr{F})$, let U be an affine open neighbourhood of x, and let $\mathscr{F}|U = \widetilde{M}$, where M is a finitely generated A-module. We know (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 1, prop. 4) that there exists a sub-module N of M such that, if one sets $P = M/N$, one has $\operatorname{Ass}(P) = \{x\}$ and $\operatorname{Ass}(N) = \operatorname{Ass}(M) - \{x\}$. Let $\mathscr{G} = \widetilde{P}$, which is a quasi-coherent $\mathscr{O}_U$-Module; let $u : j^*(\mathscr{F}) \to \mathscr{G}$ be the surjective homomorphism corresponding to the surjection $M \to P$; we obtain by composition a homomorphism $j_*(u) : j_*(j^*(\mathscr{F})) \to j_*(\mathscr{G})$, whence by composition a homomorphism

$$v : \mathscr{F} \xrightarrow{\rho_{\mathscr{F}}} j_*(j^*(\mathscr{F})) \xrightarrow{j_*(u)} j_*(\mathscr{G})$$

whose restriction u to U; we will denote by $\mathscr{F}^{(x)}$ the image of $\mathscr{F}$ by this homomorphism, which is a quasi-coherent $\mathscr{O}_X$-Module coherent (**I**, 6.1.1). We have $\operatorname{Ass}(j_*(\mathscr{G})) = \{x\}$ by virtue of (3.1.13), and *a fortiori* (3.1.7) $\operatorname{Ass}(\mathscr{F}^{(x)}) = \{x\}$, since $\mathscr{F}^{(x)} \neq 0$. Furthermore, if $\overline{\{x\}}$ is not immersed, the characterisation of $\mathscr{F}^{(x)}$ when $\overline{\{x\}}$ is not immersed follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 5, the question being local, and taking into account (**I**, 1.6.7).

$\mathscr{N}^{(x)} = \operatorname{Ker}(v)$, hence $x \notin \operatorname{Ass}(\mathscr{N}^{(x)})$, and $\mathscr{N}^{(x)}|U = \tilde{N}$, hence $x \notin \operatorname{Ass}(\mathscr{N}^{(x)})$. It follows that the homomorphism $\mathscr{F} \to \bigoplus_{x\in\operatorname{Ass}(\mathscr{F})} \mathscr{F}^{(x)}$ is injective, for its kernel $\mathscr{H}$ is contained in the intersection of the $\operatorname{Ass}(\mathscr{N}^{(x)})$, which is empty; by (3.1.5), $\mathscr{H} = 0$. Taking into account that $\operatorname{Ass}(\mathscr{F})$ is locally finite (3.1.6), it is clear that $(\mathscr{F}^{(x)})_{x\in\operatorname{Ass}(\mathscr{F})}$ is a reduced irredundant decomposition of $\mathscr{F}$ satisfying the conditions of the statement. □

IV-2 | 43

Corollary (3.2.7). — *Under the hypotheses of (3.2.6), if $\mathscr{F}$ has no associated immersed prime cycle, there exists a unique reduced irredundant decomposition of $\mathscr{F}$.*

Corollary (3.2.8). — *Let X be a Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. There exists a finite filtration $(\mathscr{F}_i)_{0\leqslant i\leqslant n}$ of $\mathscr{F}$ such that $\mathscr{F}_0 = \mathscr{F}$, $\mathscr{F}_n = 0$, formed of coherent $\mathscr{O}_X$-Modules, and such that the quotients $\mathscr{F}_i/\mathscr{F}_{i+1}$ are either zero or irredundant, and $\operatorname{Ass}(\mathscr{F}_i/\mathscr{F}_{i+1}) \subset \operatorname{Ass}(\mathscr{F})$.*

Proof. Indeed, $\mathscr{F}$ is isomorphic to a sub-$\mathscr{O}_X$-Module direct sum $\bigoplus_{j=1}^n \mathscr{G}_j$, where the $\mathscr{G}_j$ are irredundant and coherent (3.2.6); as every sub-$\mathscr{O}_X$-Module quasi-coherent of $\mathscr{G}_j$ is either zero or irredundant (3.1.7), the $\mathscr{F}_i = \mathscr{F} \cap (\bigoplus_{j=1}^{n-i} \mathscr{G}_j)$ satisfy the requirements, $\mathscr{F}_i/\mathscr{F}_{i+1}$ being isomorphic to a sub-$\mathscr{O}_X$-Module coherent of $\mathscr{G}_{n-i}$. □

3.3. Relations with flatness

Proposition (3.3.1). — *Let $f : X \to Y$ be a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent and f-flat, $\mathscr{E}$ an $\mathscr{O}_Y$-Module quasi-coherent. If, for every $y \in Y$, we set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} \boldsymbol{k}(y)$, then*

(3.3.1.1)
$$\operatorname{Ass}(\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}) \supset \bigcup_{y\in\operatorname{Ass}(\mathscr{E})} \operatorname{Ass}(\mathscr{F}_y)$$

and the two sides are equal if Y is locally Noetherian.

Proof. (Of course, $\mathscr{F}_y$ is a sheaf on the fibre $f^{-1}(y)$, and one identifies it with a sub-space of X (**I**, 3.6.1).) The question being local on X and on Y, we reduce to the case where X and Y are affine, and the proposition is then proved in Bourbaki, *Alg. comm.*, chap. IV, § 2, no 6, th. 2. □

Corollary (3.3.2). — *Let Y be a prescheme locally Noetherian without associated immersed prime cycles, $f : X \to Y$ a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent and f-flat. Then, for every $x \in \operatorname{Ass}(\mathscr{F})$, $f(x)$ is a maximal point of Y.*

Proof. It suffices to apply (3.3.1) with $\mathscr{E} = \mathscr{O}_Y$, since $\operatorname{Ass}(\mathscr{O}_Y)$ is by hypothesis the set of maximal points of Y. □

Corollary (3.3.3). — *Under the hypotheses of (3.3.1), suppose in addition that X and Y are locally Noetherian, $\mathscr{E}$ and $\mathscr{F}$ coherent. Then the following conditions are equivalent:*

label=a) *$\mathscr{E} \otimes_Y \mathscr{F}$ is without associated immersed prime cycle.*

lbbel=b) *For every point $y \in \operatorname{Ass}(\mathscr{E}) \cap f(\operatorname{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ is without associated immersed prime cycle.*

Proof. Suppose *a)* is satisfied. The hypotheses imply that $\mathscr{E} \otimes_Y \mathscr{F}$ is a coherent $\mathscr{O}_X$-Module (**I**, 5.3.11) **IV-2 | 44** and (**I**, 5.3.5); its associated prime cycles are therefore the irreducible components of $\operatorname{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$ (3.1.4), and for every maximal point x of $\operatorname{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, $f(x) = y$ is a maximal point of $\operatorname{Supp}(\mathscr{E})$ and x is a maximal point of $\operatorname{Supp}(\mathscr{F}_y)$ (2.5.5). As, by virtue of (3.3.1) and the fact that $y \in f(\operatorname{Supp}(\mathscr{F}))$

implies $\mathscr{F}_y \neq 0$ (**I**, 9.1.13), every point of $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$ is the image by f of a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, we see that condition *b)* is satisfied.

Conversely, suppose *b)* is satisfied, and let us show that if z, z' are two distinct points of $\mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F})$, neither of the two can be adherent to the other. In the first place, if $f(z) = f(z') = y$, we have $y \in \mathrm{Ass}(\mathscr{E})$ and z and z' belong to $\mathrm{Ass}(\mathscr{F}_y)$ by (3.3.1), whence $y \notin f(\mathrm{Supp}(\mathscr{F}))$; as by hypothesis neither of the two points z, z' is adherent to the other *within* $f^{-1}(y)$, neither can be adherent to the other *within* X. If $y = f(z)$ and $y' = f(z')$ are distinct, they belong to $\mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, hence neither of the two can be adherent to the other in Y; it follows from the continuity of f that neither of the points z, z' can be adherent to the other in X. ☐

Proposition (3.3.4). — *Let X, Y be two preschemes locally Noetherian, $f : X \to Y$ a morphism, $\mathscr{E}$ an $\mathscr{O}_Y$-Module coherent, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent and f-flat. Then the following conditions are equivalent:*

label=a) *$\mathscr{E} \otimes_Y \mathscr{F}$ is reduced (3.2.2).*

lbbel=b) *For every point $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ is reduced.*

Proof. Suppose *a)* is satisfied. We already know (3.3.3) that for every $y \in \mathrm{Ass}(\mathscr{E}) \cap f(\mathrm{Supp}(\mathscr{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathscr{E}$ and $\mathscr{F}_y$ is without associated immersed prime cycle. Furthermore (2.5.5), for every $x \in \mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F}) \cap f^{-1}(y)$, we have $1 = \mathrm{length}((\mathscr{E} \otimes_Y \mathscr{F})_x) = \mathrm{length}(\mathscr{E}_y) \cdot \mathrm{length}((\mathscr{F}_y)_x)$, hence $\mathrm{length}(\mathscr{E}_y) = 1$, which proves *b)*.

Conversely, suppose *b)* is satisfied; we already know that every point $x \in \mathrm{Ass}(\mathscr{E} \otimes_Y \mathscr{F})$ is a maximal point of $\mathrm{Supp}(\mathscr{E} \otimes_Y \mathscr{F})$, that $y = f(x)$ is a maximal point of $\mathrm{Supp}(\mathscr{E})$ and x a maximal point of $\mathrm{Supp}(\mathscr{F}_y)$ (3.3.1) and (3.3.3); in addition it follows from the hypothesis and from (2.5.5) that $\mathrm{length}((\mathscr{E} \otimes_Y \mathscr{F})_x) = 1$, which proves *a)*. ☐

Corollary (3.3.5). — *Let X, Y be two preschemes locally Noetherian, $f : X \to Y$ a flat morphism; if Y is reduced at the points of $f(X)$ and if $f^{-1}(y)$ is a $\boldsymbol{k}(y)$-reduced prescheme for every $y \in f(X)$, then X is reduced.*

Proof. As the Nilradical $\mathscr{N}_Y$ is coherent, the set of points where Y is reduced is open (**0**, 5.2.2), and one can restrict to the case where Y is reduced. It suffices then to apply (3.3.4) with $\mathscr{E} = \mathscr{O}_Y$ and $\mathscr{F} = \mathscr{O}_X$. ☐

Proposition (3.3.6). — *Let $f : X \to S$, $g : Y \to S$ be two morphisms, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, $\mathscr{G}$ an $\mathscr{O}_Y$-Module quasi-coherent. Suppose that: 1º $\mathscr{G}$ is g-flat; 2º X is locally Noetherian, and for every $s \in S$, $g^{-1}(s)$ is locally Noetherian (which will hold if Y is also locally Noetherian). Let $Z = X \times_S Y$; for every pair (x, y) such that $x \in X$, $y \in Y$, and $f(x) = g(y) = s$, let $\mathrm{T}_{x,y}$ be the prescheme $\mathrm{Spec}(\boldsymbol{k}(x) \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(y))$, and* **IV-2 | 45**
*let $\mathrm{I}_{x,y}$ be the image of $\mathrm{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}})$ by the canonical monomorphism $\mathrm{T}_{x,y} \to Z$ (**I**, 3.4.9). Then*

(3.3.6.1) $$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \Bigg(\bigcup_{y \in \mathrm{Ass}(\mathscr{G}_{f(x)})} \mathrm{I}_{x,y} \Bigg)$$

where for all $s \in S$, $\mathscr{G}_s = \mathscr{G} \otimes_{\mathscr{O}_S} \boldsymbol{k}(s)$.

Proof. Let $p : Z \to X$, $q : Z \to Y$ be the canonical projections, so that we have the commutative diagram

$$\begin{tikzcd} X & Z \arrow[l, "p"'] \arrow[r, "q"] & Y \\ & S \arrow[ul, "f"] \arrow[ur, "g"'] & \end{tikzcd}$$

Set $\mathscr{G}' = q^*(\mathscr{G})$, so that $\mathscr{F} \otimes_S \mathscr{G} = \mathscr{F} \otimes_X \mathscr{G}'$; as $\mathscr{G}'$ is p-flat (2.1.4), it follows from (3.3.1) that

(3.3.6.2) $$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \mathrm{Ass}(\mathscr{G}'_x)$$

with $\mathscr{G}'_x = \mathscr{G}' \otimes_{\mathscr{O}_X} \boldsymbol{k}(x)$. If $s = f(x)$, we have $\mathscr{G}'_x = \mathscr{G}_s \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(x)$, and $p^{-1}(x) = g^{-1}(s) \otimes_{\boldsymbol{k}(s)} \boldsymbol{k}(x)$; furthermore, as the field $\boldsymbol{k}(x)$ is a flat $\boldsymbol{k}(s)$-module, the morphism $p^{-1}(x) \to g^{-1}(s)$ is flat; applying

(3.3.1) to this morphism, we get

$$\mathrm{Ass}(\mathscr{G}'_x) = \bigcup_{y \in \mathrm{Ass}(\mathscr{G}_s)} \mathrm{Ass}(\mathscr{O}_{\mathrm{T}_{x,y}}) \tag{3.3.6.3}$$

whence the proposition. □

We note that if, in the statement, we suppress hypothesis 2°, we can still conclude, by virtue of (3.3.1), the relation

$$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) \supset \bigcup_{x \in \mathrm{Ass}(\mathscr{F})} \Big(\bigcup_{y \in \mathrm{Ass}(\mathscr{G}_{f(x)})} \mathrm{I}_{x,y} \Big). \tag{3.3.6.4}$$

Corollary (3.3.7). — *Under the hypotheses of (3.3.6), suppose in addition that S is also locally Noetherian and that $f(\mathrm{Ass}(\mathscr{F})) \subset \mathrm{Ass}(\mathscr{O}_S)$. Then*

$$\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G}) = \bigcup_{(x,y) \in C} \mathrm{I}_{x,y} \tag{3.3.7.1}$$

where C is the set of pairs (x, y) such that $x \in \mathrm{Ass}(\mathscr{F})$, $y \in \mathrm{Ass}(\mathscr{G})$ and $f(x) = g(y)$.

Proof. As $\mathscr{G}$ is g-flat, it follows indeed from (3.3.1) that the relation "$s \in \mathrm{Ass}(\mathscr{G}_s)$" is equivalent to "$y \in \mathrm{Ass}(\mathscr{G})$": the conclusion follows from (3.3.6.1). □

Remark (3.3.8). — We shall see later (4.2.2) that under the hypotheses of (3.3.6), $\mathrm{T}_{x,y}$ is a prescheme *without associated immersed prime cycle*; it will follow that if $\mathscr{F}$ and the $\mathscr{G}_s$ are without associated immersed prime cycle, the same is true of $\mathscr{F} \otimes_S \mathscr{G}$.

Corollary (3.3.9). — *Under the conditions of (3.3.7), we have*

$$q(\mathrm{Ass}(\mathscr{F} \otimes_S \mathscr{G})) \subset \mathrm{Ass}(\mathscr{G}) \tag{3.3.9.1}$$

(where $q : X \times_S Y \to Y$ is the canonical projection).

Proof. Indeed, if $(x, y) \in Z$, we have $q(\mathrm{I}_{x,y}) = \{y\} \subset \mathrm{Ass}(\mathscr{G})$. □

IV-2 | 46

3.4. Properties of the sheaves $\mathscr{F}/t\mathscr{F}$

Proposition (3.4.1). — *Let X be a prescheme locally Noetherian, t a section of $\mathscr{O}_X$ over X, Y the closed sub-prescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent, S the closed reduced sub-prescheme of X having as underlying space $\mathrm{Supp}(\mathscr{F})$, (S_i) the family of closed reduced sub-preschemes of X having as underlying spaces the irreducible components of S; we denote by s_i the generic point of S_i. Finally, let Z be an irreducible component of $\mathrm{Supp}(\mathscr{F}/t\mathscr{F}) = S \cap Y$, z its generic point.*

label=(i) For every i such that $Z \subset S_i$, Z is an irreducible component of $S_i \cap Y$.

lbbel=(ii) If Z is not equal to any of the S_i, then

$$\mathrm{length}((\mathscr{F}/t\mathscr{F})_z) \geqslant \sum_i \mathrm{length}(\mathscr{F}_{s_i}) \tag{3.4.1.1}$$

where the sum on the right-hand side is extended to all i such that $Z \subset S_i$.

resume,label=(i) Suppose that Z is not equal to any of the S_i. For the two sides of (3.4.1.1) to be equal, it is necessary and sufficient that the following two conditions be satisfied:

α) t_z is $\mathscr{F}_z$-regular (**0**, 15.1.4).

β) For every i such that $Z \subset S_i$, there is a unique S_i containing Z, and for this value of i, we have $\mathrm{length}(\mathscr{F}_{s_i}) = 1$; furthermore $\mathscr{O}_{S_i,z}$ is a discrete valuation ring whose uniformiser is t_z, and t_z is $\mathscr{F}_z$-regular.

Proof. (i) As in the proof of (3.4.1), we can reduce to the case where $X = \mathrm{Spec}(\mathscr{O}_z)$; the proposition then follows (taking into account (3.1.2)) from (**0**, 16.4.6.3).

(ii) and (iii) Suppose that Z is not equal to any of the S_i. For the two sides of (3.4.1.1) to be equal, it is necessary and sufficient that the following two conditions be satisfied:

label=α) t_z is $\mathscr{F}_z$-regular (**0**, 15.2.4).

lbbel=α) For every i such that $Z \subset S_i$, the canonical image of the germ t_z in $\mathscr{O}_{S_i,z}$ generates the maximal ideal of this ring (which implies that $\mathscr{O}_{S_i,z}$ is a discrete valuation ring and the image of t_z is a uniformiser of this ring).

□

Corollary (3.4.2). — *Under the general hypotheses of (3.4.1), suppose that Z is not equal to any of the S_i and that $\operatorname{length}((\mathscr{F}/t\mathscr{F})_z) = 1$. Then there is a unique S_i containing Z, and for this value of i, $\operatorname{length}(\mathscr{F}_{s_i}) = 1$; furthermore $\mathscr{O}_{S_i,z}$ is a discrete valuation ring whose uniformiser is t_z, and t_z is $\mathscr{F}_z$-regular.*

Proof. This follows from (3.4.1), the two sides of (3.4.1.1) being then equal. □

Proposition (3.4.3). — *Let X be a prescheme locally Noetherian, t a section of $\mathscr{O}_X$ over X, Y the closed sub-prescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent, T a prime cycle associated to $\mathscr{F}$, T' an irreducible component of $T \cap Y$, x the generic point of T'. Suppose that t_x is $\mathscr{F}_x$-regular; then $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$.*

Proof. As in the proof of (3.4.1), we can reduce to the case where $X = \operatorname{Spec}(\mathscr{O}_x)$; the proposition then follows from (3.1.2), as a consequence of (**0**, 16.4.6.3). □

Proposition (3.4.4). — *Let X be a prescheme locally Noetherian, t a section of $\mathscr{O}_X$ over X, Y the closed sub-prescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent, (S_i) the family of irreducible components of $\operatorname{Supp}(\mathscr{F})$. Let y be a point of Y such that t_y is $\mathscr{F}_y$-regular and such that none of the associated immersed prime cycles of $\mathscr{F}/t\mathscr{F}$ contains y. Then the irreducible components of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ that contain y are exactly the irreducible components of the $S_i \cap Y$ that contain y, and the prime cycles associated to $\mathscr{F}$ that contain y are non-immersed.*

Proof. Let us first prove the last assertion. Let $T \supset T_1$ be two prime cycles associated to $\mathscr{F}$ containing y; if x is the generic point of an irreducible component of $T \cap Y$ containing y, x is the generic point of an irreducible component of $T_1 \cap Y$ containing y, and the hypothesis that t_y is $\mathscr{F}_y$-regular implies that t_x is $\mathscr{F}_x$-regular (**0**, 15.2.4), hence, by virtue of (3.4.3), $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$. Let x_1 be the generic point of an irreducible component of $T_1 \cap Y$ containing y, and let x be the generic point of an irreducible component of $T \cap Y$ containing x_1; it follows from the preceding that x_1 and x both belong to $\operatorname{Ass}(\mathscr{F}/t\mathscr{F})$, and as $x_1 \in \overline{\{x\}}$, the hypothesis implies that $x_1 = x$. Let us still denote by T and T_1 the integral closed sub-preschemes of X having T and T_1 as underlying spaces respectively, and set $A = \mathscr{O}_{T,x}$, $A_1 = \mathscr{O}_{T_1,x}$; we thus have $A_1 = A/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A. By definition of x and x_1, A/tA and A_1/tA_1 are Artinian rings; on the other hand, since x cannot belong to $\operatorname{Ass}(\mathscr{F})$, and since A_1 is not Artinian, x cannot belong to $\operatorname{Ass}(\mathscr{F})$. We thus have $\dim A = \dim A_1$; but this implies $\mathfrak{p} = 0$ and $A = A_1$; as $\operatorname{Spec}(A)$ and $\operatorname{Spec}(A_1)$ are respectively dense in T and T_1, we have indeed $T = T_1$. **IV-2 | 49**

The S_i that contain y are therefore all the prime cycles associated to $\mathscr{F}$ containing y; if x is the generic point of an irreducible component of $S_i \cap Y$ containing y, we deduce again from (**0**, 15.2.4) that t_x is $\mathscr{F}_x$-regular, hence, by (3.4.3), $x \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F})$; this proves the first assertion (3.4.4). □

Proposition (3.4.5). — *Let X be a prescheme locally Noetherian, t a section of $\mathscr{O}_X$ over X, Y the closed sub-prescheme of X defined by the Ideal $t\mathscr{O}_X$ of $\mathscr{O}_X$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent, y a point of Y; suppose that t_y is $\mathscr{F}_y$-regular and that $\mathscr{F}/t\mathscr{F}$ is integral at the point y (3.2.4). Then $\mathscr{F}$ is integral at the point y.*

Proof. Taking into account (3.4.4), it suffices to prove that y is contained in a *unique* irreducible component of $\operatorname{Supp}(\mathscr{F})$, and that if s is the generic point of this component, $\operatorname{length}(\mathscr{F}_s) = 1$. Now, by hypothesis, y belongs to at most a unique irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$, and if z is the generic point of this component, $\operatorname{length}((\mathscr{F}/t\mathscr{F})_z) = 1$; the conclusion therefore follows from (3.4.2). □ **IV-2 | 50**

Proposition (3.4.6). — *The hypotheses being those of (3.4.1), let x be a point of Y. Suppose that Y does not contain any of the S_i containing x, and that $\mathscr{F}/t\mathscr{F}$ is reduced at the point x (3.2.2). Then t_x is $\mathscr{F}_x$-regular and $\mathscr{F}$ is reduced at the point x. Furthermore, if z_j is the generic point of an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, z_j is contained in a unique S_i, and $\mathscr{O}_{S_i,z_j}$ is a discrete valuation ring whose uniformiser is t_{z_j}.*

Proof. The fact that t_x is $\mathscr{F}_x$-regular follows from the lemma below applied to the ring $\mathscr{O}_x$: □

Lemma (3.4.6.1). — *Let A be a Noetherian ring, M a finitely generated A-module, $\mathfrak{p}_i$ the minimal elements of* $\operatorname{Supp}(M)$*, t an element of A. Suppose that t does not belong to any of the $\mathfrak{p}_i$ and that M/tM is a reduced A-module* (3.2.2)*. Then t is M-regular.*

Proof. Every prime ideal $\mathfrak{p} \in \operatorname{Supp}(M)$ contains one of the $\mathfrak{p}_i$; as t does not belong to any of the $\mathfrak{p}_i$, the homothethy of ratio t in $M_\mathfrak{p}$ is not nilpotent (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, cor. of prop. 9). Let N be the sub-module of M consisting of elements annihilated by a power of t, and set $P = M/N$; we are going to show that $N = 0$. As t is P-regular, we have an exact sequence (3.4.1.4)

$$0 \longrightarrow N/tN \longrightarrow M/tM \longrightarrow P/tP \longrightarrow 0.$$

As N is of finite type, it is annihilated by a power of t, and it suffices therefore to show that $N/tN = 0$. As N/tN is a sub-module of M/tM, it suffices to prove that $(N/tN)_\mathfrak{p} = 0$ for all $\mathfrak{p} \in \operatorname{Ass}(M/tM)$, or equivalently that $u_\mathfrak{p} : (M/tM)_\mathfrak{p} \to (P/tP)_\mathfrak{p}$ is bijective for all $\mathfrak{p} \in \operatorname{Ass}(M/tM)$. Now, we have $(P/tP)_\mathfrak{p} \neq 0$; indeed, as $\mathfrak{p} \in \operatorname{Supp}(M/tM) = \operatorname{Supp}(M) \cap V(t)$ (**0**, 1.7.5), the image of t in $A_\mathfrak{p}$ is contained in the maximal ideal $\mathfrak{p}A_\mathfrak{p}$, hence the hypothesis $P_\mathfrak{p} = tP_\mathfrak{p}$ would imply $P_\mathfrak{p} = 0$ by the lemma of Nakayama; we would then have $M_\mathfrak{p} = N_\mathfrak{p}$ and the homothethy of ratio t would be nilpotent, contradicting the remark made at the beginning, since $\mathfrak{p} \in \operatorname{Supp}(M)$. This being so, the hypothesis that M/tM is reduced implies $\operatorname{length}((M/tM)_\mathfrak{p}) = 1$, and as $(P/tP)_\mathfrak{p} \neq 0$, $u_\mathfrak{p}$ is necessarily bijective, which proves the lemma. □

By hypothesis, none of the associated immersed prime cycles of $\mathscr{F}/t\mathscr{F}$ contains x, hence none of the associated immersed prime cycles of $\mathscr{F}$ contains x, by virtue of (3.4.4). On the other hand, applying (3.4.2) to an irreducible component of $\operatorname{Supp}(\mathscr{F}/t\mathscr{F})$ containing x, we see that $\operatorname{length}(\mathscr{F}_{s_i}) = 1$ for every S_i containing x, which shows that $\mathscr{F}_x$ is reduced; finally, the last assertions are also consequences of (3.4.2).

Corollary (3.4.7). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, M a finitely generated A-module, $(x_i)_{1 \leqslant i \leqslant k}$ a family of elements of $\mathfrak{m}$ forming part of a system of parameters* (**0**, 16.3.6)*. If the A-module* **IV-2 | 51**
$N = M/(\sum_{i=1}^k x_i M)$ is integral (3.2.4)*, then M is integral and the sequence $(x_i)_{1 \leqslant i \leqslant k}$ is M-regular.*

Proof. By induction on k, we are immediately reduced to the case $k = 1$; we write x instead of x_1; the hypothesis that x forms part of a system of parameters for M implies that $\dim(N) = \dim(M) - 1$ (**0**, 16.3.7). Set $n = \dim(N)$; there is thus a unique minimal element $\mathfrak{p}$ of $\operatorname{Supp}(M)$ such that $\dim(M/\mathfrak{p}M) = n + 1$ (**0**, 16.3.4), and for every integer $j > 0$, also $\dim(M/\mathfrak{p}^j M) = n + 1$ (**0**, 16.3.5); furthermore, x also forms part of a system of parameters for $M/\mathfrak{p}^j M$ (**0**, 16.3.5), hence, if one sets $M' = M/\mathfrak{p}^j M$ and $N' = M'/xM'$, we have $\dim(N') = n$. It is clear that there exists a surjective homomorphism $v : N \to N'$; let us show that v is *bijective*. Indeed, if $P = \operatorname{Ker}(v)$, we have $\operatorname{Ass}(P) \subset \operatorname{Ass}(N)$, and since N is integral, $\operatorname{Ass}(P)$ and $\operatorname{Ass}(N)$ would both be reduced to the unique point $\mathfrak{q}$ of $\operatorname{Ass}(N) = \operatorname{Supp}(N)$; but as $\dim(N') = \dim(N)$, we have $N'_\mathfrak{q} \neq 0$, and the hypothesis $\operatorname{length}(N'_\mathfrak{q}) = 1$ implies that $N_\mathfrak{q} \to N'_\mathfrak{q}$ is surjective. We would then have $P_\mathfrak{q} = 0$, contradicting our assumption; hence our assertion. But then N', being isomorphic to N, is integral; furthermore, the support of N' (equal to the intersection of $\operatorname{Supp}(M)$ and $V(x)$) cannot contain $\operatorname{Supp}(M')$, and this last set is irreducible by construction. The hypothesis that M'/xM' is integral (hence reduced) implies then that x is M'-regular by virtue of (3.4.6). We conclude that the kernel of the homothethy $z \mapsto xz$ in M is contained in $\mathfrak{p}^j M \subset \mathfrak{m}^j M$ for every integer j, and hence this kernel is reduced to 0 (**0**, 7.3.5), which proves that x is M-regular. We can then apply (3.4.5), which proves that M is integral. □

Remark (3.4.8). — The analogous proposition to (3.4.7), where one replaces "integral" by "reduced", is no longer necessarily exact. Consider for example the polynomial ring $C = K[X, Y, Z]$ over a field K, the quotient ring $B = C/\mathfrak{p}\mathfrak{q}$, where $\mathfrak{p} = CZ$, $\mathfrak{q} = CX^2 + CY$; let A be the local ring of B corresponding to the maximal image of $CX + CY + CZ$ in B. If x, z are the canonical images of X, Z in B, it is clear that $xz \neq 0$ but $x^2 z^2 = 0$; on the other hand, as A/xA is isomorphic to $K[Y, Z]/(YZ)$, we have $\dim(A/xA) = 1$ while $\dim(A) = 2$, hence x belongs to a system of parameters of A (**0**, 16.3.4), A/xA is reduced, but A is not.

Proposition (3.4.9). — *Let A be a Noetherian ring, M a finitely generated A-module, f an M-regular element such that M/fM has no immersed associated prime ideals. If the $\mathfrak{p}_i$ $(1 \leqslant i \leqslant m)$ are the prime ideals associated to M/fM, then, for every integer $n > 0$, $f^n M$ is the intersection of the inverse images of the $f^n M_{\mathfrak{p}_i}$ by the canonical maps $M \to M_{\mathfrak{p}_i}$ $(1 \leqslant i \leqslant m)$.*

Proof. It amounts to the same to show that the $\mathfrak{p}_i$ are the prime ideals associated to $M/f^n M$, for then the $\mathfrak{p}_i$ are the sub-modules of the unique reduced primary decomposition of $f^n M$ in M. Now, we have

$$\mathrm{Ass}(f^{n-1}M/f^n M) \subset \mathrm{Ass}(M/f^n M) \subset \mathrm{Ass}(M/f^{n-1}M) \cup \mathrm{Ass}(f^{n-1}M/f^n M)$$

by (3.1.7), and as f is M-regular, $f^{n-1}M/f^n M$ is isomorphic to M/fM; it suffices therefore to reason by induction on n. □

§4. CHANGE OF BASE FIELD FOR ALGEBRAIC PRESCHEMES

4.1. If X is an algebraic prescheme over a field k, and K is an extension of k, the prescheme $X \otimes_k K$ and its underlying topological space IV-2 | 52

(4.1.1). Let k be a field, X a k-prescheme, K an extension of k. We write $X_{(K)} = X \otimes_k K$ to denote the product $X \times_k \mathrm{Spec}(K)$; recall that $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ is faithfully flat, so $X_{(K)}$ is faithfully flat over X (**I**, 3.5.2). If $\mathscr{F}$ is an $\mathscr{O}_X$-Module, we write $\mathscr{F}_{(K)} = \mathscr{F} \otimes_k K$ for the $\mathscr{O}_{X_{(K)}}$-Module inverse image of $\mathscr{F}$ under the canonical projection $X_{(K)} \to X$. Let x be a point of X and x' a point of $X_{(K)}$ above x; we have $\mathscr{O}_{X_{(K)},x'} = \mathscr{O}_{X,x} \otimes_k K'$, where K' is the local ring of $\mathrm{Spec}(\kappa(x) \otimes_k K)$ at the point that is the image of x' under the canonical isomorphism $p^{-1}(x) \xrightarrow{\sim} \mathrm{Spec}(\kappa(x) \otimes_k K)$ (where $p : X_{(K)} \to X$ is the projection) (**I**, 3.4.9).

(4.1.2). Let k be a field, K and L two extensions of k, K' a subextension of K. We write $\mathfrak{t} = \mathrm{tr.deg}_k$. Suppose that there exists a k-monomorphism $u : K' \to L$; we consider L as a K'-algebra via u, and K as a K'-algebra via the inclusion map. We have the commutative diagram

(4.1.2.1)
$$k \longrightarrow K' \longrightarrow K, \quad k \longrightarrow K' \xrightarrow{u} L$$

which gives us a commutative diagram

(4.1.2.2)
$$K \otimes_k L \xleftarrow{\sim} K \otimes_{K'} L \leftarrow L.$$

Recall the following relation (cf. Bourbaki, *Alg.*, chap. V, § 6, n° 6, th. 3 and Bourbaki, *Alg. comm.*, chap. VIII, § 1, n° 1, cor. 2 and 3 de la prop. 1):

(4.1.2.3)
$$\mathrm{deg.tr}_k(K \otimes_k L) = \mathfrak{t}\, K + \mathfrak{t}\, L - \mathfrak{t}\, K'$$

(the degree of transcendence of a ring without zero divisors over a field k being defined in Bourbaki, *Alg. comm.*, chap. VIII, § 1, n° 1). When K and L are extensions of k that are subextensions of the same extension Ω, we take $K' = K \cap L$, u the inclusion map, and (4.1.2.3) becomes

(4.1.2.4)
$$\mathrm{deg.tr}_k(K \otimes_k L) = \mathfrak{t}\, K + \mathfrak{t}\, L - \mathfrak{t}(K \cap L).$$

Recall also the following result: under the hypotheses of the first relation (4.2.1.1), let us denote further by E the field of fractions of $K \otimes_{K'} L = K \otimes_k L$, and by I the set of minimal prime ideals of $K \otimes_k L$; then $\mathrm{Card}(I) \leqslant [\mathrm{K}' : \mathrm{k}]_s$ (the degree of the largest separable algebraic extension of k contained in K') (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, cor. 1 de la prop. 1). We also have the first relation

(4.2.1.1)
$$\mathrm{deg.tr}_k\, E = \mathfrak{t}\, K + \mathfrak{t}\, L - \mathfrak{t}\, K';$$

(Bourbaki, *Alg.*, chap. IV, § 1, n° 1, cor. 3 de la prop. 1). Moreover, if in addition K' and L are algebraic extensions of k, we have

$$\mathrm{Card}(I) = [\mathrm{K}' : \mathrm{k}]_s.$$

Suppose K and L are extensions of k such that $K \cap L = k$ (inside some common overfield), $K' = k$; IV-2 | 53
let E be the field of fractions of $K \otimes_k L$, I the set of minimal prime ideals, so $\mathrm{Card}(I) = 1$ and $\mathrm{deg.tr}_k\, E = \mathfrak{t}\, K + \mathfrak{t}\, L$. In general, if we suppose only that K' is an algebraic extension of k, we have $\mathrm{Card}(I) \leqslant [K' : k]_s$, and equality holds when K' is a finite extension of k.

Now suppose further that K and L are *subextensions of the same extension* Ω of k, and let $K' = K \cap L$; then K' is a subextension of K and of L, and there is a canonical k-monomorphism $u : K' \to L$ (the inclusion map). The field of fractions of $K \otimes_k L$ is then equal to the composite field KL inside Ω.

(4.1.3). Keep the notation of (4.1.2). We have the canonical k-homomorphisms

$$k \longrightarrow K' \longrightarrow K, \quad k \longrightarrow L$$

and, setting $K' \otimes_k L = K' \otimes_k L$, the field of fractions of $K' \otimes_k L$ is $L(\mathfrak{t})$; we have the commutative diagram of canonical homomorphisms

(4.2.1.3)
$$\begin{tikzcd}
K \arrow[r] & K \otimes_k L \arrow[r] & K \otimes_{K'} L(\mathfrak{t}) \\
K' \arrow[u] \arrow[r] & K' \otimes_k L \arrow[u] \arrow[r] & L(\mathfrak{t}) \arrow[u] \\
& k \arrow[ul] \arrow[r] & L \arrow[u]
\end{tikzcd}$$

Since K is faithfully flat over K', $K \otimes_k L = K \otimes_{K'} (K' \otimes_k L)$ is faithfully flat over $K' \otimes_k L$, so $K' \otimes_k L$ is Noetherian ($\mathbf{0}_I$, 6.5.2); furthermore $K' \otimes_k L$ identifies with a subring of $K \otimes_k L$; the trace on $K' \otimes_k L$ of a prime ideal $\mathfrak{p}$ associated to $K \otimes_k L$ is the ideal $\mathfrak{o}$ of $K' \otimes_k L$ not being a zero divisor in $K \otimes_k L$ ($\mathbf{0}_I$, 6.3.4 and Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 3 de la prop. 2). Moreover, K being algebraic over K', $K \otimes_k L$ is an integral algebra over $K' \otimes_k L$, and the prime ideals of $K \otimes_k L$ inducing $\mathfrak{o}$ on $K' \otimes_k L$ are necessarily without mutual inclusion relation (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, cor. 1 de la prop. 1); this proves the first assertion (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 3, cor. 1 de la prop. 7). Furthermore, the residue field E of $\mathfrak{p}$ is algebraic over the residue field of the ideal $(\mathfrak{o})$ of $K' \otimes_k L$, that is to say $L(\mathfrak{t})$; hence $\operatorname{deg.tr}_k E = \operatorname{deg.tr}_k L(\mathfrak{t}) = \operatorname{Card}(I) = \operatorname{deg.tr}_k K$, which by switching the roles of K and L gives the second relation (4.2.1.1), whence (4.2.1.2).

IV-2 | 55

Corollary (4.2.2). — *Under the hypotheses of (3.3.6), if the preschemes $\mathrm{T}_{x,y}$ are locally Noetherian, they have no associated immersed prime cycle.*

Corollary (4.2.3). — *Under the hypotheses of (3.3.6) (resp. (3.3.7)), if the $\mathrm{T}_{x,y}$ are locally Noetherian and if $\mathscr{F}$ and the $\mathscr{G}_s$ (for $s \in S$) (resp. $\mathscr{F}$ and $\mathscr{G}$) are without associated immersed prime cycle, then so is $\mathscr{F} \otimes_{\mathscr{O}} \mathscr{G}$.*

This follows from (4.2.2), (3.3.2) and the proof of (3.3.6). In particular, since every prescheme over a field k is *flat* over k, we have thus proved assertion (i) of the following:

Proposition (4.2.4). — *Let k be a field, X and Y two locally Noetherian k-preschemes such that $X \times_k Y$ is locally Noetherian. Suppose further that X and Y are integral. Then:*

label=(i) $X \times_k Y$ has no associated immersed prime cycle; each of the irreducible components of $X \times_k Y$ dominates X and Y, and the set of these components is in bijective correspondence with the set of irreducible components of $\operatorname{Spec}(\mathrm{R}(X) \otimes_k \mathrm{R}(Y))$ (in other words, the set of minimal prime ideals of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$), where $\mathrm{R}(X)$ and $\mathrm{R}(Y)$ are the fields of rational functions of X and Y respectively.

lbbel=(ii) If a maximal point z of $X \times_k Y$ is identified with a minimal prime ideal $\mathfrak{p}$ of $\mathrm{R}(X) \otimes_k \mathrm{R}(Y)$, the local ring $\mathscr{O}_{X \times_k Y, z}$ is isomorphic to the ring of fractions $(\mathrm{R}(X) \otimes_k \mathrm{R}(Y))_{\mathfrak{p}}$. In particular, if $\mathrm{R}(X)$ or $\mathrm{R}(Y)$ is separable over k, $X \times_k Y$ is reduced.

IV-2 | 56

(iii) If in addition X and Y are locally of finite type over k, every irreducible component of $X \times_k Y$ is of dimension $\dim(X) + \dim(Y)$.

Assertion (iii) follows from (4.2.1.2), taking into account (i) and (ii). To prove (ii), we can evidently restrict to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$ are affine, A and B being integral rings with fields of fractions $K = \mathrm{R}(X)$ and $L = \mathrm{R}(Y)$ respectively; assertion (i) shows that every minimal prime ideal $\mathfrak{q}$ of $A \otimes_k B$ is the trace on $A \otimes_k B$ of a minimal prime ideal $\mathfrak{p}$ of $K \otimes_k L$; since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$, the isomorphism of the rings $(A \otimes_k B)_{\mathfrak{q}}$ and $(K \otimes_k L)_{\mathfrak{p}}$ follows from ($\mathbf{0}_I$, 1.2.5).

Finally, if for example $R(Y)$ is separable over k, we know that the ring $R(X) \otimes_k R(Y)$ is reduced (Bourbaki, *Alg.*, chap. VIII, § 7, n° 3, th. 1); it is therefore also reduced for the local rings of the maximal points of $X \times_k Y$. We conclude that $X \times_k Y$ is reduced (3.2.1).

Proposition (4.2.5). — *Let k be a field, X and Y two k-preschemes, locally Noetherian, $\mathscr{F}$ (resp. $\mathscr{G}$) an $\mathscr{O}_X$-Module (resp. an $\mathscr{O}_Y$-Module) quasi-coherent. Let (Z'_λ) (resp. (Z''_μ)) be the family of prime cycles associated to $\mathscr{F}$ (resp. $\mathscr{G}$), and denote further by Z'_λ (resp. Z''_μ) the reduced sub-prescheme of X (resp. Y) having Z'_λ (resp. Z''_μ) as underlying space. Then, if $Z'_\lambda \times_k Z''_\mu$ is locally Noetherian, the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of distinct prime cycles associated to $\mathscr{F} \otimes_k \mathscr{G}$.*

Proof. It suffices to apply (4.2.4) to the product $\mathrm{Spec}(\kappa(x)) \times_k \mathrm{Spec}(\kappa(y))$. In particular: □

Corollary (4.2.6). — *Let k be a field, X, Y two k-preschemes, locally Noetherian, such that $X \times_k Y$ is locally Noetherian. Let (Z'_λ) (resp. (Z''_μ)) be the family of the reduced sub-preschemes of X (resp. Y) having for underlying spaces the irreducible components of X (resp. Y). Then the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ, and $(Z_{\lambda\mu\nu})$ is the family of irreducible components of $X \times_k Y$.*

Proof. Indeed, we can restrict to the case where X and Y are reduced (**I**, 5.1.8); the irreducible components of X (resp. Y) are then the prime cycles associated to $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$) (3.2.1). We apply (4.2.5) to $\mathscr{F} = \mathscr{O}_X$ and $\mathscr{G} = \mathscr{O}_Y$, noting that by definition $\mathscr{O}_X \otimes_k \mathscr{O}_Y = \mathscr{O}_{X \times_k Y}$; the corollary follows from the fact that $\mathscr{O}_X \otimes_k \mathscr{O}_Y$ has no associated immersed prime cycles, since this is so for $\mathscr{O}_X$ and $\mathscr{O}_Y$ by hypothesis (4.2.3).

Applying the preceding results to the case where Y is the spectrum of an extension K of k: □

Proposition (4.2.7). — *Let k be a field, X a k-prescheme, K an extension of k such that $X \otimes_k K$ is locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, x' a point of $X \otimes_k K$, x its image in X.*

label=(i) *Let (Z_λ) be the family of reduced sub-preschemes of X having for underlying spaces the prime cycles associated to $\mathscr{F}$; then the irreducible components $Z_{\lambda\mu}$ of $Z_\lambda \otimes_k K$ are the prime cycles associated to $\mathscr{F} \otimes_k K$, and $Z_{\lambda\mu}$ dominates Z_λ; furthermore, for $Z_{\lambda\mu}$ to be immersed, it is necessary and sufficient that Z_λ be so.*

lbbel=(ii) *For x to belong to a prime cycle associated to an immersed $\mathscr{F}$, it is necessary and sufficient that x' belong to a prime cycle associated to an immersed $\mathscr{F} \otimes_k K$; for $\mathscr{F}$ to be without associated immersed prime cycle, it is necessary and sufficient that $\mathscr{F} \otimes_k K$ be so.*

IV-2 | 57

(iii) The indices λ such that $x \in Z_\lambda$ are the same as those for which there exists an index μ such that $x' \in Z_{\lambda\mu}$. In particular, if x' belongs to a single prime cycle associated to $\mathscr{F} \otimes_k K$, x belongs to a single prime cycle associated to $\mathscr{F}$.

(iv) If X is locally of finite type over k, we have $\dim(Z_{\lambda\mu}) = \dim(Z_\lambda)$.

We note that the hypothesis implies that X itself is locally Noetherian (2.2.13) and (2.2.14); assertion (i) follows from (4.2.5) and the proof of (4.2.3), for $\mathscr{G} = \mathscr{O}_Y$, with $Y = \mathrm{Spec}(K)$; (ii) and (iii) follow from (i) and (2.3.5); finally, (iv) is a particular case of (4.2.4), (iii)).

Corollary (4.2.8). — *If X is locally of finite type over k, the set of dimensions of the prime cycles associated to $\mathscr{F}$ and $\mathscr{F} \otimes_k K$; the set of dimensions of the irreducible components is the same for X and $X \otimes_k K$.*

Proposition (4.2.9). — *Suppose the hypotheses of (4.2.5) are satisfied, and suppose in addition that $\mathscr{F}$ and $\mathscr{G}$ are coherent. Let $(\mathscr{F}_\lambda)_{\lambda \in \mathrm{L}}$ and $(\mathscr{G}_\mu)_{\mu \in \mathrm{M}}$ be irredundant decompositions of $\mathscr{F}$ and $\mathscr{G}$ respectively; for every pair $(\lambda, \mu) \in \mathrm{L} \times \mathrm{M}$, let $(\mathscr{H}_{\lambda\mu\nu})_{\nu \in \mathrm{S}(\lambda,\mu)}$ be an irredundant decomposition of $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$, where $\mathrm{S}(\lambda, \mu) = \mathrm{Ass}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$ (3.2.5). Then $(\mathscr{H}_{\lambda\mu\nu})$ (for all triples (λ, μ, ν)) is an irredundant decomposition of $\mathscr{F} \otimes_k \mathscr{G}$; it is reduced if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are.*

Proof. Note that $\mathscr{F} \otimes_k \mathscr{G}$ and each of the $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ are coherent, and each of the $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$ identifies with a quotient of $\mathscr{F} \otimes_k \mathscr{G}$; each $\mathscr{H}_{\lambda\mu\nu}$ identifies by definition with a coherent quotient of $\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu$, hence also with a coherent quotient of $\mathscr{F} \otimes_k \mathscr{G}$. The family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite, since $\mathrm{Supp}(\mathscr{F}_\lambda \otimes_k \mathscr{G}_\mu)$ is contained in $\mathrm{Supp}(\mathscr{F}_\lambda) \times_k \mathrm{Supp}(\mathscr{G}_\mu)$ (where $\mathrm{Supp}(\mathscr{F}_\lambda)$ and $\mathrm{Supp}(\mathscr{G}_\mu)$ denote the corresponding reduced closed sub-preschemes), and for given λ and μ, the family of supports of the $\mathscr{H}_{\lambda\mu\nu}$ is locally finite by hypothesis; our assertion then follows from the fact that the families $(\operatorname{Supp}(\mathscr{F}_\lambda))$ and $(\operatorname{Supp}(\mathscr{G}_\mu))$ are locally finite. To show that $(\mathscr{H}_{\lambda\mu\nu})$ is an irredundant decomposition of $\mathscr{F}\otimes_k\mathscr{G}$, it suffices therefore to prove that the canonical homomorphism $\mathscr{F}\otimes_k\mathscr{G}\to\bigoplus_{\lambda,\mu,\nu}\mathscr{H}_{\lambda\mu\nu}$ is injective; or, it is the composite of the homomorphisms $\mathscr{F}\otimes_k\mathscr{G}\to\bigoplus_{\lambda,\mu}(\mathscr{F}_\lambda\otimes_k\mathscr{G}_\mu)\to\bigoplus_{\lambda,\mu,\nu}\mathscr{H}_{\lambda\mu\nu}$; the second of these homomorphisms is injective by definition, and it is the same for the first, which is nothing other than the tensor product of the canonical injective homomorphisms $\mathscr{F}\to\bigoplus_\lambda\mathscr{F}_\lambda$, $\mathscr{G}\to\bigoplus_\mu\mathscr{G}_\mu$ (remembering that $\mathscr{F}$ and $\mathscr{G}$ are flat over k). Finally, if $(\mathscr{F}_\lambda)$ and $(\mathscr{G}_\mu)$ are reduced, we can suppose that $\mathrm{L}=\operatorname{Ass}(\mathscr{F})$ and $\mathrm{M}=\operatorname{Ass}(\mathscr{G})$; the fact that the $\mathscr{H}_{\lambda\mu\nu}$ then form a reduced decomposition follows from the fact that the $\operatorname{Ass}(\mathscr{F}_\lambda\otimes_k\mathscr{G}_\mu)$ form a partition of $\operatorname{Ass}(\mathscr{F}\otimes\mathscr{G})$ by virtue of (4.2.2) (cf. (3.2.5)). □

Corollary (4.2.10). — *Under the hypotheses of (4.2.7), suppose in addition that $\mathscr{F}$ is coherent, and let $(\mathscr{F}_\lambda)_{\lambda\in\mathrm{L}}$ be an irredundant decomposition of $\mathscr{F}$; for every $\lambda\in\mathrm{L}$, let $(\mathscr{F}_{\lambda\mu})_{\mu\in\operatorname{Ass}(\mathscr{F}_\lambda\otimes_k K)}$ be an irredundant decomposition of $\mathscr{F}_\lambda\otimes_k K$. Then $(\mathscr{F}_{\lambda\mu})$ is an irredundant decomposition of $\mathscr{F}\otimes_k K$; it is reduced if $(\mathscr{F}_\lambda)$ is.*

Proof. We apply (4.2.9) with $Y=\operatorname{Spec}(K)$, $\mathscr{G}=\mathscr{O}_Y=K$, M reduced to a single element. □

4.2. Algebraic preschemes over a field, and extension of the base field The results of (4.2.2) through (4.2.10) have already been stated above in the context of the general formalism of §§ 2–3. We now apply them specifically to the case of algebraic preschemes over a field and extensions of the base field. **IV-2 | 55**

4.3. Reminders on tensor products of fields For the convenience of the reader, we recall here certain properties of tensor products of fields that we will use in the following sections; for the proofs, we refer to [**?**]. **IV-2 | 58**

(4.3.1). Recall that an extension L of a field k is said to be *primary* if the largest separable algebraic extension of k contained in L is k itself.

Proposition (4.3.2). — *Let K, L be two extensions of a field k. If L is a primary extension of k, then* $\operatorname{Spec}(L\otimes_k K)$ *is irreducible, and if ξ is its generic point, $\kappa(\xi)$ is a primary extension of K. Conversely, if for every finite separable extension K of k,* $\operatorname{Spec}(L\otimes_k K)$ *is irreducible, L is a primary extension of k.*

Proof. See the proof in [**?**], p. 14-03 to 14-06. □

Corollary (4.3.3). — *If k is separably closed (i.e. if its separable algebraic closure is radicial over k), then, for any two extensions K, L of k,* $\operatorname{Spec}(L\otimes_k K)$ *is irreducible, and conversely.*

Proof. This follows immediately from (4.3.2). □

Corollary (4.3.4). — *Let L be an extension of a field k, L_s the largest separable algebraic extension of k contained in L (sometimes called the* separable closure *of k in L); suppose L_s is of finite degree over k, and let K be a Galois extension of k containing L_s. Then* $\operatorname{Spec}(L\otimes_k K)$ *has $[\mathrm{L}_s:k]$ irreducible components, and the residue fields at the generic points of these components are primary extensions of K.*

Proof. Indeed, we have $L\otimes_k K=L\otimes_{\mathrm{L}_s}(\mathrm{L}_s\otimes_k K)$ and we know that $\mathrm{L}_s\otimes_k K$ is isomorphic to a product of $[\mathrm{L}_s:k]$ fields isomorphic to K (Bourbaki, *Alg.*, chap. VIII, § 7, n° 3, cor. 2 du th. 1); hence $L\otimes_k K$ is isomorphic to a product of $[\mathrm{L}_s:k]$ rings isomorphic to $L\otimes_{\mathrm{L}_s}K$; since L is a primary extension of L_s, the conclusion follows from (4.3.2). □

Proposition (4.3.5). — *Let K, L be two extensions of a field k. If L is a separable extension of k, the ring $L\otimes_k K$ is reduced. Conversely, if for every finite radicial extension K of k, the ring $L\otimes_k K$ is reduced, L is a separable extension of k.*

Proof. For the proof, see Bourbaki, *Alg.*, chap. VIII, § 7, n° 3, th. 1. □

Corollary (4.3.6). — *If k is perfect, then, for any two extensions K, L of k, $K\otimes_k L$ is reduced, and conversely.*

Proof. This follows immediately from (4.3.5). □

Corollary (4.3.7). — *Let L be a separable extension of k and let K be an extension of k such that K or L is finite over k; then the residue fields of the semi-local ring $L\otimes_k K$ are separable extensions of K.*

Proof. Indeed, we know that $L \otimes_k K$ is the direct sum of these residue fields E_i; for every extension K' of K, $L \otimes_k K'$ is therefore isomorphic to the direct sum of the rings $E_i \otimes_K K'$; since $L \otimes_k K'$ is reduced, the E_i are separable over K by virtue of (4.3.5). □

Corollary (4.3.8). — *If K and L are two finite separable extensions of k, $K \otimes_k L$ is a product of a finite number of separable extensions of k.*

Proposition (4.3.9). — *If k is algebraically closed, then, for any two extensions K, L of k, $L \otimes_k K$ is integral, and conversely.*

Proof. This is a consequence of (4.3.3) and (4.3.6), a perfect field being separably closed if and only if it is algebraically closed. □

4.4. Irreducible preschemes and connected preschemes over an algebraically closed field

IV-2 | 59

(4.4.1). Let k be a field, X a k-prescheme, K an extension of k. Since the morphism $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is faithfully flat, quasi-compact, and universally open (2.4.9), the same is true of the projection morphism $p : X \otimes_k K \to X$ (2.2.13). It follows therefore from (2.3.5) that every irreducible component of $X \otimes_k K$ *dominates* an irreducible component of X, and (since p is surjective) every irreducible component of X is dominated by an irreducible component of $X \otimes_k K$; we thus deduce that p is a *surjective* map from the set of irreducible components of $X \otimes_k K$ to the set of irreducible components of X. Similarly, it follows from the fact that p is continuous and surjective that every connected component of X is the image by p of a union of connected components of $X \otimes_k K$, whence a *surjective* map from the set of connected components of $X \otimes_k K$ to the set of connected components of X. We conclude that every irreducible (resp. connected) component Z' of $X \otimes_k K$ is contained in a *unique* set of the form $p^{-1}(Z)$, where Z is an irreducible (resp. connected) component of X; for every sub-prescheme Z of X having Z as underlying space (and still denoted Z), Z' is an irreducible (resp. connected) component of $Z \otimes_k K$. In particular, if $X \otimes_k K$ has a finite number n (resp. n') of irreducible (resp. connected) components (which will be the case when X is of *finite type* over k, since then $X \otimes_k K$ is of finite type over K, hence Noetherian), the number of irreducible (resp. connected) components of X is $\leqslant n$ (resp. $\leqslant n'$); to say that this number is equal to n (resp. n') means that for every reduced sub-prescheme Z of X having an irreducible (resp. connected) underlying space, $Z \otimes_k K$ is irreducible (resp. connected) (**I**, 5.1.8); the number of irreducible (resp. connected) components of $X \otimes_k K$ is therefore independent of K if and only if for every irreducible (resp. connected) component Z of X, $Z \otimes_k K$ is irreducible (resp. connected) for *every* extension K of k.

In particular, if $X \otimes_k K$ is irreducible (resp. connected), so is X (which already follows from the fact that p is surjective). Similarly, if $X \otimes_k K$ is *reduced* (resp. *integral*), so is X since p is faithfully flat (2.1.13).

In this section and the next, we will examine more closely what can be said about the irreducible (resp. connected) components of $X \otimes_k K$ as K varies; the above already shows that the number of these components is a *non-decreasing* function of K.

Lemma (4.4.2). — *Let X', X be two topological spaces, $f : X' \to X$ a continuous map. Suppose that the following conditions are satisfied:*

label=(i) f is a surjective open map (resp. such that the topological space X is canonically identified with the quotient of X' by the equivalence relation defined by f).

IV-2 | 60

(ii) For every $x \in X$, $f^{-1}(x)$ is irreducible (resp. connected).

Then for X' to be irreducible (resp. connected), it is necessary and sufficient that X be so.

Proof. The assertion concerning connectedness is nothing other than Bourbaki, *Top. gén.*, chap. I$^{\text{er}}$, 3$^{\text{e}}$ éd., § 11, n$^{\text{o}}$ 3, prop. 7. Let us prove the assertion concerning irreducibility; the condition being trivially necessary since f is surjective, let us prove that it is sufficient. Let X'_1, X'_2 be two closed subsets of X' such that $X' = X'_1 \cup X'_2$, and denote by X_i ($i = 1, 2$) the set of $x \in X$ such that $f^{-1}(x) \subset X'_i$; we have $X_i = X - f(X' - X'_i)$, and since f is open, we conclude that X_i is a closed part of X ($i = 1, 2$). On the other hand, for every $x \in X$, $f^{-1}(x)$ is irreducible by hypothesis, and is the union of the two closed subsets $X'_1 \cap f^{-1}(x)$ and $X'_2 \cap f^{-1}(x)$; one of these two subsets must therefore be equal to $f^{-1}(x)$, which means that $x \in X_1$ or $x \in X_2$; we therefore have $X = X_1 \cup X_2$, and since X is irreducible by hypothesis, we deduce $X = X_1$ or $X = X_2$, hence $X' = X'_1$ or $X' = X'_2$. □

Remark (4.4.3). — If the topology of X is not the quotient of that of X' by the equivalence relation defined by f, it can happen that X and all the fibres $f^{-1}(x)$ are irreducible, without X' being so: one can construct an example by taking for X an irreducible scheme (for example the affine line $\mathrm{Spec}(k[\mathrm{T}])$) over an algebraically closed field k and, considering a closed point x_0 of X and taking for X' the direct *sum* of $\{x_0\}$ and the open subset $X - \{x_0\}$ of X. Similarly, if one replaces the hypothesis "f open" by "f closed" (or even "f proper") (which implies, however, that the topology of X is then the quotient of that of X' by the relation defined by f), it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being so. Let us take for X the affine line over k; if x_0 is a closed point of X, t_0 a closed point of $\mathrm{P} = X \times_k \mathrm{P}$, where P is the projective line over k, $p : S \to X$, $q : S \to \mathrm{P}$ the projections, denote by X' the reduced sub-prescheme having for underlying space $p^{-1}(x_0) \cup q^{-1}(t_0)$, and let us take f to be the restriction of p to X' of p; f is proper but X' is not irreducible.

Theorem (4.4.4). — *Let k be an algebraically closed field, X a k-prescheme. If X is irreducible (resp. connected), so is $X \otimes_k K$ for every extension K of k.*

Proof. We saw indeed in (4.4.1) that the morphism $p : X \otimes_k K \to X$ is faithfully flat and open; to be able to apply the lemma (4.4.2), it suffices therefore to verify that the fibres $p^{-1}(x)$ are irreducible for every $x \in X$. But the $\kappa(x)$-prescheme $p^{-1}(x)$ is isomorphic to $\mathrm{Spec}(\kappa(x) \otimes_k K)$ (**I**, 3.6.2), hence integral by virtue of the hypothesis on k and of (4.3.9). □

Corollary (4.4.5). — *Let k be an algebraically closed field, X a k-prescheme, K an extension of k, $p : X \otimes_k K \to X$ the canonical projection. If Z is an irreducible (resp. connected) part of X, $p^{-1}(Z)$ is an irreducible (resp. connected) part; in particular, if X_0 is an irreducible (resp. connected) component of X containing Z, $p^{-1}(X_0)$ is an irreducible (resp. connected) component of $X \otimes_k K$ containing $p^{-1}(Z)$.*

Proof. The second assertion follows from the first applied to (4.4.1) replacing Z by X_0; to prove the first, let X' be the underlying space of $X \otimes_k K$, $Z' = p^{-1}(Z)$; the equivalence relation R defined by p on X' is open, hence so is the relation $R_{Z'}$ it induces on the saturated part Z', and Z identifies with the quotient space $Z'/R_{Z'}$ (Bourbaki, *Top. gén.*, chap. I^er, 3^e éd., § 5, n^o 2, prop. 4); we can therefore apply the lemma (4.4.2) to Z and Z', whence the first assertion of (4.4.5). □

IV-2 | 61

Corollary (4.4.6). — *The hypotheses and notation being those of (4.4.5), the map $Z \mapsto p^{-1}(Z)$ is a bijection from the set of irreducible (resp. connected) components of X to the set of irreducible (resp. connected) components of $X \otimes_k K$, whose inverse bijection is $Z' \mapsto p(Z')$.*

4.5. Geometrically irreducible and geometrically connected preschemes

Proposition (4.5.1). — *Let k be a field, X a k-prescheme, Ω an algebraically closed extension of k. Let n (resp. n') be the cardinal of the set of irreducible (resp. connected) components of $X \otimes_k \Omega$; this number is independent of the algebraically closed extension Ω chosen. Furthermore, for every extension K of k, the cardinal of the set of irreducible (resp. connected) components of $X \otimes_k K$ is $\leqslant n$ (resp. $\leqslant n'$).*

Proof. Indeed, two algebraically closed extensions Ω, Ω' of k can always be considered as subextensions of the same extension E of k; it suffices then to apply (4.4.6) to $X \otimes_k \Omega$ and $X \otimes_k \Omega'$ and to the common extension E of Ω and Ω' to prove the first assertion. The second is obtained by taking for Ω an algebraically closed extension of k and using (4.4.1). □

Definition (4.5.2). — The cardinal n (resp. n') of (4.5.1) is called the *geometric number of irreducible components* (resp. *of connected components*) of X (relative to k). If $n = 1$ (resp. $n' \leqslant 1$) we say that X is *geometrically irreducible* (resp. *geometrically connected*).

We thus recover a definition given for a particular case in (**III**, 4.3.4). By virtue of (4.5.1), the following properties are equivalent:

label=*a)* X is geometrically irreducible (resp. geometrically connected).
lbbel=*b)* For an algebraically closed extension Ω of k, $X \otimes_k \Omega$ is irreducible (resp. connected).
lcbel=*c)* For every extension K of k, $X \otimes_k K$ is irreducible (resp. connected).

(4.5.3). Let X be a k-prescheme, Z a locally closed subset of X. If we again denote by Z any of the sub-preschemes of X having Z as underlying space, it follows from (**I**, 5.2.1) that for an extension K of k, the number of irreducible (resp. connected) components of $Z \otimes_k K$ is independent of the sub-prescheme with underlying space Z that one has chosen; we thus define the geometric number of irreducible (resp. connected) components of Z as that of any sub-prescheme of X having Z as underlying space.

Proposition (4.5.4). — *Let X, Y be two k-preschemes, $f : X \to Y$ a surjective k-morphism. If X is geometrically irreducible (resp. geometrically connected), so is Y.*

Proof. Indeed, for every extension K of k, $f_{(K)} : X_{(K)} \to Y_{(K)}$ is surjective (**I**, 3.5.2), and $X_{(K)}$ being irreducible (resp. connected) by hypothesis, the same holds for $Y_{(K)}$. □

IV-2 | 62

Definition (4.5.5). — A morphism $f : X \to Y$ of k-preschemes is said to be *geometrically irreducible* (resp. *geometrically connected*), if, for every $y \in Y$, the $\kappa(y)$-prescheme $f^{-1}(y)$ is geometrically irreducible (resp. geometrically connected).

Proposition (4.5.6). — *label=(i) Let X be a k-prescheme, K an extension of k. Then the geometric number of irreducible (resp. connected) components of $X_{(K)}$ relative to K is equal to the geometric number of irreducible (resp. connected) components of X relative to k. In particular, for the K-prescheme $X_{(K)}$ to be geometrically irreducible (resp. geometrically connected), it is necessary and sufficient that the k-prescheme X be so.*

lbbel=(ii) Let $f : X \to Y$, $g : Y' \to Y$ be two morphisms, and set $X' = X_{(Y')} = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. If f is irreducible (resp. connected), so is f'. The converse is true if g is surjective.

Proof. label=(i) If Ω is an algebraically closed extension of k, we have $X \otimes_k \Omega = X_{(K)} \otimes_K \Omega$, whence the assertion.

lbbel=(ii) For every $y' \in Y'$, if we set $y = g(y')$, we have $f'^{-1}(y') = f^{-1}(y) \otimes_{\kappa(y)} \kappa(y')$ (**I**, 3.6.4), hence the two assertions follow from (i).

□

Proposition (4.5.7). — *Let $f : X \to Y$ be a surjective morphism that is irreducible (resp. connected), $g : Y' \to Y$ a morphism, and set $X' = X \times_Y Y'$. If in addition Y' is irreducible (resp. connected), and f is universally open (resp. flat and quasi-compact, or universally open, or universally closed), then X' is irreducible (resp. connected).*

Proof. The properties of f that one considers being all stable under base change, we can restrict to the case where $Y' = Y$; it suffices then to apply (4.4.2) (taking into account (2.3.12)). □

Corollary (4.5.8). — *Let X, Y be two k-preschemes.*

label=(i) If X is geometrically irreducible (resp. geometrically connected) and Y irreducible (resp. connected), then $X \times_k Y$ is irreducible (resp. connected).

lbbel=(ii) If X and Y are geometrically irreducible (resp. geometrically connected), so is $X \times_k Y$.

Proof. label=(i) The structural morphism $X \to \operatorname{Spec}(k)$ is surjective and universally open (resp. connected); it suffices therefore to apply (4.5.7).

lbbel=(ii) If Ω is an algebraically closed extension of k, we have $(X \times_k Y)_{(\Omega)} = X_{(\Omega)} \times_\Omega Y_{(\Omega)}$; by hypothesis $X_{(\Omega)}$ and $Y_{(\Omega)}$ are Ω-preschemes geometrically irreducible (resp. geometrically connected) (**I**, 3.3.10); by hypothesis $X_{(\Omega)}$ and $Y_{(\Omega)}$ are Ω-preschemes that are geometrically irreducible (resp. geometrically connected) (4.5.6), and it therefore suffices to apply (i).

□

Proposition (4.5.9). — *Let X be a k-prescheme. The following conditions are equivalent:*

label=a) X is geometrically irreducible (in other words, for every *extension K of k, $X \otimes_k K$ is irreducible).*

lbbel=b) For every finite separable extension K of k, $X \otimes_k K$ is irreducible.

lcbel=c) X is irreducible, and if x is its generic point, $\kappa(x)$ is a primary extension of k.

Proof. It is clear that *a)* implies *b)*. To see that *b)* implies *c)*, let us consider a finite separable extension K of k; if $p : X \otimes_k K \to X$ is the canonical projection, the maximal points of $X \otimes_k K$ are those of the fibre $p^{-1}(x) = \mathrm{Spec}(\kappa(x) \otimes_k K)$ (2.3.4) and ($\mathbf{0}_I$, 2.1.8); to say that this fibre is irreducible for every finite separable extension K of k amounts to saying that $\kappa(y)$ is a primary extension of k, by virtue of (4.3.2). Conversely, if *c)* is satisfied, the same reasoning (taking into account (4.3.2)) shows that *d)* implies *e)*. Let us show that *d)* implies *a)*. It suffices to show that $\kappa(x_\alpha) \otimes_k k'$ is a reduced ring for every finite radicial extension k' of k (4.3.5). We can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, A being a k-algebra that is reduced; then by hypothesis the ring $A \otimes_k k'$ is reduced (**I**, 5.1.4) and $\kappa(x_\alpha) \otimes_k k'$ is a ring of fractions of this ring, hence it is reduced ($\mathbf{0}_I$, 1.2.8). ☐

IV-2 | 63

Corollary (4.5.10). — *Let X be an irreducible k-prescheme, x its generic point, k' the separable algebraic closure of k in $\kappa(x)$, k'' a Galois extension of k (of finite degree or not) containing k'. Suppose k' is finite over k; then the irreducible components of $X \otimes_k k''$ are geometrically irreducible; their number is equal to $[k' : k]$ and is also the geometric number of irreducible components of X.*

Proof. As in (4.5.9), we are brought to considering the maximal points of the fibre $\mathrm{Spec}(\kappa(x) \otimes_k k'')$; by virtue of (4.3.4), they are $[k' : k]$ in number and the residue fields at these points (which are also those of $X \otimes_k k''$) are primary extensions of k'', which, taking into account (4.5.9), proves the corollary. ☐

Corollary (4.5.11). — *Let X be a k-prescheme; suppose that X has only a finite number of maximal points x_i ($1 \leqslant i \leqslant r$) and that for each i, the separable algebraic closure k'_i of k in $\kappa(x_i)$ is of finite degree over k. Then there exists a finite separable extension L of k such that the irreducible components of $X \otimes_k L$ are geometrically irreducible; their number, equal to $\sum_i [k'_i : k]$, is the geometric number of irreducible components of X.*

Proof. The maximal points of $X \otimes_k L$ are those of the various fibres $\mathrm{Spec}(\kappa(x_i) \otimes_k L)$ applied to the irreducible components of X and to their inverse images in $X \otimes_k L$. It suffices therefore to take for L a Galois extension of k finite over k containing all the k'_i and to apply (4.5.10). ☐

We note that (4.5.11) applies in particular to every k-prescheme X *of finite type* over k, since X is then Noetherian, hence has only a finite number of irreducible components, and each of the $\kappa(x_i)$ is an extension of finite type of k, hence it is also the case for the separable algebraic closure of k in $\kappa(x_i)$ ([**?**], p. 6-06, lemma 5).

Remarks (4.5.12). — label=(i) The notions defined in (4.5.2) depend on the base field k. To abbreviate, and when it is necessary to mention the base field, one may say "k-irreducible" (resp. "k-connected") instead of "geometrically irreducible (resp. connected) relative to k". We note that this terminology, in the context of schemes, is exactly the opposite of that of Weil: for Weil, "k-irreducible", "k-connected", "k-normal", etc., designate an *intrinsic* property of a prescheme X, independent of the base field k (in other words, of the morphism $X \to \mathrm{Spec}(k)$ chosen); we express these properties by saying simply that X is irreducible (resp. connected, normal, etc.). Weil says on the other hand "absolutely irreducible", "absolutely connected", "absolutely normal", etc., for the corresponding notions relative to the prescheme $X \otimes_k \Omega$, where Ω is an algebraically closed extension of k; the definition of a smaller "base field" (i.e., of a scheme over k defining V by extension of the base field from k to Ω) is for him a supplementary structure, and in a certain measure, secondary, so that his qualification of "absolute" or "intrinsic" on one hand, "relative to k" on the other, is the opposite of ours. We will therefore in the sequel avoid using the qualifier "absolute", which can lead to confusion, and whenever we use the abbreviated terminology introduced above (in conflict with the accepted terminology, we refer to (4.5.12)) to avoid any ambiguity. **IV-2** | 64

lbbel=(ii) The proposition (4.5.9) gives a "birational" criterion (in other words, depending only on the residue field at the generic point) for a k-prescheme X to be geometrically irreducible. There is no analogous criterion for X to be geometrically connected. For example, take $X = \mathrm{Spec}(\mathbf{R}[[\mathrm{T}, \mathrm{U}]]/(\mathrm{T}^2 + \mathrm{U}^2))$ (T, U indeterminates); X is an $\mathbf{R}$-scheme that is integral, and its field of rational functions $\kappa(x)$ is $\mathbf{C}((\mathrm{T}))$, as one easily verifies; $X \otimes_{\mathbf{R}} \mathbf{C}$ decomposes into two irreducible components (the two "isotropic lines") that have a common point (the maximal image in $\mathbf{C}[[\mathrm{T}, \mathrm{U}]]/(\mathrm{T}^2 + \mathrm{U}^2)$); hence $X \otimes_{\mathbf{R}} \mathbf{C}$ is connected. But let X' be the complement (open) of the closed point of X corresponding to the maximal ideal $(\mathrm{T}) + (\mathrm{U})$ of $\mathbf{R}[[\mathrm{T}, \mathrm{U}]]$; then $X' \otimes_{\mathbf{R}} \mathbf{C}$ is not connected, even though X and X' have the same field of rational functions.

Proposition (4.5.13). — *Let X, Y be two k-preschemes, $f : Y \to X$ a k-morphism. Suppose Y is geometrically connected and non-empty, and X connected. Then X is geometrically connected.*

Proof. Let Ω be an algebraic closure of k, $X' = X_{(\Omega)}$, $Y' = Y_{(\Omega)}$, $f' = f_{(\Omega)} : Y' \to X'$; it suffices to show that X' is connected. Let U' be an open and closed non-empty subset of X'; note that the morphisms $p : X' \to X$ and $q' : Y' \to Y$ are *open* (2.4.10) and *closed* since Ω is an algebraic extension of k (**II**, 6.1.10). Hence $U = p(U')$ is open and closed and non-empty in X. Since Y is non-empty, on concludes that $U' = p(U')$ is non-empty, and we deduce that $U' = X'$, without which the same reasoning applied to the open and closed subset $X' - U'$ would show that $f'^{-1}(X' - U') = Y'$, which is absurd since Y' is connected. □

Corollary (4.5.13.1). — *label=(i) Let X, Y be two k-preschemes, $f : Y \to X$ a k-morphism. If Y is geometrically connected and non-empty, the connected component X_0 of X containing $f(Y)$ is geometrically connected.*

lbbel=(ii) Let X be a k-prescheme. If Y is an irreducible component of X that is geometrically irreducible, then the connected component X_0 of X containing Y is geometrically connected.

Proof. label=(i) We can suppose Y reduced, so that f factorises as $f : Y \xrightarrow{g} X_0 \xrightarrow{j} X$, and X_0

□

denoting the reduced sub-prescheme of X having X_0 as underlying space (**I**, 5.2.2); it suffices to apply (4.5.13) to g. **IV-2** | 65

(ii) Considering a prescheme having Y as underlying space, and remarking that Y is *a fortiori* geometrically connected, it suffices to apply (i) to the injection $Y \to X$.

Corollary (4.5.14). — *Let X be a k-prescheme. If x is a point of X such that $\kappa(x)$ is a primary extension of k (in particular if x is a* rational point *of X), the connected component of x in X is geometrically connected.*

Proof. It suffices to apply (4.5.13.1) to $Y = \mathrm{Spec}(\kappa(x))$ and to the canonical morphism $Y \to X$, taking into account (4.5.9), which implies that Y is geometrically irreducible. □

Proposition (4.5.15). — *Let X be a k-prescheme, x a point of X, k' the separable algebraic closure of k in $\kappa(x)$. Suppose the following conditions are satisfied:*

label=(i) X is connected.

lbbel=(ii) k' is a finite extension of k.

Then the geometric number of connected components of X is $\leqslant [k' : k]$, and if k'' is a finite Galois extension of k containing k', the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. We note that condition (ii) is satisfied if X is *locally of finite type* over k.

There exist finite Galois extensions k'' of k containing k' by virtue of condition (ii). Set $X'' = X \otimes_k k''$; the morphism projection $p : X'' \to X$ is faithfully flat and finite, hence closed (**II**, 6.1.10) and by (2.3.6), (ii)) the image by p of every connected component X''_α of X'' is equal to X; X''_α therefore contains a point $x''_\alpha \in p^{-1}(x)$. But $p^{-1}(x)$ has a number of points equal to the number of irreducible components of $\mathrm{Spec}(\kappa(x) \otimes_k k'')$ (**I**, 3.4.9), that is to say $[k' : k]$ (4.3.4), and *a fortiori* the number of connected components of X'' is $\leqslant [k' : k]$. For every $x''_\alpha \in p^{-1}(x)$, $\kappa(x''_\alpha)$ is a primary extension of k'' by virtue of (4.3.5) and (**I**, 3.4.9). We can therefore apply the corollary (4.5.14) to x''_α and X'', which proves that all the connected components of X'' are geometrically connected. □

Corollary (4.5.16). — *Let X be a k-prescheme, (X_α) the family of its connected components, and for every α, let $x_\alpha \in X_\alpha$. Suppose the following conditions are satisfied:*

label=(i) The family (X_α) is finite.

lbbel=(ii) For every α, the separable algebraic closure k'_α of k in $\kappa(x_\alpha)$ is a finite extension of k.

Then the geometric number of connected components of X is at most $\sum_\alpha [k'_\alpha : k]$, and there exists a finite separable extension k'' of k such that all the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. Noting that the connected components of X are open by virtue of condition (i), it suffices to apply to each of the sub-preschemes of X induced on the open sets X_α the result of (4.5.15); for having an extension k'' answering the question, it suffices to take a Galois extension of k finite over k containing all the k'_α. □

Corollary (4.5.17). — *Suppose that the k-prescheme X contains a point x such that the separable algebraic closure of k in $\kappa(x)$ is finite over k. For X to be geometrically connected, it is necessary and sufficient that for every finite separable extension K of k, $X \otimes_k K$ be connected.*

IV-2 | 66

Remark (4.5.18). — We shall see in § 8 (8.4.5) that the conclusion of (4.5.17) is still valid if, instead of supposing the condition of the statement, one supposes that X is *quasi-compact*.

Proposition (4.5.19). — *Let X be a k-prescheme, Z a subset of X, k' an algebraically closed extension of k, $X' = X \otimes_k k'$, $p : X' \to X$ the canonical projection. Suppose that $Z' = p^{-1}(Z)$ is contained in a single irreducible component X'_0 of X'. Then $X_0 = p(X'_0)$ is an irreducible component of X containing Z, and furthermore X_0 is geometrically irreducible. If furthermore X'_0 is the only irreducible component of X' that meets Z', then X_0 is the only irreducible component of X that meets Z.*

To show that $X'_0 = p^{-1}(X_0)$, we shall apply the following:

Lemma (4.5.19.1). — *Let T, T' be two preschemes, $p : T' \to T$ a morphism; consider the prescheme $T'' = T' \times_T T'$, and let $p_1 : T'' \to T'$, $p_2 : T'' \to T'$ be the projections. For a subset U' of T' to be of the form $p^{-1}(U)$, where $U \subset T$, it is necessary and sufficient that $p_1^{-1}(U') = p_2^{-1}(U')$.*

Proof. The structural morphism $q : T'' \to T$ is equal to $p \circ p_1$ and to $p \circ p_2$ by definition, hence the relation is necessary. Conversely, suppose it is satisfied; let u' be a point of U', t' a point of $p^{-1}(p(u'))$; since $p(t') = p(u')$, there exists a point $t'' \in T''$ such that $p_1(t'') = u'$, $p_2(t'') = t'$ (**I**, 3.4.7); hence $t'' \in p_1^{-1}(U') = p_2^{-1}(U')$ by hypothesis, and by $t' = p_2(t'') \in U'$, which proves the lemma. □

Proof of (4.5.19). With this lemma established, to prove that $X'_0 = p^{-1}(X_0)$, we form the product $X'' = X' \times_X X'$ (relative to the morphism $p : X' \to X$); if p_1 and p_2 are the canonical projections of X'' to X', it is a matter of showing that $p_1^{-1}(X'_0) = p_2^{-1}(X'_0)$. Now, if we set $S = \operatorname{Spec}(k)$, $S' = \operatorname{Spec}(k')$, observing that if $S'' = S' \times_S S' = \operatorname{Spec}(k' \otimes_k k')$, we can also write $X'' = X' \times_{S'} S''$. If $\varphi'' : X'' \to S''$ is the structural morphism, it suffices to prove that for every $s'' \in S''$, we have $p_1^{-1}(X'_0) \cap \varphi''^{-1}(s'') = p_2^{-1}(X'_0) \cap \varphi''^{-1}(s'')$. If we set $T = \operatorname{Spec}(\kappa(s''))$, $U = X'' \times_{S'} T = X' \times_{S'} T$, and if $v : U \to X''$ is the canonical projection, it is a matter of proving that $U_1 = v^{-1}(p_1^{-1}(X'_0))$ and $U_2 = v^{-1}(p_2^{-1}(X'_0))$ are identical irreducible components of U (4.4.5), and since U_1 and U_2 both contain $w^{-1}(Z)$ (where $w = p \circ p_1 \circ v = p \circ p_2 \circ v$), the result follows from (4.4.5).

There exists a field K, common extension of k' and of $\kappa(s'')$, so that if we set $P = \operatorname{Spec}(K)$, the diagram IV-2 | 67

$$\begin{tikzcd} S' & P \arrow[l] \\ S \arrow[u] & T \arrow[l] \arrow[u] \end{tikzcd}$$

is commutative; if we set $Q = X \otimes_k K$, the diagram

$$\begin{tikzcd} X' & Q \arrow[l, "q"'] \\ X \arrow[u, "p"] & U \arrow[l, "w"] \arrow[u, "r"'] \end{tikzcd}$$

is therefore also commutative, and the two irreducible components $r^{-1}(U_1)$ and $r^{-1}(U_2)$ of Q (4.4.5) both contain $q^{-1}(Z')$; now, by virtue of (4.4.5), the images $q(r^{-1}(U_1))$ and $q(r^{-1}(U_2))$ are irreducible components of X' containing Z'; they are therefore both equal to X'_0, hence $r^{-1}(U_1) = r^{-1}(U_2)$ by virtue of (4.4.5), and finally $U_1 = U_2$.

Since p is surjective and X'_0 dominates an irreducible component of X, $X_0 = p(X'_0)$ is an irreducible component of X containing Z, and since $p^{-1}(X_0)$ is an irreducible component of X', X_0 is geometrically irreducible.

To prove the last assertion, let X_1 be a second irreducible component of X meeting Z. Taking $z \in Z \cap X_1$ (2.3.5), one sees that there exists an irreducible component of X' meeting Z' and dominating X_1; but since X'_0 is by hypothesis the only irreducible component of X' meeting Z', we necessarily have $X_1 = X_0$. □

Remark (4.5.20). — The fact that Z' is contained in a single irreducible component X'_0 of X' does *not* imply that $X_0 = p(X'_0)$ is the only irreducible component of X containing Z. Let us indeed take $k = \mathbf{R}$, $k' = \mathbf{C}$, and for X the k-scheme obtained by gluing (at a non-generic point) $X_1 = \mathrm{Spec}(\mathbf{R}[S,T]/(S^2+T^2+1))$ and $X_2 = \mathrm{Spec}(\mathbf{C}[T])$ (the local rings of these two curves at each of their closed points having residue field isomorphic to $\mathbf{C}$); X_1 and X_2 are identified with the two irreducible components of X; if x is their common point, $p^{-1}(x)$ is formed of two points distinct y', z' of X', $p^{-1}(X_1)$ is irreducible, and $p^{-1}(X_2)$ is the union of two irreducible components Y', Z' of X' such that $y' \in Y'$ and $z' \in Z'$, so that $p^{-1}(x)$ is not contained in a single irreducible component of X'.

Proposition (4.5.21). — *Let k be a separably closed field, and let k' be an algebraic closure of k. For every k-prescheme X, the canonical projection $X \otimes_k k' \to X$ is a universal homeomorphism.*

In particular, the irreducible (resp. connected) components of X are geometrically irreducible (resp. geometrically connected). **IV-2** | 68

Proof. By definition, k' is a radicial extension of k; the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is surjective, integral and radicial, hence the first assertion follows from (2.4.5), (i)); the second follows from (4.5.1). □

4.6. Geometrically reduced preschemes **IV-2** | 69

Corollary (4.6.3). — *Let X be an integral k-prescheme; for X to be geometrically reduced (resp. geometrically integral) over k, it is necessary and sufficient that its field of rational functions $\mathrm{R}(X)$ be a separable (resp. separable and primary) extension of k.*

Proof. This follows immediately from (4.5.9) and (4.6.1). □

Corollary (4.6.4). — *Let X be a reduced k-prescheme. Then, for every separable extension k' of k, $X' = X \otimes_k k'$ is reduced.*

Proof. It suffices to apply the equivalence of a) and e) in (4.6.1), replacing X by $\mathrm{Spec}(k')$ and S by X. □

Proposition (4.6.5). — *label=(i) Let X be a k-prescheme, K an extension of k. For X to be geometrically reduced (resp. geometrically integral) over k, it is necessary and sufficient that $X \otimes_k K$ be geometrically reduced (resp. geometrically integral) over K.*

lbbel=(ii) Let X, Y be two k-preschemes. If X and Y are geometrically reduced (resp. geometrically integral) over k, the same is true of $X \times_k Y$.

Proof. Assertion (i) is a trivial consequence of the definitions and of (4.4.1). To prove the part of (ii) concerning separability, we observe that if Ω is an algebraically closed extension of k, then $(X \times_k Y) \otimes_k \Omega = X \times_k (Y \otimes_k \Omega)$; since $Y \otimes_k \Omega$ is reduced, it follows that $X \times_k (Y \otimes_k \Omega)$ is reduced by virtue of (4.6.1), a)). The rest of assertion (ii) follows from what precedes and from (4.5.8), (ii)). □

Proposition (4.6.6). — *Let X be a k-prescheme of finite type. There exists a finite radicial extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k'.*

Proof. Since X can be covered by a finite number of affine opens U_i, we may restrict to the case where X is affine: indeed, if for each i, k_i is a finite radicial extension of k such that $(U_i \otimes_k k_i)_{\mathrm{red}}$ is geometrically reduced over k_i, we may suppose that the k_i are all contained in the same finite radicial extension k' of k; $(U_i \otimes_k k')_{\mathrm{red}}$ then identifies with $(U_i \otimes_k k_i)_{\mathrm{red}} \otimes_{k_i} k'$ by virtue of (**I**, 5.1.8), and hence is geometrically reduced over k' by hypothesis, and so the same is true of $(X \otimes_k k')_{\mathrm{red}}$. Suppose therefore that $X = \mathrm{Spec}(A)$, where A is a k-algebra of finite type, and let Ω be the algebraic closure of k. Denote by $\mathfrak{N}$ the nilradical of $A \otimes_k \Omega$; since $A \otimes_k \Omega$ is a Noetherian ring, $\mathfrak{N}$ is an ideal generated by a finite number of elements of the form $y_i = \sum_j x_{ij} \otimes \xi_{ij}$ where $x_{ij} \in A$, $\xi_{ij} \in \Omega$. Let K be a finite sub-extension of Ω containing the ξ_{ij}, and let $\mathfrak{N}$ be the ideal of $A \otimes_k K$ generated by the y_i; then it is clear that the y_i are nilpotent in $A \otimes_k K$; on the other hand, since $\mathfrak{N} \otimes_K \Omega = \mathfrak{N}'$ (and consequently $\mathfrak{N}' \cap (A \otimes_k K) = \mathfrak{N}$), $\mathfrak{N}$ contains the nilradical of $A \otimes_k K$, and hence is equal to it. Since $(A \otimes_k \Omega)/\mathfrak{N}' = ((A \otimes_k K)/\mathfrak{N}) \otimes_K \Omega$, we see that $(X \otimes_k K)_{\mathrm{red}} \otimes_K \Omega$ is reduced, and by (4.6.1), $(X_{(K)})_{\mathrm{red}}$ is geometrically reduced over K. Replacing K by the quasi-Galois extension over k that it generates in Ω, we see from (**I**, 5.1.8) that we may furthermore suppose K to be quasi-Galois over k, and hence a separable extension of a finite radicial extension k' of k (Bourbaki, *Alg.*, chap. V, § 10, n° 9, prop. 14). By virtue of (**I**, 5.1.8), $(X_{(k')})_{\mathrm{red}} \otimes_{k'} K$ is isomorphic to $(X_{(K)})_{\mathrm{red}}$, and hence is geometrically reduced; it follows that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced by virtue of (4.6.5), (i)). □ **IV-2** | 70

Corollary (4.6.7). — *If K is an extension of finite type of k, there exists a finite radicial extension k' of k such that the residue fields of the semi-local ring $K \otimes_k k'$ are separable over k'.*

Proof. It suffices to apply (4.6.6) to $X = \mathrm{Spec}(A)$, where A is a k-algebra of finite type of which K is the field of fractions. □

Corollary (4.6.8). — *Let X be a k-prescheme of finite type. There exists a finite extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' and the irreducible components and connected components of $X_{(k')}$ are geometrically connected.*

Proof. This follows immediately from (4.6.6), (4.5.10), and (4.5.15). □

Definition (4.6.9). — Let k be a field, X a k-prescheme, x a point of X. We say that X is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at the point x over k, if, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, X' is reduced (resp. integral) at x' (i.e., $\mathscr{O}_{X',x'}$ is reduced (resp. integral)). We say that X is *geometrically punctually integral* over k in all its points, in other words if, for every extension k' of k, all the local rings of $X \otimes_k k'$ are integral (in which case we also say that $X \otimes_k k'$ is *punctually integral*).

Let us note that, for X to be geometrically reduced over k (4.6.2), it is necessary and sufficient that it be geometrically reduced over k at every point $x \in X$.

Proposition (4.6.10). — *Let k be a field, X a k-prescheme, k' an extension of k, x' a point of $X' = X \otimes_k k'$, x its image in X. For X to be geometrically reduced (resp. geometrically punctually integral) at x over k, it is necessary and sufficient that X' be so at x' over k'. In particular, for X to be geometrically punctually integral over k, it is necessary and sufficient that X' be geometrically punctually integral over k'.*

Proof. It suffices to prove the first assertion. The condition being evidently necessary, let us prove that it is sufficient. Suppose therefore that X' is geometrically reduced (resp. geometrically punctually integral) at the point x' over k' and let us prove that X is so at the point x over k, in other words that for every extension k'' of k and every point x'' of $X'' = X \otimes_k k''$ above x, $\mathscr{O}_{X'',x''}$ is reduced (resp. integral). For this, let us note that there exists a point z of $X' \times_X X''$ which projects to x' and x'' (**I**, 3.4.7). Let s be the point of $\mathrm{Spec}(k' \otimes_k k'')$, image of z (**I**, 3.4.9); set $K = \kappa(s)$, $Z = X \otimes_k K$, so that K may be considered as a composite extension of k' and k'', and z as a point of Z whose image in X' (resp. X'') is x' (resp. x''). Since X' is geometrically reduced (resp. geometrically punctually integral) at the point x' over k', the ring $\mathscr{O}_{Z,z}$ is reduced (resp. integral), and since $\mathscr{O}_{Z,z}$ is isomorphic to a local ring of $\mathrm{Spec}(\mathscr{O}_{X'',x'}) \otimes_{k'} K$, which is isomorphic to a flat $\mathscr{O}_{X'',x'}$-module, it follows that $\mathscr{O}_{X'',x''}$ is reduced (resp. integral) ($\mathbf{0}_I$, 6.5.1). □

Corollary (4.6.11). — *Suppose k is perfect (resp. algebraically closed). For X to be geometrically reduced (resp. geometrically punctually integral) at the point x over k, it is necessary and sufficient that $\mathscr{O}_{X,x}$ be reduced (resp. integral).*

Proof. This is evidently necessary. Conversely, if this condition is satisfied, then, for every extension k' of k, $\mathrm{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$ is reduced (4.6.4) (resp. integral (4.4.4)); and hence, for every point x' of $X' = X \otimes_k k'$ above x, $\mathscr{O}_{X',x'}$, which is isomorphic to a local ring of $\mathrm{Spec}(\mathscr{O}_{X,x}) \otimes_k k'$, is reduced (resp. integral). □ **IV-2 | 71**

Proposition (4.6.12). — *Let k be a field, X a k-prescheme, x a point of X, Ω a perfect extension of k. The following conditions are equivalent:*

label=a) *X is geometrically reduced at k at the point x, in other words, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x, $\mathscr{O}_{X',x'}$ is reduced.*
lbbel=b) *The prescheme $X \otimes_k \Omega$ is reduced at a point above x.*
lcbel=c) *For every finite radicial extension k' of k, $X' = X \otimes_k k'$ is reduced at the unique point x' above x.*
ldbel=d) $\mathrm{Spec}(\mathscr{O}_{X,x})$ *is geometrically reduced over k.*
lebel=e) *$\mathscr{O}_{X,x}$ is reduced, and for every irreducible component Z of X containing x, of generic point z, $\kappa(z)$ is a separable extension of k.*

Proof. The implications $d) \Rightarrow a) \Rightarrow b) \Rightarrow c)$ are immediate, taking into account for the last (4.6.10), (4.6.11), and the fact that every radicial extension of k is isomorphic to a sub-extension of Ω. Let us prove that $c)$ implies $d)$. It suffices to show that for every finite radicial extension k' of k, $\mathscr{O}_{x,x} \otimes_k k'$ is reduced (4.6.1); but this ring is the local ring of X' at the unique point x' of X' above x (**I**, 3.5.8) and (**I**, 3.5.7); it is therefore reduced by virtue of hypothesis $c)$. Finally, the equivalence of $d)$ and $e)$ follows from the equivalence of $b)$ and $e)$ in (4.6.1), since the irreducible components of X containing x correspond bijectively to the irreducible components of $\mathrm{Spec}(\mathscr{O}_{X,x})$ (**I**, 2.4.2). □

Corollary (4.6.13). — *Under the hypotheses of (4.6.12), suppose furthermore that X is locally Noetherian. Then conditions $a)$ to $e)$ of (4.6.12) are also equivalent to the following:*

f) *There exists an open neighbourhood U of x which is geometrically reduced over k.*

Proof. Indeed, this condition trivially implies that X is geometrically reduced at the point x over k. Conversely, if this is so, since the ring $\mathscr{O}_{X,x}$ is reduced, there exists an open neighbourhood U of x in X which is reduced (**I**, 6.1.13); taking U small enough, we may furthermore suppose that U does not meet any of the irreducible components of X not containing x. The criterion (4.6.12), $e)$) then shows that U is geometrically reduced over k, taking into account the criterion (4.6.1), $e)$). □

It follows from (4.6.13) that when X is locally Noetherian, the set of $x \in X$ at which X is geometrically reduced over k is *open*, and it is the largest open subset of X which is geometrically reduced over k.

(4.6.14). It follows from (4.6.10) and (4.6.11) that if Ω is an algebraically closed extension of k, saying that X is geometrically punctually integral at the point x is the same as saying that X is geometrically punctually integral, and that the number of its connected components is equal to the number of its irreducible components (**I**, 6.1.10). It then suffices to apply (4.5.1), (4.6.1), and the first remark of (4.6.14).

For X to be geometrically punctually integral at the point x, it is necessary that X be integral at the point x, and that if z is the generic point of the unique irreducible component of X containing x, $\kappa(z)$ be a separable extension of k; this follows from (4.6.10). But these conditions are *not sufficient*, as the example (4.5.12), (ii)) shows. A sufficient but not necessary condition is that X be geometrically reduced at the point x and belong to only a single irreducible component of X, and that if one furthermore supposes X geometrically irreducible; if furthermore X is locally Noetherian, there is then an open neighbourhood of x in X which is geometrically integral. **IV-2 | 72**

Recall that if a prescheme is *locally Noetherian* and punctually integral, it is *locally integral* (**I**, 6.1.13). If X is a k-prescheme, locally of finite type over k, and geometrically punctually integral over k, it follows from this remark that, for every extension k' of k, $X \otimes_k k'$ is *locally integral* (one also says in this case that X is *geometrically locally integral*).

Proposition (4.6.15). — *label=(i) Let k be a field, X a k-prescheme of finite type. For X to be geometrically punctually integral, it is necessary and sufficient that it be geometrically reduced and that the geometric number of its irreducible components be equal to the geometric number of its connected components.*

lbbel=(ii) *Let X be a prescheme locally of finite type over k. If X is geometrically punctually integral at a point x, there exists an open neighbourhood U of x which is geometrically punctually integral. In other words, the set of $x \in X$ at which X is geometrically punctually integral is open in X.*

Proof. label=(i) Given an algebraically closed extension Ω of k, $X_{(\Omega)}$ is punctually integral (or locally integral, which amounts to the same since $X_{(\Omega)}$ is Noetherian) if and only if it is reduced and the number of its connected components is equal to the number of its irreducible components (**I**, 6.1.10). It then suffices to apply (4.5.1), (4.6.1), and the first remark of (4.6.14).

lbbel=(ii) The question being local on X, we may restrict to the case where X is a prescheme of finite type over k. Furthermore, we know that there is an open neighbourhood of x in X which is geometrically reduced (4.6.13), so we may suppose X geometrically reduced. There exists then a finite extension k' of k such that $X' = X \otimes_k k'$ is reduced and its irreducible components are geometrically irreducible (4.6.8). Let $p : X' \to X$ be the canonical projection, which is a surjective finite morphism, and let x'_j $(1 \leqslant j \leqslant r)$ be the points of $p^{-1}(x)$; by hypothesis, X' is integral at each of the x'_j, so each x'_j has an open neighbourhood V'_j in X' which is integral. Since p is closed (**II**, 6.1.10), each $p(V'_j)$ is a neighbourhood of x, and hence there exists an open neighbourhood U of x such that $p^{-1}(U)$ is contained in the union of the V'_j, and is therefore integral at each of its points, hence locally integral. The irreducible components of $p^{-1}(U)$ are open, pairwise disjoint, and since they are geometrically irreducible (4.5.9), $p^{-1}(U) = U \otimes_k k'$ is geometrically punctually integral by virtue of (i), and so it is the same for U by definition. □

Proposition (4.6.16). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. The following conditions are equivalent:*

label=a) *For every extension of finite type k' of k, if we set $X' = X \otimes_k k'$, then $\mathscr{F}' = \mathscr{F} \otimes_k k'$ is a reduced $\mathscr{O}_{X'}$-Module (3.2.2).*

lbbel=b) *For every finite radicial extension k' of k, $\mathscr{F}'$ is a reduced $\mathscr{O}_{X'}$-Module.*

lcbel=c) *$\mathscr{F}$ is reduced, and if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed sub-prescheme defined by $\mathscr{J}$ is geometrically reduced over k.*

IV-2 | 73

Proof. It is clear that a) implies b). The condition b) implies first that $\mathscr{F}$ is reduced: this means (3.2.3) that if Y is the closed sub-prescheme of X defined by the Ideal $\mathscr{J}$ coherent of $\mathscr{O}_X$, annihilator of $\mathscr{F}$, Y is reduced, and there is an $\mathscr{O}_Y$-Module coherent without torsion $\mathscr{G}$ of rank 1 on every irreducible component of Y, such that $j_*(\mathscr{G}) = \mathscr{F}$, $j : Y \to X$ being the canonical injection. Now, for every extension k' of k, the annihilator of $\mathscr{F}'$ is $\mathscr{J}' = \mathscr{J} \otimes_k k'$, the morphism projection $X' \to X$ being flat (2.1.11); the closed sub-prescheme of X' defined by $\mathscr{J}'$ is $Y' = Y \otimes_k k'$, and if $j' : Y' \to X'$ is the canonical injection, then $\mathscr{F}' = j'_*(\mathscr{G}')$. Note also that, for every extension k' of k such that $X' \otimes_k k'$ is locally Noetherian, $\mathscr{F}'$ is without associated immersed prime cycle (4.2.7). Furthermore, every maximal point y of Y is isomorphic to $\mathscr{O}_{Y,y}$, $\mathscr{G}_y$ isomorphic to $\mathscr{O}_{Y,y}$ at every maximal point y of Y; since $\mathscr{G}_y$ is isomorphic to $\mathscr{O}_{Y,y}$ at every maximal point y of Y, as it amounts to the same to say that $\mathscr{F}'$ is reduced or that the prescheme Y' is reduced, when X' is locally Noetherian. The fact that b) implies c) and that c) implies a) is therefore a consequence of (4.6.1), d) and b)); in a), the hypothesis that k' is an extension of finite type of k is not made solely to ensure that X' is locally Noetherian. When X is locally of finite type over k, X is locally Noetherian for every extension k' of k, which proves in this case the equivalence of d) and the other conditions. □

Definition (4.6.17). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.16), we say that $\mathscr{F}$ is *geometrically reduced or separable over k.*

Proposition (4.6.18). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. The following conditions are equivalent:*

label=a) *For every extension of finite type k' of k, if we set $X' = X \otimes_k k'$, then $\mathscr{F}' = \mathscr{F} \otimes_k k'$ is an integral $\mathscr{O}_{X'}$-Module (3.2.4).*

lbbel=b) *For every finite radicial extension k' of k, $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module.*

lcbel=c) *$\mathscr{F}$ is reduced (or integral), and if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ annihilating $\mathscr{F}$, the closed sub-prescheme defined by $\mathscr{J}$ is geometrically integral over k.*

Furthermore, if X is locally of finite type over k, these conditions are also equivalent to the following:

d) *For every extension k' of k (or for an algebraically closed extension k' of k), $\mathscr{F}'$ is an integral $\mathscr{O}_{X'}$-Module.*

Proof. The conditions a), b), and c) imply that $\mathscr{F}$ is integral, hence $\mathrm{Ass}(\mathscr{F})$ is reduced to a single point x, which is the generic point of $\mathrm{Supp}(\mathscr{F})$. For every extension k' of k for which X' is locally Noetherian, the prime cycles associated to $\mathscr{F}'$ are therefore non-immersed, and are the maximal points of $\mathrm{Supp}(\mathscr{F}')$, which are all above x. On the other hand, by virtue of (4.6.16), each of the conditions a), b), c) implies that $\mathscr{F}$ is geometrically reduced; it is trivial that a) implies b), and b) implies that $\mathrm{Supp}(\mathscr{F})$ is geometrically irreducible (4.5.9), hence c), since the closed sub-prescheme Y defined by $\mathscr{J}$ is geometrically reduced and geometrically irreducible, hence geometrically integral. **IV-2** | 74
Conversely, if Y is geometrically integral, for *every* extension k' of k, $\mathrm{Supp}(\mathscr{F}')$ is irreducible, and hence $\mathscr{F}'$ is integral; this shows that c) implies a), and that c) implies d) when X is locally of finite type over k. It is trivial that a) implies b), and b) implies that $\mathrm{Supp}(\mathscr{F})$ is geometrically irreducible (4.5.9), hence c). □

Definition (4.6.19). — When $\mathscr{F}$ satisfies the equivalent conditions of (4.6.18), we say that $\mathscr{F}$ is *geometrically integral over k*.

Proposition (4.6.20). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. Then, if $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) over k, $\mathscr{F} \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K.*

Proof. Indeed, for every finite extension K' of K, $(X \otimes_k K) \otimes_K K' = X \otimes_k K'$ is locally Noetherian, and we have seen in the proofs of (4.6.16) and (4.6.18) that the hypothesis on $\mathscr{F}$ implies that $\mathscr{F} \otimes_k K'$ is reduced (resp. integral), whence the conclusion. □

Proposition (4.6.21). — *Let k be a field, X, Y two k-preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent, $\mathscr{G}$ an $\mathscr{O}_Y$-Module coherent.*

label=(i) If $\mathscr{F}$ is geometrically reduced (resp. geometrically integral) and $\mathscr{G}$ reduced (resp. integral), then $\mathscr{F} \otimes_k \mathscr{G}$ is reduced (resp. integral).

lbbel=(ii) If $\mathscr{F}$ and $\mathscr{G}$ are both geometrically reduced (resp. geometrically integral), so is $\mathscr{F} \otimes_k \mathscr{G}$.

Proof. label=(i) Using (3.2.3), (4.6.16), and (4.6.18), we may restrict to the case where $\mathrm{Supp}(\mathscr{F}) = X$, $\mathrm{Supp}(\mathscr{G}) = Y$, X being geometrically reduced (resp. geometrically integral), Y reduced (resp. geometrically integral), $\mathscr{F}$ and $\mathscr{G}$ without associated immersed prime cycles, and $\mathscr{F}_x$ being isomorphic to $\mathscr{O}_x$ at every maximal point x of X, $\mathscr{G}_y$ isomorphic to $\mathscr{O}_y$ at every maximal point y of Y. We know then (4.2.3) that $\mathscr{F} \otimes_k \mathscr{G}$ is without associated immersed prime cycle, and the prime cycles associated to $\mathscr{F} \otimes_k \mathscr{G}$ are exactly the irreducible components of $X \times_k Y$ (4.2.5). We may therefore restrict to the case where X and Y are irreducible; then, if X is geometrically reduced and Y reduced, we know (4.2.4), (ii)) that $X \times_k Y$ is reduced; if X is geometrically integral and Y integral, we know that $X \times_k Y$ is furthermore irreducible (4.5.8), (i)), hence integral. Finally, every maximal point z of $X \times_k Y$ is above x and y, so the hypotheses on $\mathscr{F}$ and $\mathscr{G}$ imply, by virtue of (**I**, 9.1.12), that $(\mathscr{F} \otimes_k \mathscr{G})_z$ is isomorphic to $\mathscr{O}_z$, which completes the proof in this case.

lbbel=(ii) The argument is analogous, but here (after reduction to the case where $X = \mathrm{Supp}(\mathscr{F})$, $Y = \mathrm{Supp}(\mathscr{G})$), $X \times_k Y$ is geometrically reduced (resp. geometrically integral) by virtue of (4.6.5), (ii)).

□

The definitions (4.6.18) and (4.6.19) are localised as follows:

Definition (4.6.22). — Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. We say that $\mathscr{F}$ is *geometrically reduced or separable* (resp. *geometrically punctually integral*) *at a point $x \in X$* if, for every finite radicial (resp. finite) extension k' of k, $\mathscr{F} \otimes_k k'$ is reduced (resp. integral) at each of the points x' of $X \otimes_k k'$ above x (cf. (3.2.2) and (3.2.4)).

If $\mathscr{F}$ is reduced at x, there is an open neighbourhood U of x such that $\mathscr{F}|U$ is reduced (3.2.2). **IV-2** | 75
Reasoning as in (4.6.16), we see that saying that $\mathscr{F}$ is geometrically reduced at the point x is equivalent to saying that $\mathscr{F}$ is reduced at the point x and that the closed sub-prescheme Y defined by the annihilator $\mathscr{J}$ of $\mathscr{F}$ is *geometrically reduced at the point* x; it follows that there is an open neighbourhood U of x such that $\mathscr{F}|U$ is geometrically reduced (4.6.11). The same arguments, using (4.6.16) and (4.6.18), show that saying $\mathscr{F}$ is geometrically punctually integral at the point x is equivalent to saying that $\mathscr{F}$ is reduced at the point x and that the closed sub-prescheme Y is *geometrically punctually integral at the point* x.

4.7. Multiplicities in the primary decomposition on an algebraic prescheme

Lemma (4.7.1). — *Let A, B be two local rings, $\mathfrak{m}$ the maximal ideal of A, $\rho : A \to B$ a local homomorphism. Suppose that B is a flat A-module, and $B/\mathfrak{m}B$ a B-module of finite length. Then, for an A-module M to be of finite length, it is necessary and sufficient that $M_{(B)}$ be a B-module of finite length, and we have*

(4.7.1.1) $$\operatorname{length}_B(M_{(B)}) = \operatorname{length}_A(M) \cdot \operatorname{length}_B(B/\mathfrak{m}B).$$

Proof. This is a special case of (2.5.5.2). ☐

Corollary (4.7.2). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a homomorphism making B a flat A-module. Let $\mathfrak{q}$ be a minimal prime ideal of B, $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$, M an A-module of finite type; then $\mathfrak{p}$ is a minimal prime ideal of A, $M_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module of finite length, and we have*

(4.7.2.1) $$\operatorname{length}_{B_{\mathfrak{q}}}(M_{(B)})_{\mathfrak{q}} = (\operatorname{length}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}))(\operatorname{length}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})).$$

Proof. Indeed, $B_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$-module ($\mathbf{0}_I$, 6.3.2) and the homomorphism $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ is local, hence $B_{\mathfrak{q}}$ is a faithfully flat $A_{\mathfrak{p}}$-module ($\mathbf{0}_I$, 6.6.2); the fact that $\mathfrak{p}$ is minimal follows from A (2.3.5). Furthermore, $A_{\mathfrak{p}}$ is then an Artinian ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, prop. 9) and $M_{\mathfrak{p}}$ is therefore an $A_{\mathfrak{p}}$-module of finite length; the formula (4.7.2.1) is therefore a special case of (4.7.1.1). ☐

Proposition (4.7.3). — *Let k be a field of characteristic exponent p, X an integral locally Noetherian prescheme, $K = \mathrm{R}(X)$ the field of rational functions on X. Let k' be an extension of k, (X'_α) the family of irreducible components of $X' = X_{(k')}$, x'_α the generic point of X'_α.*

label=(i) *We suppose that X' is locally Noetherian (which holds if X is locally of finite type over k, or if k' is an extension of finite type of k). Let k'' be any extension of k', $X'' = X \otimes_k k''$, (X''_β) the family of irreducible components of X'', x''_β the generic point of X''_β. If x''_β is above x'_α, we have*

(4.7.3.1) $$\operatorname{length}(\mathscr{O}_{x''_\beta}) = \operatorname{length}(\mathscr{O}_{x'_\alpha}) \cdot \operatorname{length}(\mathscr{O}_{X'',x''_\beta})$$

where we have set $Z'' = (X'_{\mathrm{red}}) \otimes_{k'} k''$. In particular, if k'' is a separable extension of k' such that Z'' is locally Noetherian, we have $\operatorname{length}(\mathscr{O}_{x''_\beta}) = \operatorname{length}(\mathscr{O}_{x'_\alpha})$. **IV-2** | 76

lbbel=(ii) *If X is locally of finite type over k, or if k' is an extension of finite type of k, the numbers* $\operatorname{length}(\mathscr{O}_{x'_\alpha})$ *are powers of p.*

lcbel=(iii) *We suppose that X is locally of finite type over k. Then, if k' is perfect, all the numbers* $\operatorname{length}(\mathscr{O}_{x'_\alpha})$ *are equal and do not depend on the perfect extension K of k (and not on the extension k' considered). Furthermore, there exists a finite radicial extension k_1 of k such that, if x_1 is the generic point of $X_1 = X \otimes_k k_1$ (which is irreducible), we have* $\operatorname{length}(\mathscr{O}_{x'_\alpha}) = \operatorname{length}(\mathscr{O}_{x_1})$ *for all α.*

Proof. label=(i) Recall (4.2.4) that the irreducible components X'_α, which are finite in number, correspond bijectively to the minimal prime ideals $\mathfrak{p}_\alpha$ of $K \otimes_k k'$, and that $\mathscr{O}_{x'_\alpha}$ is an Artinian ring, isomorphic to $(K \otimes_k k')_{\mathfrak{p}_\alpha}$; these rings therefore depend only on the extensions K and k' of k (and are consequently the same for all integral k-preschemes having the same field of rational functions). The formula (4.7.3.1) follows from (4.7.2.1) applied to the Noetherian ring A of an affine neighbourhood of x'_α in X', and the ring B of an affine neighbourhood of x''_β in X'', with $\mathfrak{p}$ and $\mathfrak{q}$ being the minimal prime ideals corresponding to x'_α and x''_β respectively, and taking $M = A$: we will note that $A/\mathfrak{p}$ is the ring of an affine neighbourhood of x'_α in X'_{red}, and that $B = A \otimes_{k'} k''$; if $q' = q/\mathfrak{p}B$, a minimal prime ideal of $B/\mathfrak{p}B$, we have $\mathscr{O}_{X'',x''_\beta} = (B/\mathfrak{p}B)_{q'} = B_q/\mathfrak{p}B_q$, whence (4.7.3.1).

The last assertion of (i) follows from the fact that in this case Z'' is reduced (4.2.4), and hence $\mathfrak{p}B_q = qB_q$.

lbbel=(ii) If k' is an extension of finite type of k, there is a transcendental pure extension k'_0 of k such that k' is a finite extension of k'_0; since k'_0 is separable over k, it follows from (i) applied replacing k' and k'' by k and k'_0 respectively, that we may replace X by $X \otimes_k k'_0$, in other words suppose that k' is a *finite* extension of k. Suppose then that X is locally of finite type over k; then, X'' is locally Noetherian for every extension k'' of k'. Taking for k'' an algebraically closed extension of k', it follows from (i) that we are reduced, by virtue of (4.7.3.1), to proving the assertion when k' is moreover *radicial*. We know then that $K \otimes_k k_1$ is an Artinian ring having only a single prime ideal (necessarily maximal), and its quotient by that ideal is necessarily equal to $\overline{K}$; the length of $\overline{K} \otimes_k k_1$ divides that of $K \otimes_k k_1$ by virtue of (4.7.1.1). But $\overline{K} \otimes_k k_1$ is a $\overline{K}$-algebra, Artinian, having only a single prime ideal (necessarily maximal), and its quotient by that ideal is an extension finite of $\overline{K}$, hence necessarily equal to $\overline{K}$; the length of $\overline{K} \otimes_k k_1$ is therefore equal to its rank over $\overline{K}$, that is to say $[k_1 : k]$, which is a power of p.

Suppose then that X is locally of finite type over k; then, X'' is locally Noetherian for every extension k'' of k'. Taking for k'' an algebraically closed extension of k', we are reduced by virtue of (4.7.3.1) to proving the assertion when k' is *algebraically closed*. Since k' is then a separable extension of the algebraic closure $\bar{k}$ of k, it follows from (i) that we are reduced to the case where $k' = \bar{k}$, the algebraic closure. Now, we know (by replacing X by an affine open of finite type over k) that there is a finite radicial extension k_1 of k such that $(X \otimes_k k_1)_{\mathrm{red}}$ is separable over k_1 (4.6.6); hence $(X \otimes_k k_1)_{\mathrm{red}} \otimes_{k_1} \bar{k}$ is reduced (4.6.4). By virtue of (4.7.3.1), we are therefore reduced again to the case where k' is a finite radicial extension of k, and we conclude as before. **IV-2** | 77

lcbel=(iii) Let us note that two extensions k', k'' of k are contained in the same algebraically closed extension of k; to show that all the numbers length$(\mathscr{O}_{x'_\alpha})$ are equal, for k' perfect, we may therefore restrict to the case where $k' = \bar{k}$, the algebraic closure of k. To show that all the numbers length$(\mathscr{O}_{x'_\alpha})$ are equal, we may restrict to the case where $k' = \bar{k}$, the algebraic closure of k. Now, the finite radicial extension k_1 of k being then determined as in (ii), the x'_α are all above the unique generic point of $X_1 = X \otimes_k k_1$, and the relation length$(\mathscr{O}_{x'_{k_1}})$ = length$(\mathscr{O}_{x_1})$ for all α follows again from (4.7.3.1), taking into account the choice of k_1.

□

Definition (4.7.4). — Let k be a field, X a k-prescheme integral, K the field of rational functions on X. The *radicial multiplicity* of K (or of X) over k is defined as the upper bound of the lengths of the Artinian local rings $K \otimes_k k_1$, where k_1 ranges over the set of finite radicial extensions of k. The *separable multiplicity* of K (or of X) over k is defined as the number of geometric irreducible components of X if this number is finite, and $+\infty$ otherwise. Finally, the *total multiplicity* of K (or of X) over k is defined as the product of the radicial multiplicity and the separable multiplicity.

We note that if p is the characteristic exponent of k and if the radicial multiplicity of K over k is finite, it is a power $q = p^f$ of p, as follows from (4.7.3), (ii)); when $p \neq 1$, f is well-defined and called the *exponent of inseparability* of K over k; when $p = 1$, the radicial multiplicity is always equal to 1.

To say that the radicial multiplicity (resp. separable, resp. total) of X over k is 1 means that X is geometrically reduced (resp. geometrically irreducible, resp. geometrically integral) over k.

When X is locally of finite type, the radicial multiplicity of X over k is also the common length of the local rings A_i at the minimal prime ideals of the total ring of fractions A of $K \otimes_k k'$, for every perfect extension k' of k; the separable multiplicity of X over k is the number of minimal prime ideals of $A = K \otimes_k \Omega$ for every algebraically closed extension Ω of k, and finally the total multiplicity is the sum of the lengths of the local rings A_i at the minimal prime ideals of $A = K \otimes_k \Omega$; these numbers are all three finite in this case.

Definition (4.7.5). — Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, x a point of X such that $\mathscr{F}_x$ is an $\mathscr{O}_x$-module of finite length. The *geometric* *length* of $\mathscr{F}$ at x (relative to k), or *radicial multiplicity of x for $\mathscr{F}$*, is defined as the product $\lambda_x(\mathscr{F}) = (\mathrm{length}_{\mathscr{O}_x}(\mathscr{F}_x))\mu_i(\kappa(x) \mid k)$, where $\mu_i(\kappa(x) \mid k)$ denotes the radicial multiplicity of $\kappa(x)$ over k.

When X is integral, and x is the generic point of X, we have $\mathrm{length}(\mathscr{O}_x) = 1$, and hence the radicial multiplicity of x for $\mathscr{O}_X$ is none other than the radicial multiplicity of X over k, as defined in (4.7.4).

IV-2 | 78

Proposition (4.7.6). — *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of $\mathscr{O}_X$-Modules coherent. If at a point $x \in X$ the geometric length $\lambda_x(\mathscr{F})$ of $\mathscr{F}$ at this point is defined, the same is true of the geometric lengths $\lambda_x(\mathscr{F}')$ and $\lambda_x(\mathscr{F}'')$, of $\mathscr{F}'$ and $\mathscr{F}''$ at the point x, and conversely; furthermore, we have*

(4.7.6.1) $$\lambda_x(\mathscr{F}) = \lambda_x(\mathscr{F}') + \lambda_x(\mathscr{F}'').$$

Proof. The proposition follows immediately from the definition, since $\mathrm{length}(\mathscr{F}_x)$ is finite if and only if $\mathrm{length}(\mathscr{F}'_x)$ and $\mathrm{length}(\mathscr{F}''_x)$ are finite, and then $\mathrm{length}(\mathscr{F}_x) = \mathrm{length}(\mathscr{F}'_x) + \mathrm{length}(\mathscr{F}''_x)$. □

(4.7.7). Under the hypotheses of (4.7.5), the only points $x \in X$ such that $\mathscr{F}_x$ is a non-zero $\mathscr{O}_x$-module of finite length are (by virtue of (3.1.2)) the *maximal points* of $\mathrm{Supp}(\mathscr{F})$, as follows from Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 2 of prop. 7, the question being local on X. We know that there exists a closed sub-prescheme Y having $\mathrm{Supp}(\mathscr{F})$ as underlying space, and an $\mathscr{O}_Y$-Module coherent $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, $j : Y \to X$ being the canonical injection. If x is a maximal point of Y, we see that the geometric lengths of $\mathscr{F}$ and $\mathscr{G}$ at x are equal, $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,x}$ having the same residue body. We deduce from this the following characterisation of the geometric length:

Proposition (4.7.8). — *Under the hypotheses of (4.7.5), let x be a maximal point of $\mathrm{Supp}(\mathscr{F})$. Let k' be a perfect extension of k; set $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$; then the geometric length $\lambda_x(\mathscr{F})$ of $\mathscr{F}$ at x is equal to the length of the $\mathscr{O}_{x'}$-module $\mathscr{F}'_{x'}$ at every maximal point x' of $\mathrm{Supp}(\mathscr{F}')$ above x. Furthermore, there exists a finite radicial extension k_1 of k such that, setting $X_1 = X \otimes_k k_1$, $\mathscr{F}_1 = \mathscr{F} \otimes_k k_1$, $\lambda_x(\mathscr{F})$ is equal to the length of the $\mathscr{O}_{x_1}$-module $(\mathscr{F}_1)_{x_1}$ at every maximal point x_1 of $\mathrm{Supp}(\mathscr{F}_1)$ above x.*

Proof. The remark of (4.7.7) shows that we may restrict to the case where x is a maximal point of X. Furthermore, the question is local on X and we may suppose $X = \mathrm{Spec}(A)$ affine Noetherian, $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type; let $\mathfrak{p}$ (resp. $\mathfrak{q}$) be the minimal prime ideal of A (resp. B) corresponding to x (resp. x'). We have then, applying the formula (4.7.2.1),

$$\mathrm{length}_{\mathscr{O}_{x'}}(\mathscr{F}'_{x'}) = \mathrm{length}_{\mathscr{O}_x}(\mathscr{F}_x) \cdot \mathrm{length}_{B_\mathfrak{q}}(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}).$$

But since $\kappa(x) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$, the same reasoning as in (4.7.3) shows that $\mathrm{length}_{B_\mathfrak{q}}(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q})$ is equal to the length of the local ring $(B/\mathfrak{p}B)_{q'}$, where $q' = q/\mathfrak{p}B$; but $B/\mathfrak{p}B$ is a local ring of the ring $(A/\mathfrak{p}) \otimes_k k'$ for a minimal prime ideal, hence also a local ring of the total ring of fractions of $\kappa(x) \otimes_k k'$, and the length of such a ring is by definition $\mu_i(\kappa(x) \mid k)$, whence the first assertion. The second is proved in the same way, replacing k' by k_1 and using (4.7.3), (iii)). □

We also say, in the case where x is a maximal point of $\mathrm{Supp}(\mathscr{F})$, that the geometric length of $\mathscr{F}$ at x is the *radicial multiplicity of the prime cycle* $\overline{\{x\}}$ associated to $\mathscr{F}$.

Corollary (4.7.9). — *Under the hypotheses of (4.7.5), let k' be an extension of k, $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$; let Z be an irreducible component of $\mathrm{Supp}(\mathscr{F})$, and Z' an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ that dominates Z (4.2.7). Then the radicial multiplicities of Z and Z' with respect to $\mathscr{F}$ and $\mathscr{F}'$ respectively are the same.*

Proof. It suffices to consider a perfect extension k'' of k' and, setting $X'' = X \otimes_k k''$, $\mathscr{F}'' = \mathscr{F} \otimes_k k''$, a maximal point x'' of $\mathrm{Supp}(\mathscr{F}'')$ which is above the generic point of Z and Z'; it follows from (4.7.8) that $\mathrm{length}(\mathscr{F}''_{x''})$ is equal to the multiplicities of Z and of Z'. □

IV-2 | 79

Proposition (4.7.10). — *Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, x a point of X, z_i the generic points of the irreducible components of $\mathrm{Supp}(\mathscr{F})$ that contain x. For $\mathscr{F}$ to be geometrically reduced at the point x (4.6.22), it is necessary and sufficient that x does not belong to any associated immersed prime cycle of $\mathscr{F}$ and that, for all i, the geometric length $\lambda_{z_i}(\mathscr{F})$ of $\mathscr{F}$ at the point z_i be equal to 1.*

Proof. Let us first show that the conditions of the statement are sufficient. Indeed, if k' is a finite extension of k, x' a point of $X' = X \otimes_k k'$ above x, and $\mathscr{F}' = \mathscr{F} \otimes_k k'$, x' does not belong to any associated immersed prime cycle of $\mathscr{F}'$ by virtue of (4.2.7); furthermore, by virtue of (4.7.8), the geometric length of $\mathscr{F}'$ is equal to 1 at every generic point z'_j of an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ containing x', a point which is above one of the z_i by virtue of (2.3.4); *a fortiori* we have $\mathrm{length}(\mathscr{F}'_{z'_j}) = 1$, which proves that $\mathscr{F}'$ is reduced at the point x'.

Conversely, it follows first from (4.2.7) that if, for *one* finite extension k' of k, $\mathscr{F}'$ is reduced at all the points x' of X' above x, then x does not belong to any associated immersed prime cycle of $\mathscr{F}$. Furthermore, taking for k' the field k_1 of the statement of (4.7.8) (where x is replaced by one of the z_i), the hypothesis that $\mathscr{F}$ is geometrically reduced at the point x implies that $\lambda_{z_i}(\mathscr{F}) = 1$. □

Proposition (4.7.11). — *Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, k' an extension of k, $X' = X \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$, x a point of X, x' a point of X' above x. For $\mathscr{F}$ to be geometrically punctually reduced (resp. geometrically punctually integral) at the point x, it is necessary and sufficient that $\mathscr{F}'$ be geometrically punctually reduced (resp. geometrically punctually integral) at the point x'.*

Proof. The irreducible components of $\mathrm{Supp}(\mathscr{F}')$ containing x' each dominate an irreducible component of $\mathrm{Supp}(\mathscr{F})$ containing x, and conversely each of the latter is dominated by an irreducible component of $\mathrm{Supp}(\mathscr{F}')$ containing x' ((4.2.7), (i)) and (2.3.4)). The assertion concerning separability follows therefore from (4.2.7), (4.7.9), and (4.7.10). Let Y be the closed sub-prescheme of X defined by the Ideal $\mathscr{J}$ annihilator of $\mathscr{F}$, so that $Y' = Y \otimes_k k'$ is the closed sub-prescheme of X' defined by the Ideal $\mathscr{J}' = \mathscr{J} \otimes_k k'$ annihilator of $\mathscr{F}'$; saying that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically punctually reduced at the point x (resp. x') is the same as saying that $\mathscr{F}$ (resp. $\mathscr{F}'$) is geometrically punctually reduced at the point x (resp. x') and that Y (resp. Y') is geometrically punctually integral at the point x (resp. x'). By virtue of the assertion concerning separability, the hypothesis that $\mathscr{F}$ is geometrically punctually integral at the point x, or the hypothesis that $\mathscr{F}'$ is geometrically punctually integral at the point x', both imply that there exists in Y a neighbourhood of x which is separable, and the conclusion follows from (4.6.10). □

Definition (4.7.12). — Let k be a field, X a k-prescheme locally of finite type over k, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, x a maximal point of the support of $\mathscr{F}$. The *total multiplicity* of x (or of the prime cycle $\overline{\{x\}}$) **IV-2 | 80**
for $\mathscr{F}$ (relative to k) is defined as the product of the radicial multiplicity of x for $\mathscr{F}$ by the separable multiplicity of $\kappa(x)$ over k (4.7.4).

It amounts to the same to say that the total multiplicity of x for $\mathscr{F}$ is the product of the *length* $\mathrm{length}_{\mathscr{O}_x}(\mathscr{F}_x)$ by the total multiplicity of $\kappa(x)$ over k (4.7.4). With the notation of (4.7.11), the total multiplicity of x for $\mathscr{F}$ is equal to the *sum* of the total multiplicities for $\mathscr{F}'$ of the maximal points x'_i of $\mathrm{Supp}(\mathscr{F}')$ which are above x; furthermore, there exists a *finite* extension k' of k, such that the total multiplicity of x for $\mathscr{F}$ be equal to $\sum_i \mathrm{length}(\mathscr{F}'_{x'_i})$.

Proposition (4.7.13). — *The hypotheses and notation being those of (4.7.10), for $\mathscr{F}$ to be geometrically punctually integral at the point x, it suffices that x does not belong to any associated (necessarily non-immersed) prime cycle of $\mathscr{F}$ and that the total multiplicity for $\mathscr{F}$ of the generic point z of this cycle be equal to 1.*

Proof. Indeed, if k' is a finite extension of k, x' a point of $X' = X \otimes_k k'$ above x and $\mathscr{F}' = \mathscr{F} \otimes_k k'$ above x, x' cannot belong to more than a single prime cycle associated to $\mathscr{F}$, since there is only a single maximal point of $\mathrm{Supp}(\mathscr{F}')$ above x; furthermore, $\mathscr{F}$ is geometrically reduced at the point x by hypothesis; in addition, by (4.7.10), $\mathscr{F}$ is geometrically reduced at the point x by (4.7.10), whence the conclusion. □

4.8. Field of definition

(4.8.1). Given a prescheme X, we shall denote in this number by $\mathfrak{S}(X)$ the set of sub-preschemes of X, and as usual by $\mathfrak{P}(X)$ the set of subsets of the underlying space of X; for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, we shall denote by $\Phi(\mathscr{F})$ the set of quasi-coherent sub-$\mathscr{O}_X$-Modules of $\mathscr{F}$.

(4.8.2). Let k be a field, X, Y two k-preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-Modules quasi-coherent, K, K' two extensions of k such that $K' \subset K$. There are then, corresponding to the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(K')$, canonical maps

$$\Phi(\mathscr{F} \otimes_k K') \longrightarrow \Phi(\mathscr{F} \otimes_k K) \tag{4.8.2.1}$$

$$\mathrm{Hom}(\mathscr{F} \otimes_k K', \mathscr{G} \otimes_k K') \longrightarrow \mathrm{Hom}(\mathscr{F} \otimes_k K, \mathscr{G} \otimes_k K) \tag{4.8.2.2}$$

$$\mathfrak{S}(X_{(K')}) \longrightarrow \mathfrak{S}(X_{(K)}) \tag{4.8.2.3}$$

$$\mathrm{Hom}_{K'}(X_{(K')}, Y_{(K')}) \longrightarrow \mathrm{Hom}_K(X_{(K)}, Y_{(K)}) \tag{4.8.2.4}$$

$$\mathfrak{P}(X_{(K')}) \longrightarrow \mathfrak{P}(X_{(K)}) \tag{4.8.2.5}$$

If $p_X : X_{(K)} \to X_{(K')}$ is the canonical projection, the map (4.8.2.5) is $M \mapsto p_X^{-1}(M)$, and similarly (4.8.2.3) is $Z \mapsto Z \otimes_{K'} K$ (**I**, 4.4.1). The map (4.8.2.1) is $\mathscr{H} \mapsto \mathscr{H} \otimes_{\mathscr{O}_{X_{(K')}}} \mathscr{O}_{X_{(K)}}$; (4.8.2.2) is $u \mapsto p_X^*(u)$ and finally (4.8.2.4).

If we denote by $\varepsilon_{K,K'}$ any one of these canonical maps, it is clear that if K'', K', K are three extensions of k such that $K'' \subset K' \subset K$, we have

$$\varepsilon_{K,K''} = \varepsilon_{K,K'} \circ \varepsilon_{K',K''}. \tag{4.8.2.6}$$

Proposition (4.8.3). — *The canonical maps* (4.8.2.1) *to* (4.8.2.5) *are injective.*

Proof. We know indeed that p_X is faithfully flat and quasi-compact; the fact that (4.8.2.5) is injective follows from the fact that p_X is surjective. The injectivity of (4.8.2.1) follows from (2.2.2), that of (4.8.2.2) from (2.2.7), that of (4.8.2.3) from (2.2.15), and finally that of (4.8.2.4) from (2.2.16). □

IV-2 | 81

Definition (4.8.4). — With the notation of (4.8.2), we say that a sub-$\mathscr{O}_{X_{(K)}}$-Module quasi-coherent of $\mathscr{F} \otimes_k K$ (resp. a homomorphism $\mathscr{F} \otimes_k K \to \mathscr{G} \otimes_k K$, resp. a sub-prescheme of $X_{(K)}$, resp. a K-morphism $X_{(K)} \to Y_{(K)}$, resp. a subset of $X_{(K)}$) is *defined over* K' if it belongs to the image of the map (4.8.2.1) (resp. (4.8.2.2), resp. (4.8.2.3), resp. (4.8.2.4), resp. (4.8.2.5)). We say that K' is a *field of definition* of the object considered.

It is clear that K itself is a field of definition of any one of the objects under consideration in any of the five sets of images of maps (4.8.2.1) to (4.8.2.5). We will note however that for a K-prescheme Z, to say that the $\mathscr{O}_Z$-Module quasi-coherent $\mathscr{H}$ is "defined on a sub-field K' of K" has no meaning unless Z has been *given* in the form $X_{(K)}$ for a prescheme X over a sub-field k of K, and if $\mathscr{H}$ is a sub-$\mathscr{O}_Z$-Module of an $\mathscr{O}_Z$-Module which has been *given* in the form $\mathscr{F} \otimes_k K$, where $\mathscr{F}$ is an $\mathscr{O}_X$-Module quasi-coherent; there are analogous remarks for the other notions. It follows from (4.8.2.6) that if an element of one of the five sets of images of the maps (4.8.2.1) to (4.8.2.5) is *defined* on K', it is also defined on every sub-field K'' such that $K' \subset K'' \subset K$. Finally, with the notation of (4.8.2.6), if K_1 is an extension of K, and an element of one of the sets of images of $\varepsilon_{K_1,K'}$ is defined on K', it suffices that its image under $\varepsilon_{K_1,K}$ be defined on K', by virtue of the relation $\varepsilon_{K_1,K'} = \varepsilon_{K_1,K} \circ \varepsilon_{K,K'}$.

Proposition (4.8.5). — *With the notation of* (4.8.2), *let* (X_α) *be an open covering of* X. *For an element* $\mathscr{H} \in \Phi(\mathscr{F} \otimes_k K)$ *(resp.* $u \in \mathrm{Hom}(\mathscr{F} \otimes_k K, \mathscr{G} \otimes_k K)$*, resp.* $Z \in \mathfrak{S}(X_{(K)})$*, resp.* $M \in \mathfrak{P}(X_{(K)})$*) to be defined over* K'*, it is necessary and sufficient that for every* α*,* $\mathscr{H} \,|\, (X_\alpha)_{(K)}$ *(resp.* $Z \cap (X_\alpha)_{(K)}$*, resp.* $M \cap (X_\alpha)_{(K)}$*) be defined over* K'*.*

Proof. This follows immediately from the injectivity of the maps (4.8.2.1), (4.8.2.2), (4.8.2.3), and (4.8.2.5): for example, if for every α, there is a sub-$\mathscr{O}_{(X_\alpha)_{(K')}}$-Module quasi-coherent $\mathscr{H}'_\alpha$ such that $\mathscr{H} \,|\, (X_\alpha)_{(K)} = \mathscr{H}'_\alpha \otimes_{K'} K$, for two indices α, β arbitrary, we will have $\mathscr{H}'_\alpha \,|\, (X_\alpha \cap X_\beta)_{(K')} = \mathscr{H}'_\beta \,|\, (X_\alpha \cap X_\beta)_{(K')}$, hence $\mathscr{H}'_\alpha \,|\, (X_\alpha \cap X_\beta)_{(K')} = \mathscr{H}'_\beta \,|\, (X_\alpha \cap X_\beta)_{(K')}$ for any two indices α and β, and so there is a sub-$\mathscr{O}_{X_{(K')}}$-Module quasi-coherent $\mathscr{H}'$ of $\mathscr{F} \otimes_k K'$ such that $\mathscr{H} = \mathscr{H}' \otimes_{K'} K$. We argue in the same way for the other cases. □

Lemma (4.8.6). — *Let k be a field, K an extension of k, A a k-algebra, M an A-module, $\overline{N}$ a sub-$A_{(K)}$-module of $M_{(K)}$, K' a sub-extension of K. The following conditions are equivalent:*

label=a) $\overline{N}$ is of the form $N' \otimes_{K'} K$, where N' is a sub-$A_{(K')}$-module of $M_{(K')}$.
lbbel=b) If $X = \operatorname{Spec}(A)$, the $\mathscr{O}_{X_{(K)}}$-Module quasi-coherent $(\overline{N})^{\sim}$ on $X_{(K)} = \operatorname{Spec}(A_{(K)})$ is defined over K'.
lcbel=c) $\overline{N}$ is of the form $N' \otimes_{K'} K$, where N' is a sub-K'-vector subspace of $M_{(K')}$.

Proof. The equivalence of a) and b) follows trivially from (**I**, 1.6.5); it is clear that a) implies c). **IV-2 | 82**
Conversely, if c) is satisfied, we may write (up to canonical identification) $N' = \overline{N} \cap M_{(K')}$. Since $\overline{N}$ and $M_{(K')}$ are by hypothesis sub-$A_{(K')}$-modules, it follows that N' is also an $A_{(K')}$-module. □

Corollary (4.8.7). — *Under the hypotheses of (4.8.6), there exists a smallest sub-field K' of K containing k such that the conditions of (4.8.6) are satisfied.*

Proof. It suffices to see this for condition c), where this follows from Bourbaki, *Alg.*, chap. II, 3[e] éd., § 8, n[o] 6, prop. 6. □

Lemma (4.8.8). — *With the notation of (4.8.2):*

label=(i) Let $\overline{u} : \mathscr{F} \otimes_k K \to \mathscr{G} \otimes_k K$ be an $\mathscr{O}_{X_{(K)}}$-homomorphism, and let $\mathscr{H} \subset (\mathscr{F} \otimes_k K) \oplus (\mathscr{G} \otimes_k K)$ be its graph. For $\overline{u}$ to be defined over K', it is necessary and sufficient that $\mathscr{H}$ be so.
lbbel=(ii) Let $\overline{Z}$ be a closed sub-prescheme of $X_{(K)}$, and let $\overline{\mathscr{J}}$ be the quasi-coherent Ideal of $\mathscr{O}_{X_{(K)}}$ defining $\overline{Z}$. For $\overline{Z}$ to be defined over K', it is necessary and sufficient that $\overline{\mathscr{J}}$ be so.
*lcbel=(iii) Let $\overline{f}$ be a K-morphism $\overline{f} : X_{(K)} \to Y_{(K)}$ and let $\overline{Z}$ be the sub-prescheme of $X_{(K)} \times_K Y_{(K)}$, graph of $\overline{f}$ (**I**, 5.3.11); for $\overline{f}$ to be defined over K', it is necessary and sufficient that $\overline{Z}$ be so.*

Proof. Assertion (ii) follows immediately from (**I**, 4.4.5). The necessity in assertion (iii) is evident; suppose conversely that $\overline{Z} = Z'_{(K')}$, where Z' is a sub-prescheme of $(X \times_k Y)_{(K')}$, and if $p' : Z' \to X_{(K')}$ is the restriction to Z' of the first projection, we know that $p'_{(K)}$ is an isomorphism of preschemes, hence so is p' ((2.7.1), (viii)); but this means (**I**, 5.3.11) that Z' is the graph of a K'-morphism $f' : X_{(K')} \to Y_{(K')}$ (**I**, 5.3.12).

Finally, to prove (i), by virtue of (4.8.5), we may suppose that $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, $\mathscr{G} = \widetilde{N}$, where M and N are A-modules, and $\overline{u} = (\overline{v})^{\sim}$, where $\overline{v} : M_{(K)} \to N_{(K)}$ is an $A_{(K)}$-homomorphism. Let us suppose that the graph of $\overline{v}$ is of the form $P'_{(K)}$, where P' is a sub-$A_{(K')}$-module of $M_{(K')} \oplus N_{(K')}$; if $p' : P' \to M_{(K')}$ is the restriction to P' of the first projection, we know that $p' \otimes 1_K$ is an isomorphism of $A_{(K)}$-modules, hence, by faithful flatness ($\mathbf{0}_I$, 6.4.1), p' is an isomorphism of $A_{(K')}$-modules; hence P' is the graph of an $A_{(K')}$-homomorphism $v' : M_{(K')} \to N_{(K')}$ and we have evidently $\overline{v} = v' \otimes 1_K$. □

Proposition (4.8.9). — *Let k be a field, K an extension of k, X a k-prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. Let $\mathscr{H}$ be a sub-$\mathscr{O}_{X_{(K)}}$-Module quasi-coherent of $\mathscr{F} \otimes_k K$. Then there exists a smallest field of definition of $\mathscr{H}$.*

Proof. When X is affine, the assertion is simply (4.8.7). In the general case, we consider a covering (X_α) of X by affine opens; by virtue of the above, for every α there is a smallest sub-field K'_α of K containing k and such that $\mathscr{H} \mid (X_\alpha)_{(K)}$ is defined on K'_α. By virtue of (4.8.5), the sub-field of K generated by the K'_α is the smallest field of definition of $\mathscr{H}$. □

Corollary (4.8.10). — *With the notation of (4.8.9), let $\mathscr{G}$ be a second $\mathscr{O}_X$-Module quasi-coherent, and let $\overline{u} : \mathscr{F} \otimes_k K \to \mathscr{G} \otimes_k K$ be an $\mathscr{O}_{X_{(K)}}$-homomorphism; then there exists a smallest field of definition of $\overline{u}$.*

Proof. Indeed, it follows from (4.8.8) that such a field is also the smallest field of definition of the graph of $\overline{u}$, and the existence of the latter follows from (4.8.9). □ **IV-2 | 83**

Corollary (4.8.11). — *Let k be a field, X a k-prescheme, K an extension of k, $\overline{Z}$ a closed sub-prescheme of $X_{(K)}$; then there exists a smallest field of definition of $\overline{Z}$.*

Proof. Indeed, if $\overline{\mathscr{J}}$ is the quasi-coherent Ideal of $\mathscr{O}_{X_{(K)}}$ defining $\overline{Z}$, it follows from (4.8.8) that this is equivalent to the existence of a smallest field of definition of $\overline{\mathscr{J}}$, which follows from (4.8.9). □

Corollary (4.8.12). — *Let k be a field, X, Y two k-preschemes, Y being supposed separated, K an extension of k, $\overline{f} : X_{(K)} \to Y_{(K)}$ a K-morphism. Then there exists a smallest field of definition of $\overline{f}$.*

Proof. Indeed, the existence of such a field, by virtue of (4.8.8), is equivalent to that of the smallest field of definition of the graph $\overline{Z}$ of $\overline{f}$; but as $\overline{Z}$ is a sub-prescheme *closed* of $(X \times_k Y)_{(K)}$ (**I**, 5.4.3), the existence of a smallest field of definition of $\overline{Z}$ follows from (4.8.11). □

Proposition (4.8.13). — *Under the hypotheses of (4.8.11) (resp. (4.8.9), (4.8.10), (4.8.12)), suppose furthermore that X is of finite type over k (resp. that X is of finite type over k and $\mathscr{F}$ and $\mathscr{G}$ coherent, resp. that X and Y are of finite type over k). Then the smallest field of definition of $\overline{Z}$ (resp. $\mathscr{H}$, $\overline{u}$, $\overline{f}$) is an extension of finite type of k.*

Proof. We may restrict to the case of (4.8.9), and suppose X affine since X is a finite union of affine opens. The whole thing reduces then, with the notation of (4.8.6), to showing that if A is an algebra of finite type over k and M an A-module of finite type, there is a sub-extension K' of K satisfying conditions $a)$, $b)$, or $c)$ and which is of finite type over k (every sub-extension of an extension of finite type being of finite type). Now, let $(m_i)_{1 \leqslant i \leqslant n}$ be a system of generators of the A-module M; then $\overline{N}$ is an $A_{(K)}$-module of finite type, admitting a finite system of generators n_j $(1 \leqslant j \leqslant r)$ such that $n_j = \sum_i (m_i \otimes 1) b_{ij}$, where $b_{ij} \in A_{(K)}$; each b_{ij} may be written as $b_{ij} = \sum_h a_{ijh} \otimes c_{ijh}$, where $a_{ijh} \in A$ and $c_{ijh} \in K$; it is clear that the extension K' of k generated by the c_{ijh} answers the question. □

Proposition (4.8.14). — *Let k be a field, X a k-prescheme of finite type, K an extension of k containing a perfect extension of k. Then the smallest field of definition of the closed sub-prescheme $(X_{(K)})_{\text{red}}$ of $X_{(K)}$ is a finite radicial extension of k.*

Proof. If p is the characteristic exponent of k, the hypothesis implies that $k' = k^{p^{-\infty}}$ is contained in K; we know that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' (4.6.1), and hence $(X_{(k')})_{\text{red}} \otimes_{k'} K$ is reduced, and is consequently equal to $(X_{(K)})_{\text{red}}$ (**I**, 5.1.8); in other words, k' is a field of definition of $(X_{(K)})_{\text{red}}$. Since k' is an algebraic extension of k, the conclusion follows from (4.8.13). □

Remarks (4.8.15). — label=(i) The conclusion of (4.8.14) does not necessarily hold if one does not assume that K contains a perfect extension of k. For example, let k_0 be a field of characteristic $p > 0$, k the field of rational functions $k_0(s,t)$ in two indeterminates, L the field $k(s^{1/p}, t^{1/p})$ and $X = \operatorname{Spec}(L)$. Let u be a third indeterminate and K be the field $k(u, s^{1/p} + ut^{1/p})$; one easily verifies that k is algebraically closed in K and that $L \otimes_k K$ has a non-zero nilradical, and hence $X_{(K)}$ is not reduced; but as $(X_{(K)})_{\text{red}}$ is not defined on k, and K does not contain any sub-extension of finite degree over k other than k, the smallest field of definition of $(X_{(K)})_{\text{red}}$ cannot be finite over k. **IV-2 | 84**

lbbel=(ii) Given an element of one of the sets of images of the maps (4.8.2), and an extension K_1 of K, it is equivalent to say that the object considered admits a smallest field of definition, or that its image under $\varepsilon_{K_1,K}$ admits one (the two sub-fields of definition being necessarily the same), as one has seen in (4.8.4).

4.9. Field of definition of a part of a prescheme

Proposition (4.9.1). — *Let k be a field, X a k-prescheme, K an extension of k, $\overline{T}$ a closed subset of $X_{(K)}$, K' a sub-extension of K. The following conditions are equivalent:*

label=a) *$\overline{T}$ is defined over K'.*
lbbel=b) *The open subset $\overline{U} = X_{(K)} - \overline{T}$ of $X_{(K)}$ is defined over K'.*
b′) *The sub-prescheme of $X_{(K)}$ induced on the open $\overline{U}$ is defined over K'.*
c) *There exists a closed sub-prescheme $\overline{Z}$ of $X_{(K)}$ having $\overline{T}$ as underlying space, and defined over K'.*

Proof. If $g : X_{(K)} \to X_{(K')}$ is the canonical projection, saying that $\overline{T}$ is defined over K' means that there exists a closed subset T' of $X_{(K')}$ such that $\overline{T} = g^{-1}(T')$; since $g^{-1}(X_{(K')} - T') = X_{(K)} - g^{-1}(T')$ for every subset T' of $X_{(K')}$, a), b), and b') are equivalent. If $\overline{Z}$ is a closed sub-prescheme having $\overline{T}$ as underlying space, defined over K', then $\overline{Z} = Z' \otimes_{K'} K$, where Z' is a closed sub-prescheme of $X_{(K')}$ having T' as underlying space; hence a) implies c). Conversely, to see that a) implies c), it suffices to consider a closed sub-prescheme Z' of $X_{(K')}$ having T' as underlying space and to take $\overline{Z} = Z' \otimes_{K'} K$ (**I**, 5.2.1). □

Corollary (4.9.2). — *With the notation of (4.9.1), suppose that K' be a field of definition of $\overline{T}$; every extension $K'' \subset K$ of k, such that K' be a radicial extension of K'', is also a field of definition of $\overline{T}$.*

Proof. It suffices to observe that the canonical projection $g : X_{(K')} \to X_{(K'')}$ is a surjective, integral, and radicial morphism, hence a homeomorphism (2.4.5). □

Remark (4.9.3). — It follows from (4.9.2) that a closed subset $\overline{T}$ of $X_{(K)}$ does not necessarily admit a *smallest* field of definition. For example, if $K = k(t)$ where t is an indeterminate and k is a perfect field of characteristic $p > 0$, a closed subset $\overline{F}$ of $X_{(K)}$ can have no smallest field of definition: indeed, for every sub-field K' of K containing k and $\neq k$, we have $K'^p \neq K'$, K'^p contains k and K', and hence IV-2 | 85
if K' is a field of definition of $\overline{F}$, so is K'^p. However:

Proposition (4.9.4). — *With the notation of (4.9.1), suppose that K is a separable extension of K'; for K' to be a field of definition of $\overline{T}$, it is necessary and sufficient that K' be a field of definition of the reduced sub-prescheme of $X_{(K)}$ having $\overline{T}$ as underlying space.*

Proof. The condition is evidently sufficient by virtue of (4.9.1); to see that it is necessary, let us note that if $\overline{T} = g^{-1}(T')$, where T' is a closed subset of $X_{(K')}$, and if Z' is the reduced sub-prescheme of $X_{(K')}$ having T' as underlying space, then $\overline{Z} = Z' \otimes_{K'} K = \operatorname{Spec}(K) \times_{K'} Z'$ is reduced by virtue of the criterion (4.6.1), e)) and of the hypothesis on K, and has $\overline{T}$ as underlying space. □

Corollary (4.9.5). — *Suppose one of the following hypotheses is satisfied:*

label=a) *k is of characteristic 0.*
lbbel=b) *K is an algebraic extension of k and X is of finite type.*

Then every closed subset $\overline{T}$ of $X_{(K)}$ has a smallest field of definition, which is a separable extension of k (and finite over k under hypothesis b)).

Proof. Under hypothesis a), K is separable on each of its sub-fields, and it follows from (4.9.4) that the fields of definition of $\overline{T}$ are the same as those of the reduced sub-prescheme of $X_{(K)}$ having $\overline{T}$ as underlying space. The corollary follows then from (4.8.11). Under hypothesis b), we know that K is a radicial extension of a separable extension K_1 of k. Then, for every sub-extension K' of K, K' is radicial on $K' \cap K_1$, and hence, by virtue of (4.9.2), K' is a field of definition of $\overline{T}$ if and only if $K' \cap K_1$ is. This brings us back to the case where $K' = K_1$, and hence to the case where K' is a separable algebraic extension of k, and one completes the argument as in the case a). □

Proposition (4.9.6). — *Let k be a field, X a k-prescheme, K an extension of k, $\overline{U}$ a subset which is both open and closed in $X_{(K)}$, $\overline{V}$ its complement. Let K' be a sub-extension of K; the following conditions are equivalent:*

label=a) *K' is a field of definition of the subset $\overline{U}$ of $X_{(K)}$.*
a′) *K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open $\overline{U}$.*
b) *K' is a field of definition of the subset $\overline{V}$ of $X_{(K)}$.*

b′) *K' is a field of definition of the closed sub-prescheme of $X_{(K)}$ induced on the open $\overline{V}$.*

Proof. It is clear that a') implies a), and a) implies b') by virtue of (4.9.1). Since $\overline{U}$ and $\overline{V}$ play symmetric roles, b') implies b) and b) implies a'), which completes the proof. □

Corollary (4.9.7). — *Under the hypotheses of (4.9.6), there exists a smallest field of definition K' for the subset $\overline{U}$ of $X_{(K)}$, and K' is also the smallest field of definition of the subset $\overline{V}$ of $X_{(K)}$ and of the closed sub-preschemes of $X_{(K)}$ induced on the opens $\overline{U}$ and $\overline{V}$. If furthermore X is of finite type over k, K' is a finite separable extension of k.*

Proof. Taking into account (4.9.6) and (4.8.11), the only thing to prove is the last assertion. By virtue of (4.8.15), (ii)), we may restrict to the case where K is algebraically closed. By virtue of (4.8.13), we may restrict to the case where K is algebraically closed. Let $\bar{k}$ be the algebraic closure of k contained in K, and let us denote by W_i the connected components of $X_{(\bar{k})}$ ($1 \leqslant i \leqslant m$); if $p : X_{(K)} \to X_{(\bar{k})}$ is **IV-2 | 86**
the canonical projection, we know that the $p^{-1}(W_i)$ are the connected components of $X_{(K)}$ (4.4.6), and hence U is a union of a certain number of these components, and is therefore defined on $\bar{k}$; the conclusion follows by virtue of (4.9.5), b)). □

§5. Dimension and depth of locally Noetherian preschemes

5.1. Dimension of preschemes **IV-2 | 86**

(5.1.1). Let A be a ring, $\mathfrak{J}$ an ideal of A. Recall (**0**, 16.1) that the definitions of the dimension of a ring (**0**, 16.1.1) and those of the theory of the combinatorial dimension of topological spaces (**0**, 14.1.2) give the following relations:

(5.1.1.1) $$\dim(\mathrm{Spec}(A)) = \dim(A)$$

(5.1.1.2) $$\dim(\mathrm{V}(\mathfrak{J})) = \dim(A/\mathfrak{J})$$

(5.1.1.3) $$\mathrm{codim}(\mathrm{V}(\mathfrak{J}), \mathrm{Spec}(A)) = \mathrm{ht}(\mathfrak{J})$$

(5.1.1.4) $\mathrm{Spec}(A)$ is a catenary space ⇔ A is a catenary ring.

(5.1.1.5) $\mathrm{Spec}(A)$ is equidimensional ⇔ A is equidimensional ⇔ the minimal prime ideals $\mathfrak{p}_\alpha$ of A are such that the dimensions of the rings $A/\mathfrak{p}_\alpha$ are all equal.

(5.1.1.6) $\mathrm{Spec}(A)$ is equicodimensional ⇔ A is equicodimensional ⇔ all maximal ideals of A have the same height.

Recall that a Noetherian ring A is said to be *biequidimensional* if $\mathrm{Spec}(A)$ is biequidimensional, that is to say if $\mathrm{Spec}(A)$ is Noetherian and A is equidimensional, equicodimensional, catenary, and of finite dimension.

Proposition (5.1.2). — *Let X be a prescheme, Y an irreducible closed subset of X, y the generic point of Y. Then*

(5.1.2.1) $$\mathrm{codim}(Y, X) = \dim(\mathscr{O}_{X,y})$$

Proof. Indeed (**I**, 2.4.2), the irreducible closed subsets of X containing y are canonically in bijective correspondence with the irreducible closed subsets of $\mathrm{Spec}(\mathscr{O}_{X,y})$, and are therefore in bijective correspondence with the prime ideals of $\mathscr{O}_{X,y}$, and (5.1.2.1) follows from the definitions.

In particular, we thus recover the fact that the generic points of the irreducible components of X are the *only* points $x \in X$ such that $\dim(\mathscr{O}_x) = 0$ (**I**, 1.1.14). □

Corollary (5.1.3). — *For every closed subset Y of a prescheme X, we have*

(5.1.3.1) $$\mathrm{codim}(Y, X) = \inf_{y \in Y} \dim(\mathscr{O}_{X,y}).$$

Furthermore, if X is locally Noetherian, then for every $x \in Y$ we have

(5.1.3.2) $$\mathrm{codim}_x(Y, X) = \inf_{y \in Y,\, x \in \overline{\{y\}}} \dim(\mathscr{O}_{X,y}).$$

Proof. The relation (5.1.3.2) follows from (5.1.3.1) and (**0**, 14.2.6). This corollary allows us to *define*, for any subset Y of a prescheme X, the *codimension* $\mathrm{codim}(Y, X)$ of Y in X as equal to the right-hand side of (5.1.3.1). □

Proposition (5.1.4). — *For every prescheme X, we have*

(5.1.4.1) $$\dim(X) = \sup_{x \in X} \big(\dim(\mathscr{O}_x)\big).$$

If every irreducible closed subset of X contains a closed point, then we also have

(5.1.4.2) $$\dim(X) = \sup_{x \in F} \big(\dim(\mathscr{O}_x)\big)$$

where F is the set of closed points of X.

Proof. It suffices to note (by virtue of (**I**, 2.4.2)) that the chains of irreducible closed subsets of X correspond bijectively to the chains of irreducible closed subsets of a local scheme of X; when every irreducible closed subset of X contains a closed point, we can evidently restrict to the local schemes at the closed points of X.

We note that every irreducible closed subset of X contains a closed point when X is *quasi-compact* (**0**, 2.1.3); we will see a little further on (5.1.11) that the same holds when X is locally Noetherian. ☐

Corollary (5.1.5). — *For a prescheme X to be catenary, it is necessary and sufficient that, for every $x \in X$, the local ring $\mathscr{O}_x$ be catenary. If in addition every irreducible closed subset of X contains a closed point, it suffices, for X to be catenary, that $\mathscr{O}_x$ be catenary for every closed point x of X.*

Proof. The argument is the same as in (5.1.4), since it amounts to comparing chains of irreducible closed subsets having the same endpoints. ☐

(5.1.6). We will now examine properties that are more special, linked to Noetherian hypotheses.

Recall (**0**, 16.2.3) that a *local* Noetherian ring $A \neq 0$ is of *finite* dimension, equal to the minimum number of generators of an ideal of definition of A; for every prime ideal $\mathfrak{p}$ of A, the height of $\mathfrak{p}$, equal to $\dim(A_{\mathfrak{p}})$, is therefore also finite. These properties, combined with (5.1.2.1), (5.1.3.1), and (5.1.4.1), show that:

Proposition (5.1.7). — *For every non-empty closed subset Y of a locally Noetherian prescheme X,* $\operatorname{codim}(Y, X)$ *is finite. If X is Noetherian and affine and Y is an irreducible closed subset of X,* $\operatorname{codim}(Y, X)$ *is equal to the minimum number of sections s_i of $\mathscr{O}_X$ over X such that Y is an irreducible component of the set of $x \in X$ such that $s_i(x) = 0$ for all i.*

Corollary (5.1.8). — *Let X be a locally Noetherian prescheme, $\mathscr{L}$ an invertible $\mathscr{O}_X$-Module, f a section of $\mathscr{L}$ over X. Then every irreducible component of the set Z of $x \in X$ such that $f(x) = 0$ is of codimension $\leqslant 1$ in X; it is of codimension 1 if Z does not contain any irreducible component of X.*

Proof. We can restrict to the case where $\mathscr{L} = \mathscr{O}_X$. If y is the generic point of an irreducible component **IV-2 | 88**
Y of Z, the ideal (f_y) of $\mathscr{O}_{X,y}$ must be such that $\mathscr{O}_{X,y}/(f_y)$ has no prime ideal other than a single prime ideal, which means that f_y generates an ideal of definition of the local Noetherian ring $\mathscr{O}_{X,y}$; we therefore have $\operatorname{codim}(Y, X) \leqslant 1$ (5.1.7); if Z does not contain any irreducible component of X, we cannot have $\operatorname{codim}(Y, X) = 0$ by virtue of (**0**, 14.2.1). ☐

Proposition (5.1.9). — *Let X be a locally Noetherian prescheme, Y a closed subset of X. If $x \in Y$ is such that $\mathscr{O}_{X,x}$ is a catenary ring, then*

(5.1.9.1) $$\operatorname{codim}_x(Y, X) = \dim(\mathscr{O}_{X,x}) - \operatorname{codim}(\overline{\{x\}}, Y) = \dim(\mathscr{O}_{X,x}) - \dim(\mathscr{O}_{Y,x})$$

Proof. Let Y_i ($1 \leqslant i \leqslant n$) be the irreducible components of Y containing x (which are finite in number since Y is locally Noetherian), and let y_i be the generic point of Y_i. If we set $A = \mathscr{O}_{X,x}$, and if $\mathfrak{p}_i$ is the prime ideal of A corresponding to $Y_i \cap \operatorname{Spec}(A)$, the irreducible closed subsets of Y containing x correspond bijectively to the prime ideals of A containing one of the $\mathfrak{p}_i$, hence $\dim(\mathscr{O}_{Y,x}) = \sup_i \dim(A/\mathfrak{p}_i)$; on the other hand, $\mathscr{O}_{X,y_i} = A_{\mathfrak{p}_i}$, and the hypothesis on A implies $\dim(A) = \dim(A/\mathfrak{p}_i) + \dim(A_{\mathfrak{p}_i})$ (**0**, 16.1.4); the conclusion follows from the relation $\operatorname{codim}_x(Y, X) = \inf_i(\dim(\mathscr{O}_{X,y_i}))$ ((5.1.2.1) and (**0**, 14.2.6)). ☐

Proposition (5.1.10). — *label=(i) In every prescheme X, every locally constructible non-empty subset E containing a point x such that $\{x\}$ is a locally closed subset of X (which amounts to the same as saying that x is isolated in $\overline{\{x\}}$).*

lbbel=(ii) *Let X be a locally Noetherian prescheme, x a point of X such that $\{x\}$ is locally closed in X; then* $\dim(\{x\}) \leqslant 1$; *every point $y \neq x$ of $\{x\}$ is therefore closed in X.*

Proof. label=(i) This is a particular case of the following lemma: □

Lemma (5.1.10.1). — *Let X be a topological space having the following property: for every locally closed non-empty subset $Z \neq \varnothing$ of X, there exists Z', a subset of Z, locally closed in X (or, equivalently, in Z) and a point $x \in Z'$,* closed *in Z'. Then every locally closed non-empty subset $Z \neq \varnothing$ of X contains a point x isolated in $\overline{\{x\}}$.*

Proof. Indeed, let $Z' \subset Z$ be a locally closed subset of Z containing a point x such that x is closed in Z'. There is an open neighbourhood U of x in X such that $Z' \cap U$ is closed in U; this means that x is also closed in U, hence $U \cap \overline{\{x\}} = \{x\}$, in other words x is isolated in $\overline{\{x\}}$.

The lemma applies to every underlying space of a prescheme X, since Z is then also the underlying space of a sub-prescheme (**I**, 5.2.1) and it suffices to take for Z' an affine open subset, which is a Kolmogoroff quasi-compact space, and therefore contains a closed point (**0**, 2.1.3). □

Proof of (ii). Let Z be the reduced sub-prescheme of X having $\overline{\{x\}}$ as underlying space; the hypothesis implies that $\{x\}$ is open *in* Z; hence, for all $z \in Z$, the generic point x of $\operatorname{Spec}(\mathscr{O}_{Z,z})$ is *isolated* in $\operatorname{Spec}(\mathscr{O}_{Z,z})$; but the ring $A = \mathscr{O}_{Z,z}$ is a local Noetherian integral ring, and the hypothesis implies that there exists $f \in A$ such that A is a *field*; we know (**0**, 16.3.3) that this implies $\dim(A) \leqslant 1$. Since $\dim(\mathscr{O}_{Z,z}) \leqslant 1$ for all $z \in Z$, we indeed have $\dim(Z) \leqslant 1$. □

Corollary (5.1.11). — *If X is a locally Noetherian prescheme, every non-empty closed subset of X contains a closed point.*

Proof. Indeed, every closed subset of X is constructible (**0**, 9.1.1) and (9.1.5), and it suffices to apply (5.1.10). □

(5.1.12). Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent *of finite type*, $S = \operatorname{Supp}(\mathscr{F})$ its support, which is *closed* in X (**0**, 5.2.2). For every $x \in X$, we consider $\operatorname{Supp}(\mathscr{F}_x)$ as a closed subset of the local scheme $\operatorname{Spec}(\mathscr{O}_x)$; by definition (**0**, 16.1.7), $\dim(\mathscr{F}_x) = \dim(\operatorname{Supp}(\mathscr{F}_x))$; but we have

$$\operatorname{Supp}(\mathscr{F}_x) = S \cap \operatorname{Spec}(\mathscr{O}_{X,x}) = \operatorname{Spec}(\mathscr{O}_{S,x})$$

where, in the last term, S denotes any closed sub-prescheme of X having S as underlying space. If we note that the irreducible components of $\operatorname{Spec}(\mathscr{O}_{S,x})$ are the intersections of the irreducible components of $\operatorname{Spec}(\mathscr{O}_{X,x})$ and the irreducible components of S containing x, and correspond bijectively to the latter, we see that

(5.1.12.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{S,x})$$

by virtue of (5.1.1.1); it also follows from (5.1.2.1) that

(5.1.12.2) $$\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, S)$$

if X is locally Noetherian. **IV-2** | 89

We say that $\mathscr{F}$ is *equidimensional at the point $x \in X$* if $\mathscr{F}_x$ is an $\mathscr{O}_{X,x}$-module equidimensional, that is to say (**0**, 16.1.7) if $\operatorname{Supp}(\mathscr{F}_x)$ is equidimensional as a closed subset of $\operatorname{Spec}(\mathscr{O}_{X,x})$; this amounts to saying that the ring $\mathscr{O}_{S,x}$ is equidimensional.

We call the *dimension* of $\mathscr{F}$ and write $\dim(\mathscr{F})$ the dimension of the *support* $\operatorname{Supp}(\mathscr{F})$; it follows from (5.1.4.1) and (5.1.12.1) that

(5.1.12.3) $$\dim(\mathscr{F}) = \sup_{x \in X} \dim(\mathscr{F}_x).$$

If $X = \operatorname{Spec}(A)$ is an affine scheme and if $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type, then $\dim(\mathscr{F}) = \dim(M)$ by virtue of (**0**, 16.1.7) and (5.1.4.1).

Proposition (5.1.13). — *Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite type, x a point of X, x' a generization of x in X; then*

(5.1.13.1) $$\dim(\mathscr{F}_{x'}) \leqslant \dim(\mathscr{F}_x)$$

Proof. This follows immediately from (5.1.12.1) and from the definitions. □

5.2. Dimension of an algebraic prescheme

IV-2 | 90

Proposition (5.2.1). — *Let k be a field, X an irreducible prescheme locally of finite type over k, ξ its generic point. Then X is biequidimensional and* $\dim(X) = \deg.\mathrm{tr}_k\,\kappa(\xi)$.

Proof. Every local ring $\mathscr{O}_x$ is a local ring of a k-algebra of finite type, hence a quotient of a regular local ring (**0**, 17.3.9), and we know (**0**, 16.5.12) that these rings are catenary; consequently X is a catenary space (5.1.5), and since X is irreducible, it suffices (5.1.1.1) to show that for every *closed point* $x \in X$, we have

(5.2.1.1) $$\dim(\mathscr{O}_x) = \deg.\mathrm{tr}_k\,\kappa(\xi).$$

We may evidently suppose X is reduced and affine, hence an *integral* ring A, an algebra of finite type over k. We know (Bourbaki, *Alg. comm.*, chap. V, § 3, nº 1, th. 1) that there exists a sub-k-algebra $B = k[t_1, \ldots, t_n]$ of A, where the t_i are algebraically independent over k, such that A is a B-algebra *finite*. Let $\mathfrak{m} = \mathfrak{i}_x$, which is by hypothesis a maximal ideal of A; $\mathfrak{n} = B \cap \mathfrak{m}$ is therefore a maximal ideal of B (Bourbaki, *Alg. comm.*, chap. V, § 2, nº 1, prop. 1), and $A_\mathfrak{m}$ is a local ring of the $B_\mathfrak{n}$-algebra *finite* $S^{-1}A$, where $S = B - \mathfrak{n}$; since $B_\mathfrak{n}$ is integrally closed, $\dim(A_\mathfrak{m}) = \dim(B_\mathfrak{n})$ (**0**, 16.1.6). We can therefore restrict to the case where $A = k[t_1, \ldots, t_n]$; we then know that $k' = \kappa(x)$ is a *finite* extension of k (**I**, 6.4.2); let $A' = k'[t_1, \ldots, t_n]$; taking into account (**I**, 2.4.6) and (3.3.14), there exists a maximal ideal $\mathfrak{m}'$ of A' above $\mathfrak{m}$ and such that k' is the residue field of $A'_{\mathfrak{m}'}$. Since A' is integral and is a finite A-algebra, the same argument as above shows that $\dim(A'_{\mathfrak{m}'}) = \dim(A_\mathfrak{m})$, so we are reduced to the case where $A/\mathfrak{m} = k$. But in this case $\mathfrak{m}$ is generated by the polynomials $t_i - a_i$ (where $a_i \in k$, the a_i being the canonical images of the t_i in $A/\mathfrak{m}$). Replacing t_i by $t_i - a_i$, we are finally reduced to the case where $\mathfrak{m} = \sum_{i=1}^n At_i$. The completion of the local ring $A_\mathfrak{m}$ is then the formal power series ring $k[[t_1, \ldots, t_n]]$; we know that it has the same dimension as $A_\mathfrak{m}$ (**0**, 16.2.4), and on the other hand, the dimension of $k[[t_1, \ldots, t_n]]$ is equal to n (**0**, 17.1.4); whence the conclusion. □

Corollary (5.2.2). — *For a prescheme X locally of finite type over a field k,* $\dim(X)$ *coincides with the number defined in (4.1.1).*

Proof. Indeed, if X_α are the reduced sub-preschemes having as underlying spaces the irreducible components of X, we have $\dim(X) = \sup_\alpha(\dim(X_\alpha))$ (**0**, 14.2.1) and it suffices to apply (5.2.1) to the X_α. □

Corollary (5.2.3). — *Let X be a prescheme locally of finite type over a field k, x a point of X. Then*

(5.2.3.1) $$\dim_x(X) = \dim(\mathscr{O}_x) + \deg.\mathrm{tr}_k\,\kappa(x).$$

Proof. We know that there exists an open neighbourhood U of x in X such that $\dim_x(X) = \dim(U)$ (**0**, 14.1.4.1), and we can suppose that the irreducible components of U are the $X_i \cap U$, where the X_i are the irreducible components of X containing x; since $U \cap X_i$ is dense in X_i, it follows from (4.1.1.3) that $\dim(X_i) = \dim(U \cap X_i)$, hence $\dim_x(X) = \sup_i(\dim(X_i))$. Furthermore, the minimal prime ideals of $\mathscr{O}_x$ correspond to the generic points of the X_i, hence (**0**, 16.1.1.1), $\dim(\mathscr{O}_{X,x}) = \sup_i(\dim(\mathscr{O}_{X_i,x}))$. We are therefore reduced to the case where X is *irreducible*; as X is biequidimensional by (5.2.1), $\dim(X) = \dim(\overline{\{x\}}) + \mathrm{codim}(\overline{\{x\}}, X)$ (**0**, 14.3.5.1), and we know that $\dim(\overline{\{x\}}) = \deg.\mathrm{tr}_k\,\kappa(x)$ by (5.2.1) and $\mathrm{codim}(\overline{\{x\}}, X) = \dim(\mathscr{O}_x)$ by (5.1.2). □ **IV-2 | 91**

Corollary (5.2.4). — *Let X be a prescheme locally of finite type over a field k, $\mathscr{L}$ an invertible $\mathscr{O}_X$-Module, f a section of $\mathscr{L}$ over X, such that the set Y of $x \in X$ such that $f(x) = 0$ is rare in X. Then* $\dim(Y) \leqslant \dim(X) - 1$; *the two members of this inequality are equal if Y meets every irreducible component of maximum dimension of X.*

Proof. If (X_α) is the family of irreducible components of X, we have

$$\dim(Y) = \sup_\alpha \big(\dim(Y \cap X_\alpha)\big)$$

and we are thus reduced to the case where X is irreducible (since $Y \cap X_\alpha$ is rare in X_α, each X_α having a non-empty interior in X). We can then restrict to the case where $Y \neq \varnothing$; then, for every maximal point x of Y, we have (since X is biequidimensional) $\dim(\overline{\{x\}}) = \dim(X) - \mathrm{codim}(\overline{\{x\}}, X)$, and since Y is rare in X, we have

$$\mathrm{codim}(\overline{\{x\}}, X) = 1$$

by (5.1.8); whence the corollary. □

Remarks (5.2.5). — label=(i) Contrary to what holds for algebraic k-preschemes, a locally Noetherian prescheme X is not necessarily catenary (cf. (5.6.11)); however, it is so if all its local rings $\mathscr{O}_x$ are quotients of regular local rings (**0**, 16.5.12) and in particular if X is *regular* (**I**, 4.1.4). However, even a scheme (integral) $X = \operatorname{Spec}(A)$, where A is a *regular* ring, is not necessarily *biequidimensional*, in other words (**0**, 14.3.3) the codimensions in X of the various closed points of X are not necessarily the same. For example, let B be a discrete valuation ring, $\mathfrak{m} = B\pi$ its maximal ideal, $k = B/\mathfrak{m}$ its residue field, K the field of fractions of B; let A be the polynomial ring $B[T]$. There are in A maximal ideals of *height* 2, for example $(\pi) + (T)$; but there are also maximal ideals of *height* 1, for example the principal ideal $(\pi T - 1)$: indeed, a prime ideal, not the zero ideal, is of height 1 (5.1.8); on the other hand $A/(\pi T - 1)$ is isomorphic to the ring of fractions B_π (**0**, 1.2.3), which is nothing other than K, which proves that the ideal $(\pi T - 1)$ is maximal and of height 1.

lbbel=(ii) When X is a prescheme locally of finite type over a field, we have seen (4.1.3) that $\dim(U) = \dim(X)$ for every open subset U everywhere dense in X. This result *does not extend* to Noetherian regular preschemes, even if they are biequidimensional. For example, let B be a discrete valuation ring, $\mathfrak{m}$ its maximal ideal; $X = \operatorname{Spec}(A)$ has two points, (0) and $\mathfrak{m}$, the latter being the only closed point; we have $\dim(X) = 1$, but the open subset $U = \{(0)\}$ is of dimension 0 (cf. § 10).

5.3. Dimension of the support of a Module and Hilbert polynomial This section uses the results of chapter III; it will not be used in the rest of this chapter. **IV-2** | 92

Proposition (5.3.1). — *Let A be a local Artinian ring, X a projective scheme over* $\operatorname{Spec}(A)$, *$\mathscr{L}$ an $\mathscr{O}_X$-Module invertible very ample relative to A, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent $\neq 0$; we set $\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}$ for $n \in \mathbf{Z}$. Then the degree of the Hilbert polynomial $P(n) = \chi_A(\mathscr{F}(n))$ of $\mathscr{F}$ with respect to A* (**III**, 2.5.3) *is equal to the dimension of* $\operatorname{Supp}(\mathscr{F})$.

Proof. Let us argue by induction on $d = \dim(\operatorname{Supp}(\mathscr{F}))$. We know that there is a closed sub-prescheme Y of X whose $\operatorname{Supp}(\mathscr{F})$ is the underlying space, and an $\mathscr{O}_Y$-Module coherent $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, where $j : Y \to X$ is the canonical injection (**Err$_{\text{III}}$**, 30). It is immediate that the Hilbert polynomials of $\mathscr{F}$ and $\mathscr{G}$ are the same, hence we can restrict to the case where $X = \operatorname{Supp}(\mathscr{F})$. Suppose first that $d = 0$; all the points of X are closed, X is an Artinian scheme (**I**, 6.2.2), hence $\mathscr{F}(n) = \mathscr{F}$ for all integers n, and we have by (**III**, 2.5.3)

$$\chi_A(\mathscr{F}(n)) = \operatorname{length}_A(\Gamma(X, \mathscr{F}))$$

for n sufficiently large, and the degree of the Hilbert polynomial is therefore 0. Suppose now that $d > 0$, and set $Z = \operatorname{Ass}(\mathscr{F})$, which is a finite set (3.1.6); there exist an integer $m > 0$ and a section $f \in \Gamma(X, \mathscr{L}^{\otimes m})$ such that X_f is a neighbourhood of Z (**II**, 4.5.4). Multiplication by f is a homomorphism

$$\mu_f : \mathscr{F} \longrightarrow \mathscr{F}(m)$$

which is injective (3.1.8); we therefore have an exact sequence

$$0 \longrightarrow \mathscr{F} \xrightarrow{\mu_f} \mathscr{F}(m) \longrightarrow \mathscr{G} \longrightarrow 0 \tag{5.3.1.1}$$

where $\mathscr{G}$ is coherent. By Nakayama's lemma, the points $x \in \operatorname{Supp}(\mathscr{G})$ are exactly those for which $f(x) = 0$. We will now deduce that

$$\dim(\operatorname{Supp}(\mathscr{G})) = d - 1. \tag{5.3.1.2}$$

This will follow from the following lemma: □

Lemma (5.3.1.3). — *Let A be a local Artinian ring, X a projective scheme over* $\operatorname{Spec}(A)$, *$\mathscr{L}$ an $\mathscr{O}_X$-Module invertible ample relative to A; then, for every section g of $\mathscr{L}$ over X, the set X_g of $x \in X$ such that $g(x) = 0$ meets every irreducible component of X of dimension $\neq 0$.*

Proof. Indeed (**II**, 5.5.7), the set X_g is an affine open subset of X, and if it contains an irreducible component X' of X, the closed reduced sub-prescheme of X having X' as underlying space is both projective and affine, hence finite over A (**III**, 4.4.2), and is therefore an Artinian scheme, hence of dimension 0. □

Proof of (5.3.1), *continued.* This lemma being established, note that since Z contains the maximal points of X, X_f is dense, hence $\operatorname{Supp}(\mathscr{G})$ is rare in X, and the relation (5.3.1.2) follows from the lemma and from (5.2.4).

Since from the exact sequence (5.3.1.1), we deduce, for all n, the exact sequence **IV-2** | 93

$$0 \longrightarrow \mathscr{F}(n) \longrightarrow \mathscr{F}(n+m) \longrightarrow \mathscr{G}(n) \longrightarrow 0$$

and for n sufficiently large, we also have the exact sequence (**III**, 2.2.3)

$$0 \longrightarrow \Gamma(X, \mathscr{F}(n)) \longrightarrow \Gamma(X, \mathscr{F}(n+m)) \longrightarrow \Gamma(X, \mathscr{G}(n)) \longrightarrow 0$$

whence, taking into account (**III**, 2.5.3), for n sufficiently large,

$$\chi_A(\mathscr{G}(n)) = \chi_A(\mathscr{F}(n+m)) - \chi_A(\mathscr{F}(n)).$$

By virtue of (5.3.1.2) and the induction hypothesis, the degree of the polynomial $\chi_A(\mathscr{G}(n))$ is $d-1$, the preceding relation implies that the degree of $\chi_A(\mathscr{F}(n))$ is d. □

5.4. Dimension of the image of a morphism

Proposition (5.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism.*

label=(i) If f is quasi-finite, $\dim(X) \leqslant \dim(\overline{f(X)}) \leqslant \dim(Y)$.
lbbel=(ii) If f is surjective and open *(resp. surjective and* closed*),* $\dim(X) \geqslant \dim(Y)$.

Proof. label=(i) We can replace f by f_{red} (**II**, 6.2.4), hence suppose X and Y reduced; if Z is the closed reduced sub-prescheme of Y having $\overline{f(X)}$ as underlying space, $f = j \circ g$, where $j : Z \to Y$ is the canonical injection and $g : X \to Z$ is a quasi-finite morphism (**I**, 5.2.2) and (**II**, 6.2.4). We can therefore restrict to the case where $f(X)$ is dense in Y. For every $x \in X$, $\mathscr{O}_x$ is then an $\mathscr{O}_{f(x)}$-module quasi-finite (**II**, 6.2.2), and consequently $\mathfrak{m}_{f(x)}\mathscr{O}_x$ is an ideal of definition of $\mathscr{O}_x$ (**0**, 7.4.4); but we know (**0**, 16.3.10) that if $A \to B$ is a local homomorphism of local Noetherian rings, such that if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}B$ is an ideal of definition of B, then $\dim(B) \leqslant \dim(A)$; this proves (i) by virtue of (5.1.4).

lbbel=(ii) The definition of the dimension (**0**, 14.1.2) shows that it suffices to prove that for every sequence $(y_i)_{0 \leqslant i \leqslant n}$ of distinct elements of Y such that $y_i \in \overline{\{y_{i+1}\}}$ for $0 \leqslant i \leqslant n-1$, there exists a sequence $(x_i)_{0 \leqslant i \leqslant n}$ of points of X such that $x_i \in \overline{\{x_{i+1}\}}$ for $0 \leqslant i \leqslant n-1$ and $f(x_i) = y_i$ for all i. Suppose first f is surjective and *open*, and show the existence of x_i by induction on i; the existence of $x_0 \in X$ such that $f(x_0) = y_0$ follows from the fact that f is surjective. If the x_i have been determined for $i \leqslant m$ so as to satisfy $f(x_i) = y_i$ for $i \leqslant m$ and $x_i \in \overline{\{x_{i+1}\}}$ for $i < m$, we note that since f is open and y_{m+1} is a generization of y_m, there exists $x_{m+1} \in f^{-1}(y_{m+1})$ which is a generization of x_m (1.10.3) and the induction can continue.

Suppose now f is surjective and *closed*, and show the existence of x_i by *descending* induction on i, the existence of x_n such that $f(x_n) = y_n$ resulting again from the fact that f is surjective. If the x_i have been determined so as to satisfy the required conditions for $i > m$, we note that $f(\overline{\{x_{m+1}\}})$ is the closure of $\{f(x_{m+1})\} = \{y_{m+1}\}$ since f is closed (Bourbaki, *Top. gén.*, chap. I^er^, 3^e^ éd., § 5, n^o^ 4, prop. 9); there exists therefore $x_m \in \overline{\{x_{m+1}\}}$ such that $f(x_m) = y_m$ and the descending induction can continue. **IV-2** | 94

□

Corollary (5.4.2). — *If X, Y are two locally Noetherian preschemes, $f : X \to Y$ a finite morphism (hence closed), then* $\dim(X) = \dim(\overline{f(X)})$. *If furthermore f is surjective, then* $\dim(X) = \dim(Y)$.

Remarks (5.4.3). — label=(i) We have seen in (4.1.2) that if X and Y are preschemes locally of finite type over a field k and f is a k-morphism, the inequality $\dim(Y) \leqslant \dim(X)$ is already satisfied when f is *quasi-compact* and *dominant*. On the other hand, one can have $\dim(Y) > \dim(X)$ even when f is *of finite type*, *bijective*, and a *local immersion*, if we only suppose X and Y locally Noetherian. For example, let A be a discrete valuation ring, K

its field of fractions, k its residue field; if $Y = \mathrm{Spec}(A)$, and if a is the generic point of Y and b its closed point, then $\kappa(a) = K$, $\kappa(b) = k$. Let X be the prescheme *sum* of $\mathrm{Spec}(K)$ and $\mathrm{Spec}(k)$, $f : X \to Y$ the morphism which coincides with the canonical respective morphisms (**I**, 2.4.5); it is clear that f is a local bijective immersion, $\{a\}$ being open in Y; on the other hand, f is of finite type, for $K = A[\pi^{-1}]$, where π is a uniformisant of A, hence K is a finite A-algebra. However $\dim(X) = 0$ and $\dim(Y) = 1$.

lbbel=(ii) If A and B are two Noetherian rings such that $A \subset B$ and B is a finite A-algebra, the corollary (5.4.2) shows again that $\dim(B) = \dim(A)$ (**0**, 16.1.5). Suppose furthermore that A is a local Noetherian ring; then B is a Noetherian semi-local ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, n° 5, cor. 3 of prop. 9); if $\mathfrak{n}_i$ $(1 \leqslant i \leqslant r)$ are the maximal ideals of B, we have therefore

(5.4.3.1) $$\dim(A) = \sup_i \dim(B_{\mathfrak{n}_i}).$$

We observe that, under these conditions, one does not necessarily have $\dim(B_{\mathfrak{n}_i}) = \dim(A)$ for *all* i: it suffices to see this by replacing B by the direct composition of B and of the residue field k of A. But we have at the same time examples where A and B are furthermore *integral* rings of *dimension* 2 and where certain $B_{\mathfrak{n}_i}$ have dimension $< \dim(A)$ (5.6.11). We will see further on ((5.6.4) and (5.6.10)) that this last phenomenon cannot arise when A is a quotient of a regular local ring.

5.5. Dimension formula for a morphism of finite type

(5.5.1). Recall (**0**, 16.3.9) that if A, B are two local Noetherian rings, k the residue field of A, $\varphi : A \to B$ a local homomorphism, then **IV-2 | 94**

(5.5.1.1) $$\dim(B) \leqslant \dim(A) + \dim(B \otimes_A k).$$

We deduce:

Proposition (5.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, x a point of X, $y = f(x)$. Then*

(5.5.2.1) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)).$$

In particular, if x is a maximal point of the fibre $f^{-1}(y)$, then

(5.5.2.2) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y)$$

since $\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y) = \mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$ is the local ring of x in the prescheme $f^{-1}(y)$, and is of dimension 0 by hypothesis.

We will now obtain a more precise formula when we suppose that f is a morphism *of finite type*. **IV-2 | 95**

Proposition (5.5.3). — *Let A be a Noetherian ring, T an indeterminate, $\mathfrak{p}$ a prime ideal of A; then $\mathfrak{p}' = \mathfrak{p}A[T]$ is prime in $B = A[T]$ and $\mathfrak{p}' \cap A = \mathfrak{p}$. There exists an infinity of prime ideals of B distinct from $\mathfrak{p}'$, whose intersection with A is $\mathfrak{p}$; these ideals are pairwise without inclusion relation. Furthermore, if $\mathfrak{q}$ is such an ideal, then*

(5.5.3.1) $$\dim(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{p}'}) + 1 = \dim(A_{\mathfrak{p}}) + 1.$$

Proof. The first assertions reduce immediately, by replacing A by $A/\mathfrak{p}$ and observing that $A[T]/\mathfrak{p}A[T] = (A/\mathfrak{p})[T]$, in the case $\mathfrak{p} = 0$; they follow then from the fact that the prime ideals of $B = A[T]$ whose intersection with A reduces to 0 are exactly those which do not meet the multiplicative set $S = A - \{0\}$ of the *integral* ring A; but we know that there is a bijection on the set of prime ideals of $S^{-1}A[T] = K[T]$, where K is the field of fractions of A (Bourbaki, *Alg. comm.*, chap. II, § 2, n° 5, prop. 11). Furthermore, we have, from (5.5.1.1) and (5.1.2.1), $\dim(B_{\mathfrak{q}}) \leqslant \dim(A_{\mathfrak{p}}) + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})$, and if k is the field of fractions of $A/\mathfrak{p}$, $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ is canonically identified with $(k[T])_{\mathfrak{q}}$, and is therefore a discrete valuation ring, hence of dimension 1. Finally, if $\mathfrak{p} = \mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_m$ is a chain of prime ideals of A, the ideals $\mathfrak{p}_jB$ $(0 \leqslant j \leqslant m)$ are prime in B, pairwise distinct and contained in $\mathfrak{q}$; hence $\dim(B_{\mathfrak{q}}) \geqslant m + 1 = \dim(A_{\mathfrak{p}}) + 1$. This relation can also be written $\mathrm{ht}(\mathfrak{q}) = \mathrm{ht}(\mathfrak{p}) + 1$; since $\mathfrak{q} \neq \mathfrak{p}'$, $\mathrm{ht}(\mathfrak{q}) \geqslant \mathrm{ht}(\mathfrak{p}') + 1 \geqslant \mathrm{ht}(\mathfrak{p}) + 1$ by definition of the height of a prime ideal; this completes the proof of (5.5.3.1). □

Corollary (5.5.4). — *For every Noetherian ring A, we have*

(5.5.4.1) $$\dim(A[T_1, \ldots, T_r]) = \dim(A) + r$$

(T_i indeterminates).

Proof. We note by contrast that there are examples of local rings (not Noetherian) A such that $\dim(A) = 1$ and $\dim(A[T]) = 3$ [30]. □

Corollary (5.5.5). — *For every locally Noetherian prescheme X, the dimension of $X[T_1, \ldots, T_r] = X \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_r]$ (T_i indeterminates) is $\dim(X) + r$.*

Proof. This follows from (5.5.4.1) and from (**0**, 14.1.7). □

Corollary (5.5.6). — *Under the hypotheses of (5.5.3), let $\mathfrak{q}$ be a prime ideal of $B = A[T]$ such that $\mathfrak{q} \cap A = \mathfrak{p}$; if k and k' are the residue fields of $A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}$ respectively, then*

(5.5.6.1) $$\dim(A_{\mathfrak{p}}) + 1 = \dim(B_{\mathfrak{q}}) + \deg . \operatorname{tr}_k k'$$

Proof. If $\mathfrak{q} = \mathfrak{p}B$, we have, from (5.5.3.1), $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}})$, and in this case $k' = k(T)$, hence the formula (5.5.6.1) is indeed satisfied. In the opposite case, $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1$, and since $\mathfrak{q}$ corresponds to a prime ideal $\neq 0$ of $k[T]$, k' is an algebraic extension of k, hence the formula (5.5.6.1) is again the formula (5.5.6.1). □ **IV-2 | 96**

Lemma (5.5.7). — *Let A be a local Noetherian integral ring, $\mathfrak{m}$ its maximal ideal, k its residue field.*

label=(i) *For every integral ring B containing A, such that $B = A[x]$ (for some $x \in B$) and every prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{m}$, then*

(5.5.7.1) $$\dim(A) + \deg . \operatorname{tr}_A B \geqslant \dim(B_{\mathfrak{q}}) + \deg . \operatorname{tr}_k k'$$

where k' denotes the residue field of $B_{\mathfrak{q}}$ and $\deg . \operatorname{tr}_A B$ the transcendence degree of the field of fractions of B over that of A.

lbbel=(ii) *Suppose that for every maximal ideal $\mathfrak{n}$ of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, the ring $(A[T])_{\mathfrak{n}}$ is catenary; then the two members of (5.5.7.1) are equal.*

Proof. label=(i) If x is transcendental over the field of fractions of A, $\deg . \operatorname{tr}_A B = 1$ and the two members of (5.5.7.1) are equal by virtue of (5.5.6.1). In the opposite case, $B = A[T]/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal $\neq 0$ of $A[T]$, such that $\mathfrak{p} \cap A = 0$ since B contains A; we have therefore $\operatorname{ht}(\mathfrak{p}) = 1$ by virtue of (5.5.3.1). The ideal $\mathfrak{q}$ of B is of the form $\mathfrak{n}/\mathfrak{p}$, where $\mathfrak{n} \supset \mathfrak{p}$ is a prime ideal of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, and $B_{\mathfrak{q}} = (A[T])_{\mathfrak{n}}/\mathfrak{p}(A[T])_{\mathfrak{n}}$; the formula (5.5.6.1), applied to $X = \operatorname{Spec}((A[T])_{\mathfrak{n}})$ and $Y = \operatorname{Spec}(B_{\mathfrak{q}})$, gives

(5.5.7.2) $$\dim((A[T])_{\mathfrak{n}}) \geqslant \operatorname{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) + \dim(B_{\mathfrak{q}}) = 1 + \dim(B_{\mathfrak{q}})$$

since $\operatorname{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) = \operatorname{ht}(\mathfrak{p}) = 1$ by the bijective correspondence between prime ideals contained in $\mathfrak{n}$ and prime ideals of $(A[T])_{\mathfrak{n}}$. Finally, the formula (5.5.6.1) gives

(5.5.7.3) $$\dim((A[T])_{\mathfrak{n}}) = \dim(A) + 1 - \deg . \operatorname{tr}_k k'$$

which completes the proof of (5.5.7.1).

lbbel=(ii) If $(A[T])_{\mathfrak{n}}$ is catenary, the two members of (5.5.7.2) are equal (**0**, 16.1.4), hence also the two members of (5.5.7.1).

□

Theorem (5.5.8). — ***(dimension formula).*** *Let A be a local Noetherian integral ring, B an integral ring containing A and which is an A-algebra of finite type, $\mathfrak{q}$ an ideal of B such that $\mathfrak{q} \cap A$ is the maximal ideal $\mathfrak{m}$ of A, k and k' the residue fields of A and $B_{\mathfrak{q}}$ respectively. Then the inequality*

(5.5.8.1) $$\dim(A) + \deg . \operatorname{tr}_A B \geqslant \dim(B_{\mathfrak{q}}) + \deg . \operatorname{tr}_k k'.$$

Furthermore the two members are equal if, for every sub-A-algebra A' of finite type of $B[T]$ and every maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m}' \cap A = \mathfrak{m}$, $A'_{\mathfrak{m}'}$ is catenary.

Proof. Let $B = A[x_1, \dots, x_n]$ and let us argue by induction on n. Set $C = A[x_1, \dots, x_{n-1}]$ and $\mathfrak{r} = \mathfrak{q} \cap C$; $C_\mathfrak{r}$ is an integral local Noetherian ring, and if we set $S = C - \mathfrak{r}$, $\mathfrak{q}' = S^{-1}\mathfrak{q}$, $B' = S^{-1}B$, $B' = C_\mathfrak{r}[x_n]$ and $\mathfrak{q}' \cap C_\mathfrak{r} = \mathfrak{r}C_\mathfrak{r}$; furthermore the fields of fractions of B' and $C_\mathfrak{r}$ are those of B and C respectively. If k_1 is the residue field of $C_\mathfrak{r}$, we have therefore, from (5.5.7.1) **IV-2 | 97**

(5.5.8.2) $$\dim(C_\mathfrak{r}) + \deg.\mathrm{tr}_\mathfrak{c} B \geqslant \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_{k_1} k'.$$

On the other hand, the induction hypothesis gives

(5.5.8.3) $$\dim(A) + \deg.\mathrm{tr}_A C \geqslant \dim(C_\mathfrak{r}) + \deg.\mathrm{tr}_k k_1$$

whence (5.5.8.1) by adding (5.5.8.2) and (5.5.8.3) member by member. To show the second assertion, note first that any sub-A-algebra of finite type of $C[T]$ is also a sub-A-algebra of finite type of $B[T]$; by induction on n, we can therefore suppose that the two members of (5.5.8.3) are equal. On the other hand, to see that the two members of (5.5.8.2) are equal, it suffices, by virtue of (5.5.7), to verify that if $\mathfrak{n}$ is a maximal ideal of $C_\mathfrak{r}[T]$ such that $\mathfrak{n} \cap C_\mathfrak{r} = \mathfrak{r}C_\mathfrak{r}$, then $(C_\mathfrak{r}[T])_\mathfrak{n}$ is catenary; but $C_\mathfrak{r}[T] = S'^{-1}C[T]$ where $S' = C - \mathfrak{r}$, the ideal $\mathfrak{n}$ is therefore of the form $S'^{-1}\mathfrak{n}'$, where $\mathfrak{n}'$ is a prime ideal of $C[T]$ such that $\mathfrak{n}' \cap C = \mathfrak{r}$, whence $(C_\mathfrak{r}[T])_\mathfrak{n} = (C[T])_{\mathfrak{n}'}$; if $\mathfrak{m}'$ is a maximal ideal of $C[T]$ containing $\mathfrak{n}'$, $(C[T])_{\mathfrak{m}'}$ is a local ring and by hypothesis this last is catenary, hence so is $(C[T])_{\mathfrak{n}'}$ (**0**, 16.1.4). □

5.6. Dimension formula and universally catenary rings **IV-2 | 97**

Proposition (5.6.1). — *Let A be a Noetherian ring. The following properties are equivalent:*

label=a) *Every polynomial ring $A[T_1, \dots, T_n]$ is catenary.*
lbbel=b) *Every A-algebra of finite type is catenary.*
lcbel=c) *A is catenary, and for every integral A-algebra B', local, essentially of finite type (1.3.8), if we denote by $\mathfrak{p}$ the inverse image of the maximal ideal $\mathfrak{q}$ of B' in A, by A' the image of $A_\mathfrak{p}$ in B', by k and k' the residue fields of A' and B' respectively,*

(5.6.1.1) $$\dim(A') + \deg.\mathrm{tr}_{A'}(B') = \dim(B') + \deg.\mathrm{tr}_k k'.$$

Proof. The fact that b) implies c) follows from (5.5.8.1); indeed, B' is an A'-algebra local, essentially of finite type, hence of the form $B''_\mathfrak{q}$, where B'' is a sub-A'-algebra of type fini of B', $\mathfrak{q}''$ a prime ideal of B'' above the maximal ideal $\mathfrak{p}'$ of A'; in addition the fields of fractions of B' and B'' are the same, hence $\deg.\mathrm{tr}_{A'}(B') = \deg.\mathrm{tr}_{A'}(B'')$. To show (5.6.1.1), it suffices to show that (under hypothesis b)) every sub-A'-algebra A_1 of finite type of $B'[T]$ is catenary; indeed the two members of (5.5.8.1), where we replace A, B, and $\mathfrak{q}$ by A', B'', and $\mathfrak{q}''$, will then be equal, whence the equality (5.6.1.1). But hypothesis b) implies that every A-algebra essentially of finite type is catenary (**0**, 16.1.4), and A_1 is such an A-algebra (1.3.9).

It is trivial that b) implies a); conversely, a) implies b), every A-algebra of finite type being a quotient of a polynomial algebra (**0**, 16.1.4).

It remains to prove that c) implies b). Since every quotient by a prime ideal of a finite type A-algebra is a finite type integral A-algebra, if B is a finite type integral A-algebra, $\mathfrak{q}$ and $\mathfrak{q}'$ two prime ideals of B such that $\mathfrak{q}' \subset \mathfrak{q}$, by (**0**, 16.1.4.2)

(5.6.1.2) $$\dim(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) + \dim(B_{\mathfrak{q}'}) = \dim(B_\mathfrak{q}).$$

Let $\mathfrak{p}$, $\mathfrak{p}'$ be the inverse images of $\mathfrak{q}$, $\mathfrak{q}'$ respectively in A, $\mathfrak{n}$ the kernel of the homomorphism $A_\mathfrak{p} \to B_\mathfrak{q}$. The image A' of $A_\mathfrak{p}$ in $B_\mathfrak{q}$ being isomorphic to $A_\mathfrak{p}/\mathfrak{n}$, the formula (5.6.1.1) applied to A' and $B' = B_\mathfrak{q}$ gives **IV-2 | 98**

(5.6.1.3) $$\dim(A_\mathfrak{p}/\mathfrak{n}) + \deg.\mathrm{tr}_{A_\mathfrak{p}/\mathfrak{n}}(B_\mathfrak{q}) = \dim(B_\mathfrak{q}) + \deg.\mathrm{tr}_{\kappa(\mathfrak{p})} \kappa(\mathfrak{q}).$$

On the other hand, the kernel of $A_{\mathfrak{p}'} \to B_{\mathfrak{q}'}$ is $\mathfrak{n}A_{\mathfrak{p}'}$, hence, applying the formula (5.6.1.1) where we replace A' by $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and B' by $B_{\mathfrak{q}'}$, we have

(5.6.1.4) $$\dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) + \deg.\mathrm{tr}_{A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}}(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}'}) + \deg.\mathrm{tr}_{\kappa(\mathfrak{p}')} \kappa(\mathfrak{q}').$$

Finally, $B/\mathfrak{q}'$ is an integral A-algebra of finite type, the inverse image of $\mathfrak{q}/\mathfrak{q}'$ being $\mathfrak{p}'A_\mathfrak{p}$, hence, applying the formula (5.6.1.1) where we replace A' by $A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}$ and B' by $B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}$, we have

(5.6.1.5) $$\dim(A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}) + \deg.\mathrm{tr}_{A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}}(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) = \dim(B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}) + \deg.\mathrm{tr}_{\kappa(\mathfrak{p}')} \kappa(\mathfrak{q}).$$

Adding now (5.6.1.4) and (5.6.1.5) member by member, and noting that $\kappa(\mathfrak{p}')$ (resp. $\kappa(\mathfrak{q}')$) is the field of fractions of $A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}$ (resp. $B_\mathfrak{q}/\mathfrak{q}'B_\mathfrak{q}$); by the same token $A_\mathfrak{p}/\mathfrak{n}$ have the same field of fractions, and $B_\mathfrak{q}$ and $B_\mathfrak{q}$ have the same field of fractions; finally, since A is supposed catenary, by (**0**, 16.1.4.2)

$$\dim(A_\mathfrak{p}/\mathfrak{p}'A_\mathfrak{p}) + \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) = \dim(A_\mathfrak{p}/\mathfrak{n})$$

Taking into account these remarks and (5.6.1.3), we obtain the relation (5.6.1.2). □

Definition (5.6.2). — A Noetherian ring A is said to be *universally catenary* if it satisfies the equivalent conditions $a)$, $b)$, $c)$ of (5.6.1).

Remarks (5.6.3). — label=(i) If A is universally catenary, so is $S^{-1}A$ for every multiplicative subset S of A, as it follows immediately from (5.6.1), $a)$) and from the fact that every ring of fractions of a catenary ring is catenary. Conversely, if for every prime ideal $\mathfrak{p}$ of A, the ring $A_\mathfrak{p}$ is universally catenary, then A is universally catenary: this follows from (5.6.1), $c)$), noting that if $S = A - \mathfrak{p}$, $S^{-1}B$ is an $A_\mathfrak{p}$-algebra of finite type, and $B_\mathfrak{q} = (S^{-1}B)_{S^{-1}\mathfrak{q}}$.

lbbel=(ii) We say that a locally Noetherian prescheme X is *universally catenary* if, for every $x \in X$, the ring $\mathscr{O}_x$ is universally catenary. It follows from (i) that for A to be a universally catenary ring, it is necessary and sufficient that the scheme $\operatorname{Spec}(A)$ be universally catenary.

lcbel=(iii) The criterion (5.6.1), $c)$) only involves the images of A in integral A-algebras of finite type, hence the *integral quotient rings* of A. We conclude that if A is a universally catenary ring, or that X_{red} is universally catenary, it is necessary and sufficient that the scheme $\operatorname{Spec}(A)$ be so.

ldbel=(iv) The criterion (5.6.1), $b)$) and the remark (i) show that, if A is a universally catenary ring, the same is true of every A-algebra *essentially of finite type* (1.3.8).

Proposition (5.6.4). — *Every quotient of a regular ring is universally catenary.*

Proof. It amounts to seeing that a ring A is universally catenary (5.6.3), (iv)); or, we know that $A[T_1, \ldots, T_n]$ is regular (**0**, 17.3.7), hence catenary (**0**, 16.5.12), and we conclude by (5.6.1), $a)$).

In particular, it follows from the theorem of Cohen (**0**, 19.8.8) that every *complete local Noetherian ring* is universally catenary. Similarly, every algebra of finite type over a field is universally catenary. Every prescheme *locally of finite type over a field* is universally catenary.

We will see further on (6.3.7) that in (5.6.4), one can replace "regular" by "Cohen-Macaulay". □

Proposition (5.6.5). — *Let Y be an irreducible locally Noetherian prescheme, X a prescheme irreducible, $f : X \to Y$ a dominant morphism locally of finite type. Let ξ (resp. η) be the generic point of X (resp. Y), and let $e = \dim(f^{-1}(\eta)) = \deg.\operatorname{tr}_{\kappa(\eta)} \kappa(\xi)$ ("dimension of the generic fibre", cf.* (**0**, 2.1.8) *and* (4.1.1)*). For every $x \in X$, setting $y = f(x)$, we have*

(5.6.5.1) $$e + \dim(\mathscr{O}_y) \geqslant \deg.\operatorname{tr}_{\kappa(y)} \kappa(x) + \dim(\mathscr{O}_x)$$

(5.6.5.2) $$\dim(\mathscr{O}_x) \leqslant \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)) - \delta(x)$$

where $\delta(x) = \dim_x(f^{-1}(y)) - e$. **IV-2** | 99

Furthermore, if Y is universally catenary, the two members of (5.6.5.1) *(resp.* (5.6.5.2)*) are equal. If furthermore x is closed in $f^{-1}(y)$, then*

(5.6.5.3) $$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + e.$$

Proof. We can restrict to the case where X and Y are affine (hence f of finite type) and integral. Since f is dominant, $\mathscr{O}_y$ identifies with a sub-ring of $\mathscr{O}_x$, and the fields of fractions of $\mathscr{O}_x$ and $\mathscr{O}_y$ are respectively $\kappa(\xi)$ and $\kappa(\eta)$ (**I**, 8.2.7); furthermore, $\mathscr{O}_x$ is a local ring of an $\mathscr{O}_y$-algebra integral of finite type B and an ideal prime $\mathfrak{q}$ (**I**, 3.6.5); setting $A = \mathscr{O}_y$, $\mathfrak{q}$ and $\mathfrak{q}$ (5.5.8.1) to $A = \mathscr{O}_y$, we obtain (5.6.5.1). Furthermore, $\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y) = \mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$ is nothing other than the local ring of the fibre $f^{-1}(y)$ at the point x (**I**, 3.6.1); since $f^{-1}(y)$ is a prescheme of finite type over $\kappa(y)$, it follows from (5.2.3.1) that

$$\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \kappa(y)) = \dim_x(f^{-1}(y)) - \deg.\operatorname{tr}_{\kappa(y)} \kappa(x);$$

the inequality (5.6.5.2) is therefore nothing other than another form of (5.6.5.1).

The fact that the two members of (5.6.5.1) are equal when Y is universally catenary is nothing other than the equality (5.6.1.1), applied to $A = \mathscr{O}_y$ and $B' = \mathscr{O}_x$. To show that we also have (5.6.5.3) when x is closed in $f^{-1}(y)$, it suffices to remark that, in a general way, $\deg.\mathrm{tr}_{\kappa(y)}\,\kappa(x)$ is the dimension of $\overline{\{x\}} \cap f^{-1}(y)$, as it follows from (5.2.1) applied to the closed reduced sub-prescheme of $f^{-1}(y)$ having this sub-space as underlying space. □ **IV-2** | 100

Corollary (5.6.6). — *Under the general hypotheses of (5.6.5), we have*

(5.6.6.1) $$\dim(X) \leqslant \dim(Y) + e.$$

If we suppose furthermore Y universally catenary, then, for the two members of (5.6.6.1) to be equal, it is necessary and sufficient that

(5.6.6.2) $$\dim(Y) = \sup_{y \in f(X)} \dim(\mathscr{O}_y).$$

In particular, the two members of (5.6.6.1) are equal when Y is locally of finite type over a field.

Proof. The inequality (5.6.6.1) follows from (5.6.5.1) applied at a closed point x of X, taking into account (5.1.4.2) and (5.1.4.2). On the other hand, every non-empty fibre $f^{-1}(y)$ contains a closed point x (5.1.11). If we suppose Y universally catenary, we therefore have at this point the relation (5.6.5.3). Taking the upper bounds of (5.6.5.3) as x runs over X, and noting that every closed point x is *a fortiori* closed in $f^{-1}(f(x))$, the second assertion of the corollary follows from (5.1.4.1) and (5.1.4.2).

To prove the last assertion, note that there is an open affine neighbourhood U of the generic point of Y such that $f|U$ is of finite type, hence $f(U)$ is a *constructible* subset of Y (1.8.5), and therefore contains a non-empty open subset (hence everywhere dense) of Y (**0**, 9.2.2). The hypothesis that Y is locally of finite type over a field assures, on the one hand, that Y is universally catenary (5.6.4), and on the other hand that the two members of (5.6.6.2) are equal (5.2.2) and (4.1.1.3).

We note that the condition (5.6.6.2) is trivially satisfied when f is *surjective*. □

Corollary (5.6.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, n an integer. If, for every $y \in Y$, $\dim(f^{-1}(y)) \leqslant n$, then*

(5.6.7.1) $$\dim(X) \leqslant \dim(Y) + n.$$

Proof. By virtue of (**0**, 14.1.7) and (1.5.4), we can restrict to the case where X and Y are affine, hence f of finite type, X and Y reduced; let X_i ($1 \leqslant i \leqslant m$) be the sub-preschemes (integral) of X having as underlying spaces the irreducible components of X; $\dim(X) = \sup_i(\dim(X_i))$. If Z_i is the closed reduced sub-prescheme of Y having $\overline{f(X_i)}$ as underlying space, Z_i is integral (**0**, 2.1.5); the restriction $X_i \to Y$ of f factorises as $X_i \xrightarrow{g_i} Z_i \xrightarrow{j_i} Y$, where j_i is the canonical injection (**I**, 5.2.2), and g_i is dominant and of finite type (1.5.4); if z_i is the generic point of Z_i, $\dim(g_i^{-1}(z_i)) \leqslant n$ for all i by hypothesis; we see thus that to prove (5.6.7.1) it suffices to show that $\dim(X_i) \leqslant \dim(Z_i) + n$ for all i, which follows from (5.6.6.1). □ **IV-2** | 101

Corollary (5.6.8). — *Let Y be a locally Noetherian prescheme, X a prescheme irreducible, $f : X \to Y$ a morphism locally of finite type, n an integer $\geqslant 0$. Suppose that Y is universally catenary, and that for every $y \in Y$, $\dim(f^{-1}(y)) \geqslant n$ (resp. $\dim(f^{-1}(y)) = n$). Then*

(5.6.8.1) $$\dim(X) \geqslant \dim(Y) + n$$

(resp.

(5.6.8.2) $$\dim(X) = \dim(Y) + n).$$

Proof. Since $n \geqslant 0$, the hypothesis implies that f is *surjective*, hence Y is irreducible; furthermore, $\dim(f^{-1}(\eta)) \geqslant n$ (resp. $\dim(f^{-1}(\eta)) = n$). The conclusion follows from (5.6.6). □

Remarks (5.6.9). — label=(i) Even if Y is regular, X irreducible, $f : X \to Y$ dominant and of finite type, the two members of (5.6.6.1) are not necessarily equal, as is shown by the example where $Y = \mathrm{Spec}(A)$ where A is a discrete valuation ring, K the field of fractions of A, f being the canonical morphism, $X = \mathrm{Spec}(K)$ where K is the field of fractions of A.

lbbel=(ii) The example (5.4.3), (i)) shows that in (5.6.6) and (5.6.8), one cannot suppress the hypothesis that X is irreducible, the other hypotheses being satisfied. We will see however (10.6.1) that with supplementary hypotheses on Y, which are satisfied for example when Y is a prescheme locally of finite type over a field, or over a Dedekind ring having an infinity of prime ideals, such phenomena cannot arise.

Proposition (5.6.10). — *Let A be a local Noetherian integral ring universally catenary, B an integral ring containing A and which is a finite A-algebra. Then, for every maximal ideal $\mathfrak{n}$ of B, $\dim(B_{\mathfrak{n}}) = \dim(A)$.*

Proof. Indeed, $\deg . \mathrm{tr}_A B = 0$ and the residue field k' of $B_{\mathfrak{n}}$ is an algebraic extension of the field of fractions of A. The conclusion follows from the formula (5.6.1.1), every maximal ideal of B being above that of A. □

Example (5.6.11). — Let A be a local Noetherian integral ring and *integrally closed*; then, since (**0**, 16.1.6), the conclusion of (5.6.10) is valid for every integral finite A-algebra B containing A. On the contrary, we will now construct an example of a local Noetherian integral *catenary* ring A for which the conclusion of (5.6.10) will be at fault. We will use the construction of (5.2.5), (i)). Let k_0 be a field, k a pure transcendental extension $k_0(S_i)_{i\in\mathbf{N}}$ of infinite transcendence degree, V the discrete valuation ring, a local ring of the polynomial ring $k[S]$ at the prime ideal (S); finally let E be the polynomial ring $V[T]$; we have seen that in E the maximal ideal $\mathfrak{m} = (S) + (T)$ is of height 2 and the maximal ideal $\mathfrak{m}' = (ST - 1)$ is of height 1; the corresponding residue fields are $k(\mathfrak{m}) = k$ and **IV-2** | 102
$k(\mathfrak{m}') = k(S)$, field of fractions of V; by virtue of the choice of k, these fields are *isomorphic*. Let us denote by ε and ε' the canonical homomorphisms of $E/\mathfrak{m} = k(\mathfrak{m})$ and $E/\mathfrak{m}' = k(\mathfrak{m}')$ respectively; let σ be an isomorphism of $k(\mathfrak{m})$ onto $k(\mathfrak{m}')$, and consider the sub-ring C of E formed of $x \in E$ such that $\varepsilon'(x) = \sigma(\varepsilon(x))$ (this construction is a particular case of the "glueing procedures" which will be systematically studied in chap. V). It is immediate that $\mathfrak{n} = \mathfrak{m}\mathfrak{m}' = \mathfrak{m} \cap \mathfrak{m}'$ is a maximal ideal of C, $C/\mathfrak{m}\mathfrak{m}'$ identifying with the sub-field of $(E/\mathfrak{m}) \times (E/\mathfrak{m}')$ formed of the pairs $(z, \sigma(z))$. We have $E = C + C(ST - 1)$, and it is evidently the integral closure of C; we will moreover see that C is *Noetherian*: this will follow from the following lemma:

Lemma (5.6.11.1). — *Let R be a ring, S a sub-ring, $\mathfrak{R} = \mathrm{Ann}_S(R/S)$ the conductor of S in R (which is also an ideal of R).*

label=(i) For every ideal $\mathfrak{J} \subset \mathfrak{R}$ of R, there exists a strictly increasing map from the set of ideals $\mathfrak{L}$ of S such that $R.\mathfrak{L} = \mathfrak{J}$ to the set of sub-$(S/\mathfrak{R})$-modules of $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$.

lbbel=(ii) If $S/\mathfrak{R}$ and R are Noetherian and if R is a finite type S-module, then every increasing sequence of ideals of S contained in K is stationary.

We will apply this lemma by taking $R = E$, $S = C$; it is clear that $\mathfrak{n}$ is the conductor of C in E; furthermore, for every ideal $\mathfrak{a}$ of C, $\mathfrak{a}/(\mathfrak{n} \cap \mathfrak{a})$ is isomorphic to $(\mathfrak{a} + \mathfrak{n})/\mathfrak{n}$, which is a sub-$C$-module of $C/\mathfrak{n}$, hence a simple C-module; it suffices therefore to show that every ideal $\mathfrak{a} \cap \mathfrak{n}$ of C is of finite type, and this follows from (5.6.11.1).

It follows from (**0**, 16.1.5) that $\dim(C) = \dim(E) = 2$. Taking $A = C_{\mathfrak{n}}$; if $U = C - \mathfrak{n}$, the ring of fractions $B = U^{-1}E$ is therefore the integral closure of A, and it is an A-module of finite type; moreover, since $\mathfrak{m}$ and $\mathfrak{m}'$ are the only prime ideals of E containing $\mathfrak{n}$, B is a semi-local ring whose local rings at the two maximal ideals are isomorphic to $E_{\mathfrak{m}}$ and $E_{\mathfrak{m}'}$, of dimension respectively 2 and 1. This shows that $\dim(B) = 2$, hence $\dim(A) = 2$ (**0**, 16.1.5). Since A is a local integral ring, it is necessarily catenary (every prime ideal distinct from 0 and from the maximal ideal being necessarily of height 1); but it does not satisfy the conclusion of (5.6.10), and *a fortiori* is not universally catenary.

Note also that $\mathfrak{n}$ is an ideal of height 2 in C, and that for every prime ideal $\mathfrak{p}$ of height 1 in C, there is a *single* prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, necessarily of height 1, and such that $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$. Indeed, there is at least one prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, and it follows from the Cohen-Seidenberg theorem that such an ideal is necessarily of height 1; furthermore, since E is a finite C-algebra and $C_{\mathfrak{p}}$ is integrally **IV-2** | 103
closed (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 5, cor. 5 of prop. 16), this proves our assertions.

It would be interesting to know whether every local Noetherian integral ring satisfying the conclusion of (5.6.10) is universally catenary; this is what Nagata has claimed [33], but his proof does not seem complete.

5.7. Depth and property (S_k)

IV-2 | 103

Definition (5.7.1). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. We call *depth* (resp. *codepth*) of $\mathscr{F}$ at a point $x \in X$ the number $\mathrm{prof}(\mathscr{F}_x)$ (resp. $\mathrm{coprof}(\mathscr{F}_x)$) (**0**, 16.4.5) and (**0**, 16.4.9). We call the *codepth* of $\mathscr{F}$ the number

(7.5.1.1)
$$\mathrm{coprof}(\mathscr{F}) = \sup_{x\in X} \mathrm{coprof}(\mathscr{F}_x).$$

We say that $\mathscr{F}$ is an $\mathscr{O}_X$-Module de *Cohen-Macaulay* at a point x if $\mathscr{F}_x$ is an $\mathscr{O}_x$-module de Cohen-Macaulay, that is to say (**0**, 16.5.1) if $\mathrm{coprof}(\mathscr{F}_x) = 0$. We say that $\mathscr{F}$ is an $\mathscr{O}_X$-Module de *Cohen-Macaulay* if it is so at every point, that is to say if $\mathrm{coprof}(\mathscr{F}) = 0$. A point $x \in X$ such that $\mathscr{O}_x$ is a Cohen-Macaulay ring is also called a *Cohen-Macaulay point* of X.

We call *codepth* of X and write $\mathrm{coprof}(X)$ the number $\mathrm{coprof}(\mathscr{O}_X)$. We say that X is a *prescheme de Cohen-Macaulay* if $\mathscr{O}_X$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay, that is to say if $\mathrm{coprof}(X) = 0$. Every locally Noetherian prescheme of *dimension* 0 is evidently a prescheme de Cohen-Macaulay. Saying that $\mathrm{Spec}(A)$ is a Cohen-Macaulay scheme means that A is a Cohen-Macaulay ring (**0**, 16.5.13).

Definition (5.7.2). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent, k a positive or negative integer. We say that $\mathscr{F}$ satisfies property (S_k) if, for every $x \in X$,

(5.7.2.1)
$$\mathrm{prof}(\mathscr{F}_x) \geqslant \inf(k, \dim(\mathscr{F}_x))$$

We say that $\mathscr{F}$ satisfies property (S_k) at a point $x \in X$ if, for every generization x' of x in X,

(5.7.2.2)
$$\mathrm{prof}(\mathscr{F}_{x'}) \geqslant \inf(k, \dim(\mathscr{F}_{x'})).$$

We say that X satisfies property (S_k) (resp. (S_k) at a point x) if $\mathscr{O}_X$ satisfies property (S_k) (resp. (S_k) at the point x).

For $\mathscr{F}$ to satisfy property (S_k), it is necessary and sufficient that it satisfy this property at every point of X. If U is an open subset of X and $\mathscr{F}$ satisfies (S_k), the same holds for $\mathscr{F}|U$; conversely, if (U_α) is an open covering of X and $\mathscr{F}|U_\alpha$ satisfies (S_k) for all α, $\mathscr{F}$ satisfies (S_k).

Remarks (5.7.3). — label=(i) Recall that we always have $\mathrm{prof}(\mathscr{F}_x) \leqslant \dim(\mathscr{F}_x)$ (**0**, 16.4.5.1) if $\mathscr{F}_x \neq 0$. To say that $\mathscr{F}$ satisfies property (S_k) therefore means that $\mathrm{prof}(\mathscr{F}_x) \geqslant k$ *except* at points $x \in X$ such that $\dim(\mathscr{F}_x) < k$, and that at these last points we have $\dim(\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_x)$, that is to say (**0**, 16.5.1) that $\mathscr{F}$ is an $\mathscr{O}_X$-Module de Cohen-Macaulay at these points; we note that the points where $\dim(\mathscr{F}_x) = k$ are also points where $\mathrm{prof}(\mathscr{F}_x) = k$, hence $\mathscr{F}$ is again an $\mathscr{O}_X$-Module de Cohen-Macaulay at these points. To say that $\mathscr{F}$ satisfies property (S_k) *for all* k therefore means that $\mathscr{F}$ is an $\mathscr{O}_X$-Module *de Cohen-Macaulay*. It is clear that for $k' \geqslant k$, the property $(S_{k'})$ implies (S_k); for $k \leqslant 0$, *every* $\mathscr{O}_X$-Module coherent satisfies property (S_k). **IV-2 | 104**

lbbel=(ii) To verify the condition (5.7.2.1), we can restrict to the case where $x \in \mathrm{Supp}(\mathscr{F})$; in the opposite case indeed $\dim(\mathscr{F}_x) = -\infty$ (**0**, 14.1.2).

lcbel=(iii) If $X = \mathrm{Spec}(A)$, where A is a Noetherian ring, and $\mathscr{F} = \widetilde{M}$, where M is a finite type A-module, we say that M satisfies property (S_k) if $\mathscr{F}$ satisfies this property. For a locally Noetherian prescheme X and any point $x \in X$, to say $\mathscr{F}$ satisfies property (S_k) *at a point* $x \in X$ means that if we set $Y = \mathrm{Spec}(\mathscr{O}_x)$, the $\mathscr{O}_Y$-Module $\widetilde{\mathscr{F}_x}$ satisfies (S_k); we note that the condition (5.7.2.1) in general *does not* imply (5.7.2.2) for every generization x' of x, seeing as in general there is no relation between $\mathrm{prof}(\mathscr{F}_{x'})$ and $\mathrm{prof}(\mathscr{F}_x)$ (**0**, 16.4.6).

ldbel=(iv) It follows immediately from the definition that if $\mathscr{F}$ satisfies (S_k) at a point x, it satisfies also (S_k) at every generization x' of x.

lebel=(v) Property (S_k) is above all important for $k = 1$ and $k = 2$; it was introduced for $k = 2$ by Serre to express his criterion of normality (cf. (5.8.5)).

lfbel=(vi) Let X be a locally Noetherian prescheme, Y a closed sub-prescheme of X, $j : Y \to X$ the canonical injection, $\mathscr{G}$ a coherent $\mathscr{O}_Y$-Module. It is clear that for every $x \in Y$, $\dim(\mathscr{G}_x) = \dim((j_*(\mathscr{G}))_x)$ and $\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}((j_*(\mathscr{G}))_x)$, whence $\mathrm{coprof}(\mathscr{G}_x) = \mathrm{coprof}((j_*(\mathscr{G}))_x)$. For $\mathscr{G}$ to satisfy (S_k), it is necessary and sufficient that $j_*(\mathscr{G})$ satisfy (S_k).

Proposition (5.7.4). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. Setting $S = \operatorname{Supp}(\mathscr{F})$, and for every $n \geqslant 0$, denote by Z_n the set of $z \in X$ such that $\operatorname{coprof}(\mathscr{F}_z) > n$, so that $Z_n \subset S$.*

label=(i) For $\mathscr{F}$ to satisfy property (S_k), it is necessary and sufficient that, for every $n \geqslant 0$, we have

(5.7.4.1) $$\operatorname{codim}(Z_n, S) > n + k.$$

lbbel=(ii) Suppose furthermore that the Z_n are closed in X. Then, for $\mathscr{F}$ to satisfy property (S_k) at a point $x \in S$, it is necessary and sufficient that, for every $n \geqslant 0$, we have

(5.7.4.2) $$\operatorname{codim}_x(Z_n, S) > n + k.$$

Proof. label=(i) By definition (5.1.3), $\operatorname{codim}(Z_n, S) = \inf_{z \in Z_n}(\dim(\mathscr{O}_{S,z}))$ and the inequality (5.7.4.1) signifies therefore (5.1.12.2) that, for every $z \in X$, and every $n \geqslant 0$, the relation

$$\dim(\mathscr{F}_z) - \operatorname{prof}(\mathscr{F}_z) \geqslant n + 1$$

implies the relation

$$\dim(\mathscr{F}_z) \geqslant n + k + 1$$

But if we set $a = \dim(\mathscr{F}_z)$, $b = \operatorname{prof}(\mathscr{F}_z)$, we have $b \leqslant a$, and to say that for every $n \geqslant 0$, the relation $b \leqslant a - n - 1$ implies $k \leqslant a - n - 1$ is equivalent to saying that $b \geqslant \inf(k, a)$, whence the proposition.

lbbel=(ii) The argument is the same as in (i), up to restricting to $z \in X$ that are *generizations* of x, and taking into account (5.1.3.2).

□

Proposition (5.7.5). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. For $\mathscr{F}$ to satisfy (S_k) $(k \geqslant 1)$, it is necessary and sufficient that for every integer r such that $0 \leqslant r < k$, every $(\mathscr{F}|U)$-regular suite $(f_i)_{1 \leqslant i \leqslant r}$ of sections of $\mathscr{O}_X$ over an open subset U of X,* **IV-2 | 105**

$$(\mathscr{F}|U) / \Big(\sum_{i=1}^{r} f_i (\mathscr{F}|U) \Big)$$

is without associated immersed prime cycle.

Proof. We first show the proposition for $k = 1$; it is then stated as follows: *for $\mathscr{F}$ to satisfy (S_1), it is necessary and sufficient that $\mathscr{F}$ be without associated immersed prime cycle*. Indeed, to say that $\mathscr{F}$ satisfies (S_1) means that at the points $x \in \operatorname{Supp}(\mathscr{F})$ such that $\dim(\mathscr{F}_x) > 0$, $\operatorname{prof}(\mathscr{F}_x) \geqslant 1$ (since at all other points of $\operatorname{Supp}(\mathscr{F})$, $\operatorname{prof}(\mathscr{F}_x) = 0$, hence $\operatorname{prof}(\mathscr{F}_x) \geqslant 1$ signifies that $\mathfrak{m}_x$ is not associated to $\mathscr{F}_x$ (**0**, 16.4.6), (i)), or again that x is not a maximal point of a sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space. On the other hand, if $\dim(\mathscr{F}_x) > 0$, then $\dim(\mathscr{O}_{S,x}) > 0$ signifies that x is not a maximal point of $\operatorname{Supp}(\mathscr{F})$ (5.1.12.1), $\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{S,x}) = \operatorname{codim}(\overline{\{x\}}, S)$ (5.1.2), whence the conclusion.

In the second place, we show that, for k arbitrary > 1, the condition of the statement is *necessary*. We can restrict to considering the case $U = X$, and let us suppose first that $\dim(\mathscr{F}_x) \geqslant k$; then $\operatorname{prof}(\mathscr{F}_x) \geqslant k - 1$ (**0**, 15.2.4); there is then an open neighbourhood U of x and a sequence $(f_i)_{1 \leqslant i \leqslant k-1}$ of sections of $\mathscr{O}_X$ over U such that $(f_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant k - 1$. The hypothesis implies that $\mathscr{G} = (\mathscr{F}|U)/(\sum_{i=1}^{k-1} f_i(\mathscr{F}|U))$ is without associated prime cycle immersed, since $\dim(\mathscr{G}_x) = \dim(\mathscr{F}_x) - (k-1) \geqslant 1$ (**0**, 16.3.4), hence x is not associated to $\mathscr{G}$ (**0**, 16.4.6), (i)), and our assertion follows. Suppose in the second place that $\dim(\mathscr{F}_x) = r < k$; the condition of the statement requires first of all that $\dim(\mathscr{F}_x) \geqslant k$, hence $\operatorname{prof}(\mathscr{F}_x) \geqslant k - 1$; as $\mathscr{F}$ satisfies (S_{k-1}), we have $\operatorname{prof}(\mathscr{F}_x) \geqslant \inf(k-1, r) = r$, hence $\operatorname{prof}(\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_x) - (k-1)$ (**0**, 16.3.4), and our assertion follows.

Let us prove finally that for $k > 1$, the condition in the statement is *sufficient*. We proceed by induction on k; let x be a point of X, and let us suppose first $\dim(\mathscr{F}_x) \geqslant k$. The induction hypothesis implies $\operatorname{prof}(\mathscr{F}_x) \geqslant k - 1$; taking into account (**0**, 15.2.4), there is then an open neighbourhood U of x and a sequence $(f_i)_{1 \leqslant i \leqslant k-1}$ of sections of $\mathscr{O}_X$ over U, such that $(f_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant k - 1$. The hypothesis implies that $\mathscr{G} = (\mathscr{F}|U)/(\sum_{i=1}^{k-1} f_i(\mathscr{F}|U))$ is without associated prime cycle immersed; but $\dim(\mathscr{G}_x) = \dim(\mathscr{F}_x) - (k-1) \geqslant 1$ (**0**, 16.3.4), hence x is not associated to $\mathscr{G}$; we therefore have $\operatorname{prof}(\mathscr{G}_x) \geqslant 1$ (**0**, 16.4.6), and since $\operatorname{prof}(\mathscr{G}_x) = \operatorname{prof}(\mathscr{F}_x) - (k-1)$ (**0**, 16.4.6), $\operatorname{prof}(\mathscr{F}_x) \geqslant k$. Suppose in the second place that $\dim(\mathscr{F}_x) = r < k$; as $\mathscr{F}$ satisfies (S_{k-1}), $\operatorname{prof}(\mathscr{F}_x) \geqslant \inf(k-1, r) = r$, and this completes the proof. □

Corollary (5.7.6). — *Suppose $\mathscr{F}$ satisfies (S_k) $(k \geqslant 1)$; if (f_i) $(1 \leqslant i \leqslant r)$ is an $\mathscr{F}$-regular sequence of sections of $\mathscr{O}_X$ over X and if $r < k$, $\mathscr{F}/(\sum_{i=1}^r f_i\mathscr{F})$ satisfies (S_{k-r}).*

Proof. This follows immediately from (5.7.5). □

Corollary (5.7.7). — *For $\mathscr{F}$ to satisfy (S_2), it is necessary and sufficient that $\mathscr{F}$ be without associated immersed prime cycle and that for every open subset U of X and every $(\mathscr{F}|U)$-regular section f of $\mathscr{O}_X$ over U, $(\mathscr{F}|U)/f(\mathscr{F}|U)$ be without associated immersed prime cycle.*

Proof. This follows immediately from (5.7.5). □

Remarks (5.7.8). — Let X be a locally Noetherian prescheme *of dimension* 1; it then amounts to the same to say that X is a Cohen-Macaulay prescheme, or that it satisfies (S_1), or that it satisfies one of the properties (S_n) for $n \geqslant 1$, by virtue of the definitions (5.7.1) and (5.7.2). By virtue of (5.7.5), it therefore amounts to the same, for preschemes of dimension 1, to say that X is a Cohen-Macaulay prescheme or that it has no associated immersed prime cycle. For example, a locally Noetherian *reduced* prescheme of dimension 1 is a Cohen-Macaulay prescheme.

Proposition (5.7.9). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a homomorphism of rings, M a B-module such that $M_{[\mathfrak{p}]}$ is an A-module of finite type. Let $\mathfrak{p}$ be a prime ideal of A; let the prime ideals of B above $\mathfrak{p}$ and belonging to* $\operatorname{Supp}(M)$ *be finite in number, and let $(\mathfrak{q}_i)_{1 \leqslant i \leqslant n}$ be the family of these ideals. Then we have*

(5.7.9.1)
$$\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \sup_i \dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$$

(5.7.9.2)
$$\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \inf_i \operatorname{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}).$$

Proof. If $\mathfrak{b}$ is the annihilator of M, $B/\mathfrak{b}$ is identified with a sub-A-module of $\operatorname{End}_A(M_{[\mathfrak{p}]})$, hence is of finite type since A is Noetherian; there are thus only a finite number of prime ideals of B above $\mathfrak{p}$ and containing $\mathfrak{b}$, and these are precisely those which belong to $\operatorname{Supp}(M)$. Replacing B by $B/\mathfrak{b}$, which does not change the second members of (5.7.9.1) and (5.7.9.2) (by (**0**, 16.1.9) and (**0**, 16.4.8)), we may thus suppose that B is a finite A-algebra. Set $S = A - \mathfrak{p}$, and $B' = S^{-1}B$; B' is a semi-local Noetherian ring whose maximal ideals are $S^{-1}\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$, and as $M_\mathfrak{p}$ is an $A_\mathfrak{p}$-module of finite type, we have $\dim_{A_\mathfrak{p}}(M_\mathfrak{p}) = \dim_{B'}(M_\mathfrak{p})$ (**0**, 16.1.9). This being the case, the relation (5.7.9.1) becomes a particular case of (**0**, 16.1.7.4). To prove the relation (5.7.9.2), we reduce immediately, as in (**0**, 16.4.8), to the case where $\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) = 0$, and the same reasoning as in (**0**, 16.4.8) shows that $M_\mathfrak{p}$ contains a sub-B'-module of *finite length*, and hence also a simple sub-B'-module, which is necessarily isomorphic to the residue field of one of the $B_{\mathfrak{q}_i}$; thus there is at least one index i such that $\operatorname{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) = 0$ (**0**, 16.4.6), which finishes the proof. □

IV-2 | 107

Corollary (5.7.10). — *Suppose the hypotheses of* (5.7.9) *satisfied, and suppose in addition that A is a local ring; then, for M to be an A-module of Cohen-Macaulay, it is necessary and sufficient that, for all the maximal ideals $\mathfrak{q}_i$ of B above the maximal ideal of A, $M_{\mathfrak{q}_i}$ be a $B_{\mathfrak{q}_i}$-module of Cohen-Macaulay, and that in addition all the numbers* $\dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$ *be equal.*

It follows immediately from (5.7.9.1) and (5.7.9.2) that these conditions are equivalent to the relation $\dim_A(M) = \operatorname{prof}_A(M)$.

Corollary (5.7.11). — *Suppose the hypotheses of* (5.7.9) *satisfied.*

label=i) *If $M_{[\mathfrak{p}]}$ satisfies the property (S_k), so does M.*
lbbel=ii) *Suppose that for every pair of prime ideals $\mathfrak{q}$, $\mathfrak{q}'$ of B such that $\rho^{-1}(\mathfrak{q}) = \rho^{-1}(\mathfrak{q}')$, we have* $\dim_{B_\mathfrak{q}}(M_\mathfrak{q}) = \dim_{B_{\mathfrak{q}'}}(M_{\mathfrak{q}'})$. *Then, if M satisfies the property (S_k), so does $M_{[\mathfrak{p}]}$.*

This follows immediately from the relations (5.7.9.1) and (5.7.9.2) and from the definition of the property (S_k).

(5.7.12). In accordance with the definitions of (5.7.1), given a Noetherian ring A and a finite type A-module M, one defines $\mathrm{coprof}_A(M)$ as equal to $\mathrm{coprof}(M) = \sup_{x \in X}(\mathrm{coprof}_{A_x}(M_x))$, where $X = \mathrm{Spec}(A)$; we shall see later on (6.11.5) that this definition *coincides* with that of (**0**, 16.4.9) when A is a Noetherian *local* ring.

Corollary (5.7.13). — *Let A, B be two Noetherian rings, $\rho : A \to B$ a homomorphism of rings, M a B-module such that $M_{[\rho]}$ is an A-module of finite type. Then we have*

$$\mathrm{coprof}_A(M_{[\rho]}) \geqslant \mathrm{coprof}_B(M). \tag{5.7.13.1}$$

This follows from the preceding definition and from the relations (5.7.9.1) and (5.7.9.2).

5.8. Regular preschemes and property (R_k). Serre's normality criterion

(5.8.1). Recall (**0**, 4.1.4) that a ringed space $(X, \mathscr{O}_X)$ is said to be *regular* at a point $x \in X$, if $\mathscr{O}_x$ is a regular local ring. When dealing with preschemes, we shall only use this terminology in this chapter when X is *locally Noetherian*.

Definition (5.8.2). — We say that a locally Noetherian prescheme X is *regular in codimension* $\leqslant k$, or possesses the property (R_k), if the set of points where X is *not regular* is of codimension $> k$ (in other words (5.1.3)), if, for every $x \in X$, the relation $\dim(\mathscr{O}_x) \leqslant k$ implies that $\mathscr{O}_x$ is regular).

To say that X is regular means that X possesses the property (R_k) for every k.

If $X = \mathrm{Spec}(A)$, where A is a Noetherian ring, we say that A possesses the property (R_k) if X possesses this property; to say that X is regular means that the ring A is *regular* (**0**, 17.3.6). For a locally Noetherian prescheme X, we say that X *possesses the property* (R_k) *at a point* $x \in X$ if the local ring $\mathscr{O}_x$ possesses the property (R_k); this means that for every *generization* x' of x in X, the relation $\dim(\mathscr{O}_{x'}) \leqslant k$ implies that $\mathscr{O}_{x'}$ is a regular local ring. To say that X is regular at a point x amounts to saying that X satisfies (R_n) for *all* $n \geqslant 0$ at the point x, by virtue of (**0**, 17.3.6).

Proposition (5.8.3). — *If k is a field, X a prescheme locally of finite type over k, then, for every $x \in X$, there exists an open neighbourhood of x in X isomorphic to a sub-scheme of a regular k-scheme.*

Proof. Indeed, there is an open affine neighbourhood U of x isomorphic to a k-scheme of the form $\mathrm{Spec}(A)$, where A is a k-algebra of finite type; A is thus isomorphic to a quotient of a polynomial algebra $k[T_1, \ldots, T_n]$, and one knows that this latter is a regular ring (**0**, 17.3.7). □

(5.8.4). By virtue of (5.8.2), to say that X possesses the property (R_0) means that for every maximal point x of X, the ring $\mathscr{O}_x$ is a *field* (**0**, 17.1.4), in other words that X is *reduced* at this point. As the set U of $x \in X$ where X is reduced is *open* (the

nilradical of X being a coherent $\mathscr{O}_X$-Module (**I**, 6.1.1) and (**0**, 5.2.2)), this amounts to saying that X possesses the property (R_0) or that the set U is *everywhere dense*. Consequently: **IV-2** | 108

Proposition (5.8.5). — *For a locally Noetherian prescheme X to be reduced, it is necessary and sufficient that it satisfies the properties* (S_1) *and* (R_0)*.*

Proof. Taking into account (5.7.5), this follows from the preceding remark and from (3.2.1). □

Theorem (5.8.6). — (Serre's criterion). *Let X be a locally Noetherian prescheme. For X to be normal, it is necessary and sufficient that X satisfies the properties* (S_2) *and* (R_1)*; in other words, it is necessary and sufficient that, for every $x \in X$, the following properties are satisfied:*

label=i) *If $\dim(\mathscr{O}_x) \leqslant 1$, $\mathscr{O}_x$ is regular (that is to say, a field or a discrete valuation ring* (**0**, 17.1.4)*).*
lbbel=ii) *If $\dim(\mathscr{O}_x) \geqslant 2$, then $\mathrm{prof}(\mathscr{O}_x) \geqslant 2$.*

Proof. The conditions are *necessary*. Indeed, to say that X is normal means that, for every $x \in X$, $\mathscr{O}_x$ is an integral local Noetherian ring. If $\dim(\mathscr{O}_x) = 0$ (resp. $\dim(\mathscr{O}_x) = 1$), we conclude that $\mathscr{O}_x$ is a field since $\mathscr{O}_x$ is integral (resp. that $\mathscr{O}_x$ is a discrete valuation ring, by virtue of (**II**, 7.1.6)). On the other hand, for every element $f_x \neq 0$ of $\mathscr{O}_x$, one knows (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 4, prop. 8) that the prime ideals associated to $\mathscr{O}_x/f_x\mathscr{O}_x$ are not immersed, hence $\mathscr{O}_x$ satisfies (S_2) (5.7.7).

The conditions are *sufficient*. Indeed, it follows first from (5.8.5) that X is reduced. The question being local, one can further suppose that $X = \operatorname{Spec}(A)$, where A is a Noetherian integral *reduced* ring (**I**, 5.1.4); if R is the total ring of fractions of A, R is composed of a direct sum of a finite number of fields, and (taking into account (**II**, 6.3.6)), it suffices to prove that A is *integrally closed* in R. Let thus $h = f/g$ be an element of R integral over A, g and f being elements of A such that g is not a divisor of 0. We have a relation of the form

$$f^n + \sum_{i=1}^{n} a_i f^{n-i} g^i = 0 \quad \text{with} \quad a_i \in A \quad \text{for} \quad 1 \leqslant i \leqslant n. \tag{5.8.6.1}$$

Let $\mathfrak{p}$ be a prime ideal of A such that $\dim(A_\mathfrak{p}) = 1$; if $f_\mathfrak{p}$ and $g_\mathfrak{p}$ are the images of f and g in $A_\mathfrak{p}$, it follows from (5.8.6.1) that $f_\mathfrak{p}/g_\mathfrak{p}$ (which belongs to the total ring of fractions of $A_\mathfrak{p}$, since $g_\mathfrak{p}$ is not a divisor of 0 in $A_\mathfrak{p}$ by flatness (**0**, 5.3.1)) is integral over $A_\mathfrak{p}$; but as $A_\mathfrak{p}$ is regular, hence integrally closed, we have $f_\mathfrak{p}/g_\mathfrak{p} \in A_\mathfrak{p}$. In other words, we have $(fA)_\mathfrak{p} \subset (gA)_\mathfrak{p}$. But the hypothesis (S_2) implies (5.7.7), since g is not a divisor of 0 in A, that A/gA has no associated prime ideal of codimension 1; or, gA is the intersection of primary ideals $\mathfrak{q}_i$ corresponding to the $\mathfrak{p}_i$, and from what we have just seen, the $\mathfrak{q}_i$ are, by the homomorphisms $A \to A_{\mathfrak{p}_i}$, the ideals $(gA)_{\mathfrak{p}_i}$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 3, prop. 5). But by virtue of the Hauptidealsatz (**0**, 16.3.2), we have $\dim(A_{\mathfrak{p}_i}) = 1$ for $1 \leqslant i \leqslant n$, hence $(fA)_{\mathfrak{p}_i} \subset (gA)_{\mathfrak{p}_i}$ for every i by what precedes; as fA is contained in the intersection of the $(fA)_{\mathfrak{p}_i}$ $(1 \leqslant i \leqslant n)$, we have $fA \subset gA$, that is to say $f/g \in A$. □

IV-2 | 109

5.9. Z-pure and Z-closed modules Some of the notions and results of this section and of the next are special cases of notions and results developed in chapter III in the theory of local cohomology. For the convenience of the reader, we give here an independent exposition.

(5.9.1). Let X be a locally Noetherian prescheme, Z a part of X *stable by specialisation*: this means that for every finite part M of Z, the closure of M is contained in Z, and hence Z is the union of a filtered increasing family (Z_α) of closed parts of X; conversely, it is clear that such a union is stable by specialisation.

Set $U_\alpha = X - Z_\alpha$, so that $X - Z$ is the intersection of the filtered decreasing family of open sets U_α; let $i_\alpha : U_\alpha \to X$ be the canonical injection, and for $U_\alpha \supset U_\beta$, let $i_{\alpha\beta} : U_\beta \to U_\alpha$ be the canonical injection, so that we have $i_\beta = i_\alpha \circ i_{\alpha\beta}$. Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module (not necessarily quasi-coherent); we have therefore $(i_\beta)_*(\mathscr{F}|U_\beta) = (i_\alpha)_*((i_{\alpha\beta})_*(\mathscr{F}|U_\beta))$; by the canonical homomorphism (**0**, 4.4.3.2)

$$\mathscr{F}|U_\alpha \longrightarrow (i_{\alpha\beta})_*(\mathscr{F}|U_\beta)$$

we thus deduce, by application of the functor $(i_\alpha)_*$, a homomorphism

$$\rho_{\beta\alpha} : (i_\alpha)_*(\mathscr{F}|U_\alpha) \longrightarrow (i_\beta)_*(\mathscr{F}|U_\beta)$$

and one verifies immediately that $\rho_{\gamma\alpha} = \rho_{\gamma\beta} \circ \rho_{\beta\alpha}$ for $U_\alpha \supset U_\beta \supset U_\gamma$; in other words, the $\mathscr{O}_X$-Modules $(i_\alpha)_*(\mathscr{F}|U_\alpha)$ form an *inductive system* for the homomorphisms $\rho_{\beta\alpha}$. We set

$$\mathscr{H}^0_{X/Z}(\mathscr{F}) = \varinjlim_\alpha (i_\alpha)_*(\mathscr{F}|U_\alpha). \tag{5.9.1.1}$$

This $\mathscr{O}_X$-Module does not depend on the increasing family (Z_α) of closed sets whose union is Z: indeed, let V be a Noetherian open of X; one knows (**G**, II.3.10.1) that in the category of $\mathscr{O}_V$-Modules, the functor $\mathscr{G} \mapsto \Gamma(V, \mathscr{G})$ commutes with inductive limits; we thus have, by virtue of (5.9.1.1)

$$\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_\alpha \Gamma(V \cap U_\alpha, \mathscr{F}). \tag{5.9.1.2}$$

Let then (Z'_λ) be a second filtered increasing family of closed sets of X, of union Z; $V \cap Z_\alpha$ is thus the union of the $V \cap Z_\alpha \cap Z'_\lambda$; but $V \cap Z_\alpha$ is locally closed in X, hence every closed irreducible part of $V \cap Z_\alpha$ admits a generic point; as the $V \cap Z_\alpha \cap Z'_\lambda$ are closed in $V \cap Z_\alpha$ and form (for α fixed) a filtered increasing family, there exists an index λ such that $V \cap Z_\alpha \cap Z'_\lambda = V \cap Z_\alpha$ (**0**, 9.2.4), in other words $V \cap Z_\alpha \subset V \cap Z'_\lambda$. This proves that the filtered decreasing families $V \cap U_\alpha$, $V \cap U'_\lambda$ (where $U'_\lambda = X - Z'_\lambda$) are cofinal with each other, whence our assertion, by virtue of (5.9.1.2).

One notes that the set Z is not necessarily constructible: one has an example of this by taking $X = \mathrm{Spec}(A)$, where A is a Noetherian integral ring having an infinity of maximal ideals, and Z the complement of the generic point of X. **IV-2** | 110

If Z is closed and if $i : X - Z \to X$ is the canonical injection, we have

$$\mathscr{H}^0_{X/Z}(\mathscr{F}) = i_*(\mathscr{F}|X - Z) \tag{5.9.1.3}$$

and in particular, for $Z = \varnothing$, $\mathscr{H}^0_{X/Z}(\mathscr{F}) = \mathscr{F}$.

Proposition (5.9.2). — *label*=i) *The functor $\mathscr{F} \mapsto \mathscr{H}^0_{X/Z}(\mathscr{F})$ is exact on the left.*
lbbel=ii) *If $\mathscr{F}$ is quasi-coherent, so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$.*

Proof. Assertion (i) follows from the definition (5.9.1.1), from the fact that $(i_\alpha)_*$ is a functor exact on the left, and from the fact that the inductive limit preserves exactness in the category of $\mathscr{O}_X$-Modules. Assertion (ii) follows from (**I**, 9.2.2) and from the fact that an inductive limit of quasi-coherent $\mathscr{O}_X$-Modules is quasi-coherent (**I**, 1.3.9). □

Remark (5.9.3). — If $\mathscr{F}$ is an $\mathscr{O}_X$*-Algebra*, so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$ (**0**, 4.2.4); in particular $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_X$-Algebra, and for every $\mathscr{O}_X$-Module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is an $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$-Module, which is quasi-coherent if $\mathscr{F}$ is quasi-coherent (**I**, 9.6.1). More particularly, suppose that $X = \mathrm{Spec}(A)$, where A is integral and Noetherian; then $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is the $\mathscr{O}_X$-Algebra $\widetilde{B}$, where

$$B = \bigcap_{\mathfrak{p} \in X - Z} A_\mathfrak{p}. \tag{5.9.3.1}$$

This follows immediately from (5.9.1.2) and from (**I**, 8.2.1.1).

Proposition (5.9.4). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, X' a locally Noetherian prescheme, $f : X' \to X$ a flat morphism. Then $Z' = f^{-1}(Z)$ is stable by specialisation, and for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, there is a canonical isomorphism*

$$f^*(\mathscr{H}^0_{X/Z}(\mathscr{F})) \xrightarrow{\sim} \mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F})) \tag{5.9.4.1}$$

Proof. Indeed, with the notations of (5.9.1), $Z'_\alpha = f^{-1}(Z_\alpha)$ is closed in X' and Z' is the union of the Z'_α; furthermore, $(i_\alpha)_{(X')}$ is the canonical injection $i'_\alpha : U'_\alpha \to X'$, and if $U'_\alpha = X' - Z'_\alpha = f^{-1}(U_\alpha)$. As f is flat, one knows (2.3.1) that the canonical homomorphism $f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \to (i'_\alpha)_*(f^*(\mathscr{F})|U'_\alpha)$ is bijective; as, for $\alpha \leqslant \beta$, the diagram

$$\begin{tikzcd}
f^*((i_\alpha)_*(\mathscr{F}|U_\alpha)) \arrow[r, "\sim"] \arrow[d] & (i'_\alpha)_*(f^*(\mathscr{F})|U'_\alpha) \arrow[d] \\
f^*((i_\beta)_*(\mathscr{F}|U_\beta)) \arrow[r, "\sim"] & (i'_\beta)_*(f^*(\mathscr{F})|U'_\beta)
\end{tikzcd}$$

is commutative, one has therefore, passing to the limit, a canonical isomorphism $\varinjlim_\alpha (f^*((i_\alpha)_*(\mathscr{F}|U_\alpha))) \xrightarrow{\sim}$ **IV-2** | 111
$\mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$. But as the functor f^* commutes with inductive limits (**0**, 4.3.2), this gives by definition the sought isomorphism (5.9.4.1). □

Corollary (5.9.5). — *Under the hypotheses of (5.9.4), if $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent, so is $\mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$. The converse is true when f is a faithfully flat and quasi-compact morphism.*

Proof. The first assertion follows from (5.9.4.1) and from (**0**, 5.3.11); the second is equivalent to saying that if $\mathscr{H}^0_{X'/Z'}(f^*(\mathscr{F}))$ is of finite type, so is $\mathscr{H}^0_{X/Z}(\mathscr{F})$; this follows from (5.9.4.1) and from (2.5.2). □

Corollary (5.9.6). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. For every $x \in X$, setting $X_x = \operatorname{Spec}(\mathscr{O}_x)$, $Z_x = Z \cap X_x$, we have a functorial isomorphism*

(5.9.6.1)
$$(\mathscr{H}^0_{X/Z}(\mathscr{F}))_x \xrightarrow{\sim} \mathscr{H}^0_{X_x/Z_x}(\widetilde{\mathscr{F}_x})$$

Proof. It suffices to apply (5.9.4) to the canonical morphism $X_x \to X$, which is flat, and to take into account (**I**, 1.6.5). □

(5.9.7). With the notations of (5.9.1), for every α we have a canonical functorial homomorphism $\mathscr{F} \to (i_\alpha)^*(\mathscr{F}|U_\alpha)$ (**0**, 4.4.3.2), and these homomorphisms form an inductive system; by passage to the inductive limit, we thus deduce a canonical functorial homomorphism

(5.9.7.1)
$$\rho_{X/Z} : \mathscr{F} \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F})$$

Proposition (5.9.8). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ an $\mathscr{O}_X$-Module. The following properties are equivalent:*

label=a) *The homomorphism $\rho_{X/Z}$ (5.9.7.1) is injective (resp. bijective).*

lbbel=b) *For every Noetherian open V of X, the homomorphism*
$$(\rho_{X/Z})_V : \Gamma(V, \mathscr{F}) \longrightarrow \varinjlim_\alpha \Gamma(V \cap U_\alpha, \mathscr{F})$$
is injective (resp. bijective).

a′) *For every closed part $T \subset Z$ of X, the canonical homomorphism* (**0**, 4.4.3.2)
$$\mathscr{F} \longrightarrow i_*(\mathscr{F}|X - T)$$
(where $i : X - T \to X$ is the canonical injection) is injective (resp. bijective).

b′) *For every closed part $T \subset Z$ of X and every Noetherian open V of X, the homomorphism of restriction*
$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F})$$
is injective (resp. bijective).

Proof. Taking into account (5.9.1.2), the equivalence of *a)* and *b)* (resp. *a′)* and *b′)*) follows from the definition of the functor Γ and from the fact that it is exact on the left. As the homomorphism $(\rho_{X/Z})_V$ is the composite **IV-2 | 112**
$$\Gamma(V, \mathscr{F}) \longrightarrow \Gamma(V \cap (X - T), \mathscr{F}) \longrightarrow \varinjlim_\alpha \Gamma(V \cap U_\alpha, \mathscr{F})$$
for every closed part $T \subset Z$, if $(\rho_{X/Z})_V$ is injective, then so is $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap (X - T), \mathscr{F})$; on the other hand, the fact that *b′)* implies *b)* follows from the definition of an inductive limit. It remains to show that if $\rho_{X/Z}$ is bijective, then so is $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap (X - T), \mathscr{F})$, and for this it suffices, by virtue of (5.9.8.1), to see that if $U' \subset U$ are two opens contained in V and containing $V \cap Z$, the homomorphism of restriction $\Gamma(U, \mathscr{F}) \to \Gamma(U', \mathscr{F})$ is injective; but this follows from the fact that $\rho_{X/Z}$ is injective, by replacing V by U and $V \cap (X - T)$ by U'. □

Definition (5.9.9). — Under the hypotheses of (5.9.8), we say that $\mathscr{F}$ is *Z-pure* (resp. *Z-closed*) if $\rho_{X/Z}$ is injective (resp. bijective).

If $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, where M is an A-module, we say that M is Z-pure (resp. Z-closed) when $\mathscr{F}$ is Z-pure (resp. Z-closed).

We say that $\mathscr{F}$ is *Z-pure* (resp. *Z-closed*) *at a point $x \in X$* if (with the notations of (5.9.6)) $\widetilde{\mathscr{F}_x}$ is Z_x-pure (resp. Z_x-closed); this amounts to the same, by virtue of (5.9.6), to saying that the canonical homomorphism $\mathscr{F}_x \to (\mathscr{H}^0_{X/Z}(\mathscr{F}))_x$ is injective (resp. bijective).

One notes that for every $x \in X - Z$, $\mathscr{F}$ is Z-closed at the point x, by virtue of (5.9.8).

Corollary (5.9.10). — label=i) *Let (V_λ) be an open covering of X. For $\mathscr{F}$ to be Z-pure (resp. Z-closed), it is necessary and sufficient that, for every λ, $\mathscr{F}|V_\lambda$ be $(Z \cap V_\lambda)$-pure (resp. $(Z \cap V_\lambda)$-closed).*

lbbel=ii) *Let Z' be a part of Z stable by specialisation. If $\mathscr{F}$ is Z-pure (resp. Z-closed), it is Z'-pure (resp. Z'-closed).*

Proof. This follows immediately from (5.9.8), b')). □

Proposition (5.9.11). — *Under the hypotheses of (5.9.8), the $\mathscr{O}_X$-Modules $\operatorname{Ker}(\rho_{X/Z})$ and $\operatorname{Coker}(\rho_{X/Z})$ have their support contained in Z, and the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed. Furthermore, if $u : \mathscr{F} \to \mathscr{F}'$ is a homomorphism of $\mathscr{O}_X$-Modules such that $\mathscr{F}'$ is Z-closed, u factorises in a unique way as $\mathscr{F} \xrightarrow{\rho_{X/Z}} \mathscr{H}^0_{X/Z}(\mathscr{F}) \xrightarrow{v} \mathscr{F}'$. If in addition the supports of $\operatorname{Ker}(u)$ and of $\operatorname{Coker}(u)$ are contained in Z, v is an isomorphism.*

Proof. The first assertion means that, for every $x \in X - Z$, we have

$$\operatorname{Ker}(\rho_{X/Z})_x = \operatorname{Coker}(\rho_{X/Z})_x = 0,$$

i.e. that $(\rho_{X/Z})_x$ is bijective, or again that $\mathscr{F}$ is Z-pure at x, as has been indicated above (5.9.9).

To show that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, it suffices to see that for every Noetherian open V of X, $\varinjlim_\beta \Gamma(V \cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F}))$ is equal to $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$; but by definition $\Gamma(V \cap U_\beta, \mathscr{H}^0_{X/Z}(\mathscr{F})) = \varinjlim_\alpha \Gamma(V \cap U_\alpha \cap U_\beta, \mathscr{F})$. Now, the double family $(U_\alpha \cap U_\beta)$ is a filtered decreasing family, and our assertion follows from (5.9.1) and from the theorem on double inductive limits.

Let us pass to the second part of the proposition. The existence and the uniqueness of v follow **IV-2** | 113
from the fact that for every α, $(i_\alpha)_*(u)$ is the unique homomorphism making the diagram

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "u"] & \mathscr{F}' \\
(i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[u] \arrow[r, "(i_\alpha)_*(u)"] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \arrow[u]
\end{tikzcd}$$

commutative; from this there is a unique homomorphism w making all the diagrams

$$\begin{tikzcd}
(i_\alpha)_*(\mathscr{F}|U_\alpha) \arrow[r, "(i_\alpha)_*(u)"] & (i_\alpha)_*(\mathscr{F}'|U_\alpha) \\
\mathscr{H}^0_{X/Z}(\mathscr{F}) \arrow[u] \arrow[r, "w"] & \mathscr{H}^0_{X/Z}(\mathscr{F}') \arrow[u]
\end{tikzcd}$$

commutative, and finally that $\mathscr{F}'$ and $\mathscr{H}^0_{X/Z}(\mathscr{F}')$ identify canonically by hypothesis.

It remains to see that if the supports of $\operatorname{Ker}(u)$ and of $\operatorname{Coker}(u)$ are contained in Z, v is an isomorphism. It suffices to see that for every Noetherian open V of X, the corresponding homomorphism $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F})) \to \Gamma(V, \mathscr{F}')$ is an isomorphism. Now, if $t \in \Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$ has image 0 in $\Gamma(V, \mathscr{F}')$, let us note that for a given index α, we have $t \in \Gamma(V \cap U_\alpha, \mathscr{F})$, and by the hypothesis on u, $t_y = 0$ for every $y \in V \cap (X - Z)$; it follows that there is an open set containing $V \cap (X - Z)$ on which the restriction of t is zero, hence by definition t is the element 0 of $\Gamma(V, \mathscr{H}^0_{X/Z}(\mathscr{F}))$. Let us now prove that every section $s' \in \Gamma(V, \mathscr{F}')$ is the image of a section of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above V. By hypothesis, for every $x \in V \cap (X - Z)$ there exists an open neighbourhood $W^{(x)}$ of x in X, and a section $s^{(x)}$ of $\mathscr{F}$ above $W^{(x)}$, whose image by u is $s'|W^{(x)}$; $s'|W^{(x)}$ is thus also the image by v of the section $t^{(x)}$ of $\mathscr{H}^0_{X/Z}(\mathscr{F})$. Furthermore, as one has seen that v is injective, the restrictions of $t^{(x)}$ and $t^{(x')}$ to $W^{(x)} \cap W^{(x')}$ are identical for any two points x, x' of $V \cap (X - Z)$; the $t^{(x)}$ are therefore restrictions of a single section t of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above an open set U of $(X - Z) \cap V$. But as $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed, t extends in a unique way to a section of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ above V, whose image by v has the same restriction as s' to U, and hence coincides with s' for the same reason.

We say that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is the *Z-closure* of $\mathscr{F}$. □

IV-2 | 114

Remarks (5.9.12). — label=*i)* Let $\mathscr{C}(X)$ be the category of $\mathscr{O}_X$-Modules, and let $\mathscr{C}_Z(X)$ be the sub-category of $\mathscr{C}(X)$ formed of $\mathscr{O}_X$-Modules of support contained in Z; this sub-category is localisante in the sense of Gabriel, and the functor $\mathscr{H}^0_{X/Z}$ is none other than the localisation functor of Gabriel (cf. [?, 27]; this would furnish another proof of (5.9.11)). When Z is closed, the functor $i^* : \mathscr{C}(X) \to \mathscr{C}(X - Z)$ (where $i : X - Z \to X$ is the canonical injection) defines an *equivalence* of categories $\mathscr{C}(X)/\mathscr{C}_Z(X) \approx \mathscr{C}(X - Z)$.

lbbel=*ii)* It follows from (5.9.11) that the condition $\mathscr{H}^0_{X/Z}(\mathscr{F}) = 0$ is *equivalent* to $\mathrm{Supp}(\mathscr{F}) \subset Z$. Indeed, it implies that the kernel of $\rho_{X/Z}$ is equal to $\mathscr{F}$. Conversely, if $\mathrm{Supp}(\mathscr{F}) \subset Z$, it suffices to apply the second part of (5.9.11) to the unique homomorphism $u : \mathscr{F} \to 0$ to conclude that the corresponding homomorphism $v : \mathscr{H}^0_{X/Z}(\mathscr{F}) \to 0$ is an isomorphism.

lcbel=*iii)* The preceding developments hold for every annulated space locally Noetherian where every closed irreducible part admits exactly one generic point. In particular they apply, on a locally Noetherian prescheme, to any *sheaves of abelian groups* (considered as Modules over the simple sheaf associated to the constant presheaf **Z**). One has moreover for every $x \in X$ the canonical isomorphism (5.9.6.1), in which $\widetilde{\mathscr{F}_x}$ denotes the sheaf *induced* on the subspace X_x of X by the sheaf $\mathscr{F}$; the direct proof follows immediately from the definition (5.9.1.2) and from the theorem on double inductive limits.

5.10. Property (S_2) and Z-closure

(5.10.1). Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module; for every part T of X, we shall set

$$\mathrm{prof}_T(\mathscr{F}) = \inf_{z\in T} \mathrm{prof}(\mathscr{F}_z) \tag{5.10.1.1}$$

Proposition (5.10.2). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module. The following conditions are equivalent:*

label=a) *$\mathscr{F}$ is Z-pure.*

lbbel=b) $\mathrm{Ass}(\mathscr{F})$ *does not meet Z.*

If in addition $\mathscr{F}$ is coherent, these conditions are also equivalent to the following:

c) $\mathrm{prof}_Z(\mathscr{F}) \geqslant 1$.

Proof. To say that $\mathscr{F}$ is Z-pure means that for every Noetherian open V of X, and every open $U \supset X - Z$, the homomorphism of restriction $\Gamma(V, \mathscr{F}) \to \Gamma(V \cap U, \mathscr{F})$ is injective (5.9.8); but by (3.1.8), this is equivalent to $V \cap \mathrm{Ass}(\mathscr{F}) \subset U$, whence the equivalence of *a)* and *b)*. Furthermore, to say that $x \in \mathrm{Ass}(\mathscr{F})$ means that no element of $\mathfrak{m}_x$ is $\mathscr{F}_x$-regular (3.1.2), hence, when $\mathscr{F}$ is coherent, this is written $\mathrm{prof}(\mathscr{F}_x) = 0$; one deduces immediately from this the equivalence of *b)* and *c)*. ☐

Corollary (5.10.3). — *Let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ be an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules. If $\mathscr{F}$ is Z-pure, so is $\mathscr{F}'$; conversely, if $\mathscr{F}'$ and $\mathscr{F}''$ are Z-pure, so is $\mathscr{F}$.*

IV-2 | 115

Proof. This follows from the form (5.10.2), *b)*) of the condition for a quasi-coherent $\mathscr{O}_X$-Module to be Z-pure, and from (3.1.7). ☐

Corollary (5.10.4). — *Suppose $\mathscr{F}$ coherent. For $\mathscr{F}$ to be Z-pure at a point $x \in X$, it is necessary and sufficient that* $\mathrm{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 1$ *(with the notations of* (5.9.6)*).*

Proof. This follows immediately from (5.10.2) and (5.9.6). ☐

Theorem (5.10.5). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For $\mathscr{F}$ to be Z-closed, it is necessary and sufficient that* $\mathrm{prof}_Z(\mathscr{F}) \geqslant 2$.

Proof. By virtue of (5.10.2), we may restrict ourselves to the case where $\mathscr{F}$ is Z-pure and $\mathrm{prof}_Z(\mathscr{F}) \geqslant 1$; $\mathrm{prof}_{Z_\alpha}(\mathscr{F}) \geqslant 2$; and similarly, it follows from (5.9.8) that to say that $\mathscr{F}$ is Z-closed is equivalent to saying that $\mathscr{F}$ is Z_α-closed for every α. We may thus already restrict ourselves to the case where Z is *closed*. The question being local, it suffices, for every $x \in Z$, to prove the theorem for $\mathscr{F}|U$, where U is an affine open neighbourhood of x, and we can therefore restrict ourselves to the case where $X = U$ is affine. We know then that $\mathrm{Ass}(\mathscr{F})$ is finite (3.1.6), and as $\mathrm{Ass}(\mathscr{F}) \subset X - Z$, there is a section f of $\mathscr{O}_X$ above X such that $\mathrm{Ass}(\mathscr{F}) \subset X_f \subset X - Z$ (**II**, 4.5.4); we deduce from this that f is

$\mathscr{F}$-regular (3.1.9) and that for every $y \in Z$, we have $f_y \in \mathfrak{m}_y$, hence $\operatorname{prof}(\mathscr{F}_y) = 1 + \operatorname{prof}(\mathscr{F}_y / f_y \mathscr{F}_y)$ (**0**, 16.4.6). The condition $\operatorname{prof}_Z(\mathscr{F}) \geqslant 2$ is thus equivalent to $\operatorname{prof}_Z(\mathscr{F}/f\mathscr{F}) \geqslant 1$, or again (5.10.2) to the fact that $\mathscr{F}/f\mathscr{F}$ is Z-pure, and it suffices to see that this last property is equivalent to the fact that $\mathscr{F}$ is Z-closed.

Consider the exact sequence $0 \to \mathscr{F} \xrightarrow{f} \mathscr{F} \to \mathscr{F}/f\mathscr{F} \to 0$ (the homothety of ratio $f : \mathscr{F} \xrightarrow{f} \mathscr{F}$ being injective by hypothesis); if we set $W = X - Z$, we have the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(X,\mathscr{F}) \arrow[r, "f"] \arrow[d] & \Gamma(X,\mathscr{F}) \arrow[r] \arrow[d] & \Gamma(X,\mathscr{F}/f\mathscr{F}) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \Gamma(W,\mathscr{F}) \arrow[r, "f"] & \Gamma(W,\mathscr{F}) \arrow[r] & \Gamma(W,\mathscr{F}/f\mathscr{F}) & \end{tikzcd}$$

where the two rows are exact (X being affine). If the homomorphism of restriction $\Gamma(X,\mathscr{F}) \to \Gamma(W,\mathscr{F})$ is bijective, we deduce from this diagram that

$$\Gamma(X,\mathscr{F}/f\mathscr{F}) \longrightarrow \Gamma(W,\mathscr{F}/f\mathscr{F})$$

is injective, and this shows (5.9.8) that $\mathscr{F}/f\mathscr{F}$ is Z-pure. Conversely, suppose that $\mathscr{F}/f\mathscr{F}$ is Z-pure, and let s be a section of $\mathscr{F}$ above W; as $X_f \subset W$, there is an integer $n > 0$ such that $f^n(s|X_f)$ extends to a section t of $\mathscr{F}$ above X (**I**, 1.4.1); besides, the restrictions of t and of $f^n s$ to X_f being the same, it follows that the restriction of t to W is equal to $f^n s$ by virtue of the relation $\operatorname{Ass}(\mathscr{F}) \subset X_f$ (5.10.2); as f is $\mathscr{F}$-regular, it suffices to see that t is of the form $f^n t'$, where $t' \in \Gamma(X,\mathscr{F})$

to show that the homomorphism $\Gamma(X,\mathscr{F}) \to \Gamma(W,\mathscr{F})$ is surjective, hence bijective. Now, to say **IV-2** | 116
that $t = f^n t'$ means that the image of t in $\Gamma(X,\mathscr{F}/f^n\mathscr{F})$ is zero. But as $f^k\mathscr{F}/f^{k+1}\mathscr{F}$ is isomorphic to $\mathscr{F}/f\mathscr{F}$, hence Z-pure by hypothesis, we deduce from (5.10.3) by induction on n that $\mathscr{F}/f^n\mathscr{F}$ is Z-pure. But by definition the image of $t|W = f^n s$ in $\Gamma(W,\mathscr{F}/f^n\mathscr{F})$ is zero, whence the conclusion. □

Corollary (5.10.6). — *Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent. For $\mathscr{F}$ to be Z-closed at a point x, it is necessary and sufficient that* $\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}_x}) \geqslant 2$.

Proof. This follows from (5.9.6) and (5.10.5). □

Theorem (5.10.7). — (Hartshorne). *Let X be a locally Noetherian prescheme, Y a closed part of X. Suppose that for every $y \in Y$, we have* $\operatorname{prof}(\mathscr{O}_{X,y}) \geqslant 2$*; then for every connected component C of X, $C - (C \cap Y)$ is connected.*

Proof. We may restrict ourselves to the case where X is connected; it follows then from (5.10.5) that the canonical homomorphism $\mathscr{O}_X \to i_*(\mathscr{O}_X|X - Y)$ (where $i : X - Y \to X$ is the canonical injection) is bijective. Consequently, the homomorphism of restriction $\Gamma(X,\mathscr{O}_X) \to \Gamma(X - Y,\mathscr{O}_X)$ is also bijective. It suffices now to apply the lemma (**III**, 7.8.6.1). □

Corollary (5.10.8). — *Let X be a locally Noetherian prescheme, d an integer such that for every $x \in X$, the relation* $\dim(\mathscr{O}_x) \geqslant d$ *implies* $\operatorname{prof}(\mathscr{O}_x) \geqslant 2$. *Suppose X connected; then, if X', X'' are two distinct irreducible components of X, there exists a suite $(X_i)_{0 \leqslant i \leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and that, for $1 \leqslant i \leqslant n$, we have* $\operatorname{codim}(X_{i-1} \cap X_i, X) \leqslant d - 1$ *(we then say that X is* connected in codimension $\leqslant d - 1$*).*

If Y is a closed part of X such that $\operatorname{codim}(Y, X) \geqslant d$*, we have* $\dim(\mathscr{O}_{X,y}) \geqslant d$ *for every $y \in Y$* (5.1.3)*, hence* $\operatorname{prof}(\mathscr{O}_{X,y}) \geqslant 2$ *for every $y \in Y$, and it follows from* (5.10.7) *that $X - Y$ is connected. On the other hand, for* $\operatorname{codim}(Y, X) \geqslant d$*, it is necessary and sufficient that for every $y \in Y$, there exists an open neighbourhood V of y in X such that* $\operatorname{codim}(Y \cap V, V) \geqslant d$ (**0**, 14.2.3)*. Let us finally note that if $\mathfrak{Y}$ denotes the set of closed parts Y of X of codimension $\geqslant d$, the union of two sets of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$* (**0**, 14.2.5)*, and every closed set contained in a set of $\mathfrak{Y}$ also belongs to $\mathfrak{Y}$; properties which we may also express by saying that $\mathfrak{Y}$ is an* antifilter *of closed parts of X. The corollary is then the following topological lemma:*

Lemma (5.10.8.1). — *Let X be a connected locally Noetherian topological space, $\mathfrak{Y}$ an antifilter of closed parts of X. Suppose that if Y is a closed part of X such that for every $y \in Y$, there is a neighbourhood open V of y in X and a $Y_y \in \mathfrak{Y}$ such that $V \cap Y = V \cap Y_y$, then $Y \in \mathfrak{Y}$. The following conditions are then equivalent:*

*label=*a) *For every $Y \in \mathfrak{Y}$, $X - Y$ is connected.*

lbbel=b) *If X' and X'' are two distinct irreducible components of X, there exists a suite $(X_i)_{0\leqslant i\leqslant n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$ and that, for $1 \leqslant i \leqslant n$, we have $X_{i-1} \cap X_i \notin \mathfrak{Y}$.*

Proof. Suppose *b)* satisfied and let us prove that $U = X - Y$ is connected for every $Y \in \mathfrak{Y}$. If U' and U'' are two distinct irreducible components of U, there exist two irreducible components X', X'' of X such that $X' \cap U = U'$, $X'' \cap U = U''$ (**0**, 2.1.6);

let us form for these two components a suite (X_i) having the property stated in *b)* and set $U_i = X_i \cap U$ for $1 \leqslant i \leqslant n$; then U_i is an irreducible component of U (**0**, 2.1.6) and furthermore $U_i \cap U_{i-1} \neq \varnothing$ for $1 \leqslant i \leqslant n$, otherwise one would have $X_i \cap X_{i-1} \subset Y$, hence $X_i \cap X_{i-1} \in \mathfrak{Y}$, contrary to the hypothesis. This shows that U is connected. **IV-2** | 117

Let us now show that *a)* implies *b)*. Denote by Y the union of the family $(X_\alpha \cap X_\beta)$, where (X_α, X_β) runs through the set of pairs of *distinct* irreducible components of X such that $X_\alpha \cap X_\beta \in \mathfrak{Y}$. For every point $y \in Y$, in every open neighbourhood V of y in X there is only a finite number of irreducible components; this shows on the one hand that Y is closed, and on the other part that $V \cap Y$ is an intersection of V and of a set of $\mathfrak{Y}$; by the hypothesis made on $\mathfrak{Y}$, we have $Y \in \mathfrak{Y}$, hence $U = X - Y$ is connected. Let now X', X'' be two distinct irreducible components of X, U', U'' their respective traces on U; these are irreducible components of U (**0**, 2.1.6). Take the union W of the irreducible components of U such that there exists a suite $(U_i)_{0\leqslant i\leqslant n}$ of irreducible components of U for which $U_0 = U'$, $U_{i-1} \cap U_i \neq \varnothing$ for $1 \leqslant i \leqslant n$ and $U_n = W$; this is an *open and closed* set in U, since U is locally Noetherian and its irreducible components form a locally finite family of closed sets. So there is such a suite (U_i) for which $U_n = U''$; let X_i $(0 \leqslant i \leqslant n)$ be the irreducible component of X such that $X_i \cap U = U_i$ (**0**, 2.1.6); as $U_{i-1} \neq U_i$, on has $X_{i-1} \neq X_i$ for $1 \leqslant i \leqslant n$; if one had $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i such that $1 \leqslant i \leqslant n$, one would deduce $X_{i-1} \cap X_i \subset Y$, contrary to the definition of the X_i. This achieves the proof of the lemma. □

One notes that the hypothesis made on X in (5.10.8) is satisfied when X is a Cohen-Macaulay *prescheme* and $d \geqslant 2$.

Corollary (5.10.9). — *If a Noetherian local ring A satisfies* (S_2) *and is catenary, it is equidimensional.*

Proof. The hypothesis of (5.10.8) is then satisfied by $X = \mathrm{Spec}(A)$, with $d = 2$. To show that all the irreducible components of X have the same dimension, it suffices then, by virtue of (5.10.8), to show that any two such components X', X'' have the same dimension, when in addition one supposes that $\mathrm{codim}(X' \cap X'', X) = 1$. There is then an irreducible component Z of $X' \cap X''$ such that $\mathrm{codim}(Z, X) = 1$, hence $\mathrm{codim}(Z, X') = 1$, hence $\mathrm{codim}(Z, X') \geqslant 1$; of the same, $\mathrm{codim}(Z, X'') \geqslant 1$; as X is catenary, this implies $\dim(X') = \dim(X'')$. □

Proposition (5.10.10). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, and suppose that the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent. Then:*

label=i) *We have* $\mathrm{prof}_Z(\mathscr{H}^0_{X/Z}(\mathscr{F})) \geqslant 2$.

lbbel=ii) *For every point $x \in \mathrm{Ass}(\mathscr{F}) \cap (X - Z)$, we have* $\mathrm{codim}(Z \cap \overline{\{x\}}, \overline{\{x\}}) \geqslant 2$.

lcbel=iii) *The set U of $x \in X$ such that* $\mathrm{prof}_{Z_x}(\widetilde{\mathscr{O}}_x) \geqslant 2$ *and* $\mathrm{prof}_{Z_x}(\widetilde{\mathscr{F}}_x) \geqslant 2$ *(notations of* (5.9.6)*) is open in X; we have $X - U \subset Z$, and U is the largest open of X such that $\mathscr{F}|U$ is $(Z \cap U)$-closed.*

IV-2 | 118

Proof. Let us set, for brevity, $\mathscr{F}' = \mathscr{H}^0_{X/Z}(\mathscr{F})$. We know (5.9.11) that $\mathscr{F}'$ is Z-closed, and the restrictions of $\mathscr{F}$ and $\mathscr{F}'$ to $X - Z$ are canonically isomorphic (5.9.11), hence $x \in \mathrm{Ass}(\mathscr{F}')$. Consider a point $y \in Z \cap \overline{\{x\}}$; the prime ideal $\mathfrak{p}$ of $\mathscr{O}_y$ corresponding to x is associated to the $\mathscr{O}_y$-module $\mathscr{F}'_y$, hence on the one hand, by virtue of (i) and of (**0**, 16.4.6.2), $2 \leqslant \mathrm{prof}(\mathscr{F}'_y) \leqslant \dim(\mathscr{O}_y/\mathfrak{p}) = \mathrm{codim}(\overline{\{y\}}, \overline{\{x\}})$, whence (ii). Finally, to prove (iii), let us note that U is the set of $x \in X$ such that $\widetilde{\mathscr{F}}_x$ is Z_x-closed (5.10.5), or again, by virtue of (5.9.6), the set of points where the canonical homomorphism $\mathscr{F}_x \to \mathscr{F}'_x$ is bijective; it is therefore the complement of the union of the supports of $\mathrm{Ker}(\rho_{X/Z})$ and $\mathrm{Coker}(\rho_{X/Z})$, and these last are coherent $\mathscr{O}_X$-Modules by virtue of the hypothesis (**0**, 5.3.4), hence have a closed support (**0**, 5.2.2); this shows that U is open, and U is evidently the largest open such that $\mathscr{F}|U$ is $(Z \cap U)$-closed; finally the inclusion $X - U \subset Z$ follows from (5.9.11). □

We shall see further on (5.11.1) that in the most important cases assertion (ii) implies conversely that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.

(5.10.11). Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation; one has seen that $\mathscr{A} = \mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is a quasi-coherent $\mathscr{O}_X$-Algebra (5.9.3); the X-scheme $X' = \operatorname{Spec}(\mathscr{H}^0_{X/Z}(\mathscr{O}_X))$ (**II**, 1.3.1) is called the Z-*closure* of X. Furthermore, for every $\mathscr{O}_X$-Module $\mathscr{F}$, $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is an $\mathscr{A}$-Module, which is quasi-coherent if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module; in this last case, there is thus a unique $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that, denoting by $g : X' \to X$ the structural morphism

(5.10.11.1) $$\mathscr{H}^0_{X/Z}(\mathscr{F}) = g_*(\mathscr{F}').$$

Proposition (5.10.12). — *Let the notations be those of (5.10.11).*

label=i) *Let x be a point of X. For the morphism*
$$X' \times_X X_x \longrightarrow X_x \ (= \operatorname{Spec}(\mathscr{O}_{X,x}))$$
deduced from g by localisation to be an isomorphism, it is necessary and sufficient that $\mathscr{O}_X$ be Z-closed at the point x (which holds for every $x \in X - Z$).

lbbel=ii) *Set $Z' = g^{-1}(Z)$, and suppose X' locally Noetherian. Then $\mathscr{F}'$ is Z'-closed; if in addition $\mathscr{F}'$ is an $\mathscr{O}_{X'}$-Module coherent, we have $\operatorname{prof}_{Z'}(\mathscr{F}') \geqslant 2$.*

lcbel=iii) *Suppose that $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ and $\mathscr{H}^0_{X/Z}(\mathscr{F})$ are coherent. Then the morphism $g : X' \to X$ is finite; the set U of $x \in X$ such that $\operatorname{prof}_{Z_x}(\widetilde{\mathscr{O}}_x) \geqslant 2$ and $\operatorname{prof}_{Z_x}(\widetilde{\mathscr{F}}_x) \geqslant 2$ is open in X, and such that $X - U \subset Z$; furthermore U is the largest open of X such that the restriction $g^{-1}(U) \to U$ of g is an isomorphism and that the restriction $\mathscr{F}|U \to \mathscr{F}'|g^{-1}(U)$ of the g-morphism is an isomorphism.*

Proof. Assertion (i) follows from definitions, and (iii) is an immediate consequence of (5.10.10), (iii). To prove (ii), let us consider an open V of X containing $Z - Z$, and its inverse image $V' = g^{-1}(V)$; if $i : V \to X$ and $i' : V' \to X'$ are the canonical injections

the canonical homomorphism $\rho_{X'/Z'} : \mathscr{F}' \to i'_*(\mathscr{F}'|V')$ is such that $g_*(\rho_{X'/Z'})$ is the canonical **IV-2** | 119
homomorphism $\rho_{X/Z} : \mathscr{H}^0_{X/Z}(\mathscr{F}) \to i_*(\mathscr{H}^0_{X/Z}(\mathscr{F})|V)$ (**II**, 1.4.2), taking into account (5.10.11.1). As $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is Z-closed (5.9.11), $\rho_{X/Z}$ is an isomorphism, hence so is $\rho_{X'/Z'}$. As $X' - Z'$ is the intersection of the filtered family of opens containing $X - Z$, we deduce from this that $\mathscr{F}'$ is Z'-closed when X' is locally Noetherian, by virtue of (5.9.1). □

(5.10.13). We shall now apply the results above to the case where Z is one of the sets $Z^{(n)}(X)$ (or simply $Z^{(n)}$), defined as the set of $x \in X$ such that $\dim(\mathscr{O}_x) \geqslant n$; it is clear that $Z^{(n)}$ is stable by specialisation; for a closed part T of X to be contained in $Z^{(n)}$, it is necessary and sufficient that $\operatorname{codim}(T, X) \geqslant n$. We shall be interested here in the case $n = 2$.

Proposition (5.10.14). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent of support equal to X.*

label=i) *For $\mathscr{F}$ to satisfy (S_1), it is necessary and sufficient that it be $Z^{(1)}$-pure.*

lbbel=ii) *For $\mathscr{F}$ to satisfy (S_2), it is necessary and sufficient that it be $Z^{(2)}$-closed and $Z^{(1)}$-pure, or equivalently that it be $Z^{(2)}$-pure and not have an associated prime cycle of codimension 1.*

(i) *To say that $\mathscr{F}$ possesses the property (S_1) means that $\mathscr{F}$ has no associated immersed prime cycle (5.7.5), or equivalently that for every $x \in \operatorname{Ass}(\mathscr{F})$, $\dim(\mathscr{F}_x) = 0$ (3.1.4); in other words, $\dim(\mathscr{O}_x) = 0$ and $\dim(\mathscr{O}_x) \leqslant 1$ for every generization x' of x, and since $A_{x'}$ is integral, $\operatorname{prof}(A_{x'}) \geqslant 1$, hence X satisfies (S_2) at the point x; but this is equivalent to saying that $\operatorname{Ass}(\mathscr{F})$ does not meet $Z^{(1)}$.*

(ii) *We have $Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$, with the notations of (5.9.6), taking into account (**I**, 2.4.2); on the other hand, $\mathscr{F}$ not having associated prime cycles of codimension 1 implies the same hypothesis for $\widetilde{\mathscr{F}}_x$; for $\widetilde{\mathscr{F}}_x$ to satisfy (S_2), it suffices, by virtue of (5.10.14), that $\widetilde{\mathscr{F}}_x$ be $Z^{(2)}(X_x)$-closed; assertion (ii) follows from (5.10.6) and from (5.10.10), (iii).*

Corollary (5.10.15). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module of support X. Suppose that $\mathscr{F}$ has no associated prime cycles of codimension 1 and that $\mathscr{F}' = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent. Then:*

label=i) *$\mathscr{F}'$ satisfies the property* (S_2).

IV-2 | 120

lbbel=ii) *The set U of $x \in X$ such that $\mathscr{F}$ satisfies* (S_2) *at the point x* (5.7.2) *is open in X, and we have* $\operatorname{codim}(X - U, X) \geqslant 2$.

Proof. label=*i)* We know (5.9.11) that $\mathscr{F}'$ is $Z^{(2)}$-closed, and $\operatorname{Supp}(\mathscr{F}') = X$ in addition, since the maximal points of X belong to $X - Z^{(2)}$ and at these points $\mathscr{F}'_x = \mathscr{F}_x \neq 0$; hence the support of $\mathscr{F}'$ is closed, hence $\operatorname{Supp}(\mathscr{F}')$ is closed. It remains to see that $\mathscr{F}'$ has no associated prime cycles of codimension 1. But if $x \in \operatorname{Ass}(\mathscr{F}')$ and $\dim(\mathscr{F}'_x) = \dim(\mathscr{O}_x) = 1$, we have $x \in X - Z^{(2)}$, hence, since $\mathscr{F}'_x = \mathscr{F}_x$, we would have $x \in \operatorname{Ass}(\mathscr{F})$, contrary to the hypothesis. This achieves the proof of (i).

lbbel=*ii)* We have $Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$, with the notations of (5.9.6), taking into account (**I**, 2.4.2); on the other hand, $\mathscr{F}$ not having associated prime cycles of codimension 1 implies the same hypothesis for $\widetilde{\mathscr{F}_x}$; for $\widetilde{\mathscr{F}_x}$ to satisfy (S_2), it suffices, by virtue of (5.10.14), that $\widetilde{\mathscr{F}_x}$ be $Z^{(2)}(X_x)$-closed; assertion (ii) follows from (5.10.6) and from (5.10.10), (iii). ∎

Proposition (5.10.16). — *Let X be a locally Noetherian prescheme,*

$$X' = \operatorname{Spec}(\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{O}_X))$$

its $Z^{(2)}$-closure, $g : X' \to X$ the structural morphism. Suppose that X has no associated prime cycle of codimension 1.

label=i) *For a point $x \in X$ to satisfy* (S_2), *it is necessary and sufficient that the morphism $X'_x \to X_x$ deduced from g (notations of* (5.10.12)*) be an isomorphism. This condition is always satisfied if* $\operatorname{codim}(\overline{\{x\}}, X) \leqslant 1$.

lbbel=ii) *Suppose in addition that g is a finite morphism (see in* (5.11.2) *for sufficient conditions for this to be so). Then the set U of points where X satisfies* (S_2) *is open and* $\operatorname{codim}(X - U, X) \geqslant 2$; *furthermore U is the largest open of X such that the restriction $g^{-1}(U) \to U$ of g is an isomorphism.*

lcbel=iii) *Under the same hypotheses as in (ii), X' satisfies* (S_2), *and for every $x' \in X'$ such that* $\operatorname{codim}(\overline{\{x'\}}, X') \leqslant 1$, *the point $x = g(x')$ is such that* $\operatorname{codim}(\overline{\{x\}}, X) = \operatorname{codim}(\overline{\{x'\}}, X')$.

ldbel=iv) *Under the hypotheses of (ii), let $\mathscr{F}$ be an $\mathscr{O}_X$-Module coherent of support X, such that $\mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent; then the $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that $g_*(\mathscr{F}') = \mathscr{H}^0_{X/Z^{(2)}}(\mathscr{F})$ is coherent and satisfies* (S_2), *and its support is a union of irreducible components of X'.*

Proof. Assertions (i) and (ii) are in substance already proved, having been already demonstrated as a particular case of (5.10.12), (i) and (5.10.14), and (ii) is a particular case of (5.10.15), (ii).

Let us prove (iii); set $x = g(x')$; as g is finite, it is the same for the morphism $X'_x \to X_x$, hence $\dim(\mathscr{O}_{X',x'}) \leqslant \dim(X'_x) \leqslant \dim(X_x) = \dim(\mathscr{O}_{X,x})$ by virtue of (5.4.1). Suppose first that $\dim(\mathscr{O}_{X,x}) \leqslant 1$ and let us show that $\dim(\mathscr{O}_{X',x'}) \leqslant 1$; then by (5.10.12), (ii), applied to $\mathscr{O}_X$, we would have $\operatorname{prof}(\mathscr{O}_{X',x'}) \geqslant 2$, which is absurd, i.e. $x \in Z^{(2)}$; we deduce that $\operatorname{prof}(\mathscr{O}_{X',x'}) \geqslant 2$ by (5.10.12), (ii). Suppose now that $\dim(\mathscr{O}_{X,x}) \geqslant 2$, and let us show similarly that $\mathscr{O}_{X',x'}$ is isomorphic to $\mathscr{O}_{X,x}$ (5.10.12), (i), whence $\dim(\mathscr{O}_{X',x'}) = \dim(\mathscr{O}_{X,x})$. Furthermore, as X has no associated prime cycles of codimension 1, the hypothesis $\dim(\mathscr{O}_x) = 1$ implies $x \notin \operatorname{Ass}(\mathscr{O}_X)$, hence $\operatorname{prof}(\mathscr{O}_{X,x}) = 1$; but by hypothesis the relation $\dim(\mathscr{O}_x) \leqslant 1$ implies that $\mathscr{F}_x = \mathscr{F}'_x$; hence every irreducible component of $\operatorname{Supp}(\mathscr{F}')$ is a component of X', since $\mathscr{F}'$ is coherent, hence $\operatorname{Supp}(\mathscr{F}')$ is closed. This establishes the assertions of (iii).

To prove (iv), it suffices to replace $\mathscr{O}_X$ by $\mathscr{F}$ in the preceding reasoning, which establishes that $\mathscr{F}'$ IV-2 | 121
satisfies (S_2) and that if $\dim(\mathscr{F}'_{x'}) = 1$, $\mathscr{F}_x$ and $\mathscr{F}'_{x'}$ are di-isomorphic; in particular if $\dim(\mathscr{F}'_{x'}) = 0$, hence $\dim(\mathscr{F}_x) = 0$, hence $\dim(\mathscr{O}_{X,x}) = 0$ and $\dim(\mathscr{O}_{X',x'}) = 0$ since $\mathscr{F}$ has support X; every irreducible component of $\operatorname{Supp}(\mathscr{F}')$ is therefore a component of X', since $\mathscr{F}'$ is coherent, hence $\operatorname{Supp}(\mathscr{F}')$ is closed. ∎

Proposition (5.10.17). — *Let A be a Noetherian integral ring, and denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ runs through the set of prime ideals of A of height* 1. *Suppose that $A^{(1)}$ is a finite A-algebra. Then:*

label=i) *The ring $A^{(1)}$ satisfies the condition* (S_2).

lbbel=ii) *The set U of $\mathfrak{p} \in \operatorname{Spec}(A)$ such that the canonical homomorphism $A_{\mathfrak{p}} \to (A^{(1)})_{\mathfrak{p}}$ is bijective is equal to the set of $\mathfrak{p}$ such that $A_{\mathfrak{p}}$ satisfies (S_2); U is open in $X = \operatorname{Spec}(A)$ and we have* $\operatorname{codim}(X - U, X) \geqslant 2$.

lcbel=iii) *For every multiplicative subset S of A, $(S^{-1}A)^{(1)}$ is a finite $(S^{-1}A)$-algebra.*

ldbel=iv) *Let B be a finite integral A-algebra containing A. Then $B^{(1)}$ is a finite B-algebra. Furthermore, for every prime ideal $\mathfrak{q}$ of B of height 1, the prime ideal $\mathfrak{q} \cap A$ is of height 1 in A.*

Proof. If one takes into account formula (5.9.3.1), one sees that $X' = \operatorname{Spec}(A^{(1)})$ is the $Z^{(2)}$-closure of $X = \operatorname{Spec}(A)$; as A has no associated immersed prime ideals, properties (i) and (ii) are particular cases of (5.10.16), (i), (ii) and (iii). To prove (iii), it suffices to remark that $(S^{-1}A)^{(1)} = S^{-1}A^{(1)}$, which is a particular case of (5.9.4); as this is a flat A-module, the prime ideals associated to $S^{-1}A$ do not meet S, and we have $\operatorname{ht}(S^{-1}\mathfrak{p}) = \operatorname{ht}(\mathfrak{p})$ for $\mathfrak{p} \in \operatorname{Spec}(A)$ such that $S^{-1}\mathfrak{p}$ is defined, whence (iii).

To prove (iv), set $Y = \operatorname{Spec}(B)$, and let $f : Y \to X$ be the structural morphism; as f is finite, it follows from (5.4.1) that for every $y \in Y$, $\dim(\mathscr{O}_{X,f(y)}) \leqslant \dim(\mathscr{O}_{X,f(y)})$; hence, if $T = f^{-1}(Z^{(2)}(X))$, we have $T \supset Z^{(2)}(Y)$. Let us show that $\mathscr{G} = \mathscr{H}^0_{X/Z^{(2)}(X)}(f_*(\mathscr{O}_Y))$ is coherent; indeed, B being considered as an A-module; but B is a finite integral A-algebra, its field of fractions is finite over the field of fractions of A, hence B is contained in a free A-module of finite type, and hence by (5.9.2), (i) $((\mathscr{H}^0_{X/Z^{(2)}(X)}(\mathscr{O}_X))^n$ for a suitable n; this last being by hypothesis quasi-coherent of $(\mathscr{H}^0_{X/Z^{(2)}(X)}(\mathscr{O}_X))^n$, the result follows. Now, it follows from the definition (5.9.1.2) that $\mathscr{G}$ is isomorphic to $f_*(\mathscr{H}^0_{Y/T}(\mathscr{O}_Y))$; this proves *a fortiori* that $\mathscr{H}^0_{Y/T}(\mathscr{O}_Y)$ is a coherent $\mathscr{O}_Y$-Module. It follows then from (5.10.10), (ii), applied to $\mathscr{O}_Y$ and at the generic point of Y, that $\operatorname{codim}(T, Y) \geqslant 2$, that is to say $T \subset Z^{(2)}(Y)$, and finally $T = Z^{(2)}(Y)$. This proves the two assertions of (iv). □

5.11. Coherence criteria for the Modules $\mathscr{H}^0_{X/Z}(\mathscr{F})$

Proposition (5.11.1). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Denote by (x_α) the family of points of $\operatorname{Ass}(\mathscr{F}) \cap (X - Z)$ and, for every α, let Y_α be the reduced closed sub-prescheme of X having $\overline{\{x_\alpha\}}$ for underlying space, $Z_\alpha = Z \cap \overline{\{x_\alpha\}}$. Then the following two conditions are equivalent:*

label=a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-Module.*

lbbel=b) *For every α, $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$ is a coherent $\mathscr{O}_{Y_\alpha}$-Module.*

These two conditions imply the following:

c) *For every α, we have* $\operatorname{codim}(Z_\alpha, Y_\alpha) \geqslant 2$.

Moreover, the three conditions a), b), c) *are equivalent when in addition one of the following properties is satisfied:*

(i) *Every point of X admits a neighbourhood isomorphic to a sub-scheme of a regular scheme (in which case one also says that X is* locally immersible in a regular scheme*).*

(ii) *For every α, Y_α is universally catenary* (**0**, 5.6.2) *and its normalised* ((**II**, 6.3.8)) *Y'_α is finite over Y_α.*

All the properties under consideration are local on X, so one can reduce to the case where $X = \operatorname{Spec}(A)$ is affine, A is a Noetherian ring, and $\mathscr{F} = \widetilde{M}$, where M is a finite type A-module. Then, for every α, if h_α is the canonical injection $Y_\alpha \to X$, $(h_\alpha)_*(\mathscr{O}_{Y_\alpha}) = \mathscr{G}_\alpha$ is the $\mathscr{O}_X$-Module corresponding to the quotient A-module $A/\mathfrak{i}_{x_\alpha}$, and, by definition of $\operatorname{Ass}(\mathscr{F})$, this A-module is isomorphic to a sub-A-module of M. As $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$ is a sub-$\mathscr{O}_X$-Module quasi-coherent of $\mathscr{H}^0_{X/Z}(\mathscr{F})$ (5.9.2), *the hypothesis that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent implies that it is also coherent. On the other hand, it follows from the definition* (5.9.1.2) *that $\mathscr{H}^0_{X/Z}(\mathscr{G}_\alpha)$ is isomorphic to $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$; this proves that* a) *implies* b). *To see that* b) *implies* a), *it suffices to show that there is a finite filtration $(\mathscr{F}_i)_{0 \leqslant i \leqslant n}$ of $\mathscr{F}$ formed of coherent $\mathscr{O}_X$-Modules, with $\mathscr{F}_0 = \mathscr{F}$, $\mathscr{F}_n = 0$, and such that $\mathscr{H}^0_{X/Z}(\mathscr{F}_i/\mathscr{F}_{i+1})$ is coherent for every i: this follows, by induction on i, from the exact sequences*

$$0 \longrightarrow \mathscr{F}_{i+1} \longrightarrow \mathscr{F}_i \longrightarrow \mathscr{F}_i/\mathscr{F}_{i+1},$$

from the fact that $\mathscr{H}^0_{X/Z}$ is a left exact functor (5.9.2), *and finally from* (**0**, 5.3.3) *and* ((**I**, 6.1.1)). *By virtue of* (**I**, 3.2.8), *it thus suffices to prove that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent when $\mathscr{F}$ is* irredundant; *in other words, when $\operatorname{Ass}(M) = \{\mathfrak{p}\}$ is reduced to a single element. Let us now state the*

Lemma (5.11.1.1). — *Let A be a Noetherian ring, M a finite type A-module, such that $\mathrm{Ass}(M) = \{\mathfrak{p}\}$. There exists a finite filtration $(M_h)_{0 \leqslant h \leqslant m}$ of M such that $M_0 = M$, $M_m = 0$ and that M_h/M_{h+1} is isomorphic to a sub-module of $A/\mathfrak{p}$.*

Let us first note that the canonical homomorphism $M \to M_\mathfrak{p} = N$ is injective (Bourbaki, Alg. comm., *chap. IV, § 1, no 2, prop. 6). Set $B = A_\mathfrak{p}$, $\mathfrak{m} = \mathfrak{p}A_\mathfrak{p}$, maximal ideal of B; on has* $\mathrm{Ass}(N) = \{\mathfrak{m}\}$ *(loc. cit., prop. 5), and as N is a finite type B-module, there exists an integer r such that $\mathfrak{m}^r N = 0$; if one sets $N'_j = \mathfrak{m}^j N$ for $0 \leqslant j \leqslant r$, N'_j/N'_{j+1} is a $(B/\mathfrak{m})$-module of finite type, hence a direct sum of a finite number of sub-modules isomorphic to $B/\mathfrak{m}$, since $B/\mathfrak{m}$ is a field; in other words, there is a finite filtration $(N_h)_{0 \leqslant h \leqslant m}$ of N such that $N_m = 0$ and that N_h/N_{h+1} is isomorphic to $B/\mathfrak{m}$, i.e. to the field of fractions of $A/\mathfrak{p}$; the filtration of the $M_h = M \cap N_h$ satisfies the requirements, for M_h/M_{h+1} is isomorphic to a sub-$(A/\mathfrak{p})$-module of finite type of $N_h/N_{h+1} = B/\mathfrak{m}$; but it is known that such a sub-module is isomorphic to a sub-module of $A/\mathfrak{p}$.* IV-2 | 122 IV-2 | 123

The existence of the filtration (M_h) shows, by the same reasoning as above, that to prove (averaging b)*) that $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent, one can reduce to showing that $\mathscr{H}^0_{X/Z}(\mathscr{P})$ is coherent, where $\mathscr{P} = (A/\mathfrak{p})^\sim$, $\mathfrak{p}$ being an ideal associated to M. But if $\mathfrak{p} = \mathfrak{i}_y$ with $y \in Z$, the support of $\mathscr{P}$ is a closed set contained in Z, since Z is stable by specialisation; the definition (5.9.1.2) shows then that $\mathscr{H}^0_{X/Z}(\mathscr{P}) = 0$. If on the contrary $\mathfrak{p} = \mathfrak{i}_{x_\alpha}$ for some α, on has $\mathscr{P} = (h_\alpha)_*(\mathscr{O}_{Y_\alpha})$ by definition, and one has seen above that $\mathscr{H}^0_{X/Z}(\mathscr{P})$ is isomorphic to $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$, hence is coherent by virtue of the hypothesis* b).

We have already seen (5.10.10, , (ii)) that a) *implies* c)*. It remains to prove that, under one or the other of hypotheses (i), (ii),* c) *implies* b)*. Let us note that if X satisfies (i), it is the same for each Y_α. It therefore suffices (5.9.3.1) to prove the following corollary.*

Corollary (5.11.2). — *Let A be a Noetherian integral ring, satisfying one of the two hypotheses:*

(i) *A is universally catenary, and its integral closure A' is a finite A-algebra.*
(ii) *A is a quotient of a Noetherian regular ring.*

Then the ring $A^{(1)}$, intersection of the $A_\mathfrak{p}$, where $\mathfrak{p}$ runs through the set of prime ideals of A of height 1, is a finite A-algebra.

(i) *Set $X = \mathrm{Spec}(A)$. The set U of points $x \in X$ where X satisfies (S_2) is open in X satisfying hypothesis (i) (6.11.2). Furthermore, if one sets $Z = X - U$, we have $\mathrm{codim}(Z, X) \geqslant 2$: indeed, for every $x \in U$, we have $\dim(A_x) \leqslant 1$, whence $\dim(\mathscr{F}_x) = 0$, hence $\dim(\mathscr{O}_{X,x}) = 0$ and $\dim(\mathscr{O}_{X,x}) \leqslant 1$ for every generization x' of x, and as $A_{x'}$ is integral, $\mathrm{prof}(A_{x'}) \geqslant 1$, hence X satisfies (S_2) at the point x. We have therefore $Z \subset Z^{(2)}$, and $\mathscr{H}^0_{X/Z(n)}(\mathscr{O}_X) = j_*(\mathscr{H}^0_{U/Z(n)}(\mathscr{O}_U))$, where $j : U \to X$ is the canonical injection (5.9.1.2). As the prescheme U satisfies (S_2), it follows from (5.10.14) that $\mathscr{H}^0_{U/Z(n)}(\mathscr{O}_U)$ is isomorphic to $\mathscr{O}_U$; on the other hand, as $\mathrm{codim}(Z, X) \geqslant 2$, that $j_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_X$-Module, which proves the proposition in case (i).*

(ii) *The ring A' is, by virtue of hypothesis (ii), a Noetherian integral ring and integrally closed, hence (Bourbaki,* Alg. comm., *chap. VII, § 1, no 6, th. 4) intersection of its local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ runs through the set of prime ideals of height 1 of A'. Now, for such a prime ideal $\mathfrak{p}'$, if one sets $\mathfrak{p} = \mathfrak{p}' \cap A$ and $S = A - \mathfrak{p}$, $A'_{\mathfrak{p}'}$ is a local ring, a localisation $S^{-1}\mathfrak{p}'$ of $S^{-1}A'$, and $S^{-1}A'$ is by hypothesis*

a finite $A_\mathfrak{p}$-algebra; since $S^{-1}\mathfrak{p}'$ is above the maximal ideal $\mathfrak{p}A_\mathfrak{p}$ of $A_\mathfrak{p}$, this is a maximal ideal of $S^{-1}A'$; but by virtue of the hypothesis (ii) and of (5.6.3, , (i)), $A_\mathfrak{p}$ is universally catenary. One deduces therefore from (5.6.10) that $\dim(A_\mathfrak{p}) = \dim(A'_{\mathfrak{p}'}) = 1$. It follows from this that one has $A^{(1)} \subset A'$, and as A' is an A-module of finite type by hypothesis, the same holds for $A^{(1)}$ since A is Noetherian; the proposition is therefore proved in case (ii). IV-2 | 124

Remark (5.11.3). — The fact that $A^{(1)}$ is a finite A-algebra is no longer necessarily true if, in hypothesis (ii) of (5.11.2), one assumes only that A is *catenary*. An example is given by the catenary local ring A constructed in (5.6.11), whose integral closure A' (denoted B in (5.6.11)) is a finite A-algebra; if $A^{(1)}$ were also a finite A-algebra, it would be contained in the field of fractions of A, and hence also contained in A'. But on the other hand, with the notations of (5.6.11), every prime ideal of height 1 in A is of the form $\mathfrak{p}A$, where $\mathfrak{p}$ is a prime ideal of height 1 in C, and one has $A_{\mathfrak{p}A} = C_\mathfrak{p}$; on says (5.6.11) that $C_\mathfrak{p} = E_{\mathfrak{p}'}$, where $\mathfrak{p}'$ is the unique prime ideal of E above $\mathfrak{p}$; by definition, $A_{\mathfrak{p}A}$ is integrally closed and contains A'; one would therefore have $A' \subset A^{(1)}$, and the hypothesis $A^{(1)}$ is a finite A-algebra would finally imply $A^{(1)} = A'$. But this conclusion is absurd, for one of the two prime ideals of A' above the maximal ideal $\mathfrak{n}A$ of A is of height 1, whereas $\mathfrak{n}A$ is of height 2, which would contradict (5.10.17, , (iv)).

Corollary (5.11.4). — *Let X be a locally Noetherian prescheme, Z a closed part of X, $U = X - Z$, $i : U \to X$ the canonical injection, $\mathscr{F}$ a coherent $\mathscr{O}_U$-Module. For $i_*(\mathscr{F})$ to be a coherent $\mathscr{O}_X$-Module, it is necessary that for every $x \in \operatorname{Ass}(\mathscr{F})$, we have $\operatorname{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$. This condition is sufficient in each of the two following cases:*

(i) *The prescheme X is locally immersible in a regular scheme.*
(ii) *The prescheme X is universally catenary and universally Japanese (which means, by definition, that every point $x \in X$ admits an open affine neighbourhood whose ring is universally Japanese ((**0**, 23.1.1))).*

One knows ((**I**, 9.4.7)) that there exists a coherent sub-$\mathscr{O}_X$-Module $\mathscr{G}$ of $i_*(\mathscr{F})$ such that $\mathscr{G}|U = \mathscr{F}$. One evidently has $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{G}) \subset \operatorname{Ass}(i_*(\mathscr{F}))$, and as

$$\operatorname{Ass}(i_*(\mathscr{F})) = \operatorname{Ass}(\mathscr{F})$$

((3.1.13)), one has $\operatorname{Ass}(\mathscr{G}) = \operatorname{Ass}(\mathscr{F})$; it suffices then to apply (5.11.1) to the coherent $\mathscr{O}_X$-Module $\mathscr{G}$ and, when X is universally catenary and universally Japanese, hypothesis (ii) of (5.11.1) is satisfied by definition.

Corollary (5.11.5). — *Let X be a locally Noetherian prescheme, Z a part of X stable by specialisation, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module such that $\operatorname{Ass}(\mathscr{F}) \subset X - Z$. Then the condition*

a) *$\mathscr{H}^0_{X/Z}(\mathscr{F})$ is a coherent $\mathscr{O}_X$-Module*

implies the following:

d) *For every closed part T of Z in X (or only for a* filtering family *of closed parts T of Z, of union Z), $i_*(\mathscr{F}|X - T)$ (where $i : X - T \to X$ is the canonical injection) is coherent.* IV-2 | 125

When X satisfies one of the hypotheses (i), (ii) *of* (5.11.1, , a)) *and* d) *are equivalent. Let us note that one has $\operatorname{Ass}(i_*(\mathscr{F}|X - T)) = \operatorname{Ass}(\mathscr{F})$ by virtue of the hypothesis and of* ((3.1.13))*; it follows therefore from* (5.10.2) *that the canonical maps*

$$i_*(\mathscr{F}|X - T) \longrightarrow \mathscr{H}^0_{X/Z}(\mathscr{F})$$

are injective; the fact that a) *implies* d) *is therefore a consequence of this remark. Conversely, condition* d) *implies, by virtue of* (5.11.1) *that* $\operatorname{codim}(T \cap Y_\alpha, Y_\alpha) \geqslant 2$ *with the notations of* (5.11.1)*; consequently, we have* $\operatorname{codim}(Z \cap Y_\alpha, Y_\alpha) \geqslant 2$ *since Z is the union of its parts that are closed in X, and the last assertion of the corollary follows from* (5.11.1).

Corollary (5.11.6). — *Let A be a Noetherian ring, $X = \operatorname{Spec}(A)$. Consider the following properties:*

a) *For every integral quotient ring B of A, the ring $B^{(1)}$ (notation of* (5.11.2)*) is a finite B-algebra.*
b) *For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$ and every part Z of X, stable by specialisation, and such that for every $x \in \operatorname{Ass}(\mathscr{F}) \cap (X - Z)$ we have $\operatorname{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$, the $\mathscr{O}_X$-Module $\mathscr{H}^0_{X/Z}(\mathscr{F})$ is coherent.*
c) *For every closed part T of X and every coherent $\mathscr{O}_U$-Module $\mathscr{G}$ (where $U = X - T$) such that for every $x \in \operatorname{Ass}(\mathscr{G})$ we have $\operatorname{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geqslant 2$, $i_*(\mathscr{G})$ (where $i : U \to X$ is the canonical injection) is a coherent $\mathscr{O}_X$-Module.*
d) *For every integral quotient ring B of A and every ideal $\mathfrak{J}$ of height $\geqslant 2$ in B, the ring $\bigcap_{\mathfrak{p} \neq \mathfrak{J}} B_{\mathfrak{p}}$ is a finite B-algebra.*

One then has the implications

$$a) \Leftrightarrow b) \Rightarrow c) \Leftrightarrow d)$$

Furthermore, conditions a), b), c) *and* d) *are satisfied in each of the two following cases:*

(i) *A is a quotient of a regular ring.*
(ii) *A is universally catenary and universally Japanese.*

Let $B = A/\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of A, so that $Y = \mathrm{Spec}(B)$ is a closed part of X stable by specialisation; since B is integral, $\mathrm{Ass}(\mathscr{O}_{Y_\alpha})$ is reduced to the generic point $\mathfrak{q}$ of Y. If condition b) *is satisfied, one can apply it to the $\mathscr{O}_X$-Module coherent $\mathscr{F} = (A/\mathfrak{q})^\sim$ and to $Z = V(\mathfrak{I})$; this proves that $B^{(1)}$ is an A-module of finite type, and* a fortiori *a B-module of finite type. Conversely, suppose* a)*; then, if $\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module such that for every $x \in \mathrm{Ass}(\mathscr{F}) \cap (X - Z)$ we have $\mathrm{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$, one can apply (with the notations of (5.11.1)) to each of the affine schemes $Y_\alpha = \mathrm{Spec}(B_\alpha)$, where B_α is an integral quotient ring of A, the result of condition* a)*; since by hypothesis Z_α is contained in $Z^{(2)}(Y_\alpha)$, condition* a) *(taking into account (5.10.2))*

implies that $\mathscr{H}^0_{Y_\alpha/Z_\alpha}(\mathscr{O}_{Y_\alpha})$ is a coherent $\mathscr{O}_{Y_\alpha}$-Module, and b) *follows then from (5.11.1).* IV-2 | 126

To see that c) *implies* d)*, one reasons as above by applying* c) *to the case where $\mathscr{F} = (A/\mathfrak{q})^\sim$ and $Z = V(\mathfrak{I})$; conversely, one proves that* d) *implies* c) *by using the equivalence of* a) *and* b) *in (5.11.1). It is evident that* c) *is a particular case of* b)*. Finally, the fact that* a) *(and hence each of the other conditions) is satisfied when A satisfies one of the hypotheses (i), (ii) follows from (5.11.2), in view of the definition of universally catenary rings and of universally Japanese rings.*

Remarks (5.11.7). — (i) If one ignores, in (5.11.5), condition *d)*, then *a)* without any supplementary hypothesis on X; we will see further on (7.2.4) that it is indeed so when X is a *local* scheme. Similarly, we will prove that when A is a Noetherian *local* ring, the four properties *a)*, *b)*, *c)*, *d)* of (5.11.6) are equivalent (7.2.4). If one ignores whether this result extends to all Noetherian rings.

(ii) If A satisfies property *a)* of (5.11.6), the same holds for every *ring of fractions* $S^{-1}A$ and for every finite A-algebra C. Indeed, every integral quotient ring of $S^{-1}A$ is of the form $S^{-1}(A/\mathfrak{q})$, where $\mathfrak{q}$ is a prime ideal of A not meeting S; and on the other hand, if $\mathfrak{r}$ is a prime ideal of C, $\mathfrak{p}$ its inverse image in A, $C/\mathfrak{r}$ is an $(A/\mathfrak{p})$-algebra that is integral and finite containing $A/\mathfrak{p}$. Our assertions are therefore consequences of (5.10.17, , (iii) and (iv)).

5.12. Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA IV-2 | 126

We have already seen in (3.4) relations between the properties of A and of A/tA concerning the associated prime ideals, as well as the properties of being integral or reduced. One gives in this section other relations between the properties of these rings, linked to the notions of dimension and of depth.

Proposition (5.12.1). — *Let A be a Noetherian local ring, $X = \mathrm{Spec}(A)$, t an element of A forming part of a system of parameters ((**0**, 16.3.6)), X_0 the closed sub-space $V(t)$ of X, X_1 the open complement $X - X_0$. Let Z be a part of X such that every specialisation of a point of Z belongs to Z. Suppose that X is* biequidimensional *(which will be the case if A is equidimensional and a quotient of a regular local ring). Then, if one sets $Z_0 = Z \cap X_0$, $Z_1 = Z \cap X_1$, we have*

(5.12.1.1) $$\mathrm{codim}(Z_0, X_0) \leqslant \mathrm{codim}(Z, X) \leqslant \mathrm{codim}(Z_1, X_1).$$

The second inequality follows from the definition (5.1.3); to prove the first, one can reduce to proving it when Z is *closed*: indeed, by hypothesis Z is a union of closed parts Z_α, and if one has $\mathrm{codim}(X_0 \cap Z_\alpha, X_0) \leqslant \mathrm{codim}(Z_\alpha, X)$ for all α, one also has $\mathrm{codim}(Z_0, X_0) = \inf_\alpha(\mathrm{codim}(X_0 \cap Z_\alpha, X_0)) \leqslant \inf_\alpha(\mathrm{codim}(Z_\alpha, X)) = \mathrm{codim}(Z, X)$. Suppose therefore Z closed, so that one has

$$\mathrm{codim}(Z, X) = \dim(X) - \dim(Z)$$

by virtue of ((**0**, 14.3.5.1)). On the other hand, X_0 is evidently catenary, equidimensional and of dimension $\dim(X) - 1$ by virtue of ((**0**, 16.3.4)); one therefore also has IV-2 | 127

$$\mathrm{codim}(Z_0, X_0) = \dim(X_0) - \dim(Z_0) = \dim(X) - 1 - \dim(Z_0)$$

But one has $\dim(Z_0) \geqslant \dim(Z) - 1$ ((**0**, 16.3.4)), which completes the proof of the first inequality (5.12.1.1).

Proposition (5.12.2). — *Let A be a Noetherian local ring, M a finite type A-module, t an M-regular element belonging to the maximal ideal $\mathfrak{m}$ of A, k an integer $\geqslant 1$. Suppose that M/tM is equidimensional and satisfies* (S_k)*, it is of the same for M.*

Taking into account (**Err$_{\mathbf{III}}$**, 30) and (5.7.3, , (vi)), one can reduce to the case where $\mathrm{Supp}(M) = \mathrm{Spec}(A) = X$. Set $X_0 = V(t) = \mathrm{Spec}(A/tA)$; as M/tM satisfies (S_1) by hypothesis, it has no associated immersed prime ideals (3.4.4). Applying (3.4.4) to a closed point of X (and to every closed sub-prescheme of X), one sees that the irreducible components of $\operatorname{Supp}(M/tM)$ are exactly those of $Y_i \cap X_0$, where Y_i ($1 \leqslant i \leqslant r$) are the irreducible components of X. As t is by hypothesis M-regular, $V(t)$ does not contain any maximal point of X (3.1.8); each of the irreducible components of $Y_i \cap X_0$ is therefore of codimension 1 in Y_i (5.1.8); as X is catenary and that all the irreducible components of $\operatorname{Supp}(M/tM)$ have by hypothesis the same dimension, one concludes that the Y_i all have the same dimension, in other words X is equidimensional, hence *biequidimensional* since A is local and catenary. To see that M satisfies (S_k), let us apply the criterion (5.7.4): one has to verify (with the notations of (5.7.4)) that, for every integer $n \geqslant 0$, we have $\operatorname{codim}(Z_n, X) > n+k$. Now, the hypothesis on M/tM and the criterion (5.7.4) show that $\operatorname{codim}(Z_n \cap X_0, X_0) > n+k$ for every integer $n \geqslant 0$. But every specialisation of a point of Z_n still belongs to Z_n, as we shall see in (6.11.5); as X is biequidimensional and that t belongs to a system of parameters, hence also for A (the annihilator of M being nilpotent by virtue of the hypothesis $\operatorname{Supp}(M) = \operatorname{Spec}(A)$), the conclusion follows from (5.12.1).

Remark (5.12.3). — If, in the statement of (5.12.2), one assumes only that M/tM is equidimensional, it does not follow that M is equidimensional. For example, let k be a field, B the ring of polynomials $k[T,U,V]$ with 3 indeterminates, C the local ring of B at the maximal ideal $BT+BU+BV$ of B, and let $\mathfrak{p} = CU$, $\mathfrak{q} = CV + C(T+U)$; consider then the local ring $A = C/\mathfrak{p}\mathfrak{q}$ (geometrically, if X is the closed sub-prescheme of affine space with 3 dimensions over k, formed of a plane and of a line meeting the plane at a point x, A is the local ring of X at the point x). Let t, u, v be the canonical images of T, U, V in A; it is immediate that t is not a zero divisor in A, and A/tA is isomorphic to $C_0/\mathfrak{p}_0\mathfrak{q}_0$, where C_0 is
the local ring of $B_0 = k[U,V]$ at the maximal ideal $B_0U + B_0V$, and $\mathfrak{p}_0 = C_0U$, $\mathfrak{q}_0 = C_0V$ ($\mathfrak{q}_0$ being **IV-2** | 128
the ideal of C_0). It is immediate that $\operatorname{Spec}(A/tA)$ is irreducible and of dimension 1, A/tA does not satisfy (S_1) and $\operatorname{Spec}(A)$ is not equidimensional.

Corollary (5.12.4). — *Let A be a catenary Noetherian local ring, t a regular element belonging to the maximal ideal of A, k an integer $\geqslant 1$. If A/tA satisfies (S_k), it is of the same for A.*

If $k=1$, the corollary follows from (3.4.4), in view of the interpretation of condition (S_1) (5.7.5). If $k \geqslant 2$, it follows from the Hartshorne criterion (5.10.9) that A/tA is equidimensional, and one can apply (5.12.2).

Proposition (5.12.5). — *Let A be a catenary Noetherian local ring, t a regular element of A belonging to the maximal ideal of A, k an integer $\geqslant 0$. If the ring A/tA is reduced, equidimensional and satisfies (R_k), the ring A also possesses these three properties.*

One already knows (3.4.6) that A is reduced, and it follows from (5.12.2), applied for $k=1$, that A is equidimensional. Set $X = \operatorname{Spec}(A)$, $X_0 = V(t) = \operatorname{Spec}(A/tA)$, and let Z be the set of points $x \in X$ where X is not regular; by virtue of $((\mathbf{0},\ 17.3.2))$ every specialisation of a point of Z belongs to Z. It follows on the other hand from $((\mathbf{0},\ 17.1.8))$ that for every point $x \in X_0$ where X is regular, X is also regular, hence the set Z_0' of the points where X_0 is not regular contains $Z \cap X_0$; the hypothesis implies therefore that

$$k \leqslant \operatorname{codim}(Z_0', X_0) \leqslant \operatorname{codim}(Z_0, X_0).$$

But, since A is equidimensional and catenary, one has, by (5.12.1)

$$\operatorname{codim}(Z_0, X_0) \leqslant \operatorname{codim}(Z, X)$$

which proves that X satisfies (R_k).

Remark (5.12.6). — If, in the statement of (5.12.5), one assumes only that A/tA is reduced and possesses property (R_k), it does not follow necessarily that A possesses property (R_k). For example, let k be a field, $P(U,V)$ a polynomial of $k[U,V]$ irreducible such that the curve Γ defined by the principal ideal (P) in the affine plane $\operatorname{Spec}(k[U,V])$ has a singular point at the maximal ideal $(U)+(V)$ (for example $P(U,V) = U(U^2+V^2) + (U^2-V^2)$, giving a cubic with a double point). Let then B be the ring of polynomials with 4 indeterminates $k[T,U,V,W]$, C the local ring of B at the maximal ideal $BT+BU+BV+BW$, and let

$$\mathfrak{p} = CW + CP(U-T,V), \quad \mathfrak{q} = CU.$$

Consider then the local ring $A = C/\mathfrak{p}\mathfrak{q}$ (geometrically, if X is the closed sub-prescheme of affine space with 4 dimensions, formed of a hyperplane H and of a "cylinder" L with 2 dimensions having for "base" the curve Γ and whose "singular line" Y is not contained in the hyperplane H, then A is the local ring of X at the point x where Y meets H). One then sees immediately that $\operatorname{Spec}(A/tA)$ (where t is the image of T in A) has for sole singular point x, which is reduced and of dimension 2; on the contrary, the generic point y of the "singular line" Y (defined by the image in A of the ideal $C(U-T)+CV+CW$) is a singular point of X, and $\mathscr{O}_{X,y}$ is of dimension 1; in other words, A/tA is reduced and satisfies (R_1), but A does not satisfy (R_1).

Corollary (5.12.7). — *Let A be a catenary Noetherian local ring, t a regular element of A belonging to the maximal ideal of A. If A/tA is integral and integrally closed, it is of the same for A.*

By virtue of the Serre criterion (5.8.6) and the fact that the ring A is local, it suffices to show that $\operatorname{Spec}(A)$ satisfies (S_2) and (R_1). But the hypothesis on A/tA implies that A satisfies (R_1) by (5.12.5), and that A satisfies (S_2) by (5.12.4).

Proposition (5.12.8). — (Hironaka's lemma). *Let A be a Noetherian local ring, reduced, equidimensional and catenary. Suppose moreover that for every minimal prime ideal $\mathfrak{q}_i$ of A, the ring $B_i = A/\mathfrak{q}_i$ is such that $B_i^{(1)}$ (notation of (5.10.17)) is a finite B_i-algebra (which will be the case for example when A possesses one of properties* (i), (ii) *of* (5.11.6)). *Let t be an element of A, not a zero divisor in A, and such that:*

- (i) *The A-module A/tA has only one associated non-immersed prime ideal $\mathfrak{p}$ (necessarily of height* 1 *by virtue of the Hauptidealsatz* ((**0**, 16.3.2))*, but one does not assume that this is the sole prime ideal associated to A/tA).*
- (ii) *The ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$ is generated by $t/1$.*
- (iii) *The ring $A/\mathfrak{p}$ is integrally closed.*

Under these conditions, the ring A is integral and integrally closed, and one has $\mathfrak{p} = tA$.

If $Z = V(tA)$, hypothesis (i) implies that Z is an irreducible closed part of $X = \operatorname{Spec}(A)$, whose generic point z is such that $\mathfrak{i}_z = \mathfrak{p}$. Hypothesis (ii) shows that the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of the Noetherian local ring $A_{\mathfrak{p}}$ is generated by a single element, hence $A_{\mathfrak{p}}$ is a *discrete valuation ring* whose $t/1$ is a uniformiser (Bourbaki, *Alg. comm.*, chap. VI, § 3, no 6, prop. 9); $A_{\mathfrak{p}}/tA_{\mathfrak{p}}$ is the residue field of $A_{\mathfrak{p}}$, hence an $A_{\mathfrak{p}}$-module of length 1. By virtue of (3.4.2), this implies that Z is contained in only *one* irreducible component X_j of X, for every other irreducible component X_j of X, one has therefore $\dim(X_j \cap Z) < \dim(Z)$. On the other hand, as t is not a zero divisor in A, it is not contained in any of the $\mathfrak{q}_i$, and one has therefore (5.1.8) $\operatorname{codim}(X_i \cap Z, X_i) = \operatorname{codim}(X_j \cap Z, X_j) = 1$, for all $j \neq i$. As A is supposed biequidimensional, the relation $\dim(X_i) \neq \dim(X_i \cap Z) + \dim(X_j \cap Z)$ would imply $\dim(X_i) \neq \dim(X_j)$ ((**0**, 14.3.3.1)), which is absurd. One sees therefore that there is only *one minimal prime ideal* in A; as A is by hypothesis reduced, this implies that it is *integral*. Let us note now that since $A^{(1)}$ is a finite A-algebra, it is a semi-local Noetherian ring, and every non-immersed prime ideal associated to $A^{(1)}/tA^{(1)}$ is therefore of height 1 by virtue of the Hauptidealsatz ((**0**, 16.3.2)); or, for such an ideal $\mathfrak{r}$, $\mathfrak{r} \cap A$ is of height 1 by virtue of (5.10.17, , (iv)) and as it contains tA, this can only be $\mathfrak{p}$ by virtue of hypothesis (i). On the other hand, $A^{(1)}$ is contained in the integral closure A' of $A_{\mathfrak{p}}$; if one sets $S = A - \mathfrak{p}$, the integral closure of $A_{\mathfrak{p}}$ is $S^{-1}A'$ (Bourbaki, *Alg. comm.*, chap. V, § 1, no 5, prop. 17), hence $S^{-1}A' = A_{\mathfrak{p}}$ since $A_{\mathfrak{p}}$ is a discrete valuation ring; it follows (*loc. cit.*, § 2, no 1, lemma 1) that there exists only *one* prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$, and *a fortiori* only one prime $\mathfrak{p}^{(1)}$ of $A^{(1)}$ above $\mathfrak{p}$, and one has $A_{\mathfrak{p}} = A'_{\mathfrak{p}'} = A^{(1)}_{\mathfrak{p}^{(1)}}$. Let us note now that $A^{(1)}$ satisfies (S_2) (5.10.17, , (i)) and as t is not a zero divisor in $A^{(1)}$, $A^{(1)}/tA^{(1)}$ has no associated immersed prime ideals IV-2 | 129

(5.7.5). The ideal $\mathfrak{p}^{(1)}$ is therefore the sole prime ideal associated to $A^{(1)}/tA^{(1)}$, in other words, $tA^{(1)}$ is a $\mathfrak{p}^{(1)}$-primary ideal of $A^{(1)}$; this means also that the inverse image of $A^{(1)}$ of $tA^{(1)}_{\mathfrak{p}^{(1)}} = tA_{\mathfrak{p}}$, and this last is the maximal ideal of $A_{\mathfrak{p}}$ by (ii); hence $tA^{(1)} = \mathfrak{p}^{(1)}$ is prime in $A^{(1)}$. On the other hand, $A^{(1)}/\mathfrak{p}^{(1)}$ is finite over $A/\mathfrak{p}$ and has the same field of fractions (the residue field of $A^{(1)}_{\mathfrak{p}^{(1)}} = A_{\mathfrak{p}}$), hence it is identical to $A/\mathfrak{p}$ by virtue of hypothesis (iii). One can therefore write $A^{(1)} = A + \mathfrak{p}^{(1)} = A + tA^{(1)}$, and as t is contained in the radical of the semi-local Noetherian ring $A^{(1)}$, the Nakayama lemma implies $A^{(1)} = A$, whence $\mathfrak{p}^{(1)} = \mathfrak{p} = tA$. But A is catenary and A/tA IV-2 | 130 is integral and integrally closed by virtue of hypothesis (iii), one concludes by (5.12.7) that A is integrally closed.

Remark (5.12.9). — The conclusion of (5.12.8) fails if one does not further assume A equidimensional. For example, let k be a field, B the ring of polynomials $k[X, Y, Z]$ with 3 indeterminates, C the ring $B/\mathfrak{r}_1\mathfrak{r}_2$, where $\mathfrak{r}_1 = BZ$ and $\mathfrak{r}_2 = BX + BY$; let $\mathfrak{m}$ be the maximal ideal $BX + BY + BZ$, $\mathfrak{n} = \mathfrak{m}/\mathfrak{r}_1\mathfrak{r}_2$ its image in C, A the local ring $C_\mathfrak{n}$. Finally, let t_0 be the image in C of the element $Y + Z$ of B, t its image in A ($\operatorname{Spec}(C)$ is therefore formed of a plane P and of a line D not contained in this plane, $\operatorname{Spec}(C/t_0C)$ is formed of a line D' contained in P and passing through the point $D \cap P$). The ring A is reduced and catenary (being a quotient of a regular ring) and its minimal prime ideals $\mathfrak{q}_1$, $\mathfrak{q}_2$ are the images of $\mathfrak{r}_1$, $\mathfrak{r}_2$. The A-module A/tA has only one associated non-immersed prime ideal $\mathfrak{p}$, image of $BY + BZ$, $\mathfrak{p}A_\mathfrak{p}$ is generated by $t/1$ and $A/\mathfrak{p}$ is isomorphic to $k[X]$; but A is not integral.

Corollary (5.12.10). — *Let A be an integral Noetherian local ring, satisfying one of the following hypotheses:*

a) *A is a quotient of a regular ring.*
b) *A is universally catenary and universally Japanese.*

Let $(x_i)_{1\leqslant i\leqslant n}$ be a sequence of n elements of A; set $\mathfrak{I} = x_1A + \ldots + x_nA$, and suppose the following conditions are satisfied:

(i) *There exists only one non-immersed prime ideal $\mathfrak{p}$ associated to $A/\mathfrak{I}$ and $\mathfrak{p}$ is of height n.*
(ii) *The maximal ideal $\mathfrak{p}A_\mathfrak{p}$ of $A_\mathfrak{p}$ is generated by the $x_i/1$ (hence $A_\mathfrak{p}$ is a regular local ring of dimension n).*
(iii) *The ring $A/\mathfrak{p}$ is integrally closed.*

Under these conditions, for every integer i such that $0 \leqslant i \leqslant n$, the quotient ring $A/(x_1A + \ldots + x_iA)$ is integrally closed and of dimension $\dim(A) - i$. In particular, $\mathfrak{I}$ is prime, equal to $\mathfrak{p}$, A is integrally closed and $(x_i)_{1\leqslant i\leqslant n}$ is a regular A-sequence.

We proceed by induction on n, the proposition being trivial for $n = 0$ and we can therefore suppose $n \geqslant 2$. Let $\mathfrak{I}' = x_1A + \ldots + x_{n-1}A$, and let $\mathfrak{q}$ be a prime ideal minimal among those that contain $\mathfrak{I}'$; on has $\operatorname{ht}(\mathfrak{q}) \leqslant n - 1$ ((**0**, 16.3.1)); let us show that one has $\mathfrak{q} \subset \mathfrak{p}$. Indeed, if $\mathfrak{p}'$ is minimal among the prime ideals of A containing $\mathfrak{q} + x_nA$, $\mathfrak{p}'/\mathfrak{q}$ is of height 1 in $A/\mathfrak{q}$; therefore $\mathfrak{p}'$ is of height $\leqslant n$ since A is catenary ((**0**, 16.3.2)); but as it contains $\mathfrak{I}$, whence $\mathfrak{p}' = \mathfrak{I}' + x_nA$, and hence $\mathfrak{q} \subset \mathfrak{p}$. But by virtue of hypothesis (ii) and of ((**0**, 17.1.7)), $\mathfrak{I}'A_\mathfrak{p}$ is prime in $A_\mathfrak{p}$ and it is of height $n - 1$ and $\mathfrak{q}$ of height $\leqslant n - 1$; let us show that $\mathfrak{q} \cap \mathfrak{I}'A_\mathfrak{p}$ is the *unique* non-immersed prime ideal associated to $A/\mathfrak{I}'$ and it is of height $n - 1$, since $\mathfrak{I}'A_\mathfrak{p}$ is of height $n - 1$ and $\mathfrak{q}A_\mathfrak{p} = \mathfrak{I}'A_\mathfrak{p}$ and as $A_\mathfrak{q} = (A_\mathfrak{p})_{\mathfrak{q}A_\mathfrak{p}}$, the maximal ideal $\mathfrak{q}A_\mathfrak{q}$ is generated by $\mathfrak{I}'$. One sees therefore that the sequence $(x_i)_{1\leqslant i\leqslant n-1}$ already satisfies conditions (i) and (ii) of the statement. Let us show also that it satisfies (iii). Consider for this the integral ring $B = A/\mathfrak{q}$, and let t be the image of x_n in B. It is immediate (cf. (5.6.3, , (iv))) that if A satisfies hypothesis *a)* (resp. *b)*) of (5.12.10), let us show that B and t satisfy the hypotheses of Hironaka's lemma. Indeed, an ideal $\mathfrak{n}$ of B minimal among those containing $\mathfrak{q} + x_nA$, and one has seen that this unique ideal; since $B_\mathfrak{n} = A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$ and $\mathfrak{n}B_\mathfrak{n} = \mathfrak{p}A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$, the fact that $\mathfrak{q}A_\mathfrak{p}$ is generated by $\mathfrak{I}'$ implies that $\mathfrak{n}B_\mathfrak{n}$ is generated by t. Finally, $B/\mathfrak{n} = A/\mathfrak{p}$ is integrally closed. The application of (5.12.8) proves therefore that $B = A/\mathfrak{q}$ is integrally closed, and that $\mathfrak{p} = \mathfrak{q} + x_nA$. Let us now use the induction hypothesis. Considering for this the integral ring $B = A/\mathfrak{q}$, and let t be the canonical image of x_n in B. It is immediate (cf. (5.6.3, , (iv))) that if A satisfies hypothesis *a)*, then B and t satisfy the hypotheses of the lemma of Hironaka. In effect, a prime ideal $\mathfrak{n}$ of B minimal among the prime ideals of A containing $\mathfrak{q} + x_nA$, is induced on A by a prime ideal minimal among those containing $\mathfrak{q} + x_nA$; and one has shown that the only such ideal is $\mathfrak{p}$; since $B_\mathfrak{n} = A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$ and $\mathfrak{n}B_\mathfrak{n} = \mathfrak{p}A_\mathfrak{p}/\mathfrak{q}A_\mathfrak{p}$, the fact that $\mathfrak{q}A_\mathfrak{p}$ is generated by $\mathfrak{I}'$ implies that $\mathfrak{n}B_\mathfrak{n}$ is generated by t. Finally, $B/\mathfrak{n} = A/\mathfrak{p}$ is integrally closed. The application of (5.12.8) proves therefore that $B = A/\mathfrak{q}$ is integrally closed, and that $\mathfrak{p} = \mathfrak{q} + x_nA$. Let us use now the induction hypothesis, which shows that $A/(x_1A + \ldots + x_iA)$ is integrally closed and of dimension $\dim(A) - i$ for $0 \leqslant i \leqslant n - 1$ and that $\mathfrak{q} = \mathfrak{I}'$, whence $\mathfrak{p} = \mathfrak{I}' + x_nA = \mathfrak{I}$. One concludes that $A/\mathfrak{I} = A/\mathfrak{p}$ is integrally closed, and that $\dim(A/\mathfrak{p}) = \dim(A) - n$ and that $\dim(A_\mathfrak{p}) = n = \dim(A) - \dim(A_\mathfrak{p})$ since A is biequidimensional, which completes the proof.

5.13. Properties of permanence by passage to the inductive limit Throughout this paragraph, $(A_\alpha, \varphi_{\beta\alpha})$ denotes an *inductive system of rings* whose indexing set is pre-ordered and *filtered*; one sets $A = \varinjlim A_\alpha$ and denotes by φ_α the canonical homomorphism $A_\alpha \to A$. **IV-2** | 131

Proposition (5.13.1). — (i) *For every index α, let $\mathfrak{a}_\alpha$ be an ideal of A_α such that, for $\alpha \leqslant \beta$, one has $\varphi_{\beta\alpha}(\mathfrak{a}_\alpha) \subset \mathfrak{a}_\beta$. Then the $\mathfrak{a}_\alpha$ form an inductive system for the restrictions of the $\varphi_{\beta\alpha}$, $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$ is identified canonically to an ideal of A, $A/\mathfrak{a}$ to $\varinjlim(A_\alpha/\mathfrak{a}_\alpha)$. If moreover one has $\mathfrak{a}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{a}_\beta)$ for $\alpha \leqslant \beta$, then $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α.*

(ii) *Conversely, for every ideal $\mathfrak{a}$ of A, if one sets $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α, the $\mathfrak{a}_\alpha$ form an inductive system such that $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$.*

It is clear that (ii) is a particular case of (i). In (i), the fact that $\mathfrak{a}$ is identified canonically to an ideal of A, and $A/\mathfrak{a}$ to $\varinjlim(A_\alpha/\mathfrak{a}_\alpha)$ follows from the exactness of the inductive limit functor in the category of abelian groups. Moreover, $\mathfrak{a}$ is the union of the increasing family of the $\varphi_\alpha(\mathfrak{a}_\alpha)$, so that if $x_\alpha \in E_\alpha$ is such that $\varphi_\alpha(x_\alpha) \in \mathfrak{a}$, there exists $\beta \geqslant \alpha$ such that $\varphi_\alpha(x_\alpha) \in \mathfrak{a}$; but this implies the existence of an index $\beta \geqslant \alpha$ such that, if one sets $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, $y_\beta = \varphi_{\beta\alpha}(y_\alpha)$, one has $x_\beta y_\beta = 0$; since A_β is integral, one has $x_\beta = 0$ or $y_\beta = 0$ and as $x = \varphi_\beta(x_\beta)$ and $y = \varphi_\beta(y_\beta)$, this completes the proof that A is integral.

Corollary (5.13.2). — *If $\mathfrak{N}_\alpha$ is the nilradical of A_α, the nilradical of A is identified to $\varinjlim \mathfrak{N}_\alpha$, and A_{red} to $\varinjlim(A_\alpha)_{\mathrm{red}}$. In particular, if the A_α are reduced, the same holds for A.*

It is clear that $\varphi_{\beta\alpha}(\mathfrak{N}_\alpha) \subset \mathfrak{N}_\beta$ for $\alpha \leqslant \beta$, and that $\varinjlim \mathfrak{N}_\alpha$ is a nilpotent ideal contained in the nilradical $\mathfrak{N}$ of A. Conversely, if $x \in \mathfrak{N}$, one has $x^h = 0$ for an integer h; one can write $x = \varphi_\alpha(x_\alpha)$ for an index α and a $x_\alpha \in A_\alpha$; the relation $\varphi_\alpha(x_\alpha^h) = 0$ implies the existence of a $\beta \geqslant \alpha$ such that $\varphi_{\beta\alpha}(x_\alpha^h) = 0$, or else $x_\beta^h = 0$ where $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$; hence $x_\beta \in \mathfrak{N}_\beta$ and as $x = \varphi_\beta(x_\beta)$, this completes the proof of the corollary, taking into account (5.13.1). **IV-2** | 132

Proposition (5.13.3). — (i) *For every index α, let $\mathfrak{p}_\alpha$ be a prime ideal of A_α such that, for $\alpha \leqslant \beta$, one has $\varphi_{\beta\alpha}(\mathfrak{p}_\alpha) \subset \mathfrak{p}_\beta$. Then $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ is a prime ideal of A. In particular, if the A_α are integral, the same holds for A.*

(ii) *If moreover one has $\mathfrak{p}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{p}_\beta)$ for $\alpha \leqslant \beta$, then $A_\mathfrak{p}$ is identified canonically to $\varinjlim(A_\alpha)_{\mathfrak{p}_\alpha}$.*

(iii) *If A is a local ring, $\mathfrak{p}$ a maximal ideal, and if the local homomorphisms φ_α are injective, then, setting $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, one has $A = \varinjlim(A_\alpha)_{\mathfrak{p}_\alpha}$ and the homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ are injective.*

The first two assertions of (i) are in fact equivalent, since $A/\mathfrak{p} = \varinjlim(A_\alpha/\mathfrak{p}_\alpha)$. Suppose therefore all the A_α integral, and let x, y be two elements of A such that $xy = 0$. There exists an index α such that $x = \varphi_\alpha(x_\alpha)$, $y = \varphi_\alpha(y_\alpha)$ with x_α, y_α in A_α, and $\varphi_\alpha(x_\alpha y_\alpha) = 0$; but this implies the existence of an index $\beta \geqslant \alpha$ such that, if one sets $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, $y_\beta = \varphi_{\beta\alpha}(y_\alpha)$, one has $x_\beta y_\beta = 0$; as A_β is integral, one has $x_\beta = 0$ or $y_\beta = 0$ and as $x = \varphi_\beta(x_\beta)$ and $y = \varphi_\beta(y_\beta)$, this completes the proof that A is integral.

Let us now prove (ii). The hypothesis implies that the $A_\alpha - \mathfrak{p}_\alpha$ form an inductive system of multiplicative parts of the A_α, whence $A - \mathfrak{p}$ is the inductive limit; hence the local rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ form an inductive system (where the transition homomorphisms are local), and the canonical homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \to A_\mathfrak{p}$ form an inductive system of homomorphisms, whose inductive limit $\psi : \varinjlim((A_\alpha)_{\mathfrak{p}_\alpha}) \to A_\mathfrak{p}$ is a surjective homomorphism. It remains to see that ψ is injective. Now, if $x_\alpha \in A_\alpha$ and $s_\alpha \in A_\alpha - \mathfrak{p}_\alpha$ are such that $\varphi_\alpha(x_\alpha)/\varphi_\alpha(s_\alpha) = 0$ in $A_\mathfrak{p}$, there exists $s' \in A - \mathfrak{p}$ such that $s'\varphi_\alpha(x_\alpha) = 0$, and, replacing α by a larger index if needed, one can suppose that $s' = \varphi_\alpha(s'_\alpha)$ with $s'_\alpha \in A_\alpha - \mathfrak{p}_\alpha$, whence $\varphi_\alpha(s'_\alpha x_\alpha) = 0$; there is therefore a $\beta \geqslant \alpha$ such that $\varphi_{\beta\alpha}(s'_\alpha x_\alpha) = 0$ in A_β, which proves therefore that the image of x_α/s_α in $\varinjlim((A_\alpha)_{\mathfrak{p}_\alpha})$ is zero.

Finally, the first assertion of (iii) follows from (ii) and from the fact that $A_\mathfrak{p} = A$. Furthermore, the elements of $A_\alpha - \mathfrak{p}_\alpha$ are invertible in A, hence also $A_{\mathfrak{p}_\alpha} = A$, and the fact that $(A_\alpha)_{\mathfrak{p}_\alpha} \to A$ is injective follows by flatness from the hypothesis that φ_α is injective.

Corollary (5.13.4). — *Suppose that the A_α are all integral, and that the $\varphi_{\beta\alpha}$ are injective. Then, if A'_α is the integral closure of A_α, $A' = \varinjlim A'_\alpha$ is the integral closure of A. In particular, if A_α is integrally closed for every α, the same holds for A. If the A_α are local rings with local homomorphisms (resp. geometrically* *unibranch*) ((**0**, 23.2.1)) *and the* $\varphi_{\beta\alpha}$ *local injective homomorphisms* (*resp. geometrically unibranch*), A *is unibranch* (*resp. geometrically unibranch*).

If K_α is the field of fractions of A_α, it follows from (5.13.3, , (ii)) applied to $\mathfrak{p}_\alpha = 0$, that $K = \varinjlim K_\alpha$ is the field of fractions of A, and $A' = \varinjlim A'_\alpha$ is identified to

a subring of K, evidently integral over A. Conversely, let $z \in K$ be an element satisfying an integral dependence equation $z^n + \sum_{i=1}^{n} c_i z^{n-i} = 0$, where the $c_i \in A$. There exists an index α, a $z_\alpha \in K_\alpha$ and $c_{i\alpha} \in A_\alpha$ such that z (resp. the c_i) are the images of z_α (resp. of the $c_{i\alpha}$), hence one has $z_\alpha \in A'_\alpha$, which proves that $z \in A'$. It follows immediately that if the A_α are integrally closed, the same holds for A. If the A_α are local and the $\varphi_{\beta\alpha}$ local homomorphisms, one knows ((**0**, 10.3.1.3)) that A is a local ring and if k_α is the residue field of A_α, $k = \varinjlim k_\alpha$ is that of A; if the A'_α are local, A' is therefore local and A is unibranch; if moreover the residue field k'_α of A'_α is a radicial extension of k_α for every α, $k' = \varinjlim k'_\alpha$, which is the residue field of A', is a radicial extension of k, and A is therefore geometrically unibranch. **IV-2 | 133**

(5.13.5). We say that a ring A is *normal* if $\operatorname{Spec}(A)$ is normal, in other words if, for every prime ideal $\mathfrak{p}$ of A, $A_\mathfrak{p}$ is *integral and integrally closed*. One notes that this implies that A is *reduced* and that $\mathfrak{p}$ cannot contain more than *one* minimal prime ideal of A, in other words that $\mathfrak{p}$ can belong to only one *single* irreducible component of $\operatorname{Spec}(A)$; hence $\mathfrak{q}$ is the inverse image of (o) by the canonical homomorphism $A \to A_\mathfrak{p}$, and $A_\mathfrak{p}$ is therefore isomorphic to $(A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}$, which proves that the integral ring $A/\mathfrak{q}$ is *integrally closed* as intersection of integrally closed rings ((**I**, 8.2.1.1)). If A is *Noetherian* and *normal*, A is the composed direct product of a finite number of integral and integrally closed rings ((**I**, 5.1.4)).

Corollary (5.13.6). — *Suppose that the* A_β *are normal and that for* $\alpha \leqslant \beta$, *every irreducible component of* $\operatorname{Spec}(A_\beta)$ *dominates an irreducible component of* $\operatorname{Spec}(A_\alpha)$. *Then* $A = \varinjlim A_\alpha$ *is normal.*

Indeed, let $\mathfrak{p}$ be a prime ideal of A and set, for every α, $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$. By hypothesis, for every α, $\mathfrak{p}_\alpha$ contains a unique minimal ideal $\mathfrak{q}_\alpha$ of A_α, and for $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}^{-1}(\mathfrak{q}_\beta)$ is a minimal ideal of A_α contained in $\mathfrak{p}_\alpha$, hence equal to $\mathfrak{q}_\alpha$. If one sets $B_\alpha = A_\alpha/\mathfrak{q}_\alpha$, the integral rings B_α form an inductive system for the homomorphisms $\varphi'_{\beta\alpha} : \mathrm{B}_\alpha \to \mathrm{B}_\beta$ induced by the $\varphi_{\beta\alpha}$ by passage to quotients, and which are injective by definition; moreover the rings B_α are integrally closed, hence the same holds for $B = \varinjlim B_\alpha$ (5.13.4). If one sets $\mathfrak{p}'_\alpha = \mathfrak{p}_\alpha/\mathfrak{q}_\alpha$, one has moreover $(\mathrm{B}_\alpha)_{\mathfrak{p}'_\alpha} = (A_\alpha)_{\mathfrak{p}_\alpha}$; as $\varinjlim (B_\alpha)_{\mathfrak{p}'_\alpha} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha} = A_\mathfrak{p}$ is a local ring in an ideal of a prime of B (5.13.3), it is an integral ring and integrally closed, which proves the corollary.

Proposition (5.13.7). — *Suppose that the* A_α *are regular and that, for* $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}$ *makes* A_β *a flat* A_α*-module; suppose in addition that* $A = \varinjlim A_\alpha$ *is Noetherian. Then* A *is regular.*

By definition, it suffices to prove that for every prime ideal $\mathfrak{p}$ of A, the local Noetherian ring $A_\mathfrak{p}$ is regular; or, if $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, on has $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ and $A_\mathfrak{p} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$ (5.13.3); as by hypothesis the $(A_\alpha)_{\mathfrak{p}_\alpha}$ are regular and that $(A_\beta)_{\mathfrak{p}_\beta}$ is an $(A_\alpha)_{\mathfrak{p}_\alpha}$-module flat for $\alpha \leqslant \beta$ ((**0**, 6.3.2)), one sees that one is reduced to the case where the A_α and A are also *local* rings. To prove that A is regular, it suffices then to show that if M, N are two finite type A-modules, there exists i_0 such that, for $i \geqslant i_0$,

on has $\operatorname{Tor}_i^A(M, N) = 0$ ((**0**, 17.3.1) and 17.2.6). But one has the following lemma (where the A_α are again arbitrary): **IV-2 | 134**

Lemma (5.13.7.1). — *For every finite type* A*-module* M *of finite presentation, there exists an index* α *and a finite type* A_α*-module* M_α *of finite presentation such that* M *is isomorphic to* $M_\alpha \otimes_{A_\alpha} A$.

Indeed, one can suppose that $M = A^r/P$, where P is a sub-A-module of finite type of A^r. As $A^r = \varinjlim (A_\alpha^r)$ and that P is of finite type, there exists α and a sub-A_α-module of finite type P_α of A_α^r such that P is the canonical image of $P_\alpha \otimes_{A_\alpha} A$; the exactness of the tensor product shows therefore that M is isomorphic to $(A_\alpha^r/P_\alpha) \otimes_{A_\alpha} A$.

Having established this lemma, since A is Noetherian, write therefore $M = M_\alpha \otimes_{A_\alpha} A$, $N = N_\alpha \otimes_{A_\alpha} A$; by virtue of the hypothesis, A is an A_α-module flat ((**0**, 6.3.2)), hence on has $\operatorname{Tor}_i^A(M, N) = \operatorname{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \otimes_{A_\alpha} A$ an isomorphism approximately ((**III**, 6.3.9)); as by hypothesis A_α is regular, there exists i_0 such that $\operatorname{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) = 0$ for $i \geqslant i_0$.

Corollary (5.13.8). — *Suppose that the A_α are Noetherian and satisfy property* (R_k) *for an integer* $k \geqslant 0$. *Suppose furthermore that for* $\alpha \leqslant \beta$, $\varphi_{\beta\alpha}$ *makes* A_β *a flat* A_α*-module, that* $A = \varinjlim A_\alpha$ *is Noetherian and that, for every* α, ${}^a\varphi_\alpha$ *is a homeomorphism of* $\mathrm{Spec}(A)$ *over* $\mathrm{Spec}(A_\alpha)$. *Then* A *satisfies* (R_k).

Let in fact $\mathfrak{p}$ be a prime ideal of A and, for every α, set $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, so that $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ and $A_\mathfrak{p} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$ (5.13.3). It follows from ((**I**, 2.4.2)) and from the hypothesis that the morphism $\mathrm{Spec}(A_\mathfrak{p}) \to \mathrm{Spec}((A_\alpha)_{\mathfrak{p}_\alpha})$ is a surjective homeomorphism for every α; one therefore has $\dim(A_\mathfrak{p}) = \dim((A_\alpha)_{\mathfrak{p}_\alpha})$ for every α. Suppose then that $\dim(A_\mathfrak{p}) \leqslant k$, and the hypothesis implies that $(A_\alpha)_{\mathfrak{p}_\alpha}$ is a regular ring; since furthermore, for $\alpha \leqslant \beta$, $(A_\beta)_{\mathfrak{p}_\beta}$ is an $(A_\alpha)_{\mathfrak{p}_\alpha}$-module flat ((**0**, 6.3.2)), one concludes by (5.13.7) that $A_\mathfrak{p}$ is regular, which proves the corollary.

§6. Flat morphisms of locally Noetherian preschemes

Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism. For every $y \in Y$, the IV-2 | 134
fibre $f^{-1}(y) = X \times_Y \mathrm{Spec}(\boldsymbol{k}(y))$ is also a locally Noetherian prescheme, since it suffices to see this when $Y = \mathrm{Spec}(\mathrm{A})$ and $X = \mathrm{Spec}(\mathrm{B})$ are spectra of Noetherian rings, and then $\mathrm{B} \otimes_\mathrm{A} \boldsymbol{k}(y)$ is a ring of fractions of the quotient ring $\mathrm{B}/\mathfrak{J}_y\mathrm{B}$, hence is Noetherian. We propose first of all in this paragraph to relate the properties of X, of Y, and of the fibres $f^{-1}(y)$ (where y ranges over Y), under the hypothesis that the morphism f is *flat*. The questions treated will reduce to the study of relations between the Noetherian local rings $\mathscr{O}_y$, $\mathscr{O}_x$, and $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$, the homomorphism $\mathscr{O}_y \to \mathscr{O}_x$
being local and making $\mathscr{O}_x$ a flat $\mathscr{O}_y$-module. In n^os 6.11 to 6.13 (rather separate from the rest of the IV-2 | 135
paragraph, by their "absolute" rather than "relative" nature), we apply certain preceding results to find criteria allowing us to affirm that the *singular locus* (or certain analogous sets) of certain preschemes are closed sets; these criteria will play an important role in §7.

6.1. Flatness and dimension

Proposition (6.1.1). — *Let* A, B *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, $k = \mathrm{A}/\mathfrak{m}$ *its residue field,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism. Suppose that for every prime ideal* $\mathfrak{p}$ *of* A, *distinct from* $\mathfrak{m}$, *and for every minimal element* $\mathfrak{q}$ *of the set of prime ideals of* B *containing* $\mathfrak{p}\mathrm{B}$, $\varphi^{-1}(\mathfrak{q})$ *is distinct from* $\mathfrak{m}$ *(in other words, no irreducible component of* $\mathrm{Spec}(\mathrm{B}/\mathfrak{p}\mathrm{B})$ *is contained in the inverse image of the closed point* $\mathfrak{m}$ *of* $\mathrm{Spec}(\mathrm{A})$ *in* $\mathrm{Spec}(\mathrm{B})$*). Then we have*

$$\dim(\mathrm{B}) = \dim(\mathrm{A}) + \dim(\mathrm{B} \otimes_\mathrm{A} k). \tag{6.1.1.1}$$

Proof. We argue by induction on $n = \dim(\mathrm{A})$; the assertion is trivial for $n = 0$, $\mathfrak{m}\mathrm{B}$ being then contained in the nilradical of B. We may therefore suppose $n > 0$. Let $\mathfrak{q}_i$ ($1 \leqslant i \leqslant m$) be the minimal prime ideals of B, and let $\mathfrak{p}_i = \varphi^{-1}(\mathfrak{q}_i)$; for every i, we have $\mathfrak{p}_i \neq \mathfrak{m}$; otherwise (since $n > 0$) there would exist a prime ideal $\mathfrak{p} \neq \mathfrak{m}$ contained in $\mathfrak{p}_i$ and as $\mathfrak{q}_i$ is minimal among the prime ideals of B containing $\mathfrak{p}\mathrm{B}$, we would arrive at a contradiction with the hypothesis. Therefore $\mathfrak{m}$ is distinct from the union of the $\mathfrak{p}_i$ and of the minimal prime ideals $\mathfrak{p}'_j$ ($1 \leqslant j \leqslant r$) of A (Bourbaki, *Alg. comm.*, chap. II, § 1, n^o 1, prop. 2) and there exists an $a \in \mathfrak{m}$ not belonging to any of the $\mathfrak{p}_i$ nor any of the $\mathfrak{p}'_j$. Set $\mathrm{A}' = \mathrm{A}/a\mathrm{A}$, $\mathrm{B}' = \mathrm{B}/a\mathrm{B}$; we have (**0**, 16.3.4)

$$\dim(\mathrm{A}') = \dim(\mathrm{A}) - 1, \quad \dim(\mathrm{B}') = \dim(\mathrm{B}) - 1$$

by construction of a, and on the other hand $\mathrm{B}' \otimes_{\mathrm{A}'} k = \mathrm{B} \otimes_\mathrm{A} k$, hence

$$\dim(\mathrm{B}' \otimes_{\mathrm{A}'} k) = \dim(\mathrm{B} \otimes_\mathrm{A} k)$$

and it suffices to prove (6.1.1) for A' and B'; now, by virtue of the bijective correspondence between ideals of A (resp. B) containing a and ideals of A' (resp. B'), the hypotheses of the statement are also valid for A' and B'; we may therefore apply the induction hypothesis, which completes the proof. □

Corollary (6.1.2). — *Let* A, B *be two local Noetherian rings,* k *the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* $\mathrm{M} \neq 0$ *an* A*-module of finite type,* $\mathrm{N} \neq 0$ *a* B*-module of finite type. If* N *is a flat* A*-module* (*resp. if* B *is an* A*-module plat*)*, then we have*

$$\dim_\mathrm{B}(\mathrm{M} \otimes_\mathrm{A} \mathrm{N}) = \dim_\mathrm{A}(\mathrm{M}) + \dim_{\mathrm{B} \otimes_\mathrm{A} k}(\mathrm{N} \otimes_\mathrm{A} k) \tag{6.1.2.1}$$

(*resp.* (6.1.1)).

Proof. It suffices to prove the assertion concerning N; on the other hand, if $\mathfrak{b}$ is the annihilator of N, we may replace B by $B/\mathfrak{b}$, and therefore suppose that $\mathrm{Supp}(N) = \mathrm{Spec}(B)$; the hypothesis then means that the morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ corresponding to φ is *quasi-flat* (2.3.3); we conclude (2.3.4) that for every prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of A, every irreducible component of $f^{-1}(V(\mathfrak{p}))$ **IV-2 | 136** *dominates* $V(\mathfrak{p})$, and therefore the condition of (6.1.1) is satisfied, whence the conclusion. □

Corollary (6.1.3). — *Let* A, B *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, k *its residue field,* $\varphi : A \to B$ *a local homomorphism. Suppose that* $\dim(B \otimes_A k) = 0$ *(that is to say* (**0**, 16.2.1) *that* $\mathfrak{m}B$ *is an* ideal of definition *of* B*). Then we have* $\dim(B) \leqslant \dim(A)$*. If in addition there exists a* B*-module of finite type* N *that is a flat* A*-module and has support equal to* Spec(B) *(which holds in particular if* B *is an* A*-module* plat*), then* $\dim(B) = \dim(A)$.

Proof. The first assertion follows from (5.5.1) and the second from (6.1.2). □

Corollary (6.1.4). — *Let X, Y be two locally Noetherian preschemes,* $f : X \to Y$ *a surjective morphism. For every closed subset Z of Y, we have*

(6.1.4.1) $$\mathrm{codim}(f^{-1}(Z), X) \leqslant \mathrm{codim}(Z, Y)$$

Proof. In addition, if f is quasi-flat (2.3.3) the two sides of (6.1.4) are equal.

Indeed, if y is a maximal point of Z and x a maximal point of the fibre $f^{-1}(y)$, we have $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y})$ by virtue of (5.5.2); the inequality (6.1.4) then follows from (5.1.2.1) and from (5.1.3.1) and from the fact that f is surjective. If in addition f is quasi-flat, we know (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates an irreducible component of Z; the maximal points of $f^{-1}(Z)$ are therefore the maximal points of the fibres $f^{-1}(y)$, where y ranges over the set of maximal points of Z (**0**, 2.1.8), and for each of these maximal points x we have $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(x)) = 0$; as in addition $\mathscr{O}_x$ is a flat $\mathscr{O}_y$-module, we have $\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y)$ by virtue of (6.1.1), whence the equality in (6.1.4), taking into account that f is surjective. □

The proposition (6.1.1) admits the following partial converse:

Proposition (6.1.5). — *Let* A, B *be two local Noetherian rings,* k *the residue field of* A, $\varphi : A \to B$ *a local homomorphism,* M *a* B*-module of finite type. Suppose that:*

label=1° A *is a regular ring.*
lbbel=2° M *is a Cohen-Macaulay* B*-module.*
lcbel=3° $\dim_B(M) = \dim(A) + \dim_{B\otimes_A k}(M \otimes_A k)$.

Then M *is a flat* A*-module.*

Proof. We argue by induction on $n = \dim(A)$. If $n = 0$, A is a field since it is regular (**0**, 17.1.4) and the assertion is trivial. Suppose $n > 0$; let $\mathfrak{m}$ be the maximal ideal of A, and let $x \in \mathfrak{m}$ be an element not belonging to $\mathfrak{m}^2$; we know then (**0**, 17.1.8) that $A' = A/xA$ is regular and $\dim(A') = \dim(A) - 1$. Set
$$B' = B/xB, \quad M' = M/xM = M \otimes_A A';$$
then
$$B' \otimes_{A'} k = B \otimes_A k, \quad M' \otimes_{A'} k = M \otimes_A k$$
and by virtue of (5.5.1.2) we have
$$\dim_{B'}(M') \leqslant \dim(A') + \dim_{B'\otimes_{A'} k}(M' \otimes_A k);$$
we conclude therefore from (**0**, 16.3.4) that we have **IV-2 | 137**
$$\dim_B(M) \leqslant \dim_{B'}(M') + 1 \leqslant (\dim(A') + \dim_{B\otimes_A k}(M \otimes_A k)) + 1 = \dim(A) + \dim_{B\otimes_A k}(M \otimes_A k).$$

Since by hypothesis the extreme terms of these inequalities are equal, we have necessarily: (i) $\dim_{B'}(M') = \dim_B(M) - 1$, and as M is by hypothesis a Cohen-Macaulay B-module, this means that x is M-*regular* (**0**, 16.1.9) and (**0**, 16.5.5); (ii) $\dim_{B'}(M') = \dim(A') + \dim_{B'\otimes_{A'} k}(M' \otimes_{A'} k)$; as M′ is a Cohen-Macaulay B′-module by virtue of (i) and of (**0**, 16.1.9) and (**0**, 16.5.5) and A′ is regular, the induction hypothesis proves that M′ is a flat A′-module; we deduce then from (**0**, 10.2.7) that M is a flat A-module, since x is M-regular by (i). □

6.2. Flatness and projective dimension

Proposition (6.2.1). —

label=(i) *Let* A, B *be two rings,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a homomorphism such that* B *is a flat* A*-module. Then, for every* A*-module* M, *we have*

(6.2.1.1) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}) \leqslant \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}).$$

lbbel=(ii) *Suppose in addition that* A *is a Noetherian ring,* B *a faithfully flat* A*-module, and* M *an* A*-module of finite type; then*

(6.2.1.2) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}) = \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}).$$

Proof.

label=(i) We may restrict to the case where $n = \dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M})$ is finite; there therefore exists a left resolution
$$0 \longrightarrow \mathrm{P}_n \longrightarrow \mathrm{P}_{n-1} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$
of M by projective A-modules; as B is a flat A-module, the sequence
$$0 \longrightarrow \mathrm{P}_n \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{P}_{n-1} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow 0$$
is exact; on the other hand, the $\mathrm{P}_i \otimes_{\mathrm{A}} \mathrm{B}$ are projective B-modules; whence the conclusion.

lbbel=(ii) Suppose that $\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}) = m$, and consider an exact sequence
$$0 \longrightarrow \mathrm{R} \longrightarrow \mathrm{P}_{m-1} \longrightarrow \mathrm{P}_{m-2} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \longrightarrow \mathrm{M} \longrightarrow 0$$
where the P_i $(0 \leqslant i \leqslant m-1)$ are projective A-modules of finite type; as A is Noetherian, R is also an A-module of finite type. As B is a flat A-module, we also have an exact sequence
$$0 \longrightarrow \mathrm{R} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{P}_{m-1} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \ldots \longrightarrow \mathrm{P}_0 \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{B} \longrightarrow 0$$
and the hypothesis on $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}$ implies that $\mathrm{R} \otimes_{\mathrm{A}} \mathrm{B}$ is a projective B-module (**0**, 17.2.1). As B is a faithfully flat A-module, we conclude that R is a projective A-module (Bourbaki, *Alg. comm.*, chap. I$^{\mathrm{er}}$, § 3, n$^{\mathrm{o}}$ 6, prop. 12); hence $\dim.\mathrm{proj}_{\mathrm{A}}(\mathrm{M}) \leqslant m$, which completes the proof.

□

Corollary (6.2.2). — *Let* $f : X \to Y$ *be a flat morphism of preschemes,* $\mathscr{E}$ *a quasi-coherent* $\mathscr{O}_Y$*-Module. If* $\dim.\mathrm{proj}(\mathscr{E}) \leqslant n$ (**0**, 17.2.14)*, then* $\dim.\mathrm{proj}(f^*(\mathscr{E})) \leqslant n$.

Proposition (6.2.3). — *Let* A, B *be two local Noetherian rings,* k *the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* N *a* B*-module of finite type. Suppose that* B *and* N *are flat* A*-modules. Then we have*

(6.2.3.1) $$\dim.\mathrm{proj}_{\mathrm{B}}(\mathrm{N}) = \dim.\mathrm{proj}_{\mathrm{B} \otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k).$$

Proof. Consider a left resolution of N by free B-modules of finite type
$$\ldots \longrightarrow \mathrm{L}_i \longrightarrow \mathrm{L}_{i-1} \longrightarrow \ldots \longrightarrow \mathrm{L}_0 \longrightarrow \mathrm{N} \longrightarrow 0$$
As the L_i and N are flat A-modules (B being a flat A-module) it follows from (2.1.10) that the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are flat A-modules, that $\mathrm{L}_\bullet \otimes_{\mathrm{A}} k$ is a resolution of $\mathrm{N} \otimes_{\mathrm{A}} k$ and that $\mathrm{Z}_i(\mathrm{L}_\bullet \otimes_{\mathrm{A}} k) = \mathrm{Z}_i(\mathrm{L}_\bullet) \otimes_{\mathrm{A}} k$ for all i. Note that since B is Noetherian, the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are B-modules of finite type, hence the $\mathrm{Z}_i(\mathrm{L}_\bullet \otimes_{\mathrm{A}} k)$ are $(\mathrm{B} \otimes_{\mathrm{A}} k)$-modules of finite type. Now, to say that a $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module of finite type is flat is equivalent to saying that it is *free* or even *projective* (**0**, 10.1.3). As (where we set C = B) the $\mathrm{Z}_i(\mathrm{L}_\bullet)$ are flat B-modules, it follows on the other hand from (**0**, 10.2.5) that, for $\mathrm{Z}_i(\mathrm{L}_\bullet)$ to be a flat B-module, it is necessary and sufficient that $\mathrm{Z}_i(\mathrm{L}_\bullet) \otimes_{\mathrm{A}} k$ be a $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module that is flat. The smallest integer n such that $\mathrm{Z}_{n-1}(\mathrm{L}_\bullet)$ is a free B-module is therefore also the smallest integer n such that $\mathrm{Z}_{n-1}(\mathrm{L}_\bullet \otimes_{\mathrm{A}} k)$ is a free $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module, which proves the proposition (**0**, 17.2.1). □ **IV-2** | 138

6.3. Flatness and depth

Proposition (6.3.1). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* M *an* A*-module of finite type,* N *a* B*-module of finite type. If* N *is a flat* A*-module* $\neq 0$*, then we have*

(6.3.1.1) $$\mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \mathrm{prof}_{\mathrm{A}}(\mathrm{M}) + \mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k).$$

Proof. We may restrict to the case where $\mathrm{M} \neq 0$, without which both sides of (6.3.1) are all equal to $+\infty$. We will argue by induction on the integer n equal to the second member of (6.3.1.1) (which is finite by virtue of the hypotheses, of (**0**, 16.4.6.2) and of the lemma of Nakayama applied to N). Suppose first $n = 0$; then, if $\mathfrak{m}$ and $\mathfrak{n}$ denote the maximal ideals of A and B respectively, $\mathfrak{m}$ is a prime ideal associated to M and $\mathfrak{n}/\mathfrak{m}\mathrm{B}$ a prime ideal of $\mathrm{B}/\mathfrak{m}\mathrm{B} = \mathrm{B} \otimes_{\mathrm{A}} k$ associated to $\mathrm{N} \otimes_{\mathrm{A}} k$. We conclude then from (3.3.1) that $\mathfrak{n}$ is a prime ideal associated to $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$, hence (**0**, 16.4.6), (i)) that $\mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = 0$, and we distinguish two cases:

a) Suppose $\mathrm{prof}_{\mathrm{A}}(\mathrm{M}) > 0$. Let $x \in \mathfrak{m}$ be an M-regular element, and set

$$\mathrm{A}' = \mathrm{A}/x\mathrm{A}, \quad \mathrm{B}' = \mathrm{B}/x\mathrm{B}, \quad \mathrm{M}' = \mathrm{M}/x\mathrm{M}, \quad \mathrm{N}' = \mathrm{N}/x\mathrm{N};$$

then

$$\mathrm{B}' \otimes_{\mathrm{A}'} k = \mathrm{B} \otimes_{\mathrm{A}} k, \quad \mathrm{N}' \otimes_{\mathrm{A}'} k = \mathrm{N} \otimes_{\mathrm{A}} k$$

since $x \in \mathfrak{m}$, and

$$\mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' = (\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N})/x(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N});$$

in addition, as N is a flat A-module, the hypothesis that x is M-regular implies that x is also (M $\otimes_{\mathrm{A}}$ N)-regular. We thus have (**0**, 16.4.6), (ii) and (**0**, 16.4.8)) **IV-2** | 139

$$\mathrm{prof}_{\mathrm{A}'}(\mathrm{M}') = \mathrm{prof}_{\mathrm{A}}(\mathrm{M}) - 1, \quad \mathrm{prof}_{\mathrm{B}'}(\mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}') = \mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) - 1$$

and

$$\mathrm{prof}_{\mathrm{B}'\otimes_{\mathrm{A}'} k}(\mathrm{N}' \otimes_{\mathrm{A}'} k) = \mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k).$$

The equality (6.3.1.1) is therefore a consequence of the same relation for A′, B′, M′ and N′; but as N is a flat A-module, $\mathrm{N}' = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}'$ is a flat A′-module (**0**, 6.2.1); we may therefore apply the induction hypothesis, which proves (6.3.1.1) in this case.

b) Suppose $\mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k) > 0$. Consider an element $y \in \mathfrak{m}$ which is (N $\otimes_{\mathrm{A}}$ k)-regular; it follows from (**0**, 10.2.4) that y is N-regular and that we then have

$$\mathrm{N}' = \mathrm{N}/y\mathrm{N}$$

is a flat A-module *plat*, since N is supposed to be a flat A-module. Applying (**0**, 6.1.2) to the exact sequence of A-modules

$$0 \longrightarrow \mathrm{N} \xrightarrow{y} \mathrm{N} \longrightarrow \mathrm{N}' \longrightarrow 0$$

we conclude that we have isomorphisms

$$(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N})/y(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) \xrightarrow{\sim} \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}' \quad \text{and} \quad \mathrm{N}' \otimes_{\mathrm{A}} k \xrightarrow{\sim} (\mathrm{N} \otimes_{\mathrm{A}} k)/y(\mathrm{N} \otimes_{\mathrm{A}} k)$$

and in addition that y is (M $\otimes_{\mathrm{A}}$ N)-regular. We thus have

$$\mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N}' \otimes_{\mathrm{A}} k) = \mathrm{prof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k) - 1$$

and

$$\mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}') = \mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) - 1;$$

the induction hypothesis shows that the relation (6.3.1.1) is valid for A, B, M and N′, and from what precedes, we deduce (6.3.1.1) for A, B, M and N. □

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), suppose in addition* $\mathrm{M} \neq 0$ *and* $\mathrm{N} \neq 0$*; then we have*

(6.3.2.1) $$\mathrm{coprof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \mathrm{coprof}_{\mathrm{A}}(\mathrm{M}) + \mathrm{coprof}_{\mathrm{B}\otimes_{\mathrm{A}} k}(\mathrm{N} \otimes_{\mathrm{A}} k).$$

Proof. This follows immediately from (6.3.1.1) and (6.1.2) and from the definition of the codepth (**0**, 16.4.9). □

Corollary (6.3.3). — *Under the hypotheses of (6.3.1), suppose in addition* $\mathrm{M} \neq 0$ *and* $\mathrm{N} \neq 0$*; for* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ *to be a Cohen-Macaulay* B*-module, it is necessary and sufficient that* M *be a Cohen-Macaulay* A*-module and that* $\mathrm{N} \otimes_{\mathrm{A}} k$ *be a* $(\mathrm{B} \otimes_{\mathrm{A}} k)$*-module of Cohen-Macaulay.*

Proof. This follows from the corollary (6.3.2) and from the definition of Cohen-Macaulay modules by the fact that their codepth is zero (**0**, 16.5.1), taking into account that the condition $\mathrm{N} \neq 0$ is equivalent to $\mathrm{N} \otimes_{\mathrm{A}} k \neq 0$ by virtue of the lemma of Nakayama. □

Corollary (6.3.4). — *Under the hypotheses of (6.3.1), suppose in addition that* $\mathrm{N} \otimes_{\mathrm{A}} k$ *is a* B*-module of finite length; then we have*

(6.3.4.1) $$\mathrm{prof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \mathrm{prof}_{\mathrm{A}}(\mathrm{M}).$$

If in addition $\mathrm{M} \neq 0$ *and* $\mathrm{N} \neq 0$*, then we have* **IV-2** | 140

(6.3.4.2) $$\mathrm{coprof}_{\mathrm{B}}(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) = \mathrm{coprof}_{\mathrm{A}}(\mathrm{M}).$$

Proof. It amounts to the same to say that $\mathrm{N} \otimes_{\mathrm{A}} k$ is a B-module of finite length or a $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module of finite length, and we know (**0**, 16.2.3) that modules of finite length $\neq 0$ are of dimension 0 and depth 0.

The corollary (6.3.4) is applicable in particular when $\mathrm{B} \otimes_{\mathrm{A}} k$ is an Artinian ring, that is to say when $\mathfrak{m}\mathrm{B}$ is an *ideal of definition* of the ring B. □

Corollary (6.3.5). — *Let* X, Y *be two locally Noetherian preschemes,* $f : X \to Y$ *a flat morphism.*

label=(i) If, at a point $x \in X$, $\mathrm{coprof}(\mathscr{O}_x) \leqslant n$*, then, setting* $y = f(x)$*, we have* $\mathrm{coprof}(\mathscr{O}_y) \leqslant n$ *and* $\mathrm{coprof}(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)) \leqslant n$*; in particular, if* $\mathscr{O}_x$ *is a Cohen-Macaulay ring, the same is true of* $\mathscr{O}_y$ *and of* $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$*.*

lbbel=(ii) Conversely, suppose that $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ *is a Cohen-Macaulay ring. Then, if* $\mathscr{O}_y$ *is a Cohen-Macaulay ring, we have* $\mathrm{coprof}(\mathscr{O}_x) \leqslant n$ *(resp.* $\mathscr{O}_x$ *is a Cohen-Macaulay ring).*

Proof. It suffices to apply (6.3.2) for $\mathrm{M} = \mathscr{O}_y$ and $\mathrm{N} = \mathrm{B} = \mathscr{O}_x$. □

Corollary (6.3.6). — *Let* A *be a Cohen-Macaulay ring. Then* $\mathrm{A}' = \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n]$ *(*T_i *indeterminates) is a Cohen-Macaulay ring.*

Proof. Indeed, if we set $Y = \mathrm{Spec}(\mathrm{A})$, $X = \mathrm{Spec}(\mathrm{A}')$ and if $f : X \to Y$ is the canonical morphism, f is *flat* (since A' is a free A-module (2.1.2)) and for every $y \in Y$, $\boldsymbol{k}(y)[\mathrm{T}_1, \dots, \mathrm{T}_n]$ is a regular ring (**0**, 17.3.7), hence *a fortiori* a Cohen-Macaulay ring; it therefore suffices to apply (6.3.5). □

Corollary (6.3.7). — *Every quotient ring of a Cohen-Macaulay ring is universally catenary.*

Proof. This follows from (6.3.6), from (5.6.1) and from (**0**, 16.5.12). □

Proposition (6.3.8). — *Let* A *be a local Noetherian ring; suppose that there exists a Cohen-Macaulay* A*-module* M *of finite type whose support is equal to* $\mathrm{Spec}(\mathrm{A})$*. Let* B *be a quotient ring of* A*, and let* $f : \mathrm{Spec}(\widehat{\mathrm{B}}) \to \mathrm{Spec}(\mathrm{B})$ *be the canonical morphism; then, for every* $x \in \mathrm{Spec}(\mathrm{B})$, $f^{-1}(x)$ *is a Cohen-Macaulay prescheme.*

Proof. As B is a quotient of a module of finite type, on $\widehat{\mathrm{B}} = \mathrm{B} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ (**0**, 7.3.3), hence f is the restriction to $\mathrm{Spec}(\widehat{\mathrm{B}})$ of the canonical morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$, and the fibres of these two morphisms at a point of $\mathrm{Spec}(\mathrm{B})$ are the same. We are therefore reduced to proving the proposition when $\mathrm{B} = \mathrm{A}$. Let $\mathfrak{p}$ be a prime ideal *minimal* of A. Set, to simplify, $\mathrm{A}' = \widehat{\mathrm{A}}$, $\mathrm{M}' = \widehat{\mathrm{M}} = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$ (**0**, 7.3.3); we know (**0**, 16.5.2) that M' is a Cohen-Macaulay A'-module, which implies that for every prime ideal $\mathfrak{p}'$ of A', $\mathrm{M}'_{\mathfrak{p}'}$ is a Cohen-Macaulay $\mathrm{A}'_{\mathfrak{p}'}$-module (**0**, 16.5.10). Taking into account (**I**, 3.6.5), on sees that if we set $\mathrm{A}'' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_{\mathfrak{p}}$ and $\mathrm{M}'' = \mathrm{M}' \otimes_{\mathrm{A}} \mathrm{A}_{\mathfrak{p}}$, M'' is a Cohen-Macaulay A''-module. For every prime ideal $\mathfrak{p}''$ of A'' above $\mathfrak{p}$, $\mathrm{M}''_{\mathfrak{p}''} = \mathrm{M}_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{A}''_{\mathfrak{p}''}$ is a Cohen-Macaulay $\mathrm{A}''_{\mathfrak{p}''}$-module (**0**, 16.5.10); on the other hand, $\mathrm{M}_{\mathfrak{p}}$ is by hypothesis a Cohen-Macaulay $\mathrm{A}_{\mathfrak{p}}$-module and $\mathrm{A}''_{\mathfrak{p}''}$ is a flat $\mathrm{A}_{\mathfrak{p}}$-module plat since A' is a flat A-module (**0**, 7.3.3) and (6.3.2). We conclude from (6.3.3) that, if k is the residue field of $\mathrm{A}_{\mathfrak{p}}$, $k \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{A}''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. But since $\mathrm{A}_{\mathfrak{p}}$ is of dimension 0, the prime ideals of A' correspond bijectively to those of $k \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{A}''$ (**I**, 3.5.7), and if $\mathfrak{q}''$ is the prime ideal of $k \otimes_{A_\mathfrak{p}} A''$ corresponding to $\mathfrak{p}''$, the local rings $(k \otimes_{A_\mathfrak{p}} A'')_{\mathfrak{q}''}$ and $k \otimes_{A_\mathfrak{p}} A''_{\mathfrak{p}''}$ are isomorphic. Therefore (**0**, 16.5.13), the ring $k \otimes_{A_\mathfrak{p}} A''$ is a Cohen-Macaulay ring. □

IV-2 | 141

6.4. Flatness and property (S_k)

Proposition (6.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat.*

label=(i) *Let $x \in \operatorname{Supp}(\mathscr{F})$ be such that $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ has property (S_k) at the point x; then $\mathscr{E}$ has property (S_k) at the point $y = f(x)$.*

lbbel=(ii) *Suppose that for every $y \in f(\operatorname{Supp}(\mathscr{F}))$, $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ has property (S_k); then, if for a point $y \in f(\operatorname{Supp}(\mathscr{F}))$, $\mathscr{E}$ has property (S_k) at the point y, $\mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$ has property (S_k) at every point of $f^{-1}(y)$.*

Proof.

label=(i) We know (**Err**$_{\mathbf{III}}$, 30) that there exists a closed sub-prescheme X' of X having for underlying space $\operatorname{Supp}(\mathscr{F})$ and a coherent $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ such that $\mathscr{F} = j_*(\mathscr{F}')$, where j is the canonical injection $X' \hookrightarrow X$; we may replace X by X' and $\mathscr{F}$ by $\mathscr{F}'$, in other words suppose that $\operatorname{Supp}(\mathscr{F}) = X$. Let y' be a generization of y in Y; we know (2.3.4) that there is a generization x' of x in X such that $y' = f(x')$; we may in addition suppose that x' is the generic point of an irreducible component of $f^{-1}(y')$; we then have $\dim(\mathscr{F}_{x'} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y')) = 0$. By virtue of (6.1.2), we then have, setting $\mathscr{G} = \mathscr{E} \otimes_{\mathscr{O}_Y} \mathscr{F}$,

$$\dim(\mathscr{G}_{x'}) = \dim(\mathscr{E}_{y'})$$

and by virtue of (6.3.1)

$$\operatorname{prof}(\mathscr{G}_{x'}) = \operatorname{prof}(\mathscr{E}_{y'})$$

taking into account that the depth is always at most equal to the dimension. By hypothesis, we have **IV-2 | 142**

$$\operatorname{prof}(\mathscr{G}_{x'}) \geqslant \inf(k, \dim(\mathscr{G}_{x'}))$$

hence

$$\operatorname{prof}(\mathscr{E}_{y'}) \geqslant \inf(k, \dim(\mathscr{E}_{y'}))$$

which proves the first assertion.

lbbel=(ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y, we may restrict to verifying that if $x \in \operatorname{Supp}(\mathscr{G})$ and $f(x) = y$, we have $\operatorname{prof}(\mathscr{G}_x) \geqslant \inf(k, \dim(\mathscr{G}_x))$; it follows from (6.1.2) and (6.3.1) that we have

$$\dim(\mathscr{G}_x) = \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))$$

$$\operatorname{prof}(\mathscr{G}_x) = \operatorname{prof}(\mathscr{E}_y) + \operatorname{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)).$$

By hypothesis, we have

$$\operatorname{prof}(\mathscr{E}_y) \geqslant \inf(k, \dim(\mathscr{E}_y))$$

$$\operatorname{prof}(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)) \geqslant \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)))$$

whence, adding term by term,

$$\begin{aligned}\operatorname{prof}(\mathscr{G}_x) \geqslant \inf(k, \dim(\mathscr{E}_y)) + \inf(k, \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))) \geqslant \\ \geqslant \inf(k, \dim(\mathscr{E}_y) + \dim(\mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))) = \inf(k, \dim(\mathscr{G}_x))\end{aligned}$$

which proves (ii).

□

Corollary (6.4.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism, $\mathscr{E}$ a coherent $\mathscr{O}_Y$-Module. Suppose that for every $y \in Y$, the prescheme $f^{-1}(y)$ has property (S_k). Then, for $f^*(\mathscr{E})$ to have property (S_k) at a point x, it is necessary and sufficient that $\mathscr{E}$ have property (S_k) at the point $f(x)$.*

Remark (6.4.3). — Let A, B be two local Noetherian rings, k the residue field of A, $\varphi : \mathrm{A} \to \mathrm{B}$ a local homomorphism, M a B-module of finite type that is a flat A-module, and suppose in addition that the A-module A and the $(\mathrm{B} \otimes_{\mathrm{A}} k)$-module $\mathrm{M} \otimes_{\mathrm{A}} k$ have property (S_n); can one then conclude that the B-module M has property (S_n)? We do not know even when $n = 1$, $\mathrm{M} = \mathrm{B}$ and $\mathrm{B} = \widehat{\mathrm{A}}$, which is to say we do not know whether in general the *completion* of a local Noetherian ring satisfying (S_n) also satisfies (S_n), even for $n = 1$. Setting $Y = \mathrm{Spec}(\mathrm{A})$, $X = \mathrm{Spec}(\mathrm{B})$ and $\mathscr{F} = \widetilde{\mathrm{M}}$, it would suffice, by virtue of (6.4.1), (ii)), to show that for *every* $y \in Y$, $\mathscr{F}_y$ has property (S_k) (and not only when y is the closed point of Y); it would also suffice to show that the set U of $x \in X$ such that $\mathscr{F}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))$ satisfies (S_k) is *open* in X (since it contains the closed point of X by hypothesis). We will show further on (12.1.4) that U is open when B is a local ring of a finite type A-algebra. On the other hand, for $\mathrm{B} = \widehat{\mathrm{A}}$ and $\mathrm{M} = \mathrm{B}$, we have seen in (6.3.8) that the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes (in other words satisfy (S_n)) for *every* n) when we suppose that A is a quotient of a local Cohen-Macaulay ring (or more generally of a local ring satisfying the hypotheses of (6.3.8)), *for* A *to satisfy* (S_n), *it is necessary and sufficient that its completion* $\widehat{\mathrm{A}}$ *satisfy* (S_n). It would remain to see whether this property holds without restriction on A.

6.5. Flatness and property (R_k)

Proposition (6.5.1). — *Let* A, B *be two local Noetherian rings, k the residue field of* A, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism, for which* B *is a flat* A*-module. Then:*

label=(i) If B *is regular,* A *is regular.*
lbbel=(ii) If A *and* $\mathrm{B} \otimes_{\mathrm{A}} k$ *are regular,* B *is regular.*

This proposition is given here for the record, having already been proved in (**0**, 17.3.3).

Corollary (6.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

label=(i) If X is regular at a point x, Y is regular at the point $f(x)$.
lbbel=(ii) If for a $y \in f(X)$, Y is regular at the point y and if $f^{-1}(y)$ is a regular prescheme at a point x, then X is regular at the point x.

Proposition (6.5.3). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

label=(i) If X has property (R_k) at a point x (resp. has property (R_k)), Y has property (R_k) at the point $f(x)$ (resp. Y has property (R_k) at all points of $f(X)$).
lbbel=(ii) Suppose that for every $y \in f(X)$, the prescheme $f^{-1}(y)$ has property (R_k); then, if for a point $y \in f(X)$, Y has property (R_k) at the point y, X has property (R_k) at every point of $f^{-1}(y)$.

Proof.

label=(i) Set $y = f(x)$ and let y' be a generization of y; as in the proof of (6.4.1), there is a generic point x' of an irreducible component of $f^{-1}(y')$ which is a generization of x; hence $\dim(\mathscr{O}_{x'} \otimes_{\mathscr{O}_{y'}} \boldsymbol{k}(y')) = 0$ and by virtue of the hypothesis and of (6.1.2), $\dim(\mathscr{O}_{x'}) = \dim(\mathscr{O}_{y'})$. The hypothesis implies, provided that $\dim(\mathscr{O}_{x'}) > k$, that $\mathscr{O}_{x'}$ is a regular ring, and then $\mathscr{O}_{y'}$ is regular by virtue of (6.5.1).

lbbel=(ii) As for every generization x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generization of y, we may restrict to verifying that if $\dim(\mathscr{O}_x) \leqslant k$, then $\mathscr{O}_x$ is a regular ring. Now, by virtue of (6.1.2)

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))$$

hence if $\dim(\mathscr{O}_x) \leqslant k$, we have *a fortiori* $\dim(\mathscr{O}_y) \leqslant k$ and $\dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)) \leqslant k$ and the hypothesis implies that $\mathscr{O}_y$ and $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ are regular rings. We conclude then from (6.5.1) that $\mathscr{O}_x$ is a regular ring.

□

Corollary (6.5.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a flat morphism.*

label=(i) If X is normal at a point x, Y is normal at the point $f(x)$.
lbbel=(ii) If, for every $y \in f(X)$, $f^{-1}(y)$ is a normal prescheme and if Y is normal at a point $y \in f(X)$, then X is a normal prescheme at every point of $f^{-1}(y)$.

Proof. This follows immediately from the normality criterion of Serre (5.8.6) and from (6.4.1) applied for $k = 2$ and (6.5.3) applied for $k = 1$. □

Remarks (6.5.5). —

label=(i) Let A, B be two local Noetherian rings, $\varphi : \mathrm{A} \to \mathrm{B}$ a local homomorphism such that B is a flat A-module. Let k be the residue field of A, and suppose that the two rings A and $\mathrm{B} \otimes_{\mathrm{A}} k$ satisfy property (R_i) (5.8.2); then *it does not necessarily follow* that B satisfies (R_i), even in the particular case where $i = 0$ or $i = 1$ and where B is the *completion* $\widehat{\mathrm{A}}$ of A. Nagata has in fact given an example where A is *normal* (hence satisfies (R_1)) but where $\widehat{\mathrm{A}}$ is not even reduced (hence does not satisfy (R_0)) [30]. We cannot apply the proposition (6.5.3) to this case because the fibres $f^{-1}(y)$ do not necessarily satisfy property (R_i) at the points distinct from the closed point of $Y = \mathrm{Spec}(\mathrm{A})$. We will show further on (7.8.3), (v)) that such phenomena do not arise for the local Noetherian rings that one encounters most often in applications.

lbbel=(ii) The property for a prescheme of being *integral* does not at all behave like the properties we have just examined and the preceding ones: it can happen that $f : X \to Y$ is a flat morphism of finite type, that all the fibres $f^{-1}(y)$ are geometrically regular (and even geometrically regular (6.7.6)) and that Y is integral, without X being even locally integral. For example, let k be an algebraically closed field of characteristic 0, and let A be the integral ring $k[\mathrm{U},\mathrm{V}]/(\mathrm{UV} - (\mathrm{U}+\mathrm{V})^3)$ (whose spectrum is therefore a "cubic with a double point"); A is not integrally closed, and if u, v are the canonical images of U, V in A, the integral closure of A is the ring $\mathrm{C} = \mathrm{A}[t]$, with $t = (u - v)/(u + v)$, which satisfies the equation $t^2 = 1 + u + v$, whence $u = \frac{1}{2}(t^3 + t^2 - t)$, $v = \frac{1}{2}(-t^3 + t^2 + t)$ and therefore $\mathrm{C} = k[t]$, isomorphic to a polynomial ring in one indeterminate over k. If $\mathfrak{m} = \mathrm{A}u + \mathrm{A}v$, **IV-2** | 144 the maximal ideal of A (corresponding to the "double point" of the cubic), there are two maximal ideals $\mathfrak{n}_1, \mathfrak{n}_2$ of C, generated respectively by $t - 1$ and $t + 1$. Let B be the subring of the product $\mathrm{C} \times \mathrm{C}$ formed by the pairs of polynomials (f, g) such that $f(1) = g(-1)$ and $f(-1) = g(1)$ (Spec(B) is the prescheme obtained by "gluing" two copies of Spec(C), the point $\mathfrak{n}_1$ (resp. $\mathfrak{n}_2$) of one of the two copies being "glued" to the point $\mathfrak{n}_2$ (resp. $\mathfrak{n}_1$) of the other; cf. chap. V, where this operation will be discussed in a general way). There are therefore two maximal ideals $\mathfrak{r}_1, \mathfrak{r}_2$ of B above the maximal ideal $\mathfrak{m}$ of A. Moreover, as the "gluing" process commutes with localisation and completion, one easily verifies that the canonical homomorphisms $\widehat{\mathrm{A}}_{\mathfrak{m}} \to \widehat{\mathrm{B}}_{\mathfrak{r}_1}$ and $\widehat{\mathrm{A}}_{\mathfrak{m}} \to \widehat{\mathrm{B}}_{\mathfrak{r}_2}$ are bijective, and therefore $\mathrm{B}_{\mathfrak{r}_1}$ and $\mathrm{B}_{\mathfrak{r}_2}$ are flat $\mathrm{A}_{\mathfrak{m}}$-modules (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 5, prop. 10) $\mathrm{B}_{\mathfrak{r}_1}$ and $\mathrm{B}_{\mathfrak{r}_2}$ are flat $\mathrm{A}_{\mathfrak{m}}$-modules. For every other maximal ideal $\mathfrak{p}$ of A, there is a single maximal ideal $\mathfrak{q}_1, \mathfrak{q}_2$ of B above $\mathfrak{p}$, and the homomorphisms $\mathrm{A}_{\mathfrak{p}} \to \mathrm{B}_{\mathfrak{q}_1}$ and $\mathrm{A}_{\mathfrak{p}} \to \mathrm{B}_{\mathfrak{q}_2}$ are bijective. We see thus that the morphism $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ is flat and finite, and that all its fibres are geometrically regular (it is even *étale*, as we will see later (17.6.3)); however it is immediate that B is not integral.

6.6. Properties of transitivity

Proposition (6.6.1). — *For a locally Noetherian prescheme* Z*, let us denote by* $\boldsymbol{P}(\mathrm{Z})$ *any one of the following properties:*

label=a) Z *is a Cohen-Macaulay prescheme.*
lbbel=b) Z *satisfies* (S_n).
lcbel=c) Z *is regular.*
ldbel=d) Z *satisfies* (R_n).
lebel=e) Z *is normal.*
lfbel=f) Z *is reduced.*

Let then X, Y, Z *be three locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$, $g : \mathrm{Y} \to \mathrm{Z}$ *two morphisms.*

label=(i) Suppose that f *and* g *are flat and that for every* $y \in f(\mathrm{X})$ *(resp. every* $z \in g(\mathrm{Y})$*), the prescheme* $f^{-1}(y)$ *(resp.* $g^{-1}(z)$*) satisfies* $\boldsymbol{P}$*. Then* $h = g \circ f$ *is flat,* $h^{-1}(z)$ *satisfies* $\boldsymbol{P}$ *for every* $z \in h(\mathrm{X})$*, and for every* $z \in g(\mathrm{Y})$ *and every* $t \in h(\mathrm{X})$*,* $h^{-1}(z)$ *satisfies* $\boldsymbol{P}$.

lbbel=(ii) *Suppose the following conditions are satisfied: f is faithfully flat, $h = g \circ f$ is flat, for every $y \in \mathrm{Y}$, the prescheme $f^{-1}(y)$ satisfies $\boldsymbol{P}$, and for every $z \in g(\mathrm{Y})$, the prescheme $h^{-1}(z)$ satisfies $\boldsymbol{P}$. Then g is flat, and for every $z \in g(\mathrm{Y})$, $g^{-1}(z)$ satisfies $\boldsymbol{P}$.*

Proof. (i) We already know (**0**, 2.1.6) that h is flat; furthermore, for every $z \in \mathrm{Z}$, $f_z = f \otimes \mathbf{1}_{\boldsymbol{k}(z)}$: **IV-2 | 145**
$h^{-1}(z) \to g^{-1}(z)$ is flat (**0**, 2.1.4), and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4). The conclusion follows then respectively from (6.3.5), (ii), (6.4.2), (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii) and (3.3.5), (ii).

(ii) We already know that g is flat (2.2.13); in addition, for every $z \in \mathrm{Z}$, f_z is faithfully flat (2.2.13). The conclusion follows then respectively from (6.3.5), (i), (6.4.2), (6.5.2), (i), (6.5.3), (i), (6.5.4), (ii) and (2.1.13). □

Remark (6.6.2). — Suppose h flat, f faithfully flat, and suppose that for every $z \in \mathrm{Z}$, $h^{-1}(z)$ is of coprofundity $\leqslant n$ (5.7.1); then it follows from the reasoning of (6.6.1), (ii) and (6.3.2) that $g^{-1}(z)$ is of coprofundity $\leqslant n$ for every $z \in g(\mathrm{Y})$ and that for every $y \in \mathrm{Y}$, $f^{-1}(y)$ is of coprofundity $\leqslant n$.

6.7. Application to base changes in algebraic preschemes **IV-2 | 146**

Proposition (6.7.1). — *Let k be a field, X a k-prescheme locally Noetherian, $\mathscr{F}$ an $\mathscr{O}_{\mathrm{X}}$-Module coherent. Let k' be an extension of k; set $\mathrm{X}' = \mathrm{X} \otimes_k k'$, $\mathscr{F}' = \mathscr{F} \otimes_k k'$ and let $p : \mathrm{X}' \to \mathrm{X}$ be the canonical projection. We suppose either that X is locally of finite type over k, or that k' is an extension of finite type of k, so that in both cases X' is locally Noetherian. Let x' be a point of X', $x = p(x')$. Then:*

label=(i) *We have $\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{F}'_{x'})$; in particular, for $\mathscr{F}_x$ to be an $\mathscr{O}_x$-module of Cohen-Macaulay, it is necessary and sufficient that $\mathscr{F}'_{x'}$ be an $\mathscr{O}_{x'}$-module of Cohen-Macaulay.*

lbbel=(ii) *For $\mathscr{F}$ to satisfy the property (S_n) at the point x, it is necessary and sufficient that $\mathscr{F}'$ satisfy (S_n) at the point x'.*

Proof. We know that p is faithfully flat (2.2.13); the two assertions are then consequences respectively of (6.3.2) and (6.4.2), provided that one proves that $p^{-1}(x) = \mathrm{Spec}(\boldsymbol{k}(x) \otimes_k k')$ is a Cohen-Macaulay prescheme; as one of the two fields $\boldsymbol{k}(x)$, k' is an extension of finite type of k by hypothesis, we are reduced to proving the following: □

Lemma (6.7.1.1). — *Let K, L be two extensions of a field k, one of which is of finite type (so that the ring $\mathrm{K} \otimes_k \mathrm{L}$ is Noetherian). Then $\mathrm{K} \otimes_k \mathrm{L}$ is a Cohen-Macaulay ring.*

Proof. Suppose for example that L is an extension of finite type of k, so that L is extension of a pure extension $k(\mathbf{t})$ of k ($\mathbf{t}$ being a system $(t_i)_{1 \leqslant i \leqslant m}$ of indeterminates). We then set $\mathrm{A} = \mathrm{K} \otimes_k k(\mathbf{t})$, $\mathrm{B} = \mathrm{K} \otimes_k \mathrm{L}$, B is an A-module flat (**0**, 6.2.1) and of finite type; the morphism $h : \mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ is closed and finite, and as every Artinian prescheme is of Cohen-Macaulay, to prove that B is a Cohen-Macaulay ring, it suffices to prove that A is a Cohen-Macaulay ring (6.3.5). Now, if S is the set of nonzero elements of $k[\mathbf{t}]$, then $\mathrm{A} = \mathrm{S}^{-1}\mathrm{A}'$, where $\mathrm{A}' = \mathrm{K} \otimes_k k[\mathbf{t}] = \mathrm{K}[\mathbf{t}]$; by (**0**, 16.5.13), it suffices to prove that A' is regular (**0**, 17.3.7).

B) Let (L_λ) be the filtering family of sub-extensions of L which are of finite type over k; by virtue of A), each of the rings $\mathrm{C}_\lambda = \mathrm{K} \otimes_k \mathrm{L}_\lambda$ is regular; furthermore, for $\mathrm{L}_\lambda \subset \mathrm{L}_\mu$, L_μ is an L_λ-module flat, so C_μ is an C_λ-module flat (**0**, 6.2.1). Since $\mathrm{C} = \mathrm{K} \otimes_k \mathrm{L} = \varinjlim(\mathrm{K} \otimes_k \mathrm{L}_\lambda)$ is Noetherian, we can apply (5.13.7), and C is therefore regular. □ **IV-2 | 147**

Remarks (6.7.5). — (i) In the proof of the fact that $\boldsymbol{P}(\mathrm{X}, x)$ implies $\boldsymbol{P}(\mathrm{X}', x')$, one cannot dispense with the hypothesis that k' is a *separable* extension of k. We have indeed already seen (4.6.1) when $\boldsymbol{P}$ is the property d); but even when X is *geometrically integral* (4.6.2), it can happen that X is *regular* without X' being *normal*.

(ii) As recalled in (i), it follows from (4.6.1) that when X is not a geometrically reduced algebraic prescheme over k, there exist finite radicial extensions k' of k such that $\mathrm{X}' = \mathrm{X} \otimes_k k'$ is not reduced. It is interesting to give an example where X is a regular scheme over k, such that k is *algebraically closed* in the field K of rational functions of X. Let k be a field of characteristic $p > 0$, in which there exist two elements a, b forming a p-free family over k^p. Denoting again by B the ring $k[\mathrm{S}, \mathrm{T}]$, consider this time the polynomial $\mathrm{P}(\mathrm{S}, \mathrm{T}) = \mathrm{T}^p - a\mathrm{S}^p - b$; as $a\mathrm{S}^p + b$ is not a p-th power in $k(\mathrm{S})$, P is irreducible as a polynomial of $k(\mathrm{S})[\mathrm{T}]$, and the scheme $\mathrm{X}_0 = \mathrm{Spec}(\mathrm{B}/\mathrm{PB})$ is therefore an integral affine curve **IV-2 | 148** over k, whose field of rational functions is $\mathrm{K} = k(\mathrm{S})[t]$, where t is a root of P; let us show that k is *algebraically closed* in K. Suppose indeed that K contains an element z algebraic over k and not in k; one would then also have $z \notin k(\mathrm{S})$, so $[k(\mathrm{S})[z] : k(\mathrm{S})] > 1$; as $[\mathrm{K} : k(\mathrm{S})] = p$, and as K is radicial over $k(\mathrm{S})$, one would have $z^p \in k(\mathrm{S})$, so $c = z^p \in k$ since z is algebraic over k; but then one would have $t^p = a\mathrm{S}^p + b \in k^p(\mathrm{S}^p)(c)$ hence a and b would belong to $k^p(c)$, which is absurd. So X is the normalisation of the curve X_0 in K, which is therefore a normal curve (and hence *regular*) over k. If $k' = k(a^{1/p}, b^{1/p})$, it is clear that $a\mathrm{S}^p + b$ is a p-th power in $k'(\mathrm{S})$, hence $\mathrm{K} \otimes_k k'$ is *not reduced*, nor *a fortiori* the scheme $\mathrm{X} \otimes_k k'$.

Definition (6.7.6). — Let k be a field, X a k-prescheme locally Noetherian. We say that X is *geometrically regular* at a point x (resp. has the property (R_n) *geometric*) at a point x, resp. is geometrically normal at a point x) if, *for every* finite extension k' of k, $\mathrm{X}' = \mathrm{X}_{(k')} = \mathrm{X} \otimes_k k'$ is regular (resp. has the property (R_n), resp. is normal) at every point x' of X' whose projection in X is x. We say that X is *geometrically regular* (resp. has the property (R_n) *geometric*, or equivalently is geometrically regular in codimension $\leqslant n$), resp. is *geometrically normal*) if X is geometrically regular (resp. has the property (R_n) geometric, resp. is geometrically normal) at every point.

We say that an algebra A over k is a *geometrically regular ring* (resp. geometrically normal, resp. geometrically reduced, resp. having the property (R_n) geometric) if $\mathrm{Spec}(\mathrm{A})$ has the same property.

If $\mathrm{X} = \mathrm{Spec}(\mathrm{K})$, where K is an extension of k, then saying that X is geometrically regular, or geometrically normal, or *geometrically reduced*, amounts to saying that K is a *separable* extension of k: this follows from (4.6.1) and the fact that if K is a separable extension of k and k' a finite extension of k, $\mathrm{K} \otimes_k k'$ is composed of a direct product of a finite number of fields (Bourbaki, *Alg.*, chap. VIII, § 7, no 3, cor. 1 of th. 1).

IV-2 | 149

Proposition (6.7.7). — *Let k be a field,* X *a k-prescheme locally Noetherian, x a point of* X*; denote by $Q(k')$ the relation "k' is an extension of k, and $\boldsymbol{P}(\mathrm{X}_{(k')}, x')$ is true for every point x' of $\mathrm{X}_{(k')}$ whose projection in* X *is x", where $\boldsymbol{P}(\mathrm{Z}, z)$ is one of the properties* a), b), c) *of the statement* (6.7.4). *Then the following properties are equivalent:*

label=a) *$Q(k')$ is true for every finite extension k' of k.*
lbbel=b) *$Q(k')$ is true for every finite radicial extension k' of k.*
lcbel=c) *$Q(k')$ is true for every extension of finite type k' of k.*

Suppose in addition that X *is locally of finite type over k; then the three preceding properties are also equivalent to the following:*

ldbel=d) *$Q(k')$ is true for every extension k' of k.*
lebel=e) *$Q(k')$ is true for a perfect extension k' of k.*
lfbel=f) *$Q(k')$ is true for every extension $k' = k^{p^{-s}}$, where p is the characteristic exponent of k, and $s > 0$.*

Proof. To prove the equivalence of a), b) and c), it clearly suffices to show that b) implies c). Let K be then an extension of finite type of k, which is then the field of fractions of a k-algebra of finite type A; set $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$. We know (4.6.6) that there exists a finite and radicial extension k' of k such that $\mathrm{Y}' = (\mathrm{Y} \otimes_k k')_{\mathrm{red}}$ is separable over k'; if η' is the generic point of Y', $\mathrm{K}' = \boldsymbol{k}(\eta')$ is a separable extension of k' by (4.6.1). This being so, $\boldsymbol{P}(\mathrm{X}_{(k')}, x')$ is true by hypothesis for every point x' of $\mathrm{X}_{(k')}$ above x, so it follows from (6.7.4) that $\boldsymbol{P}(\mathrm{X}_{(\mathrm{K}')}, x'')$ is true for every point x'' of $\mathrm{X}_{(\mathrm{K}')}$ above x, since x'' is above a point x' of $\mathrm{X}_{(k')}$, itself above x, and that K' is separable over k'. But we also have $\mathrm{X}_{(\mathrm{K}')} = (\mathrm{X}_{(\mathrm{K})})_{(\mathrm{K}')}$ and for every $x_0 \in \mathrm{X}_{(\mathrm{K})}$ above x, there exists $x'' \in \mathrm{X}_{(\mathrm{K}')}$ above x_0; it therefore follows from (6.7.4) that $\boldsymbol{P}(\mathrm{X}_{(\mathrm{K})}, x_0)$ is true.

Suppose now that X is locally of finite type over k, so that for *every* extension K of k, $\mathrm{X}_{(\mathrm{K})}$ is locally Noetherian. As a radicial extension of k is isomorphic to a sub-extension of a perfect extension of k, it follows immediately from (6.7.4) that e) implies f); also, every finite radicial extension of k is contained in an extension $k^{p^{-s}}$, so f) implies b); d) trivially implies e), and finally e) implies d). Indeed, if K is any extension *whatever* of k, there exists an extension K' of k containing (up to k-isomorphisms) k' and K; since K' is a separable extension of k', one deduces from (6.7.4) that $Q(k')$ is true, then that $Q(\mathrm{K})$ is true, reasoning as in the first part of the proof. It therefore suffices to prove that b) *implies* e). We shall take $k' = k^{p^{-\infty}}$ in e).

The question being local on X, one can furthermore limit oneself to the case where X is affine and of finite type over k, by replacing X by a neighbourhood of x; let then $X = \mathrm{Spec}(B)$, where B is a k-algebra of finite type; furthermore, by definition of the property $\boldsymbol{P}$ in the cases considered, one can also replace X by the local scheme of X at the point x, hence suppose that B is a *local* Noetherian ring. Set $B' = B \otimes_k k'$; k' is the inductive limit of its finite sub-extensions k'_λ and if one sets $B'_\lambda = B \otimes_k k'_\lambda$, the morphism $f_\lambda : \mathrm{Spec}(B') \to \mathrm{Spec}(B'_\lambda)$ is a *homeomorphism* over $\mathrm{Spec}(B'_\lambda)$ for every λ;

indeed f_λ is bijective by virtue of (**I**, 3.5.2), (3.5.7) and (3.5.8); on the other hand, it is closed by (**II**, 6.1.10). One can now apply (5.13.6) which proves (by virtue of the hypothesis b)) that b) implies e) when $\boldsymbol{P}(Z, z)$ is the property (R_n) at the point z. The case where $\boldsymbol{P}$ is the property of being regular (i.e. of having the property (R_n) for every n) follows trivially. Finally, taking into account Serre's normality criterion (5.8.6), b) implies e) further when $\boldsymbol{P}(Z, z)$ is the property of being normal at the point z, since this is equivalent to saying that Z has at the point z the properties (S_2) and (R_1): one applies what precedes for $n = 1$, and (6.7.2), (ii) for $n = 2$. ☐ **IV-2** | 150

Corollary (6.7.8). — *Let k be a field; for a k-prescheme locally Noetherian* X *and a point $x \in X$, denote by* $\boldsymbol{P}(X, x)$ *any of the following properties:*

label=(i) $\mathrm{coprof}(\mathscr{O}_x) \leqslant n$.
lbbel=(ii) $\mathscr{O}_x$ *is a Cohen-Macaulay ring.*
lcbel=(iii) X *satisfies the property* (S_n) *at the point x.*
ldbel=(iv) X *is geometrically regular at the point x.*
lebel=(v) X *has the property* (R_n) *geometric at the point x.*
lfbel=(vi) X *is geometrically normal at the point x.*
lgbel=(vii) X *is geometrically reduced (i.e. separable) at the point x.*

Let k' be an extension of k; one supposes either that k' is of finite type, or that X *is locally of finite type over k, so that $X' = X \otimes_k k'$ is locally Noetherian. Let x' be a point of X' whose projection in* X *is x. Then, for* $\boldsymbol{P}(X, x)$ *to be true, it is necessary and sufficient that* $\boldsymbol{P}(X', x')$ *be so.*

Proof. This has already been seen for the property (vii) (4.6.11), and for the properties (i), (ii) and (iii) (6.7.1). For (iv), (v) and (vi), this follows from (6.7.7): indeed, the fact that the condition is necessary follows from the equivalence of the criteria a) and c), and the equivalence of c) and d) when X is locally of finite type over k. To see that the condition is sufficient, let k'' be a finite radicial extension of k; one can always consider k' and k'' as sub-extensions of an extension K of k; set $X'' = X \otimes_k k''$, $X_0 = X \otimes_k K$, and note that since k'' is a radicial extension of k, there is only *one* point x'' of X'' above x (**I**, 3.5.7) and (3.5.8). If then x_0 is any point of X_0 above x, it is of the same kind as $\boldsymbol{P}(X_0, x_0)$ by virtue of (6.7.7), c) and d)), and if k' is an extension of k of finite type, one can suppose that it is also locally of finite type over k', X' is also locally of finite type. One deduces then from (6.7.4) that the property $Q(k'')$ is true (with the notations of (6.7.7))), hence $\boldsymbol{P}(X, x)$ is true by (6.7.7), b)). ☐

6.8. Regular, normal, reduced, smooth morphisms **IV-2** | 150

Definition (6.8.1). — Let X, Y be two preschemes, $f : X \to Y$ a morphism such that the fibre $f^{-1}(y)$ is a locally Noetherian prescheme for every $y \in Y$, x a point of X. We say that f is respectively a morphism:

label=(i) of coprofundity $\leqslant n$ at the point x;
lbbel=(ii) of Cohen-Macaulay at the point x; **IV-2** | 151
lcbel=(iii) (S_n) at the point x;
ldbel=(iv) regular at the point x;
lebel=(v) (R_n) at the point x;
lfbel=(vi) normal at the point x;
lgbel=(vii) reduced at the point x;

if f is flat at the point x, and if in addition the corresponding property $\boldsymbol{P}(f^{-1}(f(x)), x)$ (notation of (6.7.8)) is true.

We say that f is *smooth* at the point x if it is locally of finite presentation in a neighbourhood of x and if it is regular at the point x. We say respectively that f is a morphism: of coprofundity $\leqslant n$, of Cohen-Macaulay, (S_n), regular, (R_n), normal, reduced, smooth, if it has the corresponding property at every point of X.

Proposition (6.8.2). — *Let* X, Y *be two locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a morphism, x a point of* X. *Denote by $\boldsymbol{M}(f, x)$ one of the properties (i) to (vii) of the definition (6.8.1), or the property of being smooth at the point x. Let Y' be a locally Noetherian prescheme, $g : \mathrm{Y}' \to \mathrm{Y}$ a morphism, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$. Suppose that f or g is locally of finite type. Then, for every $x' \in \mathrm{X}'$ above x, the property $\boldsymbol{M}(f, x)$ implies $\boldsymbol{M}(f', x')$.*

Proof. Set $y = f(x)$, $y' = f'(x')$; by transitivity of fibres (**I**, 3.6.4), since either $f^{-1}(y)$ or $\boldsymbol{k}(y')$ is locally of finite type over $\boldsymbol{k}(y)$ (**I**, 6.4.11), it follows from (6.7.8) that the properties $\boldsymbol{P}(f^{-1}(y), x)$ and $\boldsymbol{P}(f'^{-1}(y'), x')$ are equivalent; furthermore, since f is flat at the point x, f' is flat at the point x' (2.1.4), which proves the proposition, taking into account (1.4.3), (iii). □

Proposition (6.8.3). — *For a morphism f of locally Noetherian preschemes, let $\boldsymbol{M}(f)$ be one of the following properties: being Cohen-Macaulay, (S_n), regular, (R_n), normal, reduced.*

label=(i) Let X, Y, Z *be three locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$, $g : \mathrm{Y} \to \mathrm{Z}$ two morphisms. If $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g)$ are true, $\boldsymbol{M}(g \circ f)$ is true.*

lbbel=(ii) Conversely, if f is surjective and if $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g \circ f)$ are true, then $\boldsymbol{M}(g)$ is true.

lcbel=(iii) Let X, Y, Y′ *be three locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$, $h : \mathrm{Y}' \to \mathrm{Y}$ two morphisms; set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$. Suppose that f or h is locally of finite type. If $\boldsymbol{M}(f)$ is true, then so is $\boldsymbol{M}(f')$; the converse is true when h is faithfully flat.*

Proof. The conclusions of (i) and (iii) are also true when $\boldsymbol{M}$ is the property of being smooth and (in (iii)) h is quasi-compact.

(i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f and $g \circ f$ are flat and f is surjective for every $z \in \mathrm{Z}$, the morphism $f_z = f \otimes \mathbf{1}_{\boldsymbol{k}(z)} : u^{-1}(z) \to g^{-1}(z)$ is flat (resp. faithfully flat if f is) and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4). If $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g)$ are true, $\boldsymbol{M}(g \circ f)$ is then true for the cases where $\boldsymbol{M}$ is the property of being Cohen-Macaulay or (S_n), by virtue of (6.6.1). On the other hand, let k' be a finite extension of $\boldsymbol{k}(z)$; setting $\mathrm{Y}'_z = g^{-1}(z) \otimes_{\boldsymbol{k}(z)} k'$, $\mathrm{X}'_z = u^{-1}(z) \otimes_{\boldsymbol{k}(z)} k'$, and $f'_z = f_z \otimes \mathbf{1}_{k'} : \mathrm{X}'_z \to \mathrm{Y}'_z$; the morphism f'_z
is flat (resp. faithfully flat) and for every $y' \in \mathrm{Y}'_z$, the fibre $f_z'^{-1}(y')$ is isomorphic to $f^{-1}(y) \otimes_{\boldsymbol{k}(y)}$ **IV-2** | 152
$\boldsymbol{k}(y')$, denoting by y the image of y' in $g^{-1}(z)$ (**I**, 3.6.4). When $\boldsymbol{M}$ is the property of being regular, (R_n), normal or reduced, the hypothesis that $\boldsymbol{M}(f)$ and $\boldsymbol{M}(g)$ are true implies that Y'_z and each of the fibres $f_z'^{-1}(y')$ for $y' \in \mathrm{Y}'_z$ have the corresponding property among the properties c), d), e), f) of (6.6.1); one deduces from (6.6.1), (i) that X'_z has the same property, hence $\boldsymbol{M}(g \circ f)$ is true.

(ii) Conversely, the hypothesis that $\boldsymbol{M}(g \circ f)$ and $\boldsymbol{M}(f)$ are true and that f is surjective implies from (6.6.1), (ii)), f'_z being surjective for every z; hence $\boldsymbol{M}(g)$ is true.

(iii) The first assertion follows immediately from (6.8.2). On the other hand, if h is faithfully flat and f' flat, f is flat (2.4.1); since, with the notations of (6.8.2), the properties $\boldsymbol{P}(f^{-1}(y), x)$ and $\boldsymbol{P}(f'^{-1}(y'), x')$ are equivalent (6.7.8), one sees that $\boldsymbol{M}(f)$ and $\boldsymbol{M}(f')$ are then equivalent.

Finally, the last assertion of the proposition follows from (1.4.3), (iii) and (2.7.1), (iv). □

Remarks (6.8.4). — (i) If f is faithfully flat, g flat and if $g \circ f$ is of coprofundity $\leqslant n$, then g is of coprofundity $\leqslant n$, as follows from the reasoning of (6.6.2).

(ii) When, in (6.8.1), one takes for Y the spectrum of a field k, the notions (iv), (v) and (vi) reduce to those defined in (6.7.5). It is clear that these latter are relative to the base field k. The definition (6.8.1) thus leads to saying that a prescheme X is "regular (resp. (R_n), resp. normal) over k" instead of saying that it is "geometrically regular (resp. (R_n), resp. normal) relative to k"; one will avoid confusing this notion with the property of being regular (resp. (R_n), resp. normal) which is independent of k. One can make in this connection the same remarks as in (4.5.12).

Proposition (6.8.5). — *Let k be a field,* X, Y *two k-preschemes locally Noetherian, at least one of which is locally of finite type over k. For a k-prescheme* Z*, denote by $\boldsymbol{P}(\mathrm{Z})$ one of the properties c) to f) of (6.6.1); the property "$\boldsymbol{P}(\mathrm{Z})$ geometric" is then defined in (6.7.6) (resp. (4.6.1)). Then:*

label=(i) If X *has the property $\boldsymbol{P}$, and* Y *the property $\boldsymbol{P}$ geometric, $\mathrm{X} \times_k \mathrm{Y}$ has the property $\boldsymbol{P}$.*

lbbel=(ii) If X *and* Y *have the property $\boldsymbol{P}$ geometric, so does $\mathrm{X} \times_k \mathrm{Y}$.*

Proof. Indeed, let $f : \mathrm{X} \times_k \mathrm{Y} \to \mathrm{X}$, $g : \mathrm{X} \to \mathrm{Spec}(k)$ be the structural morphisms, which are faithfully flat (2.2.13). The hypothesis that Y has the property $\boldsymbol{P}$ geometric implies by virtue of (6.8.2) that

$\boldsymbol{M}(f)$ is true, $\boldsymbol{M}$ being the property (6.8.3) which corresponds to $\boldsymbol{P}$; under the hypothesis (ii), $\boldsymbol{M}(g)$ is also true, hence the assertion (ii) follows from (6.8.3), (i). As for the assertion (i), it follows directly from (6.6.1), (i). □

Theorem (6.8.6). — *Let* X, Y *be two locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* x *a point of* X, $y = f(x)$. *The following properties are equivalent:*

label=a) f *is smooth at the point* x.

lbbel=b) f *is regular at the point* x. IV-2 | 153

lcbel=c) $\mathscr{O}_x$ *is a formally smooth* $\mathscr{O}_y$*-algebra* (**0**, 19.3.1) *for the preadic topologies on* $\mathscr{O}_x$ *and* $\mathscr{O}_y$.

c′) $\mathscr{O}_x$ *is a formally smooth* $\mathscr{O}_y$*-algebra* (**0**, 19.3.1) *for the discrete topologies on* $\mathscr{O}_x$ *and* $\mathscr{O}_y$.

Proof. The equivalence of *a*) and *b*) follows from the definitions (6.8.1) and from the fact that for locally Noetherian preschemes, morphisms locally of finite type are locally of finite presentation (1.4.2).

In the second place, for $\mathscr{O}_x$ to be a formally smooth $\mathscr{O}_y$-algebra for the preadic topologies, it is necessary and sufficient that $\mathscr{O}_x$ be a flat $\mathscr{O}_y$-module and that $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ be a $\boldsymbol{k}(y)$-algebra formally smooth for its preadic topology (**0**, 19.7.1); but for $\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ to be a $\boldsymbol{k}(y)$-algebra formally smooth for its preadic topology, it is necessary and sufficient that it be a $\boldsymbol{k}(y)$-algebra geometrically regular (**0**, 19.6.6); this proves the equivalence of *b*) and *c*). Finally, to prove the equivalence of *c*) and *c*′), one can limit oneself to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{C})$ are affine, A being Noetherian and C a Noetherian A-algebra of finite type, which can therefore be written in the form $\mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathfrak{J}$. As here $\mathfrak{J}/\mathfrak{J}^2$ is a C-module of finite presentation, the equivalence of *c*) and *c*′) follows from (**0**, 22.6.4) applied to A, $\mathrm{B} = \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n]$ and $\mathrm{C} = \mathrm{B}/\mathfrak{J}$. □

Corollary (6.8.7). — *Let* X, Y *be two locally Noetherian preschemes,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type. Then the set of points* $x \in \mathrm{X}$ *where* f *is smooth* (*or regular*) *is open in* X.

Proof. Indeed it follows from (**0**, 22.6.5) that the set of $x \in \mathrm{X}$ satisfying the condition *c*′) of (6.8.6) is open in X, and we conclude by (6.8.6). □

Remark (6.8.8). — In (17.5.1), we shall show that the equivalence of *b*) and *c*′) in (6.8.6), as well as the corollary (6.8.7), remain valid without the Noetherian hypothesis on X and Y, provided that one restricts to morphisms *locally of finite presentation*.

6.9. The theorem of generic flatness IV-2 | 153

Theorem (6.9.1). — *Let* Y *be a locally Noetherian and integral prescheme,* $u : \mathrm{X} \to \mathrm{Y}$ *a morphism of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module. There then exists a non-empty open subset* U *of* Y *such that* $\mathscr{F}|u^{-1}(\mathrm{U})$ *is flat over* U.

Proof. One can clearly limit oneself to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is affine; then X is a finite union of affine opens X_i of finite type over Y; if, for each i, there exists a non-empty open U_i in Y such that $\mathscr{F}|(\mathrm{X}_i \cap u^{-1}(\mathrm{U}_i))$ is flat over U_i, it is clear that taking for U the intersection of the U_i, we shall have answered the question; one can therefore suppose that $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, where B is an A-algebra of finite type. We then have $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module of finite type; the theorem will follow from the following: □

Lemma (6.9.2). — *Let* A *be an integral Noetherian ring,* B *an* A*-algebra of finite type,* M *a* B*-module of finite type. Then there exists* $f \neq 0$ *in* A *such that* M_f *is an* A_f*-module free.*

Proof. Denote by K the field of fractions of A; then $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K}$ is an algebra of finite type over K and $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ is a $(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K})$-module of finite type. We shall reason by induction on the *dimension* n of the support of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$, which is either $-\infty$ or finite and $\geqslant 0$. Suppose first $n = -\infty$, i.e. $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K} = 0$; if $(m_i)_{1 \leqslant i \leqslant r}$ is a system of generators of the B-module M, there therefore exists $f \neq 0$ in A such that $f m_i = 0$ for $1 \leqslant i \leqslant r$; hence $\mathrm{M}_f = 0$ and the lemma is true in this case. Suppose now $n \geqslant 0$. We know that there exists a composition series $\mathrm{M} = \mathrm{M}_1 \supset \mathrm{M}_2 \supset \dots \supset \mathrm{M}_q = 0$ of the B-module M such that each of the quotients $\mathrm{N}_i = \mathrm{M}_i/\mathrm{M}_{i+1}$ is isomorphic to a B-module of the form $\mathrm{B}/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of B (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 4, th. 1). If the theorem is true for each of the N_i, there exists for each i an $f_i \neq 0$ in A such that $(\mathrm{N}_i)_{f_i}$ is an A_{f_i}-module free for $1 \leqslant i \leqslant q-1$. But IV-2 | 154

$(\mathrm{N}_i)_f = (\mathrm{M}_i)_f/(\mathrm{M}_{i+1})_f$ (**0**, 1.3.2) and since an extension of free modules is free, we deduce that M_f is free, setting $f = f_1 f_2 \cdots f_{q-1}$, and it therefore suffices (in the case where $\mathrm{M} = \mathrm{B}$ and B is *integral*) to show that there exists $g \neq 0$ in A and elements t_i ($1 \leqslant i \leqslant m$) algebraically independent over A and such that B_g is *integral* over $\mathrm{A}_g[t_1, \ldots, t_m]$. One can replace A by A_g, B by B_g and suppose that B is integral over $\mathrm{C} = \mathrm{A}[t_1, \ldots, t_m]$, hence a C-module of finite type and *without torsion*. We know furthermore (4.1.2) that the dimension of $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{K})$ is equal to m, hence $m = n$.

This being so, if h is the rank of the C-module without torsion B, there exists a short exact sequence of C-modules

$$0 \longrightarrow \mathrm{C}^h \longrightarrow \mathrm{B} \longrightarrow \mathrm{M}' \longrightarrow 0$$

where M' is a C-module *of torsion* of finite type; the support of M' does not therefore contain the generic point of Spec(C) (**I**, 7.4.6) and hence the support of $\mathrm{M}' \otimes_{\mathrm{A}} \mathrm{K}$ does not contain the generic point of $\mathrm{Spec}(\mathrm{C} \otimes_{\mathrm{A}} \mathrm{K})$ (**I**, 9.1.13.1); one concludes (4.1.2.1) that its dimension is $< n$. By the induction hypothesis, there exists $f \neq 0$ in A such that M'_f is an A_f-module free; hence it is the same for C^h_f, which is the same as B_f, and who in turn of (**0**, 1.3.2) is an extension of free modules. ☐

Corollary (6.9.3). — *Let* S *be a Noetherian prescheme,* $u : \mathrm{X} \to \mathrm{S}$ *a morphism of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module. There then exists a partition* $(\mathrm{S}_i)_{1 \leqslant i \leqslant n}$ *of* S *into a finite number of locally closed subsets of* S*, such that, if one still denotes by* S_i *the reduced sub-prescheme having* S_i *for underlying space, and if one sets* $\mathrm{X}_i = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}_i$*, the* $\mathscr{O}_{\mathrm{X}_i}$*-Module* $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{O}_{\mathrm{S}_i}$ *is flat over* S_i*.*

Proof. We proceed by Noetherian induction (**0**, 2.2.2) on the set of closed subsets T of S such that the reduced sub-prescheme having T for underlying space satisfies the conclusion of (6.9.3). One can therefore limit oneself to proving the corollary for S by supposing that it is true for every closed sub-prescheme reduced and having as underlying space a closed subset $\neq$ S. Since the morphism $\mathrm{S}_{\mathrm{red}} \to \mathrm{S}$ is of finite type and surjective, one can replace S by $\mathrm{S}_{\mathrm{red}}$, hence suppose S reduced and non-empty. Since S is Noetherian, the interior T of an irreducible component of S is non-empty, and the prescheme induced on T is integral; there therefore exists by virtue of (6.9.1) a non-empty open $\mathrm{U} \subset \mathrm{T}$

such that $\mathscr{F}|u^{-1}(\mathrm{U})$ is flat over U. If Y is then the sub-prescheme reduced of S having for underlying space $\mathrm{S} - \mathrm{U}$, there exists by hypothesis a partition (Y_i) of Y in locally closed subsets of Y (hence of S) such that $\mathscr{F}_i = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} \mathscr{O}_{\mathrm{Y}_i}$ is flat over Y_i for every i; it is clear that the Y_i and U form a partition answering the question. ☐ **IV-2** | 155

6.10. Dimension and depth of a Module normally flat along a closed sub-prescheme **IV-2** | 155

(6.10.1). Let X be a locally Noetherian prescheme, $\mathscr{J}$ a quasi-coherent Ideal of $\mathscr{O}_{\mathrm{X}}$, Y the closed sub-prescheme of X defined by $\mathscr{J}$, $j : \mathrm{Y} \to \mathrm{X}$ the canonical injection. For every integer $k \geqslant 0$, the $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}$ is annihilated by $\mathscr{J}$, hence of the form $j_*(\mathscr{G}_k)$, where $\mathscr{G}_k = j^*(\mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}) = j^*(\mathscr{J}^k\mathscr{F})$ is a coherent $\mathscr{O}_{\mathrm{Y}}$-Module. By abuse of language, we shall denote by $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ the graded $\mathscr{O}_{\mathrm{Y}}$-Module equal to the direct sum

$$\bigoplus_{k=0}^{\infty} \mathscr{G}_k = j^*\Big(\bigoplus_{k=0}^{\infty} \mathscr{J}^k\mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}\Big);$$

in particular, we have $\mathrm{gr}^0_{\mathscr{J}}(\mathscr{F}) = \mathscr{G}_0 = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{Y}} = j^*(\mathscr{F})$. We shall say (following Hironaka) that $\mathscr{F}$ is *normally flat along* Y if $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ is a flat $\mathscr{O}_{\mathrm{Y}}$-Module. This amounts to the same (2.1.2) and (**0**, 6.1.2) as saying that each of the $\mathscr{O}_{\mathrm{Y}}$-Modules $\mathscr{G}_k = j^*(\mathscr{J}^k\mathscr{F})$ is *locally free*.

Proposition (6.10.2). — *Let* X *be a locally Noetherian prescheme,* Y *a closed integral sub-prescheme of* X*. For every coherent* $\mathscr{O}_{\mathrm{X}}$*-Module* $\mathscr{F}$*, there exists an open* U *of* X *such that* $\mathrm{Y} \cap \mathrm{U} \neq \varnothing$ *and that* $\mathscr{F}|\mathrm{U}$ *is normally flat along* $\mathrm{Y} \cap \mathrm{U}$*.*

Proof. Indeed, let $\mathscr{J}$ be the coherent Ideal of $\mathscr{O}_{\mathrm{X}}$ defining Y; the $\mathscr{O}_{\mathrm{Y}}$-Algebra $\mathscr{B} = \mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{O}_{\mathrm{X}})$ is quasi-coherent and *of finite type* car elle est engendrée par $\mathrm{gr}^1_{\mathscr{J}}(\mathscr{O}_{\mathrm{X}})$, inverse image of $\mathscr{J}/\mathscr{J}^2$; as $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$ is a $\mathscr{B}$-module quasi-coherent generated by $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$, it is a $\mathscr{B}$-Module *of finite type*. If one sets $\mathrm{Z} = \mathrm{Spec}(\mathscr{B})$, the structural morphism $u : \mathrm{Z} \to \mathrm{Y}$ is of finite type, and if $\mathscr{H}$ is the coherent

$\mathscr{O}_Z$-Module such that $u_*(\mathscr{H}) = \mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{F})$, it suffices to apply to u and to $\mathscr{H}$ the theorem of generic flatness (6.9.1) to prove the proposition. □

Proposition (6.10.3). — *The notations being those of (6.10.1), suppose that $\mathscr{F}$ is normally flat along Y. Then:*

label=(i) *$\mathrm{Y} \cap \mathrm{Supp}(\mathscr{F})$ is a part both open and closed of Y (in other words, a union of connected components of Y).*

lbbel=(ii) *If $\mathscr{J}$ is locally nilpotent, $\mathrm{Supp}(\mathscr{F})$ is an open and closed part of X, and for every $x \in \mathrm{Supp}(\mathscr{F})$, we have*

(6.10.3.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{\mathrm{Y},x})$$

(6.10.3.2) $$\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{O}_{\mathrm{Y},x})$$

(6.10.3.3) $$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{O}_{\mathrm{Y},x}).$$

Proof. (i) With the notations of (6.10.1), we have $\mathrm{Y} \cap \mathrm{Supp}(\mathscr{F}) = \mathrm{Supp}(\mathscr{G}_0)$ (**I**, 9.1.13), and as by hypothesis $\mathscr{G}_0$ is a locally free $\mathscr{O}_\mathrm{Y}$-Module of finite type, its support is at the same time open and closed in Y.

(ii) The hypothesis that $\mathscr{J}$ is locally nilpotent implies that the underlying spaces of X and Y are the same, whence the two first assertions of (ii), taking into account (i); it remains to prove (6.10.3.2) and (6.10.3.3) by difference. Let $(f_i)_{1 \leqslant i \leqslant p}$ be a finite family of elements of the maximal ideal of $\mathscr{O}_{\mathrm{X},x}$ whose images in $\mathscr{O}_{\mathrm{Y},x} = \mathscr{O}_{\mathrm{X},x}/\mathscr{J}_x$ form a maximal $\mathscr{O}_{\mathrm{Y},x}$-regular sequence; the hypothesis on $\mathscr{F}$ and $\mathscr{J}$ implies that the $\mathscr{O}_{\mathrm{Y},x}$-module $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x)$ is free of finite type, hence the suite (f_i) is $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x)$-regular; one deduces that this suite is also $\mathscr{F}_x$-regular (**0**, 15.1.19). On the other hand, let n be the largest integer such that $\mathscr{F}_x^{(n)} = \mathscr{J}_x^n \mathscr{F}_x \neq 0$. The hypothesis implies also that $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{F}_x/\mathscr{F}_x^{(n)})$ is free of finite type, hence the suite (f_i) is also $(\mathscr{F}_x/\mathscr{F}_x^{(n)})$-regular (*loc. cit.*). Applying the lemma (3.4.1.4), one concludes by induction on i that there is an exact sequence **IV-2 | 156**

$$0 \longrightarrow \mathscr{F}_x^{(n)} \Big/ \Big(\sum_{i=1}^p f_i \mathscr{F}_x^{(n)} \Big) \longrightarrow \mathscr{F}_x \Big/ \Big(\sum_{i=1}^p f_i \mathscr{F}_x \Big).$$

But the hypothesis implies that $\mathscr{F}_x^{(n)}$ is a free $\mathscr{O}_{\mathrm{Y},x}$-module of finite type and $\neq 0$, hence $\mathscr{F}_x^{(n)}/(\sum_{i=1}^p f_i \mathscr{F}_x^{(n)})$ is isomorphic to a module of the form $(\mathscr{O}_{\mathrm{Y},x}/(\sum_{i=1}^p f_i \mathscr{O}_{\mathrm{Y},x}))^m$ with $m > 0$; as $\mathrm{prof}(\mathscr{O}_{\mathrm{Y},x}/(\sum_{i=1}^p f_i \mathscr{O}_{\mathrm{Y},x})) = 0$ (**0**, 16.4.6) one also has, by the characterisation (**0**, 16.4.6), (i) of modules of zero depth, $\mathrm{prof}(\mathscr{F}_x^{(n)}/(\sum_{i=1}^p f_i \mathscr{F}_x^{(n)})) = 0$, then $\mathrm{prof}(\mathscr{F}_x/(\sum_{i=1}^p f_i \mathscr{F}_x)) = 0$; the $(f_i)_x$ belonging to the maximal ideal of $\mathscr{O}_{\mathrm{Y},x}$ and forming a maximal $\mathscr{F}_x$-regular suite, this shows that $\mathrm{prof}(\mathscr{F}_x) = p$ (**0**, 16.4.6), (ii). □

Corollary (6.10.4). — *Let $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ (resp. $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$) be the set of $x \in \mathrm{X}$ such that $\mathscr{F}$ satisfies (S_n) (resp. such that $\mathrm{coprof}(\mathscr{F}_x) \leqslant n$). If $\mathscr{F}$ is normally flat along Y, and if $\mathscr{J}$ is locally nilpotent, then $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F}) = \mathrm{U}_{\mathrm{S}_n}(\mathscr{O}_\mathrm{Y})$ and $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F}) = \mathrm{U}_{\mathrm{C}_n}(\mathscr{O}_\mathrm{Y})$.*

Proposition (6.10.5). — *The notations being those of (6.10.1), suppose that Y is connected and non-empty, and that $\mathscr{F}$ is normally flat along Y. For every integer $n \geqslant 0$, set*

(6.10.5.1) $$r(n) = \mathrm{rg}_{\mathscr{O}_\mathrm{Y}}(\mathrm{gr}^n_{\mathscr{J}}(\mathscr{F}))$$

(the locally free $\mathscr{O}_\mathrm{Y}$-Module $\mathrm{gr}^n_{\mathscr{J}}(\mathscr{F})$ being necessarily of constant rank). Then:

label=(i) *There exists a polynomial $\mathrm{P} \in \mathbf{Q}[\mathrm{T}]$ such that $\mathrm{P}(n) = r(n)$ for all sufficiently large n.*

lbbel=(ii) *We have $\mathrm{Y} \cap \mathrm{Supp}(\mathscr{F}) = \varnothing$ (in other words $\mathscr{J} | \mathscr{J}\mathscr{F} = 0$), or $\mathrm{Y} \cap \mathrm{Supp}(\mathscr{F}) = \mathrm{Y}$ (in other words $\mathrm{Y} \subset \mathrm{Supp}(\mathscr{F})$).*

In the second case, denote $d-1$ the degree of P; for every maximal point y of Y, we have

(6.10.5.2) $$\dim(\mathscr{F}_y) = d$$

and in particular

(6.10.5.3) $$\mathrm{codim}(\mathrm{Y}, \mathrm{Supp}(\mathscr{F})) = d.$$

lcbel=(iii) *Suppose $\mathrm{Y} \subset \mathrm{Supp}(\mathscr{F})$. For every $x \in \mathrm{Y}$, we have*

(6.10.5.4) $$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{Y,x}) + d.$$

More precisely, there exist in $\mathscr{J}_x$ *elements* f_i $(1 \leqslant i \leqslant d)$ *forming part of a system of parameters for* $\mathscr{F}_x$ (**0**, 16.3.6) *and such that, in* $\mathrm{Spec}(\mathscr{O}_{X,x})$, *we have* $V(\mathscr{J}_x) = V(\sum_{i=1}^{d} f_i \mathscr{O}_{X,x}) \cap \mathrm{Supp}(\mathscr{F})$.

Proof. As Y is assumed connected, the first assertion of (ii) follows from (6.10.3), (i). If $\mathrm{Supp}(\mathscr{F}) \supset$ **IV-2** | 157
Y, and if y is a maximal point of Y; as on a module $\mathscr{F}_y$ of finite length (3.1.4); setting $s(n) = \mathrm{long}(\mathrm{gr}^n_{\mathscr{J}_y}(\mathscr{F}_y))$, we know that there exists a polynomial $R \in \mathbf{Q}[T]$ such that $s(n) = R(n)$ for n large enough, namely the Hilbert-Samuel polynomial of $\mathscr{F}_y$ for the $\mathscr{J}_y$-filtration, $\mathscr{J}_y$-preadic (**0**, 16.2.1). Since the $\mathscr{O}_{Y,y}$-modules $\mathrm{gr}^n_{\mathscr{J}_y}(\mathscr{F}_y)$ are free by hypothesis, we have $r(n) = s(n)/m$, denoting by m the length of the $\mathscr{O}_{X,y}$-module $\mathscr{O}_{Y,y} = \mathscr{O}_{X,y}/\mathscr{J}_y$; so $r(n)$ satisfies (i) in taking $P = \frac{1}{m}R$. We know furthermore (**0**, 16.2.3) that $\deg(H) = \dim(\mathscr{F}_y)$, whence the relation (6.10.5.2); the relation (6.10.5.3) follows from (5.1.12.2).

It remains to prove (iii) for any point $x \in Y$. Set $A = \mathscr{O}_{X,x}$, $\mathfrak{J} = \mathscr{J}_x$, so that $\mathscr{O}_{Y,x} = A/\mathfrak{J}$; let $S = \mathrm{gr}^\bullet_{\mathfrak{J}}(A)$, which is an $(A/\mathfrak{J})$-algebra graded of finite type, with positive degrees, such that $S_0 = A/\mathfrak{J}$, and generated by its elements of degree 1; let finally $N = \mathrm{gr}^\bullet_{\mathfrak{J}}(M)$, which is a graded S-module of finite type, each homogeneous component N_n being by hypothesis a free $(A/\mathfrak{J})$-module of length $r(n)$. Let $\mathfrak{m}$ be the maximal ideal of A, $k = A/\mathfrak{m}$ its residue field; $B = S \otimes_A k$ is a k-algebra graded of finite type, with positive degrees, such that $B_0 = k$, so that $\mathfrak{q} = B^+ = \bigoplus_{n \geqslant 1} B_n$ is the maximal ideal in B; $E = N \otimes_A k$ is a graded B-module of finite type such that $\mathrm{rg}_k(E_n) = r(n)$. Applying (**0**, 16.2.7), and noting that $f_i \in \mathfrak{J}^{n_i}$ $(1 \leqslant i \leqslant d)$ is an element whose image t_i is the image in B_{n_i}. For $n \geqslant \sup(n_i)$, consider the S-sub-module $\sum_{i=1}^{d} f_i \mathfrak{J}^{n-n_i} M$ of M; as the homogeneous component of degree n of the sub-module $\sum_i t_i E$ of E is equal to E_n as soon as n is large enough, one sees that, for n large enough, we have

$$\mathfrak{J}^n M = \sum_{i=1}^{d} f_i \mathfrak{J}^{n-n_i} M + \mathfrak{m}\mathfrak{J}^n M$$

and as $\mathfrak{J}^n M$ is an A-module of finite type, this implies, by Nakayama's lemma,

$$\mathfrak{J}^n M = \sum_{i=1}^{d} f_i \mathfrak{J}^{n-n_i} M \subset \sum_{i=1}^{d} f_i M.$$

If $\mathfrak{a}$ is the annihilator of M, we have therefore (**0**, 1.7.5) in $\mathrm{Spec}(A)$

$$V\Big(\sum_{i=1}^{d} f_i A\Big) \cap V(\mathfrak{a}) \subset V(\mathfrak{J}^n) \cap V(\mathfrak{a}) = V(\mathfrak{J}) \cap V(\mathfrak{a}) = V(\mathfrak{J})$$

since by hypothesis $Y \cap \mathrm{Spec}(A) = V(\mathfrak{J}) \cap \mathrm{Supp}(M) = V(\mathfrak{J})$. Since on the other hand the f_i belong to S, we have $V(\mathfrak{J}) \subset V(\sum_{i=1}^{d} f_i A)$, which proves the last relation of (iii). It remains to show that the f_i form part of a system of parameters for M. Replacing A by $A/\mathfrak{a}$ and the f_i by their images in $A/\mathfrak{a}$, one can limit oneself to the case where $\mathfrak{a} = 0$; one then has to show that we have $\dim(A/(\sum_{i=1}^{d} f_i A)) \leqslant \dim(A) - d$ and it remains therefore to prove (**0**, 16.3.7) that we have

$$\dim(A) \geqslant \dim(A/\mathfrak{J}) + d.$$

Now, let $\mathfrak{p}$ be a prime ideal of A containing $\mathfrak{J}$ and such that $\dim(A/\mathfrak{J}) = \dim(A/\mathfrak{p})$, so that $\mathfrak{p}$ is minimal among the prime ideals containing $\mathfrak{J}$. We have therefore $\mathfrak{p} = \mathfrak{i}_y$ where $y \in \mathrm{Spec}(A)$ is a maximal point of Y. But by virtue of (6.10.5.2) and the hypothesis $\mathrm{Spec}(A) = \mathrm{Supp}(M)$, we have $\dim(A_\mathfrak{p}) = d$; the inequality $\dim(A/\mathfrak{p}) + \dim(A_\mathfrak{p}) \leqslant \dim(A)$ (**0**, 16.1.4) completes the proof. □

Proposition (6.10.6). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module,* Y *a closed irreducible sub-prescheme of* X, *of generic point* $y \in \mathrm{Supp}(\mathscr{F})$. *There then exists a neighbourhood* U *open and non-empty of* y *in* X *such that, for every* $x \in U \cap Y$, *we have*

(6.10.6.1) $$\dim(\mathscr{F}_x) = \dim(\mathscr{F}_y) + \dim(\mathscr{O}_{Y,x})$$

(6.10.6.2) $$\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{F}_y) + \mathrm{prof}(\mathscr{O}_{Y,x})$$

(6.10.6.3) $$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{F}_y) + \mathrm{coprof}(\mathscr{O}_{Y,x}).$$

Proof. Let $Y' = Y_{\mathrm{red}}$, which is a closed integral sub-prescheme of Y having the same underlying space, and defined by a locally nilpotent Ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (**I**, 6.1.6). It follows that $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{O}_v)$ is a coherent $\mathscr{O}_{v'}$-Module, and as Y' is integral, there exists an open neighbourhood V of y in Y' such that $\mathrm{gr}^\bullet_{\mathscr{J}}(\mathscr{O}_v)|V$ is locally free (**0**, 5.2.7); otherwise said, replacing X by a neighbourhood of y, one can suppose that $\mathscr{O}_v$ is *normally flat* along Y'; one then deduces from (6.10.3) that we have **IV-2 | 158**

$$\dim(\mathscr{O}_{Y,x}) = \dim(\mathscr{O}_{Y',x}), \qquad \mathrm{prof}(\mathscr{O}_{Y,x}) = \mathrm{prof}(\mathscr{O}_{Y',x})$$

for every $x \in Y$; this allows us to limit ourselves to the case where the closed sub-prescheme Y is *integral*.

This being so, by virtue of (6.10.2), one can, replacing X by an open neighbourhood of y, suppose that $\mathscr{F}$ is *normally flat* along Y. Suppose now $p = \mathrm{prof}(\mathscr{F}_y)$; replacing at need X by an open neighbourhood of y, one can further suppose that there exist p sections f_i ($1 \leqslant i \leqslant p$) of $\mathscr{O}_X$ above X, forming an $\mathscr{F}$-regular suite, such that the $(f_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathscr{O}_X$ at the point y and that $\mathrm{prof}(\mathscr{F}_y/(\sum_{i=1}^p (f_i)_y \mathscr{F}_y)) = 0$ (**0**, 16.4.6). If $\mathfrak{J}$ is the Ideal of $\mathscr{O}_X$ defining Y, the hypothesis $\mathrm{prof}(\mathscr{G}_y) = 0$ implies that $\mathscr{G}_y$ contains a sub-module isomorphic to $\mathscr{O}_{X,y}/\mathfrak{m}_y$ (**0**, 16.4.6); reasoning as above (taking into account (**0**, 5.3.8) and (5.2.2)) shows that one can suppose that there exists a sub-$\mathscr{O}_X$-Module $\mathscr{G}'$ of $\mathscr{G}$ *isomorphic to* $\mathscr{O}_X/\mathfrak{J}$ so that $\mathscr{G}'_x = \mathscr{O}_{Y,x}$ for every $x \in Y$. Setting $\mathscr{G}'' = \mathscr{G}/\mathscr{G}'$; replacing X by an open neighbourhood of y, one can, by virtue of (6.10.2), suppose that $\mathscr{G}'$ and $\mathscr{G}''$ are *normally flat* along Y. Let then x be any point of Y, and set $q = \mathrm{prof}(\mathscr{O}_{Y,x}) = \mathrm{prof}(\mathscr{G}'_x)$; let $(g_i)_{1 \leqslant i \leqslant q}$ be a maximal $\mathscr{G}''_x$-regular suite of elements of the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_{X,x}$. Each of the homogeneous components of $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{G}''_x)$ is a flat $\mathscr{O}_{Y,x}$-module of finite type by hypothesis, hence a free $\mathscr{O}_{Y,x}$-module, and it is therefore also $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{G}''_x)$-regular, hence it is also $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{G}''_x)$-regular, and whence the result $\mathrm{gr}^\bullet_{\mathscr{J}_x}(\mathscr{G}''_x)$ is $\mathscr{G}''_x$-regular (**0**, 15.1.19). Applying the lemma (**0**, 15.1.18) to the exact sequence

$$0 \longrightarrow \mathscr{G}'_x \longrightarrow \mathscr{G}_x \longrightarrow \mathscr{G}''_x \longrightarrow 0$$

by induction on j, one concludes an exact sequence

$$0 \longrightarrow \mathscr{G}'_x \Big/ \Big(\sum_{j=1}^q g_j \mathscr{G}'_x\Big) \longrightarrow \mathscr{G}_x \Big/ \Big(\sum_{j=1}^q g_j \mathscr{G}_x\Big).$$

But by hypothesis $\mathrm{prof}(\mathscr{G}'_x/(\sum_{j=1}^q g_j \mathscr{G}'_x)) = 0$ (**0**, 16.4.6); by the characterisation of modules of zero depth, one concludes that $\mathrm{prof}(\mathscr{G}_x/(\sum_{j=1}^q g_j \mathscr{G}_x)) = 0 = \mathrm{prof}(\mathscr{F}_x/(\sum_{i=1}^p f_i \mathscr{F}_x + \sum_{j=1}^q g_j \mathscr{F}_x))$; the suite (g_i) being $\mathscr{G}''_x$-regular and $\mathscr{G}''_x$ being $\mathscr{F}_x$-regular, the suite $((f_i)_x)$ is $\mathscr{F}_x$-regular by hypothesis and is formed of elements of the maximal ideal; one deduces (**0**, 16.4.6) that $\mathrm{prof}(\mathscr{F}_x) = p + q$, which completes the proof. □

6.11. Criteria for the sets $U_{S_n}(\mathscr{F})$ or $U_{C_n}(\mathscr{F})$ to be open

Lemma (6.11.1). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module coherent; then the function* $x \mapsto \dim.\mathrm{proj}(\mathscr{F}_x)$ *is upper semi-continuous in* X.

Proof. One can restrict to the case where $X = \mathrm{Spec}(A)$ is the spectrum of a Noetherian ring and $\mathscr{F} = \widetilde{M}$, where M is an A-module of finite type. Suppose that for an $x \in X$, we have $\dim.\mathrm{proj}(M_x) = n < +\infty$ (if $n = +\infty$ there is nothing to prove); there exists a resolution

$$L_{n-1} \longrightarrow L_{n-2} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

where the L_i are free A-modules of *finite type* (A being Noetherian), whence an exact sequence **IV-2 | 159**

$$0 \longrightarrow R \longrightarrow L_{n-1} \longrightarrow L_{n-2} \longrightarrow \cdots \longrightarrow L_0 \longrightarrow M \longrightarrow 0$$

where R is an A-module of finite type; one deduces an exact sequence

$$0 \longrightarrow R_x \longrightarrow (L_{n-1})_x \longrightarrow \cdots \longrightarrow (L_0)_x \longrightarrow M_x \longrightarrow 0$$

where the $(L_i)_x$ are free A_x-modules of finite type; as by hypothesis $\dim.\mathrm{proj}(M_x) = n$, this implies that R_x is a projective A_x-module of finite type (**M**, VI, 2.1) and hence a free A_x-module of finite type ($\mathbf{0}_{\mathrm{III}}$, 10.1.3). We conclude that there exists an open neighbourhood U of x in X such that, for every $x' \in U$, $R_{x'}$ is a free $A_{x'}$-module ($\mathbf{0}_{\mathrm{I}}$, 5.2.7), whence $M_{x'}$ admits a projective resolution of length n, in other words $\dim.\mathrm{proj}(M_{x'}) \leqslant n$, which proves the lemma. □

Proposition (6.11.2). — *Let* X *be a locally Noetherian prescheme and locally immersible in a regular scheme* (5.11.1)*; let* $\mathscr{F}$ *be an* $\mathscr{O}_X$*-Module coherent. Then:*

label=(i) (M. Auslander) *The function* $x \mapsto \mathrm{coprof}(\mathscr{F}_x)$ *is upper semi-continuous in* X *(in other words, for every integer* n*, the set* $U_{C_n}(\mathscr{F})$ *of* $x \in X$ *such that* $\mathrm{coprof}(\mathscr{F}_x) \leqslant n$ *is* open*).*

lbbel=(ii) For every integer n*, the set* $U_{S_n}(\mathscr{F})$ *of* $x \in X$ *such that* $\mathscr{F}$ *has the property* (S_n) *at the point* x *is open.*

Proof. (i) The question being local on X, one can, by virtue of the hypothesis, restrict to the case where X is a closed sub-prescheme of an affine regular scheme Y. If $j : X \to Y$ is the canonical injection, and $\mathscr{G} = j_*(\mathscr{F})$, one knows that one has then, for every $x \in X$, $\dim(\mathscr{F}_x) = \dim(\mathscr{G}_x)$ and $\mathrm{prof}(\mathscr{F}_x) = \mathrm{prof}(\mathscr{G}_x)$, hence $\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{G}_x)$, and as $\mathrm{coprof}(\mathscr{G}_x) = 0$ for $y \notin X$—Y, one is reduced to proving the property for $\mathscr{G}$; in other words, one can restrict to the case where X is an affine *regular* scheme. One knows then (**0**, 17.3.4) that one has

$$\mathrm{prof}(\mathscr{F}_x) = \dim(\mathscr{O}_{X,x}) - \dim.\mathrm{proj}(\mathscr{F}_x).$$

On the other hand, if S is the unique closed sub-prescheme of X having for underlying space $\mathrm{Supp}(\mathscr{F})$, one has $\dim(\mathscr{F}_x) = \mathrm{codim}(\{x\}, S)$ (5.1.12.1), and, by virtue of (5.1.9)

$$\dim(\mathscr{F}_x) = \dim(\mathscr{O}_{X,x}) - \mathrm{codim}_x(S, X)$$

since $\mathscr{O}_{X,x}$ is a regular ring, and *a fortiori* equidimensional (**0**, 16.5.12). We can therefore write

(6.11.2.1) $$\mathrm{coprof}(\mathscr{F}_x) = \dim.\mathrm{proj}(\mathscr{F}_x) - \mathrm{codim}_x(S, X)$$

and the proposition follows then from (6.11.1) and from (**0**, 14.2.6).

(ii) As the $U_{C_n}(\mathscr{F})$ are open for every n by (i), the $Z_n = X - U_{C_n}(\mathscr{F})$ are closed in X; furthermore **IV-2 | 160**
it is clear that $Z_{n+1} \subset Z_n$, and as $\dim(\mathscr{F}_x)$ is finite for every $x \in X$ and $\mathrm{coprof}(\mathscr{F}_x) \leqslant \dim(\mathscr{F}_x)$, the intersection of the Z_n is *empty*; as one can restrict to the case where X is affine, hence quasi-compact, one can suppose that there exists an m □

such that $Z_m = \varnothing$. Now, it follows from (5.7.4) that the relation $x \in U_{S_k}(\mathscr{F})$ is equivalent to the set of relations

(6.11.2.2) $$\mathrm{codim}_x(Z_n, S) > n + k$$

for every $n \geqslant 0$; but for $n \geqslant m$ this relation is automatically satisfied, hence one need only consider the relations (6.11.2.2) for $0 \leqslant n < m$. Now (**0**, 14.2.6) the set $V_{n,k}$ of x satisfying (6.11.2.2) is open, and $U_{S_k}(\mathscr{F})$, intersection of the $V_{n,k}$ for $0 \leqslant n < m$, is also open.

Let us recall that the hypothesis that X is locally immersible in a regular scheme is always satisfied when X is a prescheme *locally of finite type over a field* k (5.8.3).

Corollary (6.11.3). — *Under the hypotheses of* (6.11.2)*, the set* $\mathrm{CM}(\mathscr{F})$ *of points* $x \in X$ *such that* $\mathscr{F}_x$ *is a Cohen-Macaulay module is open in* X*.*

Proof. Indeed, it is the set $U_{C_0}(\mathscr{F})$. □

Remarks (6.11.4). — (i) The reasoning of (6.11.2), (ii)) proves that (without any hypothesis on X) when $U_{C_n}(\mathscr{F})$ is open for *every* integer n, then $U_{S_n}(\mathscr{F})$ is open for every integer n.

(ii) We have $\mathrm{CM}(\mathscr{F}) = \bigcap U_{S_n}(\mathscr{F})$. If every point $x \in X$ admits an open neighbourhood V of *finite* dimension, the sequence of intersections $V \cap U_{S_n}(\mathscr{F})$ is *stationary* since there exists an m such that $\dim(\mathscr{F}_x) \leqslant m$ for every $x \in V$; consequently, if the $U_{S_n}(\mathscr{F})$ are open in X, then $\mathrm{CM}(\mathscr{F})$ is also open in X.

(iii) We shall write $U_{S_n}(X)$, $U_{C_n}(X)$, $\mathrm{CM}(X)$ instead of $U_{S_n}(\mathscr{O}_X)$, $U_{C_n}(\mathscr{O}_X)$, $\mathrm{CM}(\mathscr{O}_X)$.

Proposition (6.11.5). — *Let* A *be a local Noetherian ring,* M *an* A*-module of finite type. For every prime ideal* $\mathfrak{p}$ *of* A*, we have*

(6.11.5.1) $$\mathrm{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leqslant \mathrm{coprof}_A(M).$$

Proof. One can restrict to the case where $M_{\mathfrak{p}} \neq 0$. As $\widehat{A}$ is a faithfully flat A-module ($\mathbf{0}_I$, 7.3.5), there exists an ideal $\mathfrak{q}$ of $\widehat{A}$ lying over $\mathfrak{p}$ ($\mathbf{0}_I$, 6.5.1); as $\widehat{M}_{\mathfrak{q}} = (M \otimes_A \widehat{A})_{\mathfrak{q}} = M \otimes_A \widehat{A}_{\mathfrak{q}}$ and $\widehat{A}_{\mathfrak{q}}$ is a flat A-module, hence also a flat $A_{\mathfrak{p}}$-module ($\mathbf{0}_I$, 6.2.1), it follows from (6.3.2), applied to the local homomorphism $A_{\mathfrak{p}} \to \widehat{A}_{\mathfrak{q}}$, to $M_{\mathfrak{p}}$ and $\widehat{M}_{\mathfrak{q}}$, that $\operatorname{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leqslant \operatorname{coprof}_{\widehat{A}_{\mathfrak{q}}}(\widehat{M}_{\mathfrak{q}})$. On the other part, $X = \operatorname{Spec}(\widehat{A})$ is isomorphic to a sub-prescheme of a regular scheme by virtue of the Cohen structure theorem (**0**, 19.8.8, (i)). One deduces therefore from (6.11.2) that $\operatorname{coprof}_{\widehat{A}_{\mathfrak{q}}}(\widehat{M}_{\mathfrak{q}}) \leqslant \operatorname{coprof}_{\widehat{A}}(\widehat{M})$; finally, one knows that $\operatorname{coprof}_{\widehat{A}}(\widehat{M}) = \operatorname{coprof}_{A}(M)$ (**0**, 16.4.10), which finishes the proof of (6.11.5). This proposition justifies the definition of the coprofundity of an A-module when A is not a local ring, given in (5.7.12). □

Proposition (6.11.6). — *Let* X *be a locally Noetherian prescheme,* n *an integer* > 0, $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module. Suppose that for every closed integral sub-prescheme* Y *of* X*, there exists an open non-empty part* W *of* Y *such that the sub-prescheme of* X *induced on* W *satisfies* (S_n)*. Then the set* $U_{S_n}(\mathscr{F})$ *is open in* X*.*

Proof. The question being local on X, one can restrict to the case where X is Noetherian. We shall reason by induction on n: for $n = 1$, the set $U_{S_1}(\mathscr{F})$ is open in X, for the set of $x \in X$ where $\mathscr{F}$ does not satisfy (S_1) is the set of x such that $\mathscr{F}_x$ admits immersed associated prime ideals (5.7.5); if (Z_j) is the finite family of the prime cycles associated to $\mathscr{F}$ which are immersed, then $U_{S_1}(\mathscr{F}) = X - \bigcup_j Z_j$, whence our assertion since the Z_j are closed. We shall therefore suppose henceforth that $n > 1$. In the second place, one can restrict to the case where $\operatorname{Supp}(\mathscr{F}) = X$, for there is a closed sub-prescheme T of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space, and an $\mathscr{O}_T$-Module coherent $\mathscr{G}$ such that $\mathscr{F} = j_*(\mathscr{G})$, $j : T \to X$ being the canonical injection (**Err**$_{\mathbf{III}}$, 30); as it amounts to the same to prove that $\mathscr{F}$ or $\mathscr{G}$ satisfies (S_n) at a point $x \in \operatorname{Supp}(\mathscr{F})$, one can restrict to considering the case where $T = X$. Let us note finally that by definition (5.7.2), $U_{S_n}(\mathscr{F})$ is stable by generization. We shall now use the following lemma:

Lemma (6.11.6.1). — *Let* X *be a Noetherian space in which every irreducible closed subset admits a generic point. For* E *to be an open subset of* X*, it is necessary and it suffices that* E *be stable by generization, and that, for every open part* V *of* X *and every irreducible closed subset* Y *of* V*, such that* $V - Y \subset E$ *and that the generic point of* Y *belongs to* E*,* $E \cap Y$ *contains an open non-empty part of* Y*.*

Proof. We observe that this criterion implies that of ($\mathbf{0}_{\mathrm{III}}$, 9.2.6) when every irreducible closed subset admits a generic point. There is obviously nothing other than to prove the sufficiency of the conditions stated.

Consider the interior U of E; the closed set $X - U$ is a union of its irreducible components, which are finite in number and closed in X. If $E \neq U$, the hypothesis that E is stable by generization would imply that the generic point z of one of the irreducible components of $X - U$ would belong to E. Now, z belongs to only one of the irreducible components of $X - U$; if T is the union of the other irreducible components of $X - U$, $V = X - T$ is open in X, union of U and of the set $Y = \overline{\{z\}} \cap V$ closed in V and irreducible. By hypothesis $E \cap Y$ contains an open part W of Y; we conclude that $U \cup W$ is open *in* V, hence in X, which is absurd since U is supposed to be the interior of E.

By virtue of this lemma, one can suppose that there exists in X an integral sub-prescheme Y of **IV-2 | 161**
generic point y such that $\mathscr{F}$ satisfies (S_n) at the point y and at every point of $X - Y$, and restrict to verifying that there exists in X an open neighbourhood of y such that $\mathscr{F}$ satisfies (S_n) in this neighbourhood. We distinguish two cases:

1° y is a maximal point of X; as there exists an open neighbourhood of y meeting no irreducible component of X other than $\overline{\{y\}}$, one can suppose that X is irreducible, hence a *fortiori* has the same underlying space as Y, so that Y is defined by the Nilradical of $\mathscr{O}_X$, which is *nilpotent*. On the other hand, on can, replacing X by an open neighbourhood of y, suppose that $\mathscr{F}$ is *normally flat* along Y (6.9.1); it follows then from (6.10.4) that $U_{S_n}(\mathscr{F}) = U_{S_n}(\mathscr{O}_Y)$, and as the latter is by hypothesis an open neighbourhood of y in X, this proves the proposition in this case.

2° y is not a maximal point of X, in other words $\operatorname{Supp}(\mathscr{F}) = X$, $\dim(\mathscr{F}_y) \geqslant 1$, hence also $\operatorname{prof}(\mathscr{F}_y) \geqslant 1$ since $\mathscr{F}$ satisfies by hypothesis (S_n) (and *a fortiori* (S_1)) at the point y. Replacing X by an open neighbourhood of y if necessary, one can therefore suppose that there exists a section f of $\mathscr{O}_X$ over X, $\mathscr{F}$-regular and such that $f_y \in \mathfrak{m}_y$, or equivalently $f(y) = 0$ (**0**, 15.2.4); one has then also $f(x) = 0$ for every $x \in Y$ ($\mathbf{0}_{\mathrm{I}}$, 5.5.2). One knows that $\mathscr{F}/f\mathscr{F}$ satisfies (S_{n-1}) at the point y (5.7.6). Applying the induction hypothesis, and replacing X by an open neighbourhood of y, one can therefore suppose that $\mathscr{F}/f\mathscr{F}$ satisfies (S_{n-1}) at every point of X. But for every $x \in Y$, the relation $f(x) = 0$ implies that $\operatorname{prof}(\mathscr{F}_x | f_x\mathscr{F}_x) = \operatorname{prof}(\mathscr{F}_x) - 1$ and $\dim(\mathscr{F}_x | f_x\mathscr{F}_x) = \dim(\mathscr{F}_x) - 1$ (**0**, 16.3.4

and 16.4.6); the relation

$$\mathrm{prof}(\mathscr{F}_x|f_x\mathscr{F}_x) \geqslant \inf(n-1, \dim(\mathscr{F}_x|f_x\mathscr{F}_x))$$

is therefore equivalent to

$$\mathrm{prof}(\mathscr{F}_x) \geqslant \inf(n, \dim(\mathscr{F}_x)).$$

As we supposed that $\mathscr{F}$ satisfies (S_{n-1}) at every point of $\mathrm{X}-\mathrm{Y}$, this finishes the proof. □

Corollary (6.11.7). — *With the notations being those of (6.11.6):*

label=(i) The set $\mathrm{U}_{\mathrm{S}_1}(\mathscr{F})$ is open in X.

lbbel=(ii) For the set $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ to be open, it suffices that every maximal point x of $\mathrm{Supp}(\mathscr{F})$*, belonging to $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$, is interior to $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$.*

Proof. Assertion (i) was proved in the course of the proof of (6.11.6); on the other hand, for $n=2$ the case 2° of the proof of (6.11.6) is valid without any hypothesis on X, since (with the same notations) $\mathrm{U}_{\mathrm{S}_1}(\mathscr{F})$ and $\mathrm{U}_{\mathrm{S}_1}(\mathscr{F}/f\mathscr{F})$ are open in X. As for the case 1° of this proof, the hypothesis ensures precisely that it is unnecessary to consider it. □

IV-2 | 162

Proposition (6.11.8). — *Let* X *be a locally Noetherian prescheme satisfying the following property:*

(CMU) *Every closed integral sub-prescheme* Y *of* X *contains an open non-empty* W *such that the prescheme induced by* X *on* W *is a Cohen-Macaulay prescheme.*

Then, for every coherent $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{F}$, the function $x \mapsto \mathrm{coprof}(\mathscr{F}_x)$ is locally constructible and upper semi-continuous; the sets $\mathrm{U}_{\mathrm{C}_n}(\mathscr{F})$ and $\mathrm{U}_{\mathrm{S}_n}(\mathscr{F})$ are open in X.

Proof. Indeed, let Y be a closed integral sub-prescheme of X, of generic point y; by virtue of (6.10.6), there is a neighbourhood V of y in Y such that, for every $x \in \mathrm{V} \cap \mathrm{Y}$, we have

$$\mathrm{coprof}(\mathscr{F}_x) = \mathrm{coprof}(\mathscr{F}_y) + \mathrm{coprof}(\mathscr{O}_{\mathrm{Y},x}). \tag{6.11.8.1}$$

But by hypothesis there exists an open non-empty W of Y such that, for $x \in \mathrm{V}\cap\mathrm{W}$, $\mathrm{coprof}(\mathscr{O}_{\mathrm{Y},x}) = 0$, hence $\mathrm{coprof}(\mathscr{F}_x)$ is constant in a neighbourhood of y in Y, which proves that the function $x \mapsto \mathrm{coprof}(\mathscr{F}_x)$ is locally constructible ($\mathbf{0}_{\mathrm{III}}$, 9.3.2); it follows furthermore from (6.11.5) and from ($\mathbf{0}_{\mathrm{III}}$, 9.3.4) that this function is upper semi-continuous. The last assertion follows from (6.11.4), (i), or from (6.11.6). □

Remarks (6.11.9). — (i) If X satisfies the hypothesis (CMU) of (6.11.8), then, for every morphism $u : \mathrm{X}' \to \mathrm{X}$ locally of finite type, X′ also satisfies (CMU). Let indeed Y′ be a closed integral sub-prescheme of X′, y' its generic point, and let Y be the closed integral sub-prescheme of X having for underlying space $\overline{\{y\}}$; then $u|\mathrm{Y}'$ factorizes as $\mathrm{Y}' \xrightarrow{v} \mathrm{Y} \xrightarrow{j} \mathrm{X}$, where j is the canonical injection (**I**, 5.2.2), and v is locally of finite type (**I**, 6.6.6). Restricting to affine open neighbourhoods of y and y' respectively, one can therefore restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is an integral Cohen-Macaulay ring, and $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, where A′ is an A-algebra of finite type (with $f \neq 0$), **IV-2 | 163**
one can suppose furthermore that A′ contains a polynomial ring $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_n] = \mathrm{A}''$, and is an A″-algebra *finite* (Bourbaki, *Alg. comm.*, chap. V, § 3, no 1, cor. 1 of th. 1). But A″ is a Cohen-Macaulay ring (6.3.6); hence one can further restrict to the case where A′ is also an A-module *free* of finite type. But then A′ is an A-module of finite type, and as A′ is also an A-module *free* of finite type. But A′ is an A-module of finite type, hence one can further suppose that A′ is also an A-module *libre*. But then A′ is an A-module free of finite type; as A′ is an A-module of finite type, and as A′ is also an A′-module of Cohen-Macaulay (**0**, 16.5.3), hence a Cohen-Macaulay ring.

(ii) Suppose that there exists a coherent $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{F}$ such that $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$ and that $\mathscr{F}$ is an $\mathscr{O}_{\mathrm{X}}$-Module *of Cohen-Macaulay*. Then X satisfies the condition (CMU): indeed, with the notations of the proof of (6.11.8), the relation (6.11.8.1) shows that $\mathrm{coprof}(\mathscr{O}_{\mathrm{Y},x}) = 0$ in a neighbourhood (with respect to Y) of the generic point y of Y.

It is unknown whether there exist locally Noetherian preschemes of dimension $\geqslant 2$ that do not satisfy (CMU) (if $\dim(\mathrm{X}) = 1$, it is immediate that every maximal point of $\mathrm{X}_{\mathrm{red}}$ admits an open neighbourhood of dimension 1, hence a Cohen-Macaulay ring).

6.12. Nagata's criteria for $\mathrm{Reg}(X)$ to be open

(6.12.1). Given a locally Noetherian prescheme X, we call the *singular locus* of X and denote by $\mathrm{Sing}(X)$ the set of points $x \in X$ such that X is not regular at the point x (in other words, such that the local ring $\mathscr{O}_x$ is not regular); the complement $X - \mathrm{Sing}(X)$, set of $x \in X$ where X is regular, is denoted $\mathrm{Reg}(X)$. We propose in this subsection to give conditions ensuring that $\mathrm{Sing}(X)$ is *closed* (i.e. $\mathrm{Reg}(X)$ *open*).

Proposition (6.12.2). — *Let* X *be a locally Noetherian prescheme. The following conditions are equivalent:*

label=a) $\mathrm{Reg}(X)$ *is open in* X.

lbbel=b) *For every $x \in \mathrm{Reg}(X)$, there exists an open non-empty part of $\overline{\{x\}}$ contained in* $\mathrm{Reg}(X)$.

Moreover, these conditions imply the following:

lcbel=c) *For every $x \in \mathrm{Reg}(X)$, if we denote by* Y *the reduced closed sub-prescheme of* X *having $\overline{\{x\}}$ as underlying space,* $\mathrm{Reg}(Y)$ *is a neighbourhood of x in* Y.

Proof. The equivalence of $a)$ and $b)$ follows from the fact that $\mathrm{Reg}(X)$ is stable by generization ($\mathbf{0}$, 17.3.2) and from ($\mathbf{0}_{\mathrm{III}}$, 9.2.6). To prove that $c)$ implies $b)$, one can restrict to the case where $X = \mathrm{Spec}(A)$ is the spectrum of an affine ring of x; if $(t_i)_{1 \leqslant i \leqslant n}$ is a regular system of parameters ($\mathbf{0}$, 17.1.6) of the regular local ring A_x, one can suppose (replacing X by an open neighbourhood of x if necessary) that $t_i = (s_i)_x$ for every i, where $s_i \in A$ and where the family $(s_i)_{1 \leqslant i \leqslant n}$ is A-regular ($\mathbf{0}$, 15.2.4). One has then $Y = \mathrm{Spec}(A/\mathfrak{p})$, where $\mathfrak{p} = \mathfrak{j}_x$; as the t_i generate the maximal ideal $\mathfrak{m} = \mathfrak{p}A_x$ of A_x, one can further suppose (replacing X by a smaller neighbourhood of x) that the s_i generate $\mathfrak{p}$. For every $y \in Y$, the $(s_i)_y$ generate therefore $\mathfrak{p}_y$; as they form a suite A_y-regular, they also form a suite $\mathrm{gr}^0_{\mathfrak{p}_y}(\mathscr{O}^Y_y)$-regular, so they form a $\mathrm{gr}_{\mathfrak{p}_y}(\mathscr{O}^Y_y)$-regular sequence and it is thus also $\mathscr{O}^Y_y$-regular, whence the conclusion. □

Corollary (6.12.3). — *Let* X *be a locally Noetherian prescheme. The following conditions are equivalent:*

label=a) *For every sub-prescheme* Y *of* X, $\mathrm{Reg}(Y)$ *is open in* Y. **IV-2** | 164

lbbel=b) *For every closed integral sub-prescheme* Y *of* X, $\mathrm{Reg}(Y)$ *contains an open non-empty part of* Y.

Proof. It is clear that $a)$ implies $b)$. Conversely, to see that $b)$ implies $a)$, consider a closed integral sub-prescheme Z of Y of generic point z; if Y' is the sub-prescheme integral of X having for underlying space the closure $Y' = \overline{\{z\}}$ of z in X, Z is open in Y' and the sub-prescheme Z of Y', being reduced, is induced by Y' on the open subset of Z in Y'. Now the hypothesis implies that $\mathrm{Reg}(Y')$ is a neighbourhood of z in Y', hence $\mathrm{Reg}(Z)$ is a neighbourhood of z in Z; it suffices then to apply (6.12.2) by replacing X by Y and Y by Z. □

Theorem (6.12.4). — (Nagata). — *Let* A *be a Noetherian ring,* $X = \mathrm{Spec}(A)$. *The following conditions are equivalent:*

label=a) *For every prescheme X' locally of finite type over* X, $\mathrm{Reg}(X')$ *is open in* X'.

lbbel=b) *For every* A*-algebra A' finite and integral, there exists an open non-empty subset of* $X' = \mathrm{Spec}(A')$ *contained in* $\mathrm{Reg}(X')$.

lcbel=c) *For every prime ideal $\mathfrak{p}$ of* A *and every finite radicial extension K' of the field of fractions* K *of* $A/\mathfrak{p}$, *there exists a sub-*A*-algebra A' of K', finite over* A, *having K' as field of fractions, and such that there exists in $X' = \mathrm{Spec}(A')$ an open non-empty subset contained in* $\mathrm{Reg}(X')$.

Proof. It is clear that $a)$ implies $b)$. To see that $b)$ implies $c)$, it suffices to note that one can take as generators of the extension K' of K elements integral over $A/\mathfrak{p}$ (and *a fortiori* over A), and as these elements are finite in number, they generate a finite A-algebra A' whose field of fractions is K'; one can then apply $b)$ to A'. It remains therefore to prove that $c)$ implies $a)$. The question being local on X', one can first suppose that $X' = \mathrm{Spec}(A')$, where A' is an A-algebra of finite type; in view of (6.12.2), it suffices to prove that for every closed integral sub-prescheme Y' of X', $\mathrm{Reg}(Y')$ contains an open non-empty part of Y'. In other words, one can restrict to proving that if A' is an A-algebra integral *of finite type* and $X' = \mathrm{Spec}(A')$, $\mathrm{Reg}(X')$ contains an open non-empty part. Let K' be the field of fractions of A'; if $\mathfrak{p}$ is the canonical image in X of the generic point of X', K' is an extension of the field of fractions K of $A/\mathfrak{p}$, and $A/\mathfrak{p}$ is identified with a subring of A', A' being a finite $(\mathrm{A}/\mathfrak{p})$-algebra of finite type. One can obviously restrict in what follows to the proof when $\mathfrak{p} = 0$. We distinguish two cases:

I) K' *is a separable extension* of K. — One is then reduced to proving the

Lemma (6.12.4.1). — *Let* A *be a Noetherian integral ring,* A' *an* A*-algebra integral of finite type, such that the field of fractions* K' *of* A' *is a separable extension of the field of fractions* K *of* A*. If* Spec(A) *contains an open non-empty subset consisting of regular points, then so does* $\mathrm{Spec}(\mathrm{A}')$*.*

Proof. Replacing A by a ring of fractions A_f, so that $\mathrm{D}(f) \subset \mathrm{Reg}(\mathrm{Spec}(\mathrm{A}))$, one can already suppose the ring A *regular*. By hypothesis there exists in K' a system $(t_i)_{1 \leqslant i \leqslant n}$ of elements algebraically independent over K and such that K' is a finite separable algebraic extension of $\mathrm{K}(t_1, \ldots, t_n)$; considering a finite system of generators t'_j $(1 \leqslant j \leqslant m)$ of K' over $\mathrm{K}(t_1, \ldots, t_n)$, which one can **IV-2** | 165 suppose to be integral over $\mathrm{A}_1 = \mathrm{A}[t_1, \ldots, t_n]$, one sees that $\mathrm{A}'_1 = \mathrm{A}_1[t'_1, \ldots, t'_m]$ is *finite* over A_1 and has K' as field of fractions. If one sets $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}'_1 = \mathrm{Spec}(\mathrm{A}'_1)$, it follows from the fact that the fields of rational functions $\mathrm{R}(\mathrm{X}')$ and $\mathrm{R}(\mathrm{X}'_1)$ are both isomorphic to K' and from the fact that X' and X'_1 are A-preschemes of finite type, that there exists an open subset $\mathrm{U}' \subset \mathrm{X}'$ and an open $\mathrm{U}'_1 \subset \mathrm{X}'_1$ which are A-*isomorphic* (**I**, 6.5.5). One is therefore reduced to proving that $\mathrm{Reg}(\mathrm{X}'_1)$ contains a non-empty open subset, in other words one can suppose that A' is an A_1-algebra *finite*. Now, one knows (**0**, 17.3.7) that A_1 is a regular ring, and one can therefore restrict to the case where A' is an A-algebra *finite* and K' an extension *separable* and *finite* of K. Let ξ be the generic point of X, $\mathrm{A}'_\xi = \mathrm{K}'$ is then a free module over $\mathrm{A}_\xi = \mathrm{K}$, hence (Bourbaki, *Alg.*, *comm.*, chap. II, § 5, no 1, cor. of prop. 2) one can, replacing A by a ring of fractions A_f, suppose that A' is a free A-module of finite type. Let then $(x_h)_{1 \leqslant h \leqslant r}$ be a basis of this A-module, and set

$$d = \det(\mathrm{Tr}_{\mathrm{A}'/\mathrm{A}}(x_h x_k)) = \det(\mathrm{Tr}_{\mathrm{K}'/\mathrm{K}}(x_h x_k)) \in \mathrm{A}.$$

As K' is separable over K, one knows (Bourbaki, *Alg.*, chap. IX, § 2, prop. 5) that $d \neq 0$; replacing A by a ring of fractions A_z, one can therefore suppose d *invertible* in A. But then, for every $z \in \mathrm{Spec}(\mathrm{A})$, if one denotes by $\overline{x}_h$ $(1 \leqslant h \leqslant r)$ the canonical image of x_h in $\mathrm{A}'(z) = \mathrm{A}' \otimes_{\mathrm{A}} \boldsymbol{k}(z)$, one has $\det(\mathrm{Tr}_{\mathrm{A}'(z)/\boldsymbol{k}(z)}(\overline{x}_h \overline{x}_k)) = \overline{d}$, canonical image of d in $\boldsymbol{k}(z) = \mathrm{A}_z/\mathfrak{m}_z$, and as $\overline{d}$ is invertible (hence $\neq 0$) in $\boldsymbol{k}(z)$, one knows (*loc. cit.*) that $\mathrm{A}'(z)$ is a $\boldsymbol{k}(z)$-algebra *separable*, hence a composed direct of field extensions finite and separable of $\boldsymbol{k}(z)$. Such an algebra being a *regular* ring, one sees that the morphism $g : \mathrm{X}' \to \mathrm{X}$ is *flat* and that its fibres $g^{-1}(z)$ are *regular* for every $z \in \mathrm{X}$; one concludes then from (6.5.2, (ii)) that X' is *regular*, which finishes the proof in this case. □

II) *General case.* — As A' is an A-module without torsion, $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ is identified with a subring of K', hence $\mathrm{X}'' = \mathrm{Spec}(\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K})$ is a K-prescheme integral, whose field of rational functions is K'. One knows (4.6.6) that there exists a finite radicial extension K_1 of K such that $\mathrm{X}'_1 = \mathrm{Spec}(\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}_1) = \mathrm{X}'' \otimes_{\mathrm{K}} \mathrm{K}_1$, $(\mathrm{X}'_1)_{\mathrm{red}}$ is a K_1-prescheme *geometrically reduced* of finite type; furthermore, the morphism $\mathrm{Spec}(\mathrm{K}_1) \to \mathrm{Spec}(\mathrm{K})$ being radicial, finite and surjective, it is a universal homeomorphism (2.4.5), hence X'_1 is homeomorphic to X'', and consequently $(\mathrm{X}'_1)_{\mathrm{red}}$ is integral; its field of rational functions K'_1 is a finite radicial extension of K' and a *separable* extension of finite type of K_1 (4.6.1). By virtue of hypothesis *c*) of the statement, there is a sub-A-algebra *finite* A_1 of K_1, having K_1 as field of fractions and such that if one sets $\mathrm{X}_1 = \mathrm{Spec}(\mathrm{A}_1)$, $\mathrm{Reg}(\mathrm{X}_1)$ contains an open non-empty subset. Let A'_1 be the image of the canonical homomorphism $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_1 \to \mathrm{K}'_1$ and let $\mathrm{X}'_1 = \mathrm{Spec}(\mathrm{A}'_1)$; A'_1 is an integral ring which is an A'-algebra *finite* and whose field of fractions is K'_1; furthermore, as the composite homomorphism $\mathrm{A}' \to \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_1 \to \mathrm{A}'_1 \to \mathrm{K}'_1$ is identical to $\mathrm{A}' \to \mathrm{K}' \to \mathrm{K}'_1$, hence injective, the homomorphism $\mathrm{A}' \to \mathrm{A}'_1$ is injective; the morphism $g : \mathrm{X}'_1 \to \mathrm{X}'$ is therefore finite and surjective (**II**, 6.1.10). This being the hypothesis on A_1 and part I) of the proof imply that $\mathrm{Reg}(\mathrm{X}'_1)$ contains an **IV-2** | 166 open non-empty V'_1; as g is closed (**II**, 6.1.10), one can suppose that $\mathrm{V}'_1 = g^{-1}(\mathrm{V}')$, where V' is an open affine of X'; replacing A' by the ring of V' and A'_1 by that of V'_1, one can therefore suppose that X'_1 is *regular*. Furthermore, the same reasoning as in I) applied to the generic point ξ' of X' allows (replacing X' by a neighbourhood of ξ') to suppose that A'_1 is an A'-module *free*, and hence that the morphism g is *flat*. But it follows then from (6.5.2, (i)) that X' is *regular*. □

Corollary (6.12.5). — (Zariski). — *Let k be a field; for every k-prescheme* X *locally of finite type over k, the set of $x \in$* X *where* X *is regular (resp.* geometrically regular *over k) is open in* X.

Proof. The assertion of the corollary relative to the property of being regular follows from (6.12.4) by taking $\mathrm{A} = k$. The assertion relative to the property of being geometrically regular follows already from (6.8.7); one can also deduce it from (6.12.4) in the following way: set $k' = k^{p^{-\infty}}$ (p being the characteristic exponent of k); as the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is radicial, integral and surjective, it is a universal homeomorphism (2.4.5), hence the morphism projection $\mathrm{X} \otimes_k k' \to \mathrm{X}$ is a universal homeomorphism. The projection in X of $\mathrm{Reg}(\mathrm{X} \otimes_k k')$ is the set of points of X where X is geometrically regular over k, by virtue of (6.7.7, e)); hence this set is open from what precedes. □

Corollary (6.12.6). — *Let* A *be a ring having one of the following properties:*

label=(i) A *is a Dedekind ring and its field of fractions* K *is of characteristic* 0.
lbbel=(ii) A *is a semi-local Noetherian ring of dimension* $\leqslant 1$.

Then, for every prescheme X *locally of finite type over* A, $\mathrm{Reg}(\mathrm{X})$ *is open in* X.

Proof. Let us verify in both cases the condition c) of (6.12.4). In both cases, a prime ideal $\mathfrak{p}$ of A is maximal or minimal; if $\mathfrak{p}$ is maximal, an $(\mathrm{A}/\mathfrak{p})$-algebra finite and integral is a field and the condition c) of (6.12.4) is trivially satisfied. Suppose therefore $\mathfrak{p}$ non-maximal, and distinguish the two cases of the statement.

(i) If K is of characteristic 0, there is no radicial extension of K other than K itself; as a Dedekind ring is regular (**0**, 17.1.4), the condition c) of (6.12.4) is trivially satisfied.

(ii) One can then suppose A integral (6.12.2), let K be its field of fractions; if K' is a finite radicial extension of K, A' a sub-A-algebra of K' generated by a finite system of generators of K' over K, integral over A, A' is a semi-local integral ring of dimension 1 (**0**, 16.1.5), and consequently, in $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, the reduced set at the generic point is open and evidently contained in $\mathrm{Reg}(\mathrm{X}')$, which proves the condition c) of (6.12.4) in this case. □

This corollary applies notably when $\mathrm{A} = \mathbf{Z}$.

Theorem (6.12.7). — (Nagata). — *Let* A *be a local Noetherian complete ring,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$*. Then* $\mathrm{Reg}(\mathrm{X})$ *is open in* X.

Proof. In view of (6.12.2), one is reduced to the case where moreover A is *integral*, and to proving that $\mathrm{Reg}(\mathrm{X})$ contains an open non-empty part of X. We distinguish two cases:

I) *The field of fractions* K *of* A *is of characteristic* 0. — One knows then (**0**, 19.8.8) that there exists a **IV-2 | 167**
complete discrete valuation ring C and a subring B of A such that A is a B-algebra *finite* and that B is isomorphic to a formal power series ring $\mathrm{C}[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]$. As C is regular (**II**, 7.1.6), so is B (**0**, 17.3.8); furthermore, the field of fractions L of B being of characteristic 0, K is a *separable* extension finite of L, hence one can apply (6.12.4.1) to B, and this proves the proposition.

II) *The field of fractions* K *of* A *is of characteristic* $p > 0$. — Then A contains the prime field $\mathbf{F}_p$, and the theorem has been proved in this case (**0**, 22.7.6). □

Corollary (6.12.8). — *Let* A *be a local Noetherian complete ring. Then, for every prescheme* X *locally of finite type over* A, $\mathrm{Reg}(\mathrm{X})$ *is open in* X.

Proof. Let us verify the condition c) of (6.12.4). If $\mathfrak{p}$ is prime in A, $\mathrm{A}/\mathfrak{p}$ is again a local Noetherian complete ring; if K' is a finite extension of the field of fractions K of $\mathrm{A}/\mathfrak{p}$, K' is the field of fractions of a sub-A-algebra finite A' of K', generated by a finite system of generators of K' over K, integral over A. One knows then that A' is a semi-local complete ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 3 of prop. 9), hence a product of local complete rings, and as A' is integral, it is a local complete ring; and by virtue of (6.12.7), if $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{Reg}(\mathrm{X}')$ is open and non-empty, whence the conclusion. □

Proposition (6.12.9). — *Let* X *be a locally Noetherian prescheme such that* $\mathrm{Reg}(\mathrm{X})$ *is open; then, for every* $n \geqslant 0$*, the set* $\mathrm{U}_{\mathrm{R}_n}(\mathrm{X})$ *of points* $x \in \mathrm{X}$ *where* X *satisfies* (R_n) *is open.*

Proof. Indeed (5.8.2), $\mathrm{U}_{\mathrm{R}_n}(\mathrm{X})$ is the union of $\mathrm{Reg}(\mathrm{X})$ and of the set of $x \in \mathrm{Sing}(\mathrm{X})$ such that the generic points z_i of the irreducible components of $\mathrm{Sing}(\mathrm{X})$ containing x satisfy the relation $\dim(\mathscr{O}_{\mathrm{X},z_i}) \geqslant n+1$ (5.1.2); in other words, these points $x \in \mathrm{Sing}(\mathrm{X})$ are those for which $\mathrm{codim}_x(\mathrm{Sing}(\mathrm{X}), \mathrm{X}) \geqslant n+1$; as the function $x \mapsto \mathrm{codim}_x(\mathrm{Sing}(\mathrm{X}), \mathrm{X})$ is lower semi-continuous (**0**, 14.2.6), the set of these points is open in $\mathrm{Sing}(\mathrm{X})$, which finishes the proof. □

Let us also note the following elementary result:

Proposition (6.12.10). — *Let* X *be a locally Noetherian prescheme. The set* $\mathrm{U}_{\mathrm{R}_0}(\mathrm{X})$ *is open in* X. *For* $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$ *to be open in* X, *it is necessary and it suffices that every maximal* $x \in \mathrm{X}$ *such that* $x \in \mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$ *(which means that* X *is reduced at* x*) be interior to* $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$.

Proof. By definition (5.8.2), $\mathrm{U}_{\mathrm{R}_0}(\mathrm{X})$ is the set $\mathrm{X} - \bigcup_\alpha \mathrm{X}'_\alpha$, where X'_α ranges over the set of irreducible components of X such that X is not reduced at their generic point; as the set of irreducible components of X is locally finite, $\mathrm{U}_{\mathrm{R}_0}(\mathrm{X})$ is open. As for $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$, the stated condition is trivially necessary; let us prove that it is sufficient. Let X''_β be the irreducible components of X such that, at their generic point x''_β, the prescheme X is reduced; by hypothesis there exists an open $\mathrm{U} \subset \mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$ containing all the x''_β. Denote then by Z_λ the irreducible components of the closed set $\mathrm{Z} = \mathrm{X} - \mathrm{U}$ whose generic point z_λ is such that $\dim(\mathscr{O}_{\mathrm{X},z_\lambda}) \leqslant 1$ and that $\mathscr{O}_{\mathrm{X},z_\lambda}$ is non-regular. No point belonging to one of the Z_λ can belong to $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$; but conversely, if $x \in \mathrm{Z}$ does not belong to any of the Z_λ, then, for every **IV-2** | 168
generization x' of x, either x' belongs to U, or $\dim(\mathscr{O}_{\mathrm{X},x'}) \geqslant 2$, or $\dim(\mathscr{O}_{\mathrm{X},x'}) = 1$, and as x' does not belong to any Z_λ, $\overline{\{x'\}}$ is necessarily one of the irreducible components of Z, of codimension 1 in X, distinct from the Z_λ, hence $\mathscr{O}_{\mathrm{X},x'}$ is regular by definition. One concludes that $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X}) = \mathrm{X} - \bigcup_\lambda \mathrm{Z}_\lambda$; as the set of Z_λ is locally finite in Z, $\bigcup_\lambda \mathrm{Z}_\lambda$ is closed, which finishes the proof that $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$ is open in X. □

6.13. Criteria for $\mathrm{Nor}(\mathrm{X})$ to be open

(6.13.1). Given a locally Noetherian prescheme X, we shall denote by $\mathrm{Nor}(\mathrm{X})$ the set of $x \in \mathrm{X}$ where X is *normal*; this set contains $\mathrm{Reg}(\mathrm{X})$ and is contained in the (open) set of points x such that $\mathscr{O}_{\mathrm{X},x}$ is integral (i.e. the set of points x where X is reduced, and which belong to only one irreducible component of X).

Proposition (6.13.2). — *Let* X *be a locally Noetherian prescheme reduced,* X' *its normalization* (**II**, 6.3.8). *If the canonical morphism* $f : \mathrm{X}' \to \mathrm{X}$ *is of finite type,* $\mathrm{Nor}(\mathrm{X})$ *is open in* X.

Proof. Indeed, on $\mathrm{X}' = \mathrm{Spec}(f_*(\mathscr{O}_{\mathrm{X}'}))$ and $f_*(\mathscr{O}_{\mathrm{X}'})$ is a coherent $\mathscr{O}_\mathrm{X}$-Algebra by hypothesis (**II**, 6.1.3). To say that X is normal at a point x means that the canonical homomorphism $\mathscr{O}_x \to (f_*(\mathscr{O}_{\mathrm{X}'}))_x$ is bijective (**II**, 6.3.4); but the set of these points is *open* since $\mathscr{O}_\mathrm{X}$ and $f_*(\mathscr{O}_{\mathrm{X}'})$ are coherent ($\mathbf{0}_\mathrm{I}$, 5.2.7). □

Corollary (6.13.3). — *If* A *is an integral Noetherian* Japanese *ring,* $\mathrm{Nor}(\mathrm{Spec}(\mathrm{A}))$ *is open in* $\mathrm{Spec}(\mathrm{A})$.

Proposition (6.13.4). — *Let* X *be a locally Noetherian prescheme. For* $\mathrm{Nor}(\mathrm{X})$ *to be open, it is necessary and it suffices that every maximal point* x *of* X *such that* $x \in \mathrm{Nor}(\mathrm{X})$ *(which means that* X *is reduced at the point* x*) be interior to* $\mathrm{Nor}(\mathrm{X})$.

Proof. There is only to prove the sufficiency of the condition. By virtue of the remark made in (6.13.1), one can restrict (replacing X by an open of X in which X is reduced), i.e. to suppose that the maximal points of X belong to $\mathrm{U}_{\mathrm{S}_1}(\mathrm{X})$ and to $\mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$; as $\mathrm{Nor}(\mathrm{X}) = \mathrm{U}_{\mathrm{S}_1}(\mathrm{X}) \cap \mathrm{U}_{\mathrm{R}_1}(\mathrm{X})$ by virtue of Serre's criterion (5.8.6), one deduces from (6.11.7) and (6.12.10) that these two sets are open; hence $\mathrm{Nor}(\mathrm{X})$ is open in X. □

Corollary (6.13.5). — *If* X *is such that* $\mathrm{Reg}(\mathrm{X})$ *is open in* X, *then* $\mathrm{Nor}(\mathrm{X})$ *is open in* X.

Proof. Indeed, every maximal point x where X is reduced (hence $\mathscr{O}_{\mathrm{X},x}$ is a field) belongs to $\mathrm{Reg}(\mathrm{X})$, hence is interior to $\mathrm{Reg}(\mathrm{X})$ by hypothesis, and *a fortiori* is interior to $\mathrm{Nor}(\mathrm{X})$. □

Proposition (6.13.6). — (Nagata). — *Let* A *be a Noetherian integral ring,* K *its field of fractions,* K' *a finite extension of* K, A' *the integral closure of* A *in* K'. *For* A' *to be a finite* A*-algebra, it is necessary and it suffices that the two following conditions be satisfied:*

label=(i) There exists $f \neq 0$ *in* A *such that the integral closure* A'_f *of the ring of fractions* A_f *in* K' *is a finite* A_f*-algebra.*

lbbel=(ii) For every prime ideal $\mathfrak{p} \in \mathrm{Spec}(\mathrm{A})$, *the integral closure* $\mathrm{A}'_\mathfrak{p}$ *of the local ring* $\mathrm{A}_\mathfrak{p}$ *in* K' *is a finite* $\mathrm{A}_\mathfrak{p}$*-algebra.*

Proof. The conditions are necessary, for for every multiplicative subset S of A, $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K′ and is by hypothesis a $S^{-1}A$-module of finite type. To see that the conditions are sufficient, consider the filtered increasing family (B_λ) of sub-A-algebras of K′ which are A-algebras finite and have K′ as field of fractions. Set $Y = \operatorname{Spec}(A)$, $X_\lambda = \operatorname{Spec}(B_\lambda)$, denote by u_λ the morphism $X_\lambda \to Y$ and let $S_\lambda = X_\lambda - \operatorname{Nor}(X_\lambda)$, $T_\lambda = u_\lambda(S_\lambda)$. One can restrict to the B_λ which contain a finite set whose image in A'_f is a system of generators of the A_f-module A'_f; this implies, by virtue of hypothesis (i), that $(B_\lambda)_f = A'_f$ for every λ, or equivalently that $u_\lambda^{-1}(D(f))$ is contained in $\operatorname{Nor}(X_\lambda)$. By virtue of (6.13.2), S_λ is therefore *closed* in Y. But for every $\mathfrak{p} \in \operatorname{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to IV-2 | 169
$\operatorname{Nor}(X_\lambda)$; in other words T_λ is *closed* in Y, $\bigcup_\lambda T_\lambda$ is closed, and as u_λ is a finite morphism, T_λ is *closed* in Y. But for every $\mathfrak{p} \in \operatorname{Spec}(A)$, it exists by virtue of (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and consequently all points of X_λ lying over $\mathfrak{p}$ belong to $\operatorname{Nor}(X_\lambda)$; in other words $\mathfrak{p} \notin T_\lambda$; as Y is Noetherian and the T_λ are closed, there exists a λ_0 such that $T_{\lambda_0} = \varnothing$; hence B_{λ_0} is integrally closed, and as its field of fractions is K′, one has $B_{\lambda_0} = A'$. ☐

Proposition (6.13.7). — *Let* A *be a Noetherian ring,* $X = \operatorname{Spec}(A)$. *The following conditions are equivalent:*

label=a) *For every prescheme* X′ *locally of finite type over* X, $\operatorname{Nor}(X')$ *is open in* X′.
lbbel=b) *For every* A*-algebra finite and integral* A′, $\operatorname{Nor}(\operatorname{Spec}(A'))$ *is open in* $\operatorname{Spec}(A')$.
lcbel=c) *For every prime ideal* $\mathfrak{p}$ *of* A *and every finite radicial extension* K′ *of the field of fractions* K *of* $A/\mathfrak{p}$, *there exists a sub-*A*-algebra* A′ *of* K′, *finite over* A, *having* K′ *as field of fractions and such that* $\operatorname{Nor}(\operatorname{Spec}(A'))$ *is open in* $\operatorname{Spec}(A')$.

Proof. The fact that *a*) implies *b*) and that *b*) implies *c*) is proved as in (6.12.4). To show that *c*) implies *a*), one reduces as in (6.12.4) to proving that if A′ is an A-algebra integral and of finite type, the generic point of $\operatorname{Spec}(A')$ is interior to $\operatorname{Nor}(\operatorname{Spec}(A'))$. One distinguishes two cases as in (6.12.4) by proving first the

Lemma (6.13.7.1). — *Let* A *be a Noetherian integral ring,* A′ *an* A*-algebra integral of finite type, containing* A *and such that the field of fractions* K′ *of* A′ *is a separable extension of the field of fractions* K *of* A. *If* $\operatorname{Nor}(\operatorname{Spec}(A))$ *is open in* $\operatorname{Spec}(A)$, $\operatorname{Nor}(\operatorname{Spec}(A'))$ *is open in* $\operatorname{Spec}(A')$.

Proof. It suffices to prove (in view of (6.13.2)) that the generic point of $\operatorname{Spec}(A')$ is interior to $\operatorname{Nor}(\operatorname{Spec}(A'))$. The proof follows the same approach as that of (6.12.4.1), from which we conserve the notations. We note that when A is integrally closed, and then on takes $A_1 = A[t_1, \dots, t_n]$ and is integrally closed (Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, cor. 2 of prop. 13); one reduces next to the case where A′ is a free A-module of *finite type*; the reasomment then applies (replacing A by an appropriate ring of fractions A_f with $f \neq 0$) the reasoning from case I) of (6.12.4) to see that the fibres $g^{-1}(z)$ of the morphism $g : X' \to X$ are regular and *a fortiori* normal. Furthermore g is flat and X is normal, hence (6.5.4, (ii)) X′ is normal. One then passes to the general case as in (6.12.4); II)), from which we conserve further the notations; applying hypothesis *c*), one sees this time that $\operatorname{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \to X'$ is flat and surjective; one concludes by applying (6.5.4, (i)). ☐

This lemma being proved, one passes to the general case as in (6.12.4; II)), from which we conserve the notations; applying hypothesis *c*), one sees this time that $\operatorname{Nor}(X'_1)$ is open and one reduces thus to the case where X'_1 is normal and $g : X' \to X'$ flat and surjective; one concludes by applying (6.5.4, (i)). ☐

6.14. Base change and integral closure

Proposition (6.14.1). — *Let* Y, Y' *be two locally Noetherian preschemes,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a normal morphism* (6.8.1)*. Then, for every normal* Y*-prescheme* X*, the prescheme* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ *is normal.*

Proof. We note that in this statement one *does not suppose* X locally Noetherian. ☐

Lemma (6.14.1.1). — *Let* R *be a ring, composed direct of a finite number of fields.*

label=(i) For a subring A *of* R *having* R *as total ring of fractions to be normal, it is necessary and it suffices that it be integrally closed in* R.

lbbel=(ii) Let (A_λ) *be a family of normal subrings of* R*; if* $\mathrm{A} = \bigcap_\lambda \mathrm{A}_\lambda$ *admits* R *as total ring of fractions,* A *is normal.*

Proof. (i) As A is a subring of R, $\mathrm{A}_{\mathfrak{p}}$ is a subring of $\mathrm{R}_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A; furthermore $\mathrm{R}_{\mathfrak{p}}$ is a ring of fractions of $\mathrm{A}_{\mathfrak{p}}$; furthermore $\mathrm{R}_{\mathfrak{p}}$ is necessarily a composed direct of a finite number of fields, hence every non-zero divisor of 0 in $\mathrm{R}_{\mathfrak{p}}$ is invertible; this proves that $\mathrm{R}_{\mathfrak{p}}$ is the total ring of fractions of $\mathrm{A}_{\mathfrak{p}}$. If A is integrally closed in R, $\mathrm{A}_{\mathfrak{p}}$ is therefore integrally closed in $\mathrm{R}_{\mathfrak{p}}$; but if $\mathrm{R}_{\mathfrak{p}}$ is composed direct of 2 rings at least, the integral closure of $\mathrm{A}_{\mathfrak{p}}$ in $\mathrm{R}_{\mathfrak{p}}$ is composed direct of 2 rings at least non-reduced to 0, which is absurd since $\mathrm{A}_{\mathfrak{p}}$ is a local ring; hence $\mathrm{R}_{\mathfrak{p}}$ is necessarily a field and $\mathrm{A}_{\mathfrak{p}}$ is integral and integrally closed, which by definition means that A is normal. Conversely, if A is normal, $\mathrm{R}_{\mathfrak{p}}$, total ring of fractions of an integral ring $\mathrm{A}_{\mathfrak{p}}$, is a field and $\mathrm{A}_{\mathfrak{p}}$ is integrally closed in $\mathrm{R}_{\mathfrak{p}}$; hence R is an integral element over A, its image in each $\mathrm{R}_{\mathfrak{p}}$ is integral over $\mathrm{A}_{\mathfrak{p}}$, hence belongs to $\mathrm{A}_{\mathfrak{p}}$; we conclude that $x \in \mathrm{A}$ (Bourbaki, *Alg. comm.*, chap. II, § 3, no 3, cor. 1 of th. 1), and hence A is integrally closed in R.

(ii) As $\mathrm{A} \subset \mathrm{A}_\lambda \subset \mathrm{R}$ for every λ, R is also the total ring of fractions of each A_λ. In view of the **IV-2 | 170**
characterisation of normal rings having R as total ring of fractions, the assertion (ii) follows from Bourbaki, *Alg. comm.*, chap. V, § 1, no 3, prop. 12. ☐

This lemma being proved, the proof of (6.14.1) proceeds in several steps.

I) *Reduction to the case where* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, A, A' *being local Noetherian rings,* A integral, B the integral closure of A. — It suffices to prove that for every $x' \in \mathrm{X}'$, the local ring $\mathscr{O}_{\mathrm{X}',x'}$ is integral and integrally closed; let x, y, y' be the canonical images of x' in X, Y, Y' respectively; if one sets $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, $\mathrm{A}' = \mathscr{O}_{\mathrm{Y}',y'}$, $\mathrm{B} = \mathscr{O}_{\mathrm{X},x}$, the rings $\mathscr{O}_{\mathrm{X},x}$, $\mathscr{O}_{\mathrm{Y},y}$, $\mathscr{O}_{\mathrm{Y}',y'}$ for x, y, y' fixed, are the local rings at the prime ideals of the ring $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ (**I**, 3.6.5), and so it suffices to prove that B' is a *normal* ring. Let us note on the other hand that $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is a normal morphism by definition (6.8.1 and **I**, 3.6.5); one is reduced to the case where Y and Y' are local schemes, that is to say to the case where A is an integral and integrally closed local ring ($\mathbf{0}_{\mathrm{I}}$, 6.8.1 and **I**, 3.6.5); one is reduced to the case where Y and Y' are local schemes. Let us designate by (B_α) the family of *integral closures* of the sub-A-algebras *of finite type* of B; it is clear that B is the union of the filtered increasing family of the B_α. As $\varinjlim$ commutes with the tensor product, B' is therefore also the union of the filtered increasing family $\overline{\mathrm{B}'_\alpha} = \mathrm{B}_\alpha \otimes_{\mathrm{A}} \mathrm{A}'$. To prove that B' is normal, it suffices, by virtue of (5.13.6), to show that the rings B'_α are *normal* and that, for $\alpha \leqslant \beta$, every irreducible component of $\mathrm{Spec}(\mathrm{B}'_\beta)$ *dominates* one irreducible component of $\mathrm{Spec}(\mathrm{B}'_\alpha)$. But this last property follows from the hypothesis that A' is an A-module flat and from (2.3.7, (ii)).

One can therefore restrict to proving that $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ is normal when B is the integral closure of a A-algebra integral *of finite type* C; in this case, if one sets $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$, the morphism $\mathrm{Spec}(\mathrm{C}') \to \mathrm{Spec}(\mathrm{C})$ is normal (6.8.2); as $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$, one sees that one can replace A and A' by C and C' respectively, hence suppose that A is integral and that B is the *integral closure* of A. Finally, the procedure from the beginning allows to restrict to the case where A is *local* (taking into account the fact that, if B is the integral closure of A, $\mathrm{B}_{\mathfrak{p}}$ is the integral closure of $\mathrm{A}_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A).

II) *Reduction to the case where* A *is a local integral ring of dimension* 1, B *a discrete valuation ring, integral closure of* A *and the morphism* $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ *radicial.* — Let K be the field of fractions of A. One knows ($\mathbf{0}$, 23.2.7) that B is intersection of a family (V_λ) of discrete valuation rings such that, for every $x \in \mathrm{K}$, one has $x \in \mathrm{V}_\lambda$ except for a finite number of indices λ. One therefore has an exact sequence of A-modules

$$0 \longrightarrow \mathrm{B} \longrightarrow \mathrm{K} \longrightarrow \bigoplus_\lambda (\mathrm{K}/\mathrm{V}_\lambda).$$

Set $K' = K \otimes_A A'$, $V'_\lambda = V_\lambda \otimes_A A'$; by flatness, one deduces from the preceding exact sequence a new exact sequence

$$0 \longrightarrow B' \longrightarrow K' \longrightarrow \bigoplus (K'_\lambda / V'_\lambda)$$

and hence $B' = \bigcap_\lambda V'_\lambda$. Furthermore $\mathrm{Spec}(K')$ is the fibre of the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ at the generic point of $\mathrm{Spec}(A)$, hence K' is a Noetherian ring, *normal*, and hence a composed direct of a finite number of integral rings and integrally closed; the direct composite L' of the fields of fractions of these latter is the total ring of fractions of A', hence also that of B'. One can therefore apply the lemma (6.14.1.1), and if one proves that each of the V'_λ is a *normal* ring, then so is B'. **IV-2** | 171

One knows on the other hand (**0**, 23.2.7) that one can take the V_λ such that there is a sub-A-algebra finite C of K such that V_λ is the integral closure of $C_{\mathfrak{p}_\lambda}$ where $\mathfrak{p}_\lambda$ is a prime ideal of height 1 in C. If one sets $C'_{\mathfrak{p}_\lambda} = C_{\mathfrak{p}_\lambda} \otimes_A A'$, the morphism $\mathrm{Spec}(C'_{\mathfrak{p}_\lambda}) \to \mathrm{Spec}(C_{\mathfrak{p}_\lambda})$ is normal (6.8.2) and one has $V'_\lambda = V_\lambda \otimes_{C_{\mathfrak{p}_\lambda}} C'_{\mathfrak{p}_\lambda}$; one can therefore replace B by V_λ and A by $C_{\mathfrak{p}_\lambda}$, hence suppose that A is local, integral and *of dimension* 1, B its integral closure and a discrete valuation ring. There is a sub-A-algebra *finite* A_1 of B such that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A_1)$ is radicial (**0**, 23.2.5), which implies in particular that A_1 is also a local ring (obviously of dimension 1); furthermore one can suppose that B and A_1 have the same residue field (*loc. cit.*). One can therefore by the same method replace A by A_1. Applying if necessary the procedure from the beginning of I), one can suppose finally that A' is also a local ring and that the homomorphism $A \to A'$ is *local*.

III) *End of the proof.* — We shall first establish the following lemma: □

Lemma (6.14.1.2). — *Let* A *be a local Noetherian integral ring of dimension* 1, A' *a local Noetherian integral ring,* $A \to A'$ *a local homomorphism such that the corresponding morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ *is* normal. *Let* K *be the field of fractions of* A, $\mathfrak{m}$ *the maximal ideal of* A, $\boldsymbol{k} = A/\mathfrak{m}$ *its residue field.*

- label=(i) If $\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$ are the minimal ideals of A', $K' = K \otimes_A A'$ is composed direct of integral rings and integrally closed $K'_j \supset A'/\mathfrak{q}'_j$ $(1 \leqslant j \leqslant r)$, K'_j having as field of fractions the field of fractions L'_j of $A'/\mathfrak{q}'_j$, so that the total ring of fractions L' of A' is identified with the composed direct of the L'_j and A' is a subring of K'.

lbbel=(ii) The ideal $\mathfrak{p}' = \mathfrak{m}A'$ of A' is prime; the ring $A'_{\mathfrak{p}'}$ is of dimension 1 and is identified with a sub-product of the fields L'_j for the indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$.

lcbel=(iii) If $\overline{A'_{\mathfrak{p}'}}$ denotes the subring of L', product of $A'_{\mathfrak{p}'}$ and of the L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$, one has

$$A' = K' \cap \overline{A'_{\mathfrak{p}'}}. \tag{6.14.1.3}$$

Proof. Assertion (i) has already been seen in the course of the proof, independent of the hypothesis on the dimension of A. The hypothesis that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is normal implies that $A' \otimes_A \boldsymbol{k} = A'/\mathfrak{m}A'$ is a normal local ring and *integral*, which shows already that $\mathfrak{p}' = \mathfrak{m}A'$ is prime. One has, by (6.1.2), $\dim(A'_{\mathfrak{p}'}) = \dim(A) + \dim(A'_{\mathfrak{p}'}/\mathfrak{m}A'_{\mathfrak{p}'})$. But $A'_{\mathfrak{p}'}/\mathfrak{m}A'_{\mathfrak{p}'} = A'_{\mathfrak{p}'}/\mathfrak{p}'A'_{\mathfrak{p}'}$ is the residue field of $A'_{\mathfrak{p}'}$, hence $\dim(A'_{\mathfrak{p}'}) = 1$. The fact that $A'_{\mathfrak{p}'}$ is contained in the composed direct of the L'_j for these indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$, for these indices j are the minimal prime ideals of $A'_{\mathfrak{p}'}$. **IV-2** | 172

It remains to prove (6.14.1.3). Conversely, let y' be an element of this intersection; let a be a "parameter" for A, i.e. an element such that Aa is $\mathfrak{m}$-primary, and let a' be the image of a in A'; every element of K can be written x/a^n for $x \in A$ and $n > 0$ since Aa contains a power of $\mathfrak{m}$; hence one can write $y' = x'/a'^n$ with $x' \in A'$. Note now that $\mathfrak{p}'$ is the *only* prime ideal associated to $A'/a'^n A' = (A/a^n A) \otimes_A A'$: as A' is a flat A-module, this follows from (3.3.1), $\mathfrak{m}$ being the only prime ideal associated to $A/a^n A$ since A is integral. Hence, $a'^n A'$ is the inverse image in A' of $a'^n A'_{\mathfrak{p}'}$; as by hypothesis $y' \in \widehat{A'_{\mathfrak{p}'}}$, the image of x' in $A'_{\mathfrak{p}'}$ belongs to $a'^n A'_{\mathfrak{p}'}$; whence $x' \in a'^n A'$ and $y' \in A'$, which finishes the proof.

In the situation in which we have arrived at the end of II), B' is *radicial* over A' (**I**, 3.5.7) and is hence also a local ring; furthermore B' is integer over A', hence, if $\mathfrak{q}'$ is the unique prime ideal of B' lying over $\mathfrak{p}'$, $\mathfrak{q}'B'_{\mathfrak{p}'}$ is the unique maximal ideal of $B'_{\mathfrak{p}'}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no 1, prop. 1), hence $B'_{\mathfrak{q}'} = B'_{\mathfrak{p}'} = B \otimes_{A_\mathfrak{p}} A'_{\mathfrak{p}'}$. We shall first prove that $B'_{\mathfrak{p}'}$ is *Noetherian* and *normal*. $B'_{\mathfrak{p}'}$ contains $A'_{\mathfrak{p}'}$ and is contained in $K'_{\mathfrak{p}'}$, hence in the product L_1 of the L'_j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$. The composed direct L'' of the fields of fractions of these latter is also the total ring of fractions of A', hence also that of $B'_{\mathfrak{p}'}$. For every index j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$, let $\mathfrak{a}'_j$ be the product of the L'_h for these indices h such that $h \neq j$, so that $\mathfrak{a}'_j \cap A'_{\mathfrak{p}'} = \mathfrak{q}'_j A'_{\mathfrak{p}'}$; as every element of $A' - \mathfrak{p}'$ is regular in L_1, one has also

$$\mathfrak{a}'_j \cap A'_{\mathfrak{p}'} = \mathfrak{q}'_j A'_{\mathfrak{p}'}.$$

Let $\mathfrak{r}'_j = B'_{\mathfrak{p}'} \cap \mathfrak{a}'_j$, so that $B'_{\mathfrak{p}'}/\mathfrak{r}'_j$ is isomorphic to the projection of $B'_{\mathfrak{p}'}$ in L'_j; hence $B'_{\mathfrak{p}'}/\mathfrak{r}'_j$ contains the local integral ring of *dimension* 1, $A'_{\mathfrak{p}'}/\mathfrak{q}'_j A'_{\mathfrak{p}'}$, and is contained in its field of fractions L'_j; it is hence a Noetherian ring by virtue of the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5).

As the intersection of the $\mathfrak{r}'_j$ is reduced to 0, one deduces that $B'_{\mathfrak{p}'}$ itself is Noetherian, by virtue of the classical lemma: **IV-2 | 173**

Lemma (6.14.1.4). — *Let* R *be a ring,* $\mathfrak{a}$ *and* $\mathfrak{b}$ *two ideals of* R*; if* $R/\mathfrak{a}$ *and* $R/\mathfrak{b}$ *are Noetherian, so is* $R/(\mathfrak{a} \cap \mathfrak{b})$.

Proof. Indeed, let $\mathfrak{c}$ be an ideal of R such that $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{c}$; one has therefore $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{c} \subset \mathfrak{a}$; now $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{c})$ is an R-module isomorphic to $(\mathfrak{a} + \mathfrak{c})/\mathfrak{a}$, hence an ideal of $R/\mathfrak{a}$, and hence is of finite type; on the other hand $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is a sub-R-module of $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$, and this latter is isomorphic to $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$, hence is of finite type as an ideal of $R/\mathfrak{b}$; hence $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is also an R-module of finite type, and it follows finally that $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{b})$ is also an R-module of finite type. □

Let us note on the other hand that Spec(A) is made up of the closed point $\mathfrak{m}$ and of the generic point (o), of residue fields $\boldsymbol{k}$ and K respectively; as in the case where we are placed, the fields of fractions of B and its residue field are respectively isomorphic to K and $\boldsymbol{k}$, the fibres of the morphism $u : \mathrm{Spec}(B'_{\mathfrak{p}'}) \to \mathrm{Spec}(B)$ at the closed point and at the generic point of Spec(B) are respectively *isomorphic* to the fibres of the morphism $\mathrm{Spec}(A'_{\mathfrak{p}'}) \to \mathrm{Spec}(A)$ at the closed point and at the generic point of Spec(A) (**I**, 3.6.4), hence are *geometrically normal*; as on the other hand the morphism u is flat, we conclude that it is *normal* (6.8.1). But B and $B'_{\mathfrak{p}'}$ are Noetherian and B is normal, hence (6.5.4) $B'_{\mathfrak{p}'}$ is *normal*.

This being so, B is union of the filtered increasing family of its sub-A-algebras of finite type A_α; by flatness, B' is union of the filtered increasing family of the $A'_\alpha = A_\alpha \otimes_A A'$; if one sets $\mathfrak{p}'_\alpha = \mathfrak{q}' \cap A'_\alpha$, $B'_{\mathfrak{q}'}$ is also union of the filtered increasing family of the $(A'_\alpha)_{\mathfrak{p}'_\alpha}$ (5.13.3). Denote by L'' the composed direct of the fields L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$; then for every α, $(A'_\alpha)_{\mathfrak{p}'_\alpha}$ is contained in $K'_{\mathfrak{p}'}$, and the ring $\overline{B'_{\mathfrak{q}'}} = L'' \times B'_{\mathfrak{q}'}$ is therefore union of the rings $L'' \times (A'_\alpha)_{\mathfrak{p}'_\alpha} = (\overline{A'_\alpha})_{\mathfrak{p}'_\alpha}$. But each of the A_α is local, Noetherian, integral and of dimension 1, and the morphism $\mathrm{Spec}(A'_\alpha) \to \mathrm{Spec}(A_\alpha)$ is normal (6.8.2), hence one can apply (6.14.1.2) and one has

$$A'_\alpha = K' \cap (\overline{A'_\alpha})_{\mathfrak{p}'_\alpha}$$

for every α; taking the inductive limit of each of the two members, it follows that

$$B' = K' \cap \overline{B'_{\mathfrak{q}'}}.$$

But $\overline{B'_{\mathfrak{q}'}}$, composed direct of normal rings L'' and $B'_{\mathfrak{q}'}$, is normal, and as it has the same total ring of fractions as K', the lemma (6.14.1.1) shows that B' is normal. □

Corollary (6.14.2). — *Let k be a field, X a k-prescheme normal (not necessarily locally Noetherian). Then, for every separable extension k' of k, $X_{(k')} = X \otimes_k k'$ is* normal.

Proof. One knows indeed (6.7.6) that the morphism $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is then normal. □

Corollary (6.14.3). — *Let k be a field, X, Y two k-preschemes integral and normal, whose fields of rational functions* $K = R(X)$, $L = R(Y)$ *are separable extensions of k. Then the prescheme* $X \times_k Y$ *is* normal.

Proof. The question being local on X and Y, one can suppose that X = Spec(A), Y = Spec(B), where A and B are two integral rings and integrally closed, of fields of fractions K and L respectively; suppose first that A is a k-algebra of finite type, hence K is a finite type extension of k. By flatness, $A \otimes_k B$ is a sub-k-algebra of $K \otimes_k L$; now $K \otimes_k L$ is a Noetherian normal ring (by virtue of (6.7.1) or **IV-2 | 174**

(6.14.2)), hence a composed direct of integral rings C_i in finite number, so that if E_i is the field of fractions of C_i, the composed direct E of the E_i is the total ring of fractions of $K \otimes_k L$; it is also the total ring of fractions of $A \otimes_k B$, since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$. This being so, one can write $A \otimes_k B = (A \otimes_k L) \cap (K \otimes_k B)$, and it follows from (6.14.2) that $A \otimes_k L$ and $K \otimes_k B$ are normal, hence $A \otimes_k B$ is normal by virtue of (6.14.1.1).

Consider now the general case; A is then union of the filtered increasing family of its sub-k-algebras of finite type A_α, hence $A \otimes_k B$ is the inductive limit of the normal rings $A_\alpha \otimes_k B$. To prove that $A \otimes_k B$ is normal, it suffices hence (5.13.6) to prove that every irreducible component of $\mathrm{Spec}(A_\beta \otimes_k B)$ dominates one irreducible component of $\mathrm{Spec}(A_\alpha \otimes_k B)$ for $A_\alpha \subset A_\beta$, which follows immediately from the fact that B is a k-module flat, from (2.3.7, (ii)) and from the fact that A_α and A_β are integral rings. □

Proposition (6.14.4). — *Let* A *be a Noetherian ring,* A′ *a Noetherian* A*-algebra such that the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ *is normal. Let* B *be an* A*-algebra,* C *the integral closure of* A *in* B*; set* $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, C′ *being identified with a subring of* B′*; then* C′ *is the integral closure of* A′ *in* B′.

Proof. We shall proceed in several steps.

I) *Reduction to the case where* B *is reduced.* — Set $B_0 = B_{\mathrm{red}} = B/\mathfrak{N}$ where $\mathfrak{N}$ is the nilradical of B, and let C_0 be the integral closure of A in B_0. One has the following lemma:

Lemma (6.14.4.1). — *Let* A *be a ring,* B *an* A*-algebra,* $B_0 = B/\mathfrak{n}$ *the quotient of* B *by a nilideal. If* C_0 *is the integral closure of* A *in* B_0*, the inverse image* C *of* C_0 *by the canonical homomorphism* $\varphi : B \to B_0$ *is the integral closure of* A *in* B.

Proof. Indeed, if $x \in B$ is such that $\varphi(x)$ satisfies an integral dependence equation over A, one deduces that x satisfies a relation of the form $x^n + a_1 x^{n-1} + \ldots + a_n \in \mathfrak{n}$ with $a_k \in A$, whence, by raising to a sufficiently large power, an integral dependence equation for x, with coefficients in A. □

One has therefore the exact sequence of A-modules

$$0 \longrightarrow C \longrightarrow B \longrightarrow B_0/C_0$$

whence, by flatness, the exact sequence

$$0 \longrightarrow C' \longrightarrow B' \longrightarrow B'_0/C'_0$$

where one has set $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$. If one proves that C'_0 is the integral closure of A′ in B'_0, the lemma (6.14.4.1) will prove that C′ is the integral closure of A′ in B′. One can therefore restrict to the case where B is an A-algebra of finite type; let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the $B/\mathfrak{q}_i$; if C_0 is the integral closure of A in B_0, one has $C = B \cap C_0$; if one sets $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$, one has, by flatness, $C' = B' \cap C'_0$ $(\mathbf{0}_{\mathrm{I}},\ 6.1.3)$; it suffices therefore to prove that C'_0 is the integral closure of A′ in B'_0. But C_0 is composed direct of the C_i, integral closures of A in the $B_i = B/\mathfrak{q}_i$; hence C'_0 is composed direct of the $C'_i = C_i \otimes_A A'$ and it suffices to show that C'_i is the integral closure of A′ in $B'_i = B_i \otimes_A A'$. One is thus reduced to the case where B is integral and an A-algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $A \to B$, one has also $B' = B \otimes_{A/\mathfrak{p}} (A'/\mathfrak{p}A')$; as the morphism $\mathrm{Spec}(A'/\mathfrak{p}A') \to \mathrm{Spec}(A/\mathfrak{p})$ is normal, one can replace A by $A/\mathfrak{p}$ and A′ by $A'/\mathfrak{p}A'$, and suppose hence that $A \subset B$.

II) *Reduction to the case where* B *is an integral* A*-algebra of finite type containing* A. — Let (B_α) be the filtered increasing family of sub-A-algebras of finite type of B, and let C_α be the integral closure of A in B_α. It follows immediately from the definition of the integral closure that C is union of the C_α; if one sets $B'_\alpha = B_\alpha \otimes_A A'$, $C'_\alpha = C_\alpha \otimes_A A'$, C'_α is contained in B'_α by flatness, and for the same reason B′ is union of the filtered increasing family of the B'_α and C′ union of the filtered increasing family of the C'_α. If one proves that C'_α is the integral closure of A′ in B'_α, it will follow immediately that C′ is the integral closure of A′ in B′. One can therefore restrict to the case where B is an A-algebra of finite type; let $\mathfrak{q}_i$ $(1 \leqslant i \leqslant n)$ be its minimal prime ideals; as B is supposed *reduced*, it is identified with a *subring* of the product B_0 of the $B/\mathfrak{q}_i$; if C_0 is the integral closure of A in B_0, one has $C = B \cap C_0$; if one sets $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$, one has, by flatness, $C' = B' \cap C'_0$ $(\mathbf{0}_{\mathrm{I}},\ 6.1.3)$; it suffices therefore to prove that C'_0 is the integral closure of A′ in B'_0. But C_0 is composed direct of the C_i, **IV-2** | 175 integral closures of A in the $B_i = B/\mathfrak{q}_i$; hence C'_0 is composed direct of the $C'_i = C_i \otimes_A A'$ and it suffices to show that C'_i is the integral closure of A' in $B'_i = B_i \otimes_A A'$. One is thus reduced to the case where B is integral and A-algebra of finite type; if $\mathfrak{p}$ is the kernel of the homomorphism $A \to B$, one has also $B' = B \otimes_{A/\mathfrak{p}} (A'/\mathfrak{p}A')$; as the morphism $\mathrm{Spec}(A'/\mathfrak{p}A') \to \mathrm{Spec}(A/\mathfrak{p})$ is normal, one can replace A by $A/\mathfrak{p}$ and A' by $A'/\mathfrak{p}A'$, and suppose hence that $A \subset B$.

III) *Case where* A *is a field,* B *a field, finite extension of* A *and such that* A *is algebraically closed in* B. — One has then $C = A$, and A' is a Noetherian geometrically normal A-algebra, hence a composed direct of integral rings and integrally closed A'_i, the fields of fractions K'_i of the A'_i being *separable* extensions of A (4.6.1). The A'-algebra B' is therefore composed direct of the $B \otimes_A A'_i = B'_i$. One can therefore restrict to the case where A' is integral, its field of fractions K' being a separable extension of A. Now, as K' is a separable extension of A, $B \otimes_A K'$ is reduced (4.3.7), and as furthermore B is a primary extension of A, $B \otimes_A K'$ is integral (4.3.2); it is hence of the same type as B'. Let L' be the field of fractions of $B \otimes_A K'$ (which is also that of B'); it is obviously a composed field of $B(K')$ of its sub-fields B and K'; as A is algebraically closed in B, that K' is a separable extension of A, and that B and K' are linearly disjoint over A, K' is algebraically closed in $B(K') = L'$ (Bourbaki, *Alg.*, chap. V, § 9, exerc. 2); *a fortiori*, A', being integrally closed, is integrally closed in L', hence in B', which proves the proposition in this case.

IV) *Case where* A *is integral,* B *a field, finite extension of the field of fractions* K *of* A. — Then $B' = (A' \otimes_A K) \otimes_K B$ is a B-algebra Noetherian geometrically normal, and hence B' is composed direct of integral rings; the total ring of fractions L' of B' is then composed direct of a finite number of fields. This being so, as B is the field of fractions of C, B' is a ring of fractions of C' and L' is therefore also the total ring of fractions of C'. Now ($\mathbf{0}$, 6.14.1) as C is a normal ring and $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ a normal morphism of Noetherian rings, C' is a normal ring; but it follows from (6.14.1.1) that C' is then integrally closed in L', and *a fortiori* in B', and hence is the integral closure of A' in B'.

V) *End of the proof.* — By part II), one can suppose that B is an A-algebra integral of finite type containing A. Let K be the field of fractions of A, L that of B, which is a finite type extension of K. Let M be the algebraic closure of K in L, which is a finite algebraic extension of K; let C_0 be the integral closure of A in M, which is also the integral closure of A in L; if one sets $C'_0 = C \otimes_A A'$, one has hence $C' = B' \cap C'_0$ by flatness ($\mathbf{0}_{\mathrm{I}}$, 6.1.3). Now, it follows from IV) that C'_0 is the integral closure of A' in $M' = M \otimes_A A'$; furthermore, M' is a Noetherian ring and the morphism $\mathrm{Spec}(M') \to \mathrm{Spec}(M)$ is normal (as we have seen in the course of IV)); applying III) to M and M' in place of A and A' and to L in place of B, one deduces that C'_0 is the integral closure of A' in L', and $C' = C'_0 \cap B'$ is the integral closure of A' in L', and hence C' is the integral closure of A' in B'. □ **IV-2 | 176**

Corollary (6.14.5). — *Let* A *be a Noetherian ring,* A' *a Noetherian* A*-algebra such that the morphism* $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ *is normal. Let* B, C *be two* A*-algebras,* $\varphi : B \to C$ *an* A*-homomorphism,* D *the integral closure of* B *in* C. *If one sets* $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, $D' = D \otimes_A A'$, D' *is the integral closure of* B' *in* C'.

Proof. Let (B_λ) be the filtered increasing family of sub-A-algebras of finite type of B, and set $B'_\lambda = B_\lambda \otimes_A A'$, so that B' is union of the filtered increasing family of the sub-A'-algebras of finite type B'_λ. Let D_λ be the integral closure of B_λ in C, and set $D'_\lambda = D_\lambda \otimes_A A'$, so that D is union of the D_λ and D' union of the D'_λ; if we prove that D'_λ is the integral closure of B'_λ in C', it will follow that D' is the integral closure of B' in C'. Now B_λ and B'_λ are Noetherian and the morphism $\mathrm{Spec}(B'_\lambda) \to \mathrm{Spec}(B_\lambda)$ is normal (6.8.2); one can therefore apply (6.14.4) by replacing A, A' and B by B_λ, B'_λ and C respectively, whence the corollary. □

One can for example apply (6.14.5) when A is its local ring *excellent* and A' its completion $\widehat{A}$, since in this case $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is a regular morphism (7.8.2).

6.15. Geometrically unibranch preschemes

(6.15.1). We shall say that a prescheme X is *unibranch* (resp. *geometrically unibranch*) at a point x, or that the point x is unibranch (resp. geometrically unibranch) in X if the local ring $\mathscr{O}_{X,x}$ is unibranch (resp. geometrically unibranch) ($\mathbf{0}$, 23.2.1); we say that X is *unibranch* (resp. *geometrically unibranch*) if it is so at every point. It follows from this definition that, for X to be unibranch (resp. geometrically unibranch) at a point, it is necessary and it suffices that X_{red} be so at this point.

(6.15.2). One must take care that with the definitions of (6.15.1), a *local ring* A can be unibranch (resp. geometrically unibranch) *without* Spec(A) being so; in other terms, it can happen that there are prime ideals $\mathfrak{p}$ of A such that $A_{\mathfrak{p}}$ is not unibranch; it amounts to the same to say that on a prescheme, the notion of geometrically unibranch point *is not stable by generization*. We shall see this on the following example.

Let K be an algebraically closed field of characteristic 0, B the integral ring $K[U,V,W]/(U^2(U-W)-V^2(U+W))$ (U, V, W indeterminates) so that $Y = \mathrm{Spec}(B)$ is a "cone with vertex the origin, having a double generatrix". We shall designate by u, v, w the images of U, V, W in B. Let R be the field of fractions of B, and consider in R the element $t = v(u+w)/u$, which does not belong to B; let us show that $C = B[t]$ is the *integral closure* of B. One has indeed $t^2 = u^2 - w^2$, hence t is integral over B, and $v = tu/(u+w)$; the ring $C_1 = K[t,u,w]$ is integrally closed, for it is isomorphic to $K[T,U,W]/(T^2-U^2+W^2)$ and is therefore the integral closure of the integrally closed ring $K[U,W]$ in the quadratic extension $K(U,W)(\sqrt{U^2+W^2})$ of its field of fractions (Bourbaki, *Alg. comm.*, chap. V, § 1, no 6, prop. 18). The ring of fractions $K[t,u,w,1/(u+w)]$ of C_1 is hence integrally closed. In the same way, one sees that the ring $C_2 = K[t,v,w]$ is integrally closed, for t is a root of the equation $t(t-v)-w^2(2v-t)=0$, and hence $K[t,v,w,1/(t-v)] = K[t,v,w,\frac{tu}{u+w},w,\frac{u+w}{tw}]$ is integrally closed. Finally, taking into account the fact that u and w are algebraically independent over K, one proves easily that $C = K[t,u,v,w] = K[t,u,w,1/(u+w)] \cap K[t,v,w,1/(t-v)]$, which finishes showing that C is the integral closure of B. It is immediate that if $\mathfrak{m}_0$ is the maximal ideal of C lying over $\mathfrak{n}_0$, the ideal generated by u, v, w ("vertex of the cone"), there is only one maximal ideal of C lying over $\mathfrak{m}_0$, namely the ideal generated by t, u, v, w. If one sets $A = B_{\mathfrak{m}_0}$, one deduces easily that $A' = C_{\mathfrak{m}_0}$ is the integral closure of A, which is hence unibranch, and hence also geometrically unibranch since its residue field K is algebraically closed. But in A the prime ideal $\mathfrak{p}$ generated by u and v is such that the integral closure $A_{\mathfrak{p}}[t]$ of $A_{\mathfrak{p}}$ is not a local ring. **IV-2** | 177

We shall see further below (9.7.10) that when X is a locally Noetherian prescheme such that, if X′ is the normalization of X_{red}, the canonical morphism $X' \to X$ is *finite* (which will be the case if X is such that the rings of its affine open subsets are *universally Japanese* ($\mathbf{0}$, 23.1.1))), then the set of points $x \in X$ where X is geometrically unibranch is *locally constructible*.

(6.15.3). We shall say for short that a morphism $f : X \to Y$ is *radicial at a point* $y \in Y$, if $f^{-1}(y)$ is empty or reduced to a single point x and if $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(y)$. It amounts to the same to say that f is radicial at every point of Y or that f is radicial ($\mathbf{I}$, 3.5.8). If $g : f^{-1}(y) \to \mathrm{Spec}(\boldsymbol{k}(y))$ is the base change inverse image of f by the base change $\mathrm{Spec}(\boldsymbol{k}(y)) \to Y$, it amounts to the same to say that f is radicial at the point $y \in Y$, or that g is *radicial*.

Lemma (6.15.3.1). — *(i) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms. If f is radicial at a point y and g radicial at the point $z = g(y)$, then $g \circ f$ is radicial at the point $g(y)$. The converse is true if f is surjective.*

(ii) Let $f : X \to Y$, $h : Y' \to Y$ be two morphisms, and let $f' = f_{|Y'} : X \times_Y Y' \to Y'$. Let $y' \in Y'$ and $y = h(y')$; then for f to be radicial at the point y, it is necessary and it suffices that f' be radicial at the point y'.

Proof. (i) If f is radicial at the point y and g radicial at the point z, $g^{-1}(z)$ is reduced to y and $f^{-1}(g^{-1}(z)) = f^{-1}(y)$ is empty or reduced to a single point x; furthermore $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(y)$ and $\boldsymbol{k}(y)$ a radicial extension of $\boldsymbol{k}(z)$, hence $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(z)$. Conversely, suppose f surjective; if $g \circ f$ is radicial at the point $z = g(y)$, $g^{-1}(z)$ is reduced to the point y, for otherwise $f^{-1}(g^{-1}(z))$ would have at least two distinct points; furthermore, $f^{-1}(y) = f^{-1}(g^{-1}(z))$ is reduced to a single point x, and by hypothesis one has $\boldsymbol{k}(z) \subset \boldsymbol{k}(y) \subset \boldsymbol{k}(x)$, and $\boldsymbol{k}(x)$ is a radicial extension of $\boldsymbol{k}(z)$, hence $\boldsymbol{k}(y)$ is a radicial extension of $\boldsymbol{k}(z)$ and $\boldsymbol{k}(x)$ radicial over $\boldsymbol{k}(y)$. **IV-2** | 178

(ii) Let $g : f^{-1}(y) \to \mathrm{Spec}(\boldsymbol{k}(y))$, $g' : f'^{-1}(y') \to \mathrm{Spec}(\boldsymbol{k}(y'))$ be the morphisms inverse images of f and f' respectively; as $f'^{-1}(y') = f^{-1}(y) \otimes_{\boldsymbol{k}(y)} \boldsymbol{k}(y')$ (**I**, 3.6.4), g' is the inverse image of g and the assertion reduces to (2.6.1, (v)). □

Suppose then X a prescheme *reduced* not having more than a *finite* number of irreducible components, and let X′ be its *normalization* (**II**, 6.3.8); we say (*loc. cit.*) that the canonical morphism $f : \mathrm{X}' \to \mathrm{X}$ is surjective. The definition (6.15.1) shows then that for X to be geometrically unibranch at a point, it is necessary and it suffices that f be *radicial at this point*; for X to be geometrically unibranch, it is necessary and it suffices that f be *radicial*.

(6.15.4). Generalizing the definition given in (**I**, 2.2.9), we shall say, for two reduced preschemes X, Y, a morphism $f : \mathrm{X} \to \mathrm{Y}$ is *birational* if the restriction of f to the set of maximal points of X is a *bijection* of this set onto the set of maximal points of Y, and if, for every maximal point y of Y, the morphism $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Spec}(\mathscr{O}_y) \to \mathrm{Spec}(\mathscr{O}_y)$ induced by f is an isomorphism (in other words, if for the fibre $f^{-1}(y)$ reduced to a single point x (maximal in X) and if the homomorphism $\boldsymbol{k}(y) \to \boldsymbol{k}(x)$ induced by f is bijective); this notion reduces to that of (**I**, 2.2.9) when X and Y have only a *finite* number of irreducible components.

Lemma (6.15.4.1). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism, $g : \mathrm{Y}' \to \mathrm{Y}$ a flat morphism, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{|\mathrm{Y}'} : \mathrm{X}' \to \mathrm{Y}'$. If f_{red} is birational, so is f'_{red}.*

Proof. Indeed, if y' is a maximal point of Y′, one knows (2.3.4, (ii)) that $y = g(y')$ is a maximal point of Y; there is a unique point x of X in $f^{-1}(y)$ by hypothesis; since moreover $\boldsymbol{k}(x) = \boldsymbol{k}(y)$. It follows therefore from (**I**, 3.4.9) that $f'^{-1}(y')$ is reduced to a single point x' and that $\boldsymbol{k}(x') = \boldsymbol{k}(y')$. We conclude from the reasoning that, noting (2.3.7, (ii)), every irreducible component of X′ dominates one irreducible component of Y′. □

Proposition (6.15.5). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism such that f_{red} is entire and birational* (6.15.4), *hence surjective.*

label=(i) For Y to be geometrically unibranch at a point y, it is necessary and it suffices that f be radicial at the point y and that X be geometrically unibranch at the unique point x of $f^{-1}(y)$.
lbbel=(ii) For Y to be geometrically unibranch, it is necessary and it suffices that X be geometrically unibranch and that f be radicial.

Proof. One can obviously restrict to the case where X and Y are *reduced*; the fact that f is *surjective* follows from the fact that f is closed (**II**, 6.1.10) and that $f(\mathrm{X})$ contains all the maximal points of Y. This also shows immediately that (i) implies (ii). Using (**I**, 3.6.5) and (**II**, 6.1.5), one can suppose, to prove (i), that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, in other words $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ (with $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$) is a local scheme. **IV-2** | 179
Note that $\mathrm{A}' = \mathscr{O}_{\mathrm{X}',x'}$ is by hypothesis a faithfully flat A-module, hence containing A, and hence the hypothesis that X is geometrically unibranch at the point x, or the hypothesis that X′ is geometrically unibranch at the point x', both imply that A is a local ring *integral* (since A, being reduced, is also isomorphic to a subring of $\mathrm{A}'_{\mathrm{red}}$). Let K be the field of fractions of A, B the integral closure of A, and set $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : \mathrm{Y} \to \mathrm{X}$ is radicial at the point x (6.15.3). Let $\mathrm{Y}' = \mathrm{Y} \otimes_k k' = \mathrm{Y} \times_{\mathrm{X}} \mathrm{X}'$ so that one has the commutative diagram

$$\begin{tikzcd} \mathrm{Y} \arrow[r, "h"] & \mathrm{Y}' \\ \mathrm{X} \arrow[u, "g"] \arrow[r, "f"'] & \mathrm{X}' \arrow[u, "g'"'] \end{tikzcd}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y′ is geometrically unibranch (6.15.6). For X to be geometrically unibranch at the point x, it is necessary and it suffices therefore that g' be radicial at the point x' (6.15.5). Now, x is the projection of x' in X and it is equivalent to say that g is radicial at the point x or that g' is radicial at the point x' (6.15.3.1); finally, for X to be geometrically unibranch at this point, it is necessary and it suffices that g be radicial at this point, whence the proposition. □

Proposition (6.15.6). — *Let k be a field, X a k-prescheme. If X is normal, then, for every extension k' of k, $\mathrm{X}' = \mathrm{X} \otimes_k k'$ is* geometrically unibranch.

Proof. One knows that k' is an algebraic extension of a pure extension k_0 of k, and if k'' is the largest separable extension of k_0 contained in k', k' is a radicial extension of k'' and k'' a separable extension of k. One knows (6.14.2) that $\mathrm{X}'' = \mathrm{X} \otimes_k k''$ is normal; as $\mathrm{X} \otimes_k k' = \mathrm{X}'' \otimes_{k''} k'$, one sees that one can restrict to the case where the extension k' of k is *radicial*. Furthermore (**I**, 3.6.5), one can suppose that $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a *local integral and integrally closed* ring (since X is normal). The projection morphism $f : \mathrm{X}' \to \mathrm{X}$ is a universal homeomorphism, since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is a universal homeomorphism (2.4.5); as $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$ where $\mathrm{A}' = \mathrm{A} \otimes_k k'$, one sees therefore that A' is a local ring whose nilradical $\mathfrak{N}'$ is the unique minimal prime ideal, whence $\mathrm{X}'_{\mathrm{red}} = \mathrm{Spec}(\mathrm{A}'_0)$, where $\mathrm{A}'_0 = \mathrm{A}'/\mathfrak{N}'$ is a local integral ring; furthermore, if K is the field of fractions of A, the field of fractions K'_0 of A'_0 is radicial over K, since the morphism f is radicial. But A'_0 is integer over A, its integral closure B is also the integral closure of A, and one knows (Bourbaki, *Alg. comm.*, chap. V, § 2, no 3, lemma 3) that B is the set of $x \in \mathrm{K}'_0$ of which a power p^m-th (for m large enough) belongs to A (p being the characteristic exponent of K); furthermore there exists only a single prime ideal of B lying over every prime ideal of A; in particular B is a local ring and its residue field is a radicial extension of that of A, and *a fortiori* of the same type for A'_0, which proves that A'_0 is geometrically unibranch, and hence so is X' as well. □

Proposition (6.15.7). — *Let k be a field, X a k-prescheme, k' an extension of k, $\mathrm{X}' = \mathrm{X} \otimes_k k'$. Let x' be a point of X', x its projection in X. For X to be geometrically unibranch at the point x, it is necessary and it suffices that X' be geometrically unibranch at the point x'. For X to be geometrically unibranch, it is necessary and it suffices that X' be so.*

Proof. The second assertion follows from the first and from the fact that the projection $f : \mathrm{X}' \to \mathrm{X}$ is **IV-2** | 180
a surjective morphism. To prove the first assertion, one can, by virtue of (**I**, 5.1.8), restrict to the case where X is reduced, and by virtue of (**I**, 3.6.5), to the case $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ (with $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$) is a local scheme. Note that $\mathrm{A}' = \mathscr{O}_{\mathrm{X}',x'}$ is by hypothesis a faithfully flat A-module, hence containing A, and hence the hypothesis that X is geometrically unibranch at the point x, or the hypothesis that X' is geometrically unibranch at the point x', both imply that A is a local ring *integral* (since A, being reduced, is also isomorphic to a subring of $\mathrm{A}'_{\mathrm{red}}$). Let K be the field of fractions of A, B the integral closure of A, and set $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$; it amounts to the same to say that X is geometrically unibranch at the point x or that the morphism $g : \mathrm{Y} \to \mathrm{X}$ is radicial at the point x (6.15.3). Let $\mathrm{Y}' = \mathrm{Y} \otimes_k k' = \mathrm{Y} \times_{\mathrm{X}} \mathrm{X}'$ so that one has the commutative diagram

$$\begin{tikzcd} \mathrm{Y} \arrow[r, "h"] & \mathrm{Y}' \\ \mathrm{X} \arrow[u, "g"] \arrow[r, "f"'] & \mathrm{X}' \arrow[u, "g'"'] \end{tikzcd}$$

Note that f is a flat morphism; hence (6.15.4.1) the entire morphism g'_{red} is birational. As Y is normal, Y' is normal (and *a fortiori* geometrically unibranch) (6.14.1). For X' to be geometrically unibranch at x', it is necessary and it suffices therefore that g' be radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. □

Lemma (6.15.8). — *Let k be a separably closed field (in other words, such that the algebraic closure of k is* radicial *over k), X a k-prescheme locally of finite type over k, x a closed point of X. If X is unibranch at the point x, it is* geometrically unibranch *at this point.*

Proof. Indeed, one knows (**I**, 6.4.2) that $\boldsymbol{k}(x)$ is an *algebraic* extension of k; as the residue field of the integral closure $(\mathscr{O}_x)_{\mathrm{red}}$ (integral closure which is by hypothesis a local ring) is an algebraic extension of $\boldsymbol{k}(x)$, it is a radicial extension of k by hypothesis, hence also of $\boldsymbol{k}(x)$. □

Corollary (6.15.9). — *Let k be a field, X a k-prescheme locally of finite type over k, k' a separable extension close of k (in other words, such that the algebraic closure of k is radicial over k'); then, for X to be geometrically unibranch, it is necessary and it suffices that $X' = X \otimes_k k'$ be unibranch.*

Proof. In view of (6.15.7), one is reduced to proving that if X is unibranch and k separably closed, then X is geometrically unibranch. Now, X is geometrically unibranch at its closed points (6.15.8); we conclude from (10.4.9) that X is geometrically unibranch at all its points, relying on the fact (pointed out at the end of (6.15.2)) that the set of points where X is geometrically unibranch is constructible (of course, the corollary (6.15.9) will not be used before that point). □

This result justifies, to a certain extent, the terminology of "geometrically unibranch".

Proposition (6.15.10). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \to Y$ a normal morphism (6.8.1), $f : X \to Y$ a morphism; set $X' = X \times_Y Y'$ and let $p : X' \to X$ be the canonical projection, x' a point of X'. Then, if X is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point $x = p(x')$, X' is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point x'.*

Proof. By virtue of (**I**, 3.6.5), one can restrict to the case where $Y = \mathrm{Spec}(A)$, $Y' = \mathrm{Spec}(A')$, **IV-2** | 181
$X = \mathrm{Spec}(B)$, where A, A', B are local rings, the homomorphisms $A \to A'$ and $A \to B$ being local, A, A' Noetherian rings, and x being the closed point of X; it suffices to prove that if B is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) then $B' = B \otimes_A A'$ is also geometrically unibranch at the points of $p^{-1}(x)$, resp. $\mathrm{Spec}(B')$ is integral and geometrically unibranch at these points). Suppose first B reduced; B is union of the filtered increasing family of its sub-A-algebras of finite type B_λ, and by flatness, B' is union of the filtered increasing family of the sub-A'-algebras $B'_\lambda = B_\lambda \otimes_A A'$. Now, the morphism $\mathrm{Spec}(B'_\lambda) \to \mathrm{Spec}(B_\lambda)$ is normal (6.8.2), and *a fortiori* reduced; the fact that B'_λ is reduced is consequence of (3.3.5); one concludes that B' itself is reduced by (5.13.2).

Suppose now B geometrically unibranch; in view of (**I**, 5.1.8), one can furthermore suppose B reduced, hence *integral*. Let C be its integral closure, which is by hypothesis a local ring; set $Z = \mathrm{Spec}(C)$, $C' = C \otimes_A A'$, $Z' = \mathrm{Spec}(C') = Z \times_X X'$, so that one has the commutative diagram

$$\begin{tikzcd} Z \arrow[r] & Z' \\ X \arrow[u, "l"] \arrow[r, "p"'] & X' \arrow[u, "l'"'] \end{tikzcd}$$

By virtue of the first part of the reasoning, X' is *reduced*; on the other hand, as $Z' = Z \times_Y Y'$, it follows from (6.14.1) that Z' is normal (and *a fortiori* geometrically unibranch). As f is entire and birational, so is f' by (6.15.4.1), since p is flat. Finally, the hypothesis that X is geometrically unibranch at the point x implies that f is radicial at this point (6.15.5); we conclude hence from (6.15.3.1) that f' is radicial at every point $x' \in p^{-1}(x)$, and it follows then from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. □

Remarks (6.15.11). — (i) In the proof of (6.15.10), one cannot dispense with appealing to (6.14.1), even when $X = Y$, for in that case the integral closure is the ring B, which is not necessarily a Noetherian ring even if B is.

(ii) The example (6.5.5, (ii)) shows that in (6.15.10), one cannot replace "geometrically unibranch" by "integral" in the statement, even if the residue field $\boldsymbol{k}(x)$ is algebraically closed and the morphism g is étale.

§7. Formal properties of morphisms; Noetherian approximation

7.1. Formal equidimensionality and formally catenary rings **IV-2** | 183

Definition (7.1.1). — We say that a local Noetherian ring A is *formally equidimensional* if its completion $\widehat{A}$ is equidimensional (**0**, 16.1.4).

Proposition (7.1.2). — *Let* A *be a local Noetherian ring,* $\mathfrak{p}_i$ *(*$1 \leqslant i \leqslant n$*) its minimal prime ideals. For* A *to be formally equidimensional, it is necessary and sufficient that the* $A/\mathfrak{p}_i$ *be equidimensional (in other words, that the* $A/\mathfrak{p}_i$ *have the same dimension).*

Proof. Indeed, this follows from the fact that for every prime ideal $\mathfrak{p}'$ of $\widehat{A}$, $\mathfrak{p}' \cap A$ contains one of the $\mathfrak{p}_i$, hence $\mathfrak{p}'$ contains one of the $\mathfrak{p}_i\widehat{A}$, and $\widehat{A}/\mathfrak{p}_i\widehat{A} = (A/\mathfrak{p}_i) \otimes_A \widehat{A}$ is the completion of the local ring $A/\mathfrak{p}_i$ (**0**, 7.3.3), hence has the same dimension. Every maximal chain of prime ideals of $\widehat{A}$ therefore identifies canonically with a maximal chain of prime ideals of one of the $\widehat{A}/\mathfrak{p}_i\widehat{A}$, and conversely, whence the conclusion. □

Proposition (7.1.3). — *Let* A, A′ *be two local Noetherian rings,* $\mathfrak{m}$ *the maximal ideal of* A, $\varphi : A \to A'$ *a local homomorphism; suppose that* A′ *is a flat* A*-module, and that* A′ *is equidimensional and catenary. Then:*

label=i) A *is equidimensional and catenary.*

lbbel=ii) *Suppose in addition that* $\mathfrak{m}A'$ *is an ideal of definition of* A′*. Then, if* $\mathfrak{a}$ *is an ideal of* A*, for* $A'/\mathfrak{a}A'$ *to be equidimensional, it is necessary and sufficient that* $A/\mathfrak{a}A$ *be so. In particular, for every prime ideal* $\mathfrak{p}$ *of* A, $A'/\mathfrak{p}A'$ *is equidimensional.*

Proof. Let us first prove the second assertion of (ii). Set $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$, $Y = V(\mathfrak{p}) = \mathrm{Spec}(A/\mathfrak{p})$, $Y' = V(\mathfrak{p}A') = \mathrm{Spec}(A'/\mathfrak{p}A')$; if $f : X' \to X$ is the morphism corresponding to φ, Y′ is also the closed sub-prescheme $f^{-1}(Y)$ of X′. As $\mathfrak{m}A'$ is an ideal of definition of A′, we have

(7.1.3.1) $$\dim(X) = \dim(X')$$

by virtue of (6.1.3); similarly $\mathfrak{m}A'/\mathfrak{p}A'$ is an ideal of definition of $A'/\mathfrak{p}A'$ and Y′ is flat over Y (2.1.4), hence (6.1.3)

(7.1.3.2) $$\dim(Y) = \dim(Y').$$

Let y be the generic point of Y, Y'_i ($1 \leqslant i \leqslant r$) the irreducible components of Y′, y'_i the generic point of Y'_i ($1 \leqslant i \leqslant r$); we have $f(y'_i) = y$ for all i (2.3.4). On the other hand, y'_i is the maximal point of the fibre $f^{-1}(y)$ (**0**, 2.1.8), hence $\mathscr{O}_{X',y'_i} \otimes_{\mathscr{O}_{X,y}} \boldsymbol{k}(y) = \mathscr{O}_{X',y'_i}/\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is of dimension 0, in other words $\mathfrak{m}_y\mathscr{O}_{X',y'_i}$ is an ideal of definition of $\mathscr{O}_{X',y'_i}$; we can again apply (6.1.3), which gives

(7.1.3.3) $$\dim(\mathscr{O}_{X',y'_i}) = \dim(\mathscr{O}_{X,y})$$

that is to say (5.1.2)

(7.1.3.4) $$\mathrm{codim}(Y'_i, X') = \mathrm{codim}(Y, X)$$

for all i. As A′ is equidimensional and catenary, we have, by virtue of (**0**, 14.3.5) **IV-2** | 184

(7.1.3.5) $$\dim(X') = \dim(Y'_i) + \mathrm{codim}(Y'_i, X')$$

for all i; by virtue of (7.1.3.1), (7.1.3.2) and (7.1.3.4) and the inequality $\dim(Y'_i) \leqslant \dim(Y')$, we deduce

(7.1.3.6) $$\dim(X) \leqslant \dim(Y) + \mathrm{codim}(Y, X)$$

hence the two sides of this inequality are equal by (**0**, 14.2.2.2); moreover this equality implies

(7.1.3.7) $$\dim(Y'_i) = \dim(Y') = \dim(Y)$$

for all i, which proves the second assertion of (ii).

Let us now prove the first assertion of (ii). Let $\mathfrak{p}_j$ ($1 \leqslant j \leqslant n$) be the minimal prime ideals of A among those containing $\mathfrak{a}$, and let $\mathfrak{p}'_{jh}$ be the prime ideals of A′ minimal among those containing $\mathfrak{p}_jA'$ ($1 \leqslant h \leqslant m_j$); we know then (2.3.4) that the $\mathfrak{p}'_{jh}$ ($1 \leqslant j \leqslant n$, $1 \leqslant h \leqslant m_j$ for all j) are also the minimal prime ideals among those containing $\mathfrak{a}A'$. From what we have just seen, for all j, the dimensions of the rings $A'/\mathfrak{p}'_{jh}$ ($1 \leqslant h \leqslant m_j$) are all equal and are equal to $\dim(A'/\mathfrak{p}_jA')$, itself equal to $\dim(A/\mathfrak{p}_j)$ by (7.1.3.2). To say that $A'/\mathfrak{a}A'$ is equidimensional therefore means that the dimensions of the $A/\mathfrak{p}_j$ are all equal, that is to say that $A/\mathfrak{a}$ is equidimensional.

Let us finally prove (i). Let $\mathfrak{r}'$ be a prime ideal of A′ minimal among those containing $\mathfrak{m}A'$, and set $A'' = A'_{\mathfrak{r}'}$; as A″ is a flat A′-module, it is also a flat A-module; moreover A″ is equidimensional and catenary (**0**, 16.1.4), and $\mathfrak{m}A''$ is an ideal of definition of A″. We can therefore restrict ourselves to the case where $\mathfrak{m}A'$ is an ideal of definition of A′. If then $\mathfrak{p}$ and $\mathfrak{q} \subset \mathfrak{p}$ are two prime ideals of A, we can apply (ii) to $A/\mathfrak{q}$ and $A'/\mathfrak{q}A'$, which is equidimensional and flat over $A/\mathfrak{q}$; if $Z = \mathrm{Spec}(A/\mathfrak{q})$

and $Y = \mathrm{Spec}(A/\mathfrak{p})$, we have $\dim(Z) = \dim(Y) + \mathrm{codim}(Y, Z)$; moreover one can apply (7.1.3.6), which shows that $\mathrm{codim}(Y, X) = \dim X - \dim Y$; as X is equicodimensional, these relations show that it is biequidimensional by virtue of (**0**, 14.3.3). □

Corollary (7.1.4). — *Let* A *be a local Noetherian ring that is formally equidimensional. Then:*

*label=*i) A *is equidimensional and catenary (in other words, biequidimensional).*
*lbbel=*ii) *Let* $\mathfrak{a}$ *be an ideal of* A*; for* $A/\mathfrak{a}$ *to be equidimensional, it is necessary and sufficient that* $A/\mathfrak{a}$ *be formally equidimensional. In particular, for every prime ideal* $\mathfrak{p}$ *of* A*,* $A/\mathfrak{p}$ *is formally equidimensional.*

Proof. If $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}\widehat{A}$ is an ideal of definition of $\widehat{A}$. By hypothesis $\widehat{A}$ is equidimensional, and on the other hand it is known to be catenary (5.6.4); it suffices then to apply (7.1.3) with $A' = \widehat{A}$. □

Corollary (7.1.5). — *Let* A *be a local Noetherian ring such that there exists a finite type* A*-module* M *that is a Cohen-Macaulay* A*-module and for which* $\mathrm{Supp}(M) = \mathrm{Spec}(A)$ *(this holds for example if* A *is a Cohen-Macaulay ring). Then* A *is formally equidimensional, hence* (7.1.4) *for a quotient ring* B *of* A *to be formally equidimensional, it is necessary and sufficient that it be equidimensional.*

Proof. Indeed, $\widehat{M} = M \otimes_A \widehat{A}$ is a Cohen-Macaulay $\widehat{A}$-module (**0**, 16.5.2), with support equal to $\mathrm{Spec}(\widehat{A})$; hence (**0**, 16.5.4), $\widehat{A}$ is equidimensional. □ **IV-2** | 185

Remark (7.1.6). — We have seen (6.3.8) that under the hypotheses of (7.1.5), for every integral quotient ring B of A, the fibres of the canonical morphism $\mathrm{Spec}(\widehat{B}) \to \mathrm{Spec}(B)$ are preschemes of Cohen-Macaulay; hence (6.4.3) and (5.7.5), *for* B *to have no embedded associated prime ideals, it is necessary and sufficient that the same hold for* $\widehat{B}$.

Lemma (7.1.7). — *Let* A *be a local Noetherian ring,* $\mathfrak{p}_i$ *(*$1 \leqslant i \leqslant r$*) its minimal prime ideals. Suppose that each of the rings* $A/\mathfrak{p}_i$ *is formally equidimensional. Then, for every prime ideal* $\mathfrak{q}$ *of* A *and every* i *such that* $\mathfrak{p}_i \subset \mathfrak{q}$*, the ring* $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ *is formally equidimensional.*

Proof. As $A_\mathfrak{q}/\mathfrak{p}_i A_\mathfrak{q}$ is the local ring of $A/\mathfrak{p}_i$ at the prime ideal $\mathfrak{q}/\mathfrak{p}_i$, we can restrict ourselves to the case where A is integral and formally equidimensional. Set $A' = \widehat{A}$, and let $\mathfrak{q}'$ be one of the minimal prime ideals of A' among those containing $\mathfrak{q}A'$; if we set $B = A_\mathfrak{q}$, $B' = A'_{\mathfrak{q}'}$, then B' is a B-module flat (**0**, 6.3.2). Set $C = \widehat{B}$, $C' = \widehat{B'}$; as B' is a flat B-module plat, we know that C' is a flat C-module (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 4); as B' is a flat B-module, it is known that C' is a flat C-module. As A' is catenary (5.6.4) and equidimensional by hypothesis, the same holds for $B' = A'_{\mathfrak{q}'}$ (**0**, 16.1.4); moreover, as A' is isomorphic to a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), the same holds for B' (**0**, 17.3.9); we conclude therefore from (7.1.5) that C' is equidimensional. On the other hand, C' is catenary (5.6.4), hence C is equidimensional by virtue of (7.1.3), (i). □

Theorem (7.1.8). — *Let* A *be a local Noetherian ring. The following conditions are equivalent:*

*label=*a) *Every integral quotient ring of* A *is formally equidimensional.*
*lbbel=*b) *The quotient rings of* A *by its minimal prime ideals are formally equidimensional.*
*lcbel=*c) *Every local ring* B *that is an A-algebra essentially of finite type* (1.3.8) *and that is equidimensional, is formally equidimensional.*

Proof. It is trivial that $c)$ implies $a)$ and that $a)$ implies $b)$. On the other hand, as every prime ideal of A contains a minimal prime ideal of A, $b)$ implies $a)$ by virtue of (7.1.4), (ii). It remains to see that $a)$ implies $c)$.

It suffices to prove that the quotients of B by its minimal prime ideals are formally equidimensional (7.1.2); as every quotient ring of B is also an A-algebra essentially of finite type (1.3.9), we can first suppose B *integral*. If $\mathfrak{q}$ is the inverse image in A of the maximal ideal of B, we know that B is also an $A_\mathfrak{q}$-algebra essentially of finite type (1.3.10), and it follows from $a)$ and from (7.1.7) that every integral quotient of $A_\mathfrak{q}$ is formally equidimensional; we can therefore suppose that the homomorphism $A \to B$ is a *local* homomorphism. The kernel $\mathfrak{r}$ of this homomorphism is then a prime ideal of A, and by virtue of $a)$, every integral quotient ring of $A/\mathfrak{r}$ is formally equidimensional; as B is an $(A/\mathfrak{r})$-algebra essentially of finite type, we see that we can also suppose that A is *integral* and is a subring **IV-2** | 186 of B. We know (1.3.11) that B is a quotient of a local ring of the form $C_{\mathfrak{p}}$, where $C = A[T_1, \ldots, T_n]$ is a polynomial algebra and $\mathfrak{p}$ is a prime ideal of C above the maximal ideal $\mathfrak{m}$ of A. By virtue of (7.1.7), it suffices to prove that $C_{\mathfrak{p}}$ is formally equidimensional, in other words one can restrict to the case where $B = C_{\mathfrak{p}}$.

Set $A' = \widehat{A}$, and $C' = A'[T_1, \ldots, T_n] = C \otimes_A A'$; there is a unique prime ideal $\mathfrak{p}'$ of C' above the maximal ideal $\mathfrak{m}A'$ of A', hence above $\mathfrak{p}$; set $B' = C'_{\mathfrak{p}'}$; the homomorphism $B \to B'$ is local and makes B' a flat B-module. We know that $\widehat{B}'$ is a flat $\widehat{B}$-module (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 4, prop. 4); as $\widehat{B}'$ is also (7.1.3), (i)), it will suffice to prove that $\widehat{B}'$ is *equidimensional*, in order to deduce that $\widehat{B}$ is so as well, which will complete the proof.

Now, A' is a quotient of a regular ring by the theorem of Cohen (**0**, 19.8.8), hence by virtue of (7.1.5), to show that B' is formally equidimensional, it suffices to prove that B' is equidimensional, since C' is a quotient of a regular ring, hence catenary (5.6.4). But the minimal prime ideals of C' are the ideals $\mathfrak{q}'C'$, where $\mathfrak{q}'$ ranges over the minimal prime ideals of A' (5.5.3), and we have $C'/\mathfrak{q}'C' = (A'/\mathfrak{q}')[T_1, \ldots, T_n]$. As A' is equidimensional by hypothesis (since A is integral), the same holds for C' by (5.5.4). □

Definition (7.1.9). — When the equivalent conditions of (7.1.8) are satisfied, we say that A is a *formally catenary* ring.

For a local Noetherian ring, it is therefore equivalent to say that it is formally equidimensional or formally catenary and equidimensional, by virtue of (7.1.2).

Proposition (7.1.10). — *Every local quotient ring of a local Cohen-Macaulay ring is formally catenary.*

Proof. This follows immediately from (7.1.8) and (7.1.5) applied to the integral quotient rings of A. □

Proposition (7.1.11). — *label=*i) *A local Noetherian ring that is formally catenary is universally catenary.*
*lbbel=*ii) *If* A *is a local Noetherian ring that is formally catenary, every local ring* B *that is an A-algebra essentially of finite type is formally catenary.*

Proof. Assertion (ii) follows immediately from condition *c*) of (7.1.8) and from the fact that if C is a B-algebra essentially of finite type, C is also an A-algebra essentially of finite type (1.3.9). For the proof of (i), let us first note that by virtue of condition *b*) of (7.1.8), a local Noetherian ring that is formally catenary A is catenary, since the quotients of A by its minimal prime ideals are so (7.1.4). If now E is an A-algebra of finite type, it follows from what precedes and from (ii) that every local ring of E at a prime ideal is catenary, hence E is catenary. Therefore A is universally catenary. □

Remarks (7.1.12). — label=*i)* We do not know if, conversely, a local Noetherian ring that is universally catenary is always formally catenary.
lbbel=*ii)* A local Noetherian ring A of dimension 1 is necessarily equidimensional (since in this case A would have dimension 0, being its maximal ideal minimal, a case where A is of dimension 0). As $\dim(\widehat{A}) = 1$, this shows that A is even *formally equidimensional*, and *a fortiori* formally catenary (hence *universally catenary*). By contrast, the local ring of dimension 2 defined in (5.6.11), which is catenary but not universally catenary, is *a fortiori* not formally catenary.

IV-2 | 187

Corollary (7.1.13). — *Let* Y *be an irreducible prescheme locally Noetherian of dimension* 1, X *an irreducible prescheme,* $f : X \to Y$ *a dominant morphism of finite type,* ξ, η_i *the generic points of* X *and* Y *respectively. Then, for all* $y \in f(X)$, *the dimensions of the irreducible components of* $f^{-1}(y)$ *are all equal to* $\deg.\mathrm{tr}_{\boldsymbol{k}(\eta)} \boldsymbol{k}(\xi)$.

Proof. By hypothesis, $f^{-1}(\eta)$ is irreducible of generic point ξ and of dimension $\deg.\mathrm{tr}_{\boldsymbol{k}(\eta)} \boldsymbol{k}(\xi)$ (5.2.1). As Y is irreducible and of dimension 1, we have $\dim(\mathscr{O}_y) = 1$ for all $y \neq \eta$, and $\mathscr{O}_y$ is universally catenary by virtue of (7.1.12), (ii). If $y \in f(X)$ and si z is a generic point of an irreducible component Z of $f^{-1}(y)$, we have therefore, by (5.6.5)

$$\dim(Z) = 1 + \dim(f^{-1}(\eta)) - \dim(\mathscr{O}_{X,z}).$$

But we have $\dim(\mathscr{O}_{X,z}) \leqslant \dim(\mathscr{O}_y)$ (**0**, 16.3.9), and on the other hand, as z is not the generic point of X, $\dim(\mathscr{O}_{X,z}) > 0$; hence $\dim(\mathscr{O}_{X,z}) = 1$ and $\dim(Z) = \dim(f^{-1}(\eta))$. □

7.2. Strictly equidimensional rings and strictly formally catenary rings

(7.2.1). *Notations.* — For an integral Noetherian ring A, we denote by $A^{(i)}$ the intersection of the local rings $A_\mathfrak{p}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1 (5.10.17). If A is a *local* Noetherian integral ring, we denote by $A^{(a)}$ the intersection of the $A_\mathfrak{p}$ where this time $\mathfrak{p}$ ranges over the set of all the prime ideals of A *distinct from the maximal ideal*.

We will say that a local Noetherian ring A is *strictly equidimensional* if, for every prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$, we have $\dim(A/\mathfrak{p}) = \dim(A)$; this amounts to saying that A is equidimensional and *without embedded associated prime ideals*.

(7.2.1.1). *Example.* — Let A be a local Noetherian ring *of dimension* 1. Then the following conditions are equivalent:

label=*a)* A has no embedded associated prime ideals.
a′) $\widehat{A}$ has no embedded associated prime ideals.
lbbel=*b)* A is strictly equidimensional.
b′) $\widehat{A}$ is strictly equidimensional.
lcbel=*c)* A is a Cohen-Macaulay ring.

Indeed, *a*) means that the maximal ideal $\mathfrak{m}$ of A is not associated to A, hence the associated prime ideals to A are the minimal ideals of A, all distinct from $\mathfrak{m}$, and for such an ideal $\mathfrak{p}$, we necessarily have $\dim(A/\mathfrak{p}) = 1$; hence *a*) implies *b*). Conversely, *b*) implies that every associated prime ideal $\mathfrak{p} \in \mathrm{Ass}(A)$ is distinct from $\mathfrak{m}$, hence minimal, and therefore *b*) implies *a*). We already know that *a*) and *c*) are equivalent for a local ring of dimension 1 (5.7.8). As it comes to the same thing to say that A is a Cohen-Macaulay ring or that $\widehat{A}$ is a Cohen-Macaulay ring (**0**, 16.5.2), and since $\dim(\widehat{A}) = 1$, we see finally that *a′*) and *b′*) are equivalent to *c*). **IV-2 | 188**

Proposition (7.2.2). — *Let* A *be a local Noetherian integral ring; set* $A' = \widehat{A}$, $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$ *and denote by* $f : X' \to X$ *the canonical morphism. Let a be the closed point of* X, $j : X - \{a\} \to X$ *the canonical injection. Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_X$*-Module,* $\mathscr{F}' = f^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$*; then the following conditions are equivalent:*

label=a) *The* $\mathscr{O}_X$*-Module* $j_*(\mathscr{F}|X - \{a\})$ *is coherent.*
lbbel=b) *For all* $x' \in \mathrm{Ass}(\mathscr{F}')$, *we have* $\dim(\overline{\{x'\}}) \geqslant 2$.

Proof. Let a' be the closed point of X', which is the unique point of the fibre $f^{-1}(a)$, and let $j' : X' - \{a'\} \to X'$ be the canonical injection. As the morphism f is faithfully flat and quasi-compact, it is sufficient to show that $j_*(\mathscr{F}|X - \{a\})$ is coherent or that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent. On the other hand, as A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (**0**, 19.8.8), it follows from (5.11.4) that condition *b*) of the statement was equivalent to the fact that $j'_*(\mathscr{F}'|X' - \{a'\})$ is coherent. Whence the proposition.

Instead of applying (5.11.4) and using the theorem of Cohen, which (by (5.11.2)) implicitly appeals to the cohomological results of chap. III, one can also use the fact that A' is universally catenary (5.6.4) and universally Japanese by virtue of (7.6.5) (the proof of this last result not using (7.2.2)). □

Proposition (7.2.3). — *Let* A *be a local Noetherian integral ring. The notations being as in (7.2.2), the following conditions are equivalent:*

label=a) $A^{(1)}$ *is a finite A-algebra.*
lbbel=b) *For every closed subset* T *of* X *of codimension* $\geqslant 2$, *if we denote by* $i : X - T \to X$ *the canonical injection,* $i_*(\mathscr{O}_{X-T})$ *is a coherent* $\mathscr{O}_X$*-Module.*
lcbel=c) *For all* $x' \in \mathrm{Ass}(\mathscr{O}_{X'})$ *and every closed subset* T *of* X *of codimension* $\geqslant 2$, *we have* $\mathrm{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$.

Proof. Set $Z = Z^{(2)}(X)$ (5.10.13) and $Z' = f^{-1}(Z)$, which are stable parts under specialisation; the conditions *a*) and *b*) are equivalent respectively to the following properties: a_1) $\mathscr{H}^0_{X/Z}(\mathscr{O}_X)$ is coherent; b_1) $\mathscr{H}^0_{X/T}(\mathscr{O}_X)$ is coherent for every closed subset T of X of codimension $\geqslant 2$. Taking into

account (5.9.5), these two properties are equivalent respectively to: $a_1')$ $\mathscr{H}^0_{X'/Z'}(\mathscr{O}_{X'})$ is coherent; $b_1')$ setting $T' = f^{-1}(T)$, $\mathscr{H}^0_{X'/T'}(\mathscr{O}_{X'})$ is coherent for every closed subset T of X of codimension $\geqslant 2$. Now, every point of $\mathrm{Ass}(\mathscr{O}_{X'})$ projects to the generic point of X since f is a flat morphism (3.3.2); as by definition this point does not belong to Z, we see that $\mathrm{Ass}(\mathscr{O}_{X'})$ does not meet Z'; the equivalence of a_1' and b_1' therefore follows from (5.11.5), since A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (**0**, 19.8.8) (or alternatively because A' is universally Japanese (7.6.5)). **IV-2** | 189
For this reason, conditions b_1' and $c)$ are also equivalent by virtue of (5.11.4). □

Proposition (7.2.4). — *Let* A *be a local Noetherian ring and set* $X = \mathrm{Spec}(A)$. *The following conditions are equivalent:*

label=a) *For every integral quotient ring* B *of* A, *the ring* $B^{(1)}$ *is a finite B-algebra.*
lbbel=b) *For every coherent* $\mathscr{O}_X$*-Module* $\mathscr{F}$ *and every closed subset* Z, *stable under specialisation, and such that for all* $x \in \mathrm{Ass}(\mathscr{F}) \cap (X - Z)$, *we have* $\mathrm{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$, *the* $\mathscr{O}_X$*-Module* $\mathscr{H}^0_{X/Z}(\mathscr{F})$ *is coherent.*
lcbel=c) *For every closed subset* T *of* X *and every coherent* $\mathscr{O}_U$*-Module* $\mathscr{G}$ *(where* $U = X - T$*) such that, for all* $x \in \mathrm{Ass}(\mathscr{G})$, *we have* $\mathrm{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geqslant 2$, $i_*(\mathscr{G})$ *(where* $i : U \to X$ *is the canonical injection) is a coherent* $\mathscr{O}_X$*-Module.*
ldbel=d) *For every integral quotient ring* B *of* A, *and every ideal* $\mathfrak{J}$ *of height* $\geqslant 2$ *in* B, *the ring* $\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$ *is a finite B-algebra.*
lebel=e) *For every integral quotient ring* B *of* A *and every prime ideal* $\mathfrak{q}$ *of* B, *with* $C = B_{\mathfrak{q}}$ *in a prime ideal of* B, *the ring* $C^{(a)}$ *is a finite C-algebra (or, what amounts to the same, if* $Y = \mathrm{Spec}(C)$ *and* U *is the complement in* Y *of the closed point of* C *and* $j : U \to Y$ *the canonical injection,* $j_*(\mathscr{O}_U)$ *is a coherent* $\mathscr{O}_Y$*-Module).*
lfbel=f) *For every integral quotient ring* B *of* A, *every local ring* $C = B_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B, *with* $\dim(C) \geqslant 2$, *and for every ideal* $\mathfrak{t} \in \mathrm{Ass}(\widehat{C})$, *we have* $\dim(\widehat{C}/\mathfrak{t}) \geqslant 2$.

Proof. We already know (5.11.6) that $a)$ and $b)$ are equivalent, as well as that $c)$ and $a)$ imply $d)$, and that $a)$ implies $d)$. The equivalence of $a)$ and $d)$ in the present case follows from (7.2.3) applied to the condition (7.2.3), $b)$), by virtue of (5.9.3.1). The equivalence of $e)$ and $f)$ is a consequence of (7.2.2) applied to the $\mathscr{O}_Y$-Module coherent $\mathscr{O}_Y$ itself. We have already remarked (5.11.7), (ii) that if A satisfies $a)$, the same holds for every A-algebra of finite type and for every ring of fractions of A. The condition $a)$ for A therefore implies (with the notations of $e)$) that the ring C satisfies $a)$, hence also $c)$; but as C is integral and $\dim(C) \geqslant 2$, one can apply condition $c)$ by taking T the closed point of $Y = \mathrm{Spec}(C)$, which proves that $a)$ implies $e)$; on the other hand, $\dim(C) \geqslant 2$, one can apply $c)$ the condition $c)$ by taking T the closed point of Y, which gives that $a)$ implies $a)$; we can **IV-2** | 190
obviously restrict ourselves to the case where A is integral and $B = A$, and it amounts to showing that condition $e)$ implies the condition (7.2.3), $c)$). Let us consider the notations of (7.2.3), $c)$), and let $y' \in f^{-1}(T) \cap \overline{\{x'\}}$, and set $y = f(y')$ and $C = \mathscr{O}_{X,y}$; as $y \in T$, $\dim(C) \geqslant 2$, and hence (with the notations of $e)$), $j_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_Y$-Module. As the morphism f is flat, $f_z : Y' \to Y$; the morphism $f : X' \to X$ is flat, being the base change $g = f_{\{Y\}} : Y' \to Y$. The underlying space is identified with $f^{-1}(Y)$ (**I**, 3.6.5), and as A is integral and f flat, $\mathrm{Ass}(\mathscr{O}_{X'})$ is contained in the fibre of the generic point of X (3.3.2); this last being contained in Y, we have $x' \in Y'$. Let $U' = g^{-1}(U)$, and let j' be the canonical injection $U' \to Y'$; as $j_*(\mathscr{O}_U)$ is a coherent $\mathscr{O}_Y$-Module and g is flat, it follows from (5.9.4) that $j'_*(\mathscr{O}_{U'})$ is a coherent $\mathscr{O}_{Y'}$-Module. But we have $x' \in \mathrm{Ass}(\mathscr{O}_{U'})$, hence we conclude from (5.10.10) that, in Y', $\mathrm{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$ in X'. □

Theorem (7.2.5). — *Let* A *be a local Noetherian ring. The following conditions are equivalent:*

label=a) *For every integral quotient ring* B *of* A, *the completion* $\widehat{B}$ *is strictly equidimensional* (7.2.1).
lbbel=b) A *is formally catenary* (7.1.9) *and the fibres of the canonical morphism* $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ *satisfy* (S_1) *(in other words, have no embedded associated prime cycle).*
lcbel=c) A *is universally catenary* (5.6.2) *and for every integral quotient ring* B *of* A, *the ring* $B^{(1)}$ *is a finite B-algebra (cf.* (7.2.4)*).*
ldbel=d) A *is universally catenary and for every integral quotient ring* B *of* A *and every local ring* $C = B_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B, *with* $\dim(C) \geqslant 2$, *the completed ring* $C^{(a)}$ *is a finite C-algebra.*

lebel=e) A *is universally catenary and for every integral quotient ring* B *of* A *and every local ring* $C = B_{\mathfrak{q}}$ *in a prime ideal* $\mathfrak{q}$ *of* B, *with* $\dim(C) \geqslant 2$, *the completed ring* $\widehat{C}$ *is such that* $\operatorname{Spec}(\widehat{C})$ *has no associated prime cycle of dimension* 1.

Moreover, when these conditions are satisfied, for every integral quotient ring B *of* A *that is strictly equidimensional, the completion* $\widehat{B}$ *is strictly equidimensional.*

Proof. The equivalence of $c)$, $d)$ and $e)$ was proved in (7.2.4). To show that $a)$ and $b)$ are equivalent, recall (7.1.9) that to say that A is formally catenary means that for every integral quotient ring B of A, $\widehat{B}$ is equidimensional. On the other hand, for every fibre of the morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$ at a point $x \in \operatorname{Spec}(A)$, this fibre of the morphism $\operatorname{Spec}(\widehat{B}) \to \operatorname{Spec}(B)$ at the generic point x of $\operatorname{Spec}(B)$, where $B = A/\mathfrak{J}_x$; to say that this fibre has no embedded associated prime cycle is equivalent to saying that $\widehat{B}$ has no embedded associated prime ideals, by virtue of (3.3.3); this proves the equivalence of $a)$ and $b)$. The same reasoning shows that if $a)$ is satisfied, then, for every integral quotient ring B of A that has no embedded associated prime ideals, the completion $\widehat{B}$ is also without embedded associated prime ideals; on the other hand, the hypothesis that A is formally catenary implies that if B is equidimensional, it is also the case for $\widehat{B}$ (7.1.8); this establishes the last assertion of the theorem.

Let us show now that $a)$ implies $c)$. Condition $a)$ implies that A is universally catenary (7.1.11); let us show on the other hand that for every integral quotient ring B of A, $B^{(1)}$ is then a finite B-algebra. Taking into account (7.2.3) applied to B, it amounts to showing that if $X = \operatorname{Spec}(B)$, T is a closed subset of codimension $\geqslant 2$ in X, $X' = \operatorname{Spec}(\widehat{B})$, and $g : X' \to X$ is the canonical morphism, then for all $x' \in \operatorname{Ass}(\mathscr{O}_{X'})$, $\operatorname{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}) \geqslant 2$. But by hypothesis X' has no associated prime cycle of dimension 1, hence $\inf(\operatorname{codim}(g^{-1}(T) \cap \overline{\{x'\}}, \overline{\{x'\}}))$ when x' ranges over $\operatorname{Ass}(\mathscr{O}_{X'})$, is nothing other than $\operatorname{codim}(g^{-1}(T), X')$. Now, as g is a faithfully flat morphism, we have (6.1.4)

$$\operatorname{codim}(g^{-1}(T), X') = \operatorname{codim}(T, X) \geqslant 2.$$

It remains to prove that $c)$ implies $a)$. Let us reason by induction on $n = \dim(A)$, the theorem **IV-2** | 191
being trivial for $n = 0$; we can moreover restrict to the case where $A = B$ is integral. We proceed in two steps, n being $\geqslant 1$.

I) *Suppose first that* A *satisfies* (S_2). — Set $X = \operatorname{Spec}(A)$, $A' = \widehat{A}$, $X' = \operatorname{Spec}(A')$ and let $u : X' \to X$ be the canonical morphism; it amounts to showing that for all $x' \in \operatorname{Ass}(\mathscr{O}_{X'})$, we have $\dim(\overline{\{x'\}}) = n$. Let $f \neq 0$ be an element of the maximal ideal of A, and set $C = A/fA$; we know (5.7.6) that A/fA satisfies (S_1); moreover, as the prime ideals of A among those containing f are of height 1 by virtue of the Hauptidealsatz, $C = A/fA$ is *strictly equidimensional* and $\dim(C) = n - 1$. We have $\widehat{C} = C \otimes_A A' = A'/fA'$, and f is A'-regular by flatness (**0**, 6.3.4); if we set $Y' = V(fA') = \operatorname{Spec}(\widehat{C})$, then, for every maximal point y' of $Y' \cap \overline{\{x'\}}$, on has $y' \in \operatorname{Ass}(\mathscr{O}_{Y'})$ by virtue of (3.4.3); on the other hand, on has $y' \neq x'$ since $u(x')$ is the generic point of X (3.3.2) and we conclude (5.1.8) that

$$\dim(\overline{\{y'\}}) = \dim(\overline{\{x'\}}) - 1. \tag{7.2.5.1}$$

But the quotient ring $C = A/fA$ also satisfies hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii)), hence by the induction hypothesis, we have $\dim(\overline{\{y'\}}) = \dim(C) = n - 1$; the conclusion therefore follows in this case from (7.2.5.1).

II) *General case.* — As by hypothesis $A^{(1)}$ is a finite A-algebra, it is a semi-local ring, and each of its local rings $(A^{(1)})_{\mathfrak{n}}$ at a maximal ideal $\mathfrak{n}$ and moreover, as A is universally catenary, the rings $(A^{(1)})_{\mathfrak{n}}$ (in finite number) are all of *dimension* n (5.6.10); moreover, these rings satisfy hypothesis $c)$ of the statement (5.6.1) and (5.11.7), (ii). We know that the completion of $A^{(1)}$, equal to $\widehat{A} \otimes_A A^{(1)}$, is composed of the completions of the local rings $(A^{(1)})_{\mathfrak{n}}$. Set $X_1 = \operatorname{Spec}(A^{(1)})$, $X'_1 = \operatorname{Spec}((A^{(1)})\widehat{\ }) = X' \times_X X_1$; for all $x'_1 \in \operatorname{Ass}(\mathscr{O}_{X'_1})$, it follows from what precedes and from case I) that $\dim(\overline{\{x'_1\}}) = n$. Let $u_1 = u_{(X_1)} : X'_1 \to X_1$ be the canonical morphism; as A and $A^{(1)}$ have the same field of fractions, the inverse image of the generic point x of X by the projection $X_1 \to X$ reduces to the generic point x_1 of X_1, the inverse image by the projection $X'_1 \to X'$ of the generic point x' of X' induces an isomorphism of the prescheme $u_1^{-1}(x_1)$ to the prescheme $u^{-1}(x)$. This being the case, the points of $\mathrm{Ass}(\mathscr{O}_{X'})$ (resp. $\mathrm{Ass}(\mathscr{O}_{X'_1})$) are the generic points of the associated prime cycles to $u^{-1}(x)$ (resp. $u_1^{-1}(x_1)$) (3.3.1). For all $x' \in \mathrm{Ass}(\mathscr{O}_{X'})$, and if Z' (resp. Z'_1) is the closed sub-prescheme of X' (resp. X'_1) having $\overline{\{x'\}}$ (resp. $\overline{\{x'_1\}}$) as underlying space, the projection $Z'_1 \to Z'$ is a finite and surjective morphism; we conclude (5.4.2) that $\dim(\overline{\{x'\}}) = \dim(\overline{\{x'_1\}}) = n$. □

Definition (7.2.6). — When a local Noetherian ring A satisfies the equivalent conditions of (7.2.5), we say that it is *strictly formally catenary*.

Corollary (7.2.7). — *Let* A *be a local Noetherian ring such that there exists an* A*-module* M *of finite type that is a Cohen-Macaulay* A*-module and for which* $\mathrm{Supp}(M) = \mathrm{Spec}(A)$*; then* A *is strictly formally catenary.*

Proof. Indeed, A is formally catenary (7.1.5) and (7.1.9), and on the other hand the fibres of the canonical morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ are preschemes of Cohen-Macaulay (6.3.8), hence *a fortiori* satisfy (S_1). □ **IV-2 | 192**

Corollary (7.2.8). — *If* A *is a local Noetherian ring that is strictly formally catenary, every local ring* B *that is an A-algebra essentially of finite type is strictly formally catenary.*

Proof. Indeed, B is formally catenary (7.1.11) and it follows from (7.4.4)[2] that the fibres of $\mathrm{Spec}(\widehat{B}) \to \mathrm{Spec}(B)$ satisfy (S_1), whence the conclusion. □

Corollary (7.2.9). — *Every local Noetherian ring of dimension* 1 *is strictly formally catenary.*

Proof. Indeed, every *integral* quotient ring B of such a ring A is of dimension 0 or 1, hence its completion is strictly equidimensional (7.2.1.1). □

Remarks (7.2.10). — label=*i)* It follows from (7.2.7) and (7.2.8) that every quotient ring of a Cohen-Macaulay local ring is strictly formally catenary. A local Noetherian ring of dimension 2 satisfying (S_2) is strictly formally catenary, since it is a Cohen-Macaulay ring. Recall however that there exist integral local Noetherian rings of dimension 2 that are not universally catenary, nor *a fortiori* strictly formally catenary (5.6.11).

lbbel=*ii)* We do not know if a ring that is formally catenary (7.1.9) is strictly formally catenary; this is because we do not know if for a local Noetherian integral ring B, $\widehat{B}$ has no embedded associated prime ideals (6.4.3). We also do not know if, in the equivalent conditions *c*), *d*), *e*) of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is the case when A is Henselian (18.9.6).

7.3. Formal fibres of Noetherian local rings

(7.3.1). We will consider in this number and the two following properties $\boldsymbol{P}(Z,k)$ of the following form:

"Z is a locally Noetherian prescheme over a field k,
and for all $z \in Z$, we have $\boldsymbol{Q}(\mathscr{O}_z, k)$"

where $\boldsymbol{Q}(A,k)$ is a property of a local Noetherian ring A that is a k-algebra; one assumes that if k' is a field isomorphic to k, and if A' is a local Noetherian ring that is a k'-algebra di-isomorphic to A, the properties $\boldsymbol{Q}(A,k)$ and $\boldsymbol{Q}(A',k')$ are equivalent.

If X, Y are two locally Noetherian preschemes, we will say that a morphism $f : X \to Y$ is a $\boldsymbol{P}$*-morphism* if:

label=1° f is *flat*;
lbbel=2° for all $y \in Y$, the property $\boldsymbol{P}(f^{-1}(y), \boldsymbol{k}(y))$ is true.

IV-2 | 193

Lemma (7.3.2). — *Let* X, Y *be two locally Noetherian preschemes,* $f : X \to Y$ *a morphism. The following properties are equivalent:*

*label=*a) *f is a* $\boldsymbol{P}$*-morphism.*

[2] The corollary (7.2.8) is not used to prove (7.4.4).

lbbel=b) *For all $x \in X$, if we set $y = f(x)$, then $\mathscr{O}_x$ is a flat $\mathscr{O}_y$-module and the property $\boldsymbol{Q}(\mathscr{O}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y), \boldsymbol{k}(y))$ is true.*

lcbel=c) *For all $x \in X$, if we set $y = f(x)$, the morphism $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ corresponding to the homomorphism $\mathscr{O}_y \to \mathscr{O}_x$ is a* ***P**-morphism.*

c′) *For every closed point $x \in X$, the morphism $\operatorname{Spec}(\mathscr{O}_x) \to \operatorname{Spec}(\mathscr{O}_y)$ is a* ***P**-morphism.*

Proof. The equivalence of a) and b) follows from the definitions; the same holds for b) and c), taking into account (**I**, 2.4.2); finally the equivalence of b) and c') follows from the fact that, by (5.1.11), $\overline{\{x\}}$ contains a closed point. ☐

Corollary (7.3.3). — *Let A, B be two Noetherian rings, $\varphi : A \to B$ a homomorphism such that the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is a* ***P**-morphism. Then, for every multiplicative subset S of A, the morphism $\operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$ is a* ***P**-morphism.*

Proof. This follows immediately from (7.3.2) and (**I**, 1.6.2). ☐

(7.3.4). We will suppose throughout what follows that the property $\boldsymbol{Q}$ is such that the following three conditions are satisfied:

(P_I) *(transitivity).* — If $f : X \to Y$ is a regular morphism (6.8.1) and $g : Y \to Z$ is a $\boldsymbol{P}$-morphism, then $g \circ f$ is a $\boldsymbol{P}$-morphism.

(P_{II}) *(descent).* — If $f : X \to Y$ and $g : Y \to Z$ are morphisms of locally Noetherian preschemes such that f is faithfully flat and that $g \circ f$ is a $\boldsymbol{P}$-morphism, then g is a $\boldsymbol{P}$-morphism.

(P_{III}) — For every field k, the property $\boldsymbol{P}(\operatorname{Spec}(k), k)$ is true.

Remarks (7.3.5). — label=*i)* Let us note that the conditions (P_I) and (P_{III}) imply that *every regular morphism is a* ***P**-morphism*.

lbbel=*ii)* Let us note that the hypotheses of (P_I) (resp. (P_{II})) imply that $h = g \circ f$ (resp. g) is flat (2.2.13); the hypotheses of (P_I) or of (P_{II}) imply on the other hand that for all $z \in Z$, $f_z : h^{-1}(z) \to g^{-1}(z)$ is flat (2.1.4); as, for all $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (**I**, 3.6.4), this shows that it suffices, to verify conditions (P_I) and (P_{II}), to do it only when Z is the *spectrum of a field*.

lcbel=*iii)* In certain cases, the property $\boldsymbol{Q}$ will be such that the following condition is also satisfied:

(P'_I) — If $f : X \to Y$ and $g : Y \to Z$ are two $\boldsymbol{P}$-morphisms, then $g \circ f$ is a $\boldsymbol{P}$-morphism.

(7.3.6). We will say that the property $\boldsymbol{P}$ is *geometric* if it satisfies (in addition to (P_I), (P_{II}) and (P_{III})) the condition:

(P_{IV}) *(finite extension of the base field).* — If $\boldsymbol{P}(Z, k)$ is true, then, for every finite extension k' of k, $\boldsymbol{P}(Z \otimes_k k', k')$ is true.

Lemma (7.3.7). — *Let $f : X \to Y$ be a* ***P**-morphism of locally Noetherian preschemes, $g : Y' \to Y$ a morphism locally of finite type. Then, if* ***P*** *satisfies the condition* (P_{IV})*, the morphism $f' = f_{(Y')} : X \times_Y Y' \to Y'$ is a* ***P**-morphism.*

Proof. Indeed, for all $y' \in Y'$, if we set $y = g(y')$, $\boldsymbol{k}(y')$ is an extension of finite type of $\boldsymbol{k}(y)$ (**I**, 6.4.11) and $f'^{-1}(y') = f^{-1}(y) \otimes_{\boldsymbol{k}(y)} \boldsymbol{k}(y')$ (**I**, 3.6.4); it suffices therefore to apply (P_{IV}). ☐

IV-2 | 194

(7.3.8). *Examples.* — The following properties $\boldsymbol{P}(Z, k)$ satisfy the conditions (P_I), (P_{II}) and (P_{III}):

label=(i) (also denoted (i_n)) Z is of coprofundity $\leqslant n$.
lbbel=(ii) Z is a prescheme of Cohen-Macaulay.
lcbel=(iii) (also denoted (iii_n)) Z satisfies (S_n).
ldbel=(iv) Z is regular.
lebel=(v) (also denoted (v_n)) Z satisfies (R_n).
lfbel=(vi) Z is reduced.
lgbel=(vii) Z is normal.

For properties (ii) to (vii), this in fact proves the stronger condition (P'_I) which also implies the property (iv) implies all the properties (i) to (vii), and the property (iv′) implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0)

(resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

Moreover, the following are also geometric:

(iv') Z is geometrically regular.

(v') (also denoted (v'_n)) Z possesses the property (R_n) geometric.

(vi') Z is geometrically reduced.

(vii') Z is geometrically normal.

Remarks (7.3.9). — label=*i)* Each of the properties (iv'), (v_n), (vi'), (vii') implies respectively the corresponding property (iv), (v_n), (vi), (vii). The property (iv) implies all the properties (i) to (vii), and the property (iv') implies all the properties enumerated in (7.3.8). Recall also that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and of (v'_0)) is equivalent to (vi) (resp. (vi')); the conjunction of (iii_2) and of (v_1) (resp. of (iii_2) and of (v'_1)) is equivalent to (vii) (resp. (vii')) (5.8.6).

lbbel=*ii)* In all the examples of (7.3.8), the property $\boldsymbol{Q}(\mathscr{O}_z, k)$ which serves to define $\boldsymbol{P}$ is such that, for every *generization* z' of z in Z, $\boldsymbol{Q}(\mathscr{O}_z, k)$ *implies* $\boldsymbol{Q}(\mathscr{O}_{z'}, k)$: by virtue of (2.3.4), it suffices to verify this for properties (i) to (vii) by (7.3.9), (i) and even (i) to (v) from the remark (7.3.9), (i): but this is included in the definition for (iii) and (v) (5.7.2) and (5.8.2), this follows from (6.3.9) for (i), and finally from (**0**, 16.5.10) and (17.3.2) for (ii) and (iv).

(7.3.10). We will say that a property $\boldsymbol{P}(\mathrm{Z}, k)$ is of the *first type* if the property $\boldsymbol{Q}(\mathrm{A}, k)$ which serves to define it defines a property $\boldsymbol{R}(\mathrm{A})$ *not involving* k and is true when A is a *field*; we will say that $\boldsymbol{P}(\mathrm{Z}, k)$ is of the *second type* if the property $\boldsymbol{Q}(\mathrm{A}, k)$ is of the form

> "For every finite extension k' of k and every point z' of $\mathrm{Spec}(\mathrm{A} \otimes_k k')$ above the closed point of $\mathrm{Spec}(\mathrm{A})$, we have $\boldsymbol{R}(\mathscr{O}_{z'})$"

$\boldsymbol{R}(\mathrm{A})$ being assumed true when A is a field. It is clear that in the examples of (7.3.8), the properties (i) to (vii) are of the first type, the properties (iv') to (vii') of the second type. **IV-2** | 195

This being the case, if we take up the reasoning of (6.6.1) and (6.8.3), we observe that when $\boldsymbol{P}$ is of the first or second type for a property $\boldsymbol{R}(\mathrm{A})$, *the conditions* $(\mathrm{P_I})$ *and* $(\mathrm{P_{II}})$ *have the following consequences on* $\boldsymbol{R}$:

$(\mathrm{R_I})$ Let X, Y be two locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a *regular* morphism; then, for all $x \in \mathrm{X}$, the property $\boldsymbol{R}(\mathscr{O}_{f(x)})$ implies the property $\boldsymbol{R}(\mathscr{O}_x)$.

$(\mathrm{R_{II}})$ Let $\varphi : \mathrm{A} \to \mathrm{B}$ be a homomorphism of local Noetherian rings making B a flat A-module *plat*; then $\boldsymbol{R}(\mathrm{B})$ implies $\boldsymbol{R}(\mathrm{A})$.

Moreover, the property $(\mathrm{P_{III}})$ results from the fact that $\boldsymbol{R}(k)$ is true for every field k; finally, when $\boldsymbol{P}$ is of the second type, the condition $(\mathrm{P_{IV}})$ is a consequence of the definition of $\boldsymbol{P}$ and of the transitivity of finite extensions of type fini.

Remark (7.3.11). — We leave to the reader the task of formulating the property $\boldsymbol{R}$ corresponding to each of the examples of (7.3.8), and of verifying the conditions $(\mathrm{R_I})$ and $(\mathrm{R_{II}})$ in each case, with the help of the results of § 6. In fact, except for example (i) of (7.3.8), the property $\boldsymbol{R}$ satisfies in the other cases the following condition:

(R'_I) Let X, Y be two locally Noetherian preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a $\boldsymbol{P}$-morphism (for the property $\boldsymbol{P}$ of the first or second type defined by $\boldsymbol{R}$); then, for all $x \in \mathrm{X}$, the property $\boldsymbol{R}(\mathscr{O}_{f(x)})$ implies the property $\boldsymbol{R}(\mathscr{O}_x)$.

The reasoning of (6.6.1) and (6.8.3) proves that if $\boldsymbol{R}$ satisfies the conditions (R'_I) (7.3.4), (iii) and $(\mathrm{P_{II}})$, then $\boldsymbol{P}$ satisfies the conditions (P'_I) (7.3.5), (iii) and $(\mathrm{P_{II}})$.

Proposition (7.3.12). — *Let* $\boldsymbol{P}$ *be a property of the first or the second type defined from a property* $\boldsymbol{R}$ *verifying the conditions* $(\mathrm{R_I})$ *and* $(\mathrm{R_{II}})$ *(resp.* (R'_I) *and* $(\mathrm{R_{II}})$*). If for every locally Noetherian prescheme* X, $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ *denotes the set of* $x \in \mathrm{X}$ *such that* $\boldsymbol{R}(\mathscr{O}_x)$ *is true, then, for every regular morphism (resp. every* $\boldsymbol{P}$*-morphism)* $f : \mathrm{X} \to \mathrm{Y}$ *of locally Noetherian preschemes, we have*

$$\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}) = f^{-1}(\mathrm{U}_{\boldsymbol{R}}(\mathrm{Y})). \tag{7.3.12.1}$$

Proof. This is an immediate consequence of the definitions. □

(7.3.13). Given a semi-local Noetherian ring A, we will call *formal fibres* of A the fibres of the canonical morphism $f : \operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$; for every prime ideal $\mathfrak{p}$ of A, the formal fibre at $\mathfrak{p}$ is therefore the scheme $\operatorname{Spec}(\widehat{A} \otimes_A \boldsymbol{k}(\mathfrak{p}))$; as the completion of the local ring $A/\mathfrak{p}$ is $\widehat{A}/\mathfrak{p}\widehat{A} = (A/\mathfrak{p}) \otimes_A \widehat{A}$, we see that the formal fibre of A at $\mathfrak{p}$ is also the *formal fibre of* $A/\mathfrak{p}$ *at the generic point* (o) of $\operatorname{Spec}(A/\mathfrak{p})$.

The property $\boldsymbol{P}$ being defined as in (7.3.1), we will say that *the formal fibres of* A *have the property* $\boldsymbol{P}$, or that A is a $\boldsymbol{P}$*-ring*, if, for all $x \in \operatorname{Spec}(A)$, $\boldsymbol{P}(f^{-1}(x), \boldsymbol{k}(x))$ is true. As the morphism f is *flat*, this amounts to saying that f is a $\boldsymbol{P}$-morphism. **IV-2** | 196

Proposition (7.3.14). — *Let* A *be a semi-local Noetherian ring,* $\mathfrak{m}_i$ $(1 \leqslant i \leqslant r)$ *its maximal ideals, and set* $A_i = A_{\mathfrak{m}_i}$*; for* A *to be a* $\boldsymbol{P}$*-ring, it is necessary and sufficient that each of the* A_i *be so.*

Proof. Indeed, $\widehat{A}$ is the direct product of the $\widehat{A}_i$, hence the formal fibre of A at a point $x \in \operatorname{Spec}(A)$ is the *sum* of the formal fibres at x of those A_i such that $x \in \operatorname{Spec}(A_i)$. ☐

Proposition (7.3.15). — *Let* A *be a semi-local Noetherian ring.*

label=i) *If* A *is a* $\boldsymbol{P}$*-ring, every quotient ring of* A *is a* $\boldsymbol{P}$*-ring.*
lbbel=ii) *If in addition* $\boldsymbol{P}$ *satisfies the condition* $(\mathrm{P_{IV}})$ (7.3.6)*, then every finite A-algebra is a* $\boldsymbol{P}$*-ring.*

Proof. For every ideal $\mathfrak{a}$ of A, $\widehat{A} \otimes_A (A/\mathfrak{a})$ is the completion of $A/\mathfrak{a}$, and the formal fibres of $A/\mathfrak{a}$ are therefore the formal fibres of A at the points of $V(\mathfrak{a})$, whence (i). On the other hand, if B is a finite A-algebra, B is a semi-local ring, on has $\widehat{B} = B \otimes_A \widehat{A}$, and (ii) follows from (7.3.7). ☐

Proposition (7.3.16). — *Suppose that the property* $\boldsymbol{P}$ *is of the first (resp. second) type, defined from a property* $\boldsymbol{R}$ (7.3.10)*. When* $\boldsymbol{P}$ *is of the second type, suppose moreover that the relation*

"for every finite extension k' *of* k*, and every* $z' \in Z \otimes_k k'$*,* $\boldsymbol{R}(\mathscr{O}_{z'})$ *is true"*

implies $\boldsymbol{P}(Z, k)$ *(this is verified for the examples (iv') to (vii') of* (7.3.8)*, by virtue of* (6.7.7) *and* (4.6.1)*).*

Let A *be a semi-local Noetherian ring. If the property* $\boldsymbol{R}$ *satisfies the conditions* $(\mathrm{R_I})$ *and* $(\mathrm{R_{II}})$*, the following properties are equivalent:*

label=a) A *is a* $\boldsymbol{P}$*-ring.*
lbbel=b) *For every integral quotient ring* B *of* A *(resp. every finite integral A-algebra* B*) and every prime ideal* $\mathfrak{q}$ *of* $\widehat{B}$ *whose inverse image in* B *is* 0*,* $\boldsymbol{R}(\widehat{B}_{\mathfrak{q}})$ *is true.*

If in addition $\boldsymbol{R}$ *satisfies the condition* $(\mathrm{R'_I})$*, the properties* a) *and* b) *are also equivalent to:*

lcbel=c) *For every quotient ring* B *of* A *(resp. every finite A-algebra* B*), if we set* $Y = \operatorname{Spec}(B)$*,* $Y' = \operatorname{Spec}(\widehat{B})$*, and if* $g : Y' \to Y$ *is the canonical morphism, we have (with the notations of* (7.3.12)*)*

$$\mathrm{U}_{\boldsymbol{R}}(Y') = g^{-1}(\mathrm{U}_{\boldsymbol{R}}(Y)). \tag{7.3.16.1}$$

Proof. The equivalence of *a*) and *b*) is immediate when $\boldsymbol{P}$ is of the first type: indeed, for every prime ideal $\mathfrak{p}$ of A, the formal fibre of $B = A/\mathfrak{p}$ at the generic point $\mathfrak{p} \in \operatorname{Spec}(A)$ is nothing other than the formal fibre of A at the point $\mathfrak{p}$ (7.3.13). On the other hand, if we specialise *c*) to the case where B is integral and y the generic point of $Y = \operatorname{Spec}(B)$, on has $\mathscr{O}_{X,y} = \boldsymbol{k}(y)$, hence $y \in \mathrm{U}_{\boldsymbol{R}}(Y)$, hence $\boldsymbol{R}(k)$ is true for every field k; by expressing that every point $y' \in g^{-1}(y)$ belongs to $\mathrm{U}_{\boldsymbol{R}}(Y')$, we obtain the statement *b*), hence *c*) implies *b*). **IV-2** | 197

When $\boldsymbol{P}$ is of the second type, the equivalence of *a*) and *b*) follows from the more general lemma below. ☐

Lemma (7.3.16.2). — *Let* A, A′ *be two rings,* $\varphi : A \to A'$ *a homomorphism,* $f : \operatorname{Spec}(A') \to \operatorname{Spec}(A)$ *the corresponding morphism. In order that for all* $x \in \operatorname{Spec}(A)$*, every finite extension* k *of* $\boldsymbol{k}(x)$ *and every point* z *of* $f^{-1}(x) \otimes_{\boldsymbol{k}(x)} k = Z$*, the property* $\boldsymbol{R}(\mathscr{O}_{Z,z})$ *be true, it is necessary and sufficient that: for every integral finite A-algebra* B*, if* $g : \operatorname{Spec}(A' \otimes_A B) \to \operatorname{Spec}(B)$ *is the morphism deduced from* f *by base extension to* B*, and if* $T = g^{-1}(\xi)$ *is the fibre of the generic point* ξ *of* $\operatorname{Spec}(B)$*, then, for all* $t \in T$*,* $\boldsymbol{R}(\mathscr{O}_{T,t})$ *is true.*

Proof. The condition is trivially necessary since x is the point of $\operatorname{Spec}(A)$ above which ξ lies, $\boldsymbol{k}(\xi)$ is a finite extension of $\boldsymbol{k}(x)$. Conversely, let us consider a point $x \in \operatorname{Spec}(A)$ and let k be a finite extension of $\boldsymbol{k}(x)$. If we set $\mathfrak{p} = \mathfrak{J}_x$, $\boldsymbol{k}(x)$ is the field of fractions of $A/\mathfrak{p}$, and if there is a base of k over $\boldsymbol{k}(x)$ formed of integral elements over $A/\mathfrak{p}$; if B is the subring of k generated by these elements, B is an integral finite A-algebra, and k is the field $\boldsymbol{k}(\xi)$ at the generic point ξ of $\operatorname{Spec}(B)$; as x is the image of ξ in $\operatorname{Spec}(\mathrm{A})$, the fibre $g^{-1}(\xi)$ is nothing other than $f^{-1}(x) \otimes_{\boldsymbol{k}(x)} k$, which completes the proof of the lemma. ☐

Proposition (7.3.17). — *Suppose that the property* $\boldsymbol{R}$ *satisfies the conditions* $(\mathrm{R}'_{\mathrm{I}})$ *and* $(\mathrm{R}_{\mathrm{II}})$*, and that* $\boldsymbol{P}$ *is the property of the first (resp. second) type defined from* $\boldsymbol{R}$ *(7.3.10). Then, if* A *is a* $\boldsymbol{P}$*-ring, the properties* $\boldsymbol{R}(\mathrm{A})$ *and* $\boldsymbol{R}(\widehat{\mathrm{A}})$ *are equivalent.*

Proof. This follows from (7.3.12.1) applied to $\mathrm{Y} = \operatorname{Spec}(\mathrm{A})$ and $\mathrm{X} = \operatorname{Spec}(\widehat{\mathrm{A}})$. ☐

Proposition (7.3.18). — *Suppose that* $\boldsymbol{P}$ *is a property of the first (resp. second) type defined from a property* $\boldsymbol{R}$ *satisfying the conditions* $(\mathrm{R}'_{\mathrm{I}})$ *and* $(\mathrm{R}_{\mathrm{II}})$*, and also the condition following:*

$(\mathrm{R}_{\mathrm{III}})$ *For every local Noetherian complete ring* C*, if we set* $\mathrm{Z} = \operatorname{Spec}(\mathrm{C})$*, the set* $\mathrm{U}_{\boldsymbol{R}}(\mathrm{Z})$ *(7.3.12) is open in* Z*.*

Then, if A *is a* $\boldsymbol{P}$*-ring and if we set* $\mathrm{X} = \operatorname{Spec}(\mathrm{A})$*, the set* $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ *is open in* X*.*

Proof. Indeed, if $\mathrm{X}' = \operatorname{Spec}(\widehat{\mathrm{A}})$ and if $f : \mathrm{X}' \to \mathrm{X}$ is the canonical morphism, $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}') = f^{-1}(\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}))$ (7.3.12). As f is faithfully flat and quasi-compact (2.3.12), and as by hypothesis $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}')$ is open in X', the conclusion follows from (2.3.12). ☐

Remarks (7.3.19). — label=*i)* The property $\boldsymbol{R}$ which serves to define $\boldsymbol{P}$ satisfies the condition $(\mathrm{R}_{\mathrm{III}})$ in all the examples enumerated in (7.3.8). For (i), (ii) and (iii), this follows from (6.11.2) and the theorem of Cohen (**0**, 19.8.8); when $\boldsymbol{P}$ is one of the properties (iv), (iv'), (v), (v'), this follows from (6.12.7) and (6.12.9), and when $\boldsymbol{P}$ is one of the properties (vi) and (vi'), the assertion of (7.3.18) is trivial, being true for every locally Noetherian prescheme.

lbbel=*ii)* We have already noted (6.4.3) that when $\boldsymbol{P}$ is one of the properties (ii) or (iii_1) of (7.3.8), we do not know if every local Noetherian ring is a $\boldsymbol{P}$-ring. We also do not know if, in the equivalent conditions c), d), e) of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is the case when A is Henselian (18.9.6).

7.4. Permanence of properties of formal fibres

(7.4.1). Throughout the rest of this number, we suppose that the property $\boldsymbol{P}$ is of the form defined in (7.3.1), and *satisfies the conditions* $(\mathrm{P}_{\mathrm{I}})$, $(\mathrm{P}_{\mathrm{II}})$ *and* $(\mathrm{P}_{\mathrm{III}})$ of (7.3.4). Moreover, we suppose that the property $\boldsymbol{Q}$ which serves to define $\boldsymbol{P}$ is such that, for every generization z' of z in Z, $\boldsymbol{Q}(\mathscr{O}_z, k)$ implies $\boldsymbol{Q}(\mathscr{O}_{z'}, k)$.

Lemma (7.4.2). — *Let* A, A' *be two local Noetherian rings,* $\varphi : \mathrm{A} \to \mathrm{A}'$ *a local homomorphism such that* $f = {}^a\varphi : \operatorname{Spec}(\mathrm{A}') \to \operatorname{Spec}(\mathrm{A})$ *is a* $\boldsymbol{P}$*-morphism. Then, if the formal fibres of* A' *are geometrically regular,* A *is a* $\boldsymbol{P}$*-ring.*

IV-2 | 199

Proof. Consider the completed homomorphism $\widehat{\varphi} : \widehat{\mathrm{A}} \to \widehat{\mathrm{A}'}$ and the corresponding morphism $\widehat{f} = {}^a\widehat{\varphi}$; we have the commutative diagram

$$\begin{tikzcd} \operatorname{Spec}(\widehat{\mathrm{A}}) & \operatorname{Spec}(\widehat{\mathrm{A}'}) \arrow[l, "\widehat{f}"'] \\ \operatorname{Spec}(\mathrm{A}) \arrow[u, "g"] & \operatorname{Spec}(\mathrm{A}') \arrow[l, "f"] \arrow[u, "g'"'] \end{tikzcd}$$

where g and g' are the canonical morphisms. As by hypothesis f is a $\boldsymbol{P}$-morphism and g' a regular morphism, it follows from $(\mathrm{P}_{\mathrm{I}})$ that $f \circ g' = g \circ \widehat{f}$ is a $\boldsymbol{P}$-morphism. On the other hand, the hypothesis that f is a $\boldsymbol{P}$-morphism implies that f is flat, hence it is the same for $\widehat{f}$ (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, cor. of prop. 3), which is moreover a local homomorphism, hence faithfully flat (**0**, 6.6.2); it follows then from $(\mathrm{P}_{\mathrm{II}})$ that g is a $\boldsymbol{P}$-morphism. ☐

Corollary (7.4.3). — *label=*i) *Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian,* $\mathrm{A}' = \widehat{\mathrm{A}}$ *its completion. Suppose that for every prime ideal* $\mathfrak{p}'$ *of* A'*, the formal fibres of* $\mathrm{A}'_{\mathfrak{p}'}$ *are geometrically regular; then, for every prime ideal* $\mathfrak{p}$ *of* A*,* $\mathrm{A}_{\mathfrak{p}}$ *is a* $\boldsymbol{P}$*-ring.*

lbbel=ii) *Suppose that* $\boldsymbol{P}$ *satisfies the condition* $(\mathrm{P_{IV}})$ (7.3.6). *Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian*, B *an A-algebra locale essentially of finite type, and such that the homomorphism* $\mathrm{A} \to \mathrm{B}$ *is local. Set* $\mathrm{A}' = \widehat{\mathrm{A}}$*, and let* $\mathfrak{n}'$ *be the unique prime ideal of* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ *above the maximal ideal of* A'*. If the formal fibres of* $\mathrm{B}'_{\mathfrak{n}'}$ *are geometrically regular, then* B *is a* $\boldsymbol{P}$*-ring.*

Proof. label=(i) As by hypothesis $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is a $\boldsymbol{P}$-morphism, the same holds for $\mathrm{Spec}(\mathrm{A}'_{\mathfrak{p}'}) \to \mathrm{Spec}(\mathrm{A}_{\mathfrak{p}})$ by virtue of (7.3.2), *b*)), for every prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$ and every prime ideal $\mathfrak{p}$ of A. It suffices then to apply the lemma (7.4.2) to this morphism, noting that the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is surjective.

lbbel=(ii) By virtue of (7.4.2), it suffices to prove that the morphism $\mathrm{Spec}(\mathrm{B}'_{\mathfrak{n}'}) \to \mathrm{Spec}(\mathrm{B})$ is a $\boldsymbol{P}$-morphism. Now we have $\mathrm{B} = \mathrm{C}_{\mathfrak{n}}$, where C is a finite type A-algebra and $\mathfrak{n}$ is a prime ideal of C above the maximal ideal of A. If we set $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$, it follows from the hypotheses and from (7.3.7) that $\mathrm{Spec}(\mathrm{C}') \to \mathrm{Spec}(\mathrm{C})$ is a $\boldsymbol{P}$-morphism; as $\mathrm{B}'_{\mathfrak{n}'}$ is a local ring of C' at a prime ideal of C' above $\mathfrak{n}$, the conclusion follows from (7.3.2), *b*)).

□

Theorem (7.4.4). — *The hypotheses on* $\boldsymbol{P}$ *being those of* (7.4.1)*, let* A *be a* $\boldsymbol{P}$*-ring local Noetherian.*

label=i) *For every prime ideal* $\mathfrak{p}$ *of* A, $\mathrm{A}_{\mathfrak{p}}$ *is a* $\boldsymbol{P}$*-ring.*

lbbel=ii) *Suppose in addition that* $\boldsymbol{P}$ *satisfies the condition* $(\mathrm{P_{IV}})$*. Then every local ring that is an A-algebra essentially of finite type is a* $\boldsymbol{P}$*-ring.*

Proof. label=(i) Applying (7.4.3), (i), everything reduces to seeing that for every prime ideal $\mathfrak{p}'$ of $\mathrm{A}' = \widehat{\mathrm{A}}$, $\mathrm{A}'_{\mathfrak{p}'}$ has its formal fibres geometrically regular. Now, this has been proved in (**0**, 22.3.3) and (**0**, 22.5.8).

lbbel=(ii) Let B be an A-algebra essentially of finite type that is a local ring; if $\mathfrak{p}$ is the prime ideal of A above which is the maximal ideal of B, B is also an $\mathrm{A}_{\mathfrak{p}}$-algebra essentially of finite type (1.3.8); by virtue of (i), we can therefore suppose that $\mathfrak{p}$ is the maximal ideal $\mathfrak{m}$ of A. We have therefore $\mathrm{B} = \mathrm{C}_{\mathfrak{q}}$, where C is a finite type A-algebra, $\mathfrak{q}$ a prime ideal of C (necessarily above $\mathfrak{m}$); by virtue of (i), one can therefore suppose that $\mathrm{B} = \mathrm{C}_{\mathfrak{q}}$.

Let B be an A-algebra essentially of finite type that is a local ring; if $\mathfrak{p}$ is the prime ideal of A above which the maximal ideal of B lies, B is also an $\mathrm{A}_{\mathfrak{p}}$-algebra essentially of finite type (1.3.8); by virtue of (i), one can therefore suppose that the homomorphism $\mathrm{A} \to \mathrm{B}$ is local. **IV-2** | 200

Applying (7.4.3), (ii) with $\mathrm{A}' = \widehat{\mathrm{A}}$, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ is a local ring of $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ at a prime ideal above the maximal ideal of A'; as A and A' have the same residue field k, the residue field of $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ at this prime ideal is equal to k'. Moreover $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ is a finite type A'-algebra generated by a single element. Hence (7.4.3), (ii) one can restrict to proving (7.4.4.1) when $\boldsymbol{P}$ is the property (iv') of (7.3.8), that A is *complete* and C is generated by a single element t.

□

Lemma (7.4.4.1). — *Let* A *be a* $\boldsymbol{P}$*-ring local Noetherian,* k *its residue field,* C *a finite type A-algebra,* B *a local ring in a prime ideal* $\mathfrak{n}$ *of* C*, such that:* 1º *the homomorphism* $\mathrm{A} \to \mathrm{B}$ *is local;* 2º *the residue field* k' *of* B *is a finite extension of* k*. If* $\boldsymbol{P}$ *satisfies* $(\mathrm{P_{IV}})$*, then* B *is a* $\boldsymbol{P}$*-ring.*

Proof. Let $(x_i)_{1 \leqslant i \leqslant m}$ be a system of generators of the A-algebra C; let us show that we can reason by induction on m. Let C' be the sub-A-algebra of C generated by $x_1, \ldots, x_{m-1}$, and let $\mathfrak{n}' = \mathfrak{n} \cap \mathrm{C}'$. The homomorphism $\mathrm{A} \to \mathrm{C}'_{\mathfrak{n}'}$ and $\mathrm{C}'_{\mathfrak{n}'} \to \mathrm{C}_{\mathfrak{n}}$ are local homomorphisms; if k'' is the residue field of $\mathrm{C}'_{\mathfrak{n}'}$, $k \to k'' \to k'$ is factored as $k \to k'' \to k'$, hence k'' is a finite extension of k and k' a finite extension of k''. The hypothesis of induction implies that $\mathrm{C}'_{\mathfrak{n}'}$ is a $\boldsymbol{P}$-ring, and the hypothesis of induction shows again that $\mathrm{C}_{\mathfrak{n}}$ is a $\boldsymbol{P}$-ring, which reduces the problem to the case where C is an A-algebra generated by a *single* element t.

Applying (7.4.3), (ii) with $\mathrm{A}' = \widehat{\mathrm{A}}$, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ is a local ring of $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ at a prime ideal above the maximal ideal $\mathfrak{m}$; as A and A' have the same residue field k, the residue field of $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ at this prime ideal is equal to k'. Moreover $\mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ is generated by a single element over A'. It follows therefore from (7.4.3), (ii) that one can restrict to proving (7.4.4.1) when $\boldsymbol{P}$ is the property (iv') of (7.3.8), that A is *complete* and C generated by a single element t.

To show that the formal fibres of $B = C_{\mathfrak{n}}$ are then geometrically regular, let us apply the criterion (7.3.16), *b*)). Let B_1 be the integral finite B-algebra, hence generated by a finite number of integral elements over B. By multiplying these elements by an element of B, one can suppose that they are integral over C, and we can therefore write $B_1 = S^{-1}C_1$, where C_1 is a sub-C-algebra of B_1 generated by a finite number of integral elements over C, hence a finite integral C-algebra. On the other hand, B_1 is a semi-local ring, and every local ring B_2 of B_1 at a maximal ideal is a local ring of C_1, such that $A \to B_2$ is a local homomorphism; moreover, the residue field of B_2 is a finite extension of k', hence of k. We see therefore (taking into account (7.3.14) and of (i)) that we are reduced to the following question: let C be a local Noetherian integral ring, containing a sub-ring C_0 which is an A-algebra generated by a single element t and such that C is a C_0-algebra *finite*; if $\mathfrak{n}$ is a maximal ideal of C above the maximal ideal of A, and $B = C_{\mathfrak{n}}$, is the completion $\widehat{B}_{\mathfrak{q}}$ *regular* for every prime ideal $\mathfrak{q}$ of $\widehat{B}$ such that $\mathfrak{q} \cap A = 0$? **IV-2** | 201

Let us consider the case where B is a finite A-algebra; if $\mathfrak{p}$ is the prime ideal of A above which the maximal ideal of B lies, B is also an $A_{\mathfrak{p}}$-algebra essentially of finite type (1.3.8); by virtue of (i), it is enough to show that $C_{\mathfrak{n}}$ is a ***P***-ring, hence reduce to proving the lemma for the case where A is complete and C generated by a single element. In the general case, if we set $\mathfrak{p} = \mathfrak{J}_x$, $C/\mathfrak{q}$ and $B/\mathfrak{p}$ are respectively $(A/\mathfrak{r})[\bar{t}]$-algebra finite, where $\mathfrak{r} = \mathfrak{p} \cap A$; in the general case, if we set $\mathfrak{q} = \mathfrak{p} \cap C$, $C/\mathfrak{q}$ is a k-algebra of finite type, hence k' of C at the maximal ideal $\mathfrak{n}$ is an extension *finite* of the residue field k of A (**I**, 6.4.2); by virtue of (i), it suffices therefore to show that B is a ***P***-ring. This was carried out in (**0**, 22.3.3) and (**0**, 22.5.8). □

Lemma (7.4.4.2). — *Let* A *be a local Noetherian integral ring and complete, k its residue field,* C *an integral ring containing* A*, such that there exists* $t \in C$ *for which* C *is an* $A[t]$*-algebra finite. Let* $\mathfrak{n}$ *be a maximal ideal of* C *above the maximal ideal* $\mathfrak{m}$ *of* A*; set* $B = C_{\mathfrak{n}}$, $X = \operatorname{Spec}(B)$, $B' = \widehat{B}$, $X' = \operatorname{Spec}(\widehat{B})$*; if* $U = \operatorname{Reg}(X)$, $U' = \operatorname{Reg}(X')$*, and if* $f : X' \to X$ *is the canonical morphism, then* $f^{-1}(U) \subset U'$.

Proof. The assertion to prove in order to obtain (7.4.4.1) will be deduced from this lemma by observing that, since B is integral, the generic point of X belongs to U. We note that since C is an A-algebra of finite type, and $\mathfrak{n}$ a maximal ideal of C, the residue field k' of $C_{\mathfrak{n}} = B$ (hence also of B') is a finite extension of k (**I**, 6.4.11) and (6.4.2).

It follows from (6.12.8) that $\operatorname{Reg}(Y)$ is open in Y; on the other hand (6.12.7), U' is open in X', hence $S' = X' - U'$ is closed; hence $S' \cap f^{-1}(U)$ is *locally closed* in X', and everything comes down to proving that this set is *empty*. We know (5.1.10) that in the contrary case, there would exist in $S' \cap f^{-1}(U)$ a prime ideal $\mathfrak{p}'$ such that $\dim(B'/\mathfrak{p}') \leqslant 1$. Let us remark first that $\mathfrak{p}'$ cannot be the maximal ideal $\mathfrak{m}B'$ of B', where $\mathfrak{m}$ is the maximal ideal of B. Indeed, this would mean that $B = B_{\mathfrak{m}}$ would be regular, hence also $B' = \widehat{B}$ (**0**, 17.1.5), and we would have $\mathfrak{m}B' \in U'$ contrary to the hypothesis. We must therefore have $\dim(B'/\mathfrak{p}') = 1$. Set $\mathfrak{p} = \mathfrak{p} \cap C$, $\mathfrak{r} = \mathfrak{p} \cap A$, $C/\mathfrak{q}$ is a $(A/\mathfrak{r})[\bar{t}]$-algebra finite (where $\bar{t}$ is the class of t mod. $\mathfrak{q}$); as $\mathfrak{p} = \mathfrak{q}B$, $B/\mathfrak{p}$ is equal to $(C/\mathfrak{q})_{\mathfrak{n}/\mathfrak{q}}$, and $\mathfrak{n}/\mathfrak{q}$ is a maximal ideal of C (necessarily above $\mathfrak{m}/\mathfrak{r}$ of $A/\mathfrak{r}$); we see therefore that the hypotheses of (7.4.4.2) are fulfilled by $A/\mathfrak{r}$, $C/\mathfrak{q}$ and $B/\mathfrak{p}$, and as the completion of $B/\mathfrak{p}$ is $B'/\mathfrak{p}B'$, this proves our assertion. Let us suppose therefore $\mathfrak{p} = 0$. But then $A \subset V$ and V is *a fortiori* an A-algebra finite, and as the local ring A is complete, it is the same for B (Bourbaki, *Alg. comm.*, chap. IV, § 2, nº 5, cor. 3 of prop. 9), whence $B' = B$ is a field, and so by B a subring of B also reduced. □

IV-2 | 202

Corollary (7.4.5). — *Suppose that the property* ***P*** *satisfies the conditions* (P_{I}), (P_{II}), (P_{III})*. Let* A *be a Noetherian ring. The following conditions are equivalent:*

label=a) *For every prime ideal* $\mathfrak{p}$ *of* A, $A_{\mathfrak{p}}$ *is a* ***P****-ring.*
lbbel=b) *For every maximal ideal* $\mathfrak{m}$ *of* A, $A_{\mathfrak{m}}$ *is a* ***P****-ring.*

If in addition ***P*** *satisfies* (P_{IV})*, the properties a) and b) are also equivalent to:*

lcbel=c) *For every finite type A-algebra* B *and every prime ideal* $\mathfrak{q}$ *of* B, $B_{\mathfrak{q}}$ *is a* ***P****-ring.*

Proof. The equivalence of *a*) and *b*) follows from (7.4.4), (i), and that of *a*) and *c*) from (7.4.4), (ii). When condition *a*) of (7.4.5) is satisfied, we say that A is a ***P***-ring; for semi-local Noetherian rings, this definition coincides with that of (7.3.13), when the conditions (P_{I}), (P_{II}) and (P_{III}) are satisfied. The ring **Z** is a ***P***-ring (7.3.19), (iv); every local Noetherian complete ring is a ***P***-ring. □

Proposition (7.4.6). — *Suppose that the property* $\boldsymbol{P}$ *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P_{II}})$ *and* $(\mathrm{P_{III}})$. *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A, $\widehat{\mathrm{A}}$ *the separated completion of* A *for the* $\mathfrak{J}$*-preadic topology. Then, if* A *is a* $\boldsymbol{P}$*-ring* (7.4.5), *the canonical morphism* $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ *is a* $\boldsymbol{P}$*-morphism.*

Proof. Using (7.3.2), $c')$), it suffices to prove that for every maximal ideal $\mathfrak{n}$ of $\widehat{\mathrm{A}}$ with inverse image $\mathfrak{m}$ in A, the morphism $\mathrm{Spec}((\widehat{\mathrm{A}})_{\mathfrak{n}}) \to \mathrm{Spec}(\mathrm{A}_{\mathfrak{m}})$ is a $\boldsymbol{P}$-morphism. We know (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 4, prop. 8) that the canonical homomorphism $\mathrm{A}_{\mathfrak{m}} \to (\widehat{\mathrm{A}})_{\mathfrak{n}}$ is injective, that the topology $\mathfrak{m}\mathrm{A}_{\mathfrak{m}}$-preadic on $\mathrm{A}_{\mathfrak{m}}$ is induced by the topology $\mathfrak{n}(\widehat{\mathrm{A}})_{\mathfrak{n}}$-preadic, and that $\mathrm{A}_{\mathfrak{m}}$ is dense in $(\widehat{\mathrm{A}})_{\mathfrak{n}}$ for this topology, so that the completion of $\mathrm{A}_{\mathfrak{m}}$ for the $\mathfrak{m}\mathrm{A}_{\mathfrak{m}}$-preadic topology identifies with that of $(\widehat{\mathrm{A}})_{\mathfrak{n}}$ for the $\mathfrak{n}(\widehat{\mathrm{A}})_{\mathfrak{n}}$-preadic topology. We have therefore two morphisms

$$\mathrm{Spec}((\mathrm{A}_{\mathfrak{m}})\widehat{\ }) \xrightarrow{l} \mathrm{Spec}((\widehat{\mathrm{A}})_{\mathfrak{n}}) \xrightarrow{g} \mathrm{Spec}(\mathrm{A}_{\mathfrak{m}})$$

□

Corollary (7.4.7). — *Suppose that* $\boldsymbol{P}$ *satisfies the conditions* $(\mathrm{P_I})$, $(\mathrm{P'_I})$, $(\mathrm{P_{II}})$, $(\mathrm{P_{III}})$ *and* $(\mathrm{P_{IV}})$. *Then, if* A *is a* $\boldsymbol{P}$*-ring* (7.4.5), *the canonical morphism* $\mathrm{Spec}(\mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_r]]) \to \mathrm{Spec}(\mathrm{A})$ *is a* $\boldsymbol{P}$*-morphism. In particular, if in addition* A *is integral,* K *its field of fractions, and if* $\mathfrak{p}$ *is a prime ideal of* $\mathrm{B} = \mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_r]]$ *such that* $\mathfrak{p} \cap \mathrm{A} = 0$, *then the property* $\boldsymbol{P}(\mathrm{B}_{\mathfrak{p}}, \mathrm{K})$ *is true.*

Proof. Indeed, $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{A})$ factorises as **IV-2** | 203

$$\mathrm{Spec}(\mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_r]]) \xrightarrow{l} \mathrm{Spec}(\mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_r]) \xrightarrow{g} \mathrm{Spec}(\mathrm{A}).$$

It is clear that the morphism g is regular (0.17.3.7); by virtue of (7.4.5), $\mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_r]$ is a $\boldsymbol{P}$-ring, hence it follows from (7.4.6) that f is a $\boldsymbol{P}$-morphism; as g is regular, hence also a $\boldsymbol{P}$-morphism (7.3.5), (i), it follows from $(\mathrm{P'_I})$ that $g \circ f$ is a $\boldsymbol{P}$-morphism.

One will note that the conclusion is still valid if instead of supposing that $\boldsymbol{P}$ satisfies $(\mathrm{P'_I})$, one only supposes that a composite $g \circ f$ is a $\boldsymbol{P}$-morphism when g is regular and f a $\boldsymbol{P}$-morphism (symmetric condition to that of $(\mathrm{P'_I})$ in some sense). □

Remark (7.4.8). — The preceding results pose the following problems:

label=A) Let A be a Zariski *complete* ring, and let $\mathfrak{J}$ be an ideal of definition of A; if the ring $\mathrm{A}/\mathfrak{J}$ is a $\boldsymbol{P}$-ring, is A also a $\boldsymbol{P}$-ring? It would follow that for every $\boldsymbol{P}$-ring Noetherian A and every ideal $\mathfrak{J}$ of A, the separated completed $\widehat{\mathrm{A}}$ for the $\mathfrak{J}$-preadic topology would also be a $\boldsymbol{P}$-ring.

lbbel=B) Let k be a valued field, complete and non-discrete; one calls *ring of restricted formal power series* $k\{\mathrm{T}_1, \dots, \mathrm{T}_n\}$ the subring of the formal power series ring $k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ formed of the series whose coefficients *tend to* 0. Is such a ring a $\boldsymbol{P}$-ring?

lcbel=C) If A is a linearly topologised $\boldsymbol{P}$-ring, S a multiplicative subset of A, are the rings $\mathrm{A}\{S^{-1}\}$ (**0**, 7.6.1) and $\mathrm{A}_{\{S\}}$ (**0**, 7.6.15) $\boldsymbol{P}$-rings?

7.5. A criterion for $\boldsymbol{P}$-morphisms In what follows of this section, we will apply (7.5.1) to the study **IV-2** | 207
of *completed tensor products* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ of local Noetherian rings which are algebras over a field k.

Lemma (7.5.5). — *Let* k *be a field,* A, B *two local Noetherian complete rings containing* k, *the residue field of* A *being a finite extension of* k. *Let* C *be the completed tensor product* $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ (**0**, 7.7.5). *Then:*

label=(i) C *is a semi-local Noetherian complete ring.*

lbbel=(ii) C *is a flat* A*-module and a flat* B*-module.*

lcbel=(iii) If $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}\mathrm{C}$ *is contained in the radical of* C, *and* $\mathrm{C}/\mathfrak{m}\mathrm{C}$ *is isomorphic to* $(\mathrm{A}/\mathfrak{m}) \otimes_k \mathrm{B}$.

ldbel=(iv) The residue fields of the components of C *are finite extensions of the residue field of* B.

The properties (i), (iii) and the first assertion of (ii) are particular cases of (**0**, 19.7.1.2). *To prove that* C *is a flat* B*-module, let us note that for all* $h > 0$, $\mathrm{C}/\mathfrak{m}^h\mathrm{C} = (\mathrm{A}/\mathfrak{m}^h) \otimes_k \mathrm{B}$ *is a flat* B*-module, since* k *is a field; it therefore suffices to apply* (**0**, 10.2.6) *to each of the local components of* C *(whose* C *is the direct composite). Finally, if* $\mathfrak{n}$ *is the maximal ideal of* B, *the residue fields of the components of* C *are also those of the Artinian ring* $(\mathrm{A}/\mathfrak{m}) \otimes_k (\mathrm{B}/\mathfrak{n})$, *which, by hypothesis, are finite extensions of* $\mathrm{B}/\mathfrak{n}$.

Proposition (7.5.6). — *Let k be a field, A, B two local Noetherian complete rings containing k, and whose residue fields are finite extensions of k. Let $\boldsymbol{R}$ be a property satisfying the conditions (R'_i), $(\mathrm{R}_{\mathrm{III}})$ and $(\mathrm{R}_{\mathrm{IV}})$ (the property $\boldsymbol{P}$ being defined from $\boldsymbol{R}$ by* (7.5.0)*). Suppose that $\boldsymbol{R}(\mathrm{A})$ is true, as well as $\boldsymbol{P}(\mathrm{Spec}(\mathrm{B}), k)$ (which means that for every finite extension k' of k and for each of the complete local rings C_j whose direct composite is $\mathrm{B} \otimes_k k'$, $\boldsymbol{R}(\mathrm{B}_j)$ is true). Then, for each of the complete local rings C_j whose direct composite is $\mathrm{B}\widehat{\otimes}_k\mathrm{B}$, $\boldsymbol{R}(\mathrm{C}_j)$ is true.*

Proof. It follows from (7.5.5), (iii) that each of the homomorphisms $\mathrm{A} \to \mathrm{C}_j$ is local and from (7.5.5), (ii) that each of the C_j is a flat A-module; finally, by (7.5.5), (iv), the residue fields of the C_j are finite extensions of $\mathrm{A}/\mathfrak{m}$. On the other hand, the property $\boldsymbol{P}(\mathrm{Spec}(\mathrm{C} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{m})), \mathrm{A}/\mathfrak{m})$ is true, as this is the case when the residue fields of B are among those of $\mathrm{C}/\mathfrak{m}\mathrm{C} = (\mathrm{A}/\mathfrak{m}) \otimes_k \mathrm{B}$, where y ranges over $\mathrm{Spec}(\mathrm{B})$ and $x = f(y)$, from the definition of $\boldsymbol{P}$ and the hypothesis that $\boldsymbol{R}(\mathrm{A})$ is true. All the conditions of (7.5.1) are satisfied for the homomorphisms $\mathrm{A} \to \mathrm{C}_j$ and the corresponding morphisms $\mathrm{Spec}(\mathrm{C}_j) \to \mathrm{Spec}(\mathrm{A})$; the conclusion then follows from (R'_i) and the hypothesis that $\boldsymbol{R}(\mathrm{A})$ is true. ☐

Corollary (7.5.7). — *(Chevalley). — Let k be a perfect field (resp. algebraically closed), A, B two local Noetherian complete rings containing k and whose residue fields are finite extensions of k. Then, if A and B are reduced (resp. integral), the completed tensor product $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ is reduced (resp. integral).*

Proof. I) Suppose k perfect, A and B reduced; let A_i (resp. B_j) be the quotients of A (resp. B) by its prime minimal ideals ($1 \leqslant i \leqslant r$, $1 \leqslant j \leqslant s$), which are complete; the hypothesis entails that A (resp. B) identifies with a subring of the direct composite of the A_i (resp. B_j); the tensor products being taken over the base field, and as the tensor products of integral domains are subrings, $\mathrm{A} \otimes_k \mathrm{B}$ identifies with a subring of the direct composite of the $\mathrm{A}_i \otimes_k \mathrm{B}_j$. It follows that $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ identifies with a subring of the direct composite of the $\mathrm{A}_i\widehat{\otimes}_k\mathrm{B}_j$, and we are thus reduced to the case where A and B are integral. By the theorem of Nagata de finitude (**0**, 23.1.6), A' (resp. B') is an A-module (resp. B-module) of finite type and a complete local ring. Furthermore, $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ identifies with a subring of $\mathrm{A}'\widehat{\otimes}_k\mathrm{B}'$: indeed, it will suffice to prove that $\mathrm{A}\widehat{\otimes}_k\mathrm{B}$ identifies with a subring of $\mathrm{A}'\widehat{\otimes}_k\mathrm{B}$, and the latter with a subring of $\mathrm{A}'\widehat{\otimes}_k\mathrm{B}'$. This follows from the lemma below:

Lemma (7.5.7.1). — *Let A, B be two local Noetherian complete rings containing a field k and whose residue fields are finite extensions of k. Let $\mathfrak{m}$ be the maximal ideal of A. For every A-module M of finite type (equipped with the $\mathfrak{m}$-adic topology), the completed tensor product $\mathrm{M}\widehat{\otimes}_k\mathrm{B}$ (**0**, 7.7.1) identifies canonically with $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$.*

Proof. Indeed, on has a canonical isomorphism $\mathrm{M} \otimes_k \mathrm{B} \xrightarrow{\sim} \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A} \otimes_k \mathrm{B})$ hence a canonical composite homomorphism

$$\varphi : \mathrm{M}\widehat{\otimes}_k\mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$$

and it is immediate that this homomorphism is continuous for the tensor product topologies; furthermore, $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$ is separated and complete ((7.5.5) and (**0**, 7.7.8)), hence one also has by completion a continuous homomorphism

$$\widehat{\varphi} : \mathrm{M}\widehat{\otimes}_k\mathrm{B} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B})$$

It is immediate that this homomorphism is bijective when M is free of finite type; as one has an exact sequence $\mathrm{L}' \to \mathrm{L} \to \mathrm{M} \to 0$, where L and L' are free A-modules of finite type, one deduces a commutative diagram **IV-2 | 208**

$$\begin{tikzcd}
\mathrm{L}'\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & \mathrm{L}\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & \mathrm{M}\widehat{\otimes}_k\mathrm{B} \arrow[r] \arrow[d] & 0 \\
\mathrm{L}' \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & \mathrm{L} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}\widehat{\otimes}_k\mathrm{B}) \arrow[r] & 0
\end{tikzcd}$$

where the rows are exact (the first by virtue of the definition of the completed tensor product and of (**0**, 13.2.2)); the first two vertical arrows being bijections, it follows that so is the third. ☐

The fact that $A\widehat{\otimes}_k B$ identifies with a subring of $A' \otimes_A (A\widehat{\otimes}_k B)$ follows then from the fact that $A\widehat{\otimes}_k B$ is a flat A-module (7.5.5). One can thus further suppose that A and B are *integrally closed*; as moreover k is supposed *perfect*, $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$ are *geometrically normal* over k, by virtue of (6.7.7), b)); one can then apply (7.5.6) taking for $\boldsymbol{R}$ the property (vi) of (7.5.3).

II) Suppose k algebraically closed, A and B integral. The reasoning of I) brings us again to the case where A and B are *integrally closed*; it follows then from (7.5.6) that $\operatorname{Spec}(A\widehat{\otimes}_k B)$ is *normal*, hence $A\widehat{\otimes}_k B$ is a direct composite of integrally closed local complete rings, and the whole thing comes down to seeing that $A\widehat{\otimes}_k B$ is an integral *local* ring; it suffices for this to verify that if $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B, $(A/\mathfrak{m}) \otimes_k (B/\mathfrak{n})$ is an integral local ring (see the proof of (**0**, 19.7.1.2)); but as $A/\mathfrak{m}$ and $B/\mathfrak{n}$ are finite extensions of k, they are identical to k, since k is algebraically closed, whence the conclusion. □

Remark (7.5.8). — It would be interesting to determine whether, in (7.5.7), one can replace the hypothesis that k is perfect or algebraically closed by the hypothesis that $\operatorname{Spec}(A)$ or $\operatorname{Spec}(B)$ is *geometrically integral* over k, or at least by the hypothesis that A (for example) is integral and contains a subring A_0 isomorphic to a formal power series ring $k[[T_1, \ldots, T_n]]$, such that the field of fractions K of A is *separable* over the field of fractions K_0 of A_0 (one can show that this condition entails that A is geometrically integral over k, and that the two properties are equivalent when $[k : k^p]$ is finite (p being the characteristic exponent of k)). Likewise, it would be desirable to develop variants of (7.5.6) and (7.5.7) where one would weaken the hypothesis of finiteness of the residue fields of A and B, by supposing for example that only one of them is a finite extension of k, the other being arbitrary.

7.6. Applications: I. Japanese local rings

Proposition (7.6.1). — *Let* A *be a reduced local Noetherian ring whose formal fibres are geometrically normal. Then the completion* $\widehat{A}$ *is reduced, the integral closure* A' *of* A *in its total ring of fractions is a finite* A*-algebra (hence a semi-local Noetherian ring), and its completion* $\widehat{A}'$ *is isomorphic to the integral closure of* $\widehat{A}$ *in its total ring of fractions.*

Proof. **IV-2 | 209**

The formal fibres of A are *a fortiori* geometrically reduced, hence the hypothesis that A is reduced entails that so is $\widehat{A}$, by virtue of (7.3.17). Let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ be the prime minimal ideals of A, and let $B_i = A/\mathfrak{p}_i$; the formal fibres of the local rings B_i are then also geometrically normal (7.3.15), hence the $\widehat{B}_i$ are also reduced and it follows from (**0**, 23.1.7), (i) that the integral closure B_i' of B_i in its field of fractions is a B_i-module of finite type, hence an A-module of finite type; as A' is the direct composite of the B_i' (**II**, 6.3.8), one sees that A' is an A-module of finite type. Let $\mathfrak{m}_j$ $(1 \leqslant j \leqslant r)$ be the maximal ideals of the semi-local ring A'; one knows that the completion $\widehat{A}'$ of A' identifies with the direct composite of the completions $\widehat{A}'_{\mathfrak{m}_j}$ of $A'_{\mathfrak{m}_j}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, no 13, cor. de la prop. 19). Now, it follows from the hypothesis and from (7.3.15) that the formal fibres of the $A'_{\mathfrak{m}_j}$ are geometrically normal; as $\operatorname{Spec}(A')$ is normal by definition, it follows from each $\operatorname{Spec}(A'_{\mathfrak{m}_j})$, and one deduces hence from (7.3.17) that $\operatorname{Spec}(\widehat{A}'_{\mathfrak{m}_j})$ is normal for every j, hence also $\operatorname{Spec}(\widehat{A}')$. On the other hand, $\widehat{A}'$ identifies with $A' \otimes_A \widehat{A}$ since A' contains A and is contained in the total ring of fractions R of A, and that $\widehat{A}$ is a flat A-module, every regular element of A is also $\widehat{A}$-regular; finally, as $\widehat{A}$ is a flat A-module, $\widehat{A}$ contains $\widehat{A}$ and is contained in $R' = R \otimes_A \widehat{A}$; hence R' identifies canonically with a subring of the total ring of fractions R'' of $\widehat{A}$ (Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 3 de la prop. 9 and chap. III, § 3, no 4, th. 3), $\widehat{A}'$ is normal and that $\widehat{A}'$ is indeed the integral closure of $\widehat{A}$ in R''. □

Corollary (7.6.2). — *Under the hypotheses of* (7.6.1), *there is a bijective correspondence between the maximal ideals* $\mathfrak{m}_j$ *of* A' *(in other words, the set of closed points of* $\operatorname{Spec}(A')$ *above the closed point of* A*) and the set of prime minimal ideals* $\mathfrak{q}_j$ *of* $\widehat{A}$ *(in other words, the set of maximal points of* $\operatorname{Spec}(\widehat{A})$*); in this correspondence, the completion* $\widehat{A}'_{\mathfrak{m}_j}$ *of* $A'_{\mathfrak{m}_j}$ *is isomorphic to the integral closure of* $\widehat{A}/\mathfrak{q}_j$.

Proof. Indeed, one knows that the integral closure of $\widehat{A}$ in its total ring of fractions is the direct composite of the integral closures of the $\widehat{A}/\mathfrak{q}_j$, which are local complete rings (**0**, 23.1.6). □

Corollary (7.6.3). — *Under the hypotheses of (7.6.1), for $\widehat{\mathrm{A}}$ to be integral, it is necessary and sufficient that* A *be unibranch; for $\widehat{\mathrm{A}}$ to be geometrically unibranch, it is necessary and sufficient that* A *be.*

Proof. This is a particular case of (7.6.2). □

Theorem (7.6.4). — *(Zariski-Nagata). — Let* A *be a semi-local Noetherian ring. The following conditions are equivalent:*

label=a) *For every finite reduced* A*-algebra* C*, the completion $\widehat{\mathrm{C}}$ is a reduced ring.*
a′) *For every integral quotient ring* B *of* A*, with field of fractions* K*, the completion $\widehat{\mathrm{B}}$ is reduced, and the fields L_i of the total ring of fractions of $\widehat{\mathrm{B}}$ are separable extensions of* K*.*
a″) *The formal fibres are geometrically reduced (in other words, for every integral quotient ring* B *of* A*, with field of fractions* K*, the* K*-algebra $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}$ is separable* (4.6.2)*).*
lbbel=b) *Every integral quotient ring* B *of* A *is a Japanese ring.*

Proof. **IV-2 | 210**

To show that *a*) implies *b*), it suffices to verify that the L_i are separable extensions of K, or still (4.6.1) that for every finite extension K′ of K, the ring $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}'$ is reduced. It is thus when B′ is a finite integral extension of B, the nonzero elements of B′ are $\widehat{\mathrm{B}}'$-regular, which entails that $\widehat{\mathrm{B}}' \otimes_{\mathrm{B}'} \mathrm{K}'$ identifies with a subring of the total ring of fractions of $\widehat{\mathrm{B}}'$, and *a fortiori* is reduced.

To see that *a*′) implies *a*″), consider any point x of $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, and let Y be the reduced integral closed sub-prescheme of X having $\{x\}$ as underlying space; one has $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, where B is an integral quotient ring of A, and $\mathrm{Spec}(\widehat{\mathrm{B}}) = \mathrm{Y} \times_{\mathrm{X}} \mathrm{Spec}(\widehat{\mathrm{A}})$, so that the formal fibre of A at the point x is the same as that of B, or still equal to $\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K})$. As the local rings of $\mathrm{Spec}(\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K})$ are those of $\mathrm{Spec}(\widehat{\mathrm{B}})$ at the points of the fibre of x, the hypothesis implies that $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}$ is reduced, and the conditions of *a*′) imply $\widehat{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{K}$ is a separable K-algebra (4.6.1).

The condition *a*″) implies *a*), for it follows from (7.3.15) that the formal fibres of C are geometrically reduced, and if C is reduced, it follows hence that so is $\widehat{\mathrm{C}}$ (particular case of (7.3.17)).

The fact that *a*′) implies *b*) is a particular case of (**0**, 23.1.7). It remains to show, to prove the equivalence of *a*), *b*) and *c*), that *b*) implies *a*). Let us note that if C is an A-algebra finite, for every prime ideal $\mathfrak{q}$ of C, $\mathrm{C}/\mathfrak{q}$ is an $(\mathrm{A}/\mathfrak{p})$-algebra finite, where $\mathfrak{p}$ is the inverse image of $\mathfrak{q}$ in A which is a prime ideal, hence the hypothesis *b*) entails that every integral quotient ring of C is a Japanese ring. One is thus reduced to proving that under the hypothesis *b*), *the completion of every integral quotient ring of* A *is reduced*. We argue by induction on $n = \dim(\mathrm{A})$, the assertion being trivial for $n = 0$; by replacing A by the quotients of A by its prime minimal ideals $\mathfrak{p}_i$, one can restrict to the case where A is integral (every integral quotient ring of A being a quotient of one of the $\mathrm{A}/\mathfrak{p}_i$). For every prime ideal $\mathfrak{p} \neq 0$, the hypothesis of induction shows already that the completion of $\mathrm{A}/\mathfrak{p}$ is reduced, and it suffices therefore to prove that $\widehat{\mathrm{A}}$ is reduced. Furthermore, the integral closure A′ of A is by hypothesis an A-module of finite type, hence a semi-local Noetherian ring and $\widehat{\mathrm{A}}$ identifies with a subring of ((**0**, 7.3.3)) and Bourbaki, *Alg. comm.*, chap. IV, § 2, no 5, cor. 3 de la prop. 9); it will therefore suffice to prove that $\widehat{\mathrm{A}}'$ is reduced; as one has seen above, the hypothesis *b*) is also satisfied by A′, which is moreover of dimension n (**0**, 16.1.5); one can therefore restrict to the case where A is *integrally closed*. Let $t \neq 0$ be an element of the radical of A, and let $\mathfrak{q}_j$ $(1 \leqslant j \leqslant n)$ be the prime minimal ideals among those containing $t\mathrm{A}$; one has the following properties:

label=(i) t is regular.
lbbel=(ii) $\mathrm{A}/t\mathrm{A}$ has no immersed associated prime ideals.
lcbel=(iii) The $\mathrm{A}_{\mathfrak{q}_j}$ are discrete valuation rings.
ldbel=(iv) The completions of the $\mathrm{A}/\mathfrak{q}_j$ are reduced.

Indeed, (i) is trivial since A is integral and $t \neq 0$. As A is integrally closed, $\mathrm{A}/t\mathrm{A}$ satisfies (S_1), that is to say (5.7.7) is without immersed associated prime ideals (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 4, prop. 8). Since A is integrally

closed, the $\mathrm{A}_{\mathfrak{q}_j}$ are, and one knows (*loc. cit.*) that these rings are of dimension 1, hence they are discrete valuation rings, whence (iii). Finally, (iv) follows from the hypothesis of induction. The proof will be completed if we prove the **IV-2 | 211**

Lemma (7.6.4.1). — *(Zariski). — Let* A *be a semi-local Noetherian ring, t an element of its radical satisfying the conditions (i) to (iv) above; then* $\widehat{A}$ *is reduced.*

Proof. Set for simplicity $A' = \widehat{A}$, and consider t as an element of A'; one has $A'/tA' = (A/tA) \otimes_A A'$, and as A' is a flat A-module, it follows from (3.3.1) that the prime ideals of A' associated to the A'-module A'/tA' are the prime ideals $\mathfrak{q}'_{jh}$ such that $\mathfrak{q}'_{jh}$ lies above $\mathfrak{q}_j$ and is associated to the k_j-module $(A'/tA') \otimes_A k_j$, where k_j is the field of fractions of $A/\mathfrak{q}_j$. Now, $\mathrm{Spec}((A'/tA') \otimes_A k_j)$ is the formal fibre of A at the point $\mathfrak{q}_j$, or also that of $A/\mathfrak{q}_j$ (generic point of $\mathrm{Spec}(A/\mathfrak{q}_j)$). By virtue of (iv) the completion $A'/\mathfrak{q}_j A'$ of $A/\mathfrak{q}_j$ is reduced, it follows that it is of the same for the formal fibre of $A/\mathfrak{q}_j$ at the generic point of $\mathrm{Spec}(A/\mathfrak{q}_j)$; this formal fibre therefore has no prime cycle, its irreducible components being fields (3.2.1). One sees hence by (3.3.3) and the hypothesis (ii) that $\mathfrak{q}_j A'_{2_{jh}}$ is the maximal ideal of $A'_{\mathfrak{q}'_{jh}}$ for every h; as $A_{\mathfrak{q}_j}$ is a discrete valuation ring, its maximal ideal $\mathfrak{q}_j A_{\mathfrak{q}_j}$ is principal, hence the maximal ideal of the local Noetherian ring $A'_{\mathfrak{q}'_{jh}}$ is principal, which entails that this ring is a discrete valuation ring (Bourbaki, *Alg. comm.*, chap. VI, § 3, no 6, prop. 9). We have just verified the hypotheses (i) to (iv) for the *complete local ring* A'. It therefore suffices to show that if A is complete and satisfies the hypotheses (i) to (iii) ((iv) being automatically satisfied in this case),

then A is reduced. Now, the hypotheses (i) and (ii) imply that, if $\varphi : A \to \prod_{j=1}^{n} A_{\mathfrak{q}_j}$ is the canonical homomorphism, one has $t^n A = \varphi^{-1}(\varphi(t^n A))$ (3.4.9); as the canonical image of tA in the discrete valuation ring $A_{\mathfrak{q}_j}$ is a power $\mathfrak{q}_j^h A_{\mathfrak{q}_j}$ of the maximal ideal, one sees that on A the (tA)-preadic topology is the inverse image by φ of the topology induced by the product on $\prod_{j=1}^{n} A_{\mathfrak{q}_j}$. Now, as t belongs to the radical of A, the (tA)-preadic topology is separated, hence φ is *injective*. But by virtue of the hypothesis (iii), the $A_{\mathfrak{q}_j}$ are integral, hence $\prod_{j=1}^{n} A_{\mathfrak{q}_j}$ is reduced, and *a fortiori* it follows that so is A. □

□

Corollary (7.6.5). — *Let* A *be a semi-local Noetherian ring satisfying the equivalent conditions of* (7.6.4)*; every semi-local* A*-algebra essentially of finite type also satisfies the conditions of* (7.6.4).

Proof. The condition $a'')$ of (7.6.4) means indeed that A is a $\boldsymbol{P}$-ring, where $\boldsymbol{P}(Z,k)$ is the property "Z is geometrically reduced over k"; the corollary is then a consequence of the general theorem (7.4.4). □

Corollary (7.6.6). — *Let* A *be a semi-local Noetherian integral ring of dimension* 1, K *its field of fractions. For* A *to be a Japanese ring, it is necessary and sufficient that the completion* $\widehat{A}$
be reduced and that, if $\mathfrak{q}_j$ $(1 \leqslant j \leqslant n)$ *are the prime minimal ideals of* $\widehat{A}$*, the fields of fractions of the integral rings* $\widehat{A}/\mathfrak{q}_j$ *be separable extensions of* K. **IV-2** | 212

Proof. The integral quotients of A are themselves fields, hence the condition $b)$ of (7.6.4) is equivalent to the hypothesis that A is a Japanese ring, and the hypothesis $a')$ to the conditions on $\widehat{A}$, whence the conclusion. □

Remarks (7.6.7). — (i) The equivalent conditions of (7.6.6) mean again that the formal fibres of A are geometrically regular (7.3.19), (iv)). One has already observed (*loc. cit.*) that this condition is satisfied when A is a *discrete valuation ring whose field of fractions is of characteristic* 0.

(ii) The same reasoning as in the first part of the proof of (7.6.4) shows that for a semi-local Noetherian ring A, the two following conditions are equivalent:

label=*a)* For every integral quotient ring B of A, the completion $\widehat{B}$ is reduced.

a′) The formal fibres of A are reduced.

Taking into account (**0**, 23.1.7), (i), these two conditions imply the following:

b) For every integral quotient ring B of A, the integral closure of B is a finite B-algebra.

When A is a universally catenary ring, one can prove that the condition $b)$ conversely implies $a)$; we will not need to use this result.

7.7. Applications: II. Universally Japanese rings

(7.7.1). Recall (**0**, 23.1.1) that one calls a ring *universally Japanese* a ring A such that every integral A-algebra *of finite type* is a Japanese ring. This amounts to saying that for every integral A-algebra B of finite type, the integral closure of B is a finite B-algebra.

Theorem (7.7.2). — *(Nagata). — Let* A *be a Noetherian ring. The following conditions are equivalent:*

*label=*a) A *is a universally Japanese ring.*
*lbbel=*b) *Every integral quotient ring of* A *is a Japanese ring.*
*lcbel=*c) *For every maximal ideal* $\mathfrak{m}$ *of* A, *every integral quotient ring of* $A_{\mathfrak{m}}$ *is a Japanese ring, and for every integral quotient ring* B *of* A, *with field of fractions* K, *every finite radicial extension* K' *of* K *and every sub-*B*-algebra finite* B' *of* K', *with field of fractions* K', *there exists* $f' \neq 0$ *in* B' *such that* $B'_{f'}$ *is integrally closed (cf.* (6.13.7)*).*

Moreover, if A *satisfies these conditions, so does every ring of fractions* $S^{-1}A$ *of* A *and every* A*-algebra of finite type.*

Proof. Let us first show that *b*) implies *c*); every integral quotient ring of $A_{\mathfrak{m}}$ is a ring of fractions of an integral quotient ring of A, hence a Japanese ring (**0**, 23.1.1). On the other hand, every quotient B of A by one of its prime ideals is a Japanese ring (**0**, 23.1.1). One deduces that the set $\mathrm{Nor}(\mathrm{Spec}(B'))$ is open (6.13.3), hence contains a non-empty open set $D(f') = \mathrm{Spec}(B'_{f'})$.

Let us show secondly that, if A satisfies *c*), so does every A-algebra of finite type B. In the first instance, for every prime ideal $\mathfrak{q}$ of B, $B_{\mathfrak{q}}$ is a local ring in $S^{-1}B$, where $S = A - \mathfrak{p}$, $\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A; as by hypothesis every integral quotient ring of $A_{\mathfrak{p}}$ is a Japanese ring, so is every integral quotient ring of $B_{\mathfrak{q}}$, since $S^{-1}B$ is an $A_{\mathfrak{p}}$-algebra of finite type (7.6.5). On the other hand, if C is an integral quotient ring of B, with field of fractions K, K' a finite radicial extension of K and C' a sub-C-algebra finite of K', with field of fractions K', C' is an A-algebra integral of finite type, and in virtue of (6.13.7), the hypothesis *c*) on A entails that the spectrum of C' contains a non-empty open part whose points are all normal; in other words, there is a $g' \neq 0$ in C' such that $C'_{g'}$ is integrally closed, which proves our assertion. **IV-2 | 213**

This result shows that, to prove the equivalence of *a*), *b*) and *c*), it suffices to show that *c*) implies *b*), and even that for every integral quotient ring B of A, with field of fractions K, the integral closure of B is a B-module of finite type. On the other hand, if A satisfies *b*), so does every integral quotient ring of A, hence is a Japanese ring, and we have seen above that A verifies the conditions *a*), *b*), *c*); it follows that the same holds for every A-algebra of finite type. Moreover, if A satisfies *b*), every integral quotient ring of $S^{-1}A$ is an integral quotient ring of A, hence is a Japanese ring by hypothesis, and as $A/\mathfrak{p}$ is by hypothesis a Japanese ring, so is $S^{-1}(A/\mathfrak{p})$ (**0**, 23.1.1). This completes the proof of the last assertion of the statement. □

Corollary (7.7.3). — *Let* A *be a Noetherian ring satisfying the following conditions:*

label=(i) For every maximal ideal $\mathfrak{m}$ *of* A, *the formal fibres of* $A_{\mathfrak{m}}$ *are geometrically reduced.*
lbbel=(ii) The equivalent conditions of (6.12.4) *are satisfied.*

Then A *is universally Japanese.*

Proof. Indeed, it follows then from (7.6.4) that every integral quotient ring of $A_{\mathfrak{m}}$ is a Japanese ring; whence the conclusion, by virtue of (7.7.2). □

Corollary (7.7.4). — *A Dedekind ring whose field of fractions is of characteristic* 0 *(in particular* **Z***) is a universally Japanese ring. A local Noetherian ring whose formal fibres are geometrically reduced (in particular a local Noetherian complete ring) is universally Japanese.*

Proof. The first assertion follows from (7.7.3), (7.6.7), (i) and (6.12.6). The second follows from (7.6.4) and (7.7.2). □

7.8. Excellent rings

(7.8.1). The previous paragraphs of this section (as well as (6.11), (6.12) and (6.13)) have been devoted to the study of problems frequently encountered in the use of rings and *Noetherian* preschemes, and which can be grouped into the following types:

A) For a *local* Noetherian ring A, do the properties of A "transfer" to its *completion* $\widehat{A}$? For example, **IV-2** | 214
if A is reduced (resp. integral, resp. integral and integrally closed), is the same true of $\widehat{A}$? Most of these questions are related to the properties of the local *formal fibres* of A (recall that these are the fibres of the canonical morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$). An exception is formed by the properties related to the notion of dimension, in particular the notion of being equidimensional; it is then the condition of *chains* and its various refinements that play the essential role.

B) For a locally Noetherian prescheme X (in particular for an affine scheme Spec(A)), does the set of $x \in X$ at which the local ring $\mathscr{O}_x$ possesses a certain property (for example being integrally closed, or being a Cohen-Macaulay ring, or being regular) form an *open* set?

C) For an *integral* ring A, is the integral closure of A a finite extension of its field of fractions? Of course, this question can be reformulated for Noetherian preschemes (**II**, 6.3).

The problems of type B) or C) may also be posed for *local* rings, but it does not suffice in general that they be solved affirmatively for every local ring $A_{\mathfrak{p}}$ of a ring A for them to be solved for A (cf. (6.13.6)).

Let us also emphasise that in the study of these problems, we have been systematically concerned with whether the affirmative answer is *stable* for one or the other of the two most important operations in Commutative Algebra: localisation and passage to an algebra of finite type.

The results obtained in the study of these problems lead to identifying a class of Noetherian rings whose behaviour in this regard is the best possible:

Definition (7.8.2). — A ring A is called *excellent* if it is Noetherian and satisfies the following conditions:

label=(i) A is universally catenary (or, which amounts to the same (5.6.3), (i)), for every prime ideal $\mathfrak{p}$ of A, $A_{\mathfrak{p}}$ is universally catenary).

lbbel=(ii) For every prime ideal $\mathfrak{p}$ of A, the formal fibres of $A_{\mathfrak{p}}$ are geometrically regular.

lcbel=(iii) For every integral quotient ring B of A and every finite radicial extension K′ of the field of fractions K of B, there exists a sub-B-algebra finite B′ of K′, containing B, having K′ as field of fractions, and such that the set of regular points of Spec(B′) contains a non-empty open set.

With this terminology, the essential content of the results of § 7 and of part of § 6 is summarised as follows:

(7.8.3). (i) In the conditions (i) and (ii) of the definition (7.8.2) one may restrict to the maximal ideals $\mathfrak{p}$ in A. For a local Noetherian ring to be excellent, it is necessary and
sufficient that it be universally catenary and that its formal fibres be geometrically regular. **IV-2** | 215

(ii) If A is an excellent ring, so is every ring of fractions $S^{-1}A$ and every A-algebra of finite type.

(iii) A local complete ring (in particular a field or a ring of formal power series over a field) is excellent. A Dedekind ring whose field of fractions is of characteristic 0 (in particular **Z**) is excellent.

(iv) Let A be an excellent ring, $X = \operatorname{Spec}(A)$; the set $\operatorname{Reg}(X)$ (resp. $\operatorname{Nor}(X)$, $U_{R_n}(X)$) of points where X is regular (resp. normal, resp. satisfies (R_n)) is open in X. For every coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the set $U_{C_n}(\mathscr{F})$ (resp. $U_{S_n}(\mathscr{F})$, $\operatorname{CM}(\mathscr{F})$) of $x \in X$ where $\operatorname{coprof}(\mathscr{F}_x) \leqslant n$ (resp. the points where $\mathscr{F}$ satisfies (S_n), resp. the points where $\mathscr{F}_x$ is a Cohen-Macaulay $\mathscr{O}_x$-Module) is open in X.

(v) Let A be an excellent ring, $\mathfrak{J}$ an ideal of A, $\widehat{A}$ the $\mathfrak{J}$-separated completion of A for the $\mathfrak{J}$-preadic topology. Then the canonical morphism $f : \operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$ is regular (in other words, flat and with geometrically regular fibres). If we set $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(\widehat{A})$, and use the notations of (iv) and further setting $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$, we have

(7.8.3.1).

$$\operatorname{Reg}(X') = f^{-1}(\operatorname{Reg}(X)), \quad \operatorname{Nor}(X') = f^{-1}(\operatorname{Nor}(X)), \quad U_{R_n}(X') = f^{-1}(U_{R_n}(X))$$
$$U_{C_n}(\mathscr{F}') = f^{-1}(U_{C_n}(\mathscr{F})), \quad U_{S_n}(\mathscr{F}') = f^{-1}(U_{S_n}(\mathscr{F})), \quad \operatorname{CM}(\mathscr{F}') = f^{-1}(\operatorname{CM}(\mathscr{F}))$$

In particular, if $\mathfrak{J}$ is contained in the radical of A (for example if A is local and $\mathfrak{J}$ its maximal ideal), for A to be regular (resp. normal, resp. reduced, resp. to satisfy (R_n), resp. its codepth to be $\leqslant n$, resp. to satisfy (S_n), resp. to be a Cohen-Macaulay ring), it is necessary and sufficient that the same hold for $\widehat{A}$; in particular, for A to have no immersed associated primes, it is necessary and sufficient that the same hold for $\widehat{A}$.

(vi) An excellent ring is universally Japanese; in particular, if A is integral, its integral closure in every finite extension of its field of fractions is a finite A-algebra.

(vii) Let A be a reduced excellent local ring. Then the completion $\widehat{A}$ is reduced, the integral closure A' of A in its total ring of fractions is a finite A-algebra (hence a semi-local Noetherian ring), and the integral closure of $\widehat{A}$ in its total ring of fractions is isomorphic to the completion $\widehat{A}'$ of A'. Furthermore, there is a bijective canonical correspondence between the set of maximal points of $\mathrm{Spec}(\widehat{A})$ (in other words, the set of prime minimal ideals of $\widehat{A}$) and the set of closed points of $\mathrm{Spec}(A')$ (in other words, the set of maximal ideals of A'). For the local ring A to be unibranch, it is necessary and sufficient that $\widehat{A}$ be integral; for A to be geometrically unibranch, it is necessary and sufficient that $\widehat{A}$ be.

(viii) If A is an excellent integral ring, the integral closure of A is the intersection of the integral closures of the $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1.

(ix) Let A be an excellent ring, $X = \mathrm{Spec}(A)$, Z a closed part of X, $U = X - Z$, $i : U \to X$ the canonical injection, $\mathscr{F}$ an $\mathscr{O}_U$-Module coherent. For the $\mathscr{O}_X$-Module $i_*(\mathscr{F})$ to be coherent, it is necessary and sufficient that, for every $x \in \mathrm{Ass}(\mathscr{F})$, one has $\mathrm{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geqslant 2$. In particular, if A is integral and $\mathscr{F}$ torsion-free, for $i_*(\mathscr{F})$ to be coherent, it is necessary and sufficient that $\mathrm{codim}(Z, X) \geqslant 2$.

(x) Let A be an excellent local ring. For every integral quotient ring B of A, $\widehat{B}$ is equidimensional. **IV-2** | 216
For A to be equidimensional, it is necessary and sufficient that $\widehat{A}$ be.

Proof. Most of these results have already been proved.

(i) The first assertion follows from (5.6.3), (i) and (7.4.4), (i); the second from (7.4.4), (7.3.18), (6.12.7) and (6.12.4).

Assertion (ii) follows from (5.6.1), (5.6.3), (i), (7.4.4) and (6.12.4).

(iii) The first assertion has been seen in (5.6.4) and (7.4.4), taking into account (i). The second has been seen in (5.6.4) and (7.3.19), (iv), taking into account (i).

Assertion (iv) follows from (6.12.4), (6.13.5), (6.12.9) and (6.11.8) (taking into account (ii)).

The first assertion of (v) is a consequence of (7.4.6); the relations follow, taking into account respectively (6.5.1), (6.5.4), (6.5.3), (6.3.5) and (6.4.2). The last assertion of (v) is a particular case corresponding to the property (S_n) for $n = 1$ (5.7.5).

Assertion (vi) follows from (7.7.3), and assertion (vii) from (7.6.1), (7.6.2) and (7.6.3).

For assertion (viii), let us note that the ring $A^{(1)}$, intersection of the $A_{\mathfrak{p}}$ for all prime ideals $\mathfrak{p}$ of height 1 of A, is a finite A-algebra, as it follows from (vi), from (7.8.2), (i) and (5.11.2). As the integral closure A' of A is a finite A-algebra, the prime ideals of height 1 of A' are exactly those which lie above prime ideals of height 1 of A (5.10.17), (iv); the conclusion follows hence from (**0**, 23.2.9).

The first assertion of (ix) is a consequence of (5.6.3), (i) and (5.11.4); the second is a consequence of (vi) and (5.11.4), if one observes that when $\mathscr{F}$ is torsion-free, $\mathrm{Ass}(\mathscr{F})$ is reduced to the generic point of $\mathrm{Spec}(A)$. Finally, to prove (x), it suffices to observe that B is universally catenary and universally Japanese by (ii); as B is an excellent local ring by (ii), the ring $B^{(1)}$ is a finite B-algebra (5.11.2), hence $A_{\mathfrak{m}}$ is formally catenary, and the same holds for every $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A, hence also A. This shows that A is strictly formally catenary by (7.2.5), *c*)) and completes the proof of (x). ☐

Remarks (7.8.4). — (i) If one considers only the Noetherian rings A that satisfy the conditions (ii) and (iii) of the definition (7.8.2), one verifies by inspection of the preceding proofs that the assertions (ii), (iii), (iv), (v), (vi) and (vii) of (7.8.3) are still valid, replacing "excellent" by "satisfying (7.8.2), (ii) and (iii)".

(ii) The catenary local ring A constructed in (5.6.11) satisfies the conditions (ii) and (iii) (but not condition (i) of the definition (7.8.2)). Indeed, the formal fibre at the maximal ideal $\mathfrak{n}C_{\mathfrak{n}}$ of A is geometrically regular, and that at the other prime ideals $\mathfrak{p}$ of A coincides with that of the ring E at these prime ideals;

hence the condition (6.12.4), a)) is also satisfied for A by virtue of (7.8.3), (v); this proves hence that A satisfies the condition (7.8.2), (ii). On the other hand, the complement of the closed point in $\mathrm{Spec}(A)$ is isomorphic to an open affine, whose ring of affine functions is the ring of a **IV-2 | 217**

polynomial ring; by taking the body k_0 of characteristic 0 (7.8.3), (iii); consequently E is an excellent ring by (7.8.3), (ii) and this proves that A satisfies the condition (7.8.2), (ii).

(iii) One can replace the condition (i) of the definition (7.8.2) by the following (weaker in appearance, taking into account (5.6.10)):

(i′) *For every prime minimal ideal* $\mathfrak{p}_i$ *of* A $(1 \leqslant i \leqslant r)$, *let* A'_i *be the integral closure of the integral ring* $A_i = A/\mathfrak{p}_i$; *then, for every maximal ideal* $\mathfrak{m}'$ *of* A'_i, *denoting by* $\mathfrak{m}$ *the inverse image of* $\mathfrak{m}'$ *in* A,

(7.8.4.1).

$$\dim(A_{\mathfrak{m}}) = \dim(A'_{i,\mathfrak{m}'}).$$

Indeed, one has seen in a Remark (i) that under only the hypotheses (ii) and (iii) of (7.8.2), the analogues of the properties (ii), (v), (vi) and (vii) of (7.8.3) are still valid. One deduces hence from (vi) that the A'_i satisfy (ii) and (iii), and that $((A'_i)_{\mathfrak{m}'})^\wedge$ are integral and integrally closed. Let now $\mathfrak{m}$ be an arbitrary maximal ideal of A; for every i such that $\mathfrak{p}_i \subset \mathfrak{m}$, the integral closure of $A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}}$ is an $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})$-algebra finite by (vi), hence semi-local, and its local components are of the form $(A'_i)_{\mathfrak{m}'}$, where $\mathfrak{m}'$ is a prime ideal (not necessarily maximal) of A'_i above $\mathfrak{m}/\mathfrak{p}_i$; it follows then from (vii), from what precedes and from (7.8.4.1) that the quotients of the completion $(A_{\mathfrak{m}}/\mathfrak{p}_i A_{\mathfrak{m}})^\wedge$ by its prime minimal ideals all have the same dimension equal to $\dim(A_{\mathfrak{m}})$. Consequently ((7.1.9) and (7.1.8), b)), $A_{\mathfrak{m}}$ is formally catenary, and *a fortiori* (7.1.11) universally catenary; it follows that so is $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A (5.6.3), (i), hence also A.

In particular, if A is normal, or more generally if $A_{\mathfrak{m}}$ is unibranch for every maximal ideal $\mathfrak{m}$ of A, the condition (i′) is always satisfied, since $\mathfrak{m}$ can contain only a single prime ideal $\mathfrak{p}_i$ (**0**, 16.1.5); one can then omit (i) from the definition (7.8.2). More particularly, if A is a Noetherian ring, *local unibranch*, it is excellent if and only if its formal fibres are geometrically regular.

(iv) Recall that a quotient ring of a local Cohen-Macaulay ring is also regular (and *a fortiori* a quotient ring of a regular local ring) also satisfies the properties (7.8.3), (ix) and (x) by virtue of (7.2.7); but the interest of quotient rings of regular rings lies above all in their cohomological properties (chap. III, 3^e Partie).

(v) We will see further on (§ 18) that if k is a field with a complete non-discrete valuation, the ring $k\{\{T_1, \dots, T_n\}\}$ of *convergent power series* (in a neighbourhood of 0 in k^n) is an *excellent ring*.

(7.8.5). One says that a locally Noetherian prescheme X is *excellent* if for some covering (U_α) of X by affine opens, each of the rings A_α of the U_α is excellent; this property is then true for *every* covering (U_α) of X by affine opens.

Proposition (7.8.6). — *Let* X *be an excellent prescheme.*

label=(i) If $f : X' \to X$ *is a morphism locally of finite type,* X' *is excellent.* **IV-2 | 218**

lbbel=(ii) If X *is reduced, its normalised* X' *(**II**, 6.3.8) is finite over* X.

lcbel=(iii) The sets $\mathrm{Reg}(X)$, $\mathrm{Nor}(X)$, $U_{R_n}(X)$, *as well as* $U_{C_n}(\mathscr{F})$, $U_{S_n}(\mathscr{F})$ *and* $\mathrm{CM}(\mathscr{F})$ *for every coherent* $\mathscr{O}_X$*-Module* $\mathscr{F}$, *are open in* X.

Proof. This follows immediately from (7.8.3), (ii), (vi) and (iv). Let us also note that (7.8.3), (ix) is also valid without modification of the statement when X is an excellent prescheme. □

7.9. Excellent rings and resolution of singularities

(7.9.1). Given a *locally Noetherian reduced* prescheme X, one calls a *resolving morphism* for X a morphism $f : X' \to X$ such that X′ is *regular* and that f is *proper* and *birational*. When such a morphism exists, one says that one can *resolve the singularities* of X (or more simply that one can *resolve* X). For example, if A is a *Japanese* ring of dimension 1, one can resolve $X = \mathrm{Spec}(A)$ since the morphism $X' \to X$ (where X′ is the *normalised* of X) is finite (hence proper) and birational, and the local rings of X′ are integrally closed rings of dimension 1, hence discrete valuation rings, hence regular.

(7.9.2). It is clear that if one can resolve X, one can also resolve every *open* of X and every local prescheme $\mathrm{Spec}(\mathscr{O}_x)$ of X (**II**, 5.4.2). On the other hand, if one can resolve X, it is clear that there exists in X an open subset *everywhere* dense contained in $\mathrm{Reg}(X)$. These remarks show immediately that if, for every closed *integral* sub-prescheme Y of X and every Y-prescheme Y′ *integral, finite* and *radicial* over Y, one can resolve Y′, then the affine opens of X *satisfy condition* (iii) of (7.8.2).

On the other hand, for local schemes, one has the following proposition:

Proposition (7.9.3). — *Let* A *be a local Noetherian reduced ring, and suppose that one can resolve* $\mathrm{Spec}(A)$. *Then the fibres of the canonical morphism* $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ *at the maximal points of* $\mathrm{Spec}(A)$ *are regular.*

Proof. Set $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(\widehat{A})$, and let $g : X' \to X$ be the canonical morphism. Let $f : Y \to X$ be a resolving morphism; set $Y' = X' \times_X Y$, and let $f' : Y' \to X'$ and $g' : Y' \to Y$ be the canonical projections, so that one has a commutative diagram

$$\begin{tikzcd} X' \arrow[d, "g"'] & Y' \arrow[l, "f'"'] \arrow[d, "g'"] \\ X & Y \arrow[l, "f"] \end{tikzcd}$$

It will suffice to prove that the prescheme Y′ is *regular*: indeed, as f is birational and of finite type, there is an open U dense in X such that the restriction $f^{-1}(U) \to U$ of f is an isomorphism and that $f^{-1}(U)$ is everywhere dense in Y (**I**, 6.5.5); the preschemes induced respectively on the open $g^{-1}(f^{-1}(U))$ of X′ and the open $g'^{-1}(f^{-1}(U))$ of Y′ are therefore isomorphic; this proves that $g^{-1}(U)$ is regular, and *a fortiori* so are the fibres of g at the maximal points of X, which are contained in U.

Let us note that the morphism f' is proper (**II**, 5.4.2), and in particular of finite type, hence Y′ **IV-2** | 219
is Noetherian, since it is thus of X′. Let a' be the closed point of X′; to see that Y′ is regular, it will suffice to prove that *for every* $y' \in f'^{-1}(a')$, *the local ring* $\mathscr{O}_{Y',y'}$ *is regular*. Indeed, one will then have $f'^{-1}(a') \subset \mathrm{Reg}(Y')$. But as X′ is the spectrum of a complete local ring and f' is of finite type, $\mathrm{Reg}(Y')$ is *open* in Y′ (6.12.8), and as the morphism f' is *closed* and $f'^{-1}(a') \subset \mathrm{Reg}(Y')$, there exists a neighbourhood V′ of a' in X′ such that $f'^{-1}(V') \subset \mathrm{Reg}(Y')$. As $\widehat{A}$ is an integral local ring, on necessarily has $V' = X'$, hence $\mathrm{Reg}(Y') = Y'$ and the proposition will be proved.

Now, if a is the closed point of X, one has $g^{-1}(a) = \{a'\}$ and the residue fields $\boldsymbol{k}(a)$ and $\boldsymbol{k}(a')$ are isomorphic; hence $g'^{-1}(f^{-1}(a)) = f'^{-1}(a')$ and the restriction $f'^{-1}(a') \to f^{-1}(a)$ of g' is an *isomorphism* of these two fibres. Let $y = g'(y')$ and set $B = \mathscr{O}_{X,y}$; if $\mathfrak{n}$ is the maximal ideal of B, $B' = B \otimes_A \widehat{A}$ has a single maximal ideal $\mathfrak{n}'$ above $\mathfrak{n}$ of B and of the maximal ideal of $\widehat{A}$, and one has $\mathscr{O}_{Y',y'} = B'_{\mathfrak{n}'}$. To show that the local ring $B'_{\mathfrak{n}'}$ is regular, it suffices to establish that its *completion* is regular (**0**, 17.1.5), that is, or by hypothesis B is regular, hence so is its completion $\widehat{B}$. The conclusion will follow from the following lemma:

Lemma (7.9.3.1). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism such that the ring* $B' = B \otimes_A \widehat{A}$ *is Noetherian. Then, for every maximal ideal* $\mathfrak{n}'$ *of* B′ *above the maximal ideal* $\mathfrak{n}$ *of* B *and the maximal ideal of* $\widehat{A}$, *the completions of* B *and of* $B'_{\mathfrak{n}'}$ *are isomorphic.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of A and set $B'' = B'_{\mathfrak{n}'}$. For every integer $h > 0$, one has $B'/\mathfrak{m}^h B' = B \otimes_A (\widehat{A}/\mathfrak{m}^h \widehat{A})$; but $\widehat{A}/\mathfrak{m}^h \widehat{A}$ is isomorphic to $A/\mathfrak{m}^h$, hence $B'/\mathfrak{m}^h B'$ is isomorphic to $B/\mathfrak{m}^h B$, and in particular it is a local ring whose maximal ideal is $\mathfrak{n}'/\mathfrak{m}^h B'$; consequently $B''/\mathfrak{m}^h B''$, isomorphic to $(\mathrm{B}'/\mathfrak{m}^h\mathrm{B}')_{\mathfrak{n}'}$, is B-isomorphic to $\mathrm{B}'/\mathfrak{m}^h\mathrm{B}'$, and finally isomorphic to $\mathrm{B}/\mathfrak{m}^h\mathrm{B}$; in particular, the maximal ideal of B'' is equal to $\mathfrak{n}\mathrm{B}''$, and $\mathrm{B}''/\mathfrak{n}^h\mathrm{B}''$ is isomorphic to $\mathrm{B}/\mathfrak{n}^h$ from what precedes. As $\widehat{\mathrm{B}}'' = \varprojlim_h (\mathrm{B}''/\mathfrak{n}^h\mathrm{B}'')$, the lemma is proved, as was (7.9.3). □

□

Corollary (7.9.4). — *Let* A *be a local Noetherian ring.*

label=(i) *If, for every integral quotient ring* B *of* A*, one can resolve* Spec(B)*, then the formal fibres of* A *are regular.*

lbbel=(ii) *Suppose that, for every integral quotient ring* B *of* A*, and every integral ring* B′ *containing* B*, which is a finite* B*-algebra and whose field of fractions is radicial over that of* B*, one can resolve* Spec(B′)*; then the formal fibres of* A *are geometrically regular.*

Proof. Every formal fibre of A being a formal fibre of a ring at the generic point of its spectrum, it is clear that (i) is an immediate consequence of (7.9.3), and (ii) follows from (i) and (6.7.7). □

Proposition (7.9.5). — *Let* X *be a locally Noetherian prescheme, such that, for every integral closed subprescheme* Y*, finite and radicial over* X*, one can resolve* Y*. Then the rings of the affine opens of* X *satisfy the conditions (ii) and (iii) of* (7.8.2)*; if moreover* X *is universally catenary* (5.6.4)*,*

in particular if X *is* locally immersible in a regular prescheme (5.6.4)*),* X *is an* excellent prescheme. **IV-2** | 220

Proof. This follows immediately from (7.9.4) and (7.9.2). □

Remark (7.9.6). — It is possible that the converse of (7.9.5) be true, that is to say that X is reduced and that the rings of the affine opens of X satisfy the conditions (ii) and (iii) of (7.8.2) implies the possibility of resolving X (and consequently also the possibility of resolving every reduced prescheme locally of finite type over X, by virtue of (7.4.4) and (6.12.4)). This is in any case true when one limits oneself to reduced Noetherian preschemes whose residue fields are of *characteristic* 0, as shown by the recent results of Hironaka [35] (the latter states his results under hypotheses that are too restrictive, the reasoning being in fact valid under the only hypotheses (ii) and (iii) of (7.8.2) for the rings of the affine opens of X, combined with the fact that the residue fields are of characteristic 0). On the contrary, in the general case, it is known so far only how to resolve algebraic schemes of dimension 2 over a perfect field [36]. Given the importance that the resolution of singularities is acquiring in the topological study of schemes (particularly concerning their homological and homotopical properties), this gives a particular interest to the category of excellent rings and excellent preschemes.

One may also note in this regard that the less delicate part of the reasoning of Hironaka (*loc. cit.*) shows independently of all hypothesis on the characteristics of the residue fields, and using only the properties (ii) and (iii) of (7.8.2) for the local rings of X, the problem of resolution of singularities of a prescheme X reduces to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, A being a local Noetherian integral ring and *complete*. Consequently, if subsequent results were to disprove the more advanced conjecture above, and were to lead to formulating more restrictive conditions on the local rings of X, these could only be through the intermediary of restrictive conditions imposed on the local *complete* rings (probably conditions on their residue fields, perhaps the condition $[k : k^p] < +\infty$ (p being the characteristic exponent of k)).

(7.9.7). Consider on the one hand a full subcategory $\mathscr{C}$ of the category of locally Noetherian preschemes, and on the other hand a property $\boldsymbol{R}(\mathrm{A})$, subject to the following conditions:

1° For every $\mathrm{X} \in \mathscr{C}$, every prescheme locally of finite type over X belongs to $\mathscr{C}$. For every local Noetherian ring A such that $\mathrm{Spec}(\mathrm{A}) \in \mathscr{C}$ and every multiplicative subset S of A, one has $\mathrm{Spec}(\mathrm{S}^{-1}\mathrm{A}) \in \mathscr{C}$.

2° For every $\mathrm{X} \in \mathscr{C}$, the set $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X})$ of $x \in \mathrm{X}$ such that $\boldsymbol{R}(\mathscr{O}_{\mathrm{X},x})$ is true is open in X.

3° For every local Noetherian ring A such that $\mathrm{Spec}(\mathrm{A}) \in \mathscr{C}$ and every regular element t of the maximal ideal of A, $\boldsymbol{R}(\mathrm{A}/t\mathrm{A})$ implies $\boldsymbol{R}(\mathrm{A})$.

Let us denote as in (7.5.0) by $\boldsymbol{P}(\mathrm{Z}, k)$, for a field k and a k-prescheme $\mathrm{Z} \in \mathscr{C}$, the following property:

"For every finite extension k' of k, all the local rings of $\mathrm{Z} \otimes_k k'$ satisfy the property $\boldsymbol{R}$."

We will further suppose that $\boldsymbol{R}$ satisfies the following condition: **IV-2** | 221

4° If $\boldsymbol{P}(\mathrm{Z}, k)$ is true, then, for every extension *of finite type* k'' of k, all the local rings of $\mathrm{Z} \otimes_k k''$ satisfy the property $\boldsymbol{R}$.

Let us note that these conditions are satisfied when one takes for $\mathscr{C}$ the category of *excellent* preschemes and for $\boldsymbol{R}$ one of the properties (i) to (viii) of (7.5.3); for condition 1°, this follows from (7.8.3), (ii)); for condition 2°, from the reasoning of (7.5.3), and for conditions 3° and 4°, from the reasoning of (7.5.3), taking into account that every excellent ring is catenary. Finally, for condition 4°, it follows, for (i), (ii) and (iii) of (6.7.1); for (iv), (v), (vi) of (6.7.7), and for (viii) of (4.6.1); as concerns property (vii) of (7.5.3), the corresponding property $\boldsymbol{P}(\mathrm{Z}, k)$ implies that Z is locally integral (being locally Noetherian) and that each of the integral sub-preschemes of Z whose sum is geometrically integral, one also concludes the property 4° in this case.

With these notations and hypotheses:

Proposition (7.9.8). — *Let* A *be a local Noetherian ring,* k *its residue field,* B *an integral quotient ring of* A *such that* $\mathrm{Spec}(\mathrm{B}) \in \mathscr{C}$, $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism making* B *a flat* A*-module; set* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ *and let* $f : \mathrm{X} \to \mathrm{Y}$ *be the morphism corresponding to* φ*. Suppose that:*

1° the property $\boldsymbol{P}(\mathrm{B} \otimes_{\mathrm{A}} k, k)$ *is true;*

2° for every finite morphism $\mathrm{Y}_1 \to \mathrm{Y}$*, one can resolve* $(\mathrm{Y}_1)_{\mathrm{red}}$*.*

Then, for every $y \in \mathrm{Y}$*, the property* $\boldsymbol{P}(f^{-1}(y), \boldsymbol{k}(y))$ *is true.*

Proof. Let us note that condition 2° of (7.9.8) is still satisfied when one replaces Y by the spectrum of a local ring in a maximal ideal of a finite A-algebra (7.9.2); on the other hand, by virtue of condition 1° of (7.9.7), the spectrum of a ring of fractions of $\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$ also belongs to $\mathscr{C}$. The lemma (7.3.16.2) shows then (as in part I) of the reasoning of (7.5.2)) that it suffices to prove that, if Y is integral and if y is the generic point of Y, the local rings of the fibre $f^{-1}(y)$ satisfy $\boldsymbol{R}$.

This being so, there exists by hypothesis a regular prescheme Y' and a morphism proper and birational $g : \mathrm{Y}' \to \mathrm{Y}$; as Y' is Noetherian and locally integral, the hypothesis that g is birational implies that Y' is integral. As the morphism $g' : \mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}' \to \mathrm{X}$ is also birational, and let $f' : \mathrm{Y}' \to \mathrm{X}'$ and $g' : \mathrm{X}' \to \mathrm{X}'$ be the canonical projections, so that one has a commutative diagram

(7.9.8.1).

$$\begin{tikzcd} \mathrm{X} \arrow[d, "f"'] & \mathrm{X}' \arrow[l, "g'"'] \arrow[d, "f'"] \\ \mathrm{Y} & \mathrm{Y}' \arrow[l, "g"] \end{tikzcd}$$

where f' and g' are the canonical projections. Taking into account condition 2° of (7.9.7), the same reasoning as at the beginning of (7.9.3) shows that it suffices to prove

that $\mathrm{U}_{\boldsymbol{R}}(\mathrm{X}') = \mathrm{X}'$, then, designating by a the closed point of X, that $g'^{-1}(a) \subset \mathrm{U}_{\boldsymbol{R}}(\mathrm{X}')$. Now, let $x' \in g'^{-1}(a)$, and set $z' = f'(x')$, so that $b = g(z')$ is the closed point of Y. One has therefore $f'^{-1}(z') = f^{-1}(b) \otimes_{\boldsymbol{k}(b)} \boldsymbol{k}(z')$, and as $\boldsymbol{k}(z')$ is an extension *of finite type* of k (since g is proper, hence of finite type), the hypothesis implies, by virtue of condition 4° of (7.9.7), that it follows that $\boldsymbol{P}(f'^{-1}(z'), \boldsymbol{k}(z'))$ is true, and as $\mathscr{O}_{x'}$ is a regular ring and that $\mathrm{Spec}(\mathscr{O}_{x'}) \in \mathscr{C}$ by virtue of condition 1° of (7.9.7), the lemma (7.5.1.1) proves that $\boldsymbol{R}(\mathscr{O}_{\mathrm{X}',x'})$ is true, which completes the proof. ☐ **IV-2 | 222**

Corollary (7.9.9). — *Let* A, B *be two local Noetherian rings,* $\varphi : \mathrm{A} \to \mathrm{B}$ *a local homomorphism making* B *a flat* A*-module. Suppose that* A *is integral and geometrically unibranch, and that for every* A*-algebra finite integral* A' *one can resolve* $\mathrm{Spec}(\mathrm{A}')$ *(this will be the case if the formal fibres of* A *are geometrically regular and the residue field of* A *is of characteristic* 0*, by virtue of the results of Hironaka [35]). Let* k *be the residue field of* A *and suppose that* $\mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} k)$ *is geometrically punctually integral* (4.6.9)*. Then* B *is integral.*

Proof. We will apply (7.9.8) taking for $\mathscr{C}$ the category of all locally Noetherian preschemes, for $\boldsymbol{R}$ the property of being integral; the reasoning of (7.5.3) and (7.9.7) show that conditions 1°, 2°, 3° and 4° of (7.9.7) are satisfied in this case. The hypothesis on $\mathrm{B} \otimes_{\mathrm{A}} k$ and the proposition (7.9.8) show hence (with the notations of (7.9.8)) that the fibres $f^{-1}(y)$ are geometrically punctually integral for $y \in \mathrm{Y}$; *a fortiori* the $f^{-1}(y)$ for $y \in \mathrm{Y}$ are reduced preschemes, and as A is integral (and *a fortiori* reduced) (3.3.5). It remains to prove that $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ is *irreducible*. Now, the proof of (7.9.8) proves that X' is locally integral (being locally Noetherian). As the morphism $g' : \mathrm{X}' \to \mathrm{X}$ is surjective, it suffices hence to see that X' is irreducible, or (since X' is locally integral), that X' is *connected*. But it suffices for this to prove that the fibre $g'^{-1}(a)$ is *connected*. Indeed, if this point is established, X' cannot be the sum of two open non-empty preschemes X'_1, X'_2, for one would then have for example $g'^{-1}(a) \cap X'_2 = \varnothing$; but the restriction $X'_2 \to X$ of g' being proper (since X'_2 is a closed sub-prescheme of X'), $g'(X'_2)$ would be a closed non-empty part of X not containing the closed point a, which is absurd. Now, one has $g'^{-1}(a) = g^{-1}(b) \otimes_{\boldsymbol{k}(b)} \boldsymbol{k}(a)$; but $g : Y' \to Y$ is proper and birational, hence, in the Stein factorisation $Y' \xrightarrow{v} Y'' \xrightarrow{u} Y$ of the morphism g (**III**, 4.3.3), the morphism finite u is also birational and consequently Y'' is integral. As A is supposed geometrically unibranch, it follows then from (**III**, 4.3.4) that $g^{-1}(b)$ is *geometrically connected*, which completes the proof. □

Remarks (7.9.10). — (i) We will show further on (18.8.11) that for a local Noetherian reduced ring B to be geometrically unibranch, it is necessary and sufficient that for every etale morphism $h : X' \to X = \operatorname{Spec}(B)$ and every point $x' \in X'$ above the closed point of X, $\mathscr{O}_{X',x'}$ is integral. One deduces that if, in (7.9.9) one supposes, not only that $\operatorname{Spec}(B \otimes_A k)$ is geometrically punctually integral, but also geometrically unibranch, then one can conclude that B is *geometrically unibranch*.

(ii) Let X, Y be two excellent preschemes, $f : X \to Y$ a flat morphism, and suppose that for every **IV-2** | 223
finite morphism $Y_1 \to Y$, one can resolve $(Y_1)_{\text{red}}$. It follows from (7.9.8), taking for $\boldsymbol{R}$ the property of being *regular*, that the set U of $x \in X$ such that the fibre at the point $f(x)$ of the morphism

$$\operatorname{Spec}(\mathscr{O}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))) \longrightarrow \operatorname{Spec}(\boldsymbol{k}(f(x)))$$

is *geometrically regular*, is *stable by generisation*. We do not know if this set is *open* (or, which amounts to the same ($\mathbf{0}$, 9.2.5), if it is *constructible*) even in the particular case where Y is the spectrum of $\mathbf{Z}$ or of a polynomial ring in one indeterminate $k[T]$ over a field k, and where consequently the condition of "resolvability" imposed on Y is trivially satisfied.

§8. Projective limits of preschemes

8.1. Introduction **IV-3** | 5

(8.1.1). In this paragraph, we will systematically study the following situation. Let I be a filtered preordered increasing set, $(A_\lambda, \varphi_{\mu\lambda})$ an inductive system of rings having I for index set, $A = \varinjlim A_\lambda$ its inductive limit. For every $\alpha \in I$ and every A_α-prescheme X_α, consider the A_λ-preschemes $X_\lambda = X_\alpha \otimes_{A_\alpha} A_\lambda$ for $\lambda \geqslant \alpha$, and the A-prescheme $X = X_\alpha \otimes_{A_\alpha} A$; it is clear that the preschemes X_λ (for $\lambda \geqslant \alpha$) form a *projective system*, and we will see (8.2.5) that X is a *projective limit* of this system in the category of preschemes. We propose to find conditions on X_α or on the A_λ allowing us to prove properties of the following type: for X to have a property **P** (for example, the property of being proper over $S = \operatorname{Spec}(A)$, or irreducible, or connected, etc.), it is necessary and sufficient that there exist an index $\mu \geqslant \alpha$ such that, for every $\lambda \geqslant \mu$, X_λ has (with respect to $S_\lambda = \operatorname{Spec}(A_\lambda)$, as the case may be) the same property **P**. We will obtain analogous statements for properties of $\mathscr{O}_X$-Modules, of A-morphisms of A-preschemes, etc. We will also show (8.9.1) that the datum of an A-prescheme of finite presentation (1.6.1) is essentially equivalent to the datum of an A_λ-prescheme of finite presentation X_λ for some sufficiently large λ, X being then *isomorphic to* $X_\lambda \otimes_{A_\lambda} A$. There are also analogous statements for $\mathscr{O}_X$-Modules of finite presentation, their homomorphisms, the A-morphisms of A-preschemes of finite presentation, etc.

(8.1.2). The utility of such results will appear, for example, in the following questions:

label=*a)* Let Y be a prescheme, y a point of Y, (U_λ) the projective system of filtered decreasing open affine neighbourhoods of y in Y; if A_λ is the ring of U_λ, the A_λ form a filtered increasing inductive system, whose inductive limit A is the local ring $\mathscr{O}_y$; moreover, if we denote by $\mathfrak{p}_\lambda$ the prime ideal $\mathfrak{i}_y$ in the ring A_λ, the inductive system (A_λ) is cofinal in every inductive system $(A_\alpha)_f$, where f ranges over $A_\alpha - \mathfrak{p}_\alpha$ (for a fixed α), since the $D(f)$ form a **IV-3** | 6
fundamental system of neighbourhoods of y in U_α, hence in Y. The results of the present paragraph will therefore imply that the algebraic geometry of $\mathscr{O}_y$-preschemes of finite presentation (and the theory of Modules of finite presentation on these preschemes) is essentially equivalent to the algebraic geometry of preschemes of finite presentation on "sufficiently small" open neighbourhoods of y. Thus, the statement (8.10.5), (xii) implies

that if a morphism of finite presentation $f : X \to Y$ is such that $X \times_Y \operatorname{Spec}(\mathscr{O}_y)$ is proper over $\operatorname{Spec}(\mathscr{O}_y)$, it is necessary and sufficient that there exist an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U.

An important particular case, classical in a certain sense, is the one where Y is integral and where y is its generic point, so that $\mathscr{O}_y$ is nothing other than *the field* $R(Y) = K$ *of rational functions on* Y. The results of the present paragraph then amount to interpreting the algebraic geometry over K in terms of algebraic geometry over "sufficiently small" non-empty open subsets of Y, that is to say, intuitively, in terms of "families" of geometric objects indexed by the points of such an open set. This point of view has moreover been commonly used for a long time, not only in Algebraic Geometry over algebraically closed fields, but also in the arithmetic study of varieties defined over a *number field* K (a finite extension of **Q**), considering the latter as the field of fractions of its ring of integers A (the "theory of *reduction modulo* $\mathfrak{p}$" ($\mathfrak{p}$ a prime ideal of A); cf. (**I**, 3.7)). The results of §§ 8 and 9 thus provide, among other things, some of the foundations of the language of "reduction modulo $\mathfrak{p}$" in arithmetic.

We note that in the example considered here, the morphisms $S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) are the canonical *open immersions* $U_\mu \hookrightarrow U_\lambda$, and *a fortiori* are *flat* morphisms (but not faithfully flat in general), which explains the interest of the statements that appeal to such a restriction.

lbbel=*b)* Suppose that the A_λ are *fields*, so that $A = \varinjlim A_\lambda$ is also a field. This case arises generally when one starts from geometric data over an arbitrary field K, which one considers as an extension of a field k (for example the prime subfield of K). There is then interest in considering K as an inductive limit of its sub-extensions that are *of finite type over* k, which allows us to reduce many questions to the case where K is a finite type extension of k. Using also the method sketched in *a)*, one can then reduce generally to the case of a base ring A which is an *integral algebra of finite type over* k.

We note that in this example, the morphisms $S_\mu \to S_\lambda$ are *faithfully flat*.

lcbel=*c)* Suppose that we are interested in properties, local on Y, of preschemes of finite presentation over an arbitrary prescheme Y, which we can from now on suppose to be an affine scheme of ring A. There is then interest in considering A as an inductive limit of its subrings that are **Z**-*algebras of finite type*, which allows us to reduce many questions to the case where Y is the spectrum of such an algebra. This is the explication of the "Kroneckerian point of view" according to which Algebraic Geometry reduces to the Algebraic Geometry of preschemes **IV-3 | 7**
of finite type over **Z** (which one sometimes qualifies as "Absolute Algebraic Geometry"). This example shows us in particular that in most questions "relative" to a prescheme of base Y, one can reduce to the case where Y is *Noetherian*.

We note that in this example, in contrast to the previous ones, the morphisms $S_\mu \to S_\lambda$ have in general no particular regularity property.

In what follows, when we apply the results that follow to any one of the three particular situations that have just been described, we will not go into detail about the procedure that allows us to make these applications, being content to refer to what precedes.

(8.1.3). In example *a)* of (8.1.2), we saw that if Y is an integral prescheme with generic point y, $f : X \to Y$ a morphism of finite presentation, and if the generic fibre $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$, then there exists an open neighbourhood U of y in Y such that $f^{-1}(U)$ is proper over U; it follows *a fortiori* that for every $s \in U$, $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$. It turns out that one needs a converse, ensuring that if $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$, then $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$. For example, suppose that X and Y are algebraic preschemes over an algebraically closed field k (one can take for k the field **C** of complex numbers, to fix ideas); one sometimes needs to know that, for every $s \in Y$, *rational over* k, the fibre $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$, then $f^{-1}(y)$ is proper over $\boldsymbol{k}(y)$, and consequently $f^{-1}(U)$ is proper over U for a neighbourhood U of y[3]. Now this statement will follow easily from the following: the set E

[3] We note that such a statement is ultimately purely geometric in nature, in the sense that it only appeals to rational points over k, and not to generic points; for example, when $k = \mathbf{C}$, this statement has an obvious topological significance for the analyst, interpreting "proper" in the topological sense of the term, for the underlying analytic spaces formed by the points of X and Y rational over **C**.

of points $s \in \mathrm{Y}$ such that $f^{-1}(s)$ is proper over $\boldsymbol{k}(s)$ is *constructible* (and therefore identical to all of Y if it contains the closed points of Y, by the theorem of zeros of Hilbert (**I**, 10.4.8)); this amounts again to saying that if $f^{-1}(y)$ *is not proper* over $\boldsymbol{k}(y)$, then there exists an open neighbourhood U of y such that $f^{-1}(s)$ is not proper over $\boldsymbol{k}(s)$ for every $s \in \mathrm{U}$ (cf. (9.6.1), (iv))). This example illustrates the interest of systematically developing *criteria of constructibility*: this is what will be done in § 9.

8.2. Projective limits of preschemes

(8.2.1). Let S_0 be a ringed space, L a filtered preordered increasing set, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Algebras (not necessarily commutative), having L for index set. We know that, considered as an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Modules, $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ admits an inductive limit $\mathscr{A}$; let us denote by $\varphi_\lambda : \mathscr{A}_\lambda \to \mathscr{A}$ the canonical homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Modules. Let $m_\lambda : \mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \to \mathscr{A}_\lambda$ be the homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Modules that defines multiplication in the $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{A}_\lambda$; the hypothesis on the $\varphi_{\mu\lambda}$ implies that the m_λ form an inductive system of homomorphisms, and since the functor **IV-3 | 8**
$\varinjlim$ commutes with tensor products, $m = \varinjlim m_\lambda$ is a homomorphism $\mathscr{A} \otimes \mathscr{A} \to \mathscr{A}$ of $\mathscr{O}_{\mathrm{S}_0}$-Modules; by passing to the limit on the commutative diagrams expressing the associativity of m_λ and the existence of the identity section in $\mathscr{A}_\lambda$, one sees that m defines on $\mathscr{A}$ a structure of $\mathscr{O}_{\mathrm{S}_0}$-Algebra and that φ_λ is a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$-Algebras for every $\lambda \in \mathrm{L}$. Furthermore, $\mathscr{A}$ is the inductive limit of the system $(\mathscr{A}_\lambda, \varphi_{\mu\lambda})$ *in the category of* $\mathscr{O}_{\mathrm{S}_0}$*-Algebras*, in other words, for every $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{B}$, the canonical map

(8.2.1.1) $$\mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}, \mathscr{B}) \longrightarrow \varprojlim \mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}_\lambda, \mathscr{B})$$

which to every homomorphism $f : \mathscr{A} \to \mathscr{B}$ of $\mathscr{O}_{\mathrm{S}_0}$-Algebras associates the family $(f \circ \varphi_\lambda)$, is *bijective*. Indeed, we already know that it is injective and identifies $\mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Alg.}}(\mathscr{A}, \mathscr{B})$ with a subset of $\varprojlim \mathrm{Hom}_{\mathscr{O}_{\mathrm{S}_0}\text{-Mod.}}(\mathscr{A}_\lambda, \mathscr{B})$; it all comes down to seeing that if (f_λ) is an inductive system of $\mathscr{O}_{\mathrm{S}_0}$-Algebra homomorphisms, $f_\lambda : \mathscr{A}_\lambda \to \mathscr{B}$, its inductive limit $f : \mathscr{A} \to \mathscr{B}$, which is by definition a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$*-Modules*, is also a homomorphism of $\mathscr{O}_{\mathrm{S}_0}$*-Algebras*; but this results from passage to the inductive limit in the commutative diagram of $\mathscr{O}_{\mathrm{S}_0}$-Modules

$$\begin{tikzcd}
\mathscr{A}_\lambda \otimes \mathscr{A}_\lambda \arrow[r, "f_\lambda \otimes f_\lambda"] \arrow[d, "m_\lambda"'] & \mathscr{B} \otimes \mathscr{B} \arrow[d] \\
\mathscr{A}_\lambda \arrow[r, "f_\lambda"'] & \mathscr{B}
\end{tikzcd}$$

and from the fact that the functor $\varinjlim$ commutes with tensor products. We note finally that if the $\mathscr{A}_\lambda$ are commutative $\mathscr{O}_{\mathrm{S}_0}$-Algebras, the same holds for $\mathscr{A}$.

(8.2.2). Suppose now that S_0 is a *prescheme*, and that the $\mathscr{A}_\lambda$ are (commutative) *quasi-coherent* $\mathscr{O}_{\mathrm{S}_0}$-Algebras; we know then that $\mathscr{A} = \varinjlim \mathscr{A}_\lambda$ is a quasi-coherent $\mathscr{O}_{\mathrm{S}_0}$-Algebra (**I**, 4.1.1). Let us denote by S_λ (resp. S) the spectrum of the $\mathscr{O}_{\mathrm{S}_0}$-Algebra $\mathscr{A}_\lambda$ (resp. $\mathscr{A}$) (**II**, 1.3.1), and let $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ (for $\lambda \leqslant \mu$) and $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ be the S_0-morphisms corresponding to the homomorphisms $\varphi_{\mu\lambda}$ and φ_λ respectively (**II**, 1.2.7); it is clear that $(\mathrm{S}_\lambda, u_{\lambda\mu})$ is a *projective system* in the category of S_0-preschemes. We note that the $u_{\lambda\mu}$ and u_λ are *affine* morphisms (**II**, 1.6.2), hence *quasi-compact* and *separated*.

Proposition (8.2.3). — *With the notations of* (8.2.2), *the morphisms* $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ *make* S *a projective limit of the projective system* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *in the category of preschemes. Furthermore, if* $h : \mathrm{S}_0 \to \mathrm{T}$ *is a morphism, making every* S_0*-prescheme a* T*-prescheme,* S *is also a projective limit of the system* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *in the category of* T*-preschemes.*

Proof. Let us first prove the second assertion of the statement in the case $\mathrm{T} = \mathrm{S}_0$. It all comes down **IV-3 | 9**
to proving that if X is any S_0-prescheme, the canonical map

(8.2.3.1) $$\mathrm{Hom}_{\mathrm{S}_0}(\mathrm{X}, \mathrm{S}) \longrightarrow \varprojlim \mathrm{Hom}_{\mathrm{S}_0}(\mathrm{X}, \mathrm{S}_\lambda)$$

which to every S_0-morphism $v : \mathrm{X} \to \mathrm{S}$ associates the family $(u_\lambda \circ v)$, is *bijective*. Now, if $g : \mathrm{X} \to \mathrm{S}_0$ is the structural morphism and if we set $\mathscr{B} = g_*(\mathscr{O}_\mathrm{X})$, which is an $\mathscr{O}_{\mathrm{S}_0}$-Algebra, the map (8.2.3.1) identifies canonically with (8.2.1.1) (**II**, 1.2.7), and the conclusion follows from what we have seen in (8.2.1).

The other assertions of (8.2.3) are consequences of the following general lemma: □

Lemma (8.2.4). — *Let* **C** *be a category,* T *an object of* **C**, $\mathbf{C}_{\mathrm{T}}$ *the subcategory of objects of* **C** *over* T*. Let* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *be a projective system in* $\mathbf{C}_{\mathrm{T}}$*; then every projective limit of this system in* $\mathbf{C}_{\mathrm{T}}$ *is also a projective limit in* **C***, and conversely.*

Proof. Let $f_\lambda : \mathrm{S}_\lambda \to \mathrm{T}$ be the structural morphism. Suppose that S is a projective limit of $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in **C**, and let us denote by $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ the canonical corresponding morphisms. Consider a projective system of T-morphisms (in **C**) $w_\lambda : \mathrm{Y} \to \mathrm{S}_\lambda$, where $\mathrm{Y} \in \mathbf{C}_{\mathrm{T}}$. There exists by hypothesis a unique morphism (in **C**) $w : \mathrm{Y} \to \mathrm{S}$ such that $w_\lambda = u_\lambda \circ w$ for every λ. The hypothesis that the u_λ are T-morphisms implies that the morphisms $f_\lambda \circ u_\lambda : \mathrm{S} \to \mathrm{T}$ are all the same, and this composite f makes S a T-object. If $g : \mathrm{Y} \to \mathrm{T}$ is the structural morphism of Y, one has $f \circ w = f_\lambda \circ u_\lambda \circ w = f_\lambda \circ w_\lambda = g$ for every λ, which proves that w is a T-morphism. Conversely, suppose (with the same notations) that S is a projective limit of $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in $\mathbf{C}_{\mathrm{T}}$, and consider now a projective system of morphisms (in **C**) $w_\lambda : \mathrm{Y} \to \mathrm{S}_\lambda$. The composites $f_\lambda \circ w_\lambda : \mathrm{Y} \to \mathrm{T}$ are then all the same: indeed, for any two indices λ, μ, there is an index ν such that $\lambda \leqslant \nu$ and $\mu \leqslant \nu$, whence $f_\nu = f_\lambda \circ u_{\lambda\nu} = f_\mu \circ u_{\mu\nu}$; since $w_\lambda = u_{\lambda\nu} \circ w_\nu$ and $w_\mu = u_{\mu\nu} \circ w_\nu$, one has $f_\lambda \circ w_\lambda = f_\lambda \circ u_{\lambda\nu} \circ w_\nu = f_\nu \circ w_\nu$ and $f_\mu \circ w_\mu = f_\mu \circ u_{\mu\nu} \circ w_\nu = f_\nu \circ w_\nu$. If $g : \mathrm{Y} \to \mathrm{T}$ is the unique morphism thus defined, g makes Y a T-object, and the w_λ are T-morphisms; they hence have a projective limit $w : \mathrm{Y} \to \mathrm{S}$ which is a T-morphism, and *a fortiori* a morphism of **C**; moreover, the first part of the argument shows that any projective limit w' (in **C**) of the projective system (w_λ) is also a T-morphism, and as $w' : \mathrm{Y} \to \mathrm{S}'$ in $\mathbf{C}_{\mathrm{T}}$ gives $p \circ w' = w$ and $p_\lambda \circ w_\lambda = p_\lambda \circ w'_\lambda$, the definition of S'_λ as a fibre product $\mathrm{S}_\lambda \times_{\mathrm{T}} \mathrm{T}'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T′-morphism satisfying these relations, whence the lemma. □

Proposition (8.2.5). — *Under the conditions of* (8.2.2)*, let* α *be an element of* L, X_α *an* S_α*-prescheme. For every* $\lambda \geqslant \alpha$*, set* $\mathrm{X}_\lambda = \mathrm{X}_\alpha \times_{\mathrm{S}_\alpha} \mathrm{S}_\lambda$*, and for* $\alpha \leqslant \lambda \leqslant \mu$*, set* $v_{\lambda\mu} = 1_{\mathrm{X}_\alpha} \times u_{\lambda\mu}$*, so that* $(\mathrm{X}_\lambda, v_{\lambda\mu})$ *is a projective system of* X_α*-preschemes, whose index set is formed of the* $\lambda \geqslant \alpha$ *in* L*. Set likewise* $\mathrm{X} = \mathrm{X}_\alpha \times_{\mathrm{S}_\alpha} \mathrm{S}$ *and* $v_\lambda = 1_{\mathrm{X}_\alpha} \times u_\lambda$*. Then the* X_α*-morphisms* $v_\lambda : \mathrm{X} \to \mathrm{X}_\lambda$ *make* X *a projective limit of the projective system* $(\mathrm{X}_\lambda, v_{\lambda\mu})$ *in the category of* X_α*-preschemes, or in the category of all preschemes.*

Proof. This will also follow from the following general lemma: □

Lemma (8.2.6). — *Let* **C** *be a category in which fibre products exist,* $q : \mathrm{T}' \to \mathrm{T}$ *a morphism of* **C**, $\mathbf{C}_{\mathrm{T}}$ *(resp.* $\mathbf{C}_{\mathrm{T}'}$*) the category of objects of* **C** *over* T *(resp.* T′*). Let* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *be a projective system (not necessarily filtered)* **IV-3** | 10
in $\mathbf{C}_{\mathrm{T}}$*, and set* $\mathrm{S}'_\lambda = \mathrm{S}_\lambda \times_{\mathrm{T}} \mathrm{T}'$, $u'_{\lambda\mu} = u_{\lambda\mu} \times 1_{\mathrm{T}'}$*, so that* $(\mathrm{S}'_\lambda, u'_{\lambda\mu})$ *is a projective system in* $\mathbf{C}_{\mathrm{T}'}$*. Then, if* (S, u_λ) *is a projective limit of* $(\mathrm{S}_\lambda, u_{\lambda\mu})$ *in* $\mathbf{C}_{\mathrm{T}}$*,* $(\mathrm{S} \times_{\mathrm{T}} \mathrm{T}', u_\lambda \times 1_{\mathrm{T}'})$ *is a projective limit of* $(\mathrm{S}'_\lambda, u'_{\lambda\mu})$ *in* $\mathbf{C}_{\mathrm{T}'}$*.*

Proof. By hypothesis, for every λ, one has a commutative diagram

$$\begin{tikzcd} \mathrm{S}' \arrow[r, "u'_\lambda"] \arrow[d, "p"'] & \mathrm{S}'_\lambda \arrow[r, "h'_\lambda"] & \mathrm{T}' \arrow[d, "q"] \\ \mathrm{S} \arrow[r, "u_\lambda"'] & \mathrm{S}_\lambda \arrow[r, "h_\lambda"'] & \mathrm{T} \end{tikzcd}$$

where we have set $\mathrm{S}' = \mathrm{S} \times_{\mathrm{T}} \mathrm{T}'$, $u'_\lambda = u_\lambda \times 1_{\mathrm{T}'}$, $h'_\lambda = h_\lambda \times 1_{\mathrm{T}'}$. Let Y be a T′-object, $g' : \mathrm{Y} \to \mathrm{T}'$, and consider a projective system of T′-morphisms $w'_\lambda : \mathrm{Y} \to \mathrm{S}'_\lambda$. Then Y is a T-object for the morphism $g = q \circ g'$, and the $w_\lambda = p_\lambda \circ w'_\lambda$ are T-morphisms since $h_\lambda \circ w_\lambda = h_\lambda \circ p_\lambda \circ w'_\lambda = q \circ h'_\lambda \circ w'_\lambda = q \circ g'$ by hypothesis. Furthermore, one verifies immediately that (w_λ) is a projective system. There thus exists by hypothesis a unique T-morphism $w : \mathrm{Y} \to \mathrm{S}$ such that $u_\lambda \circ w = w_\lambda$ for every λ. By the definition of the fibre product, there is a unique T′-morphism $w' : \mathrm{Y} \to \mathrm{S}'$ such that $p \circ w' = w$. Then $u'_\lambda \circ w' = u_\lambda \circ w' = w_\lambda = p_\lambda \circ w'_\lambda$, and since $h'_\lambda \circ u'_\lambda \circ w' = g' = h'_\lambda \circ w'_\lambda$, the definition of S'_λ as the fibre product $\mathrm{S}_\lambda \times_{\mathrm{T}} \mathrm{T}'$ gives $u'_\lambda \circ w' = w'_\lambda$, and it is immediate that w' is the unique T′-morphism satisfying these relations, whence the lemma. □

Remark (8.2.7). — Given a ringed space S, the inductive limits with respect to a preordered set L (not necessarily filtered) exist in the category of $\mathscr{O}_{\mathrm{S}}$-Algebras, commutative or not, since the inductive limit exists by (8.2.1) and, on the other hand, for any two homomorphisms of $\mathscr{O}_{\mathrm{S}}$-Algebras $\mathscr{A} \to \mathscr{B}$, $\mathscr{A} \to \mathscr{C}$, the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is the corresponding “amalgamated sum” in this category.

When S is a prescheme, we say that the tensor product $\mathscr{B} \otimes_{\mathscr{A}} \mathscr{C}$ is a quasi-coherent $\mathscr{O}_S$-Algebra when this is so for $\mathscr{A}$, $\mathscr{B}$, and $\mathscr{C}$ (**I**, 1.3.13); from this one concludes that, in the category of quasi-coherent $\mathscr{O}_S$-Algebras, inductive limits for any preordered index set always exist. This allows us to generalise the definition of a projective limit of preschemes and propositions (8.2.3) and (8.2.5) to the case where the preordered set L is not necessarily filtered.

(8.2.8). The notations being those of (8.2.2), set $u_{\lambda\mu} = (\psi_{\lambda\mu}, \theta_{\lambda\mu})$ and $u_\lambda = (\psi_\lambda, \theta_\lambda)$; $(S_\lambda, \psi_{\lambda\mu})$ is thus a projective system of topological spaces and (ψ_λ) a projective system of continuous maps $S \to S_\lambda$.

Proposition (8.2.9). — *With the notations of (8.2.8), the projective limit of the projective system (ψ_λ) of continuous maps is a homeomorphism of the underlying space of* S *onto the projective limit of the projective system $(S_\lambda, \psi_{\lambda\mu})$ of topological spaces.*

Proof. Let T be the topological limit space of the system $(S_\lambda, \psi_{\lambda\mu})$ and set $\psi = \varprojlim \psi_\lambda : S \to T$. We can restrict to the case where $S_0 = S_\alpha$ for some $\alpha \in L$, and $\lambda \geqslant \alpha$. **IV-3** | 11

(i) Let us show first that the topology of S is the inverse image by ψ of the topology of T; in other words, if $\pi_\lambda : T \to S_\lambda$ is the canonical map, it amounts to showing that every open subset of S is a union of open subsets of the form $\psi^{-1}(\pi_\lambda^{-1}(U_\lambda))$, where U_λ is open in S_λ. Now, every open subset of S is by definition a union of open subsets obtained in the following way: one considers an open affine U_0 of S_0, of ring A_0, so that $\psi_\alpha^{-1}(U_0)$ is the open affine subset of S of ring $A = \Gamma(U_0, \mathscr{A})$, then one takes an element $f \in A$ and considers $\psi_\alpha^{-1}(U_0)$, i.e., the open set $D(f)$. These are the open subsets $D(f)$ which form a base of the topology of S (**II**, 1.3.1). Now, if one sets $A_\lambda = \Gamma(U_0, \mathscr{A}_\lambda)$, one has $A = \varinjlim A_\lambda$ (**I**, 1.3.9), hence there exists an index λ such that f is the canonical image of an element $f_\lambda \in A_\lambda$; one has then $D(f) = \psi_\lambda^{-1}(D(f_\lambda))$ (**I**, 1.2.2), and since $\psi_\lambda = \pi_\lambda \circ \psi$, our assertion is proved.

(ii) Let us now show that ψ is *bijective*, which will complete the proof. Since S is a Kolmogoroff space, it already follows from (i) that ψ is injective, and it remains to show that ψ is surjective. We can evidently replace T by an open subset $\pi_\alpha^{-1}(U_0)$, where U_0 is an open affine in $S_\alpha = S_0$, hence we can restrict to the case where the S_λ and S are affine, in other words $\mathscr{A}_\lambda$ is the sheaf of algebras associated to an A_0-algebra A_λ, and $\mathscr{A}$ the sheaf of algebras associated to $A = \varinjlim A_\lambda$; we will still denote $\varphi_{\mu\lambda} : A_\lambda \to A_\mu$ and $\varphi_\lambda : A_\lambda \to A$ the canonical homomorphisms. By definition, an element of T is a family $(\mathfrak{p}_\lambda)_{\lambda \in L}$, where $\mathfrak{p}_\lambda$ is a prime ideal of A_λ and where $\mathfrak{p}_\lambda = \varphi_{\mu\lambda}^{-1}(\mathfrak{p}_\mu)$ for $\lambda \leqslant \mu$. We know ((**0**, 5.13.3) and (5.13.1)) that there exists a prime ideal $\mathfrak{p}$ of A such that $\mathfrak{p}_\lambda = \varphi_\lambda^{-1}(\mathfrak{p})$ for every $\lambda \in L$, which completes the proof. □

In particular, we have thus proved the

Corollary (8.2.10). — *Let $(A_\lambda)_{\lambda \in L}$ be a filtered inductive system of rings, and let $\varphi_\lambda : A_\lambda \to A$ be the canonical homomorphisms. The canonical map $\mathfrak{p} \mapsto (\varphi_\lambda^{-1}(\mathfrak{p}))$ is a homeomorphism of* $\operatorname{Spec}(A)$ *onto the topological space* $\varprojlim \operatorname{Spec}(A_\lambda)$.

Corollary (8.2.11). — *With the notations of (8.2.8), for every quasi-compact open subset* U *of* S, *there exist an index λ and a quasi-compact open subset U_λ of S_λ such that $U = \psi_\lambda^{-1}(U_\lambda)$.*

Proof. This follows from the fact that, by definition of the projective limit topology, the $\psi_\lambda^{-1}(U_\lambda)$ (U_λ a quasi-compact open subset of S_λ) form a base of the topology of S, and from the fact that the index set L is filtered. □

Corollary (8.2.12). — *With the notations of (8.2.8), the inductive limit of the inductive system of homomorphisms $\theta_\lambda^* : \psi_\lambda^*(\mathscr{O}_{S_\lambda}) \to \mathscr{O}_S$ of sheaves of rings on* S *is an isomorphism*

(8.2.12.1) $$\varinjlim_\lambda \psi_\lambda^*(\mathscr{O}_{S_\lambda}) \xrightarrow{\sim} \mathscr{O}_S.$$

Proof. We can evidently suppose the S_λ affine; with the notations of the proof of (8.2.9), it all comes down to seeing that the inductive limit of the inductive system of canonical maps $(A_\lambda)_{\mathfrak{p}_\lambda} \to A_\mathfrak{p}$ is an isomorphism, which is nothing other than (**0**, 5.13.3), (ii). □

Proposition (8.2.13). — *Suppose that the morphisms $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ are open immersions, so that S_μ identifies with an open subset of S_λ for $\lambda \leqslant \mu$. Then, for every $\alpha \in \mathrm{L}$, the underlying space of the prescheme* S *identifies with the intersection of the S_λ for $\lambda \geqslant \alpha$, and the structural sheaf $\mathscr{O}_\mathrm{S}$ of the sheaf induced* (G, *II*, 1.5) *by $\mathscr{O}_{\mathrm{S}_\alpha}$ on this intersection; more generally, for every $\mathscr{O}_{\mathrm{S}_\alpha}$-Module $\mathscr{F}_\alpha$, $u_\alpha^*(\mathscr{F}_\alpha)$ identifies with the $\mathscr{O}_\mathrm{S}$-Module induced by $\mathscr{F}_\alpha$ on* S.

Proof. The first assertion follows from (8.2.9), in view of the definition of a projective limit of topological spaces; furthermore, all the $\psi_\lambda^*(\mathscr{O}_{\mathrm{S}_\lambda})$ are equal to the sheaf induced by $\mathscr{O}_{\mathrm{S}_\alpha}$ on S by definition (**0**, 1.3.7.1), and with the notations of the proof of (8.2.9), the homomorphisms $(\mathrm{A}_\lambda)_{\mathfrak{p}_\lambda} \to (\mathrm{A}_\mu)_{\mathfrak{p}_\mu}$ are bijective for a system $(\mathfrak{p}_\lambda)$ of ideals of primes corresponding to a single point of S; the assertion relative to $\mathscr{O}_\mathrm{S}$ is then deduced from (8.2.12). The last assertion then follows from the definition of $u_\alpha^*(\mathscr{F}_\alpha)$ (**0**, 1.4.3.1). □ **IV-3 | 12**

Remark (8.2.14). — The results of (8.2.9) and (8.2.12) show that S is a projective limit of the projective system $(\mathrm{S}_\lambda, u_{\lambda\mu})$ in the category of *all* ringed spaces (or of all ringed spaces with a local ringed space structure). Indeed, let Y be a ringed space, and consider a projective system of morphisms of ringed spaces $w_\lambda : \mathrm{Y} \to \mathrm{S}_\lambda$. If one sets $w_\lambda = (\rho_\lambda, \omega_\lambda)$, the ρ_λ form a projective system of continuous maps and $\rho = \varprojlim \rho_\lambda$ a continuous map $\mathrm{Y} \to \mathrm{S}$, by virtue of (8.2.9) and (8.2.12). The unique homomorphism $\omega^\sharp : \rho^*(\mathscr{O}_\mathrm{S}) \to \mathscr{O}_\mathrm{Y}$ such that $\omega_\lambda^\sharp = \omega^\sharp \circ \rho^*(\theta_\lambda^*)$ for every λ, and there is thus a unique morphism of ringed spaces $w : \mathrm{Y} \to \mathrm{S}$ satisfying these relations.

8.3. Constructible subsets of a projective limit of preschemes

(8.3.1). In what follows throughout this paragraph, we suppose the conditions of (8.2.2) are satisfied, and we keep the notations.

Theorem (8.3.2). — *For every λ, let E_λ, F_λ be two subsets of S_λ. Set*

$$\mathrm{E} = \bigcap_\lambda u_\lambda^{-1}(\mathrm{E}_\lambda), \qquad \mathrm{F} = \bigcup_\lambda u_\lambda^{-1}(\mathrm{F}_\lambda). \tag{8.3.2.1}$$

Suppose the following conditions are satisfied:

label=(i) For every λ, E_λ is pro-constructible and F_λ is ind-constructible (1.9.4).

lbbel=(ii) For $\lambda \leqslant \mu$, we have

$$\mathrm{E}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{E}_\lambda), \qquad \mathrm{F}_\mu \supset u_{\lambda\mu}^{-1}(\mathrm{F}_\lambda).$$

lcbel=(iii) There exists $\alpha \in \mathrm{L}$ such that S_α is quasi-compact (which implies that S_λ is quasi-compact for $\lambda \geqslant \alpha$).

Then the following properties are equivalent:

*label=*a) $\mathrm{E} \subset \mathrm{F}$.

*lbbel=*b) *There exists $\lambda \geqslant \alpha$ such that $u_\lambda^{-1}(\mathrm{E}_\lambda) \subset u_\lambda^{-1}(\mathrm{F}_\lambda)$ (and then we also have $u_\mu^{-1}(\mathrm{E}_\mu) \subset u_\mu^{-1}(\mathrm{F}_\mu)$ for $\mu \geqslant \lambda$).* **IV-3 | 13**

*lcbel=*c) *There exists $\lambda \geqslant \alpha$ such that $\mathrm{E}_\lambda \subset \mathrm{F}_\lambda$ (and then we also have $\mathrm{E}_\mu \subset \mathrm{F}_\mu$ for $\mu \geqslant \lambda$).*

The parenthetical remarks in b) *and* c) *follow from (ii). Set*

$$\mathrm{G}_\lambda = \mathrm{E}_\lambda \cap (\mathrm{S}_\lambda - \mathrm{F}_\lambda), \qquad \mathrm{G} = \mathrm{E} \cap (\mathrm{S} - \mathrm{F}).$$

Then G_λ is a pro-constructible subset of S_λ (1.9.5), *(i), and by virtue of* (8.3.2.1) *we have*

$$\mathrm{G}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{G}_\lambda) \qquad \textit{for } \lambda \leqslant \mu$$

$$\mathrm{G} = \bigcap_\lambda u_\lambda^{-1}(\mathrm{G}_\lambda).$$

We are thus reduced to proving the particular case of (8.3.2) *corresponding to $\mathrm{F}_\lambda = \varnothing$ for every λ:*

Corollary (8.3.3). — *For every λ, let E_λ be a pro-constructible subset of S_λ such that $\mathrm{E}_\mu \subset u_{\lambda\mu}^{-1}(\mathrm{E}_\lambda)$. Suppose that there exists $\alpha \in \mathrm{L}$ such that S_α is quasi-compact. Then the following conditions are equivalent:*

*label=*a) $\mathrm{E} = \bigcap_\lambda u_\lambda^{-1}(\mathrm{E}_\lambda) = \varnothing$.

*lbbel=*b) *There exists λ such that $u_\lambda^{-1}(\mathrm{E}_\lambda) = \varnothing$ (and then $u_\mu^{-1}(\mathrm{E}_\mu) = \varnothing$ for $\mu \geqslant \lambda$).*

*lcbel=*c) *There exists λ such that $\mathrm{E}_\lambda = \varnothing$ (and then $\mathrm{E}_\mu = \varnothing$ for $\mu \geqslant \lambda$).*

Proof. It is clear that *c)* implies *a)*. Let us prove that *a)* implies *b)*: since S_λ is quasi-compact, E_λ, being pro-constructible, is also quasi-compact, and it is the same for $u_\lambda^{-1}(E_\lambda)$ (1.9.5), (vi); then the filtered decreasing family of pro-constructible subsets $u_\lambda^{-1}(E_\lambda)$ has empty intersection, hence (1.9.9) one of them is empty.

Finally, let us show that *b)* implies *c)*. Since S_α is quasi-compact and L is filtered, we can replace S_α by an open affine, hence suppose (8.2.2) that S and the S_λ for $\lambda \geqslant \alpha$ are affine; one has then (1.9.2.1), for $\lambda \geqslant \alpha$,

$$u_\lambda(S) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu)$$

whence

$$E_\lambda \cap u_\lambda(S) = \bigcap_{\mu \geqslant \lambda} (E_\lambda \cap u_{\lambda\mu}(S_\mu)).$$

Now, since u_λ and the $u_{\lambda\mu}$ are quasi-compact, $u_\lambda(S)$ and $u_{\lambda\mu}(S_\mu)$ are pro-constructible in S_λ (1.9.5), (vii), hence the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ for $\mu \geqslant \lambda$ form a filtered decreasing family of pro-constructible subsets of S_λ. Since S_λ is quasi-compact, the hypothesis *b)* implies that the intersection of this family is empty, hence (1.9.9) one of the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ is empty, hence $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ is empty. ☐

Corollary (8.3.4). — *For every λ, let F_λ be an ind-constructible subset of S_λ such that for $\lambda \leqslant \mu$ one has $u_{\lambda\mu}^{-1}(F_\lambda) \subset F_\mu$. Suppose that there exists $\alpha \in L$ such that S_α is quasi-compact. Then the following conditions are equivalent:*

label=a) The set $F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda)$ is equal to S.
lbbel=b) There exists λ such that $u_\lambda^{-1}(F_\lambda) = S$ (and then $u_\mu^{-1}(F_\mu) = S$ for $\mu \geqslant \lambda$). **IV-3** | 14
lcbel=c) There exists λ such that $F_\lambda = S_\lambda$ (and then $F_\mu = S_\mu$ for $\mu \geqslant \lambda$).

This follows immediately from (8.3.3) by passing to complements.

Corollary (8.3.5). — *For every λ, let E_λ, F_λ be two constructible subsets of S_λ such that, for $\mu \geqslant \lambda$, one has $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ and $F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda)$. Suppose that there exists α such that S_α is quasi-compact. Then, for $E \subset F$ (resp. $E = F$), it is necessary and sufficient that there exist λ such that $E_\lambda \subset F_\lambda$ (resp. $E_\lambda = F_\lambda$), in which case one also has $E_\mu \subset F_\mu$ (resp. $E_\mu = F_\mu$) for $\mu \geqslant \lambda$.*

This is a particular case of (8.3.2).

Corollary (8.3.6). — *Suppose that there exists α such that S_α is quasi-compact. For $S = \varnothing$, it is necessary and sufficient that there exist λ such that $S_\lambda = \varnothing$.*

Corollary (8.3.7). — *On a, for every λ,*

(8.3.7.1)
$$u_\lambda(S) = \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu).$$

Proof. It is clear that the first member of (8.3.7.1) is contained in the second. Let s be a point of S_λ and set $X_\lambda = \mathrm{Spec}(\boldsymbol{k}(s))$; consider the projective system $(X_\mu, v_{\mu\nu})$ where $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ and $v_{\mu\nu} = 1 \times u_{\mu\nu}$ for $\lambda \leqslant \mu \leqslant \nu$; its projective limit is $X = X_\lambda \times_{S_\lambda} S$ (8.2.5). If $s \in \bigcap_{\mu \geqslant \lambda} u_{\lambda\mu}(S_\mu)$, this implies that $X_\mu \neq \varnothing$ for every $\mu \geqslant \lambda$ (**I**, 3.4.8); it follows then from (8.3.6) that $X \neq \varnothing$, hence $s \in u_\lambda(S)$ by (**I**, 3.4.8). ☐

Proposition (8.3.8). — *label=(i) For the morphism $u_\lambda : S \to S_\lambda$ to be dominant (resp. surjective), it is necessary and sufficient that for $\mu \geqslant \lambda$ the morphism $u_{\lambda\mu} : S_\mu \to S_\lambda$ be dominant (resp. surjective).*
lbbel=(ii) If, for some index λ, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat (resp. faithfully flat), then for every $\mu \geqslant \lambda$, the morphism $u_\lambda : S \to S_\lambda$ is flat (resp. faithfully flat).
lcbel=(iii) Suppose that the morphisms $u_\mu : S \to S_\mu$ are surjective for $\mu \geqslant \lambda$. For u_λ to be an open morphism (resp. universally open), it is necessary and sufficient that, for every $\mu \geqslant \lambda$, $u_{\lambda\mu}$ be an open morphism (resp. universally open).

Proof. label=(i) Since $u_\lambda(S) \subset u_{\lambda\mu}(S_\mu)$ for every $\mu \geqslant \lambda$, the necessity of the conditions is trivial, and it follows immediately from (8.3.7.1) that if the $u_{\lambda\mu}$ are surjective, then so is u_λ.

Suppose now that the $u_{\lambda\mu}$ are dominant for $\mu \geqslant \lambda$, and consider in S_λ an open quasi-compact non-empty subset U_λ; then the $\mathrm{U}_\mu = u_{\lambda\mu}^{-1}(\mathrm{U}_\lambda)$ for $\mu \geqslant \lambda$ form a projective system whose projective limit is $\mathrm{U} = u_\lambda^{-1}(\mathrm{U}_\lambda)$ (8.2.5). By hypothesis the U_μ are all non-empty, hence it is the same for U by (8.3.6), which proves that u_λ is dominant.

lbbel=(ii) By virtue of (i), it suffices to consider the case where the $u_{\lambda\mu}$ are flat; one can then restrict to the case where S_λ is affine, hence also the S_μ and S, and the assertion follows from (2.1.2) and (**0**, 6.2.3).

lcbel=(iii) By virtue of (8.2.5) and of (**I**, 3.5.2), it suffices to treat the case of open morphisms. Since $u_\lambda = u_{\lambda\mu} \circ u_\mu$ and u_μ is surjective, we know that if u_λ is open then so is $u_{\lambda\mu}$ (Bourbaki, *Top. gén.*, chap. I^er, 3^e ed., § 5, n^o 1, prop. 1). Conversely, to show that u_λ is open when all **IV-3 | 15** the $u_{\lambda\mu}$ are open for $\mu \geqslant \lambda$, it suffices to see that for every quasi-compact open subset V of S, $u_\lambda(\mathrm{V})$ is open in S_λ; but there exists then $\mu \geqslant \lambda$ such that $\mathrm{V} = u_\mu^{-1}(\mathrm{V}_\mu)$, where V_μ is open in S_μ (8.2.11); since u_μ is surjective, one has $\mathrm{V}_\mu = u_\mu(\mathrm{V})$ and $u_\lambda(\mathrm{V}) = u_{\lambda\mu}(u_\mu(\mathrm{V}))$ is therefore open by hypothesis.

One notes that we can arrange for u_λ to be open without the $u_{\lambda\mu}$ being open and without u_λ being surjective either. Indeed, one has an example by considering an integral ring A which is not a field, and its field of fractions K, which is the inductive limit of the A_f, where f ranges over $\mathrm{A} - \{0\}$; setting $\mathrm{S}_1 = \mathrm{Spec}(\mathrm{A})$, $\mathrm{S}_f = \mathrm{Spec}(\mathrm{A}_f)$, and $\mathrm{S} = \varprojlim \mathrm{S}_f = \mathrm{Spec}(\mathrm{K})$, the morphism $\mathrm{S} \to \mathrm{S}_1$ is not open, even though the morphisms $\mathrm{S}_f \to \mathrm{S}_1$ are open. □

(8.3.9). For every prescheme X, we will denote as usual by $\mathfrak{P}(\mathrm{X})$ the set of all subsets of the underlying set of X, by $\mathfrak{C}(\mathrm{X})$ (resp. $\mathfrak{Dc}(\mathrm{X})$, $\mathfrak{Fc}(\mathrm{X})$, $\mathfrak{LFc}(\mathrm{X})$) the set of constructible (resp. constructible and open, constructible and closed, and locally closed) subsets of X. It is clear that $(\mathfrak{P}(\mathrm{S}_\lambda), u_{\lambda\mu}^{-1})$ is an *inductive system of sets* and that the maps $u_\lambda^{-1} : \mathfrak{P}(\mathrm{S}_\lambda) \to \mathfrak{P}(\mathrm{S})$ form an inductive system of maps, whence, by passing to the inductive limit, a canonical map

$$u_{\mathfrak{P}} : \varinjlim \mathfrak{P}(\mathrm{S}_\lambda) \longrightarrow \mathfrak{P}(\mathrm{S}). \tag{8.3.9.1}$$

Furthermore, it follows from (1.8.2) that $u_{\lambda\mu}^{-1}$ sends $\mathfrak{C}(\mathrm{S}_\lambda)$ (resp. $\mathfrak{Dc}(\mathrm{S}_\lambda)$, $\mathfrak{Fc}(\mathrm{S}_\lambda)$, $\mathfrak{LFc}(\mathrm{S}_\lambda)$) into $\mathfrak{C}(\mathrm{S}_\mu)$ (resp. $\mathfrak{Dc}(\mathrm{S}_\mu)$, $\mathfrak{Fc}(\mathrm{S}_\mu)$, $\mathfrak{LFc}(\mathrm{S}_\mu)$) and that u_λ^{-1} sends $\mathfrak{C}(\mathrm{S}_\lambda)$ (resp. $\mathfrak{Dc}(\mathrm{S}_\lambda)$, $\mathfrak{Fc}(\mathrm{S}_\lambda)$, $\mathfrak{LFc}(\mathrm{S}_\lambda)$) into $\mathfrak{C}(\mathrm{S})$ (resp. $\mathfrak{Dc}(\mathrm{S})$, $\mathfrak{Fc}(\mathrm{S})$, $\mathfrak{LFc}(\mathrm{S})$). We thus have, by restriction of (8.3.9.1), canonical maps

$$\varinjlim \mathfrak{C}(\mathrm{S}_\lambda) \longrightarrow \mathfrak{C}(\mathrm{S}) \tag{8.3.9.2}$$

$$\varinjlim \mathfrak{Dc}(\mathrm{S}_\lambda) \longrightarrow \mathfrak{Dc}(\mathrm{S}) \tag{8.3.9.3}$$

$$\varinjlim \mathfrak{Fc}(\mathrm{S}_\lambda) \longrightarrow \mathfrak{Fc}(\mathrm{S}) \tag{8.3.9.4}$$

$$\varinjlim \mathfrak{LFc}(\mathrm{S}_\lambda) \longrightarrow \mathfrak{LFc}(\mathrm{S}). \tag{8.3.9.5}$$

(8.3.10). Let $g_\alpha : \mathrm{X}_\alpha \to \mathrm{S}_\alpha$ be a morphism; with the notations of (8.2.5) one has, on the one hand, a canonical map $v_{\mathfrak{P}} : \varinjlim \mathfrak{P}(\mathrm{S}_\lambda) \to \mathfrak{P}(\mathrm{X})$; on the other hand, from the morphisms of projection $g_\lambda : \mathrm{X}_\lambda \to \mathrm{S}_\lambda$ for every $\lambda \geqslant \alpha$ and a morphism of projection $g : \mathrm{X} \to \mathrm{S}$. It is clear that the $g_\lambda^{-1} : \mathfrak{P}(\mathrm{S}_\lambda) \to \mathfrak{P}(\mathrm{X}_\lambda)$ form an inductive system of maps, and that the diagrams

$$\begin{tikzcd}
\mathfrak{P}(\mathrm{S}_\lambda) \arrow[r, "u_\lambda^{-1}"] \arrow[d, "g_\lambda^{-1}"'] & \mathfrak{P}(\mathrm{S}) \arrow[d, "g^{-1}"] \\
\mathfrak{P}(\mathrm{X}_\lambda) \arrow[r, "v_\lambda^{-1}"'] & \mathfrak{P}(\mathrm{X})
\end{tikzcd}$$

are commutative; one deduces from this by passing to the inductive limit a commutative diagram **IV-3 | 16**

$$\begin{tikzcd}
\varinjlim \mathfrak{P}(\mathrm{S}_\lambda) \arrow[r, "u_{\mathfrak{P}}"] \arrow[d] & \mathfrak{P}(\mathrm{S}) \arrow[d, "g^{-1}"] \\
\varinjlim \mathfrak{P}(\mathrm{X}_\lambda) \arrow[r, "v_{\mathfrak{P}}"'] & \mathfrak{P}(\mathrm{X})
\end{tikzcd}$$

and it follows from (1.8.2) that one has analogous diagrams replacing $\mathfrak{P}$ by $\mathfrak{C}$, $\mathfrak{Dc}$, $\mathfrak{Fc}$, or $\mathfrak{LFc}$.

(8.3.10.1)

It follows from (8.3.5) that under the hypothesis that for some $\alpha \in \mathrm{L}$, S_α is *quasi-compact*, the canonical map (8.3.9.2) is *injective*. Furthermore:

Theorem (8.3.11). — *Suppose that there exists α such that S_α is quasi-compact and quasi-separated. Then the canonical maps* (8.3.9.2), (8.3.9.3), (8.3.9.4), *and* (8.3.9.5) *are* bijective.

Proof. By virtue of the preceding remark, it remains to prove that these maps are *surjective*; as every constructible subset of S is a finite union of sets of the form $\mathrm{U} \cap \complement \mathrm{V}$, where U and V are open and constructible, it will suffice to prove that (8.3.9.3) is surjective for it to be the same for (8.3.9.2) (and also for (8.3.9.4), by passing to complements). Now, since the morphisms $\mathrm{S}_\lambda \to \mathrm{S}_\alpha$ are affine and separated, the S_λ for $\lambda \geqslant \alpha$ and S are quasi-compact and quasi-separated, and one knows that the constructible open subsets of such a prescheme are none other than the quasi-compact open subsets (1.8.1). The conclusion then follows from (8.2.11) except for the map (8.3.9.5). To prove that this last map is surjective, consider a locally closed and constructible subset Z of S; Z is therefore quasi-compact ($\mathbf{0}_{\mathrm{III}}$, 9.1.4). As every point $x \in \mathrm{Z}$ admits by hypothesis an open quasi-compact neighbourhood V_x in S such that $\mathrm{Z} \cap \mathrm{V}_x$ is closed in V_x; we can cover Z by a finite number of the V_x, hence there is a quasi-compact open subset U containing Z such that Z is closed in U; in other words, Z is also constructible in U ($\mathbf{0}_{\mathrm{III}}$, 9.1.8). We know (8.2.11) that there exist λ and a quasi-compact open subset U_λ of S_λ such that $\mathrm{U} = u_\lambda^{-1}(\mathrm{U}_\lambda)$. Applying to U (which is a projective limit of $\mathrm{U}_\lambda = u_{\lambda\mu}^{-1}(\mathrm{U}_\lambda)$) the map (8.3.9.4), one sees that there exist $\mu \geqslant \lambda$ and a closed constructible subset Z_μ in U_μ such that $\mathrm{Z} = u_\mu^{-1}(\mathrm{Z}_\mu)$. But since the canonical immersion $\mathrm{U}_\mu \to \mathrm{S}_\mu$ is quasi-compact by hypothesis (1.2.7), she is of finite presentation (1.6.2), (i), and Z_μ is also a constructible subset *of* S_μ by virtue of (1.8.4) and (1.8.1); since Z_μ is evidently locally closed in S_μ, this completes the proof. □

Corollary (8.3.12). — *Suppose that there exists α such that S_α is quasi-compact, and let, for every λ, Z_λ be a constructible subset of S_λ such that $\mathrm{Z}_\mu = u_{\lambda\mu}^{-1}(\mathrm{Z}_\lambda)$ for $\mu \geqslant \lambda$. If $\mathrm{Z} = u_\lambda^{-1}(\mathrm{Z}_\lambda)$, then, for Z to be open (resp.* **IV-3 | 17**
closed, resp. locally closed) in S, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that Z_λ be so in S_λ.

Proof. Cover S_α by a finite number of open affine subsets $\mathrm{U}_\alpha^{(i)}$; then the $\mathrm{U}^{(i)} = u_\alpha^{-1}(\mathrm{U}_\alpha^{(i)})$ form an open covering of S, and for Z to be open (resp. closed, resp. locally closed) in S, it is necessary and sufficient that each of the $\mathrm{Z} \cap \mathrm{U}^{(i)}$ be so in $\mathrm{U}^{(i)}$. Since L is filtered and each $\mathrm{Z} \cap \mathrm{U}^{(i)}$ is constructible in $\mathrm{U}^{(i)}$ ($\mathbf{0}_{\mathrm{III}}$, 9.1.8), one can restrict to proving the corollary when S_α is affine, hence quasi-compact and quasi-separated; but then it follows from (8.3.11). □

Proposition (8.3.13). — *Suppose that the morphisms $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ are flat and that there exists α such that S_α is quasi-compact. For every λ, let Z'_λ, Z''_λ be two pro-constructible subsets of S_λ, such that $\mathrm{Z}'_\mu = u_{\lambda\mu}^{-1}(\mathrm{Z}'_\lambda)$ and $\mathrm{Z}''_\mu = u_{\lambda\mu}^{-1}(\mathrm{Z}''_\lambda)$ for $\mu \geqslant \lambda$; suppose further that $\overline{\mathrm{Z}'_\alpha}$ is constructible in S_α. Set $\mathrm{Z}' = u_\lambda^{-1}(\mathrm{Z}'_\lambda)$, $\mathrm{Z}'' = u_\lambda^{-1}(\mathrm{Z}''_\lambda)$; for $\mathrm{Z}'' \subset \overline{\mathrm{Z}'}$, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathrm{Z}''_\lambda \subset \overline{\mathrm{Z}'_\lambda}$.*

Proof. Indeed, one knows that $u_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ is also a flat morphism for every λ (8.3.8), (i); as Z'_λ is pro-constructible, the closure of Z'_μ for $\mu \geqslant \lambda$ (resp. of Z') in S_μ (resp. S) is equal to $u_{\lambda\mu}^{-1}(\overline{\mathrm{Z}'_\lambda})$ (resp. $u_\lambda^{-1}(\overline{\mathrm{Z}'_\lambda})$) (2.3.10). Since the $u_{\lambda\mu}^{-1}(\overline{\mathrm{Z}'_\lambda})$ and $u_\lambda^{-1}(\overline{\mathrm{Z}'_\lambda})$ are constructible (1.8.2), the conclusion follows from (8.3.2). □

8.4. Criteria of irreducibility and connectedness for projective limits of preschemes

Proposition (8.4.1). — *Suppose that there exists an index α such that S_α is quasi-compact.*

label=(i) *If S is not irreducible and if in addition the underlying space of S is Noetherian and S_α is quasi-separated, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not irreducible.*
lbbel=(ii) *If S is not connected, there exists $\lambda \geqslant \alpha$ such that, for $\mu \geqslant \lambda$, S_μ is not connected.*

Proof. Suppose that S is a union of two distinct closed subsets S', S'' of S (resp. disjoint non-empty closed subsets). In case (i), S' and S'' are constructible since the space S is Noetherian. By virtue of (8.3.11), there exist $\lambda \geqslant \alpha$ and two constructible closed subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$; since $S = S' \cup S''$, it follows from (8.3.11) that one can suppose $S_\lambda = S'_\lambda \cup S''_\lambda$; since S'_λ and S''_λ are distinct from S_λ, this proves that S_λ is not irreducible.

In case (ii), S' and S'' are open and quasi-compact, hence, by virtue of (8.2.11), there exist $\lambda \geqslant \alpha$ and two open quasi-compact subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$. Moreover, since S' and S'' are open and closed in S, they are both pro-constructible and ind-constructible (1.9.6), hence constructible (1.9.5), and it follows from (8.3.5) that one can suppose λ taken such that $S_\lambda = S'_\lambda \cup S''_\lambda$ and $S'_\lambda \cap S''_\lambda = \varnothing$, which shows that S_λ is not connected. □

Proposition (8.4.2). — *Suppose that the underlying space of S is Noetherian and that one of the two following conditions is satisfied:*

label=a) *For $\lambda \leqslant \mu$, $u_{\lambda\mu} : S_\mu \to S_\lambda$ is dominant, and there exists α such that S_α is quasi-compact.* **IV-3** | 18
lbbel=b) *There exists α such that the underlying space of S_α is Noetherian, and for $\mu \geqslant \lambda$, $u_{\lambda\mu}$ is a homeomorphism of S_μ onto a subspace of S_λ.*

Under these conditions, there exists λ such that, for every $\mu \geqslant \lambda$:

label=(i) *For every irreducible component Y_i of S ($1 \leqslant i \leqslant m$), $\overline{u_\mu(Y_i)}$ is an irreducible component of S_μ, and the map $Y_i \mapsto \overline{u_\mu(Y_i)}$ is a bijection of the set of irreducible components of S onto the set of irreducible components of S_μ.*
lbbel=(ii) *For every connected component C_j of S ($1 \leqslant j \leqslant n$), $\overline{u_\mu(C_j)}$ is a connected component of S_μ, and the map $C_j \mapsto \overline{u_\mu(C_j)}$ is a bijection of the set of connected components of S onto the set of connected components of S_μ.*

Proof. Let us first establish the

Lemma (8.4.2.1). — *Under condition* a) *or* b) *of (8.4.2), there exists λ such that, for $\mu \geqslant \lambda$, $u_\mu : S \to S_\mu$ is dominant.*

In case *a)*, this has already been proved without supposing the space S Noetherian (8.3.8), (i). In case *b)*, set $Z_\alpha = \overline{u_\alpha(S)}$; as a closed subset of the Noetherian space S_α, Z_α is constructible, and since $u_\alpha^{-1}(Z_\alpha) = S$, it follows from (8.3.5) that there exists $\lambda \geqslant \alpha$ such that $u_{\lambda\mu}^{-1}(Z_\lambda) = S_\mu$. But since $u_{\alpha\mu}$ is a homeomorphism and $u_\alpha(S)$ is dense in S_α, it follows from $S \to Z_\mu = S_\mu$.

This lemma being proved, one can suppose that for every λ, u_λ is a dominant morphism.

label=(i) Each of the S_λ is a union of the $\overline{u_\lambda(Y_i)}$, which are irreducible. On the other hand, if U_i is the open subset of S complementary to the union of the Y_j of index $j \neq i$ ($1 \leqslant i \leqslant m$), the U_i are pairwise disjoint and $Y_i = \overline{U_i}$ (**0**, 2.1.6). Since the underlying space of S is Noetherian, the U_i are quasi-compact, hence there exist an index λ and open subsets $U_{i\lambda}$ of S_λ such that $U_i = u_\lambda^{-1}(U_{i\lambda})$ for $1 \leqslant i \leqslant m$ (8.2.11). From this one concludes that if one sets $U_{i\mu} = u_{\lambda\mu}^{-1}(U_{i\lambda})$ for $\lambda \leqslant \mu$, the $U_{i\mu}$ are pairwise disjoint, the $U_{i\mu}$ are open and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant. By this, none of the closures $\overline{U_{i\mu}}$ is contained in another and $u_\mu(U_i)$ is dense in $U_{i\mu}$ since u_μ is dominant; hence $\overline{U_{i\mu}} = \overline{u_\mu(Y_i)}$, which proves that the $\overline{u_\mu(Y_i)}$ are the irreducible components of S_μ (**0**, 2.1.7) and completes the proof.
lbbel=(ii) Since the underlying space S is Noetherian, the C_j are open and closed in S and quasi-compact; the same reasoning as in (i) shows that there exist λ and open subsets $V_{j\lambda}$ of S_λ such that $C_j = u_\lambda^{-1}(V_{j\lambda})$ for $1 \leqslant j \leqslant n$. One sees also, as in (i), that if one sets

$V_{j\mu} = u_{\lambda\mu}^{-1}(V_{j\lambda})$ for $\mu \geqslant \lambda$, the $V_{j\mu}$ are pairwise disjoint, and $u_\mu(C_j)$ is dense in $V_{j\mu}$. Furthermore, it follows from (8.3.4) that for μ sufficiently large, the union of the $V_{j\mu}$ is S_μ, since every open subset of a Noetherian prescheme is ind-constructible (1.9.6). The $V_{j\mu}$ are thus the connected components of S_μ, this completes the proof.

□

We note that if the morphisms $u_{\lambda\mu}$ are *immersions*, they satisfy in particular condition *b)* of (8.4.2).

Corollary (8.4.3). — *Suppose one of the conditions* a), b) *of* (8.4.2) *is satisfied, the underlying space of* S *being Noetherian; then, for* S *to be irreducible, it is necessary and sufficient that there exist λ such that S_μ be so for every $\mu \geqslant \lambda$.*

Proposition (8.4.4). — *Suppose that there exists α such that S_α is quasi-compact and that, for $\lambda \leqslant \mu$, $u_{\lambda\mu} : S_\mu \to S_\lambda$ is dominant. Then, for* S *to be connected, it is necessary and sufficient that there exist λ such that S_μ be so for every $\mu \geqslant \lambda$.* **IV-3** | 19

Proof. The condition is sufficient by virtue of (8.4.1), (i); on the other hand, we have seen (8.3.8), (i) that $u_\lambda : S \to S_\lambda$ is dominant for λ sufficiently large, hence, if S is connected, it is the same for S_λ, since $u_\lambda(S)$ is dense in S_λ and is connected. □

Corollary (8.4.5). — *Let k be a field, X a quasi-compact k-prescheme. For X to be geometrically connected* (4.5.2)*, it is necessary and sufficient that, for every finite and separable extension* K *of k, $X \otimes_k K$ be connected.*

Proof. The condition is trivially necessary. To see that it is sufficient, we must prove that if Ω is an algebraic closure of k, $X \otimes_k \Omega$ is connected (4.5.1). Now, Ω is a filtered inductive limit of the finite sub-extensions of k, and for $k' \subset k'' \subset \Omega$, the morphism $X \otimes_k k'' \to X \otimes_k k'$ is surjective. We are thus reduced, by virtue of (8.4.4), to proving that $X \otimes_k k'$ is connected for every finite extension k' of k. But if K is the largest separable extension contained in k', the morphism $X \otimes_k k' \to X \otimes_k K$ is finite, surjective, and radicial, hence (2.4.5) a homeomorphism, and since $X \otimes_k K$ is connected by hypothesis, so is $X \otimes_k k'$. □

Remark (8.4.6). — label=(i) The proof of (8.4.2) shows that the conclusion of this proposition is valid if one supposes that the underlying space of S is Noetherian, that there exists α such that S_α is quasi-compact, and finally that the $u_\lambda : S \to S_\lambda$ are dominant.

lbbel=(ii) On the other hand, the conclusion of (8.4.2) can be at fault when the u_λ are not dominant for λ sufficiently large, even when the S_λ and S are Noetherian, as shown by the following example. Take **N** for index set, all the S_n equal to $\operatorname{Spec}(A \times K) = \operatorname{Spec}(A) \amalg \operatorname{Spec}(K)$, where K is a field, A any K-algebra, and all the morphisms $u_{n,n+1}$ equal to the same morphism corresponding to the homomorphism $(x, y) \mapsto (j(y), y)$ of $A \times K$ into itself, where $j : K \to A$ is the canonical homomorphism. One easily verifies that the inductive limit of this system of rings is the canonical homomorphism corresponding to the second projection $A \times K \to K$. One thus sees that $S = \operatorname{Spec}(K)$ is irreducible even though none of the S_n is connected.

8.5. Modules of finite presentation on a projective limit of preschemes

(8.5.1). We always keep the notations of (8.2.2); we will further restrict ourselves to the case where S_0 is one of the S_λ, to which one can always reduce.

Whenever, in this section, we consider a family $(\mathscr{F}_\lambda)$, where, for every $\lambda \in L$, $\mathscr{F}_\lambda$ is an $\mathscr{O}_{S_\lambda}$-Module, it will be understood that this family satisfies the condition

(8.5.1.1) $$\mathscr{F}_\mu = u_{\lambda\mu}^*(\mathscr{F}_\lambda) \qquad \text{for } \lambda \leqslant \mu.$$

We will then set **IV-3** | 20

(8.5.1.2) $$\mathscr{F} = u_\lambda^*(\mathscr{F}_\lambda)$$

which is an $\mathscr{O}_S$-Module not depending on the index $\lambda \in L$, by virtue of the hypothesis (8.5.1.1).

Let $(\mathscr{F}_\lambda)$, $(\mathscr{G}_\lambda)$ be two such families of $\mathscr{O}_{S_\lambda}$-Modules. It is clear that the maps $f_\lambda \mapsto u_{\lambda\mu}^*(f_\lambda)$ from $\operatorname{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda)$ to $\operatorname{Hom}_{\mathscr{O}_{S_\mu}}(\mathscr{F}_\mu, \mathscr{G}_\mu)$ define an *inductive system* of abelian groups $(\operatorname{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda), u_{\lambda\mu}^*)$, and that the maps $f_\lambda \mapsto u_\lambda^*(f_\lambda)$ form an inductive system of homomorphisms of abelian groups

$$u_\lambda^* : \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_S(\mathscr{F}, \mathscr{G});$$

whence, by passing to the inductive limit, a canonical homomorphism of abelian groups

(8.5.1.3) $$u^*_{\mathscr{F},\mathscr{G}} : \varinjlim \mathrm{Hom}_{\mathscr{O}_{S_\lambda}}(\mathscr{F}_\lambda, \mathscr{G}_\lambda) \longrightarrow \mathrm{Hom}_S(\mathscr{F}, \mathscr{G}).$$

Note that when $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$, the condition (8.5.1.1) is satisfied, and one has $\mathscr{F} = \mathscr{O}_S$; the homomorphism (8.5.1.3) thus gives (**0**, 5.1.1) a canonical homomorphism of abelian groups

(8.5.1.4) $$u_{\mathscr{G}} : \varinjlim \Gamma(S_\lambda, \mathscr{G}_\lambda) \longrightarrow \Gamma(S, \mathscr{G}).$$

Theorem (8.5.2). — *label=(i) Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated) and that, for some $\lambda \in L$, $\mathscr{F}_\lambda$ is quasi-coherent and of finite type (resp. of finite presentation) and $\mathscr{G}_\lambda$ quasi-coherent. Then the homomorphism $u^*_{\mathscr{F},\mathscr{G}}$ is injective (resp. bijective).*

lbbel=(ii) Suppose S_0 quasi-compact and quasi-separated. For every quasi-coherent $\mathscr{O}_S$-Module $\mathscr{F}$ of finite presentation, there exist $\lambda \in L$ and an $\mathscr{O}_{S_\lambda}$-Module quasi-coherent of finite presentation $\mathscr{F}_\lambda$ such that $\mathscr{F}$ is isomorphic to $u_\lambda^*(\mathscr{F}_\lambda)$.

Proof. label=(i) We can evidently restrict to the case where $S_0 = S_\lambda$ since the morphisms $u_{0\lambda} : S_\lambda \to S_0$ are affine, hence quasi-compact and separated. Consider first the case where $S_0 = \mathrm{Spec}(A_0)$ is *affine*. Then assertion (i) is equivalent to the

Lemma (8.5.2.1). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; let M_0, N_0 be two A_0-modules, and set $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$, $M = M_0 \otimes_{A_0} A = \varinjlim M_\lambda$, $N = N_0 \otimes_{A_0} A = \varinjlim N_\lambda$. If M_0 is of finite type (resp. of finite presentation), the canonical homomorphism*

(8.5.2.2) $$\varinjlim \mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \longrightarrow \mathrm{Hom}_A(M, N)$$

is injective (resp. bijective).

We know indeed (Bourbaki, *Alg.*, chap. II, 3e ed., § 5, no 1) that one has canonical functorial isomorphisms

$$\mathrm{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \xrightarrow{\sim} \mathrm{Hom}_{A_0}(M_0, N_\lambda), \qquad \mathrm{Hom}_A(M, N) \xrightarrow{\sim} \mathrm{Hom}_{A_0}(M_0, N)$$

so that the homomorphism (8.5.2.2) is none other, up to canonical isomorphisms, than the canonical homomorphism

(8.5.2.3) $$\varinjlim \mathrm{Hom}_{A_0}(M_0, N_\lambda) \longrightarrow \mathrm{Hom}_{A_0}(M_0, \varinjlim N_\lambda),$$

which, to every inductive system of homomorphisms of A_0-modules $\theta_\lambda : M_0 \to N_\lambda$, associates its inductive limit. **IV-3** | 21

Now, if M_0 is of finite type (resp. of finite presentation), one has an exact sequence $A_0^m \to M_0 \to 0$ (resp. $A_0^n \to A_0^m \to M_0 \to 0$); since it is clear that (8.5.2.3) is bijective when M_0 is of the form A_0^q, it suffices to use the left-exactness of the functor $M_0 \mapsto \mathrm{Hom}_{A_0}(M_0, P)$ and the exactness of the functor $\varinjlim$ (in the category of abelian groups) to conclude.

Let us pass to the case where S_0 is quasi-compact, and let (U_i) be a finite open affine covering of S_0; for every λ, the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form a covering of the open affine subsets of S_λ, and the $V_i = u_0^{-1}(U_i)$ a covering of the open affine subsets of S. To show that $u^*_{\mathscr{F},\mathscr{G}}$ is injective, it suffices to show that if $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ is such that $f = u_\lambda^*(f_\lambda) = 0$, then there exists $\mu \geqslant \lambda$ such that $u^*_{\lambda\mu}(f_\lambda) = 0$. By virtue of the lemma (8.5.2.1), for each i there exists $\mu_i \geqslant \lambda$ such that $f_\mu|_{U_{i\mu_i}} = 0$. It suffices therefore to take μ greater than all the λ_i.

Suppose further that S_0 is quasi-separated and $\mathscr{F}_0$ is a quasi-coherent $\mathscr{O}_{S_0}$-Module of finite presentation. Note now that S_λ is quasi-separated and $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{S_\lambda}$-Module of finite presentation (**0**, 5.2.5); since, for every pair of indices i, j, $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair of indices i, j, $u_\lambda^*(f_\lambda^{(i)}) = u_\lambda^*(f_\lambda^{(j)})$ on $U_{ij\lambda}$ by definition, it follows from what we have already seen above that there exists λ_{ij} such that $u^*_{\lambda\mu}(f_\lambda^{(i)})|_{U_{ij\lambda}} = u^*_{\lambda\mu}(f_\lambda^{(j)})|_{U_{ij\lambda}}$ for $\mu \geqslant \lambda_{ij}$;

taking μ greater than all the λ_{ij}, one sees that $u^*_{\lambda\mu}(f^{(i)}_\lambda)$ and $u^*_{\lambda\mu}(f^{(j)}_\lambda)$ coincide in $\mathrm{U}_{i\mu} \cap \mathrm{U}_{j\mu}$ for every couple (i, j), and therefore define an homomorphism $f_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ such that $f = u^*_\mu(f_\mu)$.

lbbel=(ii) Before passing to the proof of (ii), let us note the following corollaries of (i):

□

Corollary (8.5.2.4). — *Suppose* S_0 *quasi-compact,* $\mathscr{F}_\lambda$ *quasi-coherent of finite type,* $\mathscr{G}_\lambda$ *quasi-coherent. Let* $f_\lambda : \mathscr{F}_\lambda \to \mathscr{G}_\lambda$ *be a homomorphism. For* $f = u^*_\lambda(f_\lambda) : \mathscr{F} \to \mathscr{G}$ *to be an isomorphism, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u^*_{\lambda\mu}(f_\lambda) : \mathscr{F}_\mu \to \mathscr{G}_\mu$ *be so.*

Proof. We can always suppose $\mathrm{S}_\lambda = \mathrm{S}_0$, and we can further reduce to the case where S_0 is affine, hence also quasi-separated. Then the assertion is equivalent to the lemma (8.5.2.1), for there is by hypothesis an $\mathscr{O}_\mathrm{S}$-homomorphism $g : \mathscr{G} \to \mathscr{F}$ such that $g \circ f = 1_\mathscr{F}$ and $f \circ g = 1_\mathscr{G}$. Since $\mathscr{G}$ is of finite presentation, there exist by virtue of (8.5.2), (i), an index $\nu \geqslant \lambda$ and a homomorphism $g_\nu : \mathscr{G}_\nu \to \mathscr{F}_\nu$ such that $g = u^*_\nu(g_\nu)$ by virtue of (8.5.2), (i); one has therefore $u^*_\nu(g_\nu \circ f_\nu) = 1_\mathscr{F}$ and $u^*_\nu(f_\nu \circ g_\nu) = 1_\mathscr{G}$, whence the corollary. □

Corollary (8.5.2.5). — *Suppose* S_0 *quasi-compact and quasi-separated. Suppose that* $\mathscr{F}_\lambda$, $\mathscr{G}_\lambda$ *are* $\mathscr{O}_{\mathrm{S}_\lambda}$*-Modules quasi-coherent of finite presentation. For* $\mathscr{F}$ *and* $\mathscr{G}$ *to be isomorphic, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $\mathscr{F}_\mu$ *and* $\mathscr{G}_\mu$ *are isomorphic. Moreover, for every isomorphism* $f : \mathscr{F} \xrightarrow{\sim} \mathscr{G}$*, there exist* $\nu \geqslant \mu$ *and an isomorphism* $f_\nu : \mathscr{F}_\nu \xrightarrow{\sim} \mathscr{G}_\nu$ *such that* $f = u^*_\nu(f_\nu)$.

Proof. This follows from (8.5.2.4) and from (8.5.2), (i) since every homomorphism $f : \mathscr{F} \to \mathscr{G}$ is of the form $u^*_\mu(f_\mu)$ for some $\mu \geqslant \lambda$ and some homomorphism $f_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$. □ **IV-3** | 22

Proof of (8.5.2), *(ii).* Consider again first the case where $\mathrm{S}_0 = \mathrm{Spec}(\mathrm{A}_0)$ is affine. Then the assertion is equivalent to the lemma (8.5.7.1).

In the general case, we start from a finite open affine covering (U_i) of S_0, and for every λ, the $\mathrm{U}_{i\lambda} = u^{-1}_{0\lambda}(\mathrm{U}_i)$ form an open affine covering of S_λ, and the $\mathrm{V}_i = u^{-1}_0(\mathrm{U}_i)$ an open affine covering of S. By the affine case, for every i, there exist an index $\lambda(i)$ and an $\mathscr{O}_{\mathrm{U}_{i,\lambda(i)}}$-Module quasi-coherent of finite presentation $\mathscr{F}_i$ such that $u^*_{\lambda(i)}(\mathscr{F}_i) = \mathscr{F}|_{\mathrm{V}_i}$ (with the notations of (i)). Furthermore, since L is filtered, we can suppose that all the λ_i are equal to a single λ. Note now that S_λ is quasi-separated and $\mathscr{F}_\lambda$ is a quasi-coherent $\mathscr{O}_{\mathrm{S}_\lambda}$-Module of finite presentation (**0**, 5.2.5); since, for every pair of indices i, j, $\mathrm{U}_{ij\lambda} = \mathrm{U}_{i\lambda} \cap \mathrm{U}_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.5.2.5) that for every pair (i, j), there exist an index $\lambda_{ij} \geqslant \lambda$ and an isomorphism $\theta_{ij} : \mathscr{F}_i|_{\mathrm{U}_{ij\lambda}} \xrightarrow{\sim} u^*_{\lambda,\lambda_{ij}}(\mathscr{F}_j)|_{\mathrm{U}_{ij\lambda}}$ such that $u^*_{\lambda_{ij}}(\theta_{ij})$ is the identity automorphism of $\mathscr{F}|_{(\mathrm{V}_i \cap \mathrm{V}_j)}$; one can further suppose all the λ_{ij} equal to λ. Changing the notations, one can therefore suppose that it holds for every pair (i, j) an isomorphism $\theta_{ij} : \mathscr{F}_i|_{\mathrm{U}_{ij\lambda}} \xrightarrow{\sim} \mathscr{F}_j|_{\mathrm{U}_{ij\lambda}}$ such that $u^*_\lambda(\theta_{ij})$ is the identity automorphism of $\mathscr{F}|_{(\mathrm{V}_i \cap \mathrm{V}_j)}$. Finally, for any three indices i, j, k, if one sets $\mathrm{U}_{ijk,\lambda} = \mathrm{U}_{i\lambda} \cap \mathrm{U}_{j\lambda} \cap \mathrm{U}_{k\lambda}$, $\mathrm{U}_{ijk,\lambda}$ is quasi-compact, and if θ'_{ij}, θ'_{jk}, and θ'_{ik} denote the restrictions of θ_{ij}, θ_{jk}, and θ_{ik} to $\mathrm{U}_{ijk,\lambda}$, one has $u^*_\lambda(\theta'_{ij} \circ \theta'_{jk}) = u^*_\lambda(\theta'_{ik})$. There is, hence, by virtue of (i), an index $\mu \geqslant \lambda$ such that $u^*_{\lambda\mu}(\theta'_{ik}) = u^*_{\lambda\mu}(\theta'_{ij} \circ \theta'_{jk})$, hence the isomorphisms $u^*_{\lambda\mu}(\theta'_{ij}) : u^*_{\lambda\mu}(\mathscr{F}_i)|_{\mathrm{U}_{ij\mu}} \xrightarrow{\sim} u^*_{\lambda\mu}(\mathscr{F}_j)|_{\mathrm{U}_{ij\mu}}$ satisfy the gluing condition, and define on S_μ a quasi-coherent $\mathscr{O}_{\mathrm{S}_\mu}$-Module $\mathscr{F}_\mu$ of finite presentation (**0**, 3.3.1) such that $\mathscr{F}$ is isomorphic to $u^*_\mu(\mathscr{F}_\mu)$. □

(8.5.3). The result of (8.5.2) can be expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of $\mathscr{O}_\mathrm{S}$-Modules quasi-coherent of finite presentation is determined up to equivalence near the datum of the categories of $\mathscr{O}_{\mathrm{S}_\lambda}$-Modules quasi-coherent of finite presentation, of the functors $u^*_{\lambda\mu}$ between these categories, and of the transition isomorphisms $u^*_{\mu\nu} \circ u^*_{\lambda\mu} \xrightarrow{\sim} u^*_{\lambda\nu}$. In a figurative way, one can say that the datum of a quasi-coherent $\mathscr{O}_\mathrm{S}$-Module of finite presentation $\mathscr{F}$ is "functorially" equivalent to the datum of a quasi-coherent $\mathscr{O}_{\mathrm{S}_\lambda}$-Module of finite presentation $\mathscr{F}'_\lambda$ for some sufficiently large λ and that, if a quasi-coherent $\mathscr{O}_{\mathrm{S}_\mu}$-Module of finite presentation $\mathscr{F}''_\mu$ also has $\mathscr{F}$ as inverse image, then $\mathscr{F}'_\lambda$ and $\mathscr{F}''_\mu$ have the same inverse image in some suitable S_ν $(\nu \geqslant \lambda, \nu \geqslant \mu)$.

We will now interpret from this point of view various notions related to quasi-coherent $\mathcal{O}_{\mathrm{S}}$-Modules.

Corollary (8.5.4). — *Suppose* S_0 *quasi-compact and quasi-separated; then, for every quasi-coherent* $\mathcal{O}_{\mathrm{S}_\lambda}$-*Module* $\mathcal{G}_\lambda$*, the canonical homomorphism* (8.5.1.4) *is bijective.*

Proof. Indeed, it suffices to apply (8.5.2), (i) to the case where $\mathcal{F}_\lambda = \mathcal{O}_{\mathrm{S}_\lambda}$, which is of finite presentation. ☐

Proposition (8.5.5). — *Suppose* S_0 *quasi-compact, and suppose that* $\mathcal{F}_\lambda$ *is an* $\mathcal{O}_{\mathrm{S}_\lambda}$*-Module quasi-coherent of finite presentation. For* $\mathcal{F}$ *to be locally free (resp. locally free of rank* n*), it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $\mathcal{F}_\mu$ *be so.* **IV-3** | 23

Proof. The condition being trivially sufficient, let us prove that it is necessary. If $\mathcal{F}$ is locally free (resp. locally free of rank n), there exists a finite open affine covering (V_i) of S such that $\mathcal{F}|_{\mathrm{V}_i}$ is isomorphic to $\mathcal{O}_{\mathrm{S}}^{n_i}|_{\mathrm{V}_i}$ (resp. $\mathcal{O}_{\mathrm{S}}^{n}|_{\mathrm{V}_i}$) for every i. By virtue of (8.2.11), there exist $\nu \geqslant \lambda$ and for each i a quasi-compact open subset $\mathrm{U}_{i\nu}$ of S_ν such that $\mathrm{V}_i = u_\nu^{-1}(\mathrm{U}_{i\nu})$. Since S_ν is quasi-compact, each $\mathrm{U}_{i\nu}$ is a finite union of open affines; we are thus reduced to the case where S_0 is affine and $\mathcal{F} = \mathcal{O}_{\mathrm{S}}^n$; we then know that there exist $\mu \geqslant \lambda$ such that $\mathcal{F}_\mu$ is isomorphic to $\mathcal{O}_{\mathrm{S}_\mu}^n$ (8.5.2.5). ☐

Proposition (8.5.6). — *Suppose* S_0 *quasi-compact, and consider a sequence*

$$\mathcal{F}_\lambda \longrightarrow \mathcal{G}_\lambda \longrightarrow \mathcal{H}_\lambda \longrightarrow 0$$

of homomorphisms of $\mathcal{O}_{\mathrm{S}_\lambda}$*-Modules quasi-coherent, where* $\mathcal{F}_\lambda$ *and* $\mathcal{G}_\lambda$ *are of finite type and* $\mathcal{H}_\lambda$ *of finite presentation. For the corresponding sequence* $\mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ *to be exact, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that the sequence* $\mathcal{F}_\mu \to \mathcal{G}_\mu \to \mathcal{H}_\mu \to 0$ *be so (in which case it is the same for* $\nu \geqslant \mu$*).*

Proof. The fact that the condition is sufficient and the last assertion result from the fact that the functor u_λ^* (resp. $u_{\mu\nu}^*$) is right-exact. To prove that the condition is necessary, note that it follows from the hypothesis and from (8.5.2), (i) that there exists $\nu \geqslant \lambda$ such that the composite $\mathcal{F}_\nu \to \mathcal{G}_\nu \to \mathcal{H}_\nu$ is zero. If one sets $\mathcal{H}'_\nu = \mathrm{Coker}(\mathcal{F}_\nu \to \mathcal{G}_\nu)$, one has therefore a homomorphism $f_\nu : \mathcal{H}'_\nu \to \mathcal{H}_\nu$; by hypothesis, $u_\nu^*(f_\nu)$ is an isomorphism, and it follows from (8.5.2.4) that there exists $\mu \geqslant \nu$ such that $u_{\nu\mu}^*(f_\nu)$ is an isomorphism, which completes the proof. ☐

Corollary (8.5.7). — *Suppose* S_0 *quasi-compact,* $\mathcal{F}_\lambda$ *quasi-coherent of finite type, and let* $f_\lambda : \mathcal{F}_\lambda \to \mathcal{G}_\lambda$ *be a homomorphism. For* $f = u_\lambda^*(f_\lambda) : \mathcal{F} \to \mathcal{G}$ *to be surjective, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \to \mathcal{G}_\mu$ *be so.*

Proof. This is the particular case of (8.5.6) applied to the sequence $0 \to \mathcal{H}_\lambda \to 0 \to 0$, where $\mathcal{H}_\lambda = \mathrm{Coker}(f_\lambda)$, which is quasi-coherent and of finite type (taking into account the fact that one has $\mathcal{H} = \mathrm{Coker}(f)$ and $\mathcal{H}_\mu = \mathrm{Coker}(f_\mu)$, by virtue of the right-exactness of the functors u_λ^* and $u_{\lambda\mu}^*$). ☐

Corollary (8.5.8). — *Suppose* S_0 *quasi-compact and the morphisms* $u_{\lambda\mu} : \mathrm{S}_\mu \to \mathrm{S}_\lambda$ *flat. Then:*

label=(i) Let $\mathcal{F}_\lambda \xrightarrow{f_\lambda} \mathcal{G}_\lambda \xrightarrow{g_\lambda} \mathcal{H}_\lambda$ *be a sequence of homomorphisms of* $\mathcal{O}_{\mathrm{S}_\lambda}$*-Modules quasi-coherent, such that* $\mathrm{Im}\, f_\lambda$ *and* $\mathrm{Ker}\, g_\lambda$ *are of finite type. For the corresponding sequence* $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ *to be exact, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that the sequence* $\mathcal{F}_\mu \xrightarrow{f_\mu} \mathcal{G}_\mu \xrightarrow{g_\mu} \mathcal{H}_\mu$ *be so (in which case the same holds for* $\nu \geqslant \mu$*).*

lbbel=(ii) Let $f_\lambda : \mathcal{F}_\lambda \to \mathcal{G}_\lambda$ *be an* $\mathcal{O}_{\mathrm{S}_\lambda}$*-Module homomorphism quasi-coherent, and suppose that* $\mathrm{Im}\, f_\lambda$ *is of finite type. For* $f = u_\lambda^*(f_\lambda) : \mathcal{F} \to \mathcal{G}$ *to be injective, it is necessary and sufficient that there exist* $\mu \geqslant \lambda$ *such that* $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \to \mathcal{G}_\mu$ *be so.*

8.6. Sub-preschemes of finite presentation of a projective limit of preschemes

(8.6.1). Given a prescheme Y, we denote in this section by $\mathfrak{Spr}(Y)$ the ordered set (**I**, 4.1.10) of sub-preschemes of Y that are *of finite presentation* (**I**, 1.6.1), by $\mathfrak{Spro}(Y)$ (resp. $\mathfrak{Sprf}(Y)$) the subset of $\mathfrak{Spr}(Y)$ formed by the sub-preschemes induced on opens (resp. on closed sub-preschemes) of Y, of finite presentation over Y; it amounts to the same to say that a sub-prescheme Z of Y belongs to $\mathfrak{Spro}(Y)$ (resp. $\mathfrak{Sprf}(Y)$) or that it is induced on an open and that the underlying space is *retrocompact* in Y (resp. that it is closed and that the ideal $\mathscr{J}$ of $\mathscr{O}_Y$ that defines it is *of finite type*, which also means that $j_*(\mathscr{O}_Z) \in \mathfrak{Q}(\mathscr{O}_Y)$ where $j : Z \to Y$ is the canonical injection) (**I**, 1.6.1) and (**I**, 1.4.5).

(8.6.2). We always preserve the notations of (8.2.2) and suppose that S_0 is one of the S_λ. Let Y_λ be a sub-prescheme of S_λ; then $Y_\mu = u_{\mu\lambda}^{-1}(Y_\lambda)$ (resp. $Y = u_\lambda^{-1}(Y_\lambda)$) is a sub-prescheme of S_μ for $\mu \geqslant \lambda$ (resp. of S); it is induced on an open (resp. it is closed) and is of finite presentation on S_μ (resp. S) if Y_λ is of finite presentation on S_λ (resp. S) (**I**, 4.3.2). It follows that $(\mathfrak{Spr}(S_\lambda), u_{\lambda\mu}^{-1})$ (resp. $(\mathfrak{Spro}(S_\lambda), u_{\lambda\mu}^{-1})$, $(\mathfrak{Sprf}(S_\lambda), u_{\lambda\mu}^{-1})$) is an *inductive system* of sets and the maps $u_\lambda^{-1} : \mathfrak{Spr}(S_\lambda) \to \mathfrak{Spr}(S)$ (resp. $\mathfrak{Spro}(S_\lambda) \to \mathfrak{Spro}(S)$, $\mathfrak{Sprf}(S_\lambda) \to \mathfrak{Sprf}(S)$) form an inductive system of maps; whence, by passing to the inductive limit, canonical maps

$$\varinjlim \mathfrak{Spr}(S_\lambda) \longrightarrow \mathfrak{Spr}(S) \tag{8.6.2.1}$$

$$\varinjlim \mathfrak{Spro}(S_\lambda) \longrightarrow \mathfrak{Spro}(S) \tag{8.6.2.2}$$

$$\varinjlim \mathfrak{Sprf}(S_\lambda) \longrightarrow \mathfrak{Sprf}(S). \tag{8.6.2.3}$$

Recall (**I**, 4.1.10) that $\mathfrak{Spr}(X)$, for every prescheme X, is an ordered set by the relation "Z is a sub-prescheme of the sub-prescheme Y" which we write $Z \leqslant Y$. The maps $u_{\lambda\mu}^{-1}$ and u_λ^{-1} are *increasing* for the corresponding order relations in $\mathfrak{Spr}(S_\lambda)$, $\mathfrak{Spr}(S_\mu)$, $\mathfrak{Spr}(S)$. Furthermore, by writing that $\eta \leqslant \eta'$ for two elements of this set whenever there exist λ and two elements Y_λ, Y'_λ of $\mathfrak{Spr}(S_\lambda)$, of which η and η' are the canonical images, and which are such that $Y_\lambda \leqslant Y'_\lambda$; one easily verifies that this does not depend on the choice of representatives, and that one thus has indeed a *relation of order*. The fact that the u_λ^{-1} are increasing immediately implies that the canonical map (8.6.2.1) is *increasing*; the same is evidently true of (8.6.2.2) and (8.6.2.3). **IV-3** | 26

Proposition (8.6.3). — *Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Then the maps* (8.6.2.1), (8.6.2.2), (8.6.2.3) *are injective (resp. bijective).*

Proof. Taking into account the remarks of (8.6.1), the assertions relative to (8.6.2.3) follow from (8.5.11) applied to $\mathscr{F}_\lambda = \mathscr{O}_{S_\lambda}$; similarly, the assertions relative to (8.6.2.2) are special cases of (8.3.5) and (8.3.11), taking into account that the S_λ and S are quasi-compact. It remains to consider the map (8.6.2.1). Let us first prove that it is surjective when S_0 is quasi-compact and quasi-separated. Let Z be a sub-prescheme of S, of finite presentation over S; as S is of the same nature as Z, there exists an open quasi-compact U of S such that Z is a sub-prescheme of U, closed in U, of finite presentation over U. There then exist an index λ and an open quasi-compact U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$ (8.2.11); as S_λ is quasi-separated, it is of the same nature as U_λ (**I**, 1.2.7), and consequently one can restrict to the case where $U = S$; but in this case, one is reduced to the fact that (8.6.2.3) is surjective.

Finally, to see that (8.6.2.1) is injective when S_0 is quasi-compact, it will suffice to prove the following more precise result: □

Corollary (8.6.3.1). — *Suppose S_0 quasi-compact and let Z'_λ, Z''_λ be two sub-preschemes of S_λ, of finite presentation. For $Z' = u_\lambda^{-1}(Z'_\lambda)$ to be majorised by $Z'' = u_\lambda^{-1}(Z''_\lambda)$* (**I**, 4.1.10), *it is necessary and sufficient that there exist $\mu \geqslant \lambda$ such that $Z'_\mu = u_{\mu\lambda}^{-1}(Z'_\lambda)$ be majorised by $Z''_\mu = u_{\mu\lambda}^{-1}(Z''_\lambda)$.*

Proof. It is trivial that the condition is sufficient. To see that it is necessary, note first that the underlying sets Z'_λ and Z''_λ are locally constructible in S_λ by hypothesis (**I**, 1.8.4), hence the hypothesis implies the existence of a $\nu \geqslant \lambda$ such that $Z'_\lambda \subset Z''_\lambda$ (8.3.5); replacing λ by ν, we can therefore already suppose that $Z'_\lambda \subset Z''_\lambda$. Furthermore, by hypothesis (**I**, 1.6.1), the sub-spaces Z'_λ and Z''_λ of S_λ are quasi-compact; for every point $x \in Z'_\lambda$, there is an open quasi-compact neighbourhood $V(x)$ in S_λ such that $V(x) \cap Z'_\lambda$ and $V(x) \cap Z''_\lambda$ are closed in $V(x)$. By covering S_λ by a finite number of such neighbourhoods, we see that there is an open quasi-compact U_λ of S_λ containing Z'_λ and such that Z'_λ and $Z''_\lambda \cap U_\lambda$ are closed in U_λ. If we denote by Y_λ the sub-prescheme of S_λ induced by Z''_λ on $U_\lambda \cap Z''_\lambda$, it is clear that with the usual notations, Y_μ (resp. Y) is majorised by Z''_μ (resp. Z'') for $\mu \geqslant \lambda$ (**I**, 4.4.1), and as it suffices to prove that Z'_μ is majorised by Y_μ for μ large enough, we see finally that one is reduced (replacing S_λ by U_λ) to the case where Z'_λ and Z''_λ are closed sub-preschemes of S_λ. But then, this has already been proved since (8.6.2.3) is increasing and injective. □ **IV-3 | 27**

Corollary (8.6.4). — *Suppose S_0 quasi-compact, and let Z_λ be a sub-prescheme of S_λ, of finite presentation over S_λ. For $Z = u_\lambda^{-1}(Z_\lambda)$ to be a sub-prescheme induced on an open (resp. a closed sub-prescheme) of S, it is necessary and sufficient that there exist $\mu \geqslant \lambda$ such that $Z_\mu = u_{\mu\lambda}^{-1}(Z_\lambda)$ be induced on an open (resp. a closed sub-prescheme) of S_μ.*

Proof. Let $(U_\lambda^{(i)})$ be a finite affine open covering of S_λ, and set $U_\mu^{(i)} = u_{\mu\lambda}^{-1}(U_\lambda^{(i)})$ for $\mu \geqslant \lambda$ and $U^{(i)} = u_\lambda^{-1}(U_\lambda^{(i)})$. If Z is open (resp. closed) in S, $Z \cap U^{(i)}$ is open (resp. closed) in $U^{(i)}$, and conversely if each of the $Z_\mu \cap U_\mu^{(i)}$ is open (resp. closed) in $U_\mu^{(i)}$, Z_μ is open (resp. closed) in S_μ. Since L is filtered, it suffices to prove the corollary when S_λ is affine, hence quasi-separated, and the result follows from the fact that the maps (8.6.2.1), (8.6.2.2), and (8.6.2.3) are bijective. □

8.7. Criteria for a projective limit of preschemes to be a reduced (resp. integral) prescheme We preserve the hypotheses and notations of (8.2.2) and suppose always that S_0 is one of the S_λ.

Proposition (8.7.1). — *Suppose that S is not reduced. Then there exists λ_0 such that for $\lambda \geqslant \lambda_0$, S_λ is not reduced.*

Proof. The question being local on S_0, we can suppose $S_0 = \operatorname{Spec}(A_0)$ affine, whence $S = \operatorname{Spec}(A)$, where $A = \varinjlim A_\lambda$ is the inductive limit of an inductive system of A_0-algebras (A_λ). We know then (5.13.2) that the nilradical of A is the inductive limit of those of the A_λ; if it is not zero, one of the A_λ contains therefore a nilpotent element $a_\lambda \neq 0$, and the image of a_λ in the A_μ for $\mu \geqslant \lambda$ is therefore a nilpotent element $\neq 0$, and the image of a_λ in A is not zero for $\mu \geqslant \lambda$ since the image in A is not zero. □

Proposition (8.7.2). — *Suppose one of the following hypotheses satisfied:*

label=a) *S_0 is quasi-compact, the nilradical $\mathscr{N}_0$ of $\mathscr{O}_{S_0}$ is an ideal of finite type (which is satisfied for example when S_0 is Noetherian), and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
lbbel=b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Under these conditions, for S to be reduced, it is necessary and sufficient that there exist λ_0 such that S_λ be reduced for $\lambda \geqslant \lambda_0$.

Proof. Furthermore, in case b), if S is reduced, all the S_λ are reduced.

The last assertion follows from the fact that the morphism $S \to S_\lambda$ is then faithfully flat for every λ (8.3.8) and from (2.1.13). On the other hand, (8.7.1) proves that the condition of the statement is sufficient (without hypothesis on S_0 or on the $u_{\lambda\mu}$). It therefore remains to see that the condition is necessary in hypothesis a); then (8.2.13), the underlying space of S is identified with S (the S_λ being identified with sub-preschemes induced on opens of S_0), and the structural sheaf $\mathscr{O}_S$ is identified with the sheaf induced on S by all the $\mathscr{O}_{S_\lambda}$; in particular for every $s \in S$, the local ring $\mathscr{O}_s$ is the same for S and for all the S_λ. If $\mathscr{N}_\lambda$ is the nilradical of $\mathscr{O}_{S_\lambda}$, the nilradical $\mathscr{N}$ of $\mathscr{O}_S$ has at each point of S the same fibre $\mathscr{N}_s$ (induced on S by $\mathscr{N}_\lambda$). The hypothesis that S is reduced implies therefore $u_\lambda^*(\mathscr{N}_\lambda) = 0$; as $\mathscr{N}_0$ is supposed of finite type, the same is true of the $\mathscr{N}_\lambda$, and the conclusion follows then from (8.5.7). □ **IV-3 | 28**

Corollary (8.7.3). — *Suppose one of the following hypotheses satisfied:*

label=a) *S_0 is a Noetherian prescheme and the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are open immersions.*
lbbel=b) *The morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are faithfully flat.*

Then, for S to be integral, it is necessary and sufficient that there exist λ_0 such that S_λ be integral for $\lambda \geqslant \lambda_0$.

Proof. To say that a prescheme is integral means that it is both reduced and irreducible; the corollary thus follows from (8.7.2) and from (8.4.3). □

Remark (8.7.4). — If one makes no hypothesis on the $u_{\lambda\mu}$, it can happen that S is integral even though all the S_λ are neither reduced nor connected, as the example (8.4.4) shows, where one takes the ring A non reduced.

8.8. Preschemes of finite presentation over a projective limit of preschemes

(8.8.1). Always preserving the notations and hypotheses of (8.2.2), we shall suppose given in this section two S_α-preschemes X_α, Y_α, which defines (8.2.5) two projective systems of preschemes $(X_\lambda, v_{\lambda\mu})$ and $(Y_\lambda, w_{\lambda\mu})$ by setting $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, $w_{\lambda\mu} = 1_{Y_\alpha} \times u_{\lambda\mu}$ (for $\alpha \leqslant \lambda \leqslant \mu$), whose projective limits are respectively $X = X_\alpha \times_{S_\alpha} S$, $Y = Y_\alpha \times_{S_\alpha} S$, with the canonical morphisms $v_\lambda : X \to X_\lambda$ and $w_\lambda : Y \to Y_\lambda$. For $\alpha \leqslant \lambda \leqslant \mu$, there is a canonical map $\varepsilon_{\mu\lambda}$: $\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu)$, which to every S_λ-morphism $f_\lambda : X_\lambda \to Y_\lambda$ associates $f_\mu = f_\lambda \times 1_{S_\mu} : X_\lambda \times_{S_\lambda} S_\mu \to Y_\lambda \times_{S_\lambda} S_\mu$, and it is clear that $(\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda), \varepsilon_{\mu\lambda})$ is an *inductive system* of sets. Similarly, there is a canonical map $\varepsilon_\lambda : \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \to \mathrm{Hom}_S(X, Y)$ which to f_λ associates $f = f_\lambda \times 1_S : X_\lambda \times_{S_\lambda} S \to Y_\lambda \times_{S_\lambda} S$ and (ε_λ) is an inductive system of maps; whence, by passing to the inductive limit, a canonical map, functorial in S_α, X_α, and Y_α:

$$\varepsilon : \varinjlim \mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Hom}_S(X, Y). \tag{8.8.1.1}$$

Theorem (8.8.2). — *label=(i) Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and Y_α locally of finite type (resp. locally of finite presentation) over S_α. Then the map* (8.8.1.1) *is injective (resp. bijective).*

lbbel=(ii) Suppose S_0 quasi-compact and quasi-separated. For every prescheme X of finite presentation over S, there exist a $\lambda \in L$, a prescheme X_λ of finite presentation over S_λ, and an S-isomorphism $X \xrightarrow{\sim} X_\lambda \times_{S_\lambda} S$.

Proof. (i) Consider first the case where $S_0 = \mathrm{Spec}(A_0)$, $X_\alpha = \mathrm{Spec}(B_\alpha)$, and $Y_\alpha = \mathrm{Spec}(C_\alpha)$ are affine; then the $S_\lambda = \mathrm{Spec}(A_\lambda)$ and $S = \mathrm{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$, and the assertions of (i) are equivalent to the **IV-3 | 29**

Lemma (8.8.2.1). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0-algebras, $A = \varinjlim A_\lambda$; let B_α be an A_α-algebra, C_α an A_α-algebra of finite type (resp. of finite presentation). Then the canonical homomorphism*

$$\varinjlim_\lambda \mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \longrightarrow \mathrm{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \tag{8.8.2.2}$$

is injective (resp. bijective).

Proof. We know that there are functorial canonical isomorphisms

$$\mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \xrightarrow{\sim} \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A_\lambda)$$

$$\mathrm{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A) \xrightarrow{\sim} \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, B_\alpha \otimes_{A_\alpha} A)$$

by virtue of the universal property of the tensor product of two algebras. It therefore suffices to prove the □

Lemma (8.8.2.3). — *Let E be a ring, G an E-algebra of finite type (resp. of finite presentation), (F_λ) an inductive system of E-algebras. Then the canonical homomorphism*

$$\varinjlim \mathrm{Hom}_{E\text{-alg.}}(G, F_\lambda) \longrightarrow \mathrm{Hom}_{E\text{-alg.}}(G, \varinjlim F_\lambda)$$

which, to every inductive system of homomorphisms $\theta_\lambda : G \to F_\lambda$ of E-algebras, associates its inductive limit, is injective (resp. bijective).

Proof. Suppose first the E-algebra G of finite type, and let $(t_i)_{1 \leqslant i \leqslant n}$ be a system of generators of this E-algebra; let us show that if (θ'_λ), (θ''_λ) are two inductive systems of homomorphisms $G \to F_\lambda$, with $\varinjlim \theta'_\lambda = \varinjlim \theta''_\lambda$, there exists $\mu \geqslant \lambda$ such that $\theta'_\mu = \theta''_\mu$. Indeed, if $\varphi_{\mu\lambda} : F_\lambda \to F_\mu$ and $\varphi_\lambda : F_\lambda \to F = \varinjlim F_\lambda$ are the homomorphisms of the inductive system (F_λ), by hypothesis, for each index i, there exists an index λ_i such that $\varphi_{\lambda_i}(\theta'_{\lambda_i}(t_i)) = \varphi_{\lambda_i}(\theta''_{\lambda_i}(t_i))$, and one can suppose all the λ_i equal to a single λ; it follows in the same way the existence of $\mu \geqslant \lambda$ such that $\varphi_{\mu\lambda}(\theta'_\lambda(t_i)) = \varphi_{\mu\lambda}(\theta''_\lambda(t_i))$ for $1 \leqslant i \leqslant n$, that is to say $\theta'_\mu(t_i) = \theta''_\mu(t_i)$ for $1 \leqslant i \leqslant n$, whence $\theta'_\mu = \theta''_\mu$.

Suppose in the second place G of finite presentation, so that one has

$$G = E[T_1, \ldots, T_n]/\mathfrak{J},$$

where $\mathfrak{J}$ is an ideal of finite type, t_i being the class of T_i modulo $\mathfrak{J}$. Let $(P_j)_{1 \leqslant j \leqslant m}$ be a system of generators of $\mathfrak{J}$. Suppose given a homomorphism of E-algebras $\theta : G \to F$; set $b_i = \theta(t_i)$; by definition, one has therefore $P_j(b_1, b_2, \ldots, b_n) = \theta(P_j(t_1, \ldots, t_n)) = 0$ for $1 \leqslant j \leqslant m$. Now, there exist λ and elements $x_1, \ldots, x_n$ in F_λ such that $b_i = \varphi_\lambda(x_i)$ for $1 \leqslant i \leqslant n$; one has therefore $\varphi_\lambda(P_j(x_1, \ldots, x_n)) = P_j(b_1, \ldots, b_n) = 0$, and consequently there exists $\mu \geqslant \lambda$ such that **IV-3 | 30**
$\varphi_{\mu\lambda}(P_j(x_1, \ldots, x_n)) = P_j(\varphi_{\mu\lambda}(x_1), \ldots, \varphi_{\mu\lambda}(x_n)) = 0$ for $1 \leqslant j \leqslant m$; one therefore deduces for every $\nu \geqslant \mu$ a homomorphism of E-algebras $\theta_\nu = \varphi_{\nu\mu} \circ \theta_\mu$ of G into F_ν, and it is clear that θ is the inductive limit of this inductive system of homomorphisms, which completes the proof of the lemma. □

Let us now pass to the case where X_α is quasi-compact and Y_α locally of finite type over S_α. Set $Z_\alpha = X_\alpha \times_{S_\alpha} Y_\alpha$ and introduce the projective system corresponding to $Z_\lambda = X_\lambda \times_{S_\lambda} Y_\lambda$ and its limit $Z = X \times_S Y$; the canonical bijections (**I**, 3.3.14) give commutative diagrams

$$\begin{tikzcd}
\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \arrow[r, "\varepsilon_{\mu\lambda}"] \arrow[d, "\wr"] & \mathrm{Hom}_{S_\mu}(X_\mu, Y_\mu) \arrow[r, "\varepsilon_\mu"] \arrow[d, "\wr"] & \mathrm{Hom}_S(X, Y) \arrow[d, "\wr"] \\
\mathrm{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \arrow[r, "\varepsilon_{\mu\lambda}"] & \mathrm{Hom}_{X_\mu}(X_\mu, Z_\mu) \arrow[r, "\varepsilon_\mu"] & \mathrm{Hom}_X(X, Z)
\end{tikzcd}$$

and consequently we are reduced to proving that (8.8.1.1) is injective in the particular case where $S_\alpha = X_\alpha$ (taking into account (**I**, 1.3.4)). Moreover, as X_α is quasi-compact, hence a finite union of open affines, and as X is a union of the opens $f^{-1}(V_i)$, where the V_i are open affines of Y_α, every point of X has an open affine neighbourhood $W(x)$ in S_α such that f'_α and f''_α both map $W(x)$ into a single open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of open affines $W(x_k)$; this implies that for each $x \in X_\alpha$, there is an open affine neighbourhood of x such that the restrictions of f'_α and f''_α to $W(x)$ map it into the same open affine $V_{i\alpha}$. As X_α is quasi-compact, it is covered by a finite number of such open affines $W(x_k)$; by virtue of the lemma (8.8.2.1) and of the fact that L is filtered, there exists $\lambda \geqslant \alpha$ such that the restrictions of f'_λ and f''_λ to each of the opens $v_\alpha^{-1}(W(x_k))$ are equal, whence $f'_\lambda = f''_\lambda$.

Suppose now X_α quasi-compact and quasi-separated and Y_α locally of finite presentation over S_α, and let us prove that (8.8.1.1) is surjective. Suppose therefore given an X-morphism $f : X \to Y$. As X is quasi-compact, it is of the same nature as Y, which contains $f(X)$; there exists therefore $\lambda \geqslant \alpha$ and an open quasi-compact Y'_λ in Y_λ such that $Y' = w_\lambda^{-1}(Y'_\lambda)$ (8.2.11). Replacing as needed α by λ and Y_λ by Y'_λ, we can therefore restrict to the case where Y_α is *quasi-compact*, hence also Y and the Y_λ. As Y is quasi-compact, it is a finite union of open affines V_i, and X is a union of the opens $f^{-1}(V_i)$. As X is quasi-compact, we can cover X by a finite number of such open affines. And **IV-3 | 31**
since X is quasi-compact, for each i, the open quasi-compact sets in X_α may, by replacing α by an index $\lambda \geqslant \alpha$, be supposed such that $U_i = v_\alpha^{-1}(U_{i\alpha})$ where the $U_{i\alpha}$ are open quasi-compacts in X_α; moreover, using (8.3.4), on can moreover suppose the $(U_{i\alpha})$ form a *covering* of X_α.

It will then suffice to show that there exist $\lambda \geqslant \alpha$ and for each i a morphism $f_{i\lambda} : U_{i\lambda} \to V_{i\lambda}$ (where $U_{i\lambda} = v_{\alpha\lambda}^{-1}(U_{i\alpha})$ and $V_{i\lambda} = w_{\alpha\lambda}^{-1}(V_{i\alpha})$) such that the morphism $f_i = \varepsilon_\lambda(f_{i\lambda}) : U_i \to V_i$ be equal to the *restriction* of f to U_i. Indeed, if this is so, as X_α is quasi-separated (**I**, 1.2.3), the $U_{i\lambda} \cap U_{j\lambda}$ are *quasi-compact* and the uniqueness result proved above (which applies since Y_λ is locally of finite type over S_λ) proves that there exists $\mu \geqslant \lambda$ such that $f_{i\mu}$ and $f_{j\mu}$ coincide on $U_{i\mu} \cap U_{j\mu}$ for every couple (i, j), and consequently define a S_μ-morphism $f_\mu : X_\mu \to Y_\mu$ such that $\varepsilon_\mu(f_\mu) = f$.

We are thus reduced to the case where Y_α is *affine*, and as in addition we can also suppose that S_α is affine, we can also suppose $S_\alpha = \mathrm{Spec}(A_\alpha)$, $Y_\alpha = \mathrm{Spec}(C_\alpha)$, C_α being an A_α-algebra of finite presentation, $S = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(C)$, with $A = \varinjlim(C_\alpha \otimes_{A_\alpha} A_\lambda) = C_\alpha \otimes_{A_\alpha} A$. One has then

$$\mathrm{Hom}_S(X, Y) = \mathrm{Hom}_{A\text{-alg.}}(C, \Gamma(X, \mathscr{O}_X)) = \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X, \mathscr{O}_X))$$

(**I**, 2.2.4) and similarly

$$\mathrm{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) = \mathrm{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, \Gamma(X_\lambda, \mathscr{O}_{X_\lambda})) = \mathrm{Hom}_{A_\alpha\text{-alg.}}(C_\alpha, \Gamma(X_\lambda, \mathscr{O}_{X_\lambda})).$$

But as X_α is quasi-compact and quasi-separated, one knows (8.5.4) that $\varinjlim \Gamma(X_\lambda, \mathscr{O}_{X_\lambda}) = \Gamma(X, \mathscr{O}_X)$; since C_α is an A_α-algebra of finite presentation, the fact that (8.8.1.1) is bijective then follows from (8.8.2.3).

Before passing to the proof of (ii), let us note the following corollaries of (i): □

Corollary (8.8.4). — *Suppose S_0 quasi-compact, X_α of finite presentation over S_α, Y_α of finite type over S_α and quasi-separated over S_α (which is the case for example if Y_α is also of finite presentation over S_α). Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. For $f : X \to Y$ to be an isomorphism, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to Y_\lambda$ be an isomorphism.* IV-3 | 32

Proof. The condition is evidently sufficient; conversely, note that since the question is local on S_0 (since L is filtered), one can always suppose S_0 affine, hence quasi-separated. There is by hypothesis an S-morphism $g : Y \to X$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$. On the other hand, it follows also from (8.8.2), (i) and from the relations $g \circ f = 1_X$ and $f \circ g = 1_Y$ that there exists $\nu \geqslant \mu$ such that $g_\nu \circ f_\nu = 1_{X_\nu}$ and $f_\nu \circ g_\nu = 1_{Y_\nu}$, since X_α and Y_α are of finite type over S_α and quasi-compact. This means that $f_\nu : X_\nu \to Y_\nu$ is an isomorphism, whence the corollary. □

Corollary (8.8.5). — *Suppose S_0 quasi-compact and quasi-separated, X_α and Y_α of finite presentation over S_α. For every S-isomorphism $f : X \xrightarrow{\sim} Y$, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that X_λ and Y_λ be S_λ-isomorphic, and an S_μ-isomorphism $f_\mu : X_\mu \to Y_\mu$ such that $f = f_\mu \times 1_S$.*

Proof. The condition is evidently sufficient; conversely, note that since the question is local on S_0 (since L is filtered), one can always suppose S_0 affine, hence quasi-separated. There is by hypothesis an S-morphism $g : Y \to X$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$. As X_α is of finite presentation over S_α, and Y_α quasi-compact and quasi-separated, it results also from (8.8.2), (i) that f is of the form $f_\mu \times 1_S$ for some $\mu \geqslant \alpha$ and an S_μ-morphism $f_\mu : X_\mu \to Y_\mu$. Similarly for g. Applying (8.8.2.4) one obtains that there exists $\nu \geqslant \mu$ such that $g_\nu \circ f_\nu = 1_{X_\nu}$ and $f_\nu \circ g_\nu = 1_{Y_\nu}$, and the conclusion follows. □

Proof of (8.8.2), (ii). Consider first the case where $S_0 = \operatorname{Spec}(A_0)$ and $X = \operatorname{Spec}(B)$ are affine; then the $S_\lambda = \operatorname{Spec}(A_\lambda)$ and $S = \operatorname{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$. As X is of finite presentation over S, B is an A-algebra of finite presentation, hence by (8.8.2.3) applied to $G = B$ and $F_\lambda = A_\lambda$, there exist λ and an A_λ-algebra B_λ of finite presentation such that $B \simeq B_\lambda \otimes_{A_\lambda} A$. Setting $X_\lambda = \operatorname{Spec}(B_\lambda)$, we obtain an S_λ-prescheme of finite presentation such that $X \simeq X_\lambda \times_{S_\lambda} S$.

To prove (ii) in the general case, note that S is quasi-compact and quasi-separated, and X is of finite presentation over S. As X is quasi-compact, it is a finite union of open affines, and by the affine case there exist λ and affine S_λ-preschemes of finite presentation that descend each piece. By (8.8.2), (i) and the fact that L is filtered, we can glue these pieces for a large enough λ to obtain X_λ of finite presentation over S_λ with $X \simeq X_\lambda \times_{S_\lambda} S$. □

(8.8.3). The essential content of (8.8.2) is expressed again by saying that if S_0 is quasi-compact and quasi-separated, the category of S-preschemes of finite presentation is determined up to equivalence by the data of the categories of S_λ-preschemes of finite presentation, the functors $\rho_{\mu\lambda} : X_\lambda \mapsto X_\lambda \times_{S_\lambda} S_\mu$ (for $\lambda \leqslant \mu$) between these categories and the transitivity isomorphisms $\rho_{\mu\lambda} \circ \rho_{\nu\mu} \xrightarrow{\sim} \rho_{\nu\lambda}$. In the form thus stated, one can say that the data of an S_λ-prescheme of finite presentation X is equivalent "functorially" to the data of an S_μ-prescheme of finite presentation X'_μ; if an S_μ-prescheme X''_μ is such that X is S-isomorphic to $X''_\mu \times_{S_\mu} S$, there exists ν such that $\lambda \leqslant \nu$ and $X'_\lambda \times_{S_\lambda} S_\nu$ and $X''_\mu \times_{S_\mu} S_\nu$ are S_ν-isomorphic. The fact that L is filtered implies in addition that if one gives oneself a finite *family* $(X^{(i)})_{i \in I}$ of S-preschemes of finite presentation, and a finite family $(f^{(ij)})_{(i,j) \in J}$ of S-morphisms between these preschemes ($f^{(ij)}$ being therefore of the form $X^{(\sigma(j))} \to X^{(\tau(j))}$ where σ and τ are two maps from J to I), then there is an index $\lambda \in L$, a family $(X^{(i)}_\lambda)_{i \in I}$ of S_λ-preschemes of finite presentation and a family $(f^{(ij)}_\lambda)_{(i,j) \in J}$ of S_λ-morphisms such that $X^{(i)}$ is identified with $X^{(i)}_\lambda \times_{S_\lambda} S$ and $f^{(ij)}$ with $f^{(ij)}_\lambda \times 1_S$; furthermore, if one has a relation $f^{(ij)} = f^{(ik)}$, one can suppose λ chosen such that $f^{(ij)}_\lambda = f^{(ik)}_\lambda$.

Consider in particular three S-preschemes of finite presentation X, Y, Z and two S-morphisms $f : X \to Y$, $g : Y \to Z$, so that the product $X \times_Z Y$ relative to these morphisms is also an S-prescheme of finite presentation (**I**, 1.6.2). It then results from the preceding and from (**I**, 3.3.11) that one can write $X \times_Z Y = (X_\lambda \times_{Z_\lambda} Y_\lambda) \times_{S_\lambda} S$ for a convenient λ, X_λ, Y_λ, Z_λ being S_λ-preschemes of finite presentation; one can therefore say that the determination of S-preschemes of finite presentation by the data of S_λ-preschemes of finite presentation is "*compatible with fibre products*". We have seen on the other hand (8.6.3) that if g is an immersion, we can suppose the same for g_λ; identifying Y (resp. Y_λ) with a sub-prescheme of Z (resp. Z_λ), we see that one can write, for a convenient λ, $f^{-1}(Y) = f_\lambda^{-1}(Y_\lambda) \times_{S_\lambda} S$ (**I**, 4.4.1); there is therefore also "*compatibility with the formation of inverse* **IV-3** | 33 *images of sub-preschemes*". More particularly, if f_1, f_2 are two S-morphisms from X to Y, one calls the *kernel* of this pair of morphisms the inverse image N of the diagonal sub-prescheme of $Y \times_S Y$ by the S-morphism $(f_1, f_2)_S$; one deduces from what precedes that one will have for a convenient λ, $N = N_\lambda \times_{S_\lambda} S$, where N_λ is the kernel of the couple of S_λ-morphisms $(f_{1\lambda}, f_{2\lambda})$ of X_λ into Y_λ. These remarks extend to finite products and to "kernels" of arbitrary finite systems of S-morphisms from an S-prescheme of finite presentation into another; one concludes that in a general fashion the **IV-3** | 34 formation of S-preschemes of finite presentation by the data of S_λ-preschemes of finite presentation is "*compatible with finite projective limits*", such a limit being by definition the kernel of a finite system of morphisms from one S-prescheme into a product of S-preschemes (**T**, 1.8).

We shall allow ourselves in what follows to freely make the translations implied by the preceding properties (as well as by (8.3.11), (8.5.2), and (8.6.3)) without always striving to justify them as above. For example, the data of a "group prescheme" G of finite presentation over S is equivalent to the data of a group prescheme on S_λ for a large enough λ; to write indeed the associativity condition for the S-morphism "composition law" $G \times_S G \to G$ of the group prescheme amounts to writing that the kernel of the two S-morphisms from $G \times_S G \times_S G \to G$ is equal to $G \times_S G \times_S G$ (**II**, 8.3.9), and the other conditions intervening in the definition of a group prescheme are interpreted similarly.

8.9. First applications to the elimination of Noetherian hypotheses

Proposition (8.9.1). — *Let A be a ring, X an A-prescheme.*

label=(i) The following conditions are equivalent:

- *label=a) X is of finite presentation over A.*
- *lbbel=b) There exist a Noetherian ring A_0, a prescheme X_0 of finite type over A_0, a ring homomorphism $A_0 \to A$, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
- *lcbel=c) There exist a subring A_0 of A, which is a* **Z***-algebra of finite type, a prescheme X_0 of finite type over A_0, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*

lbbel=(ii) If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, one can suppose the subring A_0 such that there exists a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{A_0} A$; $\operatorname{Supp}(\mathscr{F})$ *is constructible and closed in X, and there is a closed sub-prescheme Z of X, having* $\operatorname{Supp}(\mathscr{F})$ *as underlying space, such that the canonical immersion $Z \to X$ be of finite presentation.*

lcbel=(iii) If Y is a second A-prescheme of finite presentation, and $f : X \to Y$ an A-morphism, one can suppose the subring A_0 of A such that there exists a prescheme Y_0 of finite type over A_0, an A-isomorphism $Y_0 \otimes_{A_0} A \xrightarrow{\sim} Y$, and an A_0-morphism $f_0 : X_0 \to Y_0$ (necessarily of finite type) such that f is identified with $f_0 \otimes 1_A$.

Proof. label=(i) As A is the inductive limit of its subrings which are of finite type over **Z**, *a)* implies *c)* by virtue of (8.8.2), (ii); *c)* implies *b)* since a **Z**-algebra of finite type is a Noetherian ring; finally, if A_0 is Noetherian, an A_0-prescheme of finite type is of finite presentation (**I**, 1.6.1), hence *b)* implies *a)* by virtue of (**I**, 1.6.2), (iii).

The assertion (ii) follows immediately from (8.5.2), (ii), (8.3.11), and (**I**, 1.8.2), and the assertion (iii) from (8.8.2), (i). □

(8.9.2). The proposition (8.9.1) and the results of (8.5.2) and (8.8.2) allow one to extend in numerous cases to morphisms of finite presentation $X \to Y$ the results proved by Noetherian techniques by supposing Y *locally Noetherian*. We shall see numerous examples in what follows and we shall limit **IV-3** | 35 ourselves here to giving a few results of this type.

Proposition (8.9.3). — *Let A be a ring, M an A-module of finite presentation. Then every surjective A-endomorphism of M is bijective.*

Proof. Indeed, consider A as an inductive limit of its sub-$\mathbf{Z}$-algebras of finite type. It results from (8.5.2.6) that there exists one of these sub-algebras A_0 and an A_0-module M_0 of finite presentation such that M is A-isomorphic to $M_0 \otimes_{A_0} A$; furthermore, if $f : M \to M$ is a surjective A-endomorphism, one can suppose (8.5.2), (i) that there exists an A_0-endomorphism $f_0 : M_0 \to M_0$ such that $f = f_0 \otimes 1_A$; finally (8.5.7) on can suppose that f_0 is *surjective*. But as A_0 is Noetherian and M_0 an A_0-module of finite type, M_0 is a Noetherian module, hence (Bourbaki, *Alg.*, chap. VIII, § 2, nº 2, lemma 3) f_0 is bijective, and it is the same for f by base change. □

Proposition (8.9.4). — *("generic flatness theorem"). — Let Y be an integral prescheme, $u : X \to Y$ a morphism of finite type and locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists an open non-empty U of Y such that $\mathscr{F}|u^{-1}(U)$ be flat over U.*

Proof. The reasoning from the beginning of (6.9.1) leads to proving the

Lemma (8.9.4.1). — *Let A be an integral ring, B an A-algebra of finite presentation, M a B-module of finite presentation. Then there exists $f \neq 0$ in A such that M_f be an A_f-module free.*

Proof. Indeed, by (8.9.1) there is a sub-$\mathbf{Z}$-algebra of finite type A_0 of A, a B_0-algebra of finite type and a B_0-module of finite type M_0 such that B is A-isomorphic to $B_0 \otimes_{A_0} A$ and that M is B-isomorphic to $M_0 \otimes_{A_0} A$; by (6.9.2), there exists $f_0 \neq 0$ in A_0 such that $(M_0)_{f_0}$ is an $(A_0)_{f_0}$-module free. As $M_{f_0} = (M_0)_{f_0} \otimes_{A_0} A$ and $A_{f_0} = (A_0)_{f_0} \otimes_{A_0} A$, M_{f_0} is an A_{f_0}-module free. □

□

Corollary (8.9.5). — *Let S be a quasi-compact prescheme and quasi-separated, $u : X \to S$ a morphism of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then there exists a partition $(S_i)_{1 \leqslant i \leqslant n}$ of S into a finite number of locally closed sets in S such that, for $1 \leqslant i \leqslant n$, there exists a sub-prescheme S'_i of S, having S_i as underlying space, of finite presentation over S_i, such that if one sets $X_i = X \times_S S'_i$, the $\mathscr{O}_{X_i}$-Module $\mathscr{F}_i = \mathscr{F} \otimes_{e_S} \mathscr{O}_{S'_i}$ be flat over S'_i.*

Proof. Consider a finite covering $(U_j)_{1 \leqslant j \leqslant m}$ of S by open affines and define by induction $T_1 = U_1$, $T_j = U_j - \bigcup_{h<j}(U_j \cap U_h)$ for $j \geqslant 2$; each T_j is closed in the open affine U_j, and the T_j are pairwise disjoint; furthermore the T_j are also quasi-compact since S is quasi-separated, and consequently (since S is also quasi-compact), they are *constructible* in S (**I**, 1.8.1), and hence also each of the T_j. It will therefore evidently suffice to prove the corollary for a sub-prescheme T'_j of S having T_j as underlying space and for the morphism and the Module deduced respectively from u and $\mathscr{F}$ by the base change $T'_j \to S$. Note for this that if on U_j one takes the prescheme structure induced by that of S, the open immersion $U_j \to S$ is quasi-separated since S is quasi-separated (**I**, 1.2.7), hence it is of finite presentation. Then (8.9.1), with the same notations, there exist λ, a morphism of finite type $u_\lambda : X_\lambda \to S_\lambda$ and an $\mathscr{O}_{X_\lambda}$-Module coherent $\mathscr{F}_\lambda$ such that X is isomorphic to $X_\lambda \times_{S_\lambda} S$, $\mathscr{F}$ to $\mathscr{F}_\lambda \otimes_{e_{S_\lambda}} \mathscr{O}_X$. One can then apply to S_λ, X_λ, and $\mathscr{F}_\lambda$ the result of (6.9.3) and it follows that there is a partition of S_λ into sub-preschemes $S'_{\lambda i}$ of S_λ, whose underlying sets form a partition of S_λ, and which are such that $\mathscr{F}_M = \mathscr{F}_\lambda \otimes_{e_{S_\lambda}} \mathscr{O}_{S'_{\lambda M}}$ be flat over $S'_{\lambda M}$. The $S'_i = S'_{\lambda i} \times_{S_\lambda} S$ are then sub-preschemes of S satisfying the condition by virtue of (2.1.4). □ **IV-3 | 36**

8.10. Properties of permanence of morphisms by passage to the projective limit In this section, we preserve the general hypotheses and the notations of (8.8.1).

Proposition (8.10.1). — *If there exists λ such that, for $\mu \geqslant \lambda$, f_μ is an open morphism (resp. universally open), then f is an open morphism (resp. universally open).*

Proof. As $X = X_\lambda \times_{Y_\lambda} Y$, the assertion relative to universally open morphisms is a particular case of (2.4.3), (iii). Let us therefore suppose f_μ open for $\mu \geqslant \lambda$; it suffices to see that for every open *quasi-compact* U of X, $f(U)$ is open in Y. Now, there exists $\mu \geqslant \lambda$ such that $U = v_\mu^{-1}(U_\mu)$, where U_μ is an open quasi-compact in X_μ (8.2.11); one has then $f(v_\mu^{-1}(U_\mu)) = w_\mu^{-1}(f_\mu(U_\mu))$ (**I**, 3.4.8), hence $f(U)$ is open in Y. □

Corollary (8.10.2). — *Let $f : X \to Y$ be a morphism. For f to be universally open, it suffices that, for every integer $n > 0$, if one sets $Y_n = Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_n](= \mathbf{V}_Y^n)$, and $X_n = X \times_Y Y_n$, $f_n = f \times 1_{Y_n} : X_n \to Y_n$, the projection f_n be an open morphism.*

Proof. To prove that f is universally open, it suffices to prove that it remains so for every affine U of Y; by hypothesis, if $U_n = U \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \ldots, T_n]$ is the inverse image of U in Y_n, the morphism $f_n^{-1}(U_n) \to U_n$, restriction of f_n, is open, and one sees that one can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine. Furthermore, it suffices evidently to show that for every morphism $Y' \to Y$, where $Y' = \operatorname{Spec}(A')$ is also *affine*, $f' = f_{|Y'}$ is open. Let us suppose first that A' is an A-algebra of finite type, hence a *quotient* of a polynomial algebra $A[T_1, \ldots, T_n]$; then Y' is a closed sub-prescheme of Y_n and f' the restriction of f_n to $f_n^{-1}(Y')$; but for every open V of X_n, one has $f_n(V \cap f_n^{-1}(Y')) =$ **IV-3** | 37
$f_n(V) \cap Y'$, and as by hypothesis $f_n(V)$ is open in Y_n, this shows that f' is also an open morphism. When A' is arbitrary, it can be considered as an inductive limit of its sub-A-algebras of finite type A'_λ, and the fact that f' is an open morphism follows from the preceding and from (8.10.1). □

Proposition (8.10.3). — *Suppose that there exists α such that: 1° S_α is quasi-compact; 2° the morphisms $X_\alpha \to S_\alpha$, $Y_\alpha \to S_\alpha$ are quasi-compact and the morphism $Y_\alpha \to S_\alpha$ quasi-separated; 3° for $\alpha \leqslant \lambda \leqslant \mu$, the morphisms $u_{\lambda\mu} : S_\mu \to S_\lambda$ are flat; 4° $f_\alpha(X_\alpha)$ is constructible in Y_α. Then, for f to be dominant, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ be dominant.*

Proof. The hypotheses imply that Y_α is quasi-compact and that the morphism f_α is quasi-compact (**I**, 1.2.4); hence $f_\alpha(X_\alpha) = Z_\alpha$ is pro-constructible (**I**, 1.9.5), (vii) in Y_α. If one sets $Z_\lambda = w_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geqslant \alpha$ and $Z = w_\alpha^{-1}(Z_\alpha)$, on has $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (**I**, 3.4.8). As Z_λ is pro-constructible in Y_λ (8.3.11), the canonical map $\varinjlim \mathfrak{C}(S_\lambda) \to \mathfrak{C}(S)$ is injective, which proves the theorem in the case (vi). It suffices then to apply (8.3.12) by replacing S_λ, Z'_λ, and Z''_λ by Y_λ, Y_λ, and Z_λ respectively. □

Proposition (8.10.4). — *Suppose that there exists α such that Y_α is quasi-compact of finite type and f_α of type fini and quasi-separated. For the morphism f to be separated, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ be separated.*

Proof. The question being local on Y_α (since Y_α is quasi-compact and L filtered), one can restrict to the case where Y_α is affine, hence quasi-separated, and the hypothesis implies that X_α (hence the X_λ and X) are quasi-compact and quasi-separated. Set $X'_\lambda = X_\lambda \times_{Y_\lambda} X_\lambda$ for $\lambda \geqslant \alpha$ and $X' = X \times_Y X$; the first projection $X'_\alpha \to X_\alpha$ is quasi-compact and quasi-separated by hypothesis (**I**, 1.2.3), (iii), hence X'_λ is quasi-compact and quasi-separated. Note now that if we denote by Δ_λ (resp. Δ) the diagonal of $X_\lambda \times_{Y_\lambda} X_\lambda$ (resp. $X \times_Y X$), it results from (**I**, 5.3.4) and (3.4.8) that Δ_λ is the inverse image of Δ_α by the map $v'_{\lambda\alpha} : X'_\lambda \to X'_\alpha$ (resp. $v'_\lambda : X' \to X'_\alpha$). On the other hand, Δ_α is *constructible* in X'_α: indeed, since f_α is quasi-separated, the diagonal immersion $X_\alpha \to X'_\alpha$ is quasi-compact, and locally of finite presentation since f_α is of finite type (**I**, 1.4.3) and (**I**, 1.5.4), (iii); it results then from (**I**, 1.8.4.1) that Δ_λ is locally constructible, hence constructible since X'_λ is quasi-compact. One can now apply (8.3.12) by replacing S_λ and Z_λ by X'_λ and Δ_λ respectively. □

Theorem (8.10.5). — *Suppose S_0 quasi-compact, X_α and Y_α of finite presentation over S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Consider, for a morphism, the property of being:*

label=(i) an isomorphism;
(i bis) a monomorphism;
lbbel=(ii) an immersion;
lcbel=(iii) an open immersion;
ldbel=(iv) a closed immersion;
lebel=(v) separated;
lfbel=(vi) surjective; **IV-3** | 38
lgbel=(vii) radicial;
lhbel=(viii) affine;
libel=(ix) quasi-affine;
ljbel=(x) finite;
lkbel=(xi) quasi-finite;
llbel=(xii) proper.

Then, if **P** *denotes one of the preceding properties, for f to possess the property* **P***, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ possess the property* **P** *(in which case f_μ also possesses it for $\mu \geqslant \lambda$).*

If S_0 is furthermore supposed quasi-separated, the same conclusion holds when **P** *is the property of being:*

lmbel=(xiii) projective;
lnbel=(xiv) quasi-projective.

Proof. The case where **P** is one of the properties (i) or (v) was not inserted in the statement for the record; in these cases, the theorem follows from what has been proved respectively in (8.8.4) and (8.10.4). Furthermore, taking into account (**I**, 5.4.1) and (5.3.4), (v) also results from (iv). The case (i *bis*) follows immediately from (i), by using (**I**, 5.3.8) and by noting (as we have already used in (8.10.4)) that the diagonal Δ_f is deduced from Δ_{f_λ} by the base change $S \to S_\lambda$.

We will note on the other hand that (vi), (vii), and (xi) are in fact conditions on the *fibres* $f^{-1}(y)$ of the morphisms considered, taking into account the transitivity of the fibres by base change (**I**, 3.6.4): condition (vi) means indeed that all the fibres must be non-empty, condition (vii) that they must be radicial (**I**, 3.5.8), and condition (xi) that they must be finite (**II**, 6.2.2) and (6.2.3) and (**I**, 6.4.4), taking into account that f and the f_λ are morphisms of finite type by (**I**, 1.5.4), (v). The theorem in these three cases will be yet another consequence of a general result on this type of properties, which will be established in (9.3.3); we therefore postpone until then the proof of the theorem in the case (xi) (of course, the reader will be able to verify that, except for n$^{\text{os}}$ 8.11 and 8.12, we will make no use of the theorem in these cases until (9.3.3) and that (8.11) and (8.12) will not be used before (9.3.3)).

For the cases that remain to be proved, one can restrict to proving that the condition of the statement is necessary, all the properties **P** considered being invariant by base change (see chap. I$^{\text{er}}$ and II for the references concerning each of these properties). One can furthermore suppose that $S_\alpha = S_0$ and that $Y_\alpha = S_\alpha$, hence $Y_\lambda = S_\lambda$ for every $\lambda \geqslant \alpha$. Finally, the properties (i) through (xii) are local on S_0, hence as S_0 is a finite union of open affines and L filtered, one can restrict to proving them for $S_0 = \operatorname{Spec}(A_0)$ *affine* (hence quasi-separated). We shall denote by A_λ (resp. A) the ring of S_λ (resp. S).

Cases (ii), (iii), (iv): Suppose that f is an immersion (resp. an open immersion, a closed immersion), and let X' be the sub-prescheme (resp. induced on an open, resp. closed) of S associated to f, which is therefore an S-prescheme of finite presentation. By virtue of (8.6.3), there then exists $\lambda \geqslant \alpha$ and a sub-prescheme X'_λ of S_λ, of finite presentation over S_λ, such that X' is isomorphic to $X'_\lambda \times_{S_\lambda} S$. For every $\mu \geqslant \lambda$, $X'_\mu = X'_\lambda \times_{S_\lambda} S_\mu$ is therefore a sub-prescheme (resp. induced on an open, resp. closed) of S_μ, and it results therefore from (8.8.2.4) and (8.8.5) that there exists $\mu \geqslant \lambda$ and an isomorphism $g_\mu : X_\mu \to X'_\mu$ such that $g = g_\mu \times 1_S$ is the isomorphism $X \to X'$ associated to f; whence the conclusion. **IV-3** | 39

Cases (vi) *and* (vii): We know (**I**, 1.8.1) that $Z_\alpha = f_\alpha(X_\alpha)$ is constructible in S_α; if we set $Z_\lambda = u_{\alpha\lambda}^{-1}(Z_\alpha)$ for $\lambda \geqslant \alpha$ and $Z = u_\alpha^{-1}(Z_\alpha)$, we have $Z_\lambda = f_\lambda(X_\lambda)$ and $Z = f(X)$ (**I**, 3.4.8). As Z_α is pro-constructible in S_α, and by virtue of (8.3.11) the canonical map $\varinjlim \mathfrak{C}(S_\lambda) \to \mathfrak{C}(S)$ is injective, the relation $f(X) = S$ implies the existence of $\lambda \geqslant \alpha$ such that $f_\lambda(X_\lambda) = S_\lambda$, which proves the theorem in the case (vi).

Cases (viii) *and* (ix): As $S = \operatorname{Spec}(A)$ is affine, to say that $f : X \to S$ is affine (resp. quasi-affine) means that there exists an integer r and a closed immersion (resp. an immersion) $j : X \to \mathbf{V}_S^r = \operatorname{Spec}(A[T_1, \ldots, T_r])$ of S-preschemes, since f is of finite type and S quasi-compact (**II**, 5.1.9). As $\mathbf{V}_S^r = \mathbf{V}_{S_\lambda}^r \times_{S_\lambda} S$, and $\mathbf{V}_{S_\lambda}^r$ is an S_0-prescheme of finite presentation, it results from (8.8.2), (i) applied to the S-morphism j, that there exists λ and an S_λ-morphism $j_\lambda : X_\lambda \to \mathbf{V}_{S_\lambda}^r$ such that $j = j_\lambda \times 1_S$; applying (iv) (resp. (ii)) to j, one deduces that there exists $\mu \geqslant \lambda$ such that j_μ is a closed immersion (resp. an immersion); hence f_μ is affine (resp. quasi-affine).

Case (x): By hypothesis, $X = \operatorname{Spec}(B)$, where B is an A-algebra which is an A-module of finite presentation (**I**, 1.4.7); it results therefore from (8.5.2), (ii) that there is λ and an A_λ-module of finite presentation B_λ such that the A-module B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$. The structure of A-algebra of B is defined by an A-homomorphism $m : B \otimes_A B \to B$; as $B \otimes_A B = (B_\lambda \otimes_{A_\lambda} B_\lambda) \otimes_A A$, there exists by (8.5.2), (i) $\mu \geqslant \lambda$ and an A_μ-homomorphism $m_\mu : B_\mu \otimes_{A_\mu} B_\mu \to B_\mu$ such that $m = m_\mu \otimes 1$. Considering the usual diagrams expressing the associativity and the commutativity of m, one sees by applying again (8.5.2), (i) that one can suppose ν taken large enough so that m_ν defines on B_ν an associative and commutative multiplication. In the same way one can suppose ν large enough so that B_ν thus defined admits a unit element. If $X'_\nu = \operatorname{Spec}(B_\nu)$, it is clear then that X is S-isomorphic to $X'_\nu \times_{S_\nu} S$, hence, by virtue of (i), there exists $\rho \geqslant \nu$ such that X_ρ and X'_ρ are S_ρ-isomorphic, which completes the proof in this case.

To prove the theorem in the case (xii), we shall first prove the following proposition: □

Proposition (8.10.5.1). — *(Chow's lemma for morphisms of finite presentation). — Let A be a ring, X, Y two A-preschemes of finite presentation, $f : X \to Y$* **IV-3** | 39

8.11. Application to quasi-finite morphisms We propose in this section to prove the two following theorems:

Theorem (8.11.1). — *Let $f : X \to Y$ be a proper morphism, locally of finite presentation, and quasi-finite. Then the morphism f is finite.*

Theorem (8.11.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation, quasi-finite and separated. Then the morphism f is quasi-affine, and a fortiori quasi-projective.*

Remark (8.11.3). — We will see below that one can reduce, for the proof of (8.11.1) and (8.11.2), to the case where Y is locally Noetherian; one will note that in this case one will thus obtain another proof of the theorem of Chevalley (**III**, 4.4.2).

IV-3 | 41

(8.11.4). The hypotheses and the conclusions of (8.11.1) and (8.11.2) are all local on Y (**I**, 1.6.1), (**I**, 1.2.6), (**II**, 5.1.1), (**II**, 5.4.1) and (**II**, 6.2.2)), so one can suppose $Y = \operatorname{Spec}(A)$ affine. One knows that there then exists a subring A_0 of A, which is a **Z**-algebra of finite type, and a morphism of finite type $f_0 : X_0 \to \operatorname{Spec}(A_0)$ such that X identifies with $X_0 \otimes_{A_0} A$ and f with $f_0 \times 1$ (8.9.1). Furthermore, A can be considered as inductive limit of its subrings containing A_0 and which are **Z**-algebras of finite type; using the method of (8.1.2), *c*)) as well as (8.10.5), (v), (xi) and (xii)), one sees that it suffices to prove the theorems (8.11.1) and (8.11.2) for f_0. Suppose henceforth Y Noetherian; using now the method of (8.1.2), *a*)) as well as (8.10.5), (v), (ix), (x), (xi) and (xii)) one can replace Y by $\operatorname{Spec}(\mathscr{O}_y)$, where y is an arbitrary point of Y, so one can finally suppose that $Y = \operatorname{Spec}(A)$, where A is a *local Noetherian ring*. **IV-3** | 42 Let $\mathfrak{m}$ be the maximal ideal of A, $\widehat{A}$ the completion of A for the $\mathfrak{m}$-preadic topology; one knows that $\widehat{A}$ is a local Noetherian ring and that the morphism $\operatorname{Spec}(\widehat{A}) \to \operatorname{Spec}(A)$ is faithfully flat and quasi-compact ($\mathbf{0}_I$, 7.3.5); applying (2.7.1), (i), (vii), (xiv), (xv) and (xvi)), one sees furthermore that one can restrict to the case where A is *complete*. It then follows from (**II**, 6.2.6) that $X = X' \sqcup X''$, where X' is a Y-scheme quasi-finite and X'' a Y-scheme quasi-finite such that $X'' \cap f^{-1}(y) = \varnothing$.

Let us first place ourselves in the hypotheses of (8.11.1); as f is proper, X', which is closed in X, is proper over Y (**II**, 5.4.10)), so $f(X'')$ is closed in Y; but y is not contained in $f(X'')$, and moreover $f(X'')$ is adherent to every point of Y, so $f(X'') = \varnothing$, and consequently X'' is empty and f is *finite*.

Let us now place ourselves in the hypotheses of (8.11.2) and, restricting ourselves further (as one can do from what precedes) to the case where $Y = \operatorname{Spec}(A)$ is affine and Noetherian of finite dimension, let us moreover reason by induction on the dimension of Y. Reducing as above to the case where A is furthermore local and complete, one has $\dim(\mathscr{O}_y) = \dim(A) = \dim(Y)$ and for all $z \neq y$, $\dim(\mathscr{O}_z) < \dim(\mathscr{O}_y)$, so $\dim(Y - \{y\}) < \dim(Y)$. Now, by hypothesis one has $f(X'') \subset Y - \{y\}$ and the restriction of f to X'' is evidently a quasi-finite and separated morphism; applying the hypothesis of induction to $Y - \{y\}$ and to X'', one sees that X'' is quasi-affine over $Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (**II**, 5.1.10), (ii)); as moreover X' is finite (and *a fortiori* affine) over Y, X is quasi-affine over Y (**II**, 4.6.17) and (**II**, 5.1.2), *c*')).

Proposition (8.11.5). — *Let $f : X \to Y$ be a morphism of finite presentation. The following properties are equivalent:*

*label=*a) *f is a closed immersion.*
*lbbel=*b) *f is a proper monomorphism.*
*lcbel=*c) *f is proper and for all $y \in Y$, $f^{-1}(y)$ is radicial and geometrically reduced over $k(y)$ (that is to say empty or $k(y)$-isomorphic to* $\operatorname{Spec}(k(y))$*).*

It is clear that *a)* implies *b)*. To see that *b)* implies *c)*, let us note (**I**, 3.3.12) that for all $y \in Y$, the morphism $f^{-1}(y) \to \operatorname{Spec}(k(y))$ deduced from f by extension of the base is a monomorphism, hence injective, and consequently $f^{-1}(y)$ is empty or reduced to a point, and in every case affine. Furthermore, if A is the ring of $f^{-1}(y)$, the canonical homomorphism $A \otimes_k A \to A$ is bijective (**I**, 5.3.8). This evidently implies that $A = k$, for otherwise there would exist an element $a \in A$ not in k and the images of $a \otimes 1$ and of $1 \otimes a$ in A would both be equal to a, so $a \otimes 1 = 1 \otimes a$ since 1 and a form a free system over k.

It remains to prove that *c)* implies *a)*. It follows first from (8.11.1) that f is a *finite* morphism, so $X = \operatorname{Spec}(\mathscr{A})$, where $\mathscr{A}$ is a finite $\mathscr{O}_Y$-Algebra. It suffices therefore to prove that the canonical homomorphism $\mathscr{O}_Y \to \mathscr{A}$ is *surjective* (**II**, 1.4.10)), or equivalently that for all $y \in Y$, the homomorphism $\mathscr{O}_y \to \mathscr{A}_y$ is surjective. But by hypothesis $f^{-1}(y) = \operatorname{Spec}(\mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y)$ (**II**, 1.5.5)) is such that the corresponding homomorphism $k(y) = \mathscr{O}_y/\mathfrak{m}_y \to \mathscr{A}_y/\mathfrak{m}_y\mathscr{A}_y$ is bijective; as $\mathscr{A}_y$ is an $\mathscr{O}_y$-module of finite type, the lemma of Nakayama shows that $\mathscr{O}_y \to \mathscr{A}_y$ is surjective, which completes the proof. **IV-3 | 43**

Remark (8.11.5.1). — One will note that the preceding reasoning proves that if f is a *monomorphism*, then, for all $y \in Y$, $f^{-1}(y)$ is empty or $k(y)$-isomorphic to $\operatorname{Spec}(k(y))$.

Proposition (8.11.6). — *If a morphism $f : X \to Y$ of finite presentation is a universal homeomorphism, it is finite, surjective and radicial (the converse being true by (2.4.5), (i))).*

Proof. Indeed, f being of finite type, universally closed, and separated by virtue of (2.4.4)), is proper by definition (**II**, 5.4.1)). As it is evidently quasi-finite (**II**, 6.2.3)), it is finite by (8.11.1). One knows moreover that it is radicial (2.4.4)), and evidently surjective. □

8.12. New proof and generalisation of the "Main Theorem" of Zariski

Lemma (8.12.1). — *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism, $\mathscr{C}$ an $\mathscr{O}_Y$-Algebra quasi-coherent, $Z = \operatorname{Spec}(\mathscr{C})$, which is an affine Y-prescheme over Y. Let $g : X \to Z$ be a Y-morphism, $\varphi = \mathscr{A}(g) : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism of $\mathscr{O}_Y$-Algebras* (**II**, 1.2.7). *Suppose that g is an immersion. Then, for the closed image of X by g* (**I**, 9.5.3)*) to equal Z, it is necessary and sufficient that φ be injective; g is then an open immersion.*

Proof. The hypothesis implies that $f_*(\mathscr{O}_X)$ is an $\mathscr{O}_Y$-Algebra quasi-coherent (1.7.5)); furthermore, as the canonical morphism $h : Z \to Y$ is affine, hence quasi-compact and quasi-separated (1.2.2) and (1.1.2)), $g_*(\mathscr{O}_X)$ is an $\mathscr{O}_Z$-Algebra quasi-coherent (1.7.5)). This being so, saying that the closed image of X by g equals Z signifies (**I**, 9.5.2)) that the canonical homomorphism $\theta : \mathscr{O}_Z \to g_*(\mathscr{O}_X)$ is injective. But one has $h_*(\mathscr{O}_Z) = \mathscr{C}$ by definition of Z (**II**, 1.3.1)), and $h_*(g_*(\mathscr{O}_X)) = f_*(\mathscr{O}_X)$. As Z is affine over Y, it amounts to the same to say that the homomorphism $\theta : \mathscr{O}_Z \to g_*(\mathscr{O}_X)$ is injective or that the corresponding homomorphism $\varphi = h_*(\theta) : \mathscr{C} \to f_*(\mathscr{O}_X)$ is injective (**I**, 1.3.9)). The fact that g is then an open immersion follows from (**I**, 9.5.10)) and the hypothesis. □

Lemma (8.12.2). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a quasi-separated morphism of finite type, $\mathscr{C}$ an $\mathscr{O}_Y$-Algebra quasi-coherent, $Z = \operatorname{Spec}(\mathscr{C})$. Let $g : X \to Z$ be a Y-morphism, $\varphi : \mathscr{C} \to f_*(\mathscr{O}_X)$ the corresponding $\mathscr{O}_Y$-homomorphism of $\mathscr{O}_Y$-Algebras. Let $(\mathscr{C}_\lambda)$ be the filtered increasing family of sub-$\mathscr{O}_Y$-Algebras quasi-coherent of finite type of $\mathscr{C}$ (where $\mathscr{C}$ is the union* (**I**, 9.6.6) *and* (1.7.9)*)); set $Z_\lambda = \operatorname{Spec}(\mathscr{C}_\lambda)$ and let g_λ be the morphism composed $X \xrightarrow{g} Z \to Z_\lambda$. Then the following conditions are equivalent:*

label=a) *g is an immersion.*
lbbel=b) *There exists λ such that g_λ is an immersion.*

Furthermore, when g_λ is an immersion, the same is true of g_μ for $\mu \geqslant \lambda$. **IV-3 | 44**

It suffices to apply (**II**, 3.8.4)*) by replacing $\mathscr{L}$ by $\mathscr{O}_X$ and $\mathscr{S}$ by $\mathscr{C}[T]$, and taking into account* (**II**, 3.1.7)*).*

Proposition (8.12.3). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a separated morphism of finite type. Let $\mathscr{B} = f_*(\mathscr{O}_X)$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent* (**I**, 9.2.2)*); let $\mathscr{C}$ be the integral closure of $\mathscr{O}_Y$ in $\mathscr{B}$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent* (**II**, 6.3.4)*); set $Z = \operatorname{Spec}(\mathscr{C})$, and let $g : X \to Z$ be the Y-morphism corresponding to the injection $\mathscr{C} \to \mathscr{B} = f_*(\mathscr{O}_X)$, and for all λ, let $g_\lambda : X \to Z_\lambda = \operatorname{Spec}(\mathscr{C}_\lambda)$ be the Y-morphism corresponding to the injection $\mathscr{C}_\lambda \to \mathscr{B} = f_*(\mathscr{O}_X)$. Then the following conditions are equivalent:*

label=a) *There exists a factorisation of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

a$'$) *There exists a factorisation $X \xrightarrow{f'} Y' \xrightarrow{u} Y$ of f, where f' is an open immersion and u a finite morphism.*
lbbel=b) *The morphism $g : X \to Z$ is an immersion.*
lcbel=c) *There exists λ such that $g_\lambda : X \to Z_\lambda$ is an immersion.*

Furthermore, when g_λ is an immersion, g is an open immersion, $g(X)$ is dense in Z, and there exists λ such that for $\mu \geqslant \lambda$, g_μ is an open immersion.

where f' is an open immersion and u a finite morphism.

As the homomorphism $\varphi : \mathscr{C} \to f_*(\mathscr{O}_X)$ is injective, it follows from (8.12.1) that if g is an immersion, it is an open immersion and $g(X)$ is dense in Z, and the same is true for g_λ. The fact that *a)* implies *a$'$)* follows also from (8.12.1), applied by replacing Z by Y' and g by f' (since Y' is finite over Y, hence affine over Y): indeed if Y'' is the closed image of X by f', Y'' is finite over Y and f' factorises as $X \xrightarrow{f''} Y'' \xrightarrow{j} Y'$, where j is the canonical injection, and f'' is an immersion (**I**, 4.1.10)); it follows therefore from (8.12.1) that f'' is an open immersion.

The equivalence of *b)* and *c)* follows from (8.12.2) as well as the fact that g_λ is then an immersion for λ large enough. It is clear that *c)* implies *a)*, since Z_λ is finite over Y. Let us finally show that *a)* implies *c)*. One has seen above that Y' is the closed image of X by f', and it then follows from (8.12.1) that if one sets $\mathscr{B}' = u_*(\mathscr{O}_{Y'})$, the homomorphism $\varphi' : \mathscr{B}' \to \mathscr{B} = f_*(\mathscr{O}_X)$ is *injective*. But as by hypothesis $\mathscr{B}'$ is an $\mathscr{O}_Y$-Algebra finite, it identifies by definition of $\mathscr{B}$ with one of the sub-$\mathscr{O}_Y$-Algebras $\mathscr{C}_\lambda$, which proves *c)*.

We will say that a morphism $f : X \to Y$, where Y is quasi-compact and quasi-separated, is *pseudo-finite*, if it is of finite type and satisfies condition *a)* of (8.12.3) (in which case it is necessarily separated).

Corollary (8.12.4). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \to Y$ a morphism.* **IV-3** | 45

label=(i) *Suppose f pseudo-finite. Then, for all morphisms $Y' \to Y$, where Y' is quasi-compact and quasi-separated, $f_{(Y')} : X' = X_{(Y')} \to Y'$ is pseudo-finite.*
lbbel=(ii) *Let (U_λ) be a covering of Y by quasi-compact open subsets. For f to be pseudo-finite, it is necessary and sufficient that for all λ, the restriction $f_\lambda : f^{-1}(U_\lambda) \to U_\lambda$ of f be a pseudo-finite morphism.*
lcbel=(iii) *Suppose furthermore Y Noetherian, and f of finite type. Then, for f to be pseudo-finite, it is necessary and sufficient that, for all $y \in Y$, the morphism $f_y = f \times 1 : X_y = X \times_Y \mathrm{Spec}(\mathscr{O}_y) \to \mathrm{Spec}(\mathscr{O}_y)$ be pseudo-finite.*

Proof. (i) It is clear that $f_{(Y')}$ is of type finite (1.5.4)); furthermore, a factorisation $X \xrightarrow{g} Z \xrightarrow{u} Y$ where g is an immersion and u is finite, gives a factorisation $X' \xrightarrow{g'} Z' \xrightarrow{u'} Y'$ of $f_{(Y')}$, where $Z' = Z_{(Y')}$, $g' = g_{(Y')}$ and $u' = u_{(Y')}$; g' is an immersion (**I**, 4.3.2)) and u' is finite (**II**, 6.1.5)); so $f_{(Y')}$ is pseudo-finite.

(ii) The condition is necessary by virtue of (i), the U_λ being quasi-separated since Y is. To see that it is sufficient, let us observe (with the notations of (8.12.3)) that if one sets $X_\lambda = f^{-1}(U_\lambda)$, on has $\mathscr{B}|U_\lambda = (f_\lambda)_*(\mathscr{O}_{X_\lambda})$, $\mathscr{C}|U_\lambda$ is the integral closure of $\mathscr{O}_{U_\lambda}$ in $\mathscr{B}|U_\lambda$, and consequently, if $h : Z \to Y$ is the canonical morphism, $Z_\lambda = \mathrm{Spec}(\mathscr{C}_\lambda)$ identifies with $h^{-1}(U_\lambda) = \mathrm{Spec}(\mathscr{C}|U_\lambda)$, the Y-morphism $g_\lambda : X_\lambda \to Z_\lambda = h^{-1}(U_\lambda)$ being the U_λ-morphism corresponding to f_λ. Applying the hypothesis of induction, one sees that X'' is quasi-affine over $Y - \{y\}$; but the open subset $Y - \{y\}$ being quasi-affine over Y since Y is Noetherian, X'' is also quasi-affine over Y (**II**, 5.1.10), (ii)); since on the other hand X' is finite (and *a fortiori* affine) over Y, X is quasi-affine over Y.

(iii) It suffices, in virtue of (ii), to prove that for all $y \in Y$ there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f be pseudo-finite. Designate by (U_λ) the projective filtering system of affine open neighbourhoods of y, and apply the method of (8.1.2), *a*)). Since Y is Noetherian, the restrictions $f_\lambda : f^{-1}(U_\lambda) \to U_\lambda$ of f are of finite presentation and of the same nature as f_y. By hypothesis f_y factorises as $X_y \xrightarrow{g_y} Z_y \xrightarrow{u_y} \mathrm{Spec}(\mathscr{O}_y)$, where u_y is finite and g_y is an open immersion. As $\mathscr{O}_y$ is Noetherian, it is of the same nature as Z_y, and as Z_y is of finite presentation over $\mathrm{Spec}(\mathscr{O}_y)$, there exists λ and a morphism of finite presentation $u_\lambda : Z_\lambda \to U_\lambda$ such that Z_y

identifies with $Z_\lambda \times_{U_\lambda} \operatorname{Spec}(\mathscr{O}_y)$; furthermore, one can suppose λ taken such that g_λ is an open immersion and u_λ a finite morphism (8.10.5), (ii) and (x)), which proves that f_λ is pseudo-finite. □

(8.12.5). We can now give the "Main theorem" of Zariski (**III**, 4.4.3)), a proof not using the "global" cohomological results of chap. III, but rather appealing to the finer properties of local Noetherian rings; we will furthermore generalise the statement of the theorem by removing the Noetherian hypotheses:

Theorem (8.12.6). — *("Main theorem" of Zariski). — Let Y be a prescheme quasi-compact and quasi-separated. If a morphism $f : X \to Y$ is quasi-finite, separated and of finite presentation, there exists a factorisation of f*

(8.12.6.1) $$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

IV-3 | 46

Proof. By virtue of (8.12.4), (ii)) and the local character (on Y) of the notions of quasi-finite morphism, separated and of finite presentation, one can restrict to the case where $Y = \operatorname{Spec}(A)$ is affine. Applying (8.9.1)), one can suppose that there is a subring A_0 of A, which is a $\mathbf{Z}$-algebra of finite type, and an A-isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$, f identifying by this isomorphism with $f_0 \times 1$, where $f_0 : X_0 \to \operatorname{Spec}(A_0)$ is a morphism *of finite type*; furthermore (8.10.5), (v) and (xi)) one can suppose that f_0 is separated and quasi-finite; if one proves that f_0 is pseudo-finite, the same will be true of f by (8.12.4), (i)). As A_0 is then Noetherian and the notions of morphism of finite type, separated and quasi-finite are preserved by base change, it follows from (8.12.4), (iii)) that one can even suppose that A is a local ring, *essentially of finite type over* $\mathbf{Z}$ (1.3.8)). Set $n = \dim(A)$, and let us proceed by induction on n; for $n = 0$, the theorem is evident, A being a field and the morphism f being already finite (**II**, 6.2.2)). Set $B = \Gamma(X, \mathscr{O}_X)$; denote by C the integral closure of A in B, set $Z = \operatorname{Spec}(C)$ and let $g : X \to Z$ be the Y-morphism corresponding to the canonical A-homomorphism injective $C \to B$; by virtue of (8.12.3)), it suffices to show that g is an open immersion. Let a be the closed point of Y, and $U = Y - \{a\}$; U is Noetherian and all its local rings are essentially of finite type over $\mathbf{Z}$ and of dimension $< n$; taking into account the hypothesis of induction, one sees that the restriction $f^{-1}(U) \to U$ of f is a pseudo-finite morphism. One concludes from (8.12.3)) that, if $h : Z \to Y$ is the structural morphism, the restriction $f^{-1}(U) \to h^{-1}(U)$ of g is an open immersion. Set $A' = \widehat{A}$, $Y' = \operatorname{Spec}(A')$, $X' = X_{(Y')}$, $f' = f_{(Y')} : X' \to Y'$. As the canonical morphism $u : Y' \to Y$ is flat, it follows from (2.3.1)) that $B' = \Gamma(X', \mathscr{O}_{X'})$ identifies with the A'-algebra $B \otimes_A A'$. On the other hand, as A is a local ring excellent (7.8.3)), the morphism $u : Y' \to Y$ is regular, and consequently (6.14.4)) *the integral closure of A' in B' is equal to $C' = C \otimes_A A'$.* One sees therefore that $Z' = \operatorname{Spec}(C')$ is equal to $Z_{(Y')}$ and the morphism $g' : X' \to Z'$ coming from the canonical injection $C' \to B'$ is equal to $g_{(Y')}$. As $u : Y' \to Y$ is faithfully flat and quasi-compact, to prove that g is an open immersion, it suffices to prove that g' is an open immersion (2.7.1), (x)). If $h' : Z' \to Y'$ is the canonical morphism, the fact that the restriction $f'^{-1}(U') \to h'^{-1}(U')$ of g is an open immersion implies that the same is true of the restriction $f'^{-1}(U') \to h'^{-1}(U')$ of g'. Now let us note that $u^{-1}(a)$ is reduced to the closed point a' of Y' and consequently $U' = Y' - \{a'\} = u^{-1}(U)$. If $h' : Z' \to Y'$ is the canonical morphism, it follows from (**II**, 6.2.4)); as A' is *complete*, one deduces from (**II**, 6.2.6)) that X' is Y'-isomorphic to a sum $X'_1 \sqcup X'_2$, where the restriction $f'|X'_1 : X'_1 \to Y'$ is a morphism *finite*, and $X'_2 \subset f'^{-1}(U')$. It follows from this that B' is the direct composite of the two A'-algebras $\Gamma(X'_1, \mathscr{O}_{X'_1}) = B'_1$ and $\Gamma(X'_2, \mathscr{O}_{X'_2}) = B'_2$; one concludes immediately that the integral closure C' of A' in B' is the direct composite of the integral closures C'_1, C'_2 of A' in B'_1, B'_2 respectively, whence $Z' = Z'_1 \sqcup Z'_2$, where $Z'_i = \operatorname{Spec}(C'_i)$ $(i = 1, 2)$; and the canonical morphism $g' : X' \to Z'$ is such that $g'|X'_i$ is the canonical morphism $g'_i : X'_i \to Z'_i$ $(i = 1, 2)$. But as B'_1 is already a finite A'-algebra, one has $C'_1 = B'_1$, and g'_1 is therefore an isomorphism. On the other hand, as $X'_2 \subset f'^{-1}(U')$ and is open in $f'^{-1}(U')$, one already **IV-3** | 47
knows that g'_2 is an open immersion. One concludes that g' is indeed an open immersion. □

Remark (8.12.7). — When, in (8.12.6)), one supposes that Y is an affine *scheme*, the proof by reduction to the Noetherian case shows that, in the factorisation (8.12.6.1)), the morphisms f' and u are also morphisms *of finite presentation* (1.6.2)).

Corollary (8.12.8). — *Let Y be a quasi-compact scheme such that there exists an $\mathscr{O}_Y$-Module ample* (**II**, 4.5.3)*), $f : X \to Y$ a quasi-finite and quasi-projective morphism. Then there exists a factorisation of f as*

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

Proof. The hypothesis implies that X identifies with a sub-Y-scheme quasi-compact of a Y-scheme of the form $Z = \mathbf{P}^r_Y$ (**II**, 5.3.3)). There is therefore a quasi-compact open neighbourhood U of X in Z such that X is closed in U; as Z is a scheme, the canonical injection $U \to Z$ is a morphism of finite presentation ((1.2.7) and (1.6.2))), hence the composite morphism $g : U \to Z \to Y$ is also a morphism of finite presentation (the fact that $\mathbf{P}^r_Y$ is of finite presentation over Y following immediately from the definition (**II**, 4.1.1))). Let $\mathscr{J}$ be the quasi-coherent Ideal of $\mathscr{O}_U$ defining the closed sub-prescheme X; as U is a quasi-compact scheme, $\mathscr{J}$ is the filtered inductive limit of its sub-Ideals quasi-coherent of finite type $\mathscr{J}_\lambda$ (**I**, 9.4.9)). If X_λ is the closed sub-prescheme of U defined by $\mathscr{J}_\lambda$, one therefore has $X = \bigcap X_\lambda$. For all $y \in Y$, one therefore has $f^{-1}(y) = \bigcap (X_\lambda \cap g^{-1}(y))$, and as the sets $X_\lambda \cap g^{-1}(y)$ are closed in the Noetherian space $g^{-1}(y)$, there exists for all y an index $\lambda(y)$ such that $f^{-1}(y) = X_{\lambda(y)} \cap g^{-1}(y)$. Denote by E_λ the set of $y \in Y$ such that the fibre $X_\lambda \cap g^{-1}(y)$ of the restriction of g to X_λ is a $k(y)$-prescheme finite. The hypothesis that f is quasi-finite implies, by virtue of what precedes, that $Y = \bigcup_\lambda E_\lambda$. Now, each of the X_λ is, by definition, *of finite presentation* over Y; it follows therefore from (9.2.3) and (9.2.6)) (1) that the E_λ are *constructible* sets in the scheme Y; as they form a filtered increasing family, there exists an index λ such that $E_\lambda = Y$ (1.9.9)), and for this λ, the morphism $f_\lambda : X_\lambda \to Y$, restriction of g to X_λ, is of finite presentation, finite and separated, hence one can apply (8.12.6)), and f_λ factorises as

$$X_\lambda \xrightarrow{j_\lambda} Y'_\lambda \xrightarrow{u_\lambda} Y$$

where j_λ is an open immersion and u_λ a finite morphism. As X is a closed sub-prescheme of X_λ, one has thus proved that f possesses the property (8.12.3), *a*)), whence the corollary by virtue of the equivalence of (8.12.3), *a*)) and (8.12.3), *a'*)). ☐

Corollary (8.12.9). — *Let $f : X \to Y$ be a locally quasi-finite morphism* (**Err**$_{III}$, 20)*. For all $x \in X$ there exist an open neighbourhood U of x in X, an open neighbourhood V of $y = f(x)$ in Y, such that $f(U) \subset V$ and a factorisation* **IV-3 | 48**

$$U \xrightarrow{f'} V' \xrightarrow{u} V$$

of the restriction of f to U, where f' is an open immersion and u a finite morphism.

Proof. It evidently suffices to take for V an affine neighbourhood of y in Y, for U an affine neighbourhood of x in X contained in $f^{-1}(V)$ and such that $f|U$ is quasi-finite. The morphism $U \to V$ restriction of f being then affine (hence quasi-projective), one can apply (8.12.8)). ☐

Corollary (8.12.10). — *Let Y be a prescheme integral and normal, X a prescheme integral, $f : X \to Y$ a birational morphism and locally quasi-finite* (**Err**$_{III}$, 20)*. Then f is an open immersion ouverte and furthermore it is necessary and sufficient that f be separated.*

Proof. The second assertion follows immediately from the first and from (**I**, 8.2.8)). To prove the first assertion, one can suppose X and Y affine and f quasi-finite; consider the factorisation $f = u \circ f'$ of (8.12.8)), which allows one to identify X by f' with a sub-prescheme induced on an open subset of Y'. As X is integral, on can, by virtue of (**I**, 5.2.3)), replace Y' by the reduced sub-prescheme of Y' having underlying space $\overline{X}$, hence one can also suppose that Y' is *integral*. Furthermore, since f is birational, u is also. The conclusion then follows from the following lemma: ☐

3(1) The reader will verify that the corollaries (8.12.8) to (8.12.11) are not used in § 9.

Lemma (8.12.10.1). — *Let Y' be an integral prescheme, Y an integral prescheme and normal; then a morphism $u : Y' \to Y$ finite and birational is an isomorphism.*

Proof. Setting $\mathscr{A} = u_*(\mathscr{O}_{Y'})$, so that $\mathscr{A}$ is an $\mathscr{O}_Y$-Algebra finite, Y' identifying with $\operatorname{Spec}(\mathscr{A})$ (**II**, 1.3.6)). If $R(Y)$ is the field of rational functions of Y, one therefore has, for all $y \in Y$, $\mathscr{O}_{Y,y} \subset \mathscr{A}_y \subset R(Y)$; but as the ring $\mathscr{O}_{Y,y}$ is by hypothesis integrally closed and $R(Y)$ a field of fractions, one necessarily has $\mathscr{A}_y = \mathscr{O}_{Y,y}$, whence the lemma. □

Corollary (8.12.11). — *Let Y be a prescheme integral and normal, X a prescheme integral and normal, $f : X \to Y$ a dominant and locally quasi-finite morphism. Let K and L (extension of K) be the fields of rational functions of Y and X respectively; and let Y' be the integral closure of Y relative to L (**II**, 6.3.4)); then f factorises in a unique way as $f : X \xrightarrow{f'} Y' \xrightarrow{u} Y$, where f' is birational, and correspond to the identity automorphism of L; f' is then a local isomorphism, and for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. The existence and uniqueness of the factorisation of f follow from (**II**, 6.3.9)). It follows further from (**II**, 6.2.4), (v)), reducing to the affine case, that f' is locally quasi-finite; furthermore, it follows from (**I**, 5.5.1)) that, for f' to be separated, it is necessary and sufficient that f be, since u is affine, hence separated; the last two assertions are therefore consequences of (8.12.10)) applied to f'. □

8.13. Translation in terms of pro-objects The following proposition is essentially equivalent to (8.8.2), (i)): **IV-3 | 49**

Proposition (8.13.1). — *Let S be a prescheme, $(X_\lambda, v_{\lambda\mu})$ a projective filtering system of S-preschemes; suppose that there exists α such that $v_{\alpha\lambda}$ is an affine morphism for all $\lambda \geqslant \alpha$ (which implies (**II**, 1.6.2)) that $v_{\lambda\mu}$ is affine for $\alpha \leqslant \lambda \leqslant \mu$), so that the projective limit $X = \varprojlim X_\lambda$ exists in the category of S-preschemes (8.2.3)). Let Y be an S-prescheme, and, for all $\lambda \geqslant \alpha$, let $e_\lambda : \operatorname{Hom}_S(X_\lambda, Y) \to \operatorname{Hom}_S(X, Y)$ be the map which, to every S-morphism $f_\lambda : X_\lambda \to Y$, makes correspond $f = f_\lambda \circ v_\lambda$, where $v_\lambda : X \to X_\lambda$ is the canonical morphism. The family (e_λ) is an inductive system of maps, which therefore defines a canonical map*

$$\varinjlim \operatorname{Hom}_S(X_\lambda, Y) \longrightarrow \operatorname{Hom}_S(X, Y). \tag{8.13.1.1}$$

Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and the structural morphism $Y \to S$ locally of finite type (resp. locally of finite presentation). Then the map (8.13.1.1) is injective (resp. bijective).

Proof. Setting, for $\lambda \geqslant \alpha$, $Z_\lambda = Y \times_S X_\lambda$, so that one has $Z_\lambda = Z_\alpha \times_{X_\alpha} X_\lambda$. Setting similarly $Z = Y \times_S X = Z_\alpha \times_{X_\alpha} X$; one knows then (8.2.5)) that, if one also sets $w_{\lambda\mu} = 1 \times v_{\lambda\mu} : Z_\mu \to Z_\lambda$ for $\alpha \leqslant \lambda \leqslant \mu$ and $w_\lambda = 1 \times v_\lambda : Z \to Z_\lambda$ for $\alpha \leqslant \lambda$, Z is the projective limit of the system $(Z_\lambda, w_{\lambda\mu})$ and the w_λ are the canonical morphisms. Let us note furthermore that the morphism $Z_\alpha \to X_\alpha$ is locally of finite type (resp. locally of finite presentation) (1.3.4) and (1.4.3)). Finally, one knows that one has (**I**, 3.3.14))

$$\operatorname{Hom}_S(X_\lambda, Y) = \operatorname{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \quad \text{and} \quad \operatorname{Hom}_S(X, Y) = \operatorname{Hom}_X(X, Z)$$

It suffices now to apply (8.8.2), (i)) by setting $X_\lambda = S_\lambda$ and replacing Y_λ by Z_λ. □

Corollary (8.13.2). — *With the notations of (8.13.1)), suppose X_α quasi-compact and quasi-separated, and the $v_{\alpha\lambda}$ affine for $\alpha \leqslant \lambda$; suppose furthermore that $Y = \varprojlim Y_\rho$, where $(Y_\rho, t_{\rho o})$ is a projective filtering system of S-preschemes such that, for each ρ, the structural morphism $Y_\rho \to S$ is locally of finite presentation. Then one has a canonical bijection*

$$\operatorname{Hom}_S(X, Y) \xleftarrow{\sim} \varprojlim_\rho \left(\varinjlim_\lambda \operatorname{Hom}_S(X_\lambda, Y_\rho) \right). \tag{8.13.2.1}$$

Proof. Indeed, the fact that Y is the projective limit of the Y_ρ implies in particular that the canonical map $\operatorname{Hom}_S(X, Y) \to \varprojlim \operatorname{Hom}_S(X, Y_\rho)$ is bijective; and on the other hand, the hypotheses imply, for each ρ, the existence of a canonical bijection $\operatorname{Hom}_S(X, Y_\rho) \xleftarrow{\sim} \varinjlim_\lambda \operatorname{Hom}_S(S_\lambda, Y_\rho)$ by virtue of (8.13.1)); whence the conclusion. □

(8.13.3). The preceding results allow one to interpret in the theory of preschemes the notions of "pro-variety" or of "pro-scheme" which arise in certain applications (for example in the theory of the class field body following the ideas of Serre [39] or in the theory of Néron on the reduction of abelian varieties [32]). Let us recall here rapidly the notion of *pro-object* of a category, referring to **IV-3** | 50
chap. V for more ample developments (we will moreover not use before chap. V the interpretation that follows, and the reader can therefore omit it until the end of this number). Given a category $\mathscr{C}$, the category $\mathbf{Pro}(\mathscr{C})$ of *pro-objects* of $\mathscr{C}$ has for objects the *projective systems* (in the universe where one places oneself) $\mathbf{X} = (X_\mu)_{\mu\in M}$ of objects of $\mathscr{C}$ whose index sets (depending on the projective system considered) are supposed filtered preordered. Given two such pro-objects $\mathbf{X} = (X_\mu)_{\mu\in M}$, $\mathbf{X}' = (X'_{\mu'})_{\mu'\in M'}$, the *morphisms* from $\mathbf{X}$ to $\mathbf{X}'$ are by definition the elements of the set $\varprojlim_{\mu'}(\varinjlim_\mu \operatorname{Hom}(X_\mu, X_{\mu'}))$; the verification that one can take these sets as sets of morphisms is immediate, the composition of systems of morphisms $u''_\mu : X_\mu \to X'_{\mu'}$, $u''_{\mu'} : X'_{\mu'} \to X''_{\mu''}$ which are inductive and projective with respect to the lower index and upper index respectively, being carried out "argument by argument", in other words considering the system of $u''^{\mu'}_{\mu''} \circ u''^{\mu}_{\mu'}$.

(8.13.4). Consider then a prescheme quasi-compact and quasi-separated S, and denote by $\mathscr{C}$ the full subcategory of the category $(\mathbf{Sch})_{/S}$ of S-preschemes, formed of the S-preschemes X having the following property: the structural morphism $X \to S$ factorises as $X \xrightarrow{g} X_0 \xrightarrow{f} S$, where $g : X \to X_0$ is affine and $f : X_0 \to S$ is *of finite presentation*; we will say for short that the preschemes of $\mathscr{C}$ are *essentially affine over* S. Consider on the other hand the full subcategory $\mathscr{C}'_0$ of $(\mathbf{Sch})_{/S}$ formed of the S-preschemes *of finite presentation*, and the category $\mathbf{Pro}(\mathscr{C}'_0)$ of pro-objects of $\mathscr{C}'_0$. We will say that an object $\mathbf{X} = (X_\mu)_{\mu\in M}$ of $\mathbf{Pro}(\mathscr{C}'_0)$ is *essentially affine* if there exists a $\gamma \in M$ such that for $\mu \geqslant \gamma$, the transition morphism $v_{\gamma\mu} : X_\mu \to X_\gamma$ is affine (which implies that for $\gamma \leqslant \nu \leqslant \mu$, $v_{\nu\mu} : X_\mu \to X_\nu$ is affine). One will note that an object of $\mathbf{Pro}(\mathscr{C}'_0)$ isomorphic to an essentially affine object *is not* necessarily itself essentially affine. We will denote by $\mathscr{C}'$ the full subcategory of $\mathbf{Pro}(\mathscr{C}'_0)$ formed of pro-objects essentially affine.

This being so, it follows from (8.2.2) and (8.2.3)) that for all objects $\mathbf{X} = (X_\mu)_{\mu\in M}$ of $\mathscr{C}'$, the S-prescheme $X = \varprojlim_\mu X_\mu$ exists; furthermore, as, for μ large enough, the morphism $X \to X_\mu$ is affine (8.2.2)), X is *essentially affine over* S by definition. Set $X = L(\mathbf{X})$; let us show that one has thus defined a canonical *functor*

$$L : \mathscr{C}' \longrightarrow \mathscr{C} \tag{8.13.4.1}$$

One has indeed, for two objects $\mathbf{X} = (X_\mu)$, $\mathbf{X}' = (X'_{\mu'})$ of $\mathscr{C}'$, a canonical map **IV-3** | 51

$$\varprojlim_{\mu'} \varinjlim_{\mu} \operatorname{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \operatorname{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

defined in (8.13.1.1)), and on the other hand, by definition of the projective limit, a canonical bijection

$$\varprojlim_{\mu'} \operatorname{Hom}_S(\varprojlim_\mu X_\mu, X'_{\mu'}) \xrightarrow{\sim} \operatorname{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

whence a canonical map

$$\varprojlim_{\mu'} \varinjlim_{\mu} \operatorname{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \operatorname{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'}) \tag{8.13.4.2}$$

evidently functorial in $\mathbf{X}$ and $\mathbf{X}'$, and which completes the definition of the functor L.

Proposition (8.13.5). — *The hypotheses and notations being those of* (8.13.4)*), the functor L is pleinly faithful. If furthermore S is a Noetherian prescheme (which already implies that S is quasi-compact and quasi-separated* (1.2.8)*)), L is* an equivalence of categories.

Proof. To say that L is pleinly faithful signifies that the map (8.13.4.2)) is bijective, which is a particular case of (8.13.2)). Indeed, the structural morphisms $X_\mu \to S$ being of finite presentation, are in particular quasi-compact and quasi-separated, hence the X_μ are quasi-compact and quasi-separated.

To show that when S is Noetherian L is an equivalence of categories, it suffices, since one already knows that L is pleinly faithful, to prove that every S-prescheme essentially affine X is *S-isomorphic* to an object of the form $L(\mathbf{X})$ where $\mathbf{X} \in \mathscr{C}'$ ($\mathbf{0}_{\mathrm{III}}$, 8.1.5). Now, by hypothesis there is a factorisation $X \xrightarrow{g} X_0 \xrightarrow{f} S$ of the structural morphism, f being of finite presentation and g affine. One can therefore write $X = \operatorname{Spec}(\mathscr{A})$, where $\mathscr{A}$ is an $\mathscr{O}_{X_0}$-Algebra quasi-coherent (**II**, 1.3.1)). Now, as X_0 is Noetherian (since it is of finite type over S and S is Noetherian), $\mathscr{A}$ is the filtered inductive limit of the family $(\mathscr{A}_\lambda)$ of its sub-$\mathscr{O}_{X_0}$-Algebras quasi-coherent of finite type (**I**, 9.6.6)). Set $X_\lambda = \operatorname{Spec}(\mathscr{A}_\lambda)$; the morphisms $X_\lambda \to X_0$ are of finite type, hence of finite presentation since X_0 is Noetherian, and consequently the same is true of the composite morphisms $X_\lambda \to X_0 \to S$ (1.6.2)); in other words, the X_λ belong to $\mathscr{C}'_0$ and as the morphisms $X_\lambda \to X_0$ are affine, $\mathbf{X} = (X_\lambda)$ is an object of $\mathscr{C}'$ whose projective limit exists and is S-isomorphic to X by virtue of (8.2.2)). This completes the proof. □

Remark (8.13.6). — It follows from (1.6.2) and (**II**, 1.6.2)) that if X and Y are essentially affine over S, the same is true of $X \times_S Y$. One concludes for example ($\mathbf{0}_{\mathrm{III}}$, 8.2.5) that a $\mathscr{C}$-group is nothing other than a $(\mathbf{Sch})_{/S}$-group which is a prescheme essentially affine over S. On the other hand, the finite products exist in the category $\mathscr{C}'$: indeed, if $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{Y} = (Y_\rho)_{\rho \in R}$ are two objects of $\mathscr{C}'$, the products $X_\mu \times_S Y_\rho$ are S-preschemes of finite presentation, and taking for transition morphisms $X_\nu \times_S Y_\sigma \to X_\mu \times_S Y_\rho$ the products of the transition morphisms $X_\nu \to X_\mu$ and $Y_\sigma \to Y_\rho$, one sees immediately that $(X_\mu \times_S Y_\rho)$ is a product of $\mathbf{X}$ and $\mathbf{Y}$ in $\mathbf{Pro}(\mathscr{C}'_0)$; furthermore (**II**, 1.6.2)) the transition morphisms thus defined are affine for μ and ρ large enough, hence the product $\mathbf{X} \times \mathbf{Y}$ also belongs to $\mathscr{C}'$. One concludes then as above that a $\mathscr{C}'$-group is a $\mathbf{Pro}(\mathscr{C}'_0)$-group which is essentially affine. One deduces from (8.13.5)) that the categories of the $\mathscr{C}'$-groups and the $\mathscr{C}$-groups are *equivalent* when S is *Noetherian*. It seems plausible that when S is the spectrum of a field k, the category of $\mathscr{C}$-groups is *equivalent* to that of the ***K**-groups*, where $\boldsymbol{K}$ is the category of S-preschemes *quasi-compact*; in other words, every prescheme in groups quasi-compact over k would be essentially affine. On the other hand, if one denotes by $\mathscr{C}'_0$-$\boldsymbol{gr}$ the category of the $\mathscr{C}'_0$-groups, it is very likely that the category of the $\mathscr{C}'$-groups is equivalent to the full subcategory of $\mathbf{Pro}(\mathscr{C}'_0\text{-}\boldsymbol{gr})$ formed of the "proalgebraic groups essentially affine", that is to say the projective systems $\mathbf{G} = (G_\mu)_{\mu \in M}$, where the G_μ are algebraic groups over k and the transition morphisms $G_\nu \to G_\mu$ are affine for μ large enough (which is what one can also express by saying that $\mathbf{G}$ is an extension of an algebraic group by a pro-algebraic group affine). The conjunction of these two conjectures is moreover equivalent to the following: every prescheme in groups quasi-compact over k (i.e. a prescheme in groups of finite type over k) is an extension by a prescheme in groups *affine* over k. The only pro-algebraic groups encountered in practice until now being in fact essentially affine, it would therefore be advantageous to substitute for the study of general pro-algebraic groups (introduced and studied by Serre [40]) that of schemes in groups quasi-compact over k, whose definition is conceptually much simpler. **IV-3 | 52**

8.14. Characterisation of a prescheme locally of finite presentation over another, in terms of the functor it represents

(8.14.1). Given a prescheme S, we will say still, as in (8.13.4)), that a projective filtering system $(X_\lambda, v_{\lambda\mu})$ of S-preschemes is *essentially affine* if there exists α such that $v_{\alpha\lambda}$ is an affine morphism for $\lambda \geqslant \alpha$.

The following statement, which will be especially useful in chap. V, makes precise (8.8.2), (i)) by providing it with a converse:

Proposition (8.14.2). — *Let S be a prescheme, $f : X \to S$ a morphism. For every S-prescheme T, set*

$$h_X(T) = \operatorname{Hom}_S(T, X)$$

so that h_X is a contravariant functor from the category $(\mathbf{Sch})_{/S}$ of S-preschemes to the category **Ens** *of sets* ($\mathbf{0}_{\mathrm{III}}$, 8.1.1)*, and X an object representing this functor* ($\mathbf{0}_{\mathrm{III}}$, 8.1.8)*. The following conditions are equivalent:*

label=a) f is locally of finite presentation.

lbbel=b) For every projective filtering system (Z_λ) of S-preschemes, essentially affine (8.13.4)) and formed of preschemes quasi-compact and quasi-separated, the canonical map (8.13.1.1))

$$\varinjlim h_X(Z_\lambda) \longrightarrow h_X(\varprojlim Z_\lambda) \tag{8.14.2.1}$$

is bijective.

lcbel=c) For every projective filtering system (Z_λ) of S-preschemes, such that the Z_λ are affine schemes, the map (8.14.2.1)) is bijective.

c′) *For every affine open subset U of S and every projective filtering system (Z_λ) of U-preschemes, such that the Z_λ are affine schemes, the map* (8.14.2.1)*) is bijective.*

Proof. The fact that *a)* implies *b)* is nothing other than (8.13.1)); it is trivial that *b)* implies *c)* and that *c)* implies *c′)*. It remains to see that *c′)* implies *a)*, and as the property *a)* is local on S, one can restrict to the case where S is *affine*. **IV-3** | 53

Suppose first that X is also affine; the assertion to prove is then equivalent to the following: □

Corollary (8.14.2.2). — *Let A be a ring, B an A-algebra. In order that, for every inductive filtering system (C_λ) of A-algebras, the canonical map*

$$\varinjlim \operatorname{Hom}_{A\text{-alg.}}(B, C_\lambda) \longrightarrow \operatorname{Hom}_{A\text{-alg.}}(B, \varinjlim C_\lambda) \tag{8.14.2.3}$$

be bijective, it is necessary and sufficient that B be an A-algebra of finite presentation.

Proof. It remains only to show that the condition is necessary. First, for (C_λ) take the inductive filtering system of sub-A-algebras of finite type of B, so that $\varinjlim C_\lambda = B$. The fact that (8.14.2.3)) is bijective implies in particular that the identity map $1_B : B \to B$ factorises as $B \to C_\lambda \to B$ for a suitable λ, which implies that $C_\lambda = B$, hence B is an A-algebra *of finite type*. Write then $B = C/\mathfrak{J}$, where $C = A[T_1, \ldots, T_n]$ and $\mathfrak{J}$ is an ideal of C; setting $C_\lambda = C/\mathfrak{J}_\lambda$, where $\mathfrak{J}_\lambda$ ranges over the ideals of finite type $\mathfrak{J}_\lambda \subset \mathfrak{J}$ of C; then $\mathfrak{J}$ is the filtered inductive limit of the ideals of finite type of $\mathfrak{J}$, and using the exactness of the inductive limit functor, one sees that B is still isomorphic to the inductive limit of the inductive system (C_λ). There exists therefore a λ and an A-homomorphism $u : B \to C_\lambda$ such that the composite $B \xrightarrow{u} C_\lambda \xrightarrow{p_\lambda} B$ (where p_λ is the canonical homomorphism) is the identity. Let $q_\lambda : C \to C_\lambda$ be the canonical homomorphism, and set $t_i = p_\lambda(q_\lambda(T_i))$, in other words $u(t_i) = q_\lambda(T_i)$. There exists therefore a $\mu \geqslant \lambda$ such that the n elements $u(t_i) - q_\lambda(T_i)$ belong to $\mathfrak{J}_\mu/\mathfrak{J}_\lambda$ ($1 \leqslant i \leqslant n$); if $p_{\mu\lambda} : C_\lambda \to C_\mu$ is the canonical homomorphism, replacing λ by μ and u by $p_{\mu\lambda} \circ u$, one can therefore suppose that $u(t_i) = q_\lambda(T_i)$ for all i, and if $r = p_\lambda \circ q_\lambda$ is the canonical homomorphism $C \to C/\mathfrak{J} = B$, one can therefore write $u(r(T_i)) = q_\lambda(T_i)$ for all i, whence $q_\lambda = u \circ r$. But this implies necessarily that $\mathfrak{J} = \mathfrak{J}_\lambda$, for if $z \in \mathfrak{J}$, one has $r(z) = 0$; hence on has $B = C_\lambda$.

Let us now pass to the case where S is affine and X arbitrary; what amounts to proving is that an affine open subset V of X is of finite presentation over S, and by virtue of what has just been shown, it suffices to prove that for every projective filtering system (Z_λ) of affine S-preschemes, the map

$$\varinjlim \operatorname{Hom}_S(Z_\lambda, V) \longrightarrow \operatorname{Hom}_S(\varprojlim Z_\lambda, V) \tag{8.14.2.4}$$

is bijective. It is immediate that this map is injective, for if (v_λ), (v'_λ) are two inductive systems of S-homomorphisms $v_\lambda : Z_\lambda \to V$, $v'_\lambda : Z_\lambda \to V$ such that the corresponding morphisms

$$Z \xrightarrow{v_\lambda} Z_\lambda \xrightarrow{v_\lambda} V, \qquad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V$$

are equal (u_λ being the canonical morphism), then the morphisms

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V \xrightarrow{j} X, \qquad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V \xrightarrow{j} X$$

(where j is the canonical injection) are equal, which implies $j \circ v_\lambda = j \circ v'_\lambda$ by hypothesis for a suitable λ, hence $v_\lambda = v'_\lambda$. □

§9. Constructible properties

9.1. The finite extension principle Let S be a prescheme, $f : X \to S$ a morphism of *finite presentation* (1.6.1), $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of *finite presentation*. We propose in this paragraph to give criteria ensuring, for example, that the set of $s \in S$ such that the $\mathbf{k}(s)$-prescheme $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$ has a certain property, or such that the $\mathscr{O}_{X_s}$-Module $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_X} \mathbf{k}(s)$ has a certain property, is *locally constructible*, or at least *ind-constructible* (1.9.4); we will see that this is the case for the majority of properties that arise in algebraic Geometry. Assuming only f *locally of finite presentation*, we will also give (9.9) criteria for the set of points $x \in X$ where the fibre $X_{f(x)}$ (or the $\mathscr{O}_{X_{f(x)}}$-Module $\mathscr{F}_{f(x)}$) has a certain property, to be *locally constructible*. We will see in §12 that these results, together with the additional hypothesis that f is *flat* (resp. *proper and flat*), allow us to prove that the sets considered in X (resp. in S) are even *open* in many cases.

IV-3 | 54

Proposition (9.1.1). — (Finite extension principle). *Let k be a field, $\mathscr{C}$ a set of extensions of k. Suppose that the following conditions are satisfied:*

label=(i) *If $K \in \mathscr{C}$ and if there exists a k-homomorphism $K \to K'$ (where K' belongs to the universe where we have placed ourselves), then $K' \in \mathscr{C}$.*

lbbel=(ii) *If $K \in \mathscr{C}$, there exists a sub-extension K' of K, of finite type over k, such that $K' \in \mathscr{C}$.*

lcbel=(iii) *If $K \in \mathscr{C}$ is the field of fractions of an integral k-algebra of finite type A, there exists $f \in A - \{0\}$ such that for every maximal ideal $\mathfrak{m}$ of A we have $A_f/\mathfrak{m} \in \mathscr{C}$.*

Let then Ω be an algebraically closed extension of k (in the universe considered). The following conditions are equivalent:

label=a) *$\mathscr{C}$ is non-empty.*

lbbel=b) *There exists $K' \in \mathscr{C}$ which is a finite extension of k.*

lcbel=c) *We have $\Omega \in \mathscr{C}$.*

The condition (i) obviously implies that *b)* entails *c)*, and *c)* trivially entails *a)*; let us prove that *a)* entails *b)*. By virtue of (ii) and (iii) there exists an extension $K' \in \mathscr{C}$ which is of the form $A/\mathfrak{m}$, where A is a k-algebra of finite type over k and $\mathfrak{m}$ is a maximal ideal of A. We know, by the theorem of zeros of Hilbert (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 2 du th. 1) that K' is a finite extension of k.

IV-3 | 55

Corollary (9.1.2). — *Under the hypotheses* (i), (ii), (iii) *of* (9.1.1), *if k is algebraically closed and if $\mathscr{C}$ is non-empty, $\mathscr{C}$ contains all the extensions of k in the universe considered.*

Indeed, as *a)* entails *c)*, and the conclusion follows from the hypothesis (i).

Remark (9.1.3). — In practice, one will verify the condition (ii) of (9.1.1) by noting that K is the inductive limit of its sub-extensions of finite type K_α, and by applying the results of §8, taking into account eventually that for $K_\alpha \subset K_\beta$, K_β is faithfully flat over K_α. Frequently the set $\mathscr{C}$ is formed of *fields* belonging to a set $\mathscr{C}'$ of k-algebras satisfying the following condition:

libel=(i bis) If $A \in \mathscr{C}'$ and if there exists a homomorphism of k-algebras $A \to A'$ (where A' belongs to the universe where we have placed ourselves), then $A' \in \mathscr{C}'$.

Since this is so, the condition (i) is trivially satisfied, and one will verify generally the condition (iii) of (9.1.1) by noting that the field of fractions K of A is the inductive limit of the algebras A_f (for the relation $D(f) \supset D(g)$), and by applying the results of §8, taking into account eventually that the morphisms $D(g) \to D(f)$ are open immersions.

On the other hand, when (i bis) is not satisfied, the proof of (iii) is often more delicate, and is linked to constructibility criteria that will be developed further on.

Here are some typical examples of application of the finite extension principle:

Proposition (9.1.4). — *Let k be a field, Ω an algebraically closed extension of k, X and Y two preschemes of finite type over k. The following conditions are equivalent:*

label=a) *There exists an Ω-morphism $X_{(\Omega)} \to Y_{(\Omega)}$ (resp. an Ω-morphism possessing one of the determined properties* (i) *to* (xiv) *of* (8.10.5)).

lbbel=b) *There exists a finite extension k' of k and a k'-morphism $X_{(k')} \to Y_{(k')}$ (resp. a k'-morphism possessing the property considered*).

lcbel=c) *There exists an extension K of k and a K-morphism $X_{(K)} \to Y_{(K)}$ (resp. a K-morphism possessing the property considered*).

We apply the remark (9.1.3) taking for $\mathscr{C}'$ the set of all k-algebras A (of the universe where we have placed ourselves) such that there exists an A-morphism $X \otimes_k A \to Y \otimes_k A$ (resp. a morphism having the properties of (8.10.5) that one considers). The condition (i bis) of (9.1.3) is then satisfied thanks to the fact that the properties considered in (8.10.5) are all stable under base change. The condition (iii) of (9.1.1) is thus satisfied by virtue of (8.8.2), (i)) (resp. (8.10.5))), since $\operatorname{Spec}(k)$ is quasi-compact and quasi-separated. It remains to verify the condition (ii) of (9.1.1) which also IV-3 | 56 follows from (8.8.2), (i)) and from (8.10.5), considering K as an inductive limit of its sub-extensions of finite type. We conclude therefore by (9.1.1).

In particular, if there exists an extension K of k and a K-isomorphism $X_{(K)} \xrightarrow{\sim} Y_{(K)}$, we say that X and Y are *geometrically isomorphic*.

The following corollary generalises (**II**, 6.6.5):

Corollary (9.1.5). — *Let k be a field, X a k-prescheme. If there exists an extension K of k such that $X_{(K)}$ is projective (resp. quasi-projective) over K, then X is projective (resp. quasi-projective) over k.*

The morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ being faithfully flat and quasi-compact, it follows already from (2.7.1), (v)) that X is of finite type over k. The hypothesis means that there exists a closed immersion (resp. an immersion) $X_{(K)} \to \mathbf{P}^n_K = \mathbf{P}^n_k \otimes_k K$ (**II**, 5.5.4, (ii) and 5.5.3); applying (9.1.4) for property (iv) (resp. (ii)) of (8.10.5), we deduce that for some finite extension k' of k there is a closed immersion (resp. an immersion) $X_{(k')} \to \mathbf{P}^n_{k'}$; in other words $X_{(k')}$ is projective (resp. quasi-projective) over k'. We conclude then by (**II**, 6.6.5).

Proposition (9.1.6). — *Let k be a field, Ω an algebraically closed extension of k, X a prescheme of finite type over k, $\mathscr{F}$, $\mathscr{G}$ two coherent $\mathscr{O}_X$-Modules. Suppose that there exists an isomorphism $\mathscr{F} \otimes_k \Omega \xrightarrow{\sim} \mathscr{G} \otimes_k \Omega$. Then there exists a finite extension k' of k and an isomorphism $\mathscr{F} \otimes_k k' \xrightarrow{\sim} \mathscr{G} \otimes_k k'$.*

The reasoning is the same as in (9.1.4), applying (8.5.2), (i)) (using the property (iii) of (9.1.1), the fact that the morphisms $D(g) \to D(f)$ (with the notations of (9.1.3)) are open immersions, and *a fortiori* the morphisms are flat).

9.2. Constructible and ind-constructible properties

Definition (9.2.1). — Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation. We say that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) property of algebraic preschemes if the two following conditions are satisfied:

label=(i) If k is a field, X an algebraic prescheme over k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, then for $\mathbf{P}(X, \mathscr{F}, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, \mathscr{F} \otimes_k k', k')$ be true.

lbbel=(ii) Let S be an integral Noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $s \in S$, set $X_s = u^{-1}(s) = X \times_S \mathrm{Spec}(\mathbf{k}(s))$, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true. Then one of the two sets E, $S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

IV-3 | 57

Remarks (9.2.2). — label=(i) This is of course a convention of meta-mathematical language and not a mathematical definition properly speaking. We have analogous "definitions" for relations between k, one or more algebraic k-preschemes, k-morphisms between these preschemes, coherent Modules on these preschemes or homomorphisms between these Modules.

lbbel=(ii) We will also need to consider relations involving *constructible parts* of preschemes. For example, let $\mathbf{P}(X, Z, k)$ be a relation; we will say (by abuse of language) that $\mathbf{P}$ is a constructible (resp. ind-constructible) property of the constructible part Z of X if the following two conditions are satisfied:

label=1° If k is a field, X an algebraic prescheme over k, Z a *constructible part* of X, k' an extension of k, then for $\mathbf{P}(X, Z, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, p^{-1}(Z), k')$ be true ($p : X_{(k')} \to X$ being the canonical projection).

lbbel=2° Let S be an integral Noetherian prescheme, of generic point η, $u : X \to S$ a morphism of finite type, Z a *constructible part of* X. For every $s \in S$, set $X_s = u^{-1}(s)$, $Z_s = Z \cap X_s$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, Z_s, \mathbf{k}(s))$ is true. Then one of the two sets E, $S - E$ (resp. the set E) contains a non-empty open set (and consequently is a neighbourhood of η) (resp. contains a non-empty open set if it contains η).

We will note that in condition 2° it suffices to suppose that Z is a constructible part of X for every s; the first of these two properties entails the second (1.8.2), but not conversely.

lcbel=(iii) If **P** is a constructible property, it is clear that the same is true of "not **P**". If **P**, **Q** are two constructible properties (resp. ind-constructible), the same is true of the properties "**P** or **Q**" and "**P** and **Q**"; indeed, under the hypotheses of (9.2.1), (ii)), E, E' are two parts of S and if each of E contains a non-empty open set, the same is true of $E \cup E'$, and if $S - E$ and $S - E'$ each contain a non-empty open set, the same is true of $S - (E \cup E') = (S - E) \cap (S - E')$.

ldbel=(iv) Let $\mathbf{P}(X, \mathscr{F}, k)$ be a relation satisfying the condition (9.2.1), (i)); let S be a prescheme, $u : X \to S$ a morphism of finite type; with the notations of (9.2.1), (ii)), let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true. Let on the other hand $g : S' \to S$ be any morphism, and set $X' = X_{(S')}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$; then it follows from the transitivité of fibres (**I**, 3.6.4) and from the condition (9.2.1), (i)) that *the set of* $s' \in S'$ *such that* $\mathbf{P}(X'_{s'}, \mathscr{F}'_{s'}, \mathbf{k}(s'))$ *is true is equal to* $g^{-1}(E)$. This extends immediately to the case where there appear in **P** several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules, as well as to properties of the type considered in (ii).

lebel=(v) As we will see in the remainder of this paragraph and in the rest of chap. IV, the majority of properties **P** satisfying the condition (9.2.1), (i)) also satisfy the condition (9.2.1), (ii)). As possible exceptions, let us note the property of being *projective*, or *quasi-projective*, or *affine*, or *quasi-affine* over the base field (for an algebraic prescheme); we will see (9.6.2) that these properties are *ind-constructible*, but we will give later an example where S is a non-empty open part of $\mathrm{Spec}(\mathbf{Z})$ (or a non-empty open part of a smooth curve over a finite field) and where all fibres X_s

except the generic fibre X_η are projective over $\mathbf{k}(s)$ (all preschemes X_s being of dimension 2). **IV-3** | 58

Proposition (9.2.3). — *Let* **P** *be a constructible* (*resp. ind-constructible*) *property of algebraic preschemes, S a prescheme, X a prescheme of finite presentation over S, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set E of $s \in S$ such that $\mathbf{P}(X_s, \mathscr{F}_s, \mathbf{k}(s))$ is true is locally constructible* (*resp. ind-constructible*). *Furthermore, if S is irreducible of generic point η, one of the two sets E, $S - E$ is a neighbourhood of η* (*resp. E is a neighbourhood of η if it contains this point*).

To prove these assertions, one can restrict to the case where $S = \mathrm{Spec}(A)$ is affine. We know then that there exists a subring A_0 of A which is a $\mathbf{Z}$-algebra of finite type, an A_0-prescheme of finite type X_0 and a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that X is isomorphic to $X_0 \otimes_{A_0} A$ (8.9.1). Let $p : S \to S_0 = \mathrm{Spec}(A_0)$ be the morphism corresponding to the injection $A_0 \to A$, and let E_0 be the set of $s_0 \in S_0$ such that

$$\mathbf{P}((X_0)_{s_0}, (\mathscr{F}_0)_{s_0}, \mathbf{k}(s_0))$$

is true; then, by virtue of (9.2.2), (iv)), we have $E = p^{-1}(E_0)$; on the other hand, one can restrict further (1.8.2) to a $\mathbf{Z}$-algebra of finite type, hence a Noetherian scheme. Using the criterion of ind-constructibility ($\mathbf{0}_{\mathrm{III}}$, 9.2.3) (resp. the criterion of constructibility ($\mathbf{0}_{\mathrm{III}}$, 9.2.3)) (9.2.2), (iv)) and replacing S by a closed integral sub-prescheme of S, in the case where S is Noetherian and integral, our assertion then follows from (9.2.3) applied to the ind-constructible part E of S.

We will note that when S is the spectrum of a finite ring of integers of finite type, hence a Noetherian scheme, and it is clear on the other hand that the statement of (9.2.3) also applies when there appear in **P** several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules. It applies also when there is a finite number of parts (in finite number) of the preschemes considered, provided one imposes on these parts the condition of being *locally constructible*. Indeed, the restriction to the case where S is affine shows that one can restrict to the case where these parts are *constructible*: applying (8.3.11) then (with the previous notations) one obtains a constructible part of X_0 for a suitable choice of A_0.

Corollary (9.2.4). — *Let* **P** *be a constructible* (*resp. ind-constructible*) *property of algebraic preschemes, X, Y two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. For every $s \in S$, set $X_s = X \times_S \mathrm{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \mathrm{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \to Y_s$. Then the set E of $s \in S$ such that for every $y \in Y_s$, the property $\mathbf{P}(f_s^{-1}(y), \mathbf{k}(y))$ is true, is locally constructible* (*resp. ind-constructible*).

Indeed, let Z be the set of $y \in Y$ such that $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Since the fibres $f^{-1}(y)$ and $f_s^{-1}(y)$ are isomorphic, we see that E is the set of $s \in S$ such that $Y_s \subset Z$; if $g : Y \to S$ is the structural morphism, we have therefore $E = S - g(Y - Z)$.

Now, f is of finite presentation (1.6.2), (v)), hence it follows from (9.2.3) that Z is locally constructible (resp. ind-constructible) in Y, hence $Y - Z$ is locally constructible (resp. pro-constructible) in Y. As g is of finite presentation, $g(Y - Z)$ is locally constructible (resp. pro-constructible) in S, by virtue of Chevalley's theorem (1.8.4) (resp. of (1.9.5), (vii))); hence E is locally constructible (resp. ind-constructible) in S. **IV-3** | 59

Remark (9.2.5). — We will note that if **P** is a property of algebraic preschemes for which prop. (9.2.3) is true, then **P** also satisfies the condition (9.2.1), (ii)): this results indeed from the fact that in an irreducible Noetherian space, a constructible set is either rare or contains a non-empty open set ($\mathbf{0}_{\text{III}}$, 9.2.3).

Proposition (9.2.6). — *Let* **P** *denote one of the following properties of an algebraic k-prescheme X:*

label=(i) X is empty.
lbbel=(ii) X is finite over k.
lcbel=(iii) X is radicial over k.
ldbel=(iv) $\dim(X)$ *belongs to a given part* Φ *of the set* $\overline{\mathbf{Z}} = \mathbf{Z} \cup \{-\infty\}$.

Then **P** *is constructible.*

It is clear that (i) and (ii) are particular cases of (iv), taking for Φ respectively the set $\{-\infty\}$ and the set $\{-\infty, 0\}$. We therefore need only prove (iii) and (iv). In each of these two cases the condition (i) of (9.2.1) is satisfied by virtue of (2.7.1), (xv)) and (4.1.4). On the other hand, in the case (iii), the property **P** satisfies the conclusion of (9.2.3) by virtue of (1.8.7); it therefore remains to check that the same holds in the case (iv). This will follow from the more precise proposition below:

Proposition (9.2.6.1). — *If* $f : X \to S$ *is a morphism of finite presentation, the function* $s \mapsto \dim(f^{-1}(s))$ *is locally constructible.*

The question is local on S, so one can suppose that $S = \operatorname{Spec}(A)$ is affine and prove that for every n, the set E of $s \in S$ such that $\dim(X_s) = n$ is constructible. The same reasoning as in (9.2.3) reduces to the case where A is Noetherian and integral, and it suffices then to prove the following

Corollary (9.2.6.2). — *Let S be an integral Noetherian prescheme of generic point* η, $f : X \to S$ *a morphism of finite type. Then there exists a neighbourhood of* η *in S such that the function* $s \mapsto \dim(X_s)$ *is constant in this neighbourhood.*

The images by f of the irreducible components (in finite number) of X that do not meet X_η are contained in closed parts of S not containing η (since S is integral ($\mathbf{0}$, 2.1.5)), hence (replacing S by a neighbourhood of η) one can restrict to the case where all irreducible components X_i of X meet X_η. Or, it suffices that there exists a finite covering (U_j) of X by

dense affine open sets, and the numbers $\dim((U_j)_\eta)$ are all equal to $n = \dim(X_\eta)$ (4.1.1.3); one can therefore restrict to the case where X is affine, hence also X_η. There exists then, by virtue of (4.1.2), a non-empty open set W of X such that $W \supset X_\eta$, and a $\mathbf{k}(\eta)$-morphism of finite type surjective $h : W_\eta \to \mathbf{V}^n_{\mathbf{k}(\eta)} (= \operatorname{Spec}(\mathbf{k}(\eta)[T_1, \ldots, T_n]))$; applying (8.8.2), (i)) and the method of (8.1.2), *a*)), replacing S by a neighbourhood of η if necessary, that $h = g_\eta$, where $g : W \to \mathbf{V}^n_A (= \operatorname{Spec}(A[T_1, \ldots, T_n]))$ is an S-morphism, and one can suppose this morphism of finite type and surjective by virtue of (8.10.5), (vi) and (x)). We therefore conclude that for every $s \in S$, the morphism $g_s : W_s \to \mathbf{V}^n_{\mathbf{k}(s)}$ is of finite type and surjective, hence $\dim(W_s) = n$ (4.1.2). **IV-3** | 60

9.3. Constructible properties of morphisms of algebraic preschemes

Proposition (9.3.1). — *Let $\mathbf{P}(X, k)$ be a constructible (resp. ind-constructible) property of algebraic preschemes. Let $\mathbf{P}'(f, X, Y, k)$ denote the following relation: $f : X \to Y$ is a k-morphism of algebraic preschemes such that for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), \mathbf{k}(y))$ is true. Then $\mathbf{P}'$ is a constructible (resp. ind-constructible) property.*

Indeed, as $\mathbf{P}$ satisfies the condition (9.2.1), (i)), the same is true of $\mathbf{P}'$ by virtue of the transitivity of fibres (**I**, 3.6.4); furthermore the fact that $\mathbf{P}'$ satisfies the condition (9.2.1), (ii)) follows from (9.2.4), by reason of the remark (9.2.5).

Proposition (9.3.2). — *Let $\mathbf{P}$ denote one of the following properties of a k-morphism $f : X \to Y$ of algebraic preschemes:*

label=(i) f is surjective.
lbbel=(ii) f is quasi-finite.
lcbel=(iii) f is radicial.
ldbel=(iv) For every $y \in Y$, $\dim(f^{-1}(y))$ belongs to Φ (notation of (9.2.6)).

Then $\mathbf{P}$ is constructible.

This follows immediately from (9.3.1) and (9.2.6) if one takes into account that f is of finite type (**I**, 3.5.8), and the characterisation of radicial morphisms (**II**, 6.2.2).

Proposition (9.3.3). — *Suppose that the hypotheses of (8.8.1) are satisfied, and let us keep the same notations; suppose furthermore that S_α is quasi-compact, X_α and Y_α of finite presentation over S_α, and let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism. Let $\mathbf{P}$ be an ind-constructible property of morphisms of algebraic preschemes. For every $s \in S$ (resp. $s_\lambda \in S_\lambda$) set $X_s = X \times_S \operatorname{Spec}(\mathbf{k}(s))$, $Y_s = Y \times_S \operatorname{Spec}(\mathbf{k}(s))$, $f_s = f \times 1 : X_s \to Y_s$ (resp. $X_{\lambda,s_\lambda} = X_\lambda \times_{S_\lambda} \operatorname{Spec}(\mathbf{k}(s_\lambda))$, $Y_{\lambda,s_\lambda} = Y_\lambda \times_{S_\lambda} \operatorname{Spec}(\mathbf{k}(s_\lambda))$, $f_{\lambda,s_\lambda} = f_\lambda \times 1 : X_{\lambda,s_\lambda} \to Y_{\lambda,s_\lambda}$). Then, in order that for every $s \in S$ the property $\mathbf{P}(f_s)$ be true, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that for every $s_\lambda \in S_\lambda$ we have $\mathbf{P}(f_{\lambda,s_\lambda})$.*

Indeed, let E (resp. E_λ) be the set of $s \in S$ (resp. $s_\lambda \in S_\lambda$) such that the property $\mathbf{P}(f_s)$ (resp. $\mathbf{P}(f_{\lambda,s_\lambda})$) is true; it follows from (9.2.2), (iv)) that $E_\mu = u_{\mu\lambda}^{-1}(E_\lambda)$ for $\lambda \leqslant \mu$, and $E = u_\mu^{-1}(E_\lambda)$; furthermore, by virtue of (9.2.3), E (resp. E_λ) is ind-constructible in S (resp. S_λ); the proposition then follows from (8.3.4) applied to the ind-constructible part E of S.

Remark (9.3.4). — The conjunction of (9.3.3) and of (9.3.2), (ii)) proves the assertion (8.10.5), (xi)).

Proposition (9.3.5). — *Let $\mathbf{P}(X, Y, k)$ be the property: "X and Y are two preschemes of finite type over the field k, and there exists an extension k' of k and a k'-morphism $g : X_{(k')} \to Y_{(k')}$ satisfying $\mathbf{Q}(g)$, or $\mathbf{Q}$ is one of the properties* (i) *to* (xiv) *of (8.10.5)". Then $\mathbf{P}$ is an ind-constructible property.*

The definition of $\mathbf{P}$ shows indeed that the condition (9.2.1), (i)) is satisfied, taking into account that the property $\mathbf{Q}(g)$ is stable under base change, and that two extensions of k can always be considered as sub-extensions of a third extension of k. To verify (9.2.1), (ii)), one can restrict to the case where $S = \operatorname{Spec}(A)$ is affine; if $K = \mathbf{k}(\eta)$, the field of fractions of A, there exists by hypothesis and by virtue of (9.1.4) a finite extension K' of K and a K'-morphism $g : (X_\eta)_{(K')} \to (Y_\eta)_{(K')}$, satisfying $\mathbf{Q}(g')$, and K' is evidently the field of fractions of an integral A-algebra of finite type A'. If we set $S' = \operatorname{Spec}(A')$, it follows then from (8.10.5) that there is a neighbourhood U' of the generic point η' of S' such that, if we set $X' = X \otimes_A A'$, $Y' = Y \otimes_A A'$, $Y' = Y \otimes_A A'$, there exists for every $s' \in U'$, a morphism $X'_{s'} \to Y'_{s'}$ having property $\mathbf{Q}$. But the morphism $h : S' \to S$ is of finite type, hence closed, and $h^{-1}(\eta) = \{\eta'\}$, $h(U')$ contains a neighbourhood of η in S; for every $s \in U$, there exists therefore $s' \in U'$ such that $h(s') = s$, and as $X'_{s'} = X_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, $Y'_{s'} = Y_s \otimes_{\mathbf{k}(s)} \mathbf{k}(s')$, the property $\mathbf{P}(X_s, Y_s, \mathbf{k}(s))$ is true for every $s \in U$.

(9.3.6). Example. Take for example for $\mathbf{Q}$ the property of being an isomorphism. Then, combining (9.3.5) and (9.3.3), and the hypotheses being those of (8.8.1), S_α being quasi-compact, X_α and Y_α of finite presentation over S_α, we obtain the following property: using the notations of (9.3.3), in order that, for every $s \in S$, X_s and Y_s be *geometrically isomorphic* (9.1.4), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that, for every $s_\lambda \in S_\lambda$, X_{λ,s_λ} and Y_{λ,s_λ} be geometrically isomorphic.

We have an analogous result when the preschemes considered are endowed with "composition laws" of a certain kind ($\mathbf{0}_{\mathrm{III}}$, 8.2.1), for example "preschemes in groups", "preschemes in rings", etc. ($\mathbf{0}_{\mathrm{III}}$, 8.2.3). Then the previous statement is still valid when one replaces "isomorphism" by "homomorphisms" for the composition laws considered ($\mathbf{0}_{\mathrm{III}}$, 8.2.2); it suffices here to use not only (8.10.5) but also (8.8.2), (i)) while noting that the notion of homomorphism for a composition law is expressed by saying that diagrams of morphisms of preschemes are commutative (it must be understood that the transition morphisms $X_\mu \to X_\lambda$ and $Y_\mu \to Y_\lambda$ for $\lambda \leqslant \mu$ are homomorphisms for the composition laws considered).

One can also, instead of considering morphisms of preschemes as in (9.3.5), consider homomorphisms of Modules, using (9.1.6) instead of (9.1.5).

IV-3 | 62

9.4. Constructibility of certain properties of modules

(9.4.1). Notations (9.4.1). In this n° and the following ones until the end of §9, we will systematically use the following notations: given a morphism $f : X \to S$, we will set, for every $s \in S$, $X_s = f^{-1}(s) = X \times_S \operatorname{Spec}(\mathbf{k}(s))$; for every $\mathscr{O}_X$-Module quasi-coherent $\mathscr{F}$, $\mathscr{F}_s$ will denote the $\mathscr{O}_{X_s}$-Module quasi-coherent $\mathscr{F} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$, and for every homomorphism $u : \mathscr{F} \to \mathscr{G}$ of $\mathscr{F}$ into a quasi-coherent $\mathscr{O}_X$-Module $\mathscr{G}$, $u_s : \mathscr{F}_s \to \mathscr{G}_s$ will be the morphism $p^*(u)$, designating by p the canonical projection $X_s \to X$. For every section g of $\mathscr{F}$ over X, we will denote by g_s the image of g by the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_s, \mathscr{F}_s)$. For every part Z of X, we will denote Z_s the inverse image $p^{-1}(Z) = Z \cap X_s$ (**I**, 3.6.1). Finally, if Y is a second S-prescheme and $h : X \to Y$ an S-morphism, we will denote h_s the morphism $h \times 1 : X_s \to Y_s$.

Proposition (9.4.2). — *Let S be an integral Noetherian prescheme of generic point η, $f : X \to S$ a morphism of finite presentation, $\mathscr{F}, \mathscr{G}, \mathscr{H}$ three quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to \mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules, and suppose that the sequence $\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$ is exact. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact.*

With the general notations of (9.2.1), it is a question here of the relation $\mathbf{P}(X, \mathscr{F}, \mathscr{G}, \mathscr{H}, u, v, k)$: "$X$ is an algebraic prescheme over the field k, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is an exact sequence of quasi-coherent $\mathscr{O}_X$-Modules". Since, for every extension k' of k, the canonical projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.13), the condition (9.2.1), (i)) is satisfied (2.2.7). By virtue of (9.2.3), one can restrict to the case where S is integral and Noetherian, in which case X is Noetherian, and $\mathscr{F}, \mathscr{G}, \mathscr{H}$ are coherent $\mathscr{O}_X$-Modules. The hypothesis implies that there exists a neighbourhood U of η in S such that $\mathscr{F}|f^{-1}(U) \to \mathscr{G}|f^{-1}(U) \to \mathscr{H}|f^{-1}(U)$ is *exact*, by virtue of (8.5.8), (i)) applied following the general method of (8.1.2), *a*)), and one can therefore already suppose that $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is exact; it suffices evidently to prove that $\operatorname{Ker}(v_s) = (\operatorname{Ker} v)_s$ and $\operatorname{Im}(u_s) = (\operatorname{Im} u)_s$ for s near η in S. Using (8.9.1) and (8.5.2), (i)), one sees that one can reduce to the case where S (hence also X) is Noetherian, and by consequence (since S is integral) and taking into account that $\mathscr{O}_\eta = \mathbf{k}(\eta)$ (so that **IV-3 | 63** $\operatorname{Ker}(v_s) = (\operatorname{Ker} v)_s$ and $\operatorname{Im}(u_s) = (\operatorname{Im} u)_s$), the sequence

$$0 \longrightarrow (\operatorname{Im} u)_\eta \longrightarrow (\operatorname{Ker} v)_\eta \longrightarrow (\operatorname{Ker} v / \operatorname{Im} u)_\eta \longrightarrow 0$$

is exact.

One can evidently restrict to a complex of three terms of degrees -1, 0, $+1$: $\mathscr{M} \xrightarrow{u} \mathscr{N} \xrightarrow{v} \mathscr{P}$ and to $i = 0$; the canonical homomorphism to consider is then $(\operatorname{Ker} v / \operatorname{Im} u)_s \to \operatorname{Ker}(v_s) / \operatorname{Im}(u_s)$ for every $s \in S$. As $\mathscr{F}, \mathscr{G}, \mathscr{H}$ are coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), $\operatorname{Im}(u)$ and $\operatorname{Ker}(v)$ are also coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4) and furthermore $\operatorname{Im}(u_s) = (\operatorname{Im}(u))_s$ and $\operatorname{Ker}(v_s) = (\operatorname{Ker}(v))_s$; and from the exact sequence $0 \to \operatorname{Im}(u) \to \operatorname{Ker}(v) \to \operatorname{Ker}(v) / \operatorname{Im}(u) \to 0$, the conclusion follows from (9.4.2) applied to this exact sequence (since S is integral), which completes the proof.

Corollary (9.4.3). — *Let S be an integral Noetherian prescheme, of generic point η, $f : X \to S$ a morphism of finite presentation. Let $\mathscr{L}^\bullet = (\mathscr{L}^i)_{i \in \mathbf{Z}}$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. For every i, there exist canonical homomorphisms*

$$(\mathscr{H}^i(\mathscr{L}^\bullet))_s \longrightarrow \mathscr{H}^i(\mathscr{L}^\bullet_s) \qquad (9.4.3.1)$$

which are bijective for every $s \in U$.

Proposition (9.4.4). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ three quasi-coherent $\mathscr{O}_X$-Modules of finite presentation. Let $u : \mathscr{F} \to \mathscr{G}$, $v : \mathscr{G} \to \mathscr{H}$ be two homomorphisms of $\mathscr{O}_X$-Modules. Then the set E of $s \in S$ such that the sequence $\mathscr{F}_s \xrightarrow{u_s} \mathscr{G}_s \xrightarrow{v_s} \mathscr{H}_s$ is exact is locally constructible.*

Taking into account (9.2.3), it suffices to establish that the property **P** considered in (9.4.2) is constructible. We have already remarked in the proof of (9.4.2) that the condition (9.2.1), (i)) is satisfied for this property, and it remains to verify the condition (9.2.1), (ii)). Let us therefore suppose S integral Noetherian, of generic point η, and let us prove that E or $S - E$ is a neighbourhood of η. If $\eta \in E$, our assertion follows from (9.4.2), and one can therefore restrict to the case where $\eta \notin E$, in other words, the sequence $\mathscr{F}_\eta \xrightarrow{u_\eta} \mathscr{G}_\eta \xrightarrow{v_\eta} \mathscr{H}_\eta$ is not exact. Let us then distinguish two cases:

1° Set $w = v \circ u$, and suppose first that $w_\eta = v_\eta \circ u_\eta \neq 0$. As $\mathscr{F}$, $\mathscr{G}$, $\mathscr{H}$ are coherent, the same is true of $\mathscr{N} = \operatorname{Ker}(w)$ ($\mathbf{0}_{\mathrm{I}}$, 5.3.4); it then follows from (9.4.2) applied to the suite exact $0 \to \mathscr{N} \to \mathscr{F} \xrightarrow{w} \mathscr{H}$ that there exists a neighbourhood U of η in S such that $\operatorname{Ker}(w_s) = \mathscr{N}_s$; restricting S, one can therefore suppose this relation satisfied for every $s \in S$. Let j be the canonical injection $\mathscr{N} \to \mathscr{F}$, and set $\mathscr{M} = \operatorname{Coker}(j)$; the exactness of the right of the functor $\mathscr{F} \to \mathscr{F}_s$ entails that $M_s = \operatorname{Coker}(j_s)$ for every $s \in S$. The hypothesis $w_\eta \neq 0$ means that $M_\eta \neq 0$; as M is coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4), it follows from (1.8.6) that there exists a neighbourhood U of η in S such that $M_s \neq 0$ for every $s \in U$, hence $w_s \neq 0$ for every $s \in U$, and *a fortiori* $S - E$ is a neighbourhood of η.

2° Suppose that $w_\eta = 0$; by virtue of (8.5.2), (i)), applied following the general method of (8.1.2), *a*)), there exists a neighbourhood U of η such that $w|f^{-1}(U) = 0$; replacing S by U, one can already suppose $w = 0$ in X. Then $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H}$ is a complex of three terms, to which one can apply (9.4.3); by hypothesis we have $(\mathscr{H}^0(\mathscr{L}^\bullet))_\eta = \mathscr{H}^0(\mathscr{L}^\bullet_\eta) \neq 0$, and $\mathscr{H}^0(\mathscr{L}^\bullet)$ is coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4 and 5.3.3), hence it follows from (1.8.6) that there exists a neighbourhood U of η such that $(\mathscr{H}^0(\mathscr{L}^\bullet))_s \neq 0$ for every $s \in U$; but as one can suppose that $\mathscr{H}^0(\mathscr{L}^\bullet_s) = (\mathscr{H}^0(\mathscr{L}^\bullet))_s$ for every $s \in U$ by (9.4.3), we see again that $S - E$ is a neighbourhood of η in S.

Corollary (9.4.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$, $\mathscr{G}$ two quasi-coherent $\mathscr{O}_X$-Modules of finite presentation, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\mathscr{O}_X$-Modules. Then the set of $s \in S$ such that u_s is injective* (*resp. surjective, bijective, zero*) *is locally constructible.*

It suffices to apply (9.4.4) to the sequences $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G}$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G} \to 0$, $\mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{1_{\mathscr{G}}} \mathscr{G}$.

IV-3 | 64

Corollary (9.4.6). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Let h be a section of $\mathscr{F}$ over X; for every $s \in S$, let h_s be the corresponding section of $\mathscr{F}_s$ over X_s* (*for the projection morphism $X_s \to X$*)*. Then the set of $s \in S$ such that $h_s = 0$ is locally constructible.*

It suffices to note that h corresponds to a homomorphism $u : \mathscr{O}_X \to \mathscr{F}$ and h_s to the homomorphism u_s.

Proposition (9.4.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. The set E* (*resp. E'*) *of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module locally free* (*resp. locally free of rank n*) *is locally constructible.*

If X is an algebraic prescheme over a field k, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, k' an extension of k, then, for $\mathscr{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that it be so of $\mathscr{F} \otimes_k k'$, since the projection $X_{(k')} \to X$ is a faithfully flat morphism (2.2.7). In other words, the condition (9.2.1), (i)) is satisfied for the properties we wish to prove constructible, and it remains to verify (9.2.1), (ii)); one can therefore still suppose that S is affine, Noetherian and integral. There are then four cases to consider:

1° $\eta \in E$. By virtue of (8.5.5), applied following the general method of (8.1.2), *a*)), there exists a neighbourhood U of η in S such that $\mathscr{F}|f^{-1}(U)$ is locally free (resp. locally free of rank n); *a fortiori* $\mathscr{F}_s$ is locally free (resp. locally free of rank n) for every $s \in U$.

2° $\eta \in E'$. Same reasoning as in 1°.

3° $\eta \in S - E$. As $\mathscr{F}_\eta$ is a coherent $\mathscr{O}_{X_\eta}$-Module, to say that it is not locally free means that it is not flat over $\mathscr{O}_{X_\eta}$ (Bourbaki, *Alg. comm.*, chap. II, §5, nº 2, cor. 2 du th. 1). The fact that $S - E$ is a neighbourhood of η will then follow from the more general lemma below:

Lemma (9.4.7.1). — *Let S be an integral Noetherian prescheme, of generic point η, X, Y two S-preschemes of finite type over S, $g : X \to Y$ an S-morphism, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. If $\mathscr{F}_\eta$ is not g_η-flat, then there exists a neighbourhood U of η in S such that for every $s \in U$, $\mathscr{F}_s$ is not g_s-flat.*

Taking into account (2.1.2) and Bourbaki, *Alg. comm.*, chap. Ier, §2, nº 3, Remarque 1, the hypothesis means that there exists a non-empty open V of Y_η, and an injective homomorphism $v : \mathscr{M} \to \mathscr{N}$ of $\mathscr{O}_V$-Modules *coherent*, such that the homomorphism $1 \otimes v : \mathscr{F}_\eta \otimes_{\mathscr{O}_V} \mathscr{M} \to \mathscr{F}_\eta \otimes_{\mathscr{O}_V} \mathscr{N}$ is not injective. We have $V = Y_\eta \cap W$, where W is open in Y (**I**, 3.6.1), and it follows from (8.5.2), (i) and (ii)), applied following the method of (8.1.2), *a*)), that there exists a neighbourhood U_0 of η in S, two $\mathscr{O}_Z$-Modules coherent $\mathscr{G}$, $\mathscr{H}$ (where $Z = W \cap h^{-1}(U_0)$, $h : Y \to S$ being the structural morphism) and an $\mathscr{O}_Z$-homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that $u_s : \mathscr{G}_s \to \mathscr{H}_s$ is injective (9.4.5). But for every $s \in U_0$, the homomorphism $1 \otimes u_s : \mathscr{F}_s \otimes_{\mathscr{O}_{Z_s}} \mathscr{G}_s \to \mathscr{F}_s \otimes_{\mathscr{O}_{Z_s}} \mathscr{H}_s$ is nothing but $(1 \otimes u)_s$; the hypothesis that $(1 \otimes u)_\eta$ is not injective therefore entails (9.4.5) the existence of a non-empty open $U \subset U_0$ such that for every $s \in U$, $(1 \otimes u)_s$ is not injective, and hence $\mathscr{F}_s$ is not g_s-flat for every $s \in U$. IV-3 | 65

4° $\eta \in S - E'$. It is clear that $S - E \subset S - E'$, and if $\eta \in S - E$, $S - E'$ is *a fortiori* a neighbourhood of η by 3°. Let us therefore suppose $\eta \in E$, hence $\mathscr{F}_\eta$ is locally free; these hypotheses entail that X_η is non-connected, and that the ranks of the $\mathscr{O}_{X_\eta}$-Module locally free $\mathscr{F}_\eta$ are not the same on the various connected components of X_η. Or, it follows from (8.1.2), applied following the method of (8.1.2), *a*)), that one can suppose (replacing S by a neighbourhood of η) that X and X_η have the same connected components, the connected components of X_η being the intersections of X_η and the connected components of X. The conclusion then follows from the reasoning given in 2°, applied to each of the connected components of X (which are finite in number).

Remark (9.4.7.2). — The lemma (9.4.7.1) will be later generalised and freed of Noetherian hypotheses (11.2.8).

Proposition (9.4.8). — *Let S be a locally Noetherian prescheme, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that for every $s \in S$, X_s is a locally integral prescheme. Then the set E (resp. E') of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module of torsion (resp. an $\mathscr{O}_{X_s}$-Module without torsion) is locally constructible.*

One can evidently suppose $S = \operatorname{Spec}(A)$ affine and Noetherian and prove that E (resp. E') is constructible by using the criterion ($\mathbf{0}_{\text{III}}$, 9.2.3); replacing S by the closed reduced sub-prescheme of S having as underlying space a closed irreducible part Y of S, and noting (**I**, 3.6.4) that for $s \in Y$, the fibre $(X_{(Y)})_s$ is canonically identified with X_s and the sheaf $(\mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_Y)_s$ with $\mathscr{F}_s$, we see that one is reduced to the case where S is integral of generic point η, and to proving that E or $S - E$ (resp. E' or $S - E'$) is a neighbourhood of η in S. Note furthermore that X is a finite union of affine open sets U_i, of finite type over S, and each of the $(U_i)_\eta$ is either empty (resp. non-empty), one knows that $(U_i)_s$ is also empty (resp. non-empty) in a neighbourhood of η (9.2.6). We can therefore suppose all the $(U_i)_\eta$ non-empty and integral, and to say that $\mathscr{F}_s$ is of torsion (resp. without torsion) is equivalent to saying that each $\mathscr{F}_s|(U_i)_s$ is of torsion (resp. without torsion). We are therefore reduced to the case where $X = \operatorname{Spec}(B)$ and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type; we will set $B_s = B \otimes_A \mathbf{k}(s)$, $M_s = M \otimes_A \mathbf{k}(s)$ and one can suppose B_η integral. There are then four cases to consider:

1° $\eta \in E$; M_η is therefore a B_η-module of torsion of finite type, and there is therefore an $h \neq 0$ in B_η such that $hM_\eta = 0$; by virtue of (8.5.2), (i)), applied following the general method of (8.1.2), *a*)), there exists a neighbourhood U of η in S such that $hM_s = 0$ for every $s \in U$; hence E is a neighbourhood of η.

2° $\eta \in E'$. Same reasoning as in 1°. IV-3 | 65

3° $\eta \in S - E$. As $\mathscr{F}_\eta$ is a coherent $\mathscr{O}_{X_\eta}$-Module, to say that it is not locally free is equivalent to saying that it is not flat over $\mathscr{O}_{X_\eta}$ (Bourbaki, *Alg. comm.*, chap. II, §5, nº 2, cor. 2 du th. 1). The fact that $S - E$ is a neighbourhood of η then follows from the lemma (9.4.7.1) applied to the case where g is the identity.

4° $\eta \in S - E'$. It is clear that $S - E \subset S - E'$, and if $\eta \in S - E$, $S - E'$ is *a fortiori* a neighbourhood of η by 3°. Let us therefore suppose $\eta \in E$, hence $\mathscr{F}_\eta$ is locally free; these hypotheses entail that X_η is non-connected, and that the ranks of the $\mathscr{O}_{X_\eta}$-Module locally free $\mathscr{F}_\eta$ are not the same on the various connected components of X_η. Or, it follows from (8.1.2), applied following the method of (8.1.2), *a*)), that one can suppose (replacing S by a neighbourhood of η) that X and X_η have the same connected components, the connected components of X_η being the intersections of X_η and the connected components of X. The conclusion then follows from the reasoning given in 2°, applied to each of the connected components of X (which are finite in number).

9.5. Constructibility of topological properties

Proposition (9.5.1). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. Then the set E of $s \in S$ such that $Z_s \neq \varnothing$ is locally constructible.*

Proof. Indeed, we have $E = f(Z)$, and it suffices to apply the theorem of Chevalley (**I**, 8.4). □

Corollary (9.5.2). — *If Z', Z'' are two locally constructible parts of X, the set of $s \in S$ such that $Z'_s \subset Z''_s$ (resp. $Z'_s = Z''_s$) is locally constructible.*

Proof. Indeed, the relation $Z'_s \subset Z''_s$ is equivalent to $(Z' \cap (\complement Z''))_s = \varnothing$, and $Z' \cap \complement Z''$ is locally constructible. □

Proposition (9.5.3). — *Let $f : X \to S$ be a morphism of finite presentation, Z, Z' two locally constructible parts of X such that $Z \subset Z'$. Then the set E of $s \in S$ such that Z'_s is dense in Z_s is locally constructible in S.*

Proof. We must verify the two conditions of (9.2.2), (ii). As far as the first is concerned, consider a prescheme X algebraic over a field k, two locally closed constructible parts Z, Z' of X such that $Z \subset Z'$, and an extension k' of k. Then the projection $p : X_{(k')} \to X$ is a faithfully flat and quasi-compact morphism, and we have $p^{-1}(\overline{Z}) = \overline{p^{-1}(Z)}$ and $p^{-1}(\overline{Z'}) = \overline{p^{-1}(Z')}$; by virtue of (2.3.10), as p is surjective, the relation $\overline{Z} = \overline{Z'}$ is equivalent to $\overline{p^{-1}(Z)} = \overline{p^{-1}(Z')}$.

Let us verify the second condition, and suppose therefore S affine, Noetherian, integral, of generic point η. We distinguish two cases:

label=1° $\eta \in S - E$, in other words, Z_η is not dense in Z'_η; then there exists in X an open V such that $V \cap Z_\eta = \varnothing$ and $V \cap Z'_\eta \neq \varnothing$. As X is Noetherian, V is locally constructible, and by virtue of (9.5.1), there is a neighbourhood U of η in S such that for every $s \in U$, $(V \cap Z)_s = \varnothing$ and $(V \cap Z')_s \neq \varnothing$; this implies that Z_s is not dense in Z'_s for $s \in U$, in other words $U \subset S - E$.

lbbel=2° $\eta \in E$, so Z_η is dense in Z'_η. Let us first show that we can suppose Z' *closed*. Indeed, Z_η is dense in $\overline{Z'_\eta}$ (closure taken in X_η); setting $V_\eta = X_\eta - \overline{Z_\eta}$, which is open in X_η and does not meet Z'_η, we can suppose V_η of the form $\overline{V} \cap X_\eta$, where V is open (and hence constructible) in X, and the hypothesis $V_\eta \cap Z'_\eta = \varnothing$ implies then $V_s \cap Z'_s = \varnothing$ for every s in a neighbourhood of η in S by virtue of (9.5.1). Replacing S by an open neighbourhood of η, we can therefore suppose $V \cap Z' = \varnothing$, whence $V \cap \overline{Z'} = \varnothing$ (closure taken in X), and consequently $(Z')_\eta = \overline{Z'_\eta}$, whence our assertion. The decomposition of Z' into its irreducible components being a finite union, and restricting further S to a neighbourhood of η, we can suppose that all the irreducible components Z'_i of Z' meet X_η, whence it follows that X_η contains the generic points of the Z'_i (**0**, 2.1.8). To say that Z_s is dense in Z'_s is equivalent to saying that each of the $(Z \cap Z'_i)_s$ is dense in $(Z'_i)_s$, and one is thus reduced to the case where Z' is *irreducible*. Replacing then X by the sub-prescheme having the reduced underlying space of Z', we see that we can suppose $Z' = X$ and that X is *integral* and dominates S. Finally, covering X by a finite number of affine opens W_j and replacing Z by $Z \cap W_j$, we can suppose that $X = \operatorname{Spec}(A)$, where A is a Noetherian integral ring. As X_η is Noetherian integral and Z_η is constructible and dense in X_η, Z_η contains a non-empty open of X_η of the form $(D(t))_\eta$, where t is a nonzero element of A. Replacing S by a neighbourhood of η in S if necessary, by virtue of the relation $(D(t))_\eta \subset Z_\eta$, we can suppose in addition that $D(t) \subset Z$ (9.5.2). Finally, since the homothety of ratio t_s in $\mathscr{O}_{X_s}$ is injective, it follows from (9.4.5) that for s close to η, t_s is $\mathscr{O}_{X_s}$-regular, so $(X_s)_{t_s}$ is dense in X_s, and *a fortiori* it is the same for Z_s which contains $(X_s)_{t_s}$.

□

Corollary (9.5.4). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X. The set E of $s \in S$ such that Z_s is closed (resp. open, resp. locally closed, resp. locally constructible) in X_s is locally constructible in S.*

Proof. To say that Z_s is open in X_s means that $(X - Z)_s = X_s - Z_s$ is closed in X_s, and as $X - Z$ is locally constructible, we can limit ourselves to considering the set of $s \in S$ such that Z_s is closed and the set of $s \in S$ such that Z_s is locally closed.

Let us verify in each case the two conditions of (9.2.2), (ii). The first results from the fact that $p : X_{(k')} \to X$ is faithfully flat and quasi-compact, and from (2.3.12) and (2.3.14). Let us therefore verify the second condition, S being supposed affine, Noetherian, integral, of generic point η. Set $Z' = \overline{Z}$; the same reasoning as in (9.5.3) shows that Z'_η is equal to the closure of Z_η in X_η; by virtue of (9.5.3), there is thus a neighbourhood U of η such that for $s \in U$, Z_s is dense in Z'_s, the latter being closed. To say that Z_s is closed in X_s means that $Z''_s = \varnothing$, where $Z'' = Z' - Z$; it follows from (9.5.1) that the set E of $s \in S$ where $Z''_s = \varnothing$ is such that E or $S - E$ contains a neighbourhood of η. To say that Z_s is locally closed in X_s means that Z''_s is closed in X_s; it suffices therefore to apply the preceding result replacing Z by Z'', which is locally constructible in X. □

Proposition (9.5.5). — *Let $f : X \to S$ be a morphism of finite presentation, Z a locally constructible part of X such that for every $s \in S$, Z_s is locally closed in X_s. For every $s \in S$, let $D(s) \subset \mathbf{Z} \cup \{-\infty\}$ be the set of dimensions of the irreducible components of Z_s. Then the function $s \mapsto D(s)$ is locally constructible in S.*

Proof. Let Φ be a finite part of $\mathbf{Z} \cup \{-\infty\}$; it is a question of showing that the set of $s \in S$ such that $D(s) = \Phi$ is locally constructible; taking into account (9.2.3), we have once more to verify the two conditions of (9.2.2), (ii).

As far as the first is concerned, it is a question of seeing that if X is a prescheme algebraic over a field k, Z a locally closed part of X, k' an extension of k, $p : X_{(k')} \to X$ the canonical projection, then the set of dimensions of the irreducible components of Z is the same as that of the dimensions of the irreducible components of $p^{-1}(Z)$; taking into account the existence of a sub-prescheme of X having Z as underlying space (4.2.8), this follows from (**I**, 5.2.1).

For the second condition of (9.2.2), (ii), we are in the case where S is Noetherian, integral, of generic point η, and $f : X \to S$ is a morphism of finite type, so that X is Noetherian. The sub-space Z_η of the Noetherian space X_η is by hypothesis locally closed, therefore has a finite number of irreducible components $Z_{i\eta}$, which are locally closed in X. It follows for each index i that there exists a locally closed part Z_i of X such that $(Z_i)_\eta = Z_{i\eta}$, so if $Z' = \bigcup_i Z_i$, we have $Z'_\eta = Z_\eta$. But as Z and Z' are locally constructible, we can, replacing S by a neighbourhood of η, suppose that $Z' = Z$ (9.5.2). Furthermore, for $i \neq j$, $Z_{i\eta} \cap Z_{j\eta}$ is rare and *closed* in $Z_{i\eta}$; therefore (9.5.3) and (9.5.4), we can also suppose, restricting S, that for $j \neq i$, $(Z_i)_s \cap (Z_j)_s$ is rare and closed in $(Z_i)_s$. As Z_i is locally closed in X, there is a sub-prescheme of X having Z_i as underlying space (and still denoted Z_i), which is of type fini over S (**I**, 6.3.5). Set $U_i = Z_i - \bigcup_{j \neq i}(Z_i \cap Z_j)$, which is open in Z_i and such that $(U_i)_s$ is open and everywhere dense in $(Z_i)_s$; furthermore, by construction, the U_i have no two common points. As $(Z_i)_s$ is a $\boldsymbol{k}(s)$-prescheme algebraic, the set of dimensions of the irreducible components of $Z_s = \bigcup_i (Z_i)_s$ is the same as the set of dimensions of the irreducible components of the reunion of the $(U_i)_s$ (4.1.1.3), each of these components being already an irreducible component of one of the $(U_i)_s$. We can therefore limit ourselves to the case where $Z = U_i$; furthermore, as Z_η is then irreducible, there is only a single irreducible component of Z meeting X_η, and one can evidently, restricting Z, suppose Z irreducible. We are thus finally reduced to proving the following particular case of (9.5.5): □

IV-3 | 69

Corollary (9.5.6). — *Let S be a Noetherian prescheme, integral, of generic point η, X an irreducible prescheme, $f : X \to S$ a dominant morphism of finite type. Then there exists a neighbourhood U of η in S such that, for every $s \in U$, all the irreducible components of X_s are of dimension $n = \dim(X_\eta)$.*

Proof. We can evidently reduce to the case where $S = \operatorname{Spec}(A)$ is affine, A being therefore Noetherian; replacing f by f_{red}, which is of finite type (**I**, 5.4), (vi), we can suppose A integral and X reduced, therefore integral. We know (4.1.2) that there exists an open W dense in X_η and a $\boldsymbol{k}(\eta)$-morphism *finite* and *surjective* $h : W \to \mathbf{V}((\boldsymbol{k}(\eta))^n) = \operatorname{Spec}(B')$, where $B' = \boldsymbol{k}(\eta)[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ (T_i indeterminates). If V is an open of X such that $V \cap X_\eta = W$, we know (9.5.3) that for s close to η in S, V_s is an open dense in X_s, and one can therefore (4.1.3) reduce to the case where $V = X$, $W = X_\eta$. Set $B = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ and $Y = \operatorname{Spec}(B) = \mathbf{V}^n_A$, so that $\operatorname{Spec}(B') = Y_\eta$; it follows from (8.8.2), (i) and (8.10.5), (vi) and (x), applied following the method of (8.1.2), *a*), that replacing S by a neighbourhood of η, we can suppose that $h = g_\eta$, where $g : X \to Y$ is a finite surjective morphism; in other words, we have $X = \operatorname{Spec}(C)$, where C is a finite B-algebra and the homomorphism $B \to C$ corresponding to g is *injective*; as C is an integral ring, C is therefore a B-module of finite type *without torsion*, and $C_\eta = C \otimes_A \boldsymbol{k}(\eta)$ is a module of finite type without torsion over $B_\eta = B' \otimes_A \boldsymbol{k}(\eta) = \boldsymbol{k}(\eta)[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ (being a ring of fractions whose denominators lie in $A - \{0\} \subset B - \{0\}$). It follows from (9.4.8) that there is a neighbourhood U of η in S such that for $s \in S$, $C_s = C \otimes_A \boldsymbol{k}(s)$ is a module of finite type *without torsion* over $B_s = B \otimes_A \boldsymbol{k}(s) = \boldsymbol{k}(s)[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, and in particular the homomorphism $g_s : B_s \to C_s$ is injective. As no element of B_s is a divisor of zero in C_s, for every prime ideal $\mathfrak{p}_i$ of C_s (the elements being divisors of zero of C_s), we necessarily have $\mathfrak{p}_i \cap B_s = 0$, whence the canonical homomorphism $B_s \to C_s/\mathfrak{p}_i$ is injective. We deduce that for each irreducible component $Z_i = \operatorname{Spec}(C_s/\mathfrak{p}_i)$ of X_s, the restriction to Z_i of g is a finite and dominant morphism $Z_i \to Y_s$, therefore *surjective* (**II**, 6.1.10). We conclude then from (4.1.2) that $\dim Z_i = n$, which completes the proof. □

Remark (9.5.7). — One should note that under the hypotheses of (9.5.6), it may happen that X_s is irreducible for *no* $s \neq \eta$ in a neighbourhood of η, in other words, the property "X is an irreducible k-prescheme algebraic" is *not constructible*. Take for example $S = \operatorname{Spec}(k[\mathrm{T}])$, where k is an algebraically closed field, T an indeterminate; we then have $\boldsymbol{k}(\eta) = K = k(\mathrm{T})$. Let L be a finite separable extension of K of degree > 1, and let X be the *integral closure* of S in L (**II**, 6.3.4); we then have $X = \operatorname{Spec}(B)$, where B is the integral closure of $k[\mathrm{T}]$ in L. We know that B is a Dedekind ring, and that all the maximal ideals of $k[\mathrm{T}]$, except for a finite number, are not ramified in B; as furthermore the residue field of every maximal ideal of B is necessarily k (since k is a finite extension of k), we see that for almost all the maximal ideals $\mathfrak{j}_s$ of $k[\mathrm{T}]$, $B_s = B/\mathfrak{j}_s B$ is a direct composition of $[L : K]$ copies of the field k, in other words X_s is not irreducible, although $X_\eta = \operatorname{Spec}(L)$ is. The same example shows that the property "X is an integral k-prescheme algebraic" is not constructible. Finally, the property "X is a reduced k-prescheme algebraic" is also not constructible. It suffices for this to take again $S = \operatorname{Spec}(k[\mathrm{T}])$, where this time k is an algebraically closed field of characteristic $p > 0$, and for X the integral closure of S in $L = K^{1/p}$ (where $K = k(\mathrm{T})$), so that $X = \operatorname{Spec}(B)$ with $B = k[\mathrm{T}^{1/p}]$ (k being perfect); every maximal ideal of $k[\mathrm{T}]$ is of the form $(\mathrm{T} - \alpha)$ with $\alpha \in k$; the
unique ideal of B over the ideal $(\mathrm{T} - \alpha)$ is the principal ideal $(\mathrm{T}^{1/p} - \alpha^{1/p})$ and it is immediate that **IV-3** | 71
the quotient ring $B/(\mathrm{T} - \alpha)B$ has nilpotent elements; in other words, X_s is not reduced for any $s \neq \eta$, whilst $X_\eta = \operatorname{Spec}(L)$ is integral.

We shall see a little further on (9.7) that one obtains on the other hand constructible properties when one considers the "geometric" notions corresponding to the notions of irreducible, reduced, or integral prescheme (4.5) and (4.6).

9.6. Constructibility of certain properties of morphisms

Proposition (9.6.1). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism. Let E be the set of $s \in S$ for which f_s has one of the following properties: being*

label=(i) surjective;
lbbel=(ii) dominant;
lcbel=(iii) separated;
ldbel=(iv) proper;
lebel=(v) radicial;
lfbel=(vi) finite;
lgbel=(vii) quasi-finite;
lhbel=(viii) an immersion;
libel=(ix) a closed immersion;
ljbel=(x) an open immersion;

lkbel=(xi) an isomorphism;
llbel=(xii) a monomorphism.

Then E is locally constructible in S.

Proof. The assertions concerning (i), (v), and (vii) are included only for reference, having already been established in (9.3.2).

(ii): Let us note first that f is of finite presentation (**I**, 6.2), (v), therefore, by virtue of the theorem of Chevalley (**I**, 8.4), $f(X)$ is locally constructible in Y. Furthermore, we have $f_s(X_s) = (f(X))_s$ (**I**, 3.6.1); the set of s such that f_s is dominant is the set of s such that $(f(X))_s$ is dense in Y_s; the conclusion follows therefore from (9.5.3).

(iii): As $f : X \to Y$ is of finite presentation, the diagonal immersion $\Delta_f : X \to X \times_Y X$ is of finite presentation (**I**, 6.2), (iv) and (v); it follows from (1.8.4.1) that $\Delta_f(X) = \Delta_X$ is a locally constructible part of $X \times_Y X$. To say that f_s is separated means (taking into account (**I**, 5.3.4)) that $(\Delta_X)_s$ is closed in $X_s \times_{Y_s} X_s = (X \times_Y X)_s$; the conclusion follows from (9.5.4).

(iv): Let us show that the property "X and Y are preschemes algebraic over a field k, and $f : X \to Y$ a proper k-morphism" is constructible. The condition (9.2.1), (i) is verified by virtue of (2.7.1), (vii). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η, and one must prove that E or $S - E$ is a neighbourhood of η. Suppose first that $\eta \in E$; it follows from (8.10.5), (xii), applied following the method of (8.1.2), *a*), that replacing S by a neighbourhood of η, we can suppose that f is itself a proper morphism; one then knows that it is the same for f_s for every $s \in S$ (**II**, 5.4.2), (iii). Suppose on the contrary that $\eta \in S - E$, and we distinguish two cases: **IV-3 | 72**

label=1° Suppose that f_η is not separated; then it follows from (iii) that f_s is not separated (and *a fortiori* not proper) in a neighbourhood of η.

lbbel=2° Suppose f_η separated; Y is a finite union of affine opens V_i, and for f_s to be proper, it suffices that each of its restrictions $f_s^{-1}((V_i)_s) \to (V_i)_s$ be so (**II**, 5.4.1); we can therefore reduce to the case where Y is *affine*, hence a scheme. To say that f_η is not proper means (**II**, 5.6.3) that there exists a morphism *of finite type* $h : Z \to Y_\eta$ such that the morphism $(f_\eta)_{(Z)} : X_\eta \times_{Y_\eta} Z \to Z$ is not closed. As Y is a scheme, we deduce from (8.8.2), (i) and (ii) (restricting to S if needed a neighbourhood of η) that there exists a morphism of finite type $g : Y' \to Y$ such that Z is isomorphic to Y'_η and that $g_\eta = h$; we therefore have a closed part M' of X'_η such that $f'_\eta(M')$ is not closed in Y'_η. Now M' is the trace on X'_η of a closed part N' of X'; as X' is Noetherian and f' of finite type, $f'(N')$ is constructible in Y' (**I**, 8.4) and by hypothesis $(f'(N'))_\eta = f'_\eta(N'_\eta) = f'_\eta(M')$ is not closed in Y'_η. We conclude from (9.5.4) that there exists a neighbourhood U of η in S such that for $s \in U$, $(f'(N'))_s = f'_s(N'_s)$ is not closed in Y'_s; in other words, the morphism f'_s is not closed, and *a fortiori* the morphism f_s is not proper.

(vi): The property is a conjunction of properties (iv) and (vii) (8.11.1), therefore the proposition follows from what has already been proved.

(ix): The verification of condition (9.2.1), (i) follows from (2.7.1), (xi). We can therefore reduce to the case where S is affine, Noetherian, integral, of generic point η, and it suffices to prove that E or $S - E$ is a neighbourhood of η. If $\eta \in E$, we can (replacing S by a neighbourhood of η) suppose that f is a closed immersion by virtue of (8.10.5), (iv), and then it is clear that f_s is a closed immersion for every $s \in S$ (**I**, 4.3.2). Suppose therefore that $\eta \in S - E$, and we distinguish two cases:

label=1° f_η is not a finite morphism. Then it follows from (vi) that in a neighbourhood of η, f_s is not finite, nor *a fortiori* a closed immersion.

lbbel=2° f_η is finite; then, by virtue of (8.10.5), (x), we can suppose (restricting S if needed) that f is itself a *finite* morphism. In this case $\mathscr{A} = \mathscr{A}(X) = f_*(\mathscr{O}_X)$ is an $\mathscr{O}_Y$-Module coherent (**II**, 6.1.3) and $f = \mathscr{A}(u)$, where $u : \mathscr{O}_Y \to \mathscr{A}$ is a homomorphism of $\mathscr{O}_Y$-Algebras (**II**, 1.1.2); as f_η is not a closed immersion by hypothesis, u_η is not surjective (**II**, 1.4.10), therefore (9.4.5) there exists a neighbourhood of η in which u_s is not surjective, and consequently (**I**, 4.2.3) f_s is not a closed immersion.

(viii): The verification of condition (9.2.1), (i) is done as in (ix), using this time (2.7.1), (x) and the fact that every immersion of one Noetherian prescheme into another is quasi-compact. We are therefore reduced to the case where S is affine, Noetherian, integral, of generic point η, and it suffices to prove that E or $S-E$ is a neighbourhood of η. If $\eta \in E$, we conclude as in (ix) with the help of (8.10.5), (ii). If $\eta \in S-E$, we distinguish again two cases:

label=1° $f_\eta(X_\eta)$ is not a locally closed part of Y_η. As $f(X)$ is constructible in Y (**I**, 8.4) and $f_s(X_s) = (f(X))_s$, we deduce from (9.5.4) that for s close to η, $f_s(X_s)$ is not locally closed in Y_s, and *a fortiori* f_s is not an immersion.

lbbel=2° $f_\eta(X_\eta)$ is open in Y_η but f_η is not an immersion. It follows then from (viii) that for s close to η, f_s is not an immersion, nor *a fortiori* an open immersion.

lcbel=3° $f_\eta(X_\eta)$ is locally closed in Y_η but f_η is not an immersion. As $f(X)$ is constructible in Y (**I**, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can suppose that $f(X)$ is locally closed in Y. There is then an open V of Y containing $f(X)$ and in which $f(X)$ is closed. As Y is Noetherian, V is of finite type over S, so we can reduce to the case where f is *surjective* by replacing Y by $f(X)$. But then $f_s(X_s) = (f(X))_s$ is closed in Y_s for every $s \in S$, so we can say that f_s is a closed immersion, meaning that f_s defines X as a sub-prescheme of Y_s defined by an Ideal coherent $\mathscr{J}$ of $\mathscr{O}_Y$. By hypothesis f_η is not an isomorphism, therefore $\mathscr{J}_\eta \neq 0$; we conclude (9.4.5) that in a neighbourhood of η, $\mathscr{J}_s \neq 0$, and consequently the closed immersion f_s is not open. **IV-3 | 73**

(x): Using this time (2.7.1), (ix) and (8.10.5), (iii), we are reduced to the case where S is affine, Noetherian, integral, of generic point η, and where $\eta \in S-E$. We distinguish three cases:

label=1° $f_\eta(X_\eta)$ is not open in Y_η. As $f(X)$ is constructible in Y (**I**, 8.4), we deduce from (9.5.4) that $f_s(X_s)$ is not open in Y_s for s close to η, and *a fortiori* f_s is not an open immersion.

lbbel=2° $f_\eta(X_\eta)$ is open in Y_η but f_η is not an immersion. It follows then from (viii) that for s close to η, f_s is not an immersion, nor *a fortiori* an open immersion.

lcbel=3° $f_\eta(X_\eta)$ is open in Y_η and f_η is an immersion. As $f(X)$ is constructible in Y (**I**, 8.4) and since the same holds for $\overline{f(X)}$ since Y is Noetherian, it follows from (8.3.11) that restricting to S if necessary, we can already suppose that $f(X)$ is open in Y. But then $f_s(X_s) = (f(X))_s$ is open in Y_s for every $s \in S$, so we can reduce to the case where f is *surjective* and f is a closed immersion, in other words, that X is a sub-prescheme of Y defined by an Ideal coherent $\mathscr{J}$ of $\mathscr{O}_Y$. By hypothesis f_η is not an isomorphism, hence $\mathscr{J}_\eta \neq 0$; we conclude (9.4.5) that in a neighbourhood of η, $\mathscr{J}_s \neq 0$, and consequently f_s is not an open immersion.

(xi): The property is a conjunction of properties (i) and (x) and therefore follows from what has been proved.

(xii): By virtue of (**I**, 5.3.8), to say that f_s is a monomorphism means that $\Delta_{(f)_s} = (\Delta_f)_s : X_s \to X_s \times_{Y_s} X_s = (X \times_Y X)_s$ is an isomorphism, and as $X \times_Y X$ is an S-prescheme of finite presentation (**I**, 6.2), (iv), the conclusion follows from (xi). □

Proposition (9.6.2). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ an S-morphism.*

label=I) Let E be the set of $s \in S$ for which f_s has one of the following properties: being

label=(i) affine;

lbbel=(ii) quasi-affine;

lcbel=(iii) projective;

ldbel=(iv) quasi-projective.

Then E is ind-constructible in S.

IV-3 | 74

label=II) Let $\mathscr{L}$ be an $\mathscr{O}_X$-Module invertible. Then the set E' of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to f_s, is ind-constructible in S.

Proof. I) Let us verify conditions (i) and (ii) of (9.2.1). As far as the condition (9.2.1), (i) is concerned, it follows, for properties (i) and (ii), from (2.7.1), (xiii) and (xiv); for properties (iii) and (iv), it follows from (9.1.5). Let us then verify condition (9.2.1), (ii), supposing therefore S Noetherian, integral, of generic point η, and that $f_\eta : X_\eta \to Y_\eta$ has one of the properties (i) to (iv) of the statement. Applying (8.10.5), (viii), (ix), (xiii), and (xiv) following the method of (8.1.2), *a*), we see first that there exists a neighbourhood open U of η such that, if V and W are the respective inverse images of U in X and Y, the restriction $V \to W$ of f to the one of the properties (i) to (iv) that one considers. The conclusion follows then from the stability of the properties considered under base change (**II**, 4.6.13) and (4.4.10).

II) We proceed in the same way. For condition (9.2.1), (i) it follows this time from (2.7.2). For condition (9.2.1), (ii), with the same notations as in I), it follows from (8.10.5.2) that for a neighbourhood U of η in S, the restriction $\mathscr{L}|V$ is ample (resp. very ample) relatively to f. The conclusion follows again from the stability of the properties considered under base change (**II**, 4.6.13). □

We can improve (9.6.2), II) under certain conditions:

Proposition (9.6.3). — *Let X, Y be two S-preschemes of finite presentation, $f : X \to Y$ a proper S-morphism, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible. Then the set E (resp. E') of $s \in S$ such that $\mathscr{L}_s$ is an $\mathscr{O}_{X_s}$-Module ample (resp. very ample) relatively to $f_s : X_s \to Y_s$ is locally constructible in S.*

Proof. Let k be a field, Z, T two preschemes algebraic over k, $\mathscr{M}$ an $\mathscr{O}_Z$-Module invertible, $g : Z \to T$ a k-morphism of finite type; then, if $\mathbf{P}(Z, T, \mathscr{M}, g, k)$ denotes the relation "$\mathscr{M}$ is ample (resp. very ample) relatively to g", we have already noted in (9.6.2) that $\mathbf{P}(Z, T, \mathscr{M}, g, k)$ satisfies condition (9.2.1), (i) by virtue of (2.7.2). We know on the other hand that E and E' are ind-constructible. It remains therefore to see that if S is Noetherian, integral, of generic point η and if $\eta \in S - E$ (resp. $\eta \in S - E'$), then $S - E$ (resp. $S - E'$) contains a neighbourhood of η. We shall separately consider the cases E and E'.

Case of E'. — This amounts to saying that: *either* the canonical homomorphism $(f_s)^*((f_s)_*(\mathscr{L}_s)) \to$ **IV-3** | 75
$\mathscr{L}_s$ is not surjective; *or* the preceding homomorphism is surjective and the canonical morphism $r : X_s \to \mathbf{P}((f_s)_*(\mathscr{L}_s))$ is not an immersion. Taking into account the isomorphism (9.6.3.1), these conditions are written respectively in the form: 1° the canonical homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is not surjective; 2° the preceding homomorphism is surjective and the canonical morphism $X_s \to \mathbf{P}((f_*(\mathscr{L}))_s)$ is not an immersion.

Suppose first that the canonical homomorphism $(f^*(f_*(\mathscr{L})))_\eta \to \mathscr{L}_\eta$ is not surjective. As $f_*(\mathscr{L})$ is coherent, so is $f^*(f_*(\mathscr{L}))$, and then it follows from (9.4.5) that for every s in a neighbourhood of η, the homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is not surjective, which proves in this case that $S - E'$ is a neighbourhood of η.

Suppose next that the canonical homomorphism $(f^*(f_*(\mathscr{L})))_\eta \to \mathscr{L}_\eta$ is surjective but that the morphism $X \to \mathbf{P}((f_*(\mathscr{L}))_\eta)$ is not an immersion. Then the same reasoning as above shows that for s sufficiently close to η, the homomorphism $(f^*(f_*(\mathscr{L})))_s \to \mathscr{L}_s$ is surjective; on the other hand, by virtue of (9.6.1), (viii), the morphism $X_s \to \mathbf{P}((f_*(\mathscr{L}))_s)$ is not an immersion for s in a neighbourhood of η, which completes the proof in the case of E'.

Case of E. — Consider first the particular case where $Y = S$:

Corollary (9.6.4). — *Let $f : X \to S$ be a proper morphism of finite presentation, $\mathscr{L}$ an $\mathscr{O}_X$-Module invertible. Then the set E of $s \in S$ such that $\mathscr{L}_s$ is ample (relatively to f_s) is open in S, and $\mathscr{L}|f^{-1}(E)$ is ample relatively to the restriction $f^{-1}(E) \to E$ of f.*

Proof. As condition (9.2.1), (i) is verified by the property $\mathbf{P}$ defined above, the result of (9.2.2), (iv) and the reasoning of (9.2.3) show that it suffices to reduce to the case where S is *Noetherian*; but then the result follows from (**III**, 4.7.1) and the stability of amplitude under base change (**II**, 4.6.13). □

Corollary (9.6.5). — *Under the hypotheses of (9.6.4), for $\mathscr{L}$ to be ample relatively to f, it is necessary and sufficient that, for every $s \in S$, $\mathscr{L}_s$ be ample relatively to f_s.*

(9.6.6) End of the proof of (9.6.3). — Let us return to the general case, S being Noetherian, integral, of generic point $\eta \in S - E$. As f is proper, it follows from (9.6.4) that the set V of $y \in Y$ such that $\mathscr{L}_y$ is an $\mathscr{O}_{X_y}$-Module ample, relatively to the morphism $f_y : X_y \to \operatorname{Spec}(\boldsymbol{k}(y))$, is open, and therefore $F = Y - V$ is closed in Y. As f_s is proper and, for every $y \in Y$ above s, $\mathscr{L}_y = (\mathscr{L}_s)_y$, this being the case since f_s is proper, we have $F_s \neq \varnothing$. It suffices to show that $F_s \neq \varnothing$ for every s in a neighbourhood of η in S, and as F is closed in Y and $F_\eta \neq \varnothing$, it follows from (9.5.1) that the set of $s \in S$ such that $F_s \neq \varnothing$ is a neighbourhood of η in S. □

Remarks (9.6.7). — (i) For each of the properties $\mathbf{P}$ considered in (9.6.1), proposition (9.3.3) is applicable, and these properties (for the morphisms f_s) are therefore "stable" under passage to a projective limit of a projective system of preschemes X_λ essentially affine (8.13.4) one over another.

(ii) Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is *open* in X_s, and denote by Z_s the sub-prescheme of X_s induced on the open $Z \cap X_s$. Then, in propositions (9.6.1) **IV-3** | 76
and (9.6.2), (I), we can everywhere replace f_s by its restriction $f_s|Z_s : Z_s \to Y_s$ without changing the conclusions. Indeed, the verification of (9.2.1), (i) proceeds as in (9.6.1) and (9.6.2). On the other hand, in the reduction to the Noetherian case, made in (9.2.3), if $Z = q^{-1}(Z_0)$, where Z_0 is the canonical projection and Z_0 a constructible part of X_0 (8.3.11), the fact that $(Z_0)_{s_0}$ is open in $(X_0)_{s_0}$, for $s_0 = p(s)$, follows from (2.4.10) and from the fact that the projection $X_s \to (X_0)_{s_0}$ is surjective. Now, as Z_η is open in X_η, there is an open $Z' \subset X$ such that $Z_\eta = Z'_\eta \cap X_\eta$; as X is Noetherian, Z' is constructible, and since it has the same properties as Z, we conclude from (9.5.2) and from ($\mathbf{0}_{\mathrm{III}}$, 9.2.2) that there is a neighbourhood U of η in S such that $Z_s = Z'_s$ for $s \in U$. Replacing S by U, we can therefore reduce to the case where Z is open in X, and then the result has been proved in (9.6.1) and (9.6.2).

9.7. Constructibility of properties of geometric separability, geometric irreducibility, and geometric connectedness

Lemma (9.7.1). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η has n irreducible components (resp. n' connected components), there exists a neighbourhood open U of η in S such that for every $s \in U$, X_s has at least n irreducible components (resp. n' connected components).*

Proof. We can reduce to the case where S is affine, so X is quasi-compact and quasi-separated. Let V_i ($1 \leqslant i \leqslant n$) be the interiors of the irreducible components of X_η, which are pairwise disjoint and quasi-compact, and whose reunion is dense in X_η; by virtue of (8.2.11), applied following the method of (8.1.2), *a*), there exists for each i a quasi-compact open W_i in X such that $W_i \cap X_\eta = V_i$; as X is quasi-separated, the intersections $W_i \cap W_j$ are quasi-compact (**I**, 2.7), and therefore, replacing S by a neighbourhood of η, we can suppose that $W_i \cap W_j = \varnothing$ for $i \neq j$ by virtue of (8.3.3). Furthermore, as the W_i are constructible, there is a neighbourhood U of η in S such that for every $s \in U$ the $(W_i)_s$ are non-empty (9.5.1) and (9.2.3) and the reunion of the $(W_i)_s$ is dense in X_s (9.5.3) and (9.2.3). This being the case, as the irreducible components of each $(W_i)_s$ (being finite in number) are also irreducible components of the reunion of the $(W_i)_s$ (**0**, 2.1.7), the closures in X_s of these components are the irreducible components of X_s (**0**, 2.1.6) and their number is evidently $\geqslant n$.

Let now C_j ($1 \leqslant j \leqslant n'$) be the connected components of X_η, which are quasi-compact opens of X_η, pairwise disjoint. If we replace V_i by C_j in the preceding reasoning, we see (using twice (8.3.3)) that we can suppose X a reunion of n' quasi-compact opens M_j ($1 \leqslant j \leqslant n'$), pairwise disjoint, and such that, in a neighbourhood of η, the $(M_j)_s$ are non-empty. As the reunion of the $(M_j)_s$ is X_s, the connected components of the $(M_j)_s$ are the connected components of X_s, and therefore their number is $\geqslant n'$. □

Lemma (9.7.2). — *Let S be an irreducible prescheme of generic point η, $f : X \to S$ a morphism of finite presentation. If X_η is not reduced, there exists a neighbourhood open U of η in S such that, for every $s \in U$, X_s is not reduced.*

Proof. Indeed, let $\mathscr{N}$ be the Nilradical of $\mathscr{O}_X$; it follows from (8.2.13), applied following the method of (8.1.2), *a*), that $\mathscr{N}_\eta$ is the nilradical of $\mathscr{O}_{X_\eta}$, and the hypothesis is that $\mathscr{N}_\eta \neq 0$. We conclude from (9.4.5) and (9.2.3) that there is a neighbourhood U of η such that, for every $s \in U$, $\mathscr{N}_s$ identifies with an Ideal of $\mathscr{O}_{X_s}$ and $\mathscr{N}_s \neq 0$; as $\mathscr{N}$ is evidently contained in the Nilradical of $\mathscr{O}_{X_s}$, we see that X_s is not reduced for $s \in U$. □

(9.7.3) Given a polynomial $F \in A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, where A is a ring and the T_i are indeterminates, for a ring homomorphism $\rho : A \to B$, we denote by $F_{(\rho)}$ or $F_{(B)}$ the polynomial in $B[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ obtained by replacing each coefficient of F by its image under ρ. If k is a field, $F \in k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ a non-constant polynomial, and $X = \operatorname{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]/(F))$, to say that X is integral (or that the ideal (F) is prime) means that F is *irreducible* (i.e. that in every factorisation $F = F_1F_2$ in polynomials of $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, F_1 or F_2 is of degree 0); this follows from the fact that the polynomial ring $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is factorial. We deduce immediately from (4.6.2) and (4.5.2):

Lemma (9.7.4). — *Let k be a field, Ω an algebraically closed extension of k, F a non-constant polynomial of $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$. The following conditions are equivalent:*

*label=*a) $X = \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]/(F))$ *is geometrically integral.*

*lbbel=*b) $F_{(K)}$ *is irreducible for every extension K of k.*

*lcbel=*c) $F_{(\Omega)}$ *is irreducible.*

We shall say in this case that F is geometrically irreducible.

Lemma (9.7.5). — *Let A be an integral ring, K the field of fractions of A, F a non-constant polynomial of $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ of degree d such that $F_{(K)}$ is geometrically irreducible. Then there exists $f \neq 0$ in A such that for every $x \in D(f)$, $F_{(\boldsymbol{k}(x))}$ is geometrically irreducible.*

Proof. Set $F = \sum_\alpha c_\alpha \mathrm{T}^\alpha$ as usual, with $c_\alpha = c_{\alpha_1 \alpha_2 \ldots \alpha_n}$, $\mathrm{T}^\alpha = \mathrm{T}_1^{\alpha_1} \mathrm{T}_2^{\alpha_2} \cdots \mathrm{T}_n^{\alpha_n}$, $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n \leqslant d$ (at most the c_α such that $|\alpha| = d$ being not zero). As the nonzero c_α are invertible in K, we can suppose, replacing A by A_g ($g \neq 0$ in A), that the c_α are nonzero and invertible *in* A. It then follows that for every $x \in \mathrm{Spec}(A)$, $F_{(\boldsymbol{k}(x))}$ is of degree d. □

Proof (continued). We shall prove the lemma by means of a preliminary lemma.

Lemma (9.7.5.1). — *Let A be an integral ring, $S = \mathrm{Spec}(A)$, η the generic point of S, B the polynomial ring $A[\mathrm{T}_1, \ldots, \mathrm{T}_m]$, $(P_i)_{i \in I}$ a finite family of elements of B, $\mathfrak{a}$ the ideal of B generated by the P_i. For every $s \in S$, let $\mathfrak{a}_s$ be the ideal of $\boldsymbol{k}(s)[\mathrm{T}_1, \ldots, \mathrm{T}_m]$ generated by the polynomials $(P_i)_{(\boldsymbol{k}(s))}$ ($i \in I$). Then, if $V(\mathfrak{a}_\eta) = \varnothing$, there exists a neighbourhood U of η in S such that $V(\mathfrak{a}_s) = \varnothing$ for every $s \in U$.*

Proof. Indeed, let $Y = \mathrm{Spec}(B)$, and let Z be the closed part $V(\mathfrak{a})$ of Y; as $\mathfrak{a}$ is an ideal of finite type in B, Z is constructible in Y (**0**, 9.1.5) and, with the notations introduced in (9.4.1), we have $V(\mathfrak{a}_s) = Z_s$ for every $s \in S$; as the structural morphism $Y \to S$ is of finite presentation, the conclusion of the lemma follows from (9.5.1) and (9.2.3). □

Let (p, q) be a pair of integers > 0 such that $p + q = d$; we introduce indeterminates T'_β, T''_γ for all systems of integers β, γ such that $|\beta| \leqslant p$ and $|\gamma| \leqslant q$; for every system of integers α such that $|\alpha| \leqslant d$, consider the polynomial of $B = A[\mathrm{T}'_\beta, \mathrm{T}''_\gamma \mid |\beta| \leqslant p, |\gamma| \leqslant q] : P_\alpha(\mathrm{T}'_\beta, \mathrm{T}''_\gamma) = \sum_{\delta+\gamma=\alpha} \mathrm{T}'_\delta \mathrm{T}''_\gamma - c_\alpha$. Let Ω be an algebraic closure of K; to say that there exist two polynomials F_1, F_2 of $\Omega[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, of respective degrees p and q, and such that $F_1 F_2 = F_{(\Omega)}$ means that the system of equations $P_\alpha(t'_\beta, t''_\gamma) = 0$ ($|\alpha| \leqslant d$) admits a solution (t'_β, t''_γ) formed of elements of Ω. Let $\mathfrak{a}$ be the ideal of B generated by the P_α; the preceding interpretation and the theorem of the zeros of Hilbert show that the hypothesis on $F_{(K)}$ implies that $V(\mathfrak{a}_\eta) = \varnothing$, designating by η the generic point of $\mathrm{Spec}(A)$; lemma (9.7.5.1) then proves that in a neighbourhood of η, we have $V(\mathfrak{a}_s) = \varnothing$, and consequently for these values of x, $F_{(\boldsymbol{k}(x))}$ does not admit any factorisation $F_{(\boldsymbol{k}(x))} = G_1 G_2$, where G_1, G_2 are polynomials of respective degrees p and q, whose coefficients lie in an algebraic closure of $\boldsymbol{k}(x)$. It suffices to apply this result for all pairs of integers (p, q) such that $p > 0$, $q > 0$ and $p + q = d$, to obtain the conclusion of lemma (9.7.5). □ **IV-3** | 78

Proposition (9.7.6). — *Let S be a Noetherian prescheme, integral, of generic point η, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent. If $\mathscr{F}_\eta$ is without associated prime cycle, there exists a neighbourhood U of η in S such that, for every $s \in U$, $\mathscr{F}_s$ is without associated prime cycle.*

Proof. As X_η is Noetherian, there exists an injective homomorphism $v : \mathscr{F}_\eta \to \bigoplus_{i=1}^m \mathscr{G}_i$, where each of the $\mathscr{G}_i$ is irredondant and $\mathrm{Ass}(\mathscr{F}_\eta)$ is the set of the x_i, where $\{x_i\} = \mathrm{Ass}(\mathscr{G}_i)$ (3.2.6); if Z_i is the closure of $\{x_i\}$ in X_η, we have $Z_i = \mathrm{Supp}(\mathscr{G}_i)$ (3.1.4), and by hypothesis the Z_i are the irreducible components of $Z = \mathrm{Supp}(\mathscr{F}_\eta)$. It follows from (8.5.2), (i) and (ii) (restricting to S if necessary a neighbourhood of η) and (8.5.8) that there exist $\mathscr{O}_X$-Modules coherent $\mathscr{F}_i$ and an injective homomorphism $u : \mathscr{F} \to \bigoplus_{i=1}^m \mathscr{F}_i$ such that $(\mathscr{F}_i)_\eta = \mathscr{G}_i$ for every i and $v = u_\eta$. If $Y_i = \mathrm{Supp}(\mathscr{F}_i)$, we have $(Y_i)_s = \mathrm{Supp}((\mathscr{F}_i)_s)$ for every $s \in S$ (**I**, 9.1.13) and in particular $(Y_i)_\eta = Z_i$ for every i. The hypothesis implies that for $i \neq j$, $Z_i - (Z_i \cap Z_j)$ is open and dense in Z_i; we deduce therefore from (9.5.3) and (9.5.4) that in a neighbourhood of η, $(Y_i)_s - ((Y_i)_s \cap (Y_j)_s)$ is open and dense in $(Y_i)_s$. Suppose that one has proved that each of the $(\mathscr{F}_i)_s$ is without associated prime cycle for $s \in U$. Then the elements of $\mathrm{Ass}((\mathscr{F}_i)_s)$ are the maximal points of $(Y_i)_s$; no two of them can therefore belong to a $(Y_j)_s$, for $i \neq j$, and the proposition will then be irredondant; furthermore, we can suppose that $\mathscr{F}_\eta$ is *irredondant*; in addition, we can reduce to the case where $S = \operatorname{Spec}(A)$ is affine and where the morphism $f : X \to S$ is dominant; A is then a Noetherian integral ring, *sub-ring* of a finite A-algebra B, and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type; by hypothesis, if K is the field of fractions of A, the $B_{(K)}$-module $M_{(K)}$ is irredondant. Let $\mathfrak{q}$ be the unique element of $\operatorname{Ass}(M_{(K)})$, and let $\mathfrak{p}$ be the prime ideal of B which is the inverse image of $\mathfrak{q}$ by the canonical map $B \to B_{(K)}$. We know (5.11.1.1) that there exists a finite filtration $(N_h)_{0 \leqslant h \leqslant m}$ of $M_{(K)}$ such that $N_m = 0$, and N_h/N_{h+1} is isomorphic to a nonzero $B_{(K)}$-sub-module $\neq 0$ of $B_{(K)}/\mathfrak{q}$. Let M_h be the inverse image of N_h by the canonical map $M \to M_{(K)}$. It follows from (9.4.4) that for s sufficiently close to η, $(M_h)_{(\boldsymbol{k}(s))}$ identifies with a sub-$B_{(\boldsymbol{k}(s))}$-module of $M_{(\boldsymbol{k}(s))}$, and the quotient $(M_h)_{(\boldsymbol{k}(s))}/(M_{h+1})_{(\boldsymbol{k}(s))}$ to a nonzero $B_{(\boldsymbol{k}(s))}$-sub-module $\neq 0$ of $B_{(\boldsymbol{k}(s))}/\mathfrak{p}_{(\boldsymbol{k}(s))}$. Taking into account (3.1.7), with the same notations, it remains to prove, that if B is *integral*, then $B_{(\boldsymbol{k}(s))}$ has no associated prime cycle for s close to η. Now, replacing A by A_g (where g is a nonzero element $\neq 0$ of A), we can suppose that B contains a polynomial sub-algebra $C = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ such that B is a C-module of finite type (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 1 of th. 1). As $B_{(K)}$ is a $C_{(K)}$-module without torsion, we can apply once more the preceding reasoning, replacing B, M, and $\mathfrak{q}$ by C, B, and (0) respectively, and it suffices to show that for s close to η, $C_{(\boldsymbol{k}(s))}$ has no associated prime cycle. But this is obvious since $C_{(\boldsymbol{k}(s))} = \boldsymbol{k}(s)[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is an integral ring. □ IV-3 | 79

Theorem (9.7.7). — *Let $f : X \to S$ be a morphism of finite presentation, and let E be the set of $s \in S$ for which X_s has one of the following properties: being*

label=(i) geometrically irreducible;
lbbel=(ii) geometrically connected;
lcbel=(iii) geometrically reduced;
ldbel=(iv) geometrically integral.

Then E is locally constructible in S.

Proof. To see that the properties considered are constructible, let us note first that they trivially satisfy condition (9.2.1), (i), by virtue of (4.5.6), (i) and (4.6.5), (i). It remains therefore to verify (9.2.1), (ii), so we can suppose $S = \operatorname{Spec}(A)$ affine, Noetherian, integral, of generic point η. By virtue of (4.6.8), there exists a finite extension K' of $K = \boldsymbol{k}(\eta)$ such that $(X_\eta)_{(K')}$ has its irreducible components (resp. connected components, resp. connected components) geometrically irreducible (resp. geometrically connected) and that $((X_\eta)_{(K')})_{\mathrm{red}}$ is geometrically reduced. Furthermore, there exists a base of K' over K formed of elements integral over A, the integral ring A' generated by these elements is finite over A and has K' as field of fractions. Set $S' = \operatorname{Spec}(A')$; the morphism $g : S' \to S$ is finite and dominant, therefore surjective (**II**, 6.1.10), so that $X' = X_{(S')}$; let η' be the generic point of S', $X'_{\eta'} = (X_\eta)_{(K')}$; the set E' of $s' \in S'$ such that $X'_{s'}$ has one of the properties (i), (ii), (iii), or (iv) (E corresponding well to the same property) (9.2.2), (iv); and $S - E = g(S' - E')$ and $g^{-1}(\eta) = \{\eta'\}$ since A' is integral and finite over A, therefore the image by g of every neighbourhood of η' is a neighbourhood of η. The theorem will therefore be proved if one proves that E' or $S' - E'$ is a neighbourhood of η'. In other words, we can henceforth suppose that the irreducible components (resp. connected components) of X_η are geometrically irreducible (resp. geometrically connected) and that $(X_\eta)_{\mathrm{red}}$ is geometrically reduced.

Suppose first that $\eta \in S - E$ for one of the properties (i) to (iv). If X_η is not geometrically irreducible (resp. geometrically connected), it is not irreducible (resp. connected) by virtue of the preceding hypothesis, and therefore it is the same for X_s for s in a neighbourhood of η (9.7.1), and *a fortiori* in this neighbourhood X_s is not geometrically irreducible (resp. geometrically connected). On the other hand, if X_η is not geometrically reduced, it is not reduced (since otherwise it would be equal to $(X_\eta)_{\mathrm{red}}$ which is geometrically reduced by hypothesis); therefore, in a neighbourhood of η, X_s is not reduced (9.7.2) and hence not geometrically reduced. Finally, if X_η is not geometrically integral, either it is not reduced (nor *a fortiori* integral) in a neighbourhood of η; or X_η is reduced (therefore geometrically reduced by hypothesis), and then it is not geometrically irreducible, and we have just seen above that it is then the same for X_s for s close to η; *a fortiori* X_s is not geometrically integral for these values of s. IV-3 | 80

We shall now suppose that $\eta \in E$ and examine separately each of the properties considered.

1° Suppose that X_η is geometrically integral. Let L be the field of rational functions on X_η; the hypothesis on X_η implies that L is a separable extension of K (4.6.3), therefore a finite separable extension of a pure extension $K(\mathrm{T}_1, \ldots, \mathrm{T}_n)$ (T_i indeterminates); there is then an element $x \in L$, integral over the ring $K[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, and such that $L = K(\mathrm{T}_1, \ldots, \mathrm{T}_n)(x)$; let $G \in K[\mathrm{T}_1, \ldots, \mathrm{T}_n, \mathrm{T}_{n+1}]$ be its minimal polynomial. There exists $g \neq 0$ of A such that all the coefficients $\neq 0$ of G (which all belong to K) belong to A_g; replacing A by A_g (which amounts to replacing S by a neighbourhood of η), we can therefore suppose that G has its coefficients in A; designating by F the polynomial G considered as element of $A[\mathrm{T}_1, \ldots, \mathrm{T}_n, \mathrm{T}_{n+1}]$, we then have $G = F_{(\boldsymbol{k}(\eta))}$. Set $Y = \operatorname{Spec}(A[\mathrm{T}_1, \ldots, \mathrm{T}_{n+1}]/(F))$, so that $Y_\eta = \operatorname{Spec}(K[\mathrm{T}_1, \ldots, \mathrm{T}_{n+1}]/(G))$; Y_η is an integral scheme having L as field of rational functions. As X_η and Y_η are Noetherian, there exists a non-empty open $V \subset Y_\eta$ and an immersion (necessarily dominant) $v : V \to X_\eta$ (**I**, 6.5.1), (ii) and (6.5.4), (ii). Let W be an open of Y such that $W \cap Y_\eta = V$; applying (8.8.2), (i) and (8.10.5), (iii) following the method of (8.1.2), *a*), replacing S by a neighbourhood of η, we can suppose that $v = w_\eta$ where $w : W \to X$ is an open immersion.

Since we have seen (4.6.3) that the criterion for a prescheme algebraic integral to be geometrically integral depends only on its field of rational functions; as X_η is geometrically integral by hypothesis, it is the same for Y_η, and the definition of G implies therefore that this polynomial is geometrically irreducible (9.7.4). Applying (9.7.5), we see that there is a neighbourhood U of η in S such that $F_{(\boldsymbol{k}(s))}$ is geometrically irreducible for every $s \in U$, therefore Y_s is geometrically integral for $s \in U$ (9.7.4); furthermore we can also suppose that for $s \in U$, W_s is not empty (9.5.1), and therefore is geometrically integral (4.6.3); finally, we can also suppose that for $s \in U$, $w_s : W_s \to X_s$ is an open immersion (9.6.1), (x), and that its image in X_s is everywhere dense (9.5.3). In other words, for $s \in U$, there is in X_s an open everywhere dense W'_s which is geometrically integral; the criterion (4.5.9), *c*) therefore shows that X_s is already geometrically irreducible for $s \in U$. Finally, as X_η is reduced and without associated prime cycle (3.2.1), we can also suppose that for $s \in U$, X_s has no associated prime cycle (9.7.6); this being the case (for $s \in U$ fixed), let Ω be an algebraically closed extension of $\boldsymbol{k}(s)$, and let $p : (X_s)_{(\Omega)} \to X_s$ be the canonical projection; $p^{-1}(W'_s)$ is an open dense in $(X_s)_{(\Omega)}$ (2.3.10) and is integral by hypothesis; furthermore $(X_s)_{(\Omega)}$ has no associated prime cycle (4.2.7), therefore we conclude from (3.2.1) that $(X_s)_{(\Omega)}$ is reduced; we conclude therefore from (3.2.1) and (4.6.1) that X_s is geometrically integral (4.6.1).

2° Suppose that X_η is geometrically irreducible; as X_{red} is also of finite type over S (**I**, 5.1.8), taking into account (**I**, 5.4), (vi), we can, by replacing X by X_{red}, then X_η is also integral, and as by hypothesis X_η is geometrically irreducible, it is geometrically integral, and we are brought back to the conditions of 1° and we conclude (returning to the initial hypotheses) that X_s is geometrically irreducible for s close to η.

3° Suppose X_η geometrically connected, and let Z_i ($1 \leqslant i \leqslant n$) be the irreducible components **IV-3** | 81
of X_η; there exists (by virtue of (5.10.8.1) applied to $\mathfrak{T} = \{\varnothing\}$) a *surjective* map $j \mapsto \nu(j)$ of an interval $[1, m]$ of $\mathbf{N}$ onto $[1, n]$ such that $Z_{\nu(j)} \cap Z_{\nu(j+1)} \neq \varnothing$ for $1 \leqslant j \leqslant m$. For every i, let X_i be the closure of Z_i in X, and let Y be the reunion of the X_i; as $Y_\eta = X_\eta$ by definition, we can, by virtue of (9.5.1), suppose that $Y_s = X_s$ for every $s \in S$. Now, considering the reduced sub-preschemes of X having the X_i as underlying spaces, we see by 2° that there exists a neighbourhood U of η in S such that for $s \in U$ the $(X_i)_s$ are *geometrically irreducible* (since the $(X_i)_\eta$ can be supposed to be geometrically irreducible, as we saw at the beginning). Furthermore, we can also suppose that for $s \in U$, $(X_{\nu(j)})_s \cap (X_{\nu(j+1)})_s \neq \varnothing$ (9.5.1) for $1 \leqslant j \leqslant m$; we conclude immediately that X_s is connected, therefore (4.5.13.1) geometrically connected for $s \in U$.

4° Suppose X_η geometrically reduced; let Z_i, W_i be the irreducible components of X_η, where W_i is the interior of Z_i in X_η; there is for each i an open V_i of X such that $W_i = V_i \cap X_\eta$ for every i; as the W_i are opens, pairwise disjoint, and their reunion is dense in X_η, (9.5.1), (9.5.3), and (9.5.4) allow us to suppose that for s close to η, the $(V_i)_s$ are opens, pairwise disjoint, in X_s and their reunion is dense in X_s. Furthermore, as the W_i are geometrically reduced and have been supposed at the beginning geometrically irreducible, it follows from 1° that for s close to η, the $(V_i)_s$ are geometrically integral, and *a fortiori* reduced. On the other hand, we deduce from (9.7.6) that X_s has no associated prime cycle, since it is already reduced (3.2.1); we conclude therefore from (3.2.1) and (4.6.1) that X_s is reduced, and from (4.6.1) that X_s is geometrically reduced. □

9.8. Primary decomposition in the neighbourhood of a generic fibre

Proposition (9.8.1). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. Then the set E of $s \in S$ such that $\mathscr{F}_s$ is an $\mathscr{O}_{X_s}$-Module without associated immersed prime cycle is locally constructible.* **IV-3** | 83

Proof. We know that for quasi-coherent Modules on algebraic preschemes, the property of being without associated immersed prime cycle is invariant by change of base field (4.2.7); we can thus restrict to the case where S is affine, Noetherian and integral, and to proving that in this case E or $S - E$ is a neighbourhood of η. We saw in (9.7.6) that if $\eta \in E$, E is a neighbourhood of η; it remains thus to consider the case where $\mathscr{F}_\eta$ has associated immersed prime cycles. We can restrict to the case where X is affine, and where there is a sub-$\mathscr{O}_{X_\eta}$-Module coherent $\mathscr{H}$ of $\mathscr{F}_\eta$ whose support Z' is non-empty and *rare* with respect to the support Z of $\mathscr{F}$ (3.1.3); then, by virtue of (8.5.2), (i) and (ii)) and (8.5.8), applied following the method of (8.1.2), *a*)), there exists a sub-$\mathscr{O}_X$-Module coherent $\mathscr{G}$ of $\mathscr{F}$ such that $\mathscr{G}_\eta = \mathscr{H}$; if $Y = \mathrm{Supp}(\mathscr{F})$, $Y' = \mathrm{Supp}(\mathscr{G})$, we have by extension $Y_s = \mathrm{Supp}(\mathscr{F}_s)$, $Y'_s = \mathrm{Supp}(\mathscr{G}_s)$ for all $s \in S$ (**I**, 9.1.13) and in particular $Z = Y_\eta$, $Z' = Y'_\eta$; as $Z - Z'$ is dense in Z and $Z' + \varnothing$, there is a neighbourhood U of η such that for $s \in U$, $Y_s - Y'_s$ is dense in Y_s, and $Y'_s \neq \varnothing$ (9.5.1) and (9.5.3); by considering a generic point of an irreducible component of Y'_s and a neighbourhood sufficiently small of this point in X_s, we deduce immediately from (3.1.3) that $\mathscr{F}_s$ admits associated immersed prime cycles. **IV-3** | 84

(9.8) Let S be a Noetherian prescheme, *integral*, of generic point η, $f : X \to S$ a morphism of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Consider an irredundant reduced decomposition $(\mathscr{G}_i)_{i \in I}$ of $\mathscr{F}_\eta$ (3.2.6); the $\mathscr{G}_i$ are thus quotients of $\mathscr{F}_\eta$ and there is an injective homomorphism $h : \mathscr{F}_\eta \to \bigoplus \mathscr{G}_i$; furthermore $\mathrm{Ass}(\mathscr{G}_i)$ is reduced to a point x_i. Let Z_i be the closure of $\{x_i\}$ in X, so that $(Z_i)_\eta = \mathrm{Supp}(\mathscr{G}_i)$. Denote by $\mathscr{J}_i$ the coherent Ideal of $\mathscr{O}_X$ defining the closed sub-prescheme of X of underlying space Z_i (sub-prescheme still denoted Z_i). By virtue of (8.5.2) and (8.5.8), applied following the method of (8.1.2), *a*)), we can (by restricting S to a neighbourhood of η if needed) suppose that there exist quotients $\mathscr{F}_i$ of $\mathscr{F}$ ($i \in I$) such that $\mathscr{G}_i = (\mathscr{F}_i)_\eta$ for all i, and a homomorphism $g : \mathscr{F} \to \bigoplus \mathscr{F}_i$ such that $h = g_\eta$. Furthermore (**I**, 9.3.5), there exists an integer m such that $(\mathscr{J}_i^m \mathscr{F}_i)_\eta = (\mathscr{J}_i)_\eta^m \mathscr{G}_i = 0$ and by restricting S again, we can suppose that $\mathscr{J}_i^m \mathscr{F}_i = 0$ (8.5.2.5), so that the support of $\mathscr{F}_i$ is contained in Z_i; but as it is closed and contains x_i, it is necessarily equal to Z_i. ☐

Proposition (9.8.3). — *Under the conditions of* (9.8), *for all $s \in S$, and all $i \in I$, denote by $x_{i\alpha}$ ($\alpha \in J_{s,i}$) the maximal points of $(Z_i)_s$. There exists a neighbourhood U of η in S such that, for all $s \in U$, the $x_{i\alpha}$ (for $i \in I$ and, for each i, $\alpha \in J_{s,i}$) are pairwise distinct and that $\mathrm{Ass}(\mathscr{F}_s)$ is the set of $x_{i\alpha}$ (in other words, the prime cycles associated to $\mathscr{F}_s$ are the irreducible components of $(Z_i)_s$). Furthermore, one can take U such that, for the closure of $\{x_{i\alpha}\}$ in X_s to be a maximal associated prime cycle of $\mathscr{F}_s$, it is necessary and sufficient that $(Z_i)_\eta$ (closure of $\{x_i\}$ in X_η) be a maximal associated prime cycle of $\mathscr{F}_\eta$.*

Proof. It follows from (3.1.3), *c'*)) that for each i, there exists an open W_i in X_η such that $W_i \cap (Z_i)_\eta$ is non-empty, and a coherent $\mathscr{O}_{X_\eta}$-Module $\mathscr{H}_i$, of support $W_i \cap (Z_i)_\eta$, such that there is an injective homomorphism $v_i : \mathscr{H}_i \to \mathscr{F}_\eta | W_i$. Let V_i be an open of X such that $V_i \cap X_\eta = W_i$; applying, as in (9.8.2), the results of (8.5.2) and (8.5.8), on can (by restricting S) suppose that there exists a coherent Module $\mathscr{F}'_i$, of support $V_i \cap Z_i$ and a homomorphism $u_i : \mathscr{F}'_i \to \mathscr{F} | V_i$ such that $\mathscr{H}_i = (\mathscr{F}'_i)_\eta$ and $v_i = (u_i)_\eta$. We will show that there is a neighbourhood U of η such that for $s \in U$, the following properties are satisfied:

label=(i) The $(\mathscr{F}_i)_s$ are without associated immersed prime cycle.
lbbel=(ii) The homomorphism $g_s : \mathscr{F}_s \to \bigoplus (\mathscr{F}_i)_s$ is injective.
lcbel=(iii) $(Z_i)_s \cap (V_i)_s$ is dense in $(Z_i)_s$, $(\mathscr{F}'_i)_s$ has support $(Z_i)_s \cap (V_i)_s$, and $(u_i)_s : (\mathscr{F}'_i)_s \to \mathscr{F}_s | (V_i)_s$ is injective.

IV-3 | 85

(iv) For $i \neq j$, every irreducible component of $(Z_i)_s$ is distinct from every irreducible component of $(Z_j)_s$.

Now, (i) has already been seen (9.7.6); (ii) is a particular case of (9.4.5); (iii) follows from (9.5.3) and (9.4.5). Finally, if $i \neq j$, $(Z_i)_\eta \cap (Z_j)_\eta = (Z_i \cap Z_j)_\eta$ is rare in $(Z_i)_\eta$ *or* in $(Z_j)_\eta$; suppose for example that $(Z_i \cap Z_j)_\eta$ is rare in $(Z_j)_\eta$; then it follows from (9.5.3) and (9.5.4) that for s in a neighbourhood of η, $(Z_i \cap Z_j)_s = (Z_i)_s \cap (Z_j)_s$ is rare in $(Z_j)_s$, which shows that no irreducible component of $(Z_j)_s$ can be contained in an irreducible component of $(Z_i)_s$, nor *a fortiori* be equal to it.

This being so, it follows from (ii) and (3.1.7) that for $s \in U$, we have

$$\operatorname{Ass}(\mathscr{F}_s) \subset \bigcup \operatorname{Ass}((\mathscr{F}_i)_s),$$

from (i) that $\operatorname{Ass}((\mathscr{F}_i)_s)$ is the set of $x_{i\alpha}$ ($\alpha \in J_{s,i}$). On the other hand, by virtue of (iii) and the criterion (3.1.3), c')), each of the $x_{i\alpha}$ ($\alpha \in J_{s,i}$, $i \in I$) belongs to $\operatorname{Ass}(\mathscr{F}_s)$. Finally (iv) signifies that the $x_{i\alpha}$ are pairwise distinct.

It remains to prove the last assertion of the statement. Now, it follows from (i) that for s and for given i, none of the sets $\{x_{i\alpha}\}$ can be contained in another for $\alpha \in J_{s,i}$. On the other hand, if $(Z_j)_\eta \subset (Z_i)_\eta$, we can suppose U taken such that $(Z_j)_s \subset (Z_i)_s$ for $s \in U$ (9.5.1), so each $x_{j\alpha}$ belongs to the closure of a $x_{i\alpha}$; on the contrary, if $(Z_j)_\eta \cap (Z_i)_\eta$ is rare in $(Z_j)_\eta$, we have seen by proving (iv) that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$, so none of the $x_{j\alpha}$ is adherent to a $x_{i\alpha}$. In particular, if $(Z_j)_\eta$ is maximal, this is equivalent to saying that $(Z_j)_\eta$ is rare in $(Z_i)_\eta$ for *all* $i \neq j$, and we conclude that $(Z_j)_s \cap (Z_i)_s$ is rare in $(Z_j)_s$ for all $i \neq j$, so that every $x_{j\alpha}$ ($\alpha \in J_{s,j}$) is *maximal*. □

Corollary (9.8.4). — *The notations and hypotheses being those of* (9.8)*, there exists a neighbourhood U of η in S such that, for all $s \in U$, each of the $(\mathscr{F}_i)_s$ is without associated immersed prime cycle; furthermore, if $(\mathscr{F}_{i\alpha})_{\alpha \in J_{s,i}}$ is the irredundant reduced decomposition of $(\mathscr{F}_i)_s$, then, for all $s \in U$, the family $(\mathscr{F}_{i\alpha})_{i \in I, \alpha \in J_{s,i}}$ is an irredundant reduced decomposition of $\mathscr{F}_s$.*

Proof. The first assertion follows from (9.8.1) and the definition of $\mathscr{F}_i$. On the other hand, we have seen in (9.8.3) that the homomorphism $\mathscr{F}_s \to \bigoplus(\mathscr{F}_i)_s$ is injective and by definition it is the same for each of the homomorphisms $(\mathscr{F}_i)_s \to \bigoplus \mathscr{F}_{i\alpha}$, so the homomorphism $\mathscr{F}_s \to \bigoplus_{i,\alpha} \mathscr{F}_{i\alpha}$ is injective. As we can suppose (9.4.5) that for $s \in U$ each of the $(\mathscr{F}_i)_s$ is a quotient of $\mathscr{F}_s$, the $\mathscr{F}_{i\alpha}$ are quotients of $\mathscr{F}_s$, and it remains to verify (3.2.5) that the $x_{i\alpha}$ are pairwise distinct and belong to $\operatorname{Ass}(\mathscr{F}_s)$, which has been proved in (9.8.3). □

Proposition (9.8.5). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $E(s)$ (resp. $E'(s)$) be the set (finite subset of $\mathbf{Z} \cup \{-\infty\}$) of dimensions of the maximal prime cycles associated to $\mathscr{F}_s$ (resp. of the maximal prime cycles, i.e., the irreducible components of* $\operatorname{Supp}(\mathscr{F}_s)$*). Then the functions $s \mapsto E(s)$ and $s \mapsto E'(s)$ are locally constructible in S.*

Proof. It follows from (4.2.7) and (4.2.8) that the condition (9.2.1), (i)) is satisfied for the properties **IV-3** | 86
whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that E and E' are constant in a neighbourhood of the generic point η of S; but this follows from (9.8.3) and (9.8.6). □

Proposition (9.8.6). — *With the hypotheses and notations of* (9.8)*, let $i \in I$ be such that $(Z_i)_\eta$ is maximal (in other words, an irreducible component of* $\operatorname{Supp}(\mathscr{F}_\eta)$*). Then there exists a neighbourhood U of η in S such that for all $s \in U$ and all $\alpha \in J_{s,i}$, the geometric length of $\mathscr{F}_s$ at $x_{i\alpha}$ (relative to $\mathbf{k}(s)$)* (4.7.5) *is equal to the geometric length of $\mathscr{F}_\eta$ at x_i (relative to $\mathbf{k}(\eta)$).*

Proof. We can evidently restrict to the case where $S = \operatorname{Spec}(A)$ is affine; let us first show that the sub-prescheme $(Z_i)_\eta$ of X_η, which is reduced, is *geometrically integral*. Indeed there is a finite extension K' of $K = \mathbf{k}(\eta)$ such that $(((Z_i)_\eta)_{(K')})_{\mathrm{red}}$ is geometrically reduced and that the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically irreducible (4.6.8). We proceed as in the proof of (9.7.7) by considering a sub-A-algebra A' of K' that is a field of fractions and finite over A. We set $S' = \operatorname{Spec}(A')$ and consider the finite surjective morphism $g : S' \to S$; let $X' = X_{(S')}$ and $\mathscr{F}' = \mathscr{F} \otimes_g \mathscr{O}_{S'}$ and let η' be the generic point of S'. For all $s' \in S'$, i.e., $s = g(s')$; if T is an irreducible component of $\operatorname{Supp}(\mathscr{F}_s)$, the irreducible components T'_j of $T_{(\mathbf{k}(s'))}$ are irreducible components of $\operatorname{Supp}(\mathscr{F}'_{s'})$ and dominate T (4.2.7) and the radicial multiplicities of T with respect to $\mathscr{F}_s$ and of each T'_j with respect to $\mathscr{F}'_{s'}$ are the same (4.7.9). The reasoning from the first part of (9.7.7) then shows that we can restrict to proving the proposition for X' and $\mathscr{F}'$; and by virtue of the choice of K', the reduced sub-preschemes having for underlying spaces the irreducible components of $((Z_i)_\eta)_{(K')}$ are geometrically integral (4.6.1).

Suppose now that $(Z_i)_\eta$ is geometrically integral; then (9.7.7), it is the same for (Z_i), for s in a neighbourhood of η; the definition (4.7.5) shows that it will therefore suffice to prove that the length of the $\mathcal{O}_{X_\eta,x_i}$-module $(\mathcal{F}_\eta)_{x_i}$ is equal to that of the $\mathcal{O}_{X_s,x_{i_s}}$-module $(\mathcal{F}_s)_{x_{i_s}}$ (where we have dropped the index α, now unnecessary by hypothesis). The question being evidently local on X, we can suppose (by restricting to a neighbourhood of x_i) that $\mathcal{F} = \mathcal{F}_i$, so that $\mathcal{F}_\eta$ is *irredundant* on $X = \operatorname{Spec}(B)$ affine, and one writes Z instead of Z_i, and x the generic point of Z (and of Z_η). The Noetherian ring B contains A as sub-ring, and $\mathcal{F} = \widetilde{M}$, where M is a B-module of finite type; furthermore, if K is the field of fractions of A, the $B_{(K)}$-module $M_{(K)}$ is $\neq 0$ and irredundant. Let q be the unique element of $\operatorname{Ass}(M_{(K)})$ and let $\mathfrak{p}$ be the prime ideal of B, inverse image of q. Using (5.11.1.1) as in the proof of (9.7.6), we reduce to the case where B is *integral* and M a sub-module $\neq 0$ of B; then $\mathcal{F}_\eta$ is a sub-$\mathcal{O}_{X_\eta}$-Module non-zero of $\mathcal{O}_{Z_\eta}$, and by virtue of (9.4.5), for s in a neighbourhood of η, $\mathcal{F}_s$ is isomorphic to a sub-$\mathcal{O}_{X_s}$-Module non-zero of $\mathcal{O}_{Z_s}$; as Z_s is geometrically integral, the lengths of $(\mathcal{F}_\eta)_x$ and $(\mathcal{F}_s)_{x_s}$ are both equal to 1, which finishes the proof. □

Corollary (9.8.7). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-Module of finite presentation. For all $s \in S$, let $M(s)$ be the set of pairs (d, m) such that there exists an irreducible component of* $\operatorname{Supp}(\mathcal{F}_s)$ *of dimension d and of radicial multiplicity m for $\mathcal{F}_s$ (4.7.8). Then the function $s \mapsto M(s)$ is locally constructible in S.* **IV-3 | 87**

Proof. It follows from (4.2.7) and (4.7.9) that the condition (9.2.1), (i)) is satisfied for the property whose constructibility we wish to show. We are thus reduced to the case where S is affine, Noetherian and integral, and to proving that M is constant in a neighbourhood of the generic point of S; but this follows from the proposition, namely (9.8.3) and (9.8.6). □

Proposition (9.8.8). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$-Module of finite presentation. For all $s \in S$, let $l(s)$ be the sum of the total multiplicities for $\mathcal{F}_s$ (relative to $\mathbf{k}(s)$) of the generic points of the irreducible components of* $\operatorname{Supp}(\mathcal{F}_s)$ *(4.7.12). Then the function $s \mapsto l(s)$ is locally constructible in S.*

Proof. Taking into account (4.7.12), the condition (9.2.1), (i)) is satisfied for the property whose constructibility we wish to show, and we are thus reduced to the case where S is affine, Noetherian and integral, and to showing that l is constant in a neighbourhood of the generic point η, and constant. Using the notations of (9.8.2), this follows from the definition (4.7.12), from the fact that the geometric number of irreducible components of each $(Z_i)_s$ is constant in a neighbourhood of η (9.7.8), that the geometric length of $\mathcal{F}_s$ at $x_{i\alpha}$ is also equal to that of $\mathcal{F}_\eta$ at x_i for each i such that $(Z_i)_\eta$ is maximal (9.8.6) and finally from the fact that the closure of $\{x_{i\alpha}\}$ is a maximal associated prime cycle of $\mathcal{F}_s$ if and only if $(Z_i)_\eta$ is a maximal associated prime cycle of $\mathcal{F}_\eta$ (9.8.3). □

Remark (9.8.9). — One can make precise in various ways the preceding propositions; let us limit ourselves to a statement given as an example. Let us say that a finite subset P of a k-prescheme algebraic X is *saturated* if, for every pair of points x, y of P, the generic points of the irreducible components of $\overline{\{x\}} \cap \overline{\{y\}}$ also belong to P; for every finite subset Q of X, there exists a smallest finite subset P of X containing Q and saturated; we say that P is the *saturate* of Q. For every coherent $\mathcal{O}_X$-Module $\mathcal{F}$, we will call *primary skeleton* of $\mathcal{F}$ the system (P, Q, ω, d, m) where $Q = \operatorname{Ass}(\mathcal{F})$, P is the saturate of Q, ω the ordering relation $\overline{\{x\}} \subset \overline{\{y\}}$ on P, d the function $x \mapsto \dim \overline{\{x\}}$ on P, m the function $x \mapsto \operatorname{length}_\omega \mathcal{F}_x$ defined on the set of maximal elements of Q for the relation ω. We call *virtual skeleton* any system (P, Q, ω, d, m) where P is a set, Q a subset of P, ω a relation of order on P, d a map from P to $\mathbf{N}$, m a map from $\mathbf{N}$ to the set of maximal elements of Q; one defines in an obvious way the notion of *isomorphism* of two virtual skeletons. Finally, with the preceding notations, we call *primary type* of $\mathcal{F}$ the class (for the relation of isomorphism of virtual skeletons) of the primary skeleton of $\mathcal{F}$. It follows from (4.2.6), (4.2.7), (4.2.8), (4.5.1) and (4.7.9) that if, for an algebraically closed extension k' of k, one sets $X' = X \otimes_k k'$ and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, the *primary type* of $\mathcal{F}'$ is independent of the algebraically closed extension k' of k considered; we say that this is the *geometric primary type* of $\mathcal{F}$. Given the definitions, the statement we have in view is the following: **IV-3 | 88**

(9.8) Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation. For all $s \in S$, let $T(s)$ be the geometric primary type of $\mathscr{F}_s$. Then the function $s \mapsto T(s)$ is locally constructible in S.

Taking into account the remarks that precede, we are reduced as usual to proving that if S is affine, Noetherian and integral, and of generic point η, $T(s)$ is constant in a neighbourhood of η. Reasoning as at the beginning of (9.7.7), we can suppose that all the irreducible components of X_η that intervene are geometrically irreducible, and then the proposition follows from (9.5.1), (9.5.5), (9.8.3) and (9.8.6); we leave the details to the reader. One could generalise by considering several coherent Modules and defining their "simultaneous primary skeleton", etc. The general conclusion of this paragraph is that for all the properties of the type considered (and for an S-irreducible prescheme) *the properties valid on the "generic fibre" are also valid on all neighbouring fibres*.

9.9. Constructibility of local properties of fibres

Proposition (9.9.1). — *Let $f : X \to S$ be a morphism locally of finite presentation, Z a locally constructible subset of X such that for all $s \in S$, Z_s is closed in X_s, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. Then the following subsets of X are locally constructible:* **IV-3 | 88**

label=(i) The set of $x \in X$ such that $\dim_x(Z_{f(x)})$ belongs to Φ.
lbbel=(ii) The set of $x \in X$ such that $\operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ belongs to Φ.
lcbel=(iii) The set of $x \in X$ such that the local ring $\mathscr{O}_{X_{f(x)},x}$ is equidimensional.

We note that properties (i) and (ii) can also be expressed by saying that the functions $x \mapsto \dim_x(Z_{f(x)})$ and $x \mapsto \operatorname{codim}_x(Z_{f(x)}, X_{f(x)})$ are locally constructible *in X (0_{III}, (9.3.1)).*

Proof. The questions being local on X, we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine and where f is a morphism of finite presentation; there exists then a sub-ring A_0 of A that is a $\mathbf{Z}$-algebra of finite type, an A_0-prescheme of finite type X_0 and a constructible subset Z_0 of X_0 such that $X = X_0 \otimes_{A_0} A$ and $Z = h^{-1}(Z_0)$, where $h : X \to X_0$ is the canonical projection ((8.9.1) and (8.3.11)). Furthermore, for all $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, and h_s the projection $X_s \to (X_0)_{s_0}$, we have $Z_s = h_s^{-1}((Z_0)_{s_0})$. As the morphism h_s is faithfully flat and quasi-compact, the hypothesis that Z_s is closed in X_s implies that $(Z_0)_{s_0}$ is closed in $(X_0)_{s_0}$ (2.3.12).

This being so, the transitivity of fibres (**I**, 3.6.4) and the prop. (4.2.7) imply that the set of dimensions of the irreducible components of Z_s containing x is the same as the set of dimensions of the irreducible components of $(Z_0)_{s_0}$ containing $x_0 = h_s(x)$. In particular, we have $\dim_x(Z_s) = \dim_{x_0}((Z_0)_{s_0})$. On the other hand, if $Z_s^{(\beta)}$ are the irreducible components of Z_s containing x and $X_s^{(\alpha)}$ the irreducible components of X_s containing x, we have $\operatorname{codim}_x(Z_s, X_s) =$ **IV-3 | 89**
$\inf_{\beta,\alpha}(\sup(\operatorname{codim}(Z_s^{(\beta)}, X_s^{(\alpha)})))$, (α, β) varying over the set of pairs such that $x \in Z_s^{(\beta)} \in X_s^{(\alpha)}$ (0, (14.2.6)). As the algebraic preschemes are biequidimensional (5.2.1), we can write, by virtue of (0, (14.3.3.1))

$$\operatorname{codim}_x(Z_s, X_s) = \inf_{\beta,\alpha}(\sup(\dim(X_s^{(\alpha)}) - \dim(Z_s^{(\beta)}))) \tag{9.9.1.1}$$

with the same choice of pairs (α, β). As h_s is faithfully flat and quasi-compact, for every pair formed of an irreducible component $(X_0)_{s_0}^{(\alpha)}$ of $(X_0)_{s_0}$ containing x_0 and of an irreducible component $(Z_0)_{s_0}^{(\alpha)}$ of $(Z_0)_{s_0}$ containing x_0 and contained in $(X_0)_{s_0}^{(\alpha)}$, there exists a pair $(X_s^{(\alpha)}, Z_s^{(\beta)})$ of the type described above and such that $X_s^{(\alpha)}$ dominates $(X_0)_{s_0}^{(\alpha)}$ (2.3.5). The formula (9.9.1.1) (and the analogous formula applied to $(X_0)_{s_0}$) shows then, by virtue of (4.2.7), that we have

$$\operatorname{codim}_x(Z_s, X_s) = \operatorname{codim}_{x_0}((Z_0)_{s_0}, (X_0)_{s_0}).$$

We thus see that if E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying one of the conditions (i), (ii), (iii) of the statement (resp. the same condition), we have $E = h^{-1}(E_0)$, and by virtue of (1.8.2), we see that one can restrict to considering the case where A is *Noetherian*, as well as that of (9.9.1.1), and we have (0_{III}, (9.2.3)), and also of (9.9.1), (iii).

By hypothesis Z_η is closed in X_η; as Z is constructible, it follows from (8.3.11), applied following the method of (8.1.2), *a*)), that we can, by replacing S by an open neighbourhood of η, suppose that

Z is *closed* in X. Suppose then that the irreducible components of X (resp. Z) are X_i (resp. Z_j) the irreducible components of X_η (resp. Z_η); by virtue of (0_I, (2.1.8)), the $X_i \cap X_\eta$ (resp. $Z_j \cap X_\eta$) are the irreducible components of X_η (resp. Z_η) containing x; by virtue of (9.5.1), we can suppose further, by restricting S to a neighbourhood, that for all $s \in S$, the X_i (resp. Z_j) are exactly the irreducible components of X (resp. Z) meeting $\overline{\{x\}}$ and that the $(X_i)_s \cap \overline{\{x\}}$ and $(Z_j)_s \cap \overline{\{x\}}$ are non-empty for all $s \in S$. This being so, it follows still from (9.7.1) that we can suppose, by restricting S, that the couples (i,j) such that $(Z_j)_s \subset (X_i)_s$ are the same for all $s \in S$. The conclusion follows then from (9.5.6): for all s sufficiently close to η, all the irreducible components of $(X_i)_s$ (resp. $(Z_j)_s$) have the same dimension equal to that of $(X_i)_\eta$ (resp. $(Z_j)_\eta$). Furthermore, if (i,j) is a pair such that $Z_j \neq X_i$, $(X_i)_\eta$ does not contain the generic point of $(Z_j)_\eta$, hence $\dim((X_i \cap Z_j)_\eta) < \dim((Z_j)_\eta)$; **IV-3** | 90
consequently, the common dimension of the irreducible components of $(X_i)_s \cap (Z_j)_s$ is, for all $s \in S$, strictly less than the common dimension of the irreducible components of $(Z_j)_s$, which proves that none of the irreducible components of $(Z_j)_s$ is contained in an irreducible component of $(X_i)_s$ for $s \in S$. □

Proposition (9.9.2). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$. The following subsets of X are locally constructible:*

label=(i) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is equidimensional at the point x (5.1.12).
lbbel=(ii) The set of $x \in X$ such that the geometric lengths of $\mathscr{F}_{f(x)}$ relative to $\mathbf{k}(f(x))$, at the generic points of the irreducible components of $\mathrm{Supp}(\mathscr{F}_{f(x)})$ *containing x (4.7.5), belong to Φ.*
lcbel=(iii) The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathscr{F}_{f(x)}$ and containing x belong to Φ.
ldbel=(iv) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically reduced at the point x (4.6.17).
lebel=(v) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is geometrically integral at the point x (4.6.22).
lfbel=(vi) The set of $x \in X$ such that $\dim.\mathrm{proj}((\mathscr{F}_{f(x)})_x) \in \Phi$.
lgbel=(vii) The set of $x \in X$ such that $\mathrm{coprof}((\mathscr{F}_{f(x)})_x) \in \Phi$.
lhbel=(viii) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ possesses property (S_n) at the point x (5.7.2).
libel=(ix) The set of $x \in X$ such that $\mathscr{F}_{f(x)}$ is a Cohen-Macaulay $\mathscr{O}_{X_{f(x)}}$-Module at the point x (5.7.1).

Proof. (i) The support Z of $\mathscr{F}$ is locally constructible and closed in X (8.9.1), and by considering a sub-prescheme of X having Z for underlying space and which is of finite presentation over S (8.9.1), we see that the assertion (i) is a particular case of (9.9.1), (iii)).

(ii) All the properties considered are local on X, and we will therefore restrict to the case where $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine and f a morphism of finite presentation. We keep the notations from the beginning of the proof of (9.9.1), and we suppose furthermore A_0 chosen so that there exists a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{A_0} A$. Then ((4.2.7) and (4.7.9)) the set of geometric lengths of $(\mathscr{F}_0)_{f(x_0)}$ at the generic points of the irreducible components of $\mathrm{Supp}((\mathscr{F}_0)_{f(x_0)})$ which contain x_0 is the same as the analogous set for $\mathscr{F}_{f(x)}$; in other words, if E (resp. E_0) is the set of $x \in X$ (resp. $x_0 \in X_0$) satisfying the condition (ii) of the statement, we have $E = h^{-1}(E_0)$, and by virtue of (1.8.2), we see that one can restrict to considering the case where A is *Noetherian*; in other words, say that S is *integral* and $\eta = f(x)$ its generic point. By further reducing, if S' is the sub-prescheme reduced of S having $\overline{\{f(x)\}}$ for underlying space, and if $X' = f^{-1}(S')$, the fibres of X and X' at the points of S' are the same, hence we can replace S by S' and X by X', in other words suppose that S is *integral* and $\eta = f(x)$ its generic point.

Now, if one sets $Z = \mathrm{Supp}(\mathscr{F})$, the same reasoning as in the proof of (9.9.1) shows that, if Z_i are the irreducible components of Z containing x, we can suppose that ce are exactly the irreducible components of Z meeting $\overline{\{x\}}$ and that $(Z_i)_s \cap \overline{\{x\}} \neq \varnothing$ for all $s \in S$. The conclusion follows from (9.8.3) and (9.8.6).

(iii) We reason as in (iii), using this time (4.7.11). With the same notations as in (iii), the prime cycles associated to $\mathscr{F}_s$ that are irreducible components of $(Z'_i)_s$ are immersed if and only if $(Z'_i)_\eta$ is an associated immersed prime cycle of $\mathscr{F}_\eta$. We conclude already that if x belongs to a (resp. does not belong to any) associated immersed prime cycle of $\mathscr{F}_\eta$, the set of $x' \in \overline{\{x\}}$ such that x' belongs to a (resp. does not belong to any) associated immersed prime cycle of $\mathscr{F}_{f(x')}$ is a neighbourhood of

x in $\overline{\{x\}}$. The conclusion follows from this remark, from the characterisation of the points where a Module is geometrically reduced (4.7.10), and of (ii).

(iv) Taking into account (4.7.11), we reduce still again to the case where S is Noetherian and integral and where $\eta = f(x)$ is its generic point. Let Y be a closed sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ for underlying space. Reasoning as at the beginning of the proof of (9.7.7), we see that there exists a finite integral A-algebra A' (such that if $S' = \operatorname{Spec}(A')$, the morphism $S' \to S$ is finite and surjective) such that if $Y' = Y_{(S')}$ and η' is the generic point of S', the irreducible components of $Y_{\eta'}$ are *geometrically irreducible*. On the other hand, if $X' = X_{(S')}$, the morphism projection $h : X' \to X$ is finite and surjective; consequently, if x' is a point of X' above x, we have $h(\overline{\{x'\}}) = \overline{\{x\}}$ and the image by h of a neighbourhood of x' in $\overline{\{x'\}}$ is a neighbourhood of x in $\overline{\{x\}}$. Taking into account (4.7.11) and setting $\mathscr{F}' = \mathscr{F}_{(S')}$, we are thus reduced to proving that if $\mathscr{F}_{\eta'}$ is (resp. is not) geometrically integral at the point x', the set of $y' \in \overline{\{x'\}}$ such that $\mathscr{F}'_{f'(y')}$ (where $f' = f_{(S')} : X' \to S'$) is (resp. is not) geometrically integral at the point y' is a neighbourhood of x' in $\overline{\{x'\}}$. We can thus restrict to the case where $X' = X$ and where the irreducible components of Y_η are *geometrically irreducible*; if one denotes by Z_k the closed subsets of X such that $Z_k \cap X_\eta$ are the irreducible components of Y_η, it follows from (9.7.7), (9.7.8) and (9.5.3) that for all s sufficiently close to η, the $(Z_k)_s$ are the irreducible components of Y_s and that they are *geometrically irreducible*. To say that at a point $y \in X_s$, $\mathscr{F}_s$ is geometrically integral means that $\mathscr{F}_s$ is geometrically reduced at this point and furthermore that y belongs to a *single* of the $(Z_i)_s$ (4.6.22). The conclusion follows thus from part (iv) and of (9.5.1) applied to the intersection of $\overline{\{x\}}$ and of each Z_i. **IV-3** | 91

(v) Taking into account (4.7.11), we reduce once more to the case where S is Noetherian and integral and where $\eta = f(x)$ is its generic point. Let Y be a closed sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ for underlying space. Reasoning as at the beginning of the proof of (9.7.7), we see that there exists a finite integral A-algebra A' (such that if $S' = \operatorname{Spec}(A')$, the morphism $S' \to S$ is finite and surjective) such that if $Y' = Y_{(S')}$ and if η' is the generic point of S', the irreducible components of $Y_{\eta'}$ are *geometrically irreducible*. On the other hand, if $X' = X_{(S')}$, the morphism projection $h : X' \to X$ is finite and surjective; consequently, if x' is a point of X' above x, we have $h(\overline{\{x'\}}) = \overline{\{x\}}$ and the image by h of a neighbourhood of x' in $\overline{\{x'\}}$ is a neighbourhood of x in $\overline{\{x\}}$. Taking into account (4.7.11) and setting $\mathscr{F}' = \mathscr{F}_{(S')}$, we are thus reduced to proving that if $\mathscr{F}_{\eta'}$ is (resp. is not) geometrically integral at the point x', the set of $y' \in \overline{\{x'\}}$ such that $\mathscr{F}'_{f'(y')}$ (where $f' = f_{(S')} : X' \to S'$) is (resp. is not) geometrically integral at the point y' is a neighbourhood of x' in $\overline{\{x'\}}$. We can thus restrict to the case where $X' = X$ and where the irreducible components of Y_η are *geometrically irreducible*; if one denotes by Z_k the closed subsets of X such that $Z_k \cap X_\eta$ are the irreducible components of Y_η, it follows from (9.7.7), (9.7.8) and (9.5.3) that for all s sufficiently close to η, the $(Z_k)_s$ are the irreducible components of Y_s and that they are *geometrically irreducible*. To say that at a point $y \in X_s$, $\mathscr{F}_s$ is geometrically integral means that $\mathscr{F}_s$ is geometrically reduced at this point and furthermore that y belongs to a *single* of the $(Z_i)_s$ (4.6.22). The conclusion follows thus from part (iv) and of (9.5.1) applied to the intersection of $\overline{\{x\}}$ and of each Z_i.

(vi) Keeping the same notations as in (ii), it follows from (6.2.1) that we have $\operatorname{dim.proj}((\mathscr{F}_x)_x) = \operatorname{dim.proj}((\mathscr{F}_{f(x)})_x)$; one can thus still restrict to the case where A is *Noetherian*. Furthermore, on the other hand, we reduce still to showing that, for all $x \in X$, there exists a neighbourhood V of x in $\overline{\{x\}}$ such that, for all $x' \in V$, we have $\operatorname{dim.proj}((\mathscr{F}_{f(x')})_{x'}) = \operatorname{dim.proj}((\mathscr{F}_{f(x)})_x)$; and as above, we can suppose that S is *integral* and that $\eta = f(x)$ is its generic point, in virtue of the generic flatness theorem (6.9.1), we can, by replacing S by a neighbourhood open of η, suppose that the morphism f is *flat* and that $\mathscr{F}$ is f-*flat*; one has then $\operatorname{dim.proj}((\mathscr{F}_{f(x')})_{x'}) = \operatorname{dim.proj}((\mathscr{F}_{f(x)})_x)$ for *all* $x' \in X$ by virtue of (6.2.3). This being so, on can (by replacing S by a neighbourhood open of x) suppose that $\operatorname{dim.proj}(\mathscr{F}_x) \leqslant \operatorname{dim.proj}(\mathscr{F}_x)$ for *all* $x' \in X$. On the other hand, if $\operatorname{dim.proj}(\mathscr{F}_x) = n$, there is by hypothesis a finite $\mathscr{O}_x$-module of type M such that $\operatorname{Ext}^n_{\mathscr{O}_x}(\mathscr{F}_x, M) \neq 0$ (0, (17.2.4)). Now, there exists an $\mathscr{O}_X$-Module coherent $\mathscr{G}$ such that $M = \mathscr{G}_x$ (by replacing X by a neighbourhood open of x (0_I, (5.3.8))); by virtue of (T, (4.2.2)), we have $(\mathscr{E}xt^n_{\mathscr{O}_X}(\mathscr{F},\mathscr{G}))_x \neq 0$. But $\mathscr{E}xt^n_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ is a coherent $\mathscr{O}_X$-Module (0_{III}, (12.3.3)), hence its support Y is closed (0_I, (5.2.2)); as it contains x, it contains also **IV-3** | 92

$\overline{\{x\}}$, whence we conclude (by applying again (T, (4.2.2))) that $\dim.\mathrm{proj}(\mathscr{F}_{x'}) \geqslant n$ for all $x' \in \overline{\{x\}}$, which achieves the proof of the assertion in case (vi).

(vii) As B is an A-algebra of finite type, X is S-isomorphic to a *closed sub-scheme* of an S-scheme of the form $Y = \mathrm{Spec}(A[T_1, \ldots, T_r])$; if $j : X \to Y$ is the canonical injection, and $\mathscr{G}_s = (j_s)_*(\mathscr{F}_s)$ for all $s \in S$, and, by virtue of (0, (16.4.11)), we can restrict to proving the assertion for Y and $\mathscr{G}$. In other words, we can suppose that $B = A[T_1, \ldots, T_r]$, so that the schemes $X_s = \mathrm{Spec}(\mathbf{k}(s)[T_1, \ldots, T_r])$ are *regular* (0, (17.3.7)). Let $W = \mathrm{Supp}(\mathscr{F})$ and $W_s = \mathrm{Supp}(\mathscr{F}_s)$ (**I**, 9.1.13); we have, by (6.11.2.1)

(9.9.2.1) $$\mathrm{coprof}((\mathscr{F}_{f(x)})_x) = \dim.\mathrm{proj}((\mathscr{F}_{f(x)})_x) - \mathrm{codim}_x(W_{f(x)}, X_{f(x)}).$$

But as W is constructible (8.9.1) and each W_s closed, it follows from (9.9.1), (ii)) that the two functions from the right-hand side of (9.9.2.1) are constructible, which finishes the proof of the proposition in the case (vii).

(viii) Let U_n be the set of $x \in X$ such that $\mathrm{coprof}((\mathscr{F}_{f(x)})_x) \leqslant n$, and set $Z_n = X - U_n$; it follows from (vii) that the Z_n are *constructible*; furthermore, as the function $x \mapsto \dim_x(W_{f(x)})$ takes only a finite number of values, the numbers $\dim(W_{f(x)})$ have a finite upper bound m when x ranges over X; as $\mathrm{coprof}((\mathscr{F}_{f(x)})_x) \leqslant \dim((\mathscr{F}_{f(x)})_x) \leqslant \dim(W_{f(x)})$, we have $Z_n = \varnothing$ for $n \geqslant m$. By (5.7.4), the set of $x \in X$ where $(\mathscr{F}_{f(x)})_x$ possesses property (S_k) is the set of $x \in X$ satisfying all the relations

(9.9.2.2) $$\mathrm{codim}_x((Z_n)_{f(x)}, W_{f(x)}) > n + k$$

for all $n \geqslant 0$; as this relation is automatically satisfied for $n \geqslant m$, we need only consider the relations (9.9.2.2) for $0 \leqslant n < m$. But by virtue of (9.9.1), (ii)), the set $V_{n,k}$ of x satisfying (9.9.2.2) is constructible, and so is the intersection of these sets for $0 \leqslant n < m$.

(ix) The assertion follows immediately from (vii), the set of $x \in X$ where $(\mathscr{F}_{f(x)})_x$ is a Cohen-Macaulay module being defined by the relation $\mathrm{coprof}((\mathscr{F}_{f(x)})_x) = 0$. □

Corollary (9.9.3). — *Let $f : X \to S$ be a morphism of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module* **IV-3 | 93** *of finite presentation,* **P** *one of the properties (i) to (ix) of (9.9.2). Then the set of $s \in S$ such that the property* **P** *is true at all the points $x \in X_s$, is locally constructible in S.*

Proof. Indeed, its complement is the image by f of the complement of the set E of points where **P** is true. As E is locally constructible in X, so is $X - E$, and hence $f(X - E)$ is locally constructible in S by virtue of the theorem of Chevalley (1.8.4). □

Proposition (9.9.4). — *Let $f : X \to S$ be a morphism locally of finite presentation. The set of $x \in X$ such that $X_{f(x)}$ has at the point x one of the (determined) following properties:*

label=(i) being geometrically regular;
lbbel=(ii) possessing property (R_k) geometric;
lcbel=(iii) being geometrically normal;
ldbel=(iv) being geometrically reduced (i.e. separable);
lebel=(v) being geometrically pointwise integral;

is a locally constructible subset of X.

Proof. For properties (iv) and (v), this follows from (9.9.2), (iv) and (v)) applied to $\mathscr{F} = \mathscr{O}_X$. For the other properties, taking into account (6.7.8), we reduce, as at the beginning of (9.9.2), (ii)) to the case where S is Noetherian and integral and of generic point $\eta = f(x)$. Furthermore, by virtue of the generic flatness theorem (6.9.1), we can, by replacing S by a neighbourhood open of η, suppose that f is a *flat* morphism. To say that $X_{f(y)}$ is geometrically regular at the point y means that f is *regular* at the point y (6.8.1), and we know that the set L of these points is *open* in X (6.8.7), which proves the proposition in case (i).

To prove case (ii), set $F = X - L$, which is *closed* in X. To say that X_s satisfies the geometric **IV-3 | 94** property (R_k) at the point $y \in f^{-1}(s)$ means either that $y \in L$, or that the generic points z_i of the irreducible components of the closed set F_s which contain y satisfy the relation $\dim(\mathscr{O}_{X_{z_i}, s_i}) \geqslant k + 1$ (taking into account (4.2.7) and (5.2.3))); in other words, the points $y \in F_s$ where X_s satisfies property (R_k) geometric are those such that $\mathrm{codim}_y(F_s, X_s) \geqslant k + 1$ (5.1.2). The conclusion follows thus from (i) and (9.9.1), (ii)).

Finally, in case (iii), the conclusion follows from (ii), from (9.9.2), (viii)), from the fact that property (S_k) is stable by extension of the base field (6.7.1) and finally from the criterion of Serre (5.8.6). □

Corollary (9.9.5). — *Let $f : X \to S$ be a morphism of finite presentation, and denote by* **P** *one of the properties (i) to (v) of (9.9.4). Then the set of $s \in S$ such that the property* **P** *is true at all the points $x \in X_s$, is locally constructible in S.*

Proof. The proof from (9.9.4) is the same as that of (9.9.3) from (9.9.2). □

Proposition (9.9.6). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{L}_\bullet$ a complex formed of $\mathscr{O}_X$-Modules quasi-coherent of finite presentation; for all $s \in S$, let $(\mathscr{L}_\bullet)_s$ be the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ of $\mathscr{O}_{X_s}$-Modules coherent of finite type. Then, for a given integer n, the set of $x \in X$ such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is locally constructible in X.*

Proof. We can restrict to the case where $\mathscr{L}_i = 0$ except for $i = 0, 1$ or 2, and where $n = 1$. Furthermore, the question being local on X, we can restrict to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, B being a finitely presented A-algebra. There exists then a Noetherian sub-ring A_0 of A, a finite type A_0-prescheme X_0 and a complex $\mathscr{L}_\bullet^{(0)}$ of $\mathscr{O}_{X_0}$-Modules coherent, zero except in dimensions 0, 1 and 2 and such that $X = X_0 \otimes_{A_0} A$ and $\mathscr{L}_\bullet = \mathscr{L}_\bullet^{(0)} \otimes_{A_0} A$. For all $s \in S$, if s_0 is the projection of s in $S_0 = \operatorname{Spec}(A_0)$, and h_s the projection $X_s \to (X_0)_{s_0}$, we have $(\mathscr{L}_\bullet)_s = (\mathscr{L}_\bullet^{(0)})_{s_0} \otimes_{\mathbf{k}(s_0)} \mathbf{k}(s)$, and consequently, by virtue of (1.8.2), if $x \in X$ is such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ (resp. $\neq 0$), there exists a neighbourhood open V of x in $\overline{\{x\}}$ such that, for all $x' \in V$, we have $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x')}))_{x'} = 0$ (resp. $\neq 0$). Replacing S by the sub-prescheme reduced of S having $\overline{\{f(x)\}}$ for underlying space, we can suppose that S is integral and that $f(x) = \eta$ is its generic point. Then, by restricting S to a neighbourhood open of η, we can suppose that for all $s \in S$, on a $(\mathscr{H}_n(\mathscr{L}_\bullet))_s = \mathscr{H}_n((\mathscr{L}_\bullet)_s)$ (9.4.3), and by extension, if Z is the support of $\mathscr{H}_n(\mathscr{L}_\bullet)$, the support of $\mathscr{H}_n((\mathscr{L}_\bullet)_s)$ is $Z_s = Z \cap X_s$ (**I**, 9.1.13.1). The hypothesis is that $Z_\eta \cap \overline{\{x\}} = \varnothing$ (resp. $\neq \varnothing$). As $Z \cap \overline{\{x\}}$ is closed in the Noetherian space X, we conclude from (9.5.1) that there is a neighbourhood U of η in S such that, for all $s \in U$, we have $Z_s \cap \overline{\{x\}} = \varnothing$ (resp. $\neq \varnothing$). The set $V = f^{-1}(U) \cap \overline{\{x\}}$ thus answers the question. □

§10. JACOBSON PRESCHEMES

10.1. Very dense subsets of a topological space

IV-3 | 95

(10.1.1). We say that a subset of a topological space X is *quasi-constructible* if it is a finite union of locally closed subsets of X. We say that a subset T of X is *locally quasi-constructible* if for every $x \in X$, there exists an open neighbourhood V of x such that $T \cap V$ is quasi-constructible in V. It is clear that every quasi-constructible subset of X is locally quasi-constructible; the converse is true if X is quasi-compact. The argument of (**III**, 9.1.3), where one removes the word "retrocompact", shows that the set of quasi-constructible (resp. locally quasi-constructible) subsets of X, which we will denote by $\mathfrak{Qc}(X)$ (resp. $\mathfrak{Lqc}(X)$), is stable under finite union, finite intersection, and passage to complements. If $f : X \to Y$ is a continuous map, it follows immediately from these definitions that, for every quasi-constructible (resp. locally quasi-constructible) subset Z of Y, $f^{-1}(Z)$ is quasi-constructible (resp. locally quasi-constructible) in X.

The constructible (resp. locally constructible) subsets of X are evidently quasi-constructible (resp. locally quasi-constructible); the converse is true when X is Noetherian (resp. locally Noetherian).

In what follows, we will denote respectively by $\mathfrak{O}(X)$, $\mathfrak{F}(X)$, $\mathfrak{Lf}(X)$, $\mathfrak{C}(X)$, $\mathfrak{Lc}(X)$ the set of subsets of X that are respectively open, closed, locally closed, constructible, locally constructible.

IV-3 | 96

Proposition (10.1.2). — *Let X be a topological space, X_0 a subset of X. The following conditions are equivalent:*

label=a) *For every locally closed subset $Z \neq \varnothing$ of X, we have $Z \cap X_0 \neq \varnothing$.*

a′) *For every closed subset Z of X, we have $Z = \overline{Z \cap X_0}$.*

lbbel=b) *For every locally quasi-constructible subset $Z \neq \varnothing$ of X, we have $Z \cap X_0 \neq \varnothing$.*

b′) *For every locally quasi-constructible subset Z of X, we have $Z \subset \overline{Z \cap X_0}$ (in other words, $Z \cap X_0$ is dense in Z).*

lcbel=c) *The map $U \mapsto U \cap X_0$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(X_0)$ is injective (hence bijective).*

c′) *The map $F \mapsto F \cap X_0$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is injective (hence bijective).*

c″) *The map $Z \mapsto Z \cap X_0$ of $\mathfrak{Qc}(X)$ into $\mathfrak{Qc}(X_0)$ is injective (hence bijective).*

c‴) *The map $Z \mapsto Z \cap X_0$ of $\mathfrak{Lqc}(X)$ into $\mathfrak{Lqc}(X_0)$ is injective.*

Furthermore, when these conditions are satisfied, the map $Z \mapsto Z \cap X_0$ of $\mathfrak{Lqc}(X)$ into $\mathfrak{Lqc}(X_0)$ is bijective.

Proof. Let us note that the assertions of surjectivity of $c)$ and $c')$ are trivial; they mean that every locally closed subset of X_0 is the trace on X_0 of a locally closed subset of X, and that the map $\mathfrak{Qc}(X) \to \mathfrak{Qc}(X_0)$ defined in $c'')$ is also surjective.

We will prove the implications

$$c''') \Rightarrow c'') \Rightarrow c') \Rightarrow c) \Rightarrow b) \Rightarrow b') \Rightarrow a') \Rightarrow a) \Rightarrow c''').$$

The first three are trivial. To see that $c)$ implies $b)$, let us first note that $c)$ means that X_0 is *dense* in X. Replacing X and X_0 by U and $U \cap X_0$ respectively, one can therefore restrict to the case where Z is locally closed in X, hence $Z = V \cap \complement W$, where V and W are open in X; the hypothesis $Z \neq \varnothing$ means that $V \not\subset W$, or equivalently $V \cup W \neq W$. By virtue of $c)$, we have $(V \cup W) \cap X_0 \neq W \cap X_0$, hence $V \cap X_0 \not\subset W$, and consequently $(V \cap \complement W) \cap X_0 \neq \varnothing$.

To see that $b)$ implies $b')$, it suffices to apply $b)$ to $Z \cap U$, where U is an open neighbourhood of an arbitrary point of Z. As $Z \cap \overline{X_0} \subset \overline{Z}$, it is trivial that $b')$ implies $a')$. To show that $a')$ implies $a)$, let us note that if Z is locally closed in X, we can write $Z = F - F'$ where F and F' are closed in X and $F' \subset F$; whence $Z \cap X_0 = (F \cap X_0) - (F' \cap X_0)$. If $Z \cap X_0 = \varnothing$, then, by virtue of $a')$, we would have $F \cap X_0 = F' \cap X_0$, hence $F = F'$, i.e. $Z = \varnothing$.

To see that $a)$ implies $c''')$, it suffices to show that if $Z \cap X_0 \neq \varnothing$, then $Z' \cap X_0 = Z'' \cap X_0$ is equivalent to $((Z' \cup Z'') - (Z' \cap Z'')) \cap X_0 = \varnothing$. In other words, it suffices to prove that $a)$ implies $b)$; indeed, replacing X and X_0 by U and $U \cap X_0$ respectively, one is reduced to the case where Z is locally closed in X, whence the conclusion.

It remains to show that the map $\mathfrak{Lqc}(X) \to \mathfrak{Lqc}(X_0)$ is surjective. So let Z_0 be a locally quasi-constructible subset of X_0: there is a covering (V_α) of X_0 by open subsets of X_0 such that $Z_0 \cap V_\alpha$ is quasi-constructible in V_α (and hence also in X_0). By virtue of $c)$, there is for each α a unique **IV-3 | 97**
quasi-constructible subset Z_α in X such that $Z_0 \cap V_\alpha = X_0 \cap Z_\alpha$. If α and β are any two indices, we have $Z_\beta \cap U_\alpha \cap X_0 = U_\alpha \cap X_0 = V_\alpha \cap Z_0 \cap V_\beta = Z_\alpha \cap X_0 \cap V_\beta \subset Z_0 \cap X_0$; as $Z_\beta \cap U_\alpha$ and Z_α are quasi-constructible in X, it follows from $c'')$ that $Z_\beta \cap U_\alpha \subset Z_\alpha$. If we set $Z = \bigcup_\alpha Z_\alpha$, then $Z \cap U_\alpha = Z_\alpha$ for every α; moreover, since the V_α cover X_0, it follows from $c)$ that the U_α cover X, and we see therefore that Z is locally quasi-constructible in X and $Z_0 = Z \cap X_0$. □

Definition (10.1.3). — When a subset X_0 of a topological space satisfies the equivalent conditions of (10.1.2), we say that X_0 is *very dense* in X.

We have already seen in the course of the proof of (10.1.2) that X_0 is then dense in X.

Corollary (10.1.4). — *If X_0 is very dense in X and U an open subset of X, $U \cap X_0$ is very dense in U. Conversely, if (U_α) is an open covering of X such that $U_\alpha \cap X_0$ is very dense in U_α for every α, then X_0 is very dense in X.*

As every locally closed subset of U is locally closed in X, the first assertion follows from criterion $a)$ of (10.1.2); and the same is true of the second, since if $Z \neq \varnothing$ is locally closed in X, $Z \cap U_\alpha$ is locally closed in U_α for every α, and $Z \cap U_\alpha \neq \varnothing$ for at least one α.

10.2. Quasi-homeomorphisms

Proposition (10.2.1). — *Let X_0, X be two topological spaces, $f : X_0 \to X$ a continuous map. The following conditions are equivalent:*

label=a) The map $U \mapsto f^{-1}(U)$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(X_0)$ is bijective.

a′) *The map $F \mapsto f^{-1}(F)$ of $\mathfrak{F}(X)$ into $\mathfrak{F}(X_0)$ is bijective.*

lbbel=b) The topology of X_0 is the inverse image by f of that of X, and $f(X_0)$ is very dense (10.1.3) in X.

lcbel=c) The functor $\mathscr{F} \mapsto f^*(\mathscr{F})$ from the category of sheaves of sets (resp. sheaves of abelian groups) on X to the category of sheaves of sets (resp. sheaves of abelian groups) on X_0 is an equivalence of categories.

Proof. It is clear that $a)$ and $a')$ are equivalent, and $a)$ implies that the topology of X_0 is the inverse image by f of that of X. On the other hand, if $f(X_0)$ is not very dense in X, there are two distinct open subsets U_1, U_2 of X such that $U_1 \cap f(X_0) = U_2 \cap f(X_0)$, and hence $f^{-1}(U_1) = f^{-1}(U_2)$, which completes showing that $a)$ implies $b)$. Conversely, the condition $b)$ implies that the maps $U \mapsto U \cap f(X_0)$ of $\mathfrak{O}(X)$ into $\mathfrak{O}(f(X_0))$ and $V \mapsto f^{-1}(V)$ of $\mathfrak{O}(f(X_0))$ into $\mathfrak{O}(X_0)$ are bijective, hence so is the composite $U \mapsto f^{-1}(U)$.

To see that $a)$ implies $c)$, it suffices to apply the definition of $f^*(\mathscr{F})$ (**0**, 3.7.1) and the axioms for sheaves: $a)$ means that for every open U of X, the canonical map $\Gamma(U, \mathscr{F}) \to \Gamma(f^{-1}(U), f^*(\mathscr{F}))$ is a functorial bijection in $\mathscr{F}$, whence $c)$. It remains to show that $c)$ implies $b)$.

Suppose first that $f(X_0)$ is not very dense in X; there then exist two distinct closed subsets $Y \supset Y'$ of X such that $Y \cap f(X_0) = Y' \cap f(X_0)$. Let $\mathscr{F}$ (resp. $\mathscr{F}'$) be the sheaf of abelian groups on X that is the direct image by the canonical injection $Y \to X$ (resp. $Y' \to X$) of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y (resp. Y'); the definition of f^* (**0**, 3.7.1) shows that $f^*(\mathscr{F})$ is isomorphic to $f^*(\mathscr{F}')$, but $\mathscr{F}$ is not isomorphic to $\mathscr{F}'$, hence condition $c)$ is not satisfied. (One notes that the functor f^* is not then even *faithful*, for if u and v are the identity automorphism and the zero endomorphism of $\mathscr{F}/\mathscr{F}'$, $f^*(u)$ and $f^*(v)$ are equal.) **IV-3** | 98

Let us now show that $c)$ implies that the topology of X_0 is the inverse image of that of X by f. Let us note that if condition $c)$ is satisfied for the category of sheaves of sets, it is also satisfied for the category of sheaves of abelian groups, for it follows immediately from the definitions (**III**, 8.2.5) that the category of *objects in abelian groups* in the category of sheaves of sets is none other than the category of sheaves of abelian groups. It will therefore suffice to prove our assertion supposing $c)$ satisfied for the categories of sheaves of abelian groups on X and X_0. Now, if $\mathbf{Z}_X$ denotes the simple sheaf on X associated to the constant presheaf $\mathbf{Z}$, there is an isomorphism canonical up to sign $\Gamma(X, \mathscr{F}) \xrightarrow{\sim} \mathrm{Hom}(\mathbf{Z}_X, \mathscr{F})$ functorial in $\mathscr{F}$ (by the same argument as in (**0**, 5.1.1)); as it is clear by definition (**0**, 3.7.1) that $f^*(\mathbf{Z}_X) = \mathbf{Z}_{X_0}$, the hypothesis that f^* is an equivalence means that the canonical homomorphism $\Gamma(X, \mathscr{F}) \to \Gamma(X_0, f^*(\mathscr{F}))$ is *bijective* for every sheaf of abelian groups $\mathscr{F}$ on X. Now, let Y_0 be a closed subset of X_0; let $\mathscr{F}_0$ be the sheaf on X that is the direct image by the canonical injection $Y_0 \to X_0$ of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y_0. As f^* is an equivalence, $\mathscr{F}_0$ is isomorphic to a sheaf of the form $f^*(\mathscr{F})$, where $\mathscr{F}$ is a sheaf of abelian groups on X. Consider then a section s_0 of $\mathscr{F}_0$ over X_0 such that $(s_0)_{x_0} = 1_{x_0}$ if $x_0 \in Y_0$, $(s_0)_{x_0} = 0_{x_0}$ if $x_0 \notin Y_0$. The preceding remarks show that there exists a section s of $\mathscr{F}$ over X such that $(s_0)_{x_0} = s_{f(x_0)}$ for every $x_0 \in X_0$ (the fibres $(\mathscr{F}_0)_{x_0}$ and $\mathscr{F}_{f(x_0)}$ being canonically identified (**0**, 3.7.1)); this means that Y_0 is the inverse image by f of the set Y of $x \in X$ such that $s_x \neq 0_x$, and y is closed in X. □

Definition (10.2.2). — When a continuous map $f : X_0 \to X$ satisfies the equivalent conditions of (10.2.1), we say that f is a *quasi-homeomorphism* of X_0 into X.

By virtue of (10.2.1), $b)$, to say that a subset X_0 of a topological space X is very dense in X means that the canonical injection $X_0 \to X$ is a quasi-homeomorphism.

Corollary (10.2.3). — *The composite of two quasi-homeomorphisms is a quasi-homeomorphism.*

This follows immediately from (10.2.1), $a)$.

Corollary (10.2.4). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y, $X' = f^{-1}(Y')$; then the restriction $f' = f|X' : X' \to Y'$ is a quasi-homeomorphism.*

Proof. It is clear by virtue of (10.2.1), b) that the topology induced on X' by that of X is the inverse image by f' of the topology induced on Y' by that of Y. On the other hand, let $Z \neq \varnothing$ be a closed subset *in* Y', and $z \in Z$; there is an open neighbourhood U of z in Y such that $U \cap Y'$ is the union of a finite number of closed subsets Y'_j of U; if j is an index such that $z \in Y'_j$, $Z \cap Y'_j$ is therefore closed in U. Since $f(X) \cap U$ is very dense in U (10.1.4), $Z \cap Y'_j \cap f(X)$ is not empty (10.1.2), a); but as $Z \subset Y'$, we have $Z \cap f(X) = Z \cap f'(X')$; the criterion (10.1.2), a) shows that $f'(X')$ is very dense in Y', and we conclude by (10.2.1), b). ☐ **IV-3 | 99**

Corollary (10.2.5). — *Let $f : X \to Y$ be a continuous map, (V_α) an open covering of Y. If, for every α, the restriction $f^{-1}(V_\alpha) \to V_\alpha$ of f is a quasi-homeomorphism, then f is a quasi-homeomorphism.*

This follows immediately from the criterion (10.2.1), b) and from (10.1.4).

Corollary (10.2.6). — *Let $f : X \to Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible subset of Y, $X' = f^{-1}(Y')$. For Y' to be quasi-compact (resp. Noetherian, resp. retrocompact in Y), it is necessary and sufficient that X' be quasi-compact (resp. Noetherian, resp. retrocompact in X).*

Proof. Let us prove the first two assertions; by virtue of (10.2.4), one can restrict to the case where $Y' = Y$. To say that X is quasi-compact (resp. Noetherian) means that for every increasing filtered family (U_α) in $\mathfrak{O}(X)$ (resp. for every increasing filtered family (U_α) in $\mathfrak{O}(X)$) having X as greatest element, there exists γ such that $U_\alpha = U_\gamma$ for $\alpha \geqslant \gamma$. As $U \mapsto f^{-1}(U)$ is a bijection of $\mathfrak{O}(Y)$ onto $\mathfrak{O}(X)$, our assertion follows immediately from the preceding remark.

The quasi-compact open subsets of X are therefore the sets of the form $f^{-1}(U)$ where U is an open quasi-compact subset of Y, by virtue of (10.2.1), a) and of what precedes. For X' to be retrocompact in X, it is necessary and sufficient that for every quasi-compact open U in Y, $f^{-1}(U) \cap X' = f^{-1}(U \cap Y')$ be quasi-compact (**III**, 9.1.1); the first part of the proof shows that this amounts to saying that $U \cap Y'$ is quasi-compact for every open quasi-compact U, i.e. that Y' is retrocompact in Y. ☐

Proposition (10.2.7). — *Let $f : X \to Y$ be a quasi-homeomorphism. Then the map $Z \mapsto f^{-1}(Z)$ of $\mathfrak{P}(Y)$ into $\mathfrak{P}(X)$ defines by restriction the bijections (cf. (10.1.1) for the notation):*

$$\mathfrak{O}(Y) \xrightarrow{\sim} \mathfrak{O}(X)$$
$$\mathfrak{F}(Y) \xrightarrow{\sim} \mathfrak{F}(X)$$
$$\mathfrak{Lf}(Y) \xrightarrow{\sim} \mathfrak{Lf}(X)$$
$$\mathfrak{Qc}(Y) \xrightarrow{\sim} \mathfrak{Qc}(X)$$
$$\mathfrak{Lqc}(Y) \xrightarrow{\sim} \mathfrak{Lqc}(X)$$
$$\mathfrak{C}(Y) \xrightarrow{\sim} \mathfrak{C}(X)$$
$$\mathfrak{Lc}(Y) \xrightarrow{\sim} \mathfrak{Lc}(X).$$

Proof. For the first two, this is none other than the definition (10.2.2); as the topology of X is the inverse image by f of that of $f(X)$, the five remaining maps, where one replaces Y by $f(X)$, are bijective. It therefore suffices (by (10.2.1), b)) to restrict to the case where X is a very dense subspace of Y, and that the maps $\mathfrak{Lf}(Y) \to \mathfrak{Lf}(X)$, $\mathfrak{Qc}(Y) \to \mathfrak{Qc}(X)$, $\mathfrak{Lqc}(Y) \to \mathfrak{Lqc}(X)$ are bijective has already been proved (10.1.2). Every locally constructible subset being locally quasi-constructible, the maps $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ and $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$ are injective; furthermore, for every open $U \subset Y$, the restriction $f^{-1}(U) \to U$ of f is a quasi-homeomorphism (10.2.4), hence if we show that $\mathfrak{C}(Y) \to \mathfrak{C}(X)$ is surjective, the same will be true of $\mathfrak{Lc}(Y) \to \mathfrak{Lc}(X)$. But by virtue of (10.2.6), every retrocompact open subset Z in X is of the form $f^{-1}(T)$, where T is retrocompact open in Y; this evidently proves the surjectivity of $\mathfrak{C}(Y) \to \mathfrak{C}(X)$. ☐ **IV-3 | 100**

Remarks (10.2.8). — (i) We saw in the proof of (10.2.1) that if f is a quasi-homeomorphism, the canonical map

(10.2.8.1) $$\Gamma(Y, \mathscr{F}) \longrightarrow \Gamma(X, f^*(\mathscr{F}))$$

is a functorial isomorphism in $\mathscr{F}$ in the category of sheaves of abelian groups on Y. As f^* is exact in this category (**0**, 3.7.2), this implies that the canonical homomorphisms $(T, 3.2.2)$

$$\mathrm{H}^i(Y, \mathscr{F}) \xrightarrow{\sim} \mathrm{H}^i(X, f^*(\mathscr{F}))$$

are *bijective* for every i.

(ii) If $f : X \to Y$ is a continuous map and $\mathscr{F}, \mathscr{G}$ two sheaves of abelian groups on Y, we have $f^*(\mathscr{F} \otimes_{\mathbf{Z}_Y} \mathscr{G}) = f^*(\mathscr{F}) \otimes_{\mathbf{Z}_X} f^*(\mathscr{G})$, an isomorphism canonical up to sign (**0**, 4.3.3). We conclude that if f is a quasi-homeomorphism, f^* is also an *equivalence* of the category of sheaves of rings on Y, and of the category of sheaves of rings on X; the data of a structure of ringed space on Y is therefore equivalent to that of a structure of ringed space on X.

Given two ringed spaces $(X, \mathscr{O}_X)$, $(Y, \mathscr{O}_Y)$, we will say that a morphism of ringed spaces $f = (\psi, \theta) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a *quasi-isomorphism* if ψ is a quasi-homeomorphism of X into Y and $\theta^\sharp : \psi^*(\mathscr{O}_Y) \to \mathscr{O}_X$ an isomorphism of sheaves of rings; when this is so, the ringed space $(X, \mathscr{O}_X)$ is entirely determined, up to isomorphism, by $(Y, \mathscr{O}_Y)$, the space X, and the quasi-homeomorphism ψ (which can be chosen arbitrarily). When f is a quasi-isomorphism of ringed spaces, the functor

$$\mathscr{F} \longmapsto f^*(\mathscr{F})$$

is an *equivalence* of the category of $\mathscr{O}_Y$-Modules and of that of $\mathscr{O}_X$-Modules, since $f^*(\mathscr{F})$ is canonically identified here with $\psi^*(\mathscr{F})$. We conclude for example isomorphisms of bi-∂-functors

$$\operatorname{Ext}^i_{\mathscr{O}_Y}(\mathscr{F}, \mathscr{G}) \xrightarrow{\sim} \operatorname{Ext}^i_{\mathscr{O}_X}(f^*(\mathscr{F}), f^*(\mathscr{G})).$$

In general, one can say that the usual constructions of the theory of sheaves and of homological algebra, carried out on the ringed space Y or the ringed space X, are equivalent.

10.3. Jacobson spaces IV-3 | 101

Definition (10.3.1). — We say that a topological space X is a *Jacobson space* if the set X_0 of closed points of X is very dense in X (in other words, if the canonical injection $X_0 \to X$ is a quasi-homeomorphism).

This means therefore (10.1.2) that every locally closed (or only locally quasi-constructible) subset $Z \neq \varnothing$ of X contains a closed point of X, or equivalently that *every closed subset of X is the closure of the set of its closed points*.

Proposition (10.3.2). — *Let X be a Jacobson space, Z a quasi-constructible subset of X; then the subspace Z is a Jacobson space, and for a point of Z to be closed in Z, it is necessary and sufficient that it be closed in X.*

Proof. If X_0 is the set of closed points of X, $Z \cap X_0$ is very dense in Z by virtue of (10.2.4) applied to the injection $i : X_0 \to X$; as the set Z_0 of closed points of Z evidently contains $Z \cap X_0$, it is *a fortiori* very dense in Z, hence Z is a Jacobson space. Let us now show that we have $Z_0 = Z \cap X_0$; let $x \in Z$ be a closed point in Z; let $\overline{\{x\}}$ be its closure in X and $\overline{\{x\}} \cap Z = \{x\}$, which is therefore a locally quasi-constructible subset of X, and as its intersection with X_0 is not empty (10.1.2), we have $x \in X_0$. □

Proposition (10.3.3). — *Let X be a topological space, (U_α) an open covering of X. For X to be a Jacobson space, it is necessary and sufficient that each of the subspaces U_α be a Jacobson space.*

Proof. The condition is necessary by virtue of (10.3.2). Conversely, let us show first that the hypothesis that the U_α are Jacobson spaces implies that for a point $x \in U_\alpha$ to be closed in X, it suffices that it be closed in U_α. It suffices indeed to see that this condition implies that x is also closed in each of the U_β that contain it; but $U_\alpha \cap U_\beta$ is open in U_α, hence x is closed in $U_\alpha \cap U_\beta$, and by virtue of (10.3.2), x is also closed in U_β, which completes the proof. □

10.4. Jacobson preschemes and Jacobson rings

Definition (10.4.1). — We say that a prescheme X is a *Jacobson prescheme* if the underlying topological space of X is a Jacobson space. We say that a ring A is a *Jacobson ring* if $\operatorname{Spec}(A)$ is a Jacobson prescheme.

Every closed subset of $X = \operatorname{Spec}(A)$ is of the form $Z = V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal equal to its radical, and the set X_0 of closed points of X is the set of maximal ideals of A; to say that $Z \cap X_0$ is dense in Z means therefore that $\mathfrak{a}$ is an *intersection of maximal ideals* (**I**, 1.1.4); as $\mathfrak{a}$ is an intersection of prime ideals, this amounts to saying that every prime ideal of A is an intersection of maximal ideals; by virtue of (10.3.1) and (10.1.2), the usual definition of Jacobson rings (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, def. 1) thus coincides with the definition (10.4.1).

Proposition (10.4.2). — *Let X be a prescheme, (U_α) a covering of X formed by affine open subsets. For X to be a Jacobson prescheme, it is necessary and sufficient that the rings $\Gamma(U_\alpha, \mathscr{O}_X)$ of the U_α be Jacobson rings.* **IV-3** | 102

This follows from (10.3.3) and the definition (10.4.1).

(10.4.3). A discrete prescheme is a Jacobson prescheme; an Artinian ring is therefore a Jacobson ring. Every principal ideal ring having an infinity of maximal ideals (for example **Z**) is a Jacobson ring; a Noetherian local ring A is a Jacobson ring if and only if its maximal ideal is its only prime ideal, i.e. if A is Artinian. Every sub-prescheme of a Jacobson prescheme is a Jacobson prescheme by virtue of (10.3.2).

Proposition (10.4.4). — *Let B be an integral ring. The following conditions are equivalent:*

label=a) *There exists $f \neq 0$ in B such that B_f is a field.*
lbbel=b) *The field of fractions of B is a B-algebra of finite type.*
lcbel=c) *There exists a field K containing B and which is a B-algebra of finite type.*
ldbel=d) *The generic point of* $\operatorname{Spec}(B)$ *is isolated.*

Proof. It is clear that d) is equivalent to a), since d) means that there exists $f \neq 0$ in B such that $D(f)$ is reduced to the generic point of $\operatorname{Spec}(B)$. It is trivial that a) implies b) and that b) implies c). Finally, c) implies a), by virtue of Bourbaki, *Alg. comm.*, chap. V, § 3, n° 1, cor. 2 of th. 1. □

Proposition (10.4.5). — *Let A be a ring. The following conditions are equivalent:*

label=a) *A is a Jacobson ring.*
lbbel=b) *For every non-maximal prime ideal $\mathfrak{p}$ of A and every $f \neq 0$ in $B = A/\mathfrak{p}$, B_f is not a field.*
b′) *Every A-algebra of finite type K which is a field is a finite A-algebra.*

Proof. We know that a) implies b') (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 4, cor. 3 of th. 3). Furthermore, the kernel of the homomorphism $A \to K$ is then a *maximal* ideal $\mathfrak{m}$ of A, and K is a finite extension of $A/\mathfrak{m}$ (*loc. cit.*). It is trivial that b') implies b), since $A/\mathfrak{p} = B$ is not a field and B_f is a B-algebra of finite type. It remains to show that b) implies a). We will use the following lemma: □

Lemma (10.4.5.1). — *Let X be a topological space having the following property: for every locally closed subset $Z \neq \varnothing$ of X, there exists a subset Z' of Z, locally closed in X (or in Z, which amounts to the same), and a point $x \in Z'$, closed in Z'. Then, for X to be a Jacobson space, it is necessary and sufficient that no non-closed point x of X be isolated in $\overline{\{x\}}$.*

Proof. If X is a Jacobson space, a non-closed point x of X cannot be isolated in $\overline{\{x\}}$, for this would mean that there exists an open U of X such that $U \cap \overline{\{x\}} = \{x\}$; but $U \cap \overline{\{x\}}$ is locally closed in X and would contain no closed point in X, which contradicts the hypothesis that X is a Jacobson space. Conversely, suppose the conditions of the statement are satisfied; then the set X_0 of closed points of X is identical to the set of $x \in X$ that are isolated in $\overline{\{x\}}$; but it follows from (5.1.10.1) that this set is *very dense* in X, hence X is a Jacobson space by definition. □ **IV-3** | 103

Let us recall (5.1.10) that the hypothesis made on X in (10.4.5.1) is always satisfied when the underlying space of X is a *prescheme*.

Returning to the proof of (10.4.5), condition b) means that for every non-closed point x of $\operatorname{Spec}(A)$, the generic point x of $\operatorname{Spec}(A/\mathfrak{j}_x) = \overline{\{x\}}$ is not isolated in $\overline{\{x\}}$, hence b) implies a) by the lemma (10.4.5.1) and (10.4.4).

Corollary (10.4.6). — *Every algebra of finite type B over a Jacobson ring A is a Jacobson ring, and the inverse image in A of every maximal ideal of B is a maximal ideal of A. In particular, every algebra of finite type over a field or over* $\mathbf{Z}$ *is a Jacobson ring.*

Proof. A B-algebra of finite type K which is a field is also an A-algebra of finite type, hence an A-module of finite type and *a fortiori* a B-module of finite type, whence the first assertion; the second was proved in the course of the proof of (10.4.5), applied to $K = B/\mathfrak{m}$. □

Corollary (10.4.7). — *If X is a Jacobson prescheme, $f : Y \to X$ a morphism locally of finite type, then Y is a Jacobson prescheme and the image by f of every closed point in Y is a closed point in X.*

Proof. The question being local on X and on Y, one reduces to the case where X and Y are affine, and the corollary then follows from (10.4.6). □

Corollary (10.4.8). — *If k is an algebraically closed field, X a k-prescheme locally of finite type, the set of rational points of X over k is very dense in X.*

Proof. Indeed, X is a Jacobson prescheme (10.4.7) and the closed points are exactly the rational points over k (**I**, 6.4.2). □

(10.4.9). The fact that preschemes locally of finite type over a field or over $\mathbf{Z}$ are Jacobson preschemes is particularly important, because of the possibility of reducing to this case in many questions of algebraic geometry (8.1.2), *c*). We will give two examples:

(10.4.10). Applications (10.4.10) : I : Proof of (6.15.9). Let k be a separably closed field, X a k-prescheme locally of finite type over k and *unibranch*. We know that the integral closure of a k-algebra of finite type A in an extension of finite type of its field of fractions is a finite A-algebra (Bourbaki, *Alg. comm.*, chap. V, § 3, n° 2, th. 2), hence every k-algebra of finite type is a universally Japanese ring; we conclude that the set of points $x \in X$ where X is geometrically unibranch is locally constructible (9.7.10). But the hypothesis and the lemma (6.15.8) imply that this set contains all the *closed* points of X. The conclusion therefore follows from (10.4.6), (10.3.1) and the bijectivity of the canonical map $\mathfrak{Lc}(X) \to \mathfrak{Lc}(X_0)$ (where X_0 is the set of closed points of X) (10.2.7).

(10.4.11). Applications : II : Proposition (10.4.11). Let S be a prescheme, X an S-prescheme of finite **IV-3** | 104
type. Every S-endomorphism of X which is radicial is surjective (hence bijective).

Proof. Let $f : X \to S$ be the structural morphism, $g : X \to X$ the S-endomorphism considered, and, for every $s \in S$, let g_s be the morphism deduced from g by the base change $\operatorname{Spec}(k(s)) \to S$, which is a $k(s)$-endomorphism of the fibre $f^{-1}(s) = X \times_S \operatorname{Spec}(k(s))$.

To prove that g is surjective, it suffices to prove that g_s is surjective for every $s \in S$, hence one can (by virtue of (**I**, 3.5.7)) reduce to the case where $S = \operatorname{Spec}(k)$ is the spectrum of a field k, in which case f is a morphism of finite presentation, since S is Noetherian. Applying (8.9.1) and (8.10.5), (vii), one reduces to the case where $S = \operatorname{Spec}(A)$, where A is a sub-$\mathbf{Z}$-algebra of finite type of k. But then X is a Jacobson prescheme (10.4.7), and $g(X)$ is constructible in X (**I**, 1.8.5), hence, to prove that $g(X) = X$, it suffices to show that $g(X)$ contains all the *closed* points of X (10.3.1).

Lemma (10.4.11.1). — *Let Y be a* $\mathbf{Z}$*-prescheme of finite type.*

label=(i) For a closed point $y \in Y$, it is necessary and sufficient that $k(y)$ be a finite field.
lbbel=(ii) For every prime number p and every integer $d \geqslant 1$, the set of points $y \in Y$ such that $k(y)$ is an extension of $\mathbf{F}_p$ of degree dividing d is finite.

The assertion (i) follows from the fact that the image of a closed point $y \in Y$ in $\operatorname{Spec}(\mathbf{Z})$ is a closed point (10.4.7), in other words a prime number, and from (**I**, 6.4.2). On the other hand, as Y is a finite union of affine open subsets of finite type over $\mathbf{Z}$, one can restrict to proving (ii) in the case where $Y = \operatorname{Spec}(C)$, where C is a $\mathbf{Z}$-algebra of finite type. Now, the points $y \in Y$ such that the degree of $k(y)$ over $\mathbf{F}_p$ divides d correspond bijectively to the homomorphisms $C \to \mathbf{F}_{p^d}$; but if (t_i) is a system of generators of the $\mathbf{Z}$-algebra C, every homomorphism of C is determined by its values on the elements t_i, and there are therefore only a finite number of homomorphisms of C into a finite field.

This being so, since X is a $\mathbf{Z}$-prescheme of finite type, let $T_{p,d}$ be the set of closed points $z \in X$ such that $k(z)$ is an extension of $\mathbf{F}_p$ of degree dividing d; it follows from (10.4.11.1) that the set $T_{p,d}$ is finite and that the set of closed points of X is the union of the $T_{p,d}$. Furthermore, if $z \in T_{p,d}$ and if h is any endomorphism of X, $k(h(z))$ is isomorphic to a subfield of $k(z)$, hence $h(z) \in T_{p,d}$; in other words $T_{p,d}$ is *stable* under every endomorphism of X. As g is by hypothesis injective, its restriction to $T_{p,d}$ is *bijective* since $T_{p,d}$ is finite, which completes proving the proposition.

We will see further on (17.9.7) that when one further supposes that X is an S-prescheme of finite presentation, and on the other hand that g is a monomorphism, then one can assert that g is an *automorphism* of X. □

10.5. Noetherian Jacobson preschemes

Proposition (10.5.1). — *Let B be a Noetherian integral ring. The conditions equivalent to a) through d) of (10.4.4) are then also equivalent to the following:*

*lebel=*e) Spec(B) *is finite.*

*lfbel=*f) *B is a semi-local ring of dimension $\leqslant 1$.*

Proof. It follows indeed from the Artin–Tate theorem (**0**, 16.3.3) that conditions e) and f) are equivalent. Condition f) implies that Spec(B) is the union of the set of its closed points and of its generic point, hence f) implies e) without assuming A to be Noetherian. Finally, e) implies d) without assuming A to be Noetherian, for the generic point x of X is the complement of the union of the closures $\overline{\{y\}}$, where y ranges over the set of points $y \neq x$, and as these points are finite in number, $\{x\}$ is the complement of a closed set in X. □ **IV-3** | 105

Corollary (10.5.2). — *For a Noetherian ring A to be a Jacobson ring, it is necessary and sufficient that there exist no prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is a semi-local ring of dimension 1.*

Proof. This follows immediately from (10.5.1) and from condition b) of (10.4.5), the prime ideals $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is semi-local of dimension 0 being the maximal ideals of A. □

Corollary (10.5.3). — *Let X be an irreducible Noetherian prescheme. The following conditions are equivalent:*

*label=*a) *The generic point of X is isolated.*

*lbbel=*b) *X is finite.*

*lcbel=*c) *X is of dimension $\leqslant 1$ and the set of its closed points is finite.*

Proof. There exists by hypothesis a finite number of irreducible affine open subsets U_i $(1 \leqslant i \leqslant n)$ covering X, and each of which contains the generic point of X; it suffices to prove the equivalence of a), b), and c) for each of the $(U_i)_{\text{red}}$ (taking into account (**0**, 14.1.7)). But this equivalence follows from (10.5.1). □

Remark (10.5.4). — A Noetherian prescheme X satisfying the equivalent conditions of (10.5.3) is not necessarily an affine scheme; in fact it can even be *non-separated*, as one sees by replacing, in the example (**I**, 5.5.11) of the "affine line with doubled point", X_1 and X_2 by the spectrum of the ring of the discrete valuation $(K[s])_{(s)}$, U_{12} and U_{21} by the open subset reduced to the generic point in X_1 and X_2 respectively; the non-separated prescheme X that one obtains has exactly 3 points.

Proposition (10.5.5). — *Let X be a Noetherian prescheme.*

label=(i) The set X_0 of points $x \in X$ such that $\overline{\{x\}}$ is finite is very dense in X.

lbbel=(ii) For X to be a Jacobson prescheme, it is necessary and sufficient that there exist no sub-prescheme of X isomorphic to the spectrum of a semi-local integral ring of dimension 1.

Proof. The condition that $\overline{\{x\}}$ is finite is equivalent, here (10.5.3), to the fact that x is isolated in $\overline{\{x\}}$, it being the underlying space of a sub-prescheme (Noetherian) of X; the assertion (i) therefore follows from (10.4.5.1). Similarly, taking into account (10.4.5.1), to show the assertion (ii), let us note that for a non-closed point x of X to belong to X_0, it is necessary and sufficient that the closed sub-prescheme Y of X having $\overline{\{x\}}$ as underlying space be of dimension 1 and finite, hence a finite union of sub-preschemes which are open (in Y) and affine U_i which are spectra of semi-local integral rings of dimension 1. Conversely, if there is a sub-prescheme Z of X which is the spectrum of a semi-local integral ring of dimension 1, Z is not a Jacobson prescheme (10.5.2), hence neither is X (10.3.2). □

Remark (10.5.6). — The assertion (ii) of (10.5.5) is still valid when X is *locally Noetherian*: indeed, if (V_α) is a covering of X by affine open subsets (Noetherian), every sub-prescheme of one of the V_α is a sub-prescheme of X; conversely, if a sub-prescheme Z of X is isomorphic to the spectrum of a semi-local integral ring of dimension 1, there is an α such that $V_\alpha \cap Z$ contains an affine open U of Z which is non-reduced to the generic point of Z, and which is therefore also the spectrum of a semi-local integral ring of dimension 1. We conclude by (10.3.3).

Proposition (10.5.7). — *Let X be a locally Noetherian prescheme, Y a closed subset of X such that every non-empty closed subset of X meets Y. Then the prescheme induced on the open subset $X - Y$ is a Jacobson prescheme.*

Proof. Applying the criterion (10.5.5), (ii), and supposing that there were in $X - Y$ a sub-prescheme Z which is the spectrum of a semi-local integral ring of dimension 1, the generic point z of Z being isolated in Z (or in the closure $\overline{Z}$ of Z in X), and Z being distinct from $\{z\}$. Let $y \neq z$ be a point of Z; as it does not belong to Y, it is not a closed point in X, and its closure $\overline{\{y\}}$ in X meets Y at a point $x \neq y$, which is therefore a specialisation of y. The existence of the chain $\{x\} \subset \overline{\{y\}} \subset \overline{\{z\}}$ then shows that the dimension of $\overline{\{z\}}$ would be $\geqslant 2$, and the same would be true of the dimension of $U \cap \overline{\{z\}}$, where U is an affine (hence Noetherian) open neighbourhood of x in X. But this contradicts the fact that the generic point z of $U \cap \overline{\{z\}}$ is isolated in $U \cap \overline{\{z\}}$ (10.5.3). □

Corollary (10.5.8). — *Let A be a Noetherian ring; for every element f of the radical $\mathfrak{R}$ of A, the ring A_f is a Jacobson ring, and the open subset* $\operatorname{Spec}(A) - V(\mathfrak{R})$ *is a Jacobson scheme.*

Proof. If $X = \operatorname{Spec}(A)$, $Y = V(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of A, to say that every non-empty closed subset of X meets Y is equivalent (**0**, 2.1.3) to saying that Y contains all the *closed points* of X, or equivalently that $\mathfrak{J}$ is *contained in the radical* $\mathfrak{R}$ of A. If $f \in \mathfrak{R}$, the open $D(f)$ does not meet $V(\mathfrak{R})$, and is a Jacobson space by virtue of (10.5.7). □

Corollary (10.5.9). — *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $X = \operatorname{Spec}(A)$, $Y = X - \{\mathfrak{m}\}$; then Y is a Jacobson scheme, whose closed points are the prime ideals $\mathfrak{p} \in \operatorname{Spec}(A)$ such that $\dim(A/\mathfrak{p}) = 1$.*

Proof. The first assertion is a particular case of (10.5.8); on the other hand the closed points of Y are the prime ideals $\mathfrak{p}$ of A which are maximal elements in the set of prime ideals $\neq \mathfrak{m}$, which, by definition of the dimension, means that $\dim(A/\mathfrak{p}) = 1$. □

Proposition (10.5.10). — *Let A be a Noetherian local ring, reduced, complete, and not a field. Then the finite intersections of the kernels of the local homomorphisms of A into discrete valuation rings V, which are A-algebras of finite type, form a basis of filter tending to 0 for the adic topology of A.*

Proof. It suffices (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 7, prop. 8) to prove that the intersection of the kernels considered in the statement is reduced to 0. Suppose first that A is integral and of dimension 1; by virtue of the Nagata theorem (**0**, 23.1.5) and (23.1.6), the integral closure A' of A is a complete local integral ring and integrally closed and of dimension 1; it is therefore a discrete valuation ring, and the proposition follows immediately in this case. **IV-3** | 107

Let us pass to the general case; let x be the closed point of $X = \operatorname{Spec}(A)$, and set $Y = X - \{x\}$; we know (10.5.9) that Y is a Jacobson prescheme. Let us show that the intersection of the prime ideals $\mathfrak{p}$ of A such that $\dim(A/\mathfrak{p}) = 1$ is reduced to 0. But to say that the intersection of these prime ideals is reduced to 0 means that the set of these ideals is dense in X, or equivalently in Y (since Y is dense in X), and this follows immediately from (10.5.9). □

10.6. Dimension in Jacobson preschemes The results of this section give precision in certain cases and generalise results from § 5.

Proposition (10.6.1). — *Let S be a locally Noetherian prescheme, satisfying in addition the conditions: 1° S is a Jacobson prescheme; 2° for every $s \in S$, $\mathscr{O}_{S,s}$ is universally catenary* (5.6.2); *3° every irreducible component S' of S is equicodimensional (in other words, for every closed point s of S' and every sub-prescheme S' having S' as underlying space, we have $\dim(S') = \dim(\mathscr{O}_{S',s})$). We then have the following properties:*

label=(i) *For every morphism locally of finite type $g : X \to S$, X satisfies conditions 1°, 2°, and 3° above. In particular, if X is equidimensional (for example if X is irreducible), X is biequidimensional (in other words X is catenary and for every closed point x of X, we have $\dim(\mathscr{O}_{X,x}) = \dim(X)$* (**0**, 14.3.3)*).*

lbbel=(ii) *Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism; suppose X irreducible and f dominant. If ξ (resp. η) is the generic point of X (resp. Y) and $e = \dim(f^{-1}(\eta)) = \deg.\mathrm{tr}_{k(\eta)} k(\xi)$, we have*

(10.6.1.1)
$$\dim(X) = \dim(Y) + e.$$

lcbel=(iii) *Let X, Y be two S-preschemes locally of finite type over S, $f : X \to Y$ an S-morphism, n a non-negative integer such that $\dim(f^{-1}(y)) \geqslant n$ (resp. $\dim(f^{-1}(y)) \leqslant n$) for every $y \in Y$. Then we have*

(10.6.1.2)
$$\dim(X) \geqslant \dim(Y) + n$$

(resp.

(10.6.1.3)
$$\dim(X) \leqslant \dim(Y) + n).$$

Proof. (i) The property 1° for X follows from (10.4.7). For every $x \in X$, $\mathscr{O}_{X,x}$ is the local ring of an **IV-3 | 108**
$\mathscr{O}_{S,g(x)}$-algebra of finite type at a prime ideal, and the homomorphism $\mathscr{O}_{S,g(x)} \to \mathscr{O}_{X,x}$ is local; hence (5.6.3), (iv) $\mathscr{O}_{X,x}$ is universally catenary. To show that X satisfies condition 3°, consider several cases:

a) X is a closed irreducible sub-prescheme of S; let S' be an irreducible component of S containing X, ξ the generic point of X, x a closed point of X; for every $s \in S'$, $\mathscr{O}_{S',s}$, a quotient of $\mathscr{O}_{S,s}$, is catenary (5.6.1), hence conditions 2° and 3° imply that S' is biequidimensional (**0**, 14.3.3). By virtue of (5.1.2) and of (**0**, 14.3.3.2), we therefore have

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S',x}) - \dim(\mathscr{O}_{S',\xi}) = \dim(S') - \dim(\mathscr{O}_{S',\xi})$$

which proves that $\dim(\mathscr{O}_{X,x})$ does not depend on the closed point x considered, whence the assertion in this case (5.1.4).

b) X is irreducible and g dominant. Then, for every closed point x of X, $g(x)$ is closed in S by (10.4.7); as $\mathscr{O}_{S,g(x)}$ is universally catenary, it follows from (5.6.5.3) that we have

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{S,g(x)}) + e$$

where $e = \dim(g^{-1}(\zeta))$, ζ being the generic point of S. As $\dim(S) = \dim(\mathscr{O}_{S,g(x)})$ by virtue of condition 3° for S, we have $\dim(\mathscr{O}_{X,x}) = \dim(S) + e$ for every closed point $x \in X$; this proves condition 3° for X (5.1.4), and at the same time the formula

(10.6.1.4)
$$\dim(X) = \dim(S) + e.$$

c) General case. Considering a reduced sub-prescheme X' of X having for underlying space an irreducible component of X, one reduces to the case where X is integral; using (**I**, 5.2.2) and the case a) proved above, one can then replace S by the reduced sub-prescheme having $g(X)$ as underlying space; one is then brought to case b), and this completes proving (i).

(ii) The morphism f being locally of finite type (**I**, 1.3.4), one can apply the results of (i) replacing S by Y; furthermore, as X is irreducible and f dominant, one can also replace S by Y in (10.6.1.4), which gives (10.6.1.1).

(iii) The relative assertion where $\dim(f^{-1}(y)) \leqslant n$ for every $y \in Y$ has already been proved under more general hypotheses in (5.6.7). Suppose that $\dim(f^{-1}(y)) \geqslant n$ for every $y \in Y$, and consider a generic point η of an irreducible component Y' of Y; there exists at least one irreducible component Z of $f^{-1}(\eta)$ of dimension $\geqslant n$; if ξ is the generic point of Z, ξ is also a generic point of an irreducible component X' of X such that $f(X')$ is dense in Y' (**0**, 2.1.8). Consider the reduced sub-preschemes of

X, Y having respectively for underlying spaces X', Y', and the restriction $X' \to Y'$ of f (**I**, 5.2.2); it then follows from (ii) that $\dim(X') \geqslant \dim(Y') + n$, and *a fortiori* $\dim(X) \geqslant \dim(Y') + n$; this being true for every irreducible component Y' of Y, we conclude the inequality (10.6.1.2). □

Corollary (10.6.2). — *Suppose that X satisfies conditions 1°, 2°, and 3° of (10.6.1). Then, for every dense open U in X, we have* $\dim(U) = \dim(X)$.

Proof. We know indeed that $\dim(U) \leqslant \dim(X)$ (**0**, 14.1.4); furthermore, as X is a Jacobson prescheme, U contains a closed point of every irreducible component of X, hence $\dim(U) = \dim(X)$ by virtue of (10.6.1) and (**0**, 14.1.2.1). □ **IV-3 | 109**

Proposition (10.6.3). — *Suppose that X satisfies conditions 1°, 2°, and 3° of (10.6.1), and let Y be a closed subset of X. For every $x \in Y$, and every open neighbourhood U of x in X not meeting the irreducible components of Y that do not contain x, we have*

(10.6.3.1) $$\dim_x(Y) = \dim(U \cap Y) = \dim \overline{\{x\}} + \operatorname{codim}(\overline{\{x\}}, Y) = \sup_i \dim(Y_i)$$

where Y_i ($1 \leqslant i \leqslant m$) are the irreducible components of Y containing x.

Proof. Considering a closed sub-prescheme of X having Y as underlying space, one can restrict, by virtue of (10.6.1), (i), to the case where $Y = X$. By the choice of U, U is the union of the $U \cap X_i$, hence $\dim(U) = \sup_i \dim(U \cap X_i)$; but by (10.6.2) applied to a closed sub-prescheme of X having X_i as underlying space, we have (taking into account (10.6.1), (i)), $\dim(U \cap X_i) = \dim(X_i)$; this proves that the second and the fourth terms of (10.6.3.1) are equal. As $U \cap X_i$ is biequidimensional (10.6.1), (i) (since the immersion $U \cap X_i \to X_i$ is of finite type (**I**, 6.3.5)), we have

(10.6.3.2) $$\dim(X_i) = \dim(U \cap X_i) = \dim(U \cap \overline{\{x\}}) + \operatorname{codim}(U \cap \overline{\{x\}}, U \cap X_i)$$

(**0**, 14.3.5). By (10.6.2) applied to a closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, $\dim(U \cap \overline{\{x\}}) = \dim(\overline{\{x\}})$; as we also have, for the same reasons,

(10.6.3.3) $$\dim(X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X_i)$$

we obtain

$$\dim(U) = \dim(\overline{\{x\}}) + \sup_i \operatorname{codim}(\overline{\{x\}}, X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by definition of the codimension (**0**, 14.2.1). This shows that $\dim(U)$ is *independent* of the open neighbourhood U of x satisfying the conditions of the statement, hence is equal to $\dim_x(X)$, by (**0**, 14.1.4.1). □

Corollary (10.6.4). — *Under the hypotheses of (10.6.3), let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-Module, Y its support. For every $x \in Y$, we have*

(10.6.4.1) $$\dim(\overline{\{x\}}) + \dim(\mathscr{F}_x) = \dim_x Y.$$

Proof. Indeed, this follows from (10.6.3.1) and from the formula $\dim(\mathscr{F}_x) = \operatorname{codim}(\overline{\{x\}}, Y)$ (5.1.12.2). □

10.7. Examples and counter-examples

(10.7.1). Let S be a locally Noetherian prescheme of dimension $\leqslant 1$ and suppose that S is a *prescheme de Jacobson*; when S is Noetherian, this amounts to saying that the irreducible components of S of dimension 1 are infinite, for every $x \in S$ which is not a closed point is the generic point of such a component (10.4.5) and (10.5.4). Then S also satisfies conditions 2° and 3° of (10.6.1): indeed, every local ring $\mathscr{O}_{S,s}$ is of dimension 0 or 1, and hence universally catenary (5.6.3), (7.2.9); on the other hand, an irreducible component S' of S is, either reduced to a closed point s of S', or biequidimensional.

From these remarks and from (10.6.1) it follows that every prescheme locally of finite type over S also satisfies the properties 1°, 2°, and 3°; it is in particular the case of preschemes *locally of finite type over a field or over* $\mathbf{Z}$. **IV-3** | 110

(10.7.2). Let A be a Noetherian local ring *universally catenary* and let S be the complementary of *the closed point a* in $X = \operatorname{Spec}(A)$. Then S satisfies conditions 1°, 2°, and 3° of (10.6.1): indeed, one has already seen that S is a prescheme de Jacobson (10.5.9); as A is universally catenary, so are all the local rings $A_{\mathfrak{p}}$ at the prime ideals of A (5.6.3). On the other hand, an irreducible component S' of S is the complement of a in an irreducible component X' of X; for every closed point x of S, the closure of x in X is therefore $\{x, a\}$, hence $\dim(\mathscr{O}_{X',x}) = 1$ and hence, as X' is by hypothesis biequidimensional ($\mathbf{0}$, 14.3.3), on has $\dim(\mathscr{O}_{X',x}) = \dim(X') - 1$ (5.1.2), which proves that S' is equicodimensional.

(10.7.3). Let A be a discrete valuation ring; let us show that in $B = A[T_1, \ldots, T_n]$, with n indeterminates, there exist two maximal ideals $\mathfrak{m}$, $\mathfrak{n}$ of *respective heights* n and $n+1$. We have seen (5.2.5), (i) for $n = 1$; let us prove it by induction on n. As $A[T_1, \ldots, T_{n+1}]$ is an $A[T_1, \ldots, T_n]$-module free of rank one, there are in $A[T_1, \ldots, T_{n+1}]$ two maximal ideals $\mathfrak{m}'$, $\mathfrak{n}'$ of respective heights $n+1$ and $n+2$; furthermore, by (5.5.3), these ideals are necessarily of heights $n+1$ and $n+2$, whence our assertion. Suppose $n \geqslant 2$ in what follows. Let $\mathfrak{J}$ be the ideal $\mathfrak{m} \cap \mathfrak{n} = \mathfrak{m}\mathfrak{n}$, which is contained in the *radical* of B; if one sets $B' = R^{-1}B$, the ideal $\mathfrak{J}' = R^{-1}\mathfrak{J}$ is contained in the radical of B' as a topological space $X' = \operatorname{Spec}(B')$ is identified with a subspace of $X = \operatorname{Spec}(B)$, and as the points x of X', the local rings $\mathscr{O}_{X',x}$ and $\mathscr{O}_{X,x}$ are the same (10.5.7); one knows (10.5.7) that S is a prescheme de Jacobson, evidently irreducible and Noetherian; furthermore all the local rings $\mathscr{O}_{S,s}$ are universally catenary for every $s \in S$ by virtue of (5.6.3), since A is universally catenary. However, there exist two closed points a, b of S such that $\mathscr{O}_{S,a}$ and $\mathscr{O}_{S,b}$ *do not have the same dimension*, otherwise S would not be biequidimensional (5.6.4). To see this, consider the two maximal ideals $\mathfrak{m}' = R^{-1}\mathfrak{m}$, $\mathfrak{n}' = R^{-1}\mathfrak{n}$ of B', which are of heights n and $n+1$ respectively; we have $\mathfrak{J}' = \mathfrak{m}' \cap \mathfrak{n}'$, and $\mathfrak{J}'$ is contained in no prime ideal of B' distinct from $\mathfrak{m}'$ and $\mathfrak{n}'$, which are therefore the only maximal ideals of B'. Let a', b' be the two closed points of X' corresponding to $\mathfrak{m}'$ and $\mathfrak{n}'$; the closures of a' in X' are $\{a, a'\}$; as $B'_{\mathfrak{m}'}$ is biequidimensional, $\mathfrak{p}'$ is then an ideal of B' of height n such that its closure in X' is $\{a, a'\}$; one constructs in the same way an ideal $\mathfrak{q}'$ of B' of height n such that the closure of b in X' is $\{b, b'\}$. This being so, a and b are closed points in S, hence the question is answered.

10.8. Rectified depth

Definition (10.8.1). — Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. For every $x \in X$, we call *rectified depth* of $\mathscr{F}$ at the point x and denote $\operatorname{prof}^*_x(\mathscr{F})$ the number (integer $\geqslant 0$ or $+\infty$) equal to

(10.8.1.1)
$$\operatorname{prof}^*_x(\mathscr{F}) = \operatorname{prof}(\mathscr{F}_x) + \dim(\overline{\{x\}})$$

where $\overline{\{x\}}$ is the closure of the point x in X. For every closed subset Z of X, we call *rectified depth of $\mathscr{F}$ along* Z and denote $\operatorname{prof}^*_Z(\mathscr{F})$ the number

(10.8.1.2)
$$\operatorname{prof}^*_Z(\mathscr{F}) = \inf_{x \in Z} \operatorname{prof}^*_x(\mathscr{F}).$$

In other terms, for every integer n, the relation $\operatorname{prof}^*_Z(\mathscr{F}) \geqslant n$ is equivalent to $\operatorname{prof}^*_x(\mathscr{F}) \geqslant n$ for every $x \in Z$. If $Z = X$, one writes $\operatorname{prof}^*(\mathscr{F})$ instead of $\operatorname{prof}^*_X(\mathscr{F})$.

IV-3 | 111

Remarks (10.8.2). — (i) At every closed point $x \in X$, the rectified depth equals the depth.

(ii) Let Y be a closed sub-prescheme of X, $j : Y \to X$ the canonical injection. We know (5.7.3), (vi) that $\mathrm{prof}(\mathscr{G}_x) = \mathrm{prof}(j_*(\mathscr{G})_x)$ for every $x \in Y$; we therefore deduce that $\mathrm{prof}^*_x(\mathscr{G}) = \mathrm{prof}^*_x(j_*(\mathscr{G}))$ for every $x \in Y$.

(iii) The notion of rectified depth is of interest only when it is of a *local* character, i.e. when it does not change when one replaces X by a neighbourhood open U of x. This obviously requires that x not be isolated in $\overline{\{x\}}$ when x is not a closed point of X, i.e. most often, that X be a *Jacobson prescheme* (10.4.5.1); furthermore, it will also be necessary to know that $\dim(U) = \dim(X)$ for every dense open U in X, and one must therefore also suppose that X satisfies conditions 2° and 3° of (10.6.1).

Lemma (10.8.3). — *Let X be a regular and biequidimensional prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Then, for every $x \in X$, we have*

(10.8.3.1) $$\mathrm{prof}^*_x(\mathscr{F}) = \dim(X) - \dim.\mathrm{proj}(\mathscr{F}_x).$$

Proof. Indeed, as X is biequidimensional, we have (**0**, 14.3.5.1)

$$\dim(\overline{\{x\}}) = \dim(X) - \mathrm{codim}(\overline{\{x\}}, X) = \dim(X) - \dim(\mathscr{O}_{X,x})$$

by virtue of (5.1.2). On the other hand, as X is regular, we have by (**0**, 17.3.4)

$$\mathrm{prof}(\mathscr{F}_x) = \dim(\mathscr{O}_{X,x}) - \dim.\mathrm{proj}(\mathscr{F}_x)$$

whence the lemma. □

Corollary (10.8.4). — *Under the hypotheses of (10.8.3), the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is lower semi-continuous.*

Proof. This follows from (10.8.3.1), since $x \mapsto \dim.\mathrm{proj}(\mathscr{F}_x)$ is upper semi-continuous (6.11.1). □

Proposition (10.8.5). — *Let S be a prescheme locally Noetherian, X a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose that S satisfies the conditions: 1° S is a Jacobson prescheme; 2° S is regular; 3° the irreducible components of S are equicodimensional. Then the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F})$ is semi-continuous inferiorly; in other terms, for every integer n, the set U_n of $x \in X$ such that $\mathrm{prof}^*_x(\mathscr{F}) \geqslant n$ is open.*

Proof. As the local rings $\mathscr{O}_{S,s}$ of S are regular, they are universally catenary (5.6.4); in other words, S satisfies conditions 1°, 2°, and 3° of (10.6.1), hence the same is true of X (10.6.1), (i). The notion of rectified depth being then of local character (10.8.2), (iii), one can restrict to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, A being a regular ring and B a quotient of A-algebra of finite type, hence a quotient of a polynomial ring $C = A[T_1, \ldots, T_n]$, and this last ring is regular (**0**, 17.3.7). One can therefore suppose that X is a closed sub-prescheme of a regular prescheme Y also satisfying conditions 1°, 2°, and 3° of (10.6.1); taking into account the remark (10.8.2), (ii), one is thus brought to the case where X is in addition regular and Noetherian. But as the local rings of X are then integral, the irreducible components of X are open (**I**, 6.1.10), and one can therefore **IV-3 | 112**
also suppose X irreducible. Then, as the local rings of X are catenary (**0**, 16.5.12), the hypothesis 3° of (10.6.1) implies that X is biequidimensional ((5.1.5) and (**0**, 14.3.3)); it therefore suffices to apply (10.8.4). □

One notes that if S is the spectrum of a field or of $\mathbf{Z}$, it satisfies the conditions of (10.8.5).

Corollary (10.8.6). — *The hypotheses being those of (10.8.5), for every $x \in X$, the number $\mathrm{prof}^*_x(\mathscr{F})$ is the unique integer n having the following property: there exists an open neighbourhood U of x such that for every point $x' \in U \cap \overline{\{x\}}$, closed in U, we have $\mathrm{prof}(\mathscr{F}_{x'}) = n$. In particular, for every x such that $\mathrm{prof}^*_x(\mathscr{F}) \geqslant m$ (resp. $\mathrm{prof}^*_x(\mathscr{F}) \leqslant m$), it is necessary and sufficient that there exist an open V of X such that, for every point $x' \in V \cap \overline{\{x\}}$, closed in V, we have $\mathrm{prof}(\mathscr{F}_{x'}) \geqslant m$ (resp. $\mathrm{prof}(\mathscr{F}_{x'}) \leqslant m$).*

Proof. Indeed, one can restrict to the case where x is not a closed point; if $\mathrm{prof}^*_x(\mathscr{F}) = n$, the set of $y \in X$ such that $\mathrm{prof}^*_y(\mathscr{F}) \leqslant n$ is closed by virtue of (10.8.5), hence contains $\overline{\{x\}}$, and by virtue of the lower semi-continuity of $\mathrm{prof}^*_y(\mathscr{F})$, there exists an open neighbourhood U of x such that $\mathrm{prof}^*_y(\mathscr{F}) = n$ for every $y \in U \cap \overline{\{x\}}$, hence $\mathrm{prof}(\mathscr{F}_y) = n$ if $x' \in U \cap \overline{\{x\}}$ is closed (since the notion of rectified depth is local). □

For preschemes satisfying the hypotheses of (10.8.5), the notion of rectified depth can therefore be defined by means of the values of the depth at the *closed points* of X (the latter forming a set which is very dense in every closed subset of X).

Proposition (10.8.7). — *Let S be a prescheme locally Noetherian satisfying conditions $1°$, $2°$, and $3°$ of (10.6.1). Let X be a prescheme locally of finite type over S, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, $Y = \mathrm{Supp}(\mathscr{F})$. Then, for every $x \in Y$, we have*

$$\mathrm{prof}^*_x(\mathscr{F}) = \dim_x(Y) - \mathrm{coprof}(\mathscr{F}_x). \tag{10.8.7.1}$$

Proof. Indeed, by definition, $\mathrm{coprof}(\mathscr{F}_x) = \dim(\mathscr{F}_x) - \mathrm{prof}(\mathscr{F}_x)$, and it follows from (10.6.4) that $\dim_x(Y) = \dim(\overline{\{x\}}) + \dim(\mathscr{F}_x)$; whence (10.8.7.1) by definition of $\mathrm{prof}^*_x(\mathscr{F})$. □

Corollary (10.8.8). — *Under the hypotheses on f and $\mathscr{F}$ being those of (9.9.1), the function $x \mapsto \mathrm{prof}^*_x(\mathscr{F}_{f(x)})$ is constructible.*

Proof. Let us note that for every $s \in S$, the fibre X_s being a prescheme locally of finite type over $k(s)$, is a Jacobson prescheme (10.4.7); as furthermore $\mathrm{Spec}(k(s))$ satisfies the conditions of (10.6.1), on has $\mathrm{prof}^*_x(\mathscr{F}_{f(x)}) = \dim_x(\mathrm{Supp}(\mathscr{F}_{f(x)})) - \mathrm{coprof}((\mathscr{F}_{f(x)})_x)$ (10.8.7). Now, if $Z = \mathrm{Supp}(\mathscr{F})$, we have $Z_{f(x)} = \mathrm{Supp}(\mathscr{F}_{f(x)})$ (**I**, 9.1.13) and Z is locally constructible (8.9.1); therefore the functions $x \mapsto \dim_x(\mathrm{Supp}(\mathscr{F}_{f(x)}))$ and $x \mapsto \mathrm{coprof}((\mathscr{F}_{f(x)})_x)$ are locally constructible ((9.9.1) and (9.9.3)), which proves the proposition. □

10.9. Maximal spectra and ultra-preschemes The results of this section will not be used in what follows.

(10.9.1). Let X be a Jacobson prescheme, and let $S(X)$ be the *ringed space* whose underlying space is the subspace of closed points of X, and the sheaf of rings the sheaf induced on this subspace by $\mathscr{O}_X$, in other words the sheaf $\psi^*(\mathscr{O}_X)$, where $\psi : S(X) \to X$ denotes the canonical injection. As ψ is a quasi- homeomorphism, we have seen (10.2.8), (ii) that if $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_{S(X)})$ is the **IV-3** | 113
homomorphism of sheaves of rings such that $\theta^\sharp : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{S(X)}$ is the identity, then $j_x = (\psi, \theta)$ is a *quasi-isomorphism* of ringed spaces, and $\mathscr{F} \mapsto j_X^*(\mathscr{F})$ is an equivalence of the category of $\mathscr{O}_X$-Modules and that of $\mathscr{O}_{S(X)}$-Modules. It is clear that in this equivalence, the $\mathscr{O}_X$-Modules that are locally free (resp. coherent) correspond to the $\mathscr{O}_{S(X)}$-Modules that are locally free (resp. coherent); furthermore, if $\mathscr{L}$ is an $\mathscr{O}_X$-Module locally free and if U is an open of X such that $\mathscr{L}|U$ is free, then $j_X^*(\mathscr{L})|(U \cap S(X))$ is isomorphic to $(\mathscr{O}_X|U)^n$, i.e. is isomorphic to $((\mathscr{O}_{S(X)}|(U \cap S(X)))^n$.

(10.9.2). Let X, Y be two Jacobson preschemes and $f = (p, \lambda) : X \to Y$ a morphism *locally of finite type*. One has seen (10.4.7) that $p(S(X)) \subset S(Y)$ and by restriction of p to $S(X)$, one defines therefore a continuous map $S(p) : S(X) \to S(Y)$. For every open V of Y, one defines by composition a homomorphism of rings

$$\Gamma(V \cap S(Y), \mathscr{O}_{S(Y)}) \longrightarrow \Gamma(V, \mathscr{O}_Y) \xrightarrow{\lambda_V} \Gamma(f^{-1}(V), \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(V) \cap S(X), \mathscr{O}_{S(X)})$$

where the two extreme isomorphisms have been defined in (10.9.1); it is clear that this defines a homomorphism of sheaves of rings $S(\lambda) : S(\mathscr{O}_{S(Y)}) \to \rho_*(\mathscr{O}_{S(X)})$ (recalling that the open subsets of $S(X)$ (resp. $S(Y)$) correspond bijectively to those of X (resp. Y) (10.2.1)); one obtains thus a morphism of ringed spaces $S(f) = (S(p), S(\lambda)) : (S(X), \mathscr{O}_{S(X)}) \to (S(Y), \mathscr{O}_{S(Y)})$ such that the diagram

$$\begin{tikzcd} S(X) \arrow[r, "S(f)"] \arrow[d, "j_X"'] & S(Y) \arrow[d, "j_Y"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

is commutative; furthermore, if Z is a third Jacobson prescheme and $g : Y \to Z$ a morphism locally of finite type, $S(g \circ f) = S(g) \circ S(f)$. One has thus defined a *covariant functor* $S : \mathbf{C} \to \mathbf{C}'$, where $\mathbf{C}$ is the category of *ringed spaces in local rings*, and $\mathbf{C}'$ the category whose objects are the *Jacobson preschemes* and the morphisms are the morphisms locally of finite type between Jacobson preschemes.

(10.9.3). Let us now determine the subcategory $\mathbf{C}''$ of $\mathbf{C}'$ formed by the ringed spaces isomorphic to $S(X)$. Suppose first that $X = \operatorname{Spec}(A)$, where A is a Jacobson ring; then $S(X)$ is the set of *maximal* ideals of A, equipped: 1° with the topology induced by that of X, so that a base for this topology is formed by the $D^m(h) = D(h) \cap S(X)$, i.e. the set of maximal ideals $\mathfrak{m}$ of A such that $h \notin \mathfrak{m}$; 2° with the sheaf of rings $\mathscr{O}_{S(X)}$ such that $\Gamma(D^m(h), \mathscr{O}_{S(X)}) = A_h$. We will say that this ringed space is the *maximal spectrum* of the Jacobson ring A and denote it $\operatorname{Spm}(A)$.

One notes that $j : D(h) \to X$ is the canonical injection, the ringed space induced on $D^m(h)$ by $S(X)$ and $S(D(h))$ are canonically isomorphic, and that the canonical injection $D^m(h) \to S(X)$ of ringed spaces is equal to $S(j)$.

Let B be a second Jacobson ring, $Y = \operatorname{Spec}(B)$, $\varphi : B \to A$ a ring homomorphism making A a B-algebra of *finite type*, $f = ({}^a\varphi, \widetilde{\varphi}) : X \to Y$ the corresponding morphism of preschemes, and

$$S(f) : \operatorname{Spm}(A) \longrightarrow \operatorname{Spm}(B)$$

the morphism of ringed spaces corresponding to f. It is clear that $S(f) = (\psi, \theta)$ is a morphism of ringed spaces in local rings, i.e. (Err_{11}, 1.8.2) for every $x \in \operatorname{Spm}(A)$, $\theta_x^\sharp$ is a *local* homomorphism. Conversely:

Proposition (10.9.4). — *Let A, B be two Jacobson rings. If $u = (\psi, \theta) : \operatorname{Spm}(A) \to \operatorname{Spm}(B)$ is a morphism of ringed spaces in local rings such that $\Gamma(\theta) : \mathrm{B} \to A$ makes A a B-algebra of finite type, there exists a morphism of preschemes $f : \operatorname{Spec}(A) \to \operatorname{Spec}(B)$ and a unique one such that $u = S(f)$.*

Proof. The uniqueness of f is evident, since if $f = ({}^a\varphi, \widetilde{\varphi})$, one must have $\varphi = \Gamma(\theta)$; it remains to show that $S(f)$ is defined and that $u = S(f)$. Now, the first assertion follows from the fact that φ is supposed to make A a B-algebra of finite type, the fact that $\theta_x^\sharp$ is a local homomorphism for every $x \in \operatorname{Spm}(A)$ allows one to repeat the argument of (**I**, 1.7.3), restricting to points x of $\operatorname{Spm}(A)$, to show thus successively that $\psi(x) = {}^a\varphi(x)$ for every $x \in \operatorname{Spm}(A)$, then that $\theta_x^\sharp = \varphi_x : B_{(x)} \to A_x$, which completes proving that $u = S(f)$. □

IV-3 | 114

(10.9.5). Consider now a ringed space $(X, \mathscr{O}_X)$; we will say that an open subset U of X is *ultra-affine* if the ringed space induced $(U, \mathscr{O}_X|U)$ is isomorphic to a maximal spectrum $\operatorname{Spm}(A)$, where A is a Jacobson ring. We will say that X is an *ultra-prescheme* if every point of X admits an ultra-affine open neighbourhood. One shows, as in (**I**, 2.1.3) and (2.1.4), that the ultra-affine open subsets form a *base* for the topology of X and that X is a Kolmogoroff space. If Y is a second ultra-prescheme, we will say that a morphism of ringed spaces $f : X \to Y$ is a *morphism of ultra-preschemes* if it satisfies the following conditions: 1° f is a morphism of ringed spaces in local rings; 2° for every $x \in X$, there is an ultra-affine open neighbourhood V of $f(x)$ in Y and an ultra-affine open U of x in X such that $f(U) \subset V$ and the homomorphism $\Gamma(V, \mathscr{O}_Y) \to \Gamma(U, \mathscr{O}_X)$ corresponding to f makes $\Gamma(U, \mathscr{O}_X)$ a $\Gamma(V, \mathscr{O}_Y)$-*algebra of finite type*.

It is immediate that one defines well the composite of two such morphisms, being a third such by the final remark of (10.9.3). It is clear that the category C_0'' thus defined is a subcategory of C' that contains C''; we now propose to show that $C'' = C_0''$, in other words:

Proposition (10.9.6). — *The functor $X \mapsto S(X)$ of $\mathbf{C}$ into $\mathbf{C}_0''$ is an equivalence of categories.*

Proof. 1° Let us first show that the functor $X \mapsto S(X)$ is *fully faithful*, in other words that for X, Y in $\mathbf{C}$, the canonical map

$$\operatorname{Hom}_{\mathbf{C}}(X, Y) \longrightarrow \operatorname{Hom}_{\mathbf{C}_0''}(S(X), S(Y))$$

is *bijective*. In the first place it is *injective*: let f, g be two morphisms locally of finite type of X into Y, and suppose that $S(f) = S(g)$; we have $f^{-1}(V) \cap S(X) = g^{-1}(V) \cap S(X)$ for every open V of Y, hence (10.3.1) and (10.2.7) $f^{-1}(V) = g^{-1}(V)$; it suffices therefore to prove that for every affine open V of Y, f and g coincide in $f^{-1}(V) = g^{-1}(V)$; in other words, one reduces to the case where $Y = \operatorname{Spec}(B)$ is the spectrum of a Jacobson ring B. It suffices (**I**, 2.2.4) to show that the homomorphisms of rings $B \to \Gamma(X, \mathscr{O}_X)$ corresponding to f and g are the same. Now, for every $s \in B$, the images of s under these homomorphisms are two sections of $\mathscr{O}_X$ over X which, by hypothesis, induce the *same* section over $S(X)$; we know (10.2.8) that these sections are identical, whence our assertion. In the second place let us prove that every morphism $h : S(X) \to S(Y)$ (for the category $\mathbf{C}_0''$) is of the form $S(f)$, where $f : X \to Y$ is a morphism locally of finite type. By hypothesis, there exists an ultra-affine open covering (U') (resp. (V'_μ)) of $S(X)$ (resp. $S(Y)$) such that for every λ, $h(U'_\lambda)$ is contained in one of the V'_μ and that it corresponds to the morphism $h_\lambda : U'_\lambda \to V'_\mu$ making the second ring an algebra of finite type over the first. For every couple of indices α, β, consider the open subset $U_{\alpha\beta}$ of U_α whose trace on $S(X)$ is $U'_\alpha \cap U'_\beta$; by virtue of 1°, the identical self-morphism of $U'_\alpha \cap U'_\beta$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \to U_{\beta\alpha}$ is an isomorphism of preschemes. We see immediately (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition of recollement (**0**, 4.1.7), and that this family defines a prescheme X such that we have $X' = S(X)$; it is clear then that the U_α are identified with affine open subsets of X, such that the U'_α are the restrictions of $S(X)$ to the U_α.

2° It remains to prove that every ultra-prescheme X' is of the form $S(X)$ for a Jacobson prescheme X (which will necessarily be unique to isomorphism near, by virtue of 1°). There are ultra-affine open subsets (U'_α) of X', each of which is of the form $S(U_\alpha)$, U_α being the spectrum of a Jacobson ring. For every couple of indices α, β, consider the open $U_{\alpha\beta}$ of U_α whose trace U'_α is $U'_\alpha \cap U'_\beta$; by virtue of 1°, the identical self-morphism of $U'_\alpha \cap U'_\beta$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \to U_{\beta\alpha}$ is an isomorphism of preschemes. One easily verifies (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the glueing condition (**0**, 4.1.7), and that this family defines a prescheme X; it is clear then that we have $X' = S(X)$, which completes the proof. ☐

10.10. Algebraic spaces of Serre

(10.10.1). The language introduced by Serre in (FAC) is sometimes convenient, notably in questions where the main interest is attached to the rational points over the base field k (algebraically closed by hypothesis) of the "algebraic varieties" over k that one considers. We will sketch here this language by linking it to the considerations that precede, to allow the reader to translate the statements of Serre in the language of schemes. It is moreover possible to develop the language of Serre also for preschemes over a non-algebraically closed field (and even over an Artinian ring); but this introduces considerable technical complications, and moreover, on a field of arbitrary base, the advantages (above all psychological ones) of the point of view of Serre disappear; therefore we will confine ourselves to the framework fixed by Serre. The present section, just like the preceding one, will not be used in the sequel of this Treatise, and we will therefore content ourselves with brief indications.

IV-3 | 115

(10.10.2). Given an ultra-prescheme R, one can naturally (as in the whole category) define the notion of *R-ultra-prescheme*. Consider in particular an algebraically closed field k; $\mathrm{Spm}(k)$ is then identical to $\mathrm{Spec}(k)$; we will say that a $\mathrm{Spm}(k)$-ultra-prescheme X' is a *prealgebraic space over k*: it is a ringed space in local rings whose every point has an open neighbourhood isomorphic to the maximal spectrum of a k-algebra of finite type; it follows that it is the same as saying (by (10.9.6)) that $X' = S(X)$, where X is a k-prescheme locally of finite type. If X, Y, Z are three k-preschemes locally of finite type, it is the same as to say of $X \times_k Y$ (**I**, 1.3.4), hence the same is true of $X' \times_k Y'$, in the category of prealgebraic spaces over k (and also of the "fibre product" $X' \times_{k'} Y'$, where X', Y', Z' are three prealgebraic spaces over k). One can therefore define the diagonal morphism $\Delta_{X'} : X' \to X' \times_k X'$ (which is none other than $S(\Delta_X)$); one says that X' is an *algebraic space over k* if X is a scheme, and it amounts to the same as saying that the image of $\Delta_{X'}$ is a *closed* subset of $X' \times_k X'$.

(10.10.3). The simplifications that come from the hypothesis that k is algebraically closed amount first of all to this: for every k-prescheme X locally of finite type, there is a bijective correspondence between *closed points* of X_k (i.e. the *points of X with values in k* (**I**, 3.4.4)) and *rational points of X over k* (**I**, 3.4.5), by virtue of (**I**, 6.4.2). This shows in particular (**I**, 3.4.3.1) that for two k-prealgebraic spaces X', Y', the underlying set product $X' \times Y'$ *is* identical to the underlying set of the product of the underlying topological spaces of X' and Y' (but it is well understood that the topology of the latter is not the *product* topology of the topologies of the underlying spaces, in general it is strictly finer than the latter).

On the other hand, the local rings $\mathscr{O}_x$ at the points of a prealgebraic space X' over k are k-algebras whose residue field is *isomorphic to* k; if A and B are two such k-algebras, a k-homomorphism $\varphi : A \to B$ is necessarily *local*: indeed, if an element x of the maximal ideal A were such that $\varphi(x)$ is invertible, there would exist $\lambda \in k$ nonzero such that $\varphi(x - \lambda.1)$ belongs to the maximal ideal of B, which is absurd since $x - \lambda.1$ is invertible. We conclude immediately from this that if $X' = S(X)$, $Y' = S(Y)$ are two prealgebraic spaces over k, every morphism $X' \to Y'$ of *k-ringed spaces* is also a morphism of ringed spaces *in local rings*, hence every morphism of k-preschemes $X' \to Y'$ of k-ringed spaces is also, in the notations above, of the form $S(f)$, where $f : X \to Y$ is a morphism of k-preschemes.

Finally, for every open U of X', and every $x \in U$, $s(x)$ (**0**, 5.5.1) is identified with an element of U in k, in other words a map $\bar{s} : x \to s(x)$ of U in k; as the map $h_U : s \to \bar{s}$ is evidently a homomorphism of rings $\Gamma(U, \mathscr{O}_{X'}) \to \Gamma(U, \mathscr{A}(X'))$ (the *germs of maps* of X' *in* k) and commutes with restrictions to a neighbourhood $V \subset U$, the h_U define a homomorphism of *sheaves* of rings $h : \mathscr{O}_{X'} \to \mathscr{A}(X')$. If one takes U to be ultra-affine, $\mathrm{Spm}(A)$, where A is a Jacobson ring, to say that $\bar{s} = 0$ means that for every maximal ideal $\mathfrak{m}$ of A, s belongs to $\mathfrak{m}$, or equivalently that s is in the *radical* of A; but as A is a Jacobson ring, its radical is its nilradical, hence $\bar{s} = 0$ if and only if s is nilpotent.

(10.10.4). One says that the k-prealgebraic space $X' = S(X)$ is *reduced* if it is reduced at each of its points, i.e. the set of points $x \in X$ where X is reduced is open (**0**, 5.2.2), its complement contains at least one closed point x' of X; as each of the local rings $\mathscr{O}_{x'}$ is reduced, one sees in (10.10.3) that for the homomorphism $h : \mathscr{O}_{X'} \to \mathscr{A}(X')$ to be *injective*, it is necessary and sufficient that X' be *reduced*. In (FAC), Serre restricts himself in fact to prealgebraic *reduced* spaces, which allows him to define $\mathscr{O}_{X'}$ as a *sub-sheaf* of $\mathscr{A}(X')$. One notes that if X' and Y' are two k-spaces prealgebraic reduced, it is the same of $X' \times_k Y'$: indeed, this amounts to saying that if A and B are two reduced k-algebras of finite type, $A \otimes_k B$ is also reduced; but we have just seen that the radicals of A and B are 0, and as k is algebraically closed, A and B are "separable" algebras over k in the sense of Bourbaki (*Alg.*, chap. VIII, §7, n° 5, prop. 5); hence $A \otimes_k B$ is without radical (*loc. cit.*, n° 6, cor. 3 of th. 3), and since it is a Jacobson ring, it is reduced. However, if Z' is a third prealgebraic space reduced over k, the "fibre product" $X' \times_{Z'} Y'$ of two prealgebraic reduced spaces over Z' is not in general reduced, which implies that the category of these spaces is insufficient in many questions (notably in the theory of algebraic groups). But as we have seen above, one can keep the language of Serre without restricting oneself, as this latter (who in addition considers only quasi-compact *prealgebraic* spaces), to the case of reduced prealgebraic spaces.

§11. TOPOLOGICAL PROPERTIES OF FINITELY PRESENTED FLAT MORPHISMS. FLATNESS CRITERIA

11.1. Flatness loci (Noetherian case) IV-3 | 117

Theorem (11.1.1). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *an* $\mathscr{O}_{\mathrm{X}}$*-Module coherent. Then the set* U *of* $x \in \mathrm{X}$ *such that* $\mathscr{F}$ *is* f*-flat at the point* x *is open in* X.

Proof. The question being evidently local on X and Y, we can suppose $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, A being Noetherian and B an A-algebra of finite type. We then have $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module of finite type. Applying the criterion ($\mathbf{0}_{\mathrm{III}}$, 9.2.6): it suffices therefore to prove the following assertion:

(11.1.1.1) *Let* A *be a Noetherian ring,* B *an* A*-algebra of finite type,* M *a* B*-module of finite type,* $\mathfrak{q}$ *a prime ideal of* B*,* $\mathfrak{p}$ *its inverse image in* A*. Suppose that* $\mathrm{M}_{\mathfrak{q}}$ *is a flat* $\mathrm{A}_{\mathfrak{p}}$*-module. Then there exists* $g \in \mathrm{B} - \mathfrak{q}$ *such that, for every prime ideal* $\mathfrak{q}' \supset \mathfrak{q}$ *of* B *such that* $g \notin \mathfrak{q}'$*,* $\mathrm{M}_{\mathfrak{q}'}$ *is a flat* $\mathrm{A}_{\mathfrak{p}'}$*-module, where* $\mathfrak{p}'$ *denotes the inverse image of* $\mathfrak{q}'$ *in* A *(it amounts to the same* ($\mathbf{0}_{\mathrm{I}}$, 6.3.1) *to say that* $\mathrm{M}_{\mathfrak{q}'}$ *is a flat* A*-module).*

Consider for this $\mathrm{B}_{\mathfrak{q}'}$ as an A-algebra; one has evidently $\mathfrak{p}\mathrm{B}_{\mathfrak{q}'} \subset \mathfrak{q}'\mathrm{B}_{\mathfrak{q}'}$. We know then ($\mathbf{0}_{\mathrm{III}}$, 10.2.2) that for $\mathrm{M}_{\mathfrak{q}'}$ to be a flat A-module, it is necessary and sufficient that $\mathrm{M}_{\mathfrak{q}'}/\mathfrak{p}\mathrm{M}_{\mathfrak{q}'}$ be a flat $(\mathrm{A}/\mathfrak{p})$-module and that $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}_{\mathfrak{q}'}, \mathrm{A}/\mathfrak{p}) = 0$. On the other hand, as $\mathrm{M}_{\mathfrak{q}}$ is an $\mathrm{A}_{\mathfrak{p}}$-module flat, one has $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}_{\mathfrak{q}}, \mathrm{A}/\mathfrak{p}) = 0$, which can also be written $(\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p}))_{\mathfrak{q}} = 0$. But as A and B are Noetherian, $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p})$ is a B-module of finite type, hence ($\mathbf{0}_{\mathrm{I}}$, 5.2.2) there exists $g \in \mathrm{B} - \mathfrak{q}$ such that $(\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{p}))_g = 0$ and $hg \in \mathrm{B} - \mathfrak{q}$ since $h \in \mathrm{A} - \mathfrak{p}$. Moreover, on the other hand, $(\mathrm{M}/\mathfrak{p}\mathrm{M})_{\bar{h}}$ is still a flat $(A/\mathfrak{p})$-module, car it is written $(M/\mathfrak{p}M)_{\overline{hg}}$, where $\overline{hg}$ is the canonical image of hg in $B/\mathfrak{p}B$, and it suffices to apply $(\mathbf{0}_{\mathrm{I}},\ 6.3.2)$; finally, on the other hand, $(\mathrm{Tor}_1^A(M, A/\mathfrak{p}))_{hg} = 0$ since $hg \in B - \mathfrak{q}$. Taking into account $(\mathbf{0}_{\mathrm{I}},\ 6.3.2)$, we see that we are reduced to proving the

Lemma (11.1.1.2). — *Under the conditions of* (11.1.1.1), *there exists* $g \in B - \mathfrak{q}$ *such that:*

label=i) $(M/\mathfrak{p}M)_g$ *is a flat* $(A/\mathfrak{p})$*-module;*

lbbel=ii) $(\mathrm{Tor}_1^A(M, A/\mathfrak{p}))_g = 0$.

By virtue of the generic flatness theorem (6.9.1) applied to the integral ring $A/\mathfrak{p}$, to the $(A/\mathfrak{p})$-algebra of finite type $B/\mathfrak{p}B$, and to the $(B/\mathfrak{p}B)$-module of finite type $M/\mathfrak{p}M$, there exists $h \in A - \mathfrak{p}$ such that, if $\bar{h}$ is its canonical image in $A/\mathfrak{p}$, $(M/\mathfrak{p}M)_{\bar{h}}$ be a flat $(A/\mathfrak{p})$-module. On the other hand, as $M_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$-module flat $(\mathbf{0}_{\mathrm{I}},\ 6.3.1)$, on the other hand $\mathrm{Tor}_1^A(M_{\mathfrak{q}}, A/\mathfrak{p}) = 0$, which can also be written $(\mathrm{Tor}_1^A(M, A/\mathfrak{p}))_{\mathfrak{q}} = 0$. But as A and B are Noetherian, $\mathrm{Tor}_1^A(M, A/\mathfrak{p})$ is a B-module of finite type, hence $(\mathbf{0}_{\mathrm{I}},\ 5.2.2)$ there exists $g \in B - \mathfrak{q}$ such that $(\mathrm{Tor}_1^A(M, A/\mathfrak{p}))_g = 0$ and $hg \in B - \mathfrak{q}$ since $h \in A - \mathfrak{p}$. Moreover, $(M/\mathfrak{p}M)_{hg}$ is still a flat $(A/\mathfrak{p})$-module, since it is written $(M/\mathfrak{p}M)_{\overline{hg}}$, where $\overline{hg}$ is the canonical image of hg in $B/\mathfrak{p}B$, and it suffices to apply $(\mathbf{0}_{\mathrm{I}},\ 6.3.2)$; finally, $(\mathrm{Tor}_1^A(M, A/\mathfrak{p}))_{hg} = 0$ since $hg \in B - \mathfrak{q}$. Taking into account $(\mathbf{0}_{\mathrm{I}},\ 6.3.2)$, we see that we are reduced to proving the following: □ **IV-3** | 118

Corollary (11.1.2). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism locally of finite type,* $\mathscr{F}$, $\mathscr{F}'$ *two* $\mathscr{O}_X$*-Modules coherent,* $u : \mathscr{F}' \to \mathscr{F}$ *a homomorphism of* $\mathscr{O}_X$*-Modules. Suppose that* $\mathscr{F}$ *is* f*-flat. Then the set* U *of* $x \in X$ *such that, setting* $y = f(x)$*, the homomorphism* $u_x \otimes 1 : \mathscr{F}'_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) \to \mathscr{F}_x \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$ *is injective, is open in* X.

Proof. Indeed, let $\mathscr{N}$ (resp. $\mathscr{P}$) be the kernel (resp. the cokernel) of u; applying $(\mathbf{0}_{\mathrm{III}},\ 10.2.4)$[4] to the local rings $\mathscr{O}_x$ and $\mathscr{O}_y$, and to the $\mathscr{O}_x$-modules $\mathscr{F}'_x$ and $\mathscr{F}_x$: saying that $u_x \otimes 1$ is injective is equivalent to saying that $\mathscr{P}$ is f-flat at the point x. Now as $\mathscr{N}$ and $\mathscr{P}$ are coherent, the corollary follows from (11.1.1) and from $(\mathbf{0},\ 15.2.4)$.

In particular: □

Corollary (11.1.3). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism locally of finite type,* g *a section of* $\mathscr{O}_X$ *above* X*. The set of* $x \in X$ *such that* $g_x \otimes 1$ *is a non-zero-divisor in* $\mathscr{O}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x))$ *is open in* X.

Proof. It suffices to apply (11.1.2) to the endomorphism of $\mathscr{O}_X$ defined by g. □

Corollary (11.1.4). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism locally of finite type,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module coherent and* f*-flat. Let* $(g_i)_{1 \leqslant i \leqslant n}$ *be a sequence of sections of* $\mathscr{O}_X$ *above* X*. The set* U *of* $x \in X$ *such that the sequence* $((g_i)_x \otimes 1)$ *is* $(\mathscr{F}_x \otimes_{\mathscr{O}_{f(x)}} \boldsymbol{k}(f(x)))$*-regular is open in* X.

Proof. As $\mathscr{F}$ is f-flat, the corollary follows from (11.1.1) and from $(\mathbf{0},\ 15.2.4)$. □ **IV-3** | 119

Corollary (11.1.5). — *Let* X, Y, Z *be three locally Noetherian preschemes,* $f : X \to Y$, $g : Y \to Z$ *two morphisms of finite type,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module coherent. Then the set* U *of* $y \in Y$ *such that, for every generization* y' *of* y, $\mathscr{F}$ *is* $(g \circ f)$*-flat at all the points of* $f^{-1}(y')$ *(i.e. such that* $\mathscr{F} \otimes_Y \mathrm{Spec}(\mathscr{O}_{Y,y})$ *is flat relative to the morphism* $X \times_Y \mathrm{Spec}(\mathscr{O}_{Y,y}) \to Z$*) is open in* Y.

Proof. If V is the set of $x \in X$ where $\mathscr{F}$ is $(g \circ f)$-flat, U is the set of $y \in Y$ such that for every generization y' of y, one has $f^{-1}(y') \subset V$. Now V is open (11.1.1) hence locally constructible in X, and the set W of $y \in Y$ such that $f^{-1}(y) \subset V$ is also locally constructible in Y by virtue of the Chevalley theorem (1.8.4). But it follows then from $(\mathbf{0}_{\mathrm{III}},\ 9.2.5)$ that the interior points of U are the interior points of W, whence the conclusion. □

[4]Here is a proof of $(\mathbf{0}_{\mathrm{III}},\ 10.2.4)$, which was not published in the *Algèbre commutative* of N. Bourbaki. Set $P = \mathrm{Im}(u)$, $Q = \mathrm{Coker}(u)$, $R = \mathrm{Ker}(u)$. The composite $M \otimes k \to P \otimes k \to N \otimes k$ is injective, and $M \otimes k \to P \otimes k$ is surjective, hence $P \otimes k \to N \otimes k$ is injective and $M \otimes k \to P \otimes k$ is bijective. The exact sequence $0 \to P \to N \to Q \to 0$ gives the exact sequence

$$0 = \mathrm{Tor}_1^A(N, k) \longrightarrow \mathrm{Tor}_1^A(Q, k) \longrightarrow P \otimes k \longrightarrow N \otimes k \longrightarrow Q \otimes k \longrightarrow 0$$

and since $P \otimes k \to N \otimes k$ is injective, one has $\mathrm{Tor}_1^A(Q, k) = 0$; as Q is a B-module of finite type, $(\mathbf{0}_{\mathrm{III}},\ 10.2.2)$ shows that Q is also an A-module flat by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$. The sequence $0 \to R \to M \to P \to 0$ is exact, hence by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, $R = 0$, and hence $R \otimes k = 0$; but as B is Noetherian, R is a B-module of finite type, hence $R = 0$ by virtue of Nakayama's lemma.

Corollary (11.1.6). — *Let* A *be a Noetherian ring,* B *an* A*-algebra of finite type,* C *a* B*-algebra of finite type,* M *a* C*-module of finite type. Then the set of* $\mathfrak{q} \in \mathrm{Spec}(\mathrm{B})$ *such that* $\mathrm{M}_{\mathfrak{q}}$ *is a flat* A*-module is open in* $\mathrm{Spec}(\mathrm{B})$*.*

Proof. Taking into account (2.1.2), this is a consequence of (11.1.5) applied to $\mathrm{Z} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{C})$, $\mathscr{F} = \widetilde{\mathrm{M}}$.

The results of this number will be freed from the Noetherian hypotheses in (11.3). □

11.2. Flatness of a projective limit of preschemes **IV-3** | 119

(11.2.1). Let A be a ring, M, N two A-modules, A' an A-algebra; set $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{N}' = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}'$. Recall (**III**, 6.3.8) that one defines, for every i, a canonical homomorphism of A-modules

(11.2.1.1) $$\psi_i : \mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{N}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$$

in the following way: one considers a left resolution of M by free A-modules

(11.2.1.2) $$\ldots \longrightarrow \mathrm{L}_{i+1} \xrightarrow{f_i} \mathrm{L}_i \longrightarrow \ldots \longrightarrow \mathrm{L}_0 \xrightarrow{\varepsilon} \mathrm{M} \longrightarrow 0$$

whence one deduces by tensoring with A' a complex of A'-modules

(11.2.1.3) $$\ldots \longrightarrow \mathrm{L}'_{i+1} \xrightarrow{f'_i} \mathrm{L}'_i \longrightarrow \ldots \longrightarrow \mathrm{L}'_0 \xrightarrow{\varepsilon'} \mathrm{M}' \longrightarrow 0$$

where we have set $\mathrm{L}'_i = \mathrm{L}_i \otimes_{\mathrm{A}} \mathrm{A}'$, $f'_i = f_i \otimes 1_{\mathrm{A}'}$, $\varepsilon' = \varepsilon \otimes 1_{\mathrm{A}'}$. Consider on the other hand a left resolution of M' by free A'-modules

(11.2.1.4) $$\ldots \longrightarrow \mathrm{L}''_{i+1} \xrightarrow{f'_i} \mathrm{L}''_i \longrightarrow \ldots \longrightarrow \mathrm{L}''_0 \xrightarrow{\varepsilon''} \mathrm{M}' \longrightarrow 0$$

As the L'_i are free A'-modules, we know (M, V, 1.1) that there are A'-homomorphisms u_i forming a commutative diagram **IV-3** | 120

$$\begin{tikzcd}
\ldots \arrow[r] & \mathrm{L}'_{i+1} \arrow[r, "f'_i"] \arrow[d, "u_{i+1}"] & \mathrm{L}'_i \arrow[r] \arrow[d, "u_i"] & \ldots \arrow[r] & \mathrm{L}'_0 \arrow[r, "\varepsilon'"] \arrow[d, "u_0"] & \mathrm{M}' \arrow[r] \arrow[d, "1_{\mathrm{M}'}"] & 0 \\
\ldots \arrow[r] & \mathrm{L}''_{i+1} \arrow[r, "f''_i"] & \mathrm{L}''_i \arrow[r] & \ldots \arrow[r] & \mathrm{L}''_0 \arrow[r, "\varepsilon''"] & \mathrm{M}' \arrow[r] & 0
\end{tikzcd}$$

If one composes the homomorphism $u_\bullet : \mathrm{L}'_\bullet \to \mathrm{L}''_\bullet$ of complexes thus defined with the canonical homomorphism $\mathrm{L}_\bullet \to \mathrm{L}'_\bullet$ of A-modules, one obtains a homomorphism of complexes of A-modules $\mathrm{L}_\bullet \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{L}''_\bullet \otimes_{\mathrm{A}'} \mathrm{N}'$; noting that $(\mathrm{L}_i \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{L}'_i \otimes_{\mathrm{A}'} \mathrm{N}'$, one deduces a homomorphism of complexes of A-modules $\mathrm{L}_\bullet \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{L}''_\bullet \otimes_{\mathrm{A}'} \mathrm{N}'$, whence, passing to homology, the canonical homomorphisms (11.2.1, IV.11.2.1.1). As the u_i are well-determined up to homotopy (M, V, 1.1), the homomorphisms (11.2.1, IV.11.2.1.1) do not depend on the choice of the u_i nor on the choice of the free resolutions $\mathrm{L}_\bullet$ and $\mathrm{L}''_\bullet$.

As the $\mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$ are A'-modules, one deduces canonically from (11.2.1, IV.11.2.1.1) A'-homomorphisms

(11.2.1.5) $$\varphi_i : \mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}').$$

Consider now two ring homomorphisms

$$\rho : \mathrm{A} \longrightarrow \mathrm{A}^{(1)}, \quad \sigma : \mathrm{A}^{(1)} \longrightarrow \mathrm{A}^{(2)},$$

and their composite $\sigma \circ \rho : \mathrm{A} \to \mathrm{A}^{(2)}$; set $\mathrm{M}^{(j)} = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}^{(j)}$, $\mathrm{N}^{(j)} = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}^{(j)}$ for $j = 1, 2$. Then the canonical composite homomorphism

$$\mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{N}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}^{(0)}}(\mathrm{M}^{(1)}, \mathrm{N}^{(1)}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}^{(0)}}(\mathrm{M}^{(2)}, \mathrm{N}^{(2)})$$

is the same as the canonical homomorphism deduced from $\sigma \circ \rho$; this follows from the fact that, if $\mathrm{L}_\bullet^{(j)}$ is a free resolution of $\mathrm{M}^{(j)}$, the diagram

$$\begin{tikzcd} \mathrm{L}_\bullet \arrow[r] & \mathrm{L}_\bullet \otimes_{\mathrm{A}} \mathrm{A}^{(1)} \arrow[r] \arrow[d] & \mathrm{L}_\bullet^{(1)} \arrow[d] \\ & \mathrm{L}_\bullet^{(1)} \otimes_{\mathrm{A}^{(1)}} \mathrm{A}^{(2)} \arrow[r] & \mathrm{L}_\bullet^{(2)} \end{tikzcd}$$

is commutative.

(11.2.2). The notations being those of (11.2.1), consider now an inductive system of A-algebras $(\mathrm{A}_\alpha, g_{\beta\alpha})$, and for every index α, set $\mathrm{M}_\alpha = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_\alpha$, $\mathrm{N}_\alpha = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}_\alpha$; if one denotes by $g_\alpha : \mathrm{A}_\alpha \to \mathrm{A}'$ the canonical homomorphism, one deduces canonical homomorphisms (11.2.1, IV.11.2.1.1) $\psi_{i\beta\alpha}$: $\mathrm{Tor}_i^{\mathrm{A}_\alpha}(\mathrm{M}_\alpha, \mathrm{N}_\alpha) \to \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$ which (always by virtue of (11.2.1, IV.11.2.1.1)) form an inductive system of homomorphisms; we propose to complete the result of (M, V, 9.4*) by showing that the

(11.2.2.1) $$\psi_i = \varinjlim \psi_{i\alpha} : \varinjlim_\alpha \mathrm{Tor}_i^{\mathrm{A}_\alpha}(\mathrm{M}_\alpha, \mathrm{N}_\alpha) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$$

are *isomorphisms of* A'*-modules*. To this end, we proceed as in (M, V, 9.5′), associating to each M_α its *canonical free resolution*. It amounts (taking into account the exactness of the functor $\varinjlim$) to proving the

IV-3 | 121

Lemma (11.2.2.2). — *Let* $(\mathrm{A}_\alpha, g_{\beta\alpha})$ *be an inductive system of rings,* $((\mathrm{M}_\alpha, h_{\beta\alpha})$ *an inductive system of sets,* $\mathrm{A}' = \varinjlim \mathrm{A}_\alpha$, $\mathrm{M}' = \varinjlim \mathrm{M}_\alpha$, $g_\alpha : \mathrm{A}_\alpha \to \mathrm{A}'$, $h_\alpha : \mathrm{M}_\alpha \to \mathrm{M}'$ *the canonical maps. For every* α*, let* $\mathrm{F}(\mathrm{M}_\alpha)$ *be the* A_α*-module of formal linear combinations of elements of* M_α*; or equivalently the* A'*-module of formal linear combinations of elements of* M'*; if* $h^0_{\beta\alpha} : \mathrm{F}(\mathrm{M}_\alpha) \to \mathrm{F}(\mathrm{M}_\beta)$ *(for* $\alpha \leqslant \beta$*) and* $h'_\alpha : \mathrm{F}(\mathrm{M}_\alpha) \to \mathrm{F}(\mathrm{M}')$ *are the* A_α*-homomorphisms deduced from* $h_{\beta\alpha}$ *and* h_α *respectively,* $(\mathrm{F}(\mathrm{M}_\alpha), h^0_{\beta\alpha})$ *is an inductive system of* A_α*-modules and* (h'_α) *an inductive system of homomorphisms. Then*

$$h'' = \varinjlim h'_\alpha : \varinjlim \mathrm{F}(\mathrm{M}_\alpha) \longrightarrow \mathrm{F}(\mathrm{M}') = \mathrm{F}(\varinjlim \mathrm{M}_\alpha)$$

is an isomorphism.

Proof. For the proof, see Bourbaki, *Alg.*, chap. II, 3ᵉ éd., § 6, nº 6, cor. de la prop. 10. □

(11.2.3). Let us resume the notations of (11.2.1) and consider in particular the case $i = 1$; set $\mathrm{T} = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{N})$, $\mathrm{T}' = \mathrm{T} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{T}'' = \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$; then ψ_1 is the homomorphism $\mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) / \mathrm{Im}(f_1 \otimes 1_{\mathrm{N}}) \to \mathrm{Ker}(f''_0 \otimes 1_{\mathrm{N}'}) / \mathrm{Im}(f''_1 \otimes 1_{\mathrm{N}'})$ which is deduced by passage to the quotients from the restriction of $u_1 \otimes 1 : \mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{L}''_0 \otimes_{\mathrm{A}'} \mathrm{N}'$.

Set $\mathrm{R} = \mathrm{Ker}(\varepsilon)$, $\mathrm{R}'' = \mathrm{Ker}(\varepsilon'')$, so that one has the exact sequences

(1) $$0 \longrightarrow \mathrm{R} \xrightarrow{j} \mathrm{L}_0 \xrightarrow{\varepsilon} \mathrm{M} \longrightarrow 0 \quad \text{and} \quad 0 \longrightarrow \mathrm{R}'' \xrightarrow{j''} \mathrm{L}''_0 \xrightarrow{\varepsilon''} \mathrm{M}' \longrightarrow 0,$$

whence one deduces the exact homology sequences

(11.2.3.1) $$0 = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{L}_0, \mathrm{N}) \longrightarrow \mathrm{T} \xrightarrow{\partial} \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \xrightarrow{j \otimes 1} \mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N} \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N} \longrightarrow 0$$

(11.2.3.2) $$0 = \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{L}''_0, \mathrm{N}') \longrightarrow \mathrm{T}'' \xrightarrow{\partial''} \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}' \xrightarrow{j'' \otimes 1} \mathrm{L}''_0 \otimes_{\mathrm{A}'} \mathrm{N}' \xrightarrow{\varepsilon'' \otimes 1} \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \longrightarrow 0$$

On the other hand there is a homomorphism of A'-modules

$$v : \mathrm{R}' = \mathrm{Ker}(\varepsilon) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Ker}(\varepsilon') \xrightarrow{u_0} \mathrm{Ker}(\varepsilon'') = \mathrm{R}''.$$

Let us show that the diagram IV-3 | 122

(11.2.3.3) $$\begin{tikzcd} & \mathrm{T}' \arrow[r, "\partial \otimes 1"] \arrow[d, "\varphi_1"] & \mathrm{R}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, "v \otimes 1"] & \mathrm{L}'_0 \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, "u_0 \otimes 1"] & \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & \mathrm{T}'' \arrow[r, "\partial''"] & \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & \mathrm{L}''_0 \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & \mathrm{M}' \otimes_{\mathrm{A}'} \mathrm{N}' \arrow[r] & 0 \end{tikzcd}$$

is *commutative*. For this, one verifies immediately (M, IV, 1) that the homomorphism $\partial : \mathrm{T} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$ comes (in the present case) by passage to the quotient from the homomorphism $w : \mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$, restriction of the homomorphism $g_0 \otimes 1_{\mathrm{N}} : \mathrm{L}_1 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}$, where $g_0 : \mathrm{L}_1 \to \mathrm{R}$ is such that $f_0 = j \circ g_0$; similarly for ∂''. It therefore suffices to see that the diagram

$$\begin{tikzcd}
\mathrm{Ker}(f_0 \otimes 1_{\mathrm{N}}) \arrow[r, "w"] \arrow[d] & \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \arrow[r] & \mathrm{R}' \otimes_{\mathrm{A}'} \mathrm{N}' = (\mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' \arrow[d] \\
\mathrm{Ker}(f_0'' \otimes 1_{\mathrm{N}'}) \arrow[rr] & & \mathrm{R}'' \otimes_{\mathrm{A}'} \mathrm{N}'
\end{tikzcd}$$

is commutative, which follows from the commutativity of the diagram

$$\begin{tikzcd}
\mathrm{L}_1 \arrow[r] & \mathrm{R} \\
\mathrm{L}_1'' \arrow[r] & \mathrm{R}''
\end{tikzcd}$$

Lemma (11.2.4). — *Let* A *be a ring,* C *an* A*-algebra,* M *an* A*-module,* N *a* C*-module,* A′ *an* A*-algebra. Set* $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{N}' = \mathrm{N} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{N} \otimes_{\mathrm{C}} \mathrm{C}'$. *Suppose that* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ *is a flat* C*-module. Then the canonical homomorphism*

$$\varphi_1 : \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{N}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{N}')$$

(cf. (11.2.1, IV.11.2.1.5)) is surjective.

Proof. Let us keep the notations of (11.2.3); the exactness on the right of the tensor product, applied to the hypothesis that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ is a flat C-module, shows that the sequence $\mathrm{L}_1' \xrightarrow{f_0'} \mathrm{L}_0' \xrightarrow{\varepsilon'} \mathrm{M}' \to 0$ is exact; as L_0' and L_1' are free A′-modules, *one may suppose that* $\mathrm{L}_0'' = \mathrm{L}_0'$, $\mathrm{L}_1'' = \mathrm{L}_1'$, u_0 and u_1 being the identity maps, and $f_0'' = f_0'$. As $\mathrm{R} = \mathrm{Im}(f_0)$ and $\mathrm{R}'' = \mathrm{Im}(f_0'') = \mathrm{Im}(f_0')$, the homomorphism v is surjective, and so it follows similarly for $v \otimes 1$. The proof will therefore be complete if one proves that the first line of (11.2.3, IV.11.2.3.3) is *exact*, $v \otimes 1$ being surjective and $u_0 \otimes 1$ injective (Bourbaki, *Alg. comm.*, chap. I^er, n^o 4, cor. 2 de la prop. 2). Let us use for this □

the hypothesis that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}$ is a flat C-module. Setting $\mathrm{Q} = \mathrm{Ker}(\varepsilon \otimes 1) = \mathrm{Im}(j \otimes 1)$, by the exact **IV-3** | 123
sequence (11.2.3, IV.11.2.3.1), one has the two exact sequences $0 \to \mathrm{Q} \to \mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{M} \otimes_{\mathrm{A}} \mathrm{N} \to 0$ and $\mathrm{T} \to \mathrm{R} \otimes_{\mathrm{A}} \mathrm{N} \to \mathrm{Q} \to 0$, where the homomorphisms are homomorphisms of *C-modules*; using the hypothesis of flatness, one deduces ($\mathbf{0}_{\mathrm{I}}$, 6.1.2) the exact sequence

$$0 \longrightarrow \mathrm{Q} \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow 0$$

and on the other hand, the tensor product being right exact, one has the exact sequence

$$\mathrm{T} \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow \mathrm{Q} \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow 0$$

whence finally the exact sequence

$$\mathrm{T} \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{R} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{L}_0 \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow (\mathrm{M} \otimes_{\mathrm{A}} \mathrm{N}) \otimes_{\mathrm{C}} \mathrm{C}' \longrightarrow 0$$

But by definition, for every C-module P, $\mathrm{P} \otimes_{\mathrm{C}} \mathrm{C}' = \mathrm{P} \otimes_{\mathrm{A}} \mathrm{A}'$, whence the conclusion.

Lemma (11.2.5). — *Let* A *be a ring,* $\mathfrak{J}$ *an ideal of* A*,* B *an* A*-algebra,* M *a* B*-module,* $\mathrm{A} \to \mathrm{A}'$ *a homomorphism of rings. Set* $\mathfrak{J}' = \mathfrak{J}\mathrm{A}'$, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{M} \otimes_{\mathrm{B}} \mathrm{B}'$. *Suppose verified one of the following hypotheses:*

*label=*a) $\mathfrak{J}$ *is nilpotent.*
*lbbel=*b) A′ *is Noetherian,* B′ *is an* A′*-algebra of finite type,* M′ *a* B′*-module of finite type.*

Under these conditions, suppose the two following properties are verified:

label=(i) $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *is a flat* $(\mathrm{A}/\mathfrak{J})$*-module.*
lbbel=(ii) *The canonical composite homomorphism*

$$\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/\mathfrak{J}) \xrightarrow{\psi_1} \mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{A}'/\mathfrak{J}') \xrightarrow{\theta} (\mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{A}'/\mathfrak{J}'))_{\mathfrak{p}'}$$

(where ψ_1 is the homomorphism (11.2.1, IV.11.2.1.1) and θ the canonical homomorphism of a B'-module localised at $\mathfrak{p}'$) is zero.

Then $M'_{\mathfrak{p}'}$ is a flat A'-module.

Proof. Note that under hypothesis *b*), $M'_{\mathfrak{p}'}$ is a $B'_{\mathfrak{p}'}$-module of finite type, $B'_{\mathfrak{p}'}$ is an A'-algebra Noetherian, and $\mathfrak{J}'B'_{\mathfrak{p}'}$ is contained in the radical $\mathfrak{p}'B'_{\mathfrak{p}'}$ of $B'_{\mathfrak{p}'}$; under hypothesis *a*), $\mathfrak{J}'$ is nilpotent; one can therefore apply the criterion of flatness ($\mathbf{0}_{\mathrm{III}}$, 10.2.1) or ($\mathbf{0}_{\mathrm{III}}$, 10.2.2) according to whether *a*) or *b*) is verified. In the first place, $M'_{\mathfrak{p}'} \otimes_{A'} (A'/\mathfrak{J}') = (M' \otimes_{A'} (A'/\mathfrak{J}'))_{\mathfrak{p}'} = ((M \otimes_A (A/\mathfrak{J})) \otimes_{A/\mathfrak{J}} (A'/\mathfrak{J}'))_{\mathfrak{p}'}$; hypothesis (i) implies that $M'_{\mathfrak{p}'} \otimes_{A'} (A'/\mathfrak{J}')$ is a flat $(A'/\mathfrak{J}')$-module, by ($\mathbf{0}_{\mathrm{I}}$, 6.2.1 and 6.3.2). It remains therefore to show that $\mathrm{Tor}_1^{A'}(M'_{\mathfrak{p}'}, A'/\mathfrak{J}') = 0$; now this module is equal to $(\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$ by virtue of the flatness of $B'_{\mathfrak{p}'}$ over B'. But by hypothesis (ii), the homomorphism composed $\theta \circ \psi_1 : \mathrm{Tor}_1^{A}(M, A/\mathfrak{J}) \to (\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$ is zero; moreover, by (11.2.4), applied to $C = A/\mathfrak{J}$ and $N = C$, (11.2.4, IV.11.2.4) shows (taking into account hypothesis (i)) that φ_1 is *surjective*; hence the homomorphism θ is zero, and therefore the image of θ in the $B'_{\mathfrak{p}'}$-module $(\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$, which is this last module, is zero. □

Theorem (11.2.6). — *The notations being those of* (8.5.1) *and* (8.8.1), *suppose* S_α *quasi-compact,* X_α *and* Y_α *of finite presentation over* S_α*; let* $f_\alpha : X_\alpha \to Y_\alpha$ *be an* S_α*-morphism,* $\mathscr{F}_\alpha$ *an* $\mathscr{O}_{X_\alpha}$*-Module quasi-coherent of finite presentation.*

IV-3 | 124

label=*(i) Let x be a point of* X, *x_λ its canonical projection in* X_λ*. For $\mathscr{F}$ to be f-flat at the point x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathscr{F}_\lambda$ is f_λ-flat at the point x_λ.*

lbbel=*(ii) For $\mathscr{F}$ to be f-flat, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $\mathscr{F}_\lambda$ is f_λ-flat.*

Proof. One may suppose that $S_\alpha = S_0$; as Y_0 is of finite presentation over S_0, Y_0 is quasi-compact, and $f_0 : X_0 \to Y_0$ is a morphism of finite presentation (1.6.2, (v)), hence one can also restrict to the case where $S_0 = Y_0$. Moreover, by the quasi-compactness of S_0 and of the fact that the set of indices L is filtered, one can restrict by the same reasoning to the case where $S_0 = \mathrm{Spec}(A_0)$ is affine. Furthermore X_0 is also quasi-compact, hence by the same reasoning one can also suppose $X_0 = \mathrm{Spec}(B_0)$ affine; one then has $\mathscr{F}_0 = \widetilde{M}_0$, where M_0 is a B_0-module of finite presentation, and the statement (11.2.6) is in this case equivalent to the following (by ($\mathbf{0}_{\mathrm{I}}$, 6.3.1)):

Corollary (11.2.6.1). — *Let* A_0 *be a ring,* $(A_\lambda)_{\lambda \in L}$ *an inductive filtrant system of* A_0*-algebras,* B_0 *an* A_0*-algebra of finite presentation,* M_0 *a* B_0*-module of finite presentation. Set* $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $A = \varinjlim A_\lambda$, $B = \varinjlim B_\lambda = B_0 \otimes_{A_0} A$, $M = \varinjlim M_\lambda = M_0 \otimes_{A_0} A$.

label=(i) Let $\mathfrak{p}$ be a prime ideal of B, and for every λ, let $\mathfrak{p}_\lambda$ be its inverse image in B_λ. For $M_{\mathfrak{p}}$ to be a flat A-module, it is necessary and sufficient that there exist λ such that $(M_\lambda)_{\mathfrak{p}_\lambda}$ be a flat A_λ-module.

lbbel=(ii) For M to be a flat A-module, it is necessary and sufficient that there exist λ such that M_λ be a flat A_λ-module.

One needs only prove that the conditions are necessary (2.1.4). We will proceed in several steps.

I) *Reduction to the case where the* A_λ *are Noetherian.* — By virtue of (8.9.1), there exists a subring A'_0 of A_0 that is a $\mathbf{Z}$-algebra of finite type, and a A'_0-algebra of finite type B'_0 and a B'_0-module of finite type M'_0 such that $B_0 = B'_0 \otimes_{A'_0} A_0$ and $M_0 = M'_0 \otimes_{A'_0} A_0$; as $B_\lambda = B'_0 \otimes_{A'_0} A_\lambda$ and $M_\lambda = M'_0 \otimes_{A'_0} A_\lambda$, one can replace A_0, B_0, M_0 by A'_0, B'_0, M'_0, and hence one may already suppose that A_0 is *Noetherian*. Let H be the set of couples (λ, C_λ), where C_λ is a sub-A_0-algebra of *finite type* of A_λ; order H by setting $(\lambda, C_\lambda) \leqslant (\mu, D_\mu)$ if $\lambda \leqslant \mu$ and $\varphi_{\mu\lambda}(C_\lambda) \subset D_\mu$; for this order H is filtered, since if (λ, C_λ) and (μ, D_μ) are any two elements of H, one can majorise them by a (ν, E_ν) in L, taking $\nu \geqslant \lambda$, $\nu \geqslant \mu$ and then E_ν equal to the sub-A_0-algebra of A_ν generated by $\varphi_{\nu\lambda}(C_\lambda)$ and $\varphi_{\nu\mu}(D_\mu)$. For an element $\xi = (\lambda, C_\lambda)$ of H, set $A_\xi = C_\lambda$ and if $\eta = (\mu, D_\mu)$ (hence $\lambda \leqslant \mu$ and $\varphi_{\mu\lambda}(C_\lambda) \subset D_\mu$), $\varphi_{\eta\xi} : A_\xi \to A_\eta$ will be the restriction to C_λ of $\varphi_{\mu\lambda}$, considered as a homomorphism to D_μ). One thus obtains an inductive filtrant system of A_0-algebras. Set $B_\xi = B_0 \otimes_{A_0} A_\xi$, $M_\xi = M_0 \otimes_{A_0} A_\xi$; this being so, the A_ξ are *Noetherian*; moreover, by the double limit inductive formula (Bourbaki, *Alg.*, chap. II, 3[e] éd., § 6,

n° 4, prop. 7) one has $\varinjlim_{\mathrm{H}} \mathrm{A}_\xi = \mathrm{A}$, $\varinjlim_{\mathrm{H}} \mathrm{B}_\xi = \mathrm{B}$, $\varinjlim_{\mathrm{H}} \mathrm{M}_\xi = \mathrm{M}$. Suppose (11.2.6.1) proved for the inductive system $(\mathrm{A}_\xi)_{\xi \in \mathrm{H}}$; let $\mathfrak{p}$ be a prime ideal of B, tel that $\mathrm{M}_\mathfrak{p}$ be a flat A-module; there exists then $\xi \in \mathrm{H}$ such that $(\mathrm{M}_\xi)_{\mathfrak{p}_\xi}$ be a flat A_ξ-module. Let $\xi = (\lambda, \mathrm{C}_\lambda)$, and as $\mathfrak{p}_\lambda$ is the inverse image of **IV-3** | 125
$\mathfrak{p}$ in B_λ, $\mathfrak{p}_\xi$ is the inverse image of $\mathfrak{p}_\lambda$ in B_ξ; $(\mathrm{M}_\xi)_{\mathfrak{p}_\xi} = ((\mathrm{M}_\xi \otimes_{\mathrm{C}_\lambda} \mathrm{A}_\lambda)_{\mathfrak{p}_\lambda} = ((\mathrm{M}_\xi)_{\mathfrak{p}_\xi} \otimes_{\mathrm{C}_\lambda} \mathrm{A}_\lambda)_{\mathfrak{p}_\lambda}$, hence $(\mathrm{M}_\lambda)_{\mathfrak{p}_\lambda}$ is a flat A_λ-module, by ($\mathbf{0}_\mathrm{I}$, 6.2.1 and 6.3.2). One thus treats case (ii) in a similar manner. We can therefore from now on suppose the A_λ Noetherian for $\lambda \in \mathrm{L}$ (but not necessarily A itself).

II) *Reduction of the global statement (ii) to the pointwise statement (i).* — Suppose $\mathscr{F}$ is f-flat. For every λ, let U_λ be the set of $x_\lambda \in \mathrm{X}_\lambda$ such that $\mathscr{F}_\lambda$ is f_λ-flat at the point x_λ; we know that U_λ is open in X_λ, since S_λ is Noetherian and f_λ is of finite type (11.1.1); let $\mathrm{V}_\lambda = v_\lambda^{-1}(\mathrm{U}_\lambda)$ its inverse image in X. By hypothesis, for every $x \in \mathrm{X}$, the projection of x in X_λ belongs to U_λ for some λ, which means that $\mathrm{X} = \bigcup \mathrm{V}_\lambda$. Moreover, by (2.1.4), for $\lambda \leqslant \mu$, $\mathrm{V}_\lambda \subset \mathrm{V}_\mu$, hence, since X is quasi-compact, there exists an index μ such that $\mathrm{X} = \mathrm{V}_\mu$. As the X_λ are quasi-compact, it follows from (8.3.4) that there exists a $\nu \geqslant \mu$ such that $v_{\nu\mu}^{-1}(\mathrm{U}_\mu) = \mathrm{X}_\nu$; but by (2.1.4), this implies that $\mathscr{F}_\nu$ is f_ν-flat.

III) *End of the proof.* — It remains to prove (i) supposing S_0 affine and Noetherian; if y_0 is the projection of x in S_0, one can, moreover, by virtue of (2.1.4) and de (**I**, 3.6.5), otherwise restrict to the case where A_0 is a *local* Noetherian ring, with maximal ideal $\mathfrak{m}_0 = \mathfrak{i}_{y_0}$; by definition, $\mathfrak{m}_0$ is the inverse image of the prime ideal $\mathfrak{p} = \mathfrak{i}_x$ of B, and $\mathrm{M}_\mathfrak{p}$ is supposed to be a flat A-module; one therefore has in particular $\mathrm{Tor}_1^\mathrm{A}(\mathrm{A}/\mathfrak{m}_0\mathrm{A}, \mathrm{M}) = 0$, which can also be written $\mathrm{Tor}_1^\mathrm{A}(\mathrm{A}/\mathfrak{m}_0\mathrm{A}, \mathrm{M})_\mathfrak{p} = 0$. Note now that $\mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0\mathrm{A}_\lambda, \mathrm{M}_\lambda)$ are B-modules of finite type, and the $\mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0, \mathrm{M}_0)$ is a B_0-module *of finite type*, since B_0 is Noetherian; by (11.2.2.1) $\mathrm{Tor}_1^\mathrm{A}(\mathrm{A}/\mathfrak{m}_0\mathrm{A}, \mathrm{M}) = \varinjlim \mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0\mathrm{A}_\lambda, \mathrm{M}_\lambda)$; there exists then λ such that the images of t_i^0 in $\mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0\mathrm{A}_\lambda, \mathrm{M}_\lambda)$, are zero, i.e., $h_\lambda t_i^0 = 0$ for $1 \leqslant i \leqslant m$. Or, on $h_\lambda \in \mathrm{B}_\lambda - \mathfrak{p}_\lambda$ the prime ideal of B_λ inverse image of $\mathfrak{p}$; hence the canonical images of t_i^0 in $(\mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0\mathrm{A}_\lambda, \mathrm{M}_\lambda))_{\mathfrak{p}_\lambda}$ are zero, and hence by the homomorphism $\mathrm{Tor}_1^{\mathrm{A}_0}(\mathrm{A}_0/\mathfrak{m}_0, \mathrm{M}_0) \to (\mathrm{Tor}_1^{\mathrm{A}_\lambda}(\mathrm{A}_\lambda/\mathfrak{m}_0\mathrm{A}_\lambda, \mathrm{M}_\lambda))_{\mathfrak{p}_\lambda}$ is zero. The conditions of the lemma (11.2.5) are therefore satisfied ($\mathrm{A}_0/\mathfrak{m}_0$ being a field, hence the condition (11.2.5, (i)) is trivially verified) and $(\mathrm{M}_\lambda)_{\mathfrak{p}_\lambda}$ is a flat A_λ-module, which completes the proof of the theorem.

Corollary (11.2.7). — *Let* $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ *be an affine scheme,* $f : \mathrm{X} \to \mathrm{S}$ *a morphism,* $\mathscr{F}$ *an* $\mathscr{O}_\mathrm{X}$*-Module quasi-coherent,* x *a point of* X*. The following conditions are equivalent:*

label=a) *f is a morphism of finite presentation, $\mathscr{F}$ is an $\mathscr{O}_\mathrm{X}$-Module of finite presentation and $\mathscr{F}$ is f-flat at the point x (resp. f-flat).* **IV-3** | 126

lbbel=b) *There exists an affine Noetherian scheme $\mathrm{S}_0 = \mathrm{Spec}(\mathrm{A}_0)$, a morphism of finite type $f_0 : \mathrm{X}_0 \to \mathrm{S}_0$, an $\mathscr{O}_{\mathrm{X}_0}$-Module coherent $\mathscr{F}_0$, a morphism $\mathrm{S} \to \mathrm{S}_0$ such that the S-prescheme $\mathrm{X}_0 \otimes_{\mathrm{S}_0} \mathrm{S}$ is S-isomorphic to X and that $\mathscr{F}$ is isomorphic to $\mathscr{F}_0 \otimes_{\mathscr{O}_{\mathrm{S}_0}} \mathscr{O}_\mathrm{S}$, and $\mathscr{F}_0$ is f_0-flat at the point x_0 the projection of x in X_0 (resp. f_0-flat).*

lcbel=c) *The conditions of* b) *are satisfied and in addition* A_0 *is a sub-*$\mathbf{Z}$*-algebra of finite type of* A*, the morphism* $\mathrm{S} \to \mathrm{S}_0$ *corresponding to the canonical injection* $\mathrm{A}_0 \to \mathrm{A}$.

Proof. It is clear that *c*) implies *b*) and *b*) implies *a*) by virtue of (2.1.4). On the other hand, one can consider A as the inductive limit of its sub-$\mathbf{Z}$-algebras of finite type, and by (8.9.1) we know that there is a sub-$\mathbf{Z}$-algebra A_0 and a morphism of finite type $f_0 : \mathrm{X}_0 \to \mathrm{S}_0$ with $\mathscr{F}_0$ an $\mathscr{O}_{\mathrm{X}_0}$-Module coherent such that $\mathrm{X} = \mathrm{X}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$, $\mathscr{F} = v_\lambda^*(\mathscr{F}_\lambda)$, and $\mathscr{F}_\lambda$ is f_λ-flat; this can be verified via (11.2.6), and the sub-ring A_0 of A containing A_0, and $v_\lambda : \mathrm{X} \to \mathrm{X}_\lambda$ is the canonical projection, and $\mathscr{F}_\lambda = \mathscr{F}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda$. It follows from (11.2.6) that there exists λ such that $\mathscr{F}_\lambda$ is f_λ-flat (resp. f_λ-flat at the point $x_\lambda = v_\lambda(x)$); then the sub-ring A_λ of A satisfies the conditions of *c*). □

Proposition (11.2.8). — *Let* $g : \mathrm{X} \to \mathrm{S}$, $h : \mathrm{Y} \to \mathrm{S}$ *be two morphisms of finite presentation,* $f : \mathrm{X} \to \mathrm{Y}$ *an S-morphism,* $\mathscr{F}$ *an* $\mathscr{O}_\mathrm{X}$*-Module quasi-coherent of finite presentation; for every* $s \in \mathrm{S}$*, let* $\mathrm{X}_s = g^{-1}(s) = \mathrm{X} \times_\mathrm{S} \mathrm{Spec}(\boldsymbol{k}(s))$, $\mathrm{Y}_s = h^{-1}(s) = \mathrm{Y} \times_\mathrm{S} \mathrm{Spec}(\boldsymbol{k}(s))$, f_s *the morphism* $f \times 1 : \mathrm{X}_s \to \mathrm{Y}_s$, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{S}} \boldsymbol{k}(s)$*. Then the set* E *of* $s \in \mathrm{S}$ *such that* $\mathscr{F}_s$ *is* f_s*-flat is locally constructible.*

Proof. The property that we want to prove satisfies condition (9.2.1, (i)), by virtue of (2.2.13) and (2.5.1). Taking into account (9.2.3), one can therefore restrict to the case where S is affine, Noetherian, and integral with generic point η. If $\eta \in \mathrm{S} - \mathrm{E}$, this follows immediately from the lemma (9.4.7.1). If on the contrary $\eta \in \mathrm{E}$, it follows immediately from (11.2.6), that there is a neighbourhood open U of η in S such that $\mathscr{F}|\mathrm{U}$ is $h^{-1}(\mathrm{U})$-flat; *a fortiori* (2.1.4) $\mathscr{F}_s$ is f_s-flat for every $s \in \mathrm{U}$, which completes the proof. □

Theorem (11.2.9). — *(Raynaud). — Let* A_0 *be a ring,* $(\mathrm{A}_\lambda)_{\lambda \in \mathrm{L}}$ *an inductive filtrant system of* A_0*-algebras,* B_0 *an* A_0*-algebra of finite presentation,* $\mathfrak{J}_0$ *an ideal of finite type of* B_0*,* M_0 *a* B_0*-module of finite presentation. Set* $\mathrm{B}_\lambda = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda$, $\mathfrak{J}_\lambda = \mathfrak{J}_0 \mathrm{B}_\lambda$, $\mathrm{M}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{B}_0} \mathrm{B}_\lambda$, $\mathrm{A} = \varinjlim \mathrm{A}_\lambda$, $\mathrm{B} = \varinjlim \mathrm{B}_\lambda = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$, $\mathfrak{J} = \mathfrak{J}_0 \mathrm{B}$, $\mathrm{M} = \varinjlim \mathrm{M}_\lambda = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A} = \mathrm{M}_0 \otimes_{\mathrm{B}_0} \mathrm{B}$*. For* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ *to be a flat* A*-module, it is necessary and sufficient that there exist* λ *such that* $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ *be a flat* A_λ*-module; the canonical homomorphisms*

$$(11.2.9.1) \qquad \begin{cases} \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A} \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) \\ \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}_\mu \to \mathrm{gr}^*_{\mathfrak{J}_\mu}(\mathrm{M}_\mu) \end{cases} \qquad \text{for } \lambda \leqslant \mu$$

are then bijective and $\mathrm{gr}^*_{\mathfrak{J}_\mu}(\mathrm{M}_\mu)$ *is a flat* A_μ*-module for* $\lambda \leqslant \mu$.

Proof. Let us first show that the conditions are sufficient, which amounts to proving the bijectivity of (11.2.9, IV.11.2.9.1). This follows from the following lemma:

Lemma (11.2.9.2). — *Let* A *be a ring,* B *an* A*-algebra,* M *a* B*-module,* $\mathfrak{J}$ *an ideal of* B*. Let* A' *be an* A*-algebra; set* $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}' = \mathrm{M} \otimes_{\mathrm{B}} \mathrm{B}'$, $\mathfrak{J}' = \mathfrak{J}\mathrm{B}'$*. If* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ *is a flat* A*-module, the canonical homomorphism*

$$(\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{gr}^*_{\mathfrak{J}'}(\mathrm{M}')$$

is bijective.

Indeed, by induction on k, the hypothesis that the $\mathfrak{J}^k\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}$ are flat A-modules implies first, by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$ that $\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}$ is a flat A-module; moreover $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, the sequence **IV-3 | 127**

$$0 \longrightarrow (\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow 0$$

is then exact, in other words $(\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}'$ identifies with its canonical image $\mathfrak{J}'^{k+1}\mathrm{M}'$ in $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$. On the other hand, always by $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$, the sequence

$$0 \longrightarrow (\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathfrak{J}^k\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow (\mathfrak{J}^k\mathrm{M}/\mathfrak{J}^{k+1}\mathrm{M}) \otimes_{\mathrm{A}} \mathrm{A}' \longrightarrow 0$$

is exact, which proves the lemma.

To prove the necessity of the conditions of (11.2.9), we proceed in several steps.

I) *Reduction to the case where the* A_λ *are Noetherian.* — One proceeds as in the reduction I) of (11.2.6.1), keeping the notations; it suffices simply to begin by replacing A'_0 by a sub-A'_0-algebra A''_0 of A_0 such that, if one sets $\mathrm{B}''_0 = \mathrm{B}'_0 \otimes_{\mathrm{A}'_0} \mathrm{A}''_0$, there is in B''_0 a finitely generated ideal $\mathfrak{J}''_0$ such that $\mathfrak{J}''_0 \mathrm{B}_0 = \mathfrak{J}_0$. One considers then all sub-A'_0-algebras A'_α of A_0 of type fini, which form a filtered family, and $\mathrm{B}_0 = \varinjlim \mathrm{B}'_\alpha$, where $\mathrm{B}'_\alpha = \mathrm{B}'_0 \otimes_{\mathrm{A}'_0} \mathrm{A}'_\alpha$; there is therefore an index β such that a finite system of generators of $\mathfrak{J}_0$ is formed of images in B_0 of elements of B'_β; one takes then $\mathrm{A}''_0 = \mathrm{A}'_\beta$, $\mathrm{B}''_0 = \mathrm{B}'_\beta$ and for $\mathfrak{J}''_0$ the ideal generated by these elements. One defines then as in *loc. cit.* the filtrant set H and the A_ξ, B_ξ, M_ξ for $\xi \in \mathrm{H}$; and also $\mathfrak{J}_\xi = \mathfrak{J}''_0 \mathrm{B}_\xi$ for all $\xi \in \mathrm{H}$. Suppose that one has shown that there exists $\xi = (\lambda, \mathrm{C}_\lambda)$ such that $\mathrm{gr}^*_{\mathfrak{J}_\xi}(\mathrm{M}_\xi)$ be a flat C_λ-module. As $\mathrm{M}_\lambda = \mathrm{M}_\xi \otimes_{\mathrm{C}_\lambda} \mathrm{A}_\lambda$, it follows from (11.2.9.2) that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ is a flat A_λ-module. We can therefore from now on suppose the A_λ Noetherian for $\lambda \in \mathrm{L}$ (but not necessarily A itself).

II) *Preliminary lemmas.*

Lemma (11.2.9.3). — *Let* A *be a ring,* $\mathrm{B} = \bigoplus_{i \geqslant 0} \mathrm{B}_i$ *an A-algebra graded in non-negative degrees. Suppose that* B_0 *is a local ring, and that each* M_i *is a* B_0*-module of finite type. For* M *to be a flat* A*-module, it is necessary and sufficient that, if* $\mathfrak{q}$ *is the inverse image in* A *of the maximal ideal* $\mathfrak{m}$ *of* B_0, $\mathrm{M} \otimes_{\mathrm{A}} \boldsymbol{k}(\mathfrak{q})$ *be a* $(\mathrm{B} \otimes_{\mathrm{A}} \boldsymbol{k}(\mathfrak{q}))$*-module of finite type.*

IV-3 | 128

Proof. The condition being evidently necessary, let us prove that it is sufficient. As B_0 is a local ring, so is A; by Nakayama's lemma applied on the B_0-modules M_i, if there exists a graded B-module *free* of finite base and a surjective graded homomorphism of degree 0, $u : L \to M$. Set $R = \operatorname{Ker}(u)$, which is a B-module graded; there are then a finite number of integers m_j ($1 \leqslant j \leqslant r$) such that for each integer i, R_i is the kernel of a surjective homomorphism $\bigoplus_{1\leqslant j\leqslant r} B_{i+m_j} \to M_i$; one concludes then from the hypothesis on the M_i and the B_i that R_i is a B_0-module of finite type (Bourbaki, *Alg. comm.*, chap. I, § 2, nº 8, lemma 9). To prove that R is a B-module of finite type, let us note that by virtue of the hypothesis of flatness and of ($\mathbf{0}_{\mathrm{I}}$, 6.1.2) the sequence

$$0 \longrightarrow R \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow L \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow M \otimes_A \boldsymbol{k}(\mathfrak{q}) \longrightarrow 0$$

is exact, and the hypothesis implies (Bourbaki, *loc. cit.*) that $R \otimes_A \boldsymbol{k}(\mathfrak{q})$ is a $(B \otimes_A \boldsymbol{k}(\mathfrak{q}))$-module of finite type; it therefore suffices to apply R the lemma (11.2.9.3). □

Corollary (11.2.9.4). — *Under the hypotheses of (11.2.9.3), suppose moreover that each of the B_i and each M_i is a B_0-module of finite presentation, and that M is a flat A-module. For M to be a B-module of finite type, it is necessary and sufficient that $M \otimes_A \boldsymbol{k}(\mathfrak{q})$ be a $(B \otimes_A \boldsymbol{k}(\mathfrak{q}))$-module of finite presentation.*

Proof. By virtue of (11.2.9.3), if the condition of (11.2.9.3) is verified, M is a B-module of finite type, hence (having therefore a finite base formed of homogeneous elements, and a surjective graded homomorphism of degree 0), $u : L \to M$. Set $R = \operatorname{Ker}(u)$, which is a B-module graded; as above, the R_i are B_0-modules of finite type. □

III) *Reduction to the case where the homomorphisms of transitions $\varphi_{\mu\lambda} : A_\lambda \to A_\mu$ (for $\lambda \leqslant \mu$) are injective.* — Let A'_λ be the image of A_λ by the canonical homomorphism $\varphi_\lambda : A_\lambda \to A$; it is clear that the A'_λ form an inductive system of sub-rings *Noetherian* of A, and the homomorphisms of transition $A'_\lambda \to A'_\mu$ (for $\lambda \leqslant \mu$) are injective. Set $B'_\lambda = B_0 \otimes_{A_0} A'_\lambda$, $\mathfrak{J}'_\lambda = \mathfrak{J}_0 B'_\lambda$, $M'_\lambda = M_0 \otimes_{A_0} A'_\lambda = M_\lambda \otimes_{A_\lambda} A'_\lambda$; one has still $B = \varinjlim B'_\lambda$, $M = \varinjlim M'_\lambda$. Suppose one has proved that there exists λ such that, for $\mu \geqslant \lambda$, $\operatorname{gr}^*_{\mathfrak{J}'_\mu}(M'_\mu)$ is a flat A'_μ-module. Let a_λ be the kernel of the homomorphism φ_λ, which is an ideal of the Noetherian ring A_λ. It follows from the definition of the inductive limit that there exists an index $\mu \geqslant \lambda$ such that $\varphi_{\mu\lambda}(a_\lambda) = 0$, in other words $\varphi_{\mu\lambda}$ factorises as $\varphi_{\mu\lambda} : A_\lambda \to A'_\lambda \to A_\mu$, and one can write $B_\mu = B'_\lambda \otimes_{A'_\lambda} A_\mu$, $\mathfrak{J}_\mu = \mathfrak{J}'_\lambda B_\mu$ and $M_\mu = M'_\lambda \otimes_{A'_\lambda} A_\mu$; one deduces then from (11.2.9.2) that $\operatorname{gr}^*_{\mathfrak{J}_\mu}(M_\mu)$ is a flat A_μ-module.

IV) *Reduction to the case where $B_0 = A_0[T_1, \dots, T_n]$ and $\operatorname{gr}^*_{\mathfrak{J}_0}(B_0) = B_0$.* — By virtue of the hypothesis, there is a system $(t_i)_{1\leqslant i\leqslant n}$ of generators of $\mathfrak{J}_0$ of finite type of the A_0-algebra B_0, such that the t_i for $i \leqslant m$ engender the ideal $\mathfrak{J}_0$ of B_0. Set $B'_0 = A_0[T_1, \dots, T_n]$, and let $\mathfrak{J}'_0$ be the ideal of B'_0 generated by the T_i for $i \leqslant m$; then $u_0 : B'_0 \to B_0$ is an A_0-homomorphism surjective transforming each T_i in t_i ($1 \leqslant i \leqslant n$), and such that $u_0(\mathfrak{J}'_0) = \mathfrak{J}_0$; this homomorphism allows one to consider M_0 as a B'_0-module of finite type. One then sets $B'_\lambda = B'_0 \otimes_{A_0} A_\lambda = A_\lambda[T_1, \dots, T_n]$, $\mathfrak{J}'_\lambda = \mathfrak{J}'_0 B'_\lambda$, so that $\mathfrak{J}'$ is the ideal of B' generated by the T_i for index $i \leqslant m$; moreover, it is clear that for every integer $k \geqslant 0$, one has $\mathfrak{J}'^k M = \mathfrak{J}^k M$ and $\mathfrak{J}'^k_\lambda M_\lambda = \mathfrak{J}^k_\lambda M_\lambda$ for every λ; hence $\operatorname{gr}^*_{\mathfrak{J}'}(M) = \operatorname{gr}^*_{\mathfrak{J}}(M)$ as A-module and $\operatorname{gr}^*_{\mathfrak{J}'_\lambda}(M_\lambda) = \operatorname{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda)$ as A_λ-module; one can therefore substitute B', B'_λ, $\mathfrak{J}'$, $\mathfrak{J}'_\lambda$ for B, B_λ, $\mathfrak{J}$, $\mathfrak{J}_\lambda$ respectively in the proof; one will note moreover that by construction $\operatorname{gr}^*_{\mathfrak{J}'}(B')$ identifies with B' and $\operatorname{gr}^*_{\mathfrak{J}'_\lambda}(B'_\lambda)$ with B'_λ. **IV-3 | 129**

V) *Proof that $\operatorname{gr}^*_{\mathfrak{J}}(M)$ is a $\operatorname{gr}^*_{\mathfrak{J}}(B)$-module of finite presentation.* — We thus suppose from now on A_0 and the A_λ Noetherian, the homomorphisms of transition $A_\lambda \to A_\mu$ injective, $B_0 = A_0[T_1, \dots, T_n]$, $\mathfrak{J}_0$ being generated by the T_i for index $i \leqslant m$; the ring $C_0 = \operatorname{gr}^0_{\mathfrak{J}_0}(B_0)$ identifies to B_0 and $C = \operatorname{gr}^*_{\mathfrak{J}}(B)$ identifies to B; we will use only in the sequel the fact that $C \cong C_0 \otimes_{A_0} A$.

We will need the following variant of (6.9.3):

Lemma (11.2.9.5). — *Let A_0 be a Noetherian ring, B_0 an A_0-algebra of finite type, $\mathfrak{J}_0$ an ideal of B_0, M_0 a B_0-module of finite type. There exists a sequence $(S_{0i})_{1\leqslant i\leqslant m}$ of sub-preschemes of $S_0 = \operatorname{Spec}(A_0)$ having the following properties:*

label=1º The underlying spaces of the S_{0i} are pairwise disjoint and form a covering of S_0.
lbbel=2º For each i, the set S_{0i} is open in $\bigcup_{j\geqslant i} S_{0j}$.

lcbel=3º *Each scheme* S_{0i} *is affine.*
ldbel=4º *If* A_{0i} *is the ring of* S_{0i} *and one sets* $B_{0i} = B_0 \otimes_{A_0} A_{0i}$, $\mathfrak{J}_{0i} = \mathfrak{J}_0 B_{0i}$, *then* $\mathrm{gr}^*_{\mathfrak{J}_{0i}}(M_0 \otimes_{A_0} A_{0i})$ *is a flat* A_{0i}*-module.*

Proof. One proceeds as in (6.9.3) by Noetherian induction, supposing the lemma is true when A_0 is replaced by $A'_0 = A_0/\mathfrak{a}_0$, where $\mathfrak{a}_0$ is an ideal such that $V(\mathfrak{a}_0) \neq S_0$, and considers the nilradical $\mathfrak{N}_0$ of A_0. In other words, one may suppose that A_0 is *reduced*; on the other hand, $\mathrm{gr}^*_{\mathfrak{J}_0}(B_0)$ is an A_0-algebra of finite type and $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0)$ a $\mathrm{gr}^*_{\mathfrak{J}_0}(B_0)$-module of finite type; by the generic flatness theorem (6.9.1), there exists an open $D(t_0) \subset S_0$ such that, if $A_{01} = (A_0)_{t_0}$, $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0) \otimes_{A_0} A_{01}$ is a flat A_{01}-module; but since A_{01} is a flat A_0-module, $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0 \otimes_{A_0} A_{01})$ identifies with $\mathrm{gr}^*_{\mathfrak{J}_{01}}(M_0 \otimes_{A_0} A_{01})$, where $B_{01} = B_0 \otimes_{A_0} A_{01}$, $\mathfrak{J}_{01} = \mathfrak{J}_0 B_{01}$. The complement of $D(t_0)$ in S_0 is of the form $V(\mathfrak{a}_0)$, and one concludes by applying the induction hypothesis. □

Apply this lemma to the present situation, keeping the same notations; set $U_{0i} = \bigcup_{j \leqslant i} S_{0j}$, which is an open quasi-compact of S_0. There is then a family $(f_{ik})_{1 \leqslant i \leqslant m, 1 \leqslant k \leqslant n}$ of elements of A_0 such that for each i, U_{0i} is the union of the $D(f_{ik})$ ($1 \leqslant k \leqslant n$).

The C_0-module $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0)$ is generated by a finite number of homogeneous elements of degree 1, so there is an epimorphism of C_0-graded modules $u_0 : C'_0 \to \mathrm{gr}^*_{\mathfrak{J}_0}(M_0)$. As $C = C_0 \otimes_{A_0} A$ and the canonical homomorphism $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0) \otimes_{A_0} A \to \mathrm{gr}^*_{\mathfrak{J}}(M)$ is surjective, one deduces therefore from u_0 an epimorphism of C-graded modules **IV-3 | 130**

$$u : C'' \xrightarrow{u_0 \otimes 1_{A'}} \mathrm{gr}^*_{\mathfrak{J}_0}(M_0) \otimes_{A_0} A \longrightarrow \mathrm{gr}^*_{\mathfrak{J}}(M),$$

and it comes down to seeing that the graded C-module $Q = \mathrm{Ker}(u)$ is *of finite type*.

Lemma (11.2.9.6). — *Under the preceding hypotheses, let* $\mathfrak{N}$ *be an ideal of* A_0*; set* $A'_0 = A_0/\mathfrak{N}$, $B'_0 = B_0 \otimes_{A_0} A'_0$, $\mathfrak{J}'_0 = \mathfrak{J}_0 B'_0$, $A' = A/\mathfrak{N}$, *and suppose moreover that* $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0 \otimes_{A_0} A'_0)$ *is a flat* A'_0*-module. Then there exist a finite number of elements* q_h ($1 \leqslant h \leqslant s$) *of* Q *such that, for every prime ideal* $\mathfrak{p} \supset \mathfrak{N}B$ *of* B*, the canonical images of the* q_h *in* $Q_{\mathfrak{p}}$ *engender the* $C_{\mathfrak{p}}$*-module* $Q_{\mathfrak{p}}$.

It remains to prove the lemma (11.2.9.6). Set $C' = C \otimes_A A' = C/\mathfrak{N}C$, $B' = B \otimes_A A'$, $\mathfrak{J}' = \mathfrak{J}B'$, $M' = M \otimes_A A'$. By hypothesis $\mathrm{gr}^*_{\mathfrak{J}}(M)$ is an A-module flat, hence $\mathrm{gr}^*_{\mathfrak{J}'}(M')$ identifies with $\mathrm{gr}^*_{\mathfrak{J}}(M')$ (11.2.9.2), and one has the exact sequence ($\mathbf{0}_{\mathrm{I}}$, 6.1.2)

$$0 \longrightarrow Q \otimes_A A' \longrightarrow C'' \xrightarrow{u \otimes 1_{A'}} \mathrm{gr}^*_{\mathfrak{J}'}(M') \longrightarrow 0. \tag{11.2.9.7}$$

Set $C'_0 = C_0 \otimes_{A_0} A'_0$ and consider the epimorphism of C'_0-graded modules

$$u'_0 : C''_0 \xrightarrow{u_0 \otimes 1_{A'}} \mathrm{gr}^*_{\mathfrak{J}_0}(M_0) \otimes_{A_0} A'_0 \longrightarrow \mathrm{gr}^*_{\mathfrak{J}_0}(M_0 \otimes_{A_0} A'_0).$$

Let $Q_0 = \mathrm{Ker}(u'_0)$; using the fact that $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0 \otimes_{A_0} A'_0)$ is a flat A'_0-module and that $\mathrm{gr}^*_{\mathfrak{J}_0}(M_0 \otimes_{A_0} A'_0) \otimes_{A'} A'$ identifies with $\mathrm{gr}^*_{\mathfrak{J}'}(M')$ (11.2.9.2), one sees that

$$Q \otimes_A A' \xrightarrow{\sim} Q_0 \otimes_{A_0} A'. \tag{11.2.9.8}$$

□

whence, by comparison with (11.2.9, IV.11.2.9.7), an isomorphism of C'-graded modules $Q \otimes_A A' \xrightarrow{\sim} Q_0 \otimes_{A_0} A'$. **IV-3 | 131**

As C'_0 is Noetherian, Q_0 is a C'_0-module of finite type, hence $Q_0 \otimes_{A'_0} A'$ is a C'-graded module of finite type, and so it is likewise a module of the same type of $Q \otimes_A A'$; let q_h ($1 \leqslant h \leqslant s$) be homogeneous elements of Q whose images in $Q \otimes_A A'$ engender this C'-module.

Consider now any prime ideal $\mathfrak{p} \supset \mathfrak{N}B$ of B; one has $C_{\mathfrak{p}} = \mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(B_{\mathfrak{p}})$ by flatness, and $\mathrm{gr}^0_{\mathfrak{J}_{\mathfrak{p}}}(B_{\mathfrak{p}}) = B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}}$ is reduced to 0 or is a local ring; to prove the lemma, one can evidently restrict to the second case. It is clear that for every i, $M/\mathfrak{J}^i M$ identifies with $(M_0/\mathfrak{J}_0^k M_0) \otimes_{A_0} A$, and is therefore a B-module of finite presentation. On the other hand, for every i, $M_\mu/\mathfrak{J}_\mu^i M_\mu$ for $k \leqslant i$ is the same as $M_\mu/\mathfrak{J}_\mu^{k+1} M_\mu$ is an A_μ-module flat, and also ($\mathbf{0}_{\mathrm{I}}$, 6.1.2) the quotient $\mathrm{gr}^k_{\mathfrak{J}_\mu}(M_\mu)$ is an A_μ-module flat for $k \leqslant i$. One deduces from (11.2.9.2) that each of the $(B/\mathfrak{J})$-modules $\mathrm{gr}^k_{\mathfrak{J}}(M)$ is of finite presentation.

Moreover, the images of the q_h in $\mathrm{Q}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p})$ (where $\mathrm{C}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) = \mathrm{C}_{\mathfrak{p}}/\mathfrak{p}\mathrm{C}_{\mathfrak{p}}$) engender this $(\mathrm{C}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}))$-module (since $\mathrm{C}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p})$); as on the exact sequence

$$\mathrm{Q}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow \mathrm{C}'_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow \mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(\mathrm{M}_{\mathfrak{p}}) \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}) \longrightarrow 0$$

one concludes that $\mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(\mathrm{M}_{\mathfrak{p}})$ is a $(\mathrm{C}_{\mathfrak{p}} \otimes_{\mathrm{B}_{\mathfrak{p}}} \boldsymbol{k}(\mathfrak{p}))$-module of finite presentation. Applying the lemma (11.2.9.4) where A, B, M are replaced respectively by $\mathrm{B}_{\mathfrak{p}}$, $\mathrm{C}_{\mathfrak{p}}$ and $\mathrm{gr}^*_{\mathfrak{J}_{\mathfrak{p}}}(\mathrm{M}_{\mathfrak{p}})$, one concludes that $\mathrm{Q}_{\mathfrak{p}}$ is a $\mathrm{C}_{\mathfrak{p}}$-module *of finite type*. Using now the Nakayama lemma (and the fact that $\mathrm{C} = \mathrm{B}$) one deduces that the images of the q_h in $\mathrm{Q}_{\mathfrak{p}}$ engender this $\mathrm{C}_{\mathfrak{p}}$-module. This completes the proof of the lemma (11.2.9.6) and the fact that $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ is a C-module of finite presentation.

VI) *End of the proof.* — Set, to abbreviate, $\mathrm{C}_\lambda = \mathrm{C}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda$ (equal in fact to B_λ), $\mathrm{N}_\lambda = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ and $\mathrm{N} = \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$. Note first that for each k, N identifies with $\varinjlim \mathfrak{J}^k_\lambda \mathrm{M}_\lambda$, the image of $\mathfrak{J}^k_\lambda \mathrm{M}_\lambda$ in M_μ engendering $\mathfrak{J}^k_\mu \mathrm{M}_\mu$ for $\lambda \leqslant \mu$ (resp. in M) engendering $\mathfrak{J}^k\mathrm{M}$; using again the exactness of $\varinjlim$, one concludes that N identifies canonically with $\varinjlim \mathrm{N}_\lambda$. Making this identification, we will first prove that:

(2) **(11.2.9.9)** *For λ sufficiently large, the canonical homomorphism $v_\lambda : \mathrm{N}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A} \longrightarrow \mathrm{N}$ is bijective.*

As the C_λ are Noetherian and N_λ a C_λ-module of finite type, the $\mathrm{N}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}$ are C-modules of finite presentation and form an inductive filtrant system, whose inductive limit identifies canonically with N by virtue of the fact that $\varinjlim$ commutes with tensor products. Moreover, the homomorphisms of transition $v_{\mu\lambda} : \mathrm{N}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A} \to \mathrm{N}_\mu \otimes_{\mathrm{A}_\mu} \mathrm{A}$

for $\lambda \leqslant \mu$ and the homomorphisms v_λ are *surjective*. For a fixed λ, consider the sub-C-modules **IV-3** | 132
$\mathrm{Ker}(v_{\mu\lambda}) = \mathrm{N}'_\mu$ of $\mathrm{N}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}$; by definition of the inductive limit, they form an increasing filtrant family of sub-C-modules, whose union is $\mathrm{Ker}(v_\lambda)$; but as one has seen in V) that N is a C-module of finite presentation, hence $\mathrm{Ker}(v_\lambda)$ is a C-module *of finite type*; there exists therefore an index $\mu \geqslant \lambda$ such that $\mathrm{N}'_\mu = \mathrm{Ker}(v_\lambda)$, which means (seeing that v_μ is surjective) that v_μ is *injective*; as it is surjective, this proves (11.2.9, IV.11.2.9.9).

It suffices to replace A_0 and M_0 and M_λ for λ large enough, one may suppose that the canonical homomorphism v_λ is bijective for *every* λ. Set then $\mathrm{P}_\lambda = \mathrm{N}_0 \otimes_{\mathrm{C}_0} \mathrm{C}_\lambda = \mathrm{N}_0 \otimes_{\mathrm{A}_0} \mathrm{A}_\lambda$, so that $\mathrm{N} = \varinjlim \mathrm{P}_\lambda$. As C_0 is an A_0-algebra of finite presentation and P_0 a C_0-module of finite presentation, one can apply C and N the result of (11.2.6.1) and one sees therefore that there exists an index λ such that, for $\mu \geqslant \lambda$, P_μ be a flat A_μ-module. Or, for $\mu \geqslant \lambda$, one has a commutative diagram

$$\begin{tikzcd}
\mathrm{P}_\mu \arrow[r] \arrow[d, "w_\mu"] & \mathrm{N} \arrow[d, equal] \\
\mathrm{N}_\mu \arrow[r] & \mathrm{N}
\end{tikzcd}$$

whence it follows from the definitions that w_μ is *surjective*. As, in virtue of III), the homomorphisms $\mathrm{A}_\mu \to \mathrm{A}$ are injective and that P_μ is a flat A_μ-module, the canonical homomorphism $\mathrm{P}_\mu \to \mathrm{N} = \mathrm{P}_\mu \otimes_{\mathrm{A}_\mu} \mathrm{A}$ is *injective*; one concludes therefore from the preceding commutative diagram that w_μ is also *injective*, hence bijective, and hence N_μ is an A_μ-module *flat* for $\mu \geqslant \lambda$, which completes the proof of (11.2.9). □

Remark (11.2.10). — We do not know if the generalisation of (11.2.6), (i)) analogous to the th. de Raynaud, is valid.

11.3. Application to the elimination of Noetherian hypotheses IV-3 | 132

Theorem (11.3.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then the set* U *of points $x \in X$ such that $\mathscr{F}$ is f-flat at the point x is open in* X. *If in addition* $\operatorname{Supp}(\mathscr{F}) = X$, *$f|U$ is an open morphism.*

Proof. The second assertion has already been proved (2.4.6) and is inserted here only for convenience.

The question being local on X and Y, one can suppose that X and Y are affine, and that f is a morphism of finite presentation. Let $x \in X$ be a point such that $\mathscr{F}$ is f-flat at the point x. Applying (11.2.7), one can suppose that $X = X_0 \times_{Y_0} Y$, $f = f_0 \times 1$, $\mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{Y_0}} \mathscr{O}_Y$, where Y_0 is Noetherian, $\mathscr{F}_0$ a coherent $\mathscr{O}_{X_0}$-Module; moreover, if x_0 is the projection of x in X_0, one can suppose that $\mathscr{F}_0$ is f_0-flat at the point x_0. Then, by virtue of (11.1.1), the set U_0 of points of X_0 where $\mathscr{F}_0$ is f_0-flat is a neighbourhood of x_0; hence $\mathscr{F}$ is f-flat at the points of the inverse image of U_0 in X (2.1.4), and this proves that U is a neighbourhood of x. □

IV-3 | 133

Corollary (11.3.2). — *Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation. Then the set* U *of $y \in Y$ such that, for every generization y' of y, $\mathscr{F}$ is $(g \circ f)$-flat at all the points of $f^{-1}(y')$ (i.e. such that $\mathscr{F} \otimes_Y \operatorname{Spec}(\mathscr{O}_{Y,y})$ is flat relative to the morphism $X \times_Y \operatorname{Spec}(\mathscr{O}_{Y,y}) \to Z$) is open in* Y.

Proof. The same reasoning as in (11.1.5) shows that if V is the set of $x \in X$ where $\mathscr{F}$ is $(g \circ f)$-flat, U is the set of $y \in Y$ whose every generization belongs to $E = Y - f(X - V)$. As $g \circ f$ is of finite presentation, V is open in X by virtue of (11.3.1), and hence V is ind-constructible (1.9.6), and hence $X - V$ is pro-constructible; but as f is quasi-compact, $f(X - V)$ is pro-constructible in Y (1.9.5, (vii)), and hence E is ind-constructible in Y. But it follows then from (1.10.1) that U is open in Y, hence the conclusion. □

Corollary (11.3.3). — *Let* A *be a ring,* B *an* A*-algebra of finite presentation,* C *a* B*-algebra of finite presentation,* M *a* C*-module of finite presentation. Then the set of $\mathfrak{q} \in \operatorname{Spec}(B)$ such that $M_\mathfrak{q}$ is a flat* A*-module is open in* Spec(B).

Proof. Taking into account (2.1.2), this is a consequence of (11.1.5) applied to $Z = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(C)$, $\mathscr{F} = \widetilde{M}$. □

Proposition (11.3.4). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{J}$ a quasi-coherent Ideal of finite type of $\mathscr{O}_X$,* Y *the closed sub-prescheme of* X *defined by $\mathscr{J}$, $\mathscr{F}$ an $\mathscr{O}_X$-Module of finite presentation. Suppose that $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F})$ is f-flat. Under these conditions:*

label=(i) *$\mathscr{F}$ is f-flat at the points of* Y.

lbbel=(ii) *If $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is f-flat, $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is an $(\mathscr{O}_X/\mathscr{J})$-Algebra of finite presentation and $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F})$ is an $(\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$-Module of finite presentation.*

lcbel=(iii) *Suppose $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is f-flat (which implies that $\mathscr{O}_X/\mathscr{J}$ is f-flat). Then the set* W *of $y \in Y$ such that $(\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F}))_y$ is a flat $(\mathscr{O}_X/\mathscr{J})_y$-module is open in* Y.

ldbel=(iv) *Suppose $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is f-flat. Let $S' \to S$ be any morphism; let $X' = X \times_S S'$ the closed sub-prescheme of X', $\mathscr{J}' = p^*(\mathscr{J})\mathscr{O}_{X'}$, $\mathscr{F}' = p^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$; then, if $W' = p^{-1}(W)$, one has $\operatorname{gr}^*_{\mathscr{J}'}(\mathscr{F}')|_{W'} \cong p^*(\operatorname{gr}^*_{\mathscr{J}}(\mathscr{F})|_W)$, and $\operatorname{gr}^*_{\mathscr{J}'}(\mathscr{F}')$ is a flat $(\mathscr{O}_{X'}/\mathscr{J}')$-Module at the points of W'.*

Proof. The questions being local on X and S, one may suppose that $X = \operatorname{Spec}(B)$ and $S = \operatorname{Spec}(A)$, $\mathscr{F} = \widetilde{M}$, where $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of finite type of B; by virtue of (8.9.1) and (8.5.11), there exists a sub-ring Noetherian A_0 of A, an A_0-algebra of finite type B_0, an ideal $\mathfrak{J}_0$ of B_0 such that $\mathfrak{J} = \mathfrak{J}_0 B$, a B_0-module of type fini M_0. Moreover, A is the inductive limit of its sub-A_0-algebras of finite type A_λ; and if one sets $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $\mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda$. This being so, by hypothesis $\operatorname{gr}^*_{\mathfrak{J}}(M)$ is a flat A-module; hence it follows from (11.2.9) that there exists λ such that $\operatorname{gr}^*_{\mathfrak{J}_\lambda}(M_\lambda)$ is a flat A_λ-module. To prove that $\mathscr{F}$ is f-flat at the points of Y, one may therefore restrict to the case where S is Noetherian; but then $(\mathbf{0}_\mathrm{I}, 6.1.2)$, applied by induction on n, proves that the $\mathscr{F}/\mathscr{J}^{n+1}\mathscr{F}$ are f-flat, and one concludes by $(\mathbf{0}_{\mathrm{III}}, 10.2.6)$. IV-3 | 134

The proof of (ii) reduces similarly to the case where S is Noetherian; the conclusion is then evident, using (11.2.9); in this case $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ being a $(\mathrm{B}/\mathfrak{J})$-algebra of finite type and $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$-module of finite type.

To prove (iii), one reduces as in (i) to the case where S and X are affine; with the same notations, one can suppose, by virtue of (11.2.9) that $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{B}_\lambda)$ and $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda)$ are flat A_λ-modules and that $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B}) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{B}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}$ and $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}$ for $\lambda \geqslant \alpha$, and $\mathrm{gr}^*_{\mathfrak{J}}(\mathscr{O}_\mathrm{X})$ (resp. $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathscr{O}_{\mathrm{X}_\lambda})$) is flat on S (resp. S_λ), this implies that Y is flat on S (resp. Y_λ flat on S_λ). By (11.2.9.2)), $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) = \mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathrm{M}_\lambda) \otimes_{\mathrm{A}_\lambda} \mathrm{A}$. If W^m is the set of $\mathfrak{p} \in \mathrm{Spec}(\mathrm{B}/\mathfrak{J})$ such that $\mathrm{gr}^m_{\mathfrak{J}}(\mathrm{M})$ is a flat $(\mathrm{B}/\mathfrak{J})$-module, W^m is open in $\mathrm{Spec}(\mathrm{B}/\mathfrak{J})$ (11.3.1) and one has $\mathrm{W} = \bigcap \mathrm{W}^m$, hence W is stable by generization. The assertion (iii) then follows from (11.3.2) applied by taking $\mathrm{X} = \mathrm{Spec}(\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B}))$, $\mathrm{Y} = \mathrm{Z} = \mathrm{Spec}(\mathrm{B}/\mathfrak{J})$ and $\mathscr{F} = (\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}))^\sim$.

Finally, (iv) follows immediately from (11.2.9.2).

Generalising the definition of (6.10.1), one says that for a prescheme X, a closed sub-prescheme Y of X defined by a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_\mathrm{X}$, and an $\mathscr{O}_\mathrm{X}$-Module quasi-coherent $\mathscr{F}$, $\mathscr{F}$ is *normally flat along* Y *at a point* y if $(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F}))_y$ is a flat $\mathscr{O}_{\mathrm{X},y}$-module. One says that F is *normally flat along* Y if it is so at all points of Y. □

Corollary (11.3.5). — *Under the general hypotheses of* (11.3.4)*, suppose that* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_\mathrm{X})$ *and* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ *are* f*-flat. Then the set* W *of* $y \in \mathrm{Y}$ *such that* $\mathscr{F}$ *is normally flat along* Y *at the point* y *is open in* Y*, and (with the notations of* (11.3.4)*, (iv)))* $\mathscr{F}'$ *is normally flat along* Y' *at all points of* $\mathrm{W}' = p^{-1}(\mathrm{W})$.

Proposition (11.3.6). — *The notations being those of* (8.5.1) *and* (8.8.1)*, suppose* S_α *quasi-compact,* X_α *of finite presentation over* S_α*,* Y_α *a closed sub-prescheme of* X_α *defined by a quasi-coherent Ideal* $\mathscr{J}_\alpha$ *of finite type of* $\mathscr{O}_{\mathrm{X}_\alpha}$*; suppose that* $\mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{\mathrm{X}_\alpha})$ *is flat on* S_α*; finally suppose that* $\mathscr{F}_\alpha$ *is an* $\mathscr{O}_{\mathrm{X}_\alpha}$*-Module of finite presentation. For* $\mathscr{F}$ *to be normally flat along* Y*, it is necessary and sufficient that there exist* λ *such that* $\mathscr{F}_\lambda$ *be normally flat along* Y_λ.

Proof. Note that Y (resp. Y_λ) is the closed sub-prescheme of X (resp. X_λ) defined by $\mathscr{J} = \mathscr{J}_\alpha\mathscr{O}_\mathrm{X}$ (resp. $\mathscr{J}_\lambda = \mathscr{J}_\alpha\mathscr{O}_{\mathrm{X}_\lambda}$); by virtue of (11.2.9.2), one has $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_\mathrm{X}) = \mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{\mathrm{X}_\alpha}) \otimes_{\mathscr{O}_{\mathrm{S}_\alpha}} \mathscr{O}_\mathrm{S}$ and $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{O}_{\mathrm{X}_\lambda}) = \mathrm{gr}^*_{\mathscr{J}_\alpha}(\mathscr{O}_{\mathrm{X}_\alpha}) \otimes_{\mathscr{O}_{\mathrm{S}_\alpha}} \mathscr{O}_{\mathrm{S}_\lambda}$ for $\lambda \geqslant \alpha$, and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_\mathrm{X})$ (resp. Y_λ flat on S (resp. S_λ)). If $\mathscr{F}_\lambda$ is normally flat along Y_λ, $\mathrm{gr}^*_{\mathfrak{J}_\lambda}(\mathscr{F}_\lambda)|\mathrm{Y}_\lambda$ is flat on S_λ, hence also $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is flat on S. On the other hand, if $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is flat on S, $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathscr{F}_\lambda)$ is flat on S_λ, and moreover (11.3.4), (ii)), $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathrm{B}_\lambda)$ is a $(\mathrm{B}_\lambda/\mathfrak{J}_\lambda)$-algebra of finite presentation and $\mathrm{gr}^*_{\mathscr{J}_\lambda}(\mathrm{M}_\lambda)$ a $(\mathrm{B}_\lambda/\mathfrak{J}_\lambda)$-module of finite presentation. The conclusion follows then from (11.2.6). □

Proposition (11.3.7). — *Let* $f : \mathrm{X} \to \mathrm{Y}$ *be a morphism locally of finite presentation,* $\mathscr{F}$*,* $\mathscr{F}'$ *two* $\mathscr{O}_\mathrm{X}$*-Modules quasi-coherent of finite presentation,* $u : \mathscr{F}' \to \mathscr{F}$ *an* $\mathscr{O}_\mathrm{X}$*-homomorphism,* x *a point of* X *and* $y = f(x)$*; suppose that* $\mathscr{F}$ *is* f*-flat at the point* x*. The following conditions are equivalent:*

label=a) $(\mathrm{Ker}\, u)_x = 0$ *and* $\mathrm{Coker}\, u$ *is an* $\mathscr{O}_\mathrm{X}$*-Module* f*-flat at the point* x*.*
lbbel=b) *The homomorphism* $(u \otimes 1)_x : (\mathscr{F}' \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))_x \to (\mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y))_x$ *is injective.*

If in addition $\mathscr{F}$ *is* f*-flat, the set of points* x *satisfying the preceding equivalent conditions is open in* X.

Proof. The condition a) implies b) by virtue of ($\mathbf{0}_\mathrm{I}$, 6.1.2), without hypothesis on $\mathscr{F}$. To prove the converse, one can restrict to the case where X and Y are affine, and hence, by virtue of (8.9.1) and (11.2.7), suppose that $\mathrm{X} = \mathrm{X}_0 \times_{\mathrm{Y}_0} \mathrm{Y}$, $f = f_0 \times 1$, $\mathscr{F} = \mathscr{F}'_0 \otimes_{\mathscr{O}_{\mathrm{Y}_0}} \mathscr{O}_\mathrm{Y}$, $\mathscr{F}' = \mathscr{F}'_0 \otimes_{\mathscr{O}_{\mathrm{Y}_0}} \mathscr{O}_\mathrm{Y}$, $u = u_0 \otimes 1_\mathrm{Y}$, where Y_0 is Noetherian, $f_0 : \mathrm{X}_0 \to \mathrm{Y}_0$ a morphism of finite type, $\mathscr{F}_0$, $\mathscr{F}'_0$ two $\mathscr{O}_{\mathrm{X}_0}$-Modules coherent, $u_0 : \mathscr{F}'_0 \to \mathscr{F}_0$ a homomorphism; moreover, if x_0 is the projection of x in X_0, one can suppose that $\mathscr{F}_0$ is f_0-flat at the point x_0. Set $y_0 = f_0(x_0)$. The hypothesis b) implies, by the transitivity of fibres (**I**, 3.6.4), the morphism projection $f^{-1}(y) \to f_0^{-1}(y_0)$ is faithfully flat (2.2.7), and as $(u \otimes 1)_x = ((u_0 \otimes 1)_{x_0} \otimes 1)_{k(y)}$, the hypothesis implies that $(u_0 \otimes 1)_x$ is injective and hence that $(\mathrm{Ker}\, u_0)_{x_0} = 0$ and $\mathrm{Coker}\, u_0$ is f_0-flat at x_0. One deduces first that $\mathrm{Coker}\, u$ is f-flat at x (2.1.4); moreover, by the exactness at the right of the tensor product, one has $\mathrm{Coker}\, u = (\mathrm{Coker}\, u_0) \otimes_{\mathscr{O}_\mathrm{Y}} \mathscr{O}_\mathrm{Y}$; applying then ($\mathbf{0}_\mathrm{I}$, 6.1.2) to the sequence (exact by hypothesis) $0 \to (\mathscr{F}'_0)_{x_0} \to (\mathscr{F}_0)_{x_0} \to (\mathrm{Coker}\, u_0)_{x_0} \to 0$ on en déduit que $(\mathrm{Ker}\, u)_x = 0$, thus proving condition a). **IV-3** | 135

Finally, it follows from (11.1.2) that the set U_0 of points $z_0 \in X_0$ such that the morphism $(u_0 \otimes 1)_{z_0}$ is injective is open; since $\mathscr{F}$ is f-flat, by flatness for all $z \in X$ above $z_0 \in U_0$ the morphism $(u \otimes 1)_z$ is injective, hence the set of such points contains the inverse image of U_0 in X and is a neighbourhood of x. □

Theorem (11.3.8). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent of finite presentation, $(g_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{O}_X$ above X; set $\mathscr{F}_i = \mathscr{F} / \sum_{j=1}^{i} g_j\mathscr{F}$ for $0 \leqslant i \leqslant n$ (with $\mathscr{F}_0 = \mathscr{F}$). Let x be a point of X, $y = f(x)$, and set $X_y = f^{-1}(y) = X \times_Y \mathrm{Spec}(\boldsymbol{k}(y))$, $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \boldsymbol{k}(y)$, which is an $\mathscr{O}_{X_y}$-Module. Suppose that the $(g_i)_x$ belong to the maximal ideal $\mathfrak{m}_x$. The following conditions are equivalent:*

label=a) *The sequence $((g_i)_x)$ is $\mathscr{F}_x$-regular and the $\mathscr{F}_i$ $(0 \leqslant i \leqslant n)$ are f-flat at the point x.*

lbbel=b) *The sequence $((g_i)_x)$ is $\mathscr{F}_x$-regular and $\mathscr{F}_n$ is f-flat at the point x.*

b′) *There exists a neighbourhood U of x such that the sequence $(g_i|U)$ is $(\mathscr{F}|U)$-quasi-regular, and $\mathscr{F}_n$ is f-flat at the point x.* **IV-3** | 136

lcbel=c) *$\mathscr{F}$ is f-flat at the point x, and the sequence $((g_i \otimes 1)_x)$ of elements of $\mathscr{O}_{X_y,x}$ images of the $(g_i)_x$ is $(\mathscr{F}_y)_x$-regular.*

ldbel=d) *$\mathscr{F}$ is f-flat at the point x, and for every morphism $Y' \to Y$ and every point $x' \in X' = X \times_Y Y'$ above x, if one sets $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$, the sequence $(g_i \otimes 1)_{x'}$ of elements of $\mathscr{O}_{X',x'}$ is $\mathscr{F}'_{x'}$-regular.*

Moreover, the set of $x \in X$ satisfying these conditions is open in the set Z of x such that $g_i(x) = 0$ for every i.

Proof. The fact that $a)$ implies $d)$ follows immediately from (**0**, 15.1.15), and $c)$ is a particular case of $d)$; moreover $a)$ implies $b)$ trivially. Let us show that $b)$ or $c)$ implies $a)$. The $\mathscr{F}_i$ being of finite presentation, the fact that $c)$ implies $a)$ follows immediately from (11.3.7) by induction on n, and this also shows that the set of $x \in X$ satisfying $c)$ is open in the set Z. To show that $b)$ implies $a)$, on reduces by induction on n to the case $n = 1$; we write g instead of g_1. The question being local on X and Y, one can suppose $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$ affine, $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite presentation. Moreover A is the limit inductive of its sub-**Z**-algebras of finite type. By (8.9.1) we can write $B = B_0 \otimes_{A_0} A$, $M = M_0 \otimes_{A_0} A$, where A_0 is a sub-ring Noetherian of A, B_0 an A_0-algebra of finite type and M_0 a B_0-module of finite type, B_λ is a B_λ-algebra of finite type, M_λ a B_λ-module of finite type and one has $B = \varinjlim B_\lambda$, $M = \varinjlim M_\lambda$. There exists therefore an index λ and an element $g_\lambda \in B_\lambda$ such that $g = g_\lambda \otimes 1$; returning to geometric notations and setting $Y_\lambda = \mathrm{Spec}(A_\lambda)$, $X_\lambda = \mathrm{Spec}(B_\lambda)$ and $\mathscr{F}_\lambda = \widetilde{M}_\lambda$, it will suffice to prove that for $\mu \geqslant \lambda$ sufficiently large, $(g_\mu)_{x_\mu}$ is $(\mathscr{F}_\mu)_{x_\mu}$-regular and $\mathscr{F}_\mu / g_\mu \mathscr{F}_\mu$ flat on Y_μ at the point x_μ. One deduces indeed, by (0, 15.1.16) that $\mathscr{F}_\mu$ is flat on Y_μ at the point x_μ, hence $\mathscr{F}$ flat on Y at the point x.

The fact that $b)$ implies $a)$ will follow therefore from the proposition:

Proposition (11.3.9). — *The notations and general hypotheses being those of* (8.5.1) *and* (8.8.1), *suppose that $f_\alpha : X_\alpha \to Y_\alpha$ is a morphism locally of finite presentation. Let $\mathscr{F}_\alpha$, $\mathscr{G}_\alpha$ two $\mathscr{O}_{X_\alpha}$-Modules quasi-coherent of finite presentation, $h_\alpha : \mathscr{F}_\alpha \to \mathscr{G}_\alpha$ an $\mathscr{O}_{X_\alpha}$-homomorphism, $\mathscr{H}_\alpha$ its cokernel; finally x a point of X, x_λ its projection in X_λ for $\lambda \geqslant \alpha$. For h_x to be injective and $\mathscr{H}$ to be f-flat at the point x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $(h_\lambda)_{x_\lambda}$ be injective and that $\mathscr{H}_\lambda = \mathrm{Coker}\, h_\lambda$ be f_λ-flat at the point x_λ. Moreover, the set of $x \in X$ having the preceding properties is open in X.*

Proof. Recall that by virtue of the exactness at the right of the tensor product, one has $\mathscr{H}_\mu = v_{\lambda\mu}^*(\mathscr{H}_\lambda)$ for $\lambda \leqslant \mu$ and $\mathscr{H} = v_\lambda^*(\mathscr{H}_\lambda)$, which justifies the notations.

The suffisance of the condition is necessary, let us note that $\mathscr{H}$ is of finite presentation; the question being local on X and Y, one can suppose X and Y affine, and, by virtue of (11.3.1), suppose that $\mathscr{H}$ is f-flat. □

The condition being necessary, note now the **IV-3** | 137

Lemma (11.3.9.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $\mathscr{G}$, $\mathscr{H}$ two $\mathscr{O}_X$-Modules quasi-coherent of finite presentation such that $\mathscr{H}$ is f-flat, $p : \mathscr{G} \to \mathscr{H}$ a surjective homomorphism. Then* $\mathrm{Ker}(p)$ *is an $\mathscr{O}_X$-Module of finite presentation.*

Proof. One may indeed suppose X, Y affine, and that there exists a morphism $Y \to Y_0$ where Y_0 is Noetherian, a morphism $f_0 : X_0 \to Y_0$ of finite type such that $X = X_0 \times_{Y_0} Y$, $f = (f_0)_{(Y)}$, two

$\mathscr{O}_{X_0}$-Modules coherent $\mathscr{G}_0$, $\mathscr{H}_0$ and a homomorphism $p_0 : \mathscr{G}_0 \to \mathscr{H}_0$ such that $\mathscr{G}$, $\mathscr{H}$ and p are deduced by change of base (8.9.1 and 8.5.2). One can suppose p_0 surjective (8.5.7) and $\mathscr{H}_0$ f_0-flat (11.2.7). Then, if $\mathscr{K}_0 = \mathrm{Ker}(p_0)$, $\mathscr{K}_0$ is an $\mathscr{O}_{X_0}$-Module coherent ($\mathbf{0}_{\mathrm{I}}$, 5.3.4) and by virtue of ($\mathbf{0}_{\mathrm{I}}$, 6.1.2), $\mathscr{K} = \mathrm{Ker}(p)$ is deduced from $\mathscr{K}_0$ by change of base, hence is of finite presentation. □

This lemma being proved, set $\mathscr{K} = \mathrm{Ker}(\mathscr{G} \to \mathscr{H})$, which is therefore of finite presentation. One has a canonical homomorphism $q : \mathscr{F} \to \mathscr{H}$, and by hypothesis $q_x : \mathscr{F}_x \to \mathscr{H}_x$ is an isomorphism; by (11.2.7, IV.11.2.7) (5.2.7) there exists a neighbourhood U of x such that $q|U$ is an isomorphism, and by restricting X, one may suppose that the sequence

$$0 \longrightarrow \mathscr{F} \xrightarrow{h} \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$

is *exact*. This being so, it follows from (11.2.6) that for a λ large enough, $\mathscr{H}_\lambda$ is an $\mathscr{O}_{X_\lambda}$-Module flat; set $\mathscr{K}_\lambda = \mathrm{Ker}(\mathscr{G}_\lambda \to \mathscr{H}_\lambda)$ and $\mathscr{H}_\mu = v^*_{\lambda\mu}(\mathscr{H}_\lambda)$ for $\mu \geqslant \lambda$; it follows from ($\mathbf{0}_{\mathrm{I}}$, 6.1.2) that $\mathscr{K}_\mu = \mathrm{Ker}(\mathscr{G}_\mu \to \mathscr{H}_\mu) = v^*_{\lambda\mu}(\mathscr{K}_\lambda)$. On the other hand for every $\mu \geqslant \lambda$ there is a canonical homomorphism $q_\mu : \mathscr{F}_\mu \to \mathscr{K}_\mu$ with $q_\mu = v^*_{\mu\lambda}(q_\lambda)$ for $\lambda \leqslant \mu$, and $q = v^*_\mu(q_\mu)$. As q is an isomorphism, by (11.2.7, IV.11.2.7) (5.2.7) it follows from (11.3.9.1) and (11.3.9) that q_μ is an isomorphism, and hence $u_\mu : \mathscr{F}_\mu \to \mathscr{G}_\mu$ an injective homomorphism.

Let us return to the proof of (11.3.8).

Since the set of $x \in X$ satisfying b) is open in X, it is clear that b) implies b'). Let us show finally that b') implies c). In the first place, $\mathscr{F}_n$ is f-flat in a neighbourhood of x (11.3.1), and on the other hand $\mathscr{F}_n$ is f-flat at the point x. As by definition $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ is isomorphic to $\mathscr{F}_n \otimes_{\mathscr{O}_y} \mathscr{O}_Y[T_1, \ldots, T_n]$ ($\mathbf{0}$, 15.2.2), it is f-flat, and one concludes by (11.3.4), (i)) that $\mathscr{F}$ itself is f-flat at the neighbourhood of x. Moreover, by virtue of (11.2.9.2) that, in the diagram

$$\begin{array}{ccc}((\mathrm{gr}^0_{\mathscr{J}_x}(\mathscr{F}_x))[T_1, \ldots, T_n]) \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) & \longrightarrow & \mathrm{gr}^*_{\mathscr{J}_x}(\mathscr{F}_x) \otimes_{\mathscr{O}_y} \boldsymbol{k}(y) \\ \downarrow & & \downarrow \\ (\mathrm{gr}^0_{(\mathscr{J}_y)_x}((\mathscr{F}_y)_x))[T_1, \ldots, T_n] & \longrightarrow & \mathrm{gr}^*_{(\mathscr{J}_y)_x}((\mathscr{F}_y)_x)\end{array}$$

the vertical arrows are isomorphisms; as the first line is an isomorphism, so it is the same for the second, hence the sequence $((g_i \otimes 1)_x)$ is $(\mathscr{F}_y)_x$-quasi-regular, since X_y is locally of finite type over $\boldsymbol{k}(y)$. □ **IV-3** | 138

Theorem (11.3.10). — *(Flatness criterion by fibres). — Let* S *be a prescheme,* $g : X \to S$, $h : Y \to S$ *two morphisms,* $f : X \to Y$ *an* S*-morphism,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module quasi-coherent,* x *a point of* X, $y = f(x)$, $s = h(y) = g(x)$. *Suppose verified the two hypotheses:*

label=1º S, X *and* Y *are locally Noetherian and* $\mathscr{F}$ *is coherent.*

lbbel=2º g *and* h *are locally of finite presentation and* $\mathscr{F}$ *is of finite presentation.*

Then, with the notations of (9.4.1), *if* $\mathscr{F}_z \neq 0$, *the following conditions are equivalent:*

*label=*a) $\mathscr{F}$ *is* g*-flat at the point* x *and* $\mathscr{F}_s$ *is* f_s*-flat at the point* x.

*lbbel=*b) h *is flat at the point* y *and* $\mathscr{F}$ *is* f*-flat at the point* x.

Moreover, under hypothesis 2º, the set of $x \in X$ *satisfying condition* b) *is open in* X.

Proof. The last assertion results from (11.3.1) applied to $\mathscr{O}_Y$ and to the morphism h on the one hand, and to $\mathscr{F}$ and to the morphism f (which is locally of finite presentation) on the other.

As $g = h \circ f$, it is clear that b) implies a) without assuming 1º nor 2º (2.1.6 and 2.1.4). On the other hand, a) implies b) trivially. Let us show that b) or c) implies a); under hypothesis 2º, by applying (11.2.7), one reduces to the case where in addition S is *Noetherian*, which amounts to considering the case where hypothesis 1º is satisfied. The assertion then amounts to the following lemma, which improves ($\mathbf{0}_{\mathrm{III}}$, 10.2.5):

Lemma (11.3.10.1). — *Let* $\rho : A \to B$, $\sigma : B \to C$ *be homomorphisms of local Noetherian rings,* k *the residue field of* A, M *a* C*-module* $\neq 0$ *of finite type. Then the following conditions are equivalent:*

*label=*a) M *is a flat* A*-module and* $M \otimes_A k$ *is a* $(B \otimes_A k)$*-module flat.*

*lbbel=*b) B *is a flat* A*-module and* M *is a flat* B*-module.*

We will first establish the more general following lemma:

Lemma (11.3.10.2). — *Let* A *be a commutative ring,* B *a commutative* A*-algebra,* $\mathfrak{J}$ *an ideal of* A, M *a* B*-module. On considers on one hand the following conditions:*

label=(i) $\mathfrak{J}$ *is nilpotent.*
lbbel=(ii) B *is Noetherian and* M *is ideally separated for the* $\mathfrak{J}$B*-preadic topology* $(\mathbf{0}_{\mathrm{III}},\ 10.2.1)$.
lcbel=(iii) $\mathfrak{J}$B *is contained in the radical of* B.

One considers on the other hand four properties:

label=a) M *is a flat* B*-module.*
lbbel=b) B *is a flat* A*-module.*
lcbel=c) M *is a flat* A*-module and* $\mathrm{M}/\mathfrak{J}\mathrm{M}$ *a flat* $(\mathrm{B}/\mathfrak{J}\mathrm{B})$*-module.*
ldbel=d) $\mathrm{B}/\mathfrak{J}\mathrm{B}$ *is a flat* $(\mathrm{A}/\mathfrak{J})$*-module and for every maximal ideal* $\mathfrak{m} \supset \mathfrak{J}\mathrm{B}$ *of* B *one has* $\mathfrak{m}\mathrm{M} \neq \mathrm{M}$.

Then:

label=1º *If one of the conditions (i), (ii) is verified, the conjunction of* a) *and* b) *implies* c)*, and* c) *implies the conjunction of* c) *and* d)*.*
lbbel=2º *If the condition (i) or the conjunction of (ii) and (iii) is verified, the conjunction of* c) *and* d) *implies the conjunction of* a) *and* b)*.* **IV-3** | 139

Proof. 1º The first assertion is immediate $(\mathbf{0}_{\mathrm{I}},\ 6.2.1)$. Suppose therefore $c)$ verified, and let us prove $a)$. Consider the graded rings $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$, $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$ and the graded module $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ (over both $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$ and $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$) relative to the $\mathfrak{J}$-preadic filtrations, as well as the surjective canonical maps $(\mathbf{0}_{\mathrm{III}},\ 10.1.1.2)$

$$u : \mathrm{gr}^0(\mathrm{B}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \longrightarrow \mathrm{gr}^*(\mathrm{B})$$
$$v : \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \longrightarrow \mathrm{gr}^*(\mathrm{M})$$
$$w : \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{B})} \mathrm{gr}^*(\mathrm{B}) \longrightarrow \mathrm{gr}^*(\mathrm{M})$$

It is clear there is a commutative diagram

(11.3.10.3)
$$\begin{tikzcd} \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{A})} \mathrm{gr}^*(\mathrm{A}) \arrow[rr, "v"] \arrow[dr, "1\otimes u"'] & & \mathrm{gr}^*(\mathrm{M}) \\ & \mathrm{gr}^0(\mathrm{M}) \otimes_{\mathrm{gr}^0(\mathrm{B})} \mathrm{gr}^*(\mathrm{B}) \arrow[ur, "w"'] & \end{tikzcd}$$

□

The hypothesis $c)$ implies that v is bijective $(\mathbf{0}_{\mathrm{III}},\ 10.2.1)$; as the two other maps in the diagram are surjective, they are also bijective. But as by virtue of hypothesis $c)$, $\mathrm{M}/\mathfrak{J}\mathrm{M}$ is a flat $(\mathrm{B}/\mathfrak{J}\mathrm{B})$-module, it follows from $(\mathbf{0}_{\mathrm{I}},\ 6.1.2)$. On a seen in 1º that hypothesis $c)$ implies that the three maps in the diagram (11.3.10, IV.11.3.10.3) are bijective; the fact that $\mathrm{gr}^0(\mathrm{M})$ is a flat $\mathrm{gr}^0(\mathrm{B})$-module *faithfully flat* implies therefore that u is also bijective $(\mathbf{0}_{\mathrm{I}},\ 6.4.1)$. On the other hand, the conditions (ii) and (iii) imply that B is an A-module ideally separated for the $\mathfrak{J}$-preadic filtration (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 4, prop. 2); one deduces therefore again from $(\mathbf{0}_{\mathrm{III}},\ 10.2.1)$ that B is a flat A-module, and as the image of θ by $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$ engenders the $\mathrm{B}'_{\mathfrak{p}'}$-module $(\mathrm{Tor}_1^{\mathrm{A}'}(\mathrm{M}', \mathrm{A}'/\mathfrak{J}'))_{\mathfrak{p}'}$, this last module is zero.

(11.3.10.4) Lemma (11.3.10.2) being established, one deduces (11.3.10.1) by taking for $\mathfrak{J}$ the maximal ideal of A, and by remarking that the conditions (ii) and (iii) of (11.3.10.2) are satisfied (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 4, prop. 2). This completes therefore also the proof of (11.3.10). □

Corollary (11.3.11). — *Let* $g : \mathrm{X} \to \mathrm{S}$, $h : \mathrm{Y} \to \mathrm{S}$ *be two morphisms locally of finite presentation,* $f : \mathrm{X} \to \mathrm{Y}$ *an* S*-morphism. The following conditions are equivalent:*

label=a) g *is flat and* $f_s : \mathrm{X}_s \to \mathrm{Y}_s$ *is flat for every* $s \in \mathrm{S}$.
lbbel=b) h *is flat at all the points of* $f(\mathrm{X})$ *and* f *is flat.*

Proof. It suffices to apply (11.3.10) for $\mathscr{F} = \mathscr{O}_{\mathrm{X}}$. □

Remark (11.3.12). — It would be interesting to be able to give proofs of (11.3.2) and (11.3.10) not using passage to the inductive limit; it would suffice for this to prove the following criterion:

Let A be a ring, $\mathfrak{p}$ an ideal of B containing $\mathfrak{J}$B. Then the two following conditions are equivalent:

label=*a)* $M_{\mathfrak{p}}$ is a flat A-module.

lbbel=*b)* $M_{\mathfrak{p}}/\mathfrak{J}M_{\mathfrak{p}}$ is a flat $(A/\mathfrak{J})$-module and $\mathrm{Tor}_1^A(A/\mathfrak{J}, M_{\mathfrak{p}}) = 0$.

When A is *Noetherian*, this is a consequence of $(\mathbf{0}_{\mathrm{III}}, 10.2.2)$ applied to the A-algebra Noetherian $B_{\mathfrak{p}}$; can one deduce the general statement by passage to the inductive limit?

Let A be a ring, B an A-algebra of finite presentation, M a B-module of finite presentation, $\mathfrak{J}$ an ideal of A principal and such that the intersection $\mathfrak{R} = \bigcap_{n \geqslant 0} \mathfrak{J}^{n+1}$ *is not reduced to* 0 (for instance the ring of germs of indefinitely differentiable functions in the neighbourhood of 0 in $\mathbf{R}$). Take B = A, $\mathfrak{p} = 3$, and $M = A/\mathfrak{R}$, where $\mathfrak{R}$ is a monogenic sub-module $\neq 0$ of $\mathfrak{R}$. It is clear that M is not a flat A-module of finite presentation, since being of finite presentation $\mathfrak{R} \neq 0$. However, $\mathfrak{J}^k M/\mathfrak{J}^{k+1}M$ is isomorphic to $A/\mathfrak{J}^n$ for all k and n positive integers, hence M satisfies conditions *c*) and *d*) of (11.3.10.2) when M is replaced by $M_{\mathfrak{p}}$. Taking for instance A a local ring whose maximal ideal $\mathfrak{m}$ is principal and such that the intersection $\bigcap_{n \geqslant 0} \mathfrak{m}^{n+1}$ *is not reduced to* 0 (for instance the ring of germs of indefinitely differentiable functions in the neighbourhood of 0 in $\mathbf{R}$). Take B = A, $\mathfrak{p} = 3$, and $M = A/\mathfrak{R}$. It is clear that M is not a flat A-module of finite presentation, hence it satisfies conditions *c*) and *d*) of (11.3.10.2) lorsqu'on remplace M par $M_{\mathfrak{p}}$, which is absurd since $\mathfrak{R} \neq 0$.

Proposition (11.3.13). — *Let $f : X \to Y$ be a morphism of preschemes, x a point of X; set $y = f(x)$.*

label=(i) *If X is reduced (resp. normal) at the point x, then Y is reduced (resp. normal) at the point y.*

lbbel=(ii) *Suppose in addition that f is of finite presentation at the point x. Then, if Y is reduced (resp. normal) at the point y and if the morphism f is reduced (resp. normal) at the point x* (6.8.1)*, X is reduced (resp. normal) at the point x.*

Proof. The first assertion is noted here only for convenience, having already been proved in (2.1.13). To prove (ii), one can restrict to the case where X = Spec(B), Y = Spec(A), where A is a local ring reduced (resp. integrally closed) and B an A-algebra of finite presentation. One can then write (8.9.1) $B = B_0 \otimes_{A_0} A$, where A_0 is a sub-$\mathbf{Z}$-algebra of finite type of A and B_0 an A_0-algebra of finite type. Let (C_α) be the family of sub-$\mathbf{Z}$-algebras of finite type of A, which are therefore $\mathbf{Z}$-algebras of finite type; one has $A = \varinjlim C_\alpha$. Let us distinguish the two cases:

1º Suppose A reduced and f reduced at the point x. If $\mathfrak{m}$ is the maximal ideal of A, let $\mathfrak{p}_\alpha$ be the prime ideal $\mathfrak{m} \cap C_\alpha$, and set $A'_\alpha = (C_\alpha)_{\mathfrak{p}_\alpha}$, so that one has also $A = \varinjlim A'_\alpha$ (5.13.3). Set $X_\alpha = \mathrm{Spec}(B_0 \otimes_{A_0} A'_\alpha)$ and let $f_\alpha : X_\alpha \to Y_\alpha$, x_α (resp. y_α) the projection of x (resp. y) in X_α (resp. Y_α). Since f is reduced at the point x, there exists α_0 such that f_α is reduced at the point x_α for $\alpha \geqslant \alpha_0$ ((6.7.8) and (11.2.6)); and since Y_α and X_α are Noetherian, and A'_α is reduced (5.13.2), on deduces from (3.3.5) that X_α is reduced at the point x_α. But since $B = \varinjlim (B_0 \otimes_{A_0} A'_\alpha)$, $\mathscr{O}_{X,x} = \varinjlim \mathscr{O}_{X_\alpha, x_\alpha}$ (5.13.3) and hence $\mathscr{O}_{X,x}$ is reduced (5.13.2).

2º Suppose A integrally closed and f normal at the point x. As C_α is universally Japanese (7.7.4), its integral closure C'_α is a C_α-algebra finite, evidently contained in A. Let $\mathfrak{p}'_\alpha$ be the prime ideal $\mathfrak{m} \cap C'_\alpha$, and set $A''_\alpha = (C'_\alpha)_{\mathfrak{p}'_\alpha}$, so that A''_α is a local Noetherian ring integral and integrally closed, and one has $A = \varinjlim A''_\alpha$ (5.13.3). Set $Y'_\alpha = \mathrm{Spec}(A''_\alpha)$, $X'_\alpha = \mathrm{Spec}(B_0 \otimes_{A_0} A''_\alpha)$ and let $f'_\alpha : X'_\alpha \to Y'_\alpha$, x'_α (resp. y'_α) the projection of x (resp. y) in X'_α (resp. Y'_α). Since f is normal at the point x, there exists α_0 such that, for $\alpha \geqslant \alpha_0$, f'_α is normal at the point x'_α ((6.7.8) and (11.2.6)); and hence X'_α and Y'_α are Noetherian and A''_α is integrally closed. On the other hand, as by virtue of (11.3.1), one can suppose f flat (11.3.1); on the other hand, let $Y' = \mathrm{Spec}(A')$ and let y' be the closed point of Y', so that the morphism $g : Y' \to Y$ is entire and birational. If one sets $X' = X \times_Y Y'$ and if $f' : X' \to X'$ and $g' : X' \to X$ are the projections, g' is entire and birational (6.15.3), and if the unique point of $g'^{-1}(x)$, which is above x, by (2.3.7) one can suppose that every irreducible component of X'_β dominates an irreducible component of X_β. One concludes then from (5.13.4), applied to the preschemes $\mathrm{Spec}(\mathscr{O}_{X,x_\alpha} \otimes_{A_0} A''_\alpha)$, that X is normal at the point x. □

Corollary (11.3.14). — *Let $f : X \to Y$ be a morphism of finite presentation, x a point of X, $y = f(x)$. If Y is geometrically unibranch at the point y* (6.15.1) *and if the morphism f is normal at the point x* (6.8.1)*, then X is geometrically unibranch at the point x.*

Proof. One may evidently restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, A being a local ring integral geometrically unibranch, y being the closed point of Y. Let A' the integral closure of A; set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ and let y' the closed point of Y', so that the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is entire and birational. If one sets $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ and if $f' : \mathrm{X}' \to \mathrm{Y}'$ and $g' : \mathrm{X}' \to \mathrm{X}$ are the projections, g' is entire and birational (6.15.3), and x' the unique point of $g'^{-1}(x)$, which is above x. The morphism f' is of finite presentation and normal at the point x' (6.7.8), and by the local ring $\mathscr{O}_{\mathrm{X}',x'}$ is integral and integrally closed (11.3.13). On the other hand, on can suppose f flat (11.3.1); on the other hand, g' is a morphism birational (6.15.4.1); by (11.3.1), $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is irreducible, and as $\mathscr{O}_{\mathrm{X},x}$ is reduced (11.3.13), it is integral and geometrically unibranch (6.15.5). □ IV-3 | 141

Proposition (11.3.15). — *Let* A *be a ring,* B *an* A*-algebra of finite presentation,* M *a* B*-module of finite presentation, which is a flat* A*-module. Then there exists a finite sequence* $(f_i)_{1 \leqslant i \leqslant n}$ *of elements of* A*, such that the ideal generated by the* f_i *is equal to* A*, and that, for* $0 \leqslant i < n$*,* $\mathrm{M}_{f_{i+1}}/(\sum_{j=1}^{i} f_j \mathrm{M}_{f_{i+1}})$ *is a flat* $(\mathrm{A}_{f_{i+1}}/(\sum_{j=1}^{i} f_j \mathrm{A}_{f_{i+1}}))$*-module* libre.

Proof. One may say again that the $\mathrm{D}(f_i)$ form an open covering of $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, and then, by induction, $\mathrm{Z}_{i+1} = \mathrm{D}(f_{i+1}) \cap \mathrm{V}(f_1\mathrm{A} + \ldots + f_i\mathrm{A})$, the Z_i form a *partition* of X in locally closed sets, $\mathrm{A}_{f_{i+1}}/(\sum_{j=1}^{i} f_j \mathrm{A}_{f_{i+1}})$ being the ring of an affine sub-prescheme of X having Z_{i+1} for underlying space.

To prove the proposition, one can first, by virtue of (11.2.7), suppose that there exists a sub-ring Noetherian A_0 of A, an A_0-algebra B_0 of finite type and a B_0-module of finite type M_0, *flat* on A_0, such that $\mathrm{B} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$, $\mathrm{M} = \mathrm{M}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$; it is clear that it will suffice to prove the proposition for A_0, B_0 and M_0, since if the elements $f_i \in \mathrm{A}_0$ satisfy the conditions of the statement, they will verify also these conditions for A, B, M, since $\mathrm{M}_{f_{i+1}}/(\sum_{j=1}^{i} f_j \mathrm{M}_{f_{i+1}}) = ((\mathrm{M}_0)_{f_{i+1}}/(\sum_{j=1}^{i} f_j (\mathrm{M}_0)_{f_{i+1}})) \otimes_{\mathrm{A}_0} \mathrm{A}$ (Bourbaki, *Alg. comm.*, chap. II, § 2, n° 7, prop. 18). One can therefore restrict to the case where A is Noetherian.

Let us note now that if C is a Noetherian ring, $\mathfrak{R}$ its nilradical and P a flat C-module, it follows from $(\mathbf{0}_{\mathrm{III}}, 10.1.2)$ that for P to be a free C-module libre, it is necessary and sufficient that $\mathrm{P} \otimes_{\mathrm{C}} (\mathrm{C}/\mathfrak{R})$ be a free (C/R)-module (and the ranks are equal). □

11.4. Descent of flatness by arbitrary morphisms: the case of an Artinian base

Theorem (11.4.1). — *Let* A *be a ring,* $\mathfrak{J}$ *a nilpotent ideal of* A*,* $u_\lambda : \mathrm{A} \to \mathrm{B}_\lambda$ $(\lambda \in \mathrm{L})$ *a family of homomorphisms of rings such that the intersection of the kernels of the* u_λ *is reduced to* 0*. Let* M *be an* A*-module such that for all* $\lambda \in \mathrm{L}$*,* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$ *is a* B_λ*-module free and that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *is an* $(\mathrm{A}/\mathfrak{J})$*-module free. Then* M *is an* A*-module free. If moreover the index set* L *is finite, one can replace everywhere "free" by "flat" in the preceding statement.*

Proof. In both cases it will suffice to prove that M is an A-module *flat*: indeed, when $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ is an $(\mathrm{A}/\mathfrak{J})$-module free, it will follow that M is an A-module free by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.1.2)$.

We will use the following lemma, which generalises $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$:

Lemma (11.4.1.1). — *Let* A *be a ring equipped with a finite filtration* $(\mathfrak{J}_n)_{0 \leqslant n \leqslant \mathrm{N}+1}$ *with* $\mathfrak{J}_0 = \mathrm{A}$*,* $\mathfrak{J}_{\mathrm{N}+1} = 0$*. Let* M *be an* A*-module equipped with the filtration* $(\mathfrak{J}_n \mathrm{M})_{0 \leqslant n \leqslant \mathrm{N}+1}$*, and let us denote by* $\mathrm{gr}(\mathrm{A})$ *and* $\mathrm{gr}(\mathrm{M})$ *the ring and the graded module respectively. Suppose that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}_1)$ *is an* $(\mathrm{A}/\mathfrak{J}_1)$*-module flat and that the canonical homomorphism*

$$\mathrm{gr}_0(\mathrm{M}) \otimes_{\mathrm{gr}_0(\mathrm{A})} \mathrm{gr}(\mathrm{A}) \longrightarrow \mathrm{gr}(\mathrm{M}) \tag{11.4.1.2}$$

is injective. Then M *is an* A*-module flat.*

The canonical homomorphism (11.4.1.2) is defined in the same way as that of $(\mathbf{0}_{\mathrm{III}}, 10.1.1)$ as being in degree n the homomorphism

$$(\mathrm{M}/\mathfrak{J}_1\mathrm{M}) \otimes (\mathfrak{J}_n/\mathfrak{J}_{n+1}) = (\mathrm{M} \otimes \mathfrak{J}_n)/(\mathrm{Im}(\mathrm{M} \otimes \mathfrak{J}_{n+1}) + \mathrm{Im}(\mathfrak{J}_1\mathrm{M} \otimes \mathfrak{J}_n)) \longrightarrow \mathfrak{J}_n\mathrm{M}/\mathfrak{J}_{n+1}\mathrm{M}$$

coming from the canonical homomorphism $\mathrm{M} \otimes \mathfrak{J}_n \to \mathfrak{J}_n\mathrm{M}$ by passage to quotients. The lemma is proved by induction on N, since there is nothing to prove for $\mathrm{N} = 0$. The conditions on M imply, by IV-3 | 144
the induction hypothesis, that $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}_{\mathrm{N}})$ is an $(\mathrm{A}/\mathfrak{J}_{\mathrm{N}})$-module flat. Let us now note that $\mathfrak{J}_{\mathrm{N}}^2 \subset$

$\mathfrak{J}_{N+1} = 0$ if $N \geqslant 1$, and $(M/\mathfrak{J}_1 M) \otimes (\mathfrak{J}_N/\mathfrak{J}_{N+1}) = (M/\mathfrak{J}_N M) \otimes (\mathfrak{J}_N/\mathfrak{J}_{N+1}) = (M/\mathfrak{J}_N M) \otimes \mathfrak{J}_N$; hence the canonical homomorphism

$$(M/\mathfrak{J}_N M) \otimes \mathfrak{J}_N \longrightarrow \mathfrak{J}_N M = \mathfrak{J}_N M/\mathfrak{J}_N^2 M$$

is injective. Applying $(\mathbf{0}_{\mathrm{III}},\ 10.2.1)$ to the $\mathfrak{J}_N$-preadic filtration, we conclude that M is an A-module flat.

To apply this lemma ((11.4.1)), we will denote by $\mathfrak{J}_n$ the ideal of A which is the intersection of the inverse images $u_\lambda^{-1}(\mathfrak{J}^n B_\lambda)$: it is immediate that $\mathfrak{J}_0 = A$, $\mathfrak{J}_m\mathfrak{J}_n \subset \mathfrak{J}_{m+n}$ for $m \geqslant 0$, $n \geqslant 0$; moreover, if $\mathfrak{J}^{N+1} = 0$, we also have $\mathfrak{J}_{N+1} = 0$ since the intersection of the kernels of the u_λ is reduced to 0.

Let us equip A with the filtration $(\mathfrak{J}_n)$, M with the filtration $(\mathfrak{J}_n M)$, and, for each λ, let us equip B_λ and $N_\lambda = M \otimes_A B_\lambda$ with the $\mathfrak{J}$-preadic filtrations; let us consider for each λ the commutative diagram

$$\begin{tikzcd}
\mathrm{gr}^0(M) \otimes_{\mathrm{gr}^0(A)} \mathrm{gr}(A) \arrow[r, "\varphi"] \arrow[d, "f_\lambda"'] & \mathrm{gr}(M) \arrow[d, "g_\lambda"] \\
\mathrm{gr}^0_{\mathfrak{J}}(N_\lambda) \otimes_{\mathrm{gr}^0_{\mathfrak{J}}(B_\lambda)} \mathrm{gr}_{\mathfrak{J}}(B_\lambda) \arrow[r, "\varphi_\lambda"] & \mathrm{gr}_{\mathfrak{J}}(N_\lambda)
\end{tikzcd}$$

where the horizontal arrows are the canonical homomorphisms (11.4.1.2) and the vertical arrows are deduced from the canonical homomorphisms $A \to B_\lambda$ and $M \to N_\lambda$. The hypothesis that N_λ is a B_λ-module flat implies that φ_λ is injective $(\mathbf{0}_{\mathrm{III}},\ 10.2.1)$, hence $\mathrm{Ker}(\varphi) \subset \mathrm{Ker}(f_\lambda)$. Setting $A_0 = A/\mathfrak{J}_1$, $M_0 = M \otimes_A A_0$, let us note that

$$\mathrm{gr}^0_{\mathfrak{J}}(N_\lambda) = N_\lambda/\mathfrak{J}N_\lambda = (M/\mathfrak{J}M) \otimes_A B_\lambda = (M/\mathfrak{J}_1 M) \otimes_A (B_\lambda/\mathfrak{J}B_\lambda)$$

and therefore

$$\mathrm{gr}^0_{\mathfrak{J}}(N_\lambda) \otimes_{\mathrm{gr}^0_{\mathfrak{J}}(B_\lambda)} \mathrm{gr}_{\mathfrak{J}}(B_\lambda) = M_0 \otimes_{A_0} \mathrm{gr}_{\mathfrak{J}}(B_\lambda).$$

The homomorphism f_λ can therefore be written **IV-3** | 145

$$1 \otimes \mathrm{gr}(u_\lambda) : M_0 \otimes_{A_0} \mathrm{gr}(A) \longrightarrow M_0 \otimes_{A_0} \mathrm{gr}_{\mathfrak{J}}(B_\lambda),$$

and since by hypothesis M_0 is an A_0-module *flat*, the kernel of f_λ is equal to $M_0 \otimes_{A_0} R_\lambda$, where R_λ is the kernel of $\mathrm{gr}(u_\lambda) : \mathrm{gr}(A) \to \mathrm{gr}_{\mathfrak{J}}(B_\lambda)$. It therefore suffices to prove that $\bigcap_{\lambda\in L}(M_0 \otimes_{A_0} R_\lambda) = 0$. Now, by definition of the $\mathfrak{J}_n$, the intersection of the kernels of the homomorphisms $\mathfrak{J}_n A/\mathfrak{J}_{n+1}A \to \mathfrak{J}^n B_\lambda/\mathfrak{J}^{n+1}B_\lambda$, as λ ranges over L, is reduced to 0, otherwise said $\bigcap_{\lambda\in L} R_\lambda = 0$. This being the case, suppose first that M_0 is an A_0-module free; taking a basis of M_0, we see immediately that $M_0 \otimes_{A_0} (\bigcap_{\lambda\in L} R_\lambda) = \bigcap_{\lambda\in L}(M_0 \otimes_{A_0} R_\lambda)$, whence the proposition in this case. When L is finite, the preceding formula is still true under the sole hypothesis that M_0 is an A_0-module flat $(\mathbf{0}_{\mathrm{I}},\ 6.1.3)$, which finishes the proof. □

Remark (11.4.2). — The conclusion of (11.4.1) may be inexact if L is infinite and one only supposes that $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-module flat. For example, let V be a discrete valuation ring, K its field of fractions, and let $A = V[T]/(T^2)$ (T an indeterminate); let us take for $\mathfrak{J}$ the image of (T) in A, so that $A/\mathfrak{J} = V$, and let us take $M = K$, which is an $(A/\mathfrak{J})$-module, hence equal to $M \otimes_A (A/\mathfrak{J})$; moreover M is an $(A/\mathfrak{J})$-module flat, but not an A-module flat, since $\mathfrak{J}$ is nilpotent, it would follow from $(\mathbf{0}_{\mathrm{III}},\ 10.1.2)$ that M would be an A-module free, which is absurd since $\mathfrak{J}K = 0$. On the other hand the maximal ideal $\mathfrak{m}$ of the local Noetherian ring A; on the other hand $M \otimes_A (A/\mathfrak{m}^n) = 0$ for all integers n; the $(A/\mathfrak{m}^n)$-modules $M \otimes_A (A/\mathfrak{m}^n)$ are therefore flat for all n, and the intersection of $\mathfrak{m}^n$ is reduced to 0.

Corollary (11.4.3). — *Let* A *be a semi-local ring whose radical* $\mathfrak{J}$ *is nilpotent (for example an Artinian ring),* $u_\lambda : A \to B_\lambda$ $(\lambda \in L)$ *a family of homomorphisms of rings such that the intersection of the kernels of the* u_λ *is reduced to* 0*. For an* A*-module* M *to be flat, it is necessary and sufficient that for all* $\lambda \in L$, $M \otimes_A B_\lambda$ *be a* B_λ*-module flat.*

Proof. Since $A/\mathfrak{J}$ is a direct product of a finite number of fields (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 5, prop. 16) and that $\mathfrak{J}$ is nilpotent, A is a direct product of a finite number of *local* rings A_i whose radical is nilpotent (*loc. cit.*, § 4, n° 3, cor. 1 of prop. 15) and M is therefore a direct sum of A_i-modules

M_i, each M_i being annihilated by the A_j of index $j \neq i$; for M to be a flat A-module, it is necessary and it suffices that each of the M_i be a flat A_i-module; moreover the intersection of the kernels of the homomorphisms $\mathrm{A}_i \xrightarrow{u_i} \mathrm{B}_\lambda$ is reduced to 0, and $\mathrm{M}_i \otimes_{\mathrm{A}_i} \mathrm{B}_\lambda$ is a direct factor of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$. One can therefore restrict to the case where A is also *local*. Then $\mathrm{A}/\mathfrak{J}$ is a field, hence $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ is an $(\mathrm{A}/\mathfrak{J})$-module free, and it suffices to see that for all λ, $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$ is a B_λ-module *free*, by virtue of (11.4.1). But if one sets $\mathfrak{J}_\lambda = \mathfrak{J}\mathrm{B}_\lambda$, $(\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda) \otimes_{\mathrm{B}_\lambda} (\mathrm{B}_\lambda/\mathfrak{J}_\lambda) = (\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})) \otimes_{\mathrm{A}/\mathfrak{J}} (\mathrm{B}_\lambda/\mathfrak{J}_\lambda)$ is a $(\mathrm{B}_\lambda/\mathfrak{J}_\lambda)$-module free, and $\mathfrak{J}_\lambda$ is nilpotent. The conclusion therefore follows from the hypothesis that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{B}_\lambda$ is a B_λ-module flat and from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2). □

Corollary (11.4.4). — *Let* A *be a ring,* M *an* A*-module; suppose that there exists a nilpotent ideal* $\mathfrak{J}$ *of* A *such that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *is an* $(\mathrm{A}/\mathfrak{J})$*-module free. Then the set of ideals* $\mathfrak{K}$ *of* A *such that* $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{K})$ *is an* $(\mathrm{A}/\mathfrak{K})$*-module free admits a smallest element* $\mathfrak{K}_0$ *(which is also the smallest of the ideals* $\mathfrak{K}$ *such that* **IV-3 | 146**
$\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{K})$ *is an* $(\mathrm{A}/\mathfrak{K})$*-module* flat*).*

Proof. For any homomorphism $u : \mathrm{A} \to \mathrm{A}'$ such that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$ is an A'-module free (resp. an A'-module flat), it is necessary and it suffices that u factorise through $\mathrm{A} \to \mathrm{A}/\mathfrak{K}_0 \to \mathrm{A}'$ (or equivalently that $\mathfrak{K}_0\mathrm{A}' = 0$).

The fact that the intersection $\mathfrak{K}_0$ of the ideals $\mathfrak{K}$ for which $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{K})$ is an $(\mathrm{A}/\mathfrak{K})$-module free is itself the smallest of these ideals follows from (11.4.1) applied to the ring $\mathrm{A}/\mathfrak{K}_0$, to a nilpotent ideal $\mathfrak{J}/\mathfrak{K}_0$ and to the homomorphisms $\mathrm{A}/\mathfrak{K}_0 \to \mathrm{A}/\mathfrak{K}$, whose kernels have 0 for intersection. If A' is an A-algebra such that $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$ is an A'-module free and if $\mathfrak{K}$ is the kernel of the homomorphism $\mathrm{A} \to \mathrm{A}'$, it follows from (11.4.1) applied to the ring $\mathrm{A}/\mathfrak{K}$, to the $(A/\mathfrak{K})$-module $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{K})$, to the nilpotent ideal $\mathfrak{J}\mathfrak{K}/\mathfrak{K}$ and to the *injective* homomorphism $\mathrm{A}/\mathfrak{K} \to \mathrm{A}'$, that $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{K})$ is an $(\mathrm{A}/\mathfrak{K})$-module free, hence $\mathfrak{K} \supset \mathfrak{K}_0$. The fact that one can replace "free" by "flat" in what precedes (while naturally preserving the hypothesis that $\mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ is an $(\mathrm{A}/\mathfrak{J})$-module *free*) follows from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2), as we saw at the beginning of the proof of (11.4.1). □

Proposition (11.4.5). — *Let* Y *be an irreducible prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism of finite presentation,* $\mathscr{F}$ *an* $\mathscr{O}_{\mathrm{X}}$*-Module of finite presentation. There exists an open non-empty* U *in* Y *and a closed sub-prescheme* Z *of* U*, of finite presentation over* U*, having the following property: for every base change* $\mathrm{U}' \to \mathrm{U}$*, if we set* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{U}'$*,* $f' = f_{|\mathrm{U}'}$ *and* $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{U}'}$*, then, for* $\mathscr{F}'$ *to be* f'*-flat, it is necessary and sufficient that the morphism* $\mathrm{U}' \to \mathrm{U}$ *factorise through* $\mathrm{U}' \to \mathrm{Z} \to \mathrm{U}$*. Such a scheme* Z *has the same underlying space as* U*. Suppose moreover that* Y *is affine, and let* (W_i) *be a finite affine open covering of* X*; then, one can suppose* U *chosen such that, if* $\mathrm{U}' = \mathrm{Spec}(\mathrm{A}')$ *is an affine scheme and* $\mathrm{U}' \to \mathrm{U}$ *a morphism that factorises through* $\mathrm{U}' \to \mathrm{Z} \to \mathrm{U}$*, each of the* $\Gamma(\mathrm{W}_i \times_{\mathrm{Y}} \mathrm{U}', \mathscr{F}')$ *is a free* A'*-module.*

Proof. One can evidently restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is affine. Using ($\mathbf{0}$, 8.9.1), there is a Noetherian sub-ring A_1 of A, a morphism of finite type $f_1 : \mathrm{X}_1 \to \mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}_1)$ and a coherent $\mathscr{O}_{\mathrm{X}_1}$-Module $\mathscr{F}_1$ such that $f = f_{|\mathrm{Y}}$ and $\mathscr{F} = \mathscr{F}_1 \otimes_{\mathscr{O}_{\mathrm{Y}_1}} \mathscr{O}_{\mathrm{Y}}$; one can moreover suppose that the W_i are inverse images of affine opens of X_1. Let us note moreover that Y_1 is irreducible, the morphism $\mathrm{Y} \to \mathrm{Y}_1$ being dominant ($\mathbf{I}$, 1.2.7). Suppose the proposition proved for Y_1, X_1 and $\mathscr{F}_1$, and let U_1 and Z_1 be the open set of Y_1 and the closed sub-prescheme of U_1 having the required properties. Then, if $\mathrm{U}' \to \mathrm{U}$ is a base change such that $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{U}'}$ is f'-flat, the morphism $\mathrm{U}' \to \mathrm{U}_1$ factorises through $\mathrm{U}' \to \mathrm{Z}_1 \to \mathrm{U}_1$; but since $\mathrm{U}' \to \mathrm{U} \to \mathrm{U}_1$ also factorises through $\mathrm{U}' \to \mathrm{U} \to \mathrm{U}_1$, the definition of the product of preschemes shows that $\mathrm{U}' \to \mathrm{U}$ factorises through $\mathrm{U}' \to \mathrm{Z} \to \mathrm{U}$.

One can therefore restrict to the case where A is *Noetherian*; let $\mathfrak{N}$ be its nilradical, which is nilpotent, and set $\mathrm{A}_0 = \mathrm{A}_{\mathrm{red}} = \mathrm{A}/\mathfrak{N}$, $\mathrm{Y}_0 = \mathrm{Spec}(\mathrm{A}_0) = \mathrm{Y}_{\mathrm{red}}$, $\mathrm{X}_0 = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_0$, $\mathscr{F}_0 = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathscr{O}_{\mathrm{Y}_0}$, $\mathrm{W}_{i0} = \mathrm{W}_i \times_{\mathrm{Y}} \mathrm{Y}_0$; if $\mathrm{M}_i = \Gamma(\mathrm{W}_i, \mathscr{F})$, $\mathrm{M}_{i0} = \Gamma(\mathrm{W}_{i0}, \mathscr{F}_0)$ is therefore equal to $\mathrm{M}_i \otimes_{\mathrm{A}} \mathrm{A}_0$. Since A_0 is *integral*, one can, by virtue of the generic flatness theorem (6.9.2), replacing Y by an open non-empty set of Y as needed, suppose that the M_{i0} are free A_0-modules. By virtue of (11.4.4), there is therefore for each i an ideal $\mathfrak{J}_i \subset \mathfrak{N}$ such that the A-algebras A' for which the $\mathrm{M}_i \otimes_{\mathrm{A}} \mathrm{A}'$ are A'-modules free **IV-3 | 147**
(or flat) are exactly those for which $\mathfrak{J}_i\mathrm{A}' = 0$. Let $\mathfrak{J} = \sum_i \mathfrak{J}_i$, which is an ideal contained in $\mathfrak{N}$; to say that $\mathscr{F} \otimes_{\mathscr{O}_{\mathrm{Y}}} \mathrm{A}'$ is A'-flat amounts to saying that the $\mathrm{M}_i \otimes_{\mathrm{A}} \mathrm{A}'$ are all A'-modules flat, hence that $\mathfrak{J}\mathrm{A}' = 0$. It follows that if one takes $\mathrm{Z} = \mathrm{V}(\mathfrak{J})$, one answers the question, since for any morphism $\mathrm{U}' \to \mathrm{U}$ having the property of the statement, it is necessary and it suffices that for every affine open $\mathrm{V}' \subset \mathrm{U}'$, the morphism $\mathrm{V}' \to \mathrm{U}$ have the same property. □

Corollary (11.4.6). — *Let* A *be a ring such that* $\operatorname{Spec}(A)$ *is irreducible,* $\mathfrak{p}$ *its unique minimal prime ideal,* B *an* A*-algebra of finite presentation,* M *a* B*-module of finite presentation. Suppose that* $M_{\mathfrak{p}}$ *is a flat* $A_{\mathfrak{p}}$*-module; then there exists* $t \in A - \mathfrak{p}$ *such that* M_t *is a free* A_t*-module.*

Proof. Applying (11.4.5) with $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $\mathscr{F} = \widetilde{M}$, one can (replacing A by A_h as needed, where $h \in A - \mathfrak{p}$) suppose that there exists an ideal of finite type $\mathfrak{J}$ in A such that the A-algebras A' for which $M \otimes_A A'$ is an A'-module free (or flat) are exactly those such that $\mathfrak{J}A' = 0$. In particular, the hypothesis implies that $M_{\mathfrak{p}}$ is a flat $A_{\mathfrak{p}}$-module, hence $\mathfrak{J}A_{\mathfrak{p}} = 0$, or $\mathfrak{J}_{\mathfrak{p}} = 0$. But since $\mathfrak{J}$ is of finite type, there exists $t \in A - \mathfrak{p}$ such that $\mathfrak{J}_t = 0$, or equivalently $\mathfrak{J}A_t = 0$, and therefore M_t is a free A_t-module. □

Proposition (11.4.7). — *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A, M *an* A*-module ideally separated* $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$ *for the* $\mathfrak{J}$*-preadic topology. Let* $(\mathfrak{p}_i)_{1\leqslant i\leqslant r}$ *be a finite family of prime ideals of* A *containing* $\mathfrak{J}$*; for every integer* $n \geqslant 0$*, let* $\mathfrak{p}_i^{(n)}$ *be the* n*-th symbolic power (i.e. the kernel of the homomorphism* $A \to A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i}$*); set* $\mathfrak{J}_n = \bigcap_{i=1}^r \mathfrak{p}_i^{(n)}$*, so that* $\mathfrak{J}_n \supset \mathfrak{J}^n$*, and suppose that the topology defined by the filtration* $(\mathfrak{J}_n)$ *is identical to the* $\mathfrak{J}$*-preadic topology (in other words, for every* n*, there exists* m *such that* $\mathfrak{J}^m \supset \mathfrak{J}_n$*). For* M *to be an* A*-module flat, it is necessary and sufficient that* $M \otimes_A (A/\mathfrak{J})$ *be an* $(A/\mathfrak{J})$*-module flat and that, for all* i*,* $M \otimes_A A_{\mathfrak{p}_i} = M_{\mathfrak{p}_i}$ *be a flat* $A_{\mathfrak{p}_i}$*-module.*

Proof. Since M is ideally separated, it suffices, by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$, to show that, for every $n \geqslant 0$, $M \otimes_A (A/\mathfrak{J}^n)$ is an $(A/\mathfrak{J}^n)$-module flat; since every $\mathfrak{J}^n$ contains a $\mathfrak{J}_m$, it amounts to the same to prove that for every n, $M \otimes_A (A/\mathfrak{J}_n)$ is an $(A/\mathfrak{J}_n)$-module flat. Now, since $\mathfrak{J}_1 \supset \mathfrak{J}$, $M \otimes_A (A/\mathfrak{J}_1)$ is an $(A/\mathfrak{J}_1)$-module flat; in the ring $A/\mathfrak{J}_n$, the ideal $\mathfrak{J}_1/\mathfrak{J}_n$ is nilpotent and finally the intersection of the kernels of the homomorphisms $A/\mathfrak{J}_n \to A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i}$ is zero in $A/\mathfrak{J}_n$, by definition of $\mathfrak{J}_n$. It therefore suffices, by (11.4.1), to verify that $M \otimes_A (A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i})$ is an $(A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i})$-module flat, which follows from the hypothesis that $M \otimes_A A_{\mathfrak{p}_i}$ is a flat $A_{\mathfrak{p}_i}$-module. □

Remark (11.4.8). — The hypothesis made in (11.4.7) on the topology defined by the $\mathfrak{J}_n$ is satisfied if, for every n sufficiently large, $\operatorname{Ass}(A/\mathfrak{J}^n)$ *is contained in the set of the* $\mathfrak{p}_i$. Indeed, $\mathfrak{J}^n$ is then an intersection of primary ideals for the $\mathfrak{p}_i$, each of which contains a symbolic power of $\mathfrak{p}_i$, whence the conclusion. In particular:

Corollary (11.4.9). — *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *a nilpotent ideal of* A, M *an* A*-module. For* M *to be an* A*-module flat (resp. free), it is necessary and sufficient that* $M \otimes_A (A/\mathfrak{J})$ *be an* $(A/\mathfrak{J})$*-module flat (resp. free) and that for every prime ideal* $\mathfrak{p} \in \operatorname{Ass}(A)$*,* $M_{\mathfrak{p}}$ *be a flat* $A_{\mathfrak{p}}$*-module.*

Corollary (11.4.10). — *Let* A *be a Noetherian ring,* $\mathfrak{J}$ *an ideal of* A, M *an* A*-module. Suppose that* M *is ideally separated for the* $\mathfrak{J}$*-preadic topology and that* $\operatorname{gr}_{\mathfrak{J}}(A)$ *is an* $(A/\mathfrak{J})$*-module flat. For* M *to be an* A*-module flat, it is necessary and sufficient that* $M \otimes_A (A/\mathfrak{J})$ *be an* $(A/\mathfrak{J})$*-module flat and that for every* $\mathfrak{p} \in \operatorname{Ass}(A/\mathfrak{J})$*,* $M_{\mathfrak{p}}$ *be a flat* $A_{\mathfrak{p}}$*-module.*

Proof. Taking into account (11.4.8), it suffices to show that $\operatorname{Ass}(A/\mathfrak{J}^n)$ is contained in $\operatorname{Ass}(A/\mathfrak{J})$ for every n. Now, if $a \in A$ does not belong to any $\mathfrak{p} \in \operatorname{Ass}(A/\mathfrak{J})$, the homothety of ratio a is injective in $A/\mathfrak{J}$; since each of the $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ is an $(A/\mathfrak{J})$-module flat, a is also a $(\mathfrak{J}^k/\mathfrak{J}^{k+1})$-regular element, hence a is a $(A/\mathfrak{J}^n)$-regular element for every n (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 1 of th. 1), and therefore it does not belong to any prime ideal associated to $A/\mathfrak{J}^n$, whence the corollary (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2). □ **IV-3** | 148

Proposition (11.4.11). — *Let* A *be an Artinian local ring,* $\mathfrak{m}$ *its maximal ideal,* $Y = \operatorname{Spec}(A)$*,* y *the unique point of* Y*,* $f : X \to Y$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module. Let* (A'_α) *be a family of local rings, and for each* α*,* $u_\alpha : A \to A'_\alpha$ *a homomorphism of rings (necessarily local). Set* $Y'_\alpha = \operatorname{Spec}(A'_\alpha)$*,* $X'_\alpha = X \times_Y Y'_\alpha$*,* $f'_\alpha = f_{|Y'_\alpha}$*,* $\mathscr{F}'_\alpha = \mathscr{F} \otimes_A A'_\alpha$*. Let* x *be a point of* X*, and suppose the following conditions satisfied:*

label=(i) *The intersection of the kernels of the* u_α *is reduced to* 0*.*

lbbel=(ii) *The extension* $k(x)$ *of the residue field* k *of* A *is primary* (4.3.1) *(a condition automatically satisfied if* k *is separably closed).*

lcbel=(iii) *For all* α*, there exists a point* $x'_\alpha \in X'_\alpha$ *whose projections in* X *and* Y'_α *are* x *and the closed point* y'_α *of* Y'_α*, and such that* $\mathscr{F}'_\alpha$ *is* Y'_α*-flat at the point* x'_α*.*

Under these conditions, $\mathscr{F}$ is f-flat at the point x.

Proof. The question being local on X, one can evidently restrict to the case where f is a morphism of finite type and suppose $\mathscr{F}_x \neq 0$. We will proceed in several steps.

I) *Reduction to the case where A'_α is a local ring of a* Y*-prescheme of finite type and where the residue field k'_α of A'_α is a finite extension of k.*

Since the reduction is done separately on each A'_α, one can suppress the index α in this part. Let $\mathfrak{m}'$ be the maximal ideal of A'. Let us consider A' as an inductive limit of its sub-A-algebras of finite type B_λ, and set $\mathfrak{n}_\lambda = \mathfrak{m}' \cap \mathrm{B}_\lambda$; A' is also an inductive limit of its local sub-rings $\mathrm{B}'_\lambda = (\mathrm{B}_\lambda)_{\mathfrak{n}_\lambda}$ (5.13.3), and one has evidently $\mathfrak{m}' \cap \mathrm{B}'_\lambda = \mathfrak{n}'_\lambda$, maximal ideal of B'_λ. It follows therefore (11.2.6) that in setting $\mathrm{Z}'_\lambda = \mathrm{Spec}(\mathrm{B}'_\lambda)$, $\mathscr{F} \otimes_{\mathrm{A}} \mathrm{B}'_\lambda$ is Z'_λ-flat at the point x'_λ, projection of x', the projection of x'_λ on Z'_λ being the closed point z'_λ of Z'_λ.

One can therefore suppose that there exists an affine scheme Z' of finite type over Y and a point z' of Z' such that $\mathrm{A}' = \mathscr{O}_{\mathrm{Z}',z'}$, and that, if one sets $\mathrm{T}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Z}'$, there exists a point $t' \in \mathrm{T}'$ whose projections in Z' and X are z' and x, and such that $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Z}'$ is Z'-flat at the point t'. Let Z'_1 (resp. X_1) be a closed sub-prescheme of Z' (resp. X) having for underlying space the closure of $\{z'\}$ (resp. $\{x\}$), and set $\mathrm{T}'_1 = \mathrm{X}_1 \times_{\mathrm{Y}} \mathrm{Z}'_1$. The set U of the points of T' where $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Z}'$ is Z'-flat is open in T' (11.1.1) and contains t', hence $\mathrm{V} = \mathrm{U} \cap \mathrm{T}'_1$ is open non-empty in T'_1. The ring A, being Artinian, is a ring **IV-3** | 149
of Jacobson, hence (10.4.6) Z'_1 and T'_1 are Jacobson preschemes; there exists therefore in V a point t'_0 closed in V, and its image z'_0 in Z'_1 is a *closed* point of Z'_1 (10.4.7). Let f_1 be the restriction of f to X_1, $f'_1 : \mathrm{T}'_1 \to \mathrm{Z}'_1$ the canonical projection; $\mathrm{V} \cap f_1'^{-1}(z'_0)$ is an open non-empty set in $f_1'^{-1}(z'_0)$, and since this last prescheme is flat over $k(z'_0)$ (2.3.4), a maximal point t'_1 of $\mathrm{V} \cap f_1'^{-1}(z'_0)$ is necessarily above the unique maximal point x of X_1 (2.3.4). Finally, $k(z'_0)$ is a *finite* extension of k (**I**, 6.4.2) and the homomorphism $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y} \to \mathrm{A}' = \mathscr{O}_{\mathrm{Z}',z'} \to \mathscr{O}_{\mathrm{Z}',z'_0}$ factorises as $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z}',z'} \to \mathscr{O}_{\mathrm{Z}',z'_0}$, hence its kernel is contained in that of $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z}',z'_0}$. This achieves the reduction announced.

II) *Reduction to the case where the A'_α are in finite number and are finite* A*-algebras.* — Let $\mathfrak{m}'_\alpha$ be the maximal ideal of A'_α; since A'_α is a local Noetherian ring, the intersection of the $\mathfrak{m}_\alpha'^n$ is 0 ($\mathbf{0}_{\mathrm{I}}$, 7.3.5); the intersection of the $u_\alpha^{-1}(\mathfrak{m}_\alpha'^n)$, for all indices n and α, is therefore also reduced to 0 by hypothesis (i). Since A is Artinian, there is a finite number of these ideals whose intersection is already 0; let us note the $u_\alpha^{-1}(\mathfrak{m}_i'^{n_i})$ $(1 \leqslant i \leqslant r)$. Since the A'_i are Noetherian, the $\mathfrak{m}_i'^h/\mathfrak{m}_i'^{h+1}$ are $(\mathrm{A}'_i/\mathfrak{m}'_i)$- modules of finite length, and since $\mathrm{A}'_i/\mathfrak{m}'_i$ is a k-vector space of finite rank, one sees that $\mathrm{A}''_i = \mathrm{A}'_i/\mathfrak{m}_i'^{n_i}$ is an A-algebra *finite* and a local Artinian ring. The reduction announced follows from (2.1.4).

III) *End of the proof.* — Suppose now that the A'_i $(1 \leqslant i \leqslant r)$ are in finite number and are finite A-algebras. For every i, the residue field k_i of A'_i is a finite extension of k; using hypothesis (ii), one concludes that the inverse image of x in X'_i is reduced to a *single* point x'_i (4.3.2). Let Y' be the prescheme $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, sum of the Y'_i. The hypothesis implies that $\mathscr{F}'$ is Y'-flat at the points of X'_i whose the projection in X is x and Y'. Since p is of finite type, there exists by assumption an *open set* $\mathrm{U} \ni p^{-1}(x)$ such that $\mathscr{F}'$ is Y'-flat at the points of U'(11.1.1). Since B and A' (the latter being the direct product of the A'_i); replacing X by U, one has $\mathscr{F} = \widetilde{\mathrm{M}}$, where M is a B-module, and by hypothesis $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$ is an A'-module *flat* (2.1.2); since the homomorphism $\mathrm{A} \to \mathrm{A}'$ is injective by construction, one can apply (11.4.3), which proves that M is an A-module flat. □

Proposition (11.4.12). — *Let* A *be an Artinian local ring of residue field k, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-Module. Let (A'_α) be a family of local rings, and for each α, $u_\alpha : \mathrm{A} \to \mathrm{A}'_\alpha$ a homomorphism of rings. Set $\mathrm{Y}'_\alpha = \mathrm{Spec}(\mathrm{A}'_\alpha)$, $\mathrm{X}'_\alpha = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\alpha$, $f'_\alpha = f_{|\mathrm{Y}'_\alpha}$, $\mathscr{F}'_\alpha = \mathscr{F} \otimes_{\mathrm{A}} \mathrm{A}'_\alpha$. Let x be a point of* X*, and suppose the following conditions satisfied:*

label=(i) The intersection of the kernels of the u_α is reduced to 0.

lbbel=(ii) For all α, $\mathscr{F}'_\alpha$ is f'_α-flat at every point $x'_\alpha \in \mathrm{X}'_\alpha$ whose projections in X *and Y'_α are x and the closed point y'_α of Y'_α.*

Then $\mathscr{F}$ is f-flat at the point x.

Proof. By hypothesis, the intersection of the kernels of the u_α is reduced to 0; since A is Artinian, there are already a *finite* number of these kernels whose intersection is 0, hence one can restrict to the case where the family (A'_α) is *finite*. Let k' be an algebraic closure of k; one knows $(\mathbf{0}_{\mathrm{III}}, 10.3.1)$ that there exists a local homomorphism $\mathrm{A} \to \mathrm{B}$ making B an A-module flat, *entire* over A and such that $\mathrm{B} \otimes_{\mathrm{A}} k$ is isomorphic to k'. By flatness, the kernels of the homomorphisms $\mathrm{B} \to \mathrm{B}'_\alpha = \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}'_\alpha$ are deduced from those of u_α by tensorisation with B, and since they are in finite number, their intersection is reduced to 0 $(\mathbf{0}_{\mathrm{I}}, 6.1.3)$. Let us consider the rings $\mathrm{B}'_{\alpha\beta}$, localised of B'_α at its *maximal* ideals; one knows (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. 2 of th. 1) that the intersection of the kernels of the homomorphisms $\mathrm{B}'_\alpha \to \mathrm{B}'_{\alpha\beta}$ (for a given α) is reduced to 0; one therefore concludes that the intersection of the kernels of the composite homomorphisms $v_{\alpha\beta} : \mathrm{B} \to \mathrm{B}'_\alpha \to \mathrm{B}'_{\alpha\beta}$ (α and β variable) is reduced to 0. On the other hand, since B is entire over A, B'_α is entire over A'_α, hence its maximal ideals are above the maximal ideal of A'_α. If one sets $\mathrm{Z}'_{\alpha\beta} = \mathrm{Spec}(\mathrm{B}'_{\alpha\beta})$, $\mathrm{T}_{\alpha\beta} = \mathrm{X}'_\alpha \times_{\mathrm{Y}'_\alpha} \mathrm{Z}'_{\alpha\beta}$, $\mathscr{G} = \mathscr{F} \otimes_{\mathrm{A}} \mathrm{B}$ and $\mathscr{G}'_{\alpha\beta} = \mathscr{F} \otimes_{\mathrm{B}} \mathrm{B}'_{\alpha\beta}$, one sees that hypotheses (i) and (ii) are still satisfied when one replaces A, A'_α, u_α, $\mathscr{F}$ and x by B, $\mathrm{B}'_{\alpha\beta}$, $v_{\alpha\beta}$, $\mathscr{G}$ and a point t of $\mathrm{T} = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{B}$ above x. Since the residue field of B is separably closed, one deduces from (11.4.11) that $\mathscr{G}$ is flat over B at the point t. But since B is an A-module faithfully flat, by (2.1.4), (2.2.13), from this it follows that $\mathscr{F}$ is flat over A at the point x, which proves (11.4.12). □ **IV-3 | 150**

Corollary (11.4.13). — *Let* A *be an Artinian local ring,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module. Let* (Y'_α) *be a family of* A*-preschemes, and for all* α*, set* $\mathrm{X}'_\alpha = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\alpha$, $f'_\alpha = f_{|\mathrm{Y}'_\alpha}$, $\mathscr{F}'_\alpha = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'_\alpha$*. Let* x *be a point of* X *and suppose the following hypotheses satisfied:*

label=(i) The intersection of the kernels of the homomorphisms $\mathrm{A} \to \Gamma(\mathrm{Y}'_\alpha, \mathscr{O}_{\mathrm{Y}'_\alpha})$ *corresponding to the structural morphisms is reduced to* 0*.*

lbbel=(ii) For all α*,* $\mathscr{F}'_\alpha$ *is* f'_α*-flat at every point* $x'_\alpha \in \mathrm{X}'_\alpha$ *whose projection in* X *is* x*.*

Then $\mathscr{F}$ *is* f*-flat at the point* x*.*

Proof. Indeed, for every $y'_\alpha \in \mathrm{Y}'_\alpha$, let us consider the local scheme $\mathrm{Y}''_{\alpha,y'_\alpha} = \mathrm{Spec}(\mathscr{O}_{y'_\alpha})$; by virtue of (2.1.4), $\mathscr{F}'_\alpha \otimes_{\mathscr{O}_{\mathrm{Y}'_\alpha}} \mathscr{O}_{\mathrm{Y}'_\alpha}$ is $\mathrm{Y}''_{\alpha,y'_\alpha}$-flat at the points whose projections on X and $\mathrm{Y}''_{\alpha,y'_\alpha}$ are x and the closed point of $\mathrm{Y}''_{\alpha,y'_\alpha}$, since we are immediately reduced to the case where Y'_α is affine, and it suffices then to apply Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. 2 of th. 1. Replacing the family (Y'_α) by the family of the $\mathrm{Y}''_{\alpha,y'_\alpha}$, one is therefore brought back to (11.4.12). □

11.5. Descent of flatness by arbitrary morphisms: the general case

Theorem (11.5.1). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module,* x *a point of* X*,* $y = f(x)$*. Let* (Y'_α) *be a family of local* Y*-preschemes* $\mathrm{Y}'_\alpha = \mathrm{Spec}(\mathrm{A}'_\alpha)$ *such that the images of the closed points* y'_α *of* Y'_α *are all equal to* y*. For all* α*, let* $\mathfrak{m}'_\alpha$ *be the maximal ideal of* A'_α*, and* $u_\alpha : \mathscr{O}_y \to \mathrm{A}'_\alpha$ *the canonical homomorphism* (**I**, 2.4.4)*; suppose that the intersections of the ideals* $u_\alpha^{-1}(\mathfrak{m}'^{n\alpha}_\alpha)$ *form a fundamental system of neighbourhoods of* 0 *in* $\mathscr{O}_y$*. Set* $\mathrm{X}'_\alpha = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\alpha$, $f'_\alpha = f_{|\mathrm{Y}'_\alpha}$, $\mathscr{F}'_\alpha = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'_\alpha$*, and suppose one of the following hypotheses satisfied:* **IV-3 | 151**

label=(i) For all α*,* $\mathscr{F}'_\alpha$ *is* f'_α*-flat at every point whose projection in* X *is equal to* x *and whose projection in* Y'_α *is equal to* y'_α*.*

lbbel=(ii) For all α*, there exists a point* $x'_\alpha \in \mathrm{X}'_\alpha$ *whose projection on* X *is* x *and whose projection in* Y'_α *is equal to* y'_α*, such that* $\mathscr{F}'_\alpha$ *is* f'_α*-flat at the point* x'_α*, and* $k(x)$ *is a primary extension of* $k(y)$*.*

Under these conditions, $\mathscr{F}$ *is* f*-flat at the point* x*.*

Proof. Let $\mathfrak{m}_y$ be the maximal ideal of $\mathscr{O}_y$; since $\mathscr{O}_y$ and $\mathscr{O}_x$ are Noetherian and $\mathscr{O}_y \to \mathscr{O}_x$ a homomorphism of local rings, it suffices, by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.2.2)$, to prove that for every $n > 0$, $\mathscr{F}_x / \mathfrak{m}_y^n \mathscr{F}_x$ is an $(\mathscr{O}_y/\mathfrak{m}_y^n)$-module flat. Let us denote by $(\mathfrak{J}_\lambda)$ the family of intersections of the $u_\alpha^{-1}(\mathfrak{m}'^{n\alpha}_\alpha)$; by hypothesis, there exists λ such that $\mathfrak{J}_\lambda \subset \mathfrak{m}_y^n$, and since $\mathscr{F}_x \otimes_{\mathscr{O}_y} (\mathscr{O}_y/\mathfrak{J}_\lambda) = (\mathscr{F}_x \otimes_{\mathscr{O}_y} (\mathscr{O}_y/\mathfrak{J}_\lambda)) \otimes_{\mathscr{O}_y/\mathfrak{J}_\lambda} (\mathscr{O}_y/\mathfrak{m}_y^n)$, it will suffice to prove that $\mathscr{F}_x/\mathfrak{J}_\lambda\mathscr{F}_x$ is an $(\mathscr{O}_y/\mathfrak{J}_\lambda)$-module flat. Now, for each α, such that $\mathfrak{J}_\lambda \subset u_\alpha^{-1}(\mathfrak{m}'^{n\alpha}_\alpha)$, one has, by passage to quotients, a homomorphism $u'_\alpha : \mathscr{O}_y/\mathfrak{J}_\lambda \to \mathrm{A}'_\alpha/\mathfrak{m}'^{n\alpha}_\alpha$, and the intersection of the kernels of the u'_α is reduced to 0. Taking into account (**I**, 3.6.1), one sees that one is brought back to (11.4.11) in case (ii), and to (11.4.12) in case (i). □

Corollary (11.5.2). — *Let* Y *be a locally Noetherian prescheme,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite type,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module,* x *a point of* X*,* $y = f(x)$*; set* $\mathrm{A} = \mathscr{O}_y$*. Let* $u : \mathrm{A} \to \mathrm{A}'$ *be a homomorphism of* A *into a Zariski ring* A'*, such that the image of the inverse image* $u^{-1}(\mathfrak{r}')$ *of the radical* $\mathfrak{r}'$ *of* A' *is the maximal ideal* $\mathfrak{m}$ *of* A*; suppose moreover that the homomorphism* $\widehat{u} : \widehat{\mathrm{A}} \to \widehat{\mathrm{A}}'$ *is injective. Set* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$*,* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$*,* $f' = f_{|\mathrm{Y}'}$*,* $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$*. For* $\mathscr{F}$ *to be* f*-flat at the point* x*, it is necessary and sufficient that* $\mathscr{F}'$ *be* f'*-flat at every point whose projection in* X *is equal to* x *and whose projection in* Y' *is equal to a closed point of* Y'*.*

If moreover A' *is a finite* A*-algebra, one can in what precedes replace the hypothesis that* $\widehat{u}$ *is injective by the hypothesis that* u *is injective.*

Proof. Since A (resp. A') is identified with a sub-ring of $\widehat{\mathrm{A}}$ (resp. $\widehat{\mathrm{A}}'$) (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 3, prop. 6), one sees first that u is itself injective and that $\widehat{u}$ is its extension by continuity to $\widehat{\mathrm{A}}$.

Let $(\mathfrak{m}'_\alpha)$ be the family of maximal ideals of A'; since one has

$$\mathfrak{m}'^n_\alpha \widehat{\mathrm{A}}' = \widehat{\mathfrak{m}}'^n_\alpha, \qquad \text{and} \qquad \widehat{\mathfrak{m}}'^n_\alpha \cap \mathrm{A}' = \mathfrak{m}'^n_\alpha,$$

and that the $\mathfrak{m}'^n_\alpha$ are open in A', the intersections of finitely many of the $u^{-1}(\mathfrak{m}'^n_\alpha) = \mathrm{A} \cap \widehat{u}^{-1}(\widehat{\mathfrak{m}}'^n_\alpha)$, and it will suffice to show that in $\widehat{\mathrm{A}}$, the intersections of the $\widehat{\mathfrak{m}}'^n_\alpha$ for $n > 0$ are reduced to 0 in the local **IV-3** | 152 Noetherian ring $\widehat{\mathrm{A}}_{\widehat{\mathfrak{m}}'^n_\alpha}$. On the other hand the $\widehat{\mathfrak{m}}'_\alpha$ are the maximal ideals of $\widehat{\mathrm{A}}'$, hence the canonical homomorphism $\widehat{\mathrm{A}}' \to \prod_\alpha \widehat{\mathrm{A}}'_{\widehat{\mathfrak{m}}'_\alpha}$ is injective (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 3, cor. 2 of th. 1), and since by hypothesis $\widehat{u} : \widehat{\mathrm{A}} \to \widehat{\mathrm{A}}'$ is also injective, this achieves the proof in the general case. The last assertion follows from the fact that $\widehat{\mathrm{A}}$ is an A-module faithfully flat ($\mathbf{0}_{\mathrm{I}}$, 7.3.5) and $\widehat{\mathrm{A}}' = \mathrm{A}' \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$ since A' is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, § 3, nº 4, th. 3 and chap. IV, § 2, nº 5, cor. 3 of prop. 9). □

Proposition (11.5.3). — *Let* $f : \mathrm{X} \to \mathrm{Y}$ *be a morphism locally of finite type,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_{\mathrm{X}}$*-Module of finite presentation,* x *a point of* X*,* $y = f(x)$*,* $g = (\psi, \theta) : \mathrm{Y}' \to \mathrm{Y}$ *a proper morphism of finite presentation. Suppose that:*

label=(i) *The homomorphism* $\theta_y : \mathscr{O}_y \to (g_*(\mathscr{O}_{\mathrm{Y}'}))_y$ *is injective.*
lbbel=(ii) *For every* $x' \in \mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ *whose projection in* X *is* x*,* $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$ *is* Y'*-flat at the point* x'*.*

Then $\mathscr{F}$ *is* f*-flat at the point* x*.*

Proof. The question being local on X, one can suppose f of finite presentation. Since $f' = f_{|\mathrm{Y}'} : \mathrm{X}' \to \mathrm{Y}'$ is of finite presentation (1.6.2) and $\mathscr{F}'$ is an $\mathscr{O}_{\mathrm{X}'}$-Module of finite presentation ($\mathbf{0}_{\mathrm{I}}$, 5.2.5), it follows from (11.3.1) that the set U' of points of X' where $\mathscr{F}'$ is f'-flat is open. Since U' contains $p^{-1}(x)$ by hypothesis, and that p is proper, hence *closed*, U' contains a closed set, hence closed, U' contains an open set of the form $p^{-1}(\mathrm{U})$, where U is a neighbourhood of x. Replacing X by U, one can therefore suppose that $\mathscr{F}'$ is f'-flat. On the other hand, taking into account (**I**, 3.6.5), (**II**, 5.4.2) and (2.1.4), one can also replace Y by $\mathrm{Spec}(\mathscr{O}_y)$, i.e. suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, where A is a local ring. Under these conditions, we want to prove that $\mathscr{F}$ is f-flat. By virtue of (5.13.3), A is the inductive limit of local Noetherian sub-rings A_λ such that the canonical injection $\mathrm{A}_\lambda \to \mathrm{A}$ is a *local* homomorphism. By virtue of (8.9.1), one can suppose that $\mathrm{X} = \mathrm{X}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}$, $f = f_\lambda \otimes 1_{\mathrm{A}}$, $\mathscr{F} = \mathscr{F}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}$, for a suitable λ and an $\mathscr{O}_{\mathrm{X}_\lambda}$-Module coherent. Similarly, one can suppose that $\mathrm{Y}' = \mathrm{Y}'_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}$, $g = g_\lambda \otimes 1_{\mathrm{A}}$, where $g_\lambda : \mathrm{Y}'_\lambda \to \mathrm{Y}_\lambda$ is a morphism of finite presentation; moreover (8.10.5, (xii)), one can suppose g_λ *proper*. Since by hypothesis the homomorphism $\mathrm{A} \to \Gamma(\mathrm{Y}', \mathscr{O}_{\mathrm{Y}'})$ is injective and that A_λ is a sub-ring of A, the homomorphism $\mathrm{A}_\lambda \to \Gamma(\mathrm{Y}', \mathscr{O}_{\mathrm{Y}'})$ is also injective. Finally, by virtue of (11.2.6), one can suppose λ taken such that $\mathscr{F}'_\lambda = \mathscr{F}_\lambda \otimes_{\mathrm{Y}_\lambda} \mathrm{Y}'_\lambda$ is f'_λ-flat, since $\mathscr{F}' = \mathscr{F}'_\lambda \otimes_{\mathrm{Y}'_\lambda} \mathrm{Y}'$. These remarks show that one can henceforth suppose that the ring A is *Noetherian*, the other hypotheses of (11.5.3) being verified. Let us then set $\mathrm{B} = \widehat{\mathrm{A}}$, $\mathrm{Z} = \mathrm{Spec}(\mathrm{B})$; since B is an A-module faithfully flat ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), it amounts to the same to say that $\mathscr{F}$ is f-flat or that $\mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Z}$ is Z-flat (2.1.4); similarly, if one sets $\mathrm{Z}' = \mathrm{Y}' \times_{\mathrm{Y}} \mathrm{Z}$, the morphism $\mathrm{Z}' \to \mathrm{Y}'$ is faithfully flat (2.2.13), hence if it amounts to the same to say that $\mathscr{F}'$ is f'-flat or that $\mathscr{F}' \otimes_{\mathrm{Y}} \mathrm{Z}'$ is Z'-flat; finally $h = g_{(\mathfrak{m})} : \mathrm{Z}' \to \mathrm{Z}$ is proper (**II**, 5.4.2) and of finite type (1.5.2), $\widehat{\mathrm{A}}$ is Noetherian, and if z is its closed point, the homomorphism $\mathscr{O}_z \to (h_*(\mathscr{O}_{\mathrm{Z}'}))_z$ **IV-3** | 153 is *injective*, since it follows from (2.3.1) that $h_*(\mathscr{O}_{\mathrm{Z}'}) = g_*(\mathscr{O}_{\mathrm{Y}'}) \otimes_{\mathrm{Y}} \mathrm{Z}$, and our assertion follows from hypothesis (i) and from the definition of flat modules ($\mathbf{0}_{\mathrm{I}}$, 6.1.1).

One can therefore now suppose the local Noetherian ring A *complete*; the proof will be finished if one proves that the intersection of the kernels of the homomorphisms $\mathrm{A} = \mathscr{O}_y \to \mathscr{O}_{y'}$ (where y' ranges over $g^{-1}(y)$) is reduced to 0. Indeed, the $\mathscr{O}_{y'}$ are local Noetherian rings, hence for each y' the intersection of the $\mathfrak{m}_y^n$ ($n \geqslant 0$) is reduced to 0; if $\mathfrak{a}_{n,y'}$ is the inverse image of $\mathfrak{m}_y^n$ in A, the intersections of finitely many of the $\mathfrak{a}_{n,y'}$ are neighbourhoods of 0 in A and the intersection of all the $\mathfrak{a}_{n,y'}$ is reduced to 0; the intersections $\mathfrak{a}_{n,y'}$ will therefore form a fundamental system of neighbourhoods of 0 in A (Bourbaki, *Alg. comm.*, chap. III, § 2, nº 7, prop. 8, where one can in the proof replace the decreasing sequence by an arbitrary filtered set); one will then be able to apply (11.5.1). Now, let $s \in \mathrm{A}$ be an element belonging to the kernel of each of the homomorphisms $\mathrm{A} \to \mathscr{O}_{y'}$; the image s' of s in $\Gamma(\mathrm{Y}', \mathscr{O}_{\mathrm{Y}'})$ is therefore a section of $\mathscr{O}_{\mathrm{Y}'}$ above Y' such that $s'_{y'} = 0$ for every $y' \in g^{-1}(y)$; then there is a neighbourhood V' of $g^{-1}(y)$ in Y' such that $s'|\mathrm{V}' = 0$. But since g is *closed*, V' contains a set of the form $g^{-1}(\mathrm{V})$, where V is a neighbourhood of the only closed point y of Y, i.e. $\mathrm{V}' = \mathrm{Y}'$, $s' = 0$, and since $\mathrm{A} \to \Gamma(\mathrm{Y}', \mathscr{O}_{\mathrm{Y}'})$ is injective by hypothesis, $s = 0$. □

The following particular case of (11.5.3) will be useful in chap. V:

Corollary (11.5.4). — *Let $f = (\psi, \theta) : \mathrm{X} \to \mathrm{Y}$ be a proper morphism of finite presentation, and let $p : \mathrm{X} \times_{\mathrm{Y}} \mathrm{X} \to \mathrm{X}$ be the first projection. Suppose that $\theta : \mathscr{O}_{\mathrm{Y}} \to f_*(\mathscr{O}_{\mathrm{X}})$ is injective. Then, for f to be flat, it is necessary and sufficient that p be flat.*

Proposition (11.5.5). — *Let* A *be a ring,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$*,* $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite presentation,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_{\mathrm{X}}$*-Module of finite presentation,* x *a point of* X*. Let* $\mathrm{A} \to \mathrm{A}'$ *be an injective homomorphism making* A' *an entire algebra over* A*. Set* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$*,* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$*,* $f' = f \times 1$*,* $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$*. Then, if* $\mathscr{F}'$ *is* f'*-flat at every point of* X' *whose projection in* X *is* x*,* $\mathscr{F}$ *is* f*-flat at the point* x*.*

Proof. Suppose first that A' is an A-algebra *finite* and of *finite presentation*; then the morphism $\mathrm{Y}' \to \mathrm{Y}$ is proper (**II**, 6.1.11) and of finite presentation, hence the hypotheses of (11.5.3) are satisfied, whence the conclusion. In the general case, the proposition results from this particular case, from the fact that A' is the inductive limit of its sub-A-algebras *finite* A'_λ and of the two following lemmas: □

Lemma (11.5.5.1). — *Every finite* A*-algebra* A' *is the inductive limit of* A*-algebras* A'_λ *which are finite and of finite presentation.*

Proof. One reasons as in (1.9.3.1). Indeed $\mathrm{A}' = \mathrm{B}/\mathfrak{J}$, where B is a finite A-algebra free which is an A-module free, and $\mathfrak{J}$ an ideal of B (1.4.7.1). Now, $\mathfrak{J}$ is the inductive limit of ideals $\mathfrak{J}_\lambda \subset \mathfrak{J}$ of B which are of finite type (and *a fortiori* of A-modules *of finite type*), hence, by the exactness of the functor $\varinjlim$, A' is the inductive limit of the A-algebras $\mathrm{B}/\mathfrak{J}_\lambda$; now, $\mathrm{B}/\mathfrak{J}_\lambda$ is by definition an A-module of finite presentation, hence also (1.4.7) a finite A-algebra of finite presentation. □

Lemma (11.5.5.2). — *Let* A *be a ring,* A' *an* A*-algebra,* (A'_λ) *an inductive system of* A*-algebras such that* $\mathrm{A}' = \varinjlim \mathrm{A}'_\lambda$*; set* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{Y}'_\lambda = \mathrm{Spec}(\mathrm{A}'_\lambda)$*. Let* $f : \mathrm{X} \to \mathrm{Y}$ *be a morphism of finite presentation,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_{\mathrm{X}}$*-Module of finite presentation; set* $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{|\mathrm{Y}'}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$, $\mathrm{X}'_\lambda = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'_\lambda$, $f'_\lambda = f_{|\mathrm{Y}'_\lambda}$, $\mathscr{F}'_\lambda = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'_\lambda$*. Let* x *be a point of* X*, such that* $\mathscr{F}'$ *is* f'*-flat at every point* $x' \in \mathrm{X}'$ *above* x*; then there exists* λ *such that* $\mathscr{F}'_\lambda$ *is* f'_λ*-flat at every point* $x' \in \mathrm{X}'_\lambda$ *above* x*.*

Proof. Let U' be the set of $x' \in \mathrm{X}'$ such that $\mathscr{F}'$ is f'-flat at the point x'; one knows (11.3.1) that U' is open in X' since f' is of finite type (1.6.2); similarly the set U'_λ of points of X'_λ where $\mathscr{F}'_\lambda$ is f'_λ-flat is open in X'_λ, and one knows moreover (11.2.6) that U' is the union of the $v_\lambda^{-1}(\mathrm{U}'_\lambda)$, where $v_\lambda : \mathrm{X}' \to \mathrm{X}'_\lambda$ is the canonical projection. Let us set $\mathrm{V}'_\lambda = g_\lambda^{-1}(\mathrm{U}'_\lambda)$, $\mathrm{V}' = g'^{-1}(\mathrm{U}') = \bigsqcup w_\lambda^{-1}(\mathrm{U}'_\lambda)$. Let us consider the scheme $\mathrm{T} = \mathrm{Spec}(k(x))$; set $\mathrm{T}' = \mathrm{T} \times_{\mathrm{Y}} \mathrm{Y}'$, $\mathrm{T}'_\lambda = \mathrm{T} \times_{\mathrm{Y}} \mathrm{Y}'_\lambda$, and let $w_\lambda : \mathrm{T}' \to \mathrm{T}'_\lambda$, $g' : \mathrm{T}' \to \mathrm{X}'$, $g'_\lambda : \mathrm{T}'_\lambda \to \mathrm{X}'_\lambda$ be the canonical projections. Let us set $\mathrm{V}'_\lambda = g'^{-1}_\lambda(\mathrm{U}'_\lambda)$, $\mathrm{V}' = g'^{-1}(\mathrm{U}') = \bigcup w_\lambda^{-1}(\mathrm{V}'_\lambda)$. By hypothesis on U' (and taking into account (**I**, 3.6.1)) $\mathrm{V}' = \mathrm{T}'$; as T' is quasi-compact, there exists λ such that $w_\lambda^{-1}(\mathrm{V}'_\lambda) = \mathrm{T}'$. One deduces then from (8.3.3) applied to the closed parts $\mathrm{T}'_\lambda - \mathrm{V}'_\lambda$ of T'_λ, that there exists $\mu \geqslant \lambda$ such that $\mathrm{T}'_\mu = \mathrm{V}'_\mu$; this means that $\mathscr{F}'_\mu$ is f'_μ-flat at every point of X'_λ whose the projection in X is x. □ **IV-3** | 154

11.6. Descent of flatness by arbitrary morphisms: the case of a unibranch base

Theorem (11.6.1). — *Let* A *be a local ring which is integral and geometrically unibranch* $(\mathbf{0}, 23.2.1)$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $f : \mathrm{X} \to \mathrm{Y}$ *a morphism locally of finite presentation,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module of finite presentation. Let* A' *be a local ring,* $u : \mathrm{A} \to \mathrm{A}'$ *a local injective homomorphism; set* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{|\mathrm{Y}'}$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$. *Let* x *be a point of* X *whose projection* $f(x) = y$ *is the closed point of* Y, x' *a point of* X' *whose projections in* X *and* Y' *are respectively* x *and the closed point* y' *of* Y'. *Then, if* $\mathscr{F}'$ *is* f'*-flat at the point* x', $\mathscr{F}$ *is* f*-flat at the point* x.

Proof. One can restrict to the case where f is of finite presentation, the question being local on X. We will proceed in several steps.

I) *Reduction to the case where* A *and* A' *are local rings integrally closed.*

Since u is injective and A integral, on can therefore first suppose A' integrally closed $(\mathbf{0}_{\mathrm{I}}, 1.5.8)$; the composite homomorphism $\mathrm{A} \to \mathrm{A}' \to \mathrm{A}'' = \mathrm{A}'/\mathfrak{p}'$ is therefore injective and local, and if $\mathrm{Y}'' = \mathrm{Spec}(\mathrm{A}'')$, $\mathrm{X}'' = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{A}'' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}''$, $\mathscr{F}'' = \mathscr{F} \otimes_{\mathrm{A}} \mathrm{A}''$, $\mathscr{F}''$ is Y''-flat at the points of X'' above x' (2.1.4); replacing A' by A'' as needed and taking into account (**I**, 3.4.7), one can therefore first suppose A' integral. If K' is the field of fractions of A', there is a valuation ring B' in K' that dominates A'; the composite morphism $\mathrm{A} \xrightarrow{u} \mathrm{A}' \to \mathrm{B}'$ being injective and local, the same reasoning as previously allows one to replace A' by B'; A' being a sub-local ring of A' dominated by A'. Let A_1 **IV-3** | 155
be the integral closure of A; it is clear that $\mathrm{A} \subset \mathrm{A}_1 \subset \mathrm{A}'$, and by hypothesis A_1 is a *local* ring; if $\mathfrak{m}$, $\mathfrak{m}_1$, $\mathfrak{m}'$ are the maximal ideals of A, A_1, A', one has $\mathfrak{m} \subset \mathfrak{m}_1 \subset \mathfrak{m}'$; indeed, $\mathfrak{m}_1$ is the *sole* prime ideal of A_1 above $\mathfrak{m}$, since A_1 is a local ring (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, prop. 1); since $\mathfrak{m}' \cap \mathrm{A} = \mathfrak{m}$, one has $\mathfrak{m}' \cap \mathrm{A}_1 \cap \mathrm{A} = \mathfrak{m}$, hence $\mathfrak{m}' \cap \mathrm{A}_1 = \mathfrak{m}_1$. Set $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}_1)$, $\mathrm{X}_1 = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_1$, $f_1 = f_{|\mathrm{Y}_1}$, $\mathscr{F}_1 = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}_1$, and let x_1 be the projection of x' in X_1; let us denote furthermore by y_1 the unique closed point of Y_1, so that $y_1 = f_1(x_1)$. By hypothesis the morphism $\mathrm{Spec}(k(y_1)) \to \mathrm{Spec}(k(y))$ is *radicial*, whence one concludes, by the transitivity of fibres (**I**, 3.6.4) and (**I**, 3.5.7), that the morphism $f_1^{-1}(y_1) \to f^{-1}(y)$ is radicial, and in particular x_1 is the *sole* point of X_1 whose projections in X and Y_1 are x and y_1 respectively; moreover, as one has seen, y_1 is the *sole* point of Y_1 above the projection in Y, hence x_1 is the *sole* point of X_1 whose projection in X is x. If one proves that $\mathscr{F}_1$ is f_1-flat at the point x_1, one will be able to apply (11.5.5), whence the conclusion. One is therefore brought back to the case where A is also *integrally closed*.

II) *Reduction to the case where* A *and* A' *are local rings of* $\mathbf{Z}$*-algebras of finite type integrally closed.*

One can consider A' as the inductive limit of the filtrant family of its sub-$\mathbf{Z}$-algebras (integral) of finite type B'_λ; moreover, since A' is integrally closed and the integral closure of a $\mathbf{Z}$-algebra of finite type is also a $\mathbf{Z}$-algebra of finite type (7.8.3), one sees that A' is the inductive limit of its sub-$\mathbf{Z}$-algebras of finite type B'_λ *integrally closed*; if $\mathfrak{m}'_\lambda = \mathfrak{m}' \cap \mathrm{B}'_\lambda$, A' is also the inductive limit of the local sub-rings $(\mathrm{B}'_\lambda)_{\mathfrak{m}'_\lambda} = \mathrm{A}'_\lambda$ dominated by A' (5.13.3). For every λ, $\mathrm{B}'_\lambda \cap \mathrm{A}$ is also the inductive limit of its sub-$\mathbf{Z}$-algebras of finite type $\mathrm{B}_{\alpha\lambda}$, hence $\mathrm{A} = \varinjlim_{\alpha,\lambda} \mathrm{B}_{\alpha\lambda}$, and one can moreover replace $\mathrm{B}_{\alpha\lambda}$ by the local ring $\mathrm{A}_{\alpha\lambda} = (\mathrm{B}_{\alpha\lambda})_{\mathfrak{m}_{\alpha\lambda}}$, where $\mathfrak{m}_{\alpha\lambda} = \mathfrak{m} \cap \mathrm{B}_{\alpha\lambda}$, so that $\mathrm{A}_{\alpha\lambda}$ is dominated by A'_λ and by A. Set $\mathrm{Y}_{\alpha\lambda} = \mathrm{Spec}(\mathrm{A}_{\alpha\lambda})$; it follows from (8.9.1) that there exist a pair (α, λ) large enough, a morphism $f_{\alpha\lambda} : \mathrm{X}_{\alpha\lambda} \to \mathrm{Y}_{\alpha\lambda}$ of finite type and a coherent $\mathscr{O}_{\mathrm{X}_{\alpha\lambda}}$-Module $\mathscr{F}_{\alpha\lambda}$ such that $\mathrm{X} = \mathrm{X}_{\alpha\lambda} \otimes_{\mathrm{A}_{\alpha\lambda}} \mathrm{A}$, $f = f_{\alpha\lambda} \otimes 1_{\mathrm{A}}$, $\mathscr{F} = \mathscr{F}_{\alpha\lambda} \otimes_{\mathrm{A}_{\alpha\lambda}} \mathrm{A}$; if $x_{\alpha\lambda}$ is the projection of x in $\mathrm{X}_{\alpha\lambda}$, it will suffice to show that $\mathscr{F}_{\alpha\lambda}$ is $f_{\alpha\lambda}$-flat at the point $x_{\alpha\lambda}$. Since A' is the inductive limit of the A'_μ for $\mu \geqslant \lambda$, X' is the projective limit of the $\mathrm{X}_{\alpha\lambda} \times_{\mathrm{Y}_{\alpha\lambda}} \mathrm{Y}'_\mu = \mathrm{X}'_\mu$, and one also has $\mathscr{F}' = \mathscr{F}'_{\alpha\mu} \otimes_{\mathrm{Y}_{\alpha\lambda}} \mathrm{Y}'_\mu$, where $\mathscr{F}'_{\alpha\mu} = \mathscr{F}_{\alpha\lambda} \otimes_{\mathrm{Y}_{\alpha\lambda}} \mathrm{Y}'_\mu$. Applying (11.2.6), one sees that one can take μ large enough for $\mathscr{F}'_\mu$ to be Y'_μ-flat at the point x'_μ, projection of x' in X'_μ, and moreover, by construction of the A'_μ, the projection of x'_μ in Y'_μ is the closed point of Y'_μ.

III) *Reduction to the case where the residue field* k' *of* A' *is a finite extension and radicial of the residue field* k *of* A.

One can first of all repeat the reasoning of part I) of the proof of (11.4.11), taking into account the fact that $\mathbf{Z}$ is a ring of □

11.7. Counter-examples

(11.7.1). Let us first consider the case where A is an *Artinian* local ring, and where the hypotheses of (11.4.11) are satisfied *except* for condition (ii) concerning the residue field k of A. We will see that the conclusion of (11.4.11) can then fail to hold. Let k be a field admitting a *Galois* extension k' of degree $[k' : k] > 1$, and let Γ denote the Galois group of k'. Let A be a k-algebra having a base of 3 elements $1, a, b$ with the multiplication table $a^2 = ab = ba = b^2 = 0$, so that A is an Artinian local ring whose maximal ideal $\mathfrak{m} = ka + kb$ is of square zero. Let $A' = A \otimes_k k'$, which is a k'-algebra of base $1, a, b$, an Artinian local ring of maximal ideal $\mathfrak{m}' = k'a + k'b = \mathfrak{m}A'$, of square zero; A identifies canonically with a subring of A'. Let $\mathfrak{Z}$ be the sub-k'-vector space of $\mathfrak{m}'$ generated by $a + \gamma b$, where $\gamma \in k'$ does not belong to k; it is clear that $\mathfrak{Z}$ is an ideal of A'. Let $B = A'/\mathfrak{Z}$; this is an Artinian ring which is an A-module *not flat*; indeed (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. 2 of prop. 5), B would be a free A-module, and since the canonical homomorphism $A'/\mathfrak{m}A' \to B/\mathfrak{m}B$ is bijective, the canonical homomorphism $A' \to B = A'/\mathfrak{Z}$ would itself be bijective, which is absurd. **IV-3** | 158
Alternatively, put $A_1 = B$, and $Y_1 = \operatorname{Spec}(A_1)$, and consider the canonical homomorphism $A \to A_1$, which is *local* and *injective* since $\mathfrak{Z} \cap A = 0$; let $B_1 = B \otimes_A A_1 = B \otimes_A B$, and let us see that there is a point of $X_1 = X \times_Y Y_1 = \operatorname{Spec}(B_1)$ where $\mathscr{O}_{x_1}$ is Y_1-*flat*. For this, let us remark that we have $B \otimes_A B = (A' \otimes_A A')/(\operatorname{Im}(\mathfrak{Z} \otimes_A A') + \operatorname{Im}(A' \otimes_A \mathfrak{Z}))$. Now, the structure of $C' = A' \otimes_A A'$ is easily obtained; one considers the product $C = \prod A'_\sigma$, where all the A'_σ are equal to A', and the map $\varphi : A' \otimes_A A' \to C$ such that $\varphi(x \otimes y) = (\sigma(x)y)_{\sigma \in \Gamma}$ (the group Γ operating canonically on A'); passing to quotients, one deduces a homomorphism $C'/\mathfrak{m}C' \to C/\mathfrak{m}C$ which is none other than the canonical homomorphism $k' \otimes_k k' \to \prod k'_\sigma$ (with $k'_\sigma = k'$ for all $\sigma \in \Gamma$); it is known that this last one is bijective (Bourbaki, *Alg.*, chap. VIII, § 8, prop. 4), whence it is the same for φ, since C' and C are free A-modules *libres* (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. of prop. 6). From what precedes, it follows that $B \otimes_A B$ is a semi-local A-algebra, composed as a direct product of local A-algebras $A'/((\sigma(\mathfrak{Z})) + \mathfrak{Z})$. That of these algebras corresponding to the identity of Γ is isomorphic to $A'/\mathfrak{Z}$, hence is an A_1-module *flat*, but for $\gamma \notin k$, it is clear that R is an irreductible and separable polynomial of K[T]; one deduces from this that A_1 is first of all an integral ring whence the field of fractions $K_1 = K[T]/(R)$ is a *separable* extension of K. In addition, if t is the canonical image of T in A_1, the t^j $(0 \leqslant j < n)$ form a base of the A-module A_1, and their images in k_1 a base over k; one deduces from this that $d = \det(\operatorname{Tr}_{A_1/A}(t^{i+j}))$ is an element of A whose class in k is $\neq 0$, and which is therefore invertible. The same reasoning as in (6.12.4.1), I) then proves that the morphism $\operatorname{Spec}(A_1) \to \operatorname{Spec}(A)$ is flat and has regular fibres; one concludes by (6.5.4), (ii)) that A_1 is *integrally closed*. Let us then consider the ring $A'_1 = A' \otimes_A A_1$; this is a free A'-module of finite type, whence a semi-local ring (Bourbaki, *Alg. comm.*, chap. IV, § 2, no. 5, cor. 3 of prop. 9); moreover, the maximal ideals of this finite A'-algebra are all above the maximal ideal $\mathfrak{m}'$ of A', and *a fortiori* contain $\mathfrak{m}A'_1$. But $A'_1/\mathfrak{m}A'_1 = (A'/\mathfrak{m}A') \otimes_k k_1$, and since k_1 is a separable and finite extension of k, the radical of $A'_1/\mathfrak{m}A'_1$ is equal to $(\mathfrak{m}'/\mathfrak{m}A') \otimes_k k_1$ (Bourbaki, *Alg.*, chap. VIII, § 7, no. 2, cor. 2 of prop. 3); if $\mathfrak{n}_i$ $(1 \leqslant i \leqslant r)$ are the maximal ideals of A'_1, the fields composing the algebra $k' \otimes_k k_1$, in other words the said fields are *radicial* extensions of k_1. Moreover, since $A \to A'$ is an injective homomorphism, so is $A_1 \to A'_1$, and A_1 being a flat A-module; the canonical homomorphism $A'_1 \to \prod_{i=1}^r (A'_1)_{\mathfrak{n}_i}$ being also injective (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, th. 1), the same holds for the composite $A_1 \to \prod_{i=1}^r (A'_1)_{\mathfrak{n}_i}$. But A_1 is integral, and the kernels of the homomorphisms $A_1 \to (A'_1)_{\mathfrak{n}_i}$ are finite in number; since their intersection is zero, one of them is already zero. In other terms, there is a $B_1 = (A'_1)_{\mathfrak{n}_i}$ such that the homomorphism $A_1 \to B_1$ is *injective* and *local*. Set $Y'_1 = \operatorname{Spec}(B_1)$, $X'_1 = X' \times_{Y'} Y'_1$, $\mathscr{F}'_1 = \mathscr{F}_1 \otimes_Y Y'_1$; Y'_1-flat in *all* the points of X'_1 above x'; moreover the maximal ideal of B_1 is the only one above $\mathfrak{m}'$, and hence all these points have for projection in Y'_1 the closed point y'_1. On the other hand, set $Y_1 = \operatorname{Spec}(A_1)$, $X_1 = X \times_Y Y_1$, $\mathscr{F}_1 = \mathscr{F} \otimes_Y Y_1$; if x_1 is the projection of x'_1 in X_1, and x its projection in X and y_1 is the closed point y_1. If one proves that $(\mathscr{F}_1)_{x_1}$ is a $\mathscr{O}_{y_1}$-module flat, it will follow that $\mathscr{F}_x$ is an $\mathscr{O}_y$-module flat; indeed $\mathscr{O}_{y_1}$ is a faithfully flat $\mathscr{O}_y$-module; whence (2.2.11), (iii)) $\mathscr{F}_x$ is an $\mathscr{O}_y$-module flat. As $X'_1 = X_1 \times_{Y_1} Y'_1$, $\mathscr{F}'_1 = \mathscr{F}_1 \otimes_{Y_1} Y'_1$, we have brought ourselves back to the situation of (11.6.1), with A and A' replaced by A_1 and B_1 respectively.

IV) *End of the proof.* — We are finally reduced to proving (11.6.1) under the following supplementary hypotheses:

label=(i) A and A' are local rings of **Z**-algebras of finite type (hence *excellent* rings (7.8.3)));

lbbel=(ii) A is integrally closed;
lcbel=(iii) the residue field k' of A' is a finite and radicial extension of the residue field k of A.

It is known then ((7.8.3) and (2.3.8)) that under conditions (i) and (ii), if $\mathfrak{m}$ and $\mathfrak{m}'$ are the maximal ideals of A and A' respectively, the topology on A is *induced* by the $\mathfrak{m}'$-adic topology on A'. The completion $\widehat{u} : \widehat{A} \to \widehat{A}'$ is therefore an injective homomorphism. On the other hand, since the morphism $\operatorname{Spec}(k') \to \operatorname{Spec}(k)$ is radicial, it is the same for the morphism $f'^{-1}(y') \to f^{-1}(y)$ (**I**, 3.5.7), and there is therefore only *one* point x' of X' whose projections in X and Y' are x and y' respectively. One can therefore apply the result of (11.5.2).

Corollary (11.6.2). — *Let* A *be an integral local ring, unibranch,* $Y = \operatorname{Spec}(A)$, $f : X \to Y$ *a locally of finite presentation morphism,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_X$*-Module of finite presentation. Let* A' *be a local ring,* $A \to A'$ *a local injective homomorphism; set* $Y' = \operatorname{Spec}(A')$, $X' = X \times_Y Y'$, $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$. *Let* x *be a point of* X *whose projection* $f(x) = y$ *is the closed point of* Y*; we suppose that* $\mathscr{F}'$ *is* f'*-flat in all the points* x' *of* X' *whose projections in* X *and* Y' *are respectively* x *and the closed point* y' *of* Y'*. Then* $\mathscr{F}$ *is* f*-flat at the point* x*.*

One can indeed take up again part I) of the proof of (11.6.1), which proves (with the same notations) that if $\mathscr{F}_1$ is f_1-flat in *all* the points x_1 of X_1 whose projections in X and Y_1 are x and y_1, then $\mathscr{F}$ is f-flat at the point x; one is thus brought back to the case where A is integrally closed and not geometrically unibranch. On then applies the result of (11.5.2).

(11.7.2). We will now show that the result of (11.5.4) loses its validity when one does not assume the morphism to be *proper* (and *a fortiori* (11.5.3) ceases to be exact when one does not assume g to be *proper*). Let k be a field, A_0 the ring of polynomials $k[S,T]$, A the quotient ring A_0/A_0ST; $Y = \operatorname{Spec}(A)$ is thus the reducible curve formed by the two "coordinate axes" in the affine plane $V_k^2 = \operatorname{Spec}(A_0)$. A admits two minimal prime ideals $\mathfrak{p}_1 = A_0S/A_0ST$, $\mathfrak{p}_2 = A_0T/A_0ST$, and since A is reduced, it embeds in a canonical way; moreover B_1 identifies with $k[T]$ and B_2 with $k[S]$, whence these are integral rings, integrally closed, and thus $Z = \operatorname{Spec}(B)$ is none other than the *normalised* of the prescheme Y with respect to R(Y) (**II**, 6.3.8), the *sum* of the two schemes $Z_1 = \operatorname{Spec}(B_1)$, $Z_2 = \operatorname{Spec}(B_2)$. Let us denote by y the "double point" of Y, corresponding to the maximal ideal $\mathfrak{p}_1 + \mathfrak{p}_2 = (S)$ of A, by z_1 and z_2 the points of Z that project onto y, corresponding respectively to the maximal ideals $\mathfrak{m}_1 = (T)$ and $\mathfrak{m}_2 = (S)$ of B_1 and B_2. We will denote by X the complement of z_2 in Z, and thus $X = \operatorname{Spec}(B_1 \oplus (B_2)_{\mathfrak{s}})$; it is immediate that the homomorphism $A \to A' = B_1 \oplus (B_2)_{\mathfrak{s}}$ is *injective*, and hence also the corresponding morphism $f : X \to Y$ is not a closed map (since $f(Z_2 - (z_2))$ is not closed in Y); *a fortiori* it is not proper. We will now show that f *is not flat* at the point z_1; it will suffice to show (0.6.6.2) that $\mathscr{O}_{z_1}$ is not an $\mathscr{O}_y$-module flat, but the canonical homomorphism $\mathscr{O}_y \to \mathscr{O}_{z_1}$ is not *injective*; however, the first projection $p : X \times_Y X \to X$ is an *isomorphism*: indeed, on the one hand, $B_1 \otimes_A B_1 = (A/\mathfrak{p}_1) \otimes_A (A/\mathfrak{p}_1) = A/\mathfrak{p}_1$, and $(B_2)_{\mathfrak{s}} \otimes_A (B_2)_{\mathfrak{s}} = (B_2)_{\mathfrak{s}}$, and finally $B_1 \otimes_A (B_2)_{\mathfrak{s}} = 0$, since the image of the generic element of S in B_1 is zero.

(11.7.3). The preceding example can be generalised: one considers on a field k an algebraic curve which is reduced and admits a single "ordinary double point" y (a notion that will be defined more generally later on), and its *normalised* Z, so that the morphism $g : Z \to Y$ is finite, and that the restriction of g to $Z - g^{-1}(y)$ reduces to two "simple" points z_1, z_2; moreover the prescheme $g^{-1}(y)$ is the sum of the two preschemes $\operatorname{Spec}(k(z_1))$, $\operatorname{Spec}(k(z_2))$, canonically isomorphic to $\operatorname{Spec}(k(y))$. Let X be the sub-prescheme of Z induced on the open $Z - \{z_2\}$; the morphism $f : X \to Y$, restriction of g to X, is not closed. The morphism f is *radicial*, for every $y' \in Y$, the fibre $f^{-1}(y')$ is reduced to a point x', $k(y') \to k(x')$ being an isomorphism. Without the injection $j : X \to Z$, which is not closed, the morphism $f : X \to Y$ would not be proper, and without (**II**, 5.4.3) it would also follow that it is not a finite morphism. The morphism f is *radicial*, for every $y' \in Y$, the fibre $g^{-1}(y')$ ne comprendrait qu'un seul point x', $k(y') \to k(x')$ being an isomorphism. Consider in particular the case where Z_λ is an open sub-prescheme induced on the open $Z - \{z_2\}$; the morphism $f : X \to Y$ is the restriction of g to X.

One may also interpret u'_λ as follows: it suffices to know the value of $u'_\lambda(t')$, where t' is a section of $\mathscr{F}'$ above an open subset U' of X', of the following particular type: t' is the restriction to U' of the canonical image by $\Gamma(\mathrm{U}, \mathscr{F}) \to \Gamma(g^{-1}(\mathrm{U}), \mathscr{F}')$ of a section t of $\mathscr{F}$ above an open U of X containing $g(\mathrm{U}')$ (these sections generate the $\mathscr{O}_{\mathrm{X}'}$-Module $\mathscr{F}'$ (**0**, 3.7.1)). Consider the section $u_\lambda(t)$ of $\mathscr{G}_\lambda$ above $f_\lambda^{-1}(\mathrm{U})$, and its canonical image t'' by $\Gamma(f_\lambda^{-1}(\mathrm{U}), \mathscr{G}_\lambda) \to \Gamma(g_\lambda'^{-1}(f_\lambda^{-1}(\mathrm{U})), \mathscr{G}'_\lambda)$; then $u'_\lambda(t')$ is the restriction of t'' to $f_\lambda'^{-1}(\mathrm{U}')$. Consider in particular the case where Z_λ is an open sub-prescheme of X, where $j_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ is the canonical injection, and where u_λ is the canonical homomorphism $\rho_{\mathscr{F}} : \mathscr{F} \to (j_\lambda)_*(j_\lambda^*(\mathscr{F})) = (j_\lambda)_*(\mathscr{G}_\lambda)$ (**0**, 4.4.3.2) corresponding to j_λ. Then Z'_λ is induced on an open of X′, and the preceding interpretation shows that u'_λ corresponds to the canonical injection $j'_\lambda : \mathrm{Z}'_\lambda \to \mathrm{X}'$.

(11.7.4). One will note that in the preceding example the homomorphism u is *injective* when Y is *irreducible* (one can for example take $\mathrm{Y} = \mathrm{Spec}(k[\mathrm{S}, \mathrm{T}]/(\mathrm{S}(\mathrm{S}^2 + \mathrm{T}^2) - (\mathrm{S}^2 - \mathrm{T}^2)))$, "cubique à point double"); one can indeed, replacing Y by an affine neighbourhood of y, suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, where A is integral, whence $\mathrm{Z} = \mathrm{Spec}(\mathrm{B})$, where B is the integral closure of A; as B, $\mathscr{O}_y$ and $\mathscr{O}_{z_1}$ identify then with subrings of the field of fractions of A.

(11.7.5). The examples of (11.7.2) and (11.7.3) explain the restriction to *unibranch* local rings in (11.6.1). We now wish to see that in (11.6.1), one cannot weaken the hypothesis on A by supposing only that A is *unibranch*. Consider in fact the integral complete local ring $\mathrm{A} = \mathbf{R}[[\mathrm{U}, \mathrm{V}]]/(\mathrm{U}^2 + \mathrm{V}^2)$ which is unibranch but not geometrically unibranch (6.5.11). One knows (*loc. cit.*) that if u, v are the images of U and V in A, the integral closure of A is $\bar{\mathrm{A}} = \mathrm{A}[t]$ with $t = v/u$, such that $t^2 = -1$, so that $\bar{\mathrm{A}}$ is isomorphic to $\mathbf{C}[[\mathrm{U}]]$. Set $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\bar{\mathrm{A}})$ (normalised of Y (**II**, 6.3.8)), and let y and x be the closed points of Y and X respectively; we will show that for a convenient A′-algebra, if one sets $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathscr{O}_{x'}$ is Y′-flat in an arbitrary point of X′ having these projections; it will then follow evidently that $\mathscr{O}_x$ is not Y-flat at the point x (which is moreover trivially *a priori*, $\bar{\mathrm{A}}$ not being a free A-module).

Let $\mathrm{B} = \mathrm{A} \otimes_{\mathbf{R}} \mathbf{C}$, isomorphic to $\mathbf{C}[[\mathrm{U}, \mathrm{V}]]/(\mathrm{U} + i\mathrm{V})(\mathrm{U} - i\mathrm{V})$; B has two minimal prime ideals $\mathfrak{p}'$, $\mathfrak{p}''$ generated respectively by $u + iv$ and $u - iv$, and $\mathfrak{n} = \mathfrak{p}' + \mathfrak{p}''$ is the maximal ideal of the complete local ring B. Let $\bar{\mathrm{B}} = \bar{\mathrm{A}} \otimes_{\mathbf{R}} \mathbf{C}$; $\bar{\mathrm{B}}$ is a composite direct product of two algebras isomorphic to $\mathbf{C}[[\mathrm{U}]]$, generated by the idempotents $e' = (1 \otimes 1 + i \otimes i)/2$ and $e'' = (1 \otimes 1 - i \otimes i)/2$; as the homomorphism $\mathrm{A} \to \bar{\mathrm{A}}$ is injective, so is $\mathrm{B} \to \bar{\mathrm{B}}$; one concludes immediately that $\bar{\mathrm{B}}$ identifies with $\bar{\mathrm{B}} \otimes_{\mathrm{B}} \mathrm{A}'$. Taking A′ to be the local ring $\mathrm{B}/\mathfrak{p}'$; then $\bar{\mathrm{A}} \otimes_{\mathrm{A}} \mathrm{A}'$ is isomorphic to $\mathrm{A}' \oplus (\mathrm{B}/\mathfrak{n})$. This establishes our assertion, since $\mathrm{B}/\mathfrak{n} = \mathrm{A}'/(n/\mathfrak{p}')$ is not a flat A′-module (without which it would be a free A′-module (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 2, cor. 2 of prop. 5), which is absurd).

11.8. A valuative criterion for flatness

Theorem (11.8.1). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite presentation morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_\mathrm{X}$-Module of finite presentation, x a point of* X, $y = f(x)$. *Suppose that the local ring $\mathscr{O}_y$ is integral (resp. reduced and Noetherian). For $\mathscr{F}$ to be f-flat at the point x, it is necessary and sufficient that, for every valuation ring (resp. every discrete valuation ring)* A′ *and every local homomorphism $\mathscr{O}_y \to \mathrm{A}'$, the following condition be satisfied: setting $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_\mathrm{Y} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')}$, the $\mathscr{O}_{\mathrm{X}'}$-Module $\mathscr{F}' = \mathscr{F} \otimes_\mathrm{Y} \mathrm{Y}'$ is f'-flat in all the points x' of* X′ *whose projections in* X *and* Y′ *are respectively x and the closed point y' of* Y′.

Proof. The condition being evidently necessary (2.1.4), it remains to prove that it is sufficient. One can evidently ((**I**, 3.6.5) and (**I**, 2.4.4)) restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is the spectrum of a *local* ring A and y is the closed point of Y.

(i) *Case where* A *is integral.* — Let K be the field of fractions of A, A_1 the integral closure of A; if one sets $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}_1)$, $\mathrm{X}_1 = \mathrm{X} \times_\mathrm{Y} \mathrm{Y}_1$, $f_1 = f_{(\mathrm{Y}_1)}$, it suffices, by virtue of (11.5.5), to show that $\mathscr{F}_1 = \mathscr{F} \otimes_\mathrm{Y} \mathrm{Y}_1$ is f_1-flat in *all* the points x_1 of X_1 whose projection x is in X. Now, if $f_1(x_1) = y_1$, where $\mathfrak{m}_1$ is the trace on A (and is moreover necessarily a maximal ideal), then $\mathfrak{m} = \mathfrak{i}_y$ is the trace of $\mathfrak{m}_1$ on A (which is necessarily a prime ideal of A_1 whence $\mathscr{O}_{y_1} = (\mathrm{A}_1)_{\mathfrak{m}_1}$; the local homomorphism $\mathrm{A} \to \mathscr{O}_{y_1}$ being local, so is $\mathrm{A} \to \mathrm{A}'$. There exists then at least one point $x' \in \mathrm{X}'$ whose projections in **IV-3** | 160
X_1 and Y′ are respectively x_1 and y' (**I**, 3.4.7); as the hypothesis states that $\mathscr{F}'$ is f'-flat at the point x', and $\mathscr{O}_{y_1}$ is integrally closed, one can apply (11.6.1), and one deduces that $\mathscr{F}_1$ is f_1-flat at the point x_1, whence the theorem in this case.

(ii) *Case where* A *is reduced and Noetherian.* — Let $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) be the minimal prime ideals of A; since A is reduced, it identifies canonically with a subring of the product of the $\mathrm{A}_i = \mathrm{A}/\mathfrak{p}_i$, which are local Noetherian rings; setting $Y_i = \mathrm{Spec}(A_i)$, $X_i = X \times_Y Y_i$, $f_i = f_{(Y_i)}$, it follows from (11.5.2) that it suffices to show that for each i, $\mathscr{F}_i = \mathscr{F} \otimes_Y Y_i$ is f_i-flat in every point x_i of X_i whose projections in X and Y_i are x and the closed point y_i of Y_i respectively. Now, since A_i is a local Noetherian integral ring, there exists in its field of fractions K_i a sub-ring A'_i containing A_i, which is a semi-local Noetherian ring (hence a finite A_i-algebra) and whose local rings are *geometrically unibranch* ((6.15.5) and (**0**, 23.2.5)). Since the maximal ideals of A'_i are necessarily above the maximal ideal of A_i, one deduces further from (**I**, 3.4.7) and (11.5.2) that it suffices, setting $Y'_i = \mathrm{Spec}(A'_i)$, $X'_i = X \times_Y Y'_i$, $f''_i = f_{(Y'_i)}$, to prove that $\mathscr{F}''_i = \mathscr{F} \otimes_Y Y'_i$ is f''_i-flat in every point x''_i of X''_i whose projections in X and Y'_i are x and the closed point y'_i of Y'_i respectively. Now, let A' be a discrete valuation ring for K_i that dominates A'_i, and let x' be a point of X' whose projections in X'_i and Y'_i are x''_i and y' respectively (**I**, 3.4.7); as $\mathscr{O}_{y'_i}$ is geometrically unibranch, one can again apply (11.6.1) and one deduces that $\mathscr{F}''_i$ is indeed f''_i-flat at the point x''_i. □

Remarks (11.8.2). —

label=(i) In the statement of (11.8.1), one cannot restrict to assuming that the condition on $\mathscr{F}'$ is satisfied for the *complete* valuation rings A' and that the residue field is *algebraically closed*. Indeed, it is known that every such valuation ring A' is dominated by a valuation ring A'' (**II**, 7.1.2), and that if A' is a discrete valuation ring, it is the same for A'' ($\mathbf{0}_{\mathrm{III}}$, 10.3.1).

lbbel=(ii) The proof of (11.8.1) simplifies when one supposes not only that A is integral and Noetherian, but that its completion $\widehat{A}$ is also *integral*. Replacing X by $X \otimes_A \widehat{A}$ and reasoning as in the proof of (11.5.3), one can in fact reduce to proving (11.8.1) when $A = \mathscr{O}_y$ is integral, Noetherian, and *complete*. Now, one knows (**II**, 7.1.7) that such a ring A is dominated by a discrete complete valuation ring; the conclusion follows therefore directly from (11.5.2).

11.9. Separating and universally separating families of homomorphisms of sheaves of modules

(11.9.1). Let X be a prescheme, $(f_\lambda)_{\lambda \in L}$ a family of morphisms $f_\lambda : Z_\lambda \to X$, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-Module; for every $\lambda \in L$, suppose given a quasi-coherent $\mathscr{O}_{Z_\lambda}$-Module $\mathscr{G}_\lambda$ and a homomorphism

$$u_\lambda : \mathscr{F} \longrightarrow (f_\lambda)_*(\mathscr{G}_\lambda).$$

We say that the family (u_λ) (or the corresponding family of $u^\#_\lambda : (f_\lambda)^*(\mathscr{F}) \to \mathscr{G}_\lambda$) is *separating* if the intersection of the kernels of the u_λ is *zero*. In other words, this signifies that for every open U of X, and every section t of $\mathscr{F}$ above U, such that, for *every* λ, the section $u_\lambda(t)$ (which, by definition, is a section of $\mathscr{G}_\lambda$ above $f_\lambda^{-1}(U)$) is zero, then t is itself zero. **IV-3** | 161

(11.9.2). With the notations of (11.9.1), let M be a second set of indices; for every $\lambda \in L$, let $(g_{\lambda\mu})_{\mu \in M}$ be a family of morphisms $g_{\lambda\mu} : Z'_{\lambda\mu} \to Z_\lambda$; for every pair (λ, μ), suppose given a quasi-coherent $\mathscr{O}_{Z'_{\lambda\mu}}$-Module $\mathscr{H}_{\lambda\mu}$ and a homomorphism $v_{\lambda\mu} : \mathscr{G}_\lambda \to (g_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu})$; set $h_{\lambda\mu} = f_\lambda \circ g_{\lambda\mu} : Z'_{\lambda\mu} \to X$ and consider the composite homomorphism

$$w_{\lambda\mu} : \mathscr{F} \xrightarrow{u_\lambda} (f_\lambda)_*(\mathscr{G}_\lambda) \xrightarrow{(f_\lambda)_*(v_{\lambda\mu})} (h_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu}).$$

Suppose that, for every $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu \in M}$ is separating: then, it is the same for the family $(f_\lambda)_*(v_{\lambda\mu})$ $(\mu \in M)$, as one sees at once. One concludes that, for the family (u_λ) to be separating, it is necessary and sufficient that the family $(w_{\lambda\mu})_{(\lambda,\mu) \in L \times M}$ be so.

(11.9.3). To verify that the family (u_λ) is separating (with the notations of (11.9.1)), one can evidently reduce first to the case where X is *affine*, the property being local on X. One can moreover suppose that $Z_\lambda = X$ *for every* $\lambda \in L$. Indeed, let M be an index set, sum of a family $(M_\lambda)_{\lambda \in L}$, and for every $\lambda \in L$, let $(Y_{\lambda\mu})_{\mu \in M_\lambda}$ be an open affine covering of Z_λ; let $j_{\lambda\mu} : Y_{\lambda\mu} \to Z_\lambda$ be the canonical injection and posons $\mathscr{H}_{\lambda\mu} = j^*_{\lambda\mu}(\mathscr{G}_\lambda) = \mathscr{G}_\lambda | Y_{\lambda\mu}$. If one considers the canonical homomorphism $v_{\lambda\mu} : \mathscr{G}_\lambda \to (j_{\lambda\mu})_*(j^*_{\lambda\mu}(\mathscr{G}_\lambda)) = (j_{\lambda\mu})_*(\mathscr{H}_{\lambda\mu})$ relative to $j_{\lambda\mu}$ (**0**, 4.4.3.2), it is immediate that for each $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu \in M_\lambda}$ is separating. By virtue of (11.9.2), one is therefore reduced to proving that the family of composite homomorphisms $((f_\lambda)_*(v_{\lambda\mu})) \circ u_\lambda$, where the f_λ are *affine*, is separating.

But then the $(f_\lambda)_*(\mathscr{G}_\lambda)$ are quasi-coherent $\mathscr{O}_X$-Modules (**I**, 1.6.2) and, by virtue of the definition, one can replace the Z_λ by X and the $\mathscr{G}_\lambda$ by $(f_\lambda)_*(\mathscr{G}_\lambda)$, whence our assertion.

One will note further that if L is *finite* and the f_λ are *quasi-compact*, one can, in the preceding reduction, suppose that the M_λ are also *finite*, whence in this case one is brought back to verifying that a *finite* family of homomorphisms of $\mathscr{F}$ in quasi-coherent $\mathscr{O}_X$-Modules is separating.

(11.9.4). Consider therefore the case where $Z_\lambda = X$ for every λ, and where $X = \operatorname{Spec}(A)$ is affine; then $\mathscr{F} = \widetilde{M}$ and $\mathscr{G}_\lambda = \widetilde{N}_\lambda$, where M and N_λ are A-modules, and $u_\lambda = \widetilde{\varphi}_\lambda$, where the $\varphi_\lambda : M \to N_\lambda$ are A-module homomorphisms. To say that the family (u_λ) is separating signifies then that, for every $s \in A$, the intersection of the kernels of $(\varphi_\lambda)_s : M_s \to (N_\lambda)_s$ is *separating*. One will note that if L is *finite*, it comes down to saying also that the intersection of the kernels of the φ_λ is zero. One says then also that the family (φ_λ) is *separating*. One will note that if L is *finite*, then $\bigcap_{\lambda \in L} \operatorname{Ker}((\varphi_\lambda)_s) = (\bigcap_{\lambda\in L} \operatorname{Ker}(\varphi_\lambda))_s$ (**0**, 1.3.5). But this relation is no longer exact in general when L **IV-3** | 162
is *infinite*, and the fact that the intersection of the kernels of the φ_λ is zero *does not therefore entail*, in general, that the family (φ_λ) is separating. For example, suppose A is a discrete valuation ring of maximal ideal $\mathfrak{m}$, and consider the family of homomorphisms $\varphi_k : A \to A/\mathfrak{m}^k$, whose intersection of kernels $\mathfrak{m}^k$ is reduced to 0; this family is nevertheless not separating, for the fibres of all the $\mathfrak{m}^k$ at the generic point x of $X = \operatorname{Spec}(A)$ (which is open in X) are equal to $\mathscr{O}_x = k(x)$, the field of fractions of A, and their intersection is therefore not reduced to 0.

(11.9.5). We will mainly be concerned in what follows with the problem of *change of base* for separating families. The notations being those of (11.9.1), consider a morphism $g : X' \to X$ and set $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} = g^*(\mathscr{F})$, and, for every λ, $Z'_\lambda = Z_\lambda \times_X X'$, $f'_\lambda = f_\lambda \times 1$, $\mathscr{G}'_\lambda = \mathscr{G}_\lambda \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} = g^*_\lambda(\mathscr{G}_\lambda)$, where $g'_\lambda : Z'_\lambda \to Z_\lambda$ is the projection. For every λ, we denote by $u'_\lambda : \mathscr{F}' \to (f'_\lambda)_*(\mathscr{G}'_\lambda)$ the homomorphism obtained as follows: let

$$\psi : (f_\lambda)_*(\mathscr{G}_\lambda) \longrightarrow (f_\lambda)_*((g'_\lambda)_*((g'_\lambda)^*(\mathscr{G}_\lambda))) = g_*((f'_\lambda)_*(\mathscr{G}'_\lambda))$$

be the homomorphism $(f_\lambda)_*(\rho_{g_\lambda})$, where ρ_{g_λ} is the canonical homomorphism (**0**, 4.4.3.2) corresponding to g_λ. Then u'_λ is defined as the composite

$$g^*(\mathscr{F}) \xrightarrow{g^*(u_\lambda)} g^*((f_\lambda)_*(\mathscr{G}_\lambda)) \xrightarrow{g^*(\psi)} (f'_\lambda)_*(\mathscr{G}'_\lambda)$$

where $\sigma = \sigma_{(f_\lambda)_*(\mathscr{G}'_\lambda)}$ is the canonical homomorphism (**0**, 4.4.3.3) corresponding to g. We will say for short that u'_λ is *deduced from u_λ by the change of base g*. When $f_\lambda : Z_\lambda \to X$ is an *affine* morphism, $(f_\lambda)_*(\mathscr{G}'_\lambda) = g^*((f_\lambda)_*(\mathscr{G}_\lambda))$, and in this case one has therefore simply $u'_\lambda = g^*(u_\lambda)$.

One can also interpret u'_λ in the following manner: it suffices to know the value of $u'_\lambda(t')$, where t' is a section of $\mathscr{F}'$ above an open U′ of X′, of the following particular type: t' is the restriction to U′ of the canonical image by $\Gamma(U, \mathscr{F}) \to \Gamma(g^{-1}(U), \mathscr{F}')$ of a section t of $\mathscr{F}$ above an open U of X containing $g(U')$ (these sections generate the $\mathscr{O}_{X'}$-Module $\mathscr{F}'$ (**0**, 3.7.1)). Consider the section $u_\lambda(t)$ of $\mathscr{G}_\lambda$ above $f_\lambda^{-1}(U)$, and its canonical image t'' by $\Gamma(f_\lambda^{-1}(U), \mathscr{G}_\lambda) \to \Gamma(g'^{-1}_\lambda(f_\lambda^{-1}(U)), \mathscr{G}'_\lambda)$; then $u'_\lambda(t')$ is the restriction of t'' to $f'^{-1}_\lambda(U')$. Consider in particular the case where Z_λ is an open sub-prescheme of X, where $j_\lambda : Z_\lambda \to X$ is the canonical injection, and where u_λ is the canonical homomorphism $\rho_{\mathscr{F}} : \mathscr{F} \to (j_\lambda)_*(j^*_\lambda(\mathscr{F})) = (j_\lambda)_*(\mathscr{G}_\lambda)$ (**0**, 4.4.3.2) corresponding to j_λ. Then Z'_λ is induced on an open subset of X′, and the preceding interpretation shows that u'_λ corresponds to the canonical injection $j'_\lambda : Z'_\lambda \to X'$.

(11.9.6). Under the conditions of (11.9.5), suppose that X and X′ are both affine, and that one wishes to prove that for every section t' of $\mathscr{F}'$ above X′, whose images by all the u'_λ are zero, then t' is itself zero. Then, one can also restrict to the case where $Z_\lambda = X$ for every $\lambda \in L$. Indeed, by (11.9.3) and (11.9.5), if one sets $Y'_{\lambda\mu} = g'^{-1}_\lambda(Y_{\lambda\mu})$, the homomorphism $v'_{\lambda\mu}$ deduced from $v_{\lambda\mu}$ by the change of base g is none other than the canonical homomorphism $\rho_{g'_\lambda} : \mathscr{G}'_\lambda \to (j'_{\lambda\mu})_*(j'^*_{\lambda\mu}(\mathscr{G}'_\lambda))$ corresponding **IV-3** | 163
to the canonical injection $j'_{\lambda\mu} : Y'_{\lambda\mu} \to Z'_\lambda$, as we saw in (11.9.5). The assertion follows then from the reasonings of (11.9.2) and (11.9.3), $Y_{\lambda\mu}$ and $v_{\lambda\mu}$ being replaced by $Y'_{\lambda\mu}$ and $v'_{\lambda\mu}$.

One has an analogous reduction when one wishes to prove that the family (u'_λ) is *separating* (X and X′ being affine): this follows again from (11.9.2) and (11.9.3).

Proposition (11.9.7). — *With the notations of (11.9.1) and (11.9.5), suppose* $X = \mathrm{Spec}(A)$ *and* $X' = \mathrm{Spec}(A')$ *affine, and moreover that* A' *is a projective* A*-module. Then, if the family* (u_λ) *is separating, every section* t' *of* $\mathscr{F}'$ *above* X' *whose images by all the* u'_λ *are zero, is itself zero.*

Proof. One has seen in (11.9.6) that one can restrict to the case where all the Z_λ are *equal to* X. The proposition is then a consequence of the following lemma: □

Lemma (11.9.7.1). — *Let* A *be a ring,* $(M_\lambda)_{\lambda \in L}$ *a family of* A*-modules,* M *an* A*-module, and for each* λ, $u_\lambda : M \to M_\lambda$ *a homomorphism. Suppose that the intersection of the kernels of the* u_λ *is reduced to zero. Then, for every projective* A*-module* N*, the intersection of the kernels of the homomorphisms* $u_\lambda \otimes 1 : M \otimes_A N \to M_\lambda \otimes_A N$ *is reduced to zero.*

Proof. Indeed, N is a direct factor of a free A-module L, and it suffices evidently to prove that the intersection of the kernels of the homomorphisms $v_\lambda : M \otimes_A L \to M_\lambda \otimes_A L$ is reduced to zero, since $u_\lambda \otimes 1$ is the restriction of v_λ. But the assertion follows then trivially from the hypothesis. □

Remarks (11.9.8). — We do not know if, under the hypotheses of proposition (11.9.7), the family (u'_λ) is *separating*: one would need indeed to prove (11.9.4) that, for every open $D(h') \subset X'$ (where $h' \in A'$) such that the $u'_\lambda(t')$ are all zero, then t' is itself zero. Now, one cannot apply proposition (11.9.4) to $D(h') = \mathrm{Spec}(A'_{h'})$, for the fact that A' is a projective A-module (or even a free one), does not imply that $A'_{h'}$ is a projective A-module. For example, one can take for A a discrete valuation ring and for A' a valuation ring that is an A-module free of rank 2.

One has however the following result:

Corollary (11.9.9). — *Let* X *be an Artinian prescheme,* $g : X' \to X$ *a flat morphism (note that these two conditions are satisfied if* X *is the spectrum of a* field *and* g *is an arbitrary morphism). Then, with the notations of (11.9.1) and (11.9.5), if the family* (u_λ) *is separating, it is the same for the family* (u'_λ).

Proof. One can evidently restrict to the case where $X = \mathrm{Spec}(A)$ is the spectrum of an Artinian local ring A (**I**, 6.2.2); one notes then that for every open affine $U' = \mathrm{Spec}(A')$ of X', A' is a flat A-module, hence *projective* ($\mathbf{0}_{\mathrm{III}}$, 10.1.3). It suffices therefore to apply (11.9.7) to every open affine of X' to obtain the corollary. □

Theorem (11.9.10). — *Let* X *be a prescheme,* (u_λ) *a family of homomorphisms (11.9.1),* $g : X' \to X$ *a morphism,* (u'_λ) *the family of homomorphisms deduced from* (u_λ) *by the change of base* g *(11.9.5).*

label=(i) *If* g *is a faithfully flat morphism and if the family* (u'_λ) *is separating, then the family* (u_λ) *is separating.*

lbbel=(ii) *Suppose that* g *is a flat morphism and moreover that one of the two following conditions is satisfied:*
*label=*a) L *is finite and the* f_λ *are quasi-compact.*
*lbbel=*b) *The morphism* g *is locally of finite presentation.*
Then, if the family (u_λ) *is separating, it is the same for the family* (u'_λ).

Proof. **IV-3** | 164

label=(i) By virtue of (2.2.8), it suffices to show that if a section t of $\mathscr{F}$ above an open U of X belongs to the kernel of each of the u_λ, its image t' by the canonical homomorphism $\Gamma(\rho) : \Gamma(U, \mathscr{F}) \to \Gamma(g^{-1}(U), g^*(\mathscr{F})) = \Gamma(U, g_*(g^*(\mathscr{F})))$ is zero. Now the images of t' by the $g^*(u_\lambda)$ are the images of $u_\lambda(t)$ by the homomorphism $\Gamma(U, (f_\lambda)_*(\mathscr{G}_\lambda)) \to \Gamma(g^{-1}(U), g^*((f_\lambda)_*(\mathscr{G}_\lambda)))$, whence they are zero, and *a fortiori* $u'_\lambda(t') = 0$ for every λ, whence $t' = 0$ by hypothesis, which proves (i).

lbbel=(ii) The question being local on X and X', one can restrict to the case where $X = \mathrm{Spec}(A)$ and $X' = \mathrm{Spec}(A')$ are affine, A' being a flat A-module, and to proving that, for every section z' of $\mathscr{F}'$ above X' whose images by all the u'_λ are zero, then z' is itself zero. On has seen in (11.9.3) that one can also restrict to the case where $Z_\lambda = X$ for every $\lambda \in L$. Let us distinguish now the two cases.

a) Let $\mathscr{F} = \widetilde{M}$, $\mathscr{G}_\lambda = \widetilde{N}_\lambda$, and set $M' = M \otimes_A A'$, $N'_\lambda = N_\lambda \otimes_A A'$, so that $\mathscr{F}' = \widetilde{M'}$, $\mathscr{G}'_\lambda = \widetilde{N'_\lambda}$; by abuse of language, let us still denote by u_λ the homomorphism $M \to N_\lambda$, and by u'_λ the homomorphism $M' \to N'_\lambda$. Given an element $z' \in M'$ such that $u'_\lambda(z') = 0$

for every λ; it remains to prove that $z' = 0$. Now, the hypothesis that g is flat and of finite presentation implies, by (11.3.15), that there exists a finite family $(s_i)_{1 \leqslant i \leqslant n}$ of elements of A, such that, if one sets $\mathfrak{J}_k = s_1\mathrm{A} + \ldots + s_k\mathrm{A}$, and $\mathrm{A}_i = \mathrm{A}_{s_i}/\mathfrak{J}_{i-1}\mathrm{A}_{s_i}$, the ring $\mathrm{A}'_i = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_i$ is a *free* A_i-module for $1 \leqslant i \leqslant n$, and $\mathfrak{J}_n = \mathrm{A}$. The proposition will be established if we prove the following assertion for $1 \leqslant i \leqslant n$:

libel=$(*_i)$ *There exists an integer* $m_i > 0$ *such that* $s_j^{m_i} z' = 0$ *for* $j \leqslant i$.

Indeed, setting $k = m_n$ and noting that the s_i^k $(1 \leqslant i \leqslant n)$ generate the unit ideal of A, the assertion $(*_n)$ will show that z', as a linear combination of the $s_i^k z'$, is zero.

Let us prove $(*_i)$ by induction on i, the assertion being trivially true for $i = 0$. Suppose $i > 0$ and let m be a common multiple of the m_j for $j \leqslant i-1$. We note (for $1 \leqslant h \leqslant n$) that if $\mathfrak{J}_h$ is the ideal generated by the s_j^m $(0 \leqslant j \leqslant h)$, $\mathfrak{J}_h/\mathfrak{J}_{h-1}$ is nilpotent; replacing the s_j $(1 \leqslant j \leqslant n)$ by s_j^m without changing the properties of the $\mathfrak{J}_h$ and the A'_h, and supposing $m = 1$. Then z', being annihilated by $\mathfrak{J}_{i-1}$, identifies with an element of $\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}'/\mathfrak{J}_{i-1}\mathrm{A}', \mathrm{M}')$, and this replacement comes down to replacing, for $1 \leqslant h \leqslant n$, A_h by $\mathrm{A}_h/\mathfrak{J}_{h-1}\mathrm{A}_h$, A'_h by $\mathrm{A}'_h/\mathfrak{J}_{h-1}\mathrm{A}'_h$; since A' is a flat A-module, it follows from ($\mathbf{0}_{\mathrm{III}}$, 10.1.2) that $\mathrm{B}'_h = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{B}_h$ is also a free B_h-module. One can therefore replace all the s_j $(1 \leqslant j \leqslant n)$ **IV-3** | 165
by s_j^m without changing the properties of $\mathfrak{J}_h$ and A'_h, and suppose $m = 1$. Then z', being annihilated by $\mathfrak{J}_{i-1}$, identifies with a $(\mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M}))_t$ and, analogously replacing M by N_λ since the ideal $\mathfrak{J}_{i-1}$ is of finite type ($\mathbf{0}$, 1.3.5); whence by our assertion (11.9.4). Replacing A by $\mathrm{A}/\mathfrak{J}_{i-1}$, M by N_λ, A' by $\mathrm{A}'/\mathfrak{J}_{i-1}\mathrm{A}'$, we see that one can reduce to the case where, in the initial situation, l' is the element $s = s_i \in \mathrm{A}$ such that A'_s is a free A_s-module. Let $v_\lambda = \mathrm{Hom}(1, u_\lambda) : \mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{M}) \to \mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/\mathfrak{J}_{i-1}, \mathrm{N}_\lambda)$, the homomorphism deduced from u_λ; the family (v_λ) is also *separating*. Indeed, for every $t \in \mathrm{A}$, $(\mathrm{Hom}_{\mathrm{A}}(1, u_\lambda))_t = \mathrm{Hom}_{\mathrm{A}}(\mathrm{A}/(u_\lambda)_t)$ and from this the assertion follows, since the intersection of the kernels of the $(u_\lambda)_t$ is zero and since the intersection of the kernels of $(v_\lambda)_t = \mathrm{Hom}(1, (u_\lambda)_t)$, whence our assertion (11.9.4). Replacing A by $\mathrm{A}/\mathfrak{J}_{i-1}$, M by N_λ and finally A' by $\mathrm{A}'/\mathfrak{J}_{i-1}\mathrm{A}'$, one sees that one can bring oneself to the case where, in the initial situation, the element $s = s_i \in \mathrm{A}$ is such that A'_s is a free A_s-module. Since $(u'_\lambda)_s = 0$, it follows by (11.9.7) that $z'/1 = 0$ in M'_s, which signifies that there exists an integer r such that $s^r z' = 0$ in M', which finishes the proof of $(*_i)$ by induction.

b) Let $\mathscr{F} = \widetilde{\mathrm{M}}$, $\mathscr{G}_\lambda = \widetilde{\mathrm{N}}_\lambda$, and set $\mathrm{M}' = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}'$, $\mathrm{N}'_\lambda = \mathrm{N}_\lambda \otimes_{\mathrm{A}} \mathrm{A}'$; by abuse of language, let us denote again by u_λ the homomorphism $\mathrm{M} \to \mathrm{N}_\lambda$ and by u'_λ the homomorphism $\mathrm{M}' \to \mathrm{N}'_\lambda$. Given an element $z' \in \mathrm{M}'$ such that $u'_\lambda(z') = 0$ for every λ; it is a matter of proving that $z' = 0$. Now, B is Noetherian, M and the N_λ are B-modules, the N_λ being flat A-modules. Moreover, the question being local on X and S, one can restrict to the case where $\mathrm{Z}_\lambda = \mathrm{X}$ for every λ, $\mathscr{F} = \widetilde{\mathrm{M}}$, $\mathscr{G}_\lambda = \widetilde{\mathrm{N}}_\lambda$, where B is Noetherian, M is a B-module of finite type, and the N_λ are flat A-modules, and restrict to proving that, if $t \in \mathrm{M}$ is such that $u_\lambda(t) = 0$ for every λ, then $t = 0$. To show that $t = 0$, it suffices to prove that the image $t_{\mathfrak{p}}$ of t in $\mathrm{M}_{\mathfrak{p}}$ is zero for every maximal ideal $\mathfrak{p}$ of B (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, cor. 1 of th. 1). One therefore needs also to show that the intersection of the kernels of the $(u_\lambda)_{\mathfrak{p}} : \mathrm{M}_{\mathfrak{p}} \to (\mathrm{N}_\lambda)_{\mathfrak{p}}$ deduced from (u_λ) by the change of base $\mathrm{Spec}(\mathrm{B}_{\mathfrak{p}}) \to \mathrm{Spec}(\mathrm{B})$ is reduced to 0. Otherwise stated, one is brought back to the case where B is a local Noetherian ring, M a B-module of finite type, the ideals of A whose maximal ideal $\mathfrak{m}$ is nilpotent. Suppose therefore that $\mathfrak{J}^{n+1} = 0$ (n an integer $\geqslant 0$) and let us reason by induction on n, the assertion being trivially true for $n = 0$. If $\bar{t}$ is the class of t in $\mathrm{M}/\mathfrak{J}^n\mathrm{M}$, the class of $u_\lambda(t)$ in $\mathrm{N}_\lambda/\mathfrak{J}^n\mathrm{N}_\lambda$ is zero for every λ, whence, by the hypothesis of induction, $\bar{t} = 0$, otherwise said $t \in \mathfrak{J}^n\mathrm{M}$. Now $\mathfrak{J}^n = \mathfrak{J}^n/\mathfrak{J}^{n+1}$ is a free $(\mathrm{A}/\mathfrak{J})$-module; if (e_α) is a base of this module, one can therefore write $t = \sum_\alpha e_\alpha t_\alpha^0$, with $t_\alpha^0 \in \mathrm{A}/\mathfrak{J}$, zero for all but a finite number of indices. On the other hand, since N_λ is a flat A-module, $\mathfrak{J}^n \otimes_{\mathrm{A}/\mathfrak{J}} (\mathrm{N}_\lambda/\mathfrak{J}\mathrm{N}_\lambda)$ identifies with $\mathfrak{J}^n\mathrm{N}_\lambda$ and one can therefore write $u_\lambda(t) = \sum_\alpha e_\alpha u_{\lambda 0}(t_\alpha^0) = \sum_\alpha e_\alpha \otimes u_{\lambda 0}(t_\alpha^0)$. Since by hypothesis $u_\lambda(t) = 0$, one deduces that $u_{\lambda 0}(t_\alpha^0) = 0$ for every α and every λ; whence $t_\alpha^0 = 0$ for every α, and hence $t = 0$.

□

Remark (11.9.11). — *Remark.* — It suffices to simplify to the case where $Z_\lambda = X$ for every λ. It should be noted that if the family of homomorphisms $u_\lambda : \mathscr{F} \to \mathscr{G}_\lambda$ is separating, it does *not necessarily* follow that, for every $x \in X$, the intersection of the kernels of the homomorphisms $(u_\lambda)_x : \mathscr{F}_x \to (\mathscr{G}_\lambda)_x$ is reduced to 0. For example, let X be a locally Noetherian prescheme of Jacobson, of dimension $\geqslant 1$, and $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module; for every closed point x of X, and every integer $n \geqslant 0$, $\mathfrak{m}_x^n\mathscr{F}$ is a coherent $\mathscr{O}_X$-Module with support contained in $\{x\}$. The family of canonical homomorphisms $\mathscr{F} \to \mathscr{F}/\mathfrak{m}_x^n\mathscr{F}$ (where x runs through the set X_0 of closed points of X and n the set of integers $\geqslant 0$) is *separating*: indeed, if t is a section of $\mathscr{F}$ above an open U of X whose images in the $\Gamma(U, \mathscr{F}/\mathfrak{m}_x^n\mathscr{F})$ are all zero, it follows at once that for every closed point $x \in U$, $t_x = 0$; as the set of closed points contained in U is very dense in U (10.2.1), and if $\mathscr{F} = \mathscr{O}_X$, taking $\mathscr{F} = \mathscr{O}_X$, and if y is a *non-closed* point of X, one has $(\mathscr{O}_X/\mathfrak{m}_x^n\mathscr{O}_X)_y = 0$ for every closed point x of X, but $\mathscr{O}_{X,y} \neq 0$, and the intersection of the kernels of the homomorphisms $\mathscr{O}_{X,y} \to (\mathscr{O}_X/\mathfrak{m}_x^n\mathscr{O}_X)_y$ is equal to $\mathscr{O}_{X,y}$.

Lemma (11.9.12). — *Let the notations be those of (11.9.1) and (11.9.5), and suppose the family (u_λ) separating; suppose in addition that X is an S-prescheme, where $S = \mathrm{Spec}(A)$ is affine, and that $X' = X \times_S S'$, where $S' = \mathrm{Spec}(A')$, A′ being an A-algebra; suppose finally that the $\mathscr{G}_\lambda$ are S-flat. Let $(A'_\alpha)_{\alpha \in I}$ be the filtered family of sub-A-algebras of A′ of finite type, and let $S'_\alpha = \mathrm{Spec}(A'_\alpha)$, $X''_\alpha = X \times_S S'_\alpha$, $Z''_{\alpha\lambda} = Z_\lambda \times_X X''_\alpha$, and let $\mathscr{F}''_\alpha$, $\mathscr{G}''_{\alpha\lambda}$ and $u''_{\alpha\lambda}$ be the morphisms, Modules, and homomorphisms of Modules deduced from f_λ, $\mathscr{F}$, $\mathscr{G}_\lambda$ and u_λ by means of the change of base $X''_\alpha \to X$. Then, if for every $\alpha \in I$, the family $(u''_{\alpha\lambda})_{\lambda \in L}$ is separating, it is the same for $(u'_\lambda)_{\lambda \in L}$.*

Proof. It suffices to prove that if t' is a section of $\mathscr{F}'$ above an open affine U′ of X′ such that $u'_\lambda(t') = 0$ for every $\lambda \in L$, then $t' = 0$. If $h_\alpha : X' \to X''_\alpha$ is the projection canonical, it follows from (8.2.11) that **IV-3** | 166 there exists an index α and an open quasi-compact $U''_\alpha \subset X''_\alpha$ such that $U' = h_\alpha^{-1}(U''_\alpha)$; moreover, by virtue of (8.5.2), (i)), one can suppose that t' is the canonical image of a section t''_α of $\mathscr{F}''_\alpha$ above U''_α. It suffices to replace S, X, Z_λ, $\mathscr{F}$, $\mathscr{G}_\lambda$, and u_λ by S''_α, U''_α, $f''^{-1}_\alpha(U''_\alpha)$, $\mathscr{F}''_\alpha$, $\mathscr{G}''_{\alpha\lambda}$ and $u''_{\alpha\lambda}$, and the restrictions, to be able therefore to suppose that $U' = X'$, that t' is the canonical image of a section t of $\mathscr{F}$ above X and that the homomorphism $A \to A'$ is injective, or again, if $p : S' \to S$ is the morphism corresponding, that the homomorphism $\mathscr{O}_S \to p_*(\mathscr{O}_{S'})$ in the definition of p is *injective*. It follows at once by virtue of the flatness of $\mathscr{G}_\lambda$ on S (and by reducing for example to the case where Z_λ is affine on S (**I**, 1.6.3) and (**I**, 1.6.5)) that the canonical homomorphism $\rho : \mathscr{G}_\lambda \to (g'_\lambda)_*(\mathscr{G}'_\lambda)$ is also *injective*. But the composite homomorphism

$$\Gamma(X, \mathscr{F}) \xrightarrow{\Gamma(u_\lambda)} \Gamma(Z_\lambda, \mathscr{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(Z'_\lambda, \mathscr{G}'_\lambda)$$

is equal by definition to the composite homomorphism

$$\Gamma(X, \mathscr{F}) \xrightarrow{\Gamma(\rho)} \Gamma(X', \mathscr{F}') \xrightarrow{\Gamma(u'_\lambda)} \Gamma(Z'_\lambda, \mathscr{G}'_\lambda)$$

whence the image of t by these composite homomorphisms is $u'_\lambda(t') = 0$; by virtue of the injectivity of the homomorphism $\Gamma(Z_\lambda, \mathscr{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(Z'_\lambda, \mathscr{G}'_\lambda)$, one concludes that $u_\lambda(t) = 0$ for every $\lambda \in L$, whence $t = 0$ by hypothesis, and finally $t' = 0$. □

Proposition (11.9.13). — *Let the notations be those of (11.9.1) and (11.9.5), and suppose that X is a prescheme over a field k, and that, setting $S = \mathrm{Spec}(k)$, one has $X' = X \times_S S'$, where S′ is an arbitrary k-prescheme. Then, if the family $(u_\lambda)_{\lambda \in L}$ is separating, it is the same for (u'_λ).*

Proof. One can restrict to the case where $S' = \mathrm{Spec}(A')$ is affine. If A′ is a finite k-algebra of flat and of finite presentation, and one is therefore in the conditions of application of (11.9.10), (ii), *b*)). In the general case, one considers A′ as an inductive limit of its finite sub-k-algebras A'_α of finite type, and one applies to each A'_α the result of (11.9.10), (ii), *b*)); one concludes then by the lemma (11.9.12), since the $\mathscr{G}_\lambda$ are S-flat. □

(11.9.14). Let us keep the notations of (11.9.1) and (11.9.5), and suppose that X is an S-prescheme. If for every change of base $g : X \times_S S' \to X$, where $S' \to S$ is an arbitrary morphism, the corresponding family (u'_λ) is separating, we say that the family (u_λ) is *universally separating relatively to* S. When the family (u_λ) is reduced to a single element u, we say further that u is *universally injective*, relatively to S. It is clear then that for every morphism $h : S' \to S$, the family (u'_λ) corresponding is universally separating relative to S'; conversely, if h is *faithfully flat* and if (u'_λ) is universally separating relative to S', then (u_λ) is universally separating relative to S, as follows immediately from (11.9.10), (i)) and from the fact that for every morphism $S'' \to S$, the corresponding morphism $S'' \times_S S' \to S''$ is faithfully flat.

Proposition (11.9.15). — *Let the notations be those of (11.9.1), and suppose that* X *is an* S*-prescheme, the* $\mathscr{G}_\lambda$ *being* S*-flat. Let* S_0 *be a closed sub-prescheme of* S *defined by a nilpotent quasi-coherent Ideal* $\mathscr{J}$ *of* $\mathscr{O}_S$, *such that the* $(\mathscr{O}_S/\mathscr{J})$*-Modules* $\mathscr{J}^k/\mathscr{J}^{k+1}$ *are all locally free. Let* $(u_{\lambda 0})$ *be the family of homomorphisms* **IV-3** | 167
obtained from (u_λ) *by the change of base* $X \leftarrow X_0 = X \times_S S_0$. *Then, if the family* $(u_{\lambda 0})$ *is separating (resp. universally separating relative to* S_0*), the family* (u_λ) *is separating (resp. universally separating relative to* S*).*

Proof. Let us note that if $S' \to S$ is an arbitrary change of base and $S'_0 = S' \times_S S_0$, $\mathscr{J}'$ the quasi-coherent nilpotent Ideal of $\mathscr{O}_{S'}$ such that the $(\mathscr{O}_{S'}/\mathscr{J}')$-Modules $\mathscr{J}'^k/\mathscr{J}'^{k+1}$ are all locally free for every k (2.1.8), (i)); as moreover the $\mathscr{G}'_\lambda = \mathscr{G}_\lambda \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$ are S'-flat, one sees that the assertion relative to universally separating families is a consequence of that relative to separating families. To prove this latter, one can reduce by (11.9.3) to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine, $Z_\lambda = X$ for every λ, $\mathscr{F} = \widetilde{M}$, $\mathscr{G}_\lambda = \widetilde{N}_\lambda$, where B is Noetherian, M is a B-module of finite type, and the N_λ are flat A-modules, and restrict to proving that, if $t \in M$ is such that $u_\lambda(t) = 0$ for every λ, then $t = 0$. Moreover, the question being local on X and S, it suffices to see that $t = 0$. On a $\mathscr{F} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is a nilpotent ideal of A, such that the $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ are $(A/\mathfrak{J})$-modules locally free, and the N_λ are flat A-modules. Suppose $\mathfrak{J}^{n+1} = 0$ (n an integer $\geqslant 0$) and let us reason by induction on n, the assertion being trivially true for $n = 0$. If $\bar{t}$ is the class of t in $M/\mathfrak{J}^n M$, the class of $u_\lambda(t)$ in $N_\lambda/\mathfrak{J}^n N_\lambda$ is zero for every λ, whence, by the hypothesis of induction, $\bar{t} = 0$, otherwise said $t \in \mathfrak{J}^n M$. Now $\mathfrak{J}^n = \mathfrak{J}^n/\mathfrak{J}^{n+1}$ is a free $(A/\mathfrak{J})$-module; if (e_α) is a base of this module, one can therefore write $t = \sum_\alpha e_\alpha t^0_\alpha$, with $t^0_\alpha \in A/\mathfrak{J}$, zero for all but a finite number of indices. On the other hand, since N_λ is a flat A-module, $\mathfrak{J}^n \otimes_{(A/\mathfrak{J})} (N_\lambda/\mathfrak{J}N_\lambda)$ identifies with $\mathfrak{J}^n N_\lambda$ and one can therefore write $u_\lambda(t) = \sum_\alpha e_\alpha u_{\lambda 0}(t^0_\alpha) = \sum_\alpha e_\alpha \otimes u_{\lambda 0}(t^0_\alpha)$. Since by hypothesis $u_\lambda(t) = 0$, one deduces that $u_{\lambda 0}(t^0_\alpha) = 0$ for every α and every λ; whence $t^0_\alpha = 0$ for every α, and hence $t = 0$. □

Theorem (11.9.16). — *Let the notations be those of (11.9.1), and suppose that* X *is a locally Noetherian* S*-prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module, and that the* $\mathscr{G}_\lambda$ *are* S*-flat. For every* $s \in S$, *let* $X_s = X \times_S \operatorname{Spec}(k(s))$, $(Z_\lambda)_s = Z_\lambda \times_S \operatorname{Spec}(k(s))$, $(f_\lambda)_s = (f_\lambda)_s \times 1 : (Z_\lambda)_s \to X_s$. *For the family* (u_λ) *to be universally separating relative to* S*, it is necessary and sufficient that for every* $s \in S$, *the family* $((u_\lambda)_s)$ *be separating.*

Proof. The necessity of the condition follows trivially from the definitions. Conversely, suppose the condition satisfied, and let us prove first that the family (u_λ) is separating. One can by (11.9.3) reduce to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine, $Z_\lambda = X$ for every λ, $\mathscr{F} = \widetilde{M}$, $\mathscr{G}_\lambda = \widetilde{N}_\lambda$, where B is Noetherian, M is a B-module of finite type, and the N_λ are flat A-modules, and restrict to proving that, if $t \in M$ is such that $u_\lambda(t) = 0$ for every λ, then $t = 0$. To show that $t = 0$, it suffices to prove that the image $t_\mathfrak{p}$ of t in $M_\mathfrak{p}$ is zero for every maximal ideal $\mathfrak{p}$ of B (Bourbaki, *Alg. comm.*, chap. II, § 3, no. 3, cor. 1 of th. 1). One needs therefore also show that the intersection of the **IV-3** | 168
kernels of the $(u_\lambda)_\mathfrak{p} : M_\mathfrak{p} \to (N_\lambda)_\mathfrak{p}$ deduced from (u_λ) by the change of base $\operatorname{Spec}(B_\mathfrak{p}) \to \operatorname{Spec}(B)$ is reduced to 0. Otherwise stated, one is brought back to the case where B is a local Noetherian ring of maximal ideal $\mathfrak{m}$, and that M is a B-module of finite type. Then, since $\mathfrak{m}B$ is contained in the maximal ideal of B, and M is a B-module of finite type, the intersection of $\mathfrak{m}^{n+1}M$ is reduced to 0 (**0**, 7.3.5), whence it suffices to prove that for every n, the image of t in $M/\mathfrak{m}^{n+1}M$ is zero. It suffices therefore to prove that the family deduced from (u_λ) by the change of base $\operatorname{Spec}(B/\mathfrak{m}^{n+1}B) \to \operatorname{Spec}(B)$ is separating, which signifies that one can restrict to the case where A is a local ring whose maximal ideal $\mathfrak{m}$ is nilpotent. But then the $\mathfrak{m}^k/\mathfrak{m}^{k+1}$ are $(A/\mathfrak{m})$-modules free, and, by virtue of the hypothesis on the $(u_\lambda)_s$, one is precisely in the conditions of application of (11.9.15), whence the conclusion announced.

Let us now show that $h : S' \to S$ is an arbitrary change of base, and (u'_λ) the family obtained from (u_λ) by the change of base $h' : X \times_S S' \to X$; let us prove that (u'_λ) is also separating. Suppose first that h is *locally of finite type*; then it is the same for h', and hence $X' = X \times_S S'$ is locally Noetherian; moreover, if $s' \in S'$ is above $s \in S$, it follows from (11.9.13) applied to u_α, that the ensemble E_α of the $s_\alpha \in S_\alpha$ such that $(u_\alpha)_{s_\alpha}$ is injective is constructible. Since $X_s = X \times_S \operatorname{Spec}(k(s))$, the hypothesis that

u_s is injective implies that $(u_\alpha)_{s_\alpha}$ is injective, where $s_\alpha = q_\alpha(s)$ is the image of s by the morphism $q_\alpha : \mathrm{S} \to \mathrm{S}_\alpha$, on deduces from (8.3.4) that there exists an index α such that $\mathrm{E}_\alpha = \mathrm{S}_\alpha$. But then, X_α being Noetherian, u_α is universally injective by virtue of (11.9.16), whence u is also universally injective.

Finally, if h is arbitrary, one can evidently restrict to the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{S}' = \mathrm{Spec}(\mathrm{A}')$ are affine, and consider A' as an inductive limit of its sub-A-algebras of finite type. As the $\mathscr{G}_\lambda$ are S-flat, it suffices to apply what precedes and the lemma (11.9.12) to complete the proof. □

Proposition (11.9.17). — *Let $f : \mathrm{X} \to \mathrm{S}$ be a locally of finite presentation morphism, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_\mathrm{X}$-Module of finite presentation and f-flat, U an open of X, $j : \mathrm{U} \to \mathrm{X}$ the canonical injection, $u : \mathscr{F} \to j_*(j^*(\mathscr{F}))$ the canonical homomorphism* (**0**, 4.4.3.2). *For every $s \in \mathrm{S}$, let X_s be the fibre $\mathrm{X} \times_\mathrm{S} \mathrm{Spec}(k(s))$, U_s the open $\mathrm{U} \cap \mathrm{X}_s$ of X_s, $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{S}} k(s)$, $j_s : \mathrm{U}_s \to \mathrm{X}_s$ the canonical injection, $u_s : \mathscr{F}_s \to (j_s)_*((j_s)^*(\mathscr{F}_s))$ the canonical homomorphism. For u to be universally injective relative to S, it is necessary and sufficient that u_s be injective for every $s \in \mathrm{S}$.*

Proof. There is nothing more than to prove the sufficiency of the condition. When X is locally Noetherian, the proposition is an immediate corollary of (11.9.16). We will now reduce ourselves to two stages, in the case where $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ and X are affine.

A) Case where U *is quasi-compact.* — We will use the following lemma:

Lemma (11.9.17.1). — *Under the general hypotheses of (11.9.17), and supposing in addition* S *and* X *affine and* U *quasi-compact, the set* E *of the $s \in \mathrm{S}$ such that u_s is injective is constructible.*

Proof. Indeed, the fibres X_s are locally Noetherian preschemes, whence E can also, by virtue of (5.10.2), be defined as the set of $s \in \mathrm{S}$ such that $\mathrm{Ass}(\mathscr{F}_s) \subset \mathrm{U}_s$. As $\mathrm{Ass}(\mathscr{F}_s)$ is finite (3.1.6), the hypothesis signifies therefore that there exists λ such that $s \in \mathrm{E}_\lambda$. The verification of (9.2.1), (ii)) is done easily using the study of associated prime cycles at the generic fibre (9.8.3), as well as (9.5.2) and (9.5.3). □ **IV-3** | 169

This lemma being established, one can, by virtue of (8.9.1) and (8.2.11), suppose that there exists a Noetherian sub-ring A_0 of A, a prescheme X_0 of finite type over $\mathrm{S}_0 = \mathrm{Spec}(\mathrm{A}_0)$, an open U_0 of X_0, and a coherent $\mathscr{O}_{\mathrm{X}_0}$-Module $\mathscr{F}_0$, such that $\mathrm{X} = \mathrm{X}_0 \times_{\mathrm{S}_0} \mathrm{S}$, $\mathrm{U} = \mathrm{U}_0 \times_{\mathrm{S}_0} \mathrm{S}$, and $\mathscr{F} = p_0^*(\mathscr{F}_0)$. Let (A_α) be the filtered family of sub-rings of A that are A_0-algebras of finite type, and set $\mathrm{S}_\alpha = \mathrm{Spec}(\mathrm{A}_\alpha)$, $\mathrm{X}_\alpha = \mathrm{X}_0 \times_{\mathrm{S}_0} \mathrm{S}_\alpha$, $\mathscr{F}_\alpha = \mathscr{F}_0 \otimes_{\mathrm{S}_0} \mathrm{S}_\alpha$; let u_α be the canonical homomorphism relative to $\mathscr{F}_\alpha$ and U_α, defined as in (11.9.17). For every $s \in \mathrm{S}$, the hypothesis that u_s is injective implies that $(u_\alpha)_{s_\alpha}$ is injective, where $s_\alpha = q_\alpha(s)$ is the image of s by the morphism $q_\alpha : \mathrm{S} \to \mathrm{S}_\alpha$; one deduces from (8.3.4) that there exists an index α such that $\mathrm{E}_\alpha = \mathrm{S}_\alpha$. But then, since X_α is Noetherian, u_α is universally injective by virtue of (11.9.16), whence u is also universally injective.

B) General case. — The open U is a filtered increasing union of quasi-compact opens U_λ; if $j_\lambda : \mathrm{U}_\lambda \to \mathrm{X}$ is the canonical injection and $u_\lambda : \mathscr{F} \to (j_\lambda)_*(j_\lambda^*(\mathscr{F}))$ the canonical homomorphism corresponding, it follows from the lemma (11.9.17.1) applied to u_λ, that the set E_λ of the $s \in \mathrm{S}$ such that $(u_\lambda)_s$ is injective is constructible. On the other hand, for every $s \in \mathrm{S}$, to say that u_s (resp. $(u_\lambda)_s$) is injective signifies that $\mathrm{Ass}(\mathscr{F}_s) \subset \mathrm{U} \cap \mathrm{X}_s$ (resp. $\mathrm{Ass}(\mathscr{F}_s) \subset \mathrm{U}_\lambda \cap \mathrm{X}_s$) (5.10.2). As $\mathrm{Ass}(\mathscr{F}_s)$ is finite (3.1.6), to say that u_s is injective signifies therefore that there exists λ such that $s \in \mathrm{E}_\lambda$. In virtue of (1.9.9), it follows that $\mathrm{S} = \bigcup \mathrm{E}_\lambda$, whence the conclusion follows by the first part of the reasoning that u_λ is universally injective. It follows then from the factorisation of u_λ:

$$\mathscr{F} \xrightarrow{u} j_*(\mathscr{F}|\mathrm{U}) \longrightarrow (j_\lambda)_*(\mathscr{F}|\mathrm{U}_\lambda)$$

that u is also universally injective. □

Remarks (11.9.18). — Let X be a prescheme, $\mathscr{F}$, $\mathscr{G}$ two quasi-coherent $\mathscr{O}_\mathrm{X}$-Modules of finite presentation, $\mathscr{G}$ being moreover supposed *locally free*, $u : \mathscr{F} \to \mathscr{G}$ a homomorphism. The following conditions are equivalent:

label=*a)* for every morphism $g : \mathrm{X}' \to \mathrm{X}$, the homomorphism $g^*(u) : g^*(\mathscr{F}) \to g^*(\mathscr{G})$ is injective;
lbbel=*b)* for every $x \in \mathrm{X}$, the homomorphism $u \otimes 1_{k(x)} : \mathscr{F} \otimes_{\mathscr{O}_\mathrm{X}} k(x) \to \mathscr{G} \otimes_{\mathscr{O}_\mathrm{X}} k(x)$ is injective;
lcbel=*c)* for every $x \in \mathrm{X}$, there exists a neighbourhood U of x such that $u|\mathrm{U} : \mathscr{F}|\mathrm{U} \to \mathscr{G}|\mathrm{U}$ is an isomorphism of $\mathscr{F}|\mathrm{U}$ onto a direct factor of $\mathscr{G}|\mathrm{U}$.

It is clear that $c)$ implies $a)$, and $a)$ implies $b)$. The fact that $b)$ implies $c)$ follows from (**0**, 19.1.12), (**0**, 5.2.5) and (**0**, 5.5.5).

When the preceding equivalent conditions are satisfied, one says that u is *universally injective*.

11.10. Schematically dominant families of morphisms and families of schematically dense sub-preschemes

Proposition (11.10.1). — *Let* X *be a prescheme,* $(\mathrm{Z}_\lambda)_{\lambda\in\mathrm{L}}$ *a family of preschemes, and for every* $\lambda \in \mathrm{L}$*, let* $f_\lambda = (\psi_\lambda, \theta_\lambda) : \mathrm{Z}_\lambda \to \mathrm{X}$ *be a morphism. The following conditions are equivalent:*

label=a) *The family of homomorphisms* $\theta_\lambda : \mathscr{O}_\mathrm{X} \to (f_\lambda)_*(\mathscr{O}_{\mathrm{Z}_\lambda})$ *is separating (in other words (11.9.1), the intersection of the kernels of the* θ_λ *is* zero*).*

lbbel=b) *For every open* U *of* X*, every section* t *of* $\mathscr{O}_\mathrm{X}$ *above* U *whose images by all the canonical homomorphisms*

$$(\theta_\lambda)_\mathrm{U} : \Gamma(\mathrm{U}, \mathscr{O}_\mathrm{X}) \longrightarrow \Gamma(f_\lambda^{-1}(\mathrm{U}), \mathscr{O}_{\mathrm{Z}_\lambda})$$

are zero, is itself zero.

lcbel=c) *For every open* U *of* X *and every closed sub-prescheme* Y *of* U *such that for every* $\lambda \in \mathrm{L}$*, there exists a factorisation*

$$f_\lambda^{-1}(\mathrm{U}) \xrightarrow{g_\lambda} \mathrm{Y} \xrightarrow{j} \mathrm{U}$$

of the restriction $f_\lambda^{-1}(\mathrm{U}) \to \mathrm{U}$ *of* f_λ *(where* j *is the canonical injection), one has* $\mathrm{Y} = \mathrm{U}$*.*

If moreover X is an S-prescheme, these conditions are also equivalent to the following:

ldbel=d) *For every separated morphism* $g : \mathrm{X}' \to \mathrm{S}$ *and every pair* u, v *of* S*-morphisms from an open* U *of* X *to* X'*, such that for every* λ*, the composites of* u *and* v *with the restriction of* f_λ *to* $f_\lambda^{-1}(\mathrm{U})$ *are equal, one has* $u = v$*.*

Proof. The equivalence of $a)$ and $b)$ follows from the definitions. To see that $b)$ implies $c)$, it suffices to consider the Ideal $\mathscr{J}$ of $\mathscr{O}_\mathrm{U}$ defining Y, and to note that, for every open $\mathrm{V} \subset \mathrm{U}$, the hypothesis implies that the morphism $(\theta_\lambda)_\mathrm{V}$ factorises as

$$\Gamma(\mathrm{V}, \mathscr{O}_\mathrm{U}) \longrightarrow \Gamma(\mathrm{V}, \mathscr{O}_\mathrm{U}/\mathscr{J}) \longrightarrow \Gamma(f_\lambda^{-1}(\mathrm{V}), \mathscr{O}_{\mathrm{Z}_\lambda}).$$

One concludes that every section t of $\mathscr{J}$ above V has image 0 in all the $\Gamma(f_\lambda^{-1}(\mathrm{V}), \mathscr{O}_{\mathrm{Z}_\lambda})$, whence, by virtue of $b)$, $\mathscr{J} = 0$ and $\mathrm{Y} = \mathrm{U}$. Conversely, if $c)$ is satisfied, to prove $b)$, it suffices to apply $c)$ to the closed sub-prescheme Y of U defined by the Ideal $t\mathscr{O}_\mathrm{U}$: the hypothesis that the images of t by the $(\theta_\lambda)_\mathrm{U}$ are all zero implies that one has the factorisations (11.10.1.2) for every λ (**I**, 4.1.9). To prove that $c)$ implies $d)$, it suffices to apply $c)$ to the closed sub-prescheme Y of U, the inverse image of the diagonal $\mathrm{X}' \times_\mathrm{S} \mathrm{X}'$ by (u, v), and to use (**I**, 4.4.1). Conversely, one deduces $b)$ from $d)$ by considering the S-scheme $\mathrm{X}' = \mathrm{S}[\mathrm{T}] = \mathrm{S} \otimes_\mathbf{Z} \mathbf{Z}[\mathrm{T}]$ (T indeterminate) and noting that the sections of $\mathscr{O}_\mathrm{U}$ above U correspond bijectively to the S-morphisms $\mathrm{U} \to \mathrm{S}[\mathrm{T}]$ (**I**, 3.3.15); to say that two sections of $\mathscr{O}_\mathrm{U}$ above U have the same images by all the $(\theta_\lambda)_\mathrm{U}$ is equivalent to saying that the composites of the two corresponding morphisms with all the morphisms $f_\lambda^{-1}(\mathrm{U}) \to \mathrm{U}$ are equal. □

Definition (11.10.2). — When the equivalent conditions of (11.10.1) are satisfied, one says that the family (f_λ) is *schematically dominant*. When the f_λ are the canonical immersions in X of a family (Z_λ) of sub-preschemes of X, one says also that the family (Z_λ) is *schematically dense*.

Remarks (11.10.3). —

label=(i) The notion of a schematically dominant family on X, as follows for example from a condition of the form $b)$ of (11.10.1), is *local* on X: if (W_α) is an open covering of X, the family (f_λ) is schematically dominant if and only if each of the families formed of the morphisms $f_\lambda^{-1}(\mathrm{W}_\alpha) \to \mathrm{W}_\alpha$, restrictions of the f_λ, is schematically dominant.

lbbel=(ii) If Z is the prescheme sum of the Z_λ, $f : \mathrm{Z} \to \mathrm{X}$ the morphism coinciding with f_λ on each Z_λ, it amounts to the same to say that the family (f_λ) is schematically dominant, or that the family reduced to the single element f is.

lcbel=(iii) Let M be a second set of indices, and, for every $\lambda \in \mathrm{L}$, let $(g_{\lambda\mu})_{\mu\in\mathrm{M}}$ be a family of morphisms $g_{\lambda\mu} : \mathrm{Z}'_{\lambda\mu} \to \mathrm{Z}_\lambda$; if, for every $\lambda \in \mathrm{L}$, the family $(g_{\lambda\mu})_{\mu\in\mathrm{M}}$ is schematically dominant, then, it amounts to the same to say that the family $(f_\lambda)_{\lambda\in\mathrm{L}}$ is schematically dominant, or that the family $(f_\lambda \circ g_{\lambda\mu})_{(\lambda,\mu)\in\mathrm{L}\times\mathrm{M}}$ is (11.9.2).

ldbel=(iv) Let $f : \mathrm{Z} \to \mathrm{X}$ be a morphism such that $f_*(\mathscr{O}_\mathrm{Z})$ is a quasi-coherent $\mathscr{O}_\mathrm{X}$-Module (for example a quasi-compact and quasi-separated morphism (1.7.4)). Then, to say that f is schematically dominant signifies that the *closed image* of Z *by* f (**I**, 9.5.3) *is identical to* X.

Proposition (11.10.4). — *If the family of morphisms $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ is schematically dominant, the union of the $f_\lambda(\mathrm{Z}_\lambda)$ is dense in* X. *Conversely, if this union is dense in* X *and if moreover* X *is* reduced, *the family (f_λ) is schematically dominant.*

Proof. The first assertion follows immediately from (11.10.1), *b*)). On the other hand, if X is reduced, and if the union of the $f_\lambda(\mathrm{Z}_\lambda)$ is dense in X, then, if one has the factorisations (11.10.1.2) for every $\lambda \in \mathrm{L}$, one also has factorisations $(f_\lambda^{-1}(\mathrm{U}))_{\mathrm{red}} \xrightarrow{(g_\lambda)_{\mathrm{red}}} \mathrm{Y}_{\mathrm{red}} \xrightarrow{j_{\mathrm{red}}} \mathrm{U}$, and the hypothesis implies that the underlying space of Y is identical to U, whence $\mathrm{Y} = \mathrm{Y}_{\mathrm{red}} = \mathrm{U}$ since U is reduced. □

The results of § 11.9 on separating families translate into results on schematically dominant families:

Theorem (11.10.5). — *Let (f_λ) be a family of morphisms $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$, $g : \mathrm{X}' \to \mathrm{X}$ a morphism, and set $\mathrm{Z}'_\lambda = \mathrm{Z}_\lambda \times_\mathrm{X} \mathrm{X}'$, $f'_\lambda = (f_\lambda)_{(\mathrm{X}')} : \mathrm{Z}'_\lambda \to \mathrm{X}'$.*

label=(i) If g is faithfully flat and if the family (f'_λ) is schematically dominant, then the family (f_λ) is schematically dominant.

lbbel=(ii) Suppose that g is a flat morphism, and moreover that one of the two following conditions is satisfied:

*label=*a) L *is finite and the f_λ are quasi-compact.*

*lbbel=*b) *The morphism g is locally of finite presentation.*

Then, if the family (f_λ) is schematically dominant, it is the same for the family (f'_λ).

Proposition (11.10.6). — *Let the notations be those of* (11.10.5), *and suppose that* X *is a prescheme over a field k, and that, setting $\mathrm{S} = \mathrm{Spec}(k)$, one has $\mathrm{X}' = \mathrm{X} \times_\mathrm{S} \mathrm{S}'$, where* S′ *is an arbitrary k-prescheme. Then, if the family (f_λ) is schematically dominant, it is the same for (f'_λ).* **IV-3 | 172**

Corollary (11.10.7). — *Let* X *be a prescheme over a field k, $(\mathrm{Z}_\lambda)_{\lambda\in\mathrm{L}}$ a family of k-preschemes, geometrically reduced over k, and for every $\lambda \in \mathrm{L}$, let $f_\lambda : \mathrm{Z}_\lambda \to \mathrm{X}$ be a k-morphism. Let* Y *denote the reduced sub-prescheme of* X *having for underlying space the closure of $\bigcup_{\lambda\in\mathrm{L}} f_\lambda(\mathrm{Z}_\lambda)$. Let k' be an extension of k, X', Z_λ, $f'_\lambda : \mathrm{Z}'_\lambda \to \mathrm{X}'$ the preschemes deduced from* X*, Z_λ and f_λ by extension of the base to k'. Then, if* Y′ *is the reduced sub-prescheme of* X′ *having for underlying space the closure of $\bigcup_{\lambda\in\mathrm{L}} f'_\lambda(\mathrm{Z}'_\lambda)$, one has $\mathrm{Y}' = \mathrm{Y} \otimes_k k'$. In particular,* Y *is geometrically reduced over k.*

Proof. Since the Z_λ are reduced, the morphisms f_λ factorise as $\mathrm{Z}_\lambda \xrightarrow{g_\lambda} \mathrm{Y} \xrightarrow{j} \mathrm{X}$, where j is the canonical injection (**I**, 5.2.2). It follows then from (11.10.4) that (g_λ) is a schematically dominant family. Let $\mathrm{Y}'_1 = \mathrm{Y} \otimes_k k'$, a closed sub-prescheme of X′, and let g'_λ be the morphism deduced from g_λ by extension of the base to k', so that f'_λ factorises as $\mathrm{Z}'_\lambda \xrightarrow{g'_\lambda} \mathrm{Y}'_1 \xrightarrow{j'} \mathrm{X}'$, where j' is the canonical injection. It follows from (11.10.6) that the family (g'_λ) is schematically dominant. But by hypothesis the Z'_λ are *reduced*, whence (**I**, 5.2.2) the g'_λ factorise as $\mathrm{Z}'_\lambda \to \mathrm{Y}' \to \mathrm{Y}'_1$; one concludes therefore from (11.10.1.2) that $\mathrm{Y}' = \mathrm{Y}'_1$ and $h'_\lambda = g'_\lambda$, which establishes the corollary. □

Definition (11.10.8). — Let the notations be those of (11.10.5), and suppose that X is an S-prescheme. One says that the family (f_λ) is *universally schematically dominant* (relative to S) if, for every change of base $\mathrm{S}' \to \mathrm{S}$, the family (f'_λ) corresponding to the change of base $\mathrm{X}' = \mathrm{X} \times_\mathrm{S} \mathrm{S}' \to \mathrm{X}$ is schematically dominant.

When S is the spectrum of a field, (11.10.6) then shows that a schematically dominant family is universally schematically dominant (relative to S).

When the f_λ are canonical immersions $\mathrm{Z}_\lambda \to \mathrm{X}$, one says also that the family of sub-preschemes Z_λ is *universally schematically dense in* X (relative to S) instead of saying that the family (f_λ) is universally schematically dominant (relative to S).

Theorem (11.10.9). — *Let the notations be those of (11.10.5), and suppose that* X *is a locally Noetherian* S*-prescheme and that the* Z_λ *are all* S*-flat. For every* $s \in S$*, let* $X_s = X \times_S \operatorname{Spec}(k(s))$*,* $(Z_\lambda)_s = Z_\lambda \times_S \operatorname{Spec}(k(s))$*,* $(f_\lambda)_s = (f_\lambda)_s \times 1 : (Z_\lambda)_s \to X_s$*. For the family* (f_λ) *to be universally schematically dominant relative to* S*, it is necessary and sufficient that for every* $s \in S$*, the family* $((f_\lambda)_s)$ *be schematically dominant.*

Proposition (11.10.10). — *Let* $f : X \to S$ *be a flat locally of finite presentation morphism,* U *an open of* X*. For* U *to be universally schematically dense in* X *relative to* S*, it is necessary and sufficient that, for every* $s \in S$*,* $U_s = U \cap X_s$ *be schematically dense in* X_s *(or, what amounts to the same, that one has* $\operatorname{Ass}(\mathscr{O}_{X_s}) \subset U_s$*).*

§12. STUDY OF FIBRES OF FINITELY PRESENTED FLAT MORPHISMS

12.1. Introduction *We will use throughout this paragraph the general notation of (9.4.1).* **IV-3** | 173

(12.0.1). Given a morphism *locally of finite presentation* $f : X \to Y$, we have seen (9.9) that for certain *local* properties **P** of preschemes over fields or of Modules over these preschemes, the set E of $x \in X$ such that the property **P** holds, for the *fibre* $X_{f(x)}$, at the point x of this fibre, is *locally constructible in* X. We now propose to show that, for the majority of these properties, if one further supposes that the morphism f is *flat*, then the set E is in fact *open in* X. Similarly ((9.2) to (9.8)) we have shown that if f is *of finite presentation*, and if this time **P** denotes certain *global* properties of preschemes over fields or of Modules over these preschemes, the set F of $y \in Y$ such that the property **P** holds is *locally constructible in* Y. We will show that if one further supposes that the morphism f is *proper* and *flat*, F is in fact *open in* Y.

(12.0.2). The general method of proof of the properties in question involves three stages. One first reduces to the case where Y is *affine* and X of finite presentation; then:

label=*A)* With the help of (8.9.1) (and possibly other results of § 8) and of (11.2.6), one reduces to the case where X and Y are *Noetherian*.

lbbel=*B)* One applies the results of § 9 recalled in (12.0.1) to show that E (resp. F) is *constructible*.

lcbel=*C)* To see that E is open, it suffices, by virtue of (**0**, 9.2.5), to show that if $x \in E$, then every *generization* x' of x also belongs to E. Using (**II**, 7.1.7), one sees that, since Y is *Noetherian*, there exists a spectrum of a *discrete valuation ring* Y_1 and a morphism $h : Y_1 \to X$ such that, if y_1 (resp. y'_1) is the closed point (resp. the generic point) of Y_1, one has $h(y_1) = x$, $h(y'_1) = x'$. One then performs the base change $g = f \circ h : Y_1 \to Y$; taking into account the results of §§ 4 and 6 on locally Noetherian preschemes over fields and the changes of base field, one is reduced to proving the assertion in question for a point x_1 of $X_1 = X \times_Y Y_1$ above y_1 and for a generization x'_1 of x_1 above y'_1. Since $Y_1 = \operatorname{Spec}(A)$, where A is a discrete valuation ring, with uniformiser t, one must in the end, for an A-module M, prove a property of M, knowing that M/tM has the same property and that t is M-*regular* (which follows from the flatness hypothesis); one uses for this the results of (3.4) and of (5.12). One proceeds similarly for the set $F \subset Y$.

12.2. Local properties of fibres of a flat morphism locally of finite presentation **IV-3** | 174

Theorem (12.1.1). — *Let* $f : X \to Y$ *be a morphism locally of finite presentation,* $\mathscr{F}$ *an* $\mathscr{O}_X$*-Module* f*-flat and of finite presentation,* Φ *a finite subset of* $\mathbf{Z} \cup \{\pm\infty\}$*,* k *an integer. The following subsets of* X *are open:*

label=*(i)* *The set of* $x \in X$ *such that the dimensions of the prime cycles associated to* $\mathscr{F}_{f(x)}$ *and containing* x *are elements of* Φ*.*

lbbel=*(ii)* *The set of* $x \in X$ *such that all the prime cycles associated to* $\mathscr{F}_{f(x)}$ *containing* x *have dimension* k*.*

lcbel=*(iii)* *The set of* $x \in X$ *such that* x *does not belong to any associated immersed prime cycle of* $\mathscr{F}_{f(x)}$*.*

ldbel=*(iv)* *The set of* $x \in X$ *such that* $\mathscr{F}_{f(x)}$ *is equidimensional at the point* x *and possesses the property* (S_k) *at the point* x *(5.7.2).*

lebel=*(v)* *The set of* $x \in X$ *such that* $\operatorname{coprof}((\mathscr{F}_{f(x)})_x) \leqslant k$ (**0**, 16.4.9).

lfbel=*(vi)* *The set of* $x \in X$ *such that* $\mathscr{F}_{f(x)}$ *is a Cohen–Macaulay* $\mathscr{O}_{X_{f(x)}}$*-Module at the point* x *(5.7.1).*

lgbel=*(vii)* *The set of* $x \in X$ *such that* $\mathscr{F}_{f(x)}$ *is geometrically reduced at the point* x *(4.6.22).*

lhbel=*(viii)* *The set of* $x \in X$ *such that* $\mathscr{F}_{f(x)}$ *is geometrically punctually integral at the point* x *(4.6.22).*

The questions being local on X, one first reduces to the case where $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, f a morphism of finite presentation. There is then a Noetherian subring A_0 of A, an A_0-prescheme of finite type X_0, a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ and f-flat; in addition, by virtue of (11.2.6), one can suppose that $\mathscr{F}_0$ is Y_0-flat (with $Y_0 = \operatorname{Spec}(A_0)$). If $h : X \to X_0$ is the canonical projection, the set E of points $x \in X$ where one of the properties (i) to (viii) is satisfied is equal to $h^{-1}(E_0)$, where E_0 is the set of $x_0 \in X_0$ where the corresponding property is satisfied: this follows, for properties (i) to (iii), from (4.2.7); for properties (iv) to (vi), from (6.7.1); for properties (vii) and (viii), from (4.7.11).

One is thus reduced to the case where A and B are Noetherian, f of finite type, and $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type. In virtue of (9.9.2), one knows that the set E is *constructible* for all the properties considered, and it remains in each case the step C) of (12.0.2), where one must prove that E is *stable under generization*.

(12.1.1.1). Let us begin with case (iii), which is the simplest. Let $x' \neq x$ be a generization of x in X. Set $y = f(x)$, $y' = f(x')$; there exists a spectrum of a discrete valuation ring Y_1 and a morphism $u : Y_1 \to X$ such that, if y_1 and y'_1 are the closed point and the generic point of Y_1 respectively, one has $u(y_1) = x$ and $u(y'_1) = x'$ (**II**, 7.1.7); set $g = f \circ u : Y_1 \to Y$; if $X_1 = X \times_Y Y_1$, there exists a Y_1-section $u_1 : Y_1 \to X_1$ such that $u = g_1 \circ u_1$, where $g_1 : X_1 \to X$ is the canonical projection. If $x_1 = u_1(y_1)$, $x'_1 = u_1(y'_1)$, one has $g_1(x_1) = x$ and $g_1(x'_1) = x'$, and x'_1 is a generization of x_1. Applying again (4.2.7), one sees that one is reduced to the case where $Y = \operatorname{Spec}(A)$ is the spectrum **IV-3 | 175**
of a discrete valuation ring, y being the closed point and y' the generic point of Y. If t is a uniformiser of A, the hypothesis that $\mathscr{F}$ is f-flat implies that t is $\mathscr{F}$-regular (**0**, 6.3.4). By hypothesis, none of the associated immersed prime cycles of $\mathscr{F}/t\mathscr{F}$ contains x. Then, it follows from (3.4) that x does not belong to any associated immersed prime cycle of $\mathscr{F}$, and the same is true of every generization of x. In particular, x' does not belong to any associated immersed prime cycle of $\mathscr{F}$, nor *a fortiori* to any of those associated to $\mathscr{F}_{y'}$ ($X_{y'}$ being a sub-prescheme inducing an open subset of X).

(12.1.1.2). Let us now consider cases (iv) and (v) ((vi) is only a particular case of (v)). One proceeds as above (using (6.7.1) in place of (4.2.7)) and one is reduced to the case where Y is the spectrum of a discrete valuation ring A.

The ring $C = \mathscr{O}_{X,x}$ is then the localisation of an A-algebra of finite type, hence is catenary (5.6.4), and by hypothesis the C-module M_x/tM_x satisfies (S_k) and is equidimensional (resp. on a $\operatorname{coprof}(M_x/tM_x) \leqslant k$); one deduces from (5.12.2) (resp. from (**0**, 16.4.10)) that M_x satisfies (S_k) and is equidimensional (resp. that $\operatorname{coprof}(M_x) \leqslant k$) since t is M_x-regular and belongs to the maximal ideal of C. Whence the conclusions since x' is a generization of x and $(\mathscr{F}_{y'})_{x'} = \mathscr{F}_{x'}$.

(12.1.1.3). Let us pass to cases (vii) and (viii). One can evidently replace Y by the integral sub-prescheme having $\{y'\}$ as underlying space, and X by $f^{-1}(\{y'\})$, without changing the fibres in y and y', so one can suppose A integral, with field of fractions $K = k(y')$. One knows ((4.5.11) and (4.7.8)) that there exists a finite extension K' of K such that, for the $(\mathscr{O}_X \otimes_A K')$-Module $\mathscr{F}' = \mathscr{F} \otimes_A K'$, the associated prime cycles are geometrically irreducible and the geometric lengths of $\mathscr{F}'$ at its maximal points z'_i of its support are respectively equal to the lengths of $\mathscr{F}'_{y'_1}$ at these points; it amounts therefore to the same to say that $\mathscr{F}$ is geometrically reduced (resp. geometrically punctually integral) at a point x' of $X_{y'} = X \otimes_A K$, or to say that $\mathscr{F}'$ is *reduced* (resp. *integral*), taking into account (4.2.7). Let A' be a finite A-algebra whose field of fractions is K'; it follows from (4.7.11) that at every point of $X \otimes_A A'$ above x, $\mathscr{F} \otimes_A A'$ has property (vii) (resp. (viii)). Replacing A by A' and X by $X \otimes_A A'$, one sees that one can reduce to proving that $\mathscr{F}_{f(x')}$ is *reduced* (resp. *integral*) at every generization x' of x. One proceeds then as in (12.1.1.1), and using this time (4.7.11), in the case where A is a discrete valuation ring, y being the closed point, y' the generic point of Y, and the uniformiser t of A being an *M-regular* element. Since by hypothesis $\mathscr{F}/t\mathscr{F}$ is reduced (resp. integral) at the point x, it follows from (3.4.6) (resp. (3.4.5)) that $\mathscr{F}$ is reduced (resp. integral) at the point x, hence also in a neighbourhood of x, and in particular at the point x', which finishes the proof in cases (vii) and (viii).

(12.1.1.4). It remains to examine cases (i) and (ii). Replacing Y by the integral sub-prescheme having $\{y'\}$ as underlying space, one can reduce to the case where Y is irreducible containing x', and let Z' **IV-3 | 176**
be the closure of z' in X, so that one has $x \in Z'$. To treat case (i), it will suffice to prove the

Proposition (12.1.1.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat; for every $y \in Y$, set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)$. Let z' be a point of X, $y' = f(z')$, and suppose that $z' \in \operatorname{Ass}(\mathscr{F}_{y'})$. Let Z' (resp. Y') be the reduced sub-prescheme of X (resp. Y) having the closure of $\{z'\}$ (resp. of $\{y'\}$) in Y as underlying space. Then, for every $y \in f(Z')$, the dimensions of all the irreducible components of Z'_y are equal to $\dim(Z'_{y'})$ (dimension of the closure of z' in $X_{y'}$) (which we will express more precisely later (13.2.2) by saying that the restriction $Z' \to Y'$ of f is an equidimensional morphism); furthermore, at every maximal point z of Z'_y, one has $z \in \operatorname{Ass}(\mathscr{F}_y)$.*

For this, we will now reduce to the case where Y is the spectrum of a discrete valuation ring. Let us take a spectrum of a discrete valuation ring Y_1 and a morphism $h : Y_1 \to X$ such that $h(y_1) = z$, $h(y'_1) = z'$, y_1 and y'_1 being the closed point and the generic point of Y_1 respectively (**II**, 7.1.7). Set $g = f \circ h : Y_1 \to Y$, $X_1 = X \times_Y Y_1$, and let $f_1 : X_1 \to Y_1$, $g_1 : X_1 \to X$ be the canonical projections; there is a Y_1-section $h_1 : Y_1 \to X_1$ such that $h = g_1 \circ h_1$ (**I**, 3.3.14). If $Z'_i = Z'_y \otimes_{k(y')} k(y'_1)$, one knows (4.2.6) that the irreducible components of Z'_i dominate Z'_y; let z'_i be the generic point of one of these components which contains $h_1(y'_1)$, so that $f_1(z'_i) = y'_1$ and $g_1(z'_i) = z'$. Since $h_1(y'_1)$ is a specialisation of $h_1(y_1)$, it is *a fortiori* a specialisation of z'_i; let z_1 be a generic point of one of the irreducible components of $\overline{\{z'_i\}} \cap (X_1)_{y_1}$ containing $h_1(y_1)$. Then $g_1(z_1)$ is a specialisation of $g_1(z'_i) = z'$ in X, and $z = h(y_1) = g_1(h_1(y_1))$ is a specialisation of $g_1(z_1)$; but since $f(g_1(z_1)) = y$ and z is a generic point of Z'_y, one necessarily has $z = g_1(z_1)$.

Suppose now that, if one sets $\mathscr{F}_1 = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_1}$, it will follow from (4.2.7) that $z \in \operatorname{Ass}(\mathscr{F}_y)$; furthermore, the dimensions of $\{z_1\} \cap (X_1)_{y_1}$ and of $\{z\} \cap X_y$ are equal, and of the same dimensions as $\{z'_i\} \cap (X_1)_{y'_1}$ and of $\{z'\} \cap X_{y'}$ (4.2.7); one has therefore indeed reduced, as announced, to the case where Y is the spectrum of a discrete valuation ring A, y and y' being its closed point and its generic point respectively.

Since $z' \in \operatorname{Ass}(\mathscr{F}_y)$, on a *a fortiori* $z' \in \operatorname{Ass}(\mathscr{F})$. If t is a uniformiser of A, t is $\mathscr{F}$-regular by flatness, so (3.4.3) on a $z \in \operatorname{Ass}(\mathscr{F}/t\mathscr{F}) = \operatorname{Ass}(\mathscr{F}_y)$. As for the assertion relative to dimensions, it follows from (7.1.13), applied to a closed sub-prescheme of X having Z' as underlying space.

Let us finally tackle case (ii) of (12.1.1). With the same notation, it is necessary to prove that if all the prime cycles associated to $\mathscr{F}_y$ and containing x have dimension k, the same is true of all the prime cycles associated to $\mathscr{F}_{y'}$ and containing x'. Now, one has just seen that *every* prime cycle associated to $\mathscr{F}_{y'}$ and containing x' has the same dimension as *one* of the prime cycles associated to $\mathscr{F}_y$ and containing x; this shows (ii).

Remarks (12.1.2). — label=(i) Under the conditions of (12.1.1.5) with $\mathscr{F} = \mathscr{O}_X$ (so that f is *flat*), one cannot in general assert that the restriction $Z' \to Y'$ of f is a flat morphism nor even an *open* morphism. This is what the example (6.5.5), (ii) shows, where one takes for Z' one of the two irreducible components of $X = \operatorname{Spec}(B)$. It is immediate that this morphism restriction is not open at the points of Z' above the "double point" of $Y' = Y$. **IV-3** | 177

lbbel=(ii) Under the hypotheses of (12.1.1.5), with $\mathscr{F} = \mathscr{O}_X$, one cannot weaken the condition "f is flat" to "f is universally open" (cf. (2.4.6)), as we will see further on with an example (14.4.10), (i)).

Corollary (12.1.3). — *Under the hypotheses of (12.1.1), the function $x \mapsto \operatorname{coprof}((\mathscr{F}_{f(x)})_x)$ is upper semi-continuous in X and the function $x \mapsto \operatorname{prof}^*_x(\mathscr{F}_{f(x)})$ (10.8.1) is lower semi-continuous in X.*

The first assertion is nothing other than an equivalent formulation of (12.1.1), (v). To prove the second, one can first, by taking into account (10.8.7) and (10.8.8), reduce as in (12.1.1) to the case where Y is the spectrum of a discrete valuation ring A of closed point y and of generic point y', $X = \operatorname{Spec}(B)$ being affine, $\mathscr{F} = \widetilde{M}$, where M is a B-module of finite type. Since $\mathscr{F}$ is f-flat, every irreducible component of $\operatorname{Supp}(\mathscr{F})$ which contains $x \in X_y$ dominates Y (2.3.4), in other words its generic point belongs to $X_{y'}$; the irreducible components Z of $\operatorname{Supp}(\mathscr{F})$ which contain a generization x' of x belonging to $X_{y'}$ are therefore exactly those which contain x and which, furthermore, by (12.1.1.5), $\dim(Z_y) = \dim(Z_{y'})$, whence, if $T = \operatorname{Supp}(\mathscr{F})$, $\dim_x(T_y) = \dim_{x'}(T_{y'})$. Furthermore, for a uniformiser t of A, one has seen in (12.1.1), (v) that $\operatorname{coprof}(M_x/tM_x) = \operatorname{coprof}(M_x)$, and since on the other hand $\operatorname{coprof}(M_{x'}) \leqslant \operatorname{coprof}(M_x)$ (6.11.5), one sees that one has $\operatorname{coprof}((\mathscr{F}_y)_x) \leqslant \operatorname{coprof}((\mathscr{F}_{y'})_z)$, the relation $\operatorname{prof}^*_x(\mathscr{F}_y) \leqslant \operatorname{prof}^*_{x'}(\mathscr{F}_{y'})$ follows from (10.8.7).

Remark (12.1.4). — One also deduces from (12.1.1), (i) that the function $x \mapsto \dim_x(X_{f(x)})$ is upper semi-continuous in X, but we will see further on (13.1.3) that this property is true even without supposing f flat.

Corollary (12.1.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ an $\mathscr{O}_X$-Module coherent and f-flat, $\mathscr{G}$ an $\mathscr{O}_X$-Module coherent, x a point of X, $y = f(x)$. Suppose that $\mathscr{G}$ has the property (S_k) at the point y, and that $\mathscr{F}_y$ has the property (S_k) at the point x and is equidimensional at the point x. Then:*

label=(i) $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) at the point x.
lbbel=(ii) If furthermore Y is locally immersible in a regular scheme, or is a prescheme excellent (7.8.5), there exists a neighbourhood V of x in X such that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) in this neighbourhood.

Proof. Indeed, by (12.1.1), (iv), there exists an open neighbourhood V of x such that for every $x' \in V$, $\mathscr{F}_{f(x')}$ satisfies (S_k) at the point x'. On the other hand, $\mathscr{G}$ satisfies (S_k) at all points of $\mathrm{Spec}(\mathscr{O}_y)$ (5.7.2). Replacing Y by $Y' = \mathrm{Spec}(\mathscr{O}_y)$, one is reduced (taking into account (**I**, 3.6.5)) to the case where $\mathscr{G}$ satisfies (S_k) in Y entirely and where, for every $y \in f(X)$, $\mathscr{F}_y$ has the property (S_k) at all points of $f^{-1}(y)$; since $\mathscr{F}$ is f-flat, one knows then (6.4.1) that $\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ has the property (S_k) in X, which proves (i). To prove (ii), it suffices to observe that, under the hypotheses made, $\mathscr{G}$ has the property (S_k) in a neighbourhood U of y in Y (6.11.2) and (7.8.3), (iv)); one concludes then in the same manner by replacing Y this time by U and X by $V \times_Y U$. □ **IV-3 | 178**

Theorem (12.1.6). — *Let $f : X \to Y$ be a morphism flat and locally of finite presentation, k an integer. The following subsets of X are open:*

label=(i) The set of points $x \in X$ such that $X_{f(x)}$ has the property (S_k) at the point x.
lbbel=(ii) The set of $x \in X$ such that $X_{f(x)}$ satisfies the property (R_k) geometrically at the point x, is equidimensional and also such that x does not belong to any associated immersed prime cycle of $X_{f(x)}$.
lcbel=(iii) The set of $x \in X$ such that $X_{f(x)}$ is geometrically regular (i.e. smooth) at the point x.
ldbel=(iv) The set of $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x.

Steps A) and B) of (12.0.2) are done here as in (12.1.1); for step A), one uses (6.7.8), and also (4.2.7) for (ii); for step B), on uses (9.9.2) and (9.9.4). It remains to examine step C) in each case.

Proof. label=(i) As in (12.1.1.2), one reduces (using (6.7.8)) to the case where Y is the spectrum of a discrete valuation ring A, $X = \mathrm{Spec}(B)$, where B is an A-algebra of finite type, x, x' two points of X such that $f(x) = y$ is the closed point and $y' = f(x')$ the generic point of Y, x' being in addition a generization of x. As one wishes to prove that X satisfies (S_k) at the point x', that is to say suppose the ring B local, the homomorphism $A \to B$ being local and B being an A-algebra essentially of finite type (1.3.8). Then, by hypothesis, if t is a uniformiser of A, t is a B-regular element by flatness, and B/tB satisfies (S_k). But since A is a universally catenary ring (5.6.4), B is catenary, so, by (5.12.4), it follows that B satisfies (S_k), which finishes the proof in case (i).

lbbel=(ii) Here step C) is unnecessary; since f is flat, one knows indeed (6.8.7) that when Y is locally Noetherian and f locally of finite type, the set of $x \in X$ such that $X_{f(x)}$ is geometrically regular at the point x is open in X.

lcbel=(iii) By reasoning as in (12.1.1.3), to prove that the property considered is stable under generization, one can first, by considering a finite extension K' of $k(y')$, and taking into account the definition (6.7.6) of the geometric property (R_k), as well as the invariance by change of base field of the two other properties appearing in (ii), replace in (ii) the property (R_k) geometric by the property (R_k). Proceeding then as in (i), one reduces to the case where Y is the spectrum of a discrete valuation ring A, and since A is catenary, it suffices to apply (5.12.5) to conclude.

ldbel=(iv) The set of points $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x is contained in that of $x \in X$ such that $X_{f(x)}$ is geometrically punctually integral and satisfies (S_2) at the point x, and this latter set is open in X by virtue of (12.1.1), (viii). One can therefore already suppose that, for every $x \in X$, $X_{f(x)}$ is geometrically punctually integral and satisfies (S_2) at the point x, and *a fortiori* it is equidimensional. On the other hand, by the Serre criterion (5.8.6), to say that $X_{f(x)}$ is geometrically normal at the point x means that $X_{f(x)}$ satisfies (S_2) and the property (R_1) geometrically at x; but by virtue of (ii) and the preceding remarks, this set is the intersection of two open subsets of X, hence is open in X. **IV-3 | 179**

□

Corollary (12.1.7). — *Let $f : X \to Y$ be a morphism locally of finite presentation. Then the set of points $x \in X$ where f possesses one of the following properties (6.8.1):*

label=(i) satisfy the property (S_k);
lbbel=(ii) be of coprofundity $\leqslant n$;
lcbel=(iii) be Cohen–Macaulay;
ldbel=(iv) be regular (or smooth, which amounts to the same);
lebel=(v) be normal;
lfbel=(vi) be reduced;

is open in X.

Proof. Indeed, it follows from (11.3.1) that the set of $x \in X$ where f is *flat* is *open*. One can therefore restrict to the case where f is flat, and then the corollary follows from (12.1.1), (iv), (vi) and (vii)) and from (12.1.6), (i), (ii) and (iv)). □

Remarks (12.1.8). — label=(i) In (12.1.6), (iii), one cannot suppress the hypothesis that x does not belong to any associated immersed prime cycle of $X_{f(x)}$. This is shown by the example (5.12.3): one takes $X = \mathrm{Spec}(A)$, Y being the spectrum of the local ring A_0 of $k[T]$ corresponding to the prime ideal (T) and the morphism $X \to Y$ corresponding to the injection $A_0 \to A$, deduced by localisation and passage to the quotient of the injection $k[t_0, t_1] \to k[t_0, t_1, t_2, t_3]/\mathfrak{a}$. This morphism is flat since A is an A_0-module without torsion ($\mathbf{0}$, 6.3.4). Here the fibre X_y at the closed point y of Y, equal to $\mathrm{Spec}(A/tA)$, is irreducible, of dimension 1 and satisfies the condition (R_0) geometrically since the local ring at its generic point is a field. In contrast, at the generic point y' of Y, the fibre $X_{y'}$ has two irreducible components of dimensions 0 and 1, and that of dimension 0 is not reduced, so $X_{y'}$ does not satisfy the condition (R_0).

lbbel=(ii) In (12.1.6), (ii), one cannot also suppress the hypothesis that $X_{f(x)}$ is equidimensional at the point x. One sees this with the example (5.12.6) with $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A_0)$, where A_0 is defined as in (i), the morphism $X \to Y$ also coming from the injection $k[T] \to k[T, U, V, W]$ by localisation and passage to the quotient, and being flat since A is an A_0-module without torsion ($\mathbf{0}$, 6.3.4). The fibre X_y at the closed point y of Y is reduced (hence satisfies (S_1)) and satisfies (R_1), but has two irreducible components of dimensions 2 and 1, while the fibre $X_{y'}$ at the generic point y' of Y is not reduced.

12.3. Local and global properties of fibres of a proper, flat morphism of finite presentation **IV-3 | 179**

Theorem (12.2.1). — *Let $f : X \to Y$ be a morphism* proper *of finite presentation, $\mathscr{F}$ an $\mathscr{O}_X$-Module f-flat and of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$, k an integer. The following subsets of Y are open:*

label=(i) The set of $y \in Y$ such that the dimensions of the prime cycles associated to $\mathscr{F}_y$ are contained in Φ.
lbbel=(ii) The set of $y \in Y$ such that the dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ *are contained in Φ.*
lcbel=(iii) The set of $y \in Y$ such that all the prime cycles associated to $\mathscr{F}_y$ have the same dimension equal to k.
ldbel=(iv) The set of $y \in Y$ such that $\mathscr{F}_y$ has no associated immersed prime cycle and that the set of dimensions of the irreducible components of $\mathrm{Supp}(\mathscr{F}_y)$ *is equal to Φ.* **IV-3 | 180**
lebel=(v) The set of $y \in Y$ such that $\mathscr{F}_y$ has the property (S_k) and is equidimensional at every point of X_y.
lfbel=(vi) The set of $y \in Y$ such that $\mathrm{coprof}(\mathscr{F}_y) \leqslant k$.
lgbel=(vii) The set of $y \in Y$ such that $\mathscr{F}_y$ is a Cohen–Macaulay $\mathscr{O}_{X_y}$-Module.
lhbel=(viii) The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically reduced.
libel=(ix) The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically punctually integral at every point of X_y.

ljbel=(x) *The set of $y \in Y$ such that $\mathscr{F}_y$ is geometrically integral.*
lkbel=(xi) *The set of $y \in Y$ such that $\mathscr{F}_y$ has no associated immersed prime cycle and that the sum $l(y)$ of the total multiplicities (4.7.12) of $\mathscr{F}_y$ at the maximal points of $\operatorname{Supp}(\mathscr{F}_y)$ is $\leqslant k$.*

With the exception of (ii), (iii), (iv), (x) and (xi), the properties $\mathbf{P}(y)$ considered are of the form: "for every $x \in X_y$, $\mathscr{F}_y$ has the property $\mathbf{Q}(x)$", where $\mathbf{Q}(x)$ is one of the properties (i) to (viii) of (12.1.1). It follows from (12.1.1) that the set U of $x \in X$ such that $\mathbf{Q}(x)$ is true is open, and the set of $y \in Y$ such that $\mathbf{P}(y)$ is true is nothing other than $Y - f(X - U)$; in all these cases, the theorem is therefore already true by only supposing the morphism f closed. *It remains to examine cases (ii), (iv) and (xi) separately ((x) follows by deduction immediately from (xi) and (iv)) by always applying the method described in (12.0.2).*

Proof. For (iii), one applies (i) with Φ equal to the smallest interval of $\mathbf{N}$ containing the dimensions of the prime cycles associated to $\mathscr{F}_y$, and reduces to a single element. For (ii) and (iv), it remains to examine cases (ii), (iv) and (xi) separately. □

(12.2.1.1). Case (ii): Steps A) and B) of the proof proceed as at the start of (12.1.1); for step A), one uses (8.9.1), (11.2.6) as well as (4.2.7); for step B), one uses (9.5.5) applied to $\operatorname{Supp}(\mathscr{F})$. It remains to show that if (supposing Y Noetherian) the set of dimensions of the irreducible components of $\operatorname{Supp}(\mathscr{F}_y)$ is contained in Φ, the same is true of the set of dimensions of the irreducible components of $\operatorname{Supp}(\mathscr{F}_{y'})$, for every *generization* y' of y in Y. Let Y_1 be a spectrum of a discrete valuation ring such that, if y_1 and y'_1 are the closed point and the generic point of Y_1, there exists a morphism $h : Y_1 \to Y$ such that $h(y_1) = y$, $h(y'_1) = y'$ (**II**, 7.1.7). Applying (4.2.7), one sees that one can replace Y by Y_1 and X by $X_1 = X \times_Y Y_1$, in other words, one can restrict to the case where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point.

Using (**III**, 4.7.9), one can restrict to the case where $\operatorname{Supp}(\mathscr{F}) = X$, so that f is quasi-flat (2.3.3); the irreducible components Z_i of X dominate Y then (2.3.4), in other words their generic points belong to $X_{y'}$, and the $Z_i \cap X_{y'}$ are the irreducible components of $X_{y'}$. But every irreducible component of X_y is contained in one of the Z_i; now, one knows (7.1.13) that the dimensions of all the irreducible components of $(Z_i)_y$ are equal to $\dim((Z_i)_{y'})$, which finishes the proof of (ii).

(12.2.1.2). Case (iv): The same reasoning as at the start of (12.2.1.1), using (12.1.1), (iii), shows that one can already suppose (replacing Y by an open subset of Y) that for every $y \in Y$, $\mathscr{F}_y$ has no associated immersed prime cycle. The assertion of case (iv) is then an immediate consequence of the assertions of cases (i) and (ii).

(12.2.1.3). Case (xi): For step A) of the reasoning, one uses (8.9.1), (11.2.6), (8.10.5), (xii) (to keep the hypothesis that f is proper), as well as (4.2.7), (4.5.6) and (4.7.9). For step B), one sees, as in case (iv), that one can suppose that for every $y \in Y$, $\mathscr{F}_y$ has no associated immersed prime cycle, and one applies (9.8.8) which proves that the function $z \mapsto l(z)$ is constructible. It remains therefore to see (supposing Y Noetherian and f proper) that for every generization y' of a point $y \in Y$, $l(y') \leqslant l(y)$. By reasoning as in (12.1.1.3), one sees (with the help of (4.7.8) and (4.5.11)) that one can suppose the irreducible components of $\operatorname{Supp}(\mathscr{F}_{y'})$ are geometrically irreducible and that $l(y')$ is the *sum of the lengths* of $(\mathscr{F}_{y'})_{z'_j}$ at the maximal points z'_j of $\operatorname{Supp}(\mathscr{F}_{y'})$. Applying again (4.2.7), (4.5.6) and (4.7.9), one reduces, as in (12.2.1.1), to the case where Y is a spectrum of a discrete valuation ring, y its closed point and y' its generic point. The fact that $l(y') \leqslant l(y)$ will then be a consequence of the **IV-3 | 181**

Lemma (12.2.1.4). — *Let Y be a spectrum of a discrete valuation ring, y its closed point, y' its generic point, $f : X \to Y$ a proper morphism, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module and f-flat, z_i (resp. z'_j) the maximal points of* $\operatorname{Supp}(\mathscr{F}_y)$ (*resp.* $\operatorname{Supp}(\mathscr{F}_{y'})$). *Suppose that $\mathscr{F}_y$ has no associated immersed prime cycle. Then one has*

$$\sum_j \operatorname{long}((\mathscr{F}_{y'})_{z'_j}) \leqslant \sum_i \operatorname{long}((\mathscr{F}_y)_{z_i}).$$

One has $Y = \operatorname{Spec}(A)$, where A is a discrete valuation ring, whose uniformiser we will denote by t, so that $\mathscr{F}_y = \mathscr{F}/t\mathscr{F}$. Since $\mathscr{F}$ is f-flat, the z'_j are also the maximal points of $\operatorname{Supp}(\mathscr{F})$ (2.3.4); for every z_i, denote by z'_{ik} those of the z'_j which are generizations of z_i; it follows from (3.4.1.1) that one has

$$\operatorname{long}((\mathscr{F}_y)_{z_i}) \geqslant \sum_k \operatorname{long}((\mathscr{F}_{y'})_{z'_{ik}})$$

whence by summing

$$\sum_i \operatorname{long}((\mathscr{F}_y)_{z_i}) \geqslant \sum_i \sum_k \operatorname{long}((\mathscr{F}_{y'})_{z'_{ik}}).$$

The lemma will therefore be proved if we establish that for every z'_j, there is at least one index i such that z'_j is one of the z'_{ik}. Now, since f is proper (hence closed) and $f(z'_j) = y'$, there exists $x \in X$ which is a specialisation of z'_j and is such that $f(x) = y$, in other words $\overline{\{z'_j\}} \cap X_y$ is non-empty. Since t is $\mathscr{F}$-regular by flatness, one deduces from (3.4.3) that there is at least one point of $\operatorname{Ass}(\mathscr{F}_y)$ of which z'_j is a generization; but since the points of $\operatorname{Ass}(\mathscr{F}_y)$ are the z_i, this finishes the proof of the lemma.

Corollary (12.2.2). — *Under the hypotheses of* (12.2.1):

label=(i) *The function $y \mapsto \dim(\operatorname{Supp}(\mathscr{F}_y))$ is continuous (hence locally constant) in Y.*
lbbel=(ii) *The function $y \mapsto \operatorname{coprof}(\mathscr{F}_y)$ (5.7.1) is upper semi-continuous in Y.*
lcbel=(iii) *The function $y \mapsto l(y)$ (12.2.1), (xi) is upper semi-continuous in Y when the $\mathscr{F}_y$ have no associated immersed prime cycle.* IV-3 | 182
ldbel=(iv) *The function $y \mapsto \operatorname{prof}^*(\mathscr{F}_y)$ (10.8.1) is lower semi-continuous in Y.*

Proof. label=(i) It follows from (12.2.1), (i) applied to Φ equal to the smallest interval of $\mathbf{N}$ containing the dimensions of the prime cycles associated to $\mathscr{F}_y$, that $z \mapsto \dim(\operatorname{Supp}(\mathscr{F}_z))$ is upper semi-continuous at the point y; it follows on the other hand from (12.2.1), (ii) applied to Φ equal to the set of dimensions of the irreducible components of $\operatorname{Supp}(\mathscr{F}_y)$, that this same function is lower semi-continuous at the point y, whence the conclusion. The assertions (ii) and (iii) are nothing but other formulations of (12.2.1), (vi) and (xi). Finally, by (12.1.3), the set of $x \in X$ such that $\operatorname{prof}^*_{f(x)}(\mathscr{F}_{f(x)}) \leqslant k$ is open, and the conclusion of (iv) follows by the same reasoning as at the beginning of (12.2.1).

□

Remarks (12.2.3). — label=(i) One will observe that for (12.2.1), (iii), one can dispense with the hypothesis that f is *proper*.

lbbel=(ii) In (12.2.1), (ii), one cannot replace the condition that the set $E(y)$ of dimensions of the irreducible components of $\operatorname{Supp}(\mathscr{F}_y)$ contains Φ, by the condition that $E(y)$ be *equal to* Φ. Indeed, let k be a field, and consider the projective space $P = \mathbf{P}^2_k = \operatorname{Proj}(C)$ of 3 dimensions, where $C = k[t_0, t_1, t_2, t_3]$ (t_i indeterminates); in P, let X_0 be the closed sub-prescheme "union of the line X_1 defined by $t_1 = t_3 = 0$ and of the plane X_2 defined by $t_2 - t_3 = 0$", which one describes in geometric terms by saying that X_0 is equal to $\operatorname{Proj}(C/\mathfrak{a})$, where the graded ideal $\mathfrak{a}$ is equal to $\mathfrak{b} \cdot \mathfrak{c}$, with $\mathfrak{b} = Ct_1 + Ct_3$, $\mathfrak{c} = C(t_2 - t_3)$; consider the morphism $f_0 : X_0 \to X_1$ which, in geometric terms, is the "projection of X_0 onto X_1 from the line defined by $t_0 = t_2 = 0$"; in algebraic terms, f_0 corresponds to the homomorphism of graded algebras $k[t_0, t_3] \to k[t_0, t_1, t_2, t_3]/\mathfrak{a}$ obtained by passage to the quotient from the canonical injection $k[t_0, t_3] \to k[t_0, t_1, t_2, t_3]$. It is clear that f_0 is a projective morphism, hence *proper*; furthermore f is *flat*, it suffices to note that $k[t_3]$ is a principal ring and that the ring $C' = k[t_1, t_2, t_3]/(t_2 - t_3)((t_2) + (t_3))$ is a $k[t_3]$-module without torsion ($\mathbf{0}$, 6.3.4). For every $y \in Y$ distinct from the point y_0 defined by $t_3 = 0$, $f^{-1}(y_0)$ has a single irreducible component of dimension 1, while at the point y_0 of Y, the fibre X_{y_0} has two irreducible components of dimensions 0 and 1.

lcbel=(iii) In (12.2.1), (i), one cannot also replace the condition that the set $E'(y) \supset E(y)$ of dimensions of the prime cycles associated to $\mathscr{F}_y$ be contained in Φ by the condition that it be *equal to* Φ. Indeed, let k be a field, A the polynomial ring $k[t]$, B the subring $k[t^4, t^3u, tu^3, u^4]$ of the polynomial ring $k[t, u]$ (t, u indeterminates); B is not integrally closed, the element t^2u^2 belonging to the integral closure but not being in B; if $\mathfrak{m}$ is the ideal maximal of B, generated by t^4, t^3u, tu^3 and u^4, $\mathfrak{m}$ is an associated *immersed* prime ideal of A/A^{t4}. Set $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, and let $f : X \to Y$ be the morphism corresponding to the homomorphism $A \to B$ which sends t to t^4. Since this homomorphism makes B into an A-module without torsion, f is flat ($\mathbf{0}$, 6.3.4). If y is the point of Y corresponding to the ideal tA, the prescheme $f^{-1}(y)$ is irreducible and of dimension 1, but admits a prime cycle associated immersed of dimension 0 and does not satisfy the condition (R_0) geometrically since the local ring at its generic point is a field. In contrast, at the generic point y' of Y, $f^{-1}(y')$ has no prime cycle associated immersed, being the spectrum of a $k(y')$-algebra integral of dimension 1.

ldbel=(iv) In (12.2.1), (xi), one cannot suppress the hypothesis that f be proper. Indeed, let k be a field, and the "affine line", spectrum of the polynomial ring $k[t]$ (t indeterminate), X the scheme *somme* of the closed point y of Y and of the open complement of the closed point y in Y, and the morphism $f : X \to Y$ equal to the canonical injections on each of the components of X, which is flat. But one has $l(y_0) = 1$ and $l(y) = 2$ for $y \neq y_0$ in Y.

lebel=(v) In (12.2.1), (xi), one cannot also suppress the hypothesis that $\mathscr{F}_y$ has no associated immersed prime cycle. This is shown by the example of Remark (ii): one also sees immediately that, with $\mathscr{F} = \mathscr{O}_X$, $l(y_0) = 1$ and $l(y) = 2$ for $y \neq y_0$ in Y.

lfbel=(vi) We do not know if, in (12.2.1), (xi), one can replace the hypothesis that $\mathscr{F}$ is f-flat by the hypothesis that $\mathrm{Supp}(\mathscr{F}) = X$ and that f is universally open. By taking up the proof of (12.2.1.4), one sees (using also (14.3.6)) that one would have to solve the following question: let A be a local Noetherian ring, M an A-module of finite type, t an element **IV-3** | 183
of the maximal ideal $\mathfrak{m}$ of A, which is not contained in any minimal prime ideal of A; if there exists one of these minimal prime ideals $\mathfrak{p}$ such that $\dim(A/\mathfrak{p}) = 1$, is it true that $\mathfrak{m}$ is an associated prime ideal of M/tM (t not being supposed M-regular)?

One will note moreover that one cannot, in (12.2.1), (xi), purely and simply suppress the hypothesis that $\mathscr{F}$ is f-flat. One will take here the P of the Remark (ii), then in P the closed sub-prescheme X_0 union of the three lines X_1, X_2, X_3 defined respectively by $t_1 = t_3 = 0$, $t_1 - t_0 = t_3 = 0$, $t_2 = t_3 = 0$; one defines the projection f_0 of X_0 onto X_1 as above, and its restriction $f : X \to Y$, where $Y = D_+(t_0)$ is the affine line and $X = f_0^{-1}(Y)$; to see that f is flat, it suffices to note that $k[t_3]$ is a principal ring and the ring $C' = k[t_1, t_2, t_3]/(t_2 - t_3)((t_2) + (t_3))$ is a $k[t_3]$-module without torsion (**0**, 6.3.4). For every $y \in Y$ distinct from the point y_0 defined by $t_3 = 0$, $f^{-1}(y)$ has a single irreducible component of dimension 1, while at the point y_0, $f^{-1}(y_0)$ has two irreducible components, of dimension 0 and 1. But one has $l(y_0) = 1$ and $l(y) = 2$ for $y \neq y_0$ in Y.

Theorem (12.2.4). — *Let $f : X \to Y$ be a morphism proper, flat and of finite presentation, k an integer $\geqslant 1$. The following subsets of Y are open:*

label=(i) The set of $y \in Y$ such that X_y possesses the property (S_k).

lbbel=(ii) The set of $y \in Y$ such that X_y satisfies the property (R_k) geometrically, is equidimensional at every point, and has no associated immersed prime cycle.

lcbel=(iii) The set of $y \in Y$ such that X_y is geometrically regular (i.e. smooth over $k(y)$).

ldbel=(iv) The set of $y \in Y$ such that X_y is geometrically normal.

lebel=(v) The set of $y \in Y$ such that X_y is geometrically reduced.

lfbel=(vi) The set of $y \in Y$ such that X_y is geometrically reduced and that the number of geometric connected components of X_y is equal to k.

lgbel=(vii) The set of $y \in Y$ such that X_y is geometrically punctually integral (4.6.9).

lhbel=(viii) The set of $y \in Y$ such that X_y is geometrically integral.

libel=(ix) The set of $y \in Y$ such that X_y has no associated immersed prime cycle and that the total multiplicity (4.7.4) of X_y over $k(y)$ is $\leqslant k$.

Proof. Except for (vi), these assertions are special cases of assertions of (12.2.1), or are deduced from assertions of (12.1.6) as at the start of the proof of (12.2.1). For (vi), one reduces as always (taking into account the invariance of the geometric number of connected components by base change (4.5.6)) to the case where Y is Noetherian. The set U of $y \in Y$ such that X_y is geometrically reduced is open in Y by virtue of (v); it follows then from (**III**, 7.8.7) and (**III**, 7.8.6) that for every $y \in U$, there is a neighbourhood $V \subset U$ of y and an integer m such that, for *every* $z \in V$, $\Gamma(X_z, \mathscr{O}_{X_z})$ is isomorphic to $(k(z))^m$; but by virtue of (**III**, 4.3.4), m is then the geometric number of connected components of X_z, whence the conclusion. We will give another proof of (vi) in (15.5.9). □

12.4. Local cohomological properties of fibres of a flat morphism, locally of finite presentation IV-3 | 183

Lemma (12.3.1). — *Let A be a ring, B an A-algebra of finite presentation which is a flat A-module, M a B-module of finite presentation, which is a flat A-module. Then the following conditions are equivalent:*

label=a) M is a projective B-module.
lbbel=b) M is a flat B-module.
lcbel=c) For every $y \in \operatorname{Spec}(A)$, $M \otimes_A k(y)$ is a projective $(B \otimes_A k(y))$-module.

The equivalence of *a)* and *b)* follows from Bourbaki, *Alg. comm.*, chap. II, § 5, n° 2, cor. 2 of th. 1. IV-3 | 184
Since *a)* implies *c)* trivially, it remains to prove that *c)* implies *b)*, which follows from the criterion of flatness by fibres (11.3.10), applied with $g = h$, $f = 1_X$.

Proposition (12.3.2). — *Let A be a ring, B an A-algebra of finite presentation which is a flat A-module, M a B-module of finite presentation which is a flat A-module. Then:*

label=(i) There exists a left resolution of M by B-modules free of finite type.
lbbel=(ii) One has

$$\operatorname{dim.proj}_B(M) = \sup_{y\in \operatorname{Spec}(A)} \operatorname{dim.proj}_{B\otimes_A k(y)}(M \otimes_A k(y)) = \operatorname{Tor.dim}_B(M) \tag{3}$$

where $\operatorname{Tor.dim}_B(M)$ is the smallest integer i such that $\operatorname{Tor}_j^B(M, N) = 0$ for every $j \geqslant i$ and every B-module N (and $+\infty$ if no such integer i exists).

Proof. label=(i) By virtue of (8.9.1), there exist a Noetherian subring A_0 of A, an A_0-algebra of finite type B_0 and a B_0-module of finite type M_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$ and M to $M_0 \otimes_{A_0} A$. Furthermore (11.2.7) one can suppose that B_0 and M_0 are flat A_0-modules. There is a left resolution (L_0) of M_0 formed of B_0-modules free of finite type, and since M_0 and the L_{0i} are flat A_0-modules, $(L_0) \otimes_{A_0} A$ is a resolution of M formed of B-modules free of finite type (2.1.10).

lbbel=(ii) By virtue of (**0**, 17.2.2), (ii), on a $\operatorname{Tor.dim}_B(M) \leqslant \operatorname{dim.proj}_B(M)$, and the definition of projective dimension proves immediately that $\operatorname{dim.proj}_B(M) \leqslant n$ (resp. $\operatorname{dim.proj}_{B\otimes_A k(y)}(M \otimes_A k(y)) \leqslant n$) for every $y \in \operatorname{Spec}(A)$. To prove the reverse inequalities, consider a left resolution of M by B-modules free of finite type, and suppose that $\operatorname{Tor.dim}_B(M) = n$ (resp. $\operatorname{dim.proj}_{B\otimes_A k(y)}(M \otimes_A k(y)) \leqslant n$ for every $y \in \operatorname{Spec}(A)$). Then $R = \operatorname{Im}(L_n \to L_{n-1}) = \operatorname{Ker}(L_{n-1} \to L_{n-2})$ is a B-module of finite type which is also a flat A-module, by virtue of the hypothesis on M and B and of (2.1.10). Furthermore, on a $\operatorname{Tor}_1^B(R, N) = \operatorname{Tor}_{n+1}^B(M, V, 7)$. The hypothesis $\operatorname{Tor.dim}_B(M) = n$ implies therefore $\operatorname{Tor}_1^B(R, N) = 0$ for every B-module N, that is to say R is a flat B-module, hence projective by virtue of (12.3.1); this establishes that $\operatorname{dim.proj}_B(M) \leqslant n$. The hypothesis $\operatorname{dim.proj}_{B\otimes_A k(y)}(M \otimes_A k(y)) \leqslant n$ for every $y \in \operatorname{Spec}(A)$ implies on the other hand, by tensorisation with $k(y)$, that in each of the sequences (exact by the flatness of M over A, of the L_i and of R (2.1.10))

$$0 \longrightarrow R \otimes_A k(y) \longrightarrow L_{n-1} \otimes_A k(y) \longrightarrow \cdots \longrightarrow L_0 \otimes_A k(y) \longrightarrow M \otimes_A k(y) \longrightarrow 0$$

$R \otimes_A k(y)$ is a $(B \otimes_A k(y))$-module projective (for every $y \in \operatorname{Spec}(A)$). One concludes then again from (12.3.1) that R is a projective B-module, hence $\operatorname{dim.proj}_B(M) \leqslant n$. □

Proposition (12.3.3). — *Let $f : X \to S$ be a morphism locally of finite presentation, $\mathscr{L}_\bullet$ a complex formed of $\mathscr{O}_X$-Modules quasi-coherent of finite presentation; for every $s \in S$, let $(\mathscr{L}_\bullet)_s$ be the complex $\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} k(s)$ of $\mathscr{O}_{X_s}$-Modules of finite type. Suppose that $\mathscr{L}_{n-1}$ is f-flat. Then the set U of $x \in X$ such that $(\mathscr{H}_n((\mathscr{L}_\bullet)_{f(x)}))_x = 0$ is open in X. If furthermore $\mathscr{L}_n$ is f-flat, then $\mathscr{H}_n(\mathscr{L}_\bullet)|U = 0$.*

Proof. One can evidently limit to the case where $n = 1$ and where the complex $\mathscr{L}_\bullet$ is reduced to $0 \to \mathscr{L}_2 \xrightarrow{u} \mathscr{L}_1 \xrightarrow{v} \mathscr{L}_0 \to 0$. One can first reduce to the case where $S = \operatorname{Spec}(A)$ and X are affine, IV-3 | 185
then to the case where S is *Noetherian*; indeed, by (8.9.1) and (8.5.2), one knows that there exists a Noetherian prescheme $S' = \operatorname{Spec}(A')$, where A' is a subring of A, a morphism of finite type $f' : X' \to S'$ and three coherent $\mathscr{O}_{X'}$-Modules $\mathscr{L}'_i$ $(0 \leqslant i \leqslant 2)$ such that $X = X' \times_{S'} S$, $f = f'_{(S)}$, $\mathscr{L}_i = \mathscr{L}'_i \otimes_{S'} S$, as well as two homomorphisms u', v' such that $u = u' \otimes 1$, $v = v' \otimes 1$ and $v' \circ u' = 0$.

One can furthermore suppose that $\mathscr{L}_0$ is f'-flat, and that $\mathscr{L}_1$ is f'-flat (11.2.7), and if U' is the set of $x' \in X'$ such that $(\mathscr{H}_n((\mathscr{L}'_\bullet)_{f'(x')}))_{x'} = 0$, U is the inverse image of U' by the projection $X \to X'$, which reduces us, for the first assertion, to the case where S is Noetherian. For the second assertion, one notes furthermore that A is an inductive limit of its subrings which are A'-algebras of finite type; setting $X_\lambda = X' \times_{S'} S_\lambda$, where $S_\lambda = \operatorname{Spec}(A_\lambda)$, and letting $u_\lambda = u' \otimes 1 : \mathscr{L}_2^{(\lambda)} \to \mathscr{L}_1^{(\lambda)}$, $v_\lambda = v' \otimes 1 : \mathscr{L}_1^{(\lambda)} \to \mathscr{L}_0^{(\lambda)}$, so that $\mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)}) = \operatorname{Ker}(v_\lambda)/\operatorname{Im}(u_\lambda)$. Or, since the functor $\varinjlim$ is exact in the category of commutative groups, on a $\operatorname{Ker}(v) = \varinjlim(\operatorname{Ker}(v_\lambda))$, $\operatorname{Im}(u) = \varinjlim(\operatorname{Im}(u_\lambda))$, and $\mathscr{H}_1(\mathscr{L}_\bullet) = \varinjlim \mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)})$. If one has supposed that $\mathscr{L}_1$ is f-flat and (reducing to the case where $X = U$) that $\mathscr{H}_1(\mathscr{L}_\bullet^{(\lambda)}) = 0$ for every λ, one deduces indeed $\mathscr{H}_1(\mathscr{L}_\bullet) = 0$.

I) Suppose now S Noetherian. One knows (without the flatness hypothesis on $\mathscr{L}_0$) that the set U is *constructible* in X (9.9.6). Using now (**0**, 9.2.5), it remains to show that for every *generization* x' of $x \in U$ in X, one has also $x' \in U$. The method exposed in (12.0.2) applies without change (taking into account the fact that for a prescheme Z over a field k and an extension k' of k, the projection $Z \otimes_k k' \to Z$ is faithfully flat). One can therefore suppose that S is the spectrum of a discrete valuation ring A, with uniformiser t, x being above the closed point of S and x' above the generic point of S. The hypothesis that $\mathscr{L}_0$ is f-flat implies that t is $\mathscr{L}_0$-regular (**0**, 6.3.4). One is then reduced to proving the following lemma (where B, M, N, P will be replaced by $\mathscr{O}_x$, $(\mathscr{L}_2)_x$, $(\mathscr{L}_1)_x$ and $(\mathscr{L}_0)_x$ respectively):

Lemma (12.3.3.1). — *Let B be a local Noetherian ring, M, N, P three B-modules, N being supposed of finite type, $M \xrightarrow{u} N \xrightarrow{v} P$ two homomorphisms such that $v \circ u = 0$. Let t be an element of B such that t be P-regular. Then the relation*

$$M/tM \xrightarrow{u \otimes 1_{B/tB}} N/tN \xrightarrow{v \otimes 1_{B/tB}} P/tP \tag{4}$$

is exact implies that the sequence $M \xrightarrow{u} N \xrightarrow{v} P$ is exact.

Note first that if one replaces M by its image Q in N and u by the injection $j : Q \to N$, the image of $j \otimes 1$ is the same as that of $u \otimes 1$, so one can, without changing the hypothesis nor the conclusion, suppose u injective. On the other hand, if R is the image of v and $\rho : N \to R$ the canonical surjection, t is evidently R-regular, on a $\rho \circ u = 0$, and the kernel of $\rho \otimes 1$ is contained in that of $v \otimes 1$; it is therefore equal to this, if (12.3.3.2) is exact; since on the other hand, $\operatorname{Ker}(\rho) = \operatorname{Ker}(v)$, we see from this that

$$\operatorname{Ker}(v \otimes 1)/\operatorname{Im}(u \otimes 1) = (\operatorname{Ker}(v)/\operatorname{Im}(u)) \otimes_B (B/tB) \tag{5}$$

up to canonical isomorphism.

Indeed, the hypothesis that the sequence (12.3.3.2) is exact implies then that $(\operatorname{Ker} v/\operatorname{Im} u) \otimes_B (B/tB) = 0$, and since t belongs to the maximal ideal of B and $\operatorname{Ker}(v)$ is a B-module of finite type (since B is supposed Noetherian in (12.3.3.1)), the Nakayama lemma proves that $\operatorname{Ker}(v) = \operatorname{Im}(u)$.

It remains therefore to show (12.3.3.4). Set $u' = u \otimes 1$, $v' = v \otimes 1$, $Z = \operatorname{Ker}(v)$, $H = Z/\operatorname{Im}(u)$, so that the sequences

$$0 \longrightarrow M \longrightarrow Z \longrightarrow H \longrightarrow 0$$

$$0 \longrightarrow Z \longrightarrow N \xrightarrow{v} P \longrightarrow 0$$

whence, by tensorising by B/tB and using the lemma (3.4.1.4) and the fact that t is P-regular, the exact sequences **IV-3 | 186**

$$M/tM \xrightarrow{w'} Z/tZ \longrightarrow H/tH \longrightarrow 0$$

$$0 \longrightarrow Z/tZ \longrightarrow N/tN \xrightarrow{v'} P/tP \longrightarrow 0$$

whence $\operatorname{Ker}(v') = Z/tZ$, and since $\operatorname{Im}(w') = \operatorname{Im}(u')$, one obtains (12.3.3.4).

II) Suppose now furthermore that $\mathscr{L}_1$ is f-flat; replacing in addition X by U, one can suppose that $X = U$. It is a question of seeing that for every $x \in X$, $(\mathscr{H}_1(\mathscr{L}_\bullet))_x = 0$. Set $s = f(x)$, and let $\mathfrak{n}$ be the ideal $\mathfrak{m}_s \mathscr{O}_{X,x}$, which is contained in the maximal ideal of $\mathscr{O}_{X,x}$; since $(\mathscr{H}_1(\mathscr{L}_\bullet))_x$ is an $\mathscr{O}_{X,x}$-module of finite type (X being locally Noetherian), it suffices to show that its $\mathfrak{n}$-adic separated completion is 0. Now, by virtue of (**III**, 7.4.7.2), this separated completion is equal to $\varprojlim \mathscr{H}_1(\mathscr{L}_\bullet \otimes_{\mathscr{O}_S} \mathscr{O}_{S,s}/\mathfrak{m}_s^{k+1}))$, and it will therefore suffice to prove that each of the terms of this projective system is zero; we will therefore reason by induction on k. The conclusion will follow from the following more general lemma (which we will apply for $A = \mathscr{O}_{S,s}/\mathfrak{m}_s^{k+1}$ and $\mathfrak{J}$ equal to the maximal ideal $\mathfrak{m}_s/\mathfrak{m}_s^{k+1}$ of A):

Lemma (12.3.3.5). — *Let A be a ring, $\mathfrak{J}$ a nilpotent ideal of A, $\mathrm{L}_\bullet : P \xrightarrow{u_0} Q \xrightarrow{v_0} R$ a complex of A-modules. Set $A_0 = A/\mathfrak{J}$, $\mathrm{L}_\bullet^{(0)} = \mathrm{L}_\bullet \otimes_A A_0 : P_0 \xrightarrow{u_0} Q_0 \xrightarrow{v_0} R_0$, and suppose that $\mathrm{gr}^*_{\mathfrak{J}}(A)$ is a flat A_0-module, and that Q and R are flat A-modules. Then the relation $\mathrm{H}_1(\mathrm{L}_\bullet^{(0)}) = 0$ implies $\mathrm{H}_1(\mathrm{L}_\bullet) = 0$.*

Proof. Set $\mathrm{L}_\bullet^{(k)} = \mathrm{L}_\bullet \otimes_A (A/\mathfrak{J}^{k+1}) : P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$; since there exists an integer $k \geqslant 1$ such that $\mathrm{L}_\bullet^{(k)} = \mathrm{L}_\bullet$, it will suffice to prove, by induction on k, that the sequence $P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$ is exact. This is true by hypothesis for $k = 0$. The conclusion will follow from the following lemma:

Lemma (12.3.3.3). — *Let B be a ring, M, N, P three B-modules, $u : M \to N$, $v : N \to P$ two homomorphisms such that u is injective, v surjective and $v \circ u = 0$. Let t be an element of B such that t be P-regular. Then one has*

$$\mathrm{Ker}(v \otimes 1)/\mathrm{Im}(u \otimes 1) = (\mathrm{Ker}(v)/\mathrm{Im}(u)) \otimes_B (B/tB) \tag{6}$$

up to a canonical isomorphism.

Indeed, the hypothesis that the sequence (12.3.3.2) is exact implies then that it becomes exact, and since t belongs to the maximal ideal of B and $\mathrm{Ker}(v)$ is a B-module of finite type (since B is supposed Noetherian in (12.3.3.1)), the Nakayama lemma proves that $\mathrm{Ker}(v) = \mathrm{Im}(u)$.

It remains therefore to show (12.3.3.3). Set $u' = u \otimes 1_{B/tB}$, $v' = v \otimes 1_{B/tB}$, $Z = \mathrm{Ker}(v)$, $H = Z/\mathrm{Im}(u)$, so that one has the exact sequences

$$0 \longrightarrow M \longrightarrow Z \longrightarrow H \longrightarrow 0$$

$$0 \longrightarrow Z \longrightarrow N \xrightarrow{v} P \longrightarrow 0$$

whence, by tensorising by B/tB and using the lemma (3.4.1.4) and the fact that t is P-regular, the exact sequences

$$M/tM \xrightarrow{w'} Z/tZ \longrightarrow H/tH \longrightarrow 0$$

$$0 \longrightarrow Z/tZ \longrightarrow N/tN \xrightarrow{v'} P/tP \longrightarrow 0$$

whence $\mathrm{Ker}(v') = Z/tZ$, and since $\mathrm{Im}(w') = \mathrm{Im}(u')$, one obtains (12.3.3.4b). □

□

§13. EQUIDIMENSIONAL MORPHISMS

This section is devoted to the study of the variation of the *dimension* of the fibres of a morphism **IV-3** | 187
locally of finite type $f : X \to Y$ (which has already intervened with respect to the "dimension formula" in (5.5) and (5.6)). We first prove (13.1.3) that the function $x \mapsto \dim_x f^{-1}(f(x))$ is always *upper semi-continuous* in X (Chevalley's semi-continuity theorem). We then study more particularly the morphisms, called "equidimensional", for which this function is *locally constant*. Unfortunately, **IV-3** | 188
the notion of equidimensional morphism is not stable under base change; this is why in many questions it is more convenient to work with the notion of universally open morphism, the study of which is the subject of §§ 14 and 15.

13.1. Chevalley's semi-continuity theorem

Lemma (13.1.1). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a dominant morphism of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, all the irreducible components of $f^{-1}(f(x))$ are of dimension $\geqslant e$.*

Proof. The claim is immediate when, for every $y \in f(X)$, $\mathscr{O}_y$ is a *universally catenary* ring: indeed, if x is the generic point of an irreducible component Z of $f^{-1}(y)$, it follows from (5.6.5), combined with (5.2.1), that $e + \dim(\mathscr{O}_y) = \dim(Z) + \dim(\mathscr{O}_z)$, and since by (**0**, 16.3.9), we have $\dim(\mathscr{O}_z) \leqslant \dim(\mathscr{O}_y)$, whence the conclusion in this case.

We shall now reduce the general case to this particular case. The question is evidently local on Y and, taking into account (4.1.3), it is also local on X; we can therefore restrict to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine and irreducible, B being an A-algebra of finite type. Furthermore (1.5.4), we can suppose X and Y reduced, so A and B are integral. As f is dominant, A is then a *subring* of B. Consider A as an inductive limit of its sub-$\mathbf{Z}$-algebras of finite type; it then follows from (8.9.1) that there exists such a subalgebra A_0 and an A_0-algebra of finite type B_0 such that $B = B_0 \otimes_{A_0} A$. Set $Y_0 = \operatorname{Spec}(A_0)$, $X_0 = \operatorname{Spec}(B_0)$, and let $f_0 : X_0 \to Y_0$ be the morphism corresponding to the homomorphism $A_0 \to B_0$, so that $f = f_0 \times 1_Y$. It is not *a priori* evident that the prescheme X_0 is integral, but we shall see that we can reduce to this case. Let η_0 be the generic point of Y_0, so that g is the morphism $Y \to Y_0$, we have $g(\eta) = \eta_0$; by transitivity of fibres (**I**, 3.6.4), we have $f^{-1}(\eta) = f_0^{-1}(\eta_0) \otimes_{k(\eta_0)} k(\eta)$, and as $f^{-1}(\eta)$ is irreducible by hypothesis, so is $f_0^{-1}(\eta_0)$ (4.4.1). Our assertion will then follow from the following lemma: □

Lemma (13.1.2). — *Let Y_0, Y be two irreducible preschemes of generic points η_0, η, $g : Y \to Y_0$ a dominant morphism, $f_0 : X_0 \to Y_0$ a dominant morphism such that $f_0^{-1}(\eta_0)$ is irreducible. Let X_0' be the unique irreducible component of X_0 meeting $f_0^{-1}(\eta_0)$* (**0**, 2.1.8)*, and let X_0' also denote the closed reduced sub-prescheme of X_0 having X_0' as underlying space. Suppose that the prescheme $X = X_0 \times_{Y_0} Y$ is integral; then X is isomorphic to $X_0' \times_{Y_0} Y$.*

Proof. Indeed, if $j_0 : X_0' \hookrightarrow X_0$ is the canonical injection, which is a closed immersion, $j = j_0 \times 1_Y : X_0' \times_{Y_0} Y \to X_0 \times_{Y_0} Y = X$ is a closed immersion. On the other hand, X_0' contains the fibre $f_0^{-1}(\eta_0)$, so $X_0' \times_{Y_0} Y$ contains $f^{-1}(\eta)$; note on the other hand that $f^{-1}(\eta)$ is not empty (**I**, 3.4.7), so it contains the generic point of X; by extension the image of j is necessarily X itself. But as X is integral, the only closed sub-prescheme of X having X as underlying space is X itself, so j is an isomorphism. □

This lemma being established, we can therefore suppose that X_0 is integral; for every $y_0 \in Y_0$, $\mathscr{O}_{y_0}$ is a $\mathbf{Z}$-algebra essentially of finite type, hence a universally catenary ring (5.6.4); consequently, **IV-3** | 189
for every $y_0 \in f_0(X_0)$, all the irreducible components of $f_0^{-1}(y)$ have dimension $\geqslant e$, since we know that $\dim(f_0^{-1}(\eta_0)) = e$ by transitivity of fibres (4.1.4). For every $y \in f(X)$, we then have $y_0 = g(y) \in f_0(X_0)$ and by transitivity of fibres (4.2.7), we complete the proof.

Theorem (13.1.3). — (Chevalley). — *Let $f : X \to Y$ be a morphism locally of finite type. For every integer n, the set $\mathrm{F}_n(X)$ of $x \in X$ such that $\dim_x(f^{-1}(f(x))) \geqslant n$ is closed; in other words, the function $x \mapsto \dim_x(f^{-1}(f(x)))$ is upper semi-continuous in X.*

Proof. I) Suppose first that f is locally of finite presentation. The question is evidently local on X and on Y, and we can therefore suppose $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ affine, B being an A-algebra of finite presentation. We know then (8.9.1) that there exists a Noetherian subring A_0 of A, and an A_0-algebra of finite type B_0, such that $B = B_0 \otimes_{A_0} A$. Set $Y_0 = \operatorname{Spec}(A_0)$, $X_0 = \operatorname{Spec}(B_0)$, and let $f_0 : X_0 \to Y_0$ be the morphism corresponding to the homomorphism $A_0 \to B_0$. Let $g : Y \to Y_0$ be the morphism corresponding to the canonical injection $A_0 \to A$, y a point of Y, $y_0 = g(y)$; we know that $f^{-1}(y) = f_0^{-1}(y_0) \otimes_{k(y_0)} k(y)$, and it follows from (4.2.7) that if p is the canonical projection $f^{-1}(y) \to f^{-1}(y_0)$, the irreducible components of $f^{-1}(y)$ are the irreducible components of the spaces $p^{-1}(Z_0)$, where Z_0 runs over the set of irreducible components of $f_0^{-1}(y_0)$, and each of the irreducible components of $p^{-1}(Z_0)$ dominates Z_0 and has dimension equal to $\dim(Z_0)$. Taking into account (**0**, 14.1.5), we see therefore that if $g' : X \to X_0$ is the canonical projection, x a point of X and

$x_0 = g'(x)$, we have $\dim_x(f^{-1}(f(x))) = \dim_{x_0}(f_0^{-1}(f_0(x_0)))$, whence $\mathrm{F}_n(X) = g'^{-1}(\mathrm{F}_n(X_0))$, and we are reduced to proving the theorem when Y is *Noetherian*, which we will henceforth suppose.

We can evidently suppose X and Y reduced (1.5.4). By considering the set of closed subsets Y' of Y such that the theorem is true for the closed sub-prescheme of Y having Y' as underlying space and for $X' = f^{-1}(Y')$, we can reason by Noetherian recurrence (**0**, 2.2.2), and suppose that the theorem is true for every closed subset $Y' \neq Y$ of Y. If X_i ($1 \leqslant i \leqslant m$) are the closed sub-preschemes with reduced underlying spaces that are the irreducible components of X, we have $\mathrm{F}_n(X) = \bigcup \mathrm{F}_n(X_i)$ by virtue of (**0**, 14.1.5), and we can therefore restrict to proving the theorem for each X_i, in other words, we can suppose X *irreducible*. If Z is the closed sub-prescheme of Y having $\overline{f(X)}$ as underlying space, f factorises as $X \xrightarrow{g} Z \xrightarrow{j} Y$, where j is the canonical injection of type fini (1.5.4), so it suffices to prove the theorem for Z and g; by the hypothesis of recurrence, we are thus reduced to considering only the case where $Z = Y$, in other words where Y is irreducible and f dominant. Let then η be the generic point of Y and set $e = \dim(f^{-1}(\eta))$; it follows from (13.1.1) that for $n \leqslant e$ we have $\mathrm{F}_n(X) = X$, and consequently we can restrict to the case where $n > e$. But then (9.5.6), there is an open neighbourhood U of η in Y such that $\mathrm{F}_n(X) \subset f^{-1}(Y - U)$; as $Y - U \neq Y$, the hypothesis of recurrence implies that $\mathrm{F}_n(X)$ is closed.

II) We now pass to the general case, always supposing $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ affine, B **IV-3** | 190
being an A-algebra of finite type, hence of the form $B = A[T_1, \ldots, T_n]/\mathfrak{J}$. Let $(\mathfrak{J}_\lambda)$ be the family of ideals of finite type of $A[T_1, \ldots, T_n]$ contained in $\mathfrak{J}$, so that $\mathfrak{J}$ is the filtered union of the $\mathfrak{J}_\lambda$; if X and the $X_\lambda = \mathrm{Spec}(A[T_1, \ldots, T_n]/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes of $Z = \mathrm{Spec}(A[T_1, \ldots, T_n])$, we therefore have, for the underlying spaces, $X = \bigcap X_\lambda$. If $f_\lambda : X_\lambda \to S$ is the structural morphism, we deduce that $f^{-1}(s) = \bigcap f_\lambda^{-1}(s)$ for every $s \in S$, and as the sets $f_\lambda^{-1}(s)$ are closed in the Noetherian space, there exists a λ (depending on s) such that $f^{-1}(s) = f_\lambda^{-1}(s)$. So then, for every $x \in X$, we set $d(x) = \dim_x(f^{-1}(f(x)))$, $d_\lambda(x) = \dim_x(f_\lambda^{-1}(f_\lambda(x)))$; from what precedes we see that we have $d(x) = \inf_\lambda d_\lambda(x)$; the functions d_λ being upper semi-continuous by the first part of the proof, so is d. □

Corollary (13.1.4). — *Under the hypotheses of* (13.1.3)*, the set of $x \in X$ such that x is isolated in $f^{-1}(f(x))$ is open in X.*

Proof. Indeed, it is the complement of $\mathrm{F}_1(X)$ (**0**, 14.1.10). □

We note that we thus recover, under more general hypotheses, the consequence (**III**, 4.4.10) of the "Main theorem" of Zariski.

Corollary (13.1.5). — *Under the hypotheses of* (13.1.3)*, suppose in addition that f is a closed morphism. Then, for every integer n, the set of $y \in Y$ such that $\dim(f^{-1}(y)) \geqslant n$ is closed; in other words, the map $y \mapsto \dim(f^{-1}(y))$ is upper semi-continuous; in particular, if y is a specialisation of y', we have $\dim(f^{-1}(y)) \geqslant \dim(f^{-1}(y'))$.*

Proof. Indeed, to say that $\dim(f^{-1}(y)) \geqslant n$ means that $y \in f(\mathrm{F}_n(X))$ (**0**, 14.1.6). □

Corollary (13.1.6). — *Let X, Y be two irreducible preschemes, $f : X \to Y$ a dominant morphism of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, we have $\dim_x(f^{-1}(f(x))) \geqslant e$.*

Proof. Indeed, the set of x such that $\dim_x(f^{-1}(f(x))) < e$ is open by (13.1.3), and as it cannot contain ξ, it is empty. □

Let us note finally the following easy result:

Proposition (13.1.7). — *Let Y be a quasi-compact prescheme, $f : X \to Y$ a morphism of finite type. There exists an integer n such that, for every $y \in Y$, we have $\dim(f^{-1}(y)) \leqslant n$.*

Proof. As there is a finite affine open covering (V_i) of Y such that each $f^{-1}(V_i)$ is a finite union of affine open sets, we are immediately reduced to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being an A-algebra of finite type. If B admits a system of n generators, then, for every $y \in Y$, $B \otimes_A k(y)$ is a $k(y)$-algebra admitting n generators, hence $\dim(f^{-1}(y)) \leqslant n$ by virtue of (4.1.1). □

13.2. Equidimensional morphisms: case of dominant morphisms of irreducible preschemes

(13.2.1). Let Y be an irreducible prescheme, X an irreducible prescheme, $f : X \to Y$ a *dominant* morphism locally of finite type; let η be the generic point of Y. We know (13.1.6) that for every $x \in X$, we have $\dim_x(f^{-1}(f(x))) \geqslant \dim(f^{-1}(\eta))$.

IV-3 | 191

Definition (13.2.2). — Under the hypotheses of (13.2.1), we say that f is *equidimensional at the point x* (or that X is *equidimensional over Y at the point x*) if

$$\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta)).$$

We say that f is *equidimensional* (or that X is *equidimensional over Y*) if f is equidimensional at every point $x \in X$.

It follows from Chevalley's theorem (13.1.3) that the set of $x \in X$ where f is equidimensional is *open* and non-empty. Furthermore, if f is equidimensional at the point x, all the irreducible components of $f^{-1}(f(x))$ containing x have the *same dimension*, since each of them has dimension $\geqslant \dim(f^{-1}(\eta))$ (13.1.6) and $\leqslant \dim_x(f^{-1}(f(x)))$ by virtue of (**0**, 14.1.5).

Proposition (13.2.3). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \to Y$ a dominant morphism locally of finite type; let η be the generic point of Y, x a point of X, $y = f(x)$, and suppose that we have*

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)). \tag{13.2.3.1}$$

Then f is equidimensional at the point x. The converse is true if the two members of the inequality (5.6.5.2) are equal, in particular if $\mathscr{O}_y$ is universally catenary.

Proof. This follows immediately from (5.6.5.2) and from the inequality $\delta(x) \geqslant 0$ (13.1.1). □

(13.2.4). Let now, in a general fashion, Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type, x a point of X, X' a closed irreducible subset of X containing x, $y = f(x)$, $Y' = \overline{f(X')}$, η' the generic point of Y'. Let X' and Y' also denote the closed reduced sub-preschemes of X and Y respectively having X' and Y' as underlying spaces; the restriction $X' \to Y$ of f factorises as $X' \xrightarrow{f'} Y' \xrightarrow{j} Y$, where j is the canonical injection (**I**, 5.2.2), and f' is of finite type (1.5.4). Set then, to abbreviate

$$A = \mathscr{O}_{Y,y}, \quad B = \mathscr{O}_{X,x}, \quad A' = \mathscr{O}_{Y',y}, \quad B' = \mathscr{O}_{X',x}. \tag{13.2.4.1}$$

The formula (5.6.5.2) applied to the dominant morphism f' and to the irreducible preschemes X', Y', gives

$$\dim(B') \leqslant \dim(A') + \dim(B' \otimes_{A'} k(y)) - (\dim_x(f'^{-1}(y)) - \dim(f'^{-1}(\eta'))) \tag{13.2.4.2}$$

On the other hand, the local ring A' (resp. B', $B' \otimes_{A'} k(y)$) is a quotient of A (resp. B, $B \otimes_A k(y)$), hence by (**0**, 16.1.2.1) we have

$$\dim(A') \leqslant \dim(A), \quad \dim(B') \leqslant \dim(B), \quad \dim(B' \otimes_{A'} k(y)) \leqslant \dim(B \otimes_A k(y)). \tag{13.2.4.3}$$

We deduce therefore first from (13.2.4.3) and (**0**, 16.3.9)

$$\dim(B') \leqslant \dim(A') + \dim(B' \otimes_{A'} k(y)) \leqslant \dim(A) + \dim(B \otimes_A k(y)). \tag{13.2.4.4}$$

Furthermore, by virtue of (**0**, 16.3.9), we also have the inequalities

$$\dim(B') \leqslant \dim(B) \leqslant \dim(A) + \dim(B \otimes_A k(y)). \tag{13.2.4.5}$$

Comparison of these inequalities shows therefore that: **IV-3 | 192**

Lemma (13.2.5). — *With the notations of (13.2.4), the following conditions are equivalent:*

label=a) $\dim(B') = \dim(A) + \dim(B \otimes_A k(y))$.
lbbel=b) *The following relations hold simultaneously:*
label=(i) $\dim(A') = \dim(A)$.

lbbel=(ii) $\dim(B' \otimes_{A'} k(y)) = \dim(B \otimes_A k(y))$, *in other words*

$$\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y)).$$

lcbel=(iii) $\dim_x(f'^{-1}(y)) = \dim(f'^{-1}(\eta'))$, *in other words* $f' : X' \to Y'$ *is equidimensional* (13.2.2) *at the point* x.

ldbel=(iv) *We have the equality*

$$\dim(B') = \dim(A') + \dim(B' \otimes_{A'} k(y)) - (\dim_x(f'^{-1}(y)) - \dim(f'^{-1}(\eta')))$$

(a condition always implied by (iii) when $A = \mathscr{O}_{Y,y}$ *is a universally catenary ring, by virtue of* (5.6.5)*).*

lcbel=c) *We have the equality*

label=(i) $\dim(B') = \dim(B)$.

lbbel=(ii) $\dim(B) = \dim(A) + \dim(B \otimes_A k(y))$.

(13.2.6). Recall now that the irreducible components X_i of X containing x are finite in number (5.1.2.1) and (**0**, 14.2.1.1)

$$\dim(\mathscr{O}_{X,x}) = \sup_i \dim(\mathscr{O}_{X_i,x}). \tag{13.2.6.1}$$

The equivalence of conditions $b)$ and $c)$ in (13.2.5) implies therefore, taking into account (**0**, 16.3.9) and (**0**, 14.2.1):

Proposition (13.2.7). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism of finite type,* x *a point of* X*,* $f(x) = y$*; we have*

$$\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y)). \tag{13.2.7.1}$$

For the two members of (13.2.7.1) *to be equal, it is necessary and sufficient that there exists a closed irreducible subset* X' *of* X *containing* x *and satisfying simultaneously the following conditions:*

label=(i) *If* $Y' = \overline{f(X')}$*, we have* $\dim(\mathscr{O}_{Y',y}) = \dim(\mathscr{O}_{Y,y})$.

lbbel=(ii) $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ *(in other words,* X' *contains one of the irreducible components of* $f^{-1}(y)$ *of maximal dimension among those containing* x*). This amounts to saying that* $\dim(\mathscr{O}_{X',x} \otimes_{\mathscr{O}_{Y',y}} k(y)) = \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y))$.

lcbel=(iii) X' *is equidimensional over* Y' *at the point* x*, in other words, we have*

$$\dim_x(X' \cap f^{-1}(y)) = \dim(X' \cap f^{-1}(\eta')),$$

where η' *is the generic point of* Y' *(and consequently, all the irreducible components of* $X' \cap f^{-1}(y)$ *containing* x *have the same dimension* (13.2.2)*).*

ldbel=(iv) *We have the equality*

$$\dim(\mathscr{O}_{X',x}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X',x} \otimes_{\mathscr{O}_{Y',y}} k(y))$$

IV-3 | 193

Proof. In the statement, the local rings $\mathscr{O}_{X',x}$ and $\mathscr{O}_{Y',y}$ are relative to the closed reduced sub-preschemes of X, Y respectively having X' and Y' as underlying spaces.

Furthermore, this proves that □

Corollary (13.2.8). — *If the two members of* (13.2.7.1) *are equal, the closed irreducible subsets* X' *of* X *containing* x *and satisfying conditions (i) to (iv) of* (13.2.7) *are exactly the ones for which* $\dim(\mathscr{O}_{X',x}) = \dim(\mathscr{O}_{X,x})$ *(which implies that* X' *is necessarily an irreducible component of* X*).*

Proposition (13.2.7) implies:

Corollary (13.2.9). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a morphism of finite type,* x *a point of* X*,* $f(x) = y$*. Suppose that* $\mathscr{O}_{Y,y}$ *is a universally catenary ring. Then the following conditions are equivalent:*

label=a) *The two members of* (13.2.7.1) *are equal.*

lbbel=b) *There exists an irreducible component Z of $f^{-1}(y)$ containing x, of dimension* $\dim_x(f^{-1}(y))$, *such that, for every x' in a neighbourhood of x in Z, we have*

$$\dim(\mathcal{O}_{X,x'}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x'} \otimes_{\mathcal{O}_{Y,y}} k(y)). \tag{13.2.9.1}$$

lcbel=c) *There exists an irreducible component Z of $f^{-1}(y)$ containing x, of dimension* $\dim_x(f^{-1}(y))$, *such that for the generic point z of Z, we have*

$$\dim(\mathcal{O}_{X,z}) = \dim(\mathcal{O}_{Y,y}). \tag{13.2.9.2}$$

Proof. Let us show that a) implies b). Let $\dim_x(f^{-1}(y)) = n$; by virtue of (13.2.7), there exists an irreducible component X' of X satisfying conditions (i) to (iv) of (13.2.7); let Z be an irreducible component of dimension n of $f^{-1}(y)$, containing x and contained in X'. As $f^{-1}(y)$ is locally Noetherian, there exists an open neighbourhood U of x in Z such that U meets no irreducible component of $f^{-1}(y)$ other than those containing x, hence (4.1.1.3) $\dim_x(f^{-1}(y)) = n$ for every $x' \in U$; it is then clear that the conditions (i) to (iii) of (13.2.7) are satisfied when we replace x by any point $x' \in U$, and it follows similarly from condition (iv) since $\mathcal{O}_{Y,y}$ is universally catenary; whence the conclusion by (13.2.7). The condition b) implies trivially c) by (5.1.2). Finally, if c) is satisfied and if X'' is an irreducible component of X containing Z and such that the conditions (i) to (iv) of (13.2.7) are satisfied when we replace X' by X'' and x by z, it is clear that these conditions are also satisfied when we replace X' by X'' and x by x since $\mathcal{O}_{Y,y}$ is universally catenary, hence c) implies a). □

Proposition (13.2.10). — *Let Y be an irreducible locally Noetherian prescheme, η its generic point, $f : X \to Y$ a morphism of finite type, y a point of $f(X)$. Let Z_i be the irreducible components of $f^{-1}(y)$, z_i* **IV-3 | 194** *the generic point of Z_i, and consider the following conditions:*

label=a) *For every $x \in f^{-1}(y)$, we have the relation*

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} k(y)). \tag{13.2.10.1}$$

lbbel=b) *For every i, we have*

$$\dim(\mathcal{O}_{X,z_i}) = \dim(\mathcal{O}_{Y,y}). \tag{13.2.10.2}$$

lcbel=c) *For every i, there exists an irreducible component X_i of X containing Z_i and such that* $\dim(X_i \cap f^{-1}(\eta)) = \dim(Z_i)$ *(in other words, the* closed reduced sub-prescheme X_i *of X is equidimensional over Y at the point z_i).*

Then a) implies b) and b) implies c); furthermore, if $\mathcal{O}_{Y,y}$ is universally catenary, the three conditions a), b), c) are equivalent.

Proof. The ring $\mathcal{O}_{X,z_i} \otimes_{\mathcal{O}_{Y,y}} k(y)$ being of dimension 0 (5.1.2), a) implies b) evidently; b) implies c) by virtue of (13.2.7) applied at the point z_i. Conversely, suppose that $\mathcal{O}_{Y,y}$ is universally catenary; as the condition c) implies that $f(X_i)$ is dense in Y, it follows from c) that the conditions (i) to (iv) of (13.2.7) are satisfied when we replace X' by X_i and x by z_i, hence c) implies b); finally b) implies a) by (13.2.9). □

Corollary (13.2.11). — *With the notations being those of (13.2.10), suppose that $\mathcal{O}_{Y,y}$ is universally catenary. For every $y' \in Y$, let $E(y')$ be the set of dimensions of the irreducible components of $f^{-1}(y')$ and set* $d(y') = \dim(f^{-1}(y')) = \sup(E(y'))$. *Then, if the conditions a), b), c) of (13.2.10) are satisfied, we have $E(y) \subset E(\eta)$, whence $d(y) \leqslant d(\eta)$.*

Proof. Indeed, with the notations of (13.2.10), $X_i \cap f^{-1}(\eta)$ is non-empty and is consequently an irreducible component of $f^{-1}(\eta)$ (**0**, 2.1.8). □

Remarks (13.2.12). — label=(i) Recall (6.1.2) that the relation (13.2.10.1) is always satisfied when the morphism f is *flat* at the point x.

lbbel=(ii) With the notations of (13.2.10), suppose that $\mathcal{O}_{Y,y}$ is universally catenary and in addition that the morphism f is *proper*; then, if the equivalent conditions a), b), c) of (13.2.10) are satisfied, we have $d(y) = d(\eta)$, since it follows from (13.1.5) that $d(y) \geqslant d(\eta)$.

lcbel=(iii) The morphism of (12.2.3), (ii) is *proper* and *flat* and all the local rings of Y are universally catenary; furthermore, the two irreducible components X_1, X_2 of X are *equidimensional* over Y at every point; but $E(y)$ has *two* elements for every $y \neq y_0$, whereas $E(y_0)$ is reduced to a *single* element, hence $E(y)$ is *not constant* in Y.

13.3. Equidimensional morphisms: general case

Proposition (13.3.1). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Let Y_α denote the irreducible components of Y containing y. Then the following conditions are equivalent:*

IV-3 | 195

label=a) *There exist an integer e and an open neighbourhood U of x such that the image by f of every irreducible component of U is dense in some Y_α, and such that, for every $x' \in U$, the space $U \cap f^{-1}(f(x'))$ is equidimensional and of dimension e.*

a′) *There exist an integer e and an open neighbourhood U of x such that the image by f of every irreducible component of U is dense in some Y_α, and such that, if we denote by y_α the generic point of Y_α, every irreducible component of the spaces $U \cap f^{-1}(y_\alpha)$, $U \cap f^{-1}(y_\alpha)$ is of dimension e.*

a″) *There exist an integer e and an open neighbourhood U of x such that, for each of the irreducible components U_λ of U, $f(U_\lambda)$ is dense in some Y_α and such that, for every $x' \in U_\lambda$, the irreducible components of $U_\lambda \cap f^{-1}(f(x'))$ are all of dimension e.*

lbbel=b) *There exist an integer e, an open neighbourhood U of x and a Y-morphism quasi-finite $g : U \to Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_e]$ (a prescheme we will also denote $Y[T_1, \dots, T_e]$, to abbreviate) such that the image by g of every irreducible component of U is dense in an irreducible component of $Y[T_1, \dots, T_e]$.*

Proof. It is immediate that a'') implies a), since for every $x \in U$, the irreducible components of $U \cap f^{-1}(f(x))$ are each an irreducible component of one of the $U_\lambda \cap f^{-1}(f(x))$, whence the conclusion by (**0**, 14.1.4). The condition a) trivially implies a''). Let us show that a') implies a''); we can restrict to the case where f is of finite type and X is reduced. Let U_λ be an irreducible component of U, and suppose that $f(U_\lambda)$ is dense in Y_α; by the hypothesis of a'), on can (restricting U if necessary) suppose that $f(x_\lambda)$ is one of the generic points y_α of the irreducible components of Y containing y; by virtue of (**0**, 2.1.8), $U_\lambda \cap f_\alpha^{-1}(y_\alpha)$ is the unique irreducible component of $U \cap f^{-1}(y_\alpha)$ containing x_λ and is by hypothesis of dimension e, equal to the dimensions of all the irreducible components of $U_\lambda \cap f^{-1}(y_\alpha)$; by Chevalley's theorem (13.1.3) and (13.1.1), the set V_λ of $x' \in U_\lambda$ such that $\dim_{x'}(f_\lambda^{-1}(f_\lambda(x))) = e$ is open and contains x, and it suffices to take the union of the V_λ as open set satisfying the conditions of a'').

Let us now show that a) implies b); we can restrict to the case where $Y = \operatorname{Spec}(A)$ and $U = X$. Let us first prove the following lemma:

Lemma (13.3.1.1). — *Let $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ be two affine schemes, $f : X \to Y$ a morphism of finite type, y a point of $f(X)$, and set $e = \dim(f^{-1}(y))$. Then there exists a Y-morphism $g : X \to Y[T_1, \dots, T_e]$ such that (if we set $X_y = X \times_Y \operatorname{Spec}(k(y))$), the morphism $g_y : X_y \to \operatorname{Spec}(k(y)[T_1, \dots, T_e])$ is finite. Furthermore, for any such morphism g, there exists an open neighbourhood U of $f^{-1}(y)$ in X such that $g|U$ is quasi-finite.*

Proof. Let $\mathfrak{p} = j_y$; the ring $B \otimes_A k(\mathfrak{p})$ is a $k(\mathfrak{p})$-algebra of finite type, hence by the normalisation lemma (Bourbaki, *Alg. comm.*, chap. V, § 3, nº 1, th. 1) there exist in $B \otimes_A k(\mathfrak{p})$ a finite sequence $(t_i)_{1 \leqslant i \leqslant r}$ of elements algebraically independent over $k(\mathfrak{p})$ and such that if we set $C' = k(\mathfrak{p})[t_1, \dots, t_r]$, $B \otimes_A k(\mathfrak{p})$ is a C'-algebra *finite*; we therefore have $\dim(B \otimes_A k(\mathfrak{p})) = \dim(C')$ (**0**, 16.1.5) and as $\dim(C') = r$ (5.2.1), we have $r = e$. As $B \otimes_A k(\mathfrak{p}) = (B/\mathfrak{p}B) \otimes_{A/\mathfrak{p}} k(\mathfrak{p})$, we can, by multiplying the t_i by a suitable element of $A/\mathfrak{p}$, suppose that each t_i is the canonical image in $B \otimes_A k(\mathfrak{p})$ of an element $s_i \in B$. Let then $u : A[T_1, \dots, T_e] \to B$ be the homomorphism such that $u(T_i) = s_i$ for every i, and let $g : X \to Y[T_1, \dots, T_e]$ be the corresponding morphism. It is clear that, by the choice of the s_i, g_y is a finite morphism. For every morphism $g : X \to Y[T_1, \dots, T_e]$, such that g_y is finite, it follows from (5.4.2) and (4.1.2.1) that g_y is necessarily surjective. On the other hand, by (13.1.4), the set U of $x \in X$ that are isolated in their fibre $g^{-1}(g(x))$ is open in X and contains $f^{-1}(y)$, and by definition the restriction $g|U$ is a quasi-finite morphism. □ **IV-3 | 196**

This lemma being established, it remains to show that if x_λ is a maximal point of U, $g(x_\lambda) = z_\lambda$ is a maximal point of $Z = Y[T_1, \ldots, T_e]$. By the hypothesis of a), we can (restricting U if necessary) suppose that $f(x_\lambda)$ is one of the generic points y_α of the irreducible components of Y containing y. Let $u : A[T_1, \ldots, T_e] \to B$ be the homomorphism and $h : U \cap f^{-1}(y_\alpha) \to p^{-1}(y_\alpha) = \operatorname{Spec}(k(y_\alpha)[T_1, \ldots, T_e])$ the quasi-finite morphism deduced from g, by virtue of (**0**, 2.1.8); the maximal points of Z that are images by p of the generic points of the Y_α already prove that the $f(x_\lambda)$ are generic points of the Y_α. Furthermore, if $f(x_\lambda) = y_\alpha$, the $k(y_\alpha)$-morphism $h : U \cap f^{-1}(y_\alpha) \to p^{-1}(y_\alpha)$ deduced from g is dominant and quasi-finite by hypothesis; we therefore have $\dim(U_\lambda \cap f^{-1}(y_\alpha)) = e$ (U_λ being the irreducible component of U of generic point x_λ). Similarly, for every $x' \in U_\lambda$, the morphism $U_\lambda \cap f^{-1}(f(x')) \to p^{-1}(f(x'))$ deduced from g is a $k(f(x'))$-morphism, hence (4.1.2.1) we have $\dim(f^{-1}(f(x')) \cap U_\lambda) \leqslant e$; but we know on the other hand (13.1.6) that all the irreducible components of $U_\lambda \cap f^{-1}(f(x'))$ are of dimension $\geqslant e$; we see therefore that these components are of exactly dimension e, hence b) implies a).

Let us finally prove that b) implies a). Note that the structural morphism $p : Z = Y[T_1, \ldots, T_e] \to Y$ is faithfully flat; hence (2.3.4) the maximal points of Z are the images by p of the maximal points of Y; this already proves that the $f(x_\lambda)$ are generic points of the Y_α. Furthermore, if $f(x_\lambda) = y_\alpha$, the $k(y_\alpha)$-morphism $h : U \cap f^{-1}(y_\alpha) \to p^{-1}(y_\alpha)$ deduced from g is dominant and quasi-finite by hypothesis; we therefore have $\dim(U_\lambda \cap f^{-1}(y_\alpha)) = e$. Similarly, for every $x' \in U_\lambda$, the morphism $U_\lambda \cap f^{-1}(f(x')) \to p^{-1}(f(x'))$ deduced from g is a $k(f(x'))$-morphism quasi-finite, hence (4.1.2) we have $\dim(f^{-1}(f(x')) \cap U_\lambda) \leqslant e$; but by (13.1.6) all the irreducible components of $U_\lambda \cap f^{-1}(f(x'))$ are of dimension $\geqslant e$; we see therefore that these components are of exactly dimension e, hence b) implies a). □

Definition (13.3.2). — Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. We say that f is *equidimensional at the point* x (or that X is *equidimensional over* Y *at the point* x) if the equivalent conditions of (13.3.1) are satisfied. We say that f is *equidimensional* (or that X is *equidimensional over* Y) if f is equidimensional at every point $x \in X$.

It is clear that, for f to be equidimensional at a point x, it is necessary and sufficient that f_{red} be so. Furthermore, the conditions of (13.3.1) show that the set of points where f is equidimensional is *open* in X.

We note that when X and Y are *irreducible*, to say that f is equidimen- sional at the point x means, **IV-3** | 197
by virtue of (13.3.1), a''), that f is dominant and that $\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta))$ (where η is the generic point of Y); the definition (13.3.2) thus coincides in this case with the definition (13.2.2).

Proposition (13.3.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, X_j the irreducible components of X (in finite number) containing x. For f to be equidimensional at the point x, it is necessary and sufficient that, for every j, $Y_j = \overline{f(X_j)}$ is an irreducible component of Y and, denoting by X_j and Y_j the closed sub-preschemes of X and Y with underlying spaces X_j and Y_j, by $f_j : X_j \to Y_j$ the morphism deduced from f* (**I**, 5.2.2), *by y_j the generic point of Y_j, that f_j is equidimensional at the point x and that all the numbers $\dim(f_j^{-1}(y_j))$ are equal.*

Proof. This follows immediately from (13.3.1), a')). □

Corollary (13.3.4). — *With the notations of (13.3.3), set $y = f(x)$. If $\mathscr{O}_{Y,y}$ is a universally catenary ring and if f is equidimensional at the point x, we have*

(13.3.4.1) $$\dim(\mathscr{O}_{X_j,x}) = \dim(\mathscr{O}_{Y_j,y}) + e - \deg.\mathrm{tr}_{k(y)} k(x)$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j))$.

Proof. As each of the $\mathscr{O}_{Y_j,y}$, a quotient of $\mathscr{O}_{Y,y}$, is a universally catenary ring (5.6.1), the equality is a consequence of (5.6.5). □

Corollary (13.3.5). — *With the notations of (13.3.4), suppose that f is equidimensional at the point x and that $\mathscr{O}_{Y,y}$ is a universally catenary ring. If f is also equidimensional at $\mathscr{O}_{X,x}$, then $\mathscr{O}_{X,x}$ is also equidimensional; the converse is true if the image by f of the union of the X_j is dense in a neighbourhood of y.*

Proof. Indeed, to say that $\mathscr{O}_{X,y}$ is equidimensional means that the numbers $\dim(\mathscr{O}_{X_j,y})$ are equal for all the irreducible components Y'_j of Y containing y, as it follows from (5.1.1.5) applied to the local scheme $\mathrm{Spec}(\mathscr{O}_{x,y})$; as we have (13.3.4.1), it follows by the same reasoning that $\mathscr{O}_{X,x}$ is then also equidimensional. Conversely, if $\mathscr{O}_{X,x}$ is equidimensional, all the numbers $\mathscr{O}_{Y_j,y}$ are equal by (13.3.4.1), hence $\mathscr{O}_{Y,y}$ is equidimensional if the Y_j are all the irreducible components of Y containing y, which follows from the supplementary hypothesis. ☐

Proposition (13.3.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that the ring $\mathscr{O}_y$ is equidimensional and that we have the equality*

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)) \tag{13.3.6.1}$$

(cf. (13.2.7.1)*) f is equidimensional at the point x, and the converse is true if $\mathscr{O}_y$ is a universally catenary ring.*

Proof. Keeping the notations of (13.3.3); it follows from (13.2.8) and from the hypothesis that $\mathscr{O}_x$ is equidimensional, that each Y_j is an irreducible component of Y and that each f_j is equidimensional at the point x; furthermore (13.2.8), on has (13.3.4) that

$$\dim(\mathscr{O}_{X_j,x}) = \dim(\mathscr{O}_{Y_j,y}) + \dim(\mathscr{O}_{X_j,x} \otimes_{\mathscr{O}_{Y_j,y}} k(y));$$

whence, by the inequality (5.6.5.2) and the equality (13.3.6.1):

$$\dim_x(X_j \cap f^{-1}(y)) = \dim(\mathscr{O}_{X,x}) - \dim(\mathscr{O}_{Y,y}) + \deg.\mathrm{tr}_{k(y)} k(x). \tag{13.3.6.2}$$

The first member of (13.3.6.2) is therefore independent of j; but as f_j is equidimensional at the point x, we have $\dim_x(X_j \cap f^{-1}(y)) = \dim(f_j^{-1}(y_j))$, hence the criterion of (13.3.3) shows that f is equidimensional at the point x.

Conversely, suppose that $\mathscr{O}_{Y,y}$ is a universally catenary ring and that f is equidimensional at the point x; then (13.3.5) $\mathscr{O}_{X,x}$ is equidimensional, and it follows from (13.3.4) that we have the relation

$$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + e - \deg.\mathrm{tr}_{k(y)} k(x)$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j)) = \dim_x(X_j \cap f^{-1}(y))$; but by definition e is also equal to $\dim_x(f^{-1}(y))$, whence the relation (13.3.6.1), taking into account (5.6.5.2). ☐

Proposition (13.3.7). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type and equidimensional, Z a closed subset of X. Then the function*

$$x \longmapsto \mathrm{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x))) \tag{13.3.7.1}$$

is lower semi-continuous in X.

Proof. As all the irreducible components of $f^{-1}(f(x))$ containing x have the same dimension by hypothesis (13.3.1), we have, by virtue of (5.2.1),

$$\mathrm{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x))) = \dim_x(f^{-1}(f(x))) - \dim_x(Z \cap f^{-1}(f(x))).$$

But by hypothesis the first term of the right-hand side is a continuous function of x (13.3.1) and the second is an upper semi-continuous function of x by (13.1.3); whence the conclusion. ☐

IV-3 | 198

Proposition (13.3.8). — *Let Y be a prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. Let Y' be a prescheme, $g : Y' \to Y$ a flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$; if f is equidimensional at the point x, then f' is equidimensional at every point $x' \in X'$ above x.*

Proof. The question being local on X and Y, we can restrict to the case where the irreducible component of X contains x (resp. y); as the image by f of every irreducible component of X is dense in an irreducible component of Y (13.3.1), we know (2.3.5) that the image by f' of every irreducible component of X' is dense in an irreducible component of Y'. On the other hand, by transitivity of fibres (**I**, 3.6.4), it follows from (4.2.8) that, for every $z' \in X'$, the set of dimensions of the irreducible components of $f'^{-1}(f'(z'))$ is the same as the set of dimensions of the irreducible components of $f^{-1}(f(z))$, where z is the projection of z' in X; whence the conclusion by virtue of (13.3.1), a)). ☐

Remark (13.3.9). — The property of equidimensionality of a morphism $f : X \to Y$ is not stable under arbitrary base change $g : Y' \to Y$, even when g is the canonical injection of an irreducible component Y' of Y into Y (Y' being

§14. UNIVERSALLY OPEN MORPHISMS

Sections §§ 14 and 15 are devoted to the study of the notion of *universally open* morphism (2.4.2). **IV-3** | 199
We have already seen (2.4.6) that a *flat* morphism and locally of finite presentation is universally open, the converse being inexact. In § 14, we first examine the properties of the dimensions of the fibres of a universally open morphism $f : X \to Y$; when X and Y are locally Noetherian, f behaves in this regard (14.2.1) like a flat morphism (cf. (6.1.2)), and is in particular *equidimensional* when it is locally of finite type and dominant and X is irreducible (14.2.2). Conversely, an equidimensional morphism locally of finite type $f : X \to Y$ is universally open when one further assumes that Y is *geometrically unibranch*, and in particular when Y is *normal* (Chevalley's criterion, (14.4.4)). We also show that universally open morphisms locally of finite type $f : X \to Y$ (when X and Y are locally Noetherian and Y irreducible) admit "sufficiently many" *quasi-sections*, i.e. in the neighbourhood of a closed point x of a fibre $f^{-1}(y)$, there is a closed sub-prescheme X' of X containing x and such that the restriction $X' \to Y$ of f is a *quasi-finite* morphism (hence with *discrete* fibres) and dominant (14.5.3).

In § 15 we study various properties of the fibres of universally open morphisms $f : X \to Y$, locally **IV-3** | 200
of finite type, notably when X and Y are locally Noetherian. We thus obtain in particular a criterion for a point $x \in X$ to belong to a *single* irreducible component of X, in terms of properties of the *fibre* $f^{-1}(f(x))$ of x: it suffices that Y be geometrically unibranch (for example normal) at the point $f(x)$ and that $f^{-1}(f(x))$ be geometrically punctually integral at the point x (15.3.3); if moreover Y is locally integral at the point $f(x)$, f is flat at the point x and X locally integral at the point x. We also study the variation of the *number of geometric connected components* of a fibre $f^{-1}(y)$; for example, if f is universally open and *proper*, and the fibres of f geometrically reduced, this number is *locally constant* (15.5.7). Finally, when f admits a *section* g (which will be the case when X is a *Y-scheme in groups*) and for every $y \in Y$, one denotes by X_y^0 the connected component of the fibre $X_y = f^{-1}(y)$ at the point $g(y)$ ("neutral component" in the case of groups), one studies the union X^0 of the X_y^0 for $y \in Y$, and one shows (15.6.4) that if f is universally open and the fibres X_y geometrically reduced, then X^0 is an *open* set in X.

14.1. Open morphisms

(14.1.1). Recall (1.10.2) that a continuous map $\psi : X \to Y$ is said to be *open at a point* $x \in X$ if the image by ψ of *every* neighbourhood of x in X is a neighbourhood of $\psi(x)$ in Y.

We note that this does *not* imply that there exists a fundamental system of neighbourhoods of x whose images are *open* in Y.

Proposition (14.1.2). — *Let X, Y be two topological spaces, $\psi : X \to Y$ a continuous map, x a point of X, $y = \psi(x)$.*

label=(i) If ψ is open at x, then for every subset Y' of Y containing $\psi(x)$ the restriction $\psi^{-1}(Y') \to Y'$ of ψ is open at the point x.

lbbel=(ii) Suppose that Y is the union of a locally finite family of closed subsets (Y_i) and that for every i such that $\psi(x) \in Y_i$, the restriction $\psi^{-1}(Y_i) \to Y_i$ of ψ is open at the point x; then ψ is open at the point x.

lcbel=(iii) Let $\gamma : X' \to X$ be a continuous map, x' a point of X'; if the composite map $\psi \circ \gamma : X' \to Y$ is open at the point x', ψ is open at the point $\gamma(x')$.

Proof. If U is a neighbourhood of x, we have $\psi(U \cap \psi^{-1}(Y')) = \psi(U) \cap Y'$; whence (i) follows immediately. To prove (ii), let us note that there is a neighbourhood W of $\psi(x)$ in Y meeting only finitely many of the closed subsets Y_i that *contain* $\psi(x)$; hence W is the union of the $W \cap Y_i$ for these indices. Now, if U is a neighbourhood of x such that $\psi(U) \cap Y_i$ is a neighbourhood of $\psi(x)$ in Y_i, there exists a neighbourhood $V \subset W$ of $\psi(x)$ in Y such that for all such i, $\psi(x) \in Y_i$, we have **IV-3** | 201
$V \cap Y_i \subset \psi(U) \cap Y_i$ and since the union of the $V \cap Y_i$ for these indices is V, we have $V \subset \psi(U)$, hence $\psi(U)$ is a neighbourhood of $\psi(x)$ in Y. Assertion (iii) is trivial. □

Remarks (14.1.3). — label=(i) The set Z of points $x \in X$ where a morphism is open is *not necessarily open*. For example, let K be a field, A the polynomial ring $K[S, T]$, V the affine plane $\operatorname{Spec}(A)$, X the closed sub-prescheme of V which is the "union" of the line X_1 defined by $T = 0$ and of the line X_2 defined by $S = 0$, i.e. $\operatorname{Spec}(A/\mathfrak{a})$, where $\mathfrak{a} = AST$; let $Y = X_2 = \operatorname{Spec}(A/AS)$ and for f the projection corresponding to the canonical injection $K[T] \to A/\mathfrak{a}$; then we have $Z = X_2$, which is not open in X.

lbbel=(ii) Let X, Y be two Noetherian irreducible preschemes of generic points ξ, η respectively, $f : X \to Y$ a morphism *dominant locally of finite type*; then f *is open at the point* ξ. Indeed, we can evidently restrict to the case where X and Y are reduced (hence integral) (1.5.4) and affine; by virtue of the generic flatness theorem (6.9.1), there exists a non-empty open set V of Y such that the restriction $f^{-1}(V) \to V$ of f is a *flat* morphism, and we conclude from (2.4.6) that this restriction is an open morphism. However, there are dominant morphisms of finite type $f : X \to Y$ (where X and Y are *irreducible*) that are not open (see for example (**II**, 8.1)), the set of points where such a morphism is open is not necessarily *closed* in X.

lcbel=(iii) We do not know whether, when X and Y are locally Noetherian and $f : X \to Y$ is a morphism locally of finite type, the set of points of X where f is open is open or not *locally constructible*.

Proposition (14.1.4). — *Let X, Y be two topological spaces, $\psi : X \to Y$ a continuous map. For every $y \in Y$, the set of $x \in \psi^{-1}(y)$ where ψ is open is a closed subset of $\psi^{-1}(y)$.*

Proof. Indeed, suppose that ψ is not open at a point $x \in \psi^{-1}(y)$; there exists then a neighbourhood V of x in X such that $\psi(V)$ is not a neighbourhood of y; it follows that for every $x' \in V \cap \psi^{-1}(y)$, ψ is not open at the point x'. □

Remark (14.1.5). — Even if X and Y are locally Noetherian and f a morphism *fini*, it can happen that f is open at every point of a fibre $f^{-1}(y)$, without there existing a *neighbourhood* of $f^{-1}(y)$ at every point of which f is open. Take for example K a field, A the polynomial ring $K[\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3]$, $V = \operatorname{Spec}(A)$ the affine 3-space over K, $X = \operatorname{Spec}(A/\mathfrak{a})$, where $\mathfrak{a} = \mathfrak{b}\mathfrak{c}$, with $\mathfrak{b} = (\mathrm{T}_3)$ and $\mathfrak{c} = (\mathrm{T}_1) + (\mathrm{T}_2 - \mathrm{T}_3)$ in A, so that X is the union of the plane $X_1 = \operatorname{Spec}(A/\mathfrak{b})$ ("plane with equation $\mathrm{T}_3 = 0$") and of the line $X_2 = \operatorname{Spec}(A/\mathfrak{c})$ ("line with equations $\mathrm{T}_1 = 0$, $\mathrm{T}_2 = \mathrm{T}_3$ ") which are its irreducible components. Take $Y = X_1$ and let $f : X \to Y$ be the projection corresponding to the canonical injection $K[\mathrm{T}_1, \mathrm{T}_2] \to A/\mathfrak{a}$; if y is the point common to X_1 and X_2, $f^{-1}(y)$ is reduced to y and f is open at this point but is not open at any point of X_2 in a neighbourhood of y and distinct from y.

(14.1.6). In what follows, the essential role will be played by the criterion (1.10.3) characterising the morphisms *locally of finite presentation* $f : X \to Y$ that are open at a point x by the "lifting of generizations": for every generization y' of $y = f(x)$, there exists $x' \in X$, a generization of x, such that $y' = f(x')$.

14.2. Open morphisms and the dimension formula

Theorem (14.2.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \to Y$ a morphism, x a point of X, $y = f(x)$. Suppose that f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Then we have the relation*

$$\dim(\mathscr{O}_x) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_x \otimes_{\mathscr{O}_y} k(y)). \tag{14.2.1.1}$$

Proof. Let y' be any generization of y distinct from y, and consider the closed reduced sub-prescheme Y' of Y having $\overline{\{y'\}}$ as underlying space; then no irreducible component X' of $f^{-1}(Y')$ containing x can be contained in $f^{-1}(y)$: indeed, if x' is the generic point of X', we would have $f(x') = y \neq y'$, and as x' is its only generization in $f^{-1}(Y')$, this would contradict the hypothesis that f is open at x', by virtue of the fact that *a*) implies *c*) in (1.10.3). We can therefore apply (6.1.2) to the local homomorphism of Noetherian local rings $\mathscr{O}_y \to \mathscr{O}_x$, whence the conclusion. □

Corollary (14.2.2). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Then f is equidimensional at the point x in each of the two following cases:*

label=(i) *X is irreducible and f is dominant.*
lbbel=(ii) *The rings $\mathscr{O}_x$ and $\mathscr{O}_y$ are equidimensional.*

Proof. For (i), this follows from (14.2.1) and from (13.2.3). For (ii), this follows from (14.2.1) and from (13.3.6). □

Proposition (14.2.3). — *Let Y be an irreducible locally Noetherian prescheme, η its generic point, $f : X \to Y$ a morphism locally of finite type, y a point of $f(X)$, Z an irreducible component of $f^{-1}(y)$ such that f is open at the generic point of Z. Then Z is contained in an irreducible component X' of X dominant over Y and such that $\dim(Z) = \dim(X' \cap f^{-1}(y)) = \dim(X' \cap f^{-1}(\eta))$.*

Proof. This follows indeed from (14.2.1) applied at the generic point of Z, and from (13.2.7). □

Corollary (14.2.4). — *With the notations of (14.2.3), suppose that f is open at all the points of $f^{-1}(y)$. For every $z \in Y$, let $\mathrm{E}(z)$ be the set of dimensions of the irreducible components of $f^{-1}(z)$ and $d(z) = \sup \mathrm{E}(z) = \dim(f^{-1}(z))$. Then $\mathrm{E}(y) \subset \mathrm{E}(\eta)$, whence $d(y) \leqslant d(\eta)$.*

Proof. This follows immediately from (14.2.3). □

We see that one can restrict to proving the corollary when Y is irreducible; let η be its generic point. **IV-3** | 203
It follows from (14.2.4) that $d(y) \leqslant d(\eta)$; on the other hand, as f is proper, we deduce from (13.1.5) that $z \mapsto \dim(f^{-1}(z))$ is upper semi-continuous; in particular $d(y) \geqslant d(\eta)$, hence $d(y) = d(\eta)$. Moreover, there is a neighbourhood V of y such that $d(z) \leqslant d(y)$ for every $z \in V$; but as every z is a specialisation of η, we have on the other hand $d(z) \geqslant d(\eta)$, whence finally $d(z) = d(y)$ for $z \in V$.

Corollary (14.2.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism, $y \in Y$ a point such that f is open at all the points of $f^{-1}(y)$. Then the function $z \mapsto \dim(f^{-1}(z))$ is constant in a neighbourhood of y.*

Proof. Let Y_i $(1 \leqslant i \leqslant n)$ be the closed reduced sub-preschemes of Y having as underlying spaces the irreducible components of Y containing y; if $f_i : f^{-1}(Y_i) \to Y_i$ is the restriction of f, we know that each of the f_i is proper (**II**, 5.4.5) and open at all the points of $f^{-1}(y)$ (14.1.2). As the union of the Y_i is a neighbourhood of y in Y, we see that one can restrict to proving the corollary when Y is irreducible; let η be its generic point. It follows from (14.2.4) that $d(y) \leqslant d(\eta)$; on the other hand, as f is proper, we deduce from (13.1.5) that $z \mapsto \dim(f^{-1}(z))$ is upper semi-continuous; in particular $d(y) \geqslant d(\eta)$, hence $d(y) = d(\eta)$. Moreover, there is a neighbourhood V of y such that $d(z) \leqslant d(y)$ for every $z \in V$; but as every z is a specialisation of η, we have on the other hand $d(z) \geqslant d(\eta)$, whence finally $d(z) = d(y)$ for $z \in V$. □

Corollary (14.2.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ an open morphism locally of finite type, y a point of Y. Let y_i $(1 \leqslant i \leqslant n)$ be the generic points of the irreducible components of Y containing y. Then, with the notations of (14.2.4):*

label=(i) *If for $1 \leqslant i \leqslant n$, we have $\mathrm{E}(y) = \mathrm{E}(y_i)$ (resp. $d(y) = d(y_i)$), the function E (resp. d) is constant in a neighbourhood of y.*
lbbel=(ii) *There exists a neighbourhood U of $f^{-1}(y)$ such that the function $z \mapsto \dim(U \cap f^{-1}(z))$ is constant in a neighbourhood of y.*

Proof. label=(i) The same reasoning as in (14.2.5) shows that one can restrict to considering the case where Y is irreducible, of generic point η. If y' is any generization of y, we have then (applying (14.2.4) to the restriction $f^{-1}(Y') \to Y'$ of f, where $Y' = \overline{\{y'\}}$, and using (14.1.2)) $\mathrm{E}(y) \subset \mathrm{E}(y') \subset \mathrm{E}(\eta)$ and $d(y) \leqslant d(y') \leqslant d(\eta)$; this shows that the relation $\mathrm{E}(y) = \mathrm{E}(\eta)$ (resp. $d(y) = d(\eta)$) implies $\mathrm{E}(y) = \mathrm{E}(y')$ (resp. $d(y) = d(y')$) for every generization y' of y. Now, by virtue of (9.5.5), the functions E and d are locally constructible, hence it follows from ($\mathbf{0}_{\text{III}}$, 9.2.5) applied to the set of z such that $\mathrm{E}(z) = \mathrm{E}(y)$ (resp. $d(z) = d(y)$) that this set is a neighbourhood of y.

lbbel=(ii) For each of the y_i, $f^{-1}(y_i)$ is a Noetherian space since f is of finite type; let x_{ij} $(1 \leqslant j \leqslant m_i)$ be the generic points of its irreducible components, and set $X'_{ij} = \overline{\{x_{ij}\}}$; let us show that the complement U in X of the union of the X'_{ij} that do not meet $f^{-1}(y)$ (which is evidently an open neighbourhood of $f^{-1}(y)$) answers the question. Indeed, the restriction $f|U : U \to Y$

of f is an open morphism of finite type (**I**, 6.3.5); for every pair (i, j) such that $x_{ij} \in U$, it follows from (13.1.1) applied to the restriction $X_{ij} \to Y_i = \overline{\{y_i\}}$ of f (taking into account (**II**, 5.2.2) and (1.5.4)) that all the irreducible components of $X_{ij} \cap f^{-1}(y_i)$ have dimension $\geqslant \dim(X_{ij} \cap f^{-1}(y_i))$. Taking into account (14.2.3) applied to the restriction $f^{-1}(Y_i) \to Y_i$ of f, which is an open morphism (14.1.2), $f^{-1}(y)$ is, for each i, the union of the irreducible components of $X_{ij} \cap f^{-1}(y)$ $(1 \leqslant j \leqslant m_i)$, hence $\dim(f^{-1}(y)) \geqslant \dim(U \cap f^{-1}(y_i) \cap U)$ in virtue of (14.2.4); we see therefore that one can apply (i) to $f|U$, whence the conclusion. □

Remark (14.2.7). — The example (13.2.12), (iii)) shows that under the hypotheses of (14.2.5) or (14.2.6), (ii)), one cannot, in the conclusion, replace the function $z \mapsto \dim(f^{-1}(z))$ by the function $z \mapsto \mathrm{E}(z)$; indeed, in this example, f is proper and flat (hence universally open (2.4.6)) and every neighbourhood of the unique point of $f^{-1}(y_0)$ contains the generic points of the irreducible components of $f^{-1}(\eta)$.

IV-3 | 204

14.3. Universally open morphisms

(14.3.1). Recall (2.4.2) that to say that a morphism $f : X \to Y$ is universally open means that for every morphism $g : Y' \to Y$, $f_{(Y')} : X_{(Y')} \to Y'$ is open; it suffices moreover that this be so when $Y' = Y[\mathrm{T}_1, \dots, \mathrm{T}_e]$, for every integer e (8.10.2) (and if Y is locally Noetherian, it suffices therefore that it be so for every Y' locally Noetherian).

Proposition (14.3.2). — *Let $f : X \to Y$ be a morphism of preschemes. If f is universally open, then, for every morphism $g : Y' \to Y$, where Y' is irreducible, the image by $f' = f_{(Y')} : X_{(Y')} \to Y'$ of every irreducible component of $X_{(Y')}$ is dense in Y'. Conversely, if this condition is satisfied for every Y' irreducible and every morphism of finite type $g : Y' \to Y$, and if moreover f is locally of finite presentation, then f is universally open.*

Proof. Indeed, it follows from (1.10.4) that if f is universally open, it satisfies the condition of the statement. Conversely, let us show that if f is locally of finite presentation, suppose that f is locally of finite presentation, and let us show that for every integer e, if one sets $Y'' = Y[\mathrm{T}_1, \dots, \mathrm{T}_e]$, $f_{(Y'')}$ is an open morphism. Indeed, let Y' be a closed sub-prescheme of Y'' having as underlying space a closed irreducible subset of Y''; the composite morphism $Y' \to Y'' \to Y$ is of finite type, hence every irreducible component of $X_{(Y')}$ dominates Y' by hypothesis; we deduce therefore from (1.10.4) that $f_{(Y')}$ is an open morphism.

This proposition shows that the definition (**III**, 4.3.9) coincides in the case considered with the general definition of universally open morphisms given in (2.4.2).

The notion of an open morphism at a point (14.1.1) gives rise in the same way to the following: □

Definition (14.3.3). — Let $f : X \to Y$ be a morphism of preschemes, x a point of X. We say that f is *universally open at the point* x if, for every morphism $g : Y' \to Y$, setting $X' = X \times_Y Y'$, the morphism $f' = f_{(Y')} : X' \to Y'$ is open at every point x' of X' whose projection in X is x.

Remarks (14.3.3.1). — label=(i) The reasoning of (8.10.1) shows (with the same notations) that if x is a point of X, x_λ its projection in X_λ, then, if f_μ is open at the point x_μ for every $\mu \geqslant \lambda$, f is open at the point x; it suffices to restrict to opens U of X containing x and to remark that the hypothesis implies that $f_\mu(U_\mu)$ is a neighbourhood of $f_\mu(x_\mu)$, hence $f(v_\mu^{-1}(U_\mu))$ is a neighbourhood of $f(x)$. We deduce that the statement (8.10.2) is still exact when one replaces "universally open" by "universally open at the point x", and "open morphism" by "morphism open at every point x_n of X_n whose projection in X is x": it suffices in the proof to limit to opens V containing x_n.

lbbel=(ii) The result of (14.1.4) is again valid for a morphism $f : X \to Y$, replacing "open" by "universally open". Indeed, suppose that f is not universally open at a point $x \in f^{-1}(y)$; there is then a morphism $Y' \to Y$ and a point $x' \in X'$ projecting to x, such that f' is not open at the point x'.

Now, if $y' = f'(x')$, the projection $f'^{-1}(y') \to f^{-1}(y)$ is an open morphism (2.4.10), and there is, by virtue of (14.1.4), a neighbourhood V' of x' in $f'^{-1}(y')$ where f' is not open; hence f is not universally open at any point of the image of V' in $f^{-1}(y)$, which is a neighbourhood of x in $f^{-1}(y)$. **IV-3 | 205**

Proposition (14.3.4). — *label=(i) Let $f : X \to Y$, $g : Y \to Z$ be two morphisms, x a point of X, $y = f(x)$. If f is universally open at the point x and g universally open at the point y, then $g \circ f$ is universally open at the point x. Conversely, if $g \circ f$ is universally open at the point x, g is universally open at the point y.*

lbbel=(ii) If $f : X \to Y$ is an S-morphism universally open at the point $x \in X$, then, for every base change $S' \to S$, $f_{(S')} : X_{(S')} \to Y_{(S')}$ is universally open at every point of $X_{(S')}$ above x.

lcbel=(iii) For $f : X \to Y$ to be universally open at the point $x \in X$, it is necessary and sufficient that f_{red} be so.

*ldbel=(iv) Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$; set $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$. For f to be universally open at the point x, it is necessary and sufficient that f_1 be so (recall (**I**, 3.6.5) that X_1 is canonically identified with a sub-espace of X).*

Proof. Indeed (ii) is an evident consequence of the definition (14.3.3); it also follows from the definition that in order to prove assertion (i), it suffices to do so dropping everywhere the word "universally", and this follows then from (14.1.2). Assertion (iii) follows from (ii) and from the fact that the canonical morphism $X_{\mathrm{red}} \to X$ is surjective. Finally, condition (iv) is trivially necessary. On the other hand, if it is satisfied, and if $g : Y' \to Y$ is any morphism, $X' = X_{(Y')}$, $f' = f_{(Y')}$, x' a point of X' above x, then f' is locally of finite presentation, and to see that it is open at the point x', it suffices to apply the criterion (1.10.3), *b*)). Let $y' = f'(x')$, $Y'_1 = \operatorname{Spec}(\mathscr{O}_{y'})$, $X'_1 = X' \times_{Y'} Y'_1$; since $g(y') = y$, the composite morphism $Y'_1 \to Y' \xrightarrow{g} Y$ factorises as $Y_1 \to Y_1 \to Y$ (**I**, 2.4.4), hence $X'_1 = X_1 \times_{Y_1} Y'_1$ and $f'_1 = (f_1)_{(Y_1)}$; the conclusion follows then immediately from the hypothesis that f'_1 is open at the point x' and from (1.10.3). □

Proposition (14.3.5). — *Let X, Y be two preschemes, $f : X \to Y$ a morphism, x a point of X. Let $(Y_i)_{1 \leqslant i \leqslant n}$ be a locally finite family of closed sub-preschemes of Y such that the space Y is the union of the Y_i, and suppose that for every i such that $f(x) \in Y_i$, the restriction $f^{-1}(Y_i) \to Y_i$ of f is universally open at the point x; then f is universally open at the point x.*

Proof. Taking into account the definition, this follows from the analogous proposition (14.1.2), (ii)) for open morphisms at a point. □

Proposition (14.3.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. For f to be universally open at a point $x \in X$, it is necessary and sufficient that the following condition be satisfied: for every morphism $g : Y' \to Y$, where $Y' = \operatorname{Spec}(A)$ is the spectrum of a discrete valuation ring, such that the image $g(y')$ of the closed point y' of Y' is equal to $y = f(x)$, and for every point $x' \in X' = X \times_Y Y'$ whose projections on X and Y' are x and y' respectively, there exists a generization z' of x' in X' whose projection in Y' is the generic point of Y' (in other words,* there is an irreducible component of X' containing x' and dominating Y'*).*

Moreover, one can, in the preceding condition, restrict to the case where A is complete, to an algebraically closed residue field, and where x' is rational over $k(y')$. **IV-3 | 206**

Proof. If Y' is as in the statement, the necessity of the condition follows from the fact that f' must be open at the point x', and the criterion (1.10.3). To see that the condition is sufficient, consider a morphism of finite type $g : Y'' \to U$, and let $X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \to Y''$, and x'' a point of X'' above x. Set $y'' = f''(x'')$, and let t be a generization of y'' in Y'', distinct from y''. Since Y'' is locally Noetherian, it follows from (**II**, 7.1.9) and ($\mathbf{0}_{\mathrm{III}}$, 10.3.1) that there exists a scheme $Y' = \operatorname{Spec}(A)$, where A is a discrete valuation ring, and a morphism $g : Y' \to Y''$ such that, if s and y' are the generic point and the closed point respectively, we have $g(s) = t$ and $g(y') = y''$. There is then a point x' of $X' = X'' \times_{Y''} Y'$ whose projections in X'' and Y' are x'' and y' respectively, and which is rational over $k(y')$ (**I**, 3.4.9). The hypothesis implies that there is a generization z' of x' in X' whose projection in Y' is s; if z'' is the projection of z' in X'', z'' is a generization of x'' and its projection in Y'' is t; we conclude therefore from (1.10.3) that f'' is open at the point x'', hence f is universally open at the point x (14.3.3.1), (i)). □

Corollary (14.3.7). — *The notations being those of (14.3.6):*

label=(i) Given a point $y \in Y$, for f to be universally open at all the points of $f^{-1}(y)$, it is necessary and sufficient that for every morphism $g : Y' \to Y$, where Y' is the spectrum of a discrete valuation ring and where the image $g(y')$ of the closed point y' of Y' is y, every irreducible component of $X' = X \times_Y Y'$ dominates Y'.

lbbel=(ii) For f to be universally open, it is necessary and sufficient that for every morphism $g : Y' \to Y$, where Y' is the spectrum of a discrete valuation ring, every irreducible component of $X' = X \times_Y Y'$ dominates Y'.

Proof. It is clear that it suffices to prove (i); the necessity of (i) follows from (14.3.2) and its sufficiency from (14.3.6). □

Proposition (14.3.8). — *Let Y be a locally Noetherian prescheme, irreducible, regular, and of dimension* 1 *(for example the spectrum of a Dedekind ring), $f : X \to Y$ a morphism locally of finite type, y a point of Y. The following conditions are equivalent:*

label=a) f_{red} is flat at every point of $f^{-1}(y)$.
lbbel=b) f is universally open in a neighbourhood of $(f^{-1}y)$.
lcbel=c) f is open in a neighbourhood of $f^{-1}(y)$.
ldbel=d) Every irreducible component of X meeting $f^{-1}(y)$ dominates Y.

Proof. As f_{red} is locally of finite type (1.3.4), $a)$ implies that f_{red} is flat in a neighbourhood of $f^{-1}(y)$ (11.1.1), and it suffices to apply (2.4.6) in such a neighbourhood. The implication $b) \Rightarrow c)$ is trivial, and the implication $c) \Rightarrow d)$ follows from (1.10.4) applied to a neighbourhood of $f^{-1}(y)$. It remains to see that $d)$ implies $a)$. We can evidently, by virtue of (1.3.4), restrict to the case where X is reduced. The question being moreover local on X and on Y, we can suppose $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ affine; if X_i ($1 \leqslant i \leqslant n$) are the closed sub-preschemes (integral) of X having as underlying spaces **IV-3 | 207** the irreducible components of X, then, for every $x \in X$, $\mathscr{O}_{X,x}$ is a sub-ring of the composed direct product of the $\mathscr{O}_{X_i,x}$; if $y = f(x)$, it will suffice to show that each of the $\mathscr{O}_{X_i,x}$ is an $\mathscr{O}_y$-module *without torsion*, for it will then be of the same for $\mathscr{O}_{X,x}$; as by hypothesis $\mathscr{O}_y$ is a regular local ring of dimension 1, i.e. a discrete valuation ring (**II**, 7.1.6), it will then follow from ($\mathbf{0}_1$, 6.3.4) that $\mathscr{O}_{X,x}$ is a *flat* $\mathscr{O}_y$-module. But if $X_i = \mathrm{Spec}(B_i)$, where B_i is an integral ring, the hypothesis implies that the homomorphism $A \to B_i$ is *injective* (**I**, 1.2.7); hence B_i is an A-module without torsion, and *a fortiori* $\mathscr{O}_{X_i,x}$ is an $\mathscr{O}_y$-module without torsion. □

Remarks (14.3.9). — label=(i) In the statement of (14.3.8), one cannot dispense with the hypothesis that Y is *regular*. With the notations of (11.7.5), take indeed $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(\overline{A})$, so that $f : X \to Y$ is a morphism *fini surjectif*; as A is a local ring of dimension 1, and as $\overline{A}$ is a local ring of dimension 1, it follows immediately from (1.10.4) that the morphism f is *open*. However f is not universally open (nor *a fortiori* flat) (11.7.5). One could give an analogous example by taking for Y the local scheme of an algebraic curve having an "ordinary double point" and for X the normalised of Y.

lbbel=(ii) The example of (14.1.3), (i)) shows that of a morphism f where the set of points of X where the morphism is universally open is universally open at all the points but the set of points where the morphism is universally open is not necessarily closed, for it is immediate that all the points of X *except* the morphism f is universally open (it is even a local isomorphism). It would be interesting to know if the set of points where a morphism is universally open is locally constructible.

The two following propositions have been indicated by M. Artin:

Lemma (14.3.10). — *Let A be a valuation ring (not necessarily discrete), k its residue field, K its field of fractions, and set $S =$ $\mathrm{Spec}(A)$. Let X be an irreducible prescheme, $f : X \to S$ a dominant morphism of finite type; let $X_0 = X \otimes_A k$ (resp. $X_1 = X \otimes_A K$) be the fibre of f at the closed point (resp. at the generic point) of S. Then, if $X_0 \neq \varnothing$, we have* $\dim(X_0) = \dim(X_1)$.

Proof. We can restrict to the case where X is affine, replacing if necessary X by an open affine containing a generic point of an irreducible component of X_0, of maximal dimension. Let $n = \dim(X_0) \geqslant 0$; using (4.1.1.3), it follows from (13.3.1) that there exists a neighbourhood U of X_0 in X and an S-morphism quasi-finite $g : U \to S[T_1, \ldots, T_n] = Z$ such that the morphism restriction $g_0 : U_0 = U \cap X_0 \to Z_0 = \operatorname{Spec}(k[T_1, \ldots, T_n])$ is finite and *surjective*. By base change to the open $U_1 = U \cap X_1$, we deduce a quasi-finite morphism $g_1 : U_1 \to Z_1 = \operatorname{Spec}(K[T_1, \ldots, T_n])$. As U_1 is dense in X_1 (4.1.1.3), we have $\dim(U_1) = \dim(X_1)$; if we can prove that the morphism g_1 is *dominant*, the proposition will be established, since then $\dim(U_1) \leqslant \dim(Z_1) = n$ and on the other hand $\dim(X_1) \geqslant n$ by (4.1.2). Suppose the contrary; there would then be a polynomial $F_1 \neq 0$ in $K[T_1, \ldots, T_n]$ such that $g_1(U_1) \subset V(F_1)$. If ω is a valuation on K associated to A, and if (a_α) is the family of coefficients of F_1, after multiplication of F_1 by a non-zero element of K, we can suppose that $\inf(\omega(a_\alpha)) = 0$; in other words, F_1 comes from a polynomial $F \in A[T_1, \ldots, T_n]$ (with which it is identified) such that the image F_0 of F in $k[T_1, \ldots, T_n]$ is *nonzero*. Consider then *in* Z the closed set $V(F)$; we have $V(F) \cap Z_1 = V(F_1)$, and as U_1 is dense in U (since it contains the generic point of X), **IV-3** | 208
$g(U) \subset V(F)$; in particular, $g_0(U_0) \subset V(F) \cap Z_0 = V(F_0)$; but since $F_0 \neq 0$, $V(F_0)$ is a closed subset of Z_0 distinct from Z_0, and one arrives at a contradiction. □

Proposition (14.3.11). — *Let $f : X \to S$ be a morphism of finite type, $(g_\lambda)_{\lambda \in L}$ a family of universally open morphisms $g_\lambda : Y_\lambda \to S$, and for every $\lambda \in L$, let $u_\lambda : Y_\lambda \to X$ be an S-morphism. For every $s \in S$, set $X_s = X \times_S \operatorname{Spec}(k(s))$, $(Y_\lambda)_s = Y_\lambda \times_S \operatorname{Spec}(k(s))$, and let $(u_\lambda)_s : (Y_\lambda)_s \to X_s$ be the morphism $u_\lambda \times 1$ deduced from u_λ by base change. Let $Z(s)$ be the closure in X_s of the union of the sets $(u_\lambda)_s((Y_\lambda)_s)$, and set $d(s) = \dim(Z(s))$. Then, for every generization s' of a point $s \in S$, we have $d(s) \leqslant d(s')$.*

Proof. We know (**II**, 7.1.4) that there exists a valuation ring A' and a morphism $h : S' = \operatorname{Spec}(A') \to S$ such that, if a (resp. b) is the closed point (resp. the generic point) of S', we have $h(a) = s$, $h(b) = s'$. Furthermore, the morphism projection $p : X_s \otimes_{k(s)} k(a) \to X_s$ is surjective and open (2.4.10), hence for every subset M of X_s, $\overline{p^{-1}(M)}$ is therefore equal to the closure of $X_s \otimes_{k(s)} k(a)$ by an open equivalence relation; for every M, one reasons in the same way for $X_{s'}$, and from (**I**, 3.4.8), (4.2.7) and from the fact that the g_λ are universally open, we see that one can reduce to proving the proposition on the situation obtained after base change $S' \to S$. Suppose therefore $S' = S$, s being the closed point and s' the generic point of S. The hypothesis that g_λ is open implies that every irreducible component of Y_λ dominates S (1.10.4), hence its generic point is a maximal point of $(Y_\lambda)_s$; if Z denotes the closure of X in $Z(s)$, we have therefore $u_\lambda(Y_\lambda) \subset Z$, and consequently $(u_\lambda)_s((Y_\lambda)_s) \subset Z \cap X_s$; in other words, we have $Z(s) \subset Z_s$, where $Z_s = Z \cap X_s$. But applying (14.3.10) to a reduced sub-prescheme of X having as underlying space an irreducible component of Z, we obtain $\dim(Z_s) \leqslant \dim(Z_{s'})$ (the inequality coming from the fact that there can be irreducible components of Z not meeting X_s). On the other hand, since $Z(s') \supset Z_{s'}$ by definition, we have $\dim(Z(s')) \geqslant \dim(Z_{s'})$, which completes the proof. □

Remark (14.3.12). — The case considered by M. Artin was that where $Y_\lambda = S$ for every λ, in other words the case where (u_λ) is a family of *S-sections* of X. Another useful case is that where the family (u_λ) is reduced to a single element; one can then always reduce to this case by considering the prescheme Y which is the sum of the Y_λ and the morphisms $g : Y \to S$ and $u : Y \to X$ whose restrictions to each Y_λ are respectively g_λ and u_λ.

Proposition (14.3.13). — *Let $f : X \to Y$ be a morphism locally of finite type, y a point of Y, x a maximal point of the fibre $X_y = X \times_Y \operatorname{Spec}(k(y))$. Consider the following conditions:*

label=a) *f is universally open at the point x (or even at every point of the irreducible component of X_y of generic point x (14.3.3.1), (ii))).*

lbbel=b) *For every irreducible component Y_0 of Y containing y, there exists an irreducible component Z of X containing x, dominant over Y_0 and such that $\dim_x(X_y) = \dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$, where η is the generic point of Y_0 (which implies that Z is* equidimensional *over Y_0 at the point x (13.2.2)).*

IV-3 | 209

$b')$ *For every open neighbourhood U of x in X and every generization y' of y, we have $\dim(U \cap X_{y'}) \geqslant \dim_x(U \cap X_y)$.*

Then we have the implications a) ⇒ b) ⇔ b′).

Proof. To show that b) implies b'), it suffices to remark that y' belongs to an irreducible component Y_0 of Y containing y, of generic point η; taking Z as in b) and noting that the generic point of Z (which is also that of $Z \cap X_\eta$ ($\mathbf{0}_1$, 2.1.8)) is contained in U, we have $\dim(U \cap X_{y'}) \geqslant \dim(U \cap Z \cap X_{y'})$, and, by virtue of (13.1.6), $\dim(U \cap Z \cap X_{y'}) \geqslant \dim(Z \cap X_\eta)$; whence the assertion, since $\dim_x(U \cap X_y) = \dim_x(X_y)$ (4.1.1.3).

To prove that b') implies b), one can first replace X by $f^{-1}(Y_0)$, hence suppose $Y_0 = Y$; we can restrict to the case where X and Y are affine, since $\dim_x(U \cap X_y) = \dim_x(X_y)$ (4.1.1.3). The irreducible components W_i of the Noetherian prescheme X_η are then finite in number, and the complement of the union of the $\overline{W}_i$ (closures *in* X) that do not contain x is an open neighbourhood V of x. Replacing X by V, we can therefore suppose that $x \in \overline{W}_i$ for every i (the hypothesis b') implies $\dim(V \cap X_\eta) \geqslant 0$, hence x belongs to at least one of the $\overline{W}_i$ for at least one i). Furthermore, the $\overline{W}_i$ are exactly the irreducible components of X that dominate Y ($\mathbf{0}_1$, 2.1.8); if one had, for each of these components Z, $\dim(Z \cap X_\eta) < \dim_x(X_y)$, one would conclude $\dim(X_\eta) < \dim_x(X_y)$, contrary to the hypothesis b'). The relation $\dim_x(Z \cap X_y) = \dim_x(X_y)$ follows then from (13.1.6).

It remains to show that a) implies b'). Taking into account (**II**, 7.1.4) and the invariance of the hypotheses and of the conclusion by base change (by virtue of (4.2.7)), we can restrict to the case where Y is the spectrum of a valuation ring, of closed point y and of generic point y', and where $U = X$. The hypothesis that f is open at the point x implies that there exists an irreducible component Z of X containing x and dominant over Y (1.10.3). Applying (14.3.10) to a neighbourhood of x in Z, we conclude that $\dim(Z \cap X_{y'}) = \dim_x(Z \cap X_y)$; but as x is maximal in X_y, $Z \cap X_y$ contains the irreducible component of X_y of generic point x, hence $\dim_x(Z \cap X_y) = \dim_x(X_y)$; on the other hand, we have $\dim(X_{y'}) \geqslant \dim(Z \cap X_y)$, which completes the proof of b'). □

Remark (14.3.14). — We do not know whether in (14.3.13), the conclusion subsists when one replaces the hypothesis a) by the weaker hypothesis that f is *open* at the point x. One could however easily show that it would suffice to treat the case where Y is the spectrum of an integral local ring, whose generic point is isolated, and where X is a closed sub-prescheme of the vector bundle $Y[\mathrm{T}]$.

14.4. Chevalley's criterion for universally open morphisms

Theorem (14.4.1). — *Let $f : X \to Y$ be a morphism locally of finite type, y a point of Y, x a maximal point of the fibre $X_y = X \times_Y \operatorname{Spec}(\kappa(y))$. Suppose y geometrically unibranch. Then the following conditions are equivalent:* **IV-3 | 209**

label=a) *f is universally open at x (or equivalently at every point of the irreducible component of X_y with generic point x (14.3.3.1), (ii)).* **IV-3 | 210**

lbbel=b) *If Y_0 is the unique irreducible component of Y containing y, η its generic point, there exists an irreducible component Z of X containing x, dominating Y_0 and such that $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$ (which means that Z is equidimensional over Y_0 at the point x (13.2.2)).*

b′) *For every open neighbourhood U of x in X and every generization y' of y, we have $\dim(U \cap X_{y'}) \geqslant \dim_x(U \cap X_y)$.*

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

c) *f is open at x.*

Let us first note that since $(\mathscr{O}_y)_{\mathrm{red}}$ is integral, y belongs to only one irreducible component Y_0 of Y. The fact that *b)* and *b′)* are equivalent and that *a)* implies *b′)* results from (14.3.13); on the other hand, if Y is locally Noetherian, we have seen in (14.3) that *c)* implies *b)*. It remains to show that when y is geometrically unibranch, *b)* implies *a)*.

Lemma (14.4.1.1). — *Let Y be a prescheme, $Y' = Y[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, y' a point of Y', y its image in Y. If Y is geometrically unibranch at the point y, then Y' is geometrically unibranch at the point y'.*

Indeed, for every $y \in Y$, $\operatorname{Spec}(\kappa(y)[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ is geometrically regular over $\kappa(y)$ (**0**, 17.3.7), the structural morphism $Y' \to Y$ is smooth (6.8.1), and it suffices to apply (11.3.14).

Lemma (14.4.1.2). — *Let A be a local integral ring, unibranch, B an integral ring containing A, $\mathfrak{n}$ a prime ideal of B above the maximal ideal $\mathfrak{m}$ of A. Then the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective, in other words, for every prime ideal $\mathfrak{p}$ of A, there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \subset \mathfrak{n}$ and $\mathfrak{q} \cap A = \mathfrak{p}$.*

Let K (resp. L) be the field of fractions of A (resp. B), A' the integral closure of A, B' the subring of L generated by A' and B, so that we have a commutative diagram of canonical injections

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[u] \arrow[r] & A' \arrow[u] \end{tikzcd}$$

As B' is integral over B, there exists a prime ideal $\mathfrak{n}'$ of B' such that $\mathfrak{n}' \cap B = \mathfrak{n}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, th. 1), and (for the same reason) $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is surjective. On the other hand, as A is unibranch, A' is a *local* ring, so $\mathfrak{n}' \cap A'$, which is above the maximal ideal $\mathfrak{m}$ of A, is necessarily equal to the unique maximal ideal $\mathfrak{m}'$ of A'. By virtue of the second theorem of Cohen–Seidenberg (*loc. cit.*, § 2, n° 4, th. 3) the morphism $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(A')$ is surjective, so it is the same for the composite $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(A') \to \operatorname{Spec}(A)$; but this morphism is also the composite $\operatorname{Spec}(B'_{\mathfrak{n}'}) \to \operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$, so the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective.

These lemmas being established, let us return to the proof of the implication $b) \Rightarrow a)$ in (14.4.1). By virtue of (14.3.3.1), (i), it suffices to prove that, for every integer $n \geqslant 0$ and every point x' of $X' = X[\mathrm{T}_1, \dots, \mathrm{T}_n]$ above x, the morphism $f' : X' \to Y' = Y[\mathrm{T}_1, \dots, \mathrm{T}_n]$, deduced from f by base change, is open at the point x'. Taking into account the lemma (14.4.1.1), (2.3.4) and (4.2.7), we are thus reduced to proving that f is *open* at the point x: furthermore, it suffices evidently (14.1.2), (iii) to show that the restriction of f to a closed sub-prescheme Z of X having Z as underlying space is open at the point x, so that we can restrict to the case where $X = Z$ is irreducible. Replacing X by an open neighbourhood V of x such that $V \cap X_y$ is irreducible, we may suppose, by virtue of (13.3.1), that the morphism f factorises as $X \xrightarrow{g} Y'' = Y[\mathrm{T}_1, \dots, \mathrm{T}_m] \to Y$, where g is quasi-finite, dominant and locally of finite type. As the structural morphism $Y'' \to Y$ is open (2.4.6), we are reduced to proving that $g : X \to Y''$ is open at the point x. By (14.4.1.1), $g(x)$ is a geometrically unibranch point of Y''. We are thus reduced to proving the following lemma: IV-3 | 211

Lemma (14.4.1.3). — *Let X, Y be two irreducible preschemes, $f : X \to Y$ a morphism locally quasi-finite and dominant. If $x \in X$ is such that $y = f(x)$ is unibranch over Y, then f is open at the point x.*

It suffices to prove that $f(\operatorname{Spec}(\mathscr{O}_{X,x})) = \operatorname{Spec}(\mathscr{O}_{Y,y})$ (1.10.3). One can thus restrict, by the base change $\operatorname{Spec}(\mathscr{O}_{X,y}) \to Y$, to the case where $Y = \operatorname{Spec}(A)$, where A is a local ring and y is the closed point of Y (taking into account (**I**, 3.6.5) and (**0**, 1.2.1.8)), which proves that $X \times_Y \operatorname{Spec}(\mathscr{O}_{Y,y})$ is irreducible; replacing f by f_{red}, we may suppose X and Y reduced, hence integral. Replacing X and Y eventually by affine neighbourhoods of x and y respectively, we may suppose (8.12.9) that the morphism f factorises as $X \xrightarrow{j} X_1 \xrightarrow{g} Y$, where j is an *open immersion* and g a *finite* morphism (evidently dominant); as X and X_1 are affine, j is affine, hence separated and quasi-compact, and consequently f factorises as $X \xrightarrow{u} X_2 \xrightarrow{h} X_1$, where X_2 is the closed image of X by j, h the canonical injection and u an open immersion (**I**, 9.5.3). In other terms, one can suppose that $X_1 = \operatorname{Spec}(B)$, or even that B is an A-algebra integral and finite, containing A and corresponding to the point x; the hypothesis that A is unibranch then implies (14.4.1.2) that the morphism $\operatorname{Spec}(B_{\mathfrak{n}}) \to \operatorname{Spec}(A)$ is surjective.

Corollary (14.4.2). — *Let $f : X \to Y$ be a morphism locally of finite type, y a geometrically unibranch point of Y, η the generic point of the unique irreducible component Y_0 of Y containing y. The following conditions are equivalent:*

label=a) *f is universally open at all points of X_y (or, what amounts to the same, at the maximal points of X_y* (14.3.3.1), *(ii)).*

lbbel=b) *For every $x \in X_y$, there exists an irreducible component Z of X containing x and equidimensional over Y at the point x* (13.2.2).

b′) *For every $x \in X_y$ and every open neighbourhood U of x in X, we have*

$$\dim(U \cap X_{y'}) \geqslant \dim_x(U \cap X_y).$$

b″) *For every open set U of X, we have* $\dim(U \cap X_{y'}) \geqslant \dim(U \cap X_y)$.

When moreover Y is locally Noetherian, these conditions are also equivalent to the following: IV-3 | 212

c) *f is open at all points of X_y (or, what amounts to the same (14.3.3.1), (ii), at the maximal points of X_y).*

The equivalence of a) *and* c) *when Y is locally Noetherian results from (14.4.1); the conditions* b) *or* b′), *applied to the* maximal *points of X_y, imply* a) *by virtue also of (14.3.3.1), (iii); finally,* b′) *and* b″) *are equivalent, since*

$$\dim(U \cap X_y) = \sup_{x \in X_y} (\dim_x(U \cap X_y)).$$

It remains to see that condition a) *implies* b) *and* b′) *at every point $x \in X_y$. Let $d = \dim_x(X_y)$, and let x' be the generic point of an irreducible component of X_y containing x and of dimension d. By virtue of* a) *and of (14.4.1), there is an irreducible component Z of X containing x' and equidimensional over Y at the point x', hence such that $\dim_{x'}(Z \cap X_y) = \dim(Z \cap X_\eta)$. But by construction $\dim_{x'}(Z \cap X_y) = \dim_x(X_y) = d$, and $\dim_x(Z \cap X_y) \leqslant \dim(Z \cap X_y) = d$; taking into account (13.1.6), this proves that Z is equidimensional over Y at the point x; hence* a) *implies* b). *Moreover, we have $\dim(X_\eta) \geqslant \dim(Z \cap X_\eta) = d = \dim_x(X_y)$. Replacing X by an open neighbourhood U of x, we thus see that* a) *implies* b′).

Corollary (14.4.3). — *Let Y be a prescheme geometrically unibranch, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

*label=*a) *f is universally open.*
*lbbel=*b) *For every open set U of X, every $y \in Y$ and every generization y' of y, we have*

$$\dim(U \cap X_{y'}) \geqslant \dim(U \cap X_y).$$

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

c) *f is open.*

Corollary (14.4.4). — *(Chevalley's criterion). Let $f : X \to Y$ be a morphism locally of finite type.*

label=(i) If f is equidimensional at a point $x \in X$ (13.3.2) and if $y = f(x)$ is a geometrically unibranch point of Y, f is universally open at the point x.
lbbel=(ii) If Y is geometrically unibranch, f is universally open at all the points of X where f is equidimensional, and the set of these points is open in X. In particular, if f is equidimensional, it is universally open.

Assertion (ii) is a trivial consequence of (i), since we already know that the set of points where f is equidimensional is open (13.3.2). As to assertion (i), it follows from the fact that the hypothesis implies that condition *b)* of (14.4.1) is satisfied at the generic point of every irreducible component of X_y containing x, taking into account (13.3.1); it therefore suffices to apply (14.4.1).

Remark (14.4.5). — One can prove that if Y is locally Noetherian, and if all the generizations of $y = f(x)$ are geometrically unibranch points of Y (cf. (6.15.2)), then, if f is equidimensional at the point x, it is universally open *in a neighbourhood of x*.

Corollary (14.4.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. Let y be a geometrically unibranch point of Y, and suppose furthermore that for every $x \in f^{-1}(y)$, the ring $\mathscr{O}_{X,x}$ is equidimensional. Then the following conditions are equivalent:* **IV-3 | 213**

*label=*a) *f is equidimensional (13.3.2) at all points of $f^{-1}(y)$.*
*lbbel=*b) *f is open at all points of $f^{-1}(y)$.*
*lcbel=*c) *f is universally open at all points of $f^{-1}(y)$.*

Indeed, *c)* implies trivially *b)*, and *a)* implies *c)* by virtue of (14.4.4); finally, *b)* implies *a)* by (14.2.2). More generally:

Proposition (14.4.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, such that $y = f(x)$ is geometrically unibranch. The following conditions are equivalent:*

*label=*a) *f is equidimensional (13.3.2) at the point x.*
*lbbel=*b) *The ring $\mathscr{O}_x$ is equidimensional and f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (and hence also at every point of such a component).*
*lcbel=*c) *The ring $\mathscr{O}_x$ is equidimensional, and f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (and hence also at every point of such a component).*

Furthermore, when these conditions are satisfied, for every reduced closed sub-prescheme X_i of X having as underlying space an irreducible component of X containing x, the restriction $f_i : X_i \to Y$ of f is an equidimensional morphism at the point x, and universally open at all the points of the irreducible components of $f^{-1}(y) \cap X_i$ which contain x.

Condition *a)* implies that f is equidimensional at all the generic points of the irreducible components of $f^{-1}(y)$ containing x (13.3.1), and consequently (14.4.4) universally open at these points; this proves the last assertion of the proposition, taking into account (14.3.3)) applied to each f_i. Furthermore, by (14.2.1), we have the relations

(14.4.7.1) $$\dim(\mathscr{O}_{X_i,x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{X_y,x} \otimes_{\mathscr{O}_y} \kappa(y))$$

(14.4.7.2) $$\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_y} \kappa(y))$$

and since f is equidimensional at the point x, it follows from (13.3.1) that $\dim_x(X_i \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$, hence (5.2.3) $\dim(\mathscr{O}_{X_i,x} \otimes_{\mathscr{O}_y} \kappa(y)) = \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_y} \kappa(y))$. We conclude that $\dim(\mathscr{O}_{X_i,x}) = \dim(\mathscr{O}_{X,x})$ for every i, in other words $\mathscr{O}_{X,x}$ is equidimensional, and this finishes showing that *a)* implies *c)*. It is clear that *c)* implies *b)*; finally, *b)* implies the relation (14.4.7.2) by virtue of (14.2.1); it then follows from (13.3.6) that *b)* implies *a)*.

Proposition (14.4.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, y a point of Y, x a maximal point of $X_y = f^{-1}(y)$. The following conditions are equivalent:*

*label=*a) *The morphism f is universally open at x, in other words, for every change of base $g : Y' \to Y$, the property $\mathbf{P}(Y')$ holds: for every point x' of $X' = X \times_Y Y'$ above x, the morphism $f' = f_{(Y')} : X' \to Y'$ is open at x'.* **IV-3** | 214

a$'$) *The property $\mathbf{P}(Y')$ holds for every finite morphism $g : Y' \to Y$.*

a$''$) *The property $\mathbf{P}(Y'')$ holds for the normalisation Y'' of Y_{red}* (**II**, 6.3.8).

*lbbel=*b) *For every point x'' of $X'' = X \times_Y Y''$ above x, there exists an irreducible component Z'' of X'' containing x'' and equidimensional over Y'' at the point x''.*

It is trivial that *a)* implies *a$'$)*. To show that *a$'$)* implies *a$''$)*, note that we can write $Y'' = \mathrm{Spec}(\mathscr{B})$, where $\mathscr{B}$ is a quasi-coherent $\mathscr{O}_Y$-Algebra integral over $\mathscr{O}_Y$; as Y is Noetherian, $\mathscr{B}$ is the inductive limit of its sub-$\mathscr{O}_Y$-Algebras $\mathscr{B}_\lambda$, which are quasi-coherent and of finite type (**I**, 9.6.6); but then the $\mathscr{B}_\lambda$ are *finite* $\mathscr{O}_Y$-Algebras (**II**, 6.1.2); we can therefore write $Y'' = \varprojlim_\lambda Y'_\lambda$, where $Y'_\lambda = \mathrm{Spec}(\mathscr{B}_\lambda)$, with $X'_\lambda = X \times_Y Y'_\lambda$. By virtue of *a$'$)*, the morphisms $f'_\lambda : X'_\lambda \to Y'_\lambda$ are open at all the points of X'' above x; we conclude that f'' is open at all the points of X'' above x, by (8.10.1) and (14.3.3.1), (i).

As the prescheme Y'' is *normal* by definition, the fact that *b)* applied to the equidimensional irreducible component of the statement and to the restriction of f'' to this component implies *a$''$)* and implies that *b)* is satisfied. It remains therefore to show that *a$''$)* implies *a)* and *b)*. Taking into account (1.10.3), one may restrict to the case where $Y = \mathrm{Spec}(\mathscr{O}_y)$, noting that the canonical morphism $\mathrm{Spec}(\mathscr{O}_y) \to Y$ is universally bicontinuous (**I**, 3.6.5), and on the other hand that $Y'' \times_Y \mathrm{Spec}(\mathscr{O}_y)$ is the normalisation of $(\mathrm{Spec}(\mathscr{O}_y))_{\mathrm{red}}$ as follows from the commutativity of the operations of integral closure and localisation (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 5, prop. 16). Suppose therefore that $Y = \mathrm{Spec}(A)$, where A is a local Noetherian ring, and y the closed point of Y; we know (**0**, 23.2.5) that there exists a factorisation

$$Y'' \xrightarrow{u} Y_1 \xrightarrow{v} Y$$

of the structural morphism, such that v is a *finite* morphism *surjective*, u an integral, *radicial* and dominant morphism; therefore surjective (**II**, 6.1.10) and consequently a universal *homeomorphism* (universal) (2.4.5). If we set $X_1 = X \times_Y Y_1$, the projection $X'' \to X_1$ is therefore a homeomorphism, and the hypothesis *a$''$)* implies that $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is open at all points of X_1 above x. Furthermore, this shows that in order to prove property *b)*, it suffices to prove the same property where we replace Y'', X'' and x'' by Y_1, X_1 and a point x_1 of X_1 above x. But Y_1 is Noetherian and furthermore geometrically unibranch since Y'' is normal and it is radicial (6.15.1); the property *b)* therefore results from (14.4.1). It remains to show that f is universally open at the point x, which will result from the following lemma:

Lemma (14.4.8.1). — *Let $v : Y_1 \to Y$ be a closed (resp. universally closed) and surjective morphism. For a morphism $f : X \to Y$ to be open (resp. universally open) at a point $x \in X$, it suffices that $f_1 = f_{(Y_1)} : X_1 = X \times_Y Y_1 \to Y_1$ be open (resp. universally open) at all the points of X_1 above x.*

The second assertion results trivially from the first and from the fact that for every base change $Y' \to Y$, the morphism $v_{(Y')} : Y_1 \times_Y Y' \to Y'$ is still surjective and universally closed. To prove the first assertion, consider an open neighbourhood U of x in X; as v is closed and surjective, for $f(U)$ to be an open neighbourhood of $y = f(x)$, it is necessary and sufficient that $v^{-1}(f(U))$ be a neighbourhood of $v^{-1}(y)$. But if $p : X_1 \to X$ is the canonical projection, we have $v^{-1}(f(U)) = f_1(p^{-1}(U))$ (**I**, 3.4.8), and the hypothesis implies that $f_1(p^{-1}(U))$ is a neighbourhood of $f_1(p^{-1}(x)) = v^{-1}(y)$ (**I**, 3.4.8). **IV-3** | 215

Corollary (14.4.9). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

- label=a) *f is universally open, in other words, for every base change $Y' \to Y$, the morphism $f_{(Y')} : X \times_Y Y' \to Y'$ is open.*
- a′) *For every finite morphism $Y_1 \to Y$, $f_{(Y_1)}$ is open.*
- a″) *If Y'' is the normalisation of Y_{red}, $f_{(Y'')}$ is open.*
- lbbel=b) *For every point x'' of $X'' = X \times_Y Y''$, there exists an irreducible component Z'' of X'' containing x'' and equidimensional over Y'' at the point x'' (cf. (14.4.10), (ii)).*

This results immediately from (14.4.8) and (14.1.4).

Remarks (14.4.10). — (i) The equivalence of conditions *a)* and *b)* in (14.4.8) (resp. (14.4.9)) is also valid for any prescheme Y and any morphism f locally of finite type. Indeed, *a)* implies *b)* by virtue of (14.4.1); conversely, *b)* implies that f'' is universally open at the points of X'' above x (resp. at every point of X'') by virtue of (14.4.1), and we conclude property *a)* by applying the lemma (14.4.8.1) to the integral and surjective morphism $Y'' \to Y$.

It may be that, in (14.4.1), for the equivalence of *a)* and *c)*, the supplementary hypothesis that Y is Noetherian is superfluous (cf. (14.3.14)). If so, the Noetherian hypotheses are also superfluous in (14.4.2), (14.4.3), (14.4.8) and (14.4.9).

(ii) One can give examples of morphisms $f : X \to Y$ with the following properties: Y is Noetherian, regular and of dimension 2, f is *universally open* and of finite type, X has two irreducible components X_1, X_2, but the restriction $X_i \hookrightarrow Y$ of f to one of them is *not a universally open morphism*. The principle of the construction relies on the general method of "recollement" which will be presented in chap. V, and which can therefore only be sketched here. Starting from a closed point y of Y, and considering the Y-scheme Y_1 obtained by blowing up y (**II**, 8.1.3); if $f_1 : Y_1 \to Y$ is the structural morphism, we know that the restriction of f_1 to $f_1^{-1}(Y - \{y\})$ is an *isomorphism* on $Y - \{y\}$ (*loc. cit.*), while the fibre $f_1^{-1}(y)$ is isomorphic to $\mathrm{Proj}(S)$, where $S = \bigoplus_{k=0}^{\infty} \mathfrak{m}_y^k / \mathfrak{m}_y^{k+1}$ (**II**, 3.5.3), i.e. here $\mathbf{P}^1_{\kappa(y)}$: it follows from (14.4.1) that f_1 is not open at the generic point of $f_1^{-1}(y)$. On the other hand, let us set $Y_2 = \mathbf{P}^1_Y$, and let $f_2 : Y_2 \to Y$ be the structural morphism; since f_2 is flat, hence universally open (2.4.6); it suffices then to "glue" Y_1 and Y_2 along the fibres isomorphic to $\mathbf{P}^1_{\kappa(y)}$, $f_1^{-1}(y)$ and $f_2^{-1}(y)$, this gives a morphism $f : X \to Y$ where the irreducible components of X identify with f_1 and f_2, and the restrictions of f to these components are f_1 and f_2.

Let us recall however (12.1.1.5) that if Y is locally Noetherian, $f : X \to Y$ is of finite type and *flat*, then every irreducible component of X is equidimensional over Y (and consequently the restriction of f to such a component is *universally open* if all the points of Y are *geometrically unibranch*).

(iii) Recall (12.1.2), (i) that there are morphisms $f : X \to Y$ with the following properties: Y is Noetherian (not geometrically unibranch), f is finite and flat (and even étale (17.6.3)), but the restriction of f to an irreducible component of X is *not a universally open morphism* (although f itself is so by (2.4.6)).

(iv) The Chevalley criterion (14.4.4) explains the importance of the notion of universally open morphism. This notion allows in effect, in numerous results, more or less classical, to replace a hypothesis of *normality* by the hypothesis that a certain morphism is universally open; the more general statement thus obtained will in particular apply to *flat* morphisms (2.4.6), whose importance **IV-3** | 216 in algebraic geometry is well known. One can consider that the statements involving the hypothesis that a morphism is universally open are common generalisations of statements involving a hypothesis of normality and of statements involving a hypothesis of flatness.

14.5. Universally open morphisms and quasi-sections

Lemma (14.5.1). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, such that f is equidimensional (13.3.2) at the point x; set $y = f(x)$, $e = \dim_x(f^{-1}(y))$. Let X' be a closed irreducible subspace of X containing x, n an integer such that* $\operatorname{codim}(X', X) \leqslant n$ *and* $\dim_x(X' \cap f^{-1}(y)) \leqslant e - n$. *Then necessarily* $\operatorname{codim}(X', X) = n$, $\dim_x(X' \cap f^{-1}(y)) = e - n$, *and the restriction $X' \to Y$ of f is an equidimensional morphism at x (and a fortiori dominant).*

The question being local on X, we can suppose that f is of finite type and *equidimensional*, so that for *every* $z \in Y$, *all* the irreducible components of $f^{-1}(z)$ are of dimension e (13.1.1), a''). Let x' be the generic point of X', $y' = f(x')$, $Y' = \overline{\{y'\}} = \overline{f(X')}$ and set $Z = f^{-1}(Y')$. By virtue of (**0**, 14.2.2), we have

$$\operatorname{codim}(X', Z) \leqslant n \tag{14.5.1.1}$$

and if the two sides are equal, we necessarily have $\operatorname{codim}(X', X) = n$ and Z contains an irreducible component of X, which implies (13.3.1) that $f(X')$ is dense in Y and hence $Z = X$.

On the other hand, reasoning in the reduced preschemes of Y and X having Y' and Z respectively as underlying spaces, we deduce from (5.1.2) and from (**I**, 3.6.5) that

$$\operatorname{codim}(X' \cap f^{-1}(y'), f^{-1}(y')) = \operatorname{codim}(X', Z). \tag{14.5.1.2}$$

By hypothesis, $f^{-1}(y')$ is biequidimensional (5.2.1) and of dimension e, hence (**0**, 14.3.5), by virtue of (14.5.1.2) and (14.5.1.1)

$$e' = \dim(X' \cap f^{-1}(y')) = e - \operatorname{codim}(X', Z) \geqslant e - n \tag{14.5.1.3}$$

the equality holding if and only if $\operatorname{codim}(X', X) = n$ and if $f(X')$ is dense in Y. Finally, (13.1.1), we have $\dim_x(X' \cap f^{-1}(y)) \geqslant e'$, whence, by (14.5.1.3),

$$\dim_x(X' \cap f^{-1}(y)) \geqslant e' \geqslant e - n. \tag{14.5.1.4}$$

Or, by hypothesis, we also have $\dim_x(X' \cap f^{-1}(y)) \leqslant e - n$, whence the conclusions of the proposition.

Corollary (14.5.2). — *The hypotheses on f, X, Y, x being those of (14.5.1), suppose moreover that x is not a maximal point of $f^{-1}(y)$. Then there exists an open neighbourhood U of x in X, an affine open set U of x in X, and a section $g \in \Gamma(U, \mathscr{O}_X)$ such that the set X' of $x' \in U$ such that $g(x') = 0$ contains x and does not contain any maximal point of $f^{-1}(y)$. For every g having these properties, X' is equidimensional over Y at the point x, and we have*

$$\dim_x(X' \cap f^{-1}(y)) = e - 1 \qquad \textit{and} \qquad \operatorname{codim}(X', X) = 1. \tag{14.5.2.1}$$

One can restrict to the case where $X = U$ is an open affine neighbourhood of x such that all the irreducible components of $f^{-1}(y)$ contain x. These components correspond to the minimal prime ideals of $\mathscr{O}_x/\mathfrak{m}_y\mathscr{O}_x$, and by hypothesis these ideals are distinct from $\mathfrak{m}_x/\mathfrak{m}_y\mathscr{O}_x$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2); to obtain a $g \in \Gamma(U, \mathscr{O}_X)$ satisfying the conditions of the statement, it suffices to take $g \in \mathfrak{m}_x$ such that the image of g in $\mathfrak{m}_x$ does not belong to any of the preceding prime ideals. Moreover, as $\operatorname{codim}(X', X) \leqslant 1$ (5.1.8), and as X' does not contain any of the irreducible components of $f^{-1}(y)$ of dimension e, and since these are all the irreducible components of $f^{-1}(y)$, we have (**0**, 14.2.2.2) $\dim_x(X' \cap f^{-1}(y)) \leqslant e - 1$. It then suffices to apply (14.5.1). **IV-3** | 217

Proposition (14.5.3). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X. Suppose that f is equidimensional at the point x and that x is closed in $f^{-1}(f(x))$. Then there exists a closed irreducible subspace X' of X, locally closed in X, containing x, and such that the restriction $X' \to Y$ of f (where X' is the sub-prescheme reduced of X having X' as underlying space) is a* quasi-finite and dominant *morphism.*

Indeed, with the notation of (14.5.2), the hypothesis that x is closed in $f^{-1}(y)$ implies that $e - 1 \geqslant 1$. It therefore suffices to apply (14.5.2) by descending induction on $e = \dim_x(f^{-1}(f(x)))$ until one arrives at $e = 1$; the application of (14.5.2) in this last case gives a subspace X' such that $X' \cap f^{-1}(f(x))$ is Noetherian and of dimension 0, hence finite and discrete; as X' is equidimensional over Y, $X' \cap f^{-1}(f(x'))$ is of dimension 0 for every $x' \in X'$, which implies that the restriction $X' \to Y$ of f is a quasi-finite morphism (**II**, 6.2.2).

Corollary (14.5.4). — *Under the hypotheses of (14.5.3), suppose moreover that $Y = \operatorname{Spec}(A)$, where A is a local Noetherian ring, integral and complete, and that $y = f(x)$ is the unique closed point of Y. Then there exists a local integral ring A', containing A, which is a finite A-algebra, and such that, if we set $Y' = \operatorname{Spec}(A')$ and $X' = X \times_Y Y'$, there exists a Y'-section $h : Y' \to X'$ such that the composite morphism $Y' \xrightarrow{h} X' \to X$ is an immersion whose image contains x.*

Replacing if necessary X by a reduced closed irreducible sub-prescheme of X, we may, by virtue of (14.5.3), restrict to the case where the morphism f is already quasi-finite and dominant. Using (**II**, 6.2.5), we deduce that $\mathscr{O}_x = A'$ is an integral ring which is a finite A-algebra, and that X is a disjoint sum of the closed sub-prescheme $Y' = \operatorname{Spec}(A')$ and of a sub-prescheme Y''; the scheme Y' satisfies the question, the composite morphism $Y' \to X' \to X$ being nothing other than the canonical morphism $\operatorname{Spec}(\mathscr{O}_x) \to X$.

Remark (14.5.5). — If, in the statement of (14.5.4), one does not require that the morphism $Y' \to X$ be an immersion, one can suppose that A' is moreover *integrally closed*: it suffices in effect to replace A' by its *integral closure* A_1, since we know (**0**, 23.1.5) that A_1 is an A-module of finite type.

Proposition (14.5.6). — *Let Y be a locally Noetherian prescheme, irreducible, regular and of dimension* 1*, $f : X \to Y$ a morphism locally of finite type, y a point of Y. The following conditions are equivalent:*

*label=*a) *f_{red} is flat at every point of $f^{-1}(y)$.* **IV-3** | 218
*lbbel=*b) *f is universally open in a neighbourhood of $f^{-1}(y)$.*
*lcbel=*c) *f is open in a neighbourhood of $f^{-1}(y)$.*
*ldbel=*d) *Every irreducible component of X meeting $f^{-1}(y)$ dominates Y.*
*lebel=*e) *For every point $x \in f^{-1}(y)$, closed in $f^{-1}(y)$, and every irreducible component X_i of X containing x, there exists an irreducible closed subspace X' of X, locally closed in X, containing x, contained in X_i, and such that the restriction $X' \to Y$ of f is a quasi-finite and dominant morphism.*

The equivalence of *a)*, *b)*, *c)* and *d)* has already been shown (14.3.8). It is clear that *e)* implies *d)*, since for every irreducible component X_i of X meeting $f^{-1}(y)$, there exists in $X_i \cap f^{-1}(y)$ a closed point (5.1.11). It suffices to prove that *c)* implies *e)*: consider a closed point x of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x. As the restriction $f_i : X_i \to Y$ of f to X_i is a dominant morphism, it follows from *c)* and *d)* for f_i that this morphism is open at the generic points of $X_i \cap f^{-1}(y)$, hence from (14.4.2) that f_i is equidimensional at the point x, and we conclude with the aid of (14.5.3).

Remark (14.5.7). — If, in the statement of (14.5.5), we suppose that $Y = \operatorname{Spec}(A)$, where A is a complete discrete valuation ring, and that y is the closed point of Y, we can moreover suppose that $X' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring which is a finite A-algebra, as the proof of (14.5.4) shows, and the fact that a local integral ring, regular and of dimension 1 is a discrete valuation ring (**II**, 7.1.6).

Proposition (14.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, y a point of Y. For every Y-prescheme Y', set $X(Y') = \operatorname{Hom}_Y(Y', X)$. Let x be a point of $f^{-1}(y)$; in order that f be universally open at x, it is necessary and sufficient that the following condition be satisfied:*

For every complete discrete valuation ring A, of algebraically closed residue field, every morphism $g : Y' = \operatorname{Spec}(A) \to Y$ such that the image by g of the closed point y' of Y' is equal to y, and every element $u_0 \in X(\operatorname{Spec}(\kappa(y')))$ such that $u_0(y') = x$, there exists a discrete valuation ring B, a finite A-algebra, and, *if we set $Z' = \operatorname{Spec}(B)$, an element $u \in X(Z')$ such that, if z' is the closed point of Z', the diagram*

$$\begin{tikzcd} \operatorname{Spec}(\kappa(z')) \arrow[r] \arrow[d] & Z' \arrow[d, "u"] \\ \operatorname{Spec}(\kappa(y')) \arrow[r, "u_0"] & X \end{tikzcd}$$

is commutative.

Let us note that if A and B satisfy the conditions of the statement, B is a complete discrete valuation ring (Bourbaki, *Alg. comm.*, chap. III, § 3, n° 3, prop. 7 and chap. IV, § 2, n° 2, cor. 3 of the prop. 9) of residue field isomorphic to that of A, hence algebraically closed; and since $u(z') = x$, it is in $X'' = X \times_Y Z'$ a point x'', whose projections in X and Z' are x and z' and which is rational over $\kappa(z')$; furthermore, since u is a Z'-section v of X'' such that $v(z') = x''$, the image by v of the generic point s of Z' is s; applying (14.3.6), we see that the condition of the statement is *sufficient*. Let us now prove that it is *necessary*. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. By hypothesis there is a point $x' \in X'$ above x and y' and rational over $\kappa(y')$ (**I**, 3.3.14), hence closed in $f'^{-1}(y')$. By virtue of (14.3.6), there is an irreducible component T of X' containing x' and dominant over Y', hence (14.3.8) and (14.3.13) the restriction $T \to Y'$ of f' is equidimensional at the point x. We deduce from (14.5.5) that there is a finite A-algebra B which is an integral local ring, integrally closed, and a Z'-section $v : Z' \to X''$ such that the image of v is an immersion, and setting $Z' = \operatorname{Spec}(B)$ and $X'' = X \times_Y Z' = X' \times_{Y'} Z'$, a Z'-section $v : Z' \to X''$ such that the image by v of the closed point z' of Z' is x'; the composite morphism $u : Z' \xrightarrow{v} X'' \to X'$ gives the answer to the question. **IV-3 | 219**

Proposition (14.5.9). — *Let Y be a locally Noetherian prescheme, irreducible, $f : X \to Y$ a morphism locally of finite type, y a point geometrically unibranch of Y. Then the following conditions are equivalent:*

label=a) *f is universally open at every point of $f^{-1}(y)$.*
lbbel=b) *f is open at every point of $f^{-1}(y)$.*
lcbel=c) *For every irreducible component Z of $f^{-1}(y)$, of generic point z, there exists a component of X_i of X, containing z and equidimensional over Y at the point z.*
ldbel=d) *For every closed point x of $f^{-1}(y)$, there exists an irreducible closed subspace X' of X, locally closed in X, containing x, and such that the restriction $X' \to Y$ of f is a quasi-finite and dominant morphism.*

The equivalence of *a)*, *b)* and *c)* has already been shown (14.4.2). To prove that *a)* implies *d)*, note that by (14.4.2), *a)* implies that there exists an irreducible component Z of X containing x and equidimensional over Y at the point x; the existence of X' then comes from (14.5.3) applied to the restriction $Z \to Y$ of f. Conversely, suppose *d)* is satisfied; by the Chevalley criterion (14.4.4), the restriction $X' \to Y$ of f is a universally open morphism at the point x, and *a fortiori* f is open at all the closed points of $X_y = f^{-1}(y)$. But X_y is a $\kappa(y)$-prescheme locally of finite type, hence a Jacobson prescheme; the set of closed points of X_y is therefore *dense* in X_y (10.3.1), and it follows from (14.1.4) that f is open at *all* points of X_y, which finishes proving that *d)* implies *b)*.

The following result was brought to light by D. Mumford:

Proposition (14.5.10). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a morphism universally open, surjective and locally of finite type. Then there exists a finite surjective morphism $g : Y' \to Y$ such that, if we set $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \to Y'$, every point $y' \in Y'$ admits an open neighbourhood U' such that there exists a U'-section of $f'^{-1}(U')$.* **IV-3 | 220**

We will prove the proposition in several steps.

I) Reduction to the case where Y is integral. If we have proved the proposition for each of the reduced sub-preschemes Y_i having as underlying space an irreducible component of Y, and for the inverse images $f^{-1}(Y_i)$, it is clear that the prescheme Y' which is the *sum* of the corresponding Y'_i will answer the question. We can therefore suppose Y *integral* and what follows will be limited to this case. Furthermore, in the conclusion, we will also be able to take Y' integral (up to replacing it by a suitable irreducible component).

II) Local character on Y. We will show that if one can cover Y by open sets U_j in finite number, such that, for every j, the conclusion of the proposition holds for the morphism $f^{-1}(U_j) \to U_j$, restriction of f, then the conclusion holds also for f. Indeed, we can evidently suppose the U_j affine, so that $U_j = \operatorname{Spec}(A_j)$, where A_j is an integral Noetherian ring whose field of fractions $K = R(Y)$ is the field of rational functions on Y. For every j, by hypothesis there is a finite integral A_j-algebra A'_j such that the homomorphism $A_j \to A'_j$ is injective (**I**, 1.2.7) and that the corresponding morphism $g_j : U'_j = \operatorname{Spec}(A'_j) \to \operatorname{Spec}(A_j) = U_j$ satisfies the conditions of the statement (for U_j and $f^{-1}(U_j)$). Let K' be a finite extension of K containing the fields of fractions of all the A'_j (which are finite extensions of K). Applying once more (**0**, 23.2.5) and (6.15.5), the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, and one can therefore evidently replace Y by $\operatorname{Spec}(B)$ and X by $X \otimes_A B$ in order to prove the proposition; the reduction III) permits then even to suppose from the outset that A is local, integral and $Y = \operatorname{Spec}(A)$ geometrically unibranch.

III) Reduction to the case where Y is integral, local and geometrically unibranch. Suppose first that the proposition has been proved when Y is integral and *local* (with Y' integral), and show that it is valid when Y is integral (Noetherian) affine. Indeed, by virtue of the reduction II), it suffices to prove that for every point $y \in Y$, the proposition holds for an open affine neighbourhood V of y in Y. Let $Y = \operatorname{Spec}(A)$, $\mathfrak{p} = \mathfrak{j}_y$, and set $X_1 = X \times_Y Y_1$, where $Y_1 = \operatorname{Spec}(A_\mathfrak{p})$; by hypothesis a finite surjective morphism $g_1 : Y'_1 = \operatorname{Spec}(B_1) \to Y_1$ exists, where $Y'_1 = \operatorname{Spec}(B_1)$ and $g' = g_{(Y_1)} : Y'_1 \to Y_1$, every point of Y'_1 admits an open neighbourhood U_1 such that there exists a U_1-section of $X \times_Y U'_1$. If y'_j ($1 \leqslant j \leqslant r$) are the closed points of Y'_1, there is therefore a covering of Y'_1 by open sets U'_j and a U'_j-section h'_j of $X \times_Y U'_j$ ($1 \leqslant j \leqslant r$). The $A_\mathfrak{p}$-module B_1 admits a finite system of generators of the form z_i/s (with $s \in A - \mathfrak{p}$, z_i integral over A), which one may suppose (multiplying as needed s by an element of the field of fractions of B_1, integral over A, as an element of A_s), so that if V is the open affine set $D(s) = \operatorname{Spec}(A_s) \subset Y$, Y'_1 identifies with $Y_1 \times_Y V'$, where V' is the spectrum of the finite A_s-algebra generated by the z_i/s; $g : V' \to V$ is therefore a finite surjective morphism and $g_1 = g_{(Y_1)}$. Moreover, applying the method of (8.1.2), *a*), one can suppose that each of the U'_j is the inverse image by $p : Y'_1 \to Y'_1$ of an open set W'_j of V', and that each of the sections h'_j is of the form $(v'_j)_{(Y_1)}$, where v'_j is a W'_j-section of $X \times_Y W'_j$ (8.8.2), (i). It is therefore clear that the proposition holds when $Y = \operatorname{Spec}(A)$, A being a local integral Noetherian ring. We then know that there is a finite A-algebra B, with $A \subset B$ and such that $\operatorname{Spec}(B)$ is geometrically unibranch (**0**, 23.2.5) and (6.15.5); as the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, one may therefore evidently replace Y by $\operatorname{Spec}(B)$ and X by $X \otimes_A B$ to prove the proposition; the reduction III) therefore allows even to suppose from the outset that A is local, integral and $Y = \operatorname{Spec}(A)$ geometrically unibranch. **IV-3** | 221

IV) Reduction to the case where X is integral, affine, and f quasi-finite, surjective, birational and universally open. Suppose therefore $Y = \operatorname{Spec}(A)$ integral, local and geometrically unibranch. There exists then a reduced irreducible sub-prescheme X_0 of X such that the restriction $f_0 : X_0 \to Y$ of f is a quasi-finite and *dominant* morphism and that $f_0(X_0)$ contains the closed point y of Y (14.5.9); as Y is geometrically unibranch, it follows from (14.4.1) that f_0 is still *universally open*. Furthermore we may suppose X_0 reduced, hence integral, and we see that we may, by replacing X by X_0, suppose that X is integral and f *quasi-finite and dominant*, and such that $f(X)$ contains y; as f is *open* and surjective.

Let ξ, η be the generic points of X and Y respectively; $\kappa(\xi)$ is therefore a finite extension of $K = \kappa(\eta)$. By (4.6.8), there is a *finite* extension K' of K such that $(\operatorname{Spec}(\kappa(\xi) \otimes_K K'))_{\text{red}}$ is geometrically reduced over K and that its irreducible components are geometrically irreducible; it follows (4.5.9) and (4.6.1) that the residue fields of $\operatorname{Spec}(\kappa(\xi) \otimes_K K')$ are finite extensions, primary and separable of K'. Applying once more (**0**, 23.2.5) and (6.15.5), there is a finite integral A-algebra B, having K' as field of fractions, containing A and such that $\operatorname{Spec}(B)$ is geometrically unibranch; as the morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, one may therefore again replace Y by $\operatorname{Spec}(B)$ and X by $(X \otimes_A B)_{\text{red}}$ in order to prove the proposition; the reduction III) allows then to suppose once more A local, integral and $Y = \operatorname{Spec}(A)$ geometrically unibranch, with closed point y. Furthermore, f is still a quasi-finite, *surjective* and *universally open* morphism; each of the irreducible components X_i of X ($1 \leqslant i \leqslant m$) dominates Y (1.10.4) and is *birational* over Y. So let x be a point of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x; denote also by X_i the reduced sub-prescheme (hence integral) of X having X_i as underlying space, and let $f_i : X_i \to Y$ be the restriction of f. Applying once more (14.4.1) to the quasi-finite and dominant morphism f_i, we see that f_i is universally open; if U is an affine open set of X_i containing x, $f_i(U)$ is therefore open in Y and contains y, hence is equal to Y. One may thus replace X by U in order to prove the proposition. **IV-3** | 222

V) End of the proof. We suppose therefore Y local, integral and geometrically unibranch, X integral and affine, f quasi-finite, surjective, birational and universally open; it follows that f is automatically *separated*. By virtue of the Main theorem (8.12.6), there exists a factorisation $X \xrightarrow{j} Z \xrightarrow{u} Y$, where j is an open immersion and u a *finite* morphism. Replacing moreover Z by the closed image of j (**I**, 9.5.10), we may suppose that Z is *integral*; as u is finite and birational and Y geometrically unibranch, it follows from (**III**, 4.3.5) and (4.3.4) that u (and consequently f) is a *radicial* morphism; it is therefore a universal *homeomorphism* (universal). By (2.4.5), (ii) f is a *finite* morphism. But then we satisfy the conditions of the statement with Y' *integral* by simply taking $Y' = X$ and $g = f$, the Y'-section being the diagonal morphism Δ_f.

This result can be used to develop criteria of "descent" for the properties of universally open and surjective morphisms. Let us point out in particular the following criterion, due to D. Mumford:

Corollary (14.5.11). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $g : Y' \to Y$ a morphism locally of finite type, universally open and surjective; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, for f to be affine, it is necessary and sufficient that f' be so.*

One need only prove that the condition is sufficient. The question being local on Y, we may suppose Y affine, hence Noetherian. By virtue of (14.5.10), there exists a finite surjective morphism $h : Y_1 \to Y$ such that, if we set $Y'_1 = Y' \times_Y Y_1$, and $g' = g_{(Y_1)} : Y'_1 \to Y_1$, every point of Y_1 admits an open neighbourhood U_1 such that there exists a U_1-section of $g^{-1}(U_1)$. If we set $X_1 = X \times_Y Y_1 = X' \times_{Y'} Y'_1$, the canonical projection $p : X_1 \to X$ is a finite surjective morphism, hence, if we prove that X_1 is a scheme, it will follow first of all that X is quasi-compact, hence Noetherian, then that X is affine by virtue of the theorem of Chevalley (**II**, 6.7.1). It suffices therefore to prove that the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is affine. Or, if we set

$$X'_1 = X_1 \times_{Y_1} Y'_1 = X' \times_{Y'} Y'_1 \qquad \text{and} \qquad f'_1 = (f_1)_{(Y_1)} = f'_{(Y'_1)} : X'_1 \longrightarrow Y'_1,$$

f'_1 is affine by virtue of the hypothesis. We are therefore reduced to proving the corollary in y by replacing Y, X, f and Y' by Y_1, X_1, f_1 and Y'_1; in other words, it suffices to prove that f is affine when, in the statement of (14.5.11), the hypothesis that every point of Y admits an open neighbourhood U such that there exists a U-section of $g^{-1}(U)$. Or, we have the following elementary lemma (valid in any category admitting fibre products):

Lemma (14.5.11.1). — *Let $f : X \to S$, $g : Y \to S$ be two morphisms, $p_1 : X \times_S Y \to X$, $p_2 : X \times_S Y \to Y$ the canonical projections. If $s : S \to Y$ is a section of g, then $s' = (1, s \circ f)_S$ and $s' : X \to S$ and $s' : X \to X \times_S Y$, is a section of p_1 and X, equipped with the two morphisms $f : X \to S$ and $s' : X \to X \times_S Y$, identifies with the fibre product of the Y-preschemes S and $X \times_S Y$ for the morphisms $s : S \to Y$ and $p_2 : X \times_S Y \to Y$.*

§15. STUDY OF FIBRES OF A UNIVERSALLY OPEN MORPHISM

15.1. Multiplicities of fibres of a universally open morphism

Proposition (15.1.1). — *Let Y be a locally Noetherian irreducible prescheme, of generic point η, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, y a point of Y, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-Module; we set $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{Y}} \mathbf{k}(y)$ (sheaf of modules on the fibre $f^{-1}(y)$). Let z be the generic point of an irreducible component of $\mathrm{Supp}(\mathscr{F}_y)$, and let X_i ($1 \leqslant i \leqslant n$) be those of the reduced closed sub-preschemes of X whose underlying space is an irreducible component of $\mathrm{Supp}(\mathscr{F})$ containing z, and which are such that the restriction $\mathrm{X}_i \to \mathrm{Y}$ of f is a universally open morphism at the point z; let z_i be the generic point of X_i ($1 \leqslant i \leqslant n$). If $\lambda_z(\mathscr{F}_y)$ (resp. $\lambda_i(\mathscr{F}_\eta)$) denotes the geometric length of $\mathscr{F}_y$ at a point $x \in f^{-1}(y)$ (resp. that of $\mathscr{F}_\eta$ at a point $t \in f^{-1}(\eta)$) (4.7.5), then we have* **IV-3** | 223

(15.1.1.1)
$$\lambda_z(\mathscr{F}_y) \geqslant \sum_{i=1}^{n} \lambda_{z_i}(\mathscr{F}_\eta).$$

If furthermore we suppose the ring $\mathscr{O}_y$ regular, then we have

$$\text{(15.1.1.2)} \qquad \operatorname{length}((\mathscr{F}_y)_z) \geqslant \sum_{i=1}^{n} \operatorname{length}((\mathscr{F}_\eta)_{z_i}).$$

Proof. The fibres $f^{-1}(y')$ of f (for all $y' \in \mathrm{Y}$) do not change when we replace f by f_{red}, so we can **IV-3** | 224
suppose that X and Y are reduced (2.4.3), (vi)), and hence Y integral; we will set $\mathrm{K} = \mathbf{k}(\eta)$. We can furthermore replace f by the restriction $\operatorname{Supp}(\mathscr{F}) \to \mathrm{Y}$ of f to the reduced closed sub-prescheme of X having $\operatorname{Supp}(\mathscr{F})$ as underlying space, and hence restrict to the case where $\mathrm{X} = \operatorname{Supp}(\mathscr{F})$. Finally, if $y = \eta$, there is only a single irreducible component X_i of X containing z (**0**, 2.1.8) and on a $z_i = z$, which renders (15.1.1.1) and (15.1.1.2) trivial (without hypotheses on f). We can thus restrict to the case $y \neq \eta$; it follows then from the hypothesis and from (1.10.3) that the z_i belong to $f^{-1}(\eta)$, so the second members of (15.1.1.1) and (15.1.1.2) are well-defined. Let Z_i be the reduced closed sub-prescheme of $f^{-1}(\eta)$ having $\mathrm{X}_i \cap f^{-1}(\eta)$ as underlying space; we know (4.6.6) that there exists a finite radicial extension K' of K such that, for all i, the K'-prescheme $(\mathrm{Z}_i \otimes_{\mathrm{K}} \mathrm{K}')_{\mathrm{red}}$ is geometrically reduced over K'. Furthermore, the morphism projection $f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}' \to f^{-1}(\eta)$ is finite, dominant, and radicial (**I**, 3.5.7), hence is a homeomorphism (2.4.5). So z_i' is the unique point of $f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}'$ above z_i, and it follows from (4.7.9) and (4.7.5) that we have

$$\text{(15.1.1.3)} \qquad \lambda_{z_i}(\mathscr{F}_\eta) = \lambda_{z_i'}(\mathscr{F}_\eta \otimes_{\mathrm{K}} \mathrm{K}') = \operatorname{length}((\mathscr{F}_\eta \otimes_{\mathrm{K}} \mathrm{K}')_{z_i'}).$$

Let V be a discrete valuation ring, with field of fractions K', dominating $\mathscr{O}_y$ (**II**, 7.1.7); set $\mathrm{Y}' = \operatorname{Spec}(\mathrm{V})$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $\mathscr{F}' = \mathscr{F} \otimes_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$; if $g : \mathrm{Y}' \to \mathrm{Y}$ is the structural morphism, η' the generic point of Y' and y' its closed point, then we have $g(y') = y$, $g(\eta') = \eta$; the morphism $\operatorname{Spec}(\mathbf{k}(y')) \to \operatorname{Spec}(\mathbf{k}(y))$ being faithfully flat, it is the same for the projection $g' : f'^{-1}(y') \to f^{-1}(y)$ (2.2.13), and hence (2.3.4), there is a generic point z' of an irreducible component of $f'^{-1}(y')$ such that $g'(z') = z$. By construction, we have $\mathbf{k}(\eta') = \mathrm{K}'$, hence $f'^{-1}(\eta') = f^{-1}(\eta) \otimes_{\mathrm{K}} \mathrm{K}'$; on the other hand, if we set $\mathrm{X}_i' = \mathrm{X}_i \times_{\mathrm{Y}} \mathrm{Y}'$, the restriction $\mathrm{X}_i' \to \mathrm{Y}'$ of f' is a universally open morphism at the point z', and hence (1.10.3) there exists a point $z_i'' \in f'^{-1}(\eta')$ which is a generization of z', and one can evidently suppose that z_i'' is the generic point of an irreducible component of $\mathrm{X}_i' \cap f'^{-1}(\eta')$; as the projection of $\mathrm{X}_i' \cap f'^{-1}(\eta')$ in $f^{-1}(\eta)$ is Z_i, we have $z_i'' = z_i'$. By virtue of (4.7.9) and (4.7.5), we have

$$\text{(15.1.1.4)} \qquad \lambda_z(\mathscr{F}_y) = \lambda_{z'}(\mathscr{F}_{y'}) \geqslant \operatorname{length}((\mathscr{F}_{y'})_{z'})$$

and it follows from this inequality and from (15.1.1.3) that it suffices, in order to establish (15.1.1.1), to prove the inequality

$$\text{(15.1.1.5)} \qquad \operatorname{length}((\mathscr{F}'_{y'})_z) \geqslant \sum_{i=1}^{n} \operatorname{length}((\mathscr{F}'_{\eta'})_{z_i'}).$$

Let us now consider the second inequality (15.1.1.2); we will use the following lemma:

Lemma (15.1.1.6). — *Let* A *be a regular local ring that is not a field,* K *its field of fractions, k its residue field. There exists a discrete valuation ring* V *dominating* A*, having* K *as field of fractions, and whose residue field is a pure transcendental extension of k.*

Set $\mathrm{S} = \operatorname{Spec}(\mathrm{A})$, and consider the S-scheme P obtained by blowing up the closed point s of S **IV-3** | 225
(**II**, 8.1.3); if $\mathfrak{m}$ is the maximal ideal of A, we have by definition $\mathrm{P} = \operatorname{Proj}(\mathrm{B})$, where B is the A-algebra graded by $\bigoplus_{n \geqslant 0} \mathfrak{m}^n$. If $h : \mathrm{P} \to \mathrm{S}$ is the structural morphism, the fibre $h^{-1}(s)$ is isomorphic to $\operatorname{Proj}(\mathrm{B} \otimes_{\mathrm{A}} k)$ (**II**, 2.8.10), and by definition $\mathrm{B} \otimes_{\mathrm{A}} k$ is the k-algebra graded by $\bigoplus_{n \geqslant 0} \mathfrak{m}^n/\mathfrak{m}^{n+1}$; but as A is regular, this algebra is isomorphic to a polynomial algebra $k[t_1, \ldots, t_e]$ (t_i indeterminates) (**0**, 17.1.1) and it follows that $h^{-1}(s)$ is isomorphic to $\mathbf{P}_k^{e-1}$. Let ζ be the generic point of $h^{-1}(s)$; if t_i' ($1 \leqslant i \leqslant e$) are the elements of $\mathfrak{m}$ whose classes $\bmod \mathfrak{m}^2$ are the t_i, $\mathrm{B}_{(t_j)}$ is the ring of an affine open neighbourhood of ζ in P, and we have $\mathbf{k}(\zeta) = k(t_1, \ldots, t_{e-1})$; on the other hand, as A is integrally closed, one easily verifies that B is as well (Bourbaki, *Alg. comm.*, chap. V, § 1, no 8, cor. 1 of prop. 21), so $\mathrm{B}_{(t_j)}$ is also integrally closed, and hence so is the local ring $\mathscr{O}_{\mathrm{P},\zeta}$. Finally, as $\mathrm{A} = \mathscr{O}_s$ is regular, hence a universally catenary ring (5.6.4), we have, by virtue of (5.6.5), $\dim(\mathscr{O}_{\mathrm{P},\zeta}) = \dim(\mathscr{O}_s) - (e-1)$, since h is a birational morphism, and the local ring of the fibre $h^{-1}(s)$ at its generic point ζ is of

dimension 0. But $\dim(\mathscr{O}_s) = e$, hence $\dim(\mathscr{O}_{\mathrm{P},\zeta}) = 1$, and $\mathscr{O}_{\mathrm{P},\zeta} = \mathrm{V}$ is a discrete valuation ring, Noetherian, of dimension 1 and integrally closed (**II**, 7.1.6); it clearly answers the question.

This lemma being established, one can resume the reasoning for the inequality (15.1.1.1) with $\mathrm{Y}' = \mathrm{Spec}(\mathrm{V})$; this time $\mathbf{k}(y')$ is a *separable* extension of $\mathbf{k}(y)$, and by (4.7.9), we have $\mathrm{length}((\mathscr{F}_y)_z) = \mathrm{length}((\mathscr{F}'_{y'})_z)$; as furthermore $f'^{-1}(\eta') = f^{-1}(\eta)$ and $\mathscr{F}'_{y'} = \mathscr{F}_\eta$, one is again reduced to proving the inequality (15.1.1.5). Now, if t is a uniformiser of $\mathscr{O}_{y'}$, $f'^{-1}(y')$ is the closed sub-prescheme of X' defined by the ideal $t\mathscr{O}_{\mathrm{X}'}$; on the other hand, we have $\mathscr{F}'_{y'} = \mathscr{F}'/t\mathscr{F}'$; by another part, one verifies immediately (**I**, 9.1.12) that $(\mathscr{F}'_\eta)_{z'} = \mathscr{F}'_{z'_i}$; the inequality to prove (15.1.1.5) is thus nothing but a particular case of (3.4.1.1). □

Corollary (15.1.2). — *The hypotheses being those of (15.1.1), suppose furthermore that $\lambda_z(\mathscr{F}_y) = 1$ (resp. that $\mathscr{O}_y$ is regular and $\mathrm{length}((\mathscr{F}_y)_z) = 1$). Then there exists at most one irreducible component X_i of $\mathrm{Supp}(\mathscr{F})$ containing z and such that the restriction $\mathrm{X}_i \to \mathrm{Y}$ of f is universally open at this component, and moreover, if z_i is the generic point of this component, then $\lambda_{z_i}(\mathscr{F}_\eta) = 1$ (resp. $\mathrm{length}((\mathscr{F}_\eta)_{z_i}) = 1$).*

This follows immediately from (15.1.1), noting that one has necessarily, by definition of X_i, $(\mathscr{F}_\eta)_{z_i} \neq 0$ (**I**, 9.1.13).

Corollary (15.1.3). — *Let Y be an irreducible locally Noetherian prescheme of generic point η, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-Module of support X, x a point of X, $y = f(x)$. Suppose that: 1° x belongs to a single irreducible component Z of $f^{-1}(y)$; 2° $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{Y}} \mathbf{k}(y)$ is geometrically reduced on $\mathbf{k}(y)$, in other words, if z is the generic point of Z, $\lambda_z(\mathscr{F}_y) = 1$ (resp. $\mathscr{O}_y$ is regular and $\mathrm{length}((\mathscr{F}_y)_z) = 1$). Then there exists at most one irreducible component X' of X containing x, such that $\dim_x(\mathrm{X}' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ and that the restriction $\mathrm{X}' \to \mathrm{Y}$ of f is universally open at the generic points of the irreducible components of $\mathrm{X}' \cap f^{-1}(y)$ containing x. Moreover, if such a component X' exists and if z' is its generic point, then $\lambda_{z'}(\mathscr{F}_\eta) = 1$ (resp. $\mathrm{length}((\mathscr{F}_\eta)_{z'}) = 1$).*

Indeed, an irreducible component X' of X containing x and such that $\dim_x(\mathrm{X}' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ necessarily contains Z, since an irreducible component of $\mathrm{X}' \cap f^{-1}(y)$ containing x must be of the same dimension as the sole irreducible component Z of $f^{-1}(y)$ containing x. One can then apply (15.1.2) to z. **IV-3** | 226

Remarks (15.1.4). — label=(i) In the statement of (15.1.1), one cannot replace "universally open" by "equidimensional", as the example (14.4.10), (ii)) shows, where we set $\mathscr{F} = \mathscr{O}_\mathrm{X}$; the fibres of f are then reduced Artinian schemes, and hence (with the notations introduced *loc. cit.*) we have $\lambda_{x'}(\mathscr{F}_y) = 1$, but there are two irreducible components X_1, X_2 of X containing x', and the second member of (15.1.1.1) is thus equal to 2, even though the restriction of f to each of the components X_1, X_2 is an equidimensional morphism at every point. This same example also shows that in (15.1.2) and (15.1.3), one cannot replace the hypothesis "universally open" by "equidimensional".

lbbel=(ii) It is plausible that for the validity of the inequality (15.1.1.2), one cannot suppress the hypothesis that $\mathscr{O}_y$ is *regular*.

15.2. Flatness of universally open morphisms with geometrically reduced fibres

(15.2.1). We saw in (2.4.6) that a morphism which is *flat* and locally of finite presentation is *universally open*. We have a partial converse of this result under additional hypotheses:

Theorem (15.2.2). — *Let Y be a locally Noetherian prescheme, $f : \mathrm{X} \to \mathrm{Y}$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_\mathrm{X}$-Module, x a point of X, $y = f(x)$. Suppose the following conditions are satisfied:*

label=(i) $\mathrm{Supp}(\mathscr{F}) = \mathrm{X}$.

lbbel=(ii) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x.

lcbel=(iii) $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_\mathrm{Y}} \mathbf{k}(y)$ is geometrically reduced at the point x (4.6.22).

ldbel=(iv) The ring $\mathscr{O}_y$ is reduced.

Then $\mathscr{F}$ is f-flat at the point x.

Proof. As $\mathscr{O}_y$ is reduced and Noetherian, we will apply the valuative criterion of flatness (11.8.1). Consider thus a local homomorphism $\mathscr{O}_y \to \mathrm{A}'$, where A' is a discrete valuation ring; set $\mathrm{Y}' =$

$\mathrm{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, and let x' be a point of X' whose projections in X and Y' are respectively x and the closed point y' of Y'; if we set $\mathscr{F}' = \mathscr{F} \otimes_Y Y'$, then $\mathrm{Supp}(\mathscr{F}') = X'$ by (**I**, 9.1.13); every generic point z' of an irreducible component of $f'^{-1}(y')$ containing x' has for projection in $f^{-1}(y)$ a generic point z of an irreducible component of $f^{-1}(y)$ containing x, by virtue of (4.2.6); hence f' is universally open at the point z'. Finally, it follows from (4.7.11) that $\mathscr{F}'_{y'}$ is geometrically reduced at the point x'. We see thus that we are reduced to proving the following

IV-3 | 227

Lemma (15.2.2.1). — *Let $Y = \mathrm{Spec}(A)$ be the spectrum of a discrete valuation ring, y its closed point, $f : X \to Y$ a morphism locally of finite type, x a point of $f^{-1}(y)$, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module. Suppose the following conditions are satisfied:*

label=(i) $\mathrm{Supp}(\mathscr{F}) = X$.

lbbel=(ii) f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x.

lcbel=(iii) $\mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_y} \mathbf{k}(y)$ is reduced at the point x (3.2.2).

Then $\mathscr{F}$ is f-flat at the point x.

Let us denote by t a uniformiser of A, set $B = \mathscr{O}_z$ and $\mathscr{F}_z = M$; condition (i) implies that $\mathrm{Supp}(M) = \mathrm{Spec}(B)$ (**I**, 9.1.13). Condition (ii) implies, by virtue of (14.3.2), that every irreducible component of X containing x dominates Y; in other words, the inverse image in A of every minimal prime ideal $\mathfrak{p}_i$ of B is 0, which means that M/tM is a (B/tB)-module *reduced* (3.2.2). It remains to show that under these conditions M is an A-module *without torsion* (**0**, 6.3.4), and for this it suffices to show that t is an M-*regular* element; but this follows from (3.4.6.1). □

Corollary (15.2.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose the following conditions are satisfied:*

label=(i) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x.

lbbel=(ii) $f^{-1}(y)$ is geometrically reduced (on $\mathbf{k}(y)$) at the point x (4.6.9).

lcbel=(iii) The ring $\mathscr{O}_y$ is reduced.

Under these conditions f is flat at the point x. It suffices to apply (15.2.2) with $\mathscr{F} = \mathscr{O}_X$ (4.6.22).

Remarks (15.2.4). — label=(i) The first three conditions of (15.2.2) do not change when we replace f by f_{red}; but if $\mathfrak{N}$ is the nilradical of a Noetherian local ring A, $A/\mathfrak{N}$ is flat over $A/\mathfrak{N}$ but not over A itself when $\mathfrak{N} \neq 0$ (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, cor. 2 of prop. 5). One thus sees that one cannot, in (15.2.2), suppress condition (iv).

lbbel=(ii) It follows from (2.4.6) that if the conclusion of (15.2.2) is true, and also hypothesis (i), f is universally open in a neighbourhood of x, and in particular at the generic points of the irreducible components of $f^{-1}(y)$ containing x. Even if conditions (i), (iii), and (iv) are satisfied, one cannot replace (ii) by the weaker hypothesis that f is universally open *at the point x only*. This is what the example (14.1.3), (i)) shows, where f is evidently universally open at every point of X_2, hence in particular at the point x, intersection of X_1 and X_2; conditions (ii) and (iii) of (15.2.3) are also satisfied by this example. However, $\mathscr{O}_y$ is not an $\mathscr{O}_y$-module without torsion, $\mathscr{O}_y$ identifies with a subring of $\mathscr{O}_z$ that contains the zero divisors of 0 in $\mathscr{O}_z$; hence f is not flat at the point x. IV-3 | 228

lcbel=(iii) In (15.2.3), the conclusion is not necessarily valid when we replace hypothesis (ii) by the weaker hypothesis that $f^{-1}(y)$, considered as a $\mathbf{k}(y)$-prescheme, *has no immersed associated prime cycles*. An example is provided by taking for Y a curve having a node, for example $Y = \mathrm{Spec}(K[S,T]/(S^3 - T^2))$, K being an algebraically closed field, and for X its *normalisation* (**II**, 6.3.8). If s and t are the classes of S and T in $A = K[S,T]/(S^3 - T^2)$, one verifies immediately that $X = \mathrm{Spec}(B)$, where $B = K[t, l, u]$, u being the element t/s of the field of fractions of A, and since $s = u^2$, $t = u^3$, we also have $B = K[u]$, isomorphic to a polynomial ring in one indeterminate over K. The only point x of X above the point y corresponding to the maximal ideal $(s) + (t)$ is such that, as the class $\bar{u}$ of u in $\mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ satisfies $\bar{u}^2 = 0$, $f^{-1}(y)$ is not a reduced $\mathbf{k}(y)$-prescheme. Here f is a finite morphism (2.4.5), but it is not flat at the point x, for if $\mathscr{O}_z$ were an A-module flat, it would be a free $\mathscr{O}_y$-module (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, prop. 5), which is absurd.

ldbel=(iv) Let us finally show that in (15.2.2) or (15.2.3), one cannot replace "geometrically reduced" by "reduced". We will in fact define two rings A, A′ having the following properties:

label=1) A is a Noetherian local ring of maximal ideal $\mathfrak{m}$, integral, complete, of dimension 1 and geometrically unibranch.

lbbel=2) A′ is the integral closure of A, a A-algebra whose maximal ideal is $\mathfrak{m}A'$, and the residue field k' a non-trivial finite radicial extension of the residue field $k = A/\mathfrak{m}$ of A.

One can then take $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(A')$, $\mathscr{F} = \mathscr{O}_{e_+}$, y and x being the closed points of Y and X respectively; the hypotheses (i) and (iv) of (15.2.2) are trivially satisfied; as A is of dimension 1, $f : X \to Y$ is radicial, finite, and surjective, hence a universal homeomorphism (2.4.5), which proves hypothesis (i) of (15.2.3). Finally, it is clear that $f^{-1}(y)$ is reduced. However, f is not flat at the point x, for if A′ were an A-module flat, it would be a free A-module of rank 1 (since it has the same field of fractions as A) generated by the element 1 of A (Bourbaki, *Alg. comm.*, chap. II, § 3, no 2, prop. 5), which is absurd.

To construct the rings A and A′, let us start from an imperfect field k of characteristic $p > 0$; let $K_0 = k((T))$ be the field of formal power series over k, A_0 the valuation ring corresponding to the discrete valuation of K_0; the residue field of A_0 is thus k. Let α be an element of k that is not a p-th power, and let us take $k' = k(\alpha^{1/p})$; $K = k'((T))$ is then a finite radicial extension of K_0, and $A' = A_0 \otimes_k k'$ is the discrete valuation ring for K. If $\mathfrak{m}'$ is the maximal ideal of A′, one will satisfy conditions 1) and 2) by taking $A = A_0 + \mathfrak{m}'$: indeed, as A_0 is Noetherian and A′ an A_0-module of finite type, A is a A_0-algebra finite, hence a Noetherian ring; A′ is finite over A, $\mathfrak{m}'$ is the only maximal ideal of A, which is thus a complete local ring, evidently of dimension 1 (being finite over A_0) and geometrically unibranch since A′ is integrally closed and having the same field of fractions K as A.

lebel=(v) The proof of (15.2.2) shows however that one can weaken condition (iii) by replacing "geometrically reduced" by "reduced" when one can apply the valuative criterion of flatness (11.8.1) using only discrete valuation rings A′ whose residue field k' is *separable* over $\mathbf{k}(y)$; indeed, if x' and z' have the same meanings as in (15.2.2), the radicial multiplicities $\mu_1(\mathbf{k}(z')|k')$ and $\mu_1(\mathbf{k}(z)|\mathbf{k}(y))$ are the same, by virtue of (4.7.3), hence it follows from (4.7.9) that $\operatorname{length}((\mathscr{F}_y)_z)$ and $\operatorname{length}((\mathscr{F}'_{y'})_{z'})$ are equal, and we are again reduced to the case where Y is the spectrum of a discrete valuation ring and y its closed point. For example, one will be able to obtain this result further when $\mathscr{O}_y$ is *unibranch* and dominated by a discrete valuation ring whose residue field is a separable extension of $\mathbf{k}(y)$, by virtue of (11.6.4). This is the case, for instance, when $\mathscr{O}_y$ is a *regular* ring, by (15.1.1.6). But, as Hironaka has pointed out, there are Noetherian local rings that are *integrally closed*, of dimension 2 (coming from algebraic local schemes over imperfect fields) that do not satisfy the preceding condition.

15.3. Applications: criteria for reduction and irreducibility

Proposition (15.3.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, x a point of X, $y = f(x)$. Suppose conditions* (i), (ii), *and* (iii) *of* (15.2.2) *are satisfied. Then:* **IV-3** | 228

label=(i) There exists a neighbourhood U of x in X such that $f|U$ is universally open and that, for all $x' \in U$, $\mathscr{F}_{f(x')}$ is geometrically reduced on $\mathbf{k}(f(x'))$ at the point x'.

lbbel=(ii) If furthermore Y is reduced at the point y, there exists a neighbourhood $U_1 \subset U$ of x such that $\mathscr{F}|U_1$ is f-flat and reduced.

Proof. Taking into account (**I**, 5.1.8), (4.2.7), and (4.7.11), the hypotheses do not change, neither in conclusion (i), when we replace Y by Y_{red} and X by $X \times_Y Y_{\text{red}}$, so we can suppose Y *reduced*. It follows then from (15.2.2) that $\mathscr{F}$ is f-flat at the point x, and hence also (11.1.1) in a neighbourhood of x, and *a fortiori* f is universally open in this neighbourhood (2.4.6). The fact that $\mathscr{F}_{f(x')}$ is then geometrically reduced at the point x' for all the points of a neighbourhood of x follows from (12.1.1), (vii)). Assertion (ii) follows from this and from (3.3.4). ☐ **IV-3** | 229

Proposition (15.3.2). — *The notations being those of (15.3.1), suppose that conditions* (i), (ii), (iii) *of (15.2.2) are satisfied, and furthermore that $f^{-1}(y)$ is equidimensional at the point x. Then f is equidimensional at the point x.*

Proof. We have, for all $y' \in Y$, $\mathrm{Supp}(\mathscr{F}_{y'}) = f^{-1}(y')$. By hypothesis $\mathscr{F}_y$ is reduced (and satisfies (S_1)) and is equidimensional at the point x. By virtue of (12.1.1), (ii)), one can thus suppose that $\mathscr{F}_{f(x')}$ is equidimensional and of constant dimension $e = \dim_x(f^{-1}(y))$ at the point x' for all $x' \in U$, which means that $f^{-1}(f(x'))$ is equidimensional and of dimension e at the point x'. Since moreover $\mathscr{F}|U$ is f-flat, it follows from (2.3.4) and (13.3.1), *a*)) that f is equidimensional at the point x. □

Corollary (15.3.3). — *The notations being those of (15.3.1), suppose the following conditions are satisfied:*

label=(i) $\mathrm{Supp}(\mathscr{F}) = X$.
lbbel=(ii) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x.
lcbel=(iii) $\mathscr{F}_y$ is geometrically pointwise integral on $\mathbf{k}(y)$ at the point x (4.6.22).
ldbel=(iv) Y *is geometrically unibranch at the point y.*

Then x belongs to a single irreducible component of X*. If furthermore* Y *is integral at the point y and $\mathscr{F} = \mathscr{O}_X$, then* X *is integral at the point x and f is flat at the point x.*

Proof. Note first that (i) and (iii) imply that x belongs to a single irreducible component Z of $f^{-1}(y)$. It follows thus from (14.4.7) and from (15.3.2) that for every irreducible component X_i of X containing x (and hence Z), the restriction f_i of f to X_i is an equidimensional morphism and universally open at the generic point of Z. The first assertion of the statement then follows from (15.1.3) and the second from (15.3.1). □

Remarks (15.3.4). — label=(i) The example (14.4.10), (ii)) shows that, in (15.3.3), one cannot suppress the hypothesis that Y is geometrically unibranch at the point y.
lbbel=(ii) The remark (15.2.4), (vi)) shows that the conclusion of (15.3.3) is still true under the following hypotheses: $\mathscr{O}_y$ is regular, $\mathrm{Supp}(\mathscr{F}) = X$, and $\mathscr{F}_y$ is integral at the point x. If we disregard in this statement the hypothesis that $\mathscr{O}_y$ is regular, one can replace the hypothesis that Y is geometrically unibranch by that that it is integral, or even integrally closed.

15.4. Complements on Cohen-Macaulay morphisms

Proposition (15.4.1). — *Let* Y *be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module such that* $\mathrm{Supp}(\mathscr{F}) = X$*, x a point of* X*, $y = f(x)$. We set, for all $z \in X$, $X_{f(z)} = f^{-1}(f(z))$, $\mathscr{F}_{f(z)} = \mathscr{F} \otimes_{\mathscr{O}_{f(z)}} \mathbf{k}(f(z))$. Suppose that $\mathscr{F}$ is f-flat at the point x and that $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{X_y}$-Module at the point x. Then f is equidimensional at the point x. Indeed ((11.1.1) and* **IV-3 | 230**
(12.1.1), (vi))), there exists a neighbourhood U *of x such that $\mathscr{F}|U$ is f-flat and that for all $x' \in U$, $\mathscr{F}_{f(x')}$ is a Cohen-Macaulay $\mathscr{O}_{X_{f(x')}}$-Module at the point x'; this latter property implies that $\mathscr{F}_{f(x')}$ is equidimensional at the point x'* (**0**, 16.5.4)*. Furthermore, by virtue of (12.1.1), (ii)), one can suppose that the dimensions of the irreducible components of* $\mathrm{Supp}(\mathscr{F}_{f(x')})|(U \cap X_{f(x')})$*) have a (common) value independent of x'. As* $\mathrm{Supp}(\mathscr{F}_{f(x')}) = X_{f(x')}$ *by hypothesis, it follows from (2.3.4) and (13.3.1), a)) that f is equidimensional at the point x.*

Proposition (15.4.2). — *Let* Y *be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module such that* $\mathrm{Supp}(\mathscr{F}) = X$*, x a point of* X*, $y = f(x)$. Suppose that $\mathscr{O}_y$ is regular and that $\mathscr{F}$ is a Cohen-Macaulay $\mathscr{O}_X$-Module at the point x. Then (with the notations of (15.4.1)) the following conditions are equivalent:*

*label=*a*) $\mathscr{F}$ is f-flat at the point x and $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{X_y}$-Module at the point x.*
*lbbel=*b*) $\mathscr{F}$ is f-flat at the point x.*
*lcbel=*c*) f is universally open in a neighbourhood of x.*
*ldbel=*d*) f is open at the generic points of the irreducible components of X_y containing x.*
*lebel=*e*)* $\dim(\mathscr{F}_x) = \dim(\mathscr{O}_y) + \dim((\mathscr{F}_y)_x)$.
e$'$*)* $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_y) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_y} \mathbf{k}(y))$.

As $\mathrm{Supp}(\mathscr{F}) = X$*, conditions* $e)$ *and* $e')$ *are equivalent by definition* (5.1.12)*. Condition* $a)$ *trivially implies* $b)$*;* $b)$ *implies that* $\mathscr{F}$ *is* f*-flat at the points of a neighbourhood of* x (11.1.1)*, hence* $b)$ *implies* $c)$ (2.4.6)*;* $c)$ *trivially implies* $d)$*, and* $d)$ *implies* $e')$ *by* (14.2.1)*. Finally, since* $\mathscr{O}_y$ *is regular and* $\mathscr{F}_x$ *is a Cohen-Macaulay* $\mathscr{O}_{X,x}$*-Module, it follows from* (6.1.4) *that* $e)$ *implies* $a)$*.*

Proposition (15.4.3). — *Let* $f : X \to Y$ *be a morphism locally of finite presentation,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module of finite presentation, such that* $\mathrm{Supp}(\mathscr{F}) = X$*,* x *a point of* X*,* $y = f(x)$*. Then (with the notations of* (15.4.1)*) the following conditions are equivalent:*

label=a) *$\mathscr{F}$ is f-flat at the point x and $\mathscr{F}_y$ is a Cohen-Macaulay $\mathscr{O}_{X_y}$-Module at the point x.*
lbbel=b) *There exists an open neighbourhood U of x in X, and a Y-morphism quasi-finite $g : U \to Y' = Y[T_1, \ldots, T_n]$ (T_i indeterminates), such that $\mathscr{F}|U$ is g-flat at the point x.*

Proof. Let us first show that $a)$ implies $b)$. Set $h : Y' \to Y'$; the $\mathbf{k}(y)$-prescheme $Y'_y = h^{-1}(y)$ is regular (**0**, 17.3.7); let A be the local ring of this prescheme at the point $y' = g(x)$, k its residue field, and B the local ring of X_y at the point x; set further $(\mathscr{F}_y)_x = M$, which is a B-module of finite type. The hypothesis that the morphism g is quasi-finite implies that it is the same for $g_y : X_y \to Y'_y$ (**II**, 6.2.4), hence $M \otimes_A k$ is a $(B \otimes_A k)$-module of finite length (**II**, 6.2.2); as by hypothesis M is a Cohen-Macaulay A-module flat and A a regular ring, M is a B-module Cohen-Macaulay (6.3.3). The fact that $\mathscr{F}$ is f-flat at the point x follows from that it is g-flat and from the fact that the morphism h is flat (2.1.6).

Let us now show that $a)$ implies $b)$. The question being local on X and Y, using (8.9.1) and (11.2.7), we reduce to the case where X and Y are Noetherian, taking into account (6.7.1). The hypothesis implies that f is open at the generic points of the irreducible components of X_y containing x, by (15.4.2), $e)$; it follows from (2.3.4) and (13.3.1), $a)$) that f is equidimensional at the point x. □

15.5. Separable rank of fibres of a quasi-finite and universally open morphism. Application to connected components of fibres of a proper morphism

Proposition (15.5.1). — *Let* Y *be a locally Noetherian prescheme,* $f : X \to Y$ *a separated and quasi-finite morphism. For every* $z \in Y$*, let* $n(z)$ *be the geometric number of points of* $f^{-1}(z)$ *(or* separable rank *of* $f^{-1}(z)$ *over* $\kappa(z)$*; cf.* (**I**, 6.4.8)*). We have the following properties:*

label=(i) *The function $z \mapsto n(z)$ is lower semi-continuous at every point $y \in Y$ such that f is universally open at the points of $f^{-1}(y)$.*
lbbel=(ii) *Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; if the function $z \mapsto n(z)$ is constant in a neighbourhood of y, then f is proper at the point y* (15.7.1)*.*
lcbel=(iii) *Suppose that f is universally open and that every point of Y is geometrically unibranch; then, if the function $z \mapsto n(z)$ is locally constant in Y, the irreducible components of X are pairwise disjoint.*

Proof. (i) Note that since f is quasi-finite, the number $n(z)$ is the geometric number of connected **IV-3** | 231
components of $f^{-1}(z)$; we know that the function $z \mapsto n(z)$ is locally constructible (9.7.9); taking into account ($\mathbf{0}_{\mathrm{III}}$, 9.3.4), we are reduced to proving the

Corollary (15.5.2). — *Under the general hypotheses of* (15.5.1)*, let* y *be a point of* Y *such that* f *is universally open at the points of* $f^{-1}(y)$*. Then, if* y' *is a generization of* y*, we have*

$$n(y) \leqslant n(y'). \tag{15.5.2.1}$$

For every base change $g : Z \to Y$, $f_{(Z)} : X_{(Z)} \to Z$ is separated, quasi-finite, and universally open at the points of $X_{(Z)}$ projecting into $f^{-1}(y)$; moreover, if z, z' are two points of Z such that $g(z) = y$, $g(z') = y'$ and z' is a generization of z, we have $n(z) = n(y)$ and $n(z') = n(y')$ ((**I**, 6.4.12)); it therefore suffices to prove (15.5.2.1) for a suitable g. We can evidently suppose $y \neq y'$. We will use the following lemma:

Lemma (15.5.2.2). — *Under the general hypotheses of* (15.5.1)*, if* y *is a point of* Y*,* y' *a generization of* y *distinct from* y*, there exists a spectrum of a discrete valuation ring* Z*, of closed point* z *and generic point* z'*, and a morphism* $g : Z \to Y$ *such that:* 1° $g(z) = y$*,* $g(z') = y'$*;* 2° *if one sets* $f' = f_{(Z)}$*,* $n(y)$ *is the number of points of* ${f'}^{-1}(z)$ *and* $n(y')$ *is the number of points of* ${f'}^{-1}(z')$*.*

Indeed, there exists a discrete valuation ring A_1 and a morphism g_1 from $Z_1 = \operatorname{Spec}(A_1)$ to Y such **IV-3** | 232
that if z_1 and z_1' are the closed point and the generic point of Z_1, we have $g_1(z_1) = y$ and $g_1(z_1') = y'$ ((**II**, 7.1.9)); we can therefore already suppose that Y is the spectrum of a discrete valuation ring A, y its closed point and y' its generic point. For every $x \in f^{-1}(y)$, $\kappa(x)$ is a finite extension of $\kappa(y)$ by hypothesis ((**II**, 6.2.2)); we deduce from (4.5.11) that there exists a finite algebraic extension k of $\kappa(y)$ such that the *number of points* of $f^{-1}(y) \otimes_{\kappa(y)} k$ is equal to $n(y)$. Since there exists a discrete valuation ring A_2 dominating A whose residue field is finite over $\kappa(y)$ and contains k ((**II**, 7.1.2)), we can in the second place suppose that $n(y)$ is the number of points of $f^{-1}(y)$. Finally, the same reasoning shows that there is a *finite* extension K of $\kappa(y')$ such that the number of points of $f^{-1}(y') \otimes_{\kappa(y')} K$ is equal to $n(y')$; if w is a valuation of $\kappa(y')$ associated to A, there is a discrete valuation of K extending w, and the ring A_3 of this valuation dominates A and has K as field of fractions, and satisfies the conditions of the lemma.

We can therefore henceforth suppose that Y is the spectrum of a discrete valuation ring, y its closed point, y' its generic point, and that $n(y)$ and $n(y')$ are the numbers of points of $f^{-1}(y)$ and $f^{-1}(y')$, so that for every $x \in f^{-1}(y)$ (resp. every $x' \in f^{-1}(y')$) we have $\kappa(x) = \kappa(y)$ (resp. $\kappa(x') = \kappa(y')$).

Denote then by X_i the sub-preschemes of X having for underlying spaces the irreducible components of X ($1 \leqslant i \leqslant m$). By virtue of (1.10.4), the restriction $X_i \to Y$ of f to each X_i is a *dominant* morphism, and since $f^{-1}(y')$ is discrete, each of the intersections $X_i \cap f^{-1}(y')$ is reduced to the generic point of X_i, whence (denoting by $n_i(z)$ the geometric number of points of $X_i \cap f^{-1}(z)$ for every $z \in Y$), $n_i(y') = 1$ for every i, and $n(y') = \sum_{i=1}^m n_i(y') = m$. Furthermore, $f^{-1}(y)$ is the union of the $X_i \cap f^{-1}(y)$, whence

(15.5.2.3) $$n(y) \leqslant \sum_{i=1}^m n_i(y),$$

and to prove (15.5.2.1), it suffices to establish that $n_i(y) \leqslant 1$ for every i. In other words, we can suppose X *integral*, and the morphism f birational. But then, for every $x \in f^{-1}(y)$, we have $\mathscr{O}_y \subset \mathscr{O}_z \subset K$, where $K = \kappa(y')$ is the field of fractions of $\mathscr{O}_y$. Since $\mathscr{O}_y$ is a valuation ring and $\mathscr{O}_z$ dominates $\mathscr{O}_y$, we necessarily have $\mathscr{O}_y = \mathscr{O}_z$ ((**II**, 7.1.1)). There exists then a Y-section $h : Y \to X$ such that $h(y) = x$ ((**I**, 2.4.4)); since Y is reduced and X is separated over Y, such a section is *unique* ((**I**, 7.2.2)), so the set $f^{-1}(y)$ contains only a single point, which completes the proof of (i).

(ii) As at the beginning of (i), we are reduced to showing that if the two members of (15.5.2.1) are equal for *every* generization y' of y, f is proper at the point y. Thanks to the criterion of local properness (15.7.5) and to the remarks at the beginning we can further suppose that $Y = \operatorname{Spec}(A)$ is the spectrum of a discrete valuation ring A, y its closed point and y' its generic point, and it remains to show that f is proper.

Let us use (15.5.2.2) and note that if A' is a discrete valuation ring dominating A, A' is a torsion-free A-module, so the morphism $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is faithfully flat and quasi-compact, and it therefore amounts to the same to say that f is proper or that the morphism deduced from f by the base change $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is proper (2.7.1), (vii); we can therefore suppose that $n(y)$ and **IV-3** | 233
$n(y')$ are the numbers of points of the fibres $f^{-1}(y)$ and $f'^{-1}(y')$ respectively. With the notations of (i), we have then by (15.5.2.3) and (15.5.2.1)

(15.5.2.4) $$n(y) \leqslant \sum_i n_i(y) \leqslant \sum_i n_i(y') = n(y')$$

and for the extreme members to be equal, we need $n_i(y) = n_i(y') = 1$ for every i. As we saw in (i), if $n_i(y) = 1$ for every i, $X_i \cap f^{-1}(y)$ is nonempty and the restriction $X_i \to Y$ of f is an isomorphism; since the X_i are closed sub-preschemes of X, this implies that f is proper ((**II**, 5.4.5)).

(iii) We can restrict to the case where Y is affine and reduced, and since Y is by hypothesis locally integral ((**I**, 5.1.4)), we can suppose Y integral, and the function $y \mapsto n(y)$ constant in Y. Let again X_i ($1 \leqslant i \leqslant m$) be the irreducible components of X. It follows from (1.10.4) that if y' is the generic point of Y, each of the $X_i \cap f^{-1}(y')$ is reduced to a single point; the restriction $f_i : X_i \to Y$ of f being a quasi-finite and dominant morphism and every point of Y being geometrically unibranch, it follows from the criterion of Chevalley (14.4.4) that f_i is *universally open*. So then for any point y of Y, we have $n_i(y) \leqslant n_i(y')$; on the other hand, since the $X_i \cap f^{-1}(y)$ are pairwise disjoint, $\sum_i n_i(y') = n(y')$; we therefore still have the relations (15.5.2.4). But by hypothesis the extreme members are equal, so $n(y) = \sum_i n_i(y)$, which implies that the $X_i \cap f^{-1}(y)$ are pairwise disjoint, and this completes the proof. □

Combining this proposition with the connectivity theorem of Zariski and its consequences ((**III**, 4.3)), we obtain the following results that complement ((**III**, 4.3.7) and (4.3.10)):

Proposition (15.5.3). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \to Y$ a proper morphism; for every $y \in Y$, let $n(y)$ be the geometric number of connected components (4.5.2) of $f^{-1}(y)$. Suppose that the restriction of f to every irreducible component of X is a dominant morphism in Y. Then, if y_0 is a geometrically unibranch point of Y, the function $y \mapsto n(y)$ is lower semi-continuous at the point y_0.*

Proof. Consider the Stein factorisation $f = g \circ f'$ of f ((**III**, 4.3.3)), where $g : Y' \to Y$ is a finite morphism and $f' : X \to Y'$ is a surjective morphism whose fibres are geometrically connected ((**III**, 4.3.4)). Since f' is surjective, the inverse image by f' of every irreducible component of Y' contains an irreducible component of X at least; therefore each irreducible component of Y' dominates Y. We can therefore apply the criterion of Chevalley (14.4.4) to the restrictions of g to each of these irreducible components, and we conclude that g is *universally open at every point* of $g^{-1}(y_0)$. The conclusion therefore follows from (15.5.2) and from the fact that the number of geometrically connected components of $f^{-1}(y)$ is equal to the geometric number of points of $g^{-1}(y)$ ((**III**, 4.3.4)). □

Corollary (15.5.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism; let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; then, with the notations of (15.5.3), the function $z \mapsto n(z)$ is lower semi-continuous at the point y.* **IV-3** | 234

Proof. With the notations of (15.5.3), note that in the Stein factorisation $f = g \circ f'$, f' is surjective; since f is universally open at the points of $f^{-1}(y)$, g is universally open at the points of $g^{-1}(y)$ ((14.3.4), (i)); it then suffices to apply to g the proposition (15.5.1), (i) in view of ((**III**, 4.3.4)). □

Remarks (15.5.5). — (i) Even if Y is integral and normal, X integral, f finite and surjective (hence universally open by virtue of the criterion of Chevalley (14.4.4)), the function $y \mapsto n(y)$ is not necessarily locally constant. We have an example by taking $Y = \mathrm{Spec}(k[\mathrm{T}])$ where k is algebraically closed, and $X = \mathrm{Spec}(A)$, where $A = k[\mathrm{S}, \mathrm{T}]/(\mathrm{S}^2 + \mathrm{T}^3 - 1)$; at the points y', y'' of Y corresponding to the maximal ideals $(\mathrm{T} - 1)$ and $(\mathrm{T} + 1)$, we have $n(y') = n(y'') = 1$, but $n(y) = 2$ at all other points of Y. We shall give below an additional condition ensuring that $y \mapsto n(y)$ is locally constant (15.5.7).

(ii) The example (14.4.10) (ii) shows that in (15.5.1), (iii), one cannot suppress the hypothesis that the points of Y are geometrically unibranch.

(iii) The example (11.7.5) shows that the conclusion of (15.5.1), (i) is no longer valid if one only supposes that the morphism f is *open* (even if, as is the case in the cited example, f is finite and surjective).

(iv) Finally, in (15.5.1), (iv), one cannot dispense with the hypothesis that the morphism f is *separated*, as shown by the example where X is the affine line one of whose points has been "doubled" at a point x ((**I**, 5.5.11)), and Y the affine line: here f is a local isomorphism (hence flat) and is quasi-finite, but $z \mapsto n(z)$ is not lower semi-continuous.

Lemma (15.5.6). — *Let Y be the spectrum of a discrete valuation ring, a its closed point, b its generic point. Let $f : X \to Y$ be a locally of finite type and open morphism. Suppose that X is connected and that the fibre $f^{-1}(a)$ is a reduced prescheme. Then $f^{-1}(b)$ is connected.*

Proof. We will use the following purely topological lemma (a special case of a more general result from Ch. III, 3rd Part):

Lemma (15.5.6.1). — *Let X be a locally Noetherian topological space, every closed irreducible subset of which admits a generic point. Let Z be a closed rare subset of X having the following property: for every $x \in Z$, if one denotes by T_x the set of generizations of x in X, then $\mathrm{T}_x - \{x\}$ is connected. Under these conditions, if X is connected, $X - Z$ is connected.*

Proof. Let us give an independent proof. Reasoning by the absurd, suppose that the open set $U = X - Z$, which is dense in X since Z is everywhere dense, is the union of two nonempty open sets U', U'' without common points, and denote by X' and X'' the closures in X of U' and U''. Since $X = \overline{U} = X' \cup X''$ and X is connected, we have $X' \cap X'' \neq \varnothing$, and it is clear that $X' \cap X'' \subset Z$. Let x be a generic point of an irreducible component of $X' \cap X''$. Since X is locally Noetherian, there is an open neighbourhood V of x in X such that $V \cap U'$ and $V \cap U''$ have only a finite number of irreducible components; since x is adherent to U' and to U'', it is necessarily in the closure of an irreducible component Z' of $V \cap U'$ and in the closure of an irreducible component Z'' of $V \cap U''$. But $\overline{Z'}$ (resp. $\overline{Z''}$) is closed and irreducible in X, hence admits a generic point z' (resp. z''), and we necessarily have $z' \in U'$ and $z'' \in U''$. This proves that the intersections of $\mathrm{T}_x - \{x\}$ with U' and U'' are nonempty. But by definition $(X' \cap X'') \cap (\mathrm{T}_x - \{x\}) = \varnothing$; therefore the intersections of X' and of X'' with $\mathrm{T}_x - \{x\}$ are two nonempty disjoint closed sets in $\mathrm{T}_x - \{x\}$, whose union is $\mathrm{T}_x - \{x\}$; this contradicts the hypothesis that $\mathrm{T}_x - \{x\}$ is connected. □

□

IV-3 | 235

Proof of (15.5.6). To prove (15.5.6), we apply the lemma (15.5.6.1) to X and to its closed subset $Z = f^{-1}(a)$ which is rare since f is open. Note that by hypothesis, taking into account (15.2.2.1), this implies that f is *flat* at the points of $f^{-1}(a)$. For any such point x, $\mathscr{O}_x$ is therefore an $\mathscr{O}_a$-module without torsion ($\mathbf{0}_{\mathrm{I}}$, 6.3.4), and in particular, if t is a uniformiser of $\mathscr{O}_a$, t is an $\mathscr{O}_x$-regular element; moreover $\mathscr{O}_x/t\mathscr{O}_x$ is reduced by hypothesis; if it is of depth 0, it is therefore a field ($\mathbf{0}$, 16.4.7), and as $t\mathscr{O}_x$ is then the maximal ideal of $\mathscr{O}_x$, $\operatorname{Spec}(\mathscr{O}_x) - \{x\}$ is reduced to a point, and *a fortiori* is connected. If on the contrary $\operatorname{prof}(\mathscr{O}_x/t\mathscr{O}_x) \geqslant 1$, we have $\operatorname{prof}(\mathscr{O}_x) \geqslant 2$ since t is $\mathscr{O}_x$-regular ($\mathbf{0}$, 16.4.6); it then follows from the theorem of Hartshorne (5.10.7) that $\operatorname{Spec}(\mathscr{O}_x) - \{x\}$ is connected, which completes the proof. □

Proposition (15.5.7). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism. For every $z \in Y$, let $n(z)$ be the geometric number of connected components of $f^{-1}(z)$. Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$ and that $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$* (4.6.2). *Then the function $z \mapsto n(z)$ is constant in a neighbourhood of y.*

Proof. We already know (15.5.4) that under the conditions of the statement, $z \mapsto n(z)$ is lower semi-continuous at the point y; it remains to see that it is also upper semi-continuous, and by the same reasoning as at the beginning of the proof of (15.5.1), (i), it suffices to show that if y' is a generization of y, we have

$$n(y') \leqslant n(y). \tag{15.5.7.1}$$

Using the beginning of the proof of (15.5.2) and the lemma (15.5.2.2) applied to the finite morphism g of the Stein factorisation $f = g \circ f'$ of f (recalling that the geometric number of points of connected components of $f^{-1}(y)$ is equal to that of $g^{-1}(y)$ ((**III**, 4.3.4))), we reduce to the case where $n(y)$ and $n(y')$ are respectively the number of connected components of $f^{-1}(y)$ and $f^{-1}(y')$, and where Y is the spectrum of a discrete valuation ring, y the closed point and y' the generic point of Y. We can moreover replace X by any one of its connected components, these being open and closed in X, and proper over Y. Suppose therefore that X is connected; the hypothesis that f is open at the points of $f^{-1}(y)$ implies that $f^{-1}(y) \neq \varnothing$, and we also have $f^{-1}(y') \neq \varnothing$ ($\mathbf{0}_{\mathrm{I}}$, 1.10.3); the hypothesis that f is closed implies that if $f^{-1}(y) \neq \varnothing$, we also have $f^{-1}(y') \neq 0$ ((**II**, 7.2.1)). So, if $X \neq \varnothing$ (which we can evidently suppose, the proposition being trivially true in the contrary case), $f^{-1}(y)$ and $f^{-1}(y')$ are both nonempty. Moreover, the hypothesis implies that the prescheme $f^{-1}(y)$ is *reduced* (4.6.1); since f is open at the points of $f^{-1}(y)$, the lemma (15.5.6) implies that $f^{-1}(y')$ is connected, in other words $n(y') = 1$. Since $f^{-1}(y)$ is not empty, this proves (15.5.7.1). □

Remarks (15.5.8). — The relation (15.5.7.1) is not necessarily valid if f is not supposed proper, even if it satisfies the other hypotheses of (15.5.7). With the notations of (15.5.5), (i), it suffices to consider the restriction of f to $X - \{x'\}$, where x' is one of the two points of X above a point y distinct from y' and y''; we then have $n(y) = 1$, whereas if η is the generic point of Y, $n(\eta) = 2$.

IV-3 | 236

Proposition (15.5.9). — *Let $f : X \to Y$ be a flat and locally of finite presentation morphism; for every $y \in Y$, let $n(y)$ be the geometric number of connected components of $f^{-1}(y)$.*

label=(i) If f is separated and quasi-finite, the function $y \mapsto n(y)$ (equal then to the number of geometric points of $f^{-1}(y)$) is lower semi-continuous in Y; if it is constant in a neighbourhood of y_0, f is proper at the point y_0.

lbbel=(ii) If f is proper, the function $y \mapsto n(y)$ is lower semi-continuous in Y; if in addition $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$, $n(y)$ is constant in a neighbourhood of y_0.

Proof. In all cases, the questions are local on Y, so we can suppose Y affine and f of finite presentation; using (8.9.1), (8.10.5) and (11.2.7), we reduce to the case where Y is Noetherian (and using moreover (8.2.11) for the second assertion of (i)). Moreover f is then universally open (2.4.6), it suffices to apply (15.5.1), (15.5.4) and (15.5.7) to conclude. □

Remarks (15.5.10). — We note that we have thus obtained another proof of (12.2.4), (vi). Conversely, one can deduce (15.5.7) from (12.2.4), (vi): indeed, on can restrict to the case where Y is reduced (replacing f by f_{red}), and it then follows from the hypotheses of (15.5.7) and from (15.2.3) that f is flat in an open neighbourhood of $f^{-1}(y)$; since moreover f is proper, we can suppose that this neighbourhood is of the form $f^{-1}(U)$, where U is an open neighbourhood of y in Y. We conclude then by (12.2.4), (vi). The proof of (15.5.7) given above has the advantage of exhibiting the result (15.5.6), which is of independent interest.

15.6. Connected components of fibres along a section

Proposition (15.6.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a locally of finite type morphism, $g : Y \to X$ a Y-section of X ((**I**, 2.5.5)). For every $y \in Y$, denote by X_y^0 the connected component of $f^{-1}(y)$ containing the point $g(y)$. Let y be a point of Y, v' a generization of y. Then:*

label=(i) There exists an irreducible component Z of X_y^0, of dimension equal to $\dim(X_y^0)$ and containing $g(y')$ (which will be the case if X_y^0 is irreducible); we have

$$\dim(X_{y'}^0) \geqslant \dim(X_y^0). \tag{15.6.1.1}$$

lbbel=(ii) If X_y^0 is moreover irreducible and if the two members of (15.6.1.1) are equal, then $X_y^0 \subset \overline{X_{y'}^0}$ (closure in X).

Proof. (i) Let $\overline{Z}$ be the closure of Z in X; since y is adherent to y' and g is continuous, $g(y) \in \overline{Z}$. Since $Z = \overline{Z} \cap f^{-1}(y')$, it follows from the semi-continuity theorem of Chevalley ((13.1.3)) applied to the restriction of f to the closed reduced sub-prescheme of X having $\overline{Z}$ as underlying space, that $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y)) \geqslant \dim(Z)$; but the irreducible components of $\overline{Z} \cap f^{-1}(y)$ containing $g(y)$ are evidently contained in X_y^0, whence *a fortiori* the inequality (15.6.1.1).

(ii) The reasoning of (i) shows moreover that if the two members of (15.6.1.1) are equal, we necessarily have $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y)) = \dim_{g(y)}(X_y^0)$; if X_y^0 is irreducible, $\overline{Z} \cap f^{-1}(y) = X_y^0$, so X_y^0 is contained in $\overline{Z}$, and *a fortiori* in $\overline{X_{y'}^0}$. □

IV-3 | 237

Corollary (15.6.2). — *With the notations of (15.6.1), suppose moreover that for every $y \in Y$, X_y^0 is irreducible. Then the function $z \mapsto \dim(X_y^0)$ is upper semi-continuous in Y. If moreover this function is constant in a neighbourhood of a point $y \in Y$, then $X_y^0 \subset \overline{X_{y'}^0}$ for every generization y' of y.*

Proof. Let y be a point of Y, and let U be a neighbourhood of $g(y)$ in X; then there is a neighbourhood V of y in Y such that $g(V) \subset U$, and for every $z \in V$, $U \cap X_z^0$ is dense in X_z^0, hence $\dim(X_z^0) = \dim(U \cap X_z^0)$ ((4.1.1.3)). Replacing X by U, we can suppose that f is a morphism of finite type. We then know (9.7.10) that the function $z \mapsto \dim(X_z^0)$ is locally constructible, and the assertions of the corollary follow from (15.6.1) and from ($\mathbf{0}_{\mathrm{III}}$, 9.3.4). □

Proposition (15.6.3). — *With the notations of (15.6.1), suppose that for every $y \in Y$, X_y^0 is geometrically irreducible (4.5.2). Then, for every $y \in Y$ such that the function $z \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y, f is universally open at the points of X_y^0.*

Proof. Apply the criterion (14.3.7). Let therefore $h : Y' \to Y$ be a morphism, Y' the spectrum of a discrete valuation ring, whose points we denote by t, t' with t the closed point and t' the generic point, so that $y' = h(t')$, and y' is a generization of y. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $g' = g_{(Y')} : Y' \to X'$. With the same notations as in (15.6.1), the hypothesis on X_y^0 implies that $X_t'^0 = X_y^0 \otimes_{\kappa(y)} \kappa(t)$ and $X_{t'}'^0 = X_{y'}^0 \otimes_{\kappa(y')} \kappa(t')$, and that $X_t'^0$ and $X_{t'}'^0$ are irreducible (4.4.1); since $\dim(X_y^0) = \dim(X_t'^0)$ and $\dim(X_{y'}^0) = \dim(X_{t'}'^0)$ (4.1.4), we see that it suffices to prove the proposition for f' and $X_t'^0$, in other words we can restrict to the case where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point; if x' is the generic point of $X_{y'}^0$, it follows from (15.6.2) that every point of X_y^0 is a specialisation of x', whence the conclusion. □

Proposition (15.6.4). — *Under the general hypotheses of (15.6.1), let X^0 be the union of the X_y^0 as y runs through Y. If $y \in Y$ is such that X_y^0 is geometrically reduced over $\kappa(y)$ (4.6.2) and if f is universally open at every point of X_y^0, then X^0 is a neighbourhood of X_y^0 in X. In particular, if f is universally open and if, for every $y \in Y$, X_y^0 is geometrically reduced over $\kappa(y)$, X^0 is open in X.*

Proof. Let us first show that we can reduce to the case where f is a morphism of *finite type*. Let x be a point of X_y^0; it follows from (5.10.8.1) (with $\mathfrak{F} = \{\varnothing\}$) that there is a finite sequence $(Z_i)_{0 \leqslant i \leqslant n}$ of irreducible components of $f^{-1}(y)$ such that $g(y) \in Z_0$, $x \in Z_n$ and $Z_i \cap Z_{i+1} \neq \varnothing$ for $0 \leqslant i \leqslant n-1$; the Z_i being irreducible, there is a finite sequence (U_i) $(1 \leqslant i \leqslant n)$ of open affine subsets of $f^{-1}(y)$, $x \in U_n$, U_i being an open affine neighbourhood of a point x_i of $Z_i \cap Z_{i-1}$ for $1 \leqslant i \leqslant n$. There exists for each i a quasi-compact open set W_i in X such that $U_i = f^{-1}(y) \cap W_i$; let W be the quasi-compact open set of X which is the union of the W_i. Since g is continuous, there is an open affine neighbourhood V of y in Y such that $g(V) \subset W$; if $X' = W \cap g^{-1}(V)$, the restriction of f to X' is of finite type; moreover, for every $z \in V$, $X_z'^0$ is the connected component of $X' \cap f^{-1}(z)$ containing $g(z)$, and we have $X_z'^0 \subset X_y^0$, and x belongs to X_y^0. Indeed, each of the U_i contains the two irreducible sets $U_i \cap Z_{i-1}$ and $U_i \cap Z_i$ which intersect, and the $U_i \cap Z_i$ for $1 \leqslant i \leqslant n$ is connected and contains x and $g(y)$. Since $f|X'$ is universally open at the points of $X_y'^0$, this proves our assertion, for if the proposition is proved when f is of type fini, the union X'^0 of the $X_z'^0$ for $z \in V$ will be a neighbourhood of x, and we are done since $X'^0 \subset X^0$. **IV-3** | 238

Suppose therefore that f is of finite type. Then X^0 is locally constructible (9.7.12); it suffices therefore to show that X^0 contains every generization x' of x ($\mathbf{0}_{\text{III}}$, 9.2.5). Set $y' = f(x')$; there exists a discrete valuation ring A and a morphism u from $Y' = \operatorname{Spec}(A)$ to X such that, if z is the closed point and z' the generic point of Y', we have $u(z) = x$ and $u(z') = x'$ ((**II**, 7.1.9)); so $h = f \circ u$, and therefore $h(z) = y$ and $h(z') = y'$. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, and $g' = g_{(Y')} : Y' \to X'$. If $p_y : f'^{-1}(z) \to f^{-1}(y)$ (resp. $p_y : f'^{-1}(z') \to f^{-1}(y')$) is the canonical projection, $p_y^{-1}(X_y^0)$ (resp. $p_{y'}^{-1}(X_{y'}^0)$) is the connected component of $f'^{-1}(z)$ (resp. $f'^{-1}(z')$) containing $g'(z)$ (resp. $g'(z')$), since the X_y^0 are geometrically connected (4.5.13) and it suffices to apply (4.4.1). Moreover, the morphism u corresponds to a Y'-section v of X' such that $p_Y(v(y)) = x$, $p_Y(v(y')) = x'$, and $v(y')$ is a generization of $v(y)$, and it will suffice to prove that $v(y')$ belongs to $p_{y'}^{-1}(X_y^0)$. Finally, it is clear that $p_y^{-1}(X_y^0)$ is geometrically reduced over $\kappa(z)$.

In other words, we are reduced to the case where Y is the spectrum of a discrete valuation ring, y its closed point, y' its generic point. Since $f^{-1}(y) - X_y^0$ is then closed in X, replacing X by the open set $X - (f^{-1}(y) - X_y^0)$, we can suppose that $X_y^0 = f^{-1}(y)$; similarly, one can replace X by the open set, connected component of X, which contains $g(Y)$; this connected component evidently contains X^0; in other words, we can suppose X *connected*. The proposition will then be established if we show that $X^0 = X$, or again that X_y is connected; but $f^{-1}(y')$ is geometrically reduced over $\kappa(y')$, and *a fortiori reduced*; and moreover f is open at the points of $f^{-1}(y)$, so we can apply the lemma (15.5.6), whence the conclusion. □

Corollary (15.6.5). — *Let $f : X \to Y$ be a flat morphism of finite presentation, g a Y-section of X; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the*

X_y^0 as y runs through Y. Then, for every $y \in Y$ such that X_y^0 is geometrically reduced over $\kappa(y)$, X^0 is a neighbourhood of X_y^0 in X. In particular, if f is reduced (6.8.1), X^0 is open in X.

Proof. The question being local on Y, we can suppose $Y = \mathrm{Spec}(A)$ affine. There exist therefore a Noetherian subring A_0 and a flat morphism of finite type $f_0 : X_0 \to Y_0 = \mathrm{Spec}(A_0)$ such that $X = X_0 \times_{Y_0} Y$ and $f = (f_0)_{(Y)}$ ((8.9.1) and (11.2.7)); moreover, we can suppose that there exists a Y_0-section g_0 of X_0 such that $g = (g_0)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_1 be its projection in Y_1; it follows from (4.5.13) that $(X_1)_{y_1}^0$ is geometrically connected, so, if $p : f^{-1}(y) \to f_0^{-1}(y_0)$ is the canonical projection, we have $X_y^0 = p^{-1}((X_1)_{y_1}^0)$ (4.4.1); moreover, since f is flat, the hypothesis that X_y^0 is geometrically reduced over $\kappa(y)$ implies that $(X_0)_{y_0}^0$ is geometrically reduced over $\kappa(y_0)$ (4.6.10). We are thus reduced to the case where Y is moreover *Noetherian*; since f is universally open ((11.1.1)), it suffices then to apply (15.6.4). □ **IV-3** | 239

Proposition (15.6.6). — *Let $f : X \to Y$ be a morphism such that Y is locally Noetherian and f locally of finite type, or that f is of finite presentation. Let g be a Y-section of X; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$; finally, let X^0 be the union of the X_y^0 as y runs through Y. Suppose that, for every $y \in Y$, X_y^0 is geometrically irreducible. Then:*

label=(i) *The function $y \mapsto \dim(X_y^0)$ is upper semi-continuous in Y.*

lbbel=(ii) *Let $x_0 \in X^0$, $y_0 = f(x_0)$. If the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0, there exists a neighbourhood U of y_0 such that f is universally open at all points of $X^0 \cap f^{-1}(U)$. If moreover Y is reduced and $f^{-1}(y_0)$ geometrically reduced over $\kappa(y_0)$ at the point x_0, f is flat at the point x_0.*

lcbel=(iii) *Conversely, suppose that f is universally open at the generic point of $X_{y_0}^0$ and in addition that one of the following conditions is satisfied:*

label=α) *The fibre $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$ at the point $g(y_0)$ and the ring $\mathscr{O}_{Y,y_0}$ is Noetherian.*

lbbel=α) *For every generic point y' of an irreducible component of Y containing y_0, we have $\dim(f^{-1}(y')) = \dim(X_{y'}^0)$.*

Then the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0.

ldbel=(iv) *The set W of points $x \in X^0$ such that the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of $f(x)$ and $X_{f(x)}^0$ is geometrically reduced over $\kappa(f(x))$, is open in X.*

Proof. The questions being local on Y, we can suppose that $Y = \mathrm{Spec}(A)$ is affine.

When f is of finite presentation, there exists a Noetherian subring A_1 of A and a morphism of finite type $f_1 : X_1 \to Y_1 = \mathrm{Spec}(A_1)$ such that $X = X_1 \times_{Y_1} Y$ and $f = (f_1)_{(Y)}$ (8.9.1); moreover, we can suppose that there exists a Y_1-section g_1 of X_1 such that $g = (g_1)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_1 be its projection in Y_1; it follows from (4.5.13) that $(X_1)_{y_1}^0$ is geometrically connected, so, if $p : f^{-1}(y) \to f_1^{-1}(y_1)$ is the canonical projection, $X_y^0 = p^{-1}((X_1)_{y_1}^0)$ (4.4.1). Note moreover that one can, by virtue of (9.7.11) and (9.7.12), apply (9.3.3) to the property of being geometrically irreducible; we can therefore suppose A_1 chosen so that $(X_1)_{y_1}^0$ is geometrically irreducible for every $y_1 \in Y_1$. We have $\dim(X_y^0) = \dim((X_1)_{y_1}^0)$ (4.1.4), and applying once more (9.5.5) and (9.7.12), we can suppose (restricting as needed Y to a neighbourhood of y_0) that if $\dim(X_y^0)$ is constant in Y, then $\dim((X_1)_{y_1}^0)$ is constant in Y_1. If Y is reduced, so is Y_1 since $A_1 \subset A$; moreover, if $f^{-1}(y_0)$ is geometrically reduced over $\kappa(y_0)$ at the point x_0, $f_1^{-1}(y_{01})$ at the point x_{01} (x_{01} and y_{01} being the respective projections of x_0 and y_0) in X_1 and Y_1 (4.6.10).

These remarks show that to prove (i) and (ii), we can restrict to the case where Y is Noetherian and f locally of finite type. But in this case, the assertion of (i) follows from (15.6.2) and the first assertion of (15.6.3). Moreover, if Y is reduced and $f^{-1}(y_0)$ geometrically reduced over $\kappa(y_0)$ at the point x_0, we deduce from (15.6.3) and (15.2.3) that if $\dim(X_y^0)$ is constant in a neighbourhood of y_0, f is flat at the point x_0. **IV-3** | 240

To prove (iii), let us first remark that the set X^0 is locally constructible in X (9.7.12), so, by (9.5.5), the set E of $y \in Y$ such that $\dim(X_y^0) = \dim(X_{y_0}^0)$ is locally constructible in Y. Apply then (1.10.1) to

E: it suffices to prove that $\dim(X^0_{y'}) = \dim(X^0_{y_0})$ for the generic point y' of an irreducible component of Y containing y_0; this already allows us to suppose that $Y = \operatorname{Spec}(\mathscr{O}_{X,y_0})$.

If we are in the case α), we reduce immediately, by virtue of ((**II**, 7.1.9)), and using (15.6.4), to the case where Y is the spectrum of a discrete valuation ring, y_0 its closed point and y' its generic point. But then the hypotheses imply, by virtue of (15.2.2.1), that f is *flat* at the point $g(y_0)$, and also in a neighbourhood V of this point in X ((11.1.1)). To prove our assertion, by V, for $V \cap X^0_y$ is nonempty by virtue of (2.3.4), hence is an everywhere dense open set in X^0_y and has therefore the same dimension (4.1.1.3); moreover, since X^0_y is also the connected component of $f^{-1}(y')$ containing $g(y')$, we can suppose V chosen such that V meets no other irreducible component of $f^{-1}(y_0)$ nor of $f^{-1}(y')$, in other words we can suppose that $X^0 = X$; but then the conclusion follows from (12.1.1), (i), since by hypothesis $X^0_{y_0}$ is integral.

Suppose now that we are in the case β). Apply this time ((**II**, 7.1.4)) and ((**II**, 7.1.9)) in the case α): we are then reduced to the case where Y is the spectrum of a discrete valuation ring (not necessarily discrete), y_0 its closed point and y' its generic point. By virtue of (14.3.13), there exists an irreducible component Z of X containing $X^0_{y_0}$ and dominant Y, and such that $\dim(Z \cap f^{-1}(y')) = \dim(X^0_{y_0})$; but the hypothesis β) and the fact that $\dim(Z \cap f^{-1}(y')) \leqslant \dim(f^{-1}(y'))$ show that $\dim(X^0_{y_0}) \leqslant \dim(X^0_{y'})$ whence $\dim(X^0_{y_0}) = \dim(X^0_{y'})$ by virtue of (i).

It remains to prove (iv). Note first that the sets envisaged do not change when we replace f by $f_{(Y_{\mathrm{red}})} : X \times_Y Y_{\mathrm{red}} \to Y_{\mathrm{red}}$, the projection $X \times_Y Y_{\mathrm{red}} \to X$ being a homeomorphism. In other words, we can suppose Y reduced, and then it follows from (ii) that f is *flat* at the points of W. Consider a point x_0 of W and let us prove that W is a neighbourhood of x_0; proceeding as at the beginning of the proof, and using (11.2.7), we can also suppose Y *Noetherian*; it then follows from (ii) and from (15.6.4) that X^0 is a neighbourhood of x_0 in X, and from (12.1.1), (vii)) that W is also a neighbourhood of x_0 in X. □

Corollary (15.6.7). — *Suppose the preliminary conditions of (15.6.6) satisfied on f, and suppose moreover that for every $y \in Y$, X^0_y is geometrically integral over $\kappa(y)$. Then the following conditions are equivalent:*

label=a) *The function $y \mapsto \dim(X^0_y)$ is locally constant in Y.*

lbbel=b) *The morphism $f_{(Y_{\mathrm{red}})} : X \times_Y Y_{\mathrm{red}} \to Y_{\mathrm{red}}$ deduced from f by base change, is flat at the points of X^0.*

Moreover, these conditions imply that X^0 is open in X. Finally, when Y is locally Noetherian, a) and b) are also equivalent to **IV-3** | 241

b′) *f is universally open at the points of X^0.*

Proof. The fact that *a*) implies *b*) follows from (15.6.6), (ii), together with the fact that X^0 is then open in X. If *b*) is satisfied, one can restrict to the case where Y is reduced and f flat; then, we reduce, as at the beginning of (15.6.6), and using moreover (11.2.7), to the case where Y is Noetherian, case in which the conclusion follows from (15.6.6), (iii), case α)). The equivalence of *b*) and *b*′) has already been proved when Y is locally Noetherian (15.2.2.1), taking into account the fact that the morphism $X \times_Y Y_{\mathrm{red}} \to X$ is a universal homeomorphism. □

Proposition (15.6.8). — *Let $f : X \to Y$ be a separated morphism of finite presentation, g a Y-section of X; for every $y \in Y$, let X^0_y be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the X^0_y as y runs through Y. Then, if every $y \in Y$ is such that X^0_y is proper over $\kappa(y)$, X^0 is proper over Y at the point y (i.e. (15.7.1), there exists an open neighbourhood U of y in Y such that $X^0 \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U).*

Proof. Proceeding as at the beginning of (15.6.6) (where one uses (8.10.5), (v), and where one replaces (9.6.7) by (9.6.7)), we reduce to the case where Y is Noetherian and f of finite type; moreover, the X^0_y are geometrically connected (for every $z \in Y$) by virtue of (4.5.13); finally, as $g(Y) \subset X^0$, X^0 is universally submersive over Y (15.7.8). It suffices then to apply to X^0 the criterion (15.7.9). □

Corollary (15.6.9). — *Let $f : X \to Y$ be a separated morphism such that Y is locally Noetherian and f locally of finite type, or that f is of finite presentation. Let g be a Y-section of X, and for every $y \in Y$, let X^0_y*

be the connected component of $f^{-1}(y)$ *containing* $g(y)$*. Let* y *be a point of* Y *such that* X^0_y *is proper over* $\kappa(y)$*. Then, for every generization* y' *of* y*,* $\dim(X^0_y) \geqslant \dim(X^0_{y'})$*.*

Proof. Suppose first that f is of finite presentation; then replacing as needed Y by a neighbourhood of y, we can suppose that $f|X^0$ is *proper* (15.6.8), and it suffices to apply (13.1.5) to this restriction. If Y is locally Noetherian and f locally of finite type, X is locally Noetherian; moreover the hypothesis that X^0_y is proper over $\kappa(y)$ implies that X^0_y is Noetherian; therefore, by V an open Noetherian set containing X^0_y; it exists then a neighbourhood open U of y in Y such that $g(W) \subset V$. Moreover, if z is a maximal point of X^0_y, the restriction of f to $V \cap f^{-1}(W)$ is a morphism of finite type, and applying to this morphism the result already obtained, we see that if Z is the irreducible component of X^0_y for which z is a maximal point whatever, we have $\dim(X^0_{y'}) \leqslant \dim(X^0_y)$. □

Remarks (15.6.10). — (i) The supplementary hypothesis on the existence of an irreducible component of X containing $g(y')$ and of dimension equal to $\dim(X^0_y)$, made in (15.6.1), is not superfluous, as shown by the following example. Let A be a discrete valuation ring, π a uniformiser of A, K the field of fractions of A, k its residue field, and designate by y and y' the closed point and the generic point of Y. Consider the two Y-schemes: $X_1 = \operatorname{Spec}(A[t])$, $X_2 = \operatorname{Spec}(K[u,v])$, where t, u, v are indeterminates (we note that the projection of X_2 to Y is therefore $\{y'\}$). In the ring $A[t]$, the principal ideal $(\pi t - 1)$ is *maximal*, and projects to the *generic point* y' of Y and is such that $\kappa(x_1) = K$. Consider also the closed point x_2 of X_2 corresponding to the maximal ideal $\left(u - \frac{1}{\pi}\right) + (v)$ of $K[u,v]$; **IV-3** | 242
we have also $\kappa(x_2) = K$. As one will see in Ch. V, one can therefore *reglue* X_1 and X_2 along the two sub-preschemes $\{x_1\}$ and $\{x_2\}$ having respectively $\{x_1\}$ and $\{x_2\}$ as underlying spaces, following the unique Y-isomorphism of Z_1 on Z_2. Designate by X the Y-prescheme thus obtained, by x the point of X which is the image of x_1 and x_2. The "null section" g of X_1 ((**II**, 1.7.9)) is then also a Y-section of X; $f^{-1}(y)$ is equal to $(X_1)_y$, so $f^{-1}(y)$ has two irreducible components, respectively isomorphic to $(X_1)^0_y$ and $(X_2)^0_y$, having a unique point in common, see however the prop. (15.6.9).

(ii) Consider the morphism f defined in (12.2.3), (ii), which is proper and flat; moreover, the reciprocal morphism $g : Y \to X$ is the Y-section of X. We have then $\dim(X^0_{y_0}) = 1$ whereas $\dim(X^0_y) = 0$ for $y \neq y_0$, although X^0_y is geometrically irreducible for every $y \in Y$ (since X^0_y is not reduced); one sees therefore that as in (15.6.6), (iii), one cannot suppress the hypotheses α) and β). Moreover, X^0 is not a neighbourhood of $X^0_{y_0}$ in X, so we see that as in (15.6.4), one cannot dispense with the hypothesis that X' is reduced.

(iii) In Ch. VI, we will apply the preceding results to Y-*group preschemes* locally of finite type over a locally Noetherian prescheme Y. If G is such a prescheme, there exists the "unit section" g, and one shows that G_y is always geometrically irreducible and $\dim(G^0_y) = \dim(G_y)$, setting $G_y = f^{-1}(y)$. It then follows from (15.6.2) and (15.6.3) that the function $z \mapsto \dim(G_z)$ is upper semi-continuous, and that if it is constant in a neighbourhood of a point y, f is universally open at the points of G^0_y. If moreover G_y is geometrically reduced over $\kappa(y)$ at one of its points, one shows that G_y is *smooth* (6.8.1) over $\kappa(y)$ at all its points; it will follow then from (15.6.6) that if moreover $\mathscr{O}_y$ is reduced, then also f is smooth at all the points of G^0_y (though not necessarily at all points of G_y). These results will apply for example in the theory of Picard schemes, where we shall use a general theorem ensuring that, under certain conditions, the function $z \mapsto \dim(G_z)$ is locally constant. Let us remark that it is with applications of this nature in view that the statements such as (15.6.1) are given for morphisms *locally* of finite type, and not only for morphisms of finite type.

We note also that in the case of a Y-group prescheme G, the hypothesis β) of (15.6.6) is always satisfied.

15.7. Appendix: Valuative criteria for local properness This section gives complements to the valuative criterion of properness proved in ((**II**, 7.3.10)); it is independent from the rest of § 15.

Definition (15.7.1). — Let $f : X \to Y$ be a morphism of preschemes, y a point of Y. We say that f is *proper at the point* y if there exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is a proper morphism. We say that a subset Z of X is *proper over* Y *at the point* y if there exists an open neighbourhood U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U ((**II**, 5.4.10)) (which amounts to saying that for every closed sub-prescheme of X having Z as underlying space, the restriction of f to Z is proper at the point y).

Let $g : Y' \to Y$ be any morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')}$ and, if $p : X' \to X$ is the canonical projection, $Z' = p^{-1}(Z)$. Then, if Z is proper over Y at the point y, Z' is proper over Y' at every point y' above y ((**II**, 5.4.2)).

Proposition (15.7.2). — *Let Y be an integral locally Noetherian prescheme, X an integral prescheme, $f : X \to Y$ a separated, dominant, and of finite type morphism, y a point of Y. The following conditions are equivalent:*

label=a) *f is proper at the point y.*
lbbel=b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is proper.*
lcbel=c) *Every discrete valuation ring having $R(X)$ as field of fractions and dominating a local ring $\mathscr{O}$ dominates an $\mathscr{O}_x$ where necessarily $f(x) = y$.*

IV-3 | 243

Proof. The fact that $a)$ implies $b)$ follows from ((**II**, 5.4.2)), and the implication $b) \Rightarrow c)$ follows from ((**II**, 7.3.10)). It remains therefore to show that $c)$ implies $a)$. The question being local on Y, hence on the Noetherian Y, by virtue of the lemma of Chow ((**II**, 5.6.1)), there exists an integral prescheme X', a projective morphism $p : P \to Y$, an open dominant immersion $j : X' \to P$, and a projective and birational (hence surjective) morphism $g : X' \to X$ such that the diagram

$$\begin{tikzcd} P & X' \arrow[l, "j"'] \\ Y \arrow[u, "p"] & X \arrow[l, "f"] \arrow[u, "g"'] \end{tikzcd}$$

is commutative. Since X' is integral and j dominant, P is irreducible, and one can, replacing p by p_{red}, suppose P integral ((**I**, 5.2.2) and (**II**, 5.5.5), (vi)). It amounts to proving that there exists an open neighbourhood U of y such that $p^{-1}(U) = g^{-1}(f^{-1}(U)) = V$ will be an isomorphism on $p^{-1}(U)$, hence the restriction $V \to U$ of $f \circ g$ will be proper, and as the restriction $V \to f^{-1}(U)$ of g is surjective, the restriction $f^{-1}(U) \to U$ of f will be proper ((**II**, 5.4.3)). Since $T = P - j(X')$ is closed in P, $p(T)$ is closed in Y, and it suffices to show that $y \notin p(T)$, or even that every $z' \in P$ such that $p(z') = y$ *belongs* to $j(X')$. Now, the field of rational functions $R(P)$ is equal to $R(X')$, hence to $R(X)$ by construction; since one can restrict to the case where z' is not the generic point of P, there is a discrete valuation ring A having $R(X)$ as field of fractions and dominating $\mathscr{O}_{z'}$ ((**II**, 7.1.7)); since $p(z') = y$, A dominates $\mathscr{O}_y$, hence by hypothesis there exists $x \in X$ such that $f(x) = y$ and A dominates $\mathscr{O}_x$. Since g is proper, there exists $x' \in X'$ such that $g(x') = x$ and A dominates $\mathscr{O}_{x'}$ ((**II**, 7.3.10)); therefore z' and $j(x')$ are two points of P whose local rings $\mathscr{O}_z$ and $\mathscr{O}_{j(x')} = \mathscr{O}_{z'}$ are related by ((**I**, 8.1.4)); since P is a scheme, this implies $z' = j(x')$ ((**I**, 8.2.2)), which completes the proof. $\square$

Remarks (15.7.3). — One can in (15.7.2) suppress the hypothesis that Y is locally Noetherian, replacing in condition $c)$ the discrete valuation rings by valuation rings of any kind; the proof is then unchanged, taking into account ((**II**, 7.3.10)).

Corollary (15.7.4). — *Let Y be a Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, Z a subset of X such that the maximal points of its closure $\overline{Z}$ belong to Z (which is the case when Z is finite, or when Z is constructible ($\mathbf{0}_{\mathrm{III}}$, 9.2.2)). Let y be a point of Y. The following conditions are equivalent:*

label=a) *The set $\overline{Z}$ is proper over Y at the point y.*

lbbel=b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, and if $g_1 : X_1 \to X$ is the canonical projection and $Z_1 = g_1^{-1}(Z)$, the closure $\overline{Z_1}$ of Z_1 in X_1 is proper over Y_1.*

lcbel=c) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$, such that the image $g(y')$ of the closed point y' of Y' is y, we have the following property: setting $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$, $g' = g_{(X)} : X' \to X$, $Z' = g'^{-1}(Z)$, and denoting by η' the generic point of Y', for every point $x' \in Z' \cap f'^{-1}(\eta')$ rational over $\kappa(\eta')$, there exists a Y'-section of X' containing x'.* **IV-3** | 244

Proof. The fact that a) implies b) follows from the fact that $\overline{Z_1} \subset g_1^{-1}(\overline{Z})$ and from the fact that $g_1^{-1}(\overline{Z})$ is proper over Y_1 (15.7.1). The fact that b) implies c) follows from ((**II**, 7.3.3)). It remains therefore to see that c) implies a). It clearly suffices to prove that every irreducible component of $\overline{Z}$ is proper over Y at the point y, and the hypothesis allows us therefore to restrict to the case where Z is reduced to a single point z. We can evidently also, taking into account ((**I**, 5.2.2)) and ((**II**, 5.4.6)), suppose that X and Y are reduced; then (by definition of proper part ((**II**, 5.4.10))) suppose that $\overline{Z} = X = \{z\}$ and (applying again ((**I**, 5.2.2))) that f is dominant, hence X and Y integral and Noetherian; it therefore suffices to prove that f is proper at the point y (15.7.2). Let A' be a discrete valuation ring having $R(X) = \kappa(z)$ as field of fractions and dominating $\mathscr{O}_y$; with the notations of c), we have therefore $g(y') = y$, and $g(\eta')$ is the generic point of Y. Since by definition $\kappa(\eta') = \kappa(z)$, there exists a $\kappa(\eta)$-morphism $\operatorname{Spec}(\kappa(\eta')) \to \operatorname{Spec}(\kappa(z))$ transforming η' into z, hence a $\kappa(\eta')$-section of $f'^{-1}(\eta')$ ((**I**, 3.3.14)); if z' is the image of η' by this section, z' is therefore rational over $\kappa(\eta')$ and $g'(z') = z$. We conclude therefore from hypothesis c) that there exists a Y'-section h' of X' such that $h'(\eta') = z'$; hence $h = g' \circ h' : Y' \to X$ is the Y-morphism corresponding, and since $x = h(y')$; it is clear that $f(x) = y$ and that $A' = \mathscr{O}_{y'}$ dominates $\mathscr{O}_x$, which completes the proof. □

Corollary (15.7.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, y a point of Y. The following conditions are equivalent:*

label=a) *f is proper at the point y.*

lbbel=b) *If $Y_1 = \operatorname{Spec}(\mathscr{O}_y)$, $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \to Y_1$ is proper.*

lcbel=c) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$, such that the image $g(y')$ of the closed point y' of Y' is y, and denoting by η' the generic point of Y', there exists a Y'-section of X' containing x', for every $x' \in f'^{-1}(\eta')$, rational over $\kappa(\eta')$.*

Corollary (15.7.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a locally of finite type morphism. The following conditions are equivalent:*

label=a) *There exists a proper morphism $g : X \to Z$ and an immersion $h : Z \to Y$ such that $f = h \circ g$.*

lbbel=b) *The sub-space $f(X)$ is locally closed in Y, and if Z' is the sub-prescheme of Y induced by the image $f(X)$ of Z', so that f factorises uniquely as $f = j \circ g$ (where $j : Z'' \to Y$ is the canonical injection), then $g : X \to Z''$ is proper.*

lcbel=c) *For every $y \in f(X)$, f is proper at the point y.*

ldbel=d) *For every scheme $Y' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \to Y$ such that $g(Y') \subset f(X)$, every Y'-section rationnelle ((**I**, 7.1.2)) of $X' = X \times_Y Y'$ extends uniquely to a Y'-section of X'.*

Proof. It is clear that a) implies c), let us first remark that a) implies that $h(Z)$ is locally closed in Y and $g(X)$ is closed in Z, hence $f(X)$ is locally closed. Moreover, since f is locally closed in Y; for every $y \in f(X)$, there is therefore a neighbourhood U of y in Y such that $f(X) \cap U$ is closed in U, hence $f(X)$ is locally closed. Since this last is the underlying space of Z', and f factorises as in the statement of b); the fact that g is proper results then from the local character of (15.7.5), (a). It is clear that c) implies d), by the uniqueness of the extension of a Y'-section rationnelle resulting from the hypothesis that f is separated ((**I**, 7.2.2)). Let us finally prove that d) implies c); in the first place, from d), it follows that f is separated; on the other hand, we can then apply the criterion (15.7.5), c), which shows that f is proper at every point of $f(X)$. □

Remarks (15.7.7). — (i) In (15.7.4), *c*), (15.7.5), *c*), and (15.7.6), *d*), one can restrict to the case where the discrete valuation ring A' is *complete* and admits an algebraically closed residue field. Indeed, in the proof of (15.7.4), if one considers a complete discrete valuation ring A'' dominating A' and having an algebraically closed residue field, $X'' = X' \times_{Y'} Y''$ and $f'' = f_{(Y'')}$; but $X'' = X' \times_{Y'} \times_{Y'} Y''$, and it follows then from the remark ((**II**, 7.3.9), (i)) that f' is proper; by suite Y', X' and f' also verify the hypothesis *c*) ((**II**, 7.3.8)).

(ii) Taking into account ((**II**, 7.3.8)), condition *c*) of (15.7.5) can be replaced by the condition that f' is *proper*.

(15.7.8). We say that a morphism of preschemes $f : X \to Y$ is *submersive* if it is surjective and if the topology of Y is equal to the quotient of the topology of X by the equivalence relation defined by f. We say that f is *universally submersive* if for every base change $Y' \to Y$, the morphism $f' = f_{(Y')} : X \times_Y Y' \to Y'$ is submersive. Every surjective morphism that is *universally open*, or *universally closed*, or *faithfully flat and quasi-compact* is universally submersive (2.3.12). Given a morphism $f : X \to Y$, we say that a subset Z of X is *submersive over* $f(Z)$ if the topology of the sub-space $f(Z)$ of Y is equal to the quotient of the topology of the sub-space Z of X by the equivalence relation defined by $f|Z$. We say that Z is *universally submersive over* $f(Z)$ if, for every base change $g : Y' \to Y$, the inverse image $Z' = p^{-1}(Z)$ of Z by the projection $p : X \times_Y Y' \to Y'$ is submersive over $f'(Z') = g^{-1}(f(Z))$. We note that if the morphism f admits a Y-section $h : Y \to X$, every set Z containing $h(Y)$ is universally submersive over Y.

Proposition (15.7.9). — *Let Y be a locally Noetherian prescheme, y a point of Y, $f : X \to Y$ a separated morphism of finite type. Let Z be a locally constructible subset of X having the following properties:*

label=(i) For every generization y' of y, $Z_{y'} = Z \cap f^{-1}(y')$ is a connected component of $f^{-1}(y')$ and is geometrically connected over $\kappa(y')$ (4.5.2).

lbbel=(ii) Z_y is proper over $\mathrm{Spec}(\kappa(y))$.

lcbel=(iii) Z is universally submersive over $f(Z)$ (15.7.8).

Then Z is proper over Y at the point y (in other words (15.7.1) *there exists an open neighbourhood U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper over U).*

Proof. Using the method of ((8.1.2), *a*)) and ((8.10.5), (xii)), we are reduced to proving that when $Y = \mathrm{Spec}(A)$ is the spectrum of a *local* Noetherian ring, the hypotheses (i), (ii) and (iii) imply that Z is *proper* over Y. **IV-3** | 246

Let us first note that if A' is a local Noetherian ring, $\varphi : A \to A'$ a local homomorphism and $g : Y' \to Y$ the corresponding morphism, the inverse image Z' of Z by the canonical projection $p : X \times_Y Y' \to X$ has the properties (i), (ii), (iii) of the statement vis-à-vis Y'; using (1.8.2), taking into account (2.7.1), (vii), it therefore amounts to the same to prove that Z' is proper over Y', provided that A' is a flat A-module. We shall make use of this remark by taking $A' = \widehat{A}$ ($\mathbf{0}_{\mathrm{I}}$, 7.3.5), in other words we shall now limit ourselves to the case where A is *complete*. Since Z_y is a connected component of $f^{-1}(y)$, proper over $\kappa(y)$, it follows from ((**III**, 5.5.2)) that X is the *sum* of two sub-preschemes X_0, X_1 induced on open and closed subsets of X, such that $X_0 \cap f^{-1}(y) = Z_y$. Set $Z_0 = X_0 \cap Z$, $Z_1 = X_1 \cap Z$, so that these sets form a partition of Z into two open and closed subsets. Since the Z_y are connected, we necessarily have $Z_y \subset Z_0$ or $Z_y \subset Z_1$, hence $f(Z_0) \cap f(Z_1) = \varnothing$. But since Z_0 and Z_1 are saturated for the equivalence relation defined by $f|Z$, it follows from (iii) that $f(Z_0)$ and $f(Z_1)$ are both open in $f(Z)$. But $f(Z_0)$ contains y, and every neighbourhood of y in Y is equal to Y, so $f(Z_0) = f(Z)$, and hence $f(Z_1) = \varnothing$, so $Z_1 = \varnothing$ and $Z \subset X_0$.

We can therefore henceforth suppose additionally that f is *proper*, and it remains to prove that Z is closed in X. In other words, it will suffice to prove that a *constructible* subset Z of X *proper* over Y that satisfies only the two *sole* conditions (i) and (iii), is *closed* in X. By virtue of ($\mathbf{0}_{\mathrm{III}}$, 9.2.5), it suffices therefore to prove that for every $x' \in Z$ and every specialisation x of x' in X, we have $x \in Z$. Let A_1 be a complete discrete valuation ring, u a morphism of $Y_1 = \mathrm{Spec}(A_1)$ into X such that, if y_1 and y_1' are the closed point and the generic point of Y_1, $u(y_1) = x$, $u(y_1') = x'$ ((**II**, 7.1.7)); set $g = f \circ u : Y_1 \to Y$ and $X_1 = X \times_Y Y_1$ with $f_1 = f_{(Y_1)}$, so that $u = p \circ u_1$, where $p : X_1 \to X$ is the canonical projection. Moreover, $x_1 = u_1(y_1)$, $x_1' = u_1(y_1')$, and then $p(x_1) = x$ and $p(x_1') = x'$, and x_1 is a specialisation of x_1'. Moreover, $x_1' \in Z_1 = p^{-1}(Z)$; since we have remarked that the

conditions (i) and (iii) are stable under *every* base change, it is seen that we are reduced to the following situation: Y is the spectrum of a complete discrete valuation ring, $f(x) = y$, $y' = f(x')$ its generic point, there exists a Y-section h of X such that $h(y) = x$, $h(y') = x'$, and finally $x' \in Z$; we must prove that $x \in Z$. Now, Z_y is a connected component of $f^{-1}(y)$, *proper* over $\operatorname{Spec}(\kappa(y))$ since X is proper over Y. Applying once more ((**III**, 5.5.2)), one sees, as above, that there is an open and closed subset X' of X such that $X' \cap f^{-1}(y) = Z_y$; since Z_y is connected, we can suppose X' IV-3 | 247
connected (replacing as needed X' by that of its connected components containing Z_y). But then the same reasoning as at the beginning proves that necessarily $Z \subset X'$; since $h(Y)$ is connected and contains x', it is necessarily contained in X', hence $x = h(y) \in X' \cap f^{-1}(y) = Z_y$, which completes the proof. □

Corollary (15.7.10). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a separated morphism of finite type, universally submersive (15.7.8) and such that all the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then, if every $y \in Y$ is such that X_y is proper over $\kappa(y)$, f is proper at the point y.*

Proof. Taking into account (15.7.5), we are reduced to the case where $Y = \operatorname{Spec}(\mathscr{O}_y)$ since the hypotheses are stable under base change. But in this case it suffices to apply (15.7.9) to $Z = X$. □

Corollary (15.7.11). — *Let $f : X \to Y$ be a separated morphism of finite presentation; suppose that there exists a surjective morphism of finite presentation $g : Y' \to Y$, which is either proper or flat and quasi-compact, and such that if one sets $X' = X \times_Y Y'$, there exists a Y'-section of X' (or even a Y-morphism $Y' \to X$ ((**I**, 3.3.14))). Suppose finally that the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then the set U of $y \in Y$ such that X_y is proper over $\kappa(y)$ is open in Y, and the restriction $f^{-1}(U) \to U$ of f is proper.*

Proof. The question being local on Y, we can suppose $Y = \operatorname{Spec}(A)$ affine. Applying (8.9.1) to f and to g, one sees that there exists a Noetherian sub-ring A_0 of A and two morphisms of finite type $f_0 : X_0 \to Y_0 = \operatorname{Spec}(A_0)$, $g_0 : Y'_0 \to Y_0$ such that $X = X_0 \times_{Y_0} Y$, $Y' = Y'_0 \times_{Y_0} Y$, $f = (f_0)_{(Y)}$ and $g = (g_0)_{(Y)}$; moreover, using (8.10.5), (v), (vi) and (xii) and (11.2.7), we can suppose f_0 separated and proper (resp. flat) if g is proper (resp. flat). Furthermore, we can suppose that there exists a Y'_0-section g_0 of X_0 such that $g = (g_0)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_0 be its projection in Y_0; it follows from (4.5.13) that the $f_0^{-1}(y_0)$ are geometrically connected for every $y_0 \in Y_0$. Moreover, using (2.7.1), (vii), it amounts to the same to say that $f^{-1}(y)$ is proper over $\kappa(y)$ or that $f_0^{-1}(y_0)$ is proper over $\kappa(y_0)$, where y_0 is the projection of y in Y_0. These remarks show that we are reduced to proving the corollary when Y is *Noetherian*. Let us now prove the

Lemma (15.7.11.1). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a morphism surjective, which is either proper or flat and quasi-compact. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Then, if f' is submersive (resp. universally submersive), f is also.*

Proof. We can restrict to the case where f' is submersive. Indeed, suppose prouved in this case, and let us show the lemma also holds when f' is universally submersive. Let $Y_1 \to Y$ be any morphism, and set $Y'_1 = Y' \times_Y Y_1$; if $X'_1 = X' \times_{Y'} Y'_1$, we have $f'_1 = (f')_{(Y_1)} : X'_1 \to Y'_1$; by hypothesis $g_1 : Y'_1 \to Y_1$ is surjective, and proper (resp. flat and quasi-compact) if g is. Hence, if one sets $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, it follows from the hypothesis and from the fact that $X'_1 = X_1 \times_{Y_1} Y'_1$, that f_1 is submersive, hence f universally submersive. Suppose therefore only that f' is submersive. It is clear first of all that f is surjective. Let E be a subset of Y such that $f^{-1}(E)$ is closed in X; then, if $p : X' \to X$ is the canonical projection, $p^{-1}(f^{-1}(E)) = f'^{-1}(g^{-1}(E))$ is closed in X', hence, since f' IV-3 | 248
is submersive, $g^{-1}(E)$ is closed in Y'; but as g is also universally submersive (15.7.8), E is closed in Y, which proves the lemma. □

Since there exists by hypothesis a Y'-section of X', f' is universally submersive (15.7.8), hence f is also by the lemma (15.7.11.1). It suffices then to apply (15.7.10), noting that the set of $y \in Y$ such that f is proper at y is open in Y by definition. □

§16. Differential invariants. Differentially smooth morphisms

In this paragraph we will present, in global form, some notions of differential calculus particularly IV-4 | 5
useful in algebraic geometry. We will ignore many classic developments in differential geometry

(connections, infinitesimal transformations associated to vector fields, jets, etc.), although these notions are translated in a particularly natural way for schemes. We will similarly ignore phenomena exclusive to characteristic $p > 0$ (some of which are seen, in the affine case, in (**0**, 21). For certain complements to the differential formalism for preschemes the reader may consult Exposés II and VII of [eAG64] as well as subsequent chapters of this treatise.

16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathscr{O}_X), (Y, \mathscr{O}_Y)$ be two ringed spaces and $f = (\psi, \theta) : Y \to X$ a morphism of ringed spaces (**0**, 4.1.1) such that the homomorphism

$$\theta^\# : \psi^*(\mathscr{O}_X) \longrightarrow \mathscr{O}_Y$$

is surjective, so that $\mathscr{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$. We can then endow $\psi^*(\mathscr{O}_X)$ with the $\mathscr{I}_f$-preadic filtration.

Definition (16.1.2). — The $\mathscr{O}_Y$-augmented sheaf of rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ is called the n'th *normal invariant* of f; the ringed space $(Y, \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1})$ is called the n'th *infinitesimal neighbourhood* of Y along f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathscr{O}_X)$

$$\mathscr{G}r_\bullet(f) = \bigoplus_{n \geqslant 0} (\mathscr{I}_f^n / \mathscr{I}_f^{n+1}) \tag{16.1.2.1}$$

is called the sheaf of graded rings *associated to* f. The sheaf $\mathscr{G}r_1(f) = \mathscr{I}_f/\mathscr{I}_f^2$ is called the *conormal sheaf* of f (that will be denoted by $\mathscr{N}_{Y/X}$ when there is no risk of confusion).

It is clear that the $\mathscr{O}_{Y^{(n)}} = \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ (that we also denote $\mathscr{O}_{Y_f^{(n)}}$) form a projective system **IV-4 | 6**
of sheaves of rings on Y, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ for $n \leqslant m$ identifies $\mathscr{O}_{Y^{(n)}}$ with the quotient of $\mathscr{O}_{Y^{(m)}}$ by the power $(\mathscr{I}_f/\mathscr{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathscr{O}_{Y^{(n)}}$, kernel of $\phi_{0n} : \mathscr{O}_{Y^{(n)}} \to \mathscr{O}_Y$. The $Y^{(n)}$ therefore form an inductive system of ringed spaces, all having underlying space Y, and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \to X$ equal to (ψ, θ_n), where $\theta_n^\#$ is the canonical morphism $\psi^*(\mathscr{O}_X) \to \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$. It is clear that the sheaf $\mathscr{G}r_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathscr{O}_Y = \mathscr{G}r_0(f)$ and the $\mathscr{G}r_k(f)$ of $\mathscr{O}_Y$-modules.

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathscr{O}_Y$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_\bullet(f) \tag{16.1.2.2}$$

which coincides in degrees 0 and 1 with the identities.

Examples (16.1.3). —

(i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathscr{O}_y$) and that, if $x = \psi(y)$, $\theta^\# : \mathscr{O}_x \to \mathscr{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$. So the $\mathscr{O}_{Y^{(n)}}$ are identified with the rings $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$ and $\mathscr{G}r_\bullet(f)$ with the graded ring associated with the local ring $\mathscr{O}_x$ endowed with the $\mathfrak{m}_x$-preadic filtration.

(ii) Suppose that Y is a closed subset of an open subspace U of X and that the $\mathscr{O}_Y$ is induced on Y by a quotient sheaf $\mathscr{O}_U/\mathscr{I}$, where $\mathscr{I}$ is an ideal of $\mathscr{O}_U$ such that $\mathscr{I}_x = \mathscr{O}_x$ for every $x \notin Y$; if X is a locally ringed space we also suppose that $\mathscr{I}_x \neq \mathscr{O}_x$ for $y \in Y$ so that $(Y, \mathscr{O}_Y)$ is a locally ringed space.

Let $\psi_0 : Y \to U$ be the canonical injection and denote by $\theta_0 : \mathscr{O}_U \to (\psi_0)_*(\mathscr{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathscr{O}_U) = \mathscr{O}_U|Y \to (\mathscr{O}_U/\mathscr{I})|Y$, so that $j_0 = (\psi_0, \theta_0) : Y \to U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \to X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \to X$ is the canonical injection and $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathscr{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathscr{O}_U/\mathscr{I}^{n+1})$, and we have $(\psi_0)_*(\mathscr{O}_{Y^{(n)}}) = \mathscr{O}_U/\mathscr{I}^{n+1}$, and $\mathscr{G}r_n(j) = \mathscr{G}r_n(j_0) = \psi_0^*(\mathscr{I}^n/\mathscr{I}^{n+1}) = j_0^*(\mathscr{I}^n/\mathscr{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathscr{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathscr{O}_Y$-module*, or *a fortiori* with a structure of an $\mathscr{O}_Y$-algebra. The data of such structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n}; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \to Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \to Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \to X$ be an immersion of preschemes. Then:*

(i) *$\mathscr{G}r_\bullet(f)$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra.* **IV-4** | 7

(ii) *The $Y^{(n)}$ are preschemes, canonically isomorphic to subpreschemes of X.*

(iii) *Every homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n}, makes the $\mathscr{O}_{Y^{(n)}}$ and $\mathscr{O}_{Y^{(k)}}$ for $k \leqslant n$ quasi-coherent $\mathscr{O}_Y$-algebras; the $\mathscr{O}_Y$-module structures induced from the above structures on the $\mathscr{G}r_k(f)$ for $k \leqslant n$ coincide with the ones defined in (16.1.2).*

Proof. (i) Since the question is local on X and Y, we can reduce to the case where Y is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$; since $\mathscr{O}_Y$ is the restriction to Y of $\mathscr{O}_X/\mathscr{I}$ the assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathscr{I}^{n+1}$ of $\mathscr{O}_X$. Finally, to prove (iii) we notice that the data of λ_n makes the ideal $\mathscr{I}/\mathscr{I}^n$ of the augmentation ϕ_{0n} and their quotients $\mathscr{I}/\mathscr{I}^{k+1} (1 \leqslant k \leqslant n)$ $\mathscr{O}_Y$-modules, and it suffices to prove by induction on k that the $\mathscr{I}/\mathscr{I}^{k+1}$ are quasi-coherent $\mathscr{O}_Y$-modules and the structure of quotient $\mathscr{O}_Y$-module induced on $\mathscr{I}^k/\mathscr{I}^{k+1}$ is the same as defined on (16.1.2). The second assertion is immediate, $\mathscr{I}^k/\mathscr{I}^{k+1}$ being killed by $\mathscr{I}/\mathscr{I}^{n+1}$; the first result, by induction on k, is trivial for $k = 1$ and for $\mathscr{I}/\mathscr{I}^{k+1}$ being an extension of $\mathscr{I}/\mathscr{I}^k$ by $\mathscr{I}^k/\mathscr{I}^{k+1}$ (1.4.17). □

Corollary (16.1.6). — *Under the general hypotheses of (16.1.5), if the immersion f is locally of finite presentation then the $\mathscr{G}r_n(f)$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. Indeed, with the notation from the proof of (16.1.5), $\mathscr{I}$ is an ideal of finite type of $\mathscr{O}_X$ (1.4.7), therefore the $\mathscr{I}^n/\mathscr{I}^{n+1}$ are $\mathscr{O}_Y$-modules of finite type, hence the conclusion. □

Corollary (16.1.7). — *Under the general hypotheses of (16.1.5), let $g : X \to Y$ be a morphism of preschemes, left inverse to f. Therefore, for every n, the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n}, making $\mathscr{O}_{Y^{(n)}}$ a quasi-coherent $\mathscr{O}_Y$-algebra; via these homomorphisms, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ $(n \leqslant m)$ are homomorphisms of $\mathscr{O}_Y$-algebras. Also, if g is locally of finite type, then the $\mathscr{O}_{Y^{(n)}}$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. The first assertion is an immediate result from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathscr{G}r_n(f)$ being then quasi-coherent $\mathscr{O}_Y$-modules of finite type by (16.1.6), the same goes for the $\mathscr{O}_Y$-modules $\mathscr{I}/\mathscr{I}^{n+1}$, being extensions of a finite number of the $\mathscr{G}r_k(f)$ (**III**, 1.4.17). □

Proposition (16.1.8). — *Let X be a locally Noetherian prescheme, $j : Y \to X$ an immersion; Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathscr{G}r_n(j)$ are coherent $\mathscr{O}_Y$-modules and the $\mathscr{G}r_\bullet(j)$ is a coherent sheaf of rings over the space Y.*

Proof. Everything is local on X and Y, so we reduce to the case where X is affine and j is a closed immersion and therefore all the assertions are evident except for the last, which follows from the fact that if A is a Noetherian ring and $\mathfrak{I}$ is an ideal of A, then $\mathrm{gr}^\bullet_{\mathfrak{I}}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (**0**, 5.3.7). □

Proposition (16.1.9). — *Let X be a prescheme, $j : Y \to X$ an immersion locally of finite presentation, y a point of Y. The following conditions are equivalent:* **IV-4** | 8

(a) *There exists an open neighbourhood U of y in Y such that $j|U$ is a homeomorphism of U onto an open set of X.*

(b) *There is an integer $n > 0$ such that the canonical homomorphism*

$$(\phi_{n-1,n})_y : \mathscr{O}_{Y^{(n)},y} \longrightarrow \mathscr{O}_{Y^{(n-1)},y}$$

is bijective.

(c) *There is an integer $n > 0$ such that $(\mathscr{G}r_n(j))_y = 0$.*

In addition, if the integer n satisfies (b) *or* (c), *then there is a neighbourhood V of y in Y such that $\mathscr{G}r_m(j)|V = 0$ for $m \geqslant n$ and that $\phi_{nm}|V : \mathscr{O}_{Y^{(m)}}|V \to \mathscr{O}_{Y^{(n)}}|V$ is bijective for $m \geqslant n$.*

Proof. Since the question is local on Y, we can restrict ourselves to the case where j is a closed immersion, Y being defined by a quasi-coherent ideal *of finite type* $\mathfrak{I}$ of $\mathscr{O}_X$. The equivalence of (b) and (c), for a given n, is immediate; also, since $\mathscr{I}^n/\mathscr{I}^{n+1}$ is an $\mathscr{O}_X$-module of finite type, there is an open neighbourhood U of y in X such that $\mathscr{I}^n|U = \mathscr{I}^{n+1}|U$ (**0**, 5.2.2), so we also have $\mathscr{I}^n|U = \mathscr{I}^m|U$ for $m \geqslant n$ proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the cases where the underlying space of Y is equal to the underlying space of X and where $\mathscr{I}$ is generated by a finite number of sections over X: since $\mathscr{I}$ is contained in the nilradical $\mathscr{N}$ of $\mathscr{O}_X$ (**I**, 5.1.2), it is now nilpotent which proves b). Finally, to prove that (b) implies (a), we can restrict ourselves to the case where $\mathscr{I}^n = \mathscr{I}^m$; therefore, for every $y \in Y$, since $\mathscr{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathscr{O}_{X,y}$, we must have $\mathscr{I}_y^n = 0$ because of Nakayama's lemma, since $\mathscr{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathscr{I}_x^n = 0$ is an open U of X contained in Y (**0**, 5.2.2); since on the other hand $\mathscr{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. □

Corollary (16.1.10). — *For a restriction of the immersion j to an open neighbourhood of y in Y to be an open immersion (in other words, for j to be a* local isomorphism *on the point y), it is necessary and sufficient that $(\mathscr{G}r_1(j))_y = (\mathscr{N}_{Y/X})_y = 0$.*

Proof. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. □

Remark (16.1.11). —

(i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathscr{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings over Y is called the *normal invariant of infinite order* of f, and sometimes denoted by $\mathscr{O}_{Y^{(\infty)}}$. When X is a locally Noetherian prescheme, $j : Y \to X$ a closed immersion, Y then is a closed subprescheme of X defined by a coherent ideal $\mathscr{I}$ and $\mathscr{O}_{Y^{(\infty)}}$ is exactly the *formal completion* of $\mathscr{O}_X$ along Y (**I**, 10.8.4), and $Y^{(\infty)} = (Y, \mathscr{O}_{Y^{(\infty)}})$ is the formal prescheme that is the *completion* of X along Y (**I**, 10.8.5). In all cases, we could say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (via the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n.

(ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \to X$, it can happen that the homomorphism $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Y$ is surjective without f being a local immersion and **IV-4 | 9**
without f being injective. We have an example by taking Y to be a sum of preschemes Y_λ all isomorphic to $\mathrm{Spec}(\mathscr{O}_x)$, where $x \in X$, ad taking f to be the morphism equal to the canonical morphism in each of the Y_λ.

16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \to X$ and $f' = (\psi', \theta') : Y' \to X'$ by two morphisms of ringed spaces such that $\theta^\#$ and $\theta'^\#$ are surjective; consider a commutative diagram of morphisms of ringed spaces

(16.2.1.1)
$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \end{tikzcd}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X))$ and as a result a commutative diagram of homomorphisms of sheaves of rings over Y'

$$\begin{tikzcd} \rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X)) \arrow[r, "\psi'^*(\mu^\#)"] \arrow[d, "\rho^*(\theta^\#)"'] & \psi'^*(\mathscr{O}_{X'}) \arrow[d, "\theta'^\#"] \\ \rho^*(\mathscr{O}_Y) \arrow[r, "\lambda^\#"'] & \mathscr{O}_{Y'} \end{tikzcd}$$

from which we conclude, if $\mathscr{I}$ and $\mathscr{I}'$ are the kernels of $\theta^\#$ and $\theta'^\#$, that we have $\psi'^*(\mu^\#)(\rho^*(\mathscr{I})) \subset \mathscr{I}'$, having in mind the exactness of the functor ρ^*. We deduce that, for every integer n, $\psi'^*(\mu^\#)(\rho^*(\mathscr{I}^n)) \subset \mathscr{I}'^n$, which shows that $\psi'^*(\mu^\#)$ defines, passing to the quotients, a homomorphism of sheaves of rings

(16.2.1.2)
$$\nu_n : \rho^*(\psi^*(\mathscr{O}_X)/\mathscr{I}^{n+1}) \longrightarrow \psi'^*(\mathscr{O}_{X'})/\mathscr{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, \nu_n) : Y'^{(n)} \to Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{tikzcd} Y^{(n)} \arrow[r, "h_{mn}"] & Y^{(m)} \arrow[r, "h_m"] & X \\ Y'^{(n)} \arrow[u, "w_n"] \arrow[r, "h'_{mn}"'] & Y'^{(m)} \arrow[u, "w_m"] \arrow[r, "h'_m"'] & X' \arrow[u, "v"'] \end{tikzcd} \qquad (n \leqslant m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness IV-4 | 10
of the functor ρ^*, we obtain a di-homomorphism of graded algebras (relative to the morphism $\lambda^\# : \rho^*(\mathscr{O}_Y) \to \mathscr{O}_{Y'}$)

(16.2.1.3)
$$\operatorname{gr}(u) : \rho^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f')$$

(or, if you like, a ρ-morphism (**0**, 3.5.1) $\mathscr{G}r_\bullet(f) \to \mathscr{G}r_\bullet(f')$), and in particular a di-homomorphism of conormal sheaves

$$\operatorname{gr}_1(u) : \rho^*(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_1(f').$$

It is also immediate that these homomorphisms give rise to a commutative diagram

(16.2.1.4)
$$\begin{tikzcd} \rho^*(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f))) \arrow[r] \arrow[d, "\mathbf{S}(\operatorname{gr}_1(u))"'] & \rho^*(\mathscr{G}r_\bullet(f)) \arrow[d, "\operatorname{gr}(u)"] \\ \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f')) \arrow[r] & \mathscr{G}r_\bullet(f') \end{tikzcd}$$

where the horizontal arrow are the canonical morphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \\ Y'' \arrow[u, "u'"] \arrow[r, "f''"'] & X'' \arrow[u, "v'"'] \end{tikzcd}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^{\#}$ is surjective, and if w'_n and w''_n are defined from u', v' for one and $u'' = u \circ u'$, $v'' = v \circ v'$ for the other, we have $w''_n = w_n \circ w'_n$, which follows immediately from the definitions and from (**0**, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^{*}(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathscr{G}r_\bullet(f)$ *depend functorially* on f.

Proposition (16.2.2). — *With the notation and hypotheses of (16.2.1), suppose also that f, f', u, and v are morphisms of preschemes. We have:*

(i) *The morphisms $w_n : Y'^{(n)} \to Y^{(n)}$ are morphisms of preschemes.*
(ii) *If $Y' = Y \times_X X'$, u and f' the canonical projections, and if f is an immersion or if v is flat, we have $Y'^{(n)} = Y^{(n)} \times_X X'$.*
(iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes I : \mathscr{G}r_\bullet(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_\bullet(f')$$

is bijective (resp. surjective).

Proof.

(i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho^{*}_{y'}(\theta^{\#}_{\psi'(y')})$ is a *local* homomorphism (**I**, 1.6.2), so w_n is a morphism of preschemes (**I**, 2.2.1). **IV-4 | 11**

(ii) and (iii) If f is an immersion, we can restrict ourselves to the case where f is a closed immersion, Y being defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$ and $Y^{(n)}$ by the ideal $\mathscr{I}^{n+1}$; the assertions follows from (**I**, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$ are affines, A' being a flat A-module; so $Y' = \mathrm{Spec}(B')$ where $B' = B \otimes_A A'$; in addition, if $\mathfrak{I}$ is the kernel of the homomorphism $A \to B$, the kernel $\mathfrak{I}'$ of $A' \to B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ is equal to

$$\begin{aligned} &\psi'^{*}(\sigma^{*}((\mathfrak{I}^n/\mathfrak{I}^{n+1})^{\sim}) \otimes_{\sigma^{*}(\mathscr{O}_X)} \mathscr{O}_{X'}) = \\ &\quad \psi'^{*}(\sigma^{*}((\mathfrak{I}^n/\mathfrak{I}^{n+1}))^{\sim}) \otimes_{\psi'^{*}(\sigma^{*}(\mathscr{O}_X))} \psi'^{*}(\mathscr{O}_{X'}) = \rho^{*}(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^{*}(\psi^{*}(\mathscr{O}_X))} \psi'^{*}(\mathscr{O}_{X'}) \end{aligned}$$

and in particular for $n = 0$, we have

$$\mathscr{O}_{Y'} = \rho^{*}(\mathscr{O}_Y) \otimes_{\rho^{*}(\psi^{*}(\mathscr{O}_X))} \psi'^{*}(\mathscr{O}_{X'})$$

from which we have canonical isomorphism of $\mathscr{I}'^n/\mathscr{I}'^{n+1}$ with

$$\rho^{*}(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^{*}(\mathscr{O}_Y)} \mathscr{O}_{Y'} = (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

which proves (iii). Let now $C_n = \Gamma(Y, \mathscr{O}_{Y^{(n)}}), C'_n = \Gamma(Y', \mathscr{O}_{Y'^{(n)}})$. As $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \to C_{n-1}$ (resp. $C'_n \to C'_{n-1}$) is $\Gamma(Y, \mathscr{I}^n/\mathscr{I}^{n+1})$ (resp. $\Gamma(Y, \mathscr{I}'^n/\mathscr{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathfrak{K}_n \otimes_A A' \arrow[r] \arrow[d, "r"] & C_n \otimes_A A' \arrow[r] \arrow[d, "s_n"] & C_{n-1} \otimes_A A' \arrow[r] \arrow[d, "s_{n-1}"] & 0 \\ 0 \arrow[r] & \mathfrak{K}'_n \arrow[r] & C'_n \arrow[r] & C'_{n-1} \arrow[r] & 0 \end{tikzcd}$$

where the vertical arrow of the left is bijective and the two lines are exact (A' being a flat A-module). We deduce by induction that s_n is bijective for every n, because it is true by hypothesis for $n = 0$, and is deduced by application of the five lemma for all n. That proves the second assertion of (ii).

□

Corollary (16.2.3). — *Let $g : X \to Y$, $u : Y' \to Y$ be two morphisms of preschemes, $X' = X \times_Y Y'$, $g' : X' \to Y'$ and $v : X' \to X$ by the canonical projections. Let $f : Y \to X$ by a Y-section of X (and therefore an immersion), $f' = f_{(Y')} : Y' \to X'$ the Y'-section of X' deduced from f by the base change u. We have:*

(i) *The morphism $w_n : Y'^{(n)}_{f'} \to Y^{(n)}_f$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : Y'^{(n)}_{f'} \to X'$ identifies $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_X X'$.*

(ii) *If we endow $\mathscr{O}_{Y^{(n)}_f}$ (resp. $\mathscr{O}_{Y'^{(n)}_{f'}}$) with the structure of an $\mathscr{O}_Y$-algebra defined by g (resp. with the structure of an $\mathscr{O}_{Y'}$-algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathscr{O}_{Y'}$-algebras*

(16.2.3.1) $$\rho^*(\mathscr{O}_{Y^{(n)}_f}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{O}_{Y'^{(n)}_{f'}}$$

induced by the homomorphism v_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathscr{O}_{Y'}$-modules **IV-4** | 12

(16.2.3.2) $$\mathrm{Gr}_1(u) : \mathscr{G}r_1(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_1(f')$$

is bijective.

Proof.

(i) Let us first note that $f' : Y' \to X'$ and $u : Y' \to Y$ identifies Y' with the product $Y \times_X X'$ (via the structure morphisms $f : Y \to X$ and $v : X' \to X$) (14.5.12.1). The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.

(ii) The commutative diagram

$$\begin{tikzcd} Y^{(n)}_f \arrow[d, "h_n"] & Y'^{(n)}_{f'} \arrow[l, "w_n"'] \arrow[d, "h'_n"] \\ X \arrow[d, "g"] & X' \arrow[l, "v"'] \arrow[d, "g'"] \\ Y & Y' \arrow[l, "u"'] \end{tikzcd}$$

identifies $Y'^{(n)}_{f'}$ with the product $Y^{(n)}_f \times_X X'$, so (**I**, 3.3.9) it identifies (via the morphisms $g' \circ h'_n$ and w_n) $Y'^{(n)}_{f'}$ to the product $Y^{(n)}_f \times_Y Y'$. Since $Y^{(n)}_f$ (resp. $Y'^{(n)}_{f'}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathscr{O}_Y$-algebra $\mathscr{O}_{Y^{(n)}_f}$ (resp. to the $\mathscr{O}_{Y'}$-algebra $\mathscr{O}_{Y'^{(n)}_{f'}}$), the fact that the canonical homomorphism (16.2.3.1) is bijective follows from (**II**, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathscr{O}_{Y^{(n)}_f} \to \mathscr{O}_Y$ and $\mathscr{O}_{Y'^{(n)}_{f'}} \to \mathscr{O}_{Y'}$; since $\mathscr{O}_{Y^{(n)}_f}$ is a direct sum (as an $\mathscr{O}_Y$-module) of $\mathscr{O}_Y$ and the augmentation ideal $\mathscr{I}/\mathscr{I}^{n+1}$, we can therefore see that the canonical homomorphism (16.2.3.1), restricted to $\mathscr{I}/\mathscr{I}^{n+1} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, is a bijection of the latter onto $\mathscr{I}'/\mathscr{I}'^{n+1}$. For $n = 1$ this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geqslant 2$. However:

Corollary (16.2.4). — *Under the hypotheses of (16.2.3), suppose that $u : Y' \to Y$ is a flat morphism (resp. that the $\mathscr{G}r_n(f)$ are flat $\mathscr{O}_Y$-modules for $n \leqslant m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathscr{G}r_n(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_n(f')$$

is bijective for all n (resp. for $n \leqslant m$).

Proof. If u is flat, then we deduce by base change that the same is true for $v : X' \to X$, and we already know in this case that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathscr{G}r_n(f)$ are flat for $n \leqslant m$, then we first see by induction on n that the same holds for $\mathscr{I}/\mathscr{I}^{n+1}$ for $n \leqslant m$, because of the exact sequences

$$0 \longrightarrow \mathscr{I}^n/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^n \longrightarrow 0$$

(**0**, 6.1.2); in addition, we have the commutative diagram **IV-4** | 13

$$\begin{tikzcd}
0 \arrow[r] & (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^n) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{I}'^n/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^n \arrow[r] & 0
\end{tikzcd}$$

in which the lines are exact (the first by flatness (**0**, 6.1.2)) and the two last vertical arrows are bijective by virtue of (16.2.2, (ii)); hence the conclusion. □

Remarks (16.2.5). —

(i) The reasoning of (16.2.2, (i)) still applies to (16.2.1.1) when these are morphisms of *locally ringed spaces* (**I**, 1.8.2).

(ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{I} = 0$ but the kernel $\mathfrak{I}'$ of $A' \to B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n, but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty}(\mathbf{Z}/m^h\mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the particular case of the diagram (16.2.1.1) where $X' = X$, v is the identity, X a prescheme, Y a subprescheme of X, Y' a subprescheme of Y, f, u, and $f' = f \circ u$ the canonical injections; the di-homomorphism (16.2.1.3) gives us, by tensoring with $\mathscr{O}_{Y'}$ over $\rho^*(\mathscr{O}_Y)$, a homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.1) $$u^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f').$$

On the other hand, we identify $\mathscr{O}_Y$ to $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$ and $\mathscr{O}_{Y'}$ to $\rho^*(\mathscr{O}_Y)/\mathscr{I}_u$; since ρ^* is an exact functor, we have $\rho^*(\mathscr{O}_Y) = \rho^*(\psi^*(\mathscr{O}_X))/\rho^*(\mathscr{I}_f) = \psi'^*(\mathscr{O}_X)/\rho^*(\mathscr{I}_f)$, and since $\mathscr{O}_{Y'}$ is moreover identified with $\psi'^*\mathscr{O}_X/\mathscr{I}_{f'}$, we see that $\mathscr{I}_u = \mathscr{I}_{f'}/\rho^*(\mathscr{I}_f)$. We deduce that for every integer n there is a canonical homomorphism $\mathscr{I}_{f'}^n/\mathscr{I}_{f'}^{n+1} \to \mathscr{I}_u^n/\mathscr{I}_u^{n+1}$, from which we have a canonical morphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.2) $$\mathscr{G}r_\bullet(f') \longrightarrow \mathscr{G}r_\bullet(u).$$

Proposition (16.2.7). — *Let X be a prescheme, Y a subprescheme of X, Y' a subprescheme of Y, $j : Y' \to Y$ the canonical injection. We then have an exact sequence of conormal sheaves ($\mathscr{O}_{Y'}$-modules)*

(16.2.7.1) $$j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

where the arrows are the degree 1 components of the canonical homomorphisms (16.2.6.1) and (16.2.6.2).

Proof. The problem being local, we can restrict to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/\mathfrak{I})$ and $Y' = \mathrm{Spec}(A/\mathfrak{K})$, $\mathfrak{I}$ and $\mathfrak{K}$ being ideals of A such that $\mathfrak{I} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \to \mathfrak{K}/\mathfrak{K}^2 \to (\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2 \to 0$ is exact, which is **IV-4** | 14
immediate given that the image of $\mathfrak{I}/\mathfrak{K}\mathfrak{I}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{I}+\mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2$ is identified with $\mathfrak{K}/(\mathfrak{I}+\mathfrak{K}^2)$. □

It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{I} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{I}+\mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \to S$ be a morphism of preschemes, $\Delta_f : X \to X \times_S X$ the corresponding diagonal morphism, which is an immersion (**I**, 5.3.9). We denote by $\mathscr{P}^n_f$ or $\mathscr{P}^n_{X/S}$, and call the *sheaf of principal parts of order n of the S-prescheme X*, the $\mathscr{O}_X$-augmented sheaf of rings, n-th normal invariant of Δ_f (16.1.2). We will also write $\mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S} = \varprojlim_n \mathscr{P}^n_{X/S}$, $\mathscr{G}r_n(\mathscr{P}_f) = \mathscr{G}r_n(\mathscr{P}_{X/S}) = \mathscr{G}r_n(\Delta_f)$ (16.1.2); the $\mathscr{O}_X$-module $\mathscr{G}r_1(\Delta_f)$, augmentation sheaf of ideals of $\mathscr{P}^1_{X/S}$, is denoted by Ω^1_f or $\Omega^1_{X/S}$, and is called the $\mathscr{O}_X$-module of 1-*differentials* of f, or of X with respect to S, or of the S-prescheme X.

It follows from this definition that $\mathscr{P}^0_{X/S}$ is canonically identified with $\mathscr{O}_X$ (16.1.2).

We have (16.1.2.2) a canonical surjective morphism of graded $\mathscr{O}_X$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.3.1.1}$$

And it follows from Definition (16.3.1) that for every open U of X we have $\mathscr{P}^n_{f|U} = \mathscr{P}^n_f|U$, $\mathscr{P}^\infty_{f|U} = \mathscr{P}^\infty_f|U$, $\mathscr{G}r_n(\mathscr{P}_{f|U}) = \mathscr{G}r_n(\mathscr{P}_f)|U$, $\Omega^1_{f|U} = \Omega^1_f|U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Denote by p_1, p_2 the two canonical projections of the product $X \times_S X$; since Δ_f is an X-section of $X \times_S X$ for both p_1 and p_2, *each* of these morphisms define, for all n, a homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$, right inverse of the augmentation $\mathscr{P}^n_{X/S} \to \mathscr{O}_X$ (16.1.7); we can also say that we thus define on $\mathscr{P}^n_{X/S}$ *two quasi-coherent augmented $\mathscr{O}_X$-algebra* structures; the corresponding $\mathscr{O}_X$-module structures on $\mathscr{G}r_n(\mathscr{P}^n_{X/S})$ are the same. We also have, by passing to the limit, two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^\infty_{X/S}$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \to X \times_S X$ is an *involutive automorphism* of $X \times_S X$, called the *canonical symmetry*, such that

$$p_1 \circ s = p_2, \qquad p_2 \circ s = p_1, \qquad s \circ \Delta_f = \Delta_f. \tag{16.3.3.1}$$

If we put $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ $(i = 1, 2)$, $\Delta_f = (\delta, \nu)$, $\lambda^\#$ is then an isomorphism of $\rho^*(\pi_1^*(\mathscr{O}_X))$ onto $\pi_2^*(\mathscr{O}_X)$, and $\delta^*(\lambda^\#)$ fixes $\delta^*(\mathscr{O}_{X \times_S X})$ and the kernel $\mathscr{I}$ of the homomorphism $\nu^\# : \delta^*(\mathscr{O}_{X\times_S X}) \to \mathscr{O}_X$. Therefore:

Proposition (16.3.4). — *The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the* canonical symmetry*) is an involutive automorphism of the projective system $(\mathscr{P}^n_{X/S})$ of $\mathscr{O}_X$-augmented sheaves of rings, and as a result also of the projective limit $\mathscr{P}^\infty_{X/S}$. This automorphism permutes the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$.* **IV-4** | 15

(16.3.5). In what follows, the two $\mathscr{O}_X$-algebra structures defined on the $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$ will play very different roles: *we will now agree, unless said otherwise, that when $\mathscr{P}^n_{X/S}$ or $\mathscr{P}^\infty_{X/S}$ is considered as an $\mathscr{O}_X$-algebra, it is the algebra structure induced by p_1.*

For every open U of X and every section $t \in \Gamma(U, \mathscr{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^n_{X/S})$ (resp. $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^\infty_{X/S})$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d^n_f, or $d^n_{X/S}$ (resp. d^∞_f, or $d^\infty_{X/S}$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ (resp. $\mathscr{O}_X \to \mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S}$) induced by p_2. For every open U of X, and every $t \in \Gamma(U, \mathscr{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t. We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega^1_{X/S})$, also denoted $d_{X/S}(t)$).

It follows immediately [5] from this definition that we have

$$d(t_1 t_2) = t_1 dt_2 + t_2 dt_1 \tag{16.3.6.1}$$

for every t_1, t_2 in $\Gamma(U\mathscr{O}_X)$, that is, d is a *derivation* of the ring $\Gamma(U, \mathscr{O}_X)$ in the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

[5] [Trans.] This is, locally we have (**0**, 20.1.1).

In all notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \mathrm{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine schemes, B then being an A-algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \to B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{J} = \mathfrak{J}_{B/A}$ (**0**, 20.4.1); $\mathscr{P}_f^n$ is the structure sheaf of the prescheme $\mathrm{Spec}(P_{B/A}^n)$, where

$$P_{B/A}^n = (B \otimes_A B)/\mathfrak{J}^{n+1};$$

$\mathscr{G}r_\bullet(\mathscr{P}_f)$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the graded B-module

$$\mathrm{gr}_{\mathfrak{J}}^\bullet(B \otimes_A B) = \bigoplus_{n \geqslant 0} (\mathfrak{J}^n/\mathfrak{J}^{n+1});$$

in particular $\Omega_f^1 = \Omega_{X/S}^1$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the B-module of 1-differentials of B over A, $\Omega_{B/A}^1$ (**0**, 20.4.3). The projection morphisms $p_1 : X \times_S X \to X$, $p_2 : X \times_S X \to X$ corresponding to the two homomorphisms of rings $j_1 : B \to B \otimes_A B$, $j_2 : B \to B \otimes_A B$ such that $j_1(b) = b \otimes 1$, $j_2(b) = 1 \otimes b$, so that (by the convention of (16.3.5)), $P_{B/A}^n$ is always considered as a B-algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \to P_{B/A}^n$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \to P_{B/A}^n$ is denoted by $d_{B/A}^n$ and corresponds to $d_{X/S}^n$ acting on $\Gamma(X, \mathscr{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A}t$, defined in (**0**, 20.4.6).

If $\pi_n : B \otimes_A B \to P_{B/A}^n$ is the canonical homomorphism, so we have, in light of the preceding definitions,

(16.3.7.1) $$\pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d_{B/A}^n(b') \quad \text{for } b \in B, b' \in B.$$

Proposition (16.3.8). — *The image of the canonical homomorphism $d_{X/S}^n : \mathscr{O}_X \to \mathscr{P}_{X/S}^n$ generates the* **IV-4** | 16 *$\mathscr{O}_X$-module $\mathscr{P}_{X/S}^n$.*

Proof. We immediately reduce to the case where $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine and the proposition follows from (16.3.7.1) since π_n is surjective. We note that in general $d_{X/S}^n$ *is not surjective* (even for $n = 1$). □

Proposition (16.3.9). — *Suppose that $f : X \to S$ is a morphism locally of finite type. Then the $\mathscr{P}_f^n$ and the $\mathscr{G}r_n(\mathscr{P}_f)$ are quasi-coherent $\mathscr{O}_X$-modules of finite type.*

Proof. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (**I**, 4.3.1). □

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "w"] \end{tikzcd}$$

We deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] \end{tikzcd}$$

where v is the composite homomorphism (**I**, 5.3.5) and (**I**, 5.3.15).

(16.4.1.2) $$X' \times_{S'} X' \xrightarrow{(p_1', p_2')_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

So we induce from u and v, as explained in (16.2.1), homomorphisms of augmented sheaves of rings

(16.4.1.3) $$\nu_n : \rho^*(\mathscr{P}_{X/S}^n) \longrightarrow \mathscr{P}_{X'/S'}^n$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system, and therefore give at the limit a homomorphism of sheaves of graded rings

(16.4.1.4) $$\nu_\infty : \rho^*(\mathscr{P}^\infty_{X/S}) \longrightarrow \mathscr{P}^\infty_{X'/S'};$$

on the other hand, by passing to the quotient, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

(16.4.1.5) $$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram **IV-4** | 17

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] & X'' \arrow[l, "u'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "w"] & S'' \arrow[l, "w'"] \end{tikzcd}$$

we deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] & X'' \arrow[l, "u'"'] \arrow[d, "\Delta_{f''}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] & X'' \times_{S''} X'' \arrow[l, "v'"] \end{tikzcd}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f'. We verify immediately that if $u'' = u \circ u'$, $w'' = w \circ w'$, then the composite homomorphism $v \circ v'$ is equal to the homomorphism v'' deduced from u'', v'', f, f'' as v is from u, w, f, f'. If we put $u' = (\rho', \lambda')$, $u'' = (\rho'', \lambda'')$ it follows (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathscr{P}^n_{X/S}) \to \mathscr{P}^n_{X''/S''}$ is equal to the composite

$$\rho'^*(\rho^*(\mathscr{P}^n_{X/S})) \xrightarrow{\rho'^*(\nu^\#_n)} \rho'^*(\mathscr{P}^n_{X'/S'}) \xrightarrow{\nu'_n} \mathscr{P}^n_{X''/S''},$$

and we have analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), which lets us say that the $\mathscr{P}^n_{X/S}$, $\mathscr{P}^\infty_{X/S}$ and $\mathscr{G}r_\bullet(\mathscr{P}_{X/S})$ *depend functorially on* f.

(16.4.3). We verify immediately (for example, by restricting ourselves to the affine case with help of (16.3.7)) that with the notation of (16.4.1), the diagram

(16.4.3.1) $$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d] & \mathscr{O}_{X'} \arrow[d] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

where the vertical arrows are the ones defining the algebra structure chosen in (16.3.5) (that is to say, the ones coming from the first projections) is commutative; the same goes for the diagram

(16.4.3.2) $$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d, "\rho^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

the vertical arrows defining here the algebra structure from the second projection; besides, if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have **IV-4** | 18

$$\nu_n \circ \rho^*(\sigma) = \sigma' \circ \nu_n$$

which switches one diagram with the other. We deduce from (16.4.3.1) a canonical homomorphism of *augmented* $\mathscr{O}_{X'}$*-algebras*

(16.4.3.3) $$P^n(u) : u^*(\mathscr{P}^n_{X/S}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{P}^n_{X'/S'}$$

and it follows from (16.4.3.2) that the diagram

(16.4.3.4)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "u^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ u^*(\mathscr{P}^n_{X/S}) \arrow[r, "P^n(u)"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative. We deduce a homomorphism of *graded* $\mathscr{O}_{X'}$*-algebras*

(16.4.3.5)
$$\mathrm{Gr}_\bullet(u) : u^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'})$$

and in particular a homomorphism of $\mathscr{O}_{X'}$-modules

(16.4.3.6)
$$\mathrm{Gr}_1(u) : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/S'}$$

giving rise to a commutative diagram

(16.4.3.7)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "d_{X/S}\otimes 1"'] & \mathscr{O}_{X'} \arrow[d, "d_{X'/S'}"] \\ \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] & \Omega^1_{X'/S'} \end{tikzcd}$$

(16.4.4). When $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[u] \arrow[r] & A' \arrow[u] \end{tikzcd}$$

the image of $\mathfrak{J}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathfrak{J}_{B'/A'}$, and the homomorphism ν_n corresponds to the homomorphism of rings $P^n_{B/A} \to P^n_{B'/A'}$ induced from the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (**0**, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (**0**, 20.5.4.2).

Proposition (16.4.5). — *Suppose that $X' = X \times_S S'$, f' and u the canonical projections. Then the canonical* **IV-4 | 19**
homomorphisms $P^n(u)$ (16.4.3.3) and $\mathrm{Gr}_1(u)$ (16.4.3.6) are bijective.

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)) replacing g by the first $p_1 : X \times_S X \to X$ and f by the diagonal Δ_f. □

We note that under the hypotheses of (16.4.5) the homomorphism $\mathrm{Gr}_\bullet(u)$ (16.4.3.5) is *surjective*, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — *Under the hypotheses of (16.4.5), suppose in addition that $w : S \to S'$ is flat (resp. that $\mathscr{G}r_n(\mathscr{P}^n_{X/S})$ are flat $\mathscr{O}_X$-modules for $n \leqslant m$); then the homomorphism*

$$\mathrm{Gr}_n(u) : u^*(\mathscr{G}r(\mathscr{P}^n_{X/S})) \longrightarrow \mathscr{G}r(\mathscr{P}^n_{X'/S'})$$

is bijective for each n (resp. for $n \leqslant m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \to X \times_S X$, so the conclusion follows from (16.2.4). □

(16.4.7). Let S be a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_S$-Module, and set $X = \mathbf{V}(\mathscr{E})$ (**II**, 1.7.8), the vector bundle associated to $\mathscr{E}$, equal to $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$. Let $f : X \to S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathscr{E})$, t is identified with a section of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ over U; let t' be its image in $\Gamma(f^{-1}(U), \mathscr{O}_X) = \Gamma(U, f_*(\mathscr{O}_X)) = \Gamma(U, \mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$, and set

(16.4.7.1)
$$\delta(t) = d^n_{X/S}(t') - t' \in \Gamma(f^{-1}(U), \mathscr{P}^n_{X/S});$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathscr{O}_S) \to \Gamma(f^{-1}(U), \mathscr{O}_X)$) of $\Gamma(U, \mathscr{E})$ into $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$, and therefore the image belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$. We deduce (by varying U) a canonical homomorphism of $\mathscr{O}_X$-algebras

(16.4.7.2)
$$f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{P}^n_{X/S}$$

and in view of the above remark, if $\mathscr{K}$ is the ideal kernel of augmentation $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \to \mathscr{O}_S$, the image of $\mathscr{K}^{n+1}$ by (16.4.7.2) is zero, so that by factoring by $\mathscr{K}^{n+1}$, we finally have a canonical homomorphism

(16.4.7.3)
$$\delta_n : f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})/\mathscr{K}^{n+1}) \longrightarrow \mathscr{P}^n_{X/S}.$$

Proposition (16.4.8). — *Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathscr{O}_S$-algebras*

(16.4.8.1)
$$f^*(\mathbf{S}^\bullet_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}).$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, **IV-4** | 20
since both filtrations involved in (16.4.7.3) are finite (Bourbaki, *Alg. comm.*, chap. III, §2, no 8, cor. 3 of th. 1). To do this, consider the split exact sequence of $\mathscr{O}_S$-modules

(16.4.8.2)
$$0 \longrightarrow \mathscr{E} \xrightarrow{u} \mathscr{E} \oplus \mathscr{E} \xrightarrow{v} \mathscr{E} \longrightarrow 0$$

where, for every pair of sections s, t of $\mathscr{E}$ over an open U of S, we take $u(s) = (-s, s)$ and $v(s,t) = s + t$. We have

$$X \times_S X = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}))$$

((**II**, 1.4.6) and (**II**, 1.7.11)), and the diagonal morphism $X \to X \times_S X$ corresponds (**II**, 1.2.7) to the homomorphism of $\mathscr{O}_X$-algebras $\mathbf{S}(v) : \mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}) \to \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ (**II**, 1.7.4), such that if $\mathscr{I}$ is the kernel of this homomorphism, then we have

$$\mathscr{P}^n_{X/S} = f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E})/\mathscr{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: □

Lemma (16.4.8.3). — *Let Y be a ringed space, $0 \to \mathscr{F}' \xrightarrow{u} \mathscr{F} \xrightarrow{v} \mathscr{F}'' \to 0$ an exact sequence of $\mathscr{O}_Y$-modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \to \mathscr{F}'|V \to \mathscr{F}|V \to \mathscr{F}''|V \to 0$ is split. Let $\mathscr{I}$ be the kernel ideal of $\mathbf{S}(v)$:*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}) \longrightarrow \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}''),$$

and let $\mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$ be the graded $\mathscr{O}_Y$-algebra associated to the $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ endowed with the $\mathscr{I}$-preadic filtration. *Then the homomorphism of graded $\mathscr{O}_Y$-algebras*

(16.4.8.4)
$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'') \longrightarrow \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$$

(where the first member is the graded tensor product of symmetric $\mathscr{O}_Y$-algebras endowed with the canonical gradation (**II**, 1.7.4) *and* (**II**, 2.1.2)*), induced by the canonical injection*

$$\mathscr{F}' \longrightarrow \mathscr{I} = \mathrm{gr}^1_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})),$$

is bijective.

Proof. The injection $\mathscr{F}' \to \mathscr{I}$ indeed canonically gives a homomorphism of graded $\mathscr{O}_Y$-algebras $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \to \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}))$, and since the second member is by definition a graded $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$-algebra, we induce the canonical homomorphism (16.4.8.4) by tensoring the above with $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F})$ is canonically identified with the graded tensor product $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$ (**II**, 1.7.4), and it is immediate that $\mathscr{I}$ is therefore the ideal $\mathscr{I}' \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where $\mathscr{I}'$ is the augmentation ideal of $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}')$, that is to say the (direct) sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant 1$. We conclude that $\mathscr{I}^n = \mathscr{I}'^n \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where this time $\mathscr{I}'^n$ is the direct sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant n$; we have therefore $\mathscr{I}^n/\mathscr{I}^{n+1} = \mathbf{S}^n_{\mathscr{O}_Y}(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, which proves that (16.4.8.4) is bijective. □

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is the image by f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); we can easily see that it follows from the definition of u (16.4.8.2) and of δ (16.4.7.1), given the definition of the $\mathscr{O}_X$-algebra structures of $\mathscr{P}^n_{X/S}$ and of the $d^n_{X/S}$ (16.3.5) and (16.4.3.6). **IV-4 | 21**

In particular:

Corollary (16.4.9). — *Under the conditions of (16.4.7), we have a canonical isomorphism*

(16.4.9.1)
$$\mathrm{gr}_1(\delta) : f^*(\mathscr{E}) \simeq \Omega^1_{X/S}.$$

Corollary (16.4.10). — *If $S = \mathrm{Spec}(A)$, $\mathscr{E} = \mathscr{O}_S^m$, so that*

$$X = \mathrm{Spec}(A[T_1, \ldots, T_m]),$$

then $\mathscr{P}^n_{X/S}$ is canonically identified with the $\mathscr{O}_X$-algebra corresponding to the quotient $A[T_1, \ldots, T_m]$-algebra $A[T_1, \ldots, T_m, U_1, \ldots, U_m]/\mathfrak{K}^{n+1}$, where the U_i ($1 \leqslant i \leqslant m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \ldots, U_m$.

We thus recover in particular the structure of $\Omega^1_{X/S}$ in this case (**0**, 20.5.13).

In addition, note that the $d^n_{X/S}$ then corresponds to a polynomial $F(T_1, \ldots, T_m)$, the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \ldots, T_m + U_m)$, which follows from the definition (16.4.7.1).

Proposition (16.4.11). — *Let $f : X \to S$ be a morphism, $g : S \to X$ a S-section of X, $S^{(n)}$ the n-th infinitesimal neighbourhood of S by the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathscr{O}_S$-algebras*

(16.4.11.1)
$$\varpi_n : g^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{O}_{S_g^{(n)}}$$

(via the $\mathscr{O}_S$-algebra structure on $\mathscr{O}_{S_g^{(n)}}$ defined by f (16.1.7)), making the diagram

(16.4.11.2)
$$\begin{tikzcd}
\mathscr{O}_S = g^*(\mathscr{O}_X) \arrow[rr, "\lambda_n"] \arrow[dr, "g^*(d^n_{X/S})"'] & & \mathscr{O}_{S_g^{(n)}} \\
& g^*(\mathscr{P}^n_{X/S}) \arrow[ur, "\varpi_n"'] &
\end{tikzcd}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (**I**, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g, the diagrams

(16.4.11.3)
$$\begin{tikzcd}
S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\
X \arrow[r, "(g\circ f, 1_X)_S"'] & X \times_S X
\end{tikzcd}
\qquad
\begin{tikzcd}
S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\
X \arrow[r, "(1_X, g\circ f)_S"'] & X \times_S X
\end{tikzcd}$$

identifies S with the product of the $(X \times_S X)$-preschemes X and X by the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams **IV-4 | 22**

(16.4.11.4)
$$\begin{tikzcd}
X \arrow[r, "(g\circ f, 1_X)_S"] \arrow[d, "f"'] & X \arrow[d, "p_1"] \\
S \arrow[r, "g"'] & X
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, "(1_X, g\circ f)_S"] \arrow[d, "f"'] & X \arrow[d, "p_2"] \\
S \arrow[r, "g"'] & X
\end{tikzcd}$$

identify X to the product of X-preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (particular case of the associativity formula (**I**, 3.3.9.1)). We can say that Δ_f, considered as an X-section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S-sections of X: each of these sections g in fact are deduced by *base change* $(g \circ f, 1_X)_S : X \to X \times_S X$. The definition of the homomorphism ϖ_n and the fact that it is bijective follows from the remarks of (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of the first diagram (16.4.11.4) follows also from (16.2.3, (ii)) this time applied to the second diagram (16.11.4). To explain ϖ_n, we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X, and it is clear that $g|W$ is a W-section of the morphism $U \cap f^{-1}(W)$. We can then suppose that S is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{K}$. Then the preceding definitions show that if W is an open of S, t is a section of $\mathscr{O}_X$ over $f^{-1}(W)$, $\varpi_n(d^n t|W)$ is equal to the canonical image of t in $\Gamma(W, (\mathscr{O}_X/\mathscr{K}^{n+1})|W)$. The uniqueness of ϖ_n then follows since the image of $\mathscr{O}_X$ under $d^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$ (16.3.8). □

Corollary (16.4.12). — *Let k be a field, X a k-prescheme, x a point of X* rational over k. *Then $(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$-algebra) to $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k-section g of X such that $g(\operatorname{Spec}(k)) = \{x\}$. □

Corollary (16.4.13). — *Let $f : X \to S$ be a morphism, s a point of S, $X_s = X \times_S \operatorname{Spec}(k(s))$ the fibre of f in s. If $x \in X_s$* is rational over $k(s)$, *$(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_s} k(s)$ is canonically isomorphic to $\mathscr{O}_{X_s,x}/\mathfrak{m}_x'^{\,n+1}$, where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$* (where t is a section of $\mathscr{O}_X$ over an open neighbourhood of x in X) *to the class of $t_x \otimes 1$ modulo $\mathfrak{m}_x'^{\,n+1}$.*

Proof. This follows from (16.4.5) and (16.4.12). □

The preceding corollaries justify the terminology "sheaf of principal parts of order n".

Proposition (16.4.14). — *Let $\rho : A \to B$ be a morphism of rings, S a multiplicative subset of B. Then the canonical homomorphisms*

(16.4.14.1) $$S^{-1}P^n_{B/A} \longrightarrow P^n_{S^{-1}B/A}$$

deduced from the canonical homomorphisms $P^n_{B/A} \to P^n_{S^{-1}B/A}$ (16.4.4), *form a projective system and are bijective.* **IV-4** | 23

Proof. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{I}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{I})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (**I**, 1.3.4). □

Corollary (16.4.15). — *The notation being that of* (16.4.14), *let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

(16.4.15.1) $$S^{-1}P^n_{B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

forming a projective system.

Proof. It evidently suffices to define canonical isomorphisms

(16.4.15.2) $$P^n_{S^{-1}B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

that is to say that we reduce to the case where $\rho(R)$ consists of *invertible* elements of B. But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \to B \otimes_{R^{-1}A} B$ by passing to quotients (**I**, 1.5.3). □

Corollary (16.4.16). — *Let $f : X \to S$ be a morphism of preschemes, x a point of X, $s = f(x)$. Then we have canonical isomorphisms*

(16.4.16.1) $$(\mathscr{P}^n_{X/S})_x \simeq P^n_{\mathscr{O}_x/\mathscr{O}_s}$$

forming a projective system.

We deduce from these isomorphisms of the associated graded rings, and in particular a canonical isomorphism

(16.4.16.2) $$(\Omega^1_{X/S})_x \simeq \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}.$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \dots, T_r)$. Then, for every integer n, the homomorphism of $K[U_1, \dots, U_r]$ (U_i indeterminates) into $P^n_{K/k}$ which sends U_i to $d^n T_i - T_i.1$ is surjective and defines an isomorphism from the quotient $K[U_1, \dots, U_n]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P^n_{K/k}$.*

Proof. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \ldots, T_r]$ and $S = B - \{0\}$. □

We thus recover the fact that the dT_i form a basis of the K-vector space $\Omega^1_{K/k}$ (**0**, 20.5.10).

(16.4.18). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, and consider the canonical homomorphism of augmented $\mathscr{O}_X$-algebras (16.4.3.3)

(16.4.18.1) $$g_{X/Y/Z} : \mathscr{P}^n_{X/Z} \longrightarrow \mathscr{P}^n_{X/Y}$$

(16.4.18.2) $$f_{X/Y/Z} : f^*(\mathscr{P}^n_{Y/Z}) \longrightarrow \mathscr{P}^n_{X/Z}.$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathscr{P}^n_{X/Z})$.

Proof. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and $S = Z$, $u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively. **IV-4 | 24**

We have a commutative diagram (**I**, 3.5)

(16.4.18.3) $$\begin{tikzcd} X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "j"] \arrow[d, "p"] & X \times_Z X \arrow[d, "f \times_Z f"] \\ & Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g \circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathscr{O}_X$ and $\mathscr{O}_{X\times_Y X}$ are identified respectively with $\mathscr{O}_{X\times_Z X}/\mathscr{I}$ and $\mathscr{O}_{X\times_Z X}/\mathscr{L}$, where $\mathscr{L} \supset \mathscr{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g\circ f}$ and j. The $\mathscr{O}_X$-algebra $\mathscr{P}^n_{X/Z}$ is identified with $\mathscr{O}_{X\times_Z X}/\mathscr{I}^{n+1}$, and $\mathscr{P}^n_{X/Y}$ is identified with $\mathscr{O}_{X\times_Y X}/(\mathscr{I}/\mathscr{L})^{n+1}$, which is to say with $\mathscr{O}_{X\times_Z X}/(\mathscr{I}^{n+1} + \mathscr{L})$, and therefore with the quotient of $\mathscr{P}^n_{X/Z}$ by $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$. But we know (*loc. cit*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$, so if $\mathscr{O}_Y$ is identified to $\mathscr{O}_{Y\times_Z Y}/\mathscr{K}$, where $\mathscr{K}$ is the ideal corresponding to Δ_g, $\mathscr{L}$ is equal to $(f \times_Z f)^*(\mathscr{K}).\mathscr{O}_{X\times_Z X}$ (**I**, 4.4.5). Since $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$ is the ideal of $\mathscr{P}^n_{X/Z}$ generated by the image of $\mathscr{L}$, we deduce the proposition. □

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.19.1) $$f^*(\Omega^1_{Y/Z}) \xrightarrow{f_{X/Y/Z}} \Omega^1_{X/Z} \xrightarrow{g_{X/Y/Z}} \Omega^1_{X/Y} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ a closed immersion, $\mathscr{K}$ the quasi-coherent sheaf of ideals of $\mathscr{O}_Y$ corresponding to j. It follows that $\mathscr{P}^n_{X/Y} = \mathscr{O}_X = \mathscr{O}_Y/\mathscr{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathscr{P}^n_{Y/Z}) \to \mathscr{P}^n_{X/Z}$ is surjective, and its kernel is the ideal of $j^*(\mathscr{P}^n_{Y/Z})$ generated by $j^*(\mathscr{O}_Y \cdot d^n_{Y/Z}(\mathscr{K}))$ (it should be noted that $d^n_{Y/Z}(\mathscr{K})$ is a subsheaf of abelian groups of $\mathscr{P}^n_{X/Z}$, but not an $\mathscr{O}_Y$-module in general).*

Proof. We know (**I**, 5.3.8) that the diagonal $\Delta_j : X \to X \times_Y X$ is an isomorphism, from which the first assertion follows. If ϖ_1 and ϖ_2 are the two canonical homomorphisms of algebras $\mathscr{O}_Y \to \mathscr{P}^n_{Y/Z}$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \to Y$, recall that by definition ((16.3.5) and (16.3.6)) ϖ_1 is the structure homomorphism of the $\mathscr{O}_Y$-algebra $\mathscr{P}^n_{Y/Z}$ and $\varpi_2 = d^n_{Y/Z}$. The $\mathscr{O}_X$-algebra $j^*(\mathscr{P}^n_{Y/Z})$ is therefore identified with $\mathscr{P}^n_{Y/Z}/\varpi_1(\mathscr{K})\mathscr{P}^n_{Y/Z}$ and its quotient by the ideal generated by $j^*(d^n_{Y/Z}(\mathscr{K}))$ to $\mathscr{P}^n_{Y/Z}/(\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K}))\mathscr{P}^n_{Y/Z}$. Now note that we have a commutative diagram **IV-4 | 25**

$$\begin{tikzcd} Y & X \arrow[l, "j"'] \arrow[d, "\Delta_{f\circ j}"] \\ Y \times_Z Y \arrow[from=1-1, "\Delta_f"'] & X \times_Z X \arrow[l, "j\times_Z j"] \end{tikzcd}$$

identifying X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta^n_{Y/Z}$ and $\Delta^n_{X/Z}$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f\circ j}$ respectively, then we have a diagram

$$\begin{tikzcd} \Delta^n_{Y/Z} \arrow[d] & \Delta^n_{X/Z} \arrow[l] \arrow[d] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j\times_Z j"] \end{tikzcd}$$

making $\Delta^n_{X/Z}$ the product of the $(Y \times_Z Y)$-preschemes $\Delta^n_{Y/Z}$ and $X \times_Z X$. We can also say that $\mathscr{P}^n_{X/Z}$ is identified with the sheaf of rings $\mathscr{P}^n_{Y/Z} \otimes_{\mathscr{O}_{Y\times_Z Y}} \mathscr{O}_{X\times_Z X}$. But we see immediately that (for example, by restricting to the affine case) that $\mathscr{O}_{X\times_Z X} = \mathscr{O}_{Y\times_Z Y}/(p_1^*(\mathscr{K}) + p_2^*(\mathscr{K}))\mathscr{O}_{Y\times_Z Y}$. Therefore $\mathscr{P}^n_{X/Z}$ is identified with the quotient of $\mathscr{P}^n_{Y/Z}$ by the ideal generated by the image in $\mathscr{P}^n_{Y/Z}$ of $p_1^*(\mathscr{K}) + p_2^*(\mathscr{K})$. But by definition this ideal is generated by $\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K})$. □

Corollary (16.4.21). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ an immersion. We have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.21.1) $$\mathscr{N}_{X/Y} \longrightarrow j^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{P}^n_{X/S}$ and $\Omega^1_{X/S}$ are quasi-coherent $\mathscr{O}_X$-modules of finite presentation.*

Proof. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is affine, $X = \operatorname{Spec}(B)$, where $B = A[T_1, \ldots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \ldots, T_r]$. Applying (16.4.20) where $Z = S$, $Y = \operatorname{Spec}(C)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$. Then $j^*(\mathscr{P}^n_{Y/Z})$ is a free $\mathscr{O}_X$-module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathscr{O}_Y.d^n_{Y/Z}(\mathscr{K}))$ generates a quasi-coherent $\mathscr{O}_X$-module of finite type; hence the conclusion. □

Proposition (16.4.23). — *Let X, Y be two S-preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ the canonical projections. Then the canonical homomorphism*

(16.4.23.1) $$p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega^1_{X/S}) \oplus q^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{(X\times_S Y)/S}$$

is bijective.

Proof. The commutative diagram **IV-4 | 26**

$$\begin{tikzcd} Y \arrow[d, "g"'] & X \times_S Y \arrow[l, "q"'] \arrow[d, "h"] & X \times_S Y \arrow[l, "\mathrm{id}"'] \arrow[d, "p"] \\ S & S \arrow[l, "\mathrm{id}"] & X \arrow[l, "f"] \end{tikzcd}$$

gives us a factorization of the canonical *isomorphism* $P^n(p)$ (16.4.5)

$$p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}$$

and similarly, switching X with Y, we have a factorization of the *isomorphism* $P^n(q)$

$$q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \quad (\text{resp. } q_{Z/X/S} : q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S})$$

is *injective*, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y} \quad (\text{resp. } \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X})$$

is direct summand of the image $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, this kernel is, by virtue of (16.4.18), generated by the image by $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathscr{P}^n_{Y/S})$ (resp. $p^*(\mathscr{P}^n_{X/S})$). We conclude the proposition by considering the case $n = 1$. □

We immediately generalize (16.4.23) to the case of a product of any finite number of S-preschemes.

Remarks (16.4.24). —

(i) We will see (17.2.3) that when the morphism $f : X \to Y$ in (16.4.18) is *smooth*, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally *left invertible* and in particular injective. Similarly, when the morphism $f \circ j : X \to Z$ of (16.4.20) is *smooth*, the homomorphism on the left in (16.4.21.1) is locally *left invertible* and *a fortiori* injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the "imperfection modules" studied in (**0**, 20.6), and the exact sequences where they occur.

(ii) Let X be a topological space, $\mathscr{A}$ a sheaf of rings over X and $\mathscr{B}$ an $\mathscr{A}$-algebra over X. Then it is clear that

$$U \longmapsto P^n_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{A})} \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U,\mathscr{B})$-algebras, and therefore the associated sheaf $\mathscr{P}^n_{\mathscr{B}/\mathscr{A}}$ is an augmented $\mathscr{B}$-algebra. In the particular case where X is a prescheme, $f = (\psi, \theta) : X \to S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathscr{P}^n_{X/S}$ is canonically isomorphic to $\mathscr{P}^n_{\mathscr{O}_X/\psi^*(\mathscr{O}_S)}$. It follows that the formalism developed in the present paragraph could be considered as a particular case of a differential **IV-4 | 27** formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of "varieties", the "global" constructions of the $\mathscr{P}^n$ analogous to those we have used here are also better suited for applications.

16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi, \theta) : X \to S$ be a morphism of ringed spaces. For every $\mathscr{O}_X$-module $\mathscr{F}$, we say *S-derivation* (or *(X/S)-derivation*, or *f-derivation*) *of $\mathscr{O}_X$ to $\mathscr{F}$* for every homomorphism of *sheaves of additive groups* $D : \mathscr{O}_X \to \mathscr{F}$ satisfying the following conditions:

(a) for every open V of X, and all pair of sections (t_1, t_2) of $\mathscr{O}_X$ over V, we have

$$D(t_1 t_2) = t_1 D(t_2) + D(t_1) t_2; \tag{16.5.1.1}$$

(b) for every open V of X, every section t of $\mathscr{O}_X$ over V, and every section s of $\mathscr{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

$$D((s|V)t) = (s|V)D(t). \tag{16.5.1.2}$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathscr{O}_x \to \mathscr{F}_x$ is an *$\mathscr{O}_{f(x)}$-derivation*.

Another interpretation consists of considering the $\mathscr{O}_X$-algebra $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$ as equal to $\mathscr{O}_X \oplus \mathscr{F}$, the algebra structure being defined by the condition that for every open V of X, the product of two sections of $\mathscr{O}_X$ (resp. of a section of $\mathscr{O}_X$ and a section of $\mathscr{F}$) over V is defined by the ring structure of $\Gamma(V, \mathscr{O}_X)$ (resp. the $\Gamma(V, \mathscr{O}_X)$-module structure on $\Gamma(V, \mathscr{F})$), and the product of two sections of $\mathscr{F}$ over V is chosen to be zero; then $\mathscr{F}$ is an ideal of $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, the kernel of the canonical augmentation $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F}) \to \mathscr{O}_X$, and to say that D is an S-derivation of $\mathscr{O}_X$ to $\mathscr{F}$ means that $1_{\mathscr{O}_X} + D$ is an *$\mathscr{O}_S$-homomorphism of algebras* from $\mathscr{O}_X$ to $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, which, composed with the augmentation, gives $1_{\mathscr{O}_X}$.

The S-derivations of $\mathscr{O}_X$ to $\mathscr{F}$ clearly form a *$\Gamma(X, \mathscr{O}_X)$-module* $\mathrm{Der}_{\mathscr{O}_S}(\mathscr{O}_X, \mathscr{F})$.

When $\mathscr{F} = \mathscr{O}_X$, an S-derivation of $\mathscr{O}_X$ to itself is simply called an *S-derivation of $\mathscr{O}_X$*.

Proposition (16.5.2). — *Let A be a ring, B an A-algebra, L a B-module; let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\mathscr{F} = \widetilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S-derivation D of $\mathscr{O}_X$ to $\mathscr{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ of B to L, is an isomorphism of B-modules from $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{F})$ to $\mathrm{Der}_A(B, L)$ (cf. (**0**, 20.1.2)).*

Proof. This follows immediately from the given interpretation of S-derivations in terms of homo- **IV-4** | 28
morphisms of algebras, analogous to the interpretation given in (**0**, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathscr{O}_X$-algebras and homomorphisms of B-algebras ((**I**, 1.3.13) and (**I**, 1.3.8)). □

Proposition (16.5.3). — *Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes.*

(i) *The differential $d_{X/S} : \mathscr{O}_X \to \Omega^1_{X/S}$ (16.3.6) is an S-derivation.*

(ii) *For every $\mathscr{O}_X$-module $\mathscr{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathscr{O}_X)$-modules*

$$\mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \simeq \mathrm{Der}_S(\mathscr{O}_X, \mathscr{F}). \tag{16.5.3.1}$$

Proof. The assertion (i) has already been written (16.3.6). On the other hand, it is immediate (in light of (**0**, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, considering the restrictions to a fibre $\mathscr{O}_x$ of the two sides and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S-derivation $D : \mathscr{O}_X \to \mathscr{F}$; for every affine open $V = \mathrm{Spec}(B)$ of X, such that $f(V)$ is contained in an affine open $U = \mathrm{Spec}(A)$ of S, $D_V : B \to \Gamma(V, \mathscr{F})$ is an A-derivation, and therefore there exists a *unique* B-homomorphism $u_V : \Omega^1_{B/A} \to \Gamma(V, \mathscr{F})$ such that $D_V = u_V \circ d_{B/A}$ (**0**, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V | W$, and therefore the u_V define a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{O}_X \to \mathscr{F}$ answering the question. □

(16.5.4). With the notation of (16.5.1), for every open U of X, $\mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a $\Gamma(U, \mathscr{O}_X)$-module and it is clear that the map $U \mapsto \mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathscr{O}_X$-module), in light of the pointwise characterization of S-derivations, seen in (16.5.1). This $\mathscr{O}_X$-module is denoted by $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ and is called the *sheaf of S-derivations of $\mathscr{O}_X$ in $\mathscr{F}$*, and what we have seen is further expressed in the following corollary:

Corollary (16.5.5). — *For every $\mathscr{O}_X$-module $\mathscr{F}$, the homomorphism of $\mathscr{O}_X$-modules induced by $u \mapsto u \circ d_{X/S}$*

$$\mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \longrightarrow \mathscr{D}er_S(\mathscr{O}_X, \mathscr{F}) \tag{16.5.5.1}$$

is bijective.

Corollary (16.5.6). — (i) *If the morphism $f : X \to S$ is locally of finite presentation and if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module.*

(ii) *If in addition S is locally Noetherian and if $\mathscr{F}$ is coherent, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*

Proof. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (**I**, 1.3.12); the assertion (ii) follows from (**0**, 5.3.5). □

(16.5.7). We set

$$\mathfrak{G}_{X/S} = \mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \mathscr{D}er_S(\mathscr{O}_X, \mathscr{O}_X), \tag{16.5.7.1}$$

and say that it is the *sheaf of S-derivations of $\mathscr{O}_X$*, or even the *tangent sheaf of X relative to S*: it is therefore the *dual* of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathscr{O}_X$- **IV-4** | 29
module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6).

(16.5.8). Suppose in particular that $\Omega^1_{X/S}$ is a *locally free* $\mathscr{O}_X$-module (of finite rank) (which will be the case when f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is a locally free $\mathscr{O}_X$-module of the same rank as $\Omega^1_{X/S}$ at each point. More specifically, suppose that $\Omega^1_{X/S}$ is of rank n at a point x; then there are n sections s_i ($1 \leqslant i \leqslant n$) of $\mathscr{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$ form a basis of this $k(x)$-vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathscr{O}_x$-module $(\Omega^1_{X/S})_x$, and therefore, by restricting U, we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$. So the

$\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leqslant i \leqslant n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leqslant i \leqslant n}$ the dual basis of $(ds_i)_{1 \leqslant i \leqslant n}$, so that, by (16.5.3), we have

(16.5.8.1) $$D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad \text{(Kronecker's symbol).}$$

Every $\Gamma(S, \mathscr{O}_S)$-derivation of the $\Gamma(S, \mathscr{O}_S)$-algebra $\Gamma(U, \mathscr{O}_X)$ is therefore written in an unique way as

$$D = \sum_{i=1}^{n} a_i D_i = \sum_{i=1}^{n} a_i \frac{\partial}{\partial s_i},$$

where the a_i ($1 \leqslant i \leqslant n$) are sections of $\mathscr{O}_X$ over U. For every section $g \in \Gamma(U, \mathscr{O}_X)$, if we put $dg = \sum_{i=1}^{n} c_i ds_i$, then we have $c_i = \langle D_i, dg \rangle = D_i g$ by virtue of (16.5.8.1), in other words,

(16.5.8.2) $$dg = \sum_{i=1}^{n} (D_i g) ds_i = \sum_{i=1}^{n} \frac{\partial g}{\partial s_i} ds_i.$$

(16.5.9). Let D_1, D_2 be two S-derivations of $\mathscr{O}_X$. For every open U of X, if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathscr{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation in this ring, and therefore the $\psi^*(\mathscr{O}_S)$-endomorphism of $\mathscr{O}_X$

(16.5.9.1) $$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$$

is also an S-derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{O}_X)$ a $\Gamma(S, \mathscr{O}_S)$-*Lie algebra* structure. Since the definition of this structure commutes with the restriction to an open of X, we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathscr{O}_S)$-*Lie algebra* structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is *not* $\Gamma(X, \mathscr{O}_X)$-*bilinear*.

(16.5.10). For every base change $g : S' \to S$, if we set $X' = X \times_S S'$, then we see (16.4.5) that we have a canonical isomorphism

(16.5.10.1) $$\Omega^1_{X/S} \otimes_S S' \simeq \Omega^1_{X'/S'},$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3) **IV-4** | 30

(16.5.10.2) $$\mathfrak{G}_{X/S} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'},$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) *If $g : S' \to S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism (16.5.10.2) is injective (resp. bijective).*

(ii) *If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of finite type, then the homomorphism (16.5.10.2) is bijective.*

Proof. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of finite type (resp. of finite presentation) ((16.3.9) (16.4.22)). □

(16.5.12). Since $\Omega^1_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module, we can consider the vector bundle over X defined by $\Omega^1_{X/S}$ (**II**, 1.7.8)

(16.5.12.1) $$T_{X/S} = \mathbf{V}(\Omega^1_{X/S})$$

which is called the *tangent bundle of X relative to S*. We have therefore a canonical bijection (**II**, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S-sections* of the tangent bundle of X relative to S. If $f : X \to Y$ is an S-morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega^1_{Y/S}) \to \Omega^1_{X/S}$, which, having in mind that

$$\mathbf{V}(f^*(\Omega^1_{Y/S})) = \mathbf{V}(\Omega^1_{Y/S}) \times_Y X \quad (\textbf{II}, 1.7.11)\,,$$

gives us an X-morphism $T_{X/S}(f) : T_{X/S} \to T_{Y/S} \times_Y X$. If $g : Y \to Z$ is a second S-morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (**0**, 20.5.4.1).

It follows from (16.5.10.1) and from (**II**, 1.7.11) that for every base change $g : S' \to S$ we have a canonical isomorphism

(16.5.12.2) $$T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'.$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \operatorname{Spec}(k(x))$ that are *rational over* $k(x)$, that is, the set

(16.5.13.1) $$T_{X/S}(x) = \operatorname{Hom}_{k(x)}(\Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x), k(x)),$$

which is the *dual* of the $k(x)$-vector space $\Omega^1_{\mathscr{O}_x/\mathscr{O}_s}/\mathfrak{m}_x \cdot \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}$. When $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of *finite type*, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g : S \to S'$, **IV-4** | 31
and every point $x' \in X' = X \times_S S'$ over x, we have a canonical isomorphism

(16.5.13.2) $$T_{X'/S'}(x') \simeq T_{X/S} \otimes_{k(x)} k(x').$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \to k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

(16.5.13.3) $$T_{X/S}(x) = T_{X_s/k(s)}(x) = \operatorname{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}'^2_x, k(x)),$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$. In the case where S is the spectrum of a field k, we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k, as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S-prescheme and let $g : Y \to X$ be an S-morphism; then we have a canonical homomorphism of $\mathscr{O}_Y$-modules (16.4.19)

(16.5.13.4) $$g_{Y/X/S} : g^*(\Omega^1_{X/S}) \longrightarrow \Omega^1_{Y/S}.$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y) = (\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module *of finite type*, then we can identify

$$\operatorname{Hom}_{k(y)}(g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of $k(y)$*-vector spaces*

(16.5.13.5) $$T_y(g) : T_{Y/S}(y) \longrightarrow T_{X/S}(x) \otimes_{k(x)} k(y)$$

called the *linear map tangent to g at the point y*. When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$-vector spaces $T_{Y/S}(y) \to T_{X/S}(x)$; also note that in this case, $g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y)$ is identified with $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega^1_{X/S}$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y.

(16.5.14). The interpretation of derivations of an A-algebra B to a B-module L, given in (**0**, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \to S$, $g : Y \to S$, and a closed subprescheme Y_0 of Y defined by a *zero-square* ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (so that Y and Y_0 have the *same underlying subspace*). Suppose we are given an S-morphism $u_0 : Y_0 \to X$, so that we have a commutative diagram

(16.5.14.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & Y_0 \arrow[l, "u_0"'] \arrow[d, "j"] \\ S & Y \arrow[l, "g"] \arrow[ul, dashed, "u"'] \end{tikzcd}$$

and we suggest looking for an *S-morphism* $u : Y \to X$ such that $u_0 = u \circ j$ (in other words, if it is **IV-4** | 32
possible to complete the diagram above by the dotted arrow u, keeping it *commutative*).

For that, consider an affine open $U = \mathrm{Spec}(C)$ of Y; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \mathrm{Spec}(C/\mathfrak{L})$, where $\mathfrak{L} = \Gamma(U, \mathscr{J})$, a *zero-square* ideal in C; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \mathrm{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \mathrm{Spec}(A)$ of S, so that B and C are A-algebras and $u_0|U_0$ corresponds to an A-homomorphism ψ from B to $C/\mathfrak{L}$; Let $P(U_0)$ be the set of restrictions $u|U$ of the sought homomorphisms, which corresponds canonically to A-homomorphisms of algebras $\phi : B \to C$ such that the *composite* $B \xrightarrow{\phi} C \to C/\mathfrak{L}$ *is equal to* ψ. So we know (**0**, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \mathrm{Der}_A(B, \mathfrak{L})$; when $P(U_0)$ is not empty, the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ (or even a *principal homogeneous space* (or *torsor*) *under* $\mathrm{Der}_A(B, \mathfrak{L})$).

Now notice that, since $\mathfrak{L}$ is equipped with a B-module structure via ψ, we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{L})$ onto $\mathrm{Der}_A(B, \mathfrak{L})$ (**0**, 20.4.8). Besides, as $\mathfrak{L}$ is square-zero, therefore a $(C/\mathfrak{L})$-module, every B-homomorphism $v : \Omega^1_{B/A} \to \mathfrak{L}$ can be considered as a $(C/\mathfrak{L})$-homomorphism $\Omega^1_{B/A} \otimes_B (C/\mathfrak{L}) \to \mathfrak{L}$. As $\mathscr{I}$ is square-zero, it can be considered as a quasi-coherent $\mathscr{O}_{Y_0}$-module; let's introduce the $\mathscr{O}_{Y_0}$-module

$$\mathscr{G} = \mathscr{H}om(u_0^*(\Omega^1_{X/S}), \mathscr{I}); \tag{16.5.14.2}$$

it follows from the fact that $\Omega^1_{B/A} = \Gamma(V, \Omega^1_{X/S})$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathscr{G})$.

As $P(U_0)$ is defined as a set of S-morphisms $U \to X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathscr{P}$ on Y_0. We can use this fact to prove that the map $h : \Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y, U'_0 its inverse image in Y_0, then the diagram

$$\begin{tikzcd} \Gamma(U_0, \mathscr{G}) \times P(U_0) \arrow[r, "h"] \arrow[d] & P(U_0) \arrow[d] \\ \Gamma(U'_0, \mathscr{G}) \times P(U'_0) \arrow[r, "h'"'] & P(U'_0) \end{tikzcd} \tag{16.5.14.3}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V, W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h, this follows from the commutativity of the diagram (**0**, 20.5.4.2). **IV-4** | 33

The mapping $\Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ therefore defines a homomorphism of *sheaves of sets* $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathscr{P}) \neq \varnothing$, $m_{U_0} : \Gamma(U_0, \mathscr{G}) \times \Gamma(U_0, \mathscr{P}) \to \Gamma(U_0, \mathscr{P})$ is an external law defining in $\Gamma(U_0, \mathscr{P})$ a torsor structure for the group $\Gamma(U_0, \mathscr{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathscr{P}$ over a topological space Z, a sheaf of groups $\mathscr{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathscr{P}) \neq \varnothing$, $m_U : \Gamma(U, \mathscr{G}) \times \Gamma(U, \mathscr{P}) \to \Gamma(U, \mathscr{P})$ makes $\Gamma(U, \mathscr{P})$ a *torsor* under the group $\Gamma(U, \mathscr{G})$, then we say that $\mathscr{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathscr{G}$. We say that $\mathscr{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathscr{G}$ [6] if in addition $\Gamma(U, \mathscr{P}) \neq \varnothing$ for every open $U \neq \varnothing$ in a suitable basis for the topology of Z.

For the general theory of torsors, we refer to [eAG64]; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathscr{G}$) and elements from the cohomology set $\mathrm{H}^1(Z, \mathscr{G})$. Consider a torsor $\mathscr{P}$ under $\mathscr{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathscr{P}) \neq \varnothing$ for every λ; denote by p_λ an element of $\Gamma(U_\lambda, \mathscr{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \varnothing$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathscr{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu | U_\lambda \cap U_\mu) = p_\lambda | U_\lambda \cap U_\mu$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \varnothing$, then the restrictions $\gamma'_{\lambda\mu}$, $\gamma'_{\mu\nu}$, $\gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}$, $\gamma_{\mu\nu}$, $\gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu}\gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-*cocycle* of the cover (U_λ) with

[6] [Trans.] This is nowadays more commonly called a $\mathscr{G}$-torsor rather than a torsor *under* $\mathscr{G}$.

values in $\mathscr{G}$. If, for every λ, p'_λ is a second element of $\Gamma(U_\lambda, \mathscr{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathscr{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ), an automorphism $\theta_{\lambda\mu}$ of the sheaf *of sets* $\mathscr{G}|U_\lambda \cap U_\mu$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathscr{G}|U_\lambda$ via the automorphisms $\theta_{\lambda\mu}$ (**0**, 3.3.1); we thus obtain a torsor under $\mathscr{G}$, denoted $\mathscr{P}$, and if we take for p_λ the unit section over U_λ, then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathscr{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-*coboundary*, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda \beta_\mu^{-1}$, then the torsor $\mathscr{P}$ obtained is *isomorphic to* $\mathscr{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathscr{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (**III**, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Z$-module; then every torsor over $\mathscr{G}$ is trivial.*

Returning to the problem considered in (16.5.14), we thus obtain: **IV-4** | 34

Proposition (16.5.17). — *Let X, Y be two S-preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_Y$ such that $\mathscr{I}^2 = 0$, $j : Y_0 \to Y$ the canonical injection. Let $u_0 : Y_0 \to X$ be an S-morphism, and $\mathscr{P}$ the sheaf of sets on Y such that, for every open U of Y, $\Gamma(U, \mathscr{P})$ is the set of S-morphisms $u : U \to X$ such that $u_0|U_0 = u \circ (j|U_0)$, where $U_0 = j^{-1}(U)$. Then there exists on $\mathscr{P}$ the structure of a pseudo-torsor over the $\mathscr{O}_{Y_0}$-module $\mathscr{G} = \mathscr{H}om_{\mathscr{O}_{Y_0}}(u_0^*(\Omega^1_{X/S}), \mathscr{I})$.*

In particular:

Corollary (16.5.18). — *With the notation of (16.5.16), suppose that Y is affine and $\Omega^1_{X/S}$ is of finite presentation; if there is an open cover (U_α) of Y, and, for every index α, an S-morphism $v_\alpha : U_\alpha \to X$ such that, if $U_\alpha^0 = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|U_\alpha^0) = u_0|U_\alpha^0$, then there is an S-morphism $u : Y \to X$ such that $u \circ j = u_0$.*

Proof. Indeed, $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_{Y_0}$-module (**I**, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathscr{P}$, which is by hypothesis a torsor over $\mathscr{G}$, and not only a pseudo-torsor, is *trivial*; but if w is an isomorphism from $\mathscr{G}$ to $\mathscr{P}$ (as it is a torsor over $\mathscr{G}$), the image under w of the zero section of $\mathscr{G}$ is the S-morphism we want. □

16.6. Sheaf of p-differentials and exterior differentials.

(16.6.1). Let $f : X \to S$ be a morphism of preschemes. We define the *sheaf of p-differentials of X relative to S* (p integer) to be the *p'th exterior power* (**0**, 4.1.5) of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$, denoted by

(16.6.1.1) $$\Omega^p_{X/S} = \bigwedge^p(\Omega^1_{X/S}).$$

So we have $\Omega^0_{X/S} = \mathscr{O}_X$, and $\Omega^p_{X/S} = 0$ for $p < 0$; the $\Omega^p_{X/S}$ are the homogeneous components of the exterior algebra of $\Omega^1_{X/S}$

(16.6.1.2) $$\Omega^\bullet_{X/S} = \bigwedge(\Omega^1_{X/S}) = \bigoplus_{p \in \mathbf{Z}} \bigwedge^p(\Omega^1_{X/S}),$$

which is therefore a quasi-coherent graded anti-commutative $\mathscr{O}_X$-algebra whose elements of degree 1 are square-zero. For every affine U of X, we have $\Gamma(U, \Omega^\bullet_{X/S}) = \bigwedge(\Gamma(U, \Omega^1_{X/S}))$, where $\Gamma(U, \Omega^1_{X/S})$ is considered as a $\Gamma(U, \mathscr{O}_X)$-module.

When $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines, B being then an A-algebra, we have (**0**, 4.1.5) $\Omega^p_{X/S} = (\Omega^p_{B/A})^\sim$, by putting $\Omega^p_{B/A} = \bigwedge^p \Omega^1_{B/A}$.

Theorem (16.6.2). — *There is one and only one endomorphism d of the sheaf of additive groups $\Omega^\bullet_{X/S}$ with the following properties:*

(i) $d \circ d = 0$.

(ii) *For every open set U of X and every section $f \in \Gamma(U, \mathscr{O}_X)$ we have $df = d_{X/S}f$.* **IV-4** | 35

(iii) *For every open set U of X, every pair of integers p, q and every pair of sections $\omega'_p \in \Gamma(U, \Omega^p_{X/S})$, $\omega''_q \in \Gamma(U, \Omega^q_{X/S})$, we have*

(16.6.2.1)
$$d(\omega'_p \wedge \omega''_q) = (d\omega'_p) \wedge \omega''_q + (-1)^p \omega'_p \wedge d\omega''_q.$$

Also, d is an endomorphism of graded $\psi^*(\mathscr{O}_X)$-modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d. For every affine open U of X, every section of $\Omega^p_{X/S}$ over U is (because of (ii)) a linear combination if a finite number of elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathscr{O}_X$ over U (**0**, 20.4.7). The conditions (i) and (iii) then show, by induction on p, that we necessarily have

(16.6.2.2)
$$d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p.$$

This therefore proves the *uniqueness* of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d, we can restrict ourselves to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rd ed., §10) to define an A-derivation D of degree $+1$ of an exterior algebra $\bigwedge(M)$ (where M is a B-module and an A-algebra), such derivation taking its values in a *graded anti-commutative* A-algebra $C = \bigoplus_{n=0}^{\infty} C_n$, whose elements of degree 1 are square-zero, it suffices to give arbitrarily an A-derivation D_0 of B in C_1 and an A-homomorphism D_1 of M in C_2; then it exists one and only one A-anti-derivation D of $\Lambda(M)$ in C coinciding with D_0 in B and D_1 in M.

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A-homomorphism u of $\Omega^1_{B/A}$ in $\Omega^2_{B/A}$ such that

(16.6.2.3)
$$u(g.df) = dg \wedge df$$

for whichever f, g in A; it suffices to show that there is an A-homomorphism $v : B \otimes_A \Omega^1_{B/A} \to \Omega^2_{B/A}$ such that

(16.6.2.4)
$$v(g.\omega) = dg \wedge \omega$$

for $g \in B$ and $\omega \in \Omega^1_{B/A}$. Finally, since $\Omega^1_{B/A} = \mathfrak{J}/\mathfrak{J}^2$ (where $\mathfrak{J} = \mathfrak{J}_{B/A}$ is the kernel of the canonical homomorphism $B \times_A B \to B$) and that $\Omega^1_{B/A}$ is generated by elements of the form $g.df$, it is enough to define an A-homomorphism $w : B \otimes_A (B \otimes_A B) \to \Omega^2_{B/A}$ such that

(16.6.2.5)
$$w(g' \otimes g \otimes f) = dg' \wedge (g.df)$$

and such that w is zero on the image of $B \otimes A\mathfrak{J}^2$. Or, since the second member of (16.6.2.5) is A-trilinear in g', g and f, the existence of w verifying (16.6.2.5) is immediate. Since, on the other hand, $\mathfrak{J}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Or, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes y - y \otimes x$, the formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, **IV-4** | 36
which is to say that d is a derivation.

It remains to be shown that d satisfies the condition (i). Now, the square of an anti-derivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega^\bullet_{B/A}$ is generated by $\Omega^1_{B/A}$ as a B-algebra, it is enough to verify that $d(dz) = 0$ for $z \in B$ and $x \in \Omega^1_{B/A}$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B, and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The anti-derivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — *For every base change $g : S' \to S$, if we put $X' = X \times_S S'$, the canonical morphism*

(16.6.4.1)
$$\Omega^\bullet_{X/S} \otimes_S S' \longrightarrow \Omega^\bullet_{X'/S'}$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega^{\bullet}_{X/S}$ over an open set U of X, $s \otimes 1$ its inverse image, section of $\Omega^{\bullet}_{X'/S'}$ over the inverse image U' of U in X', we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

Proof. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathscr{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of rank n in a point x, so that we have n sections $s_i \in \Gamma(U, \mathscr{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$ (16.5.8). Then, for every integer $p \geqslant 1$, the p-differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 \leqslant i_2 \leqslant \ldots \leqslant i_p$ elements of $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega^p_{X/S})$ over $\Gamma(U, \mathscr{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathscr{O}_X)$, we have

$$d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p} \tag{16.6.5.1}$$

where, in the second member, k varies in the set of the $n - p$ indexes different from the i_h, i_r being the biggest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathscr{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathscr{P}^n_{X/S}(\mathscr{F})$.

(16.7.1). Let $f : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-module. We denote by $X^{(n)}_{\Delta_f}$ the *n'th infinitesimal neighbourhood* of X via the diagonal morphism $\Delta_f : X \to X \times_S X$, by $h_n : X^{(n)}_{\Delta_f} \to X \times_S X$ the canonical morphism (16.1.2), and consider the two composite morphisms **IV-4 | 37**

$$p^{(n)}_1 : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p^{(n)}_2 : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p^{(n)}_1$ corresponds to the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ which we have chosen to define the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ (16.3.5), and $p^{(n)}_2$ to the homomorphism of sheaves of rings $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (16.3.6). Since $X^{(n)}_{\Delta_f}$ and X have the same underlying subspace, we can write

$$\mathscr{P}^n_{X/S} = (p^{(n)}_1)_*((p^{(n)}_2)^*(\mathscr{O}_X)). \tag{16.7.1.1}$$

More generally, we define

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = (p^{(n)}_1)_*((p^{(n)}_2)^*(\mathscr{F})). \tag{16.7.1.2}$$

so that $\mathscr{P}^n_{X/S} = \mathscr{P}^n_{X/S}(\mathscr{O}_X)$; by definition, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is an $\mathscr{O}_X$-module.

(16.7.2). If we come back to the definition of the inverse image of modules on ringed spaces (**0**, 4.3.1) and having in mind that $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we see that we can write the definition (16.7.1.2) in the form

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F}, \tag{16.7.2.1}$$

but where you have to be careful that, in the interpretation of the symbol $\otimes$, $\mathscr{P}^n_{X/S}$ is endowed with the structure of $\mathscr{O}_X$-module defined by *the homomorphism of sheaves of rings* $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathscr{P}^n_{X/S}(\mathscr{F})$ is canonically equipped with a $\mathscr{P}^n_{X/S}$-module structure.

Proposition (16.7.3). — (i) *The functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$ from the category of $\mathscr{O}_X$-modules to the category of $\mathscr{P}^n_{X/S}$-modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathscr{P}^n_{X/S}$ is flat.*

(ii) *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module (resp. of finite type, resp. of finite presentation), then $\mathscr{P}^n_{X/S}(\mathscr{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).*

Proof. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathscr{P}^n_{X/S}$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$. ☐

(16.7.4). The two structures of $\mathscr{O}_X$-module on $\mathscr{P}^n_{X/S}$ define in $\mathscr{P}^n_{X/S}(\mathscr{F})$ two structures of $\mathscr{O}_X$-modules, which happen to be permutable, and therefore an $\mathscr{O}_X$-bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (chosen in (16.3.5)) on the *left* and the one coming from the homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ on the *right*. On other words, for every open U of X, and every triplet $a \in \Gamma(U, \mathscr{O}_X)$, $b \in \Gamma(U, \mathscr{P}^n_{X/S})$, $t \in \Gamma(U, \mathscr{F})$, we have by definition

(16.7.4.1) $$a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b.d^n a) \otimes t = b \otimes (at) = (d^n a).(b \otimes t).$$

The $\mathscr{O}_X$-module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the *left* $\mathscr{O}_X$-module structure. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then the same is true for $\mathscr{P}^n_{X/S}(\mathscr{F})$ **IV-4** | 38
for any one of its $\mathscr{O}_X$-module structures. If also $\mathscr{F}$ is of finite type (resp. of finite presentation) and $f : X \to S$ is locally of finite type (resp. locally of finite presentation), $\mathscr{P}^n_{X/S}(\mathscr{F})$ is (for any one of its $\mathscr{O}_X$-module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

(16.7.5.1) $$d^n_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (\text{also denoted } d^n_{X/S})$$

such that, in the notations of (16.7.4), we have

(16.7.5.2) $$d^n_{X/S,\mathscr{F}}(t) = 1 \otimes t$$

and consequently, because of (16.7.4.1)

(16.7.5.3) $$d^n_{X/S,\mathscr{F}}(at) = (1 \otimes t)a = (d^n_{X/S,\mathscr{F}}(t)).a$$

(IV.16.7.5.4) $$d^n_{X/S,\mathscr{F}}(at) = (d^n_{X/S,\mathscr{F}}(a)).(1 \otimes t) = (d^n_{X/S,\mathscr{F}}(a))(d^n_{X/S,\mathscr{F}}(t)).$$

Therefore it is $\mathscr{O}_X$-linear for the structure of *right* $\mathscr{O}_X$-module on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and *semilinear* (relative to the homomorphism σ (16.3.4)) for the *left* $\mathscr{O}_X$-module structure.

Proposition (16.7.6). — *The right $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}(\mathscr{F})$ is generated by the image of $\mathscr{F}$ by the homomorphism $d_{X/S,\mathscr{F}}$.*

Proof. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathscr{F} = \mathscr{O}_X$ (16.3.8). ☐

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathscr{P}^m_{X/S} \longrightarrow \mathscr{P}^n_{X/S}$$

for $n \leqslant m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (n \leqslant m)$$

which are homomorphisms of $\mathscr{O}_X$-bimodules in light of (16.1.6) and (7.4.1); also we have commutative diagrams

$$\begin{tikzcd} \mathscr{P}^m_{X/S}(\mathscr{F}) \arrow[rr] & & \mathscr{P}^n_{X/S}(\mathscr{F}) \\ & \mathscr{F} \arrow[ul, "d^n_{X/S,\mathscr{F}}"] \arrow[ur, "d^n_{X/S,\mathscr{F}}"'] & \end{tikzcd}$$

We have therefore a projective system of $\mathscr{O}_X$-bimodules $(\mathscr{P}^n_{X/S}(\mathscr{F}))$, and we define

(16.7.7.1) $$\mathscr{P}^\infty_{X/S}(\mathscr{F}) = \varprojlim \mathscr{P}^n_{X/S}(\mathscr{F}).$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

(16.7.7.2) $$d^\infty_{X/S,\mathscr{F}} : \mathscr{F} \to \mathscr{P}^\infty_{X/S}(\mathscr{F}).$$

(16.7.8). Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) \simeq \mathscr{P}^n_{X/S}(\mathscr{F}) \otimes_{\mathscr{P}^n_{X/S}} \mathscr{P}^n_{X/S}(\mathscr{G}) \tag{16.7.8.1}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n.1, prop. 3).

We conclude in particular (or we see directly from the definition (16.7.2.1)) that if $\mathscr{F}$ has an $\mathscr{O}_X$-algebra structure (not necessarily associative), $\mathscr{P}^n_{X/S}(\mathscr{F})$ has a canonical $\mathscr{O}_X$-algebra structure; the latter is associative (resp. commutative, resp. unital, resp. a Lie algebra) if $\mathscr{F}$ is so. Also the canonical homomorphisms $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ for $n \leqslant m$ (16.7.7) are then algebra di-homomorphisms; similarly, (16.7.5.1) is then an $\mathscr{O}_X$-algebra homomorphisms when $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the $\mathscr{O}_X$-algebra structure from its structure of *right* $\mathscr{O}_X$-module.

With the same notations, we equally have a canonical homomorphisms of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{P}^n_{X/S}}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{P}^n_{X/S}(\mathscr{G})) \tag{16.7.8.2}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n.3), which is bijective when $\mathscr{P}^n_{X/S}$ is *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathscr{O}_{X'}$-bimodules

$$u^*(\mathscr{P}^n_{X/S}(\mathscr{F})) \longrightarrow \mathscr{P}^n_{X'/S'}(u^*(\mathscr{F})). \tag{16.7.9.1}$$

We leave it to the reader to extend the properties seen in (16.4) in the case $\mathscr{F} = \mathscr{O}_X$.

Remark (16.7.10). — The definition of $\mathscr{P}^n_{X/S}(\mathscr{F})$ in the form (16.7.1.2) still makes sense when $\mathscr{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (**0**, 3.7.1)); a variant of this definition allows us to define the "jet schemes" (relative to S) for any prescheme X.

16.8. Differential operators. [7]

Definition (16.8.1). — Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-modules, n an integer $\geqslant 0$. We say that a morphism $D : \mathscr{F} \to \mathscr{G}$ of sheaves of additive groups is a differential operator of order $\leqslant n$ (relative to S) if there is a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G}$ (where $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the structure of left $\mathscr{O}_X$-modules (16.7.4)) such that we have $D = u \circ d^n_{X/S,\mathscr{F}}$.

It is clear, because of the existence of canonical morphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F})$$

for $n \leqslant m$ (16.7.7), that a differential operator of order $\leqslant n$ is a differential operator of order $\leqslant m$
for $n \leqslant m$. If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, then, for every open set U of X, $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$.

We say that a homomorphism $D : \mathscr{F} \to \mathscr{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geqslant 0$ such that $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$. The *order* of a differential operator is the least upper bound of all integers n so that D is a differential operator of order $\leqslant n$ (and therefore $+\infty$ if there is no such integer); such order is always finite if X is *quasi-compact*. The differential operator of order 0 are exactly the homomorphisms of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G}$; the operators of order < 0 are *zero* by convention. For $n \geqslant 0$ a differential operator of order n is not in general a homomorphism of $\mathscr{O}_X$-modules but always a homomorphism of $\psi^*(\mathscr{O}_S)$-modules.

When $\mathscr{F} = \mathscr{O}_X$, a differential operator of order $\leqslant 1$ of $\mathscr{O}_X$ to $\mathscr{G}$ can be put in the form of $v + D$, where $v : \mathscr{O}_X \to \mathscr{G}$ is an $\mathscr{O}_X$-homomorphism, and D is an S-*derivation* (16.5.1) of $\mathscr{O}_X$ to $\mathscr{G}$: this follows from the structure of $P_{B/A}$ (**0**, 20.4.8).

[7] For a more general formalism, see the exposé VII of [eAG64] (due to P. Gabriel).

(16.8.2). To describe in a more precise manner a differential operator of order $\leqslant n$, $D : \mathscr{F} \to \mathscr{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V, to characterize the homomorphism $D = D_U : \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G})$. If we put $\Gamma(V, \mathscr{O}_S) = A$, $\Gamma(U, \mathscr{O}_X) = B$, so that B is an A-algebra, we have $\Gamma(U, \mathscr{P}^n_{X/S}(\mathscr{F})) = (B \otimes_A B)/\mathfrak{I}^{n+1}$, where we abbreviate $\mathfrak{I} = \mathfrak{I}_{B/A}$. Also put $M = \Gamma(U, \mathscr{F})$, $N = \Gamma(U, \mathscr{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A-homomorphism $D : M \to N$ factors through

$$M \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical morphism $t \mapsto 1 \otimes t$, and v is a B-homomorphism, the structure of B-module coming from the first factor (whereas we recall that in the formation of the tensor product over B, the structure of B-module on $(B \otimes_A B)/\mathfrak{I}^{n+1}$ comes from the second factor B). Note also that the B-module $((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M)/\mathfrak{I}^{n+1}(B \otimes_A M)$, where $(B \otimes_A M)$ is considered as a $(B \otimes_A B)$ module and its structure of B-module comes from $t \mapsto 1 \otimes t$ of B in $B \otimes_A B$. Let then D' be the B-homomorphism of $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; then condition of factorization of D is to say that D' must be zero on the B-module $\mathfrak{I}^{n+1}(B \otimes_A M)$.

(16.8.3). It is clear that the set of differential operators of order $\leqslant n$ from $\mathscr{F}$ to $\mathscr{G}$ forms an additive group, denoted by $\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G})$; when $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathrm{Diff}^n_{X/S}$ instead of $\mathrm{Diff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$.

We have seen (16.8.1), that given two open sets $U \supset V$ of X, we have a restriction homomorphism

$$\mathrm{Diff}^n_{X/S}(\mathscr{F}|U, \mathscr{G}|U) \longrightarrow \mathrm{Diff}^n_{X/S}(\mathscr{F}|V, \mathscr{G}|V)$$

from which we deduce that $U \mapsto \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U)$ is a presheaf of additive groups; in fact, it is actually a *sheaf*, since for an open set U varying in X, the homomorphisms $u \mapsto u \circ d^n_{U/S, \mathscr{F}|U}$ are *isomorphisms* of sheaves of additive groups **IV-4 | 41**

(16.8.3.1) $$\mathrm{Hom}_{\mathscr{O}_U}(\mathscr{P}^n_{U/S}(\mathscr{F}|U), \mathscr{G}|U) \simeq \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U),$$

because of the fact that the image of $\mathscr{F}$ by $d^n_{X/S, \mathscr{F}}$ generates $\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.6). We denote this sheaf by $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms (16.8.3.1) define an isomorphism of sheaves of additive groups*

(18.4.1) $$\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) \simeq \mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}).$$

When $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathscr{D}\mathit{iff}^n_{X/S}$ instead of $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$; it follows from (16.6.4) that $\mathscr{D}\mathit{iff}^n_{X/S}$ is the *dual* of $\mathscr{P}^n_{X/S}$; so we also write $\langle t, D\rangle$ instead of $u(t)$ if t is a section of $\mathscr{P}^n_{X/S}$ over an open set and if u is a homomorphism from $\mathscr{P}^n_{X/S}$ to $\mathscr{O}_X$ corresponding to D.

(16.8.5). When $\mathscr{P}^n_{X/S}(\mathscr{F})$ has an $\mathscr{O}_X$-bimodule structure (16.7.4), we deduce canonicallly an $\mathscr{O}_X$-bimodule structure on $\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G})$, and therefore on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ because of (16.8.4.1). More precisely, to the left $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$ corresponds, because of (16.8.1), the left $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ explained as follows: for every open set U of X, every section $a \in \Gamma(U, \mathscr{O}_X)$ and every differential operator $D : \mathscr{F}|U \to \mathscr{G}|U$, aD is the differential operator which, for every section $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.1) $$(aD)(t) = a(D(t))$$

of $\Gamma(U, \mathscr{G})$. Similarly, the right $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.2) $$(Da)(t) = D(at).$$

Proposition (16.8.6). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module for any of the structures defined in (16.8.5).*

Proof. The proposition follows from the fact that, under these hypothesis, $\mathscr{P}^n_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module of finite presentation (16.7.4) and of (**I**, 1.3.12) □

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$; we also see as in (16.8.3) that $U \mapsto \mathrm{Diff}_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$. It is immediate that $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$ is the union of the increasing filtered family of its subsheaves $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$ is similarly the union of its subgroups $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$ (16.8.1). The $\mathscr{O}_X$-bimodule structure on the $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$ induce therefore an $\mathscr{O}_X$-bimodule structure on $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$, further explained in (16.8.5.1) and (16.8.5.2). **IV-4 | 42**

Note that, for $n \leqslant m$, we have a commutative diagram

(16.8.7.1)
$$\begin{tikzcd} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] \arrow[d] & \mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G}) \arrow[d] \\ \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^m_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] & \mathscr{D}iff^m_{X/S}(\mathscr{F},\mathscr{G}) \end{tikzcd}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the horizontal arrow on the left comes from the canonical morphism $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.7). For every open set U of X, we then endow $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F})) = \varprojlim \Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$ of the projective limit topology of the discrete topologies on $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$, which defines on $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ a topological $\Gamma(U,\mathscr{O}_X)$-bimodule structure, so that $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ shows itself as a sheaf valued in the category of topological commutative groups (**0**, 3.2.6). So [God58, II, 1.11], the limit of the inductive system of sheaves of commutative groups $(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}))$ is precisely the sheaf of *continuous* germs of homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$ (the latter equipped with the discrete topology): the continuous homomorphisms $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ into the discrete group $\mathscr{G}$ indeed correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F})) \to \Gamma(U,\mathscr{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathscr{H}om.\mathrm{cont}_{\mathscr{O}_X}(\mathscr{P}^\infty_{X/S}(\mathscr{F}),\mathscr{G}) \simeq \mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$$

where the first member denotes the sheaf of germs of continuous homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$.

Proposition (16.8.8). — *Let $\mathscr{F},\mathscr{G}$ be two $\mathscr{O}_X$-modules, $D : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\psi^*(\mathscr{O}_S)$-modules, n an integer $\geqslant 0$. The following conditions are equivalent:*

(a) *D is a differential operator of order $\leqslant n$.*
(b) *For all sections a of $\mathscr{O}_X$ over an open set U, the homomorphism $D_a : \mathscr{F}|U \to \mathscr{G}|U$ such that, for every section t of $\mathscr{F}$ over an open set $V \subset U$, we have*

(16.8.8.1)
$$D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leqslant n-1$.
(c) *For every open set U of X, every family $(a_i)_{1\leqslant i\leqslant n+1}$ of $n+1$ sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, we have the identity*

(16.8.8.2)
$$\sum_{H\subset I_{n+1}} (-1)^{\mathrm{Card}(H)} \Big(\prod_{i\in H} a_i\Big) D\Big(\Big(\prod_{i\notin H} a_i\Big)t\Big) = 0$$

(where I_{n+1} is the interval $1 \leqslant i \leqslant n+1$ of $\mathbb{N}$).

IV-4 | 43

Proof. Let us prove first the equivalence of (a) and (b). By definition, to prove that D is a differential operator of order $\leqslant n$, it suffices to prove so for the restriction $D|U : \mathscr{F}|U \to \mathscr{G}|U$ to any affine open set U of X, and on the other hand the property (c) is valid for every open set U if it is so on every affine open set. We can therefore restrict ourselves to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affines. Because of (16.8.2) (where we use the same notations), the condition (a) means that the A-homomorphism $D' : B \otimes_A M \to N$ such that $D'(b \otimes t) = bD(t)$ is zero on $\mathfrak{I}^{n+1}(B \otimes_A M)$, which is, because of (**0**, 20.4.4), equivalent to saying that D' is annihilates every element of the form

$$\Big(\prod_{i=1}^{n+1}(a_i \otimes 1 - 1 \otimes a_i)\Big).(1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now, this element can be written as $\sum_{H \subset I_{n+1}} (\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i) t)$, and the image under D' of this element is the first member of (16.8.8.2), which proves the equivalence of (a) and (c).

Let us prove now the equivalence of (b) and (c). Let us reason by induction on n, the statement being trivial for $n = 0$. Writing a_{n+1} instead of a in the condition (b), we see, by the induction hypothesis, that condition (b) means that for every family $(a_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, that

$$\sum_{H' \subset I_{n+1}} (-1)^{\mathrm{Card}(H')} (\prod_{i \in H'} a_i) D_{a_{n+1}} ((\prod_{i \notin H'} a_i) t) = 0.$$

But if we replace on this relation $D_{a_{n+1}}$ by the definition (16.8.8.1), we check immediately that we have, up to sign, the first member of (16.8.8.2); from which we conclude. □

Proposition (16.8.9). — *If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, and $D' : \mathscr{G} \to \mathscr{H}$ a differential operator of order $\leqslant n'$, then $D' \circ D : \mathscr{F} \to \mathscr{H}$ is a differential operator of order $\leqslant n + n'$.*

Proof. By hypothesis, we can write $D = u \circ d^n_{X/S,\mathscr{F}}$ and $D' = v \circ d^n_{X/S,\mathscr{G}}$, where $u : \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{G}$ and $v : \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ are $\mathscr{O}_X$-homomorphisms. It all comes down to showing that the composite homomorphism of sheaves of additive groups

$$\mathscr{F} \xrightarrow{d^n_{X/S,\mathscr{F}}} \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{d^{n'}_{X/S,\mathscr{G}}} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

factors as

$$\mathscr{F} \xrightarrow{d^{n+n'}_{X/S,\mathscr{F}}} \mathscr{P}^{n+n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{w} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

where w is an $\mathscr{O}_X$-homomorphism. It suffices to prove the

Lemma (16.8.9.1). — *There is one and only one $\mathscr{O}_X$-homomorphism*

(16.8.9.2)
$$\delta : \mathscr{P}^{n+n'}_{X/S} \longrightarrow \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) = \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{P}^n_{X/S}$$

making the following diagram commute **IV-4** | 44

(16.8.9.3)
$$\begin{tikzcd}
\mathscr{O}_X \arrow[r, "d^{n+n'}_{X/S}"] \arrow[d, "d^n_{X/S}"'] & \mathscr{P}^{n+n'}_{X/S} \arrow[d, "\delta"] \\
\mathscr{P}^n_{X/S} \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S})
\end{tikzcd}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathscr{F}$

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "d^{n+n'}_{X/S,\mathscr{F}}"] \arrow[d, "d^n_{X/S,\mathscr{F}}"'] & \mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \arrow[d, "\delta\otimes 1"] \\
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}((\mathscr{F})))
\end{tikzcd}$$

and on the other hand, we verify immediately from the definition (16.7.5) that the diagram

$$\begin{tikzcd}
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "u"] \arrow[d, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{G} \arrow[d, "d^{n'}_{X/S,\mathscr{G}}"] \\
\mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}((\mathscr{F}))) \arrow[r, "1\otimes u"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{G})
\end{tikzcd}$$

is commutative. We finish the proof by taking w to be the composite $\mathscr{O}_X$-homomorphism

$$\mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \xrightarrow{\delta\otimes 1} \mathscr{P}^{n'}_{X/S}(\mathscr{P}^{n}_{X/S}(\mathscr{F})) \xrightarrow{1\otimes u} \mathscr{P}^{n'}_{X/S}(\mathscr{G}).$$

□

Proof (16.8.9.3). It remains to prove the lemma (16.8.9.3). Considering (16.7.6), which proves the uniqueness of δ, we are brought back to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines; letting $\mathfrak{I} = \mathfrak{I}_{B/A}$, it suffices to define a canonical homomorphism of B-modules

$$\phi : (B\otimes_A B)/\mathfrak{I}^{n+n'+1} \longrightarrow ((B\otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B\otimes_A B)/\mathfrak{I}^{n+1})$$

the B-module structure of the two members coming from the first B factor; recall that on tensor product of the second member, $(B\otimes_A B)/\mathfrak{I}^{n'+1}$ must be considered as a right B-module by its second B factor, and $(B\otimes_A B)/\mathfrak{I}^{n+1}$ as a left B-module by its first B factor (16.7.2). This is equivalent to defining a homomorphism of B-modules **IV-4 | 45**

$$\phi_0 : B\otimes_A B \longrightarrow ((B\otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B\otimes_A B)/\mathfrak{I}^{n+1})$$

and prove it is zero on $\mathfrak{I}^{n+n'+1}$. Now, we immediately define a homomorphism by the condition that

$$\phi_0(b\otimes b') = \pi_{n'}(b\otimes 1)\otimes \pi_n(1\otimes b') \quad \text{for } b, b' \text{ in } B$$

under the notations of (16.3.7). Also, it is immediate that ϕ_0 is a homomorphism of *rings*. Now, we can write

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes\pi_n(1\otimes 1) + \pi_{n'}(1\otimes b)\otimes\pi_n(1\otimes 1) - \pi_{n'}(1\otimes 1)\otimes\pi_n(1\otimes b)$$

and we have

$$\pi_{n'}(1\otimes b)\otimes\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)b\otimes\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes b\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes\pi_n(b\otimes 1)$$

from which, finally

(16.8.9.4) $$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes\pi_n(1\otimes 1) + \pi_{n'}(1\otimes 1)\otimes\pi_n(b\otimes 1 - 1\otimes b).$$

A product of $n+n'+1$ of terms of the form (16.8.9.4) is therefore necessarily zero, because the same is true for the product of $n+1$ terms of the form $\pi_n(b\otimes 1 - 1\otimes b)$ and of $n'+1$ terms of the form $\pi_{n'}(b\otimes 1 - 1\otimes b)$. The conclusion therefore follows from (**0**, 20.4.4). □

Corollary (16.8.10). — *The sheaf $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$ (also denoted $\mathscr{D}\!\mathit{iff}_{X/S}$) is canonically endowed with the structure of sheaf of rings, and the $\mathscr{D}\!\mathit{iff}^n_{X/S}$ form an increasing filtration compatible with such structure.*

In particular, $\mathscr{D}\!\mathit{iff}^0_{X/S}$ is a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$, which is canonically identified with $\mathscr{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathscr{O}_X$-bimodule of $\mathscr{D}\!\mathit{iff}_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathscr{O}_X$ considered as a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathscr{F} = \oplus_{\lambda\in L}\mathscr{F}_\lambda$; then it is clear (16.7.2.1) that $\mathscr{P}^n_{X/S}(\mathscr{F}) = \oplus_{\lambda\in L}\mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; since the functor $U \mapsto \Gamma(U,\mathscr{F})$ commutes with the formation of arbitrary direct sums, $d^n_{X/S,\mathscr{F}}$ is the homomorphism whose restriction to each $\mathscr{F}_\lambda$ is $d^n_{X/S,\mathscr{F}_\lambda} : \mathscr{F}_\lambda \to \mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; then we conclude immediately that we have

$$\operatorname{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\operatorname{Diff}^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}),$$

and therefore also (**0**, 3.2.6)

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}).$$

Moreover, if $\mathscr{G} = \prod_{\mu\in M}\mathscr{G}_\mu$ (**0**, 3.2.6), we have

$$\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) = \prod_{\mu\in M}\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}_\mu),$$

every homomorphism u from $\mathscr{P}^n_{X/S}(\mathscr{F})$ to $\mathscr{G}$ corresponds bijectively to the family of its composites $u_\mu : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G} \to \mathscr{G}_\mu$. We have therefore **IV-4 | 46**

$$\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\mu\in M} \mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}_\mu),$$

and consequently also

$$\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\mu\in M} \mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}_\mu).$$

(ii) So far, we have hardly encountered differential operators $\mathscr{F} \to \mathscr{G}$ where $\mathscr{F}$ and $\mathscr{G}$ are not locally free of finite rank, in which case the structure is reduced locally, because of (i), to the case of the sheaf $\mathscr{D}\mathit{iff}_{X/S}$; the latter will be studied later (16.11) in a particular case.

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathscr{I}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$, there is an open neighbourhood of x in X and a regular sequence (**0**, 15.2.2) (resp. quasi-regular (**0**, 15.2.2)) of elements of $\Gamma(U,\mathscr{O}_X)$ which generates $\mathscr{I}|U$.

We say that a regular (resp. quasi-regular) sequence of sections of $\mathscr{O}_X$ over U which generates $\mathscr{I}|U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathscr{I}|U$.

Definition (16.9.2). — Let $j : Y \to X$ be an immersion of preschemes and let U be an open set such that $j(Y) \subset U$ and that j is a closed immersion of Y in U. We say that j is regular (resp. quasi-regular) if the closed subprescheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathscr{O}_U$ (condition independent of the choice of U).

We say that a subprescheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \to X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subprescheme and $\mathscr{I}$ is the ideal of $\mathscr{O}_X$ that defines Y, it is the same as asking $\mathscr{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an *integral* ring, f and element $\neq 0$ of A, the closed subprescheme $V(f)$ of $\mathrm{Spec}(A)$ (isomorphic to $\mathrm{Spec}(A/(f))$) is *regularly immersed* in $\mathrm{Spec}(A)$.

Every regular ideal is quasi-regular (**0**, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — *Let X be a ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant m}$ a finite sequence of sections of $\mathscr{O}_X$ over X generating $\mathscr{I}$. For the (f_i) to be a quasi-regular sequence* (**0**, 15.2.2)*, it is necessary and sufficient that the following conditions are satisfied:*

(i) *The canonical images of f_i in $\mathscr{I}/\mathscr{I}^2$ form a basis of this $\mathscr{O}_X/\mathscr{I}$-module.*

(ii) *The canonical surjective homomorphism* (16.1.2.2)

$$\mathbf{S}^\bullet_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}\mathit{r}^\bullet_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Also, if this is true, then every sequence $(f'_i)_{1\leqslant i\leqslant n}$ of n sections of $\mathscr{I}$ over X which generates $\mathscr{I}$ is quasi-regular.

IV-4 | 47

Proof. The two conditions stated above only translate the ones in (**0**, 15.2.2), given the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). □

Corollary (16.9.4). — *Let X be a locally ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$. For $\mathscr{I}$ to be quasi-regular, it is necessary and sufficient that it satisfies the following conditions:*

(i) *$\mathscr{I}$ is of finite type.*

(ii) *$\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module.*

(iii) *The canonical homomorphism*

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X) \tag{16.9.4.1}$$

is bijective.

Proof. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough to show, because of (16.9.3), that if, in a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$, there is a neighbourhood U of x in X and n-sections f_i $(1 \leqslant i \leqslant n)$ of $\mathscr{I}$ over U such that the image in $\mathscr{I}/\mathscr{I}^2$ form a basis of $(\mathscr{I}/\mathscr{I}^2)|U$ over $(\mathscr{O}_X/\mathscr{I})|U$, so there will be a neighbourhood $V \subset U$ of x such that the $f_i|V$ generate $\mathscr{I}|V$. Now, by hypothesis, we have $\mathscr{I}_x \neq \mathscr{O}_x$, so that $\mathscr{I}_x$ is contained in the maximal ideal of $\mathscr{O}_X$; since $\mathscr{I}_x$ is an $\mathscr{O}_x$-module of finite type and the classes of $(f_i)_x$ in $\mathscr{I}_x/\mathscr{I}_x^2$ generate this $(\mathscr{O}_x/\mathscr{I}_x)$-module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathscr{I}_x$. Since $\mathscr{I}$ is of finite type, we conclude by (**0**, 5.2.2). □

Corollary (16.9.5). — *Let X be a locally ringed space, $\mathscr{I}$ a quasi-regular ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{I}$ over X, x a point of* $\operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$. *The following conditions are equivalent:*

- (a) *There is a neighbourhood of x in X such that $f_i|U$ form a quasi-regular sequence of elements $\Gamma(U, \mathscr{O}_X)$ generating $\mathscr{I}|U$.*
- (b) *The $(f_i)_x$ form a system of generators of $\mathscr{I}_x$ whose size is as small as possible.*
- (b′) *The $(f_i)_x$ form a minimal set of generators of $\mathscr{I}_x$.*
- (c) *If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X, \mathscr{I}/\mathscr{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathscr{O}_x/\mathscr{I}_x)$-module $\mathscr{I}_x/\mathscr{I}_x^2$.*

Proof. By hypothesis, $\mathscr{O}_x$ is a local ring, $\mathscr{I}_x$ an ideal of finite type of $\mathscr{O}_x$ contained in the maximal ideal of $\mathscr{O}_x$; the equivalence of (b), (b′) and (c) follows from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (**0**, 5.2.2) it follows that if condition (c) is satisfied (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathscr{I}/\mathscr{I}^2)|U$ has constant rank equal to n, and that the $f_i|U$ generate $\mathscr{I}|U$; it suffices now to apply the last assertion of (16.9.3) to U. □

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathscr{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathscr{O}_y/\mathscr{I}_y)$-module $(\mathscr{I}_y/\mathscr{I}_y^2)$ for all $y \in X$. We have an example by taking $X = \operatorname{Spec}(A)$, where A is a Dedekind ring, and **IV-4** | 48
$\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is a *non-principal* ideal of A; then indeed $\mathscr{I}_y/\mathscr{I}_y^2 = 0$ in every point y different from $x \in X$ corresponding to $\mathfrak{I}$, and $\mathscr{I}_x/\mathscr{I}_x^2$ has rank 1 over the field $\mathscr{O}_x/\mathscr{I}_x$; also, $\mathscr{I}$ is evidently a regular ideal.

(ii) In (16.9.5), one cannot replace "quasi-regular" by "regular", even when X is a prescheme (cf. (16.9.12)). Indeed, denote by B the ring of germs of infinitely differentiable functions on the point 0 on $\mathbb{R}$; it has a maximal ideal $\mathfrak{m}$ generated by the germ of t of the identity mapping to 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k > 0$ is not reduced to 0. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1, f_2) is *regular* in A: indeed, f_1 is not a zero divisor in A, because the relation $tP[T] \in \mathfrak{n}TB[T]$, for a polynomial $P \in B[T]$, implies that the products of t by the coefficients of P belongs to the ideal $\mathfrak{n}$, and it follows immediately that the coefficients are the same in $\mathfrak{n}$, so $P[T] \in \mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the ring of polynomials $\mathbb{R}[T]$, therefore integral, and the image of f_2 in A/f_1A, being equal to T, is not a zero divisor, so that our claim is true. However, f_2 is a zero divisor in A, since for every non-zero element $x \in \mathfrak{n}$, the image of x in A is $\neq 0$, but the image of xT is zero. We conclude that the sequence (f_2, f_1) is *not regular* in A; on the other hand, the ideal $\mathfrak{I} = f_1A + f_2A$ is distinct from A, so the conditions (b), (b′) and (c) of (16.9.5) do not imply the condition (a) when we replace "quasi-regular" by "regular".

(16.9.7). If $X = \operatorname{Spec}(A)$ is an affine scheme, we'll say that the ideal $\mathfrak{I}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathscr{J} = \widetilde{\mathfrak{I}}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular); we note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming in A a regular (resp. quasi-regular) sequence as the example (16.9.5) shows; however this is true if A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — *Let $j : Y \to X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:*

(i) j is an immersion locally of finite presentation.
(ii) The conormal sheaf $\mathscr{G}r^1(j) = \mathscr{N}_{Y/X}$ (16.1.2) is a locally free $\mathscr{O}_Y$-module.
(iii) The canonical homomorphism

$$\mathbf{S}^{\bullet}_{\mathscr{O}_Y}(\mathscr{G}r^1(j)) \longrightarrow \mathscr{G}r^{\bullet}(j)$$

(16.1.2.2) is bijective.

Proof. The problem being local on Y, we can restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X, so the translation of (16.9.4) into (16.9.8) follows from the description of $\mathscr{G}r^1(j)$ and $\mathscr{G}r^{\bullet}(j)$ in terms of the ideal $\mathscr{I}$ of $\mathscr{O}_X$ defining the subprescheme Y (16.1.3, (ii)). □

Corollary (16.9.9). — *Let Y be a prescheme, X an Y-prescheme, $j : Y \to X$ a Y-section of X, so that the n'th normal invariant $\mathscr{A}^{(n)}$ of j (16.1.2) is an augmented $\mathscr{O}_Y$-algebra (16.1.7); take $\mathscr{A}^{(\infty)} = \varprojlim \mathscr{A}^{(n)}$. For j to be a quasi-regular immersion it is necessary and sufficient that j is locally of finite presentation, and that every $y \in Y$ admits an affine open neighbourhood U of ring C such that $\mathscr{A}^{(\infty)}|U$ is isomorphic as an augmented $\mathscr{O}_U$-algebra to $\mathscr{O}_U[[T_1, \ldots, T_n]]$.*

Proof. We can reduce to the case where j is a closed immersion by restricting to a small neighbourhood of y (see the argument on (16.4.11)), and therefore $\mathscr{O}_Y$ is identified with the quotient algebra $\mathscr{O}_X/\mathscr{I}$ and the canonical surjective homomorphism $\mathscr{O}_X \to \mathscr{O}_Y$ admits a right inverse (16.1.7). We **IV-4** | 49
can therefore suppose $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ are affines, B being then an augmented A-algebra, and the augmentation ideal $\mathfrak{J}$ being of finite type. Since $\mathscr{A}^{(n)}$ is identified with $(B/\mathfrak{J}^{n+1})^{\sim}$, the corollary follows from the equivalence of (b) and (c) in (**0**, 19.5.4) when $B/\mathfrak{J} = A$. □

We note that, on the affine case considered, the fact that j is a quasi-regular immersion is further equivalent, because of (**0**, 19.5.4), to saying that B is a *formally smooth* A-algebra for the $\mathfrak{J}$-preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \to Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, $j : Y \to X$ the canonical injection, y a point of Y.*

(i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \to U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathscr{I}_y$ of the surjective homomorphism $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$.*
(ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a *coherent* ideal $\mathscr{I}$ of $\mathscr{O}_X$. The condition is clearly necessary. Conversely, if $\mathscr{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathscr{I}$ over an open neighbourhood U of y in X, we can suppose that the s_i generate $\mathscr{I}|U$ (**0**, 5.5.2) and form a regular sequence (**0**, 15.2.4), from which the claim follows.
(ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences of $\mathscr{O}_{X,y}$, formed from elements of the maximal ideal (**0**, 15.1.11)

□

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathscr{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and therefore *a fortiori* neither are quasi-regular morphisms in the sense of (6.8.1).

(ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (**0**, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\operatorname{Spec}(k(x)) \to X$ be a *regular* immersion.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, Y' a subprescheme of Y, such that the canonical injection $j : Y' \to Y$ is regular. Then the sequence of $\mathscr{O}_{Y'}$-modules*

(16.9.13.1) $$0 \longrightarrow j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U **IV-4** | 50
of the homomorphisms of (16.9.13.1) form a split exact sequence.

Let us first prove the following lemma:

Lemma (16.9.13.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A, $A' = A/\mathfrak{J}$, $(f_i)_{1 \leqslant i \leqslant r}$ a sequence of elements of A which is A'-regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{J} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geqslant n$, we have the relation*

(16.9.13.3) $$\mathfrak{J} \cap \mathfrak{K}^n = \mathfrak{J}\mathfrak{K}^n + \mathfrak{J}\mathfrak{K}^N.$$

Proof. It is clearly sufficient to prove that every element of the first is contained in the second, and, by induction on n, we reduce to the case $N = n+1$. An element of the first member of (16.9.13.3), being in $\mathfrak{K}^n$, is written as $P(f_1, \ldots, f_r)$, where $P \in A[T_1, \ldots, T_r]$ is homogeneous of degree n. If f_i' is the canonical image of f_i in A', the hypothesis $P(f_1, \ldots, f_n) \in \mathfrak{J}$ means that $P(f_1', \ldots, f_n') = 0$. But $P(f_1', \ldots, f_n') \in \mathfrak{K}'^n$, so the canonical image of $P(f_1', \ldots, f_n')$ in $\mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A'-regular implies that the canonical homomorphism $\mathbf{S}_C^n(\mathfrak{K}'/\mathfrak{K}'^2) \to \mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is bijective (**0**, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{J} + \mathfrak{K}$. It follows immediately that $P(f_1, \ldots, f_n) \in \mathfrak{J}\mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \ldots, f_n) \in \mathfrak{J}$, we finally have $P(f_1, \ldots, f_n) \in \mathfrak{J}\mathfrak{K}^n + \mathfrak{J}\mathfrak{K}^{n+1}$ which proves the lemma. □

By taking the quotient of both side of (16.9.13.3) by $\mathfrak{J}\mathfrak{K}^n$, we see that the relations (16.9.13.3) for $N \geqslant n$ entail

(16.9.13.4) $$(\mathfrak{J} \cap \mathfrak{K}^n)/\mathfrak{J}\mathfrak{K}^n \subset \bigcap_{N \geqslant n} \mathfrak{K}^N.(A/(\mathfrak{J}\mathfrak{K}^n)).$$

We deduce the

Corollary (16.9.13.5). — *Suppose that the hypotheses of (16.9.13.2) are satisfied and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A. Then for every integer $n > 0$,*

(16.9.13.6) $$\mathfrak{J} \cap \mathfrak{K}^n = \mathfrak{J}\mathfrak{K}^n.$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{J}\mathfrak{K}^n$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). □

Taking in particular $n = 2$ in (16.9.13.6), and remarking that we have $\mathfrak{L}^2 = \mathfrak{J}^2 + \mathfrak{J}\mathfrak{K} + \mathfrak{K}^2 = \mathfrak{J}\mathfrak{L} + \mathfrak{K}^2$; since $\mathfrak{J}\mathfrak{L} \subset \mathfrak{L}^2$, we deduce that

$$\mathfrak{J} \cap \mathfrak{L}^2 = \mathfrak{J}\mathfrak{L} + (\mathfrak{J} \cap \mathfrak{K}^2) = \mathfrak{J}\mathfrak{L} + \mathfrak{J}\mathfrak{K}^2 = \mathfrak{J}\mathfrak{L},$$

in other words,

(16.9.13.7) $$\mathfrak{J} \cap \mathfrak{L}^2 = \mathfrak{J}\mathfrak{L},$$

which we can also express in saying that the canonical homomorphism

$$\mathfrak{J}/\mathfrak{J}\mathfrak{L} \longrightarrow (\mathfrak{J} + \mathfrak{J}^2)/\mathfrak{J}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that the sequence of stalks of the sheaves appearing in (16.9.13.1), in a point $x \in Y'$, **IV-4** | 51
is exact. Now, if we take $A = \mathscr{O}_{X,x}$, we can write $\mathscr{O}_{Y,x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A, then $\mathscr{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an A'-*regular* sequence of elements of A', which themselves are images of elements from a A'-regular sequence of elements of A belonging to the maximal ideal of A. If $\mathfrak{K}$ is the ideal generated by the latter and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, we have $\mathscr{O}_{Y',x} = A/\mathfrak{L}$, and since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I}/\mathfrak{I}\mathfrak{L} \to (\mathfrak{I} + \mathfrak{J}^2)/\mathfrak{J}^2$ is bijective. But this shows that the sequence

$$0 \longrightarrow \mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow \mathfrak{L}/\mathfrak{L}^2 \longrightarrow (\mathfrak{I}/\mathfrak{L})/(\mathfrak{I}/\mathfrak{L})^2 \longrightarrow 0$$

is exact (see the proof of (16.2.7)), and the modules making up this sequence are precisely the stalks in x of the sheaves of (16.9.13.1). The second claim follows from the fact that $\mathscr{N}_{X/Y}$ is a locally free $\mathscr{O}_{Y'}$-module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rded., §1, nº11, prop. 21. □

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \to S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

(i) $\Omega^1_{X/S}$ is a locally projective $\mathscr{O}_X$-module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega^1_{X/S})$ is a projective $\Gamma(U, \mathscr{O}_X)$-module (not necessarily of finite type).

(ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}^\bullet_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S})$$

is bijective.

In particular, if $\Omega^1_{X/S}$ is locally free of finite rank, the $\mathscr{P}^n_{X/S}$ are locally free of finite rank $\mathscr{O}_X$-modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \to S$ is differentially smooth, since $\Omega^1_{X/S} = 0$ because of (**I**, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Let us limit ourselves to proving the following proposition:

Proposition (16.10.2). — *Let A be a ring, B a formally smooth A-algebra for the discrete topology* (**0**, 19.3.1)*. Then* $\mathrm{Spec}(B)$ *is differentially smooth over* $\mathrm{Spec}(A)$.

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B-algebra (by one or the other canonical homomorphisms $b \mapsto b \otimes 1$, $b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (**0**, 19.3.5, (iii)); therefore **IV-4** | 52
$B \otimes_A B$ is a formally smooth A-algebra with the discrete topology (**0**, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A-algebra for the $\mathfrak{I}$-preadic topology (**0**, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A-algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (**0**, 19.5.4). □

Proposition (16.10.3). — *For a morphism $f : X \to S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x, of ring A, such that $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is an augmented topological A-algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A-module and B being endowed with the B^+-preadic topology (where B^+ is the augmentation ideal). If $\Omega^1_{X/S}$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \ldots, T_n]]$.*

Proof. The notion of a differentially smooth morphism is clearly local on X, so we reduce to the case where $S = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(C)$. Consider $C \otimes_B C$ as a C-algebra (by the first factor); take $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow C with the $\mathfrak{I}$-preadic topology; we can apply to the topological C-algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (**0**, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C-algebra for the discrete topologies. The topology of $\Gamma(U, \mathscr{P}^\infty_{X/S})$ is clearly the projective limit topology of this ring (16.1.11). □

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega^1_{X/S}$ in the point x. We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fibre $f^{-1}(f(x))$.

Proposition (16.10.4). — *Let $f : X \to S$, $g : S' \to S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \to S'$.*

(i) *If f is differentially smooth, the same is true for f'.*
(ii) *Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega^1_{X'/S'}$ is a finite type $\mathscr{O}_{X'}$-module, f is differentially smooth and $\Omega^1_{X/S}$ is a finite type $\mathscr{O}_X$-module.*

Proof. Indeed, if f is differentially smooth, the $\mathscr{G}r_n(\mathscr{P}^n_{X/S})$ are *flat* $\mathscr{O}_X$-modules; therefore by (16.4.6), the homomorphism $\mathscr{G}r_n(\mathscr{P}^n_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}^n_{X'/S'})$ are bijective for every n, because of the commutative of the diagram (16.2.1.3), it follows from the definition (16.10.1) that f' is differentially smooth. On the other hand, if g is faithfully flat and quasi-compact, it follows also from (16.4.6) that $\mathscr{G}r_n(\mathscr{P}^n_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}^n_{X'/S'})$ is bijective for every n. Suppose also that f' is differentially smooth and $\Omega^1_{X'/S'}$ is of finite rank. Since the canonical projection $X' \to X$ is a faithfully flat and quasi-compact morphism, it follows first from (2.5.2) that $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. □

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \to S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \to X \times_S X$ be quasi-regular.*

IV-4 | 53

Proof. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subprescheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (**I**, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathscr{I}$ of finite type, and $\Omega^1_{X/S} = \mathscr{I}/\mathscr{I}^2$ is an $\mathscr{O}_X$-module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). □

Remark (16.10.6). — Let $f : X \to S$ be a morphism such that the $\mathscr{O}_X$-module $\Omega^1_{X/S}$ is locally free of finite rank. It follows from (**0**, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda \in L}$ of sections of $\mathscr{O}_X$ over U for which $(dz_\lambda)_{\lambda \in L}$ forms a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

16.11. Differential operators on a differentially smooth S-prescheme

(16.11.1). Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathscr{O}_X$ over U such that the dz_λ form a system of generators of $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. Let m be an integer or the symbol ∞, and take, for every λ

(16.11.1.1)
$$\zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathscr{P}^m_{X/S}).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

(16.11.1.2)
$$|\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p} = \prod_\lambda (p_\lambda !)$$

(16.11.1.3)
$$\binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}!/(\mathbf{q}!(\mathbf{p}-\mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leqslant \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \not\leqslant \mathbf{p}$,

(16.11.1.4)
$$\mathbf{z}^\mathbf{p} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^\mathbf{p} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

(16.11.1.5)
$$d^m(\mathbf{z}^\mathbf{p}) = (d^m(\mathbf{z}))^\mathbf{p} = (\boldsymbol{\zeta} + \mathbf{z})^\mathbf{p} = \sum_{\mathbf{q} \leqslant \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^\mathbf{q}$$

(16.11.1.6) $$\boldsymbol{\zeta}^{\mathbf{p}} = (d^m\mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|}\binom{\mathbf{p}}{\mathbf{q}}\mathbf{z}^{\mathbf{p}-\mathbf{q}}d^m(\mathbf{z}^{\mathbf{q}}).$$

Since the dz_λ generate $\Omega^1_{X/S}$, and are the images of δz_λ, and as the canonical morphism (16.3.1.1) is surjective, we conclude that for finite m, the δz_λ generate the $\mathscr{O}_U$-algebra $\mathscr{P}^n_{U/S}$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 du th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leqslant m$) generate the $\mathscr{O}_U$-module $\mathscr{P}^n_{U/S}$. A differential operator $D \in \mathrm{Diff}^m_{U/S}$ is consequently entirely determined by the values of $\langle\boldsymbol{\zeta}^{\mathbf{p}}, D\rangle$for $|\mathbf{p}| \leqslant m$, or, which amounts to the same by (16.11.1.5) and (16.11.1.6), by the values of the $\langle d^m(\mathbf{z}), D\rangle = D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leqslant m$; more precisely, it follows from (16.11.5) that we have **IV-4 | 54**

(16.11.1.7) $$D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}}\langle\boldsymbol{\zeta}^{\mathbf{q}}, D\rangle\mathbf{z}^{\mathbf{p}-\mathbf{q}}.$$

Theorem (16.11.2). — *Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the family $(dz_\lambda)_{\lambda\in L}$ generates $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. The following conditions are equivalent:*

(a) $f|U$ is differentially smooth and (dz_λ) is a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$.
(b) There is a family $(D_{\mathbf{p}})_{\mathbf{p}\in\mathbb{N}^{(L)}}$ of differential operators of $\mathscr{O}_U$ to itself verifying the conditions

(16.11.2.1) $$D_{\mathbf{p}}(\mathbf{z}^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}}\mathbf{z}^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}).$$

Also, when these conditions are satisfied, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

(16.11.2.2) $$D_{\mathbf{q}} \circ D_{\mathbf{p}} = D_{\mathbf{p}} \circ D_{\mathbf{q}} = \frac{(\mathbf{p}+\mathbf{q})!}{\mathbf{p}!\mathbf{q}!}D_{\mathbf{p}+\mathbf{q}} \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}).$$

Finally, if L is finite, for every integer m, the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\mathit{iff}^m_{U/S}$, in other words, every differential operator of order $\leqslant m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}|\leqslant m} a_{\mathbf{p}}D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathscr{O}_X$ over U.

Proof. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

(16.11.2.3) $$\langle\boldsymbol{\zeta}^{\mathbf{p}}, D_{\mathbf{q}}\rangle = \delta_{\mathbf{pq}} \quad \text{(Kronecker's symbol)}.$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$. Then, for every integer $m \geqslant 1$, we deduce similarly from (16.11.2.3) that the $\boldsymbol{\zeta}^{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from the definition (16.10.1), the fact that $\boldsymbol{\zeta}^{\mathbf{p}}$ form a basis of $\mathscr{P}^n_{U/S}$ for $|\mathbf{p}| \leqslant m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathscr{P}^n_{U/S} \to \mathscr{O}_U(|\mathbf{q}| \leqslant m)$ such that $\langle\zeta^{\mathbf{p}}, u_{\mathbf{q},m}\rangle = \delta\mathbf{p}, \mathbf{q}$ for $|\mathbf{p}| \leqslant m$, $|\mathbf{q}| \leqslant m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geqslant |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined, and that, if L is finite, for $|\mathbf{p}| \leqslant m$, the $D_{\mathbf{p}}$ form a basis of the dual $\mathscr{D}\mathit{iff}^m_{U/S}$ of $\mathscr{P}^n_{U/S}$. Finally, the relations (16.11.2.2) follows immediately from the expression of the values of the three operators considered on the $\mathbf{z}^{\mathbf{r}}$, and of the fact that the $\boldsymbol{\zeta}^{\mathbf{r}}$ for $|\mathbf{r}| \leqslant m$ generate $\mathscr{P}^n_{U/S}$. ☐

IV-4 | 55

Remarks (16.11.3). — (i) The fact that, because of (16.11.2.2), the $D_{\mathbf{p}}$ are pairwise permutable naturally do not imply that the $\mathscr{O}_U$-algebra $\mathscr{D}\mathit{iff}_{U/S}$ is commutative, the $D_{\mathbf{p}}$ do not commute with the sections of $\mathscr{O}_U$ unless $n = 0$.

(ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\boldsymbol{\epsilon}_\lambda = (\epsilon_{\lambda\mu})_{\mu\in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators $D_{\boldsymbol{\epsilon}_\lambda}$ are exactly the S-derivations D_i introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).

(iii) For every integer $r \geqslant 1$, we can define the notion of *differentially smooth up to order r* by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}^m_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_m(\mathscr{P}^n_{X/S})$$

are bijective *for every $m \leqslant r$*. The argument of (16.11.2) proves also that if, in the condition (a), we replace "differentially smooth" by "differentially smooth up to order r", this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leqslant r$, $|\mathbf{q}| \leqslant r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is *of characteristic p* (p equal to zero or a prime number) if, for every affine open set U of X, the ring $\Gamma(U, \mathscr{O}_X)$ is of characteristic p (**0**, 21.1.1). It follows from (**0**, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every *closed* point x of X, the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$-prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a scheme of characteristic 0, $f : X \to S$ a morphism. If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X, we can suppose there is a family (z_λ) of sections of $\mathscr{O}_X$ over X such that the (dz_λ) is a basis for the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_\lambda D_\lambda^{p_\lambda}$$

(where the D_λ are the coordinate forms corresponding to the basis (dz_λ)) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_λ are derivations. □

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \operatorname{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \operatorname{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega^1_{X/S}$ has rank 1, and that the morphism $X \to S$ **IV-4** | 56
has rank 1, and that the morphism $X \to S$ is differentially smooth up to order $p - 1$ (16.11.3, (iii)), but not of order p. However, the proof of (16.12.2) proves that if $\Omega^1_{X/S}$ is locally free, and if $n!1_{\mathscr{O}_X}$ is invertible in $\Gamma(X, \mathscr{O}_X)$, then X is differentially smooth over S up to order n.

§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.

In this paragraph, we revisit the concepts studied in ($\mathbf{0}_{\text{III}}$, 9), expressed in the geometric language of schemes from a global point of view, for preschemes locally of finite presentation over a given base.

Most of the results (except 17.7, 17.8, 17.9, 17.13, and 17.16) are reduced to various properties already encountered in ($\mathbf{0}_{\text{III}}$, 9).

For more specific results on étale morphisms, the reader should consult §18.

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y', all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathscr{J}$ of $\mathscr{O}_{Y'}$, and every morphism $Y' \to Y$, the map

(17.1.1.1) $$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

induced by the canonical map $Y'_0 \to Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y.

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

(i) Suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \to B$. According to (**0**, 19.3.1) and (**0**, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ, B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A-algebra, for the *discrete* topologies on A and B.

(ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathscr{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathscr{J}^2 = 0$, and if we have $\mathscr{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals $\mathscr{J}^{j+1}$ for $0 \leqslant j \leqslant n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leqslant j \leqslant n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). **IV** | 57

(iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* ($\mathbf{0}_{\mathbf{III}}$, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y-preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

(iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an *arbitrary* Y-prescheme Z and a closed subprescheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Z$. Then the map

(17.1.2.1) $$\mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \to Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathscr{J}|U_\alpha$ are nilpotent, and for each α, let U_α^0 be the inverse image of U_α in Z_0, which is the closed subprescheme of U_α defined by $\mathscr{J}|U_\alpha$. Let $f_0 : Z_0 \to X$ by a Y-morphism; by hypothesis, for each α, there is at most one (resp. one and only one) Y-morphism $f_\alpha : U_\alpha \to X$ whose restriction to Z_0 coincides with $f_0|U_\alpha$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|V = f_\beta|V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y-morphism $f : Z \to X$ whose restriction to Z_0 coincides with f_0.

Proposition (17.1.3). —

(i) *A monomorphism of preschemes is formally unramified; an open immersion is formally étale.*

(ii) *The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).*

(iii) *If $f : X \to Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*
(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f.*
(vi) *If $f : X \to Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$.*

Proof. According to (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \to Y$, $g : Y \to Z$, an affine scheme Z', a closed subscheme Z'_0 of Z defined by a nilpotent ideal and a morphism $Z' \to Z$. Suppose that f and g formally smooth, and consider a Z-morphism $u_0 : Z'_0 \to X$; the hypothesis on g implies that there **IV | 58**
exists a Z-morphism $v : Z' \to Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \to Z$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \to X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \to Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified.

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'', a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \to Y'$ making Y'' a Y'-prescheme; we then know by (**I**, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with $\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). □

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f.*

Proof. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \to Y$ a morphism, $j : Y'_0 \to Y'$ the canonical injection, $u_0 : Y'_0 \to Y$ a Y-morphism, such that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \to X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z-morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y-morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. □

Corollary (17.1.5). — *Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.*

Proof. This follows from (17.1.4) and (17.1.3, (ii) and (iv)). □

Proposition (17.1.6). — *Let $f : X \to Y$ be a morphism of preschemes.*

(i) *Let (U_α) be an open covering of X and, for each α, let $i_\alpha : U_\alpha \to X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.*
(ii) *Let (V_λ) be an open covering of Y. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \to V_\lambda$ of f is.*

Proof. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \to Y$ and $i_\lambda : f^{-1}(V_\lambda) \to X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \to V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if **IV | 59**
the $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f.

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathscr{J}$, which we may assume to satisfy $\mathscr{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \to Y$ be a morphism. Suppose we are given a Y-morphism $u_0 : Y'_0 \to X$; denote by W_α (resp. W^0_α) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the *same underlying topological space*). Let us first suppose that the $f \circ i_\alpha$ are *formally unramified*, and show that, if u' and u''

are two Y-morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α, we have $u'|W_\alpha = u''|W_\alpha$, since the restrictions of both Y-morphisms to W^0_α coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are *formally smooth* and prove the existence of a Y-morphism $u : Y' \to X$ whose restriction to Y'_0 is u_0. Now, since Y' is an *affine scheme*, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u. □

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y, which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \to Y$ to be formally unramified, it is necessary and sufficient that $\Omega^1_f = 0$ (what we still write $\Omega^1_{X/Y} = 0$ (16.3.1)).*

Proof. Taking into account (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the conclusion then follows from (**0**, 20.7.4) and the interpretation of $\Omega^1_{X/Y}$ in this case (16.3.7). □

Corollary (17.2.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms. For f to be formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is surjective.

Proof. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). □

Proposition (17.2.3). — *Let $f : X \to Y$ be a formally smooth morphism.*

(i) *The $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega^1_{X/Y}$ is locally free and of finite type.*

(ii) *For all morphisms $g : Y \to Z$, the sequence (16.4.19) of $\mathscr{O}_X$-modules*

(17.2.3.1) $$0 \longrightarrow f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split *exact sequence.*

IV | 60

Proof.

(i) We know (16.3.9) that if f is locally of finite type, then Ω^1_f is an $\mathscr{O}_X$-module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (**0**, 20.4.9) and (**0**, 19.2.1).

(ii) Again, we can restrict to the case where X, Y, and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (**0**, 20.5.7).

□

Corollary (17.2.4). — *If $f : X \to Y$ is formally étale, then, for all morphisms $g : Y \to Z$, the canonical homomorphism of $\mathscr{O}_X$-modules*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is bijective.

Proof. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega^1_{X/Y} = 0$ (17.2.1). □

Proposition (17.2.5). — *Let $f : X \to Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathscr{O}_X$-modules (16.4.21)*

$$0 \longrightarrow \mathscr{N}_{X'/X} \longrightarrow \Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/Y} \longrightarrow 0 \tag{17.2.5.1}$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) form a split *exact sequence.*

Proof. By virtue of (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and $X' = \operatorname{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B. The conormal sheaf $\mathscr{N}_{X'/X}$ then corresponds to the B-module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (**0**, 20.5.14). □

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

(a) *f is a monomorphism.*
(b) *f is radicial and formally unramified.*
(c) *For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$-isomorphic to* $\operatorname{Spec}(k(y))$ *(in other words, it is reduced to a single point z such that $k(y) \to \mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ is an isomorphism).*

Proof. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega^1_{X/Y} = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is quasi-coherent of finite type (16.3.9). It follows from (**I**, 9.1.13.1) that, for $(\Omega^1_{X/Y})_x = 0$, it is necessary and sufficient that if we set $Y_1 = \operatorname{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega^1_{X_1/Y_1})_x = 0$; but as the morphism $f_1 : X_1 \to Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \to X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega^1_{X/Y}$ is by definition the conormal sheaf $\mathscr{G}r_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathscr{G}r_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation IV | 61
(1.4.3.1); therefore the hypothesis $\mathscr{G}r_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (**I**, 5.3.8). □

17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \to Y$ is *smooth* (resp. *unramified*, or *net* [8] resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over* Y.

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remark (17.3.2). —

(i) Note that definition (17.3.1) can be phrased using only the functor

$$Y' \longmapsto \operatorname{Hom}_Y(Y', X)$$

considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

(ii) Let A be a ring and B an A-algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A-algebra if the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to saying that B is an A-algebra *of finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.

[8] The words "net" and "formally net" seem more preferable to the terminology used in "unramified" (resp. formally unramified") and will be used almost exclusively in Chapter V. In this chapter, we have kept the old terminology so as not to conflict with **0**, 19.10.

(iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y*.

Proposition (17.3.3). —

(i) *An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient for it to be locally of finite presentation.*
(ii) *The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).*
(iii) *If $f : X \to Y$ is a smooth (resp. unramified, resp. étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*
(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are smooth (resp. unramified, resp. étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.* **IV** | 62
(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.*

Proof. This follows from (1.4.3) and (17.1.3). □

Proposition (17.3.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f.*

Proof. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). □

Corollary (17.3.5). — *Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.*

Proof. This follows from (17.3.4) and (17.3.3, (ii)). □

Proposition (17.3.6). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism* (16.4.19)

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective.

Proof. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). □

Definition (17.3.7). — Let $f : X \to Y$ be a morphism. We say that f is *smooth* (resp. *unramified*, resp. *étale*) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y.

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over Y at the point x*.

Taking into account remark (17.3.2, (iii)), it is equivalent to saying that f is smooth (resp. unramified, resp. étale) and to say that f is smooth (resp. unramified, resp. étale) at all points of X.

It is clear that the set of points of X at which the morphism $f : X \to Y$ is smooth (resp. unramified, resp. étale) is *open* in X.

Proposition (17.3.8). — *For all preschemes Y and all locally free $\mathscr{O}_Y$-modules $\mathscr{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathscr{E})$* (**II**, 1.7.8) *associated to $\mathscr{E}$ is a smooth Y-prescheme.*

Proof. Indeed (17.3.2, (iii)), we can restrict ourselves to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{V}(\mathscr{E}) = \operatorname{Spec}(A[T_1, \ldots, T_r])$; as $A[T_1, \ldots, T_r]$ is a formally smooth A-algebra for the discrete topologies (**0**, 19.3.2), and of finite presentation, this proves the proposition (17.3.2, (ii)) □

Corollary (17.3.9). — *Under the hypotheses of* (17.3.8), *the projective prescheme $\mathbf{P}(\mathscr{E})$* (**II**, 4.1.1) *is a smooth Y-prescheme.*

Proof. We can still restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{P}(\mathscr{E}) = \mathbf{P}^r_Y$. We then know (**II**, 2.3.14) that we have a finite open cover of $\mathbf{P}^r_A$ by the $D_+(T_i)$ ($0 \leqslant i \leqslant r$) respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \ldots, T_r]$ and f for T_i; but it follows immediately from the definition of $S_{(f)}$ (**II**, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \ldots, T_{i-1}, T_{i+1}, \ldots, T_r]$; hence the corollary follows by (17.3.8). □

17.4. Characterizations of unramified morphisms. **IV** | 63

§18. SUPPLEMENT ON ÉTALE MORPHISMS. HENSELIAN LOCAL RINGS AND STRICTLY LOCAL RINGS

18.1. A remarkable equivalence of categories.

Theorem (18.1.1). — *Let $\phi : A \to B$ be a local homomorphism of local rings, $\mathfrak{m}$ the maximal ideal of A, $B_0 = B/\mathfrak{m}B$. The following conditions are equivalent:*

*label=*a) *ϕ is a formally étale homomorphism (for the preadic topologies defined by $\mathfrak{m}$ and $\mathfrak{m}B$ respectively).*
*lbbel=*b) *B is a* Noetherian *ring, B is $\mathfrak{m}B$-adically* separated and complete, *and B_0 is a formally étale $A/\mathfrak{m}$-algebra (for the discrete topologies).*

Proof. We have already seen ((**0**, 19.7.1)) that the conditions are equivalent when A is a *Noetherian* complete local ring (with B also Noetherian complete local). Now let us handle the general case. **IV-4** | 109

We first prove that *a*) implies *b*). By (**0**, 19.3.5, ii) and (**0**, 19.6.4), we see that B is Noetherian and $\mathfrak{m}B$-adically separated and complete, and it remains to show that B_0 is a formally étale $(A/\mathfrak{m})$-algebra. But if C_0 is an $(A/\mathfrak{m})$-algebra and $\mathfrak{J}_0$ a nilpotent ideal of C_0, the ring C_0 can be considered as an A-algebra, and $\mathfrak{J}_0$ as an ideal of this A-algebra; moreover, C_0 is separated and complete for the $\mathfrak{m}$-preadic topology since $\mathfrak{m}C_0 = 0$. The hypothesis on ϕ then implies that the canonical map $\mathrm{Hom}_{A\text{-alg}}(B, C_0) \to \mathrm{Hom}_{A\text{-alg}}(B, C_0/\mathfrak{J}_0)$ is bijective; but since $\mathfrak{m}C_0 = 0$, both sides identify respectively with $\mathrm{Hom}_{(A/\mathfrak{m})\text{-alg}}(B_0, C_0)$ and $\mathrm{Hom}_{(A/\mathfrak{m})\text{-alg}}(B_0, C_0/\mathfrak{J}_0)$, hence B_0 is formally étale over $A/\mathfrak{m}$.

Conversely, suppose *b*) is satisfied, and let C be an A-algebra, separated and complete for the $\mathfrak{m}$-preadic topology, $\mathfrak{J}$ an ideal of C such that $\mathfrak{J}^2 = 0$ (which is sufficient by (**0**, 19.3.6, ii)). We must show that the canonical map $\mathrm{Hom}(B, C) \to \mathrm{Hom}(B, C/\mathfrak{J})$ is bijective (the Hom being taken in the category of continuous A-algebra homomorphisms for the $\mathfrak{m}$-preadic topologies on B, C, and $C/\mathfrak{J}$). Let us pose $A_n = A/\mathfrak{m}^{n+1}$, $B_n = B/\mathfrak{m}^{n+1}B$, $C_n = C/\mathfrak{m}^{n+1}C$, $\mathfrak{J}_n$ the image of $\mathfrak{J}$ in C_n; since B and C are separated and complete, we have $B = \varprojlim B_n$ and $C = \varprojlim C_n$, and since the B_n are Artinian, and $C_n/\mathfrak{J}_n = (C/\mathfrak{J})_n$, we are reduced to proving that $\mathrm{Hom}_{A_n\text{-alg}}(B_n, C_n) \to \mathrm{Hom}_{A_n\text{-alg}}(B_n, C_n/\mathfrak{J}_n)$ is bijective for all n, i.e. that B_n is a formally étale A_n-algebra (for the discrete topologies). Now, by the already established special case ((**0**, 19.7.1)), B is formally étale over the completion $\widehat{A}$ of A; in particular, B_n is a formally étale A_n-algebra, since $A_n = \widehat{A}/\mathfrak{m}^{n+1}\widehat{A}$, which concludes the proof. □

(18.1.2). Under the hypotheses and with the notation of (18.1.1), the formally étale homomorphism $\phi : A \to B$ is also *faithfully flat* ((**0**, 19.7.1, i)). In particular, the image in B of every non-zero-divisor of A is a non-zero-divisor of B; the image in B of every element of A not belonging to $\mathfrak{m}$ is *invertible* in B; therefore ϕ extends to a homomorphism $\phi' : A_{\mathfrak{m}} \to B$, where $A_{\mathfrak{m}}$ is the localisation of A at $\mathfrak{m}$ (which coincides with A when A is already local). It is immediate that ϕ' is also a formally étale homomorphism. On the other hand, if $\mathfrak{n}$ is a maximal ideal of B, then $\mathfrak{n} \supset \mathfrak{m}B$ since $\mathfrak{m}B$ is contained in the radical of B ((**0**, 7.1.10)); the quotient $B/\mathfrak{n}$ is therefore a quotient of B_0, and conversely, any maximal ideal of B_0 is the image of a maximal ideal of B; in other words, the maximal ideals of B correspond bijectively to those of B_0. Since B_0 is a formally étale $\kappa(\mathfrak{m})$-algebra (where $\kappa(\mathfrak{m}) = A/\mathfrak{m}$), B_0 is a *finite* product of *finite separable extensions* of $\kappa(\mathfrak{m})$ ((**0**, 20.7.3)).

(18.1.3). We now specialise to the case where B is a *local* ring. Then B_0 is a finite separable extension of $\kappa(\mathfrak{m})$, therefore there exists a unique (by (**0**, 20.5.9)) polynomial $F \in A[\mathrm{T}]$ *monic* and such that B_0 is isomorphic to $\kappa(\mathfrak{m})[\mathrm{T}]/(F_0)$, where F_0 is the canonical image of F in $\kappa(\mathfrak{m})[\mathrm{T}]$; moreover, F_0 is a product of *distinct* irreducible factors in $\kappa(\mathfrak{m})[\mathrm{T}]$, the image ξ of T in B_0 is a root of F_0, and B_0 is isomorphic to $\kappa(\mathfrak{m})[\mathrm{T}]/(g_0)$ where g_0 is the minimal polynomial of ξ over $\kappa(\mathfrak{m})$, so g_0 is *separable* and divides F_0. By a well-known property of Henselian rings ((**0**, I.8.3)), B is isomorphic to $\widehat{A}[\mathrm{T}]/(g)$ where $g \in \widehat{A}[\mathrm{T}]$ is a monic polynomial lifting g_0, $\widehat{A}$ being the completion of A. **IV-4** | 110

(18.1.4). We return to the general case. To say that $B_0 = B/\mathfrak{m}B$ is a formally étale $\kappa(\mathfrak{m})$-algebra is equivalent to saying (in the notation of (18.1.2)) that the set $\mathrm{Hom}_{\kappa(\mathfrak{m})\text{-alg}}(B_0, \Omega)$ is *finite* for every algebraically closed extension Ω of $\kappa(\mathfrak{m})$, and that $\Omega^1_{B_0/\kappa(\mathfrak{m})} = 0$ ((**0**, 20.7.3)). Furthermore, the formally étale homomorphism $\phi : A \to B$ is a formally smooth homomorphism ((**0**, 19.3.1)), hence a flat homomorphism ((**0**, 19.7.1, i)).

Theorem (18.1.2). — *Let S be a prescheme, S_0 a closed sub-prescheme of S whose underlying space is identical to that of S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of S-preschemes étale over S, to the category of S_0-preschemes étale over S_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let X, Y be two S-preschemes étale over S, and set $X_0 = X \times_S S_0$, $Y_0 = Y \times_S S_0$. If $Z = X \times_S Y$, the set $\mathrm{Hom}_S(X, Y)$ is in canonical bijective correspondence with the set of X-sections $\Gamma(Z/X)$, and likewise $\mathrm{Hom}_{S_0}(X_0, Y_0)$ is in canonical bijective correspondence with $\Gamma(Z_0/X_0)$, where $Z_0 = Z \times_S S_0 = X_0 \times_{S_0} Y_0$. Now Z is étale over X, Z_0 étale over X_0, and X_0 (resp. Z_0) is a closed sub-prescheme of X (resp. Z) having the same underlying space. The open subsets of Z are therefore the *same* as the open subsets of Z_0 having the corresponding properties, and our assertion follows from (17.9.3).

To complete the proof, it suffices to show that for every S_0-prescheme X_0 étale over S_0, there exists an S-prescheme X étale over S and an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. By virtue of Proposition (18.1.1), there exists an open covering (U_α) of X_0 and for each α, an S-prescheme V_α which is étale over S, and finally an S_0-isomorphism $\theta_\alpha : U_\alpha \xrightarrow{\sim} V_\alpha \times_S S_0$. Furthermore, by virtue of the first part of the proof, there exists a unique S_0-isomorphism $\varphi_{\alpha\beta}$ from $\theta_\alpha(U_\alpha \cap U_\beta)$ to $\theta_\beta(U_\alpha \cap U_\beta)$, corresponding to the identity automorphism of $U_\alpha \cap U_\beta$, and it is immediate, for the same reason, that these isomorphisms satisfy the glueing condition ((**0**, 4.1.7)). Hence there exists an S-prescheme X such that the V_α identify canonically with sub-preschemes induced on the open subsets of X, the θ_α identify with S_0-isomorphisms which coincide on the intersections $U_\alpha \cap U_\beta$, and define an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. It is clear that X is étale over S (17.3.2), which completes the proof. □

Corollary (18.1.3). — *Let S be a prescheme, X an S-prescheme étale over S, S' an S-prescheme, S'_0 a closed sub-prescheme of S' having the same underlying space. Then the canonical map $X(S')_S \to X(S'_0)_S$* (**I**, 3.4.3) *is bijective.*

Proof. Indeed, set $X' = X \times_S S'$, $X'_0 = X \times_S S'_0$, so that $X(S')_S = \Gamma(X'/S')$ and $X(S'_0)_S = \Gamma(X'_0/S'_0)$; the corollary follows from the fact that the functor defined in (18.1.2.thm) (with S and S_0 replaced by S' and S'_0 respectively) is fully faithful (or directly from (17.9.3)). □

18.2. Coverings.

(18.2.1). Given a ring A and a commutative A-algebra B which is *finite* and a *free* A-module, recall (Bourbaki, *Alg.*, ch. VIII, § 12, no. 2) that one defines on B an *A-linear form* $\mathrm{Tr}_{B/A}$, the "trace form"; one deduces from it the definition of a *symmetric A-bilinear form* (also called "trace form")

(18.2.1.1) $$(x, y) \longmapsto \mathrm{Tr}_{B/A}(xy)$$

whose datum is equivalent to that of the associated A-linear map $\mathrm{astr}_{B/A} : B \to \check{B}$, where $\check{B}$ denotes the *dual* of the A-module B, equal to its transpose. When A is a *field*, it is equivalent to say that this bilinear form is *non-degenerate* or that B is a *separable* A-algebra (Bourbaki, *Alg.*, ch. IX, § 2, prop. 5). **IV-4** | 111

Let $f : A \to A'$ be a ring homomorphism; set $B' = B \otimes_A A'$, and let $g : B \to B'$ be the canonical homomorphism; the image by g of a base of the A-module B is then a base of the A'-module B', and it follows from the definitions that, for all $x \in B$,

(18.2.1.2) $$\mathrm{Tr}_{B'/A'}(g(x)) = f(\mathrm{Tr}_{B/A}(x)).$$

(18.2.2). Consider now a *ringed space* $(X, \mathscr{O}_X)$ and let $\mathscr{B}$ be an $\mathscr{O}_X$-Algebra which, as an $\mathscr{O}_X$-Module, is *locally free of finite rank*; then, for every open subset $U \subset X$ such that $\mathscr{B}|U$ is (as an $\mathscr{O}_U$-Module) isomorphic to $\mathscr{O}_U^n$ (for some n depending on U), $\Gamma(U, \mathscr{B})$ is a $\Gamma(U, \mathscr{O}_X)$-algebra which, as a $\Gamma(U, \mathscr{O}_X)$-module, is free of finite rank, and thus defines a $\Gamma(U, \mathscr{O}_X)$-linear form $\mathrm{Tr}_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{O}_X)}$, which we also denote by $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X,U}$; from this we deduce an associated linear map

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X,U} : \Gamma(U, \mathscr{B}) \longrightarrow \Gamma(U, \mathscr{B})^\vee = \Gamma(U, \check{\mathscr{B}}).$$

Furthermore, it follows from (18.2.1.2) that these linear maps are compatible with the restriction operations from U to a smaller open subset, and thus define an $\mathscr{O}_X$*-Module homomorphism*, also called the *trace homomorphism*:

(18.2.2.1) $$\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \mathscr{O}_X$$

and also an $\mathscr{O}_X$*-Module homomorphism*

(18.2.2.2) $$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \check{\mathscr{B}}$$

called the *homomorphism associated to the trace*, and equal to its transpose. It also follows from (18.2.1.2) that for every $x \in X$, we have

(18.2.2.3) $$(\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{Tr}_{\mathscr{B}_x|\mathscr{O}_{X,x}}$$

(18.2.2.4) $$(\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{astr}_{\mathscr{B}_x|\mathscr{O}_{X,x}}.$$

Finally, under the conditions of (18.2.1), if we set $X = \mathrm{Spec}(A)$ and $\mathscr{B} = \widetilde{B}$, the form $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X}$ (resp. the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X}$) corresponds to the form $\mathrm{Tr}_{B/A}$ (resp. the A-module homomorphism $\mathrm{astr}_{B/A}$), as also follows from (18.2.1.2).

Proposition (18.2.3). — *Let $f : X \to Y$ be a finite morphism of preschemes and set $\mathscr{B} = f_*(\mathscr{O}_X)$. The following conditions are equivalent:*

label=a) f is étale.

a′) f is a flat morphism of finite presentation, and for every $x \in X$, if we set $y = f(x)$, $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a finite separable extension of $\kappa(y)$.

lbbel=b) $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module and for every $y \in Y$, $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a finite separable $\kappa(y)$-algebra (hence the direct composite of a finite number of fields, finite separable extensions of $\kappa(y)$).

lcbel=c) $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module and the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$ *(18.2.2) is bijective.*

Proof. Since f is quasi-compact, the equivalence of a) and a') has already been shown (17.6.2). To prove the rest of the proposition, we may restrict to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being a finite A-algebra and $\mathscr{B} = \widetilde{B}$. To say that f is a morphism of finite presentation is equivalent to saying that B is an A-module of finite presentation (1.4.7). If in addition f is flat, then B is a flat A-module, so B is an A-module projective, known (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1) to be equivalent to saying that B is an A-Module *projective*; thus $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module (*loc. cit.*, no. 2, th. 1), and the converse is immediate. On the other hand, $f^{-1}(y)$ is nothing but the spectrum of the $\kappa(y)$-algebra $\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$, which completes the proof of the equivalence of a') and b). To see that b) is equivalent to c), let us note that the second assertion of b) is equivalent to the fact that $\mathrm{astr}_{\mathscr{B}(y)|\kappa(y)} : \mathscr{B}(y) \to \mathscr{B}(y)^\vee$ is bijective; since $\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$ and $\check{\mathscr{B}}_y$ are free $\mathscr{O}_{Y,y}$-modules, it follows from (18.2.2.4) and Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, cor. of prop. 6, that the homomorphism $\mathrm{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is itself bijective; the converse being evident, this completes the proof. **IV-4 | 112**

When an $\mathscr{O}_X$-Algebra $\mathscr{B}$ satisfies the equivalent conditions b) and c) of (18.2.3), we say that $\mathscr{B}$ is a *finite étale $\mathscr{O}_X$-Algebra*. When $X = \mathrm{Spec}(A)$ is affine and $\mathscr{B} = \widetilde{B}$, this amounts, by virtue of (18.2.3), to saying that B is a finite étale A-algebra, or that B is a finite A-algebra that is étale (in the sense of (17.3.2)). □

Corollary (18.2.4). — *Let $f : X \to Y$ be a morphism of finite type and of finite presentation, and set $\mathscr{B} = f_*(\mathscr{O}_X)$. Let y be a point of Y. The following conditions are equivalent:*

label=a) There exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an étale morphism.

lbbel=b) $\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module of finite type and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra.

Proof. It is clear that a) implies b) by virtue of (18.2.3). On the other hand, $\mathscr{B}$ is an $\mathscr{O}_Y$-Module of finite presentation (1.6.3) and (1.4.7), hence, if $\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module, there exists an open neighbourhood U of y in Y such that $\mathscr{B}|U$ is a locally free $\mathscr{O}_U$-Module ((**0**, 5.2.7)); moreover, by hypothesis, the homomorphism $\mathrm{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ being bijectif, one can also suppose by

(**0**, 5.2.7) that one can choose U so that the homomorphism $\operatorname{astr}_{\mathscr{B}|U,\mathscr{O}_U}$ is bijective. The fact that b) implies a) then follows from (18.2.3). □ **IV-4** | 113

Corollary (18.2.5). — *Let Y be a quasi-compact or locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type and of finite presentation; set $\mathscr{B} = f_*(\mathscr{O}_X)$. Suppose that for every* closed *point y of Y, $\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra. Then f is étale.*

Proof. Indeed, it follows from (18.2.4) that every closed point y of Y has an open neighbourhood U such that the restriction $f^{-1}(U) \to U$ of f is étale; the conclusion follows from the fact that in the two cases considered, every non-empty closed subset of Y contains a closed point ((5.1.11) and (**0**, 2.1.3)). □

Corollary (18.2.6). — *If $f : X \to Y$ is a finite étale morphism, and if we set $\mathscr{B} = f_*(\mathscr{O}_X)$, then the $\mathscr{O}_Y$-homomorphism $\operatorname{Tr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \mathscr{O}_Y$ (18.2.2) (also denoted Tr_f) is surjective.*

Proof. The question being local, we may, by virtue of (18.2.3), suppose that $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, with B a free A-module; as by virtue of (18.2.3) the bilinear form (18.2.1.1) is non-degenerate, this implies in particular that the linear form $\operatorname{Tr}_{B/A}$ is surjective. □

Remarks (18.2.7). — (i) When $f : X \to Y$ is a finite morphism such that $f_*(\mathscr{O}_X)$ is a locally free $\mathscr{O}_Y$-Module (resp. locally free of rank n), we say further that f is a *finite locally free* morphism (resp. *locally free of rank n*). This condition, by virtue of (18.2.3), is satisfied if f is a finite étale morphism, but does not imply by itself that f is étale, as shown by the example where $X = \operatorname{Spec}(K)$ and $Y = \operatorname{Spec}(k)$ are spectra of fields, K being a finite non-separable extension of k. When $f : X \to Y$ is a finite étale morphism, we say further that X is an *étale covering* of Y. One notes that in this case, f is universally open and universally closed, and in particular $f(X)$ is a subset that is both open and closed of Y.

We say that an étale covering X of Y is *trivial* if X is a *sum* of a finite number of preschemes *isomorphic* to Y. We say that an étale covering X of Y is *locally trivial* if the morphism $f : X \to Y$ is such that for every point $y \in Y$ there is an open neighbourhood U for which the covering $f^{-1}(U)$ of U is trivial.

(ii) Let $f : X \to Y$ be a finite morphism, locally free of rank n; set $\mathscr{B} = f_*(\mathscr{O}_X)$ and let $u = \operatorname{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$; we deduce from this an n-th exterior power homomorphism $\bigwedge^n u : \bigwedge^n \mathscr{B} \to \bigwedge^n \check{\mathscr{B}} = (\bigwedge^n \mathscr{B})^\vee$ of *invertible* $\mathscr{O}_Y$-Modules, and by (**0**, 5.4.2) an element

(18.2.7.1)
$$d_{X/Y} \in \Gamma(Y, (\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}}))$$

which we call the *discriminant* of X over Y. Furthermore, since $(\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}})$ is the dual of $(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B})$, $d_{X/Y}$ can also be identified with a homomorphism

(18.2.7.2)
$$(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B}) \longrightarrow \mathscr{O}_Y$$

and we denote by $\mathscr{D}_{X/Y}$ the quasi-coherent ideal of $\mathscr{O}_Y$ of finite type, *image* of the homomorphism (18.2.7.2), also called the *discriminant ideal* of X over Y.

For the homomorphism u to be bijective, it is necessary and sufficient that $\bigwedge^n u$ be bijective, or again that the section $d_{X/Y}$ have an *invertible* germ at every point $y \in Y$, which is also written $d_{X/Y}(y) \neq 0$ for every y ((**0**, 5.5.2)). It amounts to the same to say that the discriminant ideal $\mathscr{D}_{X/Y}$ is *equal* to $\mathscr{O}_Y$. **IV-4** | 114

The terminology "covering" introduced in (18.2.7), (i), is justified by the following proposition:

Proposition (18.2.8). — *Let $f : X \to Y$ be a morphism étale, separated and of finite type, and for every $y \in Y$, let $n(y)$ be the geometric number of points of $f^{-1}(y)$. Then the function $y \mapsto n(y)$ is lower semi-continuous on Y. For it to be continuous at a point y (and hence constant in a neighbourhood of y), it is necessary and sufficient that there exists an open neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a* finite *(étale) morphism.*

Proof. Since f is quasi-finite (17.6.1) and locally of finite presentation, it amounts to saying that f is finite or that f is proper (8.11.1): moreover, each fibre $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$. The conclusions therefore follow from (15.5.9), (i) and (ii) and from the fact that f is flat. □

Corollary (18.2.9). — *Let Y be a connected prescheme, $f : X \to Y$ a morphism étale, separated and of finite type. For f to be finite (in other words, for X to be an étale covering of Y), it is necessary and sufficient that all the fibres of f have the same geometric number of points.*

Remarks (18.2.10). — (i) The example of the "affine line with a doubled point" ((**I**, 5.5.11)) shows that a morphism étale of finite type of Noetherian preschemes may not be separated; for this example, the first assertion of (18.2.8) is no longer exact.

(ii) For a separated morphism of finite type and étale $f : X \to Y$ to make X a *locally trivial* covering, it is necessary and sufficient that for every $x \in X$ there exist an open neighbourhood U of $y = f(x)$ and a U-section g of $f^{-1}(U)$ such that $g(y) = x$. Indeed, the condition is obviously necessary; the fact that it is also sufficient follows from the fact that every fibre $f^{-1}(y)$ is finite ((17.6.1)) and (**I**, 6.2.2), from the characterisation of sections of a Y-scheme étale (17.9.3) and from Proposition (18.2.8).

18.3. Finite algebras and étale algebras.

Proposition (18.3.1). — *Let A be a ring, B an A-algebra of finite presentation.*

(i) For B to be an unramified A-algebra, it is necessary and sufficient that B be an A-module of finite presentation and that $(B \otimes_A B)$ be a projective *$(B \otimes_A B)$-module.*

(ii) Suppose moreover that B is a finite A-algebra. For B to be an étale A-algebra, it is necessary and sufficient that B be an A-module projective and a $(B \otimes_A B)$-module projective.

Proof. The $(B \otimes_A B)$-module structure on B is that coming from the canonical ring homomorphism $B \otimes_A B \to B$, which is surjective and has kernel $\mathfrak{J}_{B/A}$ ((**0**, 20.4.1)). **IV-4** | 115

(i) To say that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is locally of finite presentation (1.4.6). To say that B is an unramified A-algebra is equivalent to saying that it is an A-algebra of finite presentation (1.4.7). To say that B is unramified is equivalent to saying that $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is an open and closed subset of $\mathrm{Spec}(B \otimes_A B)$ and we know that for this, it is necessary and sufficient that $\mathfrak{J}_{B/A}$ be a *direct factor ideal* of $B \otimes_A B$ (Bourbaki, *Alg. comm.*, ch. II, § 4, no. 3, prop. 15); but this amounts to saying that the $(B \otimes_A B)$-module quotient $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is projective (Bourbaki, *Alg.*, ch. II, 3e éd., § 2, no. 2, prop. 4).

(ii) Recall that a flat A-module of finite presentation is projective, and conversely (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1), so the assertion (ii) follows from that of (i) and from (17.6.2). □

Proposition (18.3.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete; set $A_0 = A/\mathfrak{J}$. Then the functor*

$$B \longmapsto B \otimes_A A_0$$

is an equivalence of the category of finite étale A-algebras, to the category of finite étale A_0-algebras.

Proof. We will first prove the following lemma:

Lemma (18.3.2.1). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete.*

(i) Every projective A-module of finite type M is separated and complete for the $\mathfrak{J}$-preadic topology, hence a projective limit of the $(A/\mathfrak{J}^{n+1})$-modules projective $M/\mathfrak{J}^{n+1}M$.

(ii) Conversely, set $A_n = A/\mathfrak{J}^{n+1}$, and let (M_n) be a projective system of A_n-modules, such that, for every n, the homomorphism $M_{n+1} \otimes_{A_{n+1}} A_n \to M_n$ deduced from the di-homomorphism of transition $M_{n+1} \to M_n$ is bijective. Suppose moreover that the M_n are projective and M_0 of finite type. Then $M = \varprojlim M_n$ is a projective A-module of finite type such that the canonical homomorphism $M \otimes_A A_0 \to M_0$ is bijective.

Proof. (i) There is a free A-module of finite type L such that M is isomorphic to a direct factor of L; as L is separated for the $\mathfrak{J}$-preadic topology, so is every sub-module N of L since $\mathfrak{J}^{n+1}N \subset \mathfrak{J}^{n+1}L$; in particular M is separated for this topology, and since the surjective homomorphism $f : L \to M$ is continuous for the $\mathfrak{J}$-preadic topology, its kernel N is closed for the topology induced by that of L; since L is complete and f a strict morphism, we conclude that M is complete (Bourbaki, *Top. gén.*, ch. IX, 2e éd., § 3, no. 1, prop. 4).

(ii) It follows from Nakayama's lemma that M_0 is generated by a finite family (x_{i0}) of r elements and if for all n, x_{in} is an element of M_n whose image $x_{i,n-1}$ is the image in M_{n-1}, then (x_{in}) $(1 \leqslant i \leqslant r)$ is a system of generators of M_n (Bourbaki, *Alg. comm.*, ch. II, § 3, cor. 2 de la prop. 4). Setting, for all n, $L_n = A_n^r$; if $(e_{in})_{1 \leqslant i \leqslant r}$ is the canonical base of L_n, let $u_n : L_n \to M_n$ be the A-linear map such that $u_n(e_{in}) = x_{in}$ for all i. By hypothesis, we have a split exact sequence

$$0 \longrightarrow N_n \xrightarrow{v_n} L_n \xrightarrow{u_n} M_n \longrightarrow 0$$

and since $L_n = L_{n+1}/\mathfrak{J}^{n+1}L_{n+1}$ and $M_n = M_{n+1}/\mathfrak{J}^{n+1}M_{n+1}$, the vertical arrows in the commutative diagram **IV-4 | 116**

$$\begin{tikzcd} 0 \arrow[r] & N_{n+1} \arrow[r, "v_{n+1}"] \arrow[d] & L_{n+1} \arrow[r, "u_{n+1}"] \arrow[d] & M_{n+1} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & N_n \arrow[r, "v_n"] & L_n \arrow[r, "u_n"] & M_n \arrow[r] & 0 \end{tikzcd}$$

are all three surjective. Now we have $M = \varprojlim M_n$ and $L = A^r = \varprojlim L_n$; setting $N = \varprojlim N_n$, $u = \varprojlim u_n$, $v = \varprojlim v_n$, by (**0**, 13.2.2) the exact sequence

$$0 \longrightarrow N \xrightarrow{v} L \xrightarrow{u} M \longrightarrow 0 \tag{18.3.2.2}$$

is exact. Moreover, since for every n, v_n is left-invertible and M_n is a projective A_n-module, by virtue of ((**0**, 19.1.8)) the exact sequence (18.3.2.2) is split, which proves the lemma. □

This being established, let us first show that the functor of (18.3.2) is *fully faithful*. Set, as in the lemma, $A_n = A/\mathfrak{J}^{n+1}$; let B, C be two finite étale A-algebras, and set, for every n, $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$; by virtue of (18.3.1) and (18.3.2.1), B and C are separated and complete for the $\mathfrak{J}$-preadic topology, and we have $B = \varprojlim B_n$, $C = \varprojlim C_n$; moreover, every A-algebra homomorphism $u : B \to C$ is continuous for the $\mathfrak{J}$-preadic topologies, and hence gives a projective system of A_n-algebra homomorphisms $u_n = u \otimes 1 : B_n \to C_n$, of which it is the projective limit; the converse being evident, we have a canonical bijection

$$\mathrm{Hom}_{A\text{-alg.}}(B, C) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n).$$

But since B and C are finite étale A-algebras, it follows also from (18.1.2.thm) that the canonical map

$$\mathrm{Hom}_{A_{n+1}\text{-alg.}}(B_{n+1}, C_{n+1}) \longrightarrow \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n)$$

is *bijective* for $n \geqslant 0$, which proves that the canonical map $\mathrm{Hom}_{A\text{-alg.}}(B, C) \to \mathrm{Hom}_{A_0\text{-alg.}}(B_0, C_0)$ is bijective.

To complete the proof of (18.3.2), it suffices to show that for every finite étale A_0-algebra B_0, there exists a finite étale A-algebra B and an A_0-isomorphism $B_0 \xrightarrow{\sim} B \otimes_A A_0$. Now, by (18.1.2.thm), there is a projective system (B_n) such that B_n is an étale A_n-algebra and that the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \to B_n$ are bijective. It follows from (18.3.1) and (18.3.2.1) that the A-algebra $B = \varprojlim B_n$ is a projective A-module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is a finite étale A-algebra, it suffices, by (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A, $B_\mathfrak{m}/\mathfrak{m}B_\mathfrak{m}$ is a separable $(A/\mathfrak{m})$-algebra. Now, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)), we have $\mathfrak{J} \subset \mathfrak{m}$, so $\mathfrak{m}_0 = \mathfrak{m}/\mathfrak{J}$, we have $A_0/\mathfrak{m}_0 = A/\mathfrak{m}$ and $B_\mathfrak{m}/\mathfrak{m}B_\mathfrak{m} = (B_0)_{\mathfrak{m}_0}/\mathfrak{m}_0(B_0)_{\mathfrak{m}_0}$; the conclusion therefore follows from the fact that B_0 is a finite étale A_0-algebra (18.2.5). □ **IV-4 | 117**

(18.3.3). *Example.* — Proposition (18.3.2) applies in particular when A is a *local* ring, separated and complete, $\mathfrak{J}$ being the maximal ideal of A, so that A_0 is a field and the category of finite étale A_0-algebras is identical to that of finite separable A_0-algebras, hence isomorphic to direct composites of fields, finite separable extensions of A_0. In particular, if the field A_0 is *separably closed*, these extensions are all identical to A_0, and hence every étale covering of $\mathrm{Spec}(A)$ is *trivial* (18.2.7) by virtue of (18.3.2).

Theorem (18.3.4). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$-preadic topology, $A_0 = A/\mathfrak{J}$. Let $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A_0)$. Let X be a proper S-scheme, and set $X_0 = X \times_S S_0$. Then the functor*

$$Z \longmapsto Z \times_X X_0$$

from the category of finite étale X-schemes over X, to the category of finite étale X_0-schemes over X_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let Z' and Z'' be two finite étale X-schemes over X. Set $S_n = \mathrm{Spec}(A/\mathfrak{J}^{n+1})$, $X_n = X \times_S S_n$, $Z'_n = Z' \times_S S_n$, $Z''_n = Z'' \times_S S_n$. It follows from ((**III**, 5.4.1)) that we have a canonical bijection $\mathrm{Hom}_X(Z', Z'') \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$. Now, by virtue of (18.1.2.thm), the canonical map $\mathrm{Hom}_{X_{n+1}}(Z'_{n+1}, Z''_{n+1}) \to \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$ is bijective, which proves our assertion.

It remains to show that if $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-Algebra, there exists an $\mathscr{O}_X$-Algebra $\mathscr{B}$, finite and étale, and an isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$. It follows from (18.1.2.thm) that there is a projective system $(\mathscr{B}_n)$, where $\mathscr{B}_n$ is an $\mathscr{O}_{X_n}$-Algebra finite and étale, and the second comparison theorem ((**III**, 5.1.4)) proves the existence of a coherent $\mathscr{O}_X$-Module $\mathscr{B}$ and of a projective system of isomorphisms $\mathscr{B}_n \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_n}$. The data of an Algebra structure on $\mathscr{B}$ is equivalent to that of a homomorphism $\mathscr{F} \otimes \mathscr{F} \to \mathscr{F}$ rendering the tensor powers commutative diagrams where only tensor powers of $\mathscr{F}$ appear; by (**III**, 5.1.3), (**I**, 10.11.4) and (**I**, 10.11.7), $\mathscr{B}$ is endowed in a natural way with a structure of $\mathscr{O}_X$-Algebra for which the isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$ is an isomorphism of Algebras. Moreover, $\mathscr{B}$ is an $\mathscr{O}_X$-Module *locally free*; this also follows from (**III**, 5.1.3), (**I**, 10.11.4), (**I**, 10.11.7) and from the fact that, in the category of $\mathscr{O}_X$-Modules, the locally free $\mathscr{O}_X$-Modules $\mathscr{F}$ can be defined as those for which the functors $\mathscr{G} \mapsto \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ are *exact*. Finally, to see that $\mathscr{B}$ is a finite étale $\mathscr{O}_X$-Algebra, it suffices (18.2.5) to show that for every *closed* point $x \in X$, $\mathscr{B}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$ is a separable $\kappa(x)$-algebra. But since the structural morphism $f : X \to S$ is *proper*, $f(x)$ is a closed point of S, so it belongs to S_0, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)); the conclusion therefore follows from the fact that $X_0 = f^{-1}(S_0)$ and that $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra. □

18.4. Local structure of unramified morphisms and of étale morphisms. **IV-4** | 118

Lemma (18.4.1). — *Let A be a ring, B a finite monogenic A-algebra, u a generator of the A-algebra B, $F \in A[\mathrm{T}]$ a polynomial such that $F(u) = 0$, F' the derivative polynomial; set $u' = F'(u)$. Then the ideal of B, annihilator of $\Omega^1_{B/A}$, contains $u'B$; it is equal to $u'B$ if the ideal $\mathfrak{J}$ of $A[\mathrm{T}]$ of polynomials G such that $G(u) = 0$, is generated by F, in other words if the surjective canonical homomorphism $\varphi : A[\mathrm{T}]/F.A[\mathrm{T}] \to B$ transforming the image of T into u, is bijective.*

Proof. Set $C = A[\mathrm{T}]$, so that $B = C/\mathfrak{J}$. We have the exact sequence ((**0**, 20.5.12.1))

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega^1_{C/A} \otimes_C B \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

and $\Omega^1_{B/A}$ therefore identifies with the quotient $B/\mathfrak{J}'$, where $\mathfrak{J}'$ is the ideal generated by the elements $G'(u)$, where G ranges over a system of generators of the ideal $\mathfrak{J}$ ((**0**, 20.5.13)); whence the lemma immediately follows. □

Proposition (18.4.2). — *Let the notation be those of (18.4.1), let $\mathfrak{q}$ be a prime ideal of B. Then:*

- *(i) If $\mathfrak{q}$ does not contain u', $B_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$-algebra formally unramified ($\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A); in other terms, $\mathrm{Spec}(B_u)$ is formally unramified over $\mathrm{Spec}(A)$.*
- *(ii) Suppose moreover that F is unitaire and generates $\mathfrak{J}$. Then, for $\mathrm{Spec}(B)$ to be étale over $\mathrm{Spec}(A)$ at the point $\mathfrak{q}$, it is necessary and sufficient that $u' \notin \mathfrak{q}$.*

Proof. The hypothesis that $u' \notin \mathfrak{q}$ implies that $\Omega^1_{B_{\mathfrak{q}}/A_{\mathfrak{p}}} = 0$ ((**0**, 20.5.9)), hence (i) follows from (17.2.1). Moreover, under the hypotheses of (ii), B is a free A-module by virtue of the Euclidean division; as the annihilator $\mathfrak{J}'$ of $\Omega^1_{B/A}$ is then equal to $u'B$ by virtue of (18.4.1) and $\Omega^1_{B/A}$ is a B-module of finite presentation (16.4.22), the annihilator of $\Omega^1_{B_{\mathfrak{q}}/A_{\mathfrak{p}}}$ is equal to $u'B_{\mathfrak{q}}$ (Bourbaki, *Alg. comm.*, ch. II, § 2, no. 4, formule (g)), and (ii) follows from (i) and from the implication $c) \Rightarrow a)$ in (17.6.1). □

Corollary (18.4.3). — *Let the notation be those of (18.4.2), suppose that F is unitaire and generates $\mathfrak{J}$. Then, for B to be an étale A-algebra, it is necessary and sufficient that u' be invertible in B (or, what amounts to the same, that the ideal of $A[\mathrm{T}]$ generated by F and F' be equal to $A[\mathrm{T}]$).*

Proof. In view of (18.4.2), (ii), saying that $\mathrm{Spec}(B)$ is étale over $\mathrm{Spec}(A)$ means that u' does not belong to any prime ideal of B, i.e. that it is invertible in B. □

We say that a monic polynomial $F \in A[\mathrm{T}]$ such that the ideal of $A[\mathrm{T}]$ generated by F and F' is equal to $A[\mathrm{T}]$ is *separable*; it is immediate that this definition coincides with the usual definition (Bourbaki, *Alg.*, ch. V, § 7, no. 6) when A is a field.

Lemma (18.4.4). — *Let A be a local ring, B a finite monogenic A-algebra, u a generator of the A-algebra B. Let $\mathfrak{n}_i$ ($1 \leqslant i \leqslant r$) be the maximal ideals of B such that* $\mathrm{Spec}(B)$ *is formally unramified over* $\mathrm{Spec}(A)$ *at the points $\mathfrak{n}_i$. Then there exists a monic unitaire polynomial $F \in A[\mathrm{T}]$ such that $F(u) = 0$; such a polynomial F satisfies the conditions $F'(u) \notin \mathfrak{n}_i$ for all i ($1 \leqslant i \leqslant r$). Moreover, if k is the residue field of A, $f \in k[\mathrm{T}]$ the minimal polynomial of the image of u in $B \otimes_A k$, there exists an $F \in A[\mathrm{T}]$ such that $F(u) = 0$; such an F satisfies the conditions $F'(u) \notin \mathfrak{n}_i$ for $1 \leqslant i \leqslant r$.* **IV-4 | 119**

Proposition (18.4.5). — *Let A be a local ring, k its residue field, B an A-algebra. Suppose moreover that the residue field k is* infinite, *or else that B is a local ring. Let n be the rank of $L = B \otimes_A k$ over k. For B to be a formally unramified A-algebra (resp. étale), it is necessary and sufficient that there exist a monic separable polynomial $F \in A[\mathrm{T}]$ such that B is isomorphic to a quotient of $A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$). Moreover, one can (and necessarily must) suppose that F is of degree n (resp. of degree n).*

Proof. The conditions are sufficient by virtue of (18.4.2), without assuming either that k is infinite or that B is local. To see that the conditions are necessary, note that if B is a formally unramified A-algebra, L is a finite algebra and separable over k, hence a direct composite of a finite number of fields, finite separable extensions k_j of k ($1 \leqslant j \leqslant r$), k_j being generated by an element $\bar{\xi}_j$, of minimal polynomial f_j ($1 \leqslant j \leqslant r$) (Bourbaki, *Alg.*, ch. V, § 11, no. 4, prop. 4). Let us show that by virtue of the hypotheses made on k or B, there exists an element $\bar{\xi}$ of L generating the k-algebra L. This is immediate if B is local, since then $r = 1$. Otherwise, k being assumed infinite, one can suppose that the irreducible polynomials $f_j \in k[\mathrm{T}]$ are all *distinct* (by replacing each $\bar{\xi}_j$ by $\bar{\xi}_j + a_j$, for a suitable element $a_j \in k$); if we set $f = f_1 f_2 \ldots f_r$, it is clear that L is isomorphic to $k[\mathrm{T}]/f.k[\mathrm{T}]$ in both cases, hence generated by an element $\bar{\xi}$ of minimal polynomial $f \in k[\mathrm{T}]$ of degree n. Note now that $\mathfrak{m}B$ is contained in the radical of B; as $1, \bar{\xi}, \ldots, \bar{\xi}^{n-1}$ form a base of L over k, it follows by Nakayama's lemma that $1, u, \ldots, u^{n-1}$ generate the A-module B, and hence there exists a monic unitaire polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; moreover, since $\bar{\xi}$ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily equal to f. But then $F'(\bar{\xi})$ of F in L is $f'(\bar{\xi})$, and as $f'(\bar{\xi}) \neq \mathfrak{n}_i$ by (18.4.4), this proves that $F'(u)$ is invertible in B, hence F is a separable polynomial. Finally, if B is an étale A-algebra, B being an A-module flat of finite presentation (1.4.7), and $1, u, \ldots, u^{n-1}$ form a base of the A-module B (Bourbaki, *Alg. comm.*, ch. II, § 3, no. 2, prop. 5), in other words the A-algebra B is isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$. □

Theorem (18.4.6). — *(Chevalley).* —

(i) *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$, and set $A = \mathscr{O}_{Y,y}$. For $\mathscr{O}_{X,x}$ to be an A-algebra formally unramified (resp. formally étale), it is necessary and sufficient that there exist a monic unitaire polynomial $F \in A[\mathrm{T}]$ and a maximal ideal $\mathfrak{n}$ such that $B = A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. an algebra quotient B of $A[\mathrm{T}]/F.A[\mathrm{T}]$) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $B_{\mathfrak{n}}$ and that, if u is the image of* T *in B, one has $F'(u) \notin \mathfrak{n}$.*

(ii) *Suppose in addition that f is locally of finite presentation. Then, for f to be étale at the point x, it is necessary and sufficient that we can moreover take $B = A[\mathrm{T}]/F.A[\mathrm{T}]$.*

Proof. The conditions are sufficient by virtue of (18.4.2). To see that they are necessary, and taking into account the remark (17.1.2), when f is formally unramified and quasi-finite, since f is affine, it follows from (8.12.8) that there exists a finite A-algebra C and a maximal ideal $\mathfrak{r}$ such that $\mathscr{O}_{X,x}$ is A-isomorphic to $C_{\mathfrak{r}}$. Moreover, by (17.4.1.2) the residue field $C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}}$ is a finite separable extension of $k = A/\mathfrak{m}$. Let $\mathfrak{r}_i$ ($1 \leqslant i \leqslant h$) be the maximal ideals of the semi-local ring C *other than* $\mathfrak{r}$; there exists an element $u \in C$ belonging to all the $\mathfrak{r}_i$ and such that its image in $C/\mathfrak{r}$ equals v (Bourbaki, *Alg. comm.*, ch. II, § 1, no. 2, prop. 5). We shall now show that the sub-A-algebra $B = A[u]$ of C is finite on B and $\mathfrak{n} = \mathfrak{r} \cap B$ answers the question. **IV-4 | 120**

To handle the case where $\mathscr{O}_{X,x}$ is a formally ramified A-algebra, it suffices to prove that $B_{\mathfrak{n}}$ is isomorphic to $C_{\mathfrak{r}}$; indeed, $B_{\mathfrak{n}}$ will then be formally unramified over A and the existence of the polynomial $F \in A[\mathrm{T}]$ having the properties of the statement will follow from (18.4.4). Note now that, since f is formally unramified, $B_{\mathfrak{n}}$ is a $B'_{\mathfrak{n}}$-algebra formally étale; with the preceding notations, **IV-4 | 121** since B' does not belong to $\mathfrak{n}'$ by hypothesis, the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is étale at the point $\mathfrak{n}'$ by (18.4.2), (ii). As by hypothesis $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale at the point $\mathfrak{n}$, one concludes by (17.3.4) that $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ is étale at the point $\mathfrak{n}$; but since this morphism is an immersion, it cannot be étale at a point unless it is a local isomorphism at that point (17.9.1), hence $B_{\mathfrak{n}}$ and $B'_{\mathfrak{n}'}$ are isomorphic.

Finally, suppose that $\mathscr{O}_{X,x}$ is a formally étale A-algebra; with the preceding notations, by (17.1.5) that $B_{\mathfrak{n}}$ is a $B'_{\mathfrak{n}}$-algebra formally étale; but since the homomorphism $B'_{\mathfrak{n}'} \to B_{\mathfrak{n}}$ is surjective, this can only happen if this homomorphism is *bijective* ((**0**, 19.10.3), (i)). This completes the proof of (18.4.6). □

Corollary (18.4.7). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. For f to be formally unramified at the point x, it is necessary and sufficient that there exist an open neighbourhood U of x such that $f|U$ factorises as $U \xrightarrow{j} X' \xrightarrow{h} Y$, where h is an étale morphism and j is a closed immersion.*

Proof. We can obviously reduce to the case where $Y = \mathrm{Spec}(R)$ is affine and f is of finite type. If $A = \mathscr{O}_{Y,y}$, with $y = f(x)$, the condition for f to be formally unramified at the point x is equivalent, by virtue of (17.4.1.2), to saying that $\mathscr{O}_{X,x}$ is a formally unramified A-algebra. If it is so, we can apply (18.4.6), (i); replacing Y as needed by an affine neighbourhood of y, we may suppose (with the notation of (18.4.6)) that the polynomial F is the image in $A[\mathrm{T}]$ of a monic unitaire polynomial $G \in R[\mathrm{T}]$. We then set $X' = \mathrm{Spec}(R[\mathrm{T}]/G.R[\mathrm{T}])$; let x' be the image of the point $\mathfrak{n}$ of $\mathrm{Spec}(B)$ by the morphism corresponding to the composite homomorphism $R[\mathrm{T}]/G.R[\mathrm{T}] \to A[\mathrm{T}]/F.A[\mathrm{T}] \to B$. It follows from (18.4.2) that the canonical morphism $h : X' \to Y$ is étale at the point x, hence, restricting as needed U and X', we can suppose h étale. On the other hand, by virtue of (**I**, 6.5.1), (ii)) and (1.7.2), it follows from the fact that we have a local homomorphism $\varphi : \mathscr{O}_{X',x'} \to \mathscr{O}_{X,x}$ that this homomorphism corresponds to a morphism $j : U \to X'$ from an open neighbourhood U of x in X'; in applying (**I**, 6.5.1), (i)), suppose that $h \circ j = f|U$, hence j is of finite type ((**I**, 6.3.4), (v)); finally, the homomorphism φ is surjective by (18.4.6), (i), hence it follows from ((**I**, 6.5.4), (i)) that we can, restricting further U and X', suppose that j is a closed immersion. This proves the necessity of the condition stated; the sufficiency is immediate (17.1.3), (i) and (ii). □ **IV-4 | 122**

Corollary (18.4.8). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be non ramified (resp. étale), it is necessary and sufficient that there exist a family of flat morphisms $g_\alpha : Y'_\alpha \to Y$ and, for each α, an open subset U_α in $X'_\alpha = X \times_Y Y'_\alpha$, such that, if $g'_\alpha : X'_\alpha \to X$ and $f'_\alpha : X'_\alpha \to Y'_\alpha$ are the canonical projections, the $g'_\alpha(U_\alpha)$ form a covering of X, and that each of the composed morphisms $U_\alpha \to X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$ is a closed immersion (resp. open). One can moreover take the g_α to be étale.*

Proof. The necessity of the condition for étale morphisms is trivial, taking a single Y'_α equal to X, the morphism g_α corresponding to the diagonal, and the open subset $U \subset X \times_Y X$ being the diagonal. When f is non ramified, the necessity of the condition follows from (18.4.7); on the other hand we need to find an open covering (V_α) of X such that for each α, $f|V_\alpha$ factorises as $V_\alpha \xrightarrow{j_\alpha} X'_\alpha \xrightarrow{g_\alpha} Y$, where j_α is a closed immersion and g_α an étale morphism. Then $j_\alpha : V_\alpha \to Y'_\alpha$ is a V_α-section of X'_α, and since $f|V_\alpha$ is étale, s_α is a V_α-section of X'_α, and since g_α is étale, the morphism s_α is an immersion ouverte (17.4.1), and it suffices to take $U_\alpha = s_\alpha(V_\alpha)$ to answer the question.

The sufficiency of the conditions follows from (17.7.1); one concludes at each point x of $g'(U_\alpha)$, hence in all of X since the $g'_\alpha(U_\alpha)$ cover X. □

Proposition (18.4.9). — *Let S be a prescheme, $f : X \to S$ a morphism, $h : Y \to S$ an S-morphism, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. The following conditions are equivalent:*

*label=*a) *h is étale at the point y and g is flat at the point x.*
*lbbel=*b) *h is non ramified at the point y and f is flat at the point x.*

Proof. Since $f = h \circ g$, a) implies that f is flat at the point x (2.1.6), and it is obvious that h being non ramified at the point y, hence a) implies b).

To prove that b) implies a), we can first suppose that h is non ramified (replacing Y by an open neighbourhood of y); then, replacing S by an open neighbourhood of $s = h(y) = f(x)$, one can suppose that there exists an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ above y in Y' and an open neighbourhood V of y' in Y' such that, if the restriction of h' to V is a closed immersion (18.4.8). If one proves that h' is étale at the point y', it will follow that h is étale at the point y (17.7.1), (ii); moreover, $f' = f_{(S')} : X_{(S')} \to S'$ is flat at every point x' above x. As the projections $v : Y' \to Y$, $w : X' \to X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \to Y_{(S')}$ is flat at the point x', it will follow from (2.2.11), (iv) that g is flat at the point x. We can hence reduce to the case where h is a *closed immersion* of finite presentation, f being supposed flat at the point x and f is of finite type. Let $\mathscr{J}$ be the quasi-coherent Ideal of $\mathscr{O}_S$ of finite type defining the closed sub-prescheme Y of S. The hypothesis that f is flat at the point x implies that the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{X,x}$ is injective ((**0**, 6.5.1)); *a fortiori* the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is injective, which means that $\mathscr{J}_s = 0$, since it is the kernel of the preceding homomorphism. Since $\mathscr{J}$ is of finite type, there exists an open neighbourhood U of s in S such that $\mathscr{J}|U = 0$ ((**0**, 5.2.2)). We can hence suppose that h is an open immersion, and it is clear that g is flat at the point x. ☐ **IV-4 | 123**

Proposition (18.4.10). — *Let us denote by $\mathbf{P}(f, x)$ a property satisfying the following conditions:*

1° *For every morphism $f : X \to Y$ and every local isomorphism $h : Y \to Z$, $\mathbf{P}(f, x)$ is equivalent to $\mathbf{P}(h \circ f, x)$ for $x \in X$.*

2° *For every morphism $f : X \to Y$, every étale morphism $g : Y' \to Y$, every point $x \in X$, if we set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and if $x' \in X'$ is above x, the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(f', x')$ are equivalent ("invariance by étale base change").*

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

Proof. With the same notations as in the proof of (18.4.9) and replacing S (resp. Y) by an open neighbourhood of $s = h(y)$ (resp. y), we may find, by (18.4.8), an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ and an open neighbourhood V of y' in Y' such that h' restricted to V is an open immersion; if $x' \in X'$ is any point above x, $\mathbf{P}(f', x')$ (resp. $\mathbf{P}(g', x')$) is then equivalent to $\mathbf{P}(f, x)$ (resp. $\mathbf{P}(g, x)$). The hypothesis implies that $\mathbf{P}(g, x)$ is then equivalent to $\mathbf{P}(h \circ g, x)$, and we deduce the conclusion. ☐

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

(18.4.11). *Examples.* — One can take for the property $\mathbf{P}(f, x)$ any one of the following, taking into account (17.7.4), (ii):

(i) f is flat at the point x (2.2.11), (iv);
(ii) f is locally of finite presentation and of coprofondeur $\leqslant n$ at the point x ((6.8.1) and (6.7.8));
(iii) f is locally of finite presentation and of Cohen-Macaulay at the point x ((6.8.1) and (6.7.8));
(iv) f is locally of finite presentation and possesses the property (S_n) at the point x ((6.8.1) and (6.7.8));
(v) f is locally of finite presentation and possesses the property (R_n) at the point x ((6.8.1) and (6.7.8));
(vi) f is locally of finite presentation and normal at the point x ((6.8.1) and (6.7.8));
(vii) f is locally of finite presentation and reduced at the point x ((6.8.1) and (6.7.8));
(viii) f is non ramified at the point x (17.7.4);
(ix) f is smooth at the point x (17.7.4);
(x) f is étale at the point x (17.7.4).

Corollary (18.4.12). — *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. If f is flat and formally unramified at the point x, then, in every factorisation $f|U : U \xrightarrow{j} X' \xrightarrow{h} Y$, where U is an open neighbourhood of x, j a closed immersion and h an étale morphism, the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ corresponding to j is bijective (in particular $\mathscr{O}_{X,x}$ is an essentially étale $\mathscr{O}_{Y,f(x)}$-algebra (18.6.1)).*

(ii) For a morphism $f : X \to Y$ to be étale, it is necessary and sufficient that it be locally of finite type, formally unramified and flat.

Proof. (i) Since the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ is surjective, it suffices to prove that it is injective, IV-4 | 124
and for that it suffices to prove that $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{X',j(x)}$-module *faithfully flat*, or merely *flat* ((**0**, 6.5.1) and (6.6.2)); in other words, it amounts to showing that j is a flat morphism at the point x; but since $h \circ j = f$ is flat at the point x by hypothesis and h is étale, this follows from (18.4.10) and (18.4.11), (i).

(ii) It suffices to prove the sufficiency of the stated conditions. For every $x \in X$, we have in a neighbourhood open U of x a factorisation of $f|U$ having the properties stated in (i), and the fact that f is flat and formally non ramified at *all* the points of U. For every $x \in X$, the quasi-coherent Ideal of $\mathscr{O}_{X'}$ corresponding to the closed sub-prescheme X' associated to j, we have $\mathscr{J}_{j(x)} = 0$ for every $z \in U$, hence j is an open immersion, and f is hence étale at every point of X. □

Corollary (18.4.13). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that y admits an open neighbourhood which is a reduced prescheme and has only a finite number of irreducible components. Then, for f to be étale at the point x, it is necessary and sufficient that f be flat and formally non ramified at the point x.*

Proof. There is only the sufficiency of the condition to prove. The question being local on X and Y, one can suppose (18.4.7) that f factorises as $X \xrightarrow{j} X' \xrightarrow{h} Y$ where h is étale and j a closed immersion. Then X' is reduced (17.5.7) and, replacing X' by an open neighbourhood of $j(x)$, one can suppose that X' has only a finite number of irreducible components. Then the maximal points of X' are above the maximal points of Y (2.3.4) and since they are finite, their number is finite. It all amounts to showing, with the notations of the proof of (18.4.12), that $\mathscr{J}_{x'} = 0$ for all points x' in a neighbourhood of $j(x)$ in X', knowing that $\mathscr{J}_{j(x)} = 0$. Now, replacing X' by an affine neighbourhood of $j(x)$, we can suppose that all the irreducible components of X' contain $j(x)$; if $X' = \operatorname{Spec}(A')$, and if $\mathfrak{p}'$ is the prime ideal of A' corresponding to the point $j(x)$, the morphism $\operatorname{Spec}(A'_{\mathfrak{p}'}) \to \operatorname{Spec}(A')$ is dominant, hence the corresponding homomorphism $A' \to A'_{\mathfrak{p}'}$ is *injective* since A' is reduced (**I**, 1.2.7). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A', $\mathfrak{J}$ identifies with an ideal of A', $\mathfrak{J}_{\mathfrak{p}'} = 0$ by hypothesis, hence $\mathfrak{J} = 0$. □

Corollary (18.4.14). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, residue field k, B a finite A-algebra.*

(i) For B to be a formally unramified A-algebra, it is necessary and sufficient that $B \otimes_A k$ be a k-algebra étale.

*(ii) For B to be a finite étale A-algebra, it is necessary and sufficient that $B \otimes_A k$ be an étale k-algebra and that B be a flat A-module (which is equivalent to saying ((**0**, 6.3.3)) that for every maximal ideal $\mathfrak{n}$ of B (necessarily above $\mathfrak{m}$), $B_\mathfrak{n}$ is a flat A-module).*

Proof. It is clear that (ii) follows from (i) and (18.4.12), (ii), taking into account (17.1.2), (i). To IV-4 | 125
prove (i), let us note that if $B \otimes_A k$ is an étale k-algebra, we have $\Omega^1_{B\otimes_A k/k} = 0$ (17.2.1). Since $\Omega^1_{B\otimes_A k/k} = \Omega^1_{B/A} \otimes_B k$ ((**0**, 20.5.5)) and since B is an A-algebra of finite type, $\Omega^1_{B/A}$ is a B-module of finite type ((**0**, 20.4.7)). But since B is a finite A-algebra, $\mathfrak{m}B$ is contained in the radical of B (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1), hence the lemma of Nakayama proves that $\Omega^1_{B/A} = 0$, and hence B is a formally unramified A-algebra (17.2.1). □

18.5. Henselian local rings.

(18.5.1). Let X be a prescheme, $\mathscr{E}$ a locally free $\mathscr{O}_X$-Module of finite rank; the dual $\check{\mathscr{E}} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{O}_X)$ is therefore a locally free $\mathscr{O}_X$-Module whose rank at every point is equal to that of $\mathscr{E}$ at that point, and the canonical homomorphism $\mathscr{E} \to \mathscr{H}om_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X) = (\check{\mathscr{E}})^\vee$ is *bijective*. For every morphism $X' \to X$, set $\mathscr{E}_{(X')} = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ which is a locally free $\mathscr{O}_{X'}$-Module, and consider the *set* $\Gamma(X', \mathscr{E}_{(X')})$ of sections of this $\mathscr{O}_{X'}$-Module over X'. We shall see that this defines a *contravariant representable functor*

$$ {}^t V_{\mathscr{E}} : X' \longmapsto \Gamma(X', \mathscr{E}_{(X')}) \tag{18.5.1.1}$$

from the category of X-preschemes to the category of *sets* ((**0**, 8.1.8)). IV-4 | 126

First of all, this is indeed a well-defined functor, for if $f : X'' \to X'$ is an X-morphism, we have $\mathscr{E}_{(X'')} = f^*(\mathscr{E}_{(X')})$, whence ((**0**, 4.3.2)) a map $\Gamma(X', \mathscr{E}_{(X')}) \to \Gamma(X'', \mathscr{E}_{(X'')})$ which completes the definition of the functor ${}^tV$. Let us then show that the X-prescheme $\mathbf{V}(\check{\mathscr{E}})$ ((**II**, 1.7.8)) represents the functor ${}^tV$. It is immediate that $(\mathscr{E}_{(X')})^\vee = (\check{\mathscr{E}})_{(X')}$, hence $\mathbf{V}(\check{\mathscr{E}}) \times_X X' = \mathbf{V}((\check{\mathscr{E}})_{(X')})$; taking into account (**I**, 3.3.14), we are reduced to defining a bijection $\Gamma(\mathbf{V}(\check{\mathscr{E}}_{(X')})/X') \xrightarrow{\sim} \Gamma(X', \mathscr{E})$, and to verify that the bijection $\Gamma(\mathbf{V}(\check{\mathscr{E}})/X) \xrightarrow{\sim} \Gamma(X, \mathscr{E})$ is functorial in X'. Now, on the one hand a section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which corresponds by transposition an $\mathscr{O}_X$-homomorphism ${}^t u : \check{\mathscr{E}} \to \mathscr{O}_X$, and hence an $\mathscr{O}_X$-Algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\mathrm{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is contained in the kernel of v; it follows immediately from these definitions that

$$X' \longmapsto \mathrm{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$$

is a representable functor by $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$. If $X' = \mathrm{Spec}(A)$ is affine and such that $\mathscr{E}_{(X')}$ is isomorphic to $\mathscr{O}^n_{X'}$, $\Gamma(X', \mathscr{E}_{(X')})$ can be identified with the set A^n; in an obvious way, we can say that the object $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space over X defined by $\mathscr{E}$". We write this X-prescheme $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. We note also that if $\mathscr{J}$ is a type-finite Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-Algebra of finite presentation.

(18.5.2). Recall ((**II**, 1.7.8)) that by definition $\mathbf{V}(\check{\mathscr{E}}) = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$. Let $\mathscr{J}$ be a quasi-coherent Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, so that $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ is a closed sub-prescheme of $\mathbf{V}(\check{\mathscr{E}})$; we will interpret it as *representing a functor* from the category of X-preschemes to that of sets. Let us note for this that a section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which corresponds by transposition an $\mathscr{O}_X$-Algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\mathrm{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is *contained in the kernel* of v; if $X' = \mathrm{Spec}(A)$ is affine and such that $\mathscr{E}_{(X')}$ is isomorphic to $\mathscr{O}^n_{X'}$ and $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ is a $\mathscr{O}_{X'}$-Module of the form $\widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of the polynomial algebra $C = A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$; the set $\mathrm{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$ then identifies with the subset of A^n consisting of points $(t_1, \ldots, t_n)$ such that $\mathrm{P}(t_1, \ldots, t_n) = 0$ for all polynomials $\mathrm{P} \in \mathfrak{J}$; thus we can say that the object $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space over X defined by $\mathscr{E}$". We also write this X-prescheme $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. We note that if $\mathscr{J}$ is a type-finite Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-Algebra of finite presentation.

Lemma (18.5.3). — *Let S be a prescheme, $f : X \to S$ a finite and locally free morphism. Consider the contravariant functor from the category of S-preschemes to the category of sets*

$$S' \longmapsto \mathrm{Of}(X \times_S S') \tag{18.5.3.1}$$

where $\mathrm{Of}(X \times_S S')$ *denotes the set of subsets that are both open and closed of the underlying space of $X \times_S S'$. Then this functor is representable by an S-prescheme* $\mathbf{Of}(X)$, *which is affine, étale and of finite presentation over S.*

Proof. We have by hypothesis $X = \mathrm{Spec}(\mathscr{B})$, where $\mathscr{B} = f_*(\mathscr{O}_X)$ is a finite $\mathscr{O}_S$-Algebra and locally free. Set $X' = X \times_S S'$, $f' = f_{(S')}$, $\mathscr{B}' = \mathscr{B} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*(\mathscr{O}_{X'})$, so that $X' = \mathrm{Spec}(\mathscr{B}')$; there is then a canonical bijection, functorial in S', from the set $\mathrm{Of}(X')$ to the set $\mathrm{Id}(\mathscr{B}')$ of *idempotents* of the ring $\Gamma(S', \mathscr{B}') = \Gamma(X', \mathscr{O}_X)$. Indeed, by virtue of the equivalence of the category of affine S'-schemes and the opposite category of quasi-coherent $\mathscr{O}_{S'}$-Algebras ((**II**, 1.2.7) and (1.3.1)), there is a canonical bijective correspondence between the decompositions of X' into a *sum* $X'_1 \sqcup X'_2$ of two open subsets induced by open subsets of X' and the decompositions of $\mathscr{B}'$ into a *direct composite* of two Ideals $\mathscr{B}'_1$, $\mathscr{B}'_2$; these latter in turn form a set in canonical bijective correspondence with $\mathrm{Id}(\mathscr{B}')$. It suffices therefore to prove the lemma for the functor $S' \mapsto \mathrm{Id}(\mathscr{B}')$. **IV-4** | 127

For this, we shall show that there exists a type-finite Ideal $\mathscr{J}$ of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{B})$ such that $\mathrm{Id}(\mathscr{B}')$ is of the form $\mathrm{Al}(S', \check{\mathscr{B}}, \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'})$ (18.5.2). Note to this end that since $\mathscr{B}$ is an $\mathscr{O}_S$-Algebra, its inverse image $\mathscr{B}_{\mathbf{V}(\check{\mathscr{B}})}$ is an $\mathscr{O}_{\mathbf{V}(\check{\mathscr{B}})}$-Algebra, and we can form the "square" c^2, in this Algebra, of the canonical section c (18.5.1); it corresponds canonically to a homomorphism $u^{(2)} : \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \to \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \otimes_{\mathscr{O}_S} \mathscr{B}$, and one verifies immediately that for every affine open subset W of S, with the notations of (18.5.1), $u^{(2)}$ corresponds to the C-module homomorphism such that $u^{(2)}(1) = \sum_k (\sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j) \otimes e_k$, where (c_{ijk}) is the table of multiplication of the algebra $\Gamma(W, \mathscr{B})$. Let us show that the Ideal $\mathscr{J}$ of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{B})$ generated by the kernel of the homomorphism $u - u^{(2)}$ answers the question. Indeed, it suffices to note that the ideal $\Gamma(W, \mathscr{J})$ of C is generated by the polynomials $\mathrm{P}_k(\mathrm{T}_1, \ldots, \mathrm{T}_n) = \mathrm{T}_k - \sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j$

and that the idempotents of $\Gamma(W, \mathscr{B})$ are precisely the elements $\sum_i t_i e_i$ of this algebra such that $(t_1, \ldots, t_n)$ annihilates all the polynomials P_k ($1 \leqslant i \leqslant n$). We deduce from (18.5.2) that the S-prescheme affine $\mathbf{Of}(X) = \mathbf{Al}(\mathscr{B}, \mathscr{J})$ does represent the functor (18.5.3.1). It remains to show that $\mathbf{Of}(X)$ is étale over S, or equivalently, that it is *formally étale* over S. But if S' is an S-prescheme, S'_0 a closed sub-prescheme of S' defined by a locally nilpotent Ideal (and having the same underlying space as S'), it is clear that $X' = X \times_S S'$ and $X'_0 = X \times_S S'_0$ have the same underlying space, hence the canonical map $\mathrm{Of}(X') \to \mathrm{Of}(X'_0)$ is *bijective*, which completes the proof (17.1.1). □

Proposition (18.5.4). — *Let S be a prescheme, S_0 a closed sub-prescheme of S; consider the following properties:* **IV-4** | 128

label=a) For every finite morphism $g : S' \to S$, the canonical map

(18.5.4.1) $$\mathrm{Of}(S') \longrightarrow \mathrm{Of}(S' \times_S S_0)$$

is bijective (cf. (18.5.3)).

a′) For every finite and locally free morphism $g : S' \to S$, the map (18.5.4.1) is bijective.

lbbel=b) For every étale and separated morphism $g : S' \to S$, the canonical map

(18.5.4.2) $$\Gamma(S'/S) \longrightarrow \Gamma(S' \times_S S_0/S_0)$$

is bijective.

Condition b) implies a′); if moreover S is quasi-compact and quasi-separated, condition a) implies b).

Proof. Let us first prove that b) implies a'). Suppose therefore b) satisfied, and let $g : S' \to S$ be a finite and locally free morphism; set $S'_0 = S' \times_S S_0$, so that $g_0 = g_{(S_0)} : S'_0 \to S_0$ is finite and locally free. Then it follows from (18.5.3) that $\mathrm{P} = \mathbf{Of}(S')$ is an S-prescheme *étale and separated*; moreover, by the definition of the functor Of, we see immediately that if we set $\mathrm{P}_0 = \mathbf{Of}(S'_0)$ (for the category of S_0-preschemes), we have $\mathrm{P}_0 = \mathrm{P} \times_S S_0$. This being so, by definition we have the commutative diagram

$$\begin{tikzcd} \Gamma(\mathrm{P}/S) \arrow[r] \arrow[d, "\wr"] & \Gamma(\mathrm{P}_0/S_0) \arrow[d, "\wr"] \\ \mathrm{Of}(S') \arrow[r] & \mathrm{Of}(S'_0) \end{tikzcd}$$

where the vertical arrows are the bijections. Since hypothesis b), applied to the morphism $\mathrm{P} \to S$, implies that the first row is a bijection, the same holds for the second, which establishes our assertion.

Before proving that a) implies b) when S is quasi-compact and quasi-separated, we will establish the following lemma.

Lemma (18.5.4.3). — *If S and S_0 satisfy condition a) of (18.5.4), then, for every finite morphism $g : S' \to S$, S' is the unique open and closed neighbourhood of $S'_0 = g^{-1}(S_0) = S' \times_S S_0$ in S'.*

Proof. Indeed, it amounts to saying that if T' is a closed subset of S' such that $T' \cap S'_0 = \varnothing$, then $T' = \varnothing$. Now, if we also denote by T' the closed sub-prescheme of S' having T' as underlying space, the composite morphism $h : T' \to S' \xrightarrow{g} S$ is finite and $h^{-1}(S_0)$ is empty; condition a), applied to the morphism h, implies that T' is necessarily empty. □

This lemma being established, let us first prove that, under hypothesis a), the map (18.5.4.2) is *injective*. Indeed, if u', u'' are two S-sections of S', the fact that the morphism $S' \to S$ is non ramified implies that the prescheme of coincidences of u' and u'' is induced on an open subset U of S (17.4.6). If the restrictions to S_0 of u' and u'' are equal, then U contains S_0, hence is equal to S by virtue of the lemma (18.5.4.3) applied to the case where $S' = S$.

It remains to show that, under hypothesis a), the map (18.5.4.2) is *surjective* (when S is quasi-compact and quasi-separated). Let therefore $u_0 : S_0 \to S'_0$ be an S_0-section of S'_0; $u_0(S_0)$ being quasi-compact in S', can be covered by a finite number of open affine subsets V_i such that the restriction $g|V_i$ is a morphism of finite type; if V is the union of the V_i, we conclude that $g|V : V \to S$ is a morphism *of finite presentation*, being separated and locally of finite presentation (1.6.1). Replacing **IV-4** | 129

S' by V, one can hence suppose that g is of finite presentation. Since S is quasi-compact and quasi-separated (g quasi-finite and separated) (17.6.1), it follows from the "Main theorem" (8.12.6) that g factorises as $S' \xrightarrow{j} S'' \xrightarrow{f} S$, where j is an open immersion and f a *finite* morphism. Set $S''_0 = S'' \times_S S_0$, $j_0 = j_{(S_0)} : S'_0 \to S''_0$, which is an open immersion and $f_0 = f_{(S_0)} : S''_0 \to S_0$, which is also an S_0-section of S''_0. Then u_0 is an open immersion of S_0 into S'_0 (17.4.1), hence $g_0 : S'_0 \to S_0$ is étale, hence an open immersion of S_0 into S'_0, hence $X_0 = u_0(S_0)$ is *open and closed* in S''_0. In virtue of condition a), there exists an open and closed subset X of S'' such that $X \cap S''_0 = X_0$. Let us show that the morphism $f : S'' \to S$ is *étale at the points of* X: indeed, the set of points of X where $f|X$ is étale is open and contains $X_0 \subset S'_0$. But the lemma (18.5.4.3) applied to the finite morphism $f|X$ proves that $U = X$. On the other hand, $S' \cap X$ is open in X and contains X_0, hence by the same reasoning proves that $S' \cap X = X$, i.e. $X \subset S'$. It remains to show that, for every $s \in S$, the geometric number $n(s)$ of points of $X \cap f^{-1}(s)$ is equal to 1, for it will follow that $f|X$ is an isomorphism of the open subset $X \subset S'$ onto S, whose inverse will be the S-section sought extending u_0. But since $f|X$ is étale and finite, $s \mapsto n(s)$ is continuous on S (18.2.8), and since $X \cap S'_0 = X_0$, we have $n(s) = 1$ in S_0; the set of points $s \in S$ such that $n(s) = 1$ being open in S and containing S_0, is equal to S by (18.5.4.3). □

Remark (18.5.4.4). — One can show that the statement (18.5.4) remains valid when, in condition b), one supposes merely the morphism g étale (but not necessarily separated) [**?**, exp. XII].

Definition (18.5.5). — We say that a prescheme S and a closed sub-prescheme S_0 of S form a *Henselian couple* if they satisfy condition a) of (18.5.4).

Taking into account (**I**, 5.1.8), it amounts to the same to say that (S, S_0) is a Henselian couple or that $(S_{\mathrm{red}}, (S_0)_{\mathrm{red}})$ is one.

Proposition (18.5.6). — *(i) If (S, S_0) is a Henselian couple, then, for every finite morphism $f : S' \to S$, if S'_0 is the closed sub-prescheme $f^{-1}(S_0)$ of S', the couple (S', S'_0) is Henselian.*

(ii) Let $S = \coprod_\alpha S^{(\alpha)}$ be a sum of preschemes, S_0 a closed sub-prescheme of S, the sum of sub-preschemes $S_0^{(\alpha)}$ of the $S^{(\alpha)}$. For the couple (S, S_0) to be Henselian, it is necessary and sufficient that each of the couples $(S^{(\alpha)}, S_0^{(\alpha)})$ be Henselian.

Proof. Assertion (i) is an immediate consequence of the definition, since for every finite morphism $g : S'' \to S'$, $f \circ g : S'' \to S$ is a finite morphism. Similarly, under the conditions of (ii), for every morphism $g : S' \to S$ to be finite, it is necessary and sufficient that each of its restrictions $g^{(\alpha)} : S'^{(\alpha)} \to S^{(\alpha)}$ be finite, and the conclusion follows from (18.5.6), (ii) and the remark (18.5.7). □ **IV-4** | 130

Remark (18.5.7). — Let $S = \mathrm{Spec}(A)$ be an affine scheme, S_0 a closed sub-prescheme defined by an ideal $\mathfrak{J}$ of A. Then, if the couple (S, S_0) is Henselian, the ideal $\mathfrak{J}$ is necessarily contained in the *radical* of A. Indeed, if $\mathfrak{m}$ is a maximal ideal of A, $\mathfrak{m}$ must belong to $V(\mathfrak{J}) = S_0$, by virtue of (18.5.4.3), whence the conclusion. In particular, suppose that S_0 is *reduced to a point*, then $\mathfrak{J}$ must be the radical of A, so A must be a *local* ring, and S_0 the unique closed point of $\mathrm{Spec}(A)$.

Definition (18.5.8). — We say that a ring A is *Henselian* if it is semi-local and if, denoting by $\mathfrak{r}$ the radical of A, the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{r}))$ is Henselian. We call a *Henselian local scheme* a scheme isomorphic to the spectrum of a Henselian local ring.

Proposition (18.5.9). — *(i) For a semi-local ring A to be Henselian, it is necessary and sufficient that it be a direct composite of Henselian local rings.*

(ii) For a local ring A to be Henselian, it is necessary and sufficient that every finite A-algebra B be isomorphic to a product of local rings.

Proof. (i) Indeed, the definition, applied to $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, shows that S_0 is a finite discrete set, closed in S, and S is the union of a finite number of open and closed subsets S_i ($1 \leqslant i \leqslant n$), pairwise disjoint, each containing exactly one of the maximal ideals $\mathfrak{m}_i$ of A; the conclusion follows from (18.5.6), (ii) and the remark (18.5.7).

(ii) Every finite morphism $S' \to \mathrm{Spec}(A)$ is of the form $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$, where B is a finite A-algebra. If k is the residue field of A, $\mathrm{Spec}(B \otimes_A k)$ is a finite Artinian spectrum, hence finite and discrete. To say that the couple $(\mathrm{Spec}(A), \mathrm{Spec}(k))$ is Henselian means that B is a direct composite of rings A_i such that $\operatorname{Spec}(A_i \otimes_A k)$ is reduced to a point, i.e. that each A_i (which is a finite A-algebra) must have only a single maximal ideal (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1). □

The study of Henselian rings is thus essentially reduced to that of Henselian *local* rings.

Proposition (18.5.10). — *If A is a Henselian ring, every finite A-algebra B is a Henselian ring (hence a direct composite of Henselian local rings (18.5.9)).*

Proof. Indeed, if $\mathfrak{r}'$ is the radical of B, the inverse image of $\mathfrak{r}'$ in A is the radical $\mathfrak{r}$ of A, and every maximal ideal of B is a maximal ideal of B, hence the set $V(\mathfrak{r}')$ in $\operatorname{Spec}(B)$ is the inverse image of the set $V(\mathfrak{r})$ in $\operatorname{Spec}(A)$. The proposition is then a consequence of (18.5.6), (i). □

Theorem (18.5.11). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. The following conditions are equivalent:*

label=a) *A is Henselian, in other words, every finite A-algebra B is isomorphic to a product of local rings.*

$a')$ *Condition $a)$ is satisfied for all A-algebras B of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$, where $F \in A[\mathrm{T}]$ is a monic polynomial.*

lbbel=b) *Let $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$. For every étale morphism $g : S' \to S$, if we set $S'_0 = S' \otimes_A k$, every S_0-section u_0 of S'_0 is the restriction of an S-section u of S'.*

lcbel=c) *For every morphism locally of finite type $f : X \to S$, separated, and every point $x \in X$ such that $f(x)$ is equal to the closed point s of S and that f is quasi-finite at the point x ((**III**, Err.20)), X is the sum of two preschemes X', X'' such that $X' = \operatorname{Spec}(\mathscr{O}_{X,x})$ and that $f|X' : X' \to S$ is a finite morphism.*

$c')$ *For every morphism locally of finite type $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra over $\mathscr{O}_{S,s} = A$.*

$c'')$ *For every morphism locally of finite presentation $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra and of finite presentation over $\mathscr{O}_{S,s} = A$.*

Proof. Let us first note that $c')$ (resp. $c'')$) is equivalent to the same condition where moreover f is supposed *separated*, the question being local on X. Similarly, condition $b)$ is equivalent to the same condition where moreover g is supposed *separated*: indeed, it suffices to apply the latter to the restriction of g to an open neighbourhood affine of the point $u_0(S_0)$ in S'. Let us henceforth restrict to the case where, in $b)$, $c')$ and c'', the morphisms are separated. **IV-4** | 131

The fact that $a)$ implies $b)$ and that $b)$ implies $a')$ is a particular case of (18.5.4). Let us moreover show that $a')$ implies $a)$. It is a matter of proving that if e_0 is an idempotent of $C = B \otimes_A k$, there is an idempotent $e \in B$ whose image is e_0 in C. If b is an element of B whose image in C is e_0, the sub-A-algebra $A[b] = B'$ of B is finite, and the canonical image C' of $B' \otimes_A k$ in C contains e_0. Now, $B' \otimes_A k$ is a k-algebra, hence a direct composite of local k-algebras, and consequently e_0 is the image of an idempotent e'_0 of $B' \otimes_A k$. We are thus reduced to the case where B is monogenic, and hence by $a')$, a quotient of an algebra of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$; now, by $a')$, $A[\mathrm{T}]/F.A[\mathrm{T}]$ is a direct composite of local rings, hence so is its quotient, which proves the existence of the idempotent e.

It is immediate that $c)$ implies $c')$, and that $c')$ implies $c'')$ when B is of finite presentation. Conversely, let us show that $c'')$ implies $a')$. For if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \operatorname{Spec}(B) \to \operatorname{Spec}(A) = Y$ is finite and of finite presentation, and the preceding demonstration shows that B is a direct composite of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of $\operatorname{Spec}(A)$.

It is trivial that $c)$ implies $c')$, by virtue of the remark at the beginning. Let us prove that $c')$ implies $c'')$. Suppose $c')$ satisfied and let us prove that under the conditions of $c'')$, the set $Z = \operatorname{Spec}(\mathscr{O}_{X,x})$ identifies with an open and closed subset of X, which will establish $c'')$ (**I**, 2.4.2). In the first place, the composite morphism $Z \xrightarrow{j} X \xrightarrow{f} S$, where j is the canonical map (**I**, 2.4.1), is finite and of finite presentation by hypothesis, and since f is separated and locally of finite presentation, j is also a finite morphism closed (**II**, 6.1.10), hence this proves that Z is closed in X. Moreover, since j is a closed immersion (**I**, 4.2.2) and ((**I**, 4.2.2)), j is a closed immersion. But if $\mathscr{J}$ is the Ideal of $\mathscr{O}_X$ defining Z, we also have $\mathscr{J}_z = 0$ for every point z in a neighbourhood open V of x in X, since by hypothesis $\mathscr{J}$ is a quasi-coherent $\mathscr{O}_X$-Module of finite type ((1.4.7)) and ((**0**, 5.2.2)). Now, in an open neighbourhood V containing Z, hence Z is open in X.

Finally, $c'')$ implies $a')$: indeed, if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \mathrm{Spec}(B) \to \mathrm{Spec}(A) = Y$ IV-4 | 132
is finite and of finite presentation, and the preceding argument shows that B is a direct composite of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of $\mathrm{Spec}(A)$. □

Corollary (18.5.12). — *Let A be a semi-local ring, $\mathfrak{r}$ its radical; set $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A/\mathfrak{r})$. For A to be Henselian, it is necessary and sufficient that, for every finite morphism $f : X \to S$ and every étale and separated morphism $g : Y \to S$, if we set $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$, the canonical map*

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S_0}(X_0, Y_0)$$

is bijective.

Proof. The sufficiency of the condition follows from the equivalence of $a)$ and $b)$ in (18.5.11), by applying this condition to the case where $f = 1_S$. To see that the condition is necessary, note that if A is Henselian, the couple (X, X_0) is Henselian by (18.5.6), (i); moreover $\mathrm{Hom}_S(X, Y) = \Gamma(X \times_S Y/X)$ and $\mathrm{Hom}_{S_0}(X_0, Y_0) = \Gamma(X_0 \times_{S_0} Y_0/X_0)$; since $X \times_S Y$ is étale and separated over X, the conclusion follows from the fact that $a)$ implies $b)$ in (18.5.4). □

Remarks (18.5.13). — The equivalent conditions $a)$, $a')$, $b)$ of Theorem (18.5.11) are also equivalent to the following ("Hensel's lemma"):

a$''$) For every monic polynomial $F \in A[\mathrm{T}]$, with canonical image $F_0 \in k[\mathrm{T}]$, and every decomposition $F_0 = G_0 H_0$ of F_0 into a product of two monic polynomials that are coprime G_0, H_0 in $k[\mathrm{T}]$, there exists a unique couple (G, H) of monic polynomials of $A[\mathrm{T}]$ possessing the following properties: G_0 and H_0 are the canonical images respectively of G and H, we have $F = GH$, and the ideal of $A[\mathrm{T}]$ generated by G and H is equal to $A[\mathrm{T}]$.

Lemma (18.5.13.1). — *Let A be a local ring with residue field k, $F \in A[\mathrm{T}]$ a monic polynomial, B the A-algebra $A[\mathrm{T}]/F.A[\mathrm{T}]$. There exists a canonical bijective correspondence between the decompositions of B into direct composite of two A-algebra quotients B', B'' and the decompositions $F = GH$ of F into a product of two monic polynomials G, H of $A[\mathrm{T}]$, such that the ideal generated by G and H is equal to $A[\mathrm{T}]$; the quotient algebras B', B'' corresponding to such a couple of polynomials G, H are respectively $A[\mathrm{T}]/H.A[\mathrm{T}]$ and $A[\mathrm{T}]/G.A[\mathrm{T}]$.*

Proof. If $F = GH$ and G and H generate the ideal $A[\mathrm{T}]$, there are two polynomials P, Q of $A[\mathrm{T}]$ IV-4 | 133
such that $1 = PG + QH$. We deduce that the intersection of the principal ideals $\mathfrak{a} = G.A[\mathrm{T}]$ and $\mathfrak{b} = H.A[\mathrm{T}]$ is equal to $\mathfrak{c} = F.A[\mathrm{T}]$: indeed, if $R \in G.A[\mathrm{T}] \cap H.A[\mathrm{T}]$, we can write $R = PRG + QRH$; since RG (resp. RH) is a multiple of F since R is a multiple of H (resp. G), therefore $R \in F.A[\mathrm{T}]$. Since $A[\mathrm{T}] = \mathfrak{a} + \mathfrak{b}$, $A[\mathrm{T}]/\mathfrak{c}$ is the direct sum of ideals $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$ and $\mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$, canonically isomorphic respectively to $A[\mathrm{T}]/\mathfrak{b}$ and $A[\mathrm{T}]/\mathfrak{a}$.

Conversely, suppose given a decomposition of B into a direct composite of two A-algebras B', B'', which identify canonically with two ideals $e'B$, $e''B$ of B, corresponding to a decomposition $1 = e' + e''$ into orthogonal idempotents e', e'' of B. Set further $B_0 = B \otimes_A k = k[\mathrm{T}]/F_0.k[\mathrm{T}]$, where F_0 is the canonical image of F in $k[\mathrm{T}]$, of the same degree n as F; if e'_0, e''_0 are the canonical images of e', e'' in B_0, these are two orthogonal idempotents such that $1 = e'_0 + e''_0$, and B_0 is a direct composite $B'_0 = e'_0 B_0$ and $B''_0 = e''_0 B_0$. Let t and t_0 be the canonical images of T in B and B_0; B'_0 (resp. B''_0) is a k-algebra generated by $t'_0 = e'_0 t_0$ (resp. $t''_0 = e''_0 t_0$), and it admits a base of the form $\{e'_0, t'_0, t'^2_0, \dots, t'^{s-1}_0\}$ (resp. $\{e''_0, t''_0, t''^2_0, \dots, t''^{r-1}_0\}$) with $r + s = n$. On the other hand, B being a free A-module (of base $\{1, t, \dots, t^{n-1}\}$), B' and B'' are projective A-modules, hence free since A is a local ring (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 3, cor. de la prop. 5); it follows from the above and from Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, prop. 5, that if we set $t' = e't$, $t'' = e''t$, $\{e', t', \dots, t'^{s-1}\}$ (resp. $\{e'', t'', \dots, t''^{r-1}\}$) is a base of the A-module B' (resp. B''). There is hence a monic polynomial H (resp. G) of degree s (resp. r) in $A[\mathrm{T}]$ such that $e'H(t') = 0$ and $e''G(t'') = 0$; as $t^h = e't'^h + e''t''^h$, $t''^h = e''t''^h$ for every integer $h \geqslant 1$, and as $t'^h = e't'^h$, we also have $G(t) = e'G'(t')$ and $H(t) = e''H(t'')$, whence $G(t)H(t) = 0$; we conclude that the polynomial $G(T)H(T)$ is divisible by $F(T)$; but since these two monic polynomials have the same degree, we have $GH = F$. Moreover, B' (resp. B'') is isomorphic to $A[\mathrm{T}]/H.A[\mathrm{T}]$ (resp. $A[\mathrm{T}]/G.A[\mathrm{T}]$). Finally, there are two polynomials R, S in $A[\mathrm{T}]$ such that $e' = R(t)$ and $e'' = S(t)$; as $e'S(t') = 0$ and also $e'S(t'') = 0$, we have necessarily $e'R(t') = 0$ and $e''R(t'') = 0$, so that, by definition of G and H, $R = QH$ and $S = PG$, where P, Q belong to $A[\mathrm{T}]$;

the relation $1 = R(t) + S(t)$ in B gives $PG + QH = 1 + LF$ for some polynomial $L \in A[\mathrm{T}]$, and since $F = GH$, this proves that the ideal generated by G and H is equal to $A[\mathrm{T}]$, and completes the proof of the lemma. □

This lemma being established, it suffices to apply it, to the ring A on the one hand, to the field k on the other, to see immediately that conditions a') and a'') are equivalent.

Proposition (18.5.14). — *Every semi-local ring, separated and complete for the $\mathfrak{r}$-preadic topology (where $\mathfrak{r}$ is the radical of A) is Henselian.*

Proof. Indeed, A is a direct composite of local rings separated and complete (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19), so we are reduced to the case where A is a local ring. Let us verify criterion a') of (18.5.11). Since B is an A-module libre of finite type, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, it all amounts to seeing that the map that, to every idempotent of A, associates its class mod $\mathfrak{J}$, gives a correspondence with its image, and one verifies immediately that for every affine open subset W of S with the notations of (18.5.1), is *bijective*. Let us write $\mathrm{Idem}(A)$ for the set of idempotents of A, and for every homomorphism of rings $\varphi : A \to B$, let $\mathrm{Idem}(\varphi)$ denote the map $\mathrm{Idem}(A) \to \mathrm{Idem}(B)$ restriction of φ; it follows from the definition of the projective limit that $\mathrm{Idem}(A) = \varprojlim \mathrm{Idem}(A/\mathfrak{J}^n)$, under the maps $\psi_{nm} : \mathrm{Idem}(A/\mathfrak{J}^m) \to \mathrm{Idem}(A/\mathfrak{J}^n)$ restriction of the canonical maps $A/\mathfrak{J}^m \to A/\mathfrak{J}^n$. But since $\mathrm{Spec}(A/\mathfrak{J}^m) \to \mathrm{Spec}(A/\mathfrak{J}^n)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion. □

Proposition (18.5.15). — *Let A be a Henselian ring, $\mathfrak{r}$ its radical. Then the functor $B \mapsto B/\mathfrak{r}B$ is an equivalence of the category of finite étale A-algebras with the category of finite étale $(A/\mathfrak{r})$-algebras.* **IV-4** | 134

Proof. The fact that the functor is fully faithful is a particular case of (18.5.12). To show that this functor is an equivalence of categories, we may reduce to the case where A is local; it suffices then to apply (18.1.2.thm) (for étale morphisms), and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, reduced to a single point. □

Remarks (18.5.16). —
(i) We do not know if Proposition (18.5.15) generalises to a Henselian couple (S, S_0), even when S is affine and Noetherian.
(ii) Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is *separated and complete* for the $\mathfrak{J}$-preadic topology. Then the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{J}))$ is Henselian: indeed, for every finite A-algebra B, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything amounts to seeing that the map that to every idempotent of A associates its class mod $\mathfrak{J}$, is *bijective*. Let us write $\mathrm{Idem}(A)$ for the set of idempotents of A, and $\mathrm{Idem}(A) = \varprojlim \mathrm{Idem}(A/\mathfrak{J}^n)$ for the projective limit; but since $\mathrm{Spec}(A/\mathfrak{J}^n) \to \mathrm{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion.

Theorem (18.5.17). — *Let A be a Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point. For every smooth morphism $f : X \to S$, the canonical map*

$$\Gamma(X/S) \longrightarrow \Gamma(X_s/\kappa(s))$$

(where $X_s = f^{-1}(s)$) is surjective.

Proof. The datum of a $\kappa(s)$-section of X_s is equivalent to that of a point $x \in X$ above s whose residue field $\kappa(x)$ is isomorphic to $\kappa(s)$, and it is a matter of proving that there exists an S-section $u : S \to X$ such that $u(s) = x$. Taking into account (17.16.3), (i), we can suppose that f is étale. Then the conclusion follows from the criterion (18.5.11), b). (The reader will note that by virtue of this criterion, the validity of (18.5.17) for a local ring is necessary and sufficient for A to be Henselian.) □

Remark (18.5.18). — The conditions of (18.5.11) are also equivalent to the following:

d) For every morphism locally of finite type $f : X \to S$ and every point $x \in X$ such that $f(x)$ be equal to the closed point s of S and that $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ be a field canonically isomorphic to $k = \kappa(s)$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

Proof. It is immediate that d) implies condition b) of (18.5.11) since a closed immersion étale is an open immersion (18.9.1). Conversely, suppose the conditions of (18.5.11) satisfied, and let us prove d). The hypothesis of d) implies that f is quasi-finite at the point x ((**III**, Err.20)), hence by virtue of condition c) of (18.5.11), $\mathscr{O}_{X,x}$ is a finite A-algebra and $\operatorname{Spec}(\mathscr{O}_{X,x})$ a neighbourhood open U of x in X. Moreover, since $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ is isomorphic to $k = A/\mathfrak{m}_s$, the Nakayama lemma proves that the homomorphism $A \to \mathscr{O}_{X,x}$ is surjective, hence $f|U$ is a closed immersion. □

Proposition (18.5.19). — *Let A be a Noetherian local ring and Henselian, $S = \operatorname{Spec}(A)$, s the closed point of S, $f : X \to S$ a proper morphism; set $X_s = f^{-1}(s)$. Then the map $Y \to Y \cap X_s$ is a bijection from the set of connected components of X to the set of connected components of X_s.*

Proof. By considering a closed sub-prescheme of X having for underlying space a connected component of X, we are reduced to showing that if X is *connected and non-empty*, then X_s is *connected and non-empty*. The fact that X_s is non-empty follows from ((**II**, 7.2.1)); to show that X_s is connected, we argue by contradiction, considering the Stein factorisation $f : X \xrightarrow{f'} S' \xrightarrow{g} S$ of the proper morphism f ((**III**, 4.3.3)); by hypothesis the finite discrete set $g^{-1}(s)$ would contain at least two points. Since A is Henselian and g separated and of finite type, it would then follow from (18.5.11), c) that S' would be the sum of two non-empty preschemes S'_1, S'_2 (since the intersection of any one of them with $g^{-1}(s)$ is reduced to a single point). Since f' is surjective ((**III**, 4.3.1)), we would conclude that X is a sum of two non-empty preschemes, contradicting the hypothesis. □

18.6. Henselization.

IV-4 | 136

(18.6.1). Given a local ring A, there exists a *set* $\mathfrak{E}$ of local A-algebras that are essentially étale such that every A-algebra belonging to $\mathfrak{E}$ is A-isomorphic to an algebra belonging to $\mathfrak{E}$. It suffices indeed to observe that there exists a set $\mathfrak{F}$ of A-algebras of finite type such that every A-algebra of finite type is isomorphic to an algebra belonging to $\mathfrak{F}$; one can take $\mathfrak{F}$ to be the set of quotients of polynomial algebras $\mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ ($n \in \mathbf{N}$).

In what follows, if A and B are two local rings, we denote by $\operatorname{Hom.loc}(\mathrm{A}, \mathrm{B})$ the set of *local* homomorphisms from A to B.

Lemma (18.6.2). — *Let* A, A′ *be two local rings, k, k' their residue fields, $\phi : \mathrm{A} \to \mathrm{A}'$ a local homomorphism making* A′ *an A-algebra essentially étale* (18.6.1) *and such that the corresponding homomorphism $k \to k'$ is bijective. Then, for every local Henselian ring* B*, the canonical map*

$$\operatorname{Hom}(\phi, 1_{\mathrm{B}}) : \operatorname{Hom.loc}(\mathrm{A}', \mathrm{B}) \longrightarrow \operatorname{Hom.loc}(\mathrm{A}, \mathrm{B})$$

is bijective.

Proof. Let $\mathrm{S} = \operatorname{Spec}(\mathrm{A})$, $\mathrm{Y} = \operatorname{Spec}(\mathrm{B})$, and let s and y be the closed points of S and Y respectively. By hypothesis, A′ is isomorphic to a local ring $\mathscr{O}_{\mathrm{X},x}$ of an S-scheme X étale over S, at a point $x \in \mathrm{X}$ above s. Suppose given a local homomorphism $\psi : \mathrm{A} \to \mathrm{B}$, making Y an S-scheme; it amounts to showing that there exists *one and only one* S-morphism $f : \mathrm{Y} \to \mathrm{X}$ such that $f(y) = x$. Let $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{Y}$, and note that since $k(x) = k(s)$, there exists a unique point $x' \in \mathrm{X}$ above x and above y, and that $k(x') = k(y)$. It remains to show that there exists a unique Y-section f' of X′ such that $f'(y) = x'$. Now, the morphism $g : \mathrm{X}' \to \mathrm{Y}$ is étale and separated, and the fibre $\mathrm{X}'_0 = g^{-1}(y)$ at the point x' has the residue field $k(x') = k(y)$. If we set $\mathrm{Y}_0 = \operatorname{Spec}(k(y))$, there exists a unique Y_0-section f'_0 of X'_0 such that $f'_0(y) = x'$, and the conclusion follows from the fact that B is assumed Henselian and from (18.5.11), b). □

We say that a local A-algebra A′ satisfying the conditions of (18.6.2) is *strictly essentially étale*. Note that the criterion (18.5.11), b) means that, *for A to be Henselian, it is necessary and sufficient that every strictly essentially étale A-algebra be A-isomorphic to* A.

Lemma (18.6.3). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and strictly essentially étale.*

label=i) *There exists at most one A-homomorphism (necessarily local) from* A_1 *to* A_2.
lbbel=ii) *There exists a local A-algebra* A_3 *that is strictly essentially étale, and two A-homomorphisms* $\mathrm{A}_1 \to \mathrm{A}_3$, $\mathrm{A}_2 \to \mathrm{A}_3$.

Proof. Let $S = \mathrm{Spec}(A)$; by hypothesis there are two S-schemes étale over S, X_1, X_2, and two points $x_1 \in X_1$, $x_2 \in X_2$ above the closed point s of S and such that $A_1 = \mathscr{O}_{X_1,x_1}$, $A_2 = \mathscr{O}_{X_2,x_2}$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Let $X_3 = X_1 \times_S X_2$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Furthermore, X_3 is étale over S (17.3.3), so $A_3 = \mathscr{O}_{X_3,x_3}$ satisfies the conditions of (ii). Moreover, as X'_2 is connected, the uniqueness of f' follows from (17.4.9), whence (i). □ **IV-4 | 137**

(18.6.4). Let us denote by $\mathfrak{S}$ the subset of the set $\mathfrak{E}$ defined in (18.6.1) formed by the A-algebras that are strictly essentially étale belonging to $\mathfrak{E}$. It follows from (18.6.3) that the relation "there exists an A-homomorphism from A_1 to A_2" is a *preorder* relation on $\mathfrak{S}$, and makes $\mathfrak{S}$ a *filtered increasing* set. Indexing $\mathfrak{S}$ by itself by means of the identity map, it follows from (18.6.3), (i) that if $\lambda \leqslant \mu$ in $\mathfrak{S}$, there exists a *unique* $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ and $(A_\lambda, \phi_{\mu\lambda})$ is clearly an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms. Furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is strictly essentially étale.

Definition (18.6.5). — The *Henselization* of a local ring A is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$, denoted ${}^hA$, defined in (18.6.4).

This definition does not depend on the choice of $\mathfrak{E}$; if $\mathfrak{E}'$ is another set having the same property as $\mathfrak{E}$, and $\mathfrak{S}'$ the subset of $\mathfrak{E}'$ consisting of the strictly essentially étale A-algebras, it follows from (18.6.3), (ii) that, for the preorder relations considered, $\mathfrak{S}$ and $\mathfrak{S}'$ are *cofinal* in $\mathfrak{S}'' = \mathfrak{S} \cup \mathfrak{S}'$, so they give the same inductive limit up to an isomorphism. We shall moreover see below (18.6.6), (i) and (ii) that ${}^hA$ and the canonical homomorphism $A \to {}^hA$ are the solution of a universal problem, and are therefore determined up to a unique isomorphism.

One will observe that if A is a field, then evidently ${}^hA = A$. In the general case, ${}^hA$ is also the Henselization of all the A-algebras that are strictly essentially étale A_λ ($\lambda \in \mathfrak{S}$), since if B is an A_λ-algebra that is strictly essentially étale, B is also an A-algebra that is strictly essentially étale (18.6.1), so (up to an A-isomorphism) one of the A_μ for $\mu \geqslant \lambda$, and the A_μ such that $\mu \geqslant \lambda$ and that A_μ is an A_λ-algebra that is strictly essentially étale form a cofinal set in the preordered set of the A_λ, by virtue of (18.6.1) and (18.6.3).

Theorem (18.6.6). — *Let* A *be a local ring,* ${}^hA$ *its Henselization.*

label=i) ${}^hA$ *is a Henselian local ring and the structural homomorphism* $A \to {}^hA$ *is local.*

lbbel=ii) *For every Henselian local ring* B*, the canonical map*

$$\mathrm{Hom.loc}({}^hA, B) \longrightarrow \mathrm{Hom.loc}(A, B)$$

is bijective.

lcbel=iii) ${}^hA$ *is a faithfully flat A-module, and if* $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}.{}^hA$ *is the maximal ideal of* ${}^hA$, *and the homomorphism* $A/\mathfrak{m}A \to {}^hA/\mathfrak{m}.{}^hA$ *of residue fields is bijective.*

ldbel=iv) *If* $\widehat{A}$ *and* $({}^hA)^\wedge$ *are the separated completions of the local rings* A *and* ${}^hA$*, the homomorphism* $\widehat{A} \to ({}^hA)^\wedge$ *deduced from the structural homomorphism* $A \to {}^hA$ *by completion is bijective.*

lebel=v) *For* ${}^hA$ *to be Noetherian, it is necessary and sufficient that A be so.*

lfbel=vi) *If A is Henselian, the canonical homomorphism* $A \to {}^hA$ *is bijective.*

Proof. Let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ; the fact that, in (17.6.1), *a*) implies *c'*), that for $\lambda \leqslant \mu$, on has $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$, the homomorphism $A_\lambda/\mathfrak{m}_\lambda \to A_\mu/\mathfrak{m}_\mu$ is bijective and A_μ is a flat A_λ-module. The fact that ${}^hA$ is local, that the homomorphism $A \to {}^hA$ is local, assertion (iii) and the sufficiency of (v) follow therefore from (**0**, 10.3.1.3); the necessity of (v) follows from the fact that ${}^hA$ is a flat A-module (6.5.2). **IV-4 | 138**

To prove that ${}^hA$ is Henselian, let us apply the criterion (18.5.11), *b*). Let $S = \mathrm{Spec}({}^hA)$, $S_0 = \mathrm{Spec}({}^hA/\mathfrak{m}.{}^hA)$, and let $g : S' \to S$ be an étale morphism; reasoning as in (18.5.4), one can suppose that g is of finite presentation. It follows then from (8.8.2) and (17.7.5) that there exists an index $\lambda \in S$, an étale morphism $g_\lambda : S^{(\lambda)} \to \mathrm{Spec}(A_\lambda)$ and, if one sets $S_0^{(\lambda)} = g^{-1}(S_0^{(\lambda)})$, an $S_0^{(\lambda)}$-section $f_0^{(\lambda)} : S_0^{(\lambda)} \to S'^{(\lambda)}$ such that $S' = S'^{(\lambda)} \times_{g^{(\lambda)}} S$, $g = g^{(\lambda)} \times 1$ and $f_0 = f_0^{(\lambda)} \times 1$. Let s be the closed point of S, $x = f_0(s)$, let s_λ be the closed point of $S^{(\lambda)}$; as x_λ is above the closed point s_λ of $S^{(\lambda)}$, the local ring C_λ of $S'^{(\lambda)}$ at the point x_λ is an A_λ-algebra that is strictly essentially étale; furthermore, as $f_0^{(\lambda)}(s_\lambda) = x_\lambda$, in other words, C_λ is an A_λ-algebra that is local and strictly essentially étale, and by (18.6.1) is an A-algebra A_μ with $\mu \geqslant \lambda$. There is therefore an A_λ-homomorphism $C_\lambda \to {}^h A$, that is to say there exists a well-determined S-morphism $f : S \to S'$ of S'; by (18.5.4), there exists a neighbourhood of $\mathfrak{n}_\lambda$ in $\operatorname{Spec}(C_\lambda)$ isomorphic to $\operatorname{Spec}(A_\lambda)$ (**I**, 6.5.4); one thus concludes that there is a neighbourhood of $\mathfrak{n}$ in $\operatorname{Spec}(C)$ isomorphic to $\operatorname{Spec}(A')$, hence $C_\mathfrak{n}$ is isomorphic to A', which proves that A' is Henselian (18.6.2) and completes the proof.

To prove (ii), it suffices to note that $\operatorname{Hom.loc}({}^h A, B) \cong \varprojlim \operatorname{Hom.loc}(A_\lambda, B)$ and that by (18.6.2) the canonical homomorphisms $\operatorname{Hom.loc}(A_\lambda, B) \leftarrow \operatorname{Hom.loc}(A, B)$ are bijective.

To prove (iv), note that for every λ and every integer $n > 0$, $\mathfrak{m}_\lambda^n = \mathfrak{m}^n A_\lambda$, and $(\mathfrak{m}.{}^h A)^n = \mathfrak{m}^n.{}^h A = \varinjlim (A_\lambda/\mathfrak{m}_\lambda^n)$, and by the exactness of the functor $\varinjlim$ in the category of A-modules, that ${}^h A/(\mathfrak{m}.{}^h A)^n = \varinjlim (A_\lambda/\mathfrak{m}_\lambda^n)$, and it therefore suffices to show that, for every n and every index λ, the homomorphism $A/\mathfrak{m}^n \to A_\lambda/\mathfrak{m}_\lambda^n$ is bijective. On the other hand, since A_λ is a flat A-module, one has

$$\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1} = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_A A_\lambda = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_{A/\mathfrak{m}} (A_\lambda/\mathfrak{m}A_\lambda)$$

and as $A/\mathfrak{m} \to A_\lambda/\mathfrak{m}A_\lambda$ is bijective, the homomorphism $\mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}$ is also bijective; the conclusion then follows from Bourbaki, *Alg. comm.*, chap. III, § 2, no 8, cor. 3 du th. 1.

Finally, if A is Henselian, it follows from the remark preceding (18.6.3) that the canonical homomorphisms $A \to A_\lambda$ are bijective, which proves (vi) by definition of ${}^h A$. □

(18.6.7). Let A be a *semi-local* ring, $\mathfrak{m}_i$ ($1 \leqslant i \leqslant r$) its maximal ideals. The *Henselization* of A is the ring denoted again ${}^h A$, defined as the *product* $\prod_i {}^h(A_{\mathfrak{m}_i})$ of the Henselizations of the local rings $A_{\mathfrak{m}_i}$. It is a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_i.{}^h A$ by virtue of (18.6.6), (iii); furthermore, if $\mathfrak{r} = \bigcap \mathfrak{m}_i$ is the radical of A, it follows from the above that $\mathfrak{r}.{}^h A$ is the radical of ${}^h A$ and that the canonical map $A/\mathfrak{r} \to {}^h A/\mathfrak{r}.{}^h A$ is bijective. As the separated completion $\widehat{A}$ of A for the $\mathfrak{r}$-preadic topology is the product of the separated completions $\widehat{A}_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, no 13, prop. 18), the canonical homomorphism $\widehat{A} \to ({}^h A)^\wedge$ is bijective by (18.6.6), (iv), and it is clear by (18.6.6), (v) that for ${}^h A$ to be Noetherian, it is necessary and sufficient that A be so. **IV-4** | 138 **IV-4** | 139

To obtain the analogue of the universal property (18.6.6), (ii), when A and B are two *semi-local* rings of radicals $\mathfrak{r}$, $\mathfrak{s}$, let us agree to call a *semi-local homomorphism* ϕ from A to B any homomorphism such that $\phi(\mathfrak{r}) \subset \mathfrak{s}$. In the particular case where B is a *product* of local rings B_j ($1 \leqslant j \leqslant n$), the data of such a homomorphism amounts to the data of its projections $\phi_j : A \to B_j$, which are homomorphisms subject only to the condition that $\phi_j(\mathfrak{r}) \subset \mathfrak{n}_j$, where $\mathfrak{n}_j$ is the maximal ideal of B_j. Moreover, since $\mathfrak{m}_i$ ($1 \leqslant i \leqslant m$) are the maximal ideals of A, the prime ideal $\phi_j^{-1}(\mathfrak{n}_j)$ must contain one of the $\mathfrak{m}_i$, since it must contain their intersection $\mathfrak{r}$, hence it is equal to one of the $\mathfrak{m}_i$ and ϕ_j factorises as $A \to A_{\mathfrak{m}_i} \xrightarrow{\psi_{ij}} B_j$, where ψ_{ij} is a *local* homomorphism. If $\operatorname{Hom.sloc}(A, B)$ denotes the set of semi-local homomorphisms from A to B, one can thus identify this set with $\prod_j \left(\bigcup_i \operatorname{Hom.loc}(A_{\mathfrak{m}_i}, B_j)\right)$; in the case considered, since a semi-local Henselian ring is a direct product of Henselian local rings, one sees that the universal property (18.6.6), (ii) is still valid for a semi-local ring A, provided one replaces "local homomorphisms" by "semi-local homomorphisms" (18.6.7).

Proposition (18.6.8). — *Let* A *be a semi-local ring,* B *an A-algebra that is semi-local and entire over* A*. Then* $B \otimes_A ({}^h A)$ *is a semi-local B-algebra isomorphic to* ${}^h B$*.*

Proof. As $B \otimes_A ({}^h A)$ is entire over ${}^h A$, each of its maximal ideals lies above one of the maximal ideals of ${}^h A$, hence above a maximal ideal of A, and since B is entire over A, the prime ideals of B lying above a maximal ideal of A are maximal ideals of $B \otimes_A ({}^h A)$. Taking account of (18.6.6), (iii) and (**I**, 3.4.9), one concludes that the ring $B \otimes_A ({}^h A)$ is semi-local. Furthermore, for every Henselian local ring C, $\operatorname{Hom.sloc}(B \otimes_A ({}^h A), C)$ is in bijection with the set of pairs (ϕ, ψ) of homomorphisms $\psi : {}^h A \to C$, $\phi : B \to C$, $A \to {}^h A \xrightarrow{\psi} C$ such that the composites $A \to B \xrightarrow{\phi} C$ and $A \to {}^h A \xrightarrow{\psi} C$ are equal. But by virtue of the bijection between $\operatorname{Hom.sloc}(A, C)$ and $\operatorname{Hom.sloc}({}^h A, C)$, one sees that for every $\phi \in \operatorname{Hom.sloc}(B, C)$ there exists a unique ψ having the preceding property, hence the map

$\operatorname{Hom.sloc}(B \otimes_A ({}^h A), C) \to \operatorname{Hom.sloc}(B, C)$ is bijective, which proves the proposition by virtue of the uniqueness of the solution of a universal problem. □

Theorem (18.6.9). — *Let* A *be a semi-local ring.*

label=i) *For* ${}^h A$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*

lbbel=ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^h A)_{\mathfrak{p}}/\mathfrak{p}.({}^h A)_{\mathfrak{p}}$ *is a composed direct product of a finite number of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\operatorname{Spec}({}^h A) \to \operatorname{Spec}(A)$ *are* geometrically regular (18.6.9) *and are* discrete *spaces).*

Proof. (i) As ${}^h A$ is a faithfully flat A-module, it follows from (2.1.13) that if ${}^h A$ is reduced (resp. **IV-4** | 140
normal), so is A. To prove the converse, one can restrict to the case where A is local, so that, with the notations of (18.6.4), one has ${}^h A = \varinjlim A_\alpha$. Now, it follows from the definition of the A_α and from (17.5.7) that if A is reduced (resp. integral and integrally closed), so is each of the A_α; furthermore, the homomorphisms $A_\alpha \to A_\beta$ for $\alpha \leqslant \beta$ are injective by virtue of (**0**, 6.5.1) since A_β is a flat A_α-module; hence the morphisms $\operatorname{Spec}(A_\beta) \to \operatorname{Spec}(A_\alpha)$ are dominant, and one concludes from (5.13.2) (resp. (5.13.4)) that ${}^h A$ is reduced (resp. integral and integrally closed).

(ii) One can restrict to the case where A is local. By virtue of the fact that the functor $\varinjlim$ commutes with tensor products, the fibre of the morphism $\operatorname{Spec}({}^h A) \to \operatorname{Spec}(A)$ at a point $\mathfrak{p}$ is the projective limit of the fibres of the morphisms $\operatorname{Spec}(A_\alpha) \to \operatorname{Spec}(A)$ at this point. As ${}^h A$ is Noetherian, one is reduced to proving the following lemma: □

Lemma (18.6.9.1). — *Let* (B_α) *be an inductive system of Artinian rings,* $B = \varinjlim B_\alpha$ *its inductive limit. If* B *is Noetherian, then* B *is Artinian; if moreover, for* $\alpha \leqslant \beta$*, the homomorphisms* $\phi_{\beta\alpha} : B_\alpha \to B_\beta$ *are injective, then there exists* λ *such that, for* $\alpha \geqslant \lambda$*, the number of local components* $B_\alpha^{(i)}$ *of* B_α *is constant, that* $\phi_{\beta\alpha}(B_\alpha^{(i)}) \subset B_\beta^{(i)}$ *for every i and that the* $B^{(i)} = \varinjlim B_\alpha^{(i)}$ *are the local components of* B*.*

Proof. Indeed $Y_\alpha = \operatorname{Spec}(B_\alpha)$, $Y = \operatorname{Spec}(B)$, which, as a topological space, is equal to $\varprojlim Y_\alpha$ (8.2.10). The spaces Y_α are finite and discrete; furthermore, if the $\phi_{\beta\alpha}$ are injective, the canonical morphisms $Y_\beta \to Y_\alpha$ are dominant, hence surjective. The lemma then follows from the following purely topological lemma: □

Lemma (18.6.9.2). — *Let* $(Y_\alpha, \psi_{\alpha\beta})$ *be a projective system of finite topological spaces, and let Y be discrete. If* $Y = \varprojlim Y_\alpha$ *is Noetherian, Y is finite and discrete. If moreover the* $\psi_{\alpha\beta}$ *are surjective, then there exists* λ *such that for* $\alpha \geqslant \lambda$*, the number of elements of* Y_α *is constant and equal to the number of elements of* Y*.*

Proof. Indeed, Y being compact and Noetherian, every subset of Y is closed, hence compact, hence finite since it is compact. If moreover the $\psi_{\alpha\beta}$ are surjective, the same holds for $\psi_\alpha : Y \to Y_\alpha$, so $\operatorname{Card}(Y) \geqslant \operatorname{Card}(Y_\alpha)$, and as $\operatorname{Card}(Y_\alpha)$ is an increasing function of α, there exists λ such that for $\alpha \geqslant \lambda$, $\operatorname{Card}(Y_\alpha) = \operatorname{Card}(Y_\lambda)$; the $\psi_{\alpha\beta}$ are then bijective for $\alpha \geqslant \lambda$ and one therefore has $\operatorname{Card}(Y) = \operatorname{Card}(Y_\lambda)$. □

Corollary (18.6.10). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to possess one of the following properties:*

label=a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*

lbbel=b) *satisfying the property* (S_n) (5.7.3)*;*

lcbel=c) *being regular;*

ldbel=d) *satisfying the property* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Proof. It follows immediately from the fact that the fibres of the morphism $\operatorname{Spec}({}^h A) \to \operatorname{Spec}(A)$ are **IV-4** | 141
geometrically regular (18.6.9) and from (6.4.1), (6.5.1) and (6.5.3). □

Corollary (18.6.11). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^h A$ *to belong to* $\operatorname{Ass}({}^h A)$*, it is necessary and sufficient that* $\mathfrak{p} \cap A$ *belong to* $\operatorname{Ass}(A)$*.*

Proof. Taking account of the description of the fibres of the morphism $\operatorname{Spec}({}^h A) \to \operatorname{Spec}(A)$, this also follows immediately from (3.3.1). □

Proposition (18.6.12). — *Let* A *be a local ring; the following properties are equivalent:*

label=a) *A is unibranch* (**0**, 23.2.1) *(resp. reduced and unibranch).*
lbbel=b) *For every A-algebra* A' *that is local and strictly essentially étale,* $\mathrm{Spec}(A')$ *is irreducible (resp. integral).*
lcbel=c) $\mathrm{Spec}({}^hA)$ *is irreducible (resp. integral).*

Proof. Note first that ${}^h(A_{red}) = ({}^hA)_{red}$, since ${}^h(A_{red}) = ({}^hA) \otimes_A A_{red}$ (18.6.8), and ${}^h(A_{red})$ is reduced (18.6.9). This allows, in the proof, to consider only the case where A is reduced, hence also A' (17.5.7) and ${}^hA$ (18.6.9).

With the notations of (18.6.4), the condition $b)$ means that all the A_α are integral; one therefore concludes that $b)$ implies $c)$ (5.13.3), hence $b)$ implies $c)$.

It is clear that if ${}^hA$ is integral, so are each of the $A_\alpha \subset {}^hA$, hence $c)$ implies $b)$.

Let us show that $c)$ implies $a)$. Note first that $A \subset {}^hA$ is integral; let K and L be the fields of fractions of A and ${}^hA$ respectively. It suffices to show that for every *finite* A-algebra B, $B \subset K$, B is an anneau local, whose Henselization is therefore $B \otimes_A ({}^hA)$ (18.6.8). Now B is a semi-local ring, whose Henselization is therefore $B \otimes_A ({}^hA)$ (18.6.8). By flatness, $B \otimes_A ({}^hA)$ identifies with a subring of L, hence ${}^hB$ is integral. But as ${}^hB$ is a semi-local Henselian ring, it is a composed direct product of local rings (18.5.9), hence it can only be integral if B is local, which shows that B is local.

Let us finally prove that $a)$ implies $c)$. Let C be the integral closure of the integral ring A. As C is a local ring by hypothesis, $C \otimes_A ({}^hA)$ is the Henselization of C (18.6.8), hence is integral and integrally closed (18.6.9). As A is a subring of C, hence ${}^hA$ identifies with a subring of ${}^hC$ by flatness, hence is also integral, which completes the proof. □

Corollary (18.6.13). — *Let* A *be a local Henselian ring. Then, for A to be unibranch, it is necessary and sufficient that* A_{red} *be integral.*

Proposition (18.6.14). — label=i) *Every filtered inductive limit of Henselian local rings (where the transition homomorphisms are local) is a Henselian local ring.*
lbbel=ii) *The functor* $A \rightsquigarrow {}^hA$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
lcbel=iii) *Every Henselian local ring A is an inductive limit of an inductive filtered system of Henselian and Noetherian local rings* A_λ *(the transition homomorphisms* $A_\lambda \to A_\mu$ *being local).*

Proof. (i) Let (A_λ) be a filtered inductive system of Henselian local rings, where the transition homomorphisms are local. Let us show that $A' = \varinjlim A_\lambda$, which is local (**0**, 10.3.1.3), is Henselian. **IV-4 | 141**
Now, let C be an A'-algebra étale, $\mathfrak{n}$ a prime ideal of C above the maximal ideal $\mathfrak{m}'$ of A', such that $C_\mathfrak{n}$ is strictly essentially étale over A' (18.6.1), and in virtue of (8.8.2), (ii) and (17.7.8), there exist an index λ and an A_λ-algebra C_λ étale such that $C = C_\lambda \otimes_{A_\lambda} A'$ up to an isomorphism. Furthermore, if $\mathfrak{n}_\lambda$ is the inverse image of $\mathfrak{n}$ in C_λ, and $\mathfrak{m}_\lambda$ the maximal ideal of A_λ, one can, by virtue of (8.8.2.4) and of the transitivity of fibres (**I**, 3.6.4), suppose that $(C_\lambda)_{\mathfrak{n}_\lambda}$ is strictly essentially étale over A_λ; since A_λ is Henselian, $(C_\lambda)_{\mathfrak{n}_\lambda}$ is necessarily isomorphic to A_λ (18.5.11), b); there exists therefore a neighbourhood of $\mathfrak{n}_\lambda$ in $\mathrm{Spec}(C_\lambda)$ isomorphic to $\mathrm{Spec}(A_\lambda)$ (**I**, 6.5.4); hence $C_\mathfrak{n}$ is isomorphic to A', which proves that A' is Henselian (18.6.2) and completes the proof. **IV-4 | 142**

(ii) Suppose that $B = \varinjlim B_\lambda$, where the B_λ are local rings, the transition morphisms $B_\lambda \to B_\mu$ ($\lambda \leqslant \mu$) being local. Set $A_\lambda = {}^hB_\lambda$; the A_λ are Henselian local rings and Noetherian local rings (18.6.6), (v), and by virtue of (18.6.6), (vi), (ii) the transition morphisms $B_\lambda \to B_\mu$ uniquely determine local homomorphisms $A_\lambda \to A_\mu$, so that (A_λ) is an inductive system. It amounts to showing that $A' = \varinjlim A_\lambda$ is canonically isomorphic to $A = {}^hB$. By virtue of (18.6.6), (ii) and of the definition of inductive limits, one has, for every Henselian local ring E,

$$\mathrm{Hom.loc}(A', E) = \varprojlim \mathrm{Hom.loc}(A_\lambda, E) = \varprojlim \mathrm{Hom.loc}(B_\lambda, E) = \mathrm{Hom.loc}(B, E).$$

But as A' is Henselian by (i), this proves our assertion.

(iii) The ring A is a filtered inductive limit of its sub-local Noetherian rings B_λ, the transition homomorphisms being local (5.13.3), (iii). As the ${}^hB_\lambda$ are Henselian and Noetherian local rings (18.6.6), (v) and that $A = {}^hA$ (18.6.6), (vi), it suffices to apply (ii) to the inductive system (B_λ). □

Corollary (18.6.15). — *Let* A *be a Henselian local ring,* X *an A-prescheme of finite presentation over* A*. Then there exists a Noetherian Henselian local ring* A_0*, a local homomorphism* $A_0 \to A$*, an* A_0*-prescheme* X_0 *of finite type over* A_0 *and an A-isomorphism* $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$*.*

Proof. This follows from (18.6.14) and (8.8.2), (ii). □

18.7. Henselization and excellent rings.

(18.7.1). In this paragraph we denote by $\boldsymbol{P}(Z, k)$ a property of the form considered in (7.3.1), where we further suppose that the property $\boldsymbol{Q}(A, k)$ satisfies the following condition: **IV-4 | 142**

For every separable algebraic extension k' of k, and every k'-local algebra that is Noetherian, the property $\boldsymbol{Q}(A, k)$ is *equivalent* to $\boldsymbol{Q}(A, k')$.

It is immediate that all the properties $\boldsymbol{P}$ considered in (7.3.8) satisfy the preceding condition, since if K is a finite extension of k, $k' \otimes_k K$ is a composed direct product of a finite number of separable algebraic extensions of K.

Proposition (18.7.2). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be a* $\boldsymbol{P}$*-ring* (7.3.13)*, it is necessary and sufficient that A be so.*

Proof. Indeed, by virtue of (18.6.6), (iv), the completion $\widehat{A}$ of A is also the completion of ${}^h A$ and one therefore has $A \subset {}^h A \subset \widehat{A}$. By (18.6.9), for every $x \in \operatorname{Spec}(A)$, the fibre at x of the morphism $\operatorname{Spec}({}^h A) \to \operatorname{Spec}(A)$ is discrete, finite, and each of the points y_i of this fibre is a separable algebraic extension of $k(x)$. The formal fibre of A at the prescheme *sum* of the formal fibres of ${}^h A$ at the points y_i. The conclusion then follows from the hypothesis on $\boldsymbol{P}$ in (18.7.1). □ **IV-4 | 143**

Corollary (18.7.3). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be universally Japanese, it is necessary and sufficient that A be so.*

Proof. Indeed, for a Noetherian local ring B, it amounts to the same to say that B is universally Japanese, or that the formal fibres of B are geometrically reduced (7.6.4) and (7.7.1); the conclusion therefore follows from (18.7.2). □

Corollary (18.7.4). — *Let* A *be a local Noetherian ring. For the formal fibres of* ${}^h A$ *to be geometrically regular, it is necessary and sufficient that those of A be so.*

Proof. This is a particular case of (18.7.2). □

Proposition (18.7.5). — *Let* A *be a local Noetherian ring. If A is universally catenary* (5.6.2) *(resp. formally catenary* (7.1.9)*, resp. strictly formally catenary* (7.2.6)*), so is* ${}^h A$*.*

Proof. With the notations of (18.6.4), if A is universally catenary, the same holds for the A_α, which are A-algebras essentially of finite type (5.6.3). To prove that in this case ${}^h A$ is universally catenary, it will therefore suffice to establish the following lemma: □

Lemma (18.7.5.1). — *Let* $(B_\alpha, \phi_{\beta\alpha})$ *be an inductive system of Noetherian rings and suppose that, for* $\alpha \leqslant \beta$*,* $\phi_{\beta\alpha}$ *makes* B_β *a flat* B_α*-module and that* $B = \varinjlim B_\alpha$ *is Noetherian. Then, if the* B_α *are catenary (resp. universally catenary), the same holds for* B*.*

Proof. To say that B is universally catenary is equivalent to saying that $B[T_1, \ldots, T_n]$ is catenary for all n (5.6.2), and it is immediate that the inductive system $(B_\alpha[T_1, \ldots, T_n])$ satisfies the same conditions as (B_α); it therefore suffices to prove that the B_α are catenary. If $\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$ is a saturated chain of prime ideals, the inverse images $\mathfrak{p}_{0\alpha} \supset \mathfrak{p}_{1\alpha} \supset \cdots \supset \mathfrak{p}_{r\alpha}$ of these ideals in B_α form a chain of prime ideals for each α sufficiently large, and it suffices to show that for α sufficiently large, this chain is saturated. Now, each $\mathfrak{p}_i$ is an ideal of finite type, hence there exists λ such that, for $\alpha \geqslant \lambda$, on has $\mathfrak{p}_i = \mathfrak{p}_{i\alpha}.B$. On has therefore $1 = \operatorname{codim}(V(\mathfrak{p}_i), V(\mathfrak{p}_{i+1})) = \operatorname{codim}(V(\mathfrak{p}_{i\alpha}), V(\mathfrak{p}_{i+1,\alpha}))$ since B is a flat B_α-module (6.1.4) and (**0**, 6.2.3), which completes the proof of the lemma.

Suppose now A formally catenary. Let $\mathfrak{p}$ be a prime ideal of ${}^h A$, and $\mathfrak{q}$ its inverse image in A, so that ${}^h A/\mathfrak{q}.{}^h A$ is a quotient ring of ${}^h A$, and as ${}^h A/\mathfrak{q}.{}^h A = {}^h(A/\mathfrak{q})$ by (18.6.8), it suffices to show that if A is *integral* and formally catenary, ${}^h A$ is formally *equidimensional*. But as the completion of ${}^h A$ is equal to $\widehat{A}$, this follows from the hypothesis that A is formally catenary. □

Corollary (18.7.6). — *If a local Noetherian ring A is excellent (7.8.2), so is* ^{h}A.

Remark (18.7.7). — It can happen that ^{h}A is an excellent local ring without A being universally catenary (nor *a fortiori* excellent). To see this, let us take the example of A from (5.6.11), with the same notations. The ring E constructed in (5.6.11) is excellent when k_0 is of characteristic 0 (7.8.3), (ii) and (iii); consider two rings E_1, E_2 isomorphic to E, and "glue" $\mathrm{Spec}(\mathrm{E}_1)$ and $\mathrm{Spec}(\mathrm{E}_2)$ such that if $\mathfrak{m}_i$ and $\mathfrak{m}'_i$ are the maximal ideals of E_i $(i = 1, 2)$, corresponding to $\mathfrak{m}$ and $\mathfrak{m}'$, the point $\mathfrak{m}_1$ is "glued" to $\mathfrak{m}'_2$ and the point $\mathfrak{m}_2$ to $\mathfrak{m}'_1$; to be precise, if ε_i and ε'_i are the homomorphisms from E_i on $k(\mathfrak{m}_i)$ and $k(\mathfrak{m}'_i)$ $(i = 1, 2)$, the schema obtained is $\mathrm{Spec}(\mathrm{R})$, where R is the subring of $\mathrm{E}_1 \times \mathrm{E}_2$ formed by the pairs (x_1, x_2) such that $\varepsilon'_1(x_1) = \sigma(\varepsilon_2(x_2))$ and $\varepsilon_1(x_1) = \sigma(\varepsilon'_2(x_2))$. One verifies easily (for example with the help of (17.6.3)) that R is a finite C-algebra that is étale, and that there are two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of R above the maximal ideal $\mathfrak{n}$ of C, with completions of $\mathrm{R}_{\mathfrak{r}_1}$ and $\mathrm{R}_{\mathfrak{r}_2}$, and that the Henselization is equal to ^{h}A, having $\mathrm{A} = \mathrm{C}_{\mathfrak{n}}$; $\mathrm{R}_{\mathfrak{r}_i}$ is therefore an A-algebra that is essentially étale, and its Henselization is also equal to ^{h}A. Furthermore, one has, by (7.8.4), (ii) that the formal fibres of A are geometrically regular, hence R is universally catenary. Finally, the ring R is universally catenary, and the same holds for the quotients by its two minimal prime ideals, whence the conclusion by (5.6.3), (iii). The ring $\mathrm{R}_{\mathfrak{r}_i}$ is therefore excellent, and by the same token ^{h}A (18.7.6), while A is not universally catenary.

18.8. Strictly local rings and strict Henselization. IV-4 | 144

Proposition (18.8.1). — *Let* A *be a local ring. The following conditions are equivalent:*

label=a) *A is a Henselian ring and its residue field k is separably closed.*
lbbel=b) *A is a Henselian ring and every étale covering of* $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ *is trivial.*
lcbel=c) *For every étale morphism* $f : \mathrm{X} \to \mathrm{S}$ *and every point* $x \in \mathrm{X}$ *such that* $f(x)$ *is the closed point s of S, there exists an S-section* $u : \mathrm{S} \to \mathrm{X}$ *of X such that* $u(s) = x$.

Proof. The hypothesis $c)$ implies that for every étale morphism $f : \mathrm{X} \to \mathrm{S}$, the residue fields at the points of $f^{-1}(s)$ are all isomorphic to k, and moreover the condition $b)$ of (18.5.11) is satisfied. Hence A is Henselian, and the fact that every étale covering of S is trivial follows from (18.2.10), (ii); hence $c)$ implies $b)$. As étale coverings of $\mathrm{Spec}(k)$ are trivial if and only if k is separably closed, $b)$ implies $a)$ by virtue of (18.5.11), $b)$. Finally, it follows also from (18.5.11), $b)$ that $a)$ implies $c)$. ☐

Definition (18.8.2). — A local ring is said to be *strictly local* if it satisfies the equivalent conditions of (18.8.1). A scheme isomorphic to the spectrum of a strictly local ring is called a *strictly local scheme*.

Remark (18.8.3). — The conditions of (18.8.1) are also equivalent to the following:

d) For every morphism locally of finite type $f : \mathrm{X} \to \mathrm{S}$ and every point $x \in \mathrm{X}$ such that $f(x)$ is the closed point s of S and that $\mathscr{O}_{\mathrm{X},x}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X},x}$ is a field, a finite separable extension of $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

(One will note that the hypothesis of $d)$ is satisfied if f is *formally unramified* at the point x (17.4.1.2).)

We leave the reader the proof, which is substantially the same as that of (18.5.18).

Lemma (18.8.4). — *Let* A, A′ *be two local rings,* $\phi : \mathrm{A} \to \mathrm{A}'$ *a local homomorphism making* A′ *an A-algebra essentially étale (18.6.1). Then, for every strictly local ring* B, *every local homomorphism* $\psi : \mathrm{A} \to \mathrm{B}$ *and every homomorphism of* $k(\mathrm{A})$*-algebras* $\alpha : k(\mathrm{A}') \to k(\mathrm{B})$, *there exists one and only one local homomorphism* $\psi' : \mathrm{A}' \to \mathrm{B}$ *such that* $\psi = \psi' \circ \phi$ *and that* α *is equal to the homomorphism* $\overline{\psi}'$ *deduced from* ψ' *by passage to quotients.* IV-4 | 145

Proof. With the notations of (18.6.2), the homomorphism α corresponds to a $k(s)$-morphism $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(x))$, hence to a point x' well determined in X′ above y. The existence of the Y-section f' of X′ such that $f'(y) = x'$ then follows from (18.8.1), $c)$ since f' is étale and B strictly local, and its uniqueness follows from this and from (17.4.9). ☐

Lemma (18.8.5). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and essentially étale, K a field that is an extension of* $k(\mathrm{A})$.

label=i) *For every $k(\mathrm{A})$-homomorphism $\gamma : k(\mathrm{A}_1) \to k(\mathrm{A}_2)$, there exists at most one A-homomorphism (necessarily local) $\psi : \mathrm{A}_1 \to \mathrm{A}_2$ such that γ is the homomorphism $\overline{\psi}$ deduced from ψ by passage to quotients.*

lbbel=ii) *For every pair of local A-homomorphisms $\beta_1 : \mathrm{A}_1 \to \mathrm{K}$, $\beta_2 : \mathrm{A}_2 \to \mathrm{K}$ there exists a local A-algebra A_3 that is essentially étale, two A-homomorphisms $\phi_1 : \mathrm{A}_1 \to \mathrm{A}_3$, $\phi_2 : \mathrm{A}_2 \to \mathrm{A}_3$ and an A-homomorphism $\beta : \mathrm{A}_3 \to \mathrm{K}$ such that $\beta_1 = \beta \circ \phi_1$ and $\beta_2 = \beta \circ \phi_2$.*

Proof. With the notations of (18.6.3), the homomorphisms β_1, β_2 in (ii) correspond to two S-morphisms $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_1$, $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_2$, with respective images x_1, x_2; one thus deduces a well-determined $\mathrm{Spec}(\mathrm{K}) \to \mathrm{X}_3$ (**I**, 3.2.1) whose image x_3 lies above x_1 and x_2; the A-algebra $\mathrm{A}_3 = \mathscr{O}_{\mathrm{X}_3,x_3}$ and the homomorphism $\mathrm{A}_3 \to \mathrm{K}$ corresponding to these data then satisfy the conditions of (ii). On the other hand, the data of γ in (i) corresponds to an S-morphism $\mathrm{Spec}(k(x_2)) \to \mathrm{Spec}(k(x_1))$ or also to a point x_3' of X_3 above x_2; the uniqueness of ψ follows from this and from (17.4.9). □

(18.8.6). Let A be a local ring, $\mathfrak{m}$ its maximal ideal, k its residue field, Ω a separable closure of k. Consider the set $\mathfrak{E}$ of A-algebras essentially étale defined in (18.6.1), and denote by L the set of A-*homomorphisms* $\mathrm{A}' \to \Omega$, where A' ranges over $\mathfrak{E}$; one will note that such a homomorphism factorises as $\mathrm{A} \to k(\mathrm{A}') \xrightarrow{\omega_\lambda} \Omega$, where $\omega_\lambda : k(\mathrm{A}') \to \Omega$ is a k-homomorphism (since $\mathfrak{m}\mathrm{A}'$ is the maximal ideal of A'); conversely the data of a homomorphism ω_λ uniquely determines a homomorphism $\lambda \in \mathrm{L}$. For every $\lambda \in \mathrm{L}$, we denote by A_λ the A-algebra essentially étale belonging to $\mathfrak{E}$ on which λ is defined. The set L is *preordered* by the relation "there exists a homomorphism $\phi_{\mu\lambda} : \mathrm{A}_\lambda \to \mathrm{A}_\mu$ such that $\lambda = \mu \circ \phi_{\mu\lambda}$", and it follows from (18.8.5), (i) that this homomorphism $\phi_{\mu\lambda}$ is *unique*. Furthermore, L is *filtered increasing* for the preceding preorder relation (which one also denotes $\lambda \leqslant \mu$), by virtue of (18.8.5), (ii). It is clear that $(\mathrm{A}_\lambda, \phi_{\mu\lambda})$ is an *inductive system* of local A-algebras, **IV-4** | 146
the $\phi_{\mu\lambda}$ being local homomorphisms; furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is essentially étale. If $\overline{\phi}_{\mu\lambda} : k(\mathrm{A}_\lambda) \to k(\mathrm{A}_\mu)$ is the k-homomorphism deduced from $\phi_{\mu\lambda}$ by passage to quotients, $(k(\mathrm{A}_\lambda), \overline{\phi}_{\mu\lambda})$ is an inductive system of separable algebraic extensions of k, and the $\omega_\lambda : k(\mathrm{A}_\lambda) \to \Omega$ form an inductive system of homomorphisms of k. Furthermore, the inductive limit

$$\omega : \varinjlim k(\mathrm{A}_\lambda) \longrightarrow \Omega \tag{18.8.6.1}$$

is a k-*isomorphism*. It suffices indeed to prove that for every finite separable extension k' of k, there exists an A-algebra A' such that k' is the residue field of A' and that the homomorphism $k \to k'$ is deduced from $\mathrm{A} \to \mathrm{A}'$ by passage to quotients. But this follows from (18.1.1) applied to étale morphisms, since $\mathrm{Spec}(\mathrm{A})$ is the unique neighbourhood of its closed point.

Note now that if $\mathfrak{m}_\lambda$ is the maximal ideal of A_λ, it follows that $\mathfrak{m}_\mu = \mathfrak{m}_\lambda \mathrm{A}_\mu$ and that A_μ is a flat A_λ-module. Hence it follows from (**0**, 10.3.1.3) that the ring $\varinjlim \mathrm{A}_\lambda$ is *local*, that the homomorphism $w : \mathrm{A} \to \varinjlim \mathrm{A}_\lambda$ is local, and that one has a k-isomorphism

$$\varinjlim k(\mathrm{A}_\lambda) \xrightarrow{\sim} k(\varinjlim \mathrm{A}_\lambda); \tag{18.8.6.2}$$

identifying these two fields, one deduces a k-isomorphism

$$\omega^{-1} : \Omega \xrightarrow{\sim} k(\varinjlim \mathrm{A}_\lambda).$$

Definition (18.8.7). — Let A be a local ring, $k = k(\mathrm{A})$ its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k; the *strict Henselization* of A relative to i, denoted ${}^{hs}\mathrm{A}_{(i)}$ (or ${}^{hs}\mathrm{A}$ when this causes no confusion), is the inductive limit of the inductive system $(\mathrm{A}_\lambda, \phi_{\mu\lambda})$ defined in (18.8.6).

As one has a k-isomorphism $\omega^{-1} : \Omega \xrightarrow{\sim} k({}^{hs}\mathrm{A}_{(i)})$, Ω is endowed canonically with the structure of a ${}^{hs}\mathrm{A}_{(i)}$-algebra by the local homomorphism

$${}^{hs}\mathrm{A}_{(i)} \longrightarrow k({}^{hs}\mathrm{A}_{(i)}) \xrightarrow{\omega} \Omega.$$

As in (18.6.5) one sees that the definition given in (18.8.7) does not depend on the choice of $\mathfrak{E}$; we shall moreover see below (18.8.8), (i) and (ii) that ${}^{hs}\mathrm{A}_{(i)}$ is an object representing a functor entirely defined by the data of A and i (**0**, 8.1.8), and is hence defined as such up to an isomorphism.

Proposition (18.8.8). — *Let* A *be a local ring, k its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k, ${}^{hs}\mathrm{A}_{(i)}$ the strict Henselization of* A *corresponding to i.*

label=i) *${}^{hs}\mathrm{A}_{(i)}$ is a strictly local ring (18.8.2) and the structural homomorphism $w : \mathrm{A} \to {}^{hs}\mathrm{A}_{(i)}$ is local.*

lbbel=ii) *For every local homomorphism $u : \mathrm{A} \to \mathrm{B}$, where* B *is a strictly local ring, and every k-homomorphism $\psi : \Omega \to k(\mathrm{B})$, there exists one and only one A-homomorphism $v : {}^{hs}\mathrm{A}_{(i)} \to \mathrm{B}$ such that ψ factorises as $\Omega \xrightarrow{\omega^{-1}} k({}^{hs}\mathrm{A}_{(i)}) \xrightarrow{\overline{v}} k(\mathrm{B})$, where $\overline{v}$ is deduced from v by passage to quotients.*

lcbel=iii) *${}^{hs}\mathrm{A}_{(i)}$ is a faithfully flat A-module, and if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}.{}^{hs}\mathrm{A}_{(i)}$ is the maximal ideal of ${}^{hs}\mathrm{A}_{(i)}$, and $k({}^{hs}\mathrm{A}_{(i)})$ is a separable closure of k.*

ldbel=iv) *For ${}^{hs}\mathrm{A}_{(i)}$ to be Noetherian, it is necessary and sufficient that A be so.*

lebel=v) *If A is strictly local (hence $\Omega = k$), one has ${}^{hs}\mathrm{A} \cong \mathrm{A}$.*

lfbel=vi) *If $i' : k \to \Omega'$ is a second homomorphism from k into a separable closure of k, for every k-isomorphism $\sigma : \Omega \xrightarrow{\sim} \Omega'$, the corresponding A-homomorphism (by (ii)) $v_\sigma : {}^{hs}\mathrm{A}_{(i)} \to {}^{hs}\mathrm{A}_{(i')}$ is an isomorphism.*

Proof. We have already seen above that ${}^{hs}\mathrm{A}_{(i)}$ is a local ring and that w is a homomorphism local; the fact that ${}^{hs}\mathrm{A}_{(i)}$ is Henselian (and hence strictly local) is demonstrated as in (18.6.6). The assertions of (iii) also follow from (**0**, 10.3.1.3); the necessity of the condition (iv); the necessity of condition results from (**0**, 6.5.2). **IV-4** | 147

To prove (ii), remark, with the notations of (18.8.6), that for every $\lambda \in \mathrm{L}$, there exists one and only one local homomorphism $v_\lambda : \mathrm{A}_\lambda \to \mathrm{B}$, such that the composite $k(\mathrm{A}_\lambda) \xrightarrow{\omega_\lambda} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from v_λ by passage to quotients, by virtue of (18.8.4) and the hypothesis that B is strictly local. The uniqueness of v_λ implies moreover that the v_λ form an inductive system, whence the existence of the homomorphism v satisfying the conditions of (ii); uniqueness results from (vi) asserting the composites $v_\alpha \circ v_{\alpha-1}$ and $v_{\alpha-1} \circ v_\alpha$. Finally, if A is strictly local, and $\mathrm{A}' = \mathrm{B}_{\mathfrak{n}}$ is an A-algebra local and essentially étale, where B is an A-algebra étale, it follows from (18.8.1) that there exists a $\mathrm{Spec}(\mathrm{A})$-section of $\mathrm{Spec}(\mathrm{B})$ taking the value $\mathfrak{n}$ at the closed point of $\mathrm{Spec}(\mathrm{A})$, hence by (17.4.1) that the homomorphism $\mathrm{A} \to \mathrm{A}'$ is bijective, which proves (v). □

(18.8.8.1). Let $\mathscr{C}$ be the category of strictly local rings, with local homomorphisms as morphisms; for every $\mathrm{B} \in \mathscr{C}$, denote by $\mathrm{F}(\mathrm{B})$ the set of pairs (u, ψ) consisting of a local homomorphism $u : \mathrm{A} \to \mathrm{B}$ and a k-isomorphism $\psi : k = k(\mathrm{A}) = k \xrightarrow{\psi} \Omega \xrightarrow{\psi} k(\mathrm{B})$ such that the composite $k(\mathrm{A}) = k \xrightarrow{i} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from u by passage to quotients. One can say that the object ${}^{hs}\mathrm{A}_{(i)}$ *represents* the covariant functor $\mathrm{F} : \mathscr{C} \to \mathtt{Ens}$ (**0**, 8.1.11).

One will remark that by virtue of (18.8.8), (vi), the strict Henselizations ${}^{hs}\mathrm{A}_{(i)}$ are all A-isomorphic (for the various k-homomorphisms i of k into separable closures of k), but for given i and i', there is in general an infinity of A-isomorphisms ${}^{hs}\mathrm{A}_{(i)} \xrightarrow{\sim} {}^{hs}\mathrm{A}_{(i')}$. To be precise, the group of A-automorphisms of ${}^{hs}\mathrm{A}_{(i)}$ is isomorphic to the Galois group of Ω over k.

(18.8.9). Let now A be a *semi-local* ring, $\mathfrak{m}_j$ $(1 \leqslant j \leqslant r)$ its maximal ideals, and for every j, consider a homomorphism $i_j : k(\mathrm{A}_{\mathfrak{m}_j}) \to \Omega_j$ of the residue field into a separable closure. The *strict Henselization* of A relative to the i_j is the product $\prod_{j=1}^r({}^{hs}(\mathrm{A}_{\mathfrak{m}_j}))$, denoted ${}^{hs}\mathrm{A}$ (when there is no risk of confusion). It is therefore a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_j.{}^{hs}\mathrm{A}$ and the radical $\mathfrak{r}.{}^{hs}\mathrm{A}$ (if $\mathfrak{r}$ denotes the radical of A). Finally, the universal property (18.8.8), (ii) still holds when replacing "local homomorphism" by "semi-local homomorphism" (18.6.7), and replacing Ω by $\prod_j \Omega_j$. **IV-4** | 148

Proposition (18.8.10). — *Let* A *be a semi-local ring,* B *a finite A-algebra. Let $\mathfrak{n}_j$ $(1 \leqslant j \leqslant r)$ be the maximal ideals of* B*. Then there is, for every j, a maximal ideal $\mathfrak{q}_j$ of $\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ above $\mathfrak{n}_j$, such that ${}^{hs}\mathrm{B}$ is isomorphic to the composed direct product of local rings $(\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_j}$ $(1 \leqslant j \leqslant r)$.*

Proof. One can evidently restrict to the case where A is local, replacing A by $\mathrm{A}_{\mathfrak{m}_j}$. On the other hand, let $\mathrm{C} = \mathrm{B}_{\mathfrak{n}_j}$; by virtue of the definition of ${}^{hs}\mathrm{B}$ (18.8.1), everything amounts to seeing that ${}^{hs}\mathrm{C}$ is

C-isomorphic to $(\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}}$, where $\mathfrak{q}$ is one of the maximal ideals of $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$. Let k and K be the residue fields of A and C, Ω the separable closure of k, and let us suppose K and Ω contained in the same closure of k; as K is a finite extension of k, it is immediate that $\mathrm{K} \otimes_k \Omega$ is a composed direct product of a finite number of fields, all isomorphic to the separable closure $\mathrm{K}\Omega$ of K. On the other hand, the ring ${}^{hs}\mathrm{A}$ being Henselian, $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ is a composed direct product of local rings $(\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_i}$, where $\mathfrak{q}_i$ $(1 \leqslant i \leqslant s)$ ranges over the maximal ideals of $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$, and these local rings are Henselian (18.5.10) and have $\mathrm{K}\Omega$ for residue field, hence they are strictly local (18.8.1). Let $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ and $\mathrm{C}'_i = \mathrm{C}'_{\mathfrak{q}_i}$, which is therefore a quotient ring of C'; let $f : \mathrm{C} \to \mathrm{C}'$, $g : {}^{hs}\mathrm{A} \to \mathrm{C}'$, $\phi_i : \mathrm{C}' \to \mathrm{C}'_i$ be the canonical maps.

$$\begin{tikzcd} & \mathrm{C}'_i \arrow[r, "w"] & {}^{hs}\mathrm{C} \\ \mathrm{C} \arrow[r, "f"] & \mathrm{C}' \arrow[u, "\phi_i"] & \\ \mathrm{A} \arrow[u] \arrow[r] & {}^{hs}\mathrm{A} \arrow[u, "g"'] & \end{tikzcd}$$

Let $v : \mathrm{C} \to {}^{hs}\mathrm{C}$ be the canonical homomorphism local; there exists a unique $u : {}^{hs}\mathrm{A} \to {}^{hs}\mathrm{C}$ which, by passage to quotients, gives the canonical injection $\Omega \to \mathrm{K}\Omega$ and such that the composite $\mathrm{A} \to {}^{hs}\mathrm{A} \xrightarrow{u} {}^{hs}\mathrm{C}$ is equal to the composite $\mathrm{A} \to {}^{hs}\mathrm{A} \xrightarrow{u} {}^{hs}\mathrm{C}$ (18.8.8); hence there exists a unique local homomorphism $w_0 : \mathrm{C}' \to {}^{hs}\mathrm{C}$ such that $w_0 \circ g = u$, $w_0 \circ f = v$, and since ${}^{hs}\mathrm{C}$ is a local ring, this homomorphism factorises as $\mathrm{C}' \to \mathrm{C}'_i \xrightarrow{w} {}^{hs}\mathrm{C}$ for a well-determined index i, w being a local homomorphism. On the other hand, the ring C'_i being strictly local, there exists a local homomorphism $w' : {}^{hs}\mathrm{C} \to \mathrm{C}'_i$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$ and which is such that $w' \circ v = \phi_i \circ f$ (18.8.8). One deduces first that $w \circ w'$ is an endomorphism of ${}^{hs}\mathrm{C}$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$, hence is the identity **IV-4 | 149**
(18.8.8). On the other hand, $w' \circ w \circ \phi_i \circ f = w' \circ v = \phi_i \circ f$ and $w' \circ w \circ \phi_i \circ g = w' \circ u = \phi_i \circ g$; whence $w' \circ w \circ \phi_i = \phi_i$, which is to say $w' \circ w$ is the identity automorphism of C'_i. □

Remark (18.8.11). — The proof of (18.8.10) uses the fact that B is a *finite* A-algebra, namely to establish that $\mathrm{K} \otimes_k \Omega$ is a composed direct product of a finite number of fields. However, this last property is still satisfied when one supposes that B is an *entire* semi-local A-algebra whose maximal ideals $\mathfrak{n}_j$ have residue fields of finite separable degree $[k(\mathfrak{n}_j) : k]_s$ finite.

Proposition (18.8.12). — *Let* A *be a semi-local ring.*

label=i) *For* ${}^{hs}\mathrm{A}$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*

lbbel=ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^{hs}\mathrm{A})_{\mathfrak{p}}/\mathfrak{p}.({}^{hs}\mathrm{A})_{\mathfrak{p}}$ *is a composed direct product of a finite number of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^{hs}\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ *are* geometrically regular *and are* discrete *spaces).*

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. □

Corollary (18.8.13). — *Let* A *be a local ring. For* ${}^{hs}\mathrm{A}$ *to possess one of the following properties:*

label=a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*

lbbel=b) *satisfying the condition* (S_n) (5.7.3)*;*

lcbel=c) *being regular;*

ldbel=d) *satisfying the condition* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Corollary (18.8.14). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^{hs}\mathrm{A}$ *to belong to* $\mathrm{Ass}({}^{hs}\mathrm{A})$*, it is necessary and sufficient that* $\mathfrak{p} \cap \mathrm{A}$ *belong to* $\mathrm{Ass}(\mathrm{A})$.

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. □

Proposition (18.8.15). — *Let* A *be a local ring; the following properties are equivalent:*

label=a) *A is geometrically unibranch (resp. reduced and geometrically unibranch).*
lbbel=b) *For every A-algebra* A' *that is local and essentially étale,* $\mathrm{Spec}(A')$ *is irreducible (resp. integral).*
lcbel=c) $\mathrm{Spec}({}^{hs}A)$ *is irreducible (resp. integral).*

Proof. One reduces to the case where A is reduced, using the relation ${}^{hs}(A_{\mathrm{red}}) = ({}^{hs}A) \otimes_A A_{\mathrm{red}}$ (18.8.11), and the equivalence of *b*) and *c*) is proved as in (18.6.12); the same holds, the fact that *a*) implies *c*), taking into account the definition of a geometrically unibranch integral local ring (18.8.11). Finally, to prove that *c*) implies *a*), using the same notations as in (18.6.12), it suffices to see that ${}^{hs}B$ is integral. But, by (18.8.10), ${}^{hs}B$ is a localised ring of the semi-local ring $B \otimes_A ({}^{hs}A)$, which identifies by flatness with a subring of L, hence is integral; it follows that ${}^{hs}B$ is integral, hence also ${}^{hs}B$, which completes the proof. □

Corollary (18.8.16). — *Let* A *be a local Henselian ring (resp. strictly local). For A to be unibranch (resp. geometrically unibranch), it is necessary and sufficient that* $\mathrm{Spec}(A)$ *be irreducible.*

Proposition (18.8.17). — *Let* A *be a local Noetherian ring. If A is universally catenary, the same holds for* ${}^{hs}A$.

Proof. The proof is the same as that of (18.7.5). □

Proposition (18.8.18). — label=i) *Every filtered inductive limit of strictly local rings (where the transition homomorphisms are local) is a strictly local ring.*
lbbel=ii) *The functor* $A \rightsquigarrow {}^{hs}A$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
lcbel=iii) *Every strictly local ring A is an inductive limit of a filtered inductive system of Noetherian strictly local rings (the transition homomorphisms being local).*

Proof. The proofs are modelled on those of (18.6.14). □

IV-4 | 150

18.9. Formal fibres of Noetherian Henselian rings.

Theorem (18.9.1). — *Let* A *be a local Noetherian Henselian ring, whose formal fibres are geometrically normal. Then all the fibres of* A *are geometrically connected (hence geometrically integral).*

Proof. The two assertions stated are in fact equivalent, a prescheme that is locally Noetherian connected and normal being integral. As every finite integral A-algebra is a Henselian local ring (18.5.9) and (18.5.10), it follows from (7.3.16.2) that one is reduced to proving that if one further supposes A *integral*, then the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point of $\mathrm{Spec}(A)$ is *integral*, which follows from the □

Corollary (18.9.2). — *Let* A *be a local Noetherian Henselian integral ring, whose formal fibres are geometrically normal. Then* $\widehat{A}$ *is integral.*

Proof. Indeed, it follows from (18.8.16) that A is unibranch, hence the theorem follows from the hypothesis made on the formal fibres of A and from (7.6.3). □

Corollary (18.9.3). — *Under the hypotheses of (18.9.2), the field of fractions L of* $\widehat{A}$ *is algebraically closed in L.*

Proof. This follows from the fact that the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point is geometrically integral and from (4.3.2) and (4.3.5). □

Corollary (18.9.4). — *Let* A *be a local Noetherian ring, Henselian, and whose formal fibres are geometrically normal (for example a Noetherian Henselian local ring that is excellent), and let* $A' = \widehat{A}$*. Let* X *be an A-prescheme,* $X' = X \otimes_A A'$, $g : X' \to X$ *the canonical projection. Then the map* $U \mapsto g^{-1}(U)$ *is a bijection from the set of subsets of* X *that are both open and closed to the set of subsets of* X' *that are both open and closed.*

Proof. As g is a faithfully flat and quasi-compact morphism (2.3.12), one knows that the topology of X is the *quotient* of that of X′ by the equivalence relation defined by f. It therefore amounts to seeing that an open and closed subset U′ of X′ is *saturated* for this relation. Now, as the morphism $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ has its fibres geometrically connected (18.9.1), the fibres of g are connected, whence our assertion immediately follows.

In particular, if X is locally connected (which will be the case if X is locally Noetherian), then, for every connected component U of X (which is open and closed), $g^{-1}(U)$ is connected (Bourbaki, *Top. gén.*, chap. I^er^, 3^e^ éd., § 11, no 3, prop. 7). If one denotes by $\pi_0(X)$ the set of connected components of X, the map $\pi_0(X') \to \pi_0(X)$ canonically deduced from g is therefore *bijective*. ☐

Corollary (18.9.5). — *Under the hypotheses of* (18.9.4), *the functor*

(18.9.5.1) $$Z \longmapsto Z \otimes_A A'$$

from the category of étale preschemes over X, *to that of étale preschemes over* X′, *is fully faithful.*

Proof. Let Z_1, Z_2 be two étale preschemes over X, and set $Z'_i = Z_i \otimes_A A'$ ($i = 1, 2$); it is a matter of proving that every X′-morphism $Z'_1 \to Z'_2$ comes, by base change, from a unique X-morphism $Z_1 \to Z_2$. Suppose first that Z_2 is *separated* over X; then, as $\operatorname{Hom}_X(Z_1, Z_2)$ identifies with $\Gamma((Z_1 \times_X Z_2)/Z_1)$, and that $Z_1 \times_X Z_2$ is étale and separated over Z_1, by virtue of (17.9.3), to the set of open and closed subsets U of $Z_1 \times_X Z_2$ such that the restriction to U of the projection $p : Z_1 \times_X Z_2 \to Z_1$ is a surjective and radicial morphism. The assertion thus follows from (18.9.4) and the fact that $Z'_1 \times_{X'} Z'_2 = (Z_1 \times_X Z_2) \otimes_A A'$. IV-4 | 151

Let us pass to the general case: it will suffice to prove that the graph Γ of an X′-morphism $Z'_1 \to Z'_2$ in $Z'_1 \times_{X'} Z'_2$ is of the form $\Gamma \otimes_A A'$, where Γ is induced on an *open* subset of $Z = Z_1 \times_X Z_2$. Indeed, the restriction to Γ of the projection $p : Z \to Z_1$ is an isomorphism; by base change $A \to A'$, this restriction becomes the isomorphism restriction $p' : Z' \to Z'_1$ (with $Z' = Z \otimes_A A'$), and it suffices to apply (2.7.1), (viii); the conclusion then follows from the characterisation of graphs (**I**, 5.3).

If $q : Z' \to Z$ is the canonical projection, to prove Z′ is of the form $q^{-1}(\Gamma)$, where Γ is open in Z, it suffices, since q is a faithfully flat and quasi-compact morphism, to prove that $q^{-1}(\Gamma) = q^{-1}(U)$ for an open $U \subset Z$. Let $Z'' = Z' \times_Z Z' = Z \times_X X''$ where $X'' = X' \times_X X'$, and let $q_1 : Z'' \to Z'$, $q_2 : Z'' \to Z'$ be the canonical projections. Applying (4.5.19.1), it suffices to show that $q_1^{-1}(\Gamma') = q_2^{-1}(\Gamma')$. But this is a property which is true if and only if it is true after each base change $\operatorname{Spec}(k(x)) \to X$, where x ranges over X. In other words, it suffices to prove the corollary when X is the *spectrum of a field*; but as every X-prescheme that is étale is automatically *separated* over X (17.6.2), c'), one is reduced to the case considered at the beginning of the proof. ☐

Remarks (18.9.6). — label=*i)* It is possible that, if the residue field of A is of characteristic 0, the functor (18.9.5.1) induces even an *equivalence* of the category of étale coverings of X and the category of étale coverings of X′. One can indeed prove this when A is *excellent*, using the resolution of singularities of Hironaka (M. Artin). It is plausible that an analogous statement is still true without restriction on the characteristic of the residue field, provided one restricts to the étale coverings whose "principal Galois groups" have order prime to the residual characteristic (cf. [41]).

lbbel=*ii)* One does not know whether the formal fibres of A are geometrically connected (or even geometrically irreducible) when A is any Henselian local ring. It would suffice that, for every Noetherian Henselian local ring, Henselian and *integral* A, $\operatorname{Spec}(\widehat{A})$ be irreducible. One does not know if this is always the case.

lcbel=*iii)* One does not know whether, when A is an excellent local ring, its strict Henselization ${}^{hs}A$ has geometrically regular formal fibres. One can indeed elucidate this question, one can reduce to the case where $A = k[[T_1, \ldots, T_n]]$, k being a field of characteristic $p > 0$. Taking account of (18.8.17), the question is whether the formal fibres of ${}^{hs}A$ are geometrically regular. The answer is affirmative when $n = 1$, but is not known for $n = 2$.

The connectedness assertion made in (18.9.1) generalises in the following way:

Theorem (18.9.7). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism,* $f : \operatorname{Spec}(B) = X \to \operatorname{Spec}(A) = Y$ *the corresponding morphism. Suppose the following conditions satisfied:* IV-4 | 152

label=i) *f is flat and all the fibres $X_y = f^{-1}(y)$ ($y \in Y$) are geometrically reduced* (4.6.2).
lbbel=ii) *Either the ring* A *is geometrically unibranch* (**0**, 23.2.1), *or else the ring A is unibranch* (**0**, 23.2.1) *and $k(B)$ is a primary extension* (4.3.1) *of $k(A)$.*

Then, if η is the generic point of Y, *X_η is connected, and for every closed subset $T \subset f^{-1}(y)$, $X - T$ is connected.*

Proof. The two conclusions stated are in fact equivalent; indeed, the hypothesis that $f(T)$ is rare in Y, T does not contain any maximal point of X (2.3.4), hence is rare in X; as X is a local scheme, hence connected, to prove that $X - T$ is connected, it suffices, by virtue of (15.5.6.1), to show that, for every $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \mathrm{Spec}(B)$, B being a local Noetherian ring and $A \to B$ the corresponding local homomorphism; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), we thus have the commutative diagram

$$\begin{tikzcd} X & X_1 \arrow[l, "g_1"'] \\ Y \arrow[u, "f"] & Y_1 \arrow[l, "g"] \arrow[u, "f_1"'] \end{tikzcd}$$

□

Remark (18.9.7.1). — When the ring A is *Japanese*, one can, in the case preceding (18.9.7.2), take $A'' = A'$ by definition (**0**, 23.1.1); one sees therefore that in this case one would even be able to prove the theorem when A is *integrally closed*.

(18.9.7.2). Suppose therefore now that Y is integral and geometrically unibranch. Note that if T is a closed subset of X such that $f(T)$ is rare in Y, T contains no maximal point of X (2.3.4), hence is rare in X; as X is a local scheme, hence connected, to prove that $X - T$ is connected, it suffices, by virtue of (15.5.6.1), to show that for every $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \mathrm{Spec}(B)$, B being a local Noetherian ring and the homomorphism $A \to B$ being local; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*.

Lemma (18.9.7.3). — *Let* A, B *be two local Noetherian rings,* $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\rho : A \to B$ *a local homomorphism,* $f : X \to Y$ *the corresponding morphism. Suppose the following conditions satisfied:*

label=i) *f is flat and all the fibres X_y ($y \in Y$) are geometrically reduced.*
lbbel=ii) *A is integral, geometrically unibranch and* $\dim(A) \geqslant 1$.

Then, if b is the closed point of X, $X - \{b\}$ *is connected.*

Proof. Note first that by virtue of the theorem of Hartshorne (5.10.7), $X - \{b\}$ is connected if **IV-4** | 153
$\mathrm{prof}(B) \geqslant 2$. Now on has (6.3.1)

$$\mathrm{prof}(B) = \mathrm{prof}(A) + \mathrm{prof}(B \otimes_A k) \tag{18.9.7.4}$$

where k denotes the residue field of A. On the other hand, since A is integral and $\dim(A) \geqslant 1$, one has $\mathrm{prof}(A) \geqslant 1$, hence $\mathrm{prof}(B) \geqslant 2$ *unless* $\mathrm{prof}(B \otimes_A k) = 0$; moreover, $B \otimes_A k$ is reduced by (i), hence the relation $\mathrm{prof}(B \otimes_A k) = 0$ means that $B \otimes_A k$ is a *field* (**0**, 16.4.7). We shall henceforth suppose that this last condition is satisfied. We shall begin by treating a case where the proof is very simple.

A) *Case where A is integrally closed.* Then, if $\dim(A) \geqslant 2$, one has $\mathrm{prof}(A) \geqslant 2$ (**0**, 16.5.1), hence $\mathrm{prof}(B) \geqslant 2$. One therefore only has to consider the case where $\dim(A) = 1$ and where $B \otimes_A k$ is a field; A is then a discrete valuation ring, hence regular, and since $B \otimes_A k$ is a field, B is also a discrete valuation ring (**0**, 17.3.3); but in addition (6.1.1.1), one has $\dim(B) = \dim(A) + \dim(B \otimes_A k) = \dim(A) = 1$, hence B is also a discrete valuation ring, hence $X - \{b\}$ is reduced to a single point, whence the lemma (18.9.7.3) in this case.

One will observe that this proves the statement of the theorem (18.9.7) when one supposes the ring A *Japanese*, for by virtue of the remark (18.9.7.1), one is reduced to the case where A is *integrally* *closed*, and then, in the reduction (18.9.7.3), the ring $A_1 = \mathscr{O}_{X,y}$ is also integrally closed and one can apply the result just proved.

B) *Case where* $\dim(A) \geqslant 2$. The result of (18.9.7.3) will follow from the two following lemmas. □

Lemma (18.9.7.5). — *Let* A *be a semi-local Noetherian ring, reduced,* A' *its integral closure in its total ring of fractions,* $X = \operatorname{Spec}(A)$, $X' = \operatorname{Spec}(A')$; *let S be the set of closed points of* X *and* $U = X - S$. *Suppose that for every point* $x' \in X'$ *above a point of* U, $\dim(\mathscr{O}_{X',x'}) \geqslant 2$; *then* $A^{(1)} = \Gamma(U, \mathscr{O}_X)$ *is entire over* A *and is a sub-A-algebra isomorphic to* A'.

Proof. Recall that the $\mathfrak{p}_i$ $(1 \leqslant i \leqslant m)$ are the minimal prime ideals of A, A' is the composed direct product of the integral closures $A_i = A/\mathfrak{p}_i$, so that X' is the sum of the schemes $X'_i = \operatorname{Spec}(A'_i)$ $(1 \leqslant i \leqslant m)$. If X_0 is the sum of the schemes $X_i = \operatorname{Spec}(A_i)$ $(1 \leqslant i \leqslant m)$ and U_0 the inverse image of U in X_0, the canonical homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U_0, \mathscr{O}_{X_0})$ is injective since X is reduced and the morphism $X' \to X$ surjective. As $\Gamma(U_0, \mathscr{O}_{X_0})$ is the composed direct product of the $\Gamma(U_i, \mathscr{O}_{X_i})$ (where U_i is the inverse image of U in X_i), it suffices to prove that for each i, $\Gamma(U_i, \mathscr{O}_{X_i})$ is isomorphic to a sub-A_i-algebra of A'_i. As X is reduced and the morphism $X' \to X$ surjective, the homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U', \mathscr{O}_{X'})$ (where U' is the inverse image of U in X') is injective, and everything amounts to verifying that the canonical homomorphism $A' = \Gamma(X', \mathscr{O}_{X'}) \to \Gamma(U', \mathscr{O}_{X'})$ is *bijective*. Now (**I**, 8.2.1.1), $\Gamma(U', \mathscr{O}_{X'})$ is the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over U', and by virtue of the hypothesis, among these local rings are included all those of height 1. But the reasoning of (**0**, 23.2.7) applies also to a semi-local Noetherian integral ring, and is therefore a *semi-local Krull ring*, and is therefore the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A' (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4); *a fortiori*, A' is the intersection of the $A'_{\mathfrak{p}'}$ for $\mathfrak{p}' \in U'$, which completes the proof of the lemma. □ **IV-4 | 154**

Remark (18.9.7.6). — We know (**0**, 23.2.5) that there exists a finite sub-A-algebra A'' of K, the field of fractions of A, such that the morphism $\operatorname{Spec}(A') \to \operatorname{Spec}(A'')$ is *radicial*; as this morphism is also entire and dominant, hence surjective (**II**, 6.1.10), it is a *homeomorphism* (2.4.5). By virtue of the geometric interpretation of $\dim(\mathscr{O}_{X',x'})$ (5.1.2), one has therefore, for every point x'' of $X'' = \operatorname{Spec}(A'')$ above a point $x \in X$, $\dim(\mathscr{O}_{X'',x''}) = \dim(\mathscr{O}_{X',x'})$, whence $\dim(\mathscr{O}_{X'',x'}) \geqslant 2$ if x' is the unique point of X' above x. The same conclusion holds if $\mathscr{O}_{X,x}$ is *geometrically unibranch*, for the fibre of the morphism $X' \to X$ at the point x is then reduced to a single point, and $\mathscr{O}_{X',x'}$ is an *entire* $\mathscr{O}_{X,x}$-algebra; it suffices then to apply (**0**, 16.1.5). **IV-4 | 155**

Lemma (18.9.7.7). — *Let* A *be a local Noetherian ring, of dimension* $\geqslant 2$, *integral and geometrically unibranch,* $Y = \operatorname{Spec}(A)$, y *the closed point of* Y. *Let* X *be a locally Noetherian prescheme and* $f : X \to Y$ *a flat morphism. Then, for every closed subset* $T \subset f^{-1}(y)$, $X - T$ *is connected.*

Proof. The same reasoning as at the beginning of (18.9.7.2) reduces to proving that, for every point $x \in f^{-1}(y)$, $\operatorname{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \operatorname{Spec}(B)$, B being a local Noetherian ring and the homomorphism $A \to B$ being local; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), and setting $A^{(1)} = \Gamma(U, \mathscr{O}_X)$, $X_1 = X \times_Y Y_1 = \operatorname{Spec}(B \otimes_A A^{(1)})$, we have the commutative diagram **IV-4 | 155**

$$\begin{tikzcd}
X & X_1 \arrow[l, "g_1"'] \\
Y \arrow[u, "f"] & Y_1 \arrow[l, "g"] \arrow[u, "f_1"']
\end{tikzcd}$$

By virtue of (2.3.1) and of the definition of $A^{(1)}$, one has a canonical isomorphism $\Gamma(V, \mathscr{O}_X) \xrightarrow{\sim} B \otimes_A A^{(1)}$, and it suffices therefore to prove that this last ring is *local*, such a ring being a product of two nonzero rings (**III**, 7.8.6.1). But we have seen (18.9.7.5) that $A^{(1)}$ is a sub-A-algebra of the integral closure A' of A, hence $\dim(A) \geqslant 2$. The hypothesis that A is geometrically unibranch implies that the morphism $\operatorname{Spec}(A') \to \operatorname{Spec}(A)$ is radicial at the point y (6.15.3); *a fortiori* g_1 is radicial at the closed point of X; furthermore g_1 is an entire morphism, hence the closed points of X_1 are above x, hence there is only one closed point of X_1, which completes the proof of (18.9.7.7).

It is clear that (18.9.7.7) proves (18.9.7.3) when $\dim(A) \geqslant 2$ (even without supposing the fibres geometrically reduced).

C) *Case where* $\dim(A) = 1$. One has seen at the beginning of the proof of (18.9.7.3) that one can suppose $\dim(B \otimes_A k) = 0$, and by flatness (6.1.1.1), one therefore has

$$\dim(B) = \dim(A) + \dim(B \otimes_A k) = 1.$$

Since A is integral, Y consists of two points, the generic point η and the closed point a. Furthermore, since f is flat, every irreducible component of X dominates Y (2.3.4), hence $X - \{b\}$, which is the set of maximal points ξ_i of X, is equal to the underlying set of $f^{-1}(\eta)$, and the fibre $f^{-1}(\eta)$ is the sum of the $\mathrm{Spec}(k(\xi_i))$. The hypothesis on the X_y and the fact that these fibres are of dimension 0, show that they are *geometrically regular* (6.7.6), and *a fortiori* geometrically normal, in other words f is a *normal morphism*. But since A is integral and geometrically unibranch, it follows then from (6.15.10) that B is also integral and geometrically unibranch, hence $f^{-1}(\eta) = X - \{b\}$ is reduced to a single point, and *a fortiori* connected; this concludes the proof of (18.9.7.3) and of (18.9.7). ☐ **IV-4** | 156

Remark (18.9.7.8). — In case C) of the proof of (18.9.7.3), one can avoid making appeal to the delicate result (6.15.10) by reasoning as follows: since A is integral and unibranch of dimension 1, it follows from the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5) that its integral closure A' is a *Noetherian* ring. The same reasoning as at the beginning of the proof of (18.9.7) shows that $B' = B \otimes_A A'$ is a local ring; furthermore, by flatness, B' is contained in the total ring of fractions R of the reduced B (3.3.5). Now, the theorem of Krull-Akizuki (Bourbaki, *loc. cit.*) also applies to a local Noetherian *reduced* ring of dimension 1, and shows that for every such ring B, everything contained between B and its total ring of fractions is *Noetherian*. The ring B' being a local Noetherian ring and the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A')$ being normal, B' is an integral ring and integrally closed (6.5.4), and one concludes as at the beginning of the proof of (18.9.7).

Corollary (18.9.8). — *With the notations of* (18.9.7), *suppose verified the condition (i) of* (18.9.7) *and one of the two conditions:*

(ii′) *A is a strictly local ring* (18.8.2).

(ii″) *A is a Henselian local ring and* $k(B)$ *is a primary extension of* $k(A)$.

Then all the fibres X_y *(*$y \in Y$*) are geometrically connected.*

Proof. Let $\mathfrak{p}$ be the prime ideal of A corresponding to the point y, and set $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$, which are evidently local Noetherian rings; it is clear that the morphism $f' : \mathrm{Spec}(B') \to \mathrm{Spec}(A')$ satisfies condition (i) of (18.9.7); and as y is the generic point of $\mathrm{Spec}(A')$ and $k(A') = k(A)$, $k(B') = k(B)$, one sees that it suffices to verify that, in the case (ii′) (resp. (ii″)), A' is geometrically unibranch (resp. unibranch). But A' is integral, and in the case (ii′) (resp. (ii″)) it is strictly local (resp. Henselian) by virtue of (18.5.10). One therefore concludes that A' is geometrically unibranch (resp. unibranch) by virtue of (18.8.15) (resp. (18.6.12)). ☐

Corollary (18.9.9). — *Let* A *be a local Noetherian ring, Henselian, universally Japanese (i.e.* (7.6.4) *and* (7.7.2)*), whose formal fibres* (7.3.13) *are geometrically reduced; then all the fibres formelles of A are geometrically connected.*

Proof. It suffices to apply (18.9.8) in the case where $B = \widehat{A}$, since B is an A-module that is flat and that $k(B) = k(A)$. ☐

Remark (18.9.10). — The proofs of (18.9.4) and (18.9.5) do not use the fact that the formal fibres of A are geometrically connected. The conclusions of these two propositions are therefore still valid when one supposes the local ring A Noetherian, Henselian and universally Japanese. **IV-4** | 157

Corollary (18.9.11). — *Let* X, Y *be two locally Noetherian preschemes,* Y *being supposed geometrically unibranch and X connected. Let* $f : X \to Y$ *be a reduced morphism (i.e.* (6.8.1)*), flat and with geometrically reduced fibres. Then, for every closed subset* T *of* X *such that* $f(T)$ *is rare in* Y, $X - T$ *is connected.*

Proof. As f is flat and $f(T)$ is rare, T cannot contain any maximal point of X (2.3.4), hence is rare in X. Utilising (15.5.6.1), it suffices to show that, for every $x \in T$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. Setting $y = f(x)$, as the ring $\mathscr{O}_{Y,y}$ is geometrically unibranch, one can apply (18.9.7) to the morphism

$$f_1 : \mathrm{Spec}(\mathscr{O}_{X,x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{Y,y}),$$

which satisfies conditions (i) and (ii) of this theorem, and to the closed subset $\{x\}$ of $\operatorname{Spec}(\mathscr{O}_{X,x})$, since $f_1(\{x\}) = \{y\}$ is rare in $\operatorname{Spec}(\mathscr{O}_{Y,y})$. □

18.10. Étale preschemes over a geometrically unibranch or normal prescheme

Theorem (18.10.1). — *Let $f : X \to Y$ be a locally of finite presentation morphism, x a point of X, $y = f(x)$. Suppose that Y is integral and geometrically unibranch at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that the following two conditions be satisfied:*

*label=*i) *f is formally unramified at the point x;*
*lbbel=*ii) *the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.*

Moreover, if this is the case, X is integral and geometrically unibranch at the point x.

Finally, if $\mathscr{O}_{Y,y}$ is Noetherian, one can, in the preceding criterion, replace the condition (ii) *by the condition* (ii bis) $\dim \mathscr{O}_{Y,y} \leqslant \dim \mathscr{O}_{X,x}$.

Proof. The fact that, if f is étale, X is integral and geometrically unibranch at the point x is a particular **IV-4** | 157
case of (17.5.7). It is clear on the other hand that the conditions (i) and (ii) are necessary for f to be étale at the point x, since $\mathscr{O}_{X,x}$ is then an $\mathscr{O}_{Y,y}$-module faithfully flat (17.6.1). Let us prove that these conditions are sufficient.

Set $A = \mathscr{O}_{Y,y}$, $A' = {}^{hs}A$ (18.8.7), $Y' = \operatorname{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$. Let y' be the closed point of Y', x' a point of X' above x and y'; since the morphism $g : Y' \to Y$ is *flat* (18.8.8), it amounts to showing that f' is étale at the point x' (17.7.1),(ii)). The hypothesis (ii) implies that the morphism $\operatorname{Spec}(\mathscr{O}_{X,x}) \to \operatorname{Spec}(\mathscr{O}_{Y,y})$ is dominant (**I**, 1.2.7); as $\mathscr{O}_{Y,y}$ is integral, $\operatorname{Spec}(\mathscr{O}_{Y,y})$ has a single generic point y_1, and there is a generization x_1 of x in X above y_1. The projection morphism $X' \to X$ being flat, there exists a generization x'_1 of x' above x_1 (2.3.4); set $y'_1 = f'(x'_1)$. Now, the hypothesis on y implies that A' is *integral* (18.8.15), hence Y' has a single generic point η, necessarily above y_1 since g is flat (2.3.4); on the other hand, η is also the generic point of $g^{-1}(y_1)$ **IV-4** | 158
(**0**, 2.1.8), and one knows on the other hand that the fibres of g are discrete spaces (18.8.12), hence $g^{-1}(y_1) = \{\eta\}$, and consequently $y'_1 = \eta$. The morphism $\operatorname{Spec}(\mathscr{O}_{X',x'}) \to \operatorname{Spec}(\mathscr{O}_{Y',y'})$, restriction of f', is therefore dominant, and since $\mathscr{O}_{Y',y'} = A'$ is integral, the corresponding homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is injective (**I**, 1.2.7). On the other hand, f' is not ramified at the point x', (hence (18.8.3), *d*)), there exists a neighbourhood U of x' in X' such that $f'|U$ is a *closed immersion* (18.8.3); *a fortiori* the homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is surjective, and since it is injective, it is therefore *bijective*. But then $\mathscr{O}_{X,x'}$ is an $\mathscr{O}_{Y',y'}$-module *flat*, and f' is therefore étale at the point x' (17.6.1).

Finally, when $\mathscr{O}_{Y,y}$ is Noetherian, we know that if f is étale at the point x, we have $\dim \mathscr{O}_{Y,y} = \dim \mathscr{O}_{X,x}$ (17.6.4). Conversely, suppose the conditions (i) and (ii bis) satisfied, and let us show that they imply conditions (i) and (ii). Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$; restricting Y as needed to a neighbourhood of y, we may suppose that $\mathfrak{J} = \mathscr{J}_y$, where $\mathscr{J}$ is an Ideal of $\mathscr{O}_Y$; if Y_1 is the closed sub-prescheme of Y defined by $\mathscr{J}$, one can, restricting further X and Y to neighbourhoods of x and y respectively, suppose that f factorises as $X \xrightarrow{h} Y_1 \to Y$ (**I**, 6.5.1). It is clear that f_1 is still not ramified at the point x, hence (17.4.1) quasi-finite at this point; the demonstration of (5.4.1), (i)) then shows that $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y}/\mathfrak{J})$. But if one had $\mathfrak{J} \neq 0$, one would conclude that $\dim(\mathscr{O}_{Y,y}/\mathfrak{J}) < \dim(\mathscr{O}_{Y,y})$ since Y is integral (**0**, 16.1.2.2); one would then have $\dim \mathscr{O}_{X,x} < \dim \mathscr{O}_{Y,y}$, contradicting the hypothesis, which proves (ii). □

Remark (18.10.2). —

label=(i) When $A = \mathscr{O}_{Y,y}$ is Noetherian, and one knows that its completion $\widehat{A}$ is integral (for example if A is regular), one can, in the preceding proof, replace A' by $\widehat{A}$.

lbbel=(ii) When Y is locally Noetherian, one can give a faster proof of (18.10.1), without using strict henselisation, but invoking the rather delicate results of §§ 14 and 15. As f is quasi-finite at the point x by hypothesis (17.4.1), one can, replacing X by an open neighbourhood of x, suppose that $f^{-1}(y) = \{x\}$. The hypothesis (ii) implies, as we have seen, the existence of an irreducible component Z of X containing x and dominating the unique irreducible component Y_0 of Y containing y. The morphism $f|Z$ being quasi-finite at the point x, hence equidimensional at this point (13.2.2), it follows from the hypothesis on Y and from the Chevalley criterion (14.4.4) that $g = f|Z$ is universally open at the point x. Furthermore since f (and hence also g) is not ramified at the point x, $g^{-1}(y)$ is geometrically reduced on $k(y)$ (17.4.1); as $\mathscr{O}_{Y,y}$ is integral, one concludes from (15.2.3) that g is *flat* at the point x, hence f is *étale* at this point (18.4.9).

lcbel=(iii) With the same notations and hypotheses on Y as in (18.10.1), suppose only that f is *locally of finite type* (and not necessarily locally of finite presentation). Then, for $\mathscr{O}_{X,x}$ to be an $\mathscr{O}_{Y,y}$-algebra *essentially étale* (18.6.1), it is necessary and sufficient that f satisfy the two following conditions:

IV-4 | 159

label=1° f is formally unramified at the point x;
lbbel=2° the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.

There is nothing more than proving the sufficiency of these conditions. Taking into account (18.4.12), it suffices to show that f is *flat* at the point x. Now, by going back to the proof of (18.10.1) and the notations used in this proof, it suffices to show that f' is flat at the point x' (2.5.1); by (17.1.3) furthermore f' is formally unramified at the point x'. One can therefore restrict to carrying out the proof when $Y = \mathrm{Spec}(A)$, where A is a *strictly local* local ring and $y = f(x)$ the closed point of Y. Using then (18.8.3) we see that $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is surjective, hence *bijective* by virtue of the hypothesis, and this suffices to show the flatness of f at the point x.

Suppose furthermore that Y is *locally integral* at the point y (**I**, 2.1.8). Then the conditions 1° and 2° above (together with the fact that Y is geometrically unibranch at the point y and f locally of finite type) already imply that f is *étale at the point* x. Indeed, this follows from what precedes and from (18.4.13).

Corollary (18.10.3). — *Let* Y *be an integral prescheme, integral and geometrically unibranch, η its generic point,* X *a connected prescheme, $f : X \to Y$ a locally of finite type morphism, and such that $f^{-1}(\eta)$ is non-empty. Then, if f is formally unramified, f is étale, and* X *is integral and geometrically unibranch.*

Proof. It follows from the hypothesis and from (18.4.13) that f is étale at all the points where it is flat, and in particular at the points of $f^{-1}(\eta)$, since $\mathscr{O}_{Y,\eta} = k(\eta)$ is a field. The open set U of points of X where f is étale is therefore *non-empty*. Let us show on the other hand that for all $x \in U$, the homomorphism $\mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$ is *injective*. Indeed let s be a section of $\mathscr{O}_Y$ above an open neighbourhood of $f(x)$, that one can always suppose to be X and Y themselves (the question being local on X and Y), and suppose that its image $t \in \Gamma(Y, f_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$ has zero germ at the point x, hence vanishes in a neighbourhood V of x; then the restriction of t to the open set $U \cap V \neq \varnothing$ is zero; but the restriction $f|(U \cap V)$ is étale, hence $W = f(U \cap V)$ is open in Y and contains by construction the generic point η; the homomorphism $\mathscr{O}_{Y,\eta} \to \mathscr{O}_{X,z}$ being injective for all points $z \in f^{-1}(\eta)$, the hypothesis that $t|(U \cap V) = 0$ implies that $s_\eta = 0$, and as Y is integral, this implies $s = 0$, whence our assertion. Applying now (18.10.2), (iii)), we conclude that f is *flat* at the point x, in other words $x \in U$, hence U is both open and closed in X; as it is non-empty and X is connected, we have $U = X$.

Note now that since f is étale, hence locally quasi-finite and Y is irreducible, the maximal points of X belong to $f^{-1}(\eta)$ (2.3.4) and for all $x \in X$, there is a neighbourhood of x that contains only a finite number of these maximal points; in other words, the set of irreducible components of X is locally finite. But by (18.10.1) X is integral and geometrically unibranch at every point, hence every point of X belongs to only a single irreducible component, and since these latter form a locally finite set of open (and closed) subsets in X; since X is connected, it is integral. □

IV-4 | 160

Remark (18.10.3.1). — If $f : X \to Y$ is locally of finite presentation, dominant and quasi-compact, the fibre $f^{-1}(\eta)$ is non-empty by virtue of (**I**, 1.1.5). On the other hand, if one does not suppose f quasi-compact, *f can be non-ramified and dominant without f being étale* (always assuming the hypotheses of (18.10.3) verified for X and Y). This is what the following example shows: one takes for Y the affine plane $\mathrm{Spec}(\mathbf{C}[T, U])$; one considers on the other hand a family $(X_i)_{i \in \mathbf{Z}}$ of preschemes isomorphic to the affine line $\mathrm{Spec}(\mathbf{C}[T])$, and one forms the prescheme obtained by "gluing together" X_{2j} and X_{2j+1} at the point -1 and X_{2j+1} and X_{2j+2} at the point $+1$. Let us define on the other hand $f : X \to Y$ as being equal on X_{2j} to the closed immersion whose image is the line of equation $y = 2j$, transforming the point -1 to $(2j-1, 2j)$ and the point 1 to $(2j+1, 2j)$; on X_{2j+1}, f_j is the closed immersion whose image is the line of equation $x = 2j+1$, transforming the point -1 to $(2j+1, 2j)$ and the point 1 to $(2j+1, 2j+2)$. It is clear that f is a local immersion (hence is not ramified) and is dominant but the fibre of the generic point of Y is empty and f is not étale.

Corollary (18.10.4). — *Let $g : X \to S$, $h : X \to S$ be two locally of finite presentation morphisms, $f : X \to Y$ an S-morphism, x a point of X, $y = f(x)$, $s = g(y) = h(x)$. Suppose that h is flat at the point x and $g^{-1}(s)$ normal at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that f be formally unramified at the point x and that $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$.*

Proof. The conditions are necessary by virtue of the fact that if f is étale at the point x, it follows from this that f_s is also flat and quasi-finite at the point x (6.1.2). Conversely, if the conditions of the statement are satisfied, in order to see that f is étale at the point x, it suffices, by virtue of (17.8.3), to see that f_s is so. One can therefore restrict to the case where S is the spectrum of a field. It then follows from the hypothesis $\dim_x X = \dim_y Y$ and the fact that f is not ramified at the point x that one has (by (5.2.3) and (17.4.1)) $\dim \mathscr{O}_{X,x} = \dim \mathscr{O}_{Y,y}$ integral, and we deduce from (**0**, 7.4.4) and (**0**, 16.3.10) that the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective; it then suffices to apply (18.10.1). ☐

Corollary (18.10.5). — *With the notations of (18.10.4), suppose that h is flat and that $g^{-1}(s)$ is normal for all $s \in S$ (which will be the case for example if g is smooth). For f to be an open immersion, it is necessary and sufficient that f be a monomorphism and that, for all $x \in X$, $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$, where $y = f(x)$ and $s = g(y) = h(x)$.*

Proof. One needs only to prove the sufficiency of the conditions; now, it follows from (17.1.3) that a monomorphism locally of finite presentation is formally unramified, hence the hypotheses imply that f is étale (18.10.4); on the other hand, by (17.9.1), we conclude from these and from (18.10.4). ☐

Lemma (18.10.6). — *Let $f : X \to Y$ be a flat and locally quasi-finite morphism.*

label=(i) *The set $\mathrm{Max}(X)$ of maximal points of X is in canonical bijection with the set $\coprod_{y \in \mathrm{Max}(Y)} f^{-1}(y)$.*

lbbel=(ii) *If Y is irreducible with generic point η, the set of irreducible components of X is in canonical bijection with $f^{-1}(\eta)$, and is in particular finite if and only if $f^{-1}(\eta)$ is finite.* **IV-4** | 161

lcbel=(iii) *If the set of irreducible components of Y is locally finite, so is the set of irreducible components of X, and in particular X is locally connected.*

Proof. The assertion (i) follows immediately from (2.3.4) and (**0**, 2.1.6) and from the fact that the fibres $f^{-1}(y)$ are discrete sets. It is clear that (ii) is a particular case of (i). Finally, to prove (iii), one can, by virtue of (i), restrict to the case where f is quasi-finite and where Y is irreducible (**0**, 2.1.6) and the conclusion follows from (ii) and from the fact that the fibres of f are finite sets (**II**, 6.2.2). ☐

Proposition (18.10.7). — *Let Y be a geometrically unibranch and irreducible prescheme (resp. integral), $f : X \to Y$ an étale morphism. Then X is isomorphic to a sum of preschemes irreducible (resp. integral) X_λ, where λ ranges over the fibre $f^{-1}(\eta)$ of the generic point η of Y. The X_λ are geometrically unibranch; if Y is normal, the X_λ are normal.*

Proof. Indeed, X is geometrically unibranch (17.5.7), hence every point of X belongs to only a single irreducible component; moreover, by virtue of (18.10.6) and (17.6.1) the set of irreducible components of X is locally finite, hence (**0**, 2.1.6) these are the connected components X_λ of X ($\lambda \in f^{-1}(\eta)$), and X is the sum of the X_λ; by virtue of (17.5.7), if Y is integral (resp. normal), the X_λ are integral (resp. normal). ☐

Proposition (18.10.8). — *Let Y be a normal and integral prescheme, of generic point η, $K = k(\eta)$ its field of rational functions, $f : X \to Y$ an étale morphism. Suppose that $f^{-1}(\eta)$ is finite and non-empty, so that the K-scheme $f^{-1}(\eta)$ is the spectrum of a finite separable K-algebra L, composed as a direct product of a finite number of fields K_i ($1 \leqslant i \leqslant n$), finite separable extensions of K (17.6.2). Let Y′ be the integral closure of Y in L, isomorphic to the sum of the integral closures Y_i of Y in K_i (**II**, 6.3.6). Then the morphism f factorises uniquely as $X \xrightarrow{f'} Y' \xrightarrow{g} Y$, such that the restriction $f'_\eta : f^{-1}(\eta) \to g^{-1}(\eta)$ is the canonical isomorphism. Moreover f' is a local isomorphism; for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. By virtue of (18.10.7), one can restrict to the case where X is integral and normal. The existence and uniqueness of the factorisation considered for f follow from (**II**, 6.3.9). As f' is birational and locally quasi-finite, the final assertions follow from (8.12.10). □

Corollary (18.10.9). — *Under the hypotheses of (18.10.8), for f to be a finite morphism (in other words, for* X *to be an étale covering of* Y*), it is necessary and sufficient that* X *be isomorphic to the integral closure* Y' *of* Y *in* L*.*

Proof. If f is finite, the canonical injection $j : \mathrm{X} \to \mathrm{Y}'$ (which is an open immersion) is also a finite morphism (**II**, 6.1.5), (v)), hence a closed morphism (**II**, 6.1.10), and as $j(\mathrm{X})$ is dense in Y' by (18.10.8), we have $j(\mathrm{X}) = \mathrm{Y}'$. Conversely, as Y is normal and L is a composed direct product of finite separable extensions of K, Y' is finite over Y (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 6, cor. 1 of prop. 18), whence the corollary. □

(18.10.10). Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions. We say that a K-algebra of finite rank L is *non-ramified over* Y if: 1° L is a separable K-algebra, hence composed as a direct product of finite separable extensions K_i of K ($1 \leqslant i \leqslant n$); 2° the integral closure Y' of Y in L (sum of the preschemes that are integral closures of Y in the K_i) is *non-ramified over* Y (which, by (18.10.3), is equivalent to saying that Y' is étale over Y). When $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, where **IV-4** | 162
A is an integral ring, integrally closed, K its field of fractions, one says also (by a slight abuse of language, and when there is no confusion to be feared with the terminology of (17.3.2), (ii)) that L is non-ramified over A instead of "non-ramified over Y".

Remark (18.10.11). — Instead of saying that L is non-ramified over Y, certain authors say also that L is "non-ramified over K"; we shall not follow this usage, which lends itself to confusions.

One can then express (18.10.9) in the following form:

Corollary (18.10.12). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions. Then the functor* $\mathrm{X} \mapsto \mathrm{R}(\mathrm{X})$ *is an equivalence of the category of étale coverings of* Y *and of the category of finite* K*-algebras that are non-ramified over* Y*. One obtains a quasi-inverse functor by making correspond to every finite étale* K*-algebra* L*, non-ramified over* Y*, the integral closure* Y' *of* Y *in* L*.*

Proposition (18.10.13). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions.*

label=(i) The field K *is a* K*-algebra non-ramified over* Y*.*

lbbel=(ii) Let L *be a finite extension of* K*, non-ramified over* Y*, and let* Z *be the integral closure of* Y *in* L*. Then, if* M *is an* L*-algebra non-ramified over* Z*,* M *is also a* K*-algebra non-ramified over* Y *("transitivity" of non-ramification).*

lcbel=(iii) Let Y' *be a normal and integral prescheme,* K' *its field of rational functions,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a dominant morphism. If* L *is a* K*-algebra non-ramified over* Y*, then* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ *is a* K'*-algebra non-ramified over* Y' *("translation" property). Furthermore if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ *and* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ *are affine, and if* C *is the integral closure of* A *in* L*, then* $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ *is the integral closure of* A' *in* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$*.*

Proof. The assertion (i) is immediate, Y being its own normalisation in K. To prove (ii), let Z' be the integral closure of Z in M, which is by hypothesis an étale covering of Z. As Z is étale and separated over Y, so is Z' (17.3.3), and it is clearly the integral closure of Y in M. As M is a K-algebra, finite and separable, it follows from (18.10.10) that M is non-ramified over Y. Finally, to prove (iii), let us note that if Z is the integral closure of Y in L, $\mathrm{Z} \times_{\mathrm{Y}} \mathrm{Y}'$ is étale and separated over Y' (17.3.3), finite over Y' since Z is finite over Y, and admits for ring of rational functions $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ (**I**, 3.4.9); as Y' is normal, it follows from (17.5.7), which finishes the proof. □

Corollary (18.10.14). — *Let* Y *and* Y' *be two normal preschemes and integral,* K, K' *their respective fields of rational functions,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a dominant morphism, so that* K' *is an extension of* K*. Let* L *be a finite extension of* K *non-ramified over* Y*,* L_1 *a composite extension (Bourbaki,* Alg.*, chap. VIII, § 8, def. 1) of* L *and* K'*. Then* L_1 *is an extension of* K' *non-ramified over* Y'*. Furthermore if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ *are affine, and if* C *is the integral closure of* A *in* L*, then the subring* $\mathrm{C}_1 = \mathrm{A}[\mathrm{C}, \mathrm{A}']$ *of* L_1 *is the integral closure of* A' *in* L_1*.*

Proof. Indeed, $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ is composed as a direct product of finite separable extensions of K', and L_1 is one of these extensions; hence the integral closure of Y in L is the sum of preschemes, one of which is the integral closure of Y in L_1. The corollary therefore follows from (18.10.14). **IV-4 | 163**

When for example A is the ring of integers of an algebraic number field K, K' and L algebraic extensions of K, there are classical examples showing that the relation $\mathrm{C}_1 = \mathrm{A}[\mathrm{C}, \mathrm{A}']$ does not hold when L is not ramified over A[ ]. □

(18.10.15). Suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is affine, A being integral and integrally closed; let K be its field of fractions, L a finite separable extension of K, and we suppose that the integral closure C of A in L is a *projective* A-module of finite type (which will be the case for example if A is a Dedekind ring (Bourbaki, *Alg. comm.*, chap. VII, § 4, nº 10, prop. 22)). To say that L is non-ramified over A means that C is étale (18.10.10) and by virtue of (18.2.7), (ii)), this is equivalent to saying that the *discriminant* $d_{\mathrm{C}/\mathrm{A}}$ of Spec(C) over Spec(A) is *invertible* in A. In the particular case where C is a *free* A-module and $(x_i)_{1 \leqslant i \leqslant n}$ a basis of C over A, this means that $\det(\mathrm{Tr}_{\mathrm{C}/\mathrm{A}}(x_i x_j))$ is invertible in A.

Theorem (18.10.16). — *Let* Y *be an integral prescheme,* η *its generic point,* $f : \mathrm{X} \to \mathrm{Y}$ *a separated and locally of finite type morphism. Suppose that every irreducible component of* X *dominates* Y *and that the generic fibre* $\mathrm{X}_\eta = f^{-1}(\eta)$ *is a finite prescheme over* $\mathrm{K} = k(\eta)$*, hence* $(\mathrm{X}_\eta)_{\mathrm{red}}$ *is the spectrum of a finite separable* K*-algebra* L*, where* $\mathrm{L} = \prod_i \mathrm{L}_i$ *is a product of finite extensions* L_i *of* K*. Set*

$$n = [\mathrm{L} : \mathrm{K}] = \sum_i [\mathrm{L}_i : \mathrm{K}], \quad n_s = \sum_i [\mathrm{L}_i : \mathrm{K}]_s \quad \textit{(sum of separable degrees of the } \mathrm{L}_i\textit{)}.$$

Let y *be a geometrically unibranch point of* Y*, and denote by* $n(y)$ *the sum of the separable degrees over* $k(y)$ *of the residue fields of the isolated points of* $f^{-1}(y)$*. Then:*

label=(i) One has $n(y) \leqslant n_s \leqslant n$*.*
lbbel=(ii) Suppose X *reduced and the point* y *normal in* Y*. For* $n(y) = n$*, it is necessary and sufficient that there exist a neighbourhood* U *of* y *in* Y *such that the restriction* $f^{-1}(\mathrm{U}) \to \mathrm{U}$ *of* f *is an étale and finite morphism.*

Proof. A) Reduction to the case where f *is quasi-finite.* — One knows (13.1.4) that the set Z of isolated points $x \in \mathrm{X}$ of $f^{-1}(f(x))$ is open in X and by hypothesis Z contains $f^{-1}(\eta)$. Since $f^{-1}(\eta)$ is finite, Z is an increasing filtered union of quasi-compact open sets V_λ containing $f^{-1}(\eta)$ (dense in X by hypothesis). If $n_\lambda(y)$ is the sum of the separable degrees over $k(y)$ of the residue fields of the points of $\mathrm{V}_\lambda \cap f^{-1}(y)$, we have $n(y) = \sup n_\lambda(y)$; to prove (i), it will therefore suffice to show that $n_\lambda(y) \leqslant n_s$. On the other hand, if $n_\lambda(y) = n$, suppose that we have shown that there exists a neighbourhood U of y in Y such that the restriction $\mathrm{V}_\lambda \cap f^{-1}(\mathrm{U}) \to \mathrm{U}$ of f is an étale and finite morphism; since f is separated, the canonical injection $\mathrm{V}_\lambda \cap f^{-1}(\mathrm{U}) \to f^{-1}(\mathrm{U})$ is then also a finite morphism (**II**, 6.1.5), (v)), hence $\mathrm{V}_\lambda \cap f^{-1}(\mathrm{U})$ is closed in $f^{-1}(\mathrm{U})$; but being dense everywhere in $f^{-1}(\mathrm{U})$, it is necessarily equal to it.

B) Reduction to the case where X *is reduced,* f *finite, and* $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$*.* — We note that the hypothesis on f implies that it is also satisfied for f_{red} and that the conclusion of (i) does not concern f_{red}. One can therefore replace f everywhere by f_{red} and thus suppose X reduced.

On the other hand, by virtue of (18.12.13) ([1]), one can (after having replaced Y by an open affine neighbourhood V of y and f by its restriction $f^{-1}(\mathrm{V}) \to \mathrm{V}$) factorise f as $\mathrm{X} \xrightarrow{j} \mathrm{X}' \xrightarrow{g} \mathrm{Y}$, where g is *finite* and j an open immersion. Since X is reduced, one can replace X' by the reduced closed sub-prescheme with underlying space $\overline{j(\mathrm{X})}$ (**I**, 5.2.2), hence suppose X' reduced and such that every irreducible component of X' dominates Y. Moreover, $j(\mathrm{X})$ is an open dense set in X' and has only a finite number of irreducible components by virtue of the hypothesis; hence (**0**, 1.2.7) the irreducible components of X' are the closures of those of $j(\mathrm{X})$, and by construction the fibre $g^{-1}(\eta)$ identifies with $f^{-1}(\eta)$. If one supposes the theorem proved for f finite, one can apply it to g, and as $f^{-1}(y)$ identifies with a sub-prescheme of $g^{-1}(y)$, the relation (i) for g implies the same relation for f. Suppose on the other hand that $n(y) = n$; then, by virtue of what precedes, one can apply the conclusion of (ii) to the morphism g, and replacing Y as needed by a neighbourhood of y, one can suppose g étale; it follows in addition, on the one hand that $f^{-1}(y) = g^{-1}(y)$. As the morphism g is closed and $j(\mathrm{X})$ is open in X', there exists a neighbourhood U of y in Y such that $j(\mathrm{X}) \cap g^{-1}(\mathrm{U}) = j(f^{-1}(\mathrm{U}))$, which proves that (ii) is also true for f. **IV-4 | 164**

We are thus reduced to the case where f is moreover a *finite* morphism. It is immediate that, to prove (i), one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$. Let us show that the same is true for proving (ii). Indeed, suppose we have shown that the condition $n(y) = n$ implies that f is flat and formally unramified at the points of $f^{-1}(y)$, and that Y is *integral*, we conclude from (18.4.13) that f is *étale at the points* of $f^{-1}(y)$, hence also in a neighbourhood V of $f^{-1}(y)$ in X; but since f is a finite morphism, hence closed, V contains an open set of the form $f^{-1}(\mathrm{U})$, where U is a neighbourhood of y in Y.

C) Reduction to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ *and* $\mathscr{O}_{\mathrm{Y},y}$ *is strictly local.* — Set $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, and let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ be the strict henselisation of A (18.8.7), which is a geometrically unibranch local integral ring (18.8.15). Let $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, and let y' and η' be the closed point and the generic point of Y'; y' is above y and since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is flat (18.8.8), η' is above η (2.3.4). Set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$; f' is finite and since g is flat, every irreducible component of X' dominates Y' (2.3.7). If $n'(y')$, n' and n'_s are the numbers defined for f' in the same way as $n(y)$, n and n_s for f, we have $n'_s = n_s$ and $n'(y') = n(y)$ (**I**, 6.4.8), and it therefore amounts to proving the assertion (i) for f or for f'. Moreover, if X is reduced and y a normal point of Y (hence A integrally closed (18.8.12)). On the other hand, it follows from the definition (18.8.7), from the remark (8.1.2), *a*)) and from the theorem of the double inductive limit, that A' is the inductive limit of rings A_α such that the morphisms $\mathrm{Spec}(\mathrm{A}_\alpha) \to \mathrm{Spec}(\mathrm{A}) = \mathrm{Y}$ are étale; consequently X' is the projective limit of the $\mathrm{X}_\alpha = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{A}_\alpha$, which are étale over X, hence reduced since X is (17.5.7); by passage to the limit,
we deduce that X' is reduced (8.7.1). This being so, if $n(y) = n$, on has $n_s = n$ and consequently the residue fields at the points of X_η are necessarily separable over $k(\eta)$; the residue fields at the points of $\mathrm{X}'_{\eta'}$ are then algebraic and separable over $k(\eta')$ (4.6.1), in other words $n'_s = n' = n$. The morphism f' has therefore the same properties as f, and if one proves that f' is étale, it will follow by (17.7.1). **IV-4** | 165

D) End of the proof. — Suppose therefore A strictly local. Since A is henselian and the morphism f finite, it follows from (18.5.11) that if x_j $(1 \leqslant j \leqslant m)$ are the distinct points of $f^{-1}(y)$, X is the *sum* of open sub-preschemes $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j}) = \mathrm{X}_j$, which are the connected components of X. By hypothesis, each of them dominates Y, hence every irreducible component of X meets $f^{-1}(\eta)$, and so the number of points of $f^{-1}(\eta)$ is $\geqslant m$; *a fortiori* we have $n_s \geqslant m$. Suppose now the hypotheses of (ii) satisfied; the relations $n(y) = n_s = n$ imply first of all, since every irreducible component of X dominates Y, that each of the X_j is *irreducible*; moreover, as the X_j are *reduced*, they are *integral*; finally the relations $m = n_s = n$ imply that for the generic point ξ_j of X_j, $k(\xi_j)$ is a separable extension of $k(\eta)$ of degree 1, hence isomorphic to $k(\eta)$; in other words, the restriction $f|\mathrm{X}_j : \mathrm{X}_j \to \mathrm{Y}$ of f is *finite and birational*; moreover, X_j is *integral* and by hypothesis Y is *normal*; hence (8.12.10.1) $f|\mathrm{X}_j$ is an *isomorphism*, which proves that f is étale, and finishes the proof of (18.10.16). □

Remark (18.10.17). — The method of "étale localisation" used in the proof of (18.10.16) also allows one to improve the results of (15.5.1) by eliminating the Noetherian hypothesis. One has first of all the following result which generalises (15.5.2):

(18.10.17.1). Let $f : \mathrm{X} \to \mathrm{Y}$ be a separated and quasi-finite morphism, y a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, for every generisation y' of y, we have $n(y) \leqslant n(y')$.

Proof. One can replace Y by the reduced closed sub-prescheme $\overline{\{y'\}}$ and X by $f^{-1}(\mathrm{Z})$, hence suppose Y integral with generic point y'. Using (18.12.13) (as in (18.10.16), B)), one can suppose that X is an open dense subset of a prescheme X' and that f is the restriction to X of a finite morphism $g : \mathrm{X}' \to \mathrm{Y}$. Let us show that $g^{-1}(y') = f^{-1}(y')$. Indeed, let $i : g^{-1}(y') \to \mathrm{X}'$ be the canonical injection, which is a flat morphism since y' is the generic point of Y (**I**, 3.6.5); one can write $f^{-1}(y') = i^{-1}(\mathrm{X})$. On the other hand, the canonical injection $j : \mathrm{X} \to \mathrm{X}'$ is a morphism of finite type since the composite $g \circ j = f$ is of finite type and g is separated (1.5.4); hence X is a retrocompact open subset of X', and consequently pro-constructible in X' (**I**, 1.9.5), (v)). One concludes therefore from (2.3.10) that $i^{-1}(\overline{\mathrm{X}}) = \overline{i^{-1}(\mathrm{X})}$, in other words $f^{-1}(y')$ is dense in $g^{-1}(y')$; but as $g^{-1}(y')$ is discrete, we necessarily have $f^{-1}(y') = g^{-1}(y')$.

8(1) The reader will verify that (18.10.16) is not used in the proof of (18.12.13). If f is supposed locally of finite presentation (resp. quasi-projective), one can replace (18.12.13) by (8.12.8) (resp. (8.12.6)).

We are thus reduced to proving that the sum of the numbers $[k(x) : k(y)]_s$ where x ranges over the points of $g^{-1}(y)$ where g is universally open, is at most equal to $n(y')$ (for the morphism g). Using then the fact that the property of being universally open at a point is preserved by base change, one shows, as in (18.10.16), C)), that one can reduce to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, with $\mathscr{O}_{\mathrm{Y},y}$ *strictly local*. Applying (18.5.11), *c*)), we see that if x_j $(1 \leqslant j \leqslant n)$ are the points of $f^{-1}(y)$, X is the *sum* of n sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$; the hypothesis that g is open in each of the points x_j implies that the sub-prescheme X_j corresponding dominates Y (1.10.3); one concludes as in (18.10.16), D)). **IV-4 | 166**

We then deduce from (18.10.17.1) that *the conclusions of* (15.5.1) *are still valid when one no longer supposes* Y *locally Noetherian, but only that* f *is separated, quasi-finite, and of finite presentation*. □

Indeed, to prove assertion (i) of (15.5.1), one remarks that, since f is of finite presentation, the set E of points $z \in \mathrm{Y}$ such that $n(z) \geqslant n(y)$ is locally constructible (9.7.9); to show that y is interior to E, it suffices, by virtue of (1.10.1), to see that every generisation y' of y belongs to E, which is nothing other than (18.10.17.1).

To prove assertion (ii) of (15.5.1), one remarks that, since f is of finite presentation, using the method of (8.10.5), (xii)) applied following (8.1.2), *a*)), and one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, and the ring $\mathscr{O}_{\mathrm{Y},y}$ is *strictly local*. Applying (18.5.11), *c*)), we see then that if x_j $(1 \leqslant j \leqslant n)$ are the points of $f^{-1}(y)$, X is the *sum* of n open sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$ which are *finite* over Y, and of an open prescheme X''. If one proves that $\mathrm{X}'' = \varnothing$, we will have shown that f is a finite morphism, and the conclusion follows from the fact that, since X is open in each of the points x_j, the restriction of f to X_j, being a finite morphism (**II**, 6.1.10), is *surjective*; the hypothesis that $z \mapsto n(z)$ is constant implies (17.4.1), hence one terminates as in (18.10.16), D)).

Finally, the proof of (iii) reproduces without change that given in (15.5.1).

(18.10.17.2). By analogous methods using strict henselisation, one can, in most of the results of §§ 14 and 15, eliminate the Noetherian hypotheses.

Lemma (18.10.18). — *Let* X, Y *be two preschemes,* $f : \mathrm{X} \to \mathrm{Y}$ *a birational morphism* (1), x *a point of* X. *For* f *to be an isomorphism local at the point* x, *it is necessary and sufficient that* f *be étale at the point* x. *For* f *to be an open immersion, it is necessary and sufficient that* f *be étale and separated.*

Proof. The conditions stated being trivially necessary, all that is needed is to see that they are sufficient. For the first assertion, the question is local on X and Y, hence one can suppose X and Y affine, hence f *separated*, and we are thus reduced to proving the second assertion. By virtue of (17.9.1), it is a question of proving that f is *radicial*. Let $y \in \mathrm{Y}$, and let us prove that $f^{-1}(y)$ is radicial on $k(y)$. The base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ not changing the fact that f is étale, separated, and birational, one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, with $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$. Let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$, which is a local ring and an $\mathscr{O}_{\mathrm{Y},y}$-module faithful flat. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, so that f' is étale and separated; moreover, as the morphism $\mathrm{Y}' \to \mathrm{Y}$ is flat, the reasoning of (6.15.4.1) shows that f' is also birational. **IV-4 | 167**

Suppose therefore $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, A strictly local, f separated, étale and birational, and let us show that $f^{-1}(y)$ can contain only a single point. Indeed, if $f^{-1}(y)$ contained two distinct points x_1, x_2, there would exist two Y-sections u_1, u_2 of f such that $u_1(y) = x_1$ and $u_2(y) = x_2$ (18.8.1); but since f is étale and separated, u_1 and u_2 would be isomorphisms of Y onto two connected components (open) of X (17.9.4) and there would then be two maximal points of X at least above every maximal point of Y, which contradicts the hypothesis that f is *birational*. □

Proposition (18.10.19). — *Let* Y *be a reduced prescheme whose set of irreducible components is locally finite,* $f : \mathrm{X} \to \mathrm{Y}$ *a separated morphism of finite type and formally unramified,* g *a* Y*-rational section of* f *(i.e. a* Y*-rational map from* Y *to* X*),* U *the domain of definition of* g (**I**, 7.2.1)*, and let* Z *be the closure of* $g(\mathrm{U})$ *in* X. *Then, for all points* $y \in \mathrm{Y} - \mathrm{U}$ *such that* y *is geometrically unibranch,* $\mathrm{Z}_y = \mathrm{Z} \cap f^{-1}(y) = \varnothing$. *In particular, if* Y *is geometrically unibranch,* $g(\mathrm{U})$ *is a subset of* X *that is both open and closed. If, for every closed irreducible* T *of* X, $f(\mathrm{T})$ *is closed in* Y, g *is defined at every geometrically unibranch point of* Y.

[8](1) We understand by this the notion defined in (6.15.4), but where we do not suppose X and Y reduced.

Proof. Since Y is reduced and f separated, it follows from (**I**, 7.2.2) that there exists a U-section u of $f^{-1}(\mathrm{U})$ belonging to the class g; by (17.4.1.2), moreover, f is formally unramified, hence u is an isomorphism of U onto the prescheme induced on the open set $u(\mathrm{U})$ of X. If one denotes also by Z the reduced sub-prescheme of X having Z for underlying space, $g(\mathrm{U}) = u(\mathrm{U})$, being reduced, is induced also by Z, and by construction $f' = f|\mathrm{Z}$ is a *birational* morphism of Z to Y, moreover formally unramified since f is (17.1.3). The question being local over Y at the point y, one can restrict (replacing Y by a neighbourhood of y) to the case where Y is *integral*; then $u(\mathrm{U})$ is irreducible, hence also Z, and by construction Z is also integral. Then (**I**, 8.2.7) since f' is dominant, the homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z},x}$ is injective for all $x \in \mathrm{Z}_y$, and one concludes from (18.10.2), (iii)) that f' is an étale isomorphism local at the point x; but this would imply that the U-section u would be extendable to an open set distinct from U, contrary to the definition of the latter. We therefore necessarily have $\mathrm{Z}_y = \varnothing$, which establishes the first assertion; the second is an obvious consequence. Finally, under the last hypotheses of the statement, as Z is closed and irreducible, $f(\mathrm{Z})$ is closed in Y, hence equal to Y, i.e. $\mathrm{Z}_y \neq \varnothing$ for all $y \in \mathrm{Y}$, which finishes the proof. □

Remark (18.10.20). — Let Y be a locally Noetherian prescheme. One says that a morphism $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper* if it is locally of finite type and if, for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is essentially proper for every Y-scheme $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A})$ where A is a discrete valuation ring, the canonical map (**II**, 7.3.2.2) relative to the Y'-prescheme $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, is bijective. The reasoning of (**II**, 7.3.8) proves that this is the same as saying that f (supposed locally of finite type) is *separated* and that for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is *proper* (for Y locally Noetherian) means therefore that f is *essentially proper and of finite type* (**II**, 7.3.8); but one encounters important examples of essentially proper morphisms that are not of finite type (for example certain “Picard preschemes” or certain “preschemes of Néron-Severi” (chap. VI)). It evidently follows from (18.10.19) that if Y is *locally Noetherian, reduced and geometrically unibranch*, and if $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper and not ramified*, then every Y-rational section of f is *defined everywhere*. **IV-4 | 168**

Proposition (18.10.21). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite presentation morphism, x a point of* X, *$y = f(x)$, and suppose the ring $\mathscr{O}_{\mathrm{Y},y}$ integral and geometrically unibranch. Let r be the minimum number of generators of the $\mathscr{O}_{\mathrm{X},x}$-module $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x$ (equal to the rank of the $k(x)$-vector space $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x \otimes_{\mathscr{O}_{\mathrm{X},x}} k(x)$). Then, for f to be smooth at the point x, it is necessary and sufficient that, if y' is the generic point of the unique irreducible component of* Y *containing y, there exists a generisation x' of x such that $f(x') = y'$ and* $\dim_{x'} f^{-1}(y') \geqslant r$.

Proof. If f is smooth at the point x, it is also so in an open neighbourhood U of x, hence in every generisation x' of x, and $\dim_{x'}(f^{-1}(f(x'))) = r$ in each of these points, by virtue of (17.10.2). As f is moreover flat in U, there exists a generisation of x above every generisation of y (2.3.4), which proves that the condition is necessary. To show that it is sufficient, let us first note that by (17.5.1), it suffices to prove that the morphism deduced from f by the base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ is smooth at the point x; one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$.

The question is local on X, hence (16.4.22) and (**0**, 5.2.2)) one can suppose that X is affine, and that there exist r sections s_i $(1 \leqslant i \leqslant r)$ of $\mathscr{O}_{\mathrm{X}}$ above X such that the sections $d_{\mathrm{X}/\mathrm{Y}}(s_i)$ generate $\Omega^1_{\mathrm{X}/\mathrm{Y}}$. Let

$$g : \mathrm{X} \longrightarrow \mathrm{Z} = \mathrm{Y}[\mathrm{T}_1, \ldots, \mathrm{T}_r] = \mathbf{V}_{\mathrm{Y}}$$

be the Y-morphism corresponding to the homomorphism $\mathscr{O}_{\mathrm{Y}}^r \to f_*(\mathscr{O}_{\mathrm{X}})$ of $\mathscr{O}_{\mathrm{Y}}$-Modules defined by the s_i (considered as sections of $f_*(\mathscr{O}_{\mathrm{X}})$ above Y) (**II**, 1.2.7). We have the exact sequence

$$g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Y}} \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Z}} \longrightarrow 0.$$

As the $d\mathrm{T}_i$ $(1 \leqslant i \leqslant r)$ generate $\Omega^1_{\mathrm{Z}/\mathrm{Y}}$, it follows from the definition of g that the homomorphism $g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \to \Omega^1_{\mathrm{X}/\mathrm{Y}}$ is surjective, hence $\Omega^1_{\mathrm{X}/\mathrm{Z}} = 0$, and g is consequently *formally unramified* (17.4.1). We shall now see that g is *étale* at the point x, and taking into account (17.3.8), this will prove that f is smooth at the point x. As $\mathscr{O}_{\mathrm{Y},y}$ is supposed integral and geometrically unibranch, so is $\mathscr{O}_{\mathrm{Z},z}$ **IV-4 | 169**
by (17.5.7) where $z = g(x)$, since the structural morphism $\mathrm{Z} \to \mathrm{Y}$ is smooth (17.3.8). By virtue of (18.10.1), it suffices to show that the homomorphism $\mathscr{O}_{\mathrm{Z},z} \to \mathscr{O}_{\mathrm{X},x}$ is injective; it amounts to the

same thing to see that the morphism $\mathrm{Spec}(\mathscr{O}_{X,x}) \to \mathrm{Spec}(\mathscr{O}_{Z,z})$ corresponding is *dominant*, since $\mathscr{O}_{Z,z}$ is integral (**I**, 1.2.7). Let y' be the generic point of $Y = \mathrm{Spec}(\mathscr{O}_{Y,y})$ and let z' be the generic point of $Z = \mathrm{Spec}(\mathscr{O}_{Y,y}[T_1, \ldots, T_r])$ which is integral; if $h : Z \to Y$ is the structural morphism, on has $h(z') = y'$. It suffices to prove that there exists in $f^{-1}(y')$ a generisation x' of x such that the image of x' by the morphism $g_{y'} : f^{-1}(y') \to h^{-1}(y')$ equals z'. Moreover z' is also the generic point of $h^{-1}(y') = \mathrm{Spec}(k(y')[T_1, \ldots, T_r])$; as the morphism $g_{y'}$ is not ramified, hence quasi-finite (17.4.2), it follows from (5.4.1) that the restriction of $g_{y'}$ to every irreducible component of *dimension* $\geqslant r$ of $f^{-1}(y')$ is a dominant morphism. Now, there exists by hypothesis a component containing a generisation x' of x; *a fortiori* its generic point is a generisation of x, whence the conclusion. □

18.11. Application to Noetherian local algebras complete over a field The following lemma generalises (**0**, 21.9.1) and (**0**, 21.9.2):

Lemma (18.11.1). — *Let k be a field, A a Noetherian complete local ring which is a k-algebra, K the residue field of A.*

label=(i) If the K-vector space $\Omega^1_{K/k}$ is of finite rank, the A-module $\widehat{\Omega}^1_{A/k}$ is of finite type.

lbbel=(ii) If moreover A is a k-algebra formally smooth (for the adic topology), $\widehat{\Omega}^1_{A/k}$ is a free A-module of rank equal to

$$\dim(A) + \mathrm{rg}_K(\Omega^1_{K/k}) - \mathrm{rg}_K(\Upsilon_{K/k}).$$

Furthermore, for every subfield k_0 of k such that Ω^1_{K/k_0} is a k-vector space of finite rank, $\widehat{\Omega}^1_{A/k_0}$ is a free A-module of rank equal to

$$\dim(A) + \mathrm{rg}_K(\Omega^1_{K/k_0}) - \mathrm{rg}_K(\Upsilon_{K/k_0}) + \mathrm{rg}_K(\Omega^1_{k/k_0}).$$

Proof. The assertion (i) is noted as a reminder, having been proved in (**0**, 20.7.15). To prove (ii), note that this assertion has been in fact established by the demonstration of (**0**, 21.9.2), making only intervene the finiteness of the transcendence degree of K over k (by way of the Cartier equality (**0**, 21.7.1)). □

Lemma (18.11.2). — *Let k be a field of characteristic $p > 0$, C a k-algebra, local Noetherian, complete, whose residue field is a finite extension of k, and which is integral; let n be its dimension, L its field of fractions. Then $\Omega^1_{L/k}$ and $\Upsilon_{L/k}$ are L-vector spaces of finite rank, and one has*

(18.11.2.1) $$\mathrm{rg}_L(\Omega^1_{L/k}) - \mathrm{rg}_L(\Upsilon_{L/k}) \geqslant n.$$

Suppose furthermore that $[k : k^p] < +\infty$. Then one has the equality **IV-4** | 170

(18.11.2.2) $$\mathrm{rg}_L(\Omega^1_{L/k}) - \mathrm{rg}_L(\Upsilon_{L/k}) = n.$$

Furthermore, $\Omega^1_{C/k}$ is then isomorphic to $\widehat{\Omega}^1_{C/k}$, hence (18.11.1) is a C-module of finite type.

Proof. We know in fact (**0**, 19.8.9) that there exists a sub-k-algebra C_0 of C isomorphic to a formal power series algebra in n variables $k[[T_1, \ldots, T_n]]$ and such that C is a C_0-algebra finite. Hence, L is a *finite* extension of the field of fractions $L_0 = k((T_1, \ldots, T_n))$ of C_0. Now, one has the exact sequence of L-vector spaces obtained by applying (**0**, 20.6.17.1) to the three fields $k \subset L_0 \subset L$ (and taking into account (**0**, 20.4.13)):

$$0 \longrightarrow \Upsilon_{L/k} \otimes_{L_0} L \longrightarrow \Upsilon_{L/k} \longrightarrow \Upsilon_{L/L_0} \longrightarrow \Omega^1_{L/k} \otimes_{L_0} L \longrightarrow \Omega^1_{L/k} \longrightarrow \Omega^1_{L/L_0} \longrightarrow 0.$$

Since L is a finite extension of L_0, Ω^1_{L/L_0} and Υ_{L/L_0} are L-vector spaces of finite rank having the same rank, by the Cartier equality (**0**, 21.7.1); from the preceding exact sequence we therefore already deduce that

$$\mathrm{rg}_L(\Omega^1_{L/k}) - \mathrm{rg}_L(\Upsilon_{L/k}) = \mathrm{rg}_{L_0}(\Omega^1_{L_0/k}) - \mathrm{rg}_{L_0}(\Upsilon_{L_0/k}).$$

To prove (18.11.2.1) or (18.11.2.2), one can therefore restrict to proving these relations by replacing C and L by C_0 and L_0. As L_0 is separable over k (**0**, 21.7.1), we have $\Upsilon_{L_0/k} = 0$ (**0**, 20.6.3); one deduces hence already that in all cases $\mathrm{rg}_{L_0}(\Omega^1_{L_0/k}) \geqslant n$, with equality when $[k : k^p] < +\infty$; this already proves (18.11.2.1) and (18.11.2.2).

Finally, to see that $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ when $[k : k^p] < +\infty$, it suffices to prove that $\Omega^1_{C/k}$ is a C-module of finite type, since C is a Noetherian complete local ring (**0**, 7.3.5), taking into account

($\mathbf{0}$, 20.4.5). Since $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ and that $k[C_0^p] \subset k[C^p]$, the result follows that $\Omega^1_{C/k}$ is a C-module of finite type and C_0 a $k[C_0^p]$-module of finite type in the hypothesis $[k : k^p] < +\infty$. □

Lemma (18.11.3). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A *such that* $A_{\mathfrak{p}}$ *is a Cohen-Macaulay ring. For every system of parameters* $(t_i)_{1\leqslant i\leqslant s}$ *of* $A_{\mathfrak{p}}$*, there exists a sequence* $(x_i)_{1\leqslant i\leqslant s}$ *of elements of* $\mathfrak{p}$ *such that the* x_i *form part of a system of parameters of* A *and that, for all* i*,* $x_i/1$ *is congruent to* t_i *modulo the ideal* **IV-4** | 172
$t_1A_{\mathfrak{p}} + \ldots + t_{i-1}A_{\mathfrak{p}}$*. In particular, if* $A_{\mathfrak{p}}$ *is regular and if the* t_i *form a regular system of parameters of* $A_{\mathfrak{p}}$*, so it is also the case of the* $x_i/1$*.*

Proof. This lemma itself will be a consequence of the following: □

Lemma (18.11.3.4). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A*,* x *an element of* A*; then there exists an element* $x' \in A$ *such that* $x'/1 = x/1$ *in* $A_{\mathfrak{p}}$ *and that* x' *does not belong to any of the minimal prime ideals of* A *not contained in* $\mathfrak{p}$*.*

Proof. Let us first show how (18.11.3.4) implies (18.11.3). Multiplying the t_i if necessary by invertible elements of $A_{\mathfrak{p}}$, one can suppose that they are of the form $x_i/1$, with $x_i \in \mathfrak{p}$ for $1 \leqslant i \leqslant s$. We reason by induction on s. As t_1 forms part of a system of parameters of $A_{\mathfrak{p}}$ ($\mathbf{0}$, 16.5.5), there is no $x_1' \in A$ such that $x_1'/1 = t_1$ that can belong to a minimal prime ideal of $A_{\mathfrak{p}}$, hence it cannot belong to any of the minimal prime ideals of A contained in $\mathfrak{p}$. By virtue of (18.11.3.4), one can moreover choose x_1' such that $x_1'/1 = t_1$ (hence $x_1' \in \mathfrak{p}$) and that x_1' does not belong to *any* minimal prime ideal of A, hence x_1' forms part of a system of parameters of A ($\mathbf{0}$, 16.3.4) and (16.3.7). One reasons then by induction, considering the ring $A' = A/x_1'A$ and the prime ideal $\mathfrak{p}' = \mathfrak{p}/x_1'A$; as $A'_{\mathfrak{p}'} \cong A_{\mathfrak{p}}/t_1A_{\mathfrak{p}}$, $A'_{\mathfrak{p}'}$ is also a Cohen-Macaulay ring ($\mathbf{0}$, 16.5.5), and the images t_i' ($2 \leqslant i \leqslant s$) of the t_i in $A'_{\mathfrak{p}'}$ form a system of parameters of this ring; it suffices to apply A' and to the t_i' ($2 \leqslant i \leqslant s$) the induction hypothesis. The last assertion of (18.11.3) follows from the fact that in $A_{\mathfrak{p}}$, a system of parameters that generates the maximal ideal is a regular system of parameters ($\mathbf{0}$, 17.1.1). □

Proof of (18.11.3.4)*.* Let $(p_k')_{1\leqslant k\leqslant r}$ be the suite of minimal prime ideals of A not contained in $\mathfrak{p}$, and let $(\mathfrak{p}_j)_{1\leqslant j\leqslant n}$ be the suite of other minimal prime ideals of A; one can suppose that $r \geqslant 1$. As $\mathfrak{p}_k'$ does not contain $\bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$, there exists an element $y \in \bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$ that does not belong to any of the $\mathfrak{p}_k'$ (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2); moreover $y/1$ belongs to all the minimal prime ideals of $A_{\mathfrak{p}}$, hence is nilpotent; if $(y/1)^h = 0$, the element $x' = x + y^h$ will satisfy the requirement, since $x'/1 = x/1$ by definition of the $\mathfrak{p}_k'$, on has $x' \notin \mathfrak{p}_k'$ for $1 \leqslant k \leqslant r$, and on the other hand, if $\mathfrak{p}_j$ is a prime ideal of A not contained in $\mathfrak{p}$ but such that $x \notin \mathfrak{p}_j$, we also have $x' \notin \mathfrak{p}_j$ since $y \in \mathfrak{p}_j$. □

Proof of (18.11.3)*, continued.* We return to the proof of (18.11.3) when $p = 1$. By virtue of (18.11.3.4), there exists a regular system of parameters $(t_i)_{1\leqslant i\leqslant s}$ of $A_{\mathfrak{p}}$ where the x_i ($1 \leqslant i \leqslant s$) belong to $\mathfrak{p}$ and form part of a system of parameters $(x_j)_{1\leqslant j\leqslant n}$ of A. Set $A_0 = k[[T_1, \ldots, T_n]]$; as A is a k-algebra complete and the x_j belong to the maximal ideal $\mathfrak{m}$ of A, there exists a local k-homomorphism $u : A_0 \to A$ such that $u(T_j) = x_j$ for $1 \leqslant j \leqslant n$ (Bourbaki, *Alg. comm.*, chap. III, § 4, nº 5, prop. 6); if n is the ideal of A generated by the x_j ($1 \leqslant j \leqslant n$), it is by hypothesis an ideal of definition of A; one deduces therefore from ($\mathbf{0}$, 7.4.4) and (7.4.3) that u makes A an A_0-algebra *finite*.

Set $\mathfrak{p}_0 = \sum_{j=1}^{s} A_0T_j$, $B_0 = (A_0)_{\mathfrak{p}_0}$ and $B = A \otimes_{A_0} B_0$, so that $u_{\mathfrak{p}_0} : B_0 \to B$ makes B a B_0- **IV-4** | 173
algebra finite; moreover, if $\mathfrak{p}'$ is the ideal of B generated by $\mathfrak{p}$, on has $B_{\mathfrak{p}'} = A_{\mathfrak{p}}$; as $\mathfrak{p}'$ contains $\mathfrak{p}_0B_0$ by construction, it is above the maximal ideal $\mathfrak{p}_0B_0$ of B_0. Let us show that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(B_0)$ is *not ramified* at the point $\mathfrak{p}'$: indeed this follows from the fact that $k(\mathfrak{p}')$ is a finite extension of the field $k(\mathfrak{p}_0)$ of characteristic 0, hence is *necessarily separable*, and that $B_{\mathfrak{p}'}/\mathfrak{p}_0B_{\mathfrak{p}'} = k(\mathfrak{p}')$ by virtue of the choice of the x_i for $1 \leqslant j \leqslant s$ (17.4.1).

Let us now note the following lemma:

Lemma (18.11.3.5). — *Let* k *be a field,* R*,* S *two Noetherian complete local* k*-algebras, such that, if* K *is the residue field of* S*,* $\Omega^1_{K/k}$ *is of finite rank over* K*; let* $u : R \to S$ *be a* k*-homomorphism making* S *an* R*-algebra finite; then* $\widehat{\Omega}^1_{S/R} = \Omega^1_{S/R}$*, and the sequence*

(18.11.3.6) $$\widehat{\Omega}^1_{R/k} \otimes_R S \xrightarrow{v} \widehat{\Omega}^1_{S/k} \xrightarrow{w} \Omega^1_{S/R} \longrightarrow 0$$

(cf. ($\mathbf{0}$, 20.7.17.3)*) is exact.*

Proof. The first assertion follows from the fact that $\Omega^1_{S/R}$ is an S-module of finite type (**0**, 20.4.7) and that w is surjective; but it follows from (18.11.1) that $\widehat{\Omega}^1_{S/k}$ is an S-module of finite type; on concludes that every sub-S-module of $\widehat{\Omega}^1_{S/k}$ is closed (**0**, 7.3.5), whence the lemma. □

Applying this lemma to the case where $R = A_0$, $S = A$, and noting that $\widehat{\Omega}^1_{A_0/k}$ is an A_0-module free of rank n (**0**, 21.9.3); hence on a $\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A = \widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A$ and this is an A-module free of rank n. Since $\operatorname{Spec}(B)$ is not ramified on $\operatorname{Spec}(B_0)$ at the point $\mathfrak{p}'$, on a $(\Omega^1_{A/A_0})_{\mathfrak{p}} = \Omega^1_{B_{\mathfrak{p}'}/B_0} = 0$ (16.4.15) and (17.4.1). If one localises the exact sequence (18.11.3.6) (applied to A_0 and A) at $\mathfrak{p}$, one obtains therefore a surjective homomorphism $(\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A)_{\mathfrak{p}} \to (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$, and the conclusion follows that $\widehat{\Omega}^1_{A/k}$ is a B-module of rank n, and the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ admits a system of n generators. But the $A_{\mathfrak{p}}$-module $\widehat{\Omega}^1_{A/k}$ is of *rank n*, by virtue of (**0**, 21.9.5), which is applicable to the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ admits also a system of n generators, and since it is of rank n, this quotient is necessarily *free*, whence the conclusion. □

Lemma (18.11.4). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k. Let k' be an arbitrary extension of k, and set $A' = A\widehat{\otimes}_k k'$. Then:*

label=(i) *A' is a semi-local Noetherian ring, complete, composed as a direct product of complete local rings A'_i $(1 \leqslant i \leqslant r)$ which are A-modules faithfully flat, and whose residue fields are finite extensions of k'; if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}A'$ is an ideal of definition of A'; and $\dim(A'_i) = \dim(A)$ for all i.*

lbbel=(ii) *For all i, $\widehat{\Omega}^1_{A/k}$ is canonically isomorphic to $\widehat{\Omega}^1_{A/k} \otimes_A A'_i$.*

Proof. The first assertions follow immediately from (7.5.5) and (**0**, 6.6.2); the fact that $\mathfrak{m}A'$ is an ideal of definition of A' also follows from (7.5.5), since K is the residue field of A, $K \otimes_k k'$ is a finite k'-algebra. Finally $A'_i/\mathfrak{m}A'_i$ is one of the Artinian local rings whose direct product is $A' \otimes_k k'$; hence as A'_i is an A-module flat, the equality of dimensions of A and A'_i follows from (6.1.2). **IV-4** | 174

(ii) As $\widehat{\Omega}^1_{A/k}$ is, by the hypothesis on K, an A-module of finite type (18.11.1), $\widehat{\Omega}^1_{A/k} \otimes_A A'$ is complete and identifies with the completed tensor product $\widehat{\Omega}^1_{A/k}\widehat{\otimes}_A A'$; by virtue of the associativity of the completed tensor product (**0**, 7.7.4), $\widehat{\Omega}^1_{A/k} \otimes_A A' = \widehat{\Omega}^1_{A/k} \otimes_A (A\widehat{\otimes}_k k')$ identifies with $\widehat{\Omega}^1_{A/k}\widehat{\otimes}_k k'$. But $\Omega^1_{A/k} \otimes_k k'$ identifies with $\Omega^1_{A''/k'}$, where $A'' = A \otimes_k k'$ (**0**, 20.5.5), and as A' is by definition the separated completion of A'', the separated completion $\widehat{\Omega}^1_{A''/k'}$ identifies by construction to $\widehat{\Omega}^1_{A'/k'}$ (**0**, 20.7.4); furthermore, if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); finally, it follows that the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, and it suffices to use (**0**, 20.4.5).

The conclusion of (ii) follows now from the fact that $\Omega^1_{A'/k'}$ is a direct sum of the $\Omega^1_{A'_i/k'}$ (**0**, 20.4.13), and if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies therefore with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, hence one has a canonical isomorphism

$$\widehat{\Omega}^1_{A'/k'} \xrightarrow{\sim} \widehat{\Omega}^1_{A/k} \otimes_A A'.$$

□

Proposition (18.11.5). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal $\mathfrak{m}$, such that there exists a unique minimal prime ideal $\mathfrak{q} \subset \mathfrak{p}$ for which $\dim(A/\mathfrak{q}) = \dim(A)$ (which will in particular hold when A is equidimensional), m an integer $\geqslant 0$. The following conditions are equivalent:*

label=a) *The* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *admits a system of* m *generators.*
lbbel=b) *There exists a local* k*-homomorphism* $u : \mathrm{B} \to \mathrm{A}$*, where* $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_m]]$*, making* A *a* B*-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ *is étale at the point* $\mathfrak{p}$*.*

Proof. To prove that *b*) implies *a*), let us note that by virtue of the lemma (18.11.3.5), on has the exact sequence

$$\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A} \longrightarrow \widehat{\Omega}^1_{\mathrm{A}/k} \longrightarrow \Omega^1_{\mathrm{A}/\mathrm{B}} \longrightarrow 0$$

since A is a B-algebra finite; localising at $\mathfrak{p}$ and noting that by hypothesis $(\Omega^1_{\mathrm{A}/\mathrm{B}})_{\mathfrak{p}} = 0$ (17.4.1), one obtains a surjective homomorphism $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$, and the conclusion follows that **IV-4 | 175** $\widehat{\Omega}^1_{\mathrm{B}/k}$ is a B-module of rank m (**0**, 21.9.3).

To prove that *a*) implies *b*), let us first prove the following lemmas. □

Lemma (18.11.5.1). — *Let* k *be a field,* A *a* k*-algebra, local Noetherian and complete, whose residue field is a finite extension of* k*,* $\mathfrak{p}$ *a prime ideal of* A*,* $\mathfrak{q}$ *a minimal prime ideal of* A *contained in* $\mathfrak{p}$*. Then the minimum number of generators of the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *is at least equal to* $\dim(\mathrm{A}/\mathfrak{q})$*.*

Proof. Set $n = \dim(\mathrm{A}/\mathfrak{q})$, and let m be the minimum number of generators of the $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$, which is equal to $\mathrm{rg}_{k(\mathfrak{p})}((\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} k(\mathfrak{p}))$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 4). Note that $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{q}} = (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} \mathrm{A}_{\mathfrak{q}}$, hence the minimum number of generators of the $\mathrm{A}_{\mathfrak{q}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{q}}$ is *at most* equal to m. It suffices therefore to consider the case where $\mathfrak{p} = \mathfrak{q}$ is minimal. In second place, let us show one can restrict to the case where k is *algebraically closed*. Indeed, let k' be an algebraic closure of k, and set $\mathrm{A}' = \mathrm{A}\widehat{\otimes}_k k'$, which is composed as a direct product of k'-local algebras A'_i $(1 \leqslant i \leqslant r)$, whose residue fields are isomorphic to k', and which are A-modules faithfully flat (18.11.4). Applying (2.3.4) and (6.1.1), one sees that in a given A'_i, there exists a minimal prime ideal $\mathfrak{q}'_i$ above $\mathfrak{q}$ such that $n' = \dim(\mathrm{A}'_i/\mathfrak{q}'_i) \geqslant \dim(\mathrm{A}/\mathfrak{q}) = n$. On the other hand by (18.11.4), the minimum m' of generators of the $(\mathrm{A}'_i)_{\mathfrak{q}'_i}$-module $(\widehat{\Omega}^1_{\mathrm{A}'_i/k'})_{\mathfrak{q}'_i}$ is *at most* equal to m; whence our assertion.

Suppose therefore k algebraically closed and $\mathfrak{q} = 0$. In the case where $\mathfrak{q} = 0$, one can moreover restrict to the case where A is integral and K separable over the algebraically closed field k, hence A is a geometrically regular algebra (6.7.6); one can therefore apply (18.11.3) for $\mathfrak{p} = 0$ since $k = k^p$ and in this case $m = n$. □

Lemma (18.11.5.2). — *Let* k *be a field,* A *a* k*-algebra, local Noetherian and complete, whose residue field is a finite extension of* k*,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{p} \subset \mathfrak{m}$ *a prime ideal of* A*. Let* m *be the minimum number of generators of the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$*, and let* $(x_i)_{1 \leqslant i \leqslant m}$ *be a family of elements of* $\mathfrak{m}$ *not belonging to* $\mathfrak{p}$*. Then there exist* m *invertible elements* u_i $(1 \leqslant i \leqslant m)$ *of* A *such that if* $y_i = u_i x_i$*, the canonical images of the* dy_i *in* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *generate this* $\mathrm{A}_{\mathfrak{p}}$*-module.*

Proof. For all $x \in \mathrm{A}$, denote by $\delta(x)$ the canonical image of dx $(= d_{\mathrm{A}/k}x)$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}} \otimes_{\mathrm{A}_{\mathfrak{p}}} k(\mathfrak{p}) = \widehat{\Omega}^1_{\mathrm{A}/k} \otimes_{\mathrm{A}} k(\mathfrak{p})$; as this is a $k(\mathfrak{p})$-vector space of dimension m by hypothesis, it suffices (by virtue of the Nakayama lemma) to prove that one can determine the u_i such that the $\delta(u_i x_i)$ form a free system. We reason by induction and suppose for an integer $r < m$, we have determined the u_i $(1 \leqslant i \leqslant r)$ such that the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$ are linearly independent over $k(\mathfrak{p})$; if $\delta(x_{r+1})$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, one takes $u_{r+1} = 1$ to continue the induction. In the contrary case, note that for $u \in \mathrm{A}$, $\delta(u x_{r+1})$ is the canonical image of $(du)x_{r+1} + u(dx_{r+1})$; as $u(dx_{r+1})$ has a canonical image which is a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, there exists **IV-4 | 176** therefore $z \in \mathrm{A}$ such that $\delta(z)$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$; if $z \notin \mathfrak{m}$, one takes $u = z$; otherwise, $1 + z$ is invertible and $d(1+z) = d(z)$ since $(du)x_{r+1} = \delta(z)$; one takes then $u = 1 + z$, which finishes the proof of (18.11.5.2). □

Proof of (18.11.5), *continued.* We now return to the proof of the implication *a*) ⇒ *b*) in (18.11.5). Let us first note that since $\mathfrak{p} \neq \mathfrak{m}$, there exists a system of parameters $(x_i)_{1 \leqslant i \leqslant n}$ of A (with $n = \dim(\mathrm{A})$) such that $x_i \notin \mathfrak{p}$ for all i, without which $\mathrm{A}/\mathfrak{p}$ would be of finite length, and $\mathfrak{p}$ would be maximal, contrary to the hypothesis. But if for example $x_1 \notin \mathfrak{p}$, it suffices, for each index i such that $x_i \in \mathfrak{p}$, to replace x_i by $x_1 + x_i$, to have a system of parameters none of whose elements belong to $\mathfrak{p}$. The relation $\dim(A/\mathfrak{q}) = n$ implies, by virtue of (18.11.5.1), that $m \geqslant n$; one can therefore consider a family $(x_i)_{1 \leqslant i \leqslant m}$ of elements $x_i \notin \mathfrak{p}$, whose n first form a system of parameters of A. Multiplying furthermore the x_i by invertible elements u_i of A, one can, by (18.11.5.2), suppose that the images of dx_i $(1 \leqslant i \leqslant m)$ in $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ generate this $A_{\mathfrak{p}}$-module, and the multiplication by the u_i has not altered the fact that the x_i for $1 \leqslant i \leqslant n$ form a system of parameters. We shall consider the local k-homomorphism $u : B \to A$ such that $u(T_i) = x_i$ for $1 \leqslant i \leqslant m$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); as the x_i generate an ideal of definition of A, it follows from (**0**, 7.4.4) and (7.4.3) that u makes A a B-algebra *finite*.

But the $d_A x_i$ are the canonical images by v of the elements $d_B T_i \otimes 1$ (**0**, 20.5.2.6). If one localises the exact sequence preceding at $\mathfrak{p}$, we see therefore that $v_{\mathfrak{p}} : (\widehat{\Omega}^1_{B/k} \otimes_B A)_{\mathfrak{p}} \to (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is *surjective*, and consequently $0 = (\Omega^1_{A/B})_{\mathfrak{p}} = \Omega^1_{A_{\mathfrak{p}}/B_{\mathfrak{r}}}$ (where $\mathfrak{r}$ is the inverse image of $\mathfrak{p}$ in A, cf. (16.4.15)), thus the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is not ramified at $\mathfrak{p}$ by (17.4.1), and the property *b*) of (18.11.5) follows. □

Remark (18.11.6). — In the statement of (18.11.5), one can drop the hypothesis $\mathfrak{p} \neq \mathfrak{m}$. Indeed, for every local k-homomorphism $u : B \to A$, where $B = k[[T_1, \dots, T_r]]$ (r any integer) making A a B-algebra finite, $\mathfrak{m}$ is the only point of Spec(A) above the maximal ideal $\mathfrak{n}$ of B; the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ cannot be unramified at $\mathfrak{m}$ unless $A/\mathfrak{n}A$ is a field, an extension separable of k (17.4.1), which implies that the residue field K of A is a separable extension of k. If this condition is not satisfied, the conclusion of (18.11.5) can never be satisfied for $\mathfrak{p} = \mathfrak{m}$, whatever the integer m.

Corollary (18.11.7). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, equidimensional, whose residue field is a finite extension of k,* $\mathfrak{p}$ *a prime ideal of* A. *Then the minimum number of generators of the* $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *is at least equal to* $\dim(A)$.

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

Corollary (18.11.8). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, integral, whose residue field is a finite extension of k. If* K *is the field of fractions of* A, *one has* $\operatorname{rg}_K((\widehat{\Omega}^1_{A/k}) \otimes_A K) \geqslant \dim(A)$.

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

IV-4 | 177

Corollary (18.11.9). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, of dimension n, whose residue field is a finite extension of k. Let* $\mathfrak{p}$ *be a prime ideal of* A *distinct from the maximal ideal, and containing a minimal prime ideal* $\mathfrak{q}$ *such that* $\dim(A/\mathfrak{q}) = n$. *Suppose that the* $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *admits n generators. Then:*

label=(i) The $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *is free of rank n.*
lbbel=(ii) There exists a local k-homomorphism $u : B \to A$, *where* $B = k[[T_1, \dots, T_n]]$, *making* A *a* B*-algebra finite, and such that the corresponding morphism* $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ *is étale at the point* $\mathfrak{p}$.
lcbel=(iii) The k-algebra $A_{\mathfrak{p}}$ *is geometrically regular over k.*

Proof. Let us prove first of all (ii); by virtue of (18.11.5), there exists a local k-homomorphism $u : B = k[[T_1, \dots, T_n]] \to A$ making A a B-algebra finite, and such that the corresponding morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is not ramified at the point $\mathfrak{p}$; set $\mathfrak{r} = u^{-1}(\mathfrak{p})$. The hypothesis on $\mathfrak{q} \subset \mathfrak{p}$ and the fact that A is a quotient ring of a regular ring (**0**, 19.8.8) imply that $\dim(A_{\mathfrak{p}}) = n - \dim(A/\mathfrak{p})$ (**0**, 16.5.12). Similarly $\dim(B_{\mathfrak{r}}) = n - \dim(B/\mathfrak{r})$. Finally, since the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is not ramified at the point $\mathfrak{p}$, the fibre of this morphism at the point $\mathfrak{r}$ is of dimension 0, hence (**0**, 16.3.9) $\dim(A/\mathfrak{p}) \leqslant \dim(B/\mathfrak{r})$ and consequently $\dim(B_{\mathfrak{r}}) \leqslant \dim(A_{\mathfrak{p}})$. We conclude therefore from (18.10.1) that the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is étale at $\mathfrak{p}$.

The assertion (iii) follows from the fact that $A_{\mathfrak{p}}$ is formally smooth and that $B_{\mathfrak{r}}$ is formally smooth on k for their preadique topologies (**0**, 19.3.4) and (19.3.5)), hence $A_{\mathfrak{p}}$ is formally smooth on k for its preadique topology, and consequently geometrically regular over k (**0**, 22.5.8).

Let us finally prove (i). Let k' be an algebraically closed extension of k, and let us consider the k'-algebra semi-local $A' = A \widehat{\otimes}_k k'$. As a B-module of finite type via u, A is considered as a B-algebra finite; since the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is finite and étale at every point $\mathfrak{p}'$ above $\mathfrak{p}$ (17.3.3); furthermore, A' is composed as a direct product of local rings A'_i of dimension

n (18.11.4) and $\mathfrak{p}'$ identifies with a prime ideal $\mathfrak{p}'_i$ of one of the A'_i. The same reasoning as above proves that $(A'_i)_{\mathfrak{p}'_i}$ is geometrically regular over k', and as k' is perfect, the fact d) implies b), hence by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of finite type. But $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{p}'_i} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} (A'_i)_{\mathfrak{p}'_i}$ (18.11.4) and $(A'_i)_{\mathfrak{p}'_i}$ is a faithfully flat $A_{\mathfrak{p}}$-module; one sees that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module *free* of finite type (2.5.2); this finishes the proof. □

Theorem (18.11.10). — *Let k be a field of characteristic exponent p, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal. The following conditions are equivalent:* IV-4 | 178

label=a) For every extension k' of k, and every prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, $A_{\mathfrak{p}'}$ is a regular ring.

a') There exists a perfect extension k' of k and a prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, such that $A_{\mathfrak{p}'}$ is regular.

lbbel=b) Let n be the largest of the dimensions of the irreducible components of $\operatorname{Spec}(A)$ *to which $\mathfrak{p}$ belongs; then $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n.*

Suppose furthermore that $n = \dim(A)$ (which will be the case if A is equidimensional); then the preceding conditions are also equivalent to:

lcbel=c) There exists a local k-homomorphism $u : B \to A$, where $B = k[[T_1, \dots, T_n]]$, making A a B-algebra finite, and such that the corresponding morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ *is étale at the point $\mathfrak{p}$.*

ldbel=d) The ring $A_{\mathfrak{p}}$ is geometrically regular over k.

Furthermore, if $[k : k^p] < +\infty$, the condition d) is equivalent to a'), b), and c).

Proof. The fact that d) implies b) when $[k : k^p] < +\infty$ is nothing other than (18.11.3). It is clear that a) implies a'). Let us show c) implies a) when $\dim(A) = n$. One says that A is an A-module flat (18.11.4) and $\dim(A') = \dim(A) = n$; it follows from (2.3.4) and (6.1.1) that there exists a minimal prime ideal $\mathfrak{q}'$ of A' contained in $\mathfrak{p}'$, above $\mathfrak{q}$ and such that $\dim(A'/\mathfrak{q}') = \dim(A/\mathfrak{q}) = n$. Furthermore, by (18.11.4), $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A'_{\mathfrak{p}'}$; as k' is perfect, the hypothesis implies that it is geometrically regular over k', one can apply to A' and $\mathfrak{p}'$ the fact d) implies b), hence $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'}$ is an $A_{\mathfrak{p}'}$-module free of rank n; by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n. This shows the equivalence of a), a'), b), and c) when $\dim(A) = n$. It remains to prove that a), a'), and b) are still equivalent in the general case. Let $\mathfrak{J}$ be the kernel of A, the noyau of the canonical homomorphism $A \to A_{\mathfrak{p}}$, and set $A_1 = A/\mathfrak{J}$; we necessarily have $\mathfrak{J} \subset \mathfrak{p}$, and if we set $\mathfrak{p}_1 = \mathfrak{p}/\mathfrak{J}$, the canonical homomorphism $A_{\mathfrak{p}} \to (A_1)_{\mathfrak{p}_1}$ is *bijective*; it follows from (**I**, 6.5.4) that the canonical injection $\operatorname{Spec}(A_1) \to \operatorname{Spec}(A)$ is an *isomorphism local* at the point $\mathfrak{p}_1$, and one has $\mathfrak{J}_{\mathfrak{p}} = 0$. One sees as in (18.11.3) that the sequence is exact, and that one concludes that the condition a) (resp. a'))) for the ring A and the prime ideal $\mathfrak{p}$ is *equivalent* to the condition a) (resp. a'))) for the ring A_1 and the prime ideal $\mathfrak{p}_1$. Now, all the minimal prime ideals of A contained in $\mathfrak{p}$ contain $\mathfrak{J}$ since $\operatorname{Spec}(A_1) \to \operatorname{Spec}(A)$ is an isomorphism local at $\mathfrak{p}_1$; on the other hand the $\operatorname{Ass}_A(A/\mathfrak{J})$ are the prime ideals of $\operatorname{Ass}(A)$ which are contained in $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 2, prop. 6); hence all the minimal prime ideals of A_1 are contained in $\mathfrak{p}_1$, and consequently $\dim(A_1) = n$. It suffices then to apply to A_1 and $\mathfrak{p}_1$ which has been proved above. □

Remark (18.11.11). — IV-4 | 179

label=(i) The equivalence of conditions d) and b) in (18.11.10) is no longer valid when one no longer supposes that $[k : k^p] < +\infty$. Indeed, by the example of (**0**, 22.7.7), (ii)), the ring $B = A/\mathfrak{q}$ is integral and of dimension 1; on the other hand, the sequence

$$(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \xrightarrow{j} \widehat{\Omega}^1_{A/k} \otimes_A L \longrightarrow \widehat{\Omega}^1_{B/k} \otimes_B L \longrightarrow 0$$

is exact (same demonstration as in (18.11.3)), and we know that j *is not injective*, hence $j = 0$ since $(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L$ is of rank 1. We conclude that $\operatorname{rg}_L(\widehat{\Omega}^1_{B/k} \otimes_B L) = 2$; as $B_{\mathfrak{p}}$ is a k-algebra geometrically regular, one sees that the condition d) does not imply here the condition b).

lbbel=(ii) With the notations of (18.11.10) and supposing that $n = \dim(A)$ and $(x_i)_{1 \leqslant i \leqslant n}$ a system of parameters of A not belonging to $\mathfrak{p}$; one deduces a local k-homomorphism $u : B \to A$

such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant n$, making A a B-algebra finite. For the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ to be *étale* at the point $\mathfrak{p}$, it is necessary and sufficient that the images of $d_{\mathrm{A}}x_i$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ form a *system of generators* of this $\mathrm{A}_{\mathfrak{p}}$-module. Indeed, as one has seen in the proof of (18.11.5) that if the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, the canonical homomorphism $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is surjective and the images of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$, which generate $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}}$, are the $d_{\mathrm{A}}x_i$; whence the *necessity* of the condition. Conversely, the same reasoning shows that if this condition is satisfied, the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, and the reasoning of (18.11.9) proves in fact that this morphism is étale at $\mathfrak{p}$.

Corollary (18.11.12). — *Let k be a field of characteristic exponent p such that $[k : k^p] < +\infty$, A a k-algebra, local Noetherian and complete and integral, which is not a field, and whose residue field is a finite extension of k. Then there exists a finite radicial extension k' of k such that, setting $\mathrm{A}' = \mathrm{A} \otimes_k k'$, the k'-algebra $\mathrm{A}'_{\mathrm{red}}$ and the prime ideal $\mathfrak{o}$ of this algebra satisfy the equivalent conditions a), a′), b), c), and d) of (18.11.10). In particular, if $n = \dim(\mathrm{A}) = \dim(\mathrm{A}')$, there exists a k'-homomorphism local $\mathrm{B}' = k'[[\mathrm{T}_1, \ldots, \mathrm{T}_n]] \to \mathrm{A}'_{\mathrm{red}}$* **IV-4** | 180 *making $\mathrm{A}'_{\mathrm{red}}$ a B'-algebra finite, and such that the field of fractions K' of $\mathrm{A}'_{\mathrm{red}}$ is a finite separable extension of the field of fractions $k'((\mathrm{T}_1, \ldots, \mathrm{T}_n))$ of B'.*

Proof. The morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ being complete, radicial, and finite, A' is a local ring, complete and its nilradical is the only prime ideal above $\mathfrak{o}$ of A; by flatness, A' identifies with a sub-ring of $\mathrm{K} \otimes_k k'$, which is also the total ring of fractions of A'; on has therefore $\mathrm{K}' = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$. It is a question of proving, by the equivalence $d) \Leftrightarrow c)$, that there exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'. Or one knows (**0**, 19.8.9) that, under all the hypotheses made, there exists a sub-k-algebra $\mathrm{C} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$ of A such that A is a finite C-algebra; hence, K is a *finite* extension of the field of fractions $\mathrm{K}_1 = k((\mathrm{T}_1, \ldots, \mathrm{T}_n))$ of C, which is *separable* on k (**0**, 21.9.6.4). If p is the characteristic exponent of k, one can therefore write $\mathrm{K} \otimes_k k^{p^{-\infty}} = \mathrm{K} \otimes_{k_1} (\mathrm{K}_1 \otimes_k k^{p^{-\infty}})$ and $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$ is a *field*, a radicial extension of K_1; on concludes that $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is a finite algebra over the field $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$, hence is *Artinian*. The conclusion follows therefore from the following more general lemma: □

Lemma (18.11.12.1). — *Let k be a field of characteristic $p > 0$, K an extension of k. The following conditions are equivalent:*

label=a) The ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.
lbbel=b) There exists an extension k' of k of finite type such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a k'-algebra separable.
lcbel=c) There exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'.

Proof. Let us first show that $c)$ implies $a)$. The ring $\mathrm{A} = \mathrm{K} \otimes_k k'$ is then a local Artinian ring, and if $\mathfrak{N}$ is nilradical, the residue field $\mathrm{L} = \mathrm{A}/\mathfrak{N}$ is separable over k'; hence $\mathrm{L} \otimes_{k'} k^{p^{-\infty}}$ is separable over k', and as it is equal to $(\mathrm{A} \otimes_{k'} k^{p^{-\infty}})/(\mathfrak{N} \otimes_{k'} k^{p^{-\infty}})$, one sees that $\mathfrak{N} \otimes_{k'} k^{p^{-\infty}}$ is the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$; as $\mathfrak{N}$ is a nilpotent ideal of finite type, the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is of finite type, hence the ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.

Conversely, let us prove that $a)$ implies $c)$. Let $\mathfrak{N}$ be the nilradical of the local Artinian ring $\mathrm{B} = \mathrm{K} \otimes_k k^{p^{-\infty}}$, which is by hypothesis generated by a finite number d of elements of the form $y_i = \sum_j x_{ij} \otimes \xi_{ij}$, $x_{ij} \in \mathrm{K}$, $\xi_{ij} \in k^{p^{-\infty}}$. Let k' be the finite radicial extension of k generated by the ξ_{ij}, $\mathfrak{N}_0$ the ideal of $\mathrm{B}_0 = \mathrm{K} \otimes_k k'$ generated by the y_i; it is clear that $\mathfrak{N}_0 \otimes_{k'} k^{p^{-\infty}} = \mathfrak{N}$, and hence $\mathfrak{N} \cap \mathrm{B}_0 = \mathfrak{N}_0$, and so it is equal to $\mathrm{B}_0/\mathfrak{N}_0 = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is separable over k (4.6.1).

It is clear that $c)$ implies $b)$. Conversely, suppose $b)$ is satisfied, and let us note that there exists a separable extension k_1 of k such that k' is a finite radicial extension of k_1; set $\mathrm{K}_1 = \mathrm{K} \otimes_k k_1$, which is a field. Applying the equivalence of $a)$ and $c)$ to the extension K_1, on sees that $\mathrm{K}_1 \otimes_{k_1} k_1^{p^{-\infty}}$ is also **IV-4** | 181 Artinian (Bourbaki, *Alg. comm.*, chap. I, § 3, n° 5, cor. of prop. 8); we have thus shown that $b)$ implies $a)$, which finishes the proof. □

18.12. Applications of étale localisation to quasi-finite morphisms The results of this section were communicated by P. Deligne.

Theorem (18.12.1). — *Let $f : X \to Y$ be a locally of finite type morphism, x a point of X, $y = f(x)$. Suppose that x is an isolated point of the space $f^{-1}(y)$. Then there exist an étale morphism $Y' \to Y$, a point x' of $X' = X \times_Y Y'$ above x and a neighbourhood V' of x' in X' such that, if $f' = f_{(Y')} : X' \to Y'$, $f'|V'$ is a finite morphism. Furthermore V' is both open and closed, and X' is therefore the sum of two sub-preschemes induced on open subsets of X', of which one is finite on Y' and contains x'.*

Proof. The last assertion follows from the fact that, if f is separated, it is the same for f'; hence, if $j' : V' \to X'$ is the injection canonical, and the fact that f' is finite implies that it is the same for j' (**II**, 6.1.5), and $V' = j'(V')$ is closed in X' (**II**, 6.1.10).

The question being local on X, one can suppose $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ affine, B being an A-algebra of finite type, and one can then suppose $f^{-1}(y) = \{x\}$. Hence the question is local on Y, hence of the form $C/\mathfrak{J}$, where $C = A[T_1, \ldots, T_r]$ and $\mathfrak{J}$ is an ideal of type finite of E; we then have $A' = E'/\mathfrak{J}E'$, where $E' = C'[T_1, \ldots, T_r]$. Let $(\mathfrak{J}_\lambda)$ be the family of ideals of type fine of C contained in $\mathfrak{J}$, and denote the corresponding $X_\lambda = \operatorname{Spec}(C/\mathfrak{J}_\lambda)$ considered as closed sub-preschemes of $Z = \operatorname{Spec}(C)$, so that $\mathfrak{J}$ is the filtered union of the $\mathfrak{J}_\lambda$. If X and the $X_\lambda = \operatorname{Spec}(C/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes, $X = \bigcap X_\lambda$; if $f_\lambda : X_\lambda \to Y$ is the structural morphism, and as the sets $f_\lambda^{-1}(y)$ are closed in the Noetherian space $Z_y = \operatorname{Spec}(k(y)[T_1, \ldots, T_r])$, there exists an index λ such that $f_\lambda^{-1}(y) = f^{-1}(y) = \{x\}$. One can then suppose that for a given X_λ, the condition holds that X is at point x; if one proves the proposition for such an X_λ, it will follow for X, since x' is a point above x in $Z' = Z \times_Y Y'$, $X'_\lambda = p^{-1}(X_\lambda)$, where $p : Z' \to Z$ is the projection (from the restriction of f_λ to X_λ), the same will hold for the open set V'_λ of x' in X'_λ which will be finite over Y'. Furthermore V' and X' are sums of two open sub-preschemes, V' and X'_λ; one answers the question by taking V' as a sum of U' and V'. In particular, note that the set of isolated points of X in their fibre et open in X (13.1.4), hence f is *quasi-finite*. Set $A'' = \mathscr{O}_{Y,y}$, and set $Y'' = \operatorname{Spec}(A'')$. Set $X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \to Y''$; $X'' \to Y''$ is *separated*, quasi-finite and of presentation finite; A'' is henselian, for every $x'' \in X''$ above x an open and closed neighbourhood V'' of x'' in X'' which is finite on Y'' (18.5.11), *c*)); as the immersion $V'' \to X''$ is open and closed, it is also quasi-compact, hence $f''|V''$ is of presentation finite since $V'' \to X''$ is open, and $X'' \to Y''$ is of presentation finite. f'' is finite and quasi-compact over Y'' (1.6.2), hence $f''|V''$ is of presentation finite.

Since, by definition, A'' is the inductive limit of a filtered inductive family (A_λ) of $\mathscr{O}_{Y,y}$-algebras strictly étales ((18.6.5)) and (8.1.2), *a*))); also B_{M_λ} can be assumed to be a $\mathscr{O}_{Y,y}$-algebra of presentation finite, and it follows from (18.6.1) and (8.1.2), *a*)) and the result of (17.7.8) that B_{M_λ} is a $\mathscr{O}_{Y,y}$-algebra étale for all λ; Namely, Y'' is the inductive limit projective of a family (Y_α) of affine Y-schemes étale. If x''_α is the projection of x'' on $X'_\alpha = X \times_Y Y_\alpha$, on can suppose that $X'_\alpha \cap f'^{-1}(U)$ where V'_α is a neighbourhood of x''_α in X'_α, is finite on Y_α (8.2.11). Finally, V'' being of presentation finite over Y'', on concludes from (8.10.5), (x)) that for a suitable α, V'_α is finite over Y'_α. □

Remark (18.12.2). — In the preceding proof, one sees (taking into account (18.6.2)) that one has constructed Y' answering the question, and in addition such that if $y' = f'(X')$, the homomorphism $k(y) \to k(y')$ is bijective.

Corollary (18.12.3). — *Let $f : X \to Y$ be a locally of finite type morphism and separated, y a point of Y such that the underlying space $f^{-1}(y)$ is finite and discrete; in particular, a proper and quasi-finite morphism. Then there exist an étale morphism $Y' \to Y$, a point $y' \in Y'$ above y such that $k(y') = k(y)$, and a decomposition of $X' = X \times_Y Y'$ as a sum of two sub-preschemes X'_1, X'_2 induced on open subsets of X', such that the restriction $f' = f_{(Y')} : X'_1 \to Y'$ is a finite morphism and that $X'_2 \cap f'^{-1}(y') = 0$.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a closed morphism, hence, since X'_2 is closed in X', $f'(X'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(U') = X \times_Y U'$ is finite on Y', and the morphism $U' \to Y$ is étale; the morphism $f'^{-1}(U') \to U'$ is faithfully flat and quasi-compact, hence the conclusion. □ **IV-4** | 182

Corollary (18.12.4). — *Let $f : X \to Y$ be a finite morphism, it is necessary and sufficient that it be separated, universally closed and, for all $y \in Y$, $f^{-1}(y)$ is a finite and discrete space; in particular, a proper and quasi-finite morphism.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a closed morphism, hence, since X'_2 is closed in X', $f'(X'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(U') = X \times_Y U'$ is finite over Y', and the morphism $U' \to Y$ is étale; the morphism $U' \to Y$ is faithfully flat and quasi-compact, hence by (2.7.1), (xv)), the restriction of f to X_λ, being a finite morphism (**II**, 6.1.10), hence the conclusion results from (11.3.1), since f restricted is finite and quasi-compact. □

Remark (18.12.5). — One notes that in the proof of (18.12.4) one has not used the hypothesis (18.12.4) in its general form, but only that $f_{(Y')}$ is a universally closed morphism for every étale morphism $Y' \to Y$.

Corollary (18.12.6). — *Let $f : X \to Y$ be a morphism of preschemes. The following conditions are equivalent:*

label=a) f is a closed immersion.

lbbel=b) f is a proper monomorphism.

lcbel=c) f is proper and, for all $y \in Y$, the $k(y)$-prescheme $X_y = f^{-1}(y)$ is radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$).

Proof. This is deduced from (18.12.4) as (18.11.5) is deduced from (8.11.1). □

Proposition (18.12.7). — *Let $f : X \to Y$ be a locally of finite type morphism, y a point of Y. For there to exist a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it is necessary and sufficient that X_y be a $k(y)$-prescheme radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$) and that there exist a neighbourhood V of y such that the restriction $f^{-1}(V) \to V$ of f is a universally closed morphism.*

Proof. The conditions being evidently necessary, let us prove that they are sufficient. One can, by the second condition, already suppose f universally closed. Let us show, moreover, hence separated; this follows from the following lemma: □

Lemma (18.12.7.1). — *Let $f : X \to Y$ be a closed morphism, $y \in Y$ a point such that every neighbourhood of X_y contains an affine neighbourhood of X_y (condition always satisfied when X_y is empty or reduced to a single point). Then there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is an affine morphism.*

Proof. Indeed, let U_0 be an affine open neighbourhood of y in Y and let V be a neighbourhood of X_y contained in $f^{-1}(U_0)$ such that $f^{-1}(U) \subset V$, which is affine. As f is closed and V open, there exists a neighbourhood $U \subset U_0$ of y such that $f^{-1}(U) \subset V$. As the restriction $f^{-1}(U) \to U$ is also an affine morphism (**II**, 1.2.5). □

Suppose therefore f affine and universally closed, and let us show that for all $y \in Y$ such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it suffices to prove that f is quasi-finite (18.12.4), and since X_y is empty or reduced to a single point, f is quasi-finite (18.12.4), taking into account (18.12.6).

Proposition (18.12.8). — *Let $f : X \to Y$ be a morphism of preschemes. For f to be entire, it is necessary and sufficient that f be affine and universally closed.*

Proof. We can suppose $X = \mathrm{Spec}(B)$, and it suffices to prove that every element $b \in B$ is entire on A (**II**, 6.1.1). Let B' be the sub-A-algebra of B generated by b, which is an A-algebra of finite type; set $X' = \mathrm{Spec}(B')$, so that $f : X \to Y$ factorises as $X \xrightarrow{g} X' \xrightarrow{h} Y$, where h is of finite type, and g is *dominant* since the homomorphism $B' \to B$ is injective (**I**, 1.2.7). As h is separated and $h \circ g$ universally closed, g is universally closed (**II**, 5.4.3) and (5.4.9)); hence h is also universally closed, therefore proper since it is of finite type and separated. But then, for all $y \in Y$, the morphism $h^{-1}(y) \to \mathrm{Spec}(k(y))$ deduced from h is proper and affine, hence *finite* (**III**, 4.4.2), in other words h is *quasi-finite*; one deduces therefore from (18.12.4) that h is finite, and consequently b is entire on A. □ **IV-4** | 183

Remark (18.12.9). — One can say that a morphism $f : X \to Y$ is entire when one supposes only separated, universally closed, and such that for all $y \in Y$, the morphism $f^{-1}(y) \to \operatorname{Spec}(k(y))$ deduced from f is entire. It would suffice to prove that these conditions imply that f is affine, or again that every fibre X_y is contained in an affine open neighbourhood.

Corollary (18.12.10). — *A morphism $f : X \to Y$ which is injective and universally closed is entire.*

Proof. It suffices, by (18.12.8), to prove that f is affine, which follows from the lemma (18.12.7.1) and the hypothesis. ☐

Corollary (18.12.11). — *Let $f : X \to Y$ be a morphism of preschemes (resp. a locally of finite type morphism). For f to be a universal homeomorphism (2.4.2), it is necessary and sufficient that f be entire (resp. finite), radicial, and surjective.*

Proof. One knows that the conditions are sufficient (2.4.5); they are necessary by virtue of (18.12.10) and (18.12.4). The last proposition improves (8.11.2). ☐

Proposition (18.12.12). — *Every morphism $f : X \to Y$ which is quasi-finite and separated is quasi-affine.*

Proof. Set $\mathscr{A} = f_*(\mathscr{O}_X)$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent and separated (1.7.4); and $Z = \operatorname{Spec}(\mathscr{A})$ (**II**, 1.3.1), so that f factorises canonically as $X \xrightarrow{g} Z \xrightarrow{h} Y$, h being affine and g corresponding to the automorphism identical to $\mathscr{A}$ (**II**, 1.2.7); it suffices to prove that g is an *open immersion*, or, what comes to the same (17.9.1), that the morphism g is étale and *radicial*. It suffices evidently to prove that, for all $y \in Y$, the morphism g is étale and radicial at each point of $f^{-1}(y)$. One can apply the result to X, f the restriction and Y by X, Y, and the sections of f of f; hence f is quasi-compact and separated, and the fact that f is étale and radicial at each point of $f^{-1}(y)$ is part of (18.10.2), (iii)), quasi-separable, from X, f, and g as in (18.12.3), whose X' is the extension $Z' = Z \times_Y Y'$ a $\mathscr{O}_{Y'}$-Module $\mathscr{O}_{X'}/\mathfrak{J}E'$, so that as X and Y are both Y-schemes étale (17.3.2); hence by (**II**, 1.2.7) f factorises as $X \xrightarrow{g} Z \xrightarrow{h} Y$, where g is the canonical injection from the identity on $\mathscr{A}$, and g is an *open immersion*, and g is *radicial*, whence the result. ☐

Corollary (18.12.13). — *("Main Theorem" of Zariski). Let Y be a prescheme quasi-compact and quasi-separated. Then there exists a factorisation of f*

$$X \xrightarrow{g} Z \xrightarrow{u} Y$$

where g is an open immersion (necessarily quasi-compact) and u a finite morphism.

Proof. Indeed, it follows from (18.12.12) that f is quasi-affine, hence quasi-projective. One can therefore apply (8.12.8), where the hypothesis on $\mathscr{O}_Y$-Module ample can in fact be replaced by the sole hypothesis that Y is quasi-compact and quasi-separated: indeed, it follows from (8.12.3) that the existence of the factorisation announced is a *local* property on Y, and it suffices to prove it when Y is affine. ☐

Remark (18.12.14). — One can give a proof of (18.12.13) analogous to that of (18.12.12) (but not using (8.12.3), hence (18.5.11)), which itself uses the "Main Theorem" under its local form (8.12.9)). Keeping indeed the notations of the proof of (18.12.12), and let $\mathscr{B}$ be the $\mathscr{O}_Y$-Algebra quasi-coherente, integral closure of $\mathscr{O}_Y$ in $\mathscr{A}$ (**II**, 6.3.2). If one sets $T = \operatorname{Spec}(\mathscr{B})$, it suffices, by virtue of (8.12.3), to prove that $g : X \to T$ corresponding to the canonical injection $\mathscr{B} \to \mathscr{A}$ is an immersion, and radicial. With the notations of (18.12.12), and one can suppose that $Y = \operatorname{Spec}(C)$ and $Y' = \operatorname{Spec}(C')$ are affine, C' being a C-algebra étale (17.3.2); hence $\mathscr{A} = \overline{A}$, where A is a C-algebra, B the integral closure of C in A. The algebra A' is a C'-algebra, $A' = A \otimes_C C'$, and $\mathscr{B} = \overline{B}$, where B is the integral closure of C in A. As $Z' = Z \times_Y Y'$ and the X'_λ and X_λ are étale over X, the conclusion follows in this case.

When C is Noetherian, the demonstration of (6.14.4) no longer requires the use of (6.14.1). Indeed, the reductions successives of the proof of (6.14.4) lead back, without using (6.14.1), to the case where C is *integral*, A the field of fractions of C, and the morphism $\operatorname{Spec}(B') \to \operatorname{Spec}(B)$ is étale, it follows from (17.5.7) that B' is a normal ring, and the reasoning terminates by making appeal only to the easy lemma (6.14.1), and not to the difficult part of the proof of (6.14.1).

Proposition (18.12.15). — *Let* G *be a ring,* C′ *a* C*-algebra étale,* A *a* C*-algebra,* B *the integral closure of* C *in* A*; on sets* $A' = A \otimes_C C'$, $B' = B \otimes_C C'$, B′ *identifying to a sub-algebra of* A′*; then* B′ *is the integral closure of* C′ *in* A′*.*

Proof. Considering the filtered increasing family of sub-C-algebras of type finite of A and using the fact that A is a C-algebra of type fine, hence of the form $E/\mathfrak{J}$, where $E = C[T_1, \dots, T_r]$ and $\mathfrak{J}$ is an ideal of type fine of E; we then have $A' = E'/\mathfrak{J}E'$, where $E' = C'[T_1, \dots, T_r]$. Let $(\mathfrak{J}_\lambda)$ be the increasing filtered family of the ideals of type fine of C contained in $\mathfrak{J}$ and denote B_λ the integral closure of C in $E/\mathfrak{J}_\lambda$; if B_λ is the inductive limit of the integral closures B_λ, B is the inductive limit of the B_λ, as one sees by the reasoning of (5.13.4); same reasoning for C and the $B_\lambda \otimes_C C'$ for $\beta \geqslant \alpha$. The same, A′ is the inductive limit of the integral closures of C′ in $E'/\mathfrak{J}_\lambda E'$, hence the same reasoning as the preceding shows that $B' = B \otimes_C C'$ is the integral closure of C′ in A′ once we have reduced to the case where C is Noetherian, the proposition becomes a particular case of (6.14.4). It is necessary however to observe that, when C′ is supposed *étale* on C Noetherian, the demonstration of (6.14.4) no longer requires that one use (6.14.1). Indeed, the successive reductions in the proof of (6.14.4) lead back, *without using* (6.14.1), to the case where C is *integral*, A the field of fractions of C, hence B is a normal ring and the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(B)$ is étale, it follows from (17.5.7) that B′ is a normal ring, and the reasoning terminates by making appeal only to the elementary lemma (6.14.1), and not the difficult part of the proof of (6.14.1). □

Proposition (18.12.16). — *Let* $g : Y \to S$ *be a quasi-compact morphism,* $f : X \to Y$ *a separated and quasi-finite morphism. For every* $\mathscr{O}_Y$*-Module invertible* $\mathscr{L}$*, ample relative to* S (**II**, 4.6.1)*, the* $\mathscr{O}_X$*-Module* $f^*(\mathscr{L})$ *is ample relative to* S*.*

Proof. Indeed, by virtue of (18.12.12), the morphism f is quasi-affine, hence (**II**, 5.1.6) the $\mathscr{O}_X$-Module $\mathscr{O}_X$ is ample for f. We deduce from (**II**, 4.6.13), (iii)) that $\mathscr{O}_X \otimes_{\mathscr{O}_X} f^*(\mathscr{L}^{\otimes n}) = f^*(\mathscr{L}^{\otimes n})$ is ample relative to S for n large enough. But as $f^*(\mathscr{L}^{\otimes n}) = (f^*(\mathscr{L}))^{\otimes n}$, this implies that $f^*(\mathscr{L})$ is ample relative to S (**II**, 4.6.9), (i)). □

Corollary (18.12.17). — *Let* Z *be an affine scheme,* $h : X \to Z$ *a morphism of finite type,* $\mathscr{L}$ *an* $\mathscr{O}_X$*-Module invertible. With the notations of* (**II**, 4.5.2)*, the conditions of* (**II**, 4.5.2) *(which are equivalent to saying that* $\mathscr{L}$ *is ample) are also equivalent to each of the following:*

b″) *h is separated, one has* $G(\varepsilon) = X$ *and the canonical morphism* $u : G(\varepsilon) \to \mathrm{Proj}(S)$ *has finite and discrete fibres.*

b‴) *One has* $G(\varepsilon) = X$ *and the canonical morphism* u *is radicial.*

If $Z = \mathrm{Spec}(A)$*, the ring* S *is canonically equipped with a structure of graded* A*-algebra, hence* $h = g \circ u$*. Since every radicial morphism is separated* (**I**, 1.8.7.1)*, b‴) implies b″). As the condition b) of* (**II**, 4.5.2) *implies evidently b″), it remains to see that b″) implies* $\mathscr{L}$ *ample. Since* $h = g \circ u$ *is of finite type by hypothesis, u is also quasi-finite and separated* (**I**, 6.3.4)*, hence by the hypothesis b″) implies that u is quasi-finite and separated* (**I**, 5.5.1)*. To prove that* $\mathscr{L}$ *is ample, we shall now apply* (18.12.16)*. Set* $Y = \mathrm{Proj}(S)$*. Since* X *is quasi-compact, there exists an integer* $n > 0$ *and a finite number of elements* $f_j \in S_n$ *such that the inverse images* $u^{-1}(D_+(f_j))$ *cover* X*; one can therefore consider u as a quasi-finite and separated morphism of* X *in the open* $Y' = \bigcup D_+(f_j)$ *of* Y*. Let us consider then the* $\mathscr{O}_{Y'}$*-Module invertible* $\mathscr{L}' = \mathscr{O}_Y(n) \,|\, Y'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and consequently it is the same for* $u^*(\mathscr{L}')$ (**II**, 4.4.3)*.*

$\mathscr{L}' = \mathscr{O}_Y(n) \,|\, Y'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and as* $u^*(\mathscr{L}') = \mathscr{L}^{\otimes n}$ *by* (**II**, 4.4.3)*,* $\mathscr{L}^{\otimes n}$ *is thus ample relative to* S *(this amounts to saying that it is ample* (**II**, 4.6.6)*), and consequently the same is true of* $\mathscr{L}$ (**II**, 4.5.6)*, (i)).*

We leave it to the reader to state analogous conditions equivalent to those of (**II**, 4.6.3) *for the* $\mathscr{O}_X$*-Modules relatively ample.*

§19. Regular immersions and normal flatness

19.1. Properties of regular immersions

IV-4 | 185

(19.1.1). Let $\mathscr{J}$ be a quasi-coherent Ideal of $\mathscr{O}_X$ of finite type, Y the closed sub-prescheme of X that it defines. We say that $\mathscr{J}$ is *quasi-regular* (or *quasi-regulier*) at a point z of Y if the $(\mathscr{O}_X/\mathscr{J})_z$-module $\mathscr{J}_z/\mathscr{J}_z^2$ is free and if the canonical homomorphism from the symmetric algebra $S((\mathscr{O}_X/\mathscr{J})_z; \mathscr{J}_z/\mathscr{J}_z^2)$

to the graded ring $\mathrm{gr}(\mathscr{O}_{X,z};\mathscr{J}_z) = \bigoplus_{n\geqslant 0}\mathscr{J}_z^n/\mathscr{J}_z^{n+1}$ is bijective ((**0**, 15.1.7)). We say that $\mathscr{J}$ is *regular* at the point z if, in addition, $\mathscr{O}_{X,z}$ is a flat $\mathscr{O}_{Y,z}$-module (i.e., $\mathscr{O}_{X,z}/\mathscr{J}_z$ is a flat $\mathscr{O}_{X,z}$-module). We say that $\mathscr{J}$ is a *quasi-regular* (resp. *regular*) *Ideal* of $\mathscr{O}_X$ if it is quasi-regular (resp. regular) at every point of Y.

(19.1.2). An immersion $u : Y \to X$ of preschemes is said to be *quasi-regular* (resp. *regular*) at a point $y \in Y$ if u is a *closed* immersion in a neighbourhood of y and the Ideal $\mathscr{J}$ of $\mathscr{O}_X$ which defines $u(Y)$ in such a neighbourhood is quasi-regular (resp. regular) at the point $u(y)$. We say that u is a *quasi-regular immersion* (resp. *regular immersion*) if it is quasi-regular (resp. regular) at every point of Y. The *codimension* of u at a point $y \in Y$ is, by definition, equal to the rank of the free $(\mathscr{O}_X/\mathscr{J})_{u(y)}$-module $\mathscr{J}_{u(y)}/\mathscr{J}^2_{u(y)}$.

(19.1.3). When $\mathscr{O}_{X,z}$ is a Noetherian local ring, to say that $\mathscr{J}$ is quasi-regular at z means (by (**0**, 15.1.9)) that $\mathscr{J}_z$ is generated by a quasi-regular sequence of elements (i.e., by a sequence $(f_i)_{1\leqslant i\leqslant n}$ of elements of the maximal ideal of $\mathscr{O}_{X,z}$ which is quasi-regular for $\mathscr{O}_{X,z}$); to say that $\mathscr{J}$ is regular at z means ((**0**, 15.1.10)) that $\mathscr{J}_z$ is generated by a regular sequence of elements. In addition, when X is locally Noetherian, the set of $z \in Y$ where $\mathscr{J}$ is quasi-regular (resp. regular) is *open* in Y ((**0**, 15.1.11)).

(19.1.4). Suppose that X is locally Noetherian, and let $u : Y \to X$ be a quasi-regular closed immersion (resp. regular closed immersion) defined by an Ideal $\mathscr{J}$ of $\mathscr{O}_X$. Then for every point $z \in Y$:

label=*a)* $\mathrm{codim}^*_z(Y,X) = \mathrm{rk}_z(\mathscr{J}/\mathscr{J}^2)$ ((**0**, 16.9.5));

lbbel=*b)* $\mathrm{codim}^*_z(Y,X) = \mathrm{codim}(\mathscr{O}_{X,z}/\mathscr{J}_z,\mathscr{O}_{X,z})$, i.e., is equal to the height of $\mathscr{J}_z$ in the local ring $\mathscr{O}_{X,z}$ ((**0**, 16.9.5)), ((**0**, 16.3.3)).

In particular, the function $z \mapsto \mathrm{codim}^*_z(Y,X)$ is *locally constant* on Y. When u is a regular closed immersion, $\mathrm{codim}^*_z(Y,X) = \mathrm{prof}(\mathscr{J}_z,\mathscr{O}_{X,z})$ ((**0**, 15.1.10)). In addition, $\mathrm{codim}^*_z(Y,X)$ is also equal to the length of a maximal chain of prime ideals of $\mathscr{O}_{X,z}$ containing $\mathscr{J}_z$ ((**0**, 16.3.6)). Finally, let us mention a few supplementary properties of $\mathscr{J}$:

label=*a)* the module $\mathscr{J}^n/\mathscr{J}^{n+1}$ is a locally free $(\mathscr{O}_X/\mathscr{J})$-Module of rank $\binom{m+n-1}{n}$ if $m = \mathrm{rk}(\mathscr{J}/\mathscr{J}^2)$ ((**0**, 16.9.5));

lbbel=*b)* for every quasi-coherent $\mathscr{O}_X$-Module $\mathscr{F}$, the canonical homomorphism $\mathscr{F}/\mathscr{J}\mathscr{F}\otimes_{\mathscr{O}_X/\mathscr{J}}\mathscr{J}^n/\mathscr{J}^{n+1} \to \mathscr{J}^n\mathscr{F}/\mathscr{J}^{n+1}\mathscr{F}$ is an isomorphism ((**0**, 15.1.7));

lcbel=*c)* if u is a regular closed immersion, for every locally free $\mathscr{O}_X$-Module $\mathscr{F}$ of finite rank, the canonical map $\mathscr{F}/\mathscr{J}\mathscr{F}\otimes_{\mathscr{O}_X/\mathscr{J}}\mathscr{J}^n/\mathscr{J}^{n+1} \to \mathscr{J}^n\mathscr{F}/\mathscr{J}^{n+1}\mathscr{F}$ is again an isomorphism.

We also note the following property: if $n = \mathrm{codim}^*_z(Y,X)$, then for every y in a neighbourhood of z in Y, we have $\mathrm{codim}^*_y(Y,X) \leqslant n$; since any such point z is contained in every neighbourhood of y in X, $\mathscr{J}_z/\mathscr{J}_z^2$ is a free A-module of rank $n = \mathrm{codim}^*_y(Y,X)$; it amounts to saying that $\dim(\mathrm{A}) = n$. Now, since z is a maximal point of the sub-prescheme Y defined by $\mathscr{J}$, we have $\dim(\mathscr{O}_{X,z}/\mathscr{J}_z) = 0$, so $\mathscr{J}_z$ is an *ideal of definition* of the Noetherian local ring A; in addition, the hypothesis that $\mathscr{J}$ is quasi-regular implies that $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is an $(A/\mathscr{J}_z)$-module free of rank $\binom{m+n-1}{n-1}$; if r is the length of the Artinian ring $A/\mathscr{J}_z$, the length of $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is therefore $r\binom{m+n-1}{n-1}$; the Hilbert–Samuel polynomial of A for the $\mathscr{J}_z$-preadic filtration is thus indeed of degree n, which proves the proposition ((**0**, 16.2.3)). **IV-4** | 186

By virtue of (19.1.4), we will say henceforth "codimension" instead of "transverse codimension" when it concerns a sub-prescheme Y regularly immersed in X (even when X is not locally Noetherian).

Proposition (19.1.5). —

label=(i) For an immersion $j : Y \to X$ to be open, it is necessary and sufficient that it be regular and of codimension 0 at every point.

lbbel=(ii) Let $f : Y \to X$ be a regular (resp. quasi-regular) immersion, $g : X' \to X$ a flat morphism, $Y' = Y\times_X X'$; then the morphism $f' = f_{(X')} : Y' \to X'$ is a regular (resp. quasi-regular) immersion, and for every $y' \in Y'$, if y is the projection of y' in Y, the codimension of f' at the point y' is equal to the codimension of f at the point y. Conversely, if f is an immersion and $g : X' \to X$ a faithfully flat and quasi-compact morphism, then, if f' is a quasi-regular immersion, so is f.

lcbel=(iii) *If $f : Y \to X$ and $g : Z \to Y$ are regular immersions, then $f \circ g : Z \to X$ is a regular immersion, and for every $z \in Z$, the codimension of $f \circ g$ at the point z is the sum of the codimension of g at the point z and of the codimension of f at the point $g(z)$. Moreover, the sequence of canonical homomorphisms* (16.2.7.1)

$$0 \longrightarrow g^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Z/X} \longrightarrow \mathscr{N}_{Z/Y} \longrightarrow 0$$

(19.1.5.1) *is exact, and for every $z \in Z$, there exists a neighbourhood of z in Z in which the restrictions of the homomorphisms of this sequence form a split exact sequence.*

ldbel=(iv) *Let X be a locally Noetherian prescheme, $f : Y \to X$ and $g : Z \to Y$ two immersions, z a point of Z. The following conditions are equivalent:*

label=a) *g is a regular immersion at the point z and f is a regular immersion at the point $g(z)$.*

lbbel=b) *g and $f \circ g$ are regular immersions at the point z.*

lcbel=c) *$f \circ g$ is a regular immersion at the point z, f is a regular immersion at the point $g(z)$, and the sequence of canonical homomorphisms* (19.1.5.1)*, restricted to a sufficiently small open neighbourhood of z in Z, is exact and split.*

lebel=(v) *Let X be a locally Noetherian prescheme, $f : Y \to X$, $g : Z \to Y$ two immersions, z a point of Z; suppose that $y = g(z)$ is a regular point of Y and $x = f(g(z))$ a regular point of X (so that, by* (19.1.1)*, f is a regular immersion at the point $g(z)$); then, for $f \circ g$ to be a regular immersion at the point z, it is necessary and sufficient that g be a regular immersion at the point z.* **IV-4** | 187

Proof. Assertion (i) is another way of expressing (16.1.10). To prove the first assertion of (ii), we can restrict ourselves to the case where Y is a closed sub-prescheme of X defined by a regular (resp. quasi-regular) Ideal $\mathscr{J}$; then Y' is the closed sub-prescheme of X' defined by the Ideal $g^*(\mathscr{J})\mathscr{O}_{X'}$ which is identified here with $g^*(\mathscr{J})$ since g is flat; by replacing X by a sufficiently small affine open, we can suppose that $\mathscr{J}$ admits a (resp. quasi-regular) regular sequence of generators f_i (sections of $\mathscr{O}_X$ over X). If the sequence (f_i) is regular, so is the sequence $(f_i \otimes 1)$ (which generates $g^*(\mathscr{J})\mathscr{O}_X$) by virtue of (**0**, 15.2.5); the analogous result for quasi-regular sequences follows immediately from the flatness of g and from the criterion (16.9.4). Similarly, if g is faithfully flat and quasi-compact, and if $g^*(\mathscr{J})$ is quasi-regular, $\mathscr{J}$ is of finite type in virtue of (2.5.2); by flatness, we have $g^*(\mathscr{J})/g^*(\mathscr{J})^2 = g^*(\mathscr{J}/\mathscr{J}^2)$, hence, since $g^*(\mathscr{J}/\mathscr{J}^2)$ is locally free by hypothesis (16.9.4), it follows that $\mathscr{J}/\mathscr{J}^2$ (2.5.2). Finally, condition (iii) of (16.9.4) relative to $g^*(\mathscr{J})$ implies the same condition for $\mathscr{J}$ by virtue of the faithful flatness of g (2.2.7). We conclude from (16.9.4) that $\mathscr{J}$ is quasi-regular.

To prove (iii), we can likewise suppose that Y and Z are closed sub-preschemes of X, defined respectively by Ideals $\mathscr{J}$, $\mathscr{K}$ of $\mathscr{O}_X$ such that $\mathscr{J} \subset \mathscr{K}$; moreover, we can suppose that there exist sections f_i ($1 \leqslant i \leqslant m$) of $\mathscr{J}$ over X generating $\mathscr{J}$ and such that for $1 \leqslant i \leqslant m$, the endomorphism of $\mathscr{O}_X/(\sum_{j=1}^{i-1} f_j\mathscr{O}_X)$ defined by f_i is injective, and sections $\bar{f}_i$ ($m+1 \leqslant i \leqslant n$) of $\mathscr{K}/\mathscr{J}$ over X generating $\mathscr{K}/\mathscr{J}$ and such that for $m+1 \leqslant i \leqslant n$, the endomorphism of $(\mathscr{O}_X/\mathscr{J})/(\sum_{j=m+1}^{i-1} \bar{f}_i(\mathscr{O}_X/\mathscr{J}))$ defined by $\bar{f}_i$ is injective. For every $z \in Z$, let U be an open neighbourhood of z in X such that there exists a section f_i of $\mathscr{K}$ over U, whose class in $\Gamma(U, \mathscr{K}/\mathscr{J})$ is $\bar{f}_i|U$ for $m+1 \leqslant i \leqslant n$; then $(\mathscr{O}_X/\mathscr{J})/(\sum_{j=m+1}^{i-1} \bar{f}_j\mathscr{O}_U)$, restricted to U is isomorphic to $\mathscr{O}_U/((\mathscr{J}|U) + \sum_{j=m+1}^{i-1} f_j\mathscr{O}_U) = \mathscr{O}_U/(\sum_{j=1}^{i-1} f_j\mathscr{O}_U)$, and we see therefore that the sequence $(f_i)_{1 \leqslant i \leqslant n}$ is regular in $\mathscr{O}_U$; as it generates $\mathscr{K}|U$, this proves the first assertion of (iii). To prove the last assertion of (iii), we can restrict ourselves (with the same notations) to the case where the images t_i of the f_i in $\mathscr{K}/\mathscr{K}^2$ (for $1 \leqslant i \leqslant n$) form a basis of this $\mathscr{O}_X/\mathscr{K}$-Module (16.9.5); for the same reason, we can suppose that the t_i for $1 \leqslant i \leqslant m$ are the canonical images of elements of a basis of the $\mathscr{O}_X/\mathscr{K}$-Module $g^*(\mathscr{J}/\mathscr{J}^2)$, and that the canonical images $\bar{t}_i$ of the t_i in $(\mathscr{K}/\mathscr{J})/(\mathscr{K}/\mathscr{J})^2$ for $m+1 \leqslant i \leqslant n$ form a basis of this $\mathscr{O}_X$-$\mathscr{K}$-Module. This completes the proof of (iii).

Let us pass to the proof of (iv). The fact that *a)* implies *c)* follows from (iii). Let us prove that *c)* implies *a)*. Set again $A = \mathscr{O}_{X,f(g(z))}$, so that $\mathscr{O}_{Y,g(z)} = A/\mathfrak{J}$, $\mathscr{O}_{Z,z} = A/\mathfrak{K}$ with $\mathfrak{J} \subset \mathfrak{K}$. By hypothesis, $\mathfrak{J}$ and $\mathfrak{K}$ are regular ideals of A and we have a split exact sequence **IV-4** | 188

$$0 \longrightarrow \mathfrak{J}/\mathfrak{J}\mathfrak{K} \longrightarrow \mathfrak{K}/\mathfrak{K}^2 \longrightarrow (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \longrightarrow 0$$

(cf. (16.2.7)). This implies that there exist elements f_j ($1 \leqslant j \leqslant r+s$) of $\mathfrak{K}$ whose images f'_j in $\mathfrak{K}/\mathfrak{K}^2$ form a basis of this $(A/\mathfrak{K})$-module, and such that, in addition, the f_j of index $j \leqslant r$ belong to $\mathfrak{J}$ and are such that their images in $\mathfrak{J}/\mathfrak{J}\mathfrak{K}$ form a basis of this $(A/\mathfrak{K})$-module. The f_j for $1 \leqslant j \leqslant r+s$ generate $\mathfrak{K}$, and the f_j for $1 \leqslant j \leqslant r$ generate $\mathfrak{J}$ since A is Noetherian (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5); as $\mathfrak{K}$ is a regular ideal, it follows from (16.9.5) that the sequence $(f_j)_{1 \leqslant j \leqslant r+s}$ is regular. By definition, the images of $f_{r+1}, \ldots, f_{r+s}$ in $\mathfrak{K}/\mathfrak{J}$ form a regular sequence in this $(A/\mathfrak{J})$-module, which completes the proof that *c)* implies *a)* in (iv).

Finally, taking into account (16.9.10), it is immediate that (v), as well as the equivalence of *a)* and *b)* in (iv), follow from the □

Corollary (19.1.6). — *Let* A *be a local Noetherian ring,* $\mathfrak{J}$, $\mathfrak{K}$ *two ideals of* A *contained in its maximal ideal* $\mathfrak{m}$ *and such that* $\mathfrak{J} \subset \mathfrak{K}$, $A' = A/\mathfrak{J}$, $\mathfrak{K}' = \mathfrak{K}/\mathfrak{J}$. *Then:*

label=(i) *If* $\mathfrak{J}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{K}$.
lbbel=(ii) *If* $\mathfrak{K}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{J}$.
lcbel=(iii) *Suppose the rings* A *and* $A' = A/\mathfrak{J}$ *are regular. For the ideal* $\mathfrak{K}$ *to be regular, it is necessary and sufficient that* $\mathfrak{K}'$ *be so.*

Proof. Assertion (i) is a particular case of (19.1.5), (iii). To prove (ii), consider the canonical surjective homomorphism $u : \mathfrak{K}/\mathfrak{K}^2 \to \mathfrak{K}'/\mathfrak{K}'^2 = \mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{J})$. Since by hypothesis $\mathfrak{K}/\mathfrak{K}^2$ and $\mathfrak{K}'/\mathfrak{K}'^2$ are free $(A/\mathfrak{K})$-modules, whose ranks we denote respectively by $p+q$ and q, the kernel of u is an $(A/\mathfrak{K})$-module *projective* (Bourbaki, *Alg.*, chap. II, 3ᵉ ed., § 2, n° 2, prop. 4), hence *free* of rank p since $A/\mathfrak{K}$ is a local Noetherian ring ((**0**, 10.1.3)); in other words, there exists a basis of $\mathfrak{K}/\mathfrak{K}^2$ of which the first p elements form a basis of $(\mathfrak{K}^2 + \mathfrak{J})/\mathfrak{K}^2$. This means (by virtue of Nakayama's lemma) that there exists a system of generators $(f_i)_{1 \leqslant i \leqslant p+q}$ of $\mathfrak{K}$, such that the sequence $(f_i)_{1 \leqslant i \leqslant p}$ is formed of elements of $\mathfrak{J}$; it remains to show that this last sequence generates $\mathfrak{J}$. For this, denote by $\mathfrak{J}' \subset \mathfrak{J}$ the ideal it generates; considering the ring $A/\mathfrak{J}'$, we see that we are reduced to proving the following lemma: □

Lemma (19.1.6.1). — *Let* B *be a Noetherian local ring with maximal ideal* $\mathfrak{n}$, $(g_j)_{1 \leqslant j \leqslant q}$ *a regular sequence in* B *of* $\mathfrak{n}$ *elements,* $\mathfrak{b}$ *the ideal it generates,* $\mathfrak{c} \subset \mathfrak{b}$ *an ideal such that the canonical images of the* g_j *in* $B/\mathfrak{c}$ *form a regular sequence in this ring. Then* $\mathfrak{c} = 0$.

Proof. The lemma being trivial for $q = 0$, we reason by induction on q. The hypothesis, applied to the ring B/g_1B and to the canonical images of the g_j ($2 \leqslant j \leqslant q$) and of $\mathfrak{c}$ in this quotient ring, shows **IV-4** | 189
that $\mathfrak{c} \subset g_1B$. Let $\mathfrak{b}$ be the ideal of B consisting of the $x \in B$ such that $g_1x \in \mathfrak{c}$; we thus have $\mathfrak{c} = g_1\mathfrak{b}$; but since by hypothesis g_1 is regular in $B/\mathfrak{c}$, we necessarily have $\mathfrak{b} = \mathfrak{c}$, hence $\mathfrak{c} = g_1\mathfrak{c}$. Since $\mathfrak{c}$ is of finite type and g_1 is contained in the radical of B, Nakayama's lemma shows that $\mathfrak{c} = 0$.

Let us now prove assertion (iii) of (19.1.6). By virtue of (i) and of (19.1.2), it suffices to show that if $\mathfrak{K}$ is regular, so is $\mathfrak{J}$. Note that $\mathfrak{J}$ is regular (19.1.2), and reason by induction on the rank q of the $(A/\mathfrak{J})$-module $\mathfrak{J}/\mathfrak{J}^2$. If $q = 0$, we have $\mathfrak{J} = 0$ by Nakayama's lemma and the assertion is trivial. Since A and $A/\mathfrak{J}$ are regular, we know ((**0**, 17.1.9)) that $\mathfrak{J}$ is generated by a subset of q elements of a regular system of parameters of A, hence, if f is one of the elements of this system of generators of $\mathfrak{J}$, $A_1 = A/fA$ is regular and $f \notin \mathfrak{m}^2$ ((**0**, 17.1.8)). Let $\mathfrak{J}_1, \mathfrak{K}_1$ be the images of $\mathfrak{J}, \mathfrak{K}$ in A_1; we have $\mathfrak{J}_1 \subset \mathfrak{K}_1$, $A_1/\mathfrak{J}_1 = A/\mathfrak{J}$, hence $A_1/\mathfrak{J}_1$ is regular and by the sequence $\mathfrak{J}_1$ is a regular ideal (19.1.2). Let us show that $\mathfrak{K}_1$ is a regular ideal in A_1; since $\mathfrak{K}$ is a regular ideal in A, it suffices to show that f is part of a regular system of generators of $\mathfrak{K}$, and for this (16.9.5) it suffices to show that the image of f in $\mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{m}\mathfrak{K}) = \mathfrak{K}/\mathfrak{m}\mathfrak{K}$ is part of a basis of this $(A/\mathfrak{m})$-vector space, in other words that $f \notin \mathfrak{m}\mathfrak{K}$, which indeed follows from $f \notin \mathfrak{m}^2$. Then, as $\mathfrak{J}_1/\mathfrak{J}_1^2$ is a free $(A_1/\mathfrak{J}_1)$-module of rank $q-1$, the induction hypothesis shows that $\mathfrak{K}'_1 = \mathfrak{K}_1/\mathfrak{J}_1$ is a regular ideal of $A_1/\mathfrak{J}_1$, and since $\mathfrak{K}'$ is identified with $\mathfrak{K}'_1$ in the canonical isomorphism between $A/\mathfrak{J}$ and $A_1/\mathfrak{J}_1$, $\mathfrak{K}'$ is regular. □

Remarks (19.1.7). —

label=(i) We do not know whether the composite of two quasi-regular immersions is quasi-regular.
lbbel=(ii) In a ringed space with local rings X, for every Ideal $\mathscr{J}$ of $\mathscr{O}_X$, $\mathscr{O}_X/\mathscr{J}$ is of finite type and therefore has a closed support in X ((**0**, 5.2.2)). We can therefore define, as in ((**I**, 4.1.3) and (4.2.1)), the notions of closed ringed subspace (locally defined by an Ideal $\mathscr{J}$ of a

sheaf $\mathscr{O}_U$, where U is an open subset of X, and the notion of immersion. The definitions of quasi-regular and regular immersions, of codimension transverse, and the results of (19.1.4), (i) to (iv) are still valid, assuming in the second assertion of (i) and in (iv) that the sheaf $\mathscr{O}_X$ is coherent and the local rings $\mathscr{O}_{X,x}$ are Noetherian; as for (ii), it is necessary to *define* Y' as the ringed subspace defined by the $\mathscr{O}_X$-Ideal $g^*(\mathscr{J})$; the proofs are unchanged.

(19.1.8). We have already noted (in particular (16.9.6), (ii)) that in non-Noetherian rings, the notion of "regular sequence" of elements of the ring does not possess the properties that would make it usable. Recent research seems to show that in this case the notion that should be substituted for that of regular sequence is that of the sequence $\mathbf{f} = (f_1, \ldots, f_n)$ having the property that the complex of the exterior algebra $\mathrm{K}_0(\mathbf{f}, \mathrm{M})$ ((**III**, 1.1.3)) has *zero homology in positive degrees* (cf. (**III**, 1.1.4)); see in particular A. Grothendieck, Séminaire de géométrie algébrique, 1966, to appear.

19.2. Transversally regular immersions IV-4 | 190

Definition (19.2.1). — Let $u : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. We say that a sequence $(f_i)_{1 \leqslant i \leqslant n}$ of sections of $\mathscr{F}$ over X is *transversally $\mathscr{F}$-regular relatively to S* (or relatively to u) if the sequence (f_i) is $\mathscr{F}$-regular and such that the $\mathscr{O}_X$-Modules $\mathscr{F}/(\sum_{1 \leqslant i} f_i \mathscr{F})$ are S-flat for $0 \leqslant i \leqslant n$. We say that an Ideal $\mathscr{J}$ quasi-coherent of $\mathscr{O}_X$ is *transversally regular relatively to S* (or relatively to u) at a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{J})$ if there exists an open neighbourhood U of x and a finite sequence (f_i) of sections of $\mathscr{J}$ over U that is $\mathscr{O}_U$-regular relatively to S and that generates $\mathscr{J}|U$.

In a A-algebra B, we say that an ideal $\mathfrak{J}$ is *transversally regular relatively* to A if $\mathscr{J} = \widetilde{\mathfrak{J}}$ is transversally regular relatively to $S = \operatorname{Spec}(A)$ at every point of $V(\mathfrak{J})$ in $\operatorname{Spec}(B)$.

Definition (19.2.2). — Let S be a prescheme, X, Y two S-preschemes, $u : Y \to X$ an S-morphism which is an immersion. We say that u is a *transversally regular immersion relatively to S* at a point $y \in Y$ if there exists a neighbourhood V of $u(y)$ in X such that the sub-prescheme of X associated to $u(Y) \cap V$ is defined by an Ideal of $\mathscr{O}_U$ transversally regular relatively to S at the point $u(y)$.

It is clear that the set of points of Y where an immersion is transversally regular relatively to S is *open* in Y; if it is equal to Y, we say simply that u is a *transversally regular immersion relatively to S*.

If $S \to T$ is a flat morphism, then, if $\mathscr{J}$ is an Ideal of $\mathscr{O}_X$ transversally regular relatively to S at $x \in X$, it is also relatively to T ((**0**, 6.2.1)). A S-immersion $Y \to X$ transversally regular relatively to S at a point $y \in Y$ is therefore also a T-immersion transversally regular relatively to T at this point.

Proposition (19.2.3). — *Let S be a prescheme, X, Y two S-preschemes, $u : Y \to X$ an S-morphism which is a transversally regular immersion relatively to S at a point $y \in Y$. Then, for every change of base $S' \to S$, if we set $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' = u_{(S')} : Y' \to X'$ is transversally regular relatively to S' at every point $y' \in Y'$ above y.*

Proof. This follows immediately from the definition and from (**0**, 15.1.15). □

Proposition (19.2.4). — *Let S be a prescheme, X, Y two S-preschemes, $u : Y \to X$ an S-morphism which is an immersion. Suppose, either that S and X are locally Noetherian, or that the structural morphisms $g : X \to S$, $h : Y \to S$ are locally of finite presentation. Let y be a point of Y, s its image in S. The following conditions are equivalent:*

label=a) *u is transversally regular relatively to S at the point y.*

lbbel=b) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and if we set $X_s = g^{-1}(s)$, $Y_s = h^{-1}(s)$, the immersion $u_s : Y_s \to X_s$ is regular at the point y.*

b′) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and, for every change of base $S' \to S$, if we set $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' : Y' \to X'$ is regular at every point of Y' above y.*

lcbel=c) *The morphism h is flat in a neighbourhood of the point y, and the immersion u is regular in a neighbourhood of the point y.*

Proof. c′) The morphism h is flat in a neighbourhood of the point y, and the immersion u is quasi-regular in a neighbourhood of the point y. IV-4 | 191

We have already proved in (19.2.3) that *a)* implies *b′)* without the hypothesis of finiteness, and it is trivial that *b′)* implies *b)*. Let us show that *b)* implies *c)*. We can for this (the question being local on X and on S) suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_X$, and u the canonical injection. The hypothesis that $\mathscr{O}_X/\mathscr{J}$ is g-flat implies that the sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_g} \mathbf{k}(s) \longrightarrow \mathscr{O}_X \otimes_{\mathscr{O}_g} \mathbf{k}(s) \longrightarrow (\mathscr{O}_X/\mathscr{J}) \otimes_{\mathscr{O}_g} \mathbf{k}(s) \longrightarrow 0$$

is exact ((**0**, 6.1.2)), hence Y_s is the closed sub-prescheme of X_s defined by the Ideal $\mathscr{J}_s$ of $\mathscr{O}_{X_s} = \mathscr{O}_X \otimes_{\mathscr{O}_g} \mathbf{k}(s)$ which is identified with $\mathscr{J} \otimes_{\mathscr{O}_g} \mathbf{k}(s) = \mathscr{J}/\mathfrak{m}_s\mathscr{J}$. Consider a finite sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X, whose images in $\mathscr{J}_y/(\mathfrak{m}_s\mathscr{J}_y + \mathscr{J}_y^2)$ form a basis of this $(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y))$-module (which is free by the hypothesis *b)*)). Since X_s is locally Noetherian, it follows from (16.9.11), (16.9.5) and the hypothesis *b)*) that the images $(f_i)_s = f_i \otimes 1$, sections of $\mathscr{J}_s = \mathscr{J} \otimes_{\mathscr{O}_g} \mathbf{k}(s)$ over a neighbourhood of y in X_s, form a regular sequence generating the restriction of $\mathscr{J}_s$ at this neighbourhood. Since $\mathfrak{m}_s\mathscr{O}_{X,y}$ is contained in the radical of $\mathscr{O}_{X,y}$, and since in both cases envisaged $\mathscr{J}_y$ is an Ideal of finite type of $\mathscr{O}_{X,y}$, it follows from Nakayama's lemma that the $(f_i)_y$ also generate $\mathscr{J}_y$; since $\mathscr{J}$ is an Ideal of finite type of $\mathscr{O}_X$, there is a neighbourhood U of y in X such that the $f_i|U$ generate $\mathscr{J}|U$. The fact that *b)* implies *c)* is then a consequence of (**0**, 15.1.16) when S and X are locally Noetherian, and of (11.3.8) when g and h are locally of finite presentation; in this last case, (11.3.8) proves also the equivalence of *c)* and *c′)* (which is trivial when X and S are locally Noetherian ((**0**, 15.1.11))). Finally, if *c)* is satisfied, to prove *a)*, we can restrict ourselves to the case where Y is a closed sub-prescheme of X defined by $\mathscr{J}$, and by hypothesis there is a regular sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X such that the f_i generate $\mathscr{J}|U$ and that h is flat in U; to see that *c)* implies *a)*, it suffices to apply ((**0**, 15.1.6)) when S and X are locally Noetherian, and (11.3.8) when g and h are locally of finite presentation. □

Remarks (19.2.5). —

label=(i) When in condition *b)*, we drop the hypothesis that h is flat in a neighbourhood of y, the condition thus modified no longer implies *a)*. It suffices to take for S the spectrum of a local ring that is not a field, for s the closed point of S, and $Y = X_s$.

lbbel=(ii) Suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_X$. We proved in the course of the proof of (19.2.4) that, when the equivalent conditions of this proposition are satisfied, then, for every system (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X, such that the images of the f_i in $\mathscr{J}_y/(\mathfrak{m}_s\mathscr{J}_y + \mathscr{J}_y^2)$ form a basis of this $(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y))$-module, there exists an open neighbourhood U of y in X such that the $f_i|U$ satisfy the conditions of the definition (19.2.1).

Corollary (19.2.6). — *Suppose that S and X are locally Noetherian, that g and h are locally of finite presentation. Suppose in addition that the fibres X_s and Y_s are regular preschemes, and that the morphisms g and h are flat in a neighbourhood of $u(y)$ and y respectively. Then the immersion $Y \to X$ is transversally regular at the point x.* **IV-4** | 192

Proof. Indeed, it follows from (19.1.1) that the immersion $Y_s \to X_s$ is regular at the point x, and it suffices to apply (19.2.4), *b)*). □

Proposition (19.2.7). —

label=(i) Every open S-immersion is transversally regular relatively to S.

lbbel=(ii) Let $f : Y \to X$ be an S-immersion, $g : S' \to S$ a morphism and set $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : Y' \to X'$. If f is a transversally regular immersion relatively to S, the converse is true if g is faithfully flat, and if moreover X and Y are locally of finite presentation over S.

lcbel=(iii) If $f : Y \to X$ and $g : Z \to Y$ are transversally regular immersions relatively to S, so is $f \circ g : Z \to X$.

ldbel=(iv) Let X be an S-prescheme, $f : Y \to X$, $g : Z \to Y$ two S-immersions. Suppose that S and X are locally Noetherian, that X, Y and Z are locally of finite presentation over S. Then, if g and $f \circ g$ are transversally regular at the point $z \in Z$ relatively to S, f is transversally regular at the point $g(z)$ relatively to S.

lebel=(v) Under the general conditions of (iv), suppose that X and Y are S-flat, Y_s regular at the point $g(z)$ and X_s regular at the point $g(z)$; then, if $f \circ g$ is transversally regular at the point z, so is g.

Proof. Assertion (i) is immediate, and the first assertion of (ii) follows immediately from (19.2.3). To prove the second assertion of (ii), we apply (19.2.4): X' and Y' being locally of finite presentation over S', it follows first from this criterion that X' and Y' are flat over S', hence X and Y are flat over S (2.5.1); on the other hand, for every $s' \in S'$, if $s = g(s')$, it is a point above S, and we deduce from (19.1.5), (ii) that the same is true of $Y_s \to X_s$, since these preschemes are locally Noetherian and it thus amounts to saying that the immersion $Y_s \to X_s$ is regular or quasi-regular. To prove (iii), we reduce, as when Z is a closed sub-prescheme of Y and Y a closed sub-prescheme of X, and we form, as in (19.1.5), (iii), a system of generators of the Ideal of $\mathscr{O}_X$ defining Z. To prove (iv), we note that the hypothesis together with (19.2.4) shows first of all that X, Y and Z are S-flat in a neighbourhood of z, and it suffices moreover, if s is the image of z in S, to prove that the immersion $f_s : Y_s \to X_s$ is transversally regular at the point z, knowing that the same is true of $g_s : Z_s \to Y_s$ and $f_s \circ g_s$; but this follows from (19.1.5), (iv). Finally, for (v), we reason in the same way as for (iv); the hypothesis on $f \circ g$ implies already by (19.2.4), *b)*) that Z is S-flat in a neighbourhood of z, and it is the same for X and Y by hypothesis; we are then reduced to proving that g_s is regular, knowing that $f_s \circ g_s$ is, which follows from (19.1.5), (v), given the hypotheses made on X_s and Y_s. □

IV-4 | 193

Proposition (19.2.8). — *Let X be an S-prescheme, Y a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_X$, $u : Y \to X$ the canonical injection. If u is transversally regular relatively to S, the $\mathscr{O}_X$-Modules $\mathscr{J}^n$ and $\mathscr{J}^n/\mathscr{J}^{n+1}$ $(0 \leqslant n < m)$ are S-flat at the points of Y. For every change of base $S' \to S$, if $X' = X_{(S')}$, $Y' = Y_{(S')}$ and $u' = u_{(S')}$, we have $\mathscr{G}r_n(u') = \mathscr{G}r_n(u) \otimes_{\mathscr{O}_g} \mathscr{O}_{S'}$ as isomorphisms near for all $n > 0$.*

Proof. Indeed, $\mathscr{O}_X$ and $\mathscr{O}_X/\mathscr{J}$ are S-flat at the points of Y, and $\mathscr{J}^n/\mathscr{J}^{n+1}$ is therefore S-flat at the points of Y. The first assertion then follows from ((**0**, 6.1.2)) and the second from ((11.2.9.2)). □

Proposition (19.2.9). — *Let $f : X \to S$ be a morphism, x a point of X, $s = f(x)$. Suppose that f is of finite presentation at the point x, and is a Cohen-Macaulay morphism at the point x ((6.8.1)). Then there exists a closed sub-prescheme Y of X containing x and having the following properties:*

label=(i) The canonical injection $i : Y \to X$ is transversally regular relatively to S.
lbbel=(ii) The morphism $g = f \circ j : Y \to S$ is de Cohen-Macaulay (which amounts to saying, taking into account (i), that all the fibres $g^{-1}(g(y))$ of g are preschemes de Cohen-Macaulay).
lcbel=(iii) The point x is a maximal point of $Y_s = g^{-1}(s)$.

Proof. Set $X_s = f^{-1}(s)$; by hypothesis, the ring $\mathscr{O}_{X_s,x}$ is de Cohen-Macaulay; let $(t_i)_{1 \leqslant i \leqslant n}$ be a system of parameters of this ring, which is therefore a regular sequence ((**0**, 16.5.7)). There is a neighbourhood open U of x in X and sections h_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over U such that the t_i are the canonical images of the $(h_i)_x$ in the quotient ring $\mathscr{O}_{X_s,x}$ of $\mathscr{O}_{X,x}$. Let $\mathscr{J}$ be the Ideal of $\mathscr{O}_U$ generated by the h_i, and take for Y the closed sub-prescheme of U defined by $\mathscr{J}$. We can suppose that $f|U$ is locally of finite presentation. It follows from the definition of Y; it follows from the equivalence of *c)* and *a)* in (11.3.8) that the sequence $((h_i)_x)$ is regular in $\mathscr{O}_{X_s,x}$ and that $\mathscr{O}_{Y,x}$ is flat over $\mathscr{O}_{S,s}$; by virtue of (11.3.1) f and g are flat in a neighbourhood of x and y respectively. It follows from (19.2.4) that i is a transversally regular immersion relatively to S (in replacing Y by $Y \cap V$ for a neighbourhood open V of x in X). On thus proved (i) and we see that we can suppose that $g : Y \to S$ is a flat morphism. Since $\mathscr{O}_{Y_s,x}$, quotient of $\mathscr{O}_{X_s,x}$ by the ideal generated by the t_i, is Artinian since by construction ((**0**, 16.3.6)), x is a maximal point of Y_s. Finally, the ring $\mathscr{O}_{Y_s,x}$, being Artinian, is a ring de Cohen-Macaulay ((**0**, 16.5.1)), hence, in replacing Y by its intersection with a neighbourhood open of x in X, we can suppose that g is a Cohen-Macaulay morphism, by virtue of ((12.1.7), (iii)).

In particular, we find the existence of "quasi-sections" *flat* Y of a morphism $f : X \to S$ at a point $x \in X_s$ which is *closed* in X_s and such that f is flat at the point x and $\mathscr{O}_{X_s,x}$ is a Cohen-Macaulay ring ((17.16.1)) and we see moreover that the canonical immersion $Y \to X$ is *transversally regular* relatively to S. This last complement is also valid for the quasi-section defined in ((17.16.3), (i)). □

19.3. Relative complete intersections (flat case)

IV-4 | 194

Definition (19.3.1). — Let A be a local Noetherian ring. We say that A is a *complete intersection ring* (or an *absolute complete intersection ring*, to avoid any risk of confusion) if its completion $\widehat{A}$ is isomorphic to the quotient of a complete Noetherian regular local ring B by an ideal (i.e. (16.9.7)) *generated by a regular sequence of elements* of B.

Every regular local ring is a complete intersection ring.

We say that a locally Noetherian prescheme X is a *complete intersection* (absolute) at a point $x \in X$ if the ring $\mathscr{O}_{X,x}$ is a complete intersection ring.

Proposition (19.3.2). — *Let* B *be a local Noetherian regular ring,* $\mathfrak{J}$ *an ideal of* B*. For* $A = B/\mathfrak{J}$ *to be a complete intersection ring, it is necessary and sufficient that the ideal* $\mathfrak{J}$ *be regular (i.e.* (16.9.7) *generated by a regular sequence of elements of* B*).*

Proof. We can write $\widehat{A} = \widehat{B}/\mathfrak{J}\widehat{B}$ and since $\widehat{B}$ is a B-module faithfully flat and B Noetherian, $\mathfrak{J}\widehat{B}$ is regular if and only if $\mathfrak{J}$ is (19.1.5), (ii). This shows (by virtue of (**0**, 17.1.5), (ii)) that one part of the condition is sufficient, and on the other hand that to prove that the condition is necessary, we can restrict ourselves to the case where A and B are *complete*. By hypothesis, there exists then a complete Noetherian regular local ring B′ such that A is isomorphic to a quotient $B'/\mathfrak{J}'$, where $\mathfrak{J}'$ is a regular ideal of B′. Let us consider then the fibre product $B \times_A B' = B''$ for the surjective homomorphisms $B \to A$, $B' \to A$ ((**0**, 18.1.2)); we will first prove the following lemma: □

Lemma (19.3.2.1). — *Let* A, E, F *be three rings,* $f : E \to A$, $g : F \to A$ *two homomorphisms, and set* $G = E \times_A F$ *((**0**, 18.1.2)).*

label=(i) *If f is surjective and if* E *and* F *are Noetherian rings,* G *is Noetherian.*
lbbel=(ii) *If* A, E *and* F *are local rings and if f and g are local homomorphisms,* G *is local.*
lcbel=(iii) *If the conditions of (i) and (ii) are satisfied, if g is surjective, if* E *and* F *are complete Noetherian local rings,* G *is complete.*

Proof. label=(i) Taking into account ((**0**, 18.1.3) and (18.1.5)), G is an A-augmented ring over F, whose ideal of augmentation $\mathfrak{J}'$ is canonically identified with that of the E-module structure on $\mathfrak{J}$ of f, via the canonical homomorphism $G \to E$. If $\mathfrak{a}$ is any ideal of G, $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{J}')$ is isomorphic to $(\mathfrak{a} + \mathfrak{J}')/\mathfrak{J}'$, hence to an ideal of F, and is therefore of finite type; on the other hand, $\mathfrak{a} \cap \mathfrak{J}'$ is isomorphic to an ideal of E contained in $\mathfrak{J}$, hence also of finite type, which proves that $\mathfrak{a}$ is of finite type, hence G is Noetherian.

lbbel=(ii) Let $\mathfrak{r}$, $\mathfrak{s}$ be the maximal ideals of E and F respectively; the set $\mathfrak{m} = \mathfrak{r} \times_A \mathfrak{s}$ is evidently an ideal of G, inverse image of $\mathfrak{r}$ by the projection $G \to E$ and of $\mathfrak{s}$ by the projection $G \to F$. On the other hand, if $(x, y) \notin \mathfrak{m}$, and if for example $x \notin \mathfrak{r}$, $f(x)$ does not belong to the maximal ideal of A, hence, since $g(y) = f(x)$, and g is local, we also have $y \notin \mathfrak{s}$; we conclude that x and y are invertible, hence (x, y) is invertible, which completes the proof of (ii). IV-4 | 195

lcbel=(iii) The image of $\mathfrak{m}$ by the surjective homomorphism $G \to E$ is the maximal ideal $\mathfrak{r}$ of E, hence the image of $\mathfrak{m}^n \cap \mathfrak{J}'$ is $\mathfrak{r}^n \cap \mathfrak{J}'$, and since $\mathfrak{J}$ is closed (hence complete) for the topology induced by the $\mathfrak{r}$-adic topology of E ((**0**, 7.3.5)), $\mathfrak{J}'$ is complete for the topology induced by the $\mathfrak{m}$-preadic topology of G. On the other hand, $G/\mathfrak{J}'$, equipped with the topology induced, is isomorphic to F equipped with the $\mathfrak{s}$-adic topology, hence is complete for the quotient topology of the $\mathfrak{m}$-preadic topology of G. We deduce that G is complete for the $\mathfrak{m}$-preadic topology.

This lemma being established, it follows from the Cohen theorem ((**0**, 19.8.8)) that B″ is isomorphic to a quotient of a complete Noetherian local *regular* ring C. It then follows from (19.1.2) that the immersion $\operatorname{Spec}(B') \to \operatorname{Spec}(C)$ is regular; since by hypothesis $\operatorname{Spec}(A) \to \operatorname{Spec}(B')$, we conclude by (19.1.5), (iii) that the immersion $\operatorname{Spec}(A) \to \operatorname{Spec}(C)$ is regular; but this immersion is written also as the composite $\operatorname{Spec}(A) \to \operatorname{Spec}(B) \to \operatorname{Spec}(C)$. Now as B is regular by hypothesis, the immersion $\operatorname{Spec}(B) \to \operatorname{Spec}(C)$ is regular (19.1.2); hence it is the same for $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ by (19.1.5), (iv). □

Corollary (19.3.3). — *Let X be a locally Noetherian prescheme locally immersible in a regular prescheme. Then the set of $x \in X$ where X is a complete intersection is open in X and contains the set of regular points of X.*

Proof. We can indeed restrict ourselves to the case where $X = \mathrm{Spec}(\mathrm{B}/\mathfrak{J})$, where B is a regular ring and $\mathfrak{J}$ an ideal of B. The set of $\mathfrak{p} \supset \mathfrak{J}$ in $\mathrm{Spec}(\mathrm{B})$ such that $\mathfrak{J}_\mathfrak{p}$ is regular in $\mathrm{B}_\mathfrak{p}$ is then open ((16.9.5)), and this is the set of points of $\mathrm{Spec}(\mathrm{B})$ where X is a complete intersection by virtue of (19.3.2). ☐

Corollary (19.3.4). — *Let k be a field, X a prescheme locally of finite type over k, k' an extension of k, $X' = X \otimes_k k'$. Let x' be a point of X', x its projection in X. For X to be a complete intersection at x, it is necessary and sufficient that X' be a complete intersection at x'.*

Proof. The question being local on X, we can suppose that $X = \mathrm{Spec}(\mathrm{A})$, where A is a k-algebra of finite type, hence a quotient of a polynomial algebra $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, so X is a closed sub-prescheme of the regular scheme $Y = \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ ((**0**, 17.3.7)). If we set $Y' = Y \otimes_k k' = \mathrm{Spec}(k'[\mathrm{T}_1, \ldots, \mathrm{T}_n])$, Y' is also a regular scheme. Now, since $\mathrm{Spec}(k')$ is faithfully flat over $\mathrm{Spec}(k)$, it follows from (19.1.5), (ii) and from the fact that we are dealing with Noetherian preschemes that the immersion $Y \to X$ is regular at the point x if and only if the immersion $Y' \to X'$ is so at the point x'. The conclusion then follows from the criterion (19.3.2). ☐

Corollary (19.3.5). — *Let A, C be two Noetherian local rings, $\rho : \mathrm{A} \to \mathrm{C}$ a local homomorphism, $\mathrm{B} = \mathrm{C}/\mathfrak{J}$ a quotient of C, k the residue field of A. Suppose that ρ makes C a flat A-module, and that the ring $\mathrm{C} \otimes_\mathrm{A} k$ is regular. Then the following conditions are equivalent:*

*label=*a) *The ideal $\mathfrak{J}$ is transversally regular relatively to* A.
*lbbel=*b) B *is a flat* A*-module, and* $\mathrm{B} \otimes_\mathrm{A} k$ *is a complete intersection ring.*

Proof. By virtue of (19.2.4), condition *a)* amounts to saying that B is a flat A-module and that $\mathfrak{J} \otimes_\mathrm{A} k$ **IV-4** | 196
(which is identified with an ideal of $\mathrm{C} \otimes_\mathrm{A} k$ by virtue of the flatness of B over A ((**0**, 6.1.2))) is regular in this ring. Since the ring $\mathrm{C} \otimes_\mathrm{A} k$ is regular, it amounts to the same, by virtue of (19.3.2), to say (when B is a flat A-module) that $\mathfrak{J} \otimes_\mathrm{A} k$ is a regular ideal of $\mathrm{C} \otimes_\mathrm{A} k$, or that $\mathrm{B} \otimes_\mathrm{A} k = (\mathrm{C} \otimes_\mathrm{A} k)/(\mathfrak{J} \otimes_\mathrm{A} k)$ is a complete intersection ring; whence the corollary. ☐

Definition (19.3.6). — Let $f : X \to S$ be a flat morphism locally of finite presentation. We say that X is a *complete intersection relatively to S* at a point x, if the fibre $f^{-1}(f(x))$ is a complete intersection (absolute) at the point x. We say that X is a *complete intersection relatively to S*, or that the morphism f is a *morphism of complete intersection*, if X is a complete intersection relatively to S at each of its points.

Proposition (19.3.7). — *Let $g : X \to S$, $h : Y \to S$ be two flat morphisms locally of finite presentation, $f : Y \to X$ an S-immersion, y a point of Y, $x = f(y)$, $s = g(x) = h(y)$. Suppose that the fibre $X_s = g^{-1}(s)$ is regular at the point x. Then, for f to be a transversally regular immersion at the point y, it is necessary and sufficient that Y be a complete intersection relatively to S at the point y.*

Proof. To say that f is transversally regular at the point y amounts, by virtue of (19.2.4), to saying that $f_s : Y_s \to X_s$ is a regular immersion at the point y. But since X_s is regular at the point $x = f(y)$, this is equivalent, by virtue of (19.3.2), to saying that Y_s is a complete intersection (absolute) at the point y, whence the proposition. ☐

Corollary (19.3.8). — *Let $f : X \to S$ be a flat morphism locally of finite presentation. The set U of $x \in X$ for which X is a complete intersection relatively to S is open in X. If moreover f is proper, the set E of $s \in S$ such that $X_s = f^{-1}(s)$ is a complete intersection at each of its points is an open subset of S.*

Proof. Since $E = Y - f(X - U)$, the second assertion follows from the first and from the fact that f is a closed map. The first assertion is local on X, hence we can suppose that X is a closed sub-prescheme of a prescheme of the form $Z = S[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ (T_i indeterminate). This last being smooth over S ((17.3.8)), its fibres Z_s are regular, and it amounts to the same, by virtue of (19.3.7), to say that $x \in U$ or that the immersion $X \to Z$ is transversally regular at the point x; we conclude by the corollary by (19.2.2). ☐

Proposition (19.3.9). —

label=(i) Every open immersion is a complete intersection.

lbbel=(ii) *Let $f : X \to S$ be a flat morphism and locally of finite presentation which is a morphism of complete intersection. For every change of base $g : S' \to S$, $f' = f_{(S')} : X_{(S')} \to S'$ is a morphism of complete intersection. The converse is true if g is faithfully flat and quasi-compact.*

lcbel=(iii) *If $f : X \to Y$ and $g : Y \to Z$ are flat morphisms locally of finite presentation which are complete intersection morphisms, so is $g \circ f : X \to Z$.*

Proof. Assertion (i) follows immediately from the definition (19.3.1). To prove (ii), note first that if f is flat and locally of finite presentation, the same is true of f' ((2.5.1) and (2.7.1)); moreover, for every $s' \in S'$, if $s = g(s')$, it is equivalent to say that $f^{-1}(s)$ is a complete intersection or that $f'^{-1}(s')$ is a complete intersection (19.3.4); whence (ii), taking into account that g is surjective when it is faithfully flat. **IV-4** | 197

Finally, under the hypotheses of (iii), $g \circ f$ is flat and locally of finite presentation. By definition (19.3.6), we are reduced to the case where Z is the spectrum of a field, and we have to prove then that the local rings $\mathscr{O}_{X,x}$ are complete intersection rings, which follows from the more general result: □

Corollary (19.3.10). — *Let $f : X \to S$ be a flat morphism locally of finite presentation which is a complete intersection morphism, x a point of X, $s = f(x)$. If $\mathscr{O}_{S,s}$ is a complete Noetherian intersection ring, so is $\mathscr{O}_{X,x}$.*

Proof. We can evidently reduce to the case where $S = \operatorname{Spec}(A)$ with $A = \mathscr{O}_{S,s}$. Let us show moreover that we can restrict ourselves to the case where the local ring A is *complete*. Indeed, set $A' = \widehat{A}$, $X' = X \otimes_A A'$, and let s' be the unique closed point of $S' = \operatorname{Spec}(A')$, which is the unique point above S; if $f' = f_{(S')} : X' \to S'$, the fibre $f'^{-1}(s')$ is canonically isomorphic to $f^{-1}(s)$, since A and A' have the same residue field ((**I**, 3.6.4)); there is thus a single point $x' \in X'$ above x and above s', and we have $\mathscr{O}_{X',x'} = \mathscr{O}_{X,x} \otimes_A A'$. We deduce that the local rings $\mathscr{O}_{X,x}$ and $\mathscr{O}_{X',x'}$ have isomorphic *completions*, since in general, if E is a local ring which is a local A-algebra (the local homomorphism $A \to E$ being local), the separated completion $(E \otimes_A \widehat{A})^\wedge$ of the ring $E \otimes_A \widehat{A}$ equipped with the product topology is isomorphic to the separated completion $\widehat{E} \otimes_A \widehat{A} \cong \widehat{E}$, hence to the separated completed $\widehat{E}$ of E ((**0**, 7.7.1)), so it amounts to the same to say that $\mathscr{O}_{X,x}$ is a complete intersection ring, or that $\mathscr{O}_{X',x'}$ is one.

Suppose therefore that A is *complete*; we can suppose moreover that $X = \operatorname{Spec}(B)$, B being a quotient of a polynomial algebra $C = A[T_1, \ldots, T_r]$. Since the polynomial algebras over a field are regular rings ((**0**, 17.3.7)), it follows from (19.3.7) that the immersion $X \to Z = \operatorname{Spec}(C)$ can be supposed transversally regular relatively to S; hence we can suppose that $B = C/\mathfrak{J}$, where $\mathfrak{J}$ is generated by a regular sequence $(g_i)_{1 \leqslant i \leqslant n}$ of elements of C. On the other hand, since A is complete, we can write by hypothesis $A' \to A'/\mathfrak{R}$, where A' is a regular local ring and $\mathfrak{R}$ an ideal of A' ((**0**, 19.8.8)), and the hypothesis that A is a complete intersection ring implies that $\mathfrak{R}$ is generated by a regular sequence $(f_j)_{1 \leqslant j \leqslant m}$ (19.3.2). In the polynomial algebra $C' = A'[T_1, \ldots, T_r]$, the elements f_j $(1 \leqslant j \leqslant m)$ form a regular sequence generating the ideal $\mathfrak{R}' = \mathfrak{R}[T_1, \ldots, T_r]$ ((**0**, 15.1.4)); since $C'/\mathfrak{R}' = C$, we see that since, for each i, g'_i is an element of C' with image g_i in C, the sequence formed of the f_j $(1 \leqslant j \leqslant m)$ and the g'_i $(1 \leqslant i \leqslant n)$ is a regular sequence in C'; if $\mathfrak{J}'$ is the ideal of C' that it generates, we have $B = C'/\mathfrak{J}'$, whence the conclusion by virtue of (19.3.2), since C' is a regular ring ((**0**, 17.3.7)). □

19.4. Application: criteria of regularity and smoothness for blown-up preschemes **IV-4** | 198

(19.4.1). Let X be a prescheme, Y a closed sub-prescheme of X, defined by a quasi-coherent Ideal $\mathscr{J}$ of $\mathscr{O}_X$. Recall ((**II**, 8.1.3)) that the X-scheme X' obtained by blowing up Y is the prescheme

$$X' = \operatorname{Proj}(\mathscr{S}), \quad \text{where} \quad \mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{J}^n.$$

If $\mathscr{J}$ is of finite type, the structural morphism $f : X' \to X$ is projective. Without hypothesis on $\mathscr{J}$, the closed sub-prescheme

$$Y' = f^{-1}(Y)$$

of X' is defined by the quasi-coherent Ideal $\mathscr{J}\mathscr{O}_{X'}$ of $\mathscr{O}_{X'}$, canonically isomorphic to $\mathscr{O}_{X'}(1)$; more precisely, we have an exact sequence

$$0 \longrightarrow \mathscr{O}_{X'}(1) \xrightarrow{\widetilde{u}_0} \mathscr{O}_{X'} \longrightarrow \mathscr{O}_{Y'} \longrightarrow 0$$

(19.4.1.1) where $\mathscr{J}\mathscr{O}_{X'}$ is the image of $\widetilde{u}_0$ ((**II**, 8.1.7) and (8.1.8)). Since in a neighbourhood of a point of X', the $\mathscr{O}_{X'}$-Module $\mathscr{O}_{X'}(1)$ is free of rank 1, $\mathscr{J}\mathscr{O}_{X'}$ is generated in such a neighbourhood by an invertible section of $\mathscr{O}_{X'}$ and $\mathscr{J}\mathscr{O}_{X'}/\mathscr{J}^2\mathscr{O}_{X'}$ is therefore an $\mathscr{O}_{X'}$-Module free of rank 1. This proves the first assertion of the following lemma:

Lemma (19.4.2). — *The canonical immersion $Y' \to X'$ is regular and of codimension 1, and Y' is canonically isomorphic to the Y-scheme* $\operatorname{Proj}(\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$.

Proof. The second assertion follows from ((**II**, 3.5.3)) applied to the canonical injection $Y \to X$, taking into account that $\mathscr{O}_Y$ is isomorphic to $\mathscr{O}_X/\mathscr{J}$ and $\mathscr{J}^n \otimes_{\mathscr{O}_X} \mathscr{O}_Y = \operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ by definition, since $\mathscr{O}_Y$ is isomorphic to $\mathscr{O}_X/\mathscr{J}$ and $\mathscr{J}^n \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J})$ to $\mathscr{J}^n/\mathscr{J}^{n+1}$. □

Corollary (19.4.3). — *If the canonical immersion $Y \to X$ is quasi-regular, the restriction $g : Y' \to Y$ of the morphism f is smooth.*

Proof. Indeed, $\mathscr{N}_{Y/X} = \mathscr{J}/\mathscr{J}^2$ is then an $\mathscr{O}_Y$-Module locally free of finite rank and $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$ (16.9.8), hence Y' is Y-isomorphic to $\mathbf{P}(\mathscr{N}_{Y/X})$ and is therefore smooth over Y (17.3.9). □

We shall see later (chap. V) that g is smooth also when $Y \to X$ "has only ordinary singularities".

Proposition (19.4.4). — *With the notations of (19.4.1), suppose that X is locally Noetherian and that the morphism $g : Y' \to Y$, restriction of f, is smooth. Let x' be a point of Y', $x = g(x')$. For Y to be regular at the point x, it is necessary and sufficient that Y' be regular at the point x'; when this is so, X' is then regular at the point x'.*

Proof. The first assertion follows from (17.5.8) and the second from (19.1.1) and (19.4.2). □

Corollary (19.4.5). — *Under the general hypotheses of (19.4.4), suppose that Y is regular at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is regular at the point x_1. If* $\operatorname{Reg}(X)$ *is open (((6.12)), for example if X is excellent ((7.8.6), (iii))) and if Y is regular, then there exists an open neighbourhood U of Y in X such that X is regular at the points of $U - Y$.* **IV-4** | 199

Proof. To prove the first assertion, we can restrict ourselves to the case where X is a scheme local of closed point x; the hypothesis then implies that Y is regular ((**0**, 17.3.2)), hence it is the same for Y' by (19.4.4). Let us now note that the morphism $f : X' \to X$ is projective, hence proper ((**II**, 7.2.1)), if we consider the unique point x_1' of X' above x_1 belonging to Y'; applying (19.4.4) and using ((**0**, 17.3.2)) we conclude that X' is regular at the point x_1, hence X is regular at the point x_1' since $X' - Y'$ is isomorphic to $X - Y$ ((**II**, 8.1.3)). If Y is regular, every point of $X - Y$ which is a generization of a point of Y belongs to $\operatorname{Reg}(X)$; if $\operatorname{Reg}(X)$ is open, $U = Y \cup \operatorname{Reg}(X)$ is then locally constructible and stable under generization, hence open ((**0**, 9.2.5)), and responds to the question. □

Proposition (19.4.6). — *Let $g : X \to S$ be a morphism, Y a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$, X' the X-scheme obtained by blowing up Y, Y' the inverse image of Y in X'.*

label=(i) The following conditions are equivalent:

- *label=a) $\operatorname{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is a flat $\mathscr{O}_X$-Algebra.*
- *lbbel=b) For every $n \geqslant 0$, the $\mathscr{O}_X$-Module $\mathscr{O}_X/\mathscr{J}^{n+1}$ is g-flat (in other words, the n-th infinitesimal neighbourhood $Y^{(n)}$ of Y in X (16.1.2) is S-flat).*
- *lcbel=c) For every change of base $S_1 \to S$ and every integer $n \geqslant 1$, if we set $X_1 = X \times_S S_1$, the canonical homomorphism $\mathscr{J}^n \otimes_{\mathscr{O}_g} \mathscr{O}_{S_1} \to \mathscr{J}^n \mathscr{O}_{X_1}$ is bijective.*

When this is so, Y' is S-flat, and for every change of base $S_1 \to S$, if Y_1 is the inverse image of Y in X_1, the prescheme X_1' obtained by blowing up Y_1 in X_1 is canonically isomorphic to $X' \times_S S_1$.

lbbel=(ii) *Suppose that the equivalent conditions of (i) are satisfied and moreover that the morphisms $X \to S$ and $Y \to S$ are locally of finite presentation. Then $\mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{J}^n$ is a $\mathscr{O}_X$-Algebra of finite presentation, the morphism $X' \to X$ is of finite presentation, the canonical immersion $Y' \to X'$ is transversally regular of codimension 1 relatively to S, and the set of points of X (resp. X') where X (resp. X') is S-flat, is a neighbourhood of Y (resp. Y').*

Proof. **IV-4** | 200

label=(i) The equivalence of *a)* and *b)* follows immediately from (**0**, 6.1.2); to show the equivalence of *b)* and *c)*, we can restrict ourselves to the case where $S = \mathrm{Spec}(\mathrm{A})$ and $X = \mathrm{Spec}(\mathrm{B})$ are affine, with $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of B; the Tor exact sequence shows that condition *c)* implies that $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{B}/\mathfrak{J}^{n+1}, \mathrm{A}') = 0$ for every $n \geqslant 0$ and every A-algebra direct sum $\mathrm{A} \oplus \mathrm{M}$, where M is an A-module, where the multiplication is defined by $(a, m)(a', m') = (aa', am' + a'm)$; we see that the preceding condition is equivalent to $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{B}/\mathfrak{J}^{n+1}, \mathrm{M}) = 0$, hence to the fact that $\mathrm{B}/\mathfrak{J}^{n+1}$ is a flat A-module. If we set $\mathscr{J}_1 = \mathscr{J} \otimes_{\mathscr{O}_g} \mathscr{O}_{S_1}$, we then have $\bigoplus_{n \geqslant 0} \mathscr{J}_1^n = (\bigoplus_{n \geqslant 0} \mathscr{J}^n) \otimes_{\mathscr{O}_g} \mathscr{O}_{S_1}$ and the assertion relative to the blown-up prescheme X_1' follows immediately. Finally, to prove that Y' is S-flat, we can further restrict ourselves to the case where S and X are affine; if we set $\mathrm{C} = \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$, then every point of Y' admits, by virtue of (19.4.2) and ((**II**, 2.3.6)), a neighbourhood whose affine ring is of the form $\mathrm{C}_{(t)}$, where t is a homogeneous element of degree 1 of C; since by hypothesis C is a flat A-module, so is its ring of fractions $\mathrm{C}_{(t)}$, and of the component of degree 0, $\mathrm{C}_{(t)}$, of this A-module graded, whence our assertion.

lbbel=(ii) We can again restrict ourselves to the case where S and X are affine, B being therefore a finitely presented A-algebra, $\mathfrak{J}$ an ideal of finite type of B. By virtue of ((8.9.1)), ((8.6.3)) and (11.2.9), there exists a Noetherian subring A_0 of A, an A_0-algebra of finite type B_0, an ideal $\mathfrak{J}_0$ of type finite of B_0 such that $\mathrm{B} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}$, $\mathfrak{J} = \mathfrak{J}_0\mathrm{B}$, that $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0)$ is an A_0-module *flat* and that $\bigoplus_{n \geqslant 0} \mathfrak{J}_0^n$ is a finitely presented B_0-algebra of finite presentation. Likewise, the morphism $\mathrm{Proj}(\bigoplus \mathfrak{J}^n) \to \mathrm{Spec}(\mathrm{B}_0)$ is of type finite ((**II**, 2.7.1)), and by base change, is of finite presentation; hence the morphism $X' \to X$, which follows by base change, is of finite presentation. We already knew ((19.4.2)) that the canonical immersion $Y' \to X'$ is regular; since Y' is S-flat, as we have seen in (i), we deduce from (19.2.4) that the immersion $Y' \to X'$ is transversally regular relatively to S and that X' is S-flat at the points of Y', hence (11.3.1), X is S-flat at the points of Y, hence ((11.3.4)), X is S-flat at the points of Y in a neighbourhood of Y. On the other hand, by the $\mathscr{O}_X$-Algebra S-flat, is a flat $\mathscr{O}_X$-Algebra of finite presentation; hence the morphism $X' \to X$ is of finite presentation, hence ((11.3.1)), the fact that $\mathrm{gr}^*{}_{\mathscr{J}}(\mathscr{O}_X)$ is a flat $\mathscr{O}_X$-Algebra S-plate implies (11.3.4), that X is S-flat in a neighbourhood of Y.

□

Corollary (19.4.7). — *Let $g : X \to S$ be a morphism locally of finite presentation, Y a closed sub-prescheme of X such that the composite morphism $Y \to X \to S$ is locally of finite presentation; suppose moreover that Y is S-flat and that $\mathscr{O}_X$ is normally flat along Y ((11.3.4)). Then (with the notations of (19.4.1)), the morphism $X' \to X$ is of finite presentation, the canonical immersion $Y' \to X'$ is transversally regular of codimension 1 relatively to S, and the set of points of X (resp. X') where X (resp. X') is S-flat is a neighbourhood of Y (resp. Y').*

Proof. Indeed, by hypothesis $\mathrm{gr}^*{}_{\mathscr{J}}(\mathscr{O}_X)$ is an $\mathscr{O}_Y$-Module flat, hence is S-flat. We can therefore apply the results of (19.4.6), (i) and (ii). Comme Y' is Y-isomorphic to $\mathrm{Proj}(\mathrm{gr}^*{}_{\mathscr{J}}(\mathscr{O}_X))$ (19.4.2), the same reasoning as in (19.4.6) shows that Y' is Y-flat. □

Proposition (19.4.8). — *Suppose the hypotheses of (19.4.7) satisfied and moreover that the morphism $Y' \to Y$ is smooth. Let x' be a point of Y', x its image in Y. For Y to be smooth over S at the point x, it is necessary and sufficient that Y' be smooth over S at the point x'; when this is so, X' is then smooth over S at the point x'.*

Proof. The preceding conclusions are in particular satisfied when the morphism $X \to S$ is locally of finite presentation and the immersion $Y \to X$ is transversally regular relatively to S.

Taking into account the compatibility of the formation of a blown-up prescheme with changes of base in the case envisaged (19.4.6), (i) and of ((17.5.1)) and (17.7.1), it suffices to consider the case where S is the spectrum of an algebraically closed field. But then it amounts to the same to say that an S-prescheme locally of finite type is smooth or that it is regular at this point ((17.15.1)), hence the conclusions follow from (19.4.4). **IV-4** | 201

When $X \to S$ is locally of finite presentation, and the canonical immersion $Y' \to X$ is quasi-regular, this immersion is by definition of finite presentation since the Ideal that defines it is of finite type, hence the composite morphism $Y \to X \to S$ is locally of finite presentation. We have already seen (19.4.3) that in these conditions the morphism $Y' \to Y$ is smooth; moreover, we know then that $\mathscr{N}_{Y/X}$ is an $\mathscr{O}_Y$-Module locally free and that $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$, hence is a flat $\mathscr{O}_X$-Module. The hypotheses of (19.4.7) are thus all satisfied and we can apply the conclusions of (19.4.8). □

Corollary (19.4.9). — *Under the general hypotheses of (19.4.8), suppose moreover that Y is smooth over S at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is smooth over S at the point x_1. If Y is smooth over S, then there exists an open neighbourhood U of Y in X such that X is smooth over S at the points of $U - Y$.*

Proof. To prove the first assertion, we can restrict ourselves to the case where X is a scheme local of closed point x, and the hypothesis implies then that Y is smooth over S, every closed neighbourhood of the point being necessarily Y itself; we conclude then by the same reasoning as in (19.4.5), using the fact that the set of points where X' is smooth over S is open in X'. To prove the second assertion, we can reduce to the case where S is Noetherian, and X of finite type over S, using ((8.9.1)), ((8.6.3)), ((11.2.9)) and (17.7.6); we then conclude as in (19.4.5), using the fact that the set of points where X is smooth over S is open in X. □

Remark (19.4.10). — In (19.4.6), (ii), replace the hypothesis that X and Y are locally of finite presentation over S by the hypothesis that S and X (hence also Y) are *locally Noetherian*. Then it is clear that $\bigoplus_{n\geqslant 0} \mathscr{J}^n$ is an $\mathscr{O}_X$-Algebra of finite type, that the morphism $X' \to X$ is of finite type, and it follows again from (19.2.4) applied to locally Noetherian preschemes, that the immersion $Y' \to X'$ is transversally regular relatively to S and that X' is S-flat at the points of Y'.

Proposition (19.4.11). — *Let X be a prescheme, $\mathscr{E}$ an $\mathscr{O}_X$-Module quasi-coherent, $u : \mathscr{E} \to \mathscr{O}_X$ a homomorphism of $\mathscr{O}_X$-Modules, $\mathscr{J} = u(\mathscr{E})$ the quasi-coherent Ideal of $\mathscr{O}_X$ which is the image of u, Y the closed sub-prescheme of X defined by $\mathscr{J}$. On the other hand, let $\mathrm{P} = \mathbf{P}(\mathscr{E})$ be the projective bundle over X defined by $\mathscr{E}$ ((**II**, 4.1.1)), $p : \mathrm{P} \to X$ the structural morphism, $\alpha_1^\# : p^*(\mathscr{E}) \to \mathscr{O}_\mathrm{P}(1)$ the canonical homomorphism ((**II**, 4.1.5.1)) and $\mathscr{H}$ its kernel, so that we have an exact sequence*

$$0 \longrightarrow \mathscr{H} \longrightarrow p^*(\mathscr{E}) \xrightarrow{\alpha_1^\#} \mathscr{O}_\mathrm{P}(1) \longrightarrow 0.$$

Finally, let $\mathscr{K}$ be the quasi-coherent Ideal of $\mathscr{O}_\mathrm{P}$ which is the image of the restriction $v : \mathscr{H} \to \mathscr{O}_\mathrm{P}$ of $p^*(u)$, and let Z be the closed sub-prescheme of P defined by $\mathscr{K}$. **IV-4** | 202

*label=(i) The image of X' by the closed immersion $j : X' \to \mathbf{P}(\mathscr{E})$ corresponding to the surjective homomorphism of graded $\mathscr{O}_X$-Algebras $\mathbf{S}_{\mathscr{O}_X}(\mathscr{J}) \to \bigoplus_{n\geqslant 0} \mathscr{J}^n$ ((**II**, 3.6.2)) is a closed sub-prescheme of Z.*

lbbel=(ii) If $\mathscr{E}$ is an $\mathscr{O}_X$-Module locally free of rank m, $\mathscr{H}$ is an $\mathscr{O}_\mathrm{P}$-Module locally free of rank $m-1$.

lcbel=(iii) Suppose X is locally Noetherian, that $\mathscr{E}$ is locally free of finite type, that at every point $x \in Y$, the canonical immersion $i : Y \to X$ is regular and of codimension equal to $\mathrm{rg}_{\mathscr{O}_x}(\mathscr{E}_x)$ (which we shall express by saying that the homomorphism u is regular at the point x (cf. chap. V)). Then the image of X' by the closed immersion $j : X' \to \mathrm{P}$ is the sub-prescheme Z; at all points of $X' - Y'$ (Y' being the inverse image of Y in X') the immersion j is transversally regular relatively to X; and finally, at all points x' of Y', X' is regular, the immersion j is regular and of codimension $\mathrm{rg}_{\mathscr{O}_{P,z}}(\mathscr{H}_z)$ where $z = j(x')$ (in other words the homomorphism $v : \mathscr{H} \to \mathscr{O}_\mathrm{P}$ is regular in z).

Proof. All the questions being local on X, we can restrict ourselves to the case where $X = \mathrm{Spec}(\mathrm{A})$ is affine and $\mathscr{E} = \widetilde{\mathrm{E}}$, where E is an A-module. To verify (i), we can in addition examine what happens in an affine open of P of the form $\mathrm{D}_+(t)$, where $t \in \mathrm{E} = \mathbf{S}^1_\mathrm{A}(\mathrm{E})$ ((**II**, 2.3.14)). If we set

$S = \mathbf{S}_A(E)$, we have $\Gamma(D_+(t), \mathscr{O}_P) = S_{(t)}$ and $p^*(\mathscr{E})|D_+(t) = (E \otimes_A S_{(t)})^\sim$. If we look back at the definition of $\alpha_1^\#$ ((**II**, 2.6.2.2)), we see that for every element $a = \sum_i x^{(i)}(x_1^{(i)} \cdots x_n^{(i)})/t^n$ of $E \otimes_A S_{(t)}$ (where the $x^{(i)}$ and $x_k^{(i)}$ are in E), $\alpha_1^\#$ makes it correspond to the element $\sum_i (x^{(i)} x_1^{(i)} \cdots x_n^{(i)})/t^{n+1}$ of $(S(1))_{(t)}$. It is then a matter (to prove (i)) of showing that if this last element is zero, i.e., $a' = v(a) = \sum u(x^{(i)})u(x_1^{(i)}) \cdots u(x_n^{(i)})/(u(t))^n$ by the canonical homomorphism $S_{(t)} \to (\mathbf{S}_A(E))_{(t)} \to (\mathbf{S}_A(A))_{(t)}$. Now, this image is nothing other than the product of the element $u(t)$ and the canonical extension to $S_{(t)}$ of the homomorphism of A-modules $u : E \to A$. This proves (i).

To establish (ii), let us note that we can suppose E is an A-module free of rank m, and that $\alpha_1^\# : E \otimes_A S_{(t)} \to (S(1))_{(t)}$ is surjective and that $(S(1))_{(t)}$ is an $S_{(t)}$-module free of rank 1, the exact sequence

$$0 \longrightarrow H_{(t)} \longrightarrow E \otimes_A S_{(t)} \xrightarrow{\alpha_1^\#} (S(1))_{(t)} \longrightarrow 0$$

is split, and $H_{(t)}$ is therefore an $S_{(t)}$-module projective of rank $m-1$, whence (ii).

To prove (iii), let us examine first what happens above a point $x \in X - Y$. The hypothesis of "regularity" of u at this point made in (iii) amounts to (by virtue of (19.1.1) and of ((**0**, 17.1.7))) saying that X is regular at the point x and that the homomorphism $u_X \otimes 1 : \mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x) \to \mathfrak{m}_x/\mathfrak{m}_x^2$ is *injective*. This being the case, since the morphism $p : P \to X$ is smooth ((17.3.9)), for every $z \in P$ above x, P is regular at the point z ((17.5.8)); moreover ((**0**, 17.3.3)) the canonical homomorphism

$$(\mathfrak{m}_z/\mathfrak{m}_z^2) \otimes_{\mathbf{k}(z)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2$$

(19.4.11.1) is injective. We deduce that the homomorphism

$$(\mathscr{E}_z \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{P,z}) \otimes_{\mathscr{O}_{P,z}} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2$$

deduced from $p^*(u)$ is also injective, since it is written as the composite

$$(\mathscr{E}_z \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)) \otimes_{\mathbf{k}(z)} \mathbf{k}(z) \longrightarrow (\mathfrak{m}_z/\mathfrak{m}_z^2) \otimes_{\mathbf{k}(z)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2$$

where the first arrow is injective by flatness and the second is the injective homomorphism (19.4.11.1). **IV-4** | 203
Since $\mathscr{H}$ is (in a neighbourhood of z in P) a direct factor of $p^*(\mathscr{E})$, the homomorphism $v_z \otimes 1 : \mathscr{H}_z \otimes_{\mathscr{O}_{P,z}} \mathbf{k}(z) \to \mathfrak{m}_z/\mathfrak{m}_z^2$ is also injective, which establishes the "regularity" of the homomorphism v at the point z. It remains to see that the image by j of X' is identical to the closed sub-prescheme Z. It will suffice for this to show that the image by j of the points of $q^{-1}(x)$ (where $q : X' \to X$ is the structural morphism) are exactly those of $Z \cap p^{-1}(x)$ and that on the other hand at each of these points $z = j(x')$, the surjective homomorphism $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective. But, taking into account the hypothesis and of (19.4.3), the restriction $Y' \to Y$ of q is smooth, and since q is supposed regular, $\mathscr{O}_{Y,x'}$ is therefore also regular ((19.1.1)). To prove that the homomorphism $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective, it therefore suffices to show that the dimensions of these two regular **IV-4** | 204
rings are equal ((**0**, 17.1.9)). But by virtue of ((**0**, 17.3.3)), this is also equivalent to the equality of dimensions of the two rings $\mathscr{O}_{Z,z} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$ and $\mathscr{O}_{X',z} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$, so that we can finally replace X' and Z by the fibres $q^{-1}(x)$ and $Z \cap p^{-1}(x)$ respectively, and we are reduced to proving that the morphism $q^{-1}(x) \to Z \cap p^{-1}(x)$ deduced from j is an *isomorphism*; this we can here restrict ourselves to the case where $X = \mathrm{Spec}(B)$, $B = \mathscr{O}_{X,x}$ and $E = B^m$, whence $S = B[T_1, \ldots, T_{m-1}]$; moreover, the hypothesis implies that there exists a regular system of parameters $(t_i)_{1 \leqslant i \leqslant n}$ of B such that $u(T_i) = t_i$ for $1 \leqslant i \leqslant m$ ((**0**, 17.1.7)); let $\mathfrak{J}$ be the image of u; if $\mathfrak{m}$ is the maximal ideal of B, we have $\mathfrak{J}^n \otimes_B (B/\mathfrak{m}) = (\mathfrak{J}^n/\mathfrak{m}\mathfrak{J}^n) \otimes_B (B/\mathfrak{m})$ and by (16.9.4.1), $(\bigoplus_{n \geqslant 0} \mathfrak{J}^n) \otimes_B \mathbf{k}(x)$ is identified with $\mathbf{k}(x)[T_1, \ldots, T_m]$, whence it follows that the immersion $q^{-1}(x) \to p^{-1}(x)$ is an *isomorphism*; this also implies *a fortiori* that the closed sub-prescheme $Z \cap p^{-1}(x)$ is identical to the fibre $p^{-1}(x)$, and completes the proof of (19.4.11). □

19.5. Criteria for M-regularity The relation $H_1(\mathbf{f}, M) = 0$ implies therefore $H_0(f_n, H_1(\mathbf{f}', M)) = 0$ and $H_1(f_n, H_0(\mathbf{f}', M)) = 0$. The first of these two relations means that we have $f_n H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$ (**III**, 1.1.3.5). Now, by definition (**III**, 1.1.1), $H_1(\mathbf{f}', M)$ is isomorphic to a quotient of a submodule N of M^{n-1}; taking on M^{n-1} the filtration of the M^j for $j \leqslant n-1$, on N the induced filtration and on $H_1(\mathbf{f}', M)$ the quotient filtration from that of N, we obtain on $H_1(\mathbf{f}', M)$ a finite filtration whose quotients are isomorphic to quotients of submodules of M, hence are separated for the $\mathfrak{J}$-preadic topology by hypothesis. As the relation $f_n H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$ implies *a fortiori* $\mathfrak{J} H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$, we deduce from the hypothesis and the preceding remarks that $H_1(\mathbf{f}', M) = 0$. The induction hypothesis proves then that the sequence $\mathbf{f}'$ is M-regular. On the other hand, the relation $H_1(f_n, H_0(\mathbf{f}', M)) = 0$ means that f_n is $(M/(\sum_{i=1}^{n-1} f_i M))$-regular, hence the sequence $\mathbf{f}$ is M-regular. **IV-4 | 205**

Corollary (19.5.2). — *Suppose that* A *is a Noetherian ring, that the* f_i $(1 \leqslant i \leqslant n)$ *belong to the radical of* A, *and that* M *is a finitely generated* A*-module. Then the four conditions* a), b'), c) *of* ((19.5.1)) *are equivalent.*

Proof. We know indeed from $(\mathbf{0}_{\mathrm{I}}, 7.3.5)$ that every finitely generated A-module is separated for the $\mathfrak{J}$-preadic topology. □

Proposition (19.5.3). — *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact sequence of A*-modules,* $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant n}$ *an M-regular sequence,* $\mathfrak{J} = \sum_{i=1}^{n} f_i A$. *Consider the following properties:*

a) *The sequence* $\mathbf{f}$ *is* M''*-regular.*
b) *The* $\mathfrak{J}$*-preadic filtration of* M' *is induced on* M' *by the* $\mathfrak{J}$*-preadic filtration of* M *(in other words, the canonical homomorphism* $\mathrm{gr}^*_{\mathfrak{J}}(M') \to \mathrm{gr}^*_{\mathfrak{J}}(M)$ *is injective).*
c) *The canonical homomorphism* $M'/(\sum_{i=1}^{n} f_i M') \to M/(\sum_{i=1}^{n} f_i M)$ *is injective.*

Then we have the implications a) ⇒ b) ⇒ c), *and* a) *implies moreover that the sequence* $\mathbf{f}$ *is* M'*-regular. If every quotient of a submodule of* M'' *is separated for the* $\mathfrak{J}$*-preadic topology, the conditions* a), b) *and* c) *are equivalent.*

Proof. It is clear that *c*) is a consequence of *b*). Let us prove that under all the hypotheses of the last assertion, *c*) implies *a*): since $\mathbf{f}$ is M-regular, by (19.5.1), $H_1(\mathbf{f}, M) = 0$, whence an exact sequence, portion of the exact homology sequence

$$0 \longrightarrow H_1(\mathbf{f}, M'') \longrightarrow H_0(\mathbf{f}, M') \longrightarrow H_0(\mathbf{f}, M).$$

Now, condition *c*) expresses that the homomorphism $H_0(\mathbf{f}, M') \to H_0(\mathbf{f}, M)$ is injective (**III**, 1.1.3.5); it is therefore equivalent to $H_1(\mathbf{f}, M'') = 0$, whence, by virtue of the separation hypothesis and (19.5.1), the fact that $\mathbf{f}$ is M''-regular.

Let us show then that *a*) implies *b*), and also that the sequence $\mathbf{f}$ is M'-regular, proceeding by induction on n. For $n = 1$, f_1 being M-regular is also M'-regular and the property *c*) is none other than the lemma (3.4.1.4). For $n > 1$, the induction hypothesis shows that if we set $\mathbf{f}' = (f_1, \ldots, f_{n-1})$, the sequence $\mathbf{f}'$ is M'-regular and, by virtue of *c*), we have an exact sequence **IV-4 | 206**

$$0 \longrightarrow M'/(\textstyle\sum_{i=1}^{n-1} f_i M') \to M/(\sum_{i=1}^{n-1} f_i M) \to M''/(\sum_{i=1}^{n-1} f_i M'') \to 0.$$

By hypothesis, f_n is $(M/(\sum_{i=1}^{n-1} f_i M))$-regular and $(M''/(\sum_{i=1}^{n-1} f_i M''))$-regular, hence by the same reasoning, by virtue of (3.4.1.4), f_n is also $(M'/(\sum_{i=1}^{n-1} f_i M'))$-regular, and on the other hand, the sequence

$$0 \longrightarrow M'/(\textstyle\sum_{i=1}^{n} f_i M') \to M/(\sum_{i=1}^{n} f_i M) \to M''/(\sum_{i=1}^{n} f_i M'') \to 0$$

is exact, whence *c*).

Let us show finally that *a*) implies *b*). As the sequence $\mathbf{f}$ is then M-regular, M-quasi-regular, and M'-quasi-regular, we have canonical isomorphisms

$$\mathrm{gr}^*_{\mathfrak{J}}(M') \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(M')[T_1, \ldots, T_n], \qquad \mathrm{gr}^*_{\mathfrak{J}}(M) \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(M)[T_1, \ldots, T_n]$$

$(\mathbf{0}, 15.1.7)$, and as we saw above that $\mathrm{gr}^0_{\mathfrak{J}}(M') \to \mathrm{gr}^0_{\mathfrak{J}}(M)$ is injective, the same holds for $\mathrm{gr}^*_{\mathfrak{J}}(M') \to \mathrm{gr}^*_{\mathfrak{J}}(M)$.

We note moreover that the equivalence of conditions $a)$, $b)$, $c)$ of (19.5.3) holds in particular when A is Noetherian, the f_i belong to the radical of A, and the $\mathrm{gr}_k(\mathrm{M})$ are finitely generated A-modules, *then all conditions a) to h) are equivalent.* ☐

(19.5.4). In the rest of this number, we preserve the notations A, $\mathbf{f}$, M, $\mathfrak{J}$, and we suppose moreover M endowed with a decreasing filtration (M_k) formed of sub-A-modules, such that $\mathrm{M}_0 = \mathrm{M}$. Recall $(\mathbf{0}, 15.1.5)$ that we then define on M a second decreasing filtration formed of submodules

$$\mathrm{M}'_k = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$$

and that, if $\mathrm{gr}_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{M})$ are the graded A-modules associated to the filtrations (M_k) and (M'_k) respectively, we define a surjective graded homomorphism of degree 0 $(\mathbf{0}, 15.1.5.2)$

$$\psi_{\mathbf{M}} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$

Recall also that we have the canonical surjective homomorphism $(\mathbf{0}, 15.1.1.1)$

$$\varphi_{\mathrm{gr}_\bullet(\mathrm{M})} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{gr}_\bullet(\mathrm{M}))$$

Theorem (19.5.5). — *With the notations of ((19.5.4)), consider the following properties:*

a) *The sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) *The canonical homomorphism* $\psi_{\mathbf{M}}$ *is bijective.*
c) *The canonical homomorphism* $\varphi_{\mathrm{gr}_\bullet(\mathrm{M})}$ *is bijective (in other words* $(\mathbf{0}, 15.1.7)$*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-quasi-regular).*

Then we have the implications[9] **IV-4 | 207**

$$a) \Rightarrow b) \Rightarrow c).$$

Moreover, if for every integer $k \geqslant 0$, the quotient modules of submodules of $\mathrm{gr}_k(\mathrm{M})$ *are separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the f_i belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a), b) *and* c) are equivalent.

The implication $a) \Rightarrow b)$ has been demonstrated in $(\mathbf{0}, 15.1.8)$*; the implication $c) \Rightarrow a)$ under the separation hypothesis is a particular case of* $(\mathbf{0}, 15.1.9)$*. It remains to prove that $b)$ implies $c)$, which will be done in several steps.*

We denote by $\mathfrak{J}$ the ideal $\sum_{i=1}^n f_i\mathrm{A}$ generated by the f_i; for every sequence $\mathbf{q} = (q_1, \ldots, q_n)$ of n integers $\geqslant 0$, we set $|\mathbf{q}| = \sum_{i=1}^n q_i$ and $\mathbf{f}^{\mathbf{q}} = f_1^{q_1} f_2^{q_2} \cdots f_n^{q_n}$.

(19.5.5.1). For the map $\psi_{\mathbf{M}}$ to be injective (and hence bijective), it is necessary and sufficient that the following condition be satisfied:

$(\mathrm{I}_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \sum_{i=1}^q \mathfrak{J}^{q-i}\mathrm{M}_{p+i} + \mathrm{M}'_{p+q+1}$$

implies $x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ *for every* $\mathbf{q}$ *such that* $|\mathbf{q}| = q$*.*

We proceed as in $(\mathbf{0}, 15.1.6)$ by considering the sub-A-module Q_k (resp. Q'_k) as a term of degree k in the first (resp. second) member of $(\mathbf{0}, 15.1.5.2)$. We endow Q_k with the filtration

$$(\mathrm{Q}_k)_i = \textstyle\sum_{j \leqslant k-i} \big(\sum_{|\mathbf{l}|=j} (\mathrm{gr}_{k-j}(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))\big)\mathrm{T}^{\mathbf{l}}$$

and Q'_k the image filtration formed by $\psi_{\mathbf{M}}((\mathrm{Q}_k)_i) = (\mathrm{Q}'_k)_i$; it suffices then to prove that the homomorphisms $(\psi_{\mathbf{M}})_{ki} : \mathrm{gr}_i(\mathrm{Q}_k) \to \mathrm{gr}_i(\mathrm{Q}'_k)$ are injective. Now, we have

$$\mathrm{gr}_i(\mathrm{Q}_k) = \textstyle\sum_{|\mathbf{l}|=k-i} \big((\mathrm{M}_i/\mathrm{M}_{i+1}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})\big)\mathrm{T}^{\mathbf{l}}$$

on the other hand, $(\mathrm{Q}'_k)_{i+1}$ is the image of $\sum_{h=0}^{k-i-1} \mathfrak{J}^{k-i-h-1}\mathrm{M}_{i+h+1}$ in $\mathrm{M}'_k/\mathrm{M}'_{k+1}$. To write that $(\psi_{\mathbf{M}})_{ki}$ **IV-4 | 208**
is injective is therefore equivalent to writing the condition $(\mathrm{I}_{k-i,i})$; whence our assertion.

[9] The implication $b) \Rightarrow c)$, without the separation hypothesis on the $\mathrm{gr}_k(\mathrm{M})$, is due to P. Deligne, whose demonstration we reproduce below.

(19.5.5.2). *For the sequence* $\mathbf{f}$ *to be* $\mathrm{gr}_\bullet(\mathrm{M})$*-quasi-regular, it suffices that, for* $p \geqslant 0$ *and* $q \geqslant 0$*, the following condition be satisfied:*

$(\mathrm{S}_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation* $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$.

By definition, saying that $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{M})$-quasi-regular means that, for every $p \geqslant 0$ and every $q \geqslant 0$, the relation

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}^{q+1}\mathrm{M}_p$$

for a family $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ of elements of M_p implies $x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$. The condition $(\mathrm{S}_{q,p})$ means that there exists a family $(y^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ of elements of $\mathfrak{J}\mathrm{M}_p$ such that $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}}(x^p_{\mathbf{q}} - y^p_{\mathbf{q}}) \in \mathrm{M}_{p+1}$; it then implies $x^p_{\mathbf{q}} - y^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$, whence $x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$; hence one sees that the conditions $(\mathrm{S}_{q,p})$ imply that the sequence $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{M})$-quasi-regular.

It suffices therefore to prove the following proposition:

(19.5.5.3). *For every* $r \geqslant 0$*, if the conditions* $(\mathrm{I}_{q,p})$ *are satisfied for* $p + q < r$*, then the conditions* $(\mathrm{S}_{q,p})$ *are satisfied for* $p + q < r$*.*

Let us introduce the conditions

$(\mathrm{A}_{q,p})$ *For every family* $(x^p_{\mathbf{q}})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation* $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x^p_{\mathbf{q}} \in \mathrm{M}_{p+1}$ *implies* $x^p_{\mathbf{q}} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$*.*

It is clear that the conjunction of $(\mathrm{A}_{q,p})$ and of $(\mathrm{I}_{q,p})$ implies $(\mathrm{S}_{q,p})$. On the other hand, the conditions $(\mathrm{A}_{1,p})$ and $(\mathrm{S}_{0,p})$ are trivial. By (19.5.3), the following proposition will result:

(19.5.5.4). *For all* $p \geqslant 0$ *and* $q \geqslant 1$*, the condition* $(\mathrm{A}_{q+1,p})$ *is implied by the conjunction of the conditions* $(\mathrm{A}_{q,p})$*,* $(\mathrm{I}_{q-1,p+1})$*,* $(\mathrm{I}_{q-2,p+2})$*, ...,* $(\mathrm{I}_{0,p+q})$*.*

Indeed, once this proposition is proved, the conditions $(\mathrm{I}_{q,p})$ supposed satisfied for $p + q < r$ will imply, for each $p \leqslant r$, the condition $(\mathrm{A}_{q,p})$ for $1 \leqslant q < r - p$, by induction on q.

Let us prove (19.5.5.4). Let us note that the condition $(\mathrm{A}_{q,p})$ is also equivalent to

(19.5.5.5).

$$\mathfrak{J}^q\mathrm{M}_p \cap \mathrm{M}_{p+1} \subset \textstyle\sum_{i=1}^q \mathfrak{J}^{q-i}\mathrm{M}_{p+i}.$$

Let therefore $m \in \mathfrak{J}^{q+1}\mathrm{M}_p \cap \mathrm{M}_{p+1} \subset \mathfrak{J}^q\mathrm{M}_p \cap \mathrm{M}_{p+1}$. Applying condition $(\mathrm{A}_{q,p})$, we see that for each i such that $1 \leqslant i \leqslant q$, there exists a family $(y^{p+i}_{\mathbf{q}})_{|\mathbf{q}|=q-i}$ of elements of M_{p+i} such that

$$m = \textstyle\sum_{i=1}^q \big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y^{p+i}_{\mathbf{q}}\big).$$

Suppose that for $1 \leqslant i < j$ ($j \geqslant 1$), we have proved that

$$y^{p+i}_{\mathbf{q}} \in \mathfrak{J}\mathrm{M}_{p+i} + \mathrm{M}_{p+i+1} \qquad (\text{for } |\mathbf{q}| = q - i).$$

We deduce by definition (19.5.4)

$$m - \textstyle\sum_{i<j}\big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y^{p+i}_{\mathbf{q}}\big) \in \mathrm{M}'_{p+q+1},$$

Since by hypothesis $(\mathrm{I}_{q-j,p+j})$ is satisfied, we deduce

$$y^{p+j}_{\mathbf{q}} \in \mathfrak{J}\mathrm{M}_{p+j} + \mathrm{M}_{p+j+1}$$

and, by induction on j, one sees therefore that this condition is satisfied for *every* j such that $1 \leqslant j \leqslant q$. **IV-4 | 209**
We have therefore

$$m \in \textstyle\sum_{i=1}^q \mathfrak{J}^{q-i}(\mathfrak{J}\mathrm{M}_{p+i} + \mathrm{M}_{p+i+1}) = \sum_{i=1}^{q+1} \mathfrak{J}^{q-i+1}\mathrm{M}_{p+i}$$

and we have thus proved $(\mathrm{A}_{q+1,p})$ and completed the demonstration of (19.5.5).

Remarks (19.5.6). —

(i) The results of this number remain valid when, instead of an A-module M and elements $f_i \in \mathrm{A}$, one takes an object M of any abelian category, and n endomorphisms f_i of M that mutually commute; the demonstrations adapt without difficulty to this more general case. The hypothesis of separation for an A-module N relative to the $\mathfrak{J}$-preadic topology must here be replaced by the hypothesis that every sub-object of N contained in all the $\sum f_i^r(\mathrm{N})$ (r an arbitrary integer) is necessarily zero. The same remark applies to the results of (19.7).

(ii) Suppose that A is a *graded* ring with degrees bounded below. If M (resp. each $\mathrm{gr}_p(\mathrm{M})$) is a graded A-module with degrees bounded below, and if the f_i contain no homogeneous component of degree 0, the separation hypothesis of $(\mathbf{0}, 15.1.9)$ is *ipso facto* satisfied for M (resp. each $\mathrm{gr}_p(\mathrm{M})$), and one sees that in this case we have the implication $c) \Rightarrow a)$ in (19.5.1) (resp. (19.5.5)).

(iii) Recall that, without the separation hypothesis, the implication $c) \Rightarrow b)$ is no longer valid even for $n = 1$ $(\mathbf{0}, 15.1.12, (iii))$.

19.6. Regular sequences relative to a filtered quotient module

(19.6.1). Let A be a ring, $\mathbf{f} = (f_1, \ldots, f_n)$ a finite sequence of elements of A, $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ the ideal it generates, and consider an exact sequence of A-modules

$$0 \longrightarrow \mathrm{R} \xrightarrow{j} \mathrm{N} \xrightarrow{p} \mathrm{M} \longrightarrow 0$$

where one supposes that N is endowed with a decreasing filtration (N_k) of sub-A-modules, with $\mathrm{N}_0 = \mathrm{N}$. We set $\mathrm{R}_k = \mathrm{R} \cap \mathrm{N}_k$ (filtration induced by (N_k)), $\mathrm{M}_k = p(\mathrm{N}_k)$ (quotient filtration of (N_k)) and we denote by $\mathrm{gr}_\bullet(\mathrm{N})$, $\mathrm{gr}_\bullet(\mathrm{R})$ and $\mathrm{gr}_\bullet(\mathrm{M})$ the graded A-modules associated with these three filtrations. We set on the other hand for all k,

$$\mathrm{N}'_k = \mathrm{N}_k + \mathfrak{J}\mathrm{N}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{N}_1 + \mathfrak{J}^k\mathrm{N}_0$$

so that $\mathrm{M}'_k = p(\mathrm{N}'_k) = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$; we set on the other hand $\mathrm{R}'_k = \mathrm{R} \cap \mathrm{N}'_k$ (filtration *induced* by (N'_k)), and we denote by $\mathrm{gr}'_\bullet(\mathrm{N})$, $\mathrm{gr}'_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{R})$ the graded A-modules associated with these three filtrations; we thus have a commutative diagram of exact sequences

(19.6.1.1)
$$\begin{tikzcd}
0 \arrow[r] & \mathrm{gr}_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}(j)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}(p)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{M}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}'(j)"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}'(p)"] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[r] & 0
\end{tikzcd}$$

where the vertical arrows are the canonical homomorphisms from a graded module associated with a filtration to a graded module associated with a coarser filtration. **IV-4** | 210

We note that if one sets

$$\mathrm{R}''_k = \mathrm{R}_k + \mathfrak{J}\mathrm{R}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{R}_1 + \mathfrak{J}^k\mathrm{R}_0$$

we evidently have $\mathrm{R}''_k \subset \mathrm{R}'_k$, but the filtrations (R''_k) and (R'_k) are in general distinct; we denote by $\mathrm{gr}''_\bullet(\mathrm{R})$ the graded A-module associated with the filtration (R''_k).

(19.6.2). We defined in $(\mathbf{0}, 15.1.5.2)$ the surjective graded homomorphisms of degree 0

$$\psi_\mathrm{N} : (\mathrm{gr}_\bullet(\mathrm{N}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{N})$$
$$\psi_\mathrm{M} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$
$$\psi_\mathrm{R} : (\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}''_\bullet(\mathrm{R})$$

the first two of which are deduced from the last two vertical arrows of (19.6.1.1).

As the filtration (R''_k) is finer than (R'_k), on R we have a canonical homomorphism $\gamma : \mathrm{gr}''_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R})$; we will denote by ψ'_R the composite homomorphism $\gamma \circ \psi_\mathrm{R}$. It then follows immediately from the definitions that we have a commutative diagram

(19.6.2.1)
$$\begin{tikzcd}
 & 0 \arrow[d] \\
(\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi'_{\mathrm{R}}"] \arrow[d, "j'"] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[d, "\mathrm{gr}'(j)"] \\
(\mathrm{gr}_\bullet(\mathrm{N}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{N}}"] \arrow[d, "p'"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[d, "\mathrm{gr}'(p)"] \\
(\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{M}}"] \arrow[d] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[d] \\
0 & 0
\end{tikzcd}$$

where the columns are exact.

Proposition (19.6.3). — *With the preceding notations, suppose that the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{N})$*-regular. Consider the following properties:*

a) $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) ψ_{M} *is bijective.*
c) ψ'_{R} *is surjective (in other words* $\mathrm{gr}'_\bullet(\mathrm{R})$ *is generated as* $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{A})$*-module by* $\mathrm{Im}(\mathrm{gr}_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R}))$).
d) ψ'_{R} *is injective.*
d′) *The homomorphism* $(\psi'_{\mathrm{R}})_0 : \mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}'_\bullet(\mathrm{R})$ *obtained by restricting* ψ'_{R} *to polynomials of degree* 0*, is injective.*
e) ψ'_{R} *is bijective.*
f) j' *is injective.* IV-4 | 211
f′) *The homomorphism* $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}_\bullet(\mathrm{N}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})$ *obtained by restricting* j' *to polynomials of degree* 0*, is injective.*
g) *The* $\mathfrak{J}$*-preadic filtration of* $\mathrm{gr}_\bullet(\mathrm{R})$ *is induced by that of* $\mathrm{gr}_\bullet(\mathrm{N})$.
h) *The filtrations* (R'_k) *and* (R''_k) *on* R *are identical.*

Then we have the following implications:

$$\begin{tikzcd}
a) \arrow[r, Rightarrow] & e) \arrow[r, Rightarrow] \arrow[dr] & b) \arrow[r, Leftrightarrow] & c) \Leftrightarrow h) \\
 & & d) & \\
g) \arrow[r, Rightarrow] & d) \arrow[r, Leftrightarrow] & d') \arrow[r, Leftrightarrow] & f) \Leftrightarrow f')
\end{tikzcd}$$

Moreover, when every quotient module of a submodule of $\mathrm{gr}_k(\mathrm{M})$ *is separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a) *to* h) are equivalent.

Proof. Let us note that, from (19.5.5), the hypothesis that $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{N})$-regular implies that ψ_{N} is *bijective*; as $\mathrm{gr}'(j)$ is injective, the equivalence of d) and f) results from the diagram (19.6.2.1); as moreover f) and f') are trivially equivalent, and also d) and d'), by the five lemma this proves the equivalence of d), d'), f) and f').

As one already knows that ψ_{M} is surjective and ψ_{N} bijective, the equivalence of b) and c) also follows from the diagram (19.6.2.1) by a particular case of the five lemma.

The relation $\psi'_{\mathrm{R}} = \gamma \circ \psi_{\mathrm{R}}$ (19.6.2) and the fact that ψ_{R} is surjective show that condition c) is equivalent to saying that γ is surjective, which is also equivalent to condition h); hence we have proved the equivalence of b), c) and h).

It is trivial that e) is equivalent to the conjunction of c) and d), and hence also of b) and d). Let us note now that a) implies b) and that b) implies a) under the separation hypothesis (19.5.5). On the other hand, the implications $a) \Rightarrow g) \Rightarrow f')$, and the implication $f') \Rightarrow a)$ under the separation hypothesis, result from (19.5.3). Hence we have proved a) implies e) (equivalent to the conjunction of b) and f')), and also that all conditions are equivalent under the separation hypothesis. □

Corollary (19.6.4). — *Let $\mathfrak{N}$ be an ideal of* A, $\mathfrak{L} = \mathfrak{J} + \mathfrak{N}$. *Suppose that the filtration (hence also* (M_k)*) is the $\mathfrak{N}$-preadic filtration, so that the filtrations* (M'_k) *and* (N'_k) *are $\mathfrak{L}$-preadic filtrations. Consider* $\mathrm{gr}_\bullet(\mathrm{R})$ *(resp.* $\mathrm{gr}'_\bullet(\mathrm{R})$*) as a graded module over* $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$ *(resp.* $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$*).*

(i) *Let* S *be a part of* $\mathrm{gr}_\bullet(\mathrm{R})$ *that generates* $\mathrm{gr}_\bullet(\mathrm{R})$ *as a* $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$*-module. Then condition* c) *of* ((19.6.3)) *is equivalent to the following condition:*

c$'$) *The image of* S *in* $\mathrm{gr}'_\bullet(\mathrm{R})$ *by* ψ'_R *generates* $\mathrm{gr}'_\bullet(\mathrm{R})$ *as a* $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$*-module.*

(ii) *Suppose condition* e) *of* ((19.6.3)) *is satisfied, and also that the quotients of* $\mathrm{gr}_k(\mathrm{R})$ *are separated for the $\mathfrak{J}$-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules). Then,* for a part S of $\mathrm{gr}_\bullet(\mathrm{R})$ formed of homogeneous elements to generate $\mathrm{gr}_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$-module, it is necessary and sufficient that its image by ψ'_R generates $\mathrm{gr}'_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module.

Proof. **IV-4** | 212

(i) Consider the homomorphism of degree 0

$$\psi_\mathrm{A} : (\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{A}) = \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$$

$(\mathbf{0}, 15.1.5.2)$. The fact that this homomorphism is *surjective* implies that the sub-$\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-algebra of $\mathrm{gr}'_\bullet(\mathrm{R})$ generated by $\mathrm{S}' = \psi'_\mathrm{R}(\mathrm{S})$ is none other than $\mathrm{Im}(\psi'_\mathrm{R})$, whence the equivalence of c) and c').

(ii) Since condition e) is satisfied, it follows by (i) that it remains to prove the sufficiency of the condition in the statement. Now, as ψ'_R is bijective, to say that S' generates $\mathrm{gr}'_\bullet(\mathrm{R})$ considered as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module amounts to saying that S generates $(\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module, and by the surjectivity of ψ_A, we can in this statement replace the ring $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$ by $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$. It amounts to saying that S generates $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J})$ as a module over the ring $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J})$, or also as a $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$-module. Now, if T is the sub-$\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$-module graded of $\mathrm{gr}_\bullet(\mathrm{R})$ generated by S, and if we set $\mathrm{T}' = \mathrm{gr}_\bullet(\mathrm{R})/\mathrm{T}$, on the one hand by hypothesis $\mathfrak{J}\mathrm{T}' = 0$, or $\mathfrak{J}\mathrm{T}' = \mathrm{T}'$; but T' is a graded module, and each T'_p is a quotient of $\mathrm{gr}_p(\mathrm{R})$, hence separated for the $\mathfrak{J}$-preadic topology; the hypothesis therefore implies $\mathrm{T}' = 0$, which completes the proof of the corollary.

□

19.7. Hironaka's normal flatness criterion

Theorem (19.7.1). — (Hironaka). *Let* A *be a ring,* $\mathfrak{N}$ *an ideal of* A, $\mathbf{f} = (f_1, \ldots, f_n)$ *a sequence of elements of* A *that is* $(\mathrm{A}/\mathfrak{N})$*-regular,* $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ *the ideal generated by* $\mathbf{f}$, $\mathfrak{L} = \mathfrak{J} + \mathfrak{N}$, M *an* A*-module. In this case* $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}) = (\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{L}) = (\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$*, since* $(\mathfrak{N}^n\mathrm{M}/\mathfrak{N}^{n+1}\mathrm{M}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}) = \mathfrak{N}^n\mathrm{M}/(\mathfrak{N}^{n+1} + \mathfrak{J}\mathfrak{N}^n)\mathrm{M} = \mathfrak{N}^n\mathrm{M}/(\mathfrak{N}^{n+1} + \mathfrak{L}\mathfrak{N}^n)\mathrm{M}$*, and the last equality is a result of Bourbaki,* Alg.*, chap. II, 3^e ed., § 3, n^o 7, cor. 2 of prop. 6.*

Consider the following conditions:

a) $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{N})$*-module.*

b) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the canonical homomorphism* $(\mathbf{0}, 15.1.5.2)$

$$\psi_\mathrm{M} : ((\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$$

is bijective.

c) $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the sequence* $\mathbf{f}$ *is* $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}))$*-regular.*

d) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module, and for every* $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{N}}(\mathrm{A}/\mathfrak{L})$, $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}))_\mathfrak{p}$ *is a flat* $(\mathrm{A}/\mathfrak{N})_\mathfrak{p}$*-module.*

Then we have the implications

$$a) \Rightarrow c) \Rightarrow b) \quad \searrow \quad d)$$

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ $(k \geqslant 0)$ *is ideally separated for* $\mathfrak{J}$ *(Bourbaki,* Alg. comm.*, chap. III, § 5, n^o 1), the conditions* a), b), c) *are equivalent.*

Finally, when A *is Noetherian, the* f_i *belong to the radical of* A*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{A})$*-regular, and* M *is a finitely generated* A*-module, the conditions* a), b), c), d) *are equivalent.* **IV-4** | 213

Proof. The implication $a) \Rightarrow c)$ is immediate $(\mathbf{0}, 15.1.13)$. If $c)$ is satisfied, it results from (19.5.5) that ψ_{M} is bijective; moreover, as $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$, isomorphic to $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$, is a flat $(\mathrm{A}/\mathfrak{L})$-module, hence so is it for $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ since ψ_{M} is an $(\mathrm{A}/\mathfrak{L})$-isomorphism; hence $c)$ implies $b)$. The fact that $a)$ implies also $d)$ follows immediately.

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is ideally separated for $\mathfrak{J}$, it results from (19.5.5) that condition $b)$ implies that $\mathbf{f}$ is $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}))$-regular; moreover, as $(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$, isomorphic to $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$, is a flat $(\mathrm{A}/\mathfrak{L})$-module, it is a direct factor of this same $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$; this proves therefore that $b)$ implies $c)$. On the other hand, under the same hypothesis, $c)$ implies $a)$ by virtue of $(\mathbf{0}, 15.1.21)$ (where we can replace the Noetherian hypothesis by the hypothesis that M is ideally separated for $\mathfrak{J}$, by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$).

It remains therefore to prove that when A is Noetherian, M of finite type, and the f_i belong to the radical of A, $d)$ implies $a)$.

There exists a free A-module N of *finite type* and an exact sequence $0 \to \mathrm{R} \to \mathrm{N} \to \mathrm{M} \to 0$. It suffices to prove that $d)$ implies $b)$, since every quotient of a sub-$\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is then ideally separated for $\mathfrak{J}$, by virtue of the fact that $\mathrm{gr}^k_{\mathfrak{N}}(\mathrm{M})$ is a finitely generated A-module, that $\mathfrak{J}$ is contained in the radical of A, and that A is Noetherian (Bourbaki, *Alg. comm.*, chap. III, § 5, nº 1); in other words, it suffices to show that ψ_{M} is *bijective*.

As by hypothesis the sequence $\mathbf{f}$ is also $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N})$-regular, one can therefore apply (19.6.3), since every quotient module of a sub-$\mathrm{gr}_k(\mathrm{M})$ is separated for the $\mathfrak{J}$-preadic topology since the f_i belong to the radical of A by hypothesis. The diagram (19.6.2.1) here reads

(19.7.1.1)
$$\begin{tikzcd}
 & 0 \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{R},\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi'_{\mathrm{R}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N}) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{N}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{N}) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \arrow[r, "\psi_{\mathrm{M}}"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M}) \arrow[d] \\
0 & 0
\end{tikzcd}$$

where $\mathrm{gr}^*_{\mathfrak{N}}(\mathrm{R},\mathrm{N})$ (resp. $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N})$) is the graded associated to R for the filtration $\mathrm{R} \cap \mathfrak{N}^m\mathrm{N}$ (resp. $\mathrm{R} \cap \mathfrak{L}^m\mathrm{N}$); recall that in this diagram the columns are exact and that ψ_{N} is *bijective*. **IV-4** | 214

(19.7.1.2). Set $\mathrm{B} = \mathrm{gr}^*_{\mathfrak{N}} \mathrm{A} \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$, $\mathrm{P} = \mathrm{gr}^*_{\mathfrak{N}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{N}} (\mathrm{A}/\mathfrak{L})$, $\mathrm{Q} = \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{R},\mathrm{N})$; since ψ_{N} is bijective, $\mathrm{P}[\mathrm{T}_1, \dots, \mathrm{T}_n]$ identifies with $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{N})$, and Q is a sub-$\mathrm{B}[\mathrm{T}_1, \dots, \mathrm{T}_n]$-module of $\mathrm{P}[\mathrm{T}_1, \dots, \mathrm{T}_n]$; we will first show that we have

(19.7.1.3).
$$\mathrm{Q} = (\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \dots, \mathrm{T}_n].$$

For this, set $\mathrm{Z} = \mathrm{Q}/((\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \dots, \mathrm{T}_n])$; this is a sub-$\mathrm{B}[\mathrm{T}_1, \dots, \mathrm{T}_n]$-module of $\mathrm{P}[\mathrm{T}_1, \dots, \mathrm{T}_n]/((\mathrm{Q} \cap \mathrm{P})[\mathrm{T}_1, \dots, \mathrm{T}_n]) = (\mathrm{P}/(\mathrm{Q} \cap \mathrm{P}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$. As an $(\mathrm{A}/\mathfrak{L})$-module, it is therefore isomorphic to a submodule of a direct sum of $(\mathrm{A}/\mathfrak{L})$-modules isomorphic to $\mathrm{P}/(Q \cap \mathrm{P}) = (P+Q)/Q$. But $(P+Q)/Q$ is a $(A/\mathfrak{L})$-module isomorphic to a submodule of $\mathrm{P}[\mathrm{T}_1, \dots, \mathrm{T}_n]/\mathrm{Q}$, which is none other than $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ by the diagram (19.7.1.1).

But each $\mathrm{gr}^m_{\mathfrak{L}}(\mathrm{M})$ is by hypothesis a *flat* $(\mathrm{A}/\mathfrak{L})$-module of finite type, hence projective, and we have the inclusion $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})) \subset \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$. To see that $\mathrm{Z} = 0$, it suffices by (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 1, cor. 1 of prop. 2 and nº 3, prop. 7) to see that for every $\mathfrak{q} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$, $\mathrm{Z}_{\mathfrak{q}} = 0$. But the ideals of $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$ are of the form $\mathfrak{p}/(\mathfrak{L}/\mathfrak{N})$, where $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{N}}(\mathrm{A}/\mathfrak{L})$, and it comes down to the same to see that $\mathrm{Z}_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{N}}(\mathrm{A}/\mathfrak{L})$. Now, if $\mathfrak{p} = \mathfrak{r}/\mathfrak{N}$, where $\mathfrak{r}$ is a prime ideal of A, and $(\mathrm{A}/\mathrm{K})_{\mathfrak{p}} = \mathrm{A}_{\mathfrak{r}}/\mathrm{K}_{\mathfrak{r}}$, and the images of the f_i

in $A_{\mathfrak{r}}$ form a $\mathrm{gr}^*_{\mathfrak{R}_{\mathfrak{r}}}(M_{\mathfrak{r}})$-regular sequence $(\mathbf{0}, 15.1.14)$; applying to $A_{\mathfrak{r}}$, $\mathfrak{R}_{\mathfrak{r}}$ and $M_{\mathfrak{r}}$ the implication $a) \Rightarrow b)$ of the statement, we see from the hypothesis $d)$ that $\psi_{M_{\mathfrak{r}}}$ is bijective, hence also $\psi'_{R_{\mathfrak{r}}}$ by virtue of (19.6.3); in other words, $Q_{\mathfrak{r}} = Q'_{\mathfrak{r}}[T_1, \dots, T_n]$, where we have set $Q' = \mathrm{gr}^*_{\mathfrak{R}}(R, N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L})$, which is here a sub-B-module of P; in particular $Q'_{\mathfrak{r}} \subset Q_{\mathfrak{r}} \cap P_{\mathfrak{r}}$; hence finally $Z_{\mathfrak{r}} = Z_{\mathfrak{p}} = 0$, which completes the demonstration of (19.7.1.3).

(19.7.1.5). By virtue of (19.7.1.3), to prove that ψ'_R is surjective, it suffices to show that every homogeneous element of degree m of $Q \cap P$ is the image of an element of the same degree of P. Now, the elements of degree m of P identify (by ψ_N) with elements of $\mathrm{gr}^m_{\mathfrak{L}}(N)$; those which belong to elements of $\mathrm{gr}^m_{\mathfrak{L}}(R, N)$ identify with elements of $((R \cap \mathfrak{R}^m N) + \mathfrak{L}^{m+1}N)/\mathfrak{L}^{m+1}N$; those of $\mathrm{gr}^m_{\mathfrak{R}}(R, N)$ identify with elements of $((R \cap \mathfrak{R}^m N + \mathfrak{L}^{m+1}N)/\mathfrak{L}^{m+1}N$. It will therefore suffice to prove, for $m > 0$, the inclusion

$$((R \cap \mathfrak{L}^m N) + \mathfrak{L}^{m+1}N) \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R \cap \mathfrak{R}^m N) + \mathfrak{L}^{m+1}N.$$

As $\mathfrak{R} \subset \mathfrak{L}$, we see immediately that the first member of this relation is equal to $(R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N)) + \mathfrak{L}^{m+1}N$, so that everything reduces to proving

(19.7.1.6).
$$R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R \cap \mathfrak{R}^m N) + \mathfrak{L}^{m+1}N.$$

We will proceed by induction on m, assuming therefore (19.7.1.6) verified when m is replaced by an integer $m' < m$. We will consider on the other hand, for m fixed and for an integer $d \geqslant 0$, the relation **IV-4 | 215**

$$(*_d) \qquad R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R \cap \mathfrak{R}^d N \cap \mathfrak{L}^m N) + \mathfrak{L}^{m+1}N.$$

It is clear that it suffices to demonstrate $(*_d)$ for $d = m$, and on the other hand $(*_0)$ is trivially true. We will prove $(*_d)$ by *induction on d*; in other words, we suppose, for a $d \geqslant 0$ fixed (with $d < m$), that $(*_d)$ is true, and we wish to prove

$$(*_{d+1}) \qquad R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R \cap \mathfrak{R}^{d+1} N \cap \mathfrak{L}^m N) + \mathfrak{L}^{m+1}N.$$

Consider for this, for an $h \geqslant 0$, the relation

$$(**_{d+1,h}) \qquad R \cap (\mathfrak{R}^m N + \mathfrak{R}^d \mathfrak{L}^h N) \subset (R \cap (\mathfrak{R}^{d+1} N + \mathfrak{R}^d \mathfrak{L}^h N) \cap \mathfrak{L}^m N) + \mathfrak{L}^{m+1}N.$$

The hypothesis $(*_d)$ implies that $(**_{d+1,0})$ is true. We will prove by *induction on h* that $(**_{d+1,h})$ is true for every $h \geqslant 0$. But we have

Lemma (19.7.1.7). — *Let A be a Noetherian ring, N a finitely generated A-module, $\mathfrak{L}$ an ideal of A, E, F two submodules of N. For every $k > 0$ there exists $h > 0$ such that $E \cap (F + \mathfrak{L}^h N) \subset E \cap (E \cap F) + \mathfrak{L}^k N$.*

Proof. Indeed, let $\varphi : N \to N/(E \cap F) = N_1$ be the canonical homomorphism, and set $E_1 = \varphi(E)$, $F_1 = \varphi(F)$, so that $E_1 \cap F_1 = 0$ and $\varphi(E \cap (F + \mathfrak{L}^h N)) = E_1 \cap (F_1 + \mathfrak{L}^h N_1)$. We know $(\mathbf{0}_{\mathrm{I}}, 7.3.2)$ that there exists h such that $(E_1 + F_1) \cap \mathfrak{L}^h N_1 \subset \mathfrak{L}^k (E_1 + F_1)$; if h is thus chosen and if $x_1 \in E_1$ is such that there exists $y_1 \in F_1$ for which $x_1 - y_1 \in \mathfrak{L}^h N_1$, we deduce $x_1 - y_1 \in \mathfrak{L}^k N_1$, hence $E_1 \cap (F_1 + \mathfrak{L}^h N_1) \subset \mathfrak{L}^k N_1$; therefore $E \cap (F + \mathfrak{L}^h N) \subset E \cap F + \mathfrak{L}^k N$, which proves the lemma. ☐

It suffices then to apply this lemma with $E = R \cap \mathfrak{L}^m N$, $F = \mathfrak{R}^{d+1}N$ and to take $k = m + 1$ to deduce that $(**_{d+1,h})$ implies $(*_{d+1})$.

We suppose therefore in all that follows that $(**_{d+1,h})$ *is satisfied.*

(19.7.1.8). *Demonstration of $(**_{d+1,h+1})$; First case: $h \leqslant m - d$.*

We will first prove that, for every $h \geqslant 0$, we have

(19.7.1.9).
$$R \cap (\mathfrak{R}^{d+1}N + \mathfrak{R}^d \mathfrak{L}^h N) \subset \mathfrak{L}^h (R \cap \mathfrak{R}^d N) + \mathfrak{L}^{h+1} \mathfrak{R}^d N + \mathfrak{R}^{d+1} N.$$

Consider for this an element z of the first member of (19.7.1.9). We note that $\mathfrak{R}^{d+1}N + \mathfrak{R}^d \mathfrak{L}^h N = \mathfrak{R}^{d+1}N + \mathfrak{R}^d \mathfrak{J}^h N$ since $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$; we will consider the class $\bar{z}$ of z in $\mathrm{gr}^{h+d}_{\mathfrak{J}}(\mathrm{gr}^*_{\mathfrak{R}}(N))$, which, by virtue of (19.5.5) and the hypothesis on $\mathbf{f}$, identifies with a sub-$(A/\mathfrak{L})$-module of $\mathrm{gr}^{h+d}_{\mathfrak{L}}(N)$. Since $z \in R$, we will show that the image of $\bar{z}$ in $\mathrm{gr}^*_{\mathfrak{L}}(N)$ is zero. **IV-4 | 216**

(19.7.1.12). Indeed, in the demonstration of (19.7.1.3) that, for every prime ideal $\mathfrak{r}$ of A such that $\mathfrak{p} = \mathfrak{r}/\mathfrak{R} \in \operatorname{Ass}_{A/\mathfrak{R}}(A/\mathfrak{L})$, the map $\psi'_{R_\mathfrak{r}}$ is bijective, and therefore all the modules when localised at their primes, the image $z/1$ in $R_\mathfrak{r}$ belongs to $(R_\mathfrak{r} \cap \mathfrak{R}_\mathfrak{r}^d N_\mathfrak{r}) + \mathfrak{L}_\mathfrak{r}^{h+1} N_\mathfrak{r}$, hence the image $\bar{z}/1$ of $\bar{z}$ in $\operatorname{gr}_{\mathfrak{L}_\mathfrak{r}}(R_\mathfrak{r}, N_\mathfrak{r}) = (\operatorname{gr}^*_\mathfrak{L}(R,N))_\mathfrak{r}$, and hence $\bar{z}$ belongs to $\operatorname{gr}^{h+d}_\mathfrak{L}(R,N)$. In other words, the image of $\bar{z}$ by the canonical map $(\operatorname{gr}^*_\mathfrak{L}(N))_\mathfrak{q} \to (\operatorname{gr}^*_\mathfrak{L}(M))_\mathfrak{q}$ is zero for every $\mathfrak{q} \in \operatorname{Ass}_{A/\mathfrak{L}}(A/\mathfrak{L})$. As we saw in the demonstration of (19.7.1.3) that $\operatorname{Ass}_{A/\mathfrak{L}}(\operatorname{gr}^*_\mathfrak{L}(M)) \subset \operatorname{Ass}_{A/\mathfrak{L}}(A/\mathfrak{L})$, we deduce (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 1, cor. 1 of prop. 2 and nº 3, prop. 7) that the image of $\bar{z}$ by the canonical map $\operatorname{gr}^*_\mathfrak{L}(N) \to \operatorname{gr}^*_\mathfrak{L}(M)$ is zero, that is to say we have the relation (19.7.1.9).

(19.7.1.13). It suffices to prove that, in general, for $d < m$, we have

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d.$$

Indeed, as $\mathfrak{R} \subset \mathfrak{L}$, hence $\mathfrak{L}^{m-d+1}\mathfrak{R}^d \subset \mathfrak{L}^{m+1}$, the preceding relation implies

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{L}^{m+1} \cap \mathfrak{R}^{d+1}$$

and by induction on $d < m$, one concludes

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d.$$

(19.7.1.14). Indeed, as $\mathfrak{R} \subset \mathfrak{L}$, hence $\mathfrak{L}^{m-d+1}\mathfrak{R}^d \subset \mathfrak{L}^{m+1}$, we have

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.$$

Since $\mathfrak{R} \subset \mathfrak{L}$, $\mathfrak{L}^{m-d+1}\mathfrak{R}^d \subset \mathfrak{L}^{m+1}$, the preceding relation implies

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{L}^{m+1} \cap \mathfrak{R}^{d+1}$$

and by induction on $d < m$, one concludes that $\mathfrak{R}^{m+1} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d$.

(19.7.1.15). As $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$, the relation (19.7.1.14) also reads

$$(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \ldots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \subset \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{J}^{m-d}\mathfrak{R}^{d+1} + \ldots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1} + \mathfrak{R}^{d+1} = \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.$$

We will show that this inclusion is itself a consequence of

(19.7.1.16).

$$\mathfrak{R}^a\mathfrak{J}^b \cap \mathfrak{R}^{a+1} \subset \mathfrak{R}^{a+1}\mathfrak{J}^{b-1}$$

valid for $a \geqslant 0$, $b \geqslant 1$. Indeed, it suffices, to prove (19.7.1.15), to show that for $0 < q \leqslant d$,

(19.7.1.17).

$$(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \ldots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \subset (\mathfrak{J}^{m+1-q}\mathfrak{R}^q + \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d$$

since for $q = d$, this will imply (19.7.1.15). Now, to prove (19.7.1.17), it suffices to proceed by induction on q assuming it true for $q < d$; an element of the first member is written by hypothesis $y + z$ with $y \in \mathfrak{J}^{m+1-q}\mathfrak{R}^q$, $z \in \mathfrak{R}^d \subset \mathfrak{R}^{q+1}$; as $y + z \in \mathfrak{R}^d \subset \mathfrak{R}^{q+1}$, we deduce $y \in \mathfrak{R}^{q+1}$, since $z \in \mathfrak{R}^{q+1}$; hence

$$y + z \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}, \quad \text{and} \quad y + z \in \mathfrak{R}^d$$

by hypothesis, which proves (19.7.1.17) where q has been replaced by $q + 1$. **IV-4** | 217

It remains therefore to establish (19.7.1.16). An element of the first member is written $a_1 f_1 + \ldots + a_n f_n$, where the a_i belong to $\mathfrak{R}^a\mathfrak{J}^{b-1}$; taking into account that the sequence $\mathbf{f}$ is $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-regular, hence also $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-quasi-regular $(\mathbf{0}, 15.1.9)$, it follows from the definition $(\mathbf{0}, 15.1.7)$ that the relation $a_1 f_1 + \ldots + a_n f_n \in \mathfrak{R}^{a+1}$, where the $a_i \in \mathfrak{R}^a$, necessarily implies $a_i \in \mathfrak{R}^a\mathfrak{J}^{b-1} \cap \mathfrak{R}^{a+1}$; it suffices to argue by induction on b for $b \geqslant 1$ (the case $b = 1$ being trivial) to establish (19.7.1.16) and **IV-4** | 218
by extension also $(**_{d+1,h+1})$ for $h \leqslant m - d$.

(19.7.1.18). *Demonstration of* $(**_{d+1,h+1})$*; Second case:* $h > m - d$.

We will first prove that, for every $h \geqslant 0$, we have

(19.7.1.19).

$$R \cap (\mathfrak{R}^{d+1}N + \mathfrak{R}^d\mathfrak{L}^h N) \subset \mathfrak{L}^h(R \cap \mathfrak{R}^d N) + \mathfrak{L}^{h+1}\mathfrak{R}^d N + \mathfrak{R}^{d+1}N.$$

Consider for this an element z of the first member of (19.7.1.19). We note that $\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N} = \mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{J}^h\mathrm{N}$ since $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$; we will consider the class $\bar{z}$ of z in $\mathrm{gr}_{\mathfrak{J}}^{h+d}(\mathrm{gr}_{\mathfrak{R}}^*(\mathrm{N}))$, which, by virtue of (19.5.5) and the hypothesis on $\mathbf{f}$, identifies with a sub-$(\mathrm{A}/\mathfrak{L})$-module of $\mathrm{gr}_{\mathfrak{L}}^{h+d}(\mathrm{N})$. Since $z \in \mathrm{R}$, we will show that the image of $\bar{z}$ is zero.

(19.7.1.20).
$$\bar{z} \in \mathrm{gr}_{\mathfrak{L}}^{h+d}(\mathrm{R}, \mathrm{N}).$$

Indeed, we have noted, in the demonstration of (19.7.1.3) that, for every prime ideal $\mathfrak{r}$ of A such that $\mathfrak{p} = \mathfrak{r}/\mathfrak{R} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$, the map $\psi'_{\mathrm{R}_{\mathfrak{r}}}$ is bijective, and therefore all the modules, localised at $\mathfrak{r}$, the image $z/1$ in $\mathrm{R}_{\mathfrak{r}}$ belongs to $(\mathrm{R}_{\mathfrak{r}} \cap \mathfrak{R}_{\mathfrak{r}}^d\mathrm{N}_{\mathfrak{r}}) + \mathfrak{L}_{\mathfrak{r}}^{h+1}\mathrm{N}_{\mathfrak{r}}$; hence $\mathrm{gr}_{\mathfrak{L}_{\mathfrak{r}}}(\mathrm{R}_{\mathfrak{r}}, \mathrm{N}_{\mathfrak{r}}) = (\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{R}, \mathrm{N}))_{\mathfrak{r}}$, and hence the image of $\bar{z}$ by the canonical map $(\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{N}))_{\mathfrak{q}} \to (\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{M}))_{\mathfrak{q}}$ is zero for every $\mathfrak{q} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$. As we saw in the demonstration of (19.7.1.3) that $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{M})) \subset \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$, we deduce (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 1 of prop. 2 and n° 3, prop. 7) that the image of $\bar{z}$ by the canonical map $\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{N}) \to \mathrm{gr}_{\mathfrak{L}}^*(\mathrm{M})$ is zero, that is to say we have the relation (19.7.1.19).

(19.7.1.21). This being so, we can write by definition
$$z \equiv \textstyle\sum_{|\mathbf{p}|=h} c_{\mathbf{p}}\mathbf{f}^{\mathbf{p}} \bmod \mathfrak{R}^{d+1}\mathrm{N}$$

where one sets as usual $\mathbf{p} = (p_1, \ldots, p_n)$, $\mathbf{f}^{\mathbf{p}} = f_1^{p_1} f_2^{p_2} \cdots f_n^{p_n}$, $|\mathbf{p}| = p_1 + \ldots + p_n$ and $c_{\mathbf{p}} \in \mathfrak{R}^d\mathrm{N}$. Moreover, as ψ_{N} identifies $\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{N})$ with $(\mathrm{gr}_{\mathfrak{R}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, we have, by the preceding identification,

(19.7.1.22).
$$\bar{z} = \textstyle\sum_{|\mathbf{p}|=h} \bar{c}_{\mathbf{p}}\mathrm{T}^{\mathbf{p}}.$$

Using (19.7.1.3), we deduce that necessarily, for every $\mathbf{p}$ such that $|\mathbf{p}| = h$ ($\bar{c}_{\mathbf{p}}$ being this time identified by ψ_{N} with an element of $\mathrm{gr}_{\mathfrak{L}}^*(\mathrm{N})$),

(19.7.1.23).
$$\bar{c}_{\mathbf{p}} \in \mathrm{gr}_{\mathfrak{L}}^d(\mathrm{R}, \mathrm{N}).$$

But since $d < m$, the hypothesis that $(*_d)$ is true implies that $\bar{c}_{\mathbf{p}}$ belongs to the image of ψ'_{R}, hence we can suppose that $c_{\mathbf{p}} \in (\mathrm{R} \cap \mathfrak{R}^d\mathrm{N}) + \mathfrak{R}^d\mathfrak{L}\mathrm{N}$. The relation (19.7.1.21) then gives
$$z \in \mathfrak{J}^h(\mathrm{R} \cap \mathfrak{R}^d\mathrm{N}) + \mathfrak{J}^h\mathfrak{R}^d\mathfrak{L}\mathrm{N} + \mathfrak{R}^{d+1}\mathrm{N}$$
and as $\mathfrak{J} \subset \mathfrak{L}$, this proves (19.7.1.19).

In particular, we can therefore write
$$g = t + g_2$$
with $t \in \mathfrak{L}^h(\mathrm{R} \cap \mathfrak{R}^d\mathrm{N})$ and $g_2 \in \mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^{h+1}\mathrm{N}$; moreover, as $t \in \mathrm{R}$ and $g \in \mathrm{R}$, we have $g_2 \in \mathrm{R}$. On the other hand, using the hypothesis $h \geqslant m - d + 1$, one sees that $t \in \mathfrak{L}^{m+1}\mathrm{N}$. The element g_2 hence satisfies all the conditions stated at the start of (19.7.1.8), which completes the demonstration of (19.7.1). □

Remarks (19.7.2). —

(i) When one supposes that $\mathrm{gr}_{\mathfrak{R}}^*(\mathrm{A})$ is an $(\mathrm{A}/\mathfrak{R})$-module *flat*, the hypothesis that the sequence $\mathbf{f}$ is $(\mathrm{A}/\mathfrak{R})$-regular implies that it is also $\mathrm{gr}_{\mathfrak{R}}^*(\mathrm{A})$-regular $(\mathbf{0}, 15.1.14)$. We note that this will be the case when A and $\mathrm{A}/\mathfrak{R}$ *are regular rings* $(\mathbf{0}, 17.3.6)$: indeed, $\mathfrak{R}$ is then a regular ideal of A (19.1.2), hence quasi-regular, and localising at the maximal ideals of A containing $\mathfrak{R}$ and using $(\mathbf{0}, 15.1.7)$, one sees that $\mathrm{gr}_{\mathfrak{R}}^*(\mathrm{A})$ is a *projective* $(\mathrm{A}/\mathfrak{R})$-module.

(ii) It would be interesting to be able to prove the last conclusion of (19.7.1) under the sole hypothesis that there exists an A-algebra B which is a Noetherian ring, for which $\mathfrak{J}\mathrm{B}$ is contained in the radical of B, and that M is a finitely generated B-module.

Corollary (19.7.3). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{R}$ *an ideal of* A *such that the ring* $\mathrm{A}/\mathfrak{R}$ *is regular and of dimension* n*,* M *a finitely generated* A*-module. The following conditions are equivalent:*

a) $\mathrm{gr}_{\mathfrak{R}}^*(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{R})$*-module.*

b) *If* $\mathrm{P(T)}$ *and* $\mathrm{Q(T)}$ *are the Poincaré series of the* $(\mathrm{A}/\mathfrak{m})$*-modules graded* $\mathrm{gr}^*_{\mathfrak{m}}(\mathrm{M})$ *and* $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m})$, *we have*

$$\mathrm{Q(T)} = \mathrm{P(T)}(1-\mathrm{T})^n. \tag{19.7.3.1}$$

Let $\mathbf{f} = (f_1, \ldots, f_n)$ *be a sequence of elements of* A *such that the images of the* f_i *in* $\mathrm{A}/\mathfrak{R}$ *form a regular system of parameters of* $\mathrm{A}/\mathfrak{R}$ $(\mathbf{0}, 17.1.6)$, *so that* $\mathbf{f}$ *is an* $(\mathrm{A}/\mathfrak{R})$*-regular sequence; moreover, if* $\mathfrak{J}$ *is the ideal generated by* $\mathbf{f}$, $\mathfrak{J} + \mathfrak{R} = \mathfrak{m}$, *and as* $\mathrm{A}/\mathfrak{m}$ *is a field,* $\mathrm{gr}^*_{\mathfrak{m}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{m})$*-module. As* A *is Noetherian and* M *a finitely generated* A*-module, one can apply the equivalence of* a) *and* b) *in* (19.7.1). *Moreover,* $\mathrm{A}/\mathfrak{m}$ *being a field, the fact that* ψ_{M} *is bijective is equivalent to saying that for every* $h \geqslant 0$, $\mathrm{gr}^h_{\mathfrak{m}}(\mathrm{M})$ *and the submodule of elements of degree* h *of* $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ *have the same rank over* $\mathrm{A}/\mathfrak{m}$; *but for the second of these modules the rank in question is clearly equal to*

$\sum_{r=0}^{h} \binom{r+n-1}{n-1} \mathrm{rg}(\mathrm{gr}^{h-r}_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m}))$. *As* $(1-\mathrm{T})^{-n} = \sum_{r=0}^{\infty} \binom{r+n-1}{n-1} \mathrm{T}^r$, *this proves that the relation* (19.7.3.1) *is necessary and sufficient for* ψ_{M} *to be bijective.*

Corollary (19.7.4). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Y *regular and connected,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module normally flat* ((6.10.1)) *along* Y. *Then there exists a formal power series* $\mathrm{R} \in \mathbf{Q}[[\mathrm{T}]]$ *(independent of* $x \in \mathrm{Y}$*) such that, for every* $x \in \mathrm{Y}$, *the Poincaré series of* $\mathrm{gr}^*_{\mathfrak{m}_x}(\mathscr{F}_x)$ *is given by the formula*

$$\mathrm{P}_x(\mathscr{F})(\mathrm{T}) = \mathrm{R(T)}(1-\mathrm{T})^{-n} \qquad \textit{where } n = \dim(\mathscr{O}_{\mathrm{Y},x}). \tag{19.7.4.1}$$

IV-4 | 219

Let $\mathscr{J}$ *be the coherent Ideal of* $\mathscr{O}_{\mathrm{X}}$ *defining* Y; *the hypothesis on* $\mathscr{F}$ *means that* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ *is an* $\mathscr{O}_{\mathrm{Y}}$*-Module flat. We can therefore, for every* $x \in \mathrm{Y}$, *apply* (19.7.3) *taking* $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$, $\mathfrak{R} = \mathscr{J}_x$, $\mathrm{M} = \mathscr{F}_x$; *the hypothesis* a) *being satisfied by hypothesis, the Poincaré series* $\mathrm{P}_x(\mathscr{F})(\mathrm{T})$ *is therefore equal to* $\mathrm{R}_x(\mathrm{T})(1-\mathrm{T})^{-n}$, *where* $\mathrm{R}_x(\mathrm{T})$ *is the Poincaré series of* $\mathrm{gr}^*_{\mathscr{J}_x}(\mathscr{F}_x) \otimes_{\mathscr{O}_{\mathrm{Y},x}} k(x)$. *Now, each of the* $\mathscr{O}_{\mathrm{Y}}$*-Modules* $\mathscr{J}^h\mathscr{F} / \mathscr{J}^{h+1}\mathscr{F}$ *is flat and coherent by hypothesis, and since* Y *is supposed connected, hence each of* $\mathscr{J}^h\mathscr{F} / \mathscr{J}^{h+1}\mathscr{F}$ *is of constant rank, locally free* (2.1.12), *and since* Y *is supposed connected, each of* $\mathscr{J}^h\mathscr{F} / \mathscr{J}^{h+1}\mathscr{F}$ *is of constant rank independent of* $x \in \mathrm{Y}$, *which proves that the Poincaré series* $\mathrm{R}_x(\mathrm{T})$ *is independent of* $x \in \mathrm{Y}$.

Corollary (19.7.5). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Z *a closed sub-prescheme of* Y; *one supposes* Y *and* Z *regular. Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module.*

(i) *If* $\mathscr{F}$ *is normally flat along* Y *at the points of* Z *(cf.* (11.3.4) *for the definition of normal flatness at a point), then* $\mathscr{F}$ *is normally flat along* Z.

(ii) *Suppose moreover* Z *connected and that* $\mathscr{O}_{\mathrm{X}}$ *is normally flat along* Y *at the regular points of* Z. *If* $\mathscr{F}$ *is normally flat along* Y *and if there exists a point* $z \in \mathrm{Z}$ *such that* $\mathscr{F}$ *is normally flat along* Y, *then* $\mathscr{F}$ *is normally flat along* Y *at all the points of* Z.

Let $\mathscr{X}_z$ *and* $\mathscr{L}_z$ *be the coherent ideals of* $\mathscr{O}_{\mathrm{X}}$ *defining respectively* Y *and* Z.

(i) *The hypothesis means that* $\mathrm{gr}^*_{\mathscr{X}_z}(\mathscr{F}_z)$ *is an* $\mathscr{O}_{\mathrm{Y},z}$*-module flat; it implies therefore that* $\mathrm{gr}^*_{\mathscr{X}_z}(\mathscr{F}_z)$ *is an* $\mathscr{O}_{\mathrm{Z},z}$*-module flat, by virtue of* (19.7.1), a) *implies* b).

(ii) *The supplementary hypothesis on* X *implies that the sequence* (f_i) *is regular at every point* $z \in \mathrm{Z}$ (19.7.2). *On the other hand,* Z *being regular and connected, is integral, and if* $\mathrm{gr}^*_{\mathscr{X}_z}(\mathscr{F}_z)$ *is* $\mathscr{O}_{\mathrm{Y},z}$*-flat at a point* $z \in \mathrm{Z}$, *then, as the point generic* ζ *of* Z *is a generization of* z, *one concludes that* $\mathrm{gr}^*_{\mathscr{X}_\zeta}(\mathscr{F}_\zeta)$ *is* $\mathscr{O}_{\mathrm{Y},\zeta}$*-flat* $(\mathbf{0}_{\mathrm{I}}, 6.3.1)$. *We then deduce that if in every point* z' *of* Z, $\mathrm{gr}^*_{\mathscr{X}_{z'}}(\mathscr{F}_{z'})$ *is an* $\mathscr{O}_{\mathrm{Z},z'}$*-module flat, then* $\mathrm{gr}^*_{\mathscr{X}_{z'}}(\mathscr{F}_{z'})$ *is an* $\mathscr{O}_{\mathrm{Y},z'}$*-module flat, since we can apply the equivalence of* a) *and* d) *in* (19.7.1).

19.8. Properties of passage to the projective limit In this number, the notations and conventions on projective limits are those of (8.5.1) and (8.8.1). **IV-4** | 219

Proposition (19.8.1). — *Suppose that the transition morphisms $\mathrm{S}_\mu \to \mathrm{S}_\lambda$ ($\lambda \leqslant \mu$) are flat, and moreover that one of the two following hypotheses is satisfied:*

1° *The preschemes S_λ are locally Noetherian.*

2° *The transition morphisms $\mathrm{S}_\mu \to \mathrm{S}_\lambda$ are surjective (and hence faithfully flat).*

Under these conditions: **IV-4** | 220

(i) *Suppose S_α quasi-compact. Let $\mathscr{F}_\alpha$ be a quasi-coherent $\mathscr{O}_{\mathrm{S}_\alpha}$-Module, which one supposes moreover of finite type when hypothesis 1° is satisfied. Let $(f_{i\alpha})_{1\leqslant i\leqslant n}$ be a finite sequence of sections of $\mathscr{F}_\alpha$ above S_α, the sequence of sections f_i of the $\mathscr{O}_{\mathrm{S}}$-Module $\mathscr{F}$ above S, corresponding to the $f_{i\alpha}$. For the sequence of sections f_i ($1 \leqslant i \leqslant n$) to be $\mathscr{F}$-regular, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that the sequence of sections $f_{i\lambda}$ of $\mathscr{F}_\lambda$ above S_λ is $\mathscr{F}_\lambda$-regular.*

(ii) *Suppose X_α quasi-compact, and let $j_\alpha : \mathrm{Y}_\alpha \to \mathrm{X}_\alpha$ be an immersion, which one supposes locally of finite presentation when hypothesis 2° is satisfied. Then, for the corresponding immersion $j : \mathrm{X} \to \mathrm{S}$ to be regular, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $j_\lambda : \mathrm{X}_\lambda \to \mathrm{S}_\lambda$ is regular.*

Proof. (i) If $p_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ is the canonical projection, we have $\mathscr{F}/(\sum_{j=1}^{i-1} f_j\mathscr{F}) = p_\lambda^*(\mathscr{F}_\lambda/(\sum_{j=1}^{i-1} f_{j\lambda}\mathscr{F}_\lambda))$, hence we are reduced to the case $n = 1$, in which case on omits the index i. The fact that the condition is sufficient results from the fact that p_λ is flat (8.3.8) and from $(\mathbf{0}, 15.1.5)$. In case 2°, p_λ is faithfully flat (8.3.8) and the necessity of the condition results from $(\mathbf{0}, 15.2.5)$, with $\lambda = \alpha$. In case 1°, we denote by $\mathscr{N}_\lambda$ (resp. $\mathscr{N}$) the kernel of the homomorphism $\mathscr{F}_\lambda \to \mathscr{F}_\lambda$ (resp. $\mathscr{F} \to \mathscr{F}$), multiplication by f_λ (resp. by f); since $\mathscr{F}_\lambda$ is coherent by hypothesis, the hypothesis $\mathscr{N} = 0$ implies $\mathscr{N}_\lambda = 0$ for $\lambda \geqslant \alpha$ by virtue of (8.5.8, (ii)).

(ii) As p_λ is flat, the sufficiency of the condition results from ((19.1.5), (ii)). To prove its necessity, let us note that, since $j_\alpha(\mathrm{X}_\alpha)$ is quasi-compact and the transition morphisms $(\mathrm{X}_\mu)_{s_\mu} \to (\mathrm{X}_\lambda)_{s_\lambda}$ are flat (since the structural morphisms $\mathrm{X}_\lambda \to \mathrm{S}_\lambda$ are flat ((11.2.6)) and (8.2.11)), we can apply the argument (19.8.1, (i)) to the images of a finite generating set of sections of $\mathscr{J}_{\mathrm{Y}_\alpha}$ above each of a finite cover, to deduce that there exists $\lambda \geqslant \alpha$ such that j_λ is regular. In case 1° and in particular when hypothesis 2° is satisfied, the image of X_α in S_α is locally Noetherian and the transition morphisms are flat; by the argument of (19.2.4), there exists an open neighbourhood V_λ of the image x_λ of x in X_λ such that the restrictions to V_λ of $f_{i\lambda}$ form a $(\mathscr{F}_\lambda|\mathrm{U}_\lambda)$-transversally regular sequence relative to S_λ; the conclusion then follows. □

Proposition (19.8.2). — *Let X_α be an S_α-prescheme quasi-compact and locally of finite presentation.*

(i) *Let $\mathscr{F}_\alpha$ be a quasi-coherent $\mathscr{O}_{\mathrm{X}_\alpha}$-Module of finite presentation, $(f_{i\alpha})_{1\leqslant i\leqslant n}$ a finite sequence of sections of $\mathscr{O}_{\mathrm{X}}$ above X_α, corresponding to sections f_i of $\mathscr{O}_{\mathrm{X}}$ above X. For the sequence of sections $f_{i\alpha}$ to be $\mathscr{F}$-transversally regular relative to S ((19.2.1)), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that the sequence of sections $f_{i\lambda}$ of $\mathscr{O}_{\mathrm{X}_\lambda}$ above X_λ is $\mathscr{F}_\lambda$-transversally regular relative to S_λ.*

(ii) *Let Y_α be an S_α-prescheme quasi-compact, $j_\alpha : \mathrm{Y}_\alpha \to \mathrm{X}_\alpha$ an S_α-immersion locally of finite presentation. For the corresponding S-immersion $j : \mathrm{Y} \to \mathrm{X}$ to be transversally regular relative to S, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $j_\lambda : \mathrm{Y}_\lambda \to \mathrm{X}_\lambda$ is transversally regular relative to S_λ.*

Proof. (i) The sufficiency of the condition results from $(\mathbf{0}, 15.1.15)$. To prove its necessity, let us note that, since $j_\alpha(\mathrm{X}_\alpha)$ is quasi-compact, we may limit ourselves to the case where S_α is also quasi-compact, and j_α a closed immersion, so that the image of X_α is defined by a quasi-coherent Ideal $\mathscr{J}_\alpha$ of $\mathscr{O}_{\mathrm{S}_\alpha}$ of finite type (in cases 1° and 2°); the image of X_λ (resp. X) is then defined by the Ideal $\mathscr{J}_\lambda = \mathscr{J}_\alpha\mathscr{O}_{\mathrm{S}_\lambda}$ (resp. $\mathscr{J} = \mathscr{J}_\alpha\mathscr{O}_{\mathrm{S}}$) which is of finite type. To see that for $\lambda \geqslant \alpha$ sufficiently large, the sections $f_{i\lambda}$ of $\mathscr{J}_\lambda$ above X_λ form an $(\mathscr{F}_\lambda|\mathrm{U}_\lambda)$-transversally regular sequence relative to S_λ, it suffices to apply (19.8.1), (i) to the restrictions of $f_{i\alpha}$ to the elements U_α of a finite affine open cover of X_α, after having verified that the corresponding covers have the required properties. As X_α is quasi-compact and of finite presentation over S_α, the transition morphisms $(\mathrm{X}_\mu)_{s_\mu} \to (\mathrm{X}_\lambda)_{s_\lambda}$ are flat (since the structural morphisms $\mathrm{X}_\lambda \to \mathrm{S}_\lambda$ and $\mathrm{S}_\mu \to \mathrm{S}_\lambda$ are flat ((11.2.6)) and (8.2.11)), and we may apply (19.8.1) **IV-4** | 221 to deduce that for $\lambda \geqslant \alpha$ sufficiently large, one can suppose $\mathscr{F}_\lambda$ is S_λ-flat at the point x_λ. It results then from ((11.3.8)) that there exists an open neighbourhood V_λ of x_λ in X_λ satisfying the required conditions.

Moreover, using (19.2.7), (ii), the sufficiency of the condition results from ((19.2.7), (ii)). To prove its necessity, it suffices to show that for every point $x \in j(Y)$ there exists an open neighbourhood $V(x)$ of x and an index $\lambda = \lambda(x) \geqslant \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ, such that the sequence of restrictions of $f_{i\lambda}$ to V_λ be $(\mathscr{F}_\lambda|V_\lambda)$-transversally regular relative to S_λ: indeed, by the condition of being quasi-compact, one deduces from this the existence of an open neighbourhood V_λ of x_λ satisfying the conditions.

Let s be the image of x in S, and for every $\lambda \geqslant \alpha$, let s_λ be the image of s in S_λ. As X_s (resp. Y_s) is the projective limit of the fibres $(X_\lambda)_{s_\lambda}$ (resp. $(Y_\lambda)_{s_\lambda}$), and we can apply to these fibres the conditions of (19.8.1), (ii) (since by hypothesis the immersion $Y_s \to X_s$ is regular), it follows that there exists an index $\lambda(s) \geqslant \alpha$ such that for $\lambda \geqslant \lambda(s)$ the immersion $(Y_\lambda)_{s_\lambda} \to (X_\lambda)_{s_\lambda}$ is regular. Moreover ((11.2.6)), one can suppose that $\mathscr{F}_\lambda$ is S_λ-flat at x_λ. It results then from ((11.3.8)) that there exists an open neighbourhood V_λ of x_λ in X_λ satisfying the required conditions, together with open sets $W(x_\lambda)$ in X such that the structural morphisms $W(x_\lambda) \to S$ and $Y \cap W(x_\lambda) \to S$ are flat; applying ((11.2.6)) and (8.2.11), there exists an open neighbourhood V_λ of x_λ in X_λ such that the restrictions to V_λ and to $Y_\lambda \cap V_\lambda$ of the structural morphisms $X_\lambda \to S_\lambda$ and $Y_\lambda \to S_\lambda$ are flat. We conclude therefore from (19.2.4) that there exists an open neighbourhood V_λ of x_λ satisfying the required conditions. □

Remark (19.8.3). — The preceding demonstrations show that the statements of (19.8.1) and (19.8.2) subsist when one replaces the conditions concerning regularity or transversal regularity throughout X (resp. X_λ) by these conditions in a neighbourhood of a point $x \in X$ (resp. in a neighbourhood of x_λ, projection of x).

19.9. $\mathscr{F}$-regular sequences and depth

IV-4 | 222

(19.9.1). Recall that if X is a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module, T a part of X, one sets (5.10.1)

(19.9.1.1) $$\operatorname{prof}_T(\mathscr{F}) = \inf_{x \in T} \operatorname{prof}(\mathscr{F}_x).$$

We set on the other hand, for every point $t \in T$,

(19.9.1.2) $$\operatorname{prof}_{T,t}(\mathscr{F}) = \inf_{x \in T \cap \operatorname{Spec}(\mathscr{O}_{X,t})} \operatorname{prof}(\mathscr{F}_x),$$

and we say that $\operatorname{prof}_{T,t}(\mathscr{F})$ is the *T-depth of $\mathscr{F}$ at the point t*; it is clear that we have

(19.9.1.3) $$\operatorname{prof}_T(\mathscr{F}) = \inf_{t \in T} \operatorname{prof}_{T,t}(\mathscr{F}).$$

Lemma (19.9.2). — *Let* A *be a Noetherian ring,* M *a finitely generated* A*-module,* $\mathfrak{J}$ *an ideal of* A*. In order that, for every prime ideal* $\mathfrak{p} \supset \mathfrak{J}$*, we have* $\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \geqslant r$*, it is necessary and sufficient that there exist an M-regular sequence of* r *elements belonging to* $\mathfrak{J}$*.*

Proof. The sufficiency of the condition resulting immediately from $(\mathbf{0}, 16.4.5)$, let us prove its necessity. We argue by induction on r (there is nothing to prove if $r = 0$): since $\operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) \geqslant r - 1$ for every $\mathfrak{p} \supset \mathfrak{J}$, there exists by hypothesis an M-regular sequence $(g_i)_{1 \leqslant i \leqslant r-1}$ of elements of $\mathfrak{J}$. Set $N = M/(g_1 M + \ldots + g_{r-1} M)$; for every $\mathfrak{p} \supset \mathfrak{J}$, the images of the g_i in $A_\mathfrak{p}$ belong to the maximal ideal $\mathfrak{p}A_\mathfrak{p}$, and form an $M_\mathfrak{p}$-regular sequence $(\mathbf{0}, 15.1.14)$; we conclude therefore from $(\mathbf{0}, 16.4.6)$ that $\operatorname{prof}_{A_\mathfrak{p}}(N_\mathfrak{p}) = \operatorname{prof}_{A_\mathfrak{p}}(M_\mathfrak{p}) - (r-1) \geqslant 1$, and one sees that this reduces to proving the lemma for $r = 1$. Now the hypothesis means then that, for every $\mathfrak{p} \supset \mathfrak{J}$, $\mathfrak{p}A_\mathfrak{p}$ is not associated to M $(\mathbf{0}, 16.4.6)$, hence (Bourbaki, *Alg. comm.*, chap. IV, § 1, nº 2, prop. 5), $\mathfrak{p}$ is not contained in any of the prime ideals of Ass(M), hence it is not contained in their union (Bourbaki, *Alg. comm.*, chap. II, § 1, nº 1, prop. 2). But as this union is the set of elements that are not M-regular (Bourbaki, *Alg. comm.*, chap. IV, § 1, cor. 2 of prop. 2), this proves the lemma. □

Proposition (19.9.3). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module,* Y *a closed subset of* X*,* y *a point of* Y*,* n *an integer* > 0*. The following conditions are equivalent:*

a) *There exists an open neighbourhood* U *of* y *in* X *such that* $\mathrm{prof}_{Y\cap U}(\mathscr{F}|U) \geqslant n$*.*
b) *There exists an open neighbourhood* U *of* y *in* X *and a suite* $(\mathscr{F}|U)$*-regular sequence of* n *sections* f_i *of* $\mathscr{O}_U$ *above* U*, such that* $f_i(x) = 0$ $(\mathbf{0}_I, 5.5.1)$ *for* $1 \leqslant i \leqslant n$ *at every point* $x \in U \cap Y$*.*
c) $\mathrm{prof}_{Y,y}(\mathscr{F}) \geqslant n$*.*

Moreover, the set of $y \in Y$ *satisfying these conditions is open in* Y*.*

Proof. The last assertion results trivially from a). The equivalence of a) and b) follows from (19.9.2) applied to an affine open neighbourhood of y in X and from $(\mathbf{0}, 15.1.14)$. It is clear that a) implies c), since every open neighbourhood of y in X containing $\mathrm{Spec}(\mathscr{O}_{X,y})$; conversely that c) implies b). It results from c) and the definition (19.9.1) that we can apply (19.9.2) to the $\mathscr{O}_{X,y}$-module $\mathscr{F}_y$ and the ideal $\mathscr{J}_y$ of $\mathscr{O}_{X,y}$, hence there exists a $\mathscr{F}_y$-regular sequence $(t_i)_{1\leqslant i\leqslant n}$ of n elements of $\mathscr{J}_y$. There is therefore an open neighbourhood U of y in X and a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{J}$ above U such that the germs t_i are the germs of f_i at the point y; using $(\mathbf{0}, 15.2.4)$, one deduces that there exists an open neighbourhood $U' \subset U$ of y in X such that the $f_i|U'$ form a $(\mathscr{F}|U')$-regular sequence. □ **IV-4 | 223**

Corollary (19.9.4). — *With the notations of* ((19.9.3))*, the function* $y \mapsto \mathrm{prof}_{Y,y}(\mathscr{F})$ *is lower semi-continuous on* Y*.*

Proposition (19.9.5). — *With the notations of* ((19.9.3))*, let* X′ *be a locally Noetherian prescheme,* $g : X' \to X$ *a flat morphism,* $Y' = g^{-1}(Y)$*,* $\mathscr{F}' = g^*(\mathscr{F})$*. Then, for every point* $y' \in Y'$ *above* y*, we have* $\mathrm{prof}_{Y',y'}(\mathscr{F}') = \mathrm{prof}_{Y,y}(\mathscr{F})$*.*

Proof. Indeed, for every $z' \in \mathrm{Spec}(\mathscr{O}_{X',y'})$, it results from (6.3.1) that $\mathrm{prof}(\mathscr{F}_{z'}) \geqslant \mathrm{prof}(\mathscr{F}_{g(z')})$, and if z'' is a maximal point of the fibre $g^{-1}(g(z'))$, generization of z' (hence belonging also to $\mathrm{Spec}(\mathscr{O}_{X',y'})$), we have (*loc. cit.*) $\mathrm{prof}(\mathscr{F}_{z''}) = \mathrm{prof}(\mathscr{F}_{g(z')})$. The proposition then results from the fact that $g(\mathrm{Spec}(\mathscr{O}_{X',y'})) = \mathrm{Spec}(\mathscr{O}_{X,g(y')})$ since g is flat (2.3.4). □

Proposition (19.9.6). — *Let* $f : X \to S$ *be a morphism locally of finite presentation,* Y *a closed subset of* X*,* $\mathscr{F}$ *a coherent* $\mathscr{O}_X$*-Module of finite presentation and* f*-flat,* n *an integer* > 0*. For every point* $y \in Y$*, denoting by* $X_{f(y)}$ *the fibre of* f *at the point* $f(y)$*, by* $\mathscr{F}_{f(y)}$ *the inverse image of* $\mathscr{F}$ *in* $X_{f(y)}$*, and setting* $Y_{f(y)} = X_{f(y)} \cap Y$*:*

a) $\mathrm{prof}_{f(y),y}(\mathscr{F}_{f(y)}) \geqslant n$*.*
b) *There exists an open neighbourhood* U *of* y *in* X *and a sequence* $(g_i)_{1\leqslant i\leqslant n}$ *of* n *sections of* $\mathscr{O}_X$ *above* U*, that is transversally* $\mathscr{F}$*-regular relative to* S *(cf.* (19.2.1) *for the terminology) and such that* $g_i(x) = 0$ *for* $1 \leqslant i \leqslant n$ *at every point* $x \in U \cap Y$*.*

The set V_n *of* $y \in Y$ *satisfying these conditions is open and constructible in* X $(\mathbf{0}_{III}, 9.1.1)$*; if moreover* Y *is locally constructible in* X*,* V_n *is also locally constructible.*

Proof. By virtue of (19.9.3), condition a) is equivalent to the existence of an open neighbourhood U of y in X and of a sequence $(\mathscr{F}_{f(y)}|(U \cap X_{f(y)}))$-regular of n sections h_i of $\mathscr{O}_{X_{f(y)}}$ above $U \cap X_{f(y)}$, such that $h_i(z) = 0$ for $1 \leqslant i \leqslant n$ at every point $z \in U \cap Y_{f(y)}$. If we denote by Y a closed sub-prescheme of X having Y as underlying space and by $\mathscr{J}$ the quasi-coherent Ideal of $\mathscr{O}_X$ defining Y, one can further say that $h_i \in \mathscr{J}_{f(y)}$ for $1 \leqslant i \leqslant n$. As one can replace U by an affine open neighbourhood smaller than y, one can suppose that the h_i are the images of a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{F}$ above U, so that the germs $(g_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathscr{O}_{X,y}$. The equivalence of a) and b) results then from (11.3.8), which proves also that the set V_n is open in Y.

It remains to prove the last assertion; as this involves local properties on X, we can suppose $S = \mathrm{Spec}(A)$ affine and X of finite presentation over S; there exists then a Noetherian sub-ring A_0 of A, a prescheme X_0 of finite type over $S_0 = \mathrm{Spec}(A_0)$ and a coherent $\mathscr{O}_{X_0}$-Module $\mathscr{F}_0$ such that $X = X_0 \times_{S_0} S$ and $\mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_X$ (8.9.1); one can moreover suppose that, if $f_0 : X_0 \to S_0$ is the structural morphism, $\mathscr{F}_0$ is f_0-flat ((11.2.6)). Since Y is constructible, we can moreover (8.3.11) suppose that Y is locally constructible in X_0, so Y_0 is the inverse image of a locally constructible Y_0 in X_0; we use the argument of (19.9.5) and the fact that prof is invariant under flat base change. The assertion on V_n being constructible follows from the criterion $(\mathbf{0}_{III}, 9.1.1)$. □

§20. Meromorphic functions and pseudo-morphisms

20.0. Introduction Most of the concepts and results of §§20 and 21 directly relate to Chapter I, and hardly depend on Chapters I and IV, except for the occasional usage of the notion of depth and of a local regular ring (in (20.6), (21.11), (21.13), and (21.15)), of Zariski's "Main theorem" in (20.4) and (21.12), and the properties of transversely regular immersions in (20.6) and (21.15).

In §20, we introduce several variants of the concept of a rational map, already studied in (**I**, 7) from **IV-4** | 226
a point of view still fairly close to the classical point of view, and for this reason quite ill-suited to the case of not necessarily reduced preschemes. The notions and results of §20 are used in §21 (n$^{\text{os}}$(21.1) and (21.7)) to develop the general notion of a divisor and its most elementary properties. This notion is especially convenient when the local rings of the preschemes considered are Noetherian and integrally closed, and especially when they are also *factorial* ((21.6) and (21.7)), because of their identification in the latter case with the notion of a 1-*codimensional cycle* (a linear combination of irreducible subpreschemes of codimension 1). In (21.9), we determine the divisors on a Noetherian prescheme of dimension 1 but not necessarily normal, which is useful for various applications. The (21.11) and (21.12) give two important theorems, due respectively to Auslander–Buchsbaum and Van der Waerden, and relate the notion of a factorial ring (the n$^{\text{os}}$(21.9), (21.11), and (21.12) are independent of each other). In the n$^{\text{os}}$(21.13) and (21.14), also independent of the previous three, we study a useful variant of the notion of a local factorial ring, that of a local *parafactorial* ring, which is introduced in particular **[?]** in the development of the comparison theorem of the Picard group of a projective prescheme X over a field k and a "hyperplane section". We will see in (21.14.1) (Ramanujam–Samuel theorem) that the local parafactorial rings are much more numerous than one would have expected *a priori*.

In (20.5), (20.6), and (21.15), we review the previous notions but from a point of view "relative" to a fixed base prescheme. For the moment these notions are used only relatively rarely; in particular, the concept of a relative divisor is hardly used except when it comes to positive divisors, and in this case it is explained advantageously without the help of the notion of a relative meromorphic functions, using the notion of a transversely regular immersion of codimension 1. It will therefore be advantageous to omit these sections on a first reading.

20.1. Meromorphic functions

(20.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space, and let $\mathscr{S}$ be a subsheaf *of sets* of $\mathscr{O}_X$. For every open U of X, consider the *ring of fractions* $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (Bourbaki, *Alg. comm.*, chap. II, §2, n$^{\text{o}}$1). It is immediate that the map $U \mapsto \Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ is a *presheaf of rings* ((**0**, 1.5.1) and (**0**, 1.5.7)). We denote by $\mathscr{O}_X[\mathscr{S}^{-1}]$ the *sheaf of rings* associated to this presheaf and we say that this is the *sheaf of rings of fractions of $\mathscr{O}_X$ with denominators in $\mathscr{S}$*; this is a *flat* $\mathscr{O}_X$-module. It is immediate that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{O}_X[\mathscr{S}^{-1}])_x \simeq \mathscr{O}_x[\mathscr{S}_x^{-1}], \tag{20.1.1.1}$$

since the reasoning of (**0**, 1.4.5) generalizes immediately in the case where we have an inductive system $(A_\alpha, \phi_{\beta\alpha})$ of rings, and for each index α a subset S_α of A_α such that $\phi_{\beta\alpha}(S_\alpha) \subset S_\beta$ for $\alpha \leqslant \beta$; **IV-4** | 227
we then take for S the inductive limit in $A = \varinjlim A_\alpha$ of the inductive limit of the subsets (S_α).

(20.1.2). Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module. We then set

$$\mathscr{F}[\mathscr{S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X[\mathscr{S}^{-1}], \tag{20.1.2.1}$$

and we say that this is the *sheaf of modules of fractions of $\mathscr{F}$ with denominators in $\mathscr{S}$*; it is immediate that it is associated to the presheaf of modules $U \mapsto \Gamma(U, \mathscr{F})[\Gamma(U, \mathscr{S})^{-1}]$, and that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{F}[\mathscr{S}^{-1}])_x \simeq \mathscr{F}_x[\mathscr{S}_x^{-1}]. \tag{20.1.2.2}$$

(20.1.3). We will focus here on the case where $\mathscr{S}$ is the subsheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{O}_X$ such that for every open U, $\Gamma(U, \mathscr{S})$ is the *set of regular elements* of the ring $\Gamma(U, \mathscr{O}_X)$; it is immediate that it is a sheaf (and not only a presheaf), the regularity of a section of $\mathscr{O}_X$ over U being checked "fibre by fibre" (i.e.

meaning that the germ of the section in x is regular in $\mathscr{O}_X$ for all $x \in U$), in other words $\mathscr{S}(\mathscr{O}_X)_x$ is none other than the set of regular elements of $\mathscr{O}_{X,x}$. The corresponding sheaf of rings

$$\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$$

is called the *sheaf of germs of meromorphic functions on* X, and the sections of $\mathscr{M}_X$ over X are called the *meromorphic functions on* X; they form a ring which we denote by $M(X)$. For every $\mathscr{O}_X$*-module* $\mathscr{F}$,

$$\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{F}[\mathscr{S}^{-1}]$$

is also denoted $\mathscr{M}_X(\mathscr{F})$ and called the *sheaf of germs of meromorphic sections of* $\mathscr{F}$; its sections over X form an $M(X)$-module denoted by $M(X, \mathscr{F})$, whose elements are called *meromorphic sections of* $\mathscr{F}$ *over* X. These definitions imply that for every open U of X, we have a canonical isomorphism $\mathscr{M}_X(\mathscr{F})|U \simeq \mathscr{M}_U(\mathscr{F}|U)$, in particular $\mathscr{M}_X|U \simeq \mathscr{M}_U$.

(20.1.3.1). If X is a *reduced prescheme*, then we note that if an element $s \in \Gamma(U, \mathscr{O}_X)$ is such that $s_\xi \neq 0$ for every maximal point ξ of U, then s is *regular*. Indeed, if $st = 0$ for a $t \in \Gamma(U, \mathscr{O}_X)$, then we have $s_\xi t_\xi = 0$, so $t_\xi = 0$ since $\mathscr{O}_{X,\xi}$ is a field, and say that $t_\xi = 0$ for every maximal point ξ of X means that $t = 0$: we are immediately reduced to the case where U is affine, and an element of a reduced ring which belongs to all the minimal prime ideals is zero by definition. The converse is true if the set of irreducible components of X is *locally finite*. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine; if $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ are the minimal prime ideals of A and if $s \in \mathfrak{p}_i$ for an index i, then there exists $t \in A$ such that $t \in \mathfrak{p}_j$ for $j \neq i$ and $t \notin \mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, n°1, Prop. 1); so we have $st \in \mathfrak{p}_i$ for all i, and as a result $st = 0$ since A is reduced; s is therefore nonregular.

(20.1.4). For every open U of X, the homomorphism $t \mapsto t/1$ from $\Gamma(U, \mathscr{O}_X)$ to $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (which is none other than the *total ring of fractions* of $\Gamma(U, \mathscr{O}_X)$) is injective; these homomorphisms **IV-4** | 228
thus define a *canonical injective homomorphism*

$$i : \mathscr{O}_X \longrightarrow \mathscr{M}_X \tag{20.1.4.1}$$

which allows us to identify $\mathscr{O}_X$ with a subsheaf of $\mathscr{M}_X$. Given a meromorphic function $\phi \in M(X)$, we say that ϕ is *defined* on an open U of X if $\phi|U$ is a *section of* $\mathscr{O}_U$ over U; the axioms of sheaves show that there is, for a given section ϕ, a *largest* open on which ϕ is defined; we call this the *domain of definition* of ϕ and denote it by $\operatorname{dom}(\phi)$.

(20.1.5). For every $\mathscr{O}_X$-module $\mathscr{F}$, we obtain from (20.1.4.1) a di-homomorphism consisting of i and the homomorphism of sheaves of additive groups

$$1_{\mathscr{F}} \otimes i : \mathscr{F} \longrightarrow \mathscr{M}_X(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X. \tag{20.1.5.1}$$

We note that the latter is not injective in general; when it is injective, we say that $\mathscr{F}$ is *strictly torsion-free*: this means that for every open U of X and every section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element in this ring the homothety $z \mapsto sz$ of $\Gamma(U, \mathscr{F})$ is injective; this condition is evidently satisfied if $\mathscr{F}$ is *locally free*.

Proposition (20.1.6). — *Let* X *be a locally Noetherian prescheme,* $\mathscr{F}$ *a quasi-coherent* $\mathscr{O}_X$*-module. For* $\mathscr{F}$ *to be strictly torsion-free, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{O}_X)$.

Proof. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, and we know that the elements s of A belonging to an ideal of $\operatorname{Ass}(M)$ are exactly those for which the homothety $z \mapsto sz$ is not injective (Bourbaki, *Alg. comm.*, chap. IV, §1, n°1, cor. 2 of prop. 2). □

(20.1.7). If u is a section of $\mathscr{M}_X(\mathscr{F})$ over X, then we say that u is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X such that $u|V$ is the image of a section of $\mathscr{F}$ over V under the di-homomorphism (20.1.5.1). We say that u is *defined* on an open U of X if it is defined at every point in U; there is still a larger open in which u is defined, called the *domain of definition* of u and denoted $\operatorname{dom}(u)$. When $\mathscr{F}$ is strictly torsion-free, such that $\mathscr{F}$ is identified by (20.1.5.1) with a subsheaf of $\mathscr{M}_X(\mathscr{F})$, then saying u is defined on U means that $u|V$ is a *section of* $\mathscr{F}$ *over* U.

(20.1.8). In accordance with the general notation of ($\mathbf{0_I}$, 5.4.7), we denote by $\mathscr{M}_X^*$ the sheaf of multiplicative groups such that $\Gamma(U, \mathscr{M}_X^*)$ is (for every open U of X) the group of *invertible elements* of $\Gamma(U, \mathscr{M}_X)$. This sheaf is none other than the sheaf $\mathscr{S}(\mathscr{M}_X)$ defined in (20.1.3): indeed, if

$s \in \Gamma(U, \mathscr{S}(\mathscr{M}_X))$, then for every $x \in U$, there exists an open neighbourhood $V \subset U$ of x such that $s|V$ is a regular element in the *total ring of fractions* of $\Gamma(V, \mathscr{O}_X)$, and we know that such an element is necessarily invertible in this ring of fractions. We say that the sections of $\mathscr{M}_X^*$ over X are the *regular meromorphic functions* (note that we are deviating here from the terminology followed by certain authors, who call "regular" meromorphic functions those which are *sections of* $\mathscr{O}_X$, identified with a subsheaf of $\mathscr{M}_X$).

Let $\mathscr{L}$ be an *invertible* $\mathscr{O}_X$*-module* ($\mathbf{0_I}$, 5.4.1); then it is clear that $\mathscr{M}_X(\mathscr{L}) = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}_X$ is an **IV-4** | 229
invertible $\mathscr{M}_X$*-module*. Let U be an open such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$; as every automorphism of $\mathscr{M}_U$ is multiplication by an invertible element of $\Gamma(U, \mathscr{M}_X)$ ($\mathbf{0_I}$, 5.4.7), it is equivalent to saying that a section $s \in \Gamma(U, \mathscr{M}_X(\mathscr{L}))$ has an invertible image in $\Gamma(U, \mathscr{M}_X)$ under *an* isomorphism or by *any* isomorphism on $\Gamma(U, \mathscr{M}_X)$; we say in this case that s is a *regular meromorphic section of* $\mathscr{L}$ over U; a section s of $\mathscr{L}$ over X is called a *regular meromorphic section of* $\mathscr{L}$ if, for every open U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $s|U$ is a regular meromorphic section of $\mathscr{L}$ over U. We denote by $(\mathscr{M}_X(\mathscr{L}))^*$ the subsheaf of $\mathscr{M}_X(\mathscr{L})$ such that for every open U, $\Gamma(U, (\mathscr{M}_X(\mathscr{L}))^*)$ is the set of regular meromorphic sections of $\mathscr{L}$ over U. Let s be a meromorphic section of $\mathscr{L}$ over X (i.e. a section of $\mathscr{M}_X(\mathscr{L})$); it defines a homomorphism $h_s : \mathscr{M}_X \to \mathscr{M}_X(\mathscr{L})$ which sends every section t of $\mathscr{M}_X$ over an open U to $(s|U)t$. It follows immediately from the above that for s to be *regular*, it is necessary and sufficient for h_s to be *injective*, and in fact h_s is then a *bijective* homomorphism from $\mathscr{M}_X$ to $\mathscr{M}_X(\mathscr{L})$, and its restriction to $\mathscr{M}_X^*$ is a bijection to $(\mathscr{M}_X(\mathscr{L}))^*$. We conclude that the homothety $t \mapsto ts$ is an isomorphism from $M(X)$ to $M(X, \mathscr{L})$.

(20.1.9). Let s be a regular meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; then for every $\mathscr{O}_X$-module $\mathscr{F}$, s similarly defines a homomorphism $h_s \otimes 1_{\mathscr{F}} : \mathscr{M}_X(\mathscr{F}) \to \mathscr{M}_X(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L})$, which is again *bijective*.

(20.1.10). Let s be a meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; for s to be regular, it is necessary and sufficient for there to exist a meromorphic section s' of $\mathscr{L}^{-1}$ over X such that the canonical image of $s \otimes s'$ in $\mathscr{M}_X$ ($\mathbf{0_I}$, 5.4.3) is the unit section, and this section s' is then unique: indeed, the necessity of the local existence of such a section is evident, and its local uniqueness implies its global (and unique) existence; moreover, the existence of s' is trivially sufficient for s to be regular. We will take $s' = s^{-1}$.

Finally, if $\mathscr{L}'$ is a second invertible $\mathscr{O}_X$-module, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, then $s \otimes s'$ is evidently a regular meromorphic section of $\mathscr{L} \otimes \mathscr{L}'$ over X.

(20.1.11). If $f : X' \to X$ is a morphism of ringed spaces, then there is in general no natural map sending a meromorphic function on X to a meromorphic function on X'. For example, if X is the spectrum of a local integral domain A, X' its residue field k, then there is no natural homomorphism from the field of fractions K of A to k, and we can only send an element of K to an element of k if it is already in A.

In general, if $f = (\psi, \theta)$, then for every open U of X, denote by $\mathscr{S}_f(U)$ the set of *regular* sections $s \in \Gamma(U, \mathscr{O}_X)$ such that the image of s under

$$\Gamma(\theta^\#) : \Gamma(U, \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathscr{O}_{X'})$$

is a *regular* section. It is immediate that $U \mapsto \mathscr{S}_f(U)$ is a *subsheaf* of the sheaf of sets $\mathscr{S}(\mathscr{O}_X)$, which we denote by $\mathscr{S}_f$. We set $\mathscr{M}_f = \mathscr{O}_X[\mathscr{S}_f^{-1}]$; this is a subsheaf of rings of $\mathscr{M}_X$, and we canonically **IV-4** | 230
obtain from $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ extending $\theta^\#$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1, prop. 2); hence, recalling that $f^*(\mathscr{M}_f) = \psi^*(\mathscr{M}_f) \otimes_{\psi^*(\mathscr{O}_X)} \mathscr{O}_{X'}$, we get a canonical homomorphism of $\mathscr{O}_{X'}$-algebras

$$f^*(\mathscr{M}_f) \longrightarrow \mathscr{M}_{X'}. \tag{20.1.11.1}$$

For every meromorphic function ϕ on X that *is a section of* $\mathscr{M}_f$, $\Gamma(\theta'^\#)(\phi)$ is a meromorphic function on X', called the *inverse image of* ϕ *under* f, and denoted by $\phi \circ f$ if there is no cause for confusion.

Similarly, if $\mathscr{F}$ is an $\mathscr{O}_X$-module, then we set $\mathscr{M}_f(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_f$, and we immediately obtain from $\theta'^\#$ a canonical homomorphism (which is also written as $u \mapsto u \circ f$)

$$\Gamma(X, \mathscr{M}_f(\mathscr{F})) \longrightarrow \Gamma(X', \mathscr{M}_{X'}(f^*(\mathscr{F}))).$$

In addition, if $u \in \Gamma(X, \mathscr{M}_f(\mathscr{F}))$ is defined (20.1.7) at a point x, then u coincides, on a neighbourhood U of x, with a section of the form $\sum_i h_i \otimes (t_i/s_i)$, where the h_i belong to $\Gamma(U, \mathscr{F})$, the t_i to $\Gamma(U, \mathscr{O}_X)$, and the s_i to $\Gamma(U, \mathscr{S}_f)$. As by hypothesis the images of the s_i in $\Gamma(f^{-1}(U), \mathscr{O}_{X'})$ are regular, we see that $u \circ f$ is defined at every point of $f^{-1}(U)$; in other words, we have

$$f^{-1}(\mathrm{dom}(u)) \subset \mathrm{dom}(u \circ f). \tag{20.1.11.2}$$

We will see later (20.6.5, (i)) examples (with $\mathscr{F} = \mathscr{O}_X$) where the two sides of (20.1.11.2) can be different.

Consider in particular the case where $\mathscr{M}_f = \mathscr{M}_X$; then, if $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module, the image in $\mathscr{M}_{X'}(f^*(\mathscr{L}))$, under $\Gamma(\theta'^{\#})$, of a *regular* meromorphic section of $\mathscr{L}$ over X (20.1.8) is a *regular* meromorphic section of $f^*(\mathscr{L})$ over X', as it follows immediately from the definition of its sections, and from the fact that a homomorphism of rings sends an invertible element to an invertible element.

Let $f' : X'' \to X'$ be a second morphism of ringed spaces, and suppose that $\mathscr{M}_f = \mathscr{M}_X$ and $\mathscr{M}_{f'} = \mathscr{M}_{X'}$; then, if we set $f'' = f \circ f'$, we also have $\mathscr{M}_{f''} = \mathscr{M}_X$, and we immediately see that for every meromorphic section u of $\mathscr{F}$ over X, we have $u \circ f'' = (u \circ f) \circ f'$.

Proposition (20.1.12). — *If the morphism $f : X' \to X$ is flat* ($\mathbf{0_I}$, 6.7.1), *then we have $\mathscr{M}_f = \mathscr{M}_X$, and the homomorphism $\phi \mapsto \phi \circ f$ is defined on all of $M(X)$. In addition, if f is a (flat) morphism of locally ringed spaces, then we have* $\mathrm{dom}(\phi \circ f) = f^{-1}(\mathrm{dom}(\phi))$; *if in addition f is surjective (thus faithfully flat), then the homomorphism $\phi \mapsto \phi \circ f$ is injective.*

Proof. The first assertion follows from the fact that, if B is an A-algebra which is a flat A-module, then every element of A not a divisor of 0 in A is not a divisor of 0 in B ($\mathbf{0_I}$, 6.3.4). To prove the other assertions, note that, for every $x' \in X'$, if $x = f(x')$, then $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, and as the homomorphism $\mathscr{O}_{X,x} \to \mathscr{O}_{X',x'}$ is *local* by hypothesis, it is injective (($\mathbf{0_I}$, 6.5.1) and ($\mathbf{0_I}$, 6.6.2)); if we set $A = \mathscr{O}_{X,x}$, $B = \mathscr{O}_{X',x'}$, such that A identifies with a subring of B, then $(f^*(\mathscr{M}_X))_{x'}$ is equal to $S^{-1}A \otimes_A B = S^{-1}B$, where S is the set of regular elements of A, $(\mathscr{M}_{X'})_{x'}$ is equal to $T^{-1}B$, where T is the set of regular elements of B, and as we have seen that $S \subset T$, the homomorphism $S^{-1}B \to T^{-1}B$ **IV-4** | 231
is injective; in other words, this proves that the homomorphism (20.1.11.1) $f^*(\mathscr{M}_X) \to \mathscr{M}_{X'}$ is *injective* (hence the last assertion of the statement). The quotient $f^*(\mathscr{M}_X)/\mathscr{O}_{X'}$ identifies with an $\mathscr{O}_{X'}$-submodule of $\mathscr{M}_{X'}/\mathscr{O}_{X'}$, and $(f^*(\mathscr{M}_X)/\mathscr{O}_{X'})_{x'}$ identifies with $(\mathscr{M}_X/\mathscr{O}_X)_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{X',x'}$. Then suppose that $x \notin \mathrm{dom}(\phi)$; the image of ϕ_x in $(\mathscr{M}_X/\mathscr{O}_X)_x$ is therefore $\neq 0$; by faithful flatness, we deduce that it is the same for the image of $(\phi \circ f)_{x'}$, so $x' \notin \mathrm{dom}(\phi \circ f)$, which finishes the proof. □

Remark (20.1.13). — Let X be a *reduced* complex analytic space; then the notion of a meromorphic function on X defined above coincides with the usual notion. Consider on the other hand a prescheme Y, locally of finite type over the field $\mathbf{C}$; we then know that we can associate to Y an analytic space Y^{an} having the same underlying topological space, and the canonical morphism $f : Y^{\mathrm{an}} \to Y$ is *flat* [**?**]; by virtue of (20.1.12), the canonical homomorphism $u \mapsto u \circ f$ from $M(Y)$ to $M(Y^{\mathrm{an}})$ is therefore always defined and is injective; but it is not *surjective* in general. For example, when $Y = \mathbf{V}_0^r$ ($\mathbf{Err_{III}}$, 14) is the affine space of dimension r over $\mathbf{C}$, $M(Y)$ canonically identifies with the field $R(Y)$ of rational functions on Y (20.2.13, (i)), while $M(Y^{\mathrm{an}})$ is the field of usual meromorphic functions on $\mathbf{C}^r$. Because of this fact, it is often preferable, in algebraic geometry, to abstain from the terminology introduced in this section, and to use the equivalent terminology of "pseudo-function" which will be defined below.

20.2. Pseudo-morphisms and pseudo-functions

§21. Divisors

21.1. Divisors on a ringed space

(21.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space. We retain the notation of (20.1): $\mathscr{S}(\mathscr{O}_X)$ denotes the subsheaf of $\mathscr{O}_X$ whose sections over an open U are the *regular* elements of $\Gamma(U, \mathscr{O}_X)$; $\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}(\mathscr{O}_X)^{-1}]$ the sheaf of germs of meromorphic functions on X; $\mathscr{M}_X^*$ the sheaf of *invertible* elements of $\mathscr{M}_X$, which is also equal to the sheaf $\mathscr{S}(\mathscr{M}_X)$ of regular elements of $\mathscr{M}_X$ (20.1.8); $\mathscr{O}_X^*$ the sheaf of invertible elements of $\mathscr{O}_X$, which is a *subsheaf of groups* of $\mathscr{M}_X^*$.

(21.1.2). We call the quotient sheaf $\mathscr{M}_X^*/\mathscr{O}_X^*$ the *sheaf of germs of divisors on* X and denote it by $\mathscr{D}iv_X$; it is a sheaf of *commutative groups* (written additively); the sections of $\mathscr{D}iv_X$ over X are called the *divisors on* X and form a commutative group denoted $\operatorname{Div}(X)$. For every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, the image of f in $\Gamma(X, \mathscr{D}iv_X) = \operatorname{Div}(X)$ is a divisor, called the *divisor of* f and denoted $\operatorname{div}(f)$; by definition, for every open U of X and every $x \in U$, we have $\operatorname{div}(f)|U = 0$ **IV-4** | 255
if and only if $f|U \in \Gamma(U, \mathscr{O}_X^*)$; by the definition of quotient sheaves, for every $x \in X$ there exists an open neighbourhood U of x such that $\operatorname{div}(f)|U = \operatorname{div}_U(g)$ for some $g \in \Gamma(U, \mathscr{M}_X^*)$, and more precisely $g = f|U$; conversely, every divisor D on X is *locally* of the form $\operatorname{div}(g)$ for a regular meromorphic function g.

(21.1.2.1). For every point $x \in X$, the stalk $(\mathscr{D}iv_X)_x$ of the sheaf $\mathscr{D}iv_X$ at x is the quotient group $\mathscr{M}_{X,x}^*/\mathscr{O}_{X,x}^*$; the image of a divisor D on X in $(\mathscr{D}iv_X)_x$ is denoted D_x; to say that $D_x = 0$ means that there exists an open neighbourhood U of x such that $D|U = 0$.

The *support* of a divisor D is the closed set of $x \in X$ such that $D_x \neq 0$. We denote it $\operatorname{Supp}(D)$. **IV-4** | 256

For every open U of X, we evidently have $\mathscr{M}_X^*|U = \mathscr{M}_U^*$, $\mathscr{O}_X^*|U = \mathscr{O}_U^*$, hence $\mathscr{D}iv_X|U = \mathscr{D}iv_U$, and consequently the sheaf $\mathscr{D}iv_X$ is equal to the presheaf $U \mapsto \operatorname{Div}(U)$.

When $X = \operatorname{Spec}(A)$ is affine, we write $\operatorname{Div}(A)$ instead of $\operatorname{Div}(\operatorname{Spec}(A))$.

(21.1.3). We will always denote *additively* the group $\operatorname{Div}(X)$ of divisors on X. For two regular meromorphic functions f, g on X, we thus have

(21.1.3.1) $$\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g),$$

(21.1.3.2) $$\operatorname{div}(f^{-1}) = -\operatorname{div}(f).$$

By definition, for every regular meromorphic function f on X, we have the equivalence

(21.1.3.3) $$\operatorname{div}(f) = 0 \iff f \in \Gamma(X, \mathscr{O}_X^*),$$

whence, for two regular meromorphic functions f, g on X, we have

(21.1.3.4) $$\operatorname{div}(f) = \operatorname{div}(g) \iff fg^{-1} \in \Gamma(X, \mathscr{O}_X^*).$$

(21.1.4). Now let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, and let s be a *regular meromorphic section* (20.1.8) of $\mathscr{L}$ over X. Every $x \in X$ admits a neighbourhood U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, hence $\mathscr{M}_X(\mathscr{L})|U$ is isomorphic to $\mathscr{M}_X|U$; by one of these isomorphisms, $s|U$ corresponds to a section $f \in \Gamma(U, \mathscr{M}_X^*)$, and since two isomorphisms differ only by multiplication by an element of $\Gamma(U, \mathscr{O}_X^*)$ ($\mathbf{0_I}$, 5.4.3), the element $\operatorname{div}_U(f)$ of $\Gamma(U, \mathscr{D}iv_X)$ is *independent* of the isomorphism chosen; it is clear that these elements (for U variable) are the restrictions of a section of $\mathscr{D}iv_X$ over X, which we call the *divisor of* s and denote $\operatorname{div}(s)$ (such a divisor is not necessarily of the form $\operatorname{div}(g)$ for a regular meromorphic function g on X; see (21.2.9)). For $\mathscr{L} = \mathscr{O}_X$, the definition of $\operatorname{div}(s)$ coincides with that of (21.1.2). If $\mathscr{L}, \mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, we immediately have

(21.1.4.1) $$\operatorname{div}(s \otimes s') = \operatorname{div}(s) + \operatorname{div}(s')$$

(21.1.4.2) $$\operatorname{div}(s^{\otimes n}) = n.\operatorname{div}(s) \quad \text{for all} \quad n \in \mathbf{Z}$$

(s^{-1} being the regular meromorphic section of $\mathscr{L}^{-1}$ over X defined in (20.1.10)) and, for two regular meromorphic sections s, s' of $\mathscr{L}$ over X, we have the relation

(21.1.4.3) $$\operatorname{div}(s) = \operatorname{div}(s') \iff s' = ts \quad \text{with} \quad t \in \Gamma(X, \mathscr{O}_X^*).$$

(21.1.5). The sheaf $\mathscr{S}(\mathscr{O}_X)$ (20.1.3) whose sections over an open U of X are the regular elements of $\Gamma(U, \mathscr{O}_X)$ is a *sub-sheaf of monoids* of $\mathscr{M}_X^*$; we can write

(21.1.5.1) $$\mathscr{S}(\mathscr{O}_X) = \mathscr{O}_X \cap \mathscr{M}_X^*.$$

If we denote by $\mathscr{S}(\mathscr{O}_X)^{-1}$ the sheaf whose sections over U are the inverses in $\Gamma(U, \mathscr{M}_X^*)$ of the elements of $\Gamma(U, \mathscr{S}(\mathscr{O}_X))$, it is clear that $\Gamma(U, \mathscr{S}(\mathscr{O}_X)) \cap \Gamma(U, \mathscr{S}(\mathscr{O}_X)^{-1}) = \Gamma(U, \mathscr{O}_X^*)$, hence **IV-4 | 257**

(21.1.5.2) $$\mathscr{S}(\mathscr{O}_X) \cap \mathscr{S}(\mathscr{O}_X)^{-1} = \mathscr{O}_X^*.$$

Definition (21.1.6). — The sub-sheaf of sets of $\mathscr{D}iv_X$, the canonical image of the sub-sheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{M}_X^*$, is denoted $\mathscr{D}iv_X^+$; its sections over X are called *positive divisors* and their set is denoted $\mathrm{Div}^+(X)$.

Since $\mathscr{S}(\mathscr{O}_X)$ is a sheaf of monoids (multiplicative), we have

(21.1.6.1) $$\mathrm{Div}^+(X) + \mathrm{Div}^+(X) \subset \mathrm{Div}^+(X)$$

and on the other hand, by virtue of (21.1.5.2) and (21.1.3.3)

(21.1.6.2) $$\mathrm{Div}^+(X) \cap (-\mathrm{Div}^+(X)) = \{0\}.$$

These two relations show that $\mathrm{Div}^+(X)$ is the set of positive elements for an *order structure* on the group $\mathrm{Div}(X)$, compatible with this group structure; we denote this order relation by $D \leqslant D'$, in other words, we have

(21.1.6.3) $$D \geqslant 0 \iff D \in \mathrm{Div}^+(X).$$

We will always suppose in what follows that $\mathrm{Div}(X)$ is equipped with this order structure.

By definition, for every regular meromorphic function f on X, we have the relation

(21.1.6.4) $$\mathrm{div}(f) \geqslant 0 \iff f \in \Gamma(X, \mathscr{S}(\mathscr{O}_X))$$

in other words, $\mathrm{div}(f) \geqslant 0$ means that f is a *regular section* of $\mathscr{O}_X$, or equivalently a section of $\mathscr{O}_X$ *invertible in* $M(X)$.

More generally, given a divisor D on X, the relation $\mathrm{div}(f) \geqslant D$ is equivalent to the following: for every open U of X such that $D|U = \mathrm{div}(g)$, where $g \in \Gamma(U, \mathscr{M}_X^*)$, there exists a regular element t of $\Gamma(U, \mathscr{O}_X)$ such that $f|U = tg$.

(21.1.7). Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, s a regular meromorphic section of $\mathscr{L}$ over X; we have the relation

(21.1.7.1) $$\mathrm{div}(s) \geqslant 0 \iff s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*)$$

as follows immediately from the definitions (21.1.4) and (21.1.6).

Proposition (21.1.8). — *Let X be a locally Noetherian prescheme, D a divisor on X. Suppose that for every $x \in X$ such that* $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$ *we have $D_x \geqslant 0$ (resp. $D_x = 0$). Then $D \geqslant 0$ (resp. $D = 0$).*

Proof. The question being local on X, we can suppose that $D = \mathrm{div}(f)$, f being a regular meromorphic function on X; the hypothesis means that if $T = X - \mathrm{dom}(f)$, then $\mathrm{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in T$ (since $\mathrm{dom}(f)$ contains the maximal points of X). By (5.10.5), the restriction homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X - T, \mathscr{O}_X)$ is *bijective*, which shows that there exists a section s of $\mathscr{O}_X$ over X such that $f = s|(X - T)$. But by definition of T, this implies $T = \varnothing$, hence $f = s$ and $D \geqslant 0$, and we immediately deduce the conclusion by applying the preceding to $-D$, by virtue of (21.1.6.2). ☐ **IV-4 | 258**

Corollary (21.1.9). — *Let X be a locally Noetherian prescheme, D a divisor on X. Let S be the support of D. Then, for every maximal point x of S,* $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$.

Proof. Set $X_1 = \mathrm{Spec}(\mathscr{O}_{X,x})$; taking into account (20.2.11) and (20.3.6), the sheaf $\mathscr{M}_{X_1}$ is induced on X_1 by $\mathscr{M}_X$, hence we can restrict to the case where $X = X_1$, in which case $x \in S$ and x is the maximal point of S, so necessarily $S = \{x\}$. If $\mathrm{prof}(\mathscr{O}_{X,x}) \neq 1$, we would conclude, by virtue of (21.1.8), that $D = 0$, which contradicts the definition of S. ☐

Proposition (21.1.10). — *Let A be a local Noetherian ring; in order that* $\mathrm{Div}(A) = 0$, *it is necessary and sufficient that* $\mathrm{prof}(A) = 0$, *in other words, that the maximal ideal $\mathfrak{m}$ of A be associated to A* (**0**, *16.4.6).*

Proof. Indeed, to say that $\mathrm{Div}(A) = 0$ means that in A every regular element is invertible, or equivalently that all the elements of $\mathfrak{m}$ are divisors of zero, which means that $\mathfrak{m} \in \mathrm{Ass}(A)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 3 of prop. 2). ☐

21.2. Divisors and invertible fractionary Ideals

(21.2.1). Let $(X, \mathscr{O}_X)$ be a ringed space. We call *fractionary Ideal* on X a sub-$\mathscr{O}_X$-Module of the $\mathscr{O}_X$-Module $\mathscr{M}_X$ of germs of meromorphic functions on X. A fractionary Ideal $\mathscr{J}$ on X which is an invertible $\mathscr{O}_X$-Module is called an *invertible fractionary Ideal.*

Proposition (21.2.2). — *For an invertible fractionary Ideal $\mathscr{J}$ on X to be invertible, it is necessary and sufficient that for every $x \in X$, there exists a neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{M}_X^*)$ such that $\mathscr{J}|U = \mathscr{O}_U.f$.*

Proof. The condition is evidently sufficient, since the map $s \mapsto s(f|V)$ of $\Gamma(V, \mathscr{O}_X)$ is evidently bijective for every open $V \subset U$. To see that it is necessary, let us note that there exists by hypothesis a neighbourhood U of x and an isomorphism of $\mathscr{O}_X$-Modules $\mathscr{O}_U \xrightarrow{\sim} \mathscr{J}|U$. If f is the image of the section $1 \in \Gamma(U, \mathscr{O}_X)$ by this isomorphism, on can suppose, restricting U, that $f = u/s$, where $u \in \Gamma(U, \mathscr{O}_X)$ and $s \in \Gamma(U, \mathscr{S}(\mathscr{O}_X))$, and the isomorphism considered then corresponds, for every section $v \in \Gamma(V, \mathscr{O}_X)$ (where V is an open contained in U) the section $v(u|V)/(s|V)$; to say that this map is bijective means that $u|V$ is a regular element of $\Gamma(V, \mathscr{O}_X)$, hence $f \in \Gamma(U, \mathscr{M}_X^*)$. □

We note that for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$ with $f \in \Gamma(U, \mathscr{M}_X^*)$, the section f is **IV-4** | 259
determined *up to multiplication* by an element of $\Gamma(U, \mathscr{O}_X^*)$, since the multiplication by these elements gives all the automorphisms of the $\mathscr{O}_U$-Module $\mathscr{O}_U$ ($\mathbf{0_I}$, 5.4.3).

Corollary (21.2.3). — *label=(i) Let $\mathscr{J}$ be an invertible fractionary Ideal; then the invertible $\mathscr{O}_X$-Module $\mathscr{J}^{-1}$ is canonically identified with the fractionary Ideal $\mathscr{J}'$ (the* transporter *of $\mathscr{J}$ into $\mathscr{O}_X$) defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, we have $\mathscr{J}'|U = \mathscr{O}_U.f^{-1}$.*

lbbel=(ii) If $\mathscr{J}_1$ and $\mathscr{J}_2$ are two invertible fractionary Ideals, the canonical map $\mathscr{J}_1 \otimes \mathscr{J}_2 \to \mathscr{J}_1\mathscr{J}_2$ is an isomorphism of $\mathscr{O}_X$-Modules.

Proof. The assertion (ii) follows immediately from (21.2.2). On the other hand, the remark made at the end of (21.2.2) shows that there is indeed a unique fractionary Ideal $\mathscr{J}'$ and a unique isomorphism defined by the condition of the statement; the isomorphism of $\mathscr{J}'$ on $\mathscr{J}^{-1} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{O}_X)$ is obtained by making correspond, for every section $s(f^{-1}|V)$ of $\Gamma(V, \mathscr{J}')$ (where V is an open contained in U and $s \in \Gamma(V, \mathscr{O}_X)$) the homomorphism $t(f|V) \mapsto st$ of $\Gamma(V, \mathscr{J})$ to $\Gamma(V, \mathscr{O}_X)$. □

By virtue of (21.2.3), (i), we will identify invertible $\mathscr{O}_X$-Modules and their inverses $\mathscr{J}$, $\mathscr{J}^{-1}$ considered as sub-$\mathscr{O}_X$-Modules of $\mathscr{M}_X$.

(21.2.4). It follows from (21.2.3) that the set $\mathrm{Id.\,inv}(X)$ of invertible fractionary Ideals on X is equipped with a structure of *commutative group* for the law of composition $(\mathscr{J}_1, \mathscr{J}_2) \mapsto \mathscr{J}_1\mathscr{J}_2$, the neutral element of this group being $\mathscr{O}_X$. It is clear that for every open U of X, we have $(\mathscr{J}_1\mathscr{J}_2)|U = (\mathscr{J}_1|U)(\mathscr{J}_2|U)$, hence the restriction map $\mathscr{J} \mapsto \mathscr{J}|U$ is a homomorphism of groups $\mathrm{Id.\,inv}(X) \to \mathrm{Id.\,inv}(U)$; we thus define a *presheaf of commutative groups* $U \mapsto \mathrm{Id.\,inv}(U)$; it is immediate that in fact this presheaf is a *sheaf of commutative groups*, which we denote $\mathscr{J}d.inv_X$.

(21.2.5). For every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, it follows from (21.2.2) that $\mathscr{J}(f) = \mathscr{O}_X.f$ is an invertible fractionary Ideal, and we evidently have $\mathscr{J}(f_1f_2) = \mathscr{J}(f_1)\mathscr{J}(f_2)$, in other words the map $f \mapsto \mathscr{J}(f)$ is a homomorphism of the commutative group $\Gamma(X, \mathscr{M}_X^*)$ to the commutative group $\mathrm{Id.\,inv}(X)$. Replacing X by an arbitrary open U of X and noting that the homomorphisms thus obtained are compatible with the restriction operations, we obtain a canonical homomorphism of sheaves of commutative groups:

(21.2.5.1) $$I_0 : \mathscr{M}_X^* \longrightarrow \mathscr{J}d.inv_X.$$

Let us note that if $f \in \Gamma(X, \mathscr{O}_X^*)$, then $\mathscr{J}(f) = \mathscr{O}_X$; we deduce immediately that the homomorphism I_0 factorises as

(21.2.5.2) $$\mathscr{M}_X^* \longrightarrow \mathscr{M}_X^*/\mathscr{O}_X^* = \mathscr{D}iv_X \xrightarrow{I} \mathscr{J}d.inv_X$$

where I is a homomorphism of the sheaf of additive groups $\mathscr{D}iv_X$ to the sheaf of multiplicative groups $\mathscr{J}d.inv_X$; we thus have for every open U of X a homomorphism $\mathscr{J}_U : \mathrm{Div}(U) \to \mathrm{Id.\,inv}(U)$

of commutative groups, such that, for every section $f \in \Gamma(U, \mathscr{M}_X^*)$, we have

$$\mathscr{J}_U(\mathrm{div}_U(f)) = \mathscr{O}_U.f. \tag{21.2.5.3}$$

(21.2.5.4). We deduce that $f \in \Gamma(X, \mathscr{J}_X(D))$, for every divisor $D \in \mathrm{Div}(X)$, we have the relation

$$f \in \Gamma(X, \mathscr{J}_X(D)) \iff \mathrm{div}(f) \geqslant D. \tag{21.2.5.4}$$

Proposition (21.2.6). — *The homomorphism $I : \mathscr{D}iv_X \to \mathscr{J}d.inv_X$ is bijective.*

Proof. We define indeed a homomorphism I'_X of $\mathrm{Id.\,inv}(X)$ to $\mathrm{Div}(X)$ by making correspond to every invertible fractionary Ideal $\mathscr{J}$ on X the divisor $I'_X(\mathscr{J})$ defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$ (21.2.2), we take $I'_X(\mathscr{J})|U = \mathrm{div}_U(f)$; by virtue of the remark following (21.2.2), this definition is independent of the generator f chosen for $\mathscr{J}|U$, and determines a divisor on X. Moreover, this definition immediately shows that the homomorphisms $\mathscr{J}_X$ and I'_X are inverses of one another. Replacing X by an arbitrary open U, we deduce the definition of the isomorphism $I' : \mathscr{J}d.inv_X \to \mathscr{D}iv_X$, inverse of I, whence the proposition. We set $I'_X(\mathscr{J}) = \mathrm{div}(\mathscr{J})$, so that for every regular meromorphic function f on X

$$\mathrm{div}(\mathscr{O}_X.f) = \mathrm{div}(f). \tag{21.2.6.1}$$

□

(21.2.7). We will often identify the sheaves $\mathscr{D}iv_X$ and $\mathscr{J}d.inv_X$ (resp. the groups $\mathrm{Div}(X)$ and $\mathrm{Id.\,inv}(X)$) by means of the isomorphisms I and I' (resp. $\mathscr{J}_X$ and I'_X) above. We note that we have the relation

$$D \geqslant 0 \iff \mathscr{J}_X(D) \subset \mathscr{O}_X \quad \text{for} \quad D \in \mathrm{Div}(X) \tag{21.2.7.1}$$

as follows immediately from the definitions (21.1.6) and (21.1.5.1); in other terms, the image $\mathscr{J}_X(\mathrm{Div}^+(X))$ is the set of *Ideals* of $\mathscr{O}_X$ (also sometimes called *integral Ideals*) which are invertible $\mathscr{O}_X$-Modules: such an Ideal $\mathscr{J}$ is further characterised by the fact that for every $x \in X$, there is a neighbourhood U of x in X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where f is a regular element of $\Gamma(U, \mathscr{O}_X)$. The set $\mathscr{J}_X(\mathrm{Div}^+(X))$ of these Ideals is thus a sub-monoid of $\mathrm{Id.\,inv}(X)$, equal to the set of positive elements for an order relation on $\mathrm{Id.\,inv}(X)$, and it is immediate that this relation is none other than the relation *opposite* to the inclusion; in other words, we have

$$\mathscr{J}_1 \subset \mathscr{J}_2 \iff \mathrm{div}(\mathscr{J}_1) \geqslant \mathrm{div}(\mathscr{J}_2) \tag{21.2.7.2}$$

for $\mathscr{J}_1$, $\mathscr{J}_2$ in $\mathrm{Id.\,inv}(X)$. **IV-4** | 260

(21.2.8). For every divisor D on X, we set

$$\mathscr{O}_X(D) = (\mathscr{J}_X(D))^{-1}; \tag{21.2.8.1}$$

$\mathscr{O}_X(D)$ is thus an invertible fractionary Ideal, defined as follows: for every open U of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, $\mathscr{O}_X(D)|U$ is the frac- tionary Ideal $\mathscr{O}_U.f^{-1}$; by virtue of **IV-4** | 261
(21.1.6), for every regular meromorphic function f, we have the relation

$$f \in \Gamma(X, \mathscr{O}_X(D)) \iff \mathrm{div}(f) \geqslant -D. \tag{21.2.8.2}$$

Moreover, it is clear that we have canonical isomorphisms (21.2.3)

$$\begin{cases} \mathscr{O}_X(0) = \mathscr{O}_X, \quad \mathscr{O}_X(D+D') = \mathscr{O}_X(D)\mathscr{O}_X(D') \xrightarrow{\sim} \mathscr{O}_X(D) \otimes \mathscr{O}_X(D') \\ \mathscr{O}_X(nD) = (\mathscr{O}_X(D))^n \xrightarrow{\sim} (\mathscr{O}_X(D))^{\otimes n} \end{cases} \tag{21.2.8.3}$$

for every integer $n \in \mathbf{Z}$, and for any two divisors D, D' on X.

(21.2.9). Let $\mathscr{J}$ be an invertible fractionary Ideal on X. The canonical injection $\mathscr{J} \hookrightarrow \mathscr{M}_X$ defines by tensorisation a homomorphism of $\mathscr{O}_X$-Modules

$$\mathscr{M}_X(\mathscr{J}) = \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{M}_X \longrightarrow \mathscr{M}_X \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{M}_X \tag{21.2.9.1}$$

which is an *isomorphism*: indeed, if U is an open of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, the homomorphism (21.2.9.1) restricted to U is none other than the isomorphism which, for every section t of $\mathscr{M}_X(\mathscr{J})|U = \mathscr{M}_X|U$ over $V \subset U$, makes correspond the section $t(f|V)^{-1}$ of the same sheaf. In the isomorphism (21.2.9.1), the regular meromorphic sections of $\mathscr{J}$ over X correspond to the regular meromorphic functions on X.

Consider in particular the case where $\mathscr{J} = \mathscr{O}_X(D)$, where D is a divisor on X; we then have a canonical isomorphism

(21.2.9.2) $$\mathscr{M}_X(\mathscr{O}_X(D)) \xrightarrow{\sim} \mathscr{M}_X$$

and we denote by s_D the regular meromorphic section of $\mathscr{O}_X(D)$ over X which corresponds by this isomorphism to the section 1 of $\mathscr{M}_X$. If U is an open of X such that $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, then $s_D|U = 1$ in $\Gamma(U, \mathscr{M}_X)$; as $D|U = \operatorname{div}_U(f)$, we deduce from (21.1.4) that

(21.2.9.3) $$\operatorname{div}(s_D) = D.$$

On the other hand, we also deduce from the canonical isomorphisms (21.2.8.3) the formulas

(21.2.9.4) $$s_0 = 1, \quad s_{D+D'} = s_D \otimes s_{D'}, \quad s_{nD} = s_D^{\otimes n} \quad (n \in \mathbf{Z}).$$

(21.2.10). Consider, between two pairs $(\mathscr{L}, s)$, $(\mathscr{L}', s')$, where $\mathscr{L}$ and $\mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, the relation: "there exists an isomorphism $u : \mathscr{L} \xrightarrow{\sim} \mathscr{L}'$ such that $\overline{u}(s) = s'$", where $\overline{u} : \Gamma(X, \mathscr{M}_X(\mathscr{L})) \xrightarrow{\sim} \Gamma(X, \mathscr{M}_X(\mathscr{L}'))$ is the isomorphism derived from u (we note that the isomorphism u satisfying this condition is then determined in a unique way). It is clear that this is an equivalence relation, and since there exists a set of invertible $\mathscr{O}_X$-Modules such that every invertible $\mathscr{O}_X$-Module is isomorphic to an element of this set ($\mathbf{0_I}$, 5.4.7), we can speak of the *set* $D(X)$ of *equivalence classes* of pairs $(\mathscr{L}, s)$ for the above relation. For every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ and every regular mero- morphic section s of $\mathscr{L}$ over X, **IV-4** | 262
we denote by $\operatorname{cl}(\mathscr{L}, s)$ the element of $D(X)$ corresponding to the pair $(\mathscr{L}, s)$. It follows from (21.2.10) that if s, s' are two regular meromorphic sections of $\mathscr{L}$ over X, the relation $\operatorname{cl}(\mathscr{L}, s) = \operatorname{cl}(\mathscr{L}, s')$ is equivalent to the existence of a section $t \in \Gamma(X, \mathscr{O}_X^*)$ such that $s' = ts$.

It is immediate that if $(\mathscr{L}, s)$ is equivalent to $(\mathscr{L}_1, s_1)$ and $(\mathscr{L}', s')$ to $(\mathscr{L}_1', s_1')$, the pairs $(\mathscr{L} \otimes \mathscr{L}', s \otimes s')$ and $(\mathscr{L}_1 \otimes \mathscr{L}_1', s_1 \otimes s_1')$ are equivalent; we thus define in $D(X)$ a law of composition by setting

(21.2.10.1) $$\operatorname{cl}(\mathscr{L}, s) \operatorname{cl}(\mathscr{L}', s') = \operatorname{cl}(\mathscr{L} \otimes \mathscr{L}', s \otimes s');$$

it is immediate that this is the law of a *commutative group*, whose neutral element is $\operatorname{cl}(\mathscr{O}_X, 1)$ and where the inverse of $\operatorname{cl}(\mathscr{L}, s)$ is $\operatorname{cl}(\mathscr{L}^{-1}, s^{\otimes(-1)})$.

Proposition (21.2.11). — *The maps*

(21.2.11.1) $$D \longmapsto \operatorname{cl}(\mathscr{O}_X(D), s_D), \qquad \operatorname{cl}(\mathscr{L}, s) \longmapsto \operatorname{div}(s)$$

are inverse isomorphisms of $\operatorname{Div}(X)$ *to* $D(X)$ *and of* $D(X)$ *to* $\operatorname{Div}(X)$ *respectively.*

Proof. Taking into account (21.2.8.3), (21.2.9.4), and (21.2.10.1), it suffices to verify that the composites of these two maps are the identities in $\operatorname{Div}(X)$ and $D(X)$ respectively. The first assertion is none other than (21.2.9.3). On the other hand, let s be a regular meromorphic section of $\mathscr{L}$ over X, and let U be an open of X such that there exists an isomorphism of $\mathscr{L}|U$ onto $\mathscr{O}_U$, transforming $s|U$ to $f \in \Gamma(U, \mathscr{M}_X^*)$, so that $D|U = \operatorname{div}_U(f)$, $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$ and $s_D|U$ is the unit element of $\Gamma(U, \mathscr{M}_X)$. There is thus an isomorphism $v_U : \mathscr{L}|U \to \mathscr{O}_X(D)|U$ such that $\overline{v}_U$ (notation of (21.2.10)) transforms $s|U$ to $f.f^{-1} = 1$; on sees immediately that these isomorphisms are compatible with the restriction operations, and thus define an isomorphism $v : \mathscr{L} \to \mathscr{O}_X(D)$ such that $\overline{v}(s) = s_D$. □

We can *transport* by the first of the isomorphisms (21.2.11.1) the structure of ordered group of $\operatorname{Div}(X)$ to $D(X)$; the elements $\geqslant 0$ of $D(X)$ are thus the classes $\operatorname{cl}(\mathscr{L}, s)$ such that $\operatorname{div}(s) \geqslant 0$, that is to say (21.1.7.1) such that
$$s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*).$$

(21.2.12). Let D be a *positive* divisor on a *prescheme* X; the fractionary Ideal $\mathscr{J}_X(D)$ is an Ideal of $\mathscr{O}_X$, which is an invertible $\mathscr{O}_X$-Module; let $Y(D)$ be the sub-prescheme of X that it defines. For every $x \in Y(D)$, there is by hypothesis a neighbourhood U of x in X and a regular section $t \in \Gamma(U, \mathscr{O}_X)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U.t$ (21.2.7); in other terms, the canonical immersion $Y(D) \to X$ is *regular and of codimension* 1 (19.1.4). Conversely, if Y is a sub-prescheme of X, regularly immersed in X and of codimension 1 at every point of Y, there exists *a unique positive divisor* D such that $Y(D) = Y$, since there is a neighbourhood U of x in X such that $Y \cap U$ is defined by an Ideal of the form $\mathscr{O}_U.t$, where t is a regular element of $\Gamma(U, \mathscr{O}_X)$. We note that $\operatorname{Supp}(D) = Y(D)$, since to say that $D_x \neq 0$ means that t_x is not invertible, that is to say $x \in Y(D)$.

21.3. Linear equivalence of divisors

(21.3.1). We say that a divisor D on X is *principal* if it is of the form $\operatorname{div}(f)$, where f is a regular meromorphic function on X; the meromorphic functions f' such that $\operatorname{div}(f') = D$ are then all of the form tf, where $t \in \Gamma(X, \mathscr{O}_X^*)$ (21.1.3.4). The set of principal divisors is a subgroup of $\operatorname{Div}(X)$, denoted $\operatorname{Div.princ}(X)$, isomorphic to $\Gamma(X, \mathscr{M}_X^*)/\Gamma(X, \mathscr{O}_X^*)$. Two divisors D, D' are said to be *linearly equivalent* if $D - D'$ is a principal divisor; the principal divisors are thus the divisors linearly equivalent to 0.

IV-4 | 263

(21.3.2). Recall ($\mathbf{0_I}$, 5.4.7) that we can speak of the set of equivalence classes of invertible $\mathscr{O}_X$-Modules for the isomorphism relation; we denote this set by $\operatorname{Pic}(X)$, and for every invertible $\mathscr{O}_X$-Module $\mathscr{L}$, we denote by $\operatorname{cl}(\mathscr{L})$ the class of equivalence of $\mathscr{O}_X$-Modules isomorphic to $\mathscr{L}$; moreover, $\operatorname{Pic}(X)$ is a commutative group for the multiplication defined by $\operatorname{cl}(\mathscr{L})\operatorname{cl}(\mathscr{L}') = \operatorname{cl}(\mathscr{L} \otimes \mathscr{L}')$. It is clear that the map

(21.3.2.1) $$r : \operatorname{cl}(\mathscr{L}, s) \longmapsto \operatorname{cl}(\mathscr{L})$$

is a homomorphism of the group $D(X)$ (21.2.10) to the group $\operatorname{Pic}(X)$. By composition, we thus deduce a homomorphism

(21.3.2.2) $$l : \operatorname{Div}(X) \xrightarrow{\sim} D(X) \xrightarrow{r} \operatorname{Pic}(X)$$

(also denoted l_X) such that, for every divisor D, we have

(21.3.2.3) $$l(D) = \operatorname{cl}(\mathscr{O}_X(D)).$$

Let us note finally that, if $u : X' \to X$ is a morphism of ringed spaces, $\mathscr{L}_1$, $\mathscr{L}_2$ two invertible isomorphic $\mathscr{O}_X$-Modules, the invertible $\mathscr{O}_{X'}$-Modules $u^*(\mathscr{L}_1)$ and $u^*(\mathscr{L}_2)$ ($\mathbf{0_I}$, 5.4.5) are isomorphic; as moreover, for any two invertible $\mathscr{O}_X$-Modules $\mathscr{L}_1$, $\mathscr{L}_2$, we have $u^*(\mathscr{L}_1 \otimes \mathscr{L}_2) = u^*(\mathscr{L}_1) \otimes u^*(\mathscr{L}_2)$ up to canonical isomorphism ($\mathbf{0_I}$, 4.3.3), we see that the morphism u canonically defines a homomorphism of commutative groups

(21.3.2.4) $$\operatorname{Pic}(u) : \operatorname{Pic}(X) \longrightarrow \operatorname{Pic}(X').$$

Proposition (21.3.3). — *label=(i) The kernel of the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is the subgroup $\operatorname{Div.princ}(X)$, in other words, for $\mathscr{O}_X(D)$ and $\mathscr{O}_X(D')$ to be isomorphic, it is necessary and sufficient that D and D' be linearly equivalent. There is thus a canonical injective homomorphism*

(21.3.3.1) $$\operatorname{Div}(X)/\operatorname{Div.princ}(X) \longrightarrow \operatorname{Pic}(X)$$

derived from l.

lbbel=(ii) For an invertible $\mathscr{O}_X$-Module $\mathscr{L}$ to be such that $\operatorname{cl}(\mathscr{L})$ is of the form $l(D)$, or equivalently that $\mathscr{L}$ is isomorphic to an $\mathscr{O}_X$-Module of the form $\mathscr{O}_X(D)$, it is necessary and sufficient that there exists a regular meromorphic section of $\mathscr{L}$.

Proof. The proposition follows immediately from the definitions and from (21.2.10). □ **IV-4 | 264**

Proposition (21.3.4). — *Let X be a prescheme satisfying one of the two following hypotheses:*

label=a) X is locally Noetherian and $\operatorname{Ass}(\mathscr{O}_X)$ is contained in an affine open of X.
lbbel=b) X is reduced and the set of its irreducible components is locally finite.

Then the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is surjective, and gives by passage to the quotient an isomorphism

$$\operatorname{Div}(X)/\operatorname{Div.princ}(X) \xrightarrow{\sim} \operatorname{Pic}(X).$$

Proof. It suffices to show that every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ admits a regular meromorphic section over X (21.3.3). In both cases, it will suffice, thanks to (20.2.11), (ii), to define a section s of $(\mathscr{M}_X(\mathscr{L}))^*$ over a *schematically dense* open U of X, or else, in case *a*), over an open U containing $\mathrm{Ass}(\mathscr{O}_X)$ (20.2.13), (iv). Indeed, let V be an arbitrary open of X such that $\mathscr{L}|V$ is isomorphic to $\mathscr{O}_V$, so that there exists an isomorphism of $\mathscr{M}_X(\mathscr{L})|V$ onto $\mathscr{M}_X|V$, transforming $s|(U\cap V)$ to a section f_V of $\mathscr{M}_X^*$ over $U\cap V$. As $U\cap V$ is schematically dense in V, it follows from (20.2.11) that there exists a unique regular meromorphic function g_V such that $g_V|(U\cap V) = f_V$, and this section corresponds by the isomorphism considered to a regular meromorphic section u_V of $\mathscr{L}$ over V such that $u_V|(U\cap V) = s|(U\cap V)$. Moreover, if V' is a second open of X such that $\mathscr{L}|V'$ is isomorphic to $\mathscr{O}_{V'}$, the restrictions of u_V and $u_{V'}$ to $V\cap V'$ are equal, since they correspond by isomorphism to two meromorphic functions which coincide on an open schematically dense $U\cap V\cap V'$, and we conclude again by (20.2.11); the u_V are thus well the restrictions of a single regular meromorphic section of $\mathscr{L}$ over X.

In case *b*), we take for each of the maximal points x_λ of X an open U_λ containing x_λ, distinct from $\overline{\{x_\lambda\}}$ and such that $\mathscr{L}|U_\lambda$ is isomorphic to $\mathscr{O}_{U_\lambda}$; we take for s the section such that $s|U_\lambda$ is the section of $\mathscr{L}|U_\lambda$ which corresponds by the preceding isomorphism to the unit section of $\mathscr{O}_{U_\lambda}$.

In case *a*), we can take U to be an affine open (hence Noetherian), in other words, suppose that $X = \mathrm{Spec}(A)$, where A is Noetherian, and $\mathscr{L} = \widetilde{P}$, where P is a projective A-module of rank 1. If S is the set of regular elements of A, we have $\Gamma(X, \mathscr{M}_X) = S^{-1}A$ (20.2.12) and $\Gamma(X, \mathscr{M}_X(\mathscr{L})) = S^{-1}P$; now S is the set of elements not belonging to any of the ideals associated to A, hence $S^{-1}A$ is a semi-local ring whose maximal ideals come from the maximal elements of $\mathrm{Ass}(A)$, and $S^{-1}P$ is a projective $S^{-1}A$-module of rank 1, hence a *free* $S^{-1}A$-module of rank 1 (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 3, prop. 5); an element forming a base of this $S^{-1}A$-module is thus (20.1.8) a regular meromorphic section of $\mathscr{L}$ over X. □

Corollary (21.3.5). — *If X is a Noetherian prescheme, such that there exists an invertible $\mathscr{O}_X$-Module ample (**II**, 4.5.3) (for example a quasi-projective prescheme over a Noetherian ring spectre (**II**, 5.3.1 and 4.6.6)), the canonical homomorphism $l : \mathrm{Div}(X) \to \mathrm{Pic}(X)$ is surjective.*

Proof. Indeed (**II**, 4.5.4), there exists then a neighbourhood of $\mathrm{Ass}(\mathscr{O}_X)$ which is an affine open of the finite set $\mathrm{Ass}(\mathscr{O}_X)$. □

IV-4 | 265

Remark (21.3.6). — Recall ($\mathbf{0_I}$, 5.4.7) that there is a canonical isomorphism $\pi : \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\sim} \mathrm{Pic}(X)$ defined as follows. We start from a 1-cocycle $(c_{\alpha\beta})$ with values in $\mathscr{O}_X^*$ corresponding to a covering (U_α) of X, $c_{\alpha\beta}$ being an element of $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X^*)$, and we associate to it the class of the invertible $\mathscr{O}_X$-Module obtained by gluing the $\mathscr{O}_{U_\alpha}$ along the isomorphisms $\mathscr{O}_{U_\alpha}|(U_\alpha\cap U_\beta) \xrightarrow{\sim} \mathscr{O}_{U_\beta}|(U_\alpha\cap U_\beta)$ defined by multiplication by $c_{\alpha\beta}$. On the other hand, we deduce from the exact sequence of sheaves of commutative groups

$$1 \longrightarrow \mathscr{O}_X^* \longrightarrow \mathscr{M}_X^* \longrightarrow \mathscr{D}iv_X \longrightarrow 0$$

the boundary homomorphism of the exact cohomology sequence

$$\partial : \mathrm{Div}(X) \longrightarrow \mathrm{H}^1(X, \mathscr{O}_X^*). \tag{21.3.6.1}$$

Let us show that the composite homomorphism

$$\mathrm{Div}(X) \xrightarrow{\partial} \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\pi} \mathrm{Pic}(X)$$

is the homomorphism l defined in (21.3.2.2). Indeed, we must start from a covering (U_α) of X and a divisor D such that $D|U_\alpha = \mathrm{div}_{U_\alpha}(g_\alpha)$, where g_α is a regular meromorphic function over U_α; $\partial(D)$ is the cohomology class of the cocycle $(c_{\alpha\beta})$, where $c_{\alpha\beta} = g_\alpha|_\beta g_\beta|_\alpha^{-1}$, $g_{\alpha|\beta}$ denoting the restriction of g_α to $U_\alpha\cap U_\beta$. It is clear that the image by π of this cohomology class is the class of the invertible fractional Ideal $\mathscr{L}$ such that for every α, $\mathscr{L}|U_\alpha = \mathscr{O}_{U_\alpha}.g_\alpha^{-1}$, which is none other by definition than $\mathscr{O}_X(D)$ (21.2.8).

21.4. Inverse images of divisors

(21.4.1). Let $f : X' \to X$ be a morphism of ringed spaces; we propose to give conditions allowing us to associate to a divisor D on X a divisor D' on X', the *inverse image* of D by f. Let us first note that for every section $t \in \Gamma(X, \mathscr{O}_X^*)$, the image of t by the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is still invertible, in other words belongs to $\Gamma(X', \mathscr{O}_{X'}^*)$. Consider then D as given by the class of equivalence of a pair $(\mathscr{L}, s)$, where $\mathscr{L}$ is an invertible $\mathscr{O}_X$-Module and s a regular meromorphic section of $\mathscr{L}$ over X (21.2.11). Form the invertible $\mathscr{O}_{X'}$-Module $f^*(\mathscr{L}) = \mathscr{L}'$; to say that the inverse image $s_0 f$ of s by f exists (20.1.11) and is a regular meromorphic section of $\mathscr{L}'$ over X' amounts to saying that the inverse images by f of s and $s^{\otimes(-1)}$ exist, in other words that $s \in \Gamma(X, \mathscr{M}_f(\mathscr{L}))$ and $s^{\otimes(-1)} \in \Gamma(X, \mathscr{M}_f(\mathscr{L}^{-1}))$; the remark made above shows that if $(\mathscr{L}_1, s_1)$ is a pair equivalent to $(\mathscr{L}, s)$ in the sense of (21.2.10), the inverse image $s_1 \circ f$ exists and is a regular meromorphic section of $\mathscr{L}_1' = f^*(\mathscr{L}_1)$ over X', and the pairs $(\mathscr{L}', s_0 f)$ and $(\mathscr{L}_1', s_1 o f)$ are equivalent. We can therefore pose the following definition:

Definition (21.4.2). — Given a morphism $f : X' \to X$ of ringed spaces, we say that the *inverse image by* f of a divisor D on X exists if $s_D \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(D)))$ and $s_{-D} \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(-D)))$ (cf. (20.1.11)). We call then the *inverse image* of D by f and denote $f^*(D)$ the divisor on X' which corresponds canonically (21.2.11) to the class of the pair $(f^*(\mathscr{O}_X(D)), s_D \circ f)$.

It follows immediately from this definition that if D and D' have inverse images by f, so does $-D$ and $D + D'$ (taking into account (21.2.9.4)) and we have $f^*(D + D') = f^*(D) + f^*(D')$. In other **IV-4** | 266
words, the set $\mathrm{Div}^f(X)$ of divisors on X whose inverse image by f exists is a *subgroup* of $\mathrm{Div}(X)$, and the map $D \mapsto f^*(D)$ is an *increasing homomorphism* of the ordered subgroup $\mathrm{Div}^f(X)$ to the ordered group $\mathrm{Div}(X')$, making commutative the diagram

(21.4.2.1)
$$\begin{tikzcd} \mathrm{Div}^f(X) \arrow[r, "l_X"] \arrow[d, "f^*"'] & \mathrm{Pic}(X) \arrow[d, "\mathrm{Pic}(f)"] \\ \mathrm{Div}(X') \arrow[r, "l_{X'}"] & \mathrm{Pic}(X') \end{tikzcd}$$

(21.4.3). The definition (21.4.2) also shows immediately that, for $f^*(D)$ to exist, it suffices that for every open U of X such that $D|U = \mathrm{div}(g)$ (restriction of f to $f^{-1}(U) \to U$), where g is a regular meromorphic function on X, to say that the inverse image of s_D by f exists and is a regular meromorphic section on X'. We deduce also immediately a second description of $\mathrm{Div}^f(X)$ and of $f^*(D)$: we consider the sub-sheaf of groups of $\mathscr{M}_X^*$, denoted $\mathscr{M}_f^*$, formed of germs of meromorphic functions on an open of X whose inverse image exists and is regular on the inverse image open (20.1.11). Then if $f = (\psi, 0)$, the canonical homomorphism (20.1.11) $\psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ gives by restriction a homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*) \to \mathscr{M}_{X'}^*$. Setting $\mathscr{D}iv_X' = \mathscr{M}_f^* / \mathscr{O}_X^*$, we have $\mathrm{Div}^f(X) = \Gamma(X, \mathscr{D}iv_X')$, and the map $D \mapsto f^*(D)$ corresponds to the homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*)/\psi^*(\mathscr{O}_X^*) \to \mathscr{M}_{X'}^*/\mathscr{O}_{X'}^* = \mathscr{D}iv_{X'}$, derived from the preceding by passage to quotients.

(21.4.4). It follows also from the preceding definitions that if $f' : X'' \to X'$ is a second morphism of ringed spaces, D a divisor on X such that the inverse images $f^*(D)$ and $f'^*(f^*(D))$ exist, then $(f \circ f')^*(D)$ exists and is equal to $f'^*(f^*(D))$.

Proposition (21.4.5). — *Let $f : X' \to X$ be a morphism of ringed spaces. In each of the three following cases, the inverse image by f of every divisor on X is defined:*

label=(i) f is flat.
lbbel=(ii) X and X' are locally Noetherian preschemes and $f(\mathrm{Ass}(\mathscr{O}_{X'})) \subset \mathrm{Ass}(\mathscr{O}_X)$ and $g'(\mathrm{Ass}(\mathscr{O}_{X'})) \subset \mathrm{Ass}(\mathscr{O}_X)$.
lcbel=(iii) X and X' are preschemes, the set of irreducible components of X is locally finite, X' is reduced and every irreducible component of X' dominates an irreducible component of X.

Proof. It suffices indeed to show that in all three cases we have $\mathscr{M}_f = \mathscr{M}_X$; in case (i), this follows from (20.1.12). In case (iii), we can restrict to the case where $X = \mathrm{Spec}(A)$ and $X' = \mathrm{Spec}(A')$ are affine; if $s \in A$ is regular, it does not belong to any minimal prime ideal of A (20.1.3.1), hence the hypothesis implies that its image in A' does not belong to any minimal prime ideal of A', and is therefore a regular element of A' (20.1.3.1). In case (ii) the meromorphic functions on X' are identified with the pseudo-functions on X' (20.2.11), and the hypothesis, combined with (20.2.13), (iv), ensures that the inverse image by f of every schematically dense open in X is a schematically dense open in X'; we conclude therefore by (20.3.12). □ **IV-4** | 267

Corollary (21.4.6). — *Let X be a prescheme having one of the following properties:*

label=(i) X is locally Noetherian.

lbbel=(ii) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{D}iv_X)_x \xrightarrow{\sim} \mathrm{Div}(\mathscr{O}_{X,x}). \tag{21.4.6.1}$$

Proof. This follows indeed from (20.2.11), (20.3.7), and from the fact that $(\mathscr{O}_X^*)_x$ is identified with the group of invertible elements of the ring $\mathscr{O}_{X,x}$. □

(21.4.7). Let X, X' be two preschemes, $f : X' \to X$ a morphism. If D is a *positive* divisor on X such that the inverse image $f^*(D)$ is defined (21.4.2), then the sub-prescheme $Y(f^*(D))$ of X' is none other than the inverse image $f^{-1}(Y(D))$; this follows also from the definitions (21.4.2) and (21.2.12).

Proposition (21.4.8). — *Let X, Y be two preschemes, $f : X \to Y$ a faithfully flat morphism. Then, if D is a divisor on Y such that $f^*(D) \geqslant 0$ (the existence of $f^*(D)$ resulting from (21.4.5)), we have $D \geqslant 0$. In particular, the map $D \mapsto f^*(D)$ of $\mathrm{Div}(Y)$ to $\mathrm{Div}(X)$ is injective.*

Proof. The question being local on Y, we can restrict to the case where $D = \mathrm{div}(w)$, with $w = uv^{-1}$, u and v being two regular sections of $\mathscr{O}_Y$ over Y. By hypothesis we have $u_y\mathscr{O}_{X,x} \subset v_y\mathscr{O}_{X,x}$, hence, for every $x \in X$, setting $y = f(x)$, we have $u_y\mathscr{O}_{Y,y} \subset v_y\mathscr{O}_{Y,y}$; we conclude that $u_y\mathscr{O}_{Y,y} \subset v_y\mathscr{O}_{Y,y}$ by virtue of the hypothesis that $\mathscr{O}_{X,x}$ is a faithfully flat $\mathscr{O}_{Y,y}$-module, and since f is surjective, we have $u\mathscr{O}_Y \subset v\mathscr{O}_Y$, whence $D \geqslant 0$. □

21.5. Direct images of divisors

(21.5.1). Let X, X' be two preschemes, $f : X' \to X$ a morphism. In this section, we shall give sufficient conditions for one to associate to a divisor D' on X' a divisor on X, the *direct image* of D' by f. We will restrict ourselves to the case where f is a *finite* morphism (for more general conditions, see the chapter of this Treatise devoted to the theory of intersections).

Lemma (21.5.2). — *Let A be a ring, E an A-module free of finite rank. For an endomorphism u of E to be injective, it is necessary and sufficient that* $\det(u)$ *be a regular element of A.*

Proof. This is proved in Bourbaki, *Alg.*, chap. III, 3e éd., § 8, no 2, prop. 3. □

(21.5.3). Suppose now that the morphism $f : X' \to X$ is *finite*, and in addition that f satisfies one of the two following properties:

label=I) f is a finite morphism *locally free*, in other words (18.2.7) the $\mathscr{O}_X$-Module $f_*(\mathscr{O}_{X'})$ is *locally free*.

lbbel=II) X is a locally Noetherian reduced prescheme, the $\mathscr{R}(X)$-Module $f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is locally free, and for every section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$ (U open in X), $\mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ is a **IV-4** | 268 section of $\mathscr{O}_X$ over U (cf. (**II**, 6.5.1)). (We recall that condition (II) is satisfied for every *finite morphism* f when X is a locally Noetherian normal prescheme (*loc. cit.*).)

We know then (**II**, 6.5.5) that for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ we define the *norm* $\mathscr{L} = \mathrm{N}_{X'/X}(\mathscr{L}')$ (which we also write $\mathrm{N}(\mathscr{L}')$), which is an invertible $\mathscr{O}_X$-Module. Moreover, for every open U of X and every *regular* section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$, the norm $\mathrm{N}_{X'/X}(s') = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ (which we also write $\mathrm{N}(s')$) is a regular element of $\Gamma(U, \mathscr{O}_X)$; we are indeed immediately reduced to the case where $U = X$ is affine, and then the conclusion follows from (21.5.2) in hypothesis (I); on the other hand, in hypothesis (II), the fact that $\mathscr{R}(X)$ is a flat $\mathscr{O}_X$-Module implies that the section $s' \otimes 1$ of $\Gamma(U, f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X))$ is also regular ($\mathbf{0_I}$, 6.1.2), and the conclusion follows again from (21.5.2) applied to the ring $\Gamma(U, \mathscr{R}(X))$, taking into account the definition of the norm (**II**, 6.5.3).

This being so, let u' be a section of $\mathscr{L}'$ over X'; the morphism f being affine, every point $x \in X$ admits a neighbourhood U such that $u'|f^{-1}(U)$ is of the form t'/s', where $t' \in \Gamma(f^{-1}(U), \mathscr{L}')$ and s' is a regular section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$; the element $\mathrm{N}(t')/\mathrm{N}(s')$ (where $\mathrm{N}(t')$ is the section of $\mathscr{L}$ defined in (**II**, 6.5.3)) is then a section of $\mathscr{L}$ over U, and from what precedes and the multiplicative properties of the norm (**II**, 6.5.3.1) this section does not depend on the choice of $u'|f^{-1}(U)$ but only on its representation in the form t'/s'; for the same reason, the sections of $\mathscr{L}$ over U thus defined are the restrictions of a single meromorphic section of $\mathscr{L}$ over X, which we call the *norm* of u' and denote $\mathrm{N}_{X'/X}(u')$ (or simply $\mathrm{N}(u')$). The map

(21.5.3.1) $$\mathrm{N}_{X'/X} : \Gamma(X', \mathscr{M}_{X'}(\mathscr{L}')) \longrightarrow \Gamma(X, \mathscr{M}_X(\mathrm{N}_{X'/X}(\mathscr{L}')))$$

extends the norm defined in (**II**, 6.5.5.3); if u' is a *regular* meromorphic section of $\mathscr{L}'$ over X', it follows immediately from what precedes that $\mathrm{N}(u')$ is a regular meromorphic section of $\mathscr{L}$ over X, since (with the same notation) $\mathrm{N}(t')$ is regular if t' is. Finally, if $\mathscr{L}'_1$, $\mathscr{L}'_2$ are two invertible $\mathscr{O}_{X'}$-Modules, s'_1 (resp. s'_2) a regular meromorphic section of $\mathscr{L}'_1$ (resp. $\mathscr{L}'_2$) over X', we have, by virtue of what precedes and of (**II**, 6.5.3.1)

(21.5.3.2) $$\mathrm{N}(s'_1 \otimes s'_2) = \mathrm{N}(s'_1) \otimes \mathrm{N}(s'_2).$$

(21.5.4). Suppose always that f satisfies one of hypotheses I), II) of (21.5.3). If $(\mathscr{L}'_1, s'_1)$, $(\mathscr{L}'_2, s'_2)$ are two pairs each formed of an invertible $\mathscr{O}_{X'}$-Module and a regular meromorphic section of this Module over X', which are moreover *equivalent* in the sense of (21.2.10), then the pairs $(\mathrm{N}(\mathscr{L}'_1), \mathrm{N}(s'_1))$ and $(\mathrm{N}(\mathscr{L}'_2), \mathrm{N}(s'_2))$ are also equivalent, since an isomorphism of invertible $\mathscr{O}_{X'}$-Modules has for norm an isomorphism of their norms (**II**, 6.5.4), and we have already seen above that $\mathrm{N}_{X'/X}$ transforms the sections of $\Gamma(X', \mathscr{O}^*_{X'})$ to sections of $\Gamma(X, \mathscr{O}^*_X)$. We can therefore pose the following definition:

Definition (21.5.5). — Given a finite morphism $f : X' \to X$ of preschemes, satisfying one of conditions I), II) of (21.5.3), we call the *direct image* (or *norm*) of a divisor D' on X' by f, and denote $f_*(D')$ (or $\mathrm{N}_{X'/X}(D')$), the divisor on X which corresponds canonically (21.2.11) to the class of the pair $(\mathrm{N}_{X'/X}(\mathscr{O}_{X'}(D')), \mathrm{N}_{X'/X}(s_{D'}))$.

It follows immediately from this definition, taking into account (21.2.9.4) and (21.5.3.2), that if D'_1, D'_2, D' are divisors on X', we have

(21.5.5.1) $$f_*(D'_1 + D'_2) = f_*(D'_1) + f_*(D'_2)$$

and $D' \geqslant 0$ implies $f_*(D') \geqslant 0$, in other words, $D' \mapsto f_*(D')$ is an *increasing homomorphism* of the ordered group $\mathrm{Div}(X')$ to the ordered group $\mathrm{Div}(X)$. The definition (21.5.5) also shows immediately that for every open U of X, we have $f^U_*(D'|f^{-1}(U)) = f_*(D')|U$ (f^U being the restriction $f^{-1}(U) \to U$ of f), and the homomorphisms f^U_*, for U variable, thus define a *homomorphism of sheaves of ordered groups* **IV-4 | 269**

(21.5.5.2) $$\mathrm{N}_{X'/X} : f_*(\mathscr{D}iv_{X'}) \longrightarrow \mathscr{D}iv_X.$$

Moreover, for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ and every regular meromorphic section s' of $\mathscr{L}'$ over $f^{-1}(U)$, we have, from the preceding definitions and (21.1.4)

(21.5.5.3) $$\mathrm{div}_U(\mathrm{N}(s')) = f^U_*(\mathrm{div}_{f^{-1}(U)}(s')).$$

Proposition (21.5.6). — *Let $f : X' \to X$ be a finite morphism locally free and suppose that $f_*(\mathscr{O}_{X'})$ is of constant rank n. Then, for every divisor D on X, $f_*(D)$ is defined and we have*

(21.5.6.1) $$f_*(f^*(D)) = nD.$$

Proof. The first assertion follows from the fact that f is flat (21.4.5), and the second is an immediate consequence of the definitions and of (**II**, 6.5.3.2). □

Proposition (21.5.7). — *Let $f : X' \to X$ be a finite morphism satisfying one of hypotheses I), II) of (21.5.3), $f' : X'' \to X'$ a finite morphism locally free of constant rank n. Then $f'' = f \circ f' : X'' \to X$ satisfies the same hypothesis as f, and for every divisor D'' on X'', we have*

(21.5.7.1) $$f''_*(D'') = f_*(f'_*(D'')).$$

Proof. Taking into account the definition (21.5.5), it suffices to prove the following: □

Lemma (21.5.7.2). — *Under the hypotheses of (21.5.7), we have a functorial isomorphism*

(21.5.7.3) $$\mathrm{N}_{X''/X}(\mathscr{L}'') \xrightarrow{\sim} \mathrm{N}_{X'/X}(\mathrm{N}_{X''/X'}(\mathscr{L}''))$$

in the category of invertible $\mathscr{O}_{X''}$-Modules.

Proof. Indeed, taking into account the definition of the norm of a section $\mathscr{L}''$ (**II**, 6.5.3) and of the definition (21.5.5), we will obtain (21.5.7.3), and it suffices to prove that for every section s of $f''_*(\mathscr{O}_{X''}) = f_*(f'_*(\mathscr{O}_{X''}))$ over an open U of X, we have

(21.5.7.4) $$\mathrm{N}_{f''_*(\mathscr{O}_{X''})|\mathscr{O}_X}(s) = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(\mathrm{N}_{f'_*(\mathscr{O}_{X''})|\mathscr{O}_{X'}}(s))$$

which follows from the transitivity of the norm (Bourbaki, *Alg.*, chap. VIII, § 12, nº 2, prop. 7) when **IV-4** | 270
f satisfies hypothesis (I) and A is Noetherian and reduced, one can suppose that A' is an A-module free of rank m, and then A'' is an A-module projective of rank mn, and restricting X to a suitable open, one can suppose that A'' is an A-module free of rank mn; the formula (21.5.7.4) follows then again from the transitivity of the norms. When f satisfies hypothesis (II), A is Noetherian and reduced, its total ring of fractions $A' \otimes_A \mathrm{R}$ is an R-module projective of rank m, hence $A'' \otimes_A \mathrm{R} = A'' \otimes_{A'} \mathrm{R}$ is an R-module projective of rank mn; the proposition also follows then from the transitivity of the norms. □

Proposition (21.5.8). — *Let $f : X' \to X$ be a finite morphism, $g : Y \to X$ a morphism; set $Y' = X' \times_X Y$, $f' = f_{(Y)} : Y' \to Y$, $g' = g_{(X)} : Y' \to X'$. Suppose satisfying one of the following conditions:*

label=(i) *f is locally free and g is flat.*

lbbel=(ii) *f is locally free, X and Y are locally Noetherian, $g(\mathrm{Ass}(\mathscr{O}_Y)) \subset \mathrm{Ass}(\mathscr{O}_X)$ and $g'(\mathrm{Ass}(\mathscr{O}_{Y'})) \subset \mathrm{Ass}(\mathscr{O}_{X'})$.*

lcbel=(iii) *f satisfies hypothesis* II) *of (21.5.3), Y is locally Noetherian, Y and Y' are reduced and every irreducible component of Y (resp. Y') dominates an irreducible component of X (resp. X').*

Then, for every divisor D' on X', $g^*(D')$ is defined, $g^*(f_*(D'))$ is defined and we have

(21.5.8.1) $$g^*(f_*(D')) = f'_*(g'^*(D')).$$

Proof. Indeed, in all cases, it follows from (**II**, 6.5.8) that there is a functorial isomorphism

$$g^*(\mathrm{N}_{X'/X}(\mathscr{L}')) \xrightarrow{\sim} \mathrm{N}_{Y'/Y}(g'^*(\mathscr{L}'))$$

in the category of invertible $\mathscr{O}_{X'}$-Modules; moreover (**II**, 6.5.4), if s' is the corresponding section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$, s'' the corresponding section of $\mathscr{O}_{Y'}$ over $g'^{-1}(f^{-1}(U))$ (where U is open of X), $\mathrm{N}_{Y'/Y}(s'')$ is a section of $\mathscr{O}_Y$ over U. The formula (21.5.8.1) will thus follow from the definitions, provided one proves that $g^*(D')$ and $g'^*(D')$ are both defined, for arbitrary divisors D' on X' and D on X. As far as D is concerned, this results from the hypotheses made and from (21.4.5). As far as D' is concerned, in case (i) g' is flat, hence the conclusion by (21.4.5); in what concerns D', in case (ii), the conclusion results again from (21.4.5) by the hypotheses; in case (iii), g' is flat, hence the conclusion again by (21.4.5). □

21.6. 1-codimensional cycle associated to a divisor **IV-4** | 270

(21.6.1). We call *multiplicity of* Z *at the point* x and denote by $\mathrm{mult}_x(\mathrm{Z})$ the integer n_x (positive or negative). To say that $\mathrm{Z} \geqslant 0$ means that $\mathrm{mult}_x(\mathrm{Z}) \geqslant 0$ for all $x \in \mathrm{X}$. We call the *support* of Z and denote by $\mathrm{Supp}(\mathrm{Z})$ the union of the $\overline{\{x\}}$ such that $\mathrm{mult}_x(\mathrm{Z}) \neq 0$; by definition of $\mathfrak{Z}(\mathrm{X})$, this is a closed part of X, as the union of a locally finite family of closed parts. We call the *dimension* (resp. *codimension*) of Z and denote by $\dim(\mathrm{Z})$ (resp. $\mathrm{codim}(\mathrm{Z})$) the dimension (resp. codimension in X) of $\mathrm{Supp}(\mathrm{Z})$.

(21.6.2). We say that a closed subspace Y of X is *purely of codimension* p (in X) if *each* of its irreducible components is of codimension p in X. We say that a cycle is *purely of codimension* p, or (by abuse of language) is a *p-codimensional cycle*, if its support is purely of codimension p. We denote by $\mathrm{X}^{(p)}$ the set of $x \in \mathrm{X}$ such that $\mathrm{codim}(\overline{\{x\}}, \mathrm{X}) = p$, or, what amounts to the same (5.1.2), $\dim(\mathscr{O}_{\mathrm{X},x}) = p$: the **IV-4** | 271
cycles purely of codimension p form a subgroup $\mathfrak{Z}^p(\mathrm{X})$ of $\mathfrak{Z}(\mathrm{X})$, isomorphic to the sub-group of $\mathbf{Z}^{\mathrm{X}^{(p)}}$ consisting of the $(n_x)_{x \in \mathrm{X}^{(p)}}$ such that the set of $\overline{\{x\}}$ (or the set of x) for which $n_x \neq 0$ is locally finite; this sub-group contains the free group $\mathbf{Z}^{(\mathrm{X}^{(p)})}$ and is identical to it when X is Noetherian. We denote

by $\mathfrak{Z}^{p+}(\mathrm{X})$ the set of elements $\geqslant 0$ of $\mathfrak{Z}^p(\mathrm{X})$. It is clear that the ordered group $\mathfrak{Z}(\mathrm{X})$ contains the direct sum of the ordered sub-groups $\mathfrak{Z}^p(\mathrm{X})$, and is identical to this direct sum when X is Noetherian; in the latter case, we consider $\mathfrak{Z}(\mathrm{X})$ as *graded* by the $\mathrm{Z}^p(\mathrm{X})$ for $p \in \mathbf{N}$.

(21.6.3). Let $\mathrm{Z} = \sum_{x\in\mathrm{X}} n_x.\overline{\{x\}}$ be a cycle on X, U an open subset of X; we call the *restriction* of Z to U and denote by Z|U the cycle $\sum_{x\in\mathrm{U}} n_x.(\mathrm{U}\cap\overline{\{x\}})$ on U; we have $\mathrm{Supp}(\mathrm{Z}|\mathrm{U}) = \mathrm{Supp}(\mathrm{Z})\cap\mathrm{U}$. It is clear that $\mathrm{Z} \hookrightarrow \mathrm{Z}|\mathrm{U}$ is a homomorphism of ordered groups from $\mathfrak{Z}(\mathrm{X})$ to $\mathfrak{Z}(\mathrm{U})$ (resp. from $\mathfrak{Z}^p(\mathrm{X})$ to $\mathfrak{Z}^p(\mathrm{U})$), and that if V is an open subset contained in U, we have $\mathrm{Z}|\mathrm{V} = (\mathrm{Z}|\mathrm{U})|\mathrm{V}$; the map $\mathrm{U} \hookrightarrow \mathfrak{Z}^+(\mathrm{U})$ (resp. $\mathrm{U} \hookrightarrow \mathfrak{Z}^{p+}(\mathrm{U})$) is therefore a *presheaf* on X of ordered commutative groups. In fact, this presheaf is a *sheaf*, as a direct sum of sheaves $(i_x)_*(\mathbf{Z}_{\overline{x}})$, where x ranges over X (resp. $\mathrm{X}^{(p)}$) and for each $x \in \mathrm{X}$, i_x is the canonical injection $\overline{\{x\}} \to \mathrm{X}$ and $\mathbf{Z}_{\overline{x}}$ is the simple sheaf associated to the constant sheaf $\mathbf{Z}$ on the space $\overline{\{x\}}$: this follows at once from the description of the sections of a direct sum of sheaves of commutative groups (G, II, 2.7). We denote this sheaf by $\mathscr{Z}_\mathrm{X}$ (resp. $\mathscr{Z}^p_\mathrm{X}$). We denote by $\mathscr{Z}^+_\mathrm{X}$ (resp. $\mathscr{Z}^{p+}_\mathrm{X}$) the sub-sheaf of monoids $\mathrm{U} \hookrightarrow \mathfrak{Z}^+(\mathrm{U})$ (resp. $\mathrm{U} \hookrightarrow \mathfrak{Z}^{p+}(\mathrm{U})$) of $\mathscr{Z}_\mathrm{X}$ (resp. **IV-4** | 272
$\mathscr{Z}^p_\mathrm{X}$). The sheaf $\mathscr{Z}_\mathrm{X}$ is evidently a *direct sum* of sheaves $\mathscr{Z}^p_\mathrm{X}$.

Let us note finally that the sheaves $\mathscr{Z}^p_\mathrm{X}$ (and therefore also $\mathscr{Z}_\mathrm{X}$) are *flasque*. Indeed, suppose given a section Z of $\mathscr{Z}^p_\mathrm{X}$ over an open U, so that the set S of $x \in \mathrm{U}$ such that $\mathrm{mult}_x(\mathrm{Z}) \neq 0$ is locally finite *in* U; this set is also locally finite *in* X, since for all $z \in \mathrm{X}$ and every open Noetherian neighbourhood V of z, $\mathrm{U}\cap\mathrm{V}$ is Noetherian, and thus contains only a finite number of points of S. We thus define a section Z' of $\mathscr{Z}^p_\mathrm{X}$ over X by setting $\mathrm{Z}' = \sum_{x\in\mathrm{U}} \mathrm{mult}_x(\mathrm{Z}).\overline{\{x\}}$ and since the closure of x in U is $\overline{\{x\}}\cap\mathrm{U}$, we have $\mathrm{Z}'|\mathrm{U} = \mathrm{Z}$, whence our assertion.

(21.6.4). We now propose to define a canonical homomorphism

$$c : \mathscr{D}iv_\mathrm{X} \longrightarrow \mathscr{Z}^1_\mathrm{X} \tag{21.6.4.1}$$

of sheaves of commutative groups. It clearly suffices to define a homomorphism of $\mathscr{M}^*_\mathrm{X}$ to $\mathscr{Z}^1_\mathrm{X}$, which vanishes on $\mathscr{O}^*_\mathrm{X}$ (21.1.2); or, by definition, $\mathscr{M}^*_\mathrm{X}$ is the *symmetrised* monoid sheaf of $\mathscr{O}_\mathrm{X}\cap\mathscr{M}^*_\mathrm{X}$, and a homomorphism $\mathscr{M}^*_\mathrm{X} \to \mathscr{Z}^1_\mathrm{X}$ is uniquely determined by the data of its restriction $\mathscr{S}(\mathscr{O}_\mathrm{X}) \to \mathscr{Z}^1_\mathrm{X}$, which is a homomorphism of sheaves of *monoids* subject to the sole condition of being zero on $\mathscr{O}^*_\mathrm{X}$; it therefore amounts to the same thing to say that, in order to define c, it suffices to define a homomorphism of *sheaves of monoids*

$$c : \mathscr{D}iv^+_\mathrm{X} \longrightarrow \mathscr{Z}^1_\mathrm{X}. \tag{21.6.4.2}$$

(21.6.5). Now, we have seen that every positive divisor D on X gives rise to the closed sub-prescheme Y(D) of X, defined by the Ideal $\mathscr{J}_\mathrm{X}(\mathrm{D}) \subset \mathscr{O}_\mathrm{X}$, and which is regularly immersed and of codimension 1. At each of the maximal points x of Y(D), we have ((19.1.4) and (5.1.2)) $\mathrm{codim}(\overline{\{x\}},\mathrm{X}) = \dim(\mathscr{O}_{\mathrm{X},x}) = 1$, in other words $x \in \mathrm{X}^{(1)}$; moreover the set of these maximal points is locally finite in X (3.1.6), and $\mathscr{O}_{\mathrm{Y(D)},x}$ is an *Artinian* ring. At every point $x \in \mathrm{X}^{(1)}$ which is not a maximal point of Y(D), we necessarily have $x \notin \mathrm{Y(D)}$, so $\mathscr{O}_{\mathrm{Y(D)},x} = 0$. Set

$$\mathrm{cyc}(\mathrm{D}) = \sum_{x\in\mathrm{X}^{(1)}} \mathrm{length}(\mathscr{O}_{\mathrm{Y(D)},x}).\overline{\{x\}} \tag{21.6.5.1}$$

which is thus an element of $\mathfrak{Z}^1(\mathrm{X})$.

Proposition (21.6.6). — *The map* $\mathrm{D} \mapsto \mathrm{cyc}(\mathrm{D})$ *is a homomorphism of the monoid* $\mathrm{Div}^+(\mathrm{X})$ *to* $\mathfrak{Z}^1(\mathrm{X})$.

Proof. It amounts to showing that for two positive divisors D, D′, we have

$$\mathrm{cyc}(\mathrm{D}+\mathrm{D}') = \mathrm{cyc}(\mathrm{D}) + \mathrm{cyc}(\mathrm{D}').$$

Now, by (21.2.5) $\mathscr{J}_\mathrm{X}(\mathrm{D}+\mathrm{D}') = \mathscr{J}_\mathrm{X}(\mathrm{D})\mathscr{J}_\mathrm{X}(\mathrm{D}')$; it thus amounts to showing that if $x \in \mathrm{X}^{(1)}$, setting $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$, and if t and t' are two regular elements of A, then $\mathrm{length}(\mathrm{A}/tt'\mathrm{A}) = \mathrm{length}(\mathrm{A}/t\mathrm{A}) + \mathrm{length}(\mathrm{A}/t'\mathrm{A})$; but since t is regular, $t\mathrm{A}/tt'\mathrm{A}$ is isomorphic to $\mathrm{A}/t'\mathrm{A}$, whence the proposition at once. □

It follows at once from the definitions that for every open $\mathrm{U} \subset \mathrm{X}$, we have

$$\mathrm{Y}(\mathrm{D}|\mathrm{U}) = \mathrm{Y}(\mathrm{D})\cap\mathrm{U},$$

whence $\mathrm{cyc}(D|U) = \mathrm{cyc}(D)|U$, and the homomorphisms $\mathrm{cyc} : \mathrm{Div}^+(U) \to \mathfrak{Z}^1(U)$ define a homomorphism of sheaves of monoids (21.6.4.2), whence the sought-after homomorphism (21.6.4.1) of sheaves of groups. IV-4 | 273

We have $\mathrm{Supp}(\mathrm{cyc}(D)) \subset \mathrm{Supp}(D)$ for every divisor D and

(21.6.6.2).

$$\mathrm{Supp}(\mathrm{cyc}(D)) = \mathrm{Supp}(D) \quad \text{when} \quad D \geqslant 0;$$

we have indeed already seen the second relation (21.2.12); when D is arbitrary, the relation $D_x = 0$ implies the existence of an open neighbourhood U of x such that $D|U = 0$, whence $\mathrm{cyc}(D)|U = 0$, which proves our assertion.

(21.6.7). The homomorphism (21.6.4.1) gives, by passage to groups of sections over X, a *homomorphism of ordered groups* $\mathrm{Div}(X) \to \mathfrak{Z}^1(X)$, which we will also denote $D \mapsto \mathrm{cyc}(D)$; the number $\mathrm{mult}_x(\mathrm{cyc}(D))$ for $x \in X^{(1)}$ is also denoted $\mathrm{mult}_x(D)$ or $\mathrm{mult}_{\overline{\{x\}}}(D)$ and is called the *multiplicity of the divisor* D *at the point* x, or the *multiplicity of the prime cycle* $\overline{\{x\}}$ *in* D; it is a positive or negative integer, equal as we have seen to $\mathrm{length}(\mathscr{O}_{Y(D),x})$ when D is a *positive* divisor; we thus have by definition

$$\mathrm{cyc}(D) = \sum_{x \in X^{(1)}} \mathrm{mult}_x(D).\overline{\{x\}}, \tag{21.6.7.1}$$

and by virtue of the fact that $D \mapsto \mathrm{cyc}(D)$ is a homomorphism, we have

$$\mathrm{mult}_x(-D) = -\mathrm{mult}_x(D), \qquad \mathrm{mult}_x(D + D') = \mathrm{mult}_x(D) + \mathrm{mult}_x(D') \tag{21.6.7.2}$$

for any two divisors D, D′.

For every regular meromorphic function f on X, we set in particular, for $x \in X^{(1)}$

$$o_x(f) = \mathrm{mult}_x(\mathrm{div}(f)) \tag{21.6.7.3}$$

and we say that $o_x(f)$ (a positive or negative integer) is the *order of f at the point $x \in X^{(1)}$*. If $f_x \in \mathscr{O}_{X,x}$ (a regular element of this local ring by hypothesis), then we have

$$o_x(f) = \mathrm{length}(\mathscr{O}_{X,x}/f_x\mathscr{O}_{X,x}); \tag{21.6.7.4}$$

for two regular meromorphic functions f, f' on X, we have

$$o_x(ff') = o_x(f) + o_x(f'), \qquad o_x(f^{-1}) = -o_x(f) \tag{21.6.7.5}$$

for all $x \in X^{(1)}$. The 1-codimensional cycles

$$Z^+(f) = \sum_{x \in X^{(1)}} (o_x(f))^+.\overline{\{x\}}, \qquad Z^-(f) = \sum_{x \in X^{(1)}} (o_x(f))^-.\overline{\{x\}}$$

are called respectively the *cycle of zeros* (or *polar cycle*) of f, and we evidently have $\mathrm{cyc}(\mathrm{div}(f)) = Z^+(f) - Z^-(f)$. The 1-codimensional cycles of the form $\mathrm{cyc}(\mathrm{div}(f))$ are called *principal* (or *linearly equivalent to zero*) and form a subgroup $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ of $\mathfrak{Z}^1(X)$. The sections over X of the image $c(\mathscr{D}iv_X) \subset \mathscr{Z}^1_X$ are called *locally principal cycles*; such a cycle Z is therefore characterised by the fact that, for all $y \in X$, there is an open neighbourhood U of y in X and a regular meromorphic function f on U such that $Z|U = \sum_{x \in U \cap X^{(1)}} o_x(f)(\overline{\{x\}} \cap U)$. If $Z = \sum_{x \in X^{(1)}} n_x.\overline{\{x\}}$, setting $T_y = \mathrm{Spec}(\mathscr{O}_{X,y})$ and $Z_y = \sum_{x \in X^{(1)} \cap T_y} n_x(\overline{\{x\}} \cap T_y)$, Z_y is *principal*, otherwise said there exists a regular meromorphic function g on T_y such that $Z_y = \mathrm{cyc}(\mathrm{div}(g))$. This follows at once from the definition above and from (20.3.7) which establishes a bijective correspondence between the regular meromorphic functions on T_y and the germs of regular meromorphic functions on the open neighbourhoods of y in X when X is locally Noetherian. We express this further by saying that Z_y is principal in saying that Z is *principal at the point* y. The set of points where Z is principal is evidently *open*, by virtue of what precedes. IV-4 | 274

We deduce from the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ a homomorphism $\mathrm{Div}(X)/\mathrm{Div.princ}(X) \to \mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X)$, by virtue of the definition of $\mathfrak{Z}^1_{\mathrm{princ}}(X)$. We say that the group $\mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is the *group of classes of* 1*-codimensional cycles* of X and we denote it $\mathfrak{Cl}(X)$. Two elements of $\mathfrak{Z}^1(X)$ having the same image in $\mathfrak{Cl}(X)$ are called *linearly equivalent*.

(21.6.8). Let us consider in particular the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a *Noetherian integral and integrally closed* ring. Then $\mathrm{X}^{(1)}$ is the set of *prime ideals of height* 1 of A, and $\mathfrak{Z}^1(\mathrm{X})$ is identified with the *group of divisors* (in the sense of N. Bourbaki) of the Krull ring A (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 3, cor. du th. 2 et n° 6, th. 3).

On the other hand, the regular meromorphic functions on X are identified with the nonzero elements of the field of fractions K of A, the map $f \mapsto \mathrm{cyc}(\mathrm{div}(f))$ in $\mathrm{M}(\mathrm{X})$ is identified in $\mathfrak{Z}^1(\mathrm{X})$ with the map denoted $f \mapsto \mathrm{div}(f)$ in Bourbaki (*loc. cit.*, § 1, n° 1); $\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$ is identified with the *group of principal divisors* of A in the sense of Bourbaki, and $\mathfrak{Cl}(\mathrm{X})$ with the *group of classes of divisors* of A in the sense of Bourbaki (*loc. cit.*, § 1, n° 2 et n° 10).

Theorem (21.6.9). — *Let* X *be a locally Noetherian normal prescheme.*

label=(i) The canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ *is injective and its image is formed of locally principal cycles.*

lbbel=(ii) The following conditions are equivalent:

*label=*a) *The homomorphism* $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ *is bijective.*

*lbbel=*b) *Every* 1*-codimensional cycle is locally principal.*

*lcbel=*c) *For all* $x \in \mathrm{X}$*, the local ring* $\mathscr{O}_{\mathrm{X},x}$ *is factorial.*

(We then say that X *is a* locally factorial *prescheme.)*

Proof. label=(i) It suffices to show that

$$\mathrm{cyc}^{-1}(\mathfrak{Z}^{1+}(\mathrm{X})) = \mathrm{Div}^+(\mathrm{X}), \tag{21.6.9.1}$$

since we have $\mathrm{Div}^+(\mathrm{X}) \cap (-\mathrm{Div}^+(\mathrm{X})) = 0$ and $\mathfrak{Z}^{1+}(\mathrm{X}) \cap (-\mathfrak{Z}^{1+}(\mathrm{X})) = 0$. We are thus reduced to proving that if D is a divisor on X such that $\mathrm{mult}_x(\mathrm{D}) \geqslant 0$ for all $x \in \mathrm{X}^{(1)}$, then $\mathrm{D} \geqslant 0$. Now, for all $x \notin \mathrm{X}^{(1)}$, the local ring $\mathscr{O}_{\mathrm{X},x}$ is integral and integrally closed and of **IV-4 | 275**
dimension 0 or $\geqslant 2$, therefore on the one hand $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 0$ or $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ (**0**, 16.5.1). On the other hand, for points $x \in \mathrm{X}^{(1)}$, the ring $\mathscr{O}_{\mathrm{X},x}$ is a discrete valuation ring ((**II**, 7.1.6)), whence $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 1$ (**0**, 16.5.1); the only points of X such that $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 1$ are thus the points of $\mathrm{X}^{(1)}$, and the conclusion follows from (21.1.8).

lbbel=(ii) The equivalence of *a)* and *b)* is clear by virtue of (i). From the characterisation of locally principal cycles given in (21.6.7), and the relation between 1-codimensional cycles on $\mathrm{Spec}(\mathrm{A})$ and divisors (in the sense of Bourbaki) of the ring A when A is a Noetherian integrally closed ring (21.6.8), condition *b)* is equivalent to saying that for all $x \in \mathrm{X}$, every divisor of the ring $\mathscr{O}_{\mathrm{X},x}$ is principal, in other words that the ring $\mathscr{O}_{\mathrm{X},x}$ is factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 1), whence the equivalence of *b)* and *c)*. □

(21.6.9.2). When A is a Noetherian *factorial* ring, it is clear that $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$ is locally factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 3, prop. 3). In this case, if one writes an element $f \neq 0$ of the field of fractions K of A in the form r/s, where r and s are two *coprime* elements of A, whose divisors are well-determined (*loc. cit.*, § 3, n° 3), these divisors are identified respectively with the *divisor of zeros* and the *divisor of poles* of f (21.6.7).

Corollary (21.6.10). — *Let* X *be a locally Noetherian normal prescheme.*

label=(i) There exists a canonical injective homomorphism

$$\mathrm{Pic}(\mathrm{X}) \longrightarrow \mathfrak{Cl}(\mathrm{X}). \tag{21.6.10.1}$$

lbbel=(ii) If X *is locally factorial, the homomorphism* (21.6.10.1) *is bijective, and conversely.*

Proof. We have seen (21.6.7) that the image of $\mathrm{Div.\,princ}(\mathrm{X})$ by the homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ is $\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$; we thus deduce from (21.6.9) that the homomorphism $\mathrm{Div}(\mathrm{X}) / \mathrm{Div.\,princ}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X}) / \mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X}) = \mathfrak{Cl}(\mathrm{X})$ deduced from cyc is injective, and that it is bijective if and only if X is locally factorial. The conclusion follows from the fact that, when X is locally Noetherian and reduced, the canonical homomorphism $\mathrm{Div}(\mathrm{X}) / \mathrm{Div.\,princ}(\mathrm{X}) \to \mathrm{Pic}(\mathrm{X})$ (21.3.3.1) is bijective (21.3.4), (ii). □

Corollary (21.6.11). — *Let* X *be a locally Noetherian prescheme which is locally factorial. Then the sheaf* $\mathscr{D}iv_{\mathrm{X}}$ *is flasque, and for every open* U *of* X*, the canonical homomorphism* $\mathrm{Pic}(\mathrm{X}) \to \mathrm{Pic}(\mathrm{U})$ *is surjective.*

Proof. Taking into account (21.6.9), (ii), the first assertion is equivalent to saying that the sheaf $\mathscr{Z}^1_X$ is flasque, which was seen above (21.6.3). For every open U of X, the canonical homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U)$ is thus surjective; as in (21.6.10), the homomorphism $\mathrm{Pic}(X) \to \mathrm{Pic}(U)$ is identified canonically with $\mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X) \to \mathfrak{Z}^1(U)/\mathfrak{Z}^1_{\mathrm{princ}}(U)$, it is also surjective. □

Proposition (21.6.12). — *Let X be a Noetherian reduced prescheme. Let $(U_\lambda)_{\lambda \in L}$ be a decreasing filtered family of opens of X satisfying the following conditions:*

1° *If $Y_\lambda = X - U_\lambda$, then $\mathrm{codim}(Y_\lambda, X) \geqslant 2$ for all $\lambda \in L$.*
2° *For all $x \in \bigcap_{\lambda \in L} U_\lambda$, the ring $\mathscr{O}_{X,x}$ is factorial.*

Then there exist canonical isomorphisms

(21.6.12.1)
$$\varinjlim \mathrm{Div}(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X), \qquad \varinjlim \mathrm{Pic}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$$

such that, for every open V of X, the diagrams

(21.6.12.2)
$$\begin{tikzcd} \varinjlim \mathrm{Div}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Z}^1(X) \arrow[d] \\ \varinjlim \mathrm{Div}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Z}^1(V) \end{tikzcd} \qquad \begin{tikzcd} \varinjlim \mathrm{Pic}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Cl}(X) \arrow[d] \\ \varinjlim \mathrm{Pic}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Cl}(V) \end{tikzcd}$$

are commutative.

Proof. The hypothesis 1° on U_λ implies that for all $\lambda \in L$, we have $X^{(1)} \subset U_\lambda$ (5.1.3.1) and therefore the restriction homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U_\lambda)$ is bijective and consequently we have an isomorphism $\varinjlim \mathfrak{Z}^1(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X)$. The canonical homomorphisms $\mathrm{cyc} : \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(U_\lambda)$ thus define, by passage to the inductive limit, the first of the canonical homomorphisms (21.6.12.1), and the second is deduced by passage to quotients. Furthermore, it follows from condition 1° that the U_λ are schematically dense in X since X is reduced (11.10.4), and therefore every meromorphic function on X is entirely determined by its restriction to each U_λ; we deduce from this that in the isomorphism $\mathfrak{Z}^1(X) \xrightarrow{\sim} \mathfrak{Z}^1(U_\lambda)$, the image of $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is $\mathfrak{Z}^1_{\mathrm{princ}}(U_\lambda)$, and therefore we also have a canonical isomorphism $\varinjlim \mathfrak{Cl}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$. The second of the canonical homomorphisms (21.6.12.1) will therefore be an isomorphism if the first is, and it remains to prove this last point, the commutativity of the diagrams (21.6.12.2) being trivial.

Let us first show that the homomorphism $\varinjlim \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is *injective*. Set $T = \bigcap_\lambda U_\lambda$, and **IV-4 | 276**
note that the U_λ form a *fundamental system of neighbourhoods* of T. Indeed, for every open $V \supset T$, $X - V$ is a closed subspace of X, thus a Noetherian space whose every irreducible closed subspace admits a generic point; as the sets $(X - V) \cap (X - U_\lambda)$ are closed in $X - V$, they form an increasing filtered family and their union is $X - V$; our assertion follows from (**0**, 9.2.4). This being so, it will suffice to show that if $D \in \mathrm{Div}(U_\lambda)$ is such that $\mathrm{cyc}(D) = 0$ in $\mathfrak{Z}^1(U_\lambda)$, then there exists $\mu \geqslant \lambda$ such that $D|U_\mu = 0$. By virtue of what precedes, it will suffice to show that for all $x \in T$, $D_x = 0$; indeed, if this holds, there will be an open neighbourhood $W(x)$ of x in X such that $D|W(x) = 0$, and the union of the $W(x)$ for $x \in T$ contains a U_μ. Taking into account (21.4.6), we are thus reduced to the case where $X = \mathrm{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but by hypothesis $\mathscr{O}_{X,x}$ is factorial, thus integral and integrally closed, and $\mathrm{Spec}(\mathscr{O}_{X,x})$ is therefore normal; whence the conclusion follows from (21.6.9), (i).

To prove that the homomorphism $\varinjlim \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is bijective, it will suffice to prove that **IV-4 | 277**
for all $Z \in \mathfrak{Z}^1(\overline{X})$ and all $x \in T$, there is a neighbourhood $W(x)$ of x in X and a divisor $D^{(x)}$ on $W(x)$ such that $\mathrm{cyc}(D^{(x)}) = Z|W(x)$. Indeed, we can then cover the quasi-compact set T by a finite number of the $W(x_i)$, and by the first part of the proof, there will exist an index λ such that in each of the $W(x_i) \cap W(x_j) \cap U_\lambda$, the restrictions of $D^{(x_i)}$ and $D^{(x_j)}$ coincide; since we can moreover assume that U_λ is contained in the union of the $W(x_i)$, we see that the restrictions $D^{(x_i)}|(W(x_i) \cap U_\lambda)$ are the restrictions of a single divisor $D \in \mathrm{Div}(U_\lambda)$ which will be such that $\mathrm{cyc}(D) = Z|U_\lambda$. We are thus reduced again to the case where $X = \mathrm{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but as $\mathscr{O}_{X,x}$ is factorial, it is Noetherian, integral and integrally closed, and it suffices to apply (21.6.9), (ii). □

Corollary (21.6.13). — *Let* A *be a Noetherian local ring which is integral and integrally closed and of dimension* $\geqslant 2$*. Set* $X = \mathrm{Spec}(A)$*, and let* a *be the closed point of* X*,* $U = X - \{a\}$*. For* A *to be factorial, it is necessary and sufficient that* U *is locally factorial and that* $\mathrm{Pic}(U) = 0$*.*

Proof. Indeed, to say that A is factorial is equivalent to saying that $\mathfrak{Cl}(X) = 0$ (Bourbaki, *Alg. comm.*, chap. VII, § 1, nº 4, cor. du th. 2 et § 3, th. 1); it therefore suffices to use the existence of the second isomorphism (21.6.12.1), taking the family (U_λ) restricted to the single open U. □

Corollary (21.6.14). — *Let* A *be a Noetherian local ring of dimension* $\geqslant 2$, $X = \mathrm{Spec}(A)$*,* a *the closed point of* X*,* $U = X - \{a\}$*. The following conditions are equivalent:*

label=a) A *is factorial.*

lbbel=b) $\mathrm{Pic}(U) = 0$ *and* $\mathrm{prof}(A) \geqslant 2$ *(conditions which we will express later* (21.13) *by saying that the ring* A *is* parafactorial*), and* U *is locally factorial.*

Proof. It is clear that if A is factorial, it is normal, whence $\dim(A) \geqslant 2$ (**0**, 16.5.1) and (21.6.13) shows that *a)* implies *b)*. Conversely, if *b)* is satisfied, it suffices to see that A is integrally closed to conclude by (21.6.13) that *b)* implies *a)*. By virtue of the Serre criterion (5.8.6), it suffices to verify for X the conditions (R_1) and (S_2). Now, U being locally factorial satisfies these conditions, and the hypothesis $\mathrm{prof}(A) \geqslant 2$ implies that X also satisfies them. □

21.7. Interpretation of positive 1-codimensional cycles in terms of sub-preschemes

(21.7.1). Let X be a locally Noetherian prescheme, $C = \sum_{x \in X^{(1)}} n_x . \overline{\{x\}}$ a positive 1-codimensional cycle (so that $n_x \geqslant 0$ for all $x \in X^{(1)}$, and $n_x = 0$ except at a locally finite set of points). Let Y(C) denote the closed sub-prescheme of X, *closed image* (**I**, 9.5.3) and (9.5.1) of the sub-prescheme *sum* $Y'(C) = \coprod_{x \in X^{(1)}} \mathrm{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n_x})$ by the canonical morphism, and let $\mathscr{J}_X(C)$ (or $\mathscr{J}(C)$) denote the Ideal of $\mathscr{O}_X$ defining Y(C).

Proposition (21.7.2). — *Let* X *be a locally Noetherian prescheme satisfying the condition* (R_1) (5.8.2)*. For a closed sub-prescheme* Y *of* X *to be of the form* Y(C)*, where* C *is a positive* 1*-codimensional cycle, it is necessary and sufficient that it satisfies the two following conditions:*

label=(i) Y *is purely of codimension* 1*.* **IV-4** | 278

lbbel=(ii) Y *satisfies the condition* (S_1)*, in other words* (5.7.5) *has no associated prime cycle.*

We then have

$$\mathrm{mult}_x(C) = \mathrm{length}\ \mathscr{O}_{Y,x}. \tag{21.7.2.1}$$

The map $C \mapsto Y(C)$ *is a bijection from* $\mathfrak{Z}^{1+}(X)$ *onto the set of closed sub-preschemes of* X *satisfying conditions (i) and (ii).*

Proof. The conditions are necessary (without assuming that X satisfies (R_1)). Indeed, since the question is local on X, we can assume that $X = \mathrm{Spec}(A)$, where A is a Noetherian ring; we then have $Y(C) = \mathrm{Spec}(A/\mathfrak{J})$, where $\mathfrak{J}$ is, by definition (**I**, 9.5.1) the kernel of the canonical homomorphism $A \to \bigoplus A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the prime ideals corresponding to points $x \in X^{(1)}$ for which $n_x \geqslant 0$, and we have set $n_i = n_x$. The inverse image $\mathfrak{q}_i$ of $\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$ in A is a $\mathfrak{p}_i$-primary ideal and we have $\mathfrak{J} = \bigcap_i \mathfrak{q}_i$. Moreover, as the $x \in X^{(1)}$ are such that $\overline{\{x\}}$ is of codimension 1, no point of $X^{(1)}$ can be adherent to another point of $X^{(1)}$, so the $\mathfrak{p}_i$ are minimal primes of $A/\mathfrak{J}$ and the set of $\mathfrak{p}_i$ is equal to $\mathrm{Ass}(A/\mathfrak{J})$, which proves conditions (i) and (ii).

The conditions are sufficient; let $\mathscr{J}$ denote the Ideal of $\mathscr{O}_X$ defining Y, and hypothesis (ii) implies that $\mathrm{Ass}(\mathscr{O}_X/\mathscr{J})$ is identical to the set F of maximal points of Y, and hypothesis (i) implies that F is contained in $X^{(1)}$; hence (3.2.6) $\mathscr{O}_X/\mathscr{J}$ is an $\mathscr{O}_X$-Module which is irredundant such that $\mathrm{Ass}(\mathscr{S}(x)) = \{x\}$. Now, since X satisfies (R_1), for each $x \in F$, $\mathscr{O}_{X,x}$ is a discrete valuation ring and so the only primary ideals are the powers $\mathfrak{m}_x^{n_x}$ of the maximal ideal, where the $\mathfrak{p}_i$ correspond to the points of F. We see then that Y is of the form Y(C). □

Corollary (21.7.3). — *Let* X *be a locally Noetherian prescheme such that* X *satisfies* (R_1)*. For every positive divisor* D *on* X*, the maximal points of the closed sub-prescheme* Y(D) (21.2.12) *majorise the closed sub-prescheme* Y(cyc(D)) (21.7.1)*, and this has the same underlying space; for these two sub-preschemes* *to be equal, it is necessary and sufficient that* $\mathrm{Y}(\mathrm{D})$ *satisfies the condition* (S_1), *or else, for every* $z \in \mathrm{Y}(\mathrm{D})$ *distinct from a maximal point of* $\mathrm{Y}(\mathrm{D})$, *that* $\operatorname{prof}(\mathscr{O}_{\mathrm{X},z}) \geqslant 2$ *(a condition always satisfied when* X *is* normal*).*

Proof. Indeed, the question is local, and for all $x \in \mathrm{T}$, $\mathrm{D}_x = 0$ implies the existence of an open neighbourhood U of x such that $\mathrm{D}|\mathrm{U} = 0$, whence $\operatorname{cyc}(\mathrm{D})|\mathrm{U} = 0$, which proves our assertion. We may suppose $\mathrm{X} = \operatorname{Spec}(\mathrm{A})$, and that t is a regular section of $\mathscr{O}_\mathrm{X}$ over X, such that $\mathscr{J}_\mathrm{X}(\mathrm{D}) = t\mathscr{O}_\mathrm{X}$, meaning that $\mathrm{Y}(\mathrm{D})$ satisfies (S_1), at every maximal point x of $\mathrm{Y}(\mathrm{D})$, $\mathscr{O}_{\mathrm{X},x}$ is by hypothesis a discrete valuation ring, therefore $t_x\mathscr{O}_{\mathrm{X},x} = \mathfrak{m}_x^{n_x}$, where $n_x = \operatorname{mult}_x(\mathrm{D})$. We can assume $\mathrm{X} = \operatorname{Spec}(\mathrm{A})$, and then, if $\mathscr{J}_\mathrm{X}(\operatorname{cyc}(\mathrm{D})) = \mathscr{J}'$, it follows from what precedes and from (21.7.1) that $\mathfrak{J}$ and $\mathfrak{J}'$ have the same *non-immersed primary ideals*, whence $\mathfrak{J}' \subset \mathfrak{J}$, since $\mathfrak{J}$ is the intersection of these primary ideals **IV-4** | 279 (21.7.2). This proves that $\mathrm{Y}(\mathrm{D})$ majorises $\mathrm{Y}(\operatorname{cyc}(\mathrm{D}))$ and that these two sub-preschemes are equal if and only if $\mathrm{Y}(\mathrm{D})$ has no associated immersed prime cycle. Since for all $x \in \mathrm{Y}(\mathrm{D})$, it is a condition on the local ring, and since by hypothesis $\operatorname{prof}(\mathscr{O}_{\mathrm{X},z}) \geqslant 2$ at all non-maximal points $z \in \mathrm{Y}(\mathrm{D})$, the condition (S_1) is then also satisfied. □

(21.7.3.1). We see therefore that when X is *normal*, $\operatorname{Div}^+(\mathrm{X})$ is identified canonically with the set of closed sub-preschemes Y of X satisfying conditions (i) and (ii) of (21.7.2) and *regularly immersed* in X.

Proposition (21.7.4). — *Let* X *be a locally Noetherian prescheme, reduced and with isolated points. The following conditions are equivalent:*

label=a) *The canonical homomorphism* $c : \mathscr{D}iv_\mathrm{X} \to \mathscr{Z}^1_\mathrm{X}$ (21.6.4) *is an isomorphism of sheaves of ordered groups, and we have* $\operatorname{Div}(\mathrm{X}) = \operatorname{Div.princ}(\mathrm{X})$.

a′) *The canonical homomorphism* $c : \mathscr{D}iv_\mathrm{X} \to \mathscr{Z}^1_\mathrm{X}$ *is injective, and every prime* 1*-codimensional cycle on* X *is the image by* cyc *of a positive principal divisor, and the canonical homomorphism* $c : \mathscr{D}iv_\mathrm{X} \to \mathscr{Z}^1_\mathrm{X}$ *is injective.*

a″) *For every integral closed sub-prescheme* Y *of* X*, of codimension* 1*, the canonical immersion* $\mathrm{Y} \to \mathrm{X}$ *is regular.*

lbbel=b) X *is normal and the homomorphism* $\operatorname{cyc} : \operatorname{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ *is bijective.*

lcbel=c) X *is locally factorial and* $\operatorname{Pic}(\mathrm{X}) = 0$.

ldbel=d) X *is normal, and for every open* U *of* X*, we have* $\operatorname{Pic}(\mathrm{U}) = 0$.

Proof. The equivalence of *a), a′), a″), b)* and *c)* follows at once from (21.7.4) and from (21.6.11). Furthermore, it follows at once that every non-empty open U of X also satisfies these conditions, in other words *c)* implies *d)*. It remains to show that *d)* implies *b)*. Now, let us denote by U_λ the open complements of finite unions of sets of the form $\overline{\{x\}}$, where $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$; it is clear that the U_λ form a decreasing filtered family and that for all $x \in \bigcap_\lambda \mathrm{U}_\lambda$, we have $\dim(\mathscr{O}_{\mathrm{X},x}) \leqslant 1$, so $\mathscr{O}_{\mathrm{X},x}$ is a principal ring and therefore one can apply to the family (U_λ) the proposition (21.6.12), which shows that $\mathfrak{Cl}(\mathrm{X})$ is isomorphic to $\varinjlim \operatorname{Pic}(\mathrm{U}_\lambda)$, hence $\mathfrak{Cl}(\mathrm{X}) = 0$ by virtue of the hypothesis, which proves *b)* by definition. □

Remark (21.7.5). — We cannot replace in (21.7.4) the condition *a′)* by the weaker condition that every prime 1-codimensional cycle on X is the image by cyc of a positive divisor. This is shown by the example where X is the affine scheme defined in (14.1.5) ("the union of a plane and a line that intersects it") *which is not normal*; with the notation introduced in (14.1.5), the prime 1-codimensional cycles of X are those of the plane X_2 and the line X_3; if t_1, t_2, t_3 are the images of $\mathrm{T}_1, \mathrm{T}_2, \mathrm{T}_3$ in $\mathrm{A}/\mathfrak{a}$, the 1-codimensional prime ideals of X are defined by the monogenic prime ideals in $\mathrm{A}/\mathfrak{a}$ (P irreducible in $\mathrm{K}[\mathrm{T}_1, \mathrm{T}_2]$) and $t_3 - a$, with $a \neq 0$ in K; these elements are indeed regular elements in $\mathrm{A}/\mathfrak{a}$, so every 1-codimensional cycle is indeed the canonical image of a positive divisor.

Corollary (21.7.6). — *Let* X *be a locally Noetherian prescheme, reduced and with isolated points. The following conditions are equivalent:*

label=a) *The canonical homomorphism* $c : \mathscr{D}iv_\mathrm{X} \to \mathscr{Z}^1_\mathrm{X}$ *is an isomorphism of sheaves of ordered groups, and we have* $\operatorname{Div}(\mathrm{X}) = \operatorname{Div.princ}(\mathrm{X})$.

lbbel=b) X *is normal and every* 1*-codimensional cycle on* X *is principal.*

lcbel=c) X *is locally factorial and* $\operatorname{Pic}(\mathrm{X}) = 0$.

ldbel=d) X *is normal, and for every open* U *of* X*, we have* $\operatorname{Pic}(\mathrm{U}) = 0$.

IV-4 | 280

Remark (21.7.7). — When $X = \mathrm{Spec}(A)$, where A is a Noetherian integral ring, the conditions of (21.7.6) are equivalent to saying that A is a *factorial* ring.

21.8. Divisors and normalization

Lemma (21.8.1). — *Let $f : X' \to X$ be an integral morphism of preschemes. For every $\mathscr{O}_X$-Module locally free $\mathscr{E}'$ of constant rank n and every $x \in X$ such that, if we set $U' = f^{-1}(U)$, $\mathscr{E}'|U'$ is isomorphic to $\mathscr{O}_{U'}^n$.*

Proof. The question being local on X, we can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, in which case $X' = \mathrm{Spec}(A')$, where A' is an A-algebra that is integral. Then A' is the inductive limit of its *finite* sub-A-algebras A'_λ. Setting $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ and denoting by $p_\lambda : X' \to X'_\lambda$ the structural morphism, it follows from (8.5.2) and (8.5.5) that there exists an index λ and an $\mathscr{O}_{X'_\lambda}$-Module locally free $\mathscr{E}'_\lambda$ of rank n such that $\mathscr{E}' = p_\lambda^*(\mathscr{E}'_\lambda)$, and it will clearly suffice to prove the lemma for X'_λ and $\mathscr{E}'_\lambda$; in other words, we may assume that the morphism f is *finite*. Set $B = \mathscr{O}_{X,x}$ and let B' be the ring of the affine scheme $X'_0 = X' \times_X \mathrm{Spec}(\mathscr{O}_{X,x})$; as B is a local ring and B' is a B-algebra which is finite, B' is a semi-local ring (Bourbaki, *Alg. comm.*, chap. V, § 2, nº 1, prop. 3); we conclude that the $\mathscr{O}_{X'_0}$-Module locally free $\mathscr{E}'_0 = \mathscr{E}' \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{X,x}$ is isomorphic to $\mathscr{O}_{X'_0}^n$ (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 3, prop. 5). Considering X'_0 as the projective limit of the preschemes induced by X' on the opens $f^{-1}(U)$, where U ranges over the set of affine open neighbourhoods of x in X, following the method of (8.1.2), *a)*), and applying (8.5.2.5), we obtain the sought conclusion. □ **IV-4 | 281**

Corollary (21.8.2). — *Let $f : X' \to X$ be an integral morphism of preschemes. Then:*

label=(i) *We have* $R^1 f_*(\mathscr{O}_{X'}^*) = 0$ (**0**, 12.2.1).
lbbel=(ii) *The group* $\mathrm{Pic}(X')$ *is canonically isomorphic to* $H^1(X, f_*(\mathscr{O}_{X'}^*))$.

Proof. label=(i) (i) is the sheaf associated to the presheaf of commutative groups $U \mapsto H^1(f^{-1}(U), \mathscr{O}_{X'}^*)$ on X (*loc. cit.*); the fibre of this sheaf at a point x can be identified with the commutative group $\varinjlim \mathrm{Pic}(f^{-1}(U))$, where U ranges over the open neighbourhoods of x in X, and for every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, for every open neighbourhood U' of x in X, and every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, the transition homomorphisms $\mathrm{Pic}(f^{-1}(U')) \to \mathrm{Pic}(f^{-1}(U))$ being defined by (21.3.2.4). Now, for all $x \in X$, every open neighbourhood U' of x in X, and every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, it follows from (21.8.1) that there is an open neighbourhood $U \subset U'$ of x such that the image of ζ in $\mathrm{Pic}(f^{-1}(U))$ at the point x is zero. Therefore the fibre of $R^1 f_*(\mathscr{O}_{X'}^*)$ at the point x is zero.

lbbel=(ii) (ii) The Leray spectral sequence for the composed functor $\mathscr{F}' \mapsto \Gamma(X, f_*(\mathscr{F}'))$ of sheaves of commutative groups on X' (T, 2.4) gives the exact sequence of low degree

$$0 \longrightarrow H^1(X, f_*(\mathscr{O}_{X'}^*)) \longrightarrow H^1(X', \mathscr{O}_{X'}^*) \longrightarrow H^0(X, R^1 f_*(\mathscr{O}_{X'}^*))$$

and the conclusion follows from (i) and from the isomorphism of $\mathrm{Pic}(X')$ and $H^1(X', \mathscr{O}_{X'}^*)$ (**0**, 5.4.7).

□

Proposition (21.8.3). — *Let $f = (\psi, \theta) : X' \to X$ be an integral morphism of preschemes; suppose that, for every open U of X, the homomorphism $\Gamma(\theta) : \Gamma(U, \mathscr{O}_X) \to \Gamma(U, f_*(\mathscr{O}_{X'}))$ sends regular elements to regular elements, which allows us to canonically extend the homomorphism $\theta : \mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ to a homomorphism of sheaves of rings $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$, whence homomorphisms of sheaves of multiplicative groups $\theta^* : \mathscr{O}_X^* \to f_*(\mathscr{O}_{X'}^*)$ and $\theta'' : \mathscr{M}_X^* \to f_*(\mathscr{M}_{X'}^*)$, giving by passage to quotients a homomorphism $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$. We then have a commutative diagram*

(21.8.3.1)
$$\begin{tikzcd} 1 \arrow[r] & \mathscr{O}_X^* \arrow[r] \arrow[d, "\theta^*"] & \mathscr{M}_X^* \arrow[r] \arrow[d, "\theta''"] & \mathscr{D}iv_X \arrow[r] \arrow[d, "\theta'''"] & 0 \\ 1 \arrow[r] & f_*(\mathscr{O}_{X'}^*) \arrow[r] & f_*(\mathscr{M}_{X'}^*) \arrow[r] & f_*(\mathscr{D}iv_{X'}) \arrow[r] & 0 \end{tikzcd}$$

in which the two rows are exact.

Proof. The only point to verify is the exactness of the second row of the diagram, which follows from the application of the exact sequence of cohomology for the functor f_* to the exact sequence of sheaves of commutative groups on X'

$$1 \longrightarrow \mathscr{O}^*_{X'} \longrightarrow \mathscr{M}^*_{X'} \longrightarrow \mathscr{D}iv_{X'} \longrightarrow 0$$

and from (21.8.2). □

Corollary (21.8.4). — *If, in addition to the hypotheses of* (21.8.3)*, the homomorphism θ' is an isomorphism of sheaves of rings, then $\theta^* : \mathscr{O}^*_X \to f_*(\mathscr{O}^*_{X'})$ is injective, $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective,* $\mathrm{Ker}(\theta''')$ *is isomorphic to* $\mathrm{Coker}(\theta^*)$.

Proof. This is an immediate consequence of the snake diagram (Bourbaki, *Alg. comm.*, chap. I, § 2, n° 4, prop. 2) applied to the diagram (21.8.3.1). □

IV-4 | 282

Proposition (21.8.5). — *Let X, X' be two locally Noetherian preschemes, $f : X' \to X$ an integral morphism; suppose that there exists a schematically dense open U in X such that $f^{-1}(U)$ is schematically dense in X' (cf.* (20.3.5)*), and that the morphism $f^{-1}(U) \to U$, restriction of f, is an isomorphism. Then:*

label=(i) *The homomorphism $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$ is defined and bijective, the homomorphism*

$$\theta^* : \mathscr{O}^*_X \longrightarrow f_*(\mathscr{O}^*_{X'})$$

is injective, the homomorphism $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective, $\mathrm{Ker}(\theta''')$ *is isomorphic to $\mathscr{N} = \mathrm{Coker}(\theta^*)$ and the support of the sheaf of multiplicative groups $\mathscr{N}$ is contained in $S = X - U$.*

lbbel=(ii) *Suppose moreover that the set S is discrete and set, for brevity, $\mathscr{O}^*_X = f_*(\mathscr{O}_{X'})$. Then, on a commutative diagram*

(21.8.5.1)
$$\begin{tikzcd}
1 \arrow[r] & \prod_{s\in S} \mathscr{O}^{*\prime\prime}_{X,s}/\mathscr{O}^*_{X,s} \arrow[r, "j"] \arrow[d] & \mathrm{Div}(X) \arrow[r] \arrow[d, "k_X"] & \mathrm{Div}(X') \arrow[r] \arrow[d, "k_{X'}"] & 0 \\
1 \arrow[r] & (\prod_{s\in S} \mathscr{O}^{*\prime\prime}_{X,s}/\mathscr{O}^*_{X,s})/\operatorname{Im}\Gamma(X', \mathscr{O}^*_{X'}) \arrow[r] & \mathrm{Pic}(X) \arrow[r] & \mathrm{Pic}(X') \arrow[r] & 0
\end{tikzcd}$$

where the two rows are exact and where the left vertical arrow is the canonical homomorphism.

Proof. label=(i) The hypothesis implies that we have a canonical isomorphism $\mathscr{M}'_X \xrightarrow{\sim} f_*(\mathscr{M}_{X'})$ for the sheaves of germs of pseudo-functions (20.2.2), whence the assertion relative to θ', given the existence of the canonical isomorphisms $\mathscr{M}_X \xrightarrow{\sim} \mathscr{M}'_X$ and $\mathscr{M}_{X'} \xrightarrow{\sim} \mathscr{M}'_{X'}$ (20.2.11). The rest of assertion (i) follows from (21.8.4), except what concerns the support of $\mathscr{N}$, which follows directly from the hypothesis on U.

lbbel=(ii) (ii) Applying the exact sequence of cohomology to the two exact sequences of sheaves of commutative groups

(21.8.5.2)
$$1 \longrightarrow \mathscr{N} \longrightarrow \mathscr{D}iv_X \longrightarrow f_*(\mathscr{D}iv_{X'}) \longrightarrow 0$$

(21.8.5.3)
$$1 \longrightarrow \mathscr{O}^*_X \longrightarrow f_*(\mathscr{O}^*_{X'}) \longrightarrow \mathscr{N} \longrightarrow 1$$

we get respectively the exact sequences

$$1 \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{j} \mathrm{Div}(X) \longrightarrow \mathrm{Div}(X') \longrightarrow H^1(X, \mathscr{N})$$

$$\Gamma(X', \mathscr{O}^*_{X'}) \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{\partial} H^1(X, \mathscr{O}^*_X)$$

taking into account the canonical isomorphism $\mathrm{Pic}(X') \xrightarrow{\sim} H^1(X, f_*(\mathscr{O}^*_{X'}))$ (21.8.2). **IV-4 | 283**
Now, as the support of $\mathscr{N}$ is contained in the discrete set S, closed in X, we have $H^1(X, \mathscr{N}) = H^1(S, \mathscr{N}|S) = \prod_{s\in S} H^1(\{s\}, \mathscr{N}_s) = 0$ by definition of the cohomology (G, II, 4.9.2) and $H^1(S, \mathscr{N}|S) = \prod_{s\in S} H^1(\{s\}, \mathscr{N}_s) = 0$ by definition of the cohomology (G, II, 4.4). Similarly, we have $\Gamma(X, \mathscr{N}) = \prod_{s\in S} \mathscr{N}_s$ and $\mathscr{N}_s = \mathscr{O}^{*\prime\prime}_{X,s}/\mathscr{O}^*_{X,s}$. By the second exact sequence (21.8.5.3), the image of t by $\partial : \Gamma(X, \mathscr{N}) \to H^1(X, \mathscr{O}^*_X)$ comes from the cocycle obtained as follows: we describe a section t of $\mathscr{N}$ over X, and we describe a

covering of X formed of U and of opens U_i such that $U_i \cap S$ reduces to a single point s_i and that $U_i \cap U_j \subset U$ for $i \neq j$, then considering for each i a section t_i of $\mathscr{N}$ over U_i, which may be considered as a section of $\mathscr{M}^*_{X'}$ over $f^{-1}(U_i)$, thus also as a section of $\mathscr{M}^*_X$ over U_i, and finally considering a section u_i of $\mathscr{O}^*_{X'}$ over $f^{-1}(U_i)$, the germ $(t_i)_{s_i}$ comes from a section of $\mathscr{O}^*_{X'}$ over $f^{-1}(U_i)$; it canonically corresponds to this section a section $d_i \in \Gamma(U_i, \mathscr{D}iv_X)$ the restriction of u_i in $U \cap U_i$ to $(u_j|(U_i \cap U_j))(u_i|(U_i \cap U_j))^{-1}$ in $U_i \cap U_j$, whence the expression of $j'(t)$ and the commutativity of the diagram (21.8.5.1). □

Remark (21.8.6). — label=(i) The conditions of application of (21.8.5), (i) are in particular satisfied when X is a locally Noetherian prescheme which is *reduced* with only a finite number of irreducible components, X′ is *normalisied* and that X′ is also locally Noetherian ((**II**, 6.3.8)); the $\mathscr{O}_X$-Module $\mathscr{D}iv_X$ is then an *extension* of $f_*(\mathscr{D}iv_{X'})$ by the cokernel $\mathscr{N}$ of $\mathscr{O}^*_X \to f_*(\mathscr{O}^*_{X'})$, which we can consider as known. If moreover X is an algebraic curve (reduced) over a field k, we are in the conditions of application of (21.8.5), (ii).

lbbel=(ii) When the conditions of application of (21.8.5), (ii) are satisfied and in addition that the canonical homomorphism globally $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is *bijective*, we see that the kernels of the surjective homomorphisms $\mathrm{Div}(X) \to \mathrm{Div}(X')$ and $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ are *isomorphic*.

lcbel=(iii) When the conditions of application of (21.8.5), (ii) are satisfied, we see that the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ cannot be injective (in which case it is bijective) unless $\mathscr{O}^{*''}_{X,s} = \mathscr{O}^*_{X,s}$ for all $s \in S$. For an $s \in S$ such that $\mathscr{O}_{X,s}$ is *local* (which, since $X' = \mathrm{Spec}(\mathscr{O}^*_X)$ (**II**, 1.3), means that there exists no point $s' \in X'$ above s, i.e. the residue fields $\boldsymbol{k}(s)$ and $\boldsymbol{k}(s')$ are equal; moreover, if this condition is satisfied, it is furthermore needed that the multiplicative group $1 + \mathfrak{m}_s$ has for image $1 + \mathfrak{m}_s$, or again $\mathfrak{m}_s = \mathfrak{r}$, which implies that $\mathscr{O}^{*''}_{X,s} \otimes_{\mathscr{O}_{X,s}} \boldsymbol{k}(s)$ is a composed direct of fields isomorphic to $\boldsymbol{k}(s)$ (which, when the morphism f is *finite*, implies in the general case, by (**0**, 23.2.6), the canonical homomorphism $\mathscr{O}^{*''}_{X,s} \to \mathscr{O}^*_{X,s}$ defines, by passage to quotients, a homomorphism of the multiplicative group $(\boldsymbol{k}(s))^*$ into the product of the multiplicative groups $(\boldsymbol{k}(s'_j))^*$, the s'_j being the points (in number > 1) of X′ above s. It is immediate that this homomorphism cannot be bijective if $\boldsymbol{k}(s)$ and all the $\boldsymbol{k}(s'_j)$ are equal to the field $\mathbf{F}_2$ with 2 elements; moreover, if this condition is satisfied, it is furthermore needed that the multiplicative group $1 + \mathfrak{m}_s$ has for image $1 + \mathfrak{m}_s$, in other words that $\mathfrak{m}_s = \mathfrak{r}$, which implies that $\mathscr{O}^{*''}_{X,s}$ is *non ramified* at the points s'_j (17.4.1). If for example *no* residue field of X is isomorphic to $\mathbf{F}_2$, the canonical homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ cannot be bijective if f is an *isomorphism*. In the case where the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is bijective, we conclude from this and from (ii) that in the preceding considerations, one can replace the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ by the homomorphism $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$. **IV-4** | 284

21.9. Divisors on preschemes of dimension 1

(21.9.1). Let X be a topological space, x a point of X, $i_x : \{x\} \to X$ the canonical injection. If $A(x)$ is a commutative group, we can consider the commutative group on the space $\{x\}$ reduced to a single point, and take its direct image sheaf of commutative groups on X (**0**, 3.4.1); it follows at once from the definitions that for every open U of X, $\Gamma(U, (i_x)_*(A(x))) = A(x)$ if $x \in U$, and is reduced to 0 in the contrary case, and for $y \in \overline{\{x\}}$, $((i_x)_*(A(x)))_y = A(x)$, and for $y \notin \overline{\{x\}}$, $((i_x)_*(A(x)))_y = 0$.

If now $\mathscr{F}$ is a sheaf of commutative groups on X and Y a part of X, for every open U of X, we have a canonical homomorphism

$$\Gamma(U, \mathscr{F}) \longrightarrow \prod_{x \in U \cap Y} \mathscr{F}_x = \prod_{x \in Y} \Gamma(U, (i_x)_*(\mathscr{F}_x)) \tag{21.9.1.1}$$

and since these homomorphisms commute with the restriction operators, they define a canonical homomorphism of *sheaves*

$$j_Y : \mathscr{F} \longrightarrow \prod_{x \in Y} (i_x)_*(\mathscr{F}_x). \tag{21.9.1.2}$$

Lemma (21.9.2). — *Let* X *be a locally Noetherian topological space,* X_0 *the set of its closed points,* $\mathscr{F}$ *a sheaf of commutative groups on* X*. The following conditions are equivalent:*

label=a) *The canonical homomorphism* $j_{X_0} : \mathscr{F} \to \prod_{x \in X_0}(i_x)_*(\mathscr{F}_x)$ *is injective and has for image* $\bigoplus_{x\in X_0}(i_x)_*(\mathscr{F}_x)$*.*

a′) *There exists a family of commutative groups* $(A(x))_{x\in X_0}$ *such that* $\mathscr{F}$ *is isomorphic to* $\bigoplus_{x\in X_0}(i_x)_*(A(x))$*.*

lbbel=b) *Every section of* $\mathscr{F}$ *over an open* U *has a discrete support contained in* X_0*.*

When this is the case, for all $x \in X_0$*, the group* $A(x)$ *is determined to a unique isomorphism close to* $\mathscr{F}_x$*. Furthermore, the sheaf* $\mathscr{F}$ *is then flasque.*

Proof. As the points of X_0 are closed in X, we have $((i_x)_*(A(x)))_y = 0$ for $y \neq x \in X_0$; this remark **IV-4** | 285
proves the uniqueness assertion relative to the groups $A(x)$, and it is clear moreover that *a)* implies *a′)*. To see that *a′)* implies *b)*, we may restrict to the case where U is Noetherian; then (G, II, 3.10) we have

$$\Gamma(U, \bigoplus_{x\in X_0}(i_x)_*(A(x))) = \bigoplus_{x\in X_0}\Gamma(U,(i_x)_*(A(x))),$$

and thus every section s of $\mathscr{F}$ over U is a direct sum of a *finite* number of sections $s_k \in \Gamma(U,(i_{x_k})_*(A(x_k)))$ $(1 \leqslant k \leqslant n)$ with $x_k \in X_0$. Since x_k is closed in X, the support of s_k is reduced to $\{x_k\}$, so the support of s is contained in the finite set of the x_k, which is evidently discrete. Let us show that *b)* implies *a)*; for every open Noetherian U, every support of a section of $\mathscr{F}$ over U is *finite*. We deduce that for every open U which is discrete and quasi-compact, the image of the homomorphism (21.9.1.1) for $Y = X_0$ is contained in $\bigoplus_{x\in X_0}\Gamma(U,(i_x)_*(\mathscr{F}_x))$ and that this homomorphism is injective, which establishes the hypothesis *b)*.

Finally, to show that $\mathscr{F}$ is flasque, consider a section s of $\mathscr{F}$ over an open U of X; as the support of s is a discrete and closed subspace in X, we extend s to a section s' of $\mathscr{F}$ over X by setting $s'_x = 0$ in $X - U$. □

Remark (21.9.3). — label=(i) In the condition *b)* of (21.9.2), it does not suffice to assume that the support of every section of $\mathscr{F}$ over any open of X is *discrete*; this is shown by the example where we take for X the spectrum of a discrete valuation ring, with a closed point b and a generic point a, and for $\mathscr{F}$ the sheaf of commutative groups such that $\mathscr{F}_b = 0$ and $\mathscr{F}_a = \mathbf{Z}$.

lbbel=(ii) The conditions of (21.9.2) being supposed satisfied, let E be a discrete part of X_0, and for each $x \in E$, let $a(x)$ be an element of $A(x)$. There then exists a unique section t of X such that $t_x = a(x)$ for all $x \in E$ and that the support of t is contained in E. Indeed, for every Noetherian open U, $E \cap U$ is finite, and it suffices to take for t the section $\bigoplus_{x\in E}(i_x)_*(A(x))$ whose restriction to each Noetherian open U is the sum of the $a(x)$ for $x \in E \cap U$.

Proposition (21.9.4). — *Let* X *be a locally Noetherian prescheme,* $X^{(1)}$ *the set of points* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) \leqslant 1$*. We then have a canonical isomorphism*

$$\mathscr{D}iv_X \xrightarrow{\sim} \bigoplus_{x\in X^{(1)}}(i_x)_*(\mathrm{Div}(\mathscr{O}_{X,x})) \qquad (21.9.4.1)$$

and $\mathscr{D}iv_X$ *is flasque.*

Proof. Taking into account the isomorphism (21.4.6.1), the homomorphism (21.9.4.1) is defined by (21.9.1.1): let us prove that it is a bijection. As $\dim(X) \leqslant 1$, the points of $X^{(1)}$ are the *non-isolated closed points* of X. We are thus reduced to proving that: 1° $\mathscr{D}iv_X$ satisfies the condition *b)* of (21.9.2); 2° for every isolated point $x \in X$, we have $(\mathscr{D}iv_X)_x = 0$. The second point follows from the fact that $\mathscr{O}_{X,x}$ is an Artinian ring and that every regular element of $\mathscr{O}_{X,x}$ is therefore invertible. For the 1°, it **IV-4** | 286
suffices to note that for every open U of X and every divisor $D \in \mathrm{Div}(U)$, the maximal points of the support S of D are such that $\mathrm{prof}(\mathscr{O}_{X,x}) \leqslant 1$ (for S is equal to S, and S is therefore discrete, the set of irreducible components of S being locally finite. □

Corollary (21.9.5). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$*. For every discrete part* $E \subset X^{(1)}$ *and every family* $(D(x))_{x\in E}$ *with* $D(x) \in \mathrm{Div}(\mathscr{O}_{X,x})$*, there exists a unique divisor* D *on* X *such that the support of* D *is contained in* E *and that* $D_x = D(x)$ *for all* $x \in E$*.*

Proof. This follows from (21.9.4) and from (21.9.3), (ii). □

Corollary (21.9.6). — *Let X be a locally Noetherian prescheme of dimension $\leqslant 1$. Every divisor D on X is of the form $D' - D''$, where D', D'' are two divisors $\geqslant 0$ with supports contained in that of D.*

Proof. By virtue of (21.9.5), it is at once reduced to the case where $X = \mathrm{Spec}(A)$, where A is a Noetherian local ring; it then suffices to observe that $M(X)$ is the total ring of fractions of A, and that a section of $\mathscr{M}_X^*$ over X is written by definition as the quotient b/a of two regular elements of A. □

Corollary (21.9.7). — *Let X be a locally Noetherian prescheme of dimension $\leqslant 1$, without isolated point, and U a dense open in X. There then exists a divisor $D \geqslant 0$ on X, with support contained in U and meeting each of the irreducible components of X.*

Proof. We can assume that U is the union of disjoint non-empty opens U_α, each of which is contained in a single irreducible component of X; it suffices then to take in each U_α a point x_α closed in X (there exists one since the unique generic point of U_α cannot be isolated by hypothesis), and to apply (21.9.5) to the discrete set of the x_α. □

The interest of this corollary is that it will allow us to prove that a separated algebraic curve X over a field k is quasi-projective, the divisor $D \geqslant 0$ defined in (21.9.7) being then necessarily *ample* by virtue of the Riemann–Roch theorem for curves (chap. V).

(21.9.8). For locally Noetherian preschemes X of dimension $\leqslant 1$ the proposition (21.9.4) brings back the determination of $\mathscr{D}iv_X$ to the case where $X = \mathrm{Spec}(A)$, A being a Noetherian local ring of dimension 1.

When A is a regular local ring of dimension 1 (in other words, a discrete valuation ring), the group $\mathrm{Div}(A)$ is canonically identified with $\mathbf{Z}$ (21.6.8); by (21.9.2):

Proposition (21.9.9). — *Let X be a locally Noetherian regular prescheme of dimension $\leqslant 1$. Then the sheaf $\mathscr{D}iv_X$ is canonically isomorphic to $\bigoplus_{x \in X^{(1)}} (i_x)_*(\mathbf{Z}(x))$, where $\mathbf{Z}(x)$ is the additive group $\mathbf{Z}$ considered as a sheaf of groups on the space $\{x\}$.*

(21.9.10). Suppose only that the Noetherian local ring A of dimension 1 is *reduced*; then, if A' is the *normalised* of A (the integral closure of A in its total ring of fractions), A' is Noetherian by virtue of the Krull–Akizuki theorem (Bourbaki, *Alg. comm.*, chap. VII, § 2, n° 5, prop. 5), and we have seen in (21.8.6) how $\mathrm{Div}(A)$ can be obtained as an *extension* of $\mathrm{Div}(A')$ and this last group is of the form $\mathbf{Z}'$. **IV-4** | 287

Proposition (21.9.11). — *Let X be a Noetherian prescheme, X_0 a closed sub-prescheme of X having the following properties:*

1° $\dim(X_0) \leqslant 1$.

2° *For every locally closed part Y of X such that $Y \cap X_0$ is discrete, there exists a part Y' of Y, closed in X and open in Y, containing $Y \cap X_0$.*

Under these conditions:

label=(i) *Let Z_0 be the union of the sets $\overline{\{x\}} \cap X_0$ when x ranges over the part of $\mathrm{Ass}(\mathscr{O}_X)$ formed of points such that $\overline{\{x\}} \cap X_0$ is finite. Then, for every divisor D_0 on X_0, whose support does not meet Z_0, there exists a divisor D on X whose inverse image by the injection $X_0 \to X$ exists* (21.4) *and is equal to D_0; if moreover $D_0 \geqslant 0$, we can assume $D \geqslant 0$.*

lbbel=(ii) *Suppose in addition that there exists in X_0 an affine open U_0 containing $(\mathrm{Ass}(\mathscr{O}_{X_0})) \cup Z_0$. Then the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ *is surjective.*

Proof. label=(i) Taking into account (21.9.6), we can restrict to showing the assertion relative to divisors $D_0 \geqslant 0$; the support T of D_0 is a finite discrete and closed set in X_0, that is $T \neq \varnothing$, there is nothing to show. For all $x \in T$, there exists an element $s_x \in \mathscr{O}_{X_0,x}$ which is not a divisor of zero and whose image in $(\mathscr{D}iv_X)_x$ is $(D_0)_x$. There exists an open affine neighbourhood $U^{(x)}$ of x in X and a section $g^{(x)}$ of $\mathscr{O}_X$ over $U^{(x)}$ such that s_x is the image of the germ $(g^{(x)})_x \in \mathscr{O}_{X,x}$; taking $U^{(x)}$ small enough, we can arrange that $g^{(x)}$ is not a zero divisor in $\Gamma(U^{(x)}, \mathscr{O}_X)$, in other words (3.1.9), writing $V^{(x)}$ for the closed part of X defined by $g^{(x)} = 0$, that $V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X) = \varnothing$. Now, if $z \in \mathrm{Ass}(\mathscr{O}_X)$, the hypothesis $x \notin \overline{\{z\}}$, that is $x \notin \overline{\{z\}} \cap X_0$, either x is not isolated in $\overline{\{z\}} \cap X_0$. Now, if y is a generization of x such that $\overline{\{y\}}$ is of codimension 1 in X, the sub-prescheme reduced Y of X having

$\overline{\{y\}}$ for underlying space, $X_0 \cap Y$ being by hypothesis a finite closed part of X and open in Y, containing $Y \cap X_0$, the condition (S_1) is then also satisfied.

We can define a divisor $D'^{(x)} = \mathrm{div}(g^{(x)})$ on $U^{(x)}$ by $D'^{(x)} = \mathrm{div}(g^{(x)})$, and $D'^{(x)}|(X - V^{(x)}) = 0$, which has a meaning since $U^{(x)} \cap (X - V^{(x)})$ is the complement of the zeros of $g^{(x)}$ in $U^{(x)}$; by the hypothesis $x \notin Z_0$ we have $x \notin \overline{\{z\}}$; as $V^{(x)} \cap X_0$ is reduced to $\{x\}$, $V^{(x)}$ is closed in X. On can then define a divisor $D = \sum_{x \in T} D'^{(x)}$, which has a meaning since T is finite.

IV-4 | 288

lbbel=(ii) Taking into account the commutative diagram (21.4.2.1), the conclusion will follow if we prove that every $\mathscr{O}_{X_0}$-Module invertible $\mathscr{L}_0$ is isomorphic to an $\mathscr{O}_{X_0}$-Module of the form $\mathscr{O}_{X_0}(D_0)$ (21.2.8), where D_0 is a divisor on X_0 whose support does not meet Z_0. Now, as U_0 is schematically dense in X_0 (20.2.13), (iv), it suffices to prove that there exists a section $t \in \Gamma(U_0, \mathscr{L}_0)$ such that $t(z_0) \neq 0$ at the points of $(\mathrm{Ass}(\mathscr{O}_{X_0})) \cup Z_0$, the conclusion results from ((**II**, 5.1.4)) and, as U_0 is affine, $\mathscr{L}_0$ is ample (**II**, 4.5.4), the conclusion results from (**II**, 5.3.1) and (**II**, 5.5.3).

□

Corollary (21.9.12). — *Let* A *be a Henselian local ring* (18.5.8), $S = \mathrm{Spec}(A)$, s_0 *the closed point of* S, $f : X \to S$ *a separated morphism of finite presentation, and suppose that the set* $X_0 = f^{-1}(s_0)$ *is of dimension* $\leqslant 1$. *Then, for every closed sub-scheme* X'_0 *of* X, *having* X_0 *as underlying set and of finite presentation over* S, *the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(X) \to \mathrm{Pic}(X'_0)$ *is surjective.*

Proof. Let us first show that it suffices to prove the corollary when the local ring A is moreover Noetherian. Indeed (18.6.15) we can write $A = \varinjlim A_\lambda$, where the A_λ are local Noetherian and Henselian rings, the homomorphisms $A_\lambda \to A$ being local; there exists in addition an index α and a separated morphism of finite presentation $f_\alpha : X_\alpha \to S_\alpha = \mathrm{Spec}(A_\alpha)$ such that X and f are deduced from X_α and f_α by base change $S \to S_\alpha$ (8.10.5), (v); with the usual notation (8.8.1), the morphisms $f_\lambda : X_\lambda \to S_\lambda$ will be separated for $\lambda \geqslant \alpha$ and we will have $X = X_\lambda \times_{S_\lambda} S$ (8.6.3); since $\dim(X_0) \leqslant 1$ by hypothesis and (4.1.4), we thus have $\dim(X'_{0\lambda}) = \dim(X'_0) \leqslant 1$ by transitivity of the fibres and (4.1.4). It amounts to seeing that if we have proved that the homomorphism $\mathrm{Pic}(X_\lambda) \to \mathrm{Pic}(X'_{0\lambda})$ is surjective for all $\lambda \geqslant \alpha$, we have the same conclusion for X and X'_0. Moreover, we can assume that if $s_{0\alpha}$ is the unique closed point of S_α, there is a closed sub-scheme $X'_{0\alpha}$ of X_α, such that $X'_0 = X'_{0\alpha} \times_{S_\alpha} S$ (8.6.3); since $\dim(X'_0) \leqslant 1$ by hypothesis and $\dim(X'_{0\lambda}) = \dim(X'_0) \leqslant 1$ by transitivity of the fibres and (4.1.4). We evidently have the commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}(X_\lambda) & \longrightarrow & \mathrm{Pic}(X) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X'_{0\lambda}) & \longrightarrow & \mathrm{Pic}(X'_0) \end{array}$$

For every invertible $\mathscr{O}_{X'_0}$-Module $\mathscr{L}'_0$, there exists $\lambda \geqslant \alpha$ and an invertible $\mathscr{O}_{X'_{0\lambda}}$-Module $\mathscr{L}'_{0\lambda}$ such that $\mathscr{L}'_0$ is deduced by base change $S \to S_\lambda$ (8.5.2) and (8.5.5); it suffices then to take the $\mathscr{O}_X$-Module invertible $\mathscr{L}$ deduced from $\mathscr{L}_\lambda$ by base change $S \to S_\lambda$, and by virtue of the commutativity of the preceding diagram, $\mathrm{cl}(\mathscr{L}'_0)$ will be the image of $\mathrm{cl}(\mathscr{L})$.

Suppose therefore that A is Noetherian, and verify that X and X_0 satisfy the conditions of (21.9.11), (ii). Since by hypothesis $\dim(X'_0) \leqslant 1$; on the other hand, for verifying the condition 2° of (21.9.11), consider a sub-prescheme Y of X having Y as underlying set, the morphism $g : Y \to S$, restriction of f, being quasi-finite at each of the points x_i of $Y \cap X_0$ $(1 \leqslant i \leqslant n)$, we can apply (18.5.11), *c)*) and we see that Y is the *sum* of sub-preschemes *open* $Y_i = \mathrm{Spec}(\mathscr{O}_{Y,x_i})$ $(1 \leqslant i \leqslant n)$ and of a prescheme Y'' such that $Y'' \cap X_0 = \varnothing$, and in addition that the canonical injections $j_i : Y_i \to X$ are such that $f \circ j_i$ is a finite morphism, thus Y_i is *closed* in X; on the question by taking $Y' = \bigcup_{1 \leqslant i \leqslant n} Y_i$.

It remains to verify the supplementary hypothesis of (21.9.11), (ii). Now, X'_0 being a separated curve over the field $\boldsymbol{k}(s_0)$, is a $\boldsymbol{k}(s_0)$-scheme quasi-projective (chap. V) ([1]), thus there exists an $\mathscr{O}_{X'_0}$-Module (**II**, 5.3.1), and this achieves the proof. □

Remark (21.9.13). — Under the conditions of (21.9.12), supposing in addition f proper, the morphism f is even *projective*: indeed, if $\mathscr{L}_0$ is an $\mathscr{O}_{X'_0}$-Module ample, it exists, by virtue of (21.9.12) an $\mathscr{O}_X$-Module invertible $\mathscr{L}$ whose inverse image in X'_0 is isomorphic to $\mathscr{L}_0$, hence is ample and, as every neighbourhood of s_0 in S is necessarily the whole of S, we deduce then from (9.6.4) that $\mathscr{L}$ is an $\mathscr{O}_X$-Module ample, whence the conclusion ((**II**, 5.3.1) and (**II**, 5.5.3)).

21.10. Inverse images and direct images of 1-codimensional cycles

(21.10.1). Let X, X′ be two locally Noetherian preschemes, $f : X' \to X$ a morphism, T a closed rare subset of X contained in $f(X')$, and let Z be a 1-codimensional cycle on X with support contained in T. We propose to define, under suitable hypotheses on f, a 1-codimensional cycle $f^*(Z)$ on X′, the *inverse image* of Z by f. **IV-4 | 289**

Let us first suppose that $Z \mapsto f^*(Z)$ is a *homomorphism* of ordered groups from the subgroup of $\mathfrak{Z}^1(X)$ formed by cycles with support contained in T, into the ordered group $\mathfrak{Z}^1(X')$. For this, let

(21.10.1.1) $$Z = \sum_{x \in T \cap X^{(1)}} n_x . \overline{\{x\}}$$

where the family of the $x \in T \cap X^{(1)}$ such that $n_x \neq 0$ is locally finite. For all $x' \in X'^{(1)}$, we define an integer $n_{x'}$ in the following way, setting $x = f(x')$:

label=1° if $x \notin T$, we set $n_{x'} = 0$;

lbbel=2° if $x \in X^{(1)}$ and if $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, we know (6.1.1) that $\dim(\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}) = 0$, in other words $\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}$ is an $\mathscr{O}_{X',x'}$-module of finite length $\lambda_{x'}$, and we set $n_{x'} = \lambda_{x'} n_x$;

lcbel=3° if $\mathscr{O}_{X,x}$ is factorial and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, we know (21.6.9) that the canonical homomorphism cyc : $\mathrm{Div}(\mathscr{O}_{X,x}) \to \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x}))$ is bijective, and on the other hand since $\dim(\mathscr{O}_{X',x'}) = 1$ and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, $\mathrm{Ass}(\mathscr{O}_{X',x'})$ consists solely of the maximal points of $\mathrm{Spec}(\mathscr{O}_{X',x'})$, so the hypothesis on f implies that $f(\mathrm{Ass}(\mathscr{O}_{X',x'})) \subset \mathrm{Ass}(\mathscr{O}_{X,x})$ and it follows from (21.4.5), (ii) that the homomorphism $f^* : \mathrm{Div}(\mathscr{O}_{X,x}) \to \mathrm{Div}(\mathscr{O}_{X',x'})$ is defined; finally, x' being the unique closed point of $\mathrm{Spec}(\mathscr{O}_{X',x'})$, $\mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'}))$ is canonically isomorphic to **Z**.

We have thus defined a canonical composite homomorphism

(21.10.1.2) $$\mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x})) \xleftarrow{\mathrm{cyc}^{-1}} \mathrm{Div}(\mathscr{O}_{X,x}) \xrightarrow{f'} \mathrm{Div}(\mathscr{O}_{X',x'}) \xrightarrow{\mathrm{cyc}^0} \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'})) \xrightarrow{\sim} \mathbf{Z}$$

If Z_x is the cycle $\sum_{y \in \mathrm{Spec}(\mathscr{O}_{X,x}) \cap X^{(1)}} n_y . (\overline{\{y\}} \cap \mathrm{Spec}(\mathscr{O}_{X,x}))$ on $\mathrm{Spec}(\mathscr{O}_{X,x})$, we take for $n_{x'}$ the *image* of Z_x by the homomorphism (21.10.1.2).

IV-4 | 290

(21.10.2). We propose to show that:

label=*A)* When two of the conditions 1°, 2°, 3° of (21.10.1) are *simultaneously* satisfied, the corresponding values of $n_{x'}$ *coincide*.

lbbel=*B)* The set of $x' \in X'^{(1)}$ such that $n_{x'} \neq 0$ is *locally finite* in X′.

To prove A), let us first suppose that $x \notin T$ and that x satisfies one of the conditions 2° or 3° of (21.10.1); then $n_x = 0$ and if one is in case 2°, we have $n_{x'} = 0$; if one is in case 3°, we have $Z_x = 0$ since $\mathrm{Supp}(Z) \subset T$, so again $n_{x'} = 0$. It remains to consider the case where one is simultaneously in cases 2° and 3°; then, since $\dim(\mathscr{O}_{X,x}) = 1$, $\mathscr{O}_{X,x}$ is a discrete valuation ring; if t is a uniformiser of this ring, the divisor corresponding to Z_x is $\mathrm{div}(t^{n_x})$ in $\mathrm{Div}(\mathscr{O}_{X,x})$, and its image in $\mathrm{Div}(\mathscr{O}_{X',x'})$ is the divisor of t'^{n_x}, where t' is the image of t. One can evidently restrict to the case where $n_x = 1$, and then by the definition (21.6.5.1) the image of Z_x by (21.10.1.1) is the number $\lambda_{x'}$, which finishes the proof of A).

Let us now prove B). Set $T_0 = \mathrm{Supp}(Z) \subset T$, $T'_0 = f^{-1}(T_0)$; it suffices to prove that the relation $n_{x'} \neq 0$ implies that x' belongs to the set of maximal points of T'_0, this last being locally finite in X′. It is immediate that if x' was not maximal in the closed set T'_0, there would exist a generisation y' of x' in T'_0, distinct from x', and since $x' \in X'^{(1)}$, y' would necessarily be a maximal point of X′; as a **IV-4 | 291** result $y = f(y')$ would be a maximal point of X by hypothesis; but this is absurd since $y \in T_0$ and T_0 is purely of codimension 1 in X.

(21.10.3). We can now set

$$f^*(Z) = \sum_{x' \in X'^{(1)}} n_{x'} \overline{\{x'\}},$$

the sum of the second member having a meaning by virtue of what we proved in (21.10.2); we say that the 1-codimensional cycle $f^*(Z)$ is the *inverse image* of Z by f. It is immediate that the map f^* thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, V an open of X′ such that $f(V) \subset U$ and $f' : V \to U$ the restriction of f, it follows immediately from the definitions that one has

(21.10.3.1) $$f'^*(Z|U) = f^*(Z)|V.$$

Let us denote by $\Gamma_T(\mathscr{Z}^1_X)$ the largest sub-sheaf of commutative groups of $\mathscr{Z}^1_X$ with support contained in T; it follows from the relation (21.10.3.1) that the maps $\Gamma(U, \Gamma_T(\mathscr{Z}^1_X)) \to \Gamma(V, \mathscr{Z}^1_{X'})$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X′

$$\psi^*(\Gamma_T(\mathscr{Z}^1_X)) \longrightarrow \mathscr{Z}^1_{X'}$$

where ψ is the continuous map underlying the morphism f.

Proposition (21.10.4). — *Suppose satisfied the conditions of* (21.10.1)*. Then, for every divisor* D *on* X *such that* Supp(D) ⊂ T *and such that* $f^*(D)$ *is defined* (21.4.2)*, we have*

(21.10.4.1) $$\mathrm{cyc}(f^*(D)) = f^*(\mathrm{cyc}(D)).$$

Proof. The question being local on X, we can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, where $D = \mathrm{div}(t)$, where t is a regular non-invertible element of A, and where the sub-prescheme Y(D) (21.2.12) has only a single maximal point y, so that $\mathrm{cyc}(D) = n_y.\overline{\{y\}}$, where n_y is the length of $\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}$ (21.6.5.1). We saw in the proof of (21.10.2), B) that the points $x' \in X'$ such that $\mathrm{mult}_{x'}(f^*(\mathrm{cyc}(D))) \neq 0$ are maximal points of $f^{-1}(\overline{\{y\}})$. If the point x' is in case 3° of (21.10.1), the equality of the multiplicities in x' of the two members of (21.10.4.1) follows from the definition of $n_{x'}$ by means of the homomorphism (21.10.1.1). Suppose on the contrary that x' is in case 2° of (21.10.1), and let $x = f(x')$; since $x \in X^{(1)} \cap \overline{\{y\}}$, we have necessarily $x = y$. Note now that $f^*(D) = \mathrm{div}(t')$, where t' is the image of t in $\Gamma(X', \mathscr{O}_{X'})$, and $Y(f^*(D)) = f^{-1}(Y(D))$; the multiplicity of $f^*(D)$ at the point x' is therefore the length $n_{x'}$ of the $\mathscr{O}_{X',x'}$-module $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'} = (\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}) \otimes_{\mathscr{O}_{X,y}} \mathscr{O}_{X',x'}$; it follows from (4.7.1) that $n_{x'} = \lambda_{x'} n_y$, so the multiplicities at the point x' of the two members of (21.10.4.1) are again equal, which finishes the proof. □

(21.10.5). Suppose now that $f : X' \to X$ is a morphism of locally Noetherian preschemes, transforming every maximal point of X′ into a maximal point of X, and suppose furthermore that for every *closed rare* subset T of X, f satisfies the conditions of (21.10.1); this means that for all $x' \in X'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) of (21.10.1). Noting that every 1-codimensional cycle with rare support in X is defined for *every* 1-codimensional cycle Z on X; by virtue of (21.10.3.1), we have thus defined a homomorphism of sheaves of ordered commutative groups **IV-4 | 292**

$$\psi^*(\mathscr{Z}^1_X) \longrightarrow \mathscr{Z}^1_{X'}.$$

If furthermore, for every divisor D on X, $f^*(D)$ is defined (21.4.5), the fact that the support of D is rare in X (21.6.6) implies that (21.10.4) is applicable, and we therefore have the formula (21.10.4.1) for every divisor D on X. In particular:

Proposition (21.10.6). — *Let* X, X′ *be two locally Noetherian preschemes,* $f : X' \to X$ *a flat morphism. Then* $f^*(Z)$ *is defined for every* 1*-codimensional cycle* Z *on* X*,* $f^*(D)$ *is defined for every divisor* D *on* X *and we have the relation* (21.10.4.1)*.*

Proof. Indeed, if $x' \in X'^{(1)}$ is such that $x = f(x')$ is not maximal, it follows from (6.1.1) that we have necessarily $x \in X^{(1)}$, so one is in case (ii) of (21.10.1). We can therefore apply (21.10.5), taking into account (21.4.5) and (2.3.4). □

Remark (21.10.7). — The existence of $f^*(\mathrm{Z})$ for every 1-codimensional cycle Z on X already follows from the hypothesis that f is *flat at all points x' of* X$'$ *of codimension* $\leqslant 1$ (i.e. such that $\dim \mathscr{O}_{\mathrm{X}',x'} \leqslant 1$); indeed, for every maximal $z' \in \mathrm{X}'$, it follows from (6.1.1) that $z = f(z')$ is a maximal point of X since $\dim \mathscr{O}_{\mathrm{X},z} \leqslant \dim \mathscr{O}_{\mathrm{X}',z'} = 0$. Moreover, if $x' \in \mathrm{X}'^{(1)}$, $x = f(x')$ is maximal or belongs to $\mathrm{X}^{(1)}$ by (6.1.1); one can again apply (21.10.5).

Proposition (21.10.8). — *Let* $\mathrm{X}, \mathrm{X}', \mathrm{X}''$ *be three locally Noetherian preschemes,* $f : \mathrm{X}' \to \mathrm{X}$, $g : \mathrm{X}'' \to \mathrm{X}'$ *two morphisms; suppose that f (resp. g) is flat at every point of codimension $\leqslant 1$ in* X$'$ *(resp.* X$''$*). Then $f \circ g$ is flat at every point of codimension $\leqslant 1$ in* X$''$ *and for every* 1*-codimensional cycle* Z *on* X, *we have* $(f \circ g)^*(\mathrm{Z}) = g^*(f^*(\mathrm{Z}))$.

Proof. The first assertion follows from (6.1.1) and (2.1.6). The second follows from the fact that f (resp. g, resp. $f \circ g$) transforms maximal points of X (resp. X$'$, resp. X$''$) into maximal points of X (resp. X, resp. X) and from the fact that, if $x'' = g(x'') \in \mathrm{X}''^{(1)}$ and $x = f(x') \in \mathrm{X}^{(1)}$ is such that $x' = g(x'') \in \mathrm{X}'^{(1)}$, we have

$$\mathrm{length}(\mathscr{O}_{\mathrm{X}'',x''}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}'',x''}) = \mathrm{length}(\mathscr{O}_{\mathrm{X}'',x''}/\mathfrak{m}_{x'}\mathscr{O}_{\mathrm{X}'',x''}) \cdot \mathrm{length}(\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'})$$

Indeed, if we set $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$, $\mathrm{A}' = \mathscr{O}_{\mathrm{X}',x'}$, $\mathrm{A}'' = \mathscr{O}_{\mathrm{X}'',x''}$, $\mathfrak{m} = \mathfrak{m}_x$, $\mathfrak{m}' = \mathfrak{m}_{x'}$, we have $\mathrm{A}''/\mathfrak{m}\mathrm{A}'' = (\mathrm{A}'/\mathfrak{m}\mathrm{A}') \otimes_{\mathrm{A}'} \mathrm{A}''$, and the formula

$$\mathrm{length}_{\mathrm{A}''}(\mathrm{A}''/\mathfrak{m}\mathrm{A}'') = \mathrm{length}_{\mathrm{A}'}(\mathrm{A}'/\mathfrak{m}\mathrm{A}') \cdot \mathrm{length}_{\mathrm{A}''}(\mathrm{A}''/\mathfrak{m}'\mathrm{A}'')$$

follows from (4.7.1) and the hypothesis of flatness of A$''$ over A$'$. □

(21.10.9). Let X be a locally Noetherian prescheme, Λ a commutative group, written additively, and considered as a $\mathbf{Z}$-module; we will again denote by $\mathbf{Z}$ and Λ the simple sheaves on X associated to the constant presheaves respectively equal to $\mathbf{Z}$ and Λ ($\mathbf{0}$, 3.6). We call *sheaf of germs of* 1*-codimensional cycles with coefficients in* Λ the sheaf of commutative groups $\mathscr{Z}^1_{\mathrm{X}} \otimes_{\mathbf{Z}} \Lambda$. If $\Lambda = \mathbf{Q}$, we say that the sections of $\mathscr{Z}^1_{\mathrm{X}} \otimes_{\mathbf{Z}} \mathbf{Q}$ over X are the 1*-codimensional cycles with rational coefficients*. Since the fibres of **IV-4 | 293**
$\mathscr{Z}^1_{\mathrm{X}}$ are torsion-free $\mathbf{Z}$-modules (21.6.3), the canonical homomorphism $\mathscr{Z}^1_{\mathrm{X}} \to \mathscr{Z}^1_{\mathrm{X}} \otimes_{\mathbf{Z}} \mathbf{Q}$ is *injective*, so that the 1-codimensional cycles are identified with 1-codimensional cycles with rational coefficients.

(21.10.10). We will now see that under certain conditions, we can enlarge the definition of $f^*(\mathrm{Z})$ given in (21.10.3) for a 1-codimensional cycle Z on X, but to take for $f^*(\mathrm{Z})$ a 1-codimensional cycle *with rational coefficients* on X$'$. The most general case where we place ourselves is the one where f transforms every maximal point of X$'$ into a maximal point of X, and where, at every point $x' \in \mathrm{X}'^{(1)}$, we have one of the conditions (i), (ii), (iii) of (21.10.1) *or a fourth condition* (setting $x = f(x')$):

(iv) $x \in \mathrm{X}^{(1)}$, $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X},x})$ *and furthermore, if we set* $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$, $\mathrm{A}' = \widehat{\mathscr{O}}_{\mathrm{X}',x'}$, *and if* $\mathrm{K} = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ *denotes the total ring of fractions of* A, *then* A$'$ *is a finite* A*-algebra and* $\mathrm{K}' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ *is a free* K*-module.*

Let us note then $r_{x'}$ the rank of the free K-module K$'$, $q_{x'}$ the degree of $\boldsymbol{k}(x')$ over $\boldsymbol{k}(x) = \boldsymbol{k}(\mathrm{A})$, and set $\mu_{x'} = r_{x'}/q_{x'}$. For a 1-codimensional cycle with support contained in T, given by (21.10.1.1) and for all $x' \in \mathrm{X}'^{(1)}$, we define $c_{x'} \in \mathbf{Q}$ to equal $n_{x'}$ when one is in one of the cases 1°, 2°, 3° of (21.10.1); but there remains here a fourth possibility:

4° if x' satisfies condition (iv) above, we set $c_{x'} = \mu_{x'} n_x \in \mathbf{Q}$.

(21.10.11). We must still prove that when condition 4° of (21.10.10) is satisfied *at the same time* as one **IV-4 | 294**
of the conditions 1°, 2°, 3° of (21.10.1), we have $n_{x'} = c_{x'}$. This is evident if $x \notin \mathrm{T}$ since then $n_x = 0$. To study the other two cases, let us note that $\mathfrak{m}_{x'}\mathscr{O}_{\mathrm{X}',x'}$ is closed for the $\mathfrak{m}_{x'}$-preadic topology, so the completion of $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ for this last topology is $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$. If one is simultaneously in cases 2° and 4°, $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ is discrete for the $\mathfrak{m}_{x'}$-preadic topology, so isomorphic to $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$. Since A$'$ is a finite A-algebra and flat ($\mathbf{0}_{\mathrm{III}}$, 10.2.3), it is a free A-module (*ibid.*, 10.1.3), and the rank of $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ over $\mathrm{A}/\mathfrak{m}_x\mathrm{A} = \boldsymbol{k}(x)$ is equal to the rank of A$'$ over A, hence also to the rank of K$'$ over K on A, hence also $\lambda_{x'}$ of the $\mathscr{O}_{\mathrm{X}',x'}$-module $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ (or of $\boldsymbol{k}(x')$-module) by the rank $[\boldsymbol{k}(x') : \boldsymbol{k}(x)] = q_{x'}$, which proves the relation $\mu_{x'} = r_{x'}/q_{x'} = \lambda_{x'}$ in this case.

Suppose finally that one is simultaneously in cases 3° and 4°. Then, since $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$, $\mathscr{O}_{\mathrm{X},x}$ is a discrete valuation ring, hence regular. On the other hand, $\mathscr{O}_{\mathrm{X}',x'}$ is of dimension 1, and since $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X}',x'})$, $\mathscr{O}_{\mathrm{X}',x'}$ is a Cohen-Macaulay ring ($\mathbf{0}$, 16.4.6); finally, since A$'$ is an A-module of finite type, $A'/\mathfrak{m}_x A'$ is a $\boldsymbol{k}(x)$-vector space of finite rank; since $\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}$ is contained in $A'/\mathfrak{m}_x A'$, it is also a $\boldsymbol{k}(x)$-vector space of finite rank, hence an Artinian ring. The application of (6.1.5) then shows that $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, so one is also in case 2°, and we conclude by what was seen above.

This being the case, in the case we have considered, we will set

$$f^*(Z) = \sum_{x' \in X'^{(1)}} c_{x'}.\overline{\{x'\}}$$

which is thus a 1-codimensional cycle on X' with *rational* coefficients. We have thus again defined a homomorphism f^* of ordered groups, satisfying (21.10.3.1), and hence a homomorphism of sheaves of commutative ordered groups

$$\psi^*(\mathscr{Z}^1_X) \longrightarrow \mathscr{Z}^1_{X'} \otimes_{\mathbf{Z}} \mathbf{Q}$$

whence, by tensorisation, a homomorphism of sheaves of $\mathbf{Q}$-vector spaces ordered

(21.10.11.1) $$\psi^*(\mathscr{Z}^1_X \otimes_{\mathbf{Z}} \mathbf{Q}) \longrightarrow \mathscr{Z}^1_{X'} \otimes_{\mathbf{Z}} \mathbf{Q}.$$

When f satisfies the preceding conditions for every *closed rare* subset T in X, i.e. that for all $x' \in X'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) or (iv), $f^*(Z)$ is then defined for *every* 1-codimensional cycle Z on X, and we have defined a homomorphism of sheaves of commutative ordered groups

$$\psi^*(\mathscr{Z}^1_X) \longrightarrow \mathscr{Z}^1_{X'} \otimes_{\mathbf{Z}} \mathbf{Q}$$

Remark (21.10.12). — When one is in the situation of (21.10.10), one can effectively, for 1-codimensional cycles Z on X, have for $f^*(Z)$ cycles 1-codimensional with *non-integer* coefficients; in other words, the numbers $\mu_{x'}$ can be non-integers. We have an example by taking the integral ring and its integral closure A' of (**0**, 6.15.11, (ii)) and (21.10.10), (iv) of $\operatorname{Spec}(A')$ and one has $\mu_{x'} = \frac{1}{2}$.

Lemma (21.10.13). — *Let* A *be a local Noetherian ring of dimension* 1*,* t *a regular element of* A *belonging to the maximal ideal* $\mathfrak{m}$ *(which implies that* $\mathfrak{m} \notin \operatorname{Ass}(A)$*).*

label=(i) *For every A-module* M *of finite type, the kernel* $N_t(M)$ *and the cokernel* $P_t(M)$ *of the homothety* $t_M : M \to M$ *of ratio* t *are of finite length. We set* $d_t(M) = \operatorname{length} P_t(M) - \operatorname{length} N_t(M)$*.*

lbbel=(ii) *If* $0 \to M' \to M \to M'' \to 0$ *is an exact sequence of A-modules of finite type, we have* $d_t(M) = d_t(M') + d_t(M'')$*.*

lcbel=(iii) *We have* $d_t(M) \geqslant 0$ *for every A-module* M *of finite type; for* $d_t(M) = 0$*, it is necessary and sufficient that* M *be of finite length.*

ldbel=(iv) *If* K *is the total ring of fractions of* A *and if* $M \otimes_A K$ *is a free* K*-module of rank* n*, we have* $d_t(M) = n \cdot d_t(A) = n \cdot \operatorname{length}(A/tA)$*.*

lebel=(v) *If* M *satisfies the hypotheses of (iv) and if furthermore* t *is* M*-regular, we have* $\operatorname{length}(M/tM) = n \cdot \operatorname{length}(A/tA)$*.*

Proof. (i) $\operatorname{Spec}(A)$ is formed of the point $\mathfrak{m}$ and the minimal prime ideals $\mathfrak{p}_i$; since by hypothesis $t \notin \mathfrak{p}_i$ for all i (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 1, cor. 3 of prop. 2), the image of t in each of the $A_{\mathfrak{p}_i}$ is invertible, and the supports of the A-modules of finite type $N_t(M)$ and $P_t(M)$ are therefore empty or reduced to $\mathfrak{m}$; we conclude from this (**0**, 16.1.10) that these modules are of finite length. **IV-4 | 295**

(ii) Since t is regular, we have an exact sequence

$$0 \longrightarrow A \xrightarrow{t_A} A \longrightarrow A/tA \longrightarrow 0$$

and since $\operatorname{Tor}_i^A(M, A) = 0$ for $i \geqslant 1$, the Tor exact sequence gives on one hand the exact sequence

$$0 \longrightarrow \operatorname{Tor}_1^A(M, A/tA) \longrightarrow M \xrightarrow{t_M} M \longrightarrow M/tM \longrightarrow 0$$

and on the other hand, for $i \geqslant 2$,

$$0 = \operatorname{Tor}_i^A(M, A) \longrightarrow \operatorname{Tor}_i^A(M, A/tA) \longrightarrow \operatorname{Tor}_{i-1}^A(M, A) = 0$$

hence $N_t(M) = \operatorname{Tor}_1^A(M, A/tA)$ and $\operatorname{Tor}_i^A(M, A/tA) = 0$ for $i \geqslant 2$; the Tor exact sequence gives the exact sequence

$$0 \longrightarrow \operatorname{Tor}_1^A(M', A/tA) \longrightarrow \operatorname{Tor}_1^A(M, A/tA) \longrightarrow \operatorname{Tor}_1^A(M'', A/tA) \longrightarrow$$

$$\mathrm{M}' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M}'' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow 0$$

and this sequence can be written, in view of what precedes

$$0 \longrightarrow \mathrm{N}_t(\mathrm{M}') \longrightarrow \mathrm{N}_t(\mathrm{M}) \longrightarrow \mathrm{N}_t(\mathrm{M}'') \longrightarrow \mathrm{P}_t(\mathrm{M}') \longrightarrow \mathrm{P}_t(\mathrm{M}) \longrightarrow \mathrm{P}_t(\mathrm{M}'') \longrightarrow 0$$

which proves (ii).

(iii) To prove (iii), we note that for M there is a composition series (M_h) of M such that the quotients $\mathrm{M}_h/\mathrm{M}_{h+1}$ are isomorphic to $\mathrm{A}/\mathfrak{m}$ or to one of the $\mathrm{A}/\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, th. 1), and that for M to be of finite length, it is necessary and sufficient that all these quotients be isomorphic to $\mathrm{A}/\mathfrak{m}$. It all comes down therefore (by virtue of (ii)) to proving that $d_t(\mathrm{A}/\mathfrak{m}) = 0$ and $d_t(\mathrm{A}/\mathfrak{p}_i) > 0$. Now, the image of t in $\mathrm{A}/\mathfrak{m}$ being 0, we have $\mathrm{N}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$, whence the first assertion; on the other hand, the image of t in $\mathrm{A}/\mathfrak{p}_i$ is regular, so $\mathrm{N}_t(\mathrm{A}/\mathfrak{p}_i) = 0$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{p}_i) = \mathrm{A}/(t\mathrm{A} + \mathfrak{p}_i)$ which is not reduced to 0, whence the second assertion.

(iv) There is a basis of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ of the form $(x_j/s)_{1 \leqslant j \leqslant n}$, where s is a regular element of A and $x_j \in \mathrm{M}$. Consider the homomorphism $u : \mathrm{A}^n \to \mathrm{M}$ which transforms the element e_j $(1 \leqslant j \leqslant n)$ of the canonical basis of A^n into x_j and let us show that $\mathrm{Ker}(u)$ and $\mathrm{Coker}(u)$ are of *finite length*; indeed, for each i, the image of s in $\mathrm{A}_{\mathfrak{p}_i}$ is invertible, and since $\mathrm{K}_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_i}$, the images of the x_j/s in $\mathrm{M}_{\mathfrak{p}_i}$ form a basis of this $\mathrm{A}_{\mathfrak{p}_i}$-module; hence $u_{\mathfrak{p}_i} : \mathrm{A}^n_{\mathfrak{p}_i} \to \mathrm{M}_{\mathfrak{p}_i}$ is bijective. As in (i), we conclude that the supports of $\mathrm{Ker}(u)$ and of $\mathrm{Coker}(u)$ are contained in $\{\mathfrak{m}\}$, so these modules are of finite length. This being so, it follows from (ii) and (iii) that $d_t(\mathrm{M}) = d_t(\mathrm{A}^n) = n \cdot d_t(\mathrm{A}) = n \cdot \mathrm{length}(\mathrm{A}/t\mathrm{A})$.

Finally, it is clear that (v) follows immediately from (iv), since then $\mathrm{N}_t(\mathrm{M}) = 0$. □

This lemma allows us to generalise (21.10.4):

Proposition (21.10.13*). — *Suppose that f satisfies the conditions of (21.10.10). Then, for every divisor* D *on* X *such that* $\mathrm{Supp}(\mathrm{D}) \subset \mathrm{T}$ *and such that* $f^*(\mathrm{D})$ *is defined, we have*

(21.10.13.1) $$\mathrm{cyc}(f^*(\mathrm{D})) = f^*(\mathrm{cyc}(\mathrm{D})).$$

Proof. Reasoning as in (21.10.4), it all amounts to seeing (with the same notation) that if x' is in case 4° of (21.10.10) and if n_y is the length of $\mathscr{O}_{\mathrm{X},y}/t_y\mathscr{O}_{\mathrm{X},y}$, $\mu_{x'}$ being the rational number defined in (21.10.10), then the length $n_{x'}$ of $\mathscr{O}_{\mathrm{X}',x'}/t'_{x'}\mathscr{O}_{\mathrm{X}',x'}$ is equal to $\mu_{x'}n_y$. Since $\mathscr{O}_{\mathrm{X}',x'}/t'_{x'}\mathscr{O}_{\mathrm{X}',x'}$ is of finite length, it has the same length as its $\mathfrak{m}_{x'}$-preadic completion $\mathrm{A}'/t'_{x'}\mathrm{A}'$, which can also be written $\mathrm{A}'/t_y\mathrm{A}'$; moreover, since $t'_{x'}$ is regular by hypothesis in $\mathscr{O}_{\mathrm{X}',x'}$, it is also regular in A′ by flatness ($\mathbf{0}_{\mathrm{I}}$, 6.3.4), and when A′ is considered as an A-module, we can also say that t_y is A′-regular. Since A is of dimension 1 and A′ is an A-module of finite type such that $\mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ is a free K-module of rank $r_{x'}$, we can apply (21.10.13), (v) to $\mathrm{M} = \mathrm{A}'$ and to t_y, and the length of $\mathrm{A}'/t_y\mathrm{A}'$ as an A′-module is therefore $r_{x'}n_y$. Since $\boldsymbol{k}(x')$ is a $\boldsymbol{k}(x)$-vector space of rank $q_{x'}$, the length of $\mathrm{A}'/t_y\mathrm{A}'$ as an A′-*module* is therefore $r_{x'}n_y/q_{x'} = \mu_{x'}n_y$, which finishes the proof. □ **IV-4 | 296**

(21.10.14). Let now X, X′ be two locally Noetherian preschemes, $f : \mathrm{X}' \to \mathrm{X}$ a morphism having the *two* following properties:

label=*a)* f is finite;
lbbel=*b)* the image by f of every maximal point of X′ is a maximal point of X.

For all $x \in \mathrm{X}^{(1)}$, the points $x' \in f^{-1}(x)$ all belong to $\mathrm{X}'^{(1)}$, as a consequence of hypothesis *b)* and of the inequality ($\mathbf{0}$, 16.3.9.1), the fibre $f^{-1}(x)$ being discrete. Let then

$$Z' = \sum_{x' \in \mathrm{X}'^{(1)}} n_{x'}.\overline{\{x'\}}$$

be a 1-codimensional cycle on X′. For all $x \in \mathrm{X}^{(1)}$, we define an integer n_x by the formula

$$n_x = \sum_{x' \in f^{-1}(x)} n_{x'} \cdot [\boldsymbol{k}(x') : \boldsymbol{k}(x)]$$

which has a meaning, the points of $f^{-1}(x)$ being in finite number and $\boldsymbol{k}(x')$ being a field of finite degree over $\boldsymbol{k}(x)$ (**I**, 6.4.4). Furthermore, the set of $x \in \mathrm{X}^{(1)}$ such that $n_x \neq 0$ is *locally finite* in X, since it is contained in the image by f of the set of $x' \in \mathrm{X}'^{(1)}$ such that $n_{x'} \neq 0$, and the conclusion follows from the fact that the morphism f is quasi-compact. We can therefore define a 1-codimensional cycle on X by setting

$$f_*(\mathrm{Z}') = \sum_{x \in \mathrm{X}^{(1)}} n_x.\overline{\{x\}} \tag{21.10.14.1}$$

and we say that $f_*(\mathrm{Z}')$ is the *direct image* of Z' by f. It is clear that the map $f_* : \mathfrak{Z}^1(\mathrm{X}') \to \mathfrak{Z}^1(\mathrm{X})$ thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, $f_{\mathrm{U}} : f^{-1}(\mathrm{U}) \to \mathrm{U}$ the restriction of f, it follows immediately from the definitions that we have

$$(f_{\mathrm{U}})_*(\mathrm{Z}'|f^{-1}(\mathrm{U})) = f_*(\mathrm{Z}')|\mathrm{U} \tag{21.10.14.2}$$

hence, denoting by ψ the continuous map underlying the morphism f, the maps $\Gamma(f^{-1}(\mathrm{U}), \mathscr{Z}^1_{\mathrm{X}'}) \to \Gamma(\mathrm{U}, \mathscr{Z}^1_{\mathrm{X}})$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X

$$\psi_*(\mathscr{Z}^1_{\mathrm{X}'}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}}.$$

Proposition (21.10.15). — *Let* $\mathrm{X}, \mathrm{X}', \mathrm{X}''$ *be three locally Noetherian preschemes,* $f : \mathrm{X}' \to \mathrm{X}$, $g : \mathrm{X}'' \to \mathrm{X}'$ *two morphisms satisfying conditions* a) *and* b) *of (21.10.14). Then* $f \circ g$ *satisfies the same conditions, and for every* 1*-codimensional cycle* Z'' *on* X''*, we have* $(f \circ g)_*(\mathrm{Z}'') = f_*(g_*(\mathrm{Z}''))$. **IV-4 | 297**

Proof. This follows immediately from the definitions. □

Proposition (21.10.16). — *Let* $\mathrm{X}, \mathrm{X}', \mathrm{X}_1$ *be three locally Noetherian preschemes,* $f : \mathrm{X}' \to \mathrm{X}$ *a morphism satisfying conditions* a) *and* b) *of (21.10.14),* $g : \mathrm{X}_1 \to \mathrm{X}$ *a flat morphism. Set* $\mathrm{X}'_1 = \mathrm{X}' \times_{\mathrm{X}} \mathrm{X}_1$ *(so that* X'_1 *is locally Noetherian) and let* $f_1 : \mathrm{X}'_1 \to \mathrm{X}_1$ *and* $g_1 : \mathrm{X}'_1 \to \mathrm{X}'$ *be the canonical projections. Then* f_1 *satisfies conditions* a) *and* b) *of (21.10.14), and for every* 1*-codimensional cycle* Z' *on* X'*, we have*

$$g^*(f_*(\mathrm{Z}')) = (f_1)_*(g_1^*(\mathrm{Z}')). \tag{21.10.16.1}$$

Proof. It is clear that f_1 is finite, and it satisfies condition $b)$ of (21.10.14) by virtue of (2.3.7). To prove (21.10.16.1), we are immediately reduced to the case where X, X' and X_1 are spectra of local Noetherian rings of dimension 1, A, A′ and A_1, A′ being a finite flat A-module. Denoting by x, x' and x_1 the closed points of X, X' and X_1 respectively, it is a matter of seeing that we have

$$\sum_{x'_1} \lambda_{x'_1}[\boldsymbol{k}(x'_1) : \boldsymbol{k}(x_1)] = \lambda_{x_1}[\boldsymbol{k}(x') : \boldsymbol{k}(x)] \tag{21.10.16.2}$$

where x'_1 ranges over the closed points of X'_1 (i.e. the points lying above both x' and x_1) and where $\lambda_{x'_1} = \mathrm{length}(\mathscr{O}_{\mathrm{X}_1,x'_1}/\mathfrak{m}_{x'}\mathscr{O}_{\mathrm{X}_1,x'_1})$ and $\lambda_{x_1} = \mathrm{length}(\mathrm{A}_1/\mathfrak{m}_x\mathrm{A}_1)$. Since we have

$$[\boldsymbol{k}(x') : \boldsymbol{k}(x)]\, \mathrm{length}_{\mathrm{A}'}(\mathrm{A}'_1/\mathfrak{m}_{x'}\mathrm{A}'_1) = \mathrm{length}_{\mathrm{A}_1}(\mathrm{A}'_1/\mathfrak{m}_{x'}\mathrm{A}'_1)$$

where we have set $\mathrm{A}'_1 = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{A}_1$. We have thus $\mathrm{A}'_1/\mathfrak{m}_{x'}\mathrm{A}'_1 = (\mathrm{A}'/\mathfrak{m}_{x'}\mathrm{A}') \otimes_{\mathrm{A}} \mathrm{A}_1$, and as A_1 is a flat A-module, by (4.7.1),

$$\mathrm{length}_{\mathrm{A}_1}(\mathrm{A}'_1/\mathfrak{m}_{x'}\mathrm{A}'_1) = \mathrm{length}_{\mathrm{A}_1}(\mathrm{A}_1/\mathfrak{m}_x\mathrm{A}_1)\, \mathrm{length}_{\mathrm{A}_1}(\mathrm{A}'_1/\mathfrak{m}\mathrm{A}'_1) = [\boldsymbol{k}(x') : \boldsymbol{k}(x)] \cdot \lambda_{x_1}$$

which finishes the proof. □

Proposition (21.10.17). — *Let* X, X' *be two locally Noetherian preschemes,* $f : \mathrm{X}' \to \mathrm{X}$ *a finite locally free morphism. Then* f *satisfies condition* b) *of (21.10.14), and for every divisor* D' *on* X'*, we have*

$$f_*(\mathrm{cyc}(\mathrm{D}')) = \mathrm{cyc}(f_*(\mathrm{D}')). \tag{21.10.17.1}$$

Proof. Since f is flat and finite, condition $b)$ of (21.10.14) and the relation $f(\mathrm{X}'^{(1)}) \subset \mathrm{X}^{(1)}$ follow from (6.1.1). The definition (21.10.14.1) shows that we can reduce to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A}) = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ for an $x \in \mathrm{X}^{(1)}$; then $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$, where A′ is a free A-module of finite rank; furthermore, we can suppose that $\mathrm{D}' = \mathrm{div}(t')$, where t' is a regular element of A′; we then have $f_*(\mathrm{D}') = \mathrm{div}(t)$, where $t = \mathrm{N}_{\mathrm{A}'/\mathrm{A}}(t')$ is a regular element of A (21.5.2). We can restrict to the case where t' is not invertible in A′, the ring $\mathrm{A}/t\mathrm{A}$ is then of finite length, and $\mathrm{A}'/t'\mathrm{A}'$ is a direct sum of local Artinian rings A'_i $(1 \leqslant i \leqslant r)$, the residue field of A'_i being $\boldsymbol{k}(x'_i)$, where x'_i $(1 \leqslant i \leqslant r)$ are the points of X′ lying above x. If t'_i $(1 \leqslant i \leqslant r)$ is the image of t' in A'_i, $\mathrm{A}'/t'\mathrm{A}'$ is a direct sum of the $\mathrm{A}'_i/t'_i\mathrm{A}'_i$; since the product $\mathrm{length}_{\mathrm{A}'_i}(\mathrm{A}'_i/t'_i\mathrm{A}'_i) \cdot [\boldsymbol{k}(x'_i) : \boldsymbol{k}(x)]$ is equal to $\mathrm{length}_{\mathrm{A}}(\mathrm{A}'_i/t'_i\mathrm{A}'_i)$, we see that the multiplicity at point x of the first member of (21.10.17.1) is $\operatorname{length}_{\mathrm{A}}(\mathrm{A}'/t'\mathrm{A}')$, so that the formula to prove reduces to

(21.10.17.2) $$\operatorname{length}_{\mathrm{A}}(\mathrm{A}'/t'\mathrm{A}') = \operatorname{length}_{\mathrm{A}}(\mathrm{A}/t\mathrm{A}).$$

This relation follows from the following more general lemma: □

Lemma (21.10.17.3). — *Let* A *be a local Noetherian ring of dimension* 1, M *a free A-module of finite rank, u an injective endomorphism of* M. *Then we have* **IV-4** | 298

(21.10.17.4) $$\operatorname{length}_{\mathrm{A}}(\operatorname{Coker} u) = \operatorname{length}_{\mathrm{A}}(\mathrm{A}/(\det u)\mathrm{A}).$$

Proof. We distinguish several cases.

I) A *is a discrete valuation ring*. Let π be a uniformiser of A, and note $\operatorname{length}_{\mathrm{A}}(\mathrm{A}/\pi^k\mathrm{A}) = k$; if n is the rank of M and if π^{m_i} ($1 \leqslant i \leqslant n$) are the invariant factors of u, $\operatorname{Coker}(u)$ is a direct sum of A-modules $\mathrm{A}/\pi^{m_i}\mathrm{A}$, so has length $m = \sum_{i=1}^n m_i$, and $\det(u)$ is the product of an invertible element and of t^m, whence the conclusion in this case (Bourbaki, *Alg.*, chap. VII, §4, nº 5, cor. 1 of prop. 4).

II) A *is a complete integral ring (of dimension* 1*)*. We know then (**0**, 19.8.8, (ii)) that there is a subring B of A, which is a discrete valuation ring, such that $\mathrm{B} \to \mathrm{A}$ is a local homomorphism making A a B-module of finite type; since this B-module is evidently torsion-free, it is *free* (Bourbaki, *Alg. comm.*, chap. VI, §3, nº 6, lemma 1). Let M′ denote the set M equipped with its structure of B-module (free), by u' the endomorphism u considered as a B-endomorphism. It follows from I) that

(21.10.17.5) $$\operatorname{length}_{\mathrm{B}}(\operatorname{Coker} u') = \operatorname{length}_{\mathrm{B}}(\mathrm{B}/(\det u')\mathrm{B}).$$

But $\det u' = \mathrm{N}_{\mathrm{A}/\mathrm{B}}(\det u)$, therefore, by applying (21.10.17.4) to the homothety $x \mapsto (\det u)x$ of the free B-module A, we get

$$\operatorname{length}_{\mathrm{B}}(\mathrm{B}/(\det u')\mathrm{B}) = \operatorname{length}_{\mathrm{B}}(\mathrm{A}/(\det u)\mathrm{A}),$$

whence, substituting in (21.10.17.5) and dividing by $[\boldsymbol{k}(\mathrm{A}) : \boldsymbol{k}(\mathrm{B})]$, the length of the B-module $\boldsymbol{k}(\mathrm{A})$, we obtain (21.10.17.4) in the case considered.

III) A *is a complete ring*. Let us note that we can further suppose that $\mathfrak{m} \notin \operatorname{Ass}(\mathrm{A})$; indeed, since $\det(u)$ is a regular element of A, if we had $\mathfrak{m} \in \operatorname{Ass}(\mathrm{A})$, then $\det(u)$ would be invertible, u an automorphism of M, and the formula (21.10.17.4) would then be trivial, both members being zero. **IV-4** | 299

In what follows, for an endomorphism v of a module N over a ring R, such that $\operatorname{Ker} v$ and $\operatorname{Coker} v$ are of *finite length*, we set

$$\chi(\mathrm{N}, v) = \operatorname{length}_{\mathrm{R}}(\operatorname{Ker}(v)) - \operatorname{length}_{\mathrm{R}}(\operatorname{Coker}(v)).$$

We note that the hypothesis on v amounts to saying that the complex

$$\mathrm{K}^0 : 0 \longrightarrow \mathrm{N} \xrightarrow{v} \mathrm{N} \longrightarrow 0$$

has its cohomology modules of finite length and that $\chi(\mathrm{N}, v) = \chi(\mathrm{H}^0(\mathrm{K}^0))$ ($\mathbf{0}_{\mathrm{III}}$, 11.10). We deduce that if N′, N, N″ are three R-modules, and if there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{N}' \arrow[r, "r"] \arrow[d, "v'"] & \mathrm{N} \arrow[r, "s"] \arrow[d, "v"] & \mathrm{N}'' \arrow[r] \arrow[d, "v''"] & 0 \\
0 \arrow[r] & \mathrm{N}' \arrow[r, "r"'] & \mathrm{N} \arrow[r, "s"'] & \mathrm{N}'' \arrow[r] & 0
\end{tikzcd}$$

whose rows are exact, and if two of the three numbers $\chi(\mathrm{N}, v)$, $\chi(\mathrm{N}', v')$ and $\chi(\mathrm{N}'', v'')$ are defined, then so is the third and one has

(21.10.17.6) $$\chi(\mathrm{N}, v) = \chi(\mathrm{N}', v') + \chi(\mathrm{N}'', v'').$$

This follows indeed immediately from the cohomology exact sequence.

Finally, if N is an R-module of *finite length*, we have $\chi(\mathrm{N}, v) = 0$.

With this notation, we have the following lemma: □

Lemma (21.10.17.7). — *Let* A *be a local Noetherian ring of dimension* 1 *whose maximal ideal* $\mathfrak{m}$ *is such that* $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$*; let* $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ *be the minimal prime ideals of* A*. Let* M *be a free* A*-module of finite type,* u *an endomorphism of* M *such that* $\chi(\mathrm{M}, u)$ *is defined; for each* i*, set* $\mathrm{M}_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{p}_i)$ *and let* u_i *be the endomorphism* $u \otimes 1_{\mathrm{A}/\mathfrak{p}_i}$ *of* M_i*; then if* $\chi(\mathrm{M}_i, u_i)$ *is defined for each* i*, we have*

(21.10.17.8)
$$\chi(\mathrm{M}, u) = \sum_{i=1}^{n} \mathrm{length}(\mathrm{A}_{\mathfrak{p}_i}) \cdot \chi(\mathrm{M}_i, u_i).$$

Proof. Since $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$, we have a reduced primary decomposition $(0) = \bigcap_{1 \leqslant i \leqslant n} \mathfrak{q}_i$, where the ideal $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary for $1 \leqslant i \leqslant n$. If we set $\mathrm{M}'_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{q}_i)$, we have therefore an exact sequence **IV-4 | 300**

$$0 \longrightarrow \mathrm{M} \longrightarrow \bigoplus_i \mathrm{M}'_i \longrightarrow \mathrm{M}'' \longrightarrow 0$$

of A-modules, where M'' is of *finite length*: indeed by localising the preceding exact sequence at each of the $\mathfrak{p}_i$, we obtain $\mathrm{M}''_{\mathfrak{p}_i} = 0$, since $(\mathfrak{q}_i)_{\mathfrak{p}_i} = 0$ and $(\mathfrak{q}_j)_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_j}$ for $j \neq i$ (Bourbaki, *Alg. comm.*, chap. IV, §2, n° 4, prop. 6), so $(\mathrm{M}'_i)_{\mathfrak{p}_i} = \mathrm{M}_{\mathfrak{p}_i}$ and $(\mathrm{M}'_j)_{\mathfrak{p}_i} = 0$; the support of M'' is therefore reduced to $\mathfrak{m}$, hence M'' is of finite length $(\mathbf{0}, 16.1.10)$. If we set $u'_i = u \otimes 1_{\mathrm{A}/\mathfrak{q}_i}$, and if u'' is the endomorphism of M'' deduced from $\bigoplus u'_i$ by passing to quotients, we deduce from (21.10.17.6) that $\chi(\mathrm{M}, u) + \chi(\mathrm{M}'', u'') = \sum_i \chi(\mathrm{M}'_i, u'_i)$, and since M'' is of finite length, $\chi(\mathrm{M}'', u'') = 0$. To prove (21.10.17.8), we can therefore reduce to the case where $n = 1$. We will then denote by $\mathfrak{p}$ the unique minimal prime ideal of A; if $\mathrm{A}_0 = \mathrm{A}/\mathfrak{p}$, $\mathrm{M}_0 = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_0$, $u_0 = u \otimes 1_{\mathrm{A}_0}$, it is a matter of proving that if $\chi(\mathrm{M}_0, u_0)$ is defined, then

(21.10.17.9)
$$\chi(\mathrm{M}, u) = \mathrm{length}\,\mathrm{A}_{\mathfrak{p}} \cdot \chi(\mathrm{M}_0, u_0).$$

Let $\mathfrak{n}_j$ $(0 \leqslant j \leqslant r)$ be the "j-th symbolic power" of $\mathfrak{p}$, the inverse image of $(\mathfrak{p}\mathrm{A}_{\mathfrak{p}})^j$ in A $(1 \leqslant j \leqslant r)$, with $\mathfrak{n}_0 = \mathrm{A}$ and $\mathfrak{n}_r = 0$; set

$$\mathrm{M}_j = \mathfrak{n}_j\mathrm{M}/\mathfrak{n}_{j+1}\mathrm{M} \qquad (0 \leqslant j \leqslant r-1),$$

and denote by v_j the endomorphism of M_j deduced from u by restriction to $\mathfrak{n}_j\mathrm{M}$ and passage to quotients. The first assertion will imply the second, by applying (21.10.17.6) to each of the exact sequences

$$0 \longrightarrow \mathfrak{n}_j\mathrm{M}/\mathfrak{n}_{j+1}\mathrm{M} \longrightarrow \mathrm{M}/\mathfrak{n}_{j+1}\mathrm{M} \longrightarrow \mathrm{M}/\mathfrak{n}_j\mathrm{M} \longrightarrow 0.$$

and that each of the numbers $\chi(\mathrm{M}_j, v_j)$ is defined and one has

(21.10.17.10)
$$\chi(\mathrm{M}, u) = \sum_j \chi(\mathrm{M}_j, v_j).$$

To prove the first assertion, we note that if m is the rank of the free A-module M, M_j is isomorphic to $(\mathfrak{n}_j/\mathfrak{n}_{j+1})^m$, or since $\mathfrak{p}$ annihilates each of the quotients (since $\mathfrak{p} \cdot \mathfrak{n}_j \subset \mathfrak{n}_{j+1}$), M_j is an A_0-module isomorphic to $\mathrm{M}_0 \otimes_{\mathrm{A}_0} (\mathfrak{n}_j/\mathfrak{n}_{j+1})$. Let us denote by l_j the rank of the A_0-module $\mathfrak{n}_j/\mathfrak{n}_{j+1}$; since the field of fractions K_0 of A_0 is the residue field of $\mathrm{A}_{\mathfrak{p}}$, l_j is also the length of the $\mathrm{A}_{\mathfrak{p}}$-module $(\mathfrak{p}\mathrm{A}_{\mathfrak{p}})^j/(\mathfrak{p}\mathrm{A}_{\mathfrak{p}})^{j+1}$. There is a system of generators of the A_0-module M_j which contains a basis of $\mathrm{M}_j \otimes_{\mathrm{A}_0} \mathrm{K}_0$; hence there is an A_0-homomorphism

$$w_j : \mathrm{M}_0^{l_j} \longrightarrow \mathrm{M}_j$$

whose localisation at the ideal (0) of A_0 is an isomorphism, so that the support of $\mathrm{Ker}(w_j)$ and of $\mathrm{Coker}(w_j)$ is reduced to the maximal ideal $\mathfrak{m}/\mathfrak{p}$ of A_0; $\mathrm{Ker}(w_j)$ and $\mathrm{Coker}(w_j)$ are therefore A_0-modules of finite length $(\mathbf{0}, 16.1.10)$. Since by hypothesis $\chi(\mathrm{M}_0, u_0)$ is defined, it follows from the same of $\chi(\mathrm{M}_0^{l_j}, u_0^{l_j}) = l_j\chi(\mathrm{M}_0, u_0)$ and by virtue of (21.10.17.6) and the fact that $\mathrm{Ker}(w_j)$ and $\mathrm{Coker}(w_j)$ are of finite length, we see that $\chi(\mathrm{M}_j, v_j)$ is defined and equal to $l_j\chi(\mathrm{M}_0, u_0)$; the relation (21.10.17.10) then gives

$$\chi(\mathrm{M}, u) = (\sum_j l_j)\chi(\mathrm{M}_0, u_0),$$

and by virtue of a preceding remark, $\sum_j l_j$ is none other than the length of $\mathrm{A}_{\mathfrak{p}}$, which finishes the proof of the lemma (21.10.17.7). □

To apply this lemma when A is a complete ring and $\mathfrak{m} \notin \mathrm{Ass}(A)$, we observe that if u is injective, so is each of the u_i (with the notation of the lemma): indeed, $\det u$ is then a regular element of A, so is not contained in any of the $\mathfrak{p}_i$, and its image $\det u_i$ in $A/\mathfrak{p}_i$ is therefore a nonzero element of this integral ring, which proves that u_i is injective. Since $\mathrm{Coker}(u_i)$ is an image of $\mathrm{Coker}(u)$, it is also of finite length and $\chi(M_i, u_i)$ is therefore defined for all i; we have therefore, by the formula (21.10.17.8), $\chi(M, u) + \chi(M'', u'') = \sum_i \chi(M'_i, u'_i)$ and, since M'' is of finite length, $\chi(M'', u'') = 0$. On the other hand, since $\det(u)$ is a regular element of A, it is not contained in any of the $\mathfrak{p}_i$; the ideal $(\det u)A$ is therefore $\mathfrak{m}$-primary and the quotient $A/(\det u)A$ is of finite length. Applying the same reasoning above to the injective homothety $t : \xi \to (\det u)\xi$ of A and to its images $t_i = \det u_i$ in the $A/\mathfrak{p}_i$, we get IV-4 | 301

$$\chi(A, t) = \sum_i \mathrm{length}(A_{\mathfrak{p}_i}) \cdot \chi(A/\mathfrak{p}_i, t_i).$$

But all the rings $A/\mathfrak{p}_i$ are integral and complete, and by applying the result of II) to each of them, one obtains again the relation (21.10.17.4) for M and u.

IV) *General case*. Set $A' = \widehat{A}$, $M' = M \otimes_A A'$, $u' = u \otimes 1_{A'}$; we have $\det(u') = \det(u)$, and by flatness, $\mathrm{Coker}(u') = (\mathrm{Coker}(u))_{(A')}$ and $A'/(\det u)A' = (A/(\det u)A)_{(A')}$; since the formula (21.10.17.4) is true for A' and u' by virtue of III), it is also true for A and u.

This finishes the proof of (21.10.17).

Proposition (21.10.18). — *Let* X, X′ *be two locally Noetherian preschemes, $f : X' \to X$ a finite morphism, transforming every maximal point of* X′ *into a maximal point of* X, *and satisfying for all $x' \in X'$ one of the conditions (ii), (iii), (iv) of* (21.10.10). *Suppose furthermore that there exists an integer n such that, for every maximal point x of* X, *$(f_*(\mathscr{O}_{X'}))_x$ is a free $\mathscr{O}_{X,x}$-module of rank n. Then, for every* 1*-codimensional cycle* Z *on* X, *we have*

(21.10.18.1) $$f_*(f^*(Z)) = n \cdot Z$$

("projection formula").

Proof. By virtue of the definitions, we are immediately reduced to the case of a local Noetherian ring A of dimension 1, of closed point x, and where $Z = \overline{\{x\}}$; set $X' = \mathrm{Spec}(A')$ where A' is a finite A-algebra and for every minimal prime ideal $\mathfrak{p}_i$ of A, $A'_{\mathfrak{p}_i}$ is a free $A_{\mathfrak{p}_i}$-module of rank n. Let us show that we can furthermore restrict to the case where A is *complete*; let us make the base change $h : Y \to X$, where $Y = \mathrm{Spec}(B)$, $B = \widehat{A}$, and set $Y' = X' \times_X Y = \mathrm{Spec}(B \otimes_A A')$ and $g = f_{(Y)} : Y' \to Y$; the morphism g is then finite, and it satisfies the same conditions as f; if we prove the formula (21.10.18.1) for g and $\overline{\{x\}}$, we verify from (21.10.16.1). We can therefore suppose that A is complete, and hence so is A' which is also decomposed as a direct product of complete local rings; one can therefore also restrict ourselves to the case where A' is local, and verify (21.10.18.1) for each of the conditions (ii), (iii), (iv) of X'. IV-4 | 302

In case (ii), A' is a flat A-module of finite type, of rank n by hypothesis, hence by definition (21.10.1) and (21.10.3) $f^*(Z) = \lambda_{x'} \cdot \overline{\{x'\}}$, where $\lambda_{x'}$ is the length of the A'-module $A'/\mathfrak{m}_x A'$, hence $f_*(f^*(Z)) = (\lambda_{x'}[\boldsymbol{k}(x') : \boldsymbol{k}(x)]) \cdot \overline{\{x\}}$; but $\lambda_{x'}[\boldsymbol{k}(x') : \boldsymbol{k}(x)]$ is the length of $A'/\mathfrak{m}_x A'$ as A-module, or its rank as $\boldsymbol{k}(x)$-vector space, which is equal to n.

In case (iii), A is a discrete valuation ring, hence regular, and the hypothesis $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$ implies that A' is a Cohen-Macaulay ring (**0**, 16.4.6); since $\dim(A') = \dim(A) = 1$ that $A'/\mathfrak{m}_x A'$ is an Artinian ring, it follows from (6.1.5) that A' is a flat A-module and one is reduced to case (ii).

In case (iv), if K is the total ring of fractions of A, $K' = A' \otimes_A K$ is by hypothesis a free K-module of rank n and by definition (21.10.10), we have $f^*(Z) = (n/[\boldsymbol{k}(x') : \boldsymbol{k}(x)]) \cdot \overline{\{x'\}}$, whence $f_*(f^*(Z)) = n \cdot \overline{\{x\}}$.

We note that the formula (21.10.18.1) is applicable in particular when the morphism f is *finite* and *flat* and such that for every maximal point x of X, $(f_*(\mathscr{O}_{X'}))_x$ is a free $\mathscr{O}_{X,x}$-module of rank n. □

Corollary (21.10.19). — *Under the hypotheses of* (21.10.18), *let* D *be a divisor on* X *such that $f^*(D)$ is defined* (21.4.5); *then we have*

(21.10.19.1) $$f_*(\mathrm{cyc}(f^*(D))) = n \cdot \mathrm{cyc}(D).$$

Proof. This follows from (21.10.18) and (21.10.13bis). □

21.11. Factoriality of regular local rings

Theorem (21.11.1). — (Auslander-Buchsbaum) — *A regular local Noetherian ring is factorial.* **IV-4** | 302

Proof. The proof that follows is due to I. Kaplansky.

Let A be a regular local ring of dimension n; we will reason by induction on n. For $n = 0$, A is a field, and for $n = 1$, a discrete valuation ring, hence principal (and *a fortiori* factorial). Suppose therefore $n \geqslant 2$ and the theorem proved for regular rings of dimension $< n$. Set $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, denote by a the closed point of X and set $\mathrm{U} = \mathrm{X} - \{a\}$. At every point $y \in \mathrm{U}$, we have $\dim(\mathscr{O}_{\mathrm{X},y}) \leqslant n-1$, and since A is regular, the rings $\mathscr{O}_{\mathrm{X},y}$ are also regular (**0**, 17.3.2), hence by the induction hypothesis they are factorial. Moreover, since $\mathrm{prof}(\mathrm{A}) = \dim(\mathrm{A}) \geqslant 2$ since A is regular, hence Cohen-Macaulay (**0**, 17.1.3). Using (21.6.14), we are reduced to proving that $\mathrm{Pic}(\mathrm{U}) = 0$. Consider therefore an **IV-4** | 303 invertible $\mathscr{O}_\mathrm{U}$-Module $\mathscr{L}$, and let us prove that it is isomorphic to $\mathscr{O}_\mathrm{U}$. It follows from (**I**, 9.4.5) that there exists a *coherent* $\mathscr{O}_\mathrm{X}$-Module $\mathscr{F}$ such that $\mathscr{F}|\mathrm{U} = \mathscr{L}$. Since A is regular, hence of finite cohomological dimension (**0**, 17.3.1), there exists a *finite* free resolution of $\mathscr{F}$:

$$0 \leftarrow \mathscr{F} \leftarrow \mathscr{O}_\mathrm{X}^{n_1} \leftarrow \mathscr{O}_\mathrm{X}^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_\mathrm{X}^{n_h} \leftarrow 0$$

(**0**, 17.2.8 and **0**, 17.2.2, (iii)). By restriction to U, we have therefore a finite resolution

$$0 \leftarrow \mathscr{L} \leftarrow \mathscr{O}_\mathrm{U}^{n_1} \leftarrow \mathscr{O}_\mathrm{U}^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_\mathrm{U}^{n_h} \leftarrow 0. \tag{21.11.1.1}$$

□

The theorem will follow from the following general considerations. On a ringed space X, let $\mathscr{E}$ be an $\mathscr{O}_\mathrm{X}$-Module locally free of finite rank; we will denote by $\Lambda^{\max}\mathscr{E}$ the invertible $\mathscr{O}_\mathrm{X}$-Module which, in a neighbourhood of each point of X, is equal to $\Lambda^p\mathscr{E}$, denoting by p the rank of $\mathscr{E}$ in this neighbourhood (which can vary with the connected component). With this notation, we have the

Lemma (21.11.2). — *Let* X *be a ringed space in local rings, and*

$$0 \leftarrow \mathscr{E}_0 \xleftarrow{u_0} \mathscr{E}_1 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

an exact sequence of $\mathscr{O}_\mathrm{X}$-Modules locally free of finite rank; then the invertible $\mathscr{O}_\mathrm{X}$-Module $\bigotimes_{0\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^i}$ is isomorphic to $\mathscr{O}_\mathrm{X}$.

Proof. Let us show how this lemma will allow us to conclude the proof of (21.11.1). It suffices to note that, for every integer n, $\Lambda^{\max}(\mathscr{O}_\mathrm{U}^n) = \Lambda^n\mathscr{O}_\mathrm{U}^n$ is isomorphic to $\mathscr{O}_\mathrm{U}$, and on the other hand $\Lambda^{\max}\mathscr{L} = \mathscr{L}$ for every invertible $\mathscr{O}_\mathrm{U}$-Module $\mathscr{L}$; the lemma (21.11.2), applied to the exact sequence (21.11.1.1), shows that $\mathscr{L}$ is isomorphic to $\mathscr{O}_\mathrm{U}$.

It remains to prove (21.11.2); we proceed by induction on h, the lemma being trivial for $h = 1$. For $h > 1$, $\mathscr{N} = \mathrm{Ker}(u_0)$ is an $\mathscr{O}_\mathrm{X}$-Module locally free of finite rank ($\mathbf{0}_\mathrm{I}$, 5.5.5), and we have the two exact sequences

$$0 \leftarrow \mathscr{E}_0 \leftarrow \mathscr{E}_1 \leftarrow \mathscr{N} \leftarrow 0$$
$$0 \leftarrow \mathscr{N} \leftarrow \mathscr{E}_2 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

By virtue of the induction hypothesis, $(\Lambda^{\max}\mathscr{N}) \otimes (\bigotimes_{2\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^{i-1}})$ is isomorphic to $\mathscr{O}_\mathrm{X}$; it will therefore suffice to define a canonical isomorphism $(\Lambda^{\max}\mathscr{N}) \otimes (\Lambda^{\max}\mathscr{E}_0) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1$ to finish the proof. Now, there exists an open covering (U_α) of X such that in each U_α, $\mathscr{E}_1|\mathrm{U}_\alpha$ is a direct sum of $\mathscr{N}|\mathrm{U}_\alpha$ and of a free $\mathscr{O}_{\mathrm{U}_\alpha}$-Module $\mathscr{M}_\alpha$ ($\mathbf{0}_\mathrm{I}$, 5.5.5), whence a canonical isomorphism $v_\alpha : \mathscr{M}_\alpha \xrightarrow{\sim} \mathscr{E}_0|\mathrm{U}_\alpha$. Since we have a canonical isomorphism

$$r_\alpha : (\Lambda^{\max}\mathscr{N}|\mathrm{U}_\alpha) \otimes (\Lambda^{\max}\mathscr{M}_\alpha) \xrightarrow{\sim} (\Lambda^{\max}\mathscr{E}_1)|\mathrm{U}_\alpha$$

we deduce, by means of v_α, an isomorphism

$$u_\alpha : (\Lambda^{\max}\mathscr{N}|\mathrm{U}_\alpha) \otimes (\Lambda^{\max}\mathscr{E}_0|\mathrm{U}_\alpha) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1|\mathrm{U}_\alpha$$

and it is a matter of showing that u_α and u_β coincide in $\mathrm{U}_\alpha \cap \mathrm{U}_\beta$ for any two indices α, β. Now, if **IV-4** | 304 v'_α and v'_β are the restrictions to $\mathrm{U}_\alpha \cap \mathrm{U}_\beta$ of v_α and v_β respectively, we have $v_\alpha = v'_\beta \circ w_{\beta\alpha}$, where $w_{\beta\alpha} : \mathscr{M}_\alpha|(\mathrm{U}_\alpha \cap \mathrm{U}_\beta) \xrightarrow{\sim} \mathscr{M}_\beta|(\mathrm{U}_\alpha \cap \mathrm{U}_\beta)$ is the "parallel projection" such that for every section $s \in \Gamma(\mathrm{U}_\alpha \cap \mathrm{U}_\beta, \mathscr{M}_\alpha)$, $w_{\beta\alpha}(s) = s + t$ with $t \in \Gamma(\mathrm{U}_\alpha \cap \mathrm{U}_\beta, \mathscr{N})$; the identity of u_α and u_β follows immediately from this fact and from the definition of the canonical isomorphism r_α (Bourbaki, *Alg.*, chap. III, 3[e] ed.). □

21.12. The Van der Waerden purity theorem for the ramification locus of a birational morphism

(21.12.1). Let X and U be two preschemes, $f : \mathrm{U} \to \mathrm{X}$ a quasi-compact and quasi-separated morphism, so that $f_*(\mathscr{O}_\mathrm{U})$ is a quasi-coherent $\mathscr{O}_\mathrm{X}$-Algebra (1.7.4). We call *affine envelope* of the X-prescheme U the X-prescheme affine over X

$$\mathrm{U}^0 = \mathrm{Aff}(\mathrm{U}/\mathrm{X}) = \mathrm{Spec}(f_*(\mathscr{O}_\mathrm{U})) = \mathrm{Spec}(\mathscr{A}(\mathrm{U})) \quad (\mathbf{II},\ 1.1.1).$$

If $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is the structural morphism, we have therefore by definition

$$\mathscr{A}(\mathrm{U}^0) = f^0_*(\mathscr{O}_{\mathrm{U}^0}) = f_*(\mathscr{O}_\mathrm{U}),$$

and at the identity isomorphism of $f_*(\mathscr{O}_\mathrm{U})$, there corresponds by $(\mathbf{II}, 1.2.7)$ a canonical X-morphism

(21.12.1.1) $$i_\mathrm{U} : \mathrm{U} \longrightarrow \mathrm{U}^0.$$

For i_U to be an *isomorphism*, it is necessary and sufficient that the morphism $f : \mathrm{U} \to \mathrm{X}$ be *affine*. For every affine X-prescheme V, the map $u \mapsto u \circ i_\mathrm{U}$ is a *bijection*

$$\mathrm{Hom}_\mathrm{X}(\mathrm{U}^0, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V})$$

functorial in V: this follows from the existence of canonical bijections

$$\mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_x}(\mathscr{A}(\mathrm{V}), \mathscr{A}(\mathrm{U}))$$

and $\mathrm{Hom}_\mathrm{X}(\mathrm{U}^0, \mathrm{V}) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_x}(\mathscr{A}(\mathrm{V}), \mathscr{A}(\mathrm{U}^0))$ $(\mathbf{II}, 1.2.7)$. We can therefore say that U^0 *represents* the covariant functor $\mathrm{V} \mapsto \mathrm{Hom}_\mathrm{X}(\mathrm{U}, \mathrm{V})$ in the category of X-preschemes affine over X. We deduce from this $(\mathbf{0}_\mathrm{III}, 8.1.7)$ that for a fixed X, $\mathrm{U} \mapsto \mathrm{Aff}(\mathrm{U}/\mathrm{X})$ is a covariant functor from the category of X-preschemes quasi-compact and quasi-separated over X, into the category of X-preschemes affine over X; more precisely, if U_1, U_2 are two X-preschemes quasi-compact and quasi-separated over X, to every X-morphism $g : \mathrm{U}_1 \to \mathrm{U}_2$ corresponds the unique X-morphism $g^0 : \mathrm{U}_1^0 \to \mathrm{U}_2^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U}_1 \arrow[r, "g"] \arrow[d, "i_{\mathrm{U}_1}"'] & \mathrm{U}_2 \arrow[d, "i_{\mathrm{U}_2}"] \\ \mathrm{U}_1^0 \arrow[r, "g^0"'] & \mathrm{U}_2^0 \end{tikzcd}$$

More generally, consider a commutative diagram **IV-4** | 305

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "f"'] & \mathrm{U}' \arrow[d, "f'"] \\ \mathrm{X} \arrow[r, "u"'] & \mathrm{X}' \end{tikzcd}$$

where the morphisms f, f' are quasi-compact and quasi-separated and the morphism u *affine*. If $h = u \circ f$, we have $h_*(\mathscr{O}_\mathrm{U}) = u_*(f_*(\mathscr{O}_\mathrm{U})) = u_*(f^0_*(\mathscr{O}_{\mathrm{U}^0}))$ and $u \circ f^0$ is an affine morphism; hence $\mathrm{U}^0 = \mathrm{Aff}(\mathrm{U}/\mathrm{X}')$ (relative to the morphism h), whence a unique X′-morphism $v^0 : \mathrm{U}^0 \to \mathrm{U}'^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "i_\mathrm{U}"'] & \mathrm{U}' \arrow[d, "i_{\mathrm{U}'}"] \\ \mathrm{U}^0 \arrow[r, "v^0"'] & \mathrm{U}'^0 \end{tikzcd}$$

Proposition (21.12.2). — *Let* $f : \mathrm{U} \to \mathrm{X}$ *be a quasi-compact and quasi-separated morphism,* $h : \mathrm{X}' \to \mathrm{X}$ *a flat morphism; set* $\mathrm{U}' = \mathrm{U} \times_\mathrm{X} \mathrm{X}'$, $f' = f_{\{\mathrm{X}'\}} : \mathrm{U}' \to \mathrm{X}'$. *Then there is a canonical* X′*-isomorphism*

(21.12.2.1) $$\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') \xrightarrow{\sim} \mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_\mathrm{X} \mathrm{X}'.$$

Proof. Indeed, we have $\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') = \mathrm{Spec}(f'_*(\mathscr{O}_{\mathrm{U}'}))$, and $\mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_{\mathrm{X}} \mathrm{X}' = \mathrm{Spec}(h^*(f_*(\mathscr{O}_{\mathrm{U}})))$; the isomorphism of the assertion comes from the canonical isomorphism $h^*(f_*(\mathscr{O}_{\mathrm{U}})) \xrightarrow{\sim} f'_*(\mathscr{O}_{\mathrm{U}'})$ (2.3.1). □

Corollary (21.12.3). — *For every quasi-compact and quasi-separated morphism $f : \mathrm{U} \to \mathrm{X}$ and all $x \in \mathrm{X}$, there is a canonical isomorphism up to canonical isomorphism*

$$\mathrm{U}^0 \times_{\mathrm{X}} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) = (\mathrm{U} \times_{\mathrm{X}} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))^0.$$

It also follows from (21.12.2) that, up to canonical isomorphism, for every open V of X

(21.12.4) $$(f^{-1}(\mathrm{V}))^0 = (f^0)^{-1}(\mathrm{V}).$$

(21.12.5). We will consider in particular the case where $f : \mathrm{U} \to \mathrm{X}$ is an *open immersion*, U thus being identified with an open of X. Since the morphism $f^{-1}(\mathrm{U}) \to \mathrm{U}$ is the identity, it follows from (21.12.4) that the morphism $(f^0)^{-1}(\mathrm{U}) \to \mathrm{U}$ restriction of f^0 is an isomorphism, $i_{\mathrm{U}} : \mathrm{U} \to \mathrm{U}^0$ being thus an *open immersion* allowing us to identify U with an open of U^0. More generally, for an open V of X, the restriction $(f^0)^{-1}(\mathrm{V}) \to \mathrm{V}$ of f^0 is an isomorphism of $(f^0)^{-1}(\mathrm{V})$ onto the prescheme induced on the open $\mathrm{V} \cap \mathrm{U}$ if and only if the open immersion $\mathrm{U} \cap \mathrm{V} \to \mathrm{V}$ is an *affine* morphism. It is clear (**II**, 1.2.1) that the union of these opens V is the *largest* of them, U_1, which contains U; by virtue **IV-4** | 306
of what precedes, U_1 is also the largest open not meeting the set

$$\mathrm{Daf}(\mathrm{U}/\mathrm{X}) = f^0(\mathrm{U}^0) - \mathrm{U}$$

("affinity defect" of the open U relative to X, which is empty if and only if U is affine over X); in other words, the closed set $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ is the *closure* of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$.

We note that for every *flat* morphism $h : \mathrm{X}' \to \mathrm{X}$, if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, we have

(21.12.5.1) $$\mathrm{Daf}(\mathrm{U}'/\mathrm{X}') = h^{-1}(\mathrm{Daf}(\mathrm{U}/\mathrm{X}))$$

as follows immediately from (21.12.2.1) and from (**I**, 3.4.8). In particular, for every open V of X, we have

(21.12.5.2) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{V})/\mathrm{V}) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{V}$$

and for all $x \in \mathrm{X}$,

(21.12.5.3) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}).$$

We will, when X is locally Noetherian, give precisions on the nature of the set $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$, which will show for example that when U is not everywhere dense in X, $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ is not any rare closed subset:

Theorem (21.12.6). — *Let X be a locally Noetherian prescheme, U a non-empty open, $f : \mathrm{U} \to \mathrm{X}$ the canonical injection. Then:*

label=(i) *The closure $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ is of codimension $\geqslant 2$ in X.*
lbbel=(ii) *If $\mathrm{T} = \mathrm{X} - \mathrm{U} \supset \mathrm{Z}$ is of codimension $\geqslant 2$, the morphism $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is surjective.*

Proof. (i) Let us first show that for every point $x \in \mathrm{Daf}(\mathrm{U}/\mathrm{X})$ we have necessarily $\dim(\mathscr{O}_{\mathrm{X},x}) > 1$. Indeed, x cannot evidently be a maximal point of X, not being contained in $\overline{\mathrm{U}}$; it is therefore a matter of seeing that we cannot have $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$. Now, this relation would imply, by (21.12.5.3), $x \in \mathrm{Daf}(\mathrm{U}^{(a)}/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))$, where $\mathrm{U}^{(a)} = \mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. But the only open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ itself and the parts of the (finite) set of maximal points of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. Now, the open set reduced to a single maximal point of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine, by definition of preschemes; we conclude that *all* open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are affine, hence $\mathrm{Daf}(\mathrm{U}^{(a)}/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \varnothing$, contradicting the hypothesis.

To prove (i), it remains to show more, namely that if $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$ is such that $x \in \mathrm{X}$, there exists a neighbourhood open W of x in X such that W does not meet $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$, i.e. such that the canonical injection $f_{\mathrm{W}} : \mathrm{U} \cap \mathrm{W} \to \mathrm{W}$ is affine. But, with the same notations as above, we see that the injection $f^{(a)} : \mathrm{U}^{(a)} \to \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine. We can evidently restrict to the case where X is Noetherian, and x the closed point of X. For this, it suffices to make the change of base $h : \mathrm{X}' = \mathrm{Spec}(\mathrm{A}) \to \mathrm{X}$, where $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$; if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, $f' = f_{\{\mathrm{X}'\}}$ the canonical injection $\mathrm{U}' \to \mathrm{X}'$, and since the morphism

h is *flat*, it follows from (21.12.2) that if we prove that x belongs to $f'^0(U'^0)$, then $x \in f^0(U^0)$. By virtue of (6.1.1), we have made the desired reduction.

Let then X_1 be a closed reduced sub-prescheme of X having as underlying space an irreducible component of X, of *maximal* dimension, among those that contain T, and set $U_1 = U \cap X_1$, $T_1 = T \cap X_1 = X_1 - U_1$; we have therefore $\operatorname{codim}(T_1, X_1) \geqslant 2$ (**0**, 14.2.1), and the pair (U_1, X_1) satisfies the same hypotheses that the pair (U, X) (X_1 being the spectrum of a quotient of A, hence a local Noetherian complete ring). We have on the other hand a commutative diagram **IV-4** | 307

$$\begin{tikzcd} U_1 \arrow[r, "i"] \arrow[d, "h"'] & U \arrow[d, "f"] \\ X_1 \arrow[r, "i"'] & X \end{tikzcd}$$

where i and j are the canonical injections, i being thus an *affine* morphism; we deduce from (21.12.1) the existence of a morphism $j^0 : U_1^0 \to U^0$ making commutative the diagram

$$\begin{tikzcd} U_1^0 \arrow[r, "j^0"] \arrow[d, "f_1^0"'] & U^0 \arrow[d, "f^0"] \\ X_1 \arrow[r, "i"'] & X \end{tikzcd}$$

and hence, to prove that $x \in f^0(U^0)$, it suffices to prove that $x \in f_1^0(U_1^0)$. We can therefore replace X by X_1, and we can hence suppose that the ring A is *integral*. But A is a quotient of a regular Noetherian ring by the Cohen structure theorem (**0**, 19.8.8, (i)) and since it is integral, we can apply (5.11.1) with the family (x_a) reduced to the single maximal point of X; since $\operatorname{codim}(T, X) \geqslant 2$ by hypothesis, we see that $f_*(\mathscr{O}_U)$ is a *coherent* $\mathscr{O}_X$-Module, hence the morphism $f^0 : U^0 \to X$ is *finite*; since this morphism is dominant (U being everywhere dense), it is surjective (**II**, 6.1.10), and hence $x \in f^0(U^0)$.

(ii) It is a matter of proving that for every point $x \in T$, we have $x \in f^0(U^0)$; we will first show that we can reduce to the case where $X = \operatorname{Spec}(A)$, where A is a *complete* local Noetherian ring, and x the closed point of X. For this, it suffices to make the change of base $h : X' = \operatorname{Spec}(A) \to X$, where $A = \widehat{\mathscr{O}}_{X,x}$; if we set $U' = h^{-1}(U)$, $f' = f_{\{X'\}}$ the canonical injection $U' \to X'$, and since the morphism h is *flat*, it follows from (21.12.2) that if we prove that x belongs to $f'^0(U'^0)$, then $x \in f^0(U^0)$. By virtue of (6.1.1), we have made the desired reduction.

Let then X_1 be a closed reduced sub-prescheme of X having as underlying space an irreducible component of X, of *maximal* dimension, among those that contain T, and set $U_1 = U \cap X_1$, $T_1 = T \cap X_1 = X_1 - U_1$; we have therefore $\operatorname{codim}(T_1, X_1) \geqslant 2$ (**0**, 14.2.1), and the pair (U_1, X_1) satisfies the same hypotheses (the pair (U, X) (X_1 being the spectrum of a quotient of A, hence a local Noetherian complete ring). On the other hand, by the commutative diagram above we see that $x \in f^0(U^0)$ as we have seen, and we can hence suppose that the ring A is integral. But A is a quotient of a regular Noetherian ring by the Cohen structure theorem (**0**, 19.8.8, (i)) and since it is integral, we can apply (5.11.1) with the family (x_a) reduced to the single maximal point of X; since $\operatorname{codim}(T, X) \geqslant 2$ by hypothesis, we see that $f_*(\mathscr{O}_U)$ is a *coherent* $\mathscr{O}_X$-Module, hence the morphism $f^0 : U^0 \to X$ is *finite*; since this morphism is dominant (U being everywhere dense), it is surjective (**II**, 6.1.10), and hence $x \in f^0(U^0)$. □

Corollary (21.12.7). — *Let* X *be a locally Noetherian prescheme,* U *a sub-prescheme induced on an open of* X, $j : U \to X$ *the canonical immersion; suppose that* j *is an affine morphism. Then every irreducible component of* $T = X - U$ *is of codimension* $\leqslant 1$ *(and hence everywhere dense).*

Proof. Suppose indeed that one of the irreducible components T_1 of T is of codimension $\geqslant 2$ in X. Replacing X if needed by a neighbourhood open of the generic point of T_1, we can suppose T irreducible and of codimension $\geqslant 2$. But then the hypothesis that $j : U \to X$ is affine implies that U^0 is identified with U and j^0 with j, in other words $j^0(U^0) = U$; but this contradicts the conclusion of (21.12.6) which, under the hypothesis of dimension, says that j^0 must be surjective. □ **IV-4** | 308

(21.12.8). Let A be a local Noetherian ring, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, y the unique closed point of Y, $\mathrm{Y}' = \mathrm{Y} - \{y\}$. Consider the following condition:

(W) *For every open* U *of* Y *contained in* Y′, *containing no irreducible component of* Y′ *and such that the canonical injection* $\mathrm{U} \to \mathrm{Y}'$ *is affine,* U *is itself an affine open.*

This condition is implied by the following:

(W′) *For every closed subset* T *of* Y *whose every irreducible component is of codimension* 1 *and such that for every irreducible component* Y_i *of* Y, $\mathrm{Y}_i \cap \mathrm{T} \neq \{y\}$, *the open* $\mathrm{U} = \mathrm{Y} - \mathrm{T}$ *is affine.*

Indeed, if (W′) is satisfied and if U is an open of Y satisfying the hypothesis of (W), it follows from (21.12.7) that no irreducible component of $\mathrm{T} = \mathrm{Y} - \mathrm{U}$ can be of codimension $\geqslant 2$; since on the other hand U contains no irreducible component of $\mathrm{Y}_i - \{y\}$ of Y, condition (W′) on Y, shows that U is affine.

We note that condition (W′) simplifies when Y is *irreducible* and is then equivalent to the following:

(W″) *For every closed irreducible subset* T *of* Y, *of codimension* 1 *in* Y, $\mathrm{Y} - \mathrm{T}$ *is an affine open.*

Indeed, it is clear that (W′) implies (W″) when Y is irreducible, and the converse is true by considering the irreducible components of T and of Y and noting that the intersection of a finite number of affine open subsets is an affine open (**I**, 5.5.6).

(21.12.9). *Examples.* — If A is a local Noetherian *factorial* ring, it satisfies (W′) (and *a fortiori* (W)), since every prime ideal of height 1 is principal (**I**, 1.3.6). But there are local Noetherian rings which are not factorial, for example those of dimension $\leqslant 1$: indeed, as remarked in the proof of (21.12.6), (i), that *all* opens of Y are affine. One can on the other hand prove by using the theory of local duality (chap. III, 3ᵉ partie) that every local Noetherian ring of dimension 2 satisfies (W′).

The interest of condition (W) resides in the following result:

Proposition (21.12.10). — *Let* X, Y *be two locally Noetherian preschemes,* $g : \mathrm{X} \to \mathrm{Y}$ *a morphism,* y *a closed point of* Y, $\mathrm{X}' = g^{-1}(Y')$; *suppose that the morphism* $g' : \mathrm{X}' \to \mathrm{Y}'$ *restriction of* g *is an open immersion, and that the local ring* $\mathscr{O}_{\mathrm{Y},y}$ *satisfies the condition* (W) *of* (21.12.8). *Then for every irreducible* Z *of* $\mathrm{X}_y = g^{-1}(y)$, *either* Z *is of codimension* $\leqslant 1$ *in* X, *or its generic point is isolated in* Z. *If* Z *is locally of finite type over* $\boldsymbol{k}(y)$, *the second alternative implies that* Z *is reduced to a single point.*

Proof. The last assertion of the statement results from the fact that Z is then a prescheme of Jacobson (**I**, 10.4.7), and since the set of closed points of Z is then dense in Z, the generic point of Z cannot be isolated in Z unless Z is reduced to a single point.

Suppose that there exists an irreducible component Z of X_y whose generic point z is not isolated **IV-4 | 309**
in Z, and such that $\mathrm{codim}(\mathrm{Z}, \mathrm{X}) \geqslant 2$. The question being local on X and Y, suppose X and Y affine; since on the other hand $z \in \mathrm{X}_y$ is the generic point of T, we have $g(\mathrm{T}) \subset \overline{\{y\}} = \{y\}$, i.e. $\mathrm{T} = \mathrm{X}_y$. We are then exactly in the conditions of application of (21.12.10), in other words $\mathrm{T} = \mathrm{X}_y$. Set $\mathrm{X}_i = \overline{\{x_i\}}$. By hypothesis, z is not isolated in X_y, and since z is not a maximal point of X (this would imply $\mathrm{codim}(\mathrm{Z}, \mathrm{X}) = 0$), it follows that there exists x_i in X_y with $x_i \neq z$. Since $\mathrm{X}_i \neq \mathrm{Z}$, there exists X_i a point t_i such that if we set $\mathrm{T}_i = \overline{\{t_i\}}$, we have $\dim(\mathscr{O}_{\mathrm{T}_i,z}) = 1$ (10.5.9); this implies by hypothesis $t_i \notin \mathrm{Z}$, so t_i is not a generisation of z. Replacing X by $\mathrm{X} - \bigcup_i \mathrm{T}_i$, we see that g of $\mathrm{X}' \to \mathrm{X}'_y$ does not contain the images $g(t_i)$, so does not contain y. This being so, since X and Y are affines, the morphism $g : \mathrm{X} \to \mathrm{Y}$ is affine, hence $g' : \mathrm{X}' \to \mathrm{Y}'$ is also an affine morphism of the restriction $g' : \mathrm{X}' \to \mathrm{Y}'$. Set $\mathrm{Y}_1 = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, $\mathrm{Y}'_1 = \mathrm{Y}_1 - \{y\} = \mathrm{Y}' \cap \mathrm{Y}_1$, $\mathrm{U}_1 = \mathrm{U} \cap \mathrm{X}_1$, $\mathrm{T}_1 = \mathrm{T} \cap \mathrm{X}_1 = \mathrm{X}_1 - \mathrm{U}_1$; we have therefore $\mathrm{codim}(\mathrm{T}_1, \mathrm{X}_1) \geqslant 2$ (**0**, 14.2.1), and the pair $(\mathrm{U}_1, \mathrm{X}_1)$ satisfies the same hypotheses as the pair (U, X). This implies that U_1 is an open affine in Y_1 by virtue of (W); one would conclude from this that the open $\mathrm{X} - \mathrm{X}_y$ of X, isomorphic to U, would be affine, which contradicts (21.12.7); the proposition is thus proved. □

Theorem (21.12.12). — (van der Waerden) — *Let* X, Y *be two locally Noetherian integral preschemes,* **IV-4 | 310**
$g : \mathrm{X} \to \mathrm{Y}$ *a birational morphism and locally of finite type. Suppose furthermore that* Y *is normal and that for all* $y \in \mathrm{Y}$, *the local ring* $\mathscr{O}_{\mathrm{Y},y}$ *satisfies the condition* (W) *of* (21.12.8) *(conditions satisfied in particular when* Y *is* locally factorial (21.12.9)*). If* V *is the largest open of* X *such that the restriction* $g|\mathrm{V} : \mathrm{V} \to \mathrm{Y}$ *is a local isomorphism, all the irreducible components of* $\mathrm{T} = \mathrm{X} - \mathrm{V}$ *are of codimension* 1 *in* X.

Proof. We note that since g is birational, the open V is not empty; it suffices therefore to prove that a maximal point z of T cannot be isolated in X_y, where $y = g(z)$. But, up to restringing to an open neighbourhood of z, we can restrict to the case $T = X_y$; then all the fibres $g^{-1}(y')$ ($y' \in Y$) would be empty or reduced to a single point, and it would follow from the "Main theorem" (8.12.10) that g would be a local isomorphism, contradicting the hypothesis $z \in T$. □

Corollary (21.12.13). — *Suppose satisfied the hypotheses of* (21.12.12) *and in addition, suppose that g is quasi-finite at each of the points of* $X^{(1)}$ *(recall that these are the points where* $\dim(\mathscr{O}_{X,x}) = 1$*); if furthermore g is separated, g is an open immersion.*

Proof. It suffices to prove the first assertion, the second being a consequence of the first and of (**I**, 8.2.8). It all comes down to proving, with the notations of (21.12.12), that $T = \varnothing$; in the contrary case, a generic point z of an irreducible component of T would belong to $X^{(1)}$ by virtue of (21.12.12), and by hypothesis it would be isolated in X_y with $y = g(z)$; but as we saw in the proof of (21.12.12) this is not possible. □

Remarks (21.12.14). — (i) The conclusion of (21.12.12) applies when Y is regular and integral, by virtue of the Auslander-Buchsbaum theorem (21.11.1). On the contrary, one knows examples where X and Y are normal algebraic schemes of dimension 3, over an algebraically closed field of any characteristic, and where the conclusion of (21.12.12) is no longer valid.

(ii) The set T of (21.12.12) is also the set of points where g is *ramified*. Indeed, if g is unramified at a point $x \in X$, it is also unramified in an open neighbourhood of x (17.3.7), hence $g|U$ is a separated, quasi-finite and birational morphism. Since Y is normal, we conclude from the "Main theorem" (**III**, 4.4.9) that g is an isomorphism local at the point x, hence $x \in X - T$. Conversely, g is evidently unramified at every point where it is a local isomorphism. This justifies the title given to this section.

(iii) Without supposing that Y is locally factorial, but instead assuming by contrast that X is **IV-4 | 311** intersection complete relative to Y (chap. V), we will see in chap. V that one has a more precise result than (21.12.12), expressing T as a 1-codimensional cycle of an $\mathscr{O}_X$-Module invertible, canonically associated to g. This applies notably when X and Y are both regular, or when g is a S-morphism, X and Y being smooth preschemes over S.

(iv) One may ask whether (21.12.10) admits a converse; in other words, for a local Noetherian ring A, set $Y = \operatorname{Spec}(A)$, y denoting the closed point of Y, and for every locally Noetherian prescheme X and every morphism $g : X \to Y$ such that $g' : X' \to Y'$ is an open immersion, is every irreducible component of X_y of codimension $\leqslant 1$ in X or reduced to a single point, is it true that A satisfies condition (W) of (21.12.8)?

(v) Let Y be an integral *regular* prescheme, X a locally Noetherian *normal* prescheme, $g : X \to Y$ a morphism locally of finite type; suppose furthermore that, if η is the generic point of Y, the fibre X_η is *étale* over $\boldsymbol{k}(\eta)$, and let T' be the complement of the largest open V' of X such that $g|V' : V' \to Y$ is *étale*; is it true that all the irreducible components of T' are of codimension 1? This is indeed the case as well when one moreover supposes that g is *locally quasi-finite* ("purity theorem" of Zariski-Nagata). One can show that the preceding conjecture would be a consequence of the following (and it would even be equivalent): for a regular local ring A contained in a local integral integrally closed ring B, such that B is a finite A-module, the open U of the points of Spec(B) for which Spec(B) is unramified over Spec(A) is affine, or equivalently (or also by the same argument (18.10.1), étale on Spec(A)) is *affine*.

Proposition (21.12.15). — *Let S be a prescheme, $g : X \to S$ a flat morphism locally of finite presentation, whose fibres $X_s = g^{-1}(s)$ are geometrically irreducible* (4.5.2)*, $h : Y \to S$ a smooth morphism; denote by $Y_s = h^{-1}(s)$ the fibres of this morphism. Let $f : X \to Y$ be a proper S-morphism such that, for every maximal point η of S, the morphism $f_\eta : X_\eta \to Y_\eta$ is an isomorphism. Then f is an isomorphism.*

Proof. Since h is flat, every maximal point of Y lies above a maximal point of S (2.3.4), hence the **IV-4 | 312** union of the Y_η, as η ranges over the set of maximal points of S, is dense in Y; since the f_η are isomorphisms, f is dominant and hence *surjective* since it is proper. We must therefore show that f is an *open immersion*. For every $s \in S$, by (17.9.5), it suffices to prove that for all $s \in S$, $f_s : X_s \to Y_s$ is an open immersion. Since every $s \in S$ is a generisation of a maximal point η, by making the change of base $\operatorname{Spec}(\mathscr{O}_{S,s}) \to S$ we can reduce to the case where S is a local integral scheme of closed point s and generic point η. Since X_s and X_η are geometrically irreducible, the irreducible components of X_s and Y_s are of the same dimension. But by hypothesis X_s is irreducible, and since f_η is an isomorphism, X is irreducible and reduced $(\mathbf{I}, 3.2.1)$ and by flatness $(\mathbf{I}, 3.3.2)$, hence X is reduced (*ibid.*, 3.2.1) and by flatness, X is integral. Since $g : X \to S$ is flat, the maximal points of X are contained in X_η (2.3.4) and since f_η is an isomorphism, X is irreducible and the local ring at its generic point is a field; moreover, by flatness (3.3.2), X has no prime cycles of dimension 1, hence X is reduced (*ibid.*, 3.2.1). To prove that f is an open immersion, we note that the assertion is evident at the points of $X^{(1)}$ that belong to X_η; the only point of $X^{(1)}$ belonging to X_s is the generic x of X_s, by virtue of (6.1.1). Now, it follows from (2.4.6) and (14.2.2) that the morphisms g and h are equidimensional; as the irreducible components of X_s and Y_s are all of the same dimension. But by hypothesis X_s is irreducible, and since f_η is surjective, it follows from the same of $f_s : X_s \to Y_s$, and $\dim(f_s^{-1}(y)) = 0$, i.e. f is well *quasi-finite* at the point x, and the conclusion follows from the fact that we can apply the criterion (21.12.13); to see that f is quasi-finite at the points of $X^{(1)}$, we remark that the assertion is evident at those that belong to X_η; the only point of $X^{(1)}$ belonging to X_s is the generic x of X_s, by virtue of (6.1.1). Now, it follows from (2.4.6) and (14.2.2) that the morphisms g and h are equidimensional; as X_η and Y_η are isomorphic, the irreducible components of X_s and of Y_s all have the same dimension. But by hypothesis X_s is irreducible, and since f_η is surjective, hence so is f_s, and $\dim(f_s^{-1}(y)) = 0$, i.e. f is well *quasi-finite* at the point x, and the conclusion follows from (21.12.13). □

Corollary (21.12.16). — *Let $g : X \to S$ be a proper, flat morphism of finite presentation, $h : Y \to S$ a smooth, proper morphism of finite presentation, $f : X \to Y$ an S-morphism. Suppose that the fibres $X_s = g^{-1}(s)$ of g are geometrically irreducible. Let U be the set of $s \in S$ such that $f_s : X_s \to Y_s$ is an isomorphism. Then U is open and closed in S, and the morphism $g^{-1}(U) \to h^{-1}(U)$, restriction of f, is an isomorphism.*

Proof. The last assertion follows from the first and from (17.9.5). We already know (9.6.1), (xi) that U is locally constructible in S. By virtue of (1.10.1), it suffices to prove that U is stable by *specialisation* and by *generisation*. To prove these assertions, one can, by a change of base $\operatorname{Spec}(\mathscr{O}_{S,s}) \to S$, reduce to the case where S is a local integral scheme of closed point s and generic point η.

Stability by specialisation follows from (21.12.15) (noting that the projection of the closed point of **IV-4** | 313
S is the closed point of S_0) in order to reduce to the case where moreover S is *Noetherian*; but then the conclusion follows from (**III**, 4.6.7, (ii)).

Stability by generisation follows from (21.12.15). □

Remarks (21.12.17). — (i) The conclusion of the proposition (21.12.15) is no longer valid if we **IV-4** | 313
eliminate the hypothesis that the fibres X_s are irreducible. Take indeed $S = \operatorname{Spec}(A)$, where A is a discrete valuation ring, $Y = \mathbb{P}^1_S$, which is proper and smooth over S (17.3.9). Denote moreover s and η the closed point and the generic point of S; let z be a closed point of Y_s, for example a rational point $k = \boldsymbol{k}(s)$, and let X be the Y-prescheme obtained by blowing up the point z. As in the proof of (15.1.6), if $f : X \to Y$ is the structural morphism, $f^{-1}(z)$ in X_s is isomorphic to the complement of Z_1, so $Z_1 = f^{-1}(z)$ and Z_2, the closure in X_s of the complement of Z_1, are the two irreducible components of X_s. On the other hand, f is evidently proper and $g = h \circ f$ is flat, since X is integral $(\mathbf{II}, 8.1.4)$ and for every affine open U of X, $\Gamma(U, \mathscr{O}_X)$ is a torsion-free A-module, hence flat since A is a discrete valuation ring $(\mathbf{0}, 6.3.4)$. It is clear that $f_\eta : X_\eta \to Y_\eta$ is an isomorphism, but f_s is not an isomorphism, although $f_s : X_s \to Y_s$ is not reduced to a single point for the closed point z, f'' is not an isomorphism.

(ii) The conclusion of (21.12.15) also fails if, under the hypotheses, we suppress the hypothesis that f is *proper*. It suffices for this to see to define S and Y as in (i), but to replace X by the complement X'_2 of Z_2 in the prescheme X defined in (i), f by the restriction $f' : X' \to Y$ of f; the morphism $g' : X' \to S$, restriction of g, is again flat, and this time X'_2 is geometrically irreducible; X'_2 is moreover an affine scheme isomorphic to the complement of a closed point in $\mathbb{P}^1_k$; since its image under f'_s is not reduced to the closed point z, f' is not a proper morphism and f'_s is not an isomorphism, although f'_η is.

(iii) It is possible that the statement of the proposition (21.12.15) remains valid when we replace the word "isomorphism" by "étale morphism" (cf. (21.12.14), (v)). It will then also be the same of (21.12.16).

21.13. Parafactorial pairs. Parafactorial local rings

Definition (21.13.1). — Let X be a ringed space, Y a closed subset of X, $U = X - Y$. We say that the pair (X, Y) is *parafactorial* if, for every open V of X, the restriction functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$, from the category of $\mathscr{O}_V$-Modules invertibles into that of $\mathscr{O}_{U\cap V}$-Modules invertibles, is an *equivalence of categories*.

To say that the pair (X, Y) is parafactorial means therefore that, for every open V of X, the following conditions are satisfied:

label=1º the functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$ is *fully faithful*, in other words, for two $\mathscr{O}_V$-Modules invertibles $\mathscr{L}$, $\mathscr{L}'$, the map of restriction

$$\mathrm{Hom}_{\mathscr{O}_V}(\mathscr{L}, \mathscr{L}') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{U\cap V}}(\mathscr{L}|(U \cap V), \mathscr{L}'|(U \cap V))$$

is *bijective*;

lbbel=2º the functor $\mathscr{L} \mapsto \mathscr{L}|(U \cap V)$ is *essentially surjective*, in other words, for every $\mathscr{O}_{U\cap V}$-Module invertible $\mathscr{L}_0$, there exists an $\mathscr{O}_V$-Module invertible $\mathscr{L}$ such that $\mathscr{L}_0$ is isomorphic to $\mathscr{L}|(U \cap V)$; this is again expressed by saying that the canonical homomorphism (21.3.2.4) **IV-4 | 314**

$$\mathrm{Pic}(V) \longrightarrow \mathrm{Pic}(U \cap V)$$

is *surjective*.

Lemma (21.13.2). — *Let $f : X' \to X$ be a morphism of ringed spaces; for every open V of X, denote $f_V : f^{-1}(V) \to V$ the restriction of f. The following conditions are equivalent:*

label=a) For every open V of X, the functor $\mathscr{E} \mapsto f_V^*(\mathscr{E})$ from the category of $\mathscr{O}_V$-Modules locally free of finite rank, into the category of $\mathscr{O}_{f^{-1}(V)}$-Modules locally free of finite rank, is faithful (resp. fully faithful).

a′) For every open V of X, the functor $\mathscr{L} \mapsto f_V^*(\mathscr{L})$ from the category of $\mathscr{O}_V$-Modules locally free of rank 1 into the category of $\mathscr{O}_{f^{-1}(V)}$-Modules locally free of rank 1 is faithful (resp. fully faithful).

lbbel=b) For every open V of X, the canonical homomorphism ($\mathbf{0}_I$, 4.4.3.2) $\mathscr{E} \to (f_V)_*(f_V^*(\mathscr{E}))$ is injective (resp. an isomorphism) for every $\mathscr{O}_V$-Module locally free of finite rank $\mathscr{E}$.

b′) The canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is a monomorphism (resp. an isomorphism) for every $\mathscr{O}_V$-Module locally free of finite rank $\mathscr{E}$.

Suppose that the canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is bijective. Then, for every $\mathscr{O}_{X'}$-Module locally free of finite rank $\mathscr{F}$, $\mathscr{F}$ is of the form $f^*(\mathscr{E})$, where $\mathscr{E}$ is an $\mathscr{O}_X$-Module locally free of finite rank, if and only if the two following conditions are satisfied:

label=(i) $f_*(\mathscr{E}')$ is an $\mathscr{O}_X$-Module locally free;

lbbel=(ii) the canonical homomorphism ($\mathbf{0}_I$, 4.4.3.3) $f^*(f_*(\mathscr{E}')) \to \mathscr{E}'$ is an isomorphism.

When these two conditions are satisfied, $\mathscr{E}$ is isomorphic to $f_*(\mathscr{E}')$.

Proof. To say that the functor $\mathscr{E} \mapsto (j_V)_*(\mathscr{E}|(U \cap V))$ is injective (resp. bijective) for every open V of X, the canonical homomorphism $\mathscr{E} \to (j_V)_*(j_V^*(\mathscr{E}))$ is injective (resp. bijective); since this must take place in replacing V by an open $W \subset V$ and $\mathscr{E}_1$, $\mathscr{E}_2$ by $\mathscr{E}_1|W$, $\mathscr{E}_2|W$, and since $\mathrm{Hom}_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2|W) = \Gamma(W, \mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2))$, the condition also means that the canonical homomorphism of sheaves **IV-4 | 315**

$$\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2) \longrightarrow (f_V)_*(f_V^*(\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2))) \tag{21.13.2.1}$$

is injective (resp. bijective). But since $\mathscr{E}_1$ and $\mathscr{E}_2$ are locally free of finite rank, $\mathscr{H}om_{\mathscr{O}_V}(\mathscr{E}_1, \mathscr{E}_2)$, isomorphic to $\mathscr{E}_1 \otimes_{\mathscr{O}_V} \mathscr{E}_2$ ($\mathbf{0}_I$, 5.4.2) is also locally free of finite rank; this already shows that b) implies a), and b') is a particular case of a'). It is clear that a') is a particular case of a). Conversely, if b) is a local property, it can, for the verification, be restricted to the case where $\mathscr{E} = \mathscr{O}_V^n$, and this shows that b') implies b).

If the canonical homomorphism $\mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ is bijective, and if the conditions (i) and (ii) are satisfied, it is clear that $\mathscr{E}' = f^*(\mathscr{E})$ with $\mathscr{E} = f_*(\mathscr{E}')$, up to canonical isomorphism. □

In this paragraph, we will apply the preceding lemma to a canonical injection $j : \mathrm{U} \to \mathrm{X}$, where U is the ringed space induced by X on an *open* of X. With these notations, (21.13.2) translates as:

Corollary (21.13.3). — *Let X be a ringed space, Y a closed subset of X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ the canonical injection. The following conditions are equivalent:*

label=a) *For every open V of X, the functor restriction $\mathscr{E} \mapsto \mathscr{E}|(\mathrm{U} \cap \mathrm{V})$ from the category of $\mathscr{O}_{\mathrm{V}}$-Modules locally free of finite rank into the category of $\mathscr{O}_{\mathrm{U}\cap\mathrm{V}}$-Modules locally free of finite rank is faithful (resp. fully faithful).*

lbbel=b) *For every open V of X, the canonical homomorphism $\mathscr{E} \to (j_{\mathrm{V}})_*((\mathscr{E}|(\mathrm{U} \cap \mathrm{V}))$ is injective (resp. bijective) for every $\mathscr{O}_{\mathrm{V}}$-Module locally free of finite rank $\mathscr{E}$.*

b′) *The canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is injective (resp. bijective).*

Suppose that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is bijective. Then, for an $\mathscr{O}_{\mathrm{U}}$-Module locally free of finite rank $\mathscr{F}$ to be of the form $\mathscr{E}|\mathrm{U}$, where $\mathscr{E}$ is an $\mathscr{O}_{\mathrm{X}}$-Module locally free of finite rank, it is necessary and sufficient that $j_*(\mathscr{F})$ is an $\mathscr{O}_{\mathrm{X}}$-Module locally free of finite rank, and in this case, we can take $\mathscr{E} = j_*(\mathscr{F})$.

Lemma (21.13.4). — *Let X be a locally Noetherian prescheme, Y a closed subset of X. For the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ to be injective (resp. bijective), it suffices that $\mathrm{prof}_{\mathrm{Y}}(\mathscr{O}_{\mathrm{X}}) \geqslant 1$ (resp. $\mathrm{prof}_{\mathrm{Y}}(\mathscr{O}_{\mathrm{X}}) \geqslant 2$) (in other words, that $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$) for all $x \in \mathrm{Y}$.*

Proof. These assertions are indeed particular cases of (5.10.2) and (5.10.5). □

Proposition (21.13.5). — *Let X be a ringed space, Y a closed subset of X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ the canonical injection. For the pair (X, Y) to be parafactorial, it is necessary and sufficient that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is bijective, and that for every open V of X and every $\mathscr{O}_{\mathrm{U}\cap\mathrm{V}}$-Module invertible $\mathscr{L}$, $(j_{\mathrm{V}})_*(\mathscr{L})$ is an $\mathscr{O}_{\mathrm{V}}$-Module invertible (notation of (21.13.3)).*

Proof. This is an immediate consequence of the definition (21.13.1) and of (21.13.3). □

Corollary (21.13.6). — *Let X be a ringed space, Y a closed subset of X.*

label=(i) *If the pair (X, Y) is parafactorial, then so is $(\mathrm{W}, \mathrm{Y} \cap \mathrm{W})$ for every open W of X. Conversely, if (W_α) is an open covering of X such that each of the pairs $(\mathrm{W}_\alpha, \mathrm{Y} \cap \mathrm{W}_\alpha)$ is parafactorial, then the pair (X, Y) is parafactorial.*

lbbel=(ii) *If the pair (X, Y) is parafactorial, then so is $(\mathrm{X}, \mathrm{Y}')$ for every closed subset Y' of Y.*

lcbel=(iii) *Suppose that X is a prescheme, and let X' be a prescheme, $f : \mathrm{X}' \to \mathrm{X}$ a faithfully flat and quasi-compact morphism; set $\mathrm{Y}' = f^{-1}(\mathrm{Y})$. Suppose that the pair $(\mathrm{X}', \mathrm{Y}')$ is parafactorial and that the open $\mathrm{U} = \mathrm{X} - \mathrm{Y}$ is retrocompact in X* ($\mathbf{0}_{\mathrm{III}}$, 9.1.1). *Then the pair (X, Y) is parafactorial.*

Proof. (i) Since the fact that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is bijective is a local property on X, it all comes down to seeing (by virtue of (21.13.5)) by denoting by j_α the injection canonical $\mathrm{U} \cap \mathrm{W}_\alpha \to \mathrm{W}_\alpha$, that for all α and every $\mathscr{O}_{\mathrm{U}}$-Module invertible $\mathscr{L}$, $(j_\alpha)_*(\mathscr{L}|(\mathrm{U} \cap \mathrm{W}_\alpha))$ is a $\mathscr{O}_{\mathrm{W}_\alpha}$-Module invertible. But the property of being an $\mathscr{O}_{\mathrm{X}}$-Module invertible is local on X; since $j_*(\mathscr{L})|\mathrm{W}_\alpha = (j_\alpha)_*(\mathscr{L}|(\mathrm{U} \cap \mathrm{W}_\alpha))$, this proves our assertion. **IV-4 | 316**

(ii) Set $\mathrm{U}' = \mathrm{X} - \mathrm{Y}'$ and let $j'' : \mathrm{U}' \to \mathrm{X}$ be the canonical injection. To say that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j''_*(\mathscr{O}_{\mathrm{U}'})$ is bijective amounts to saying that for every open V of X, the homomorphism $\Gamma(\mathrm{V}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{V} \cap \mathrm{U}', \mathscr{O}_{\mathrm{X}})$ is bijective; but the homomorphism composed

$$\Gamma(\mathrm{V}, \mathscr{O}_{\mathrm{X}}) \longrightarrow \Gamma(\mathrm{V} \cap \mathrm{U}', \mathscr{O}_{\mathrm{X}}) \longrightarrow \Gamma(\mathrm{V} \cap \mathrm{U}, \mathscr{O}_{\mathrm{X}})$$

is bijective by hypothesis (by replacing V by $\mathrm{V} \cap \mathrm{U}'$, we see that $\Gamma(\mathrm{V} \cap \mathrm{U}', \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{V} \cap \mathrm{U}, \mathscr{O}_{\mathrm{X}})$ is bijective. Noting then that if $j'' : \mathrm{U}' \to \mathrm{X}$ is the canonical injection and if $\mathscr{L}'$ is an $\mathscr{O}_{\mathrm{U}}$-Module invertible, then $j''_*(\mathscr{L}')|\mathrm{U}$ is an $\mathscr{O}_{\mathrm{X}}$-Module invertible. If we denote by $\mathscr{L}_1$ the $\mathscr{O}_{\mathrm{V}_1}$-Module invertible equal to $\mathscr{L}''$ in W and to $\mathscr{L}'$ in $\mathrm{V} \cap \mathrm{U}'$, on concludes from the fact that $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is bijective and from (21.13.2), *b*)) that $j_*(\mathscr{L})|V_1$ is an $\mathscr{O}_{\mathrm{X}}$-Module invertible. It suffices then to replace X by an open V and X' by $f^{-1}(\mathrm{V})$ in what precedes.

(iii) Set $\mathrm{U}' = \mathrm{X}' - \mathrm{Y}' = f^{-1}(\mathrm{U})$ and note that since it is a question of preschemes, on can write $\mathrm{U}' = \mathrm{U} \times_{\mathrm{X}} \mathrm{X}'$; let $f_{\mathrm{U}} : f^{-1}(\mathrm{U}) \to \mathrm{U}$ be the restriction of f and let $j' : \mathrm{U}' \to \mathrm{X}'$ be the canonical injection, which we also write $j_{(\mathrm{X}')}$. Let us first show that the canonical homomorphism $\rho : \mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is bijective; for this, it is necessary that for every open V of X, the canonical homomorphism $\mathscr{O}_{\mathrm{X}',x'} \to j'_*(\mathscr{O}_{\mathrm{U}'})$ is bijective; since j is quasi-compact and separated by hypothesis, the morphism f is quasi-compact and separated, hence by (2.3.1), $\mathscr{O}_{X'} = f^*(\mathscr{O}_X)$, we see that ρ is bijective since $f^*(\rho)$ is bijective and f is faithfully flat (2.2.7). Let then $\mathscr{L}' = f_U^*(\mathscr{L})$ be a $\mathscr{O}_U$-Module invertible $\mathscr{L}$. By hypothesis $j'_*(\mathscr{L}') = j'_*(f_U^*(\mathscr{L}))$ is an $\mathscr{O}_{X'}$-Module invertible, which is isomorphic to $f^*(j_*(\mathscr{L}))$ by (2.3.1) as $\mathscr{O}_{X'} = f^*(\mathscr{O}_X)$; we conclude then that ρ is bijective since f is faithfully flat. Since f is faithfully flat and quasi-compact, this implies that $j_*(\mathscr{L})$ is an $\mathscr{O}_X$-Module locally free (2.5.2). For the proof, it suffices to replace X by an open V and X′ by $f^{-1}(V)$ in what precedes. □

Definition (21.13.7). — We say that a *local ring* A is parafactorial if, setting $X = \mathrm{Spec}(A)$ and denoting by a the closed point of X, the pair $(X, \{a\})$ is parafactorial.

Proposition (21.13.8). — *The notations being those of (21.13.7), set $U = X - \{a\}$. For A to be parafactorial, it is necessary and sufficient that it satisfies the following conditions:*

label=(i) The canonical homomorphism $A = \Gamma(X, \mathscr{O}_X) \to \Gamma(U, \mathscr{O}_X)$ is bijective.

lbbel=(ii) $\mathrm{Pic}(U) = 0$.

If furthermore A is Noetherian, condition (i) is equivalent to **IV-4** | 317

(i bis) $\mathrm{prof}(A) \geqslant 2$.

Proof. Indeed, the only open of X containing a is X itself, and hence every $\mathscr{O}_X$-Module invertible is isomorphic to $\mathscr{O}_X$, i.e. $\mathrm{Pic}(X) = 0$; the first assertion follows therefore from the definition (21.13.1) and from (21.13.3). The second assertion is nothing but a particular case of (21.13.4). □

(21.13.9). *Examples.* — (i) A local Noetherian parafactorial ring is necessarily of dimension $\geqslant 2$ by virtue of (21.13.8); in other words a local Noetherian ring of dimension $\leqslant 1$ is not parafactorial.

(ii) A local Noetherian *factorial* ring A of dimension $\geqslant 2$ is parafactorial, as follows from (21.13.8) and (21.6.14).

(iii) If A is a local Noetherian ring of dimension $\geqslant 3$ and parafactorial, it is not necessarily normal nor even reduced. Consider a regular local ring B of dimension $\geqslant 3$, and let $A = D_B(B)$ (**0**, 18.2.3), isomorphic to $B[T]/(T^2)$; we see immediately that $\mathrm{prof}(A) = \mathrm{prof}(B) \geqslant 3$. To show that A is parafactorial, it suffices therefore, with the notations of (21.13.8), to prove that $\mathrm{Pic}(U) = 0$. Let $\mathfrak{J}$ be the ideal of the augmentation $A \to B$, which is such that $\mathfrak{J}^2 = 0$ and which, as a B-module, is isomorphic to B. Since $B = A/\mathfrak{J}$, $X = \mathrm{Spec}(A)$ and $X_0 = \mathrm{Spec}(B)$ have the same underlying space; if $\mathscr{J} = \widetilde{\mathfrak{J}}$, X_0 is the sub-prescheme defined by $\mathscr{J}$; we will denote U_0 the prescheme induced by X_0 on the open U. For every $z \in \mathfrak{J}$, set $\varphi(z) = 1 + z$; since $(1+z)(1-z) = 1$, φ is the kernel of the surjective canonical homomorphism of groups of multiplicative groups $A^* \to B^*$; in other words, we have an exact sequence of commutative groups

$$0 \longrightarrow \mathfrak{J} \xrightarrow{\varphi} A^* \longrightarrow B^* \longrightarrow 1$$

(the three last groups being multiplicative, the two first being additive). For the same reason, for every $t \in A$, we have the exact sequence

$$0 \longrightarrow \mathfrak{J}_t \xrightarrow{\varphi_t} (A_t)^* \longrightarrow (B_{t_0})^* \longrightarrow 1$$

denoting by t_0 the image of t in B, since $B_t = A_t/\mathfrak{J}_t$; in other words, we have an exact sequence of sheaves of commutative groups on the topological space X

$$0 \longrightarrow \mathscr{J} \xrightarrow{u} \mathscr{O}_X^* \longrightarrow \mathscr{O}_{X_0}^* \longrightarrow 1$$

whence by restriction to the open U, an exact sequence

$$0 \longrightarrow \mathscr{J}|U \longrightarrow \mathscr{O}_U^* \longrightarrow \mathscr{O}_{U_0}^* \longrightarrow 1.$$

By the cohomology exact sequence, we deduce an exact sequence

(21.13.9.1) $$H^1(U, \mathscr{J}) \longrightarrow H^1(U, \mathscr{O}_U^*) \xrightarrow{u} H^1(U, \mathscr{O}_{U_0}^*).$$

But since $\mathfrak{J}$ is a B-module isomorphic to B, we have $H^1(U_0, \mathscr{O}_{U_0}) = H^1(U_0, \mathscr{O}_{U_0})$, and it follows from the chap. III, 3e partie (see also [?, III, Exemple III-1]) that we have $H^1(U_0, \mathscr{O}_{U_0}) = 0$ by reason of the **IV-4** | 318
relation $\mathrm{prof}(B) \geqslant 3$. On the other hand, since B is factorial, $H^1(U, \mathscr{O}_{U_0}^*) = \mathrm{Pic}(U_0) = 0$; we obtain therefore from the preceding exact sequence $\mathrm{Pic}(U) = H^1(U, \mathscr{O}_U^*) = 0$.

(iv) There also exist local Noetherian rings of dimension 3 which are integral, integrally closed and parafactorial, but not factorial. Let indeed B be a local Noetherian ring, complete, integral, integrally closed, of dimension $\geqslant 2$ and not factorial (for example the completion of the localised polynomial algebra $k[\mathrm{U},\mathrm{V},\mathrm{W}]/(\mathrm{U}^2-\mathrm{UV})$ at the ideal image of $(\mathrm{U})+(\mathrm{V})+(\mathrm{W})$). Then it follows from what we will see below (21.14.2) that the local ring $\mathrm{A}=\mathrm{B}[[\mathrm{T}]]$ of formal power series in T over B is parafactorial, but it is not factorial, without which B would be so by virtue of (21.13.12) below.

(v) One can show that an intersection complete local ring of absolute dimension (19.3.1) of dimension $\geqslant 4$, is parafactorial (cf. [?, XI, 3.13 (i)]).

(vi) We have seen (Remark (ii)) that every local Noetherian factorial ring A of dimension 2 is parafactorial. But there are local Noetherian rings of dimension 2 which are parafactorial and not factorial. One can show that, for an arbitrary local Noetherian ring A of dimension 2 to be factorial and not parafactorial, it is necessary and sufficient that the following three conditions be satisfied:

label=1° A is a Cohen-Macaulay ring (in other words $\operatorname{prof}(\mathrm{A})=2$).

lbbel=2° A is integral and if A' is its integral closure, A' is factorial and is a finite A-algebra.

lcbel=3° Let $\mathfrak{J}$ be the conductor of A in A', or equivalently the largest ideal of A' contained in A); set $\mathrm{B}=\mathrm{A}/\mathfrak{J}$, $\mathrm{B}'=\mathrm{A}'/\mathfrak{J}$; then $\dim(\mathrm{B})=1$ (which implies $\mathrm{A}'\neq\mathrm{A}$, i.e. A is not integrally closed), and the canonical map $\operatorname{Div}(\mathrm{B})\to\operatorname{Div}(\mathrm{B}')$ (21.4.5) is *surjective*.

One can furthermore show that these conditions imply the following property: 4° the ring B' is also reduced, and the morphism $\operatorname{Spec}(\mathrm{B}')\to\operatorname{Spec}(\mathrm{B})$ is bijective.

If we set $\mathrm{X}=\operatorname{Spec}(\mathrm{A})$, $\mathrm{X}'=\operatorname{Spec}(\mathrm{A}')$, $\mathrm{Y}=\operatorname{Spec}(\mathrm{B})$, $\mathrm{Y}'=\operatorname{Spec}(\mathrm{B}')$, so that Y (resp. Y') is defined by the ideal $\mathfrak{J}$ of A (resp. A'), the structural morphism $f:\mathrm{X}'\to\mathrm{X}$ is an isomorphism on $\mathrm{X}'-\mathrm{Y}'$ over $\mathrm{X}-\mathrm{Y}$ since f is *bijective* and in other words, X is a *local* Noetherian *unibranch* prescheme (and in particular A' is a *local* Noetherian ring); in general X is not geometrically unibranch. The space Y, of dimension 1, is formed of the closed point x of X and the maximal points y_i ($1\leqslant i\leqslant r$) of Y, and the unique point y_i' of Y' above y_i is also a maximal point of Y'; since B and B' are reduced, we deduce from this that $\mathfrak{m}_{\mathscr{O}_{\mathrm{X}',y_i'}}=\mathfrak{m}_{y_i}$; for $z=y_i$ and z' the unique point of $f^{-1}(z)$, hence for all $z\in\mathrm{U}=\mathrm{X}-(x)$ and all z' the unique point of $f^{-1}(z)$; this relation is also verified obviously for $z\notin\mathrm{Y}$ and z' the unique point of $f^{-1}(z)$ (which is not characteristic 0 (21.1.1)) on U (but *not* étale in general).

We will now limit ourselves to demonstrating that conditions 1°, 2°, 3° above are *sufficient* for A to be parafactorial. Since A' is factorial by virtue of condition 1° and (21.13.8), it suffices to show that $\operatorname{Pic}(\mathrm{U})=0$. Since A' is factorial, by virtue of 2°, we obtain and we deduce from (21.8.5), (ii) an exact sequence **IV-4 | 319**

$$1\longrightarrow\Big(\prod_{s\in\mathrm{S}}\mathscr{O}^*_{\mathrm{U},s}/\operatorname{Im}\Gamma(\mathrm{U}',\mathscr{O}^*_{\mathrm{U}'})\Big)\longrightarrow\operatorname{Pic}(\mathrm{U})\longrightarrow\operatorname{Pic}(\mathrm{U}')\longrightarrow 0$$

and it is a matter of showing that the second term of this exact sequence is reduced to the neutral element, i.e. for the set y_i ($1\leqslant i\leqslant r$) of A and A' corresponding to points y_i, y_i', and we note that $\mathfrak{p}_i\cap\mathrm{A}=\mathfrak{p}_i$; denoting for each i an element inversible a_i'/s_i of $\mathrm{A}'_{\mathfrak{p}_i}$, with $a_i'\in\mathrm{A}'$, $s_i\notin\mathfrak{p}_i$; since we have only to examine that the quotient groups $\mathrm{A}^*_{\mathfrak{p}_i}/\mathrm{A}^*_{\mathfrak{p}_i}$ are the same; we can suppose $s_i=1$ for all i; let $b_i'\in\mathrm{B}_i'$ be the canonical image of a_i', so that $b_i'/1$ is an element $\neq 0$ of $\boldsymbol{k}(y_i')=\mathscr{O}_{v',y_i'}$. The open $\mathrm{U}\cap\mathrm{Y}$ formed by the y_i is an open affine of the form $\mathrm{D}(t)$ in Y, with $t\in\mathfrak{m}$, the maximal ideal of A; it exists therefore an invertible element $b'\in\mathrm{B}'$ such that the canonical images of $b_i'/1$ are the images in $\mathrm{A}'_{\mathfrak{p}_i}/\mathrm{A}_{\mathfrak{p}_i}$ are the same, which finishes the proof.

Proposition (21.13.10). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X*. For the pair* (X, Y) *to be parafactorial, it is necessary and sufficient that, for all* $y\in\mathrm{Y}$*, the local ring* $\mathscr{O}_{\mathrm{X},y}$ *is parafactorial.*

Proof. Set $\mathrm{U}=\mathrm{X}-\mathrm{Y}$ and let $j:\mathrm{U}\to\mathrm{X}$ be the canonical injection. **IV-4 | 320**

Let us first show that if the pair (X, Y) is parafactorial, each of the rings $\mathscr{O}_{\mathrm{X},y}$ ($y\in\mathrm{Y}$) is parafactorial. Taking into account the preceding remark, it all amounts to seeing that if $\mathrm{T}_y=\operatorname{Spec}(\mathscr{O}_{\mathrm{X},y})$ and $\mathrm{U}_y=\mathrm{T}_y-\{y\}$, every $\mathscr{O}_{\mathrm{U}_y}$-Module invertible $\mathscr{L}_0$ is isomorphic to $\mathscr{O}_{\mathrm{U}_y}$. Now, when V ranges over the open neighbourhoods of y in X, it follows from (8.2.13) and (1.2.4.2) that the prescheme U_y is projective limit of the preschemes induced by X on the opens $\mathrm{V}-(\mathrm{V}\cap\overline{\{y\}})$, which are quasi-compact since X is Noetherian. Since the $U_y = V - (V \cap \overline{\{y\}})$ are separated, it follows then from (8.5.2), (ii) and (8.5.5) that there is a neighbourhood open affine V of y in X and an $\mathscr{O}_V$-Module invertible $\mathscr{L}'_V$ such that the sheaf induced by $j_*(\mathscr{L})$ on U_y is $\mathscr{L}_0$. But $\mathscr{O}_{X,y}$ is parafactorial, so the pair $(V, V \cap \overline{\{y\}})$ is parafactorial by (21.13.6), (i)), so there exists an $\mathscr{O}_V$-Module invertible $\mathscr{L}_V$ inducing $\mathscr{L}'_V$ on U_y. But as T_y is projective limit, by replacing V by a smaller neighbourhood open of y, one can suppose that $\mathscr{L}_V$ is isomorphic to $\mathscr{O}_V$, whence $\mathscr{L}_0$ is isomorphic to $\mathscr{O}_{U_y}$, hence invertible. But this contradicts the definition of V, and concludes the proof of (21.13.10).

Conversely, let us prove that if all the $\mathscr{O}_{X,y}$ $(y \in Y)$ are parafactorial, the pair (X, Y) is parafactorial. This is evidently reduced, taking into account the remark at the beginning, to showing that for every $\mathscr{O}_U$-Module invertible $\mathscr{L}$, $j_*(\mathscr{L})$ is an $\mathscr{O}_X$-Module invertible. The set of $x \in X$ such that the restriction of $j_*(\mathscr{L})$ to a neighbourhood open of x in X is invertible is evidently open in X; it is a matter of showing that $V = X$. For this, suppose the contrary and set $Z = X - V \neq \varnothing$. Let y be a *maximal* point of Z; since $Z \subset Y$, $\mathscr{O}_{X,y}$ is by hypothesis a parafactorial ring. Replacing X if needed by a neighbourhood open of y, we can suppose that the restriction of $j_*(\mathscr{L})$ to $W - (W \cap \overline{\{y\}})$ is equal. If we set $V_1 = V \cup W$ and if we denote by $\mathscr{L}_1$ the $\mathscr{O}_{V_1}$-Module equal to $j_*(\mathscr{L})|W$ in W, $\mathscr{L}_V$ in V, we have $U \subset V_1$ and $\mathscr{L}_1|U = \mathscr{L}$. We conclude from the fact that $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective and from (21.13.2), *b*)) that $j_*(\mathscr{L})|V_1$ is an $\mathscr{O}_X$-Module invertible. But this contradicts the definition of V, and concludes the proof of (21.13.10). □

Corollary (21.13.11). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X, $U = X - Y$. *Suppose that the pair* (X, Y) *is parafactorial and that* U *is locally factorial; in other terms* (21.13.10), *for all* $x \in X$, *the local ring* $\mathscr{O}_{X,x}$ *is:* 1° *parafactorial if* $x \in Y$; 2° *factorial if* $x \notin Y$. *Then* X *is locally factorial (in other words,* $\mathscr{O}_{X,x}$ *is in fact factorial for* all $x \in X$).

Proof. Suppose indeed the contrary, and let y be a point of Y such that $\mathscr{O}_{X,y}$ is not factorial and has a minimal dimension among all the points of Y having this property. Then, if we set $T_y = \mathrm{Spec}(\mathscr{O}_{X,y})$, $U_y = T_y - \{y\}$, the hypothesis that $\mathscr{O}_{X,y}$ is parafactorial implies $\mathrm{Pic}(U_y) = 0$ and $\mathrm{prof}(\mathscr{O}_{X,y}) \geqslant 2$ (21.13.8); moreover by the choice of y every U_y is locally factorial, contradicting the hypothesis. But then (21.6.14) proves that $\mathscr{O}_{X,y}$ is factorial, contradicting the hypothesis. □

Proposition (21.13.12). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism making* B *a flat A-module. Then, if* B *is factorial, so is* A.

Proof. We already know that A is integral and integrally closed (6.5.4), so we can restrict ourselves by induction on $n = \dim(A)$. Let $X = \mathrm{Spec}(A)$, a the closed point of X, $X' = \mathrm{Spec}(B)$, $f : X' \to X$ the structural morphism, $U = X - \{a\}$, $U' = f^{-1}(U)$. The local rings $\mathscr{O}_{X',x'}$ at points $x' \in f^{-1}(a)$ are factorial and of dimension $\geqslant 2$ (6.1.2), hence parafactorial (21.13.9), (ii)); we conclude (21.13.10) that the pair $(X', f^{-1}(a))$ is parafactorial. By virtue of (21.13.6), (iii), this shows that the ring A is parafactorial, hence $\mathrm{prof}(A) \geqslant 2$ and $\mathrm{Pic}(U) = 0$ (21.13.8). Finally, the local rings of U being of dimension $< n$, the induction hypothesis proves that U is *locally factorial*; the conclusion follows then from (21.6.14). □

(21.13.12.1). We note that the statement of (21.13.12) where one replaces "factorial" by "parafactorial" is no longer exact, as shown by the example where $A = k$ is a field and B the local factorial ring $k[[T_1, T_2]]$. However, it follows from (21.13.6), (iii) that, under the conditions of (21.13.12), if $\mathfrak{m}$ denotes the maximal ideal of A and if $\mathfrak{m}B$ is an *ideal of definition* of B (i.e. $\dim(B/\mathfrak{m}B) = 0$), then, when B is parafactorial, so is A. In particular, if the completion $\widehat{A}$ of a local Noetherian ring is parafactorial, so is A. IV-4 | 321

Remark (21.13.13). — Part of the definitions and results above relates to more general considerations on the cohomology of sheaves of commutative groups, developed in chap. III, 3[e] partie, and which we have given here an exposition independent for the convenience of the reader. Indeed, if X is a ringed space, there is a canonical *equivalence* between the category of the $\mathscr{O}_X$-Modules invertibles and that of *principal homogeneous sheaves under the sheaf of groups* $\mathscr{O}_X^*$ (16.5.15). This equivalence is defined by making correspond to every $\mathscr{O}_X$-Module invertible $\mathscr{L}$ the sheaf $\mathscr{L}^* = \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{L})$ of germs of isomorphisms of $\mathscr{O}_X$ onto $\mathscr{L}$; it is immediate that $\mathscr{L}^*$ is canonically equipped with a structure of principal homogeneous sheaf under $\mathscr{O}_X^* \cong \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{O}_X)$. The verification of this equivalence of categories is immediate. This being said, consider in general on a topological space X and a sheaf of groups $\mathscr{G}$, and let Y be a closed subset of X, and let i be an integer such that $0 \leqslant i \leqslant 2$; we say that the pair (X, Y) is *i-pure* for $\mathscr{G}$ if, for every open V of X, the restriction functor $\mathscr{P} \mapsto \mathscr{P}|(U \cap V)$, from the category of principal sheaves homogeneous under the sheaf of groups $\mathscr{G}|V$, into the analogous category under $\mathscr{G}|(U \cap V)$, is *faithful* ($i = 0$), resp. *fully faithful* ($i = 1$), resp. an *equivalence of categories* ($i = 2$). In the case where X is a ringed space and $\mathscr{G} = \mathscr{O}_X^*$, to say that (X, Y) is 0-pure therefore means for $i = 0$, that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is *injective*; for $i = 1$, that this homomorphism is *bijective*; and finally, for $i = 2$, that the pair (X, Y) is *parafactorial* (21.13.3).

Returning to the general case, recall that the morphisms of the category of principal sheaves are by definition isomorphisms. To say that the pair (X, Y) is 0-pure means therefore that for every open V of X, the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is injective; to say that the pair (X, Y) is 1-pure means that the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is bijective (and so also the canonical homomorphism $H^1(V, \mathscr{G}) \to H^1(U \cap V, \mathscr{G})$ is injective); finally, one can show that for the pair (X, Y) to be 2-pure, it is necessary and sufficient that the homomorphisms $H^i(V, \mathscr{G}) \to H^i(U \cap V, \mathscr{G})$ be bijective for $i = 0$ and $i = 1$. When $\mathscr{G}$ is a sheaf of commutative groups, by introducing the cohomology sheaves $\mathscr{H}_Y^i(\mathscr{G})$ defined in chap. III, 3^e partie, saying that (X, Y) is i-pure for $i \leqslant 2$ means that $\mathscr{H}_Y^i(\mathscr{G}) = 0$ for $p \leqslant i$; under this form, the notion generalises immediately for every integer i.

Proposition (21.13.14). — *Let* X *be a locally Noetherian* normal *prescheme,* $(U_\lambda)_{\lambda \in L}$ *a decreasing filtered family of opens of* X*. The following conditions are equivalent:*

label=a) *Every* 1*-codimensional cycle* Z *on* X *such that there exists an index* λ *for which* $Z|U_\lambda$ *is locally principal* (21.6.7)*, is locally principal.*

lbbel=b) *For all* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) \geqslant 2$ *and such that* x *does not belong to* $\bigcap_{\lambda \in L} U_\lambda$*, the local ring* $\mathscr{O}_{X,x}$ *is parafactorial.*

Proof. It is clear that the set S′ of X generisations of points of S is the intersection of the open neighbourhoods of S. We can restrict to the case where X is Noetherian (the properties a) and b) being local on X); as every 1-codimensional cycle on X which is principal at the points of S is also principal at the points of a neighbourhood open of S, we see that condition a) of (21.13.15) amounts to condition a) of (21.13.14) applied to the family (U_λ) of open neighbourhoods of S. The corollary follows then from the equivalence of a) and b) in (21.13.14). **IV-4** | 322

Let us note on the other hand that b') implies b) by virtue of (21.13.10); conversely, if b') is satisfied and if $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and one has $\operatorname{codim}(Y, X) \geqslant 2$, hence $\mathscr{O}_{X,x}$ is parafactorial, which proves the equivalence of a') and b').

Let us prove that a) implies c), i.e. a') implies b''). First we prove that a'') implies b''). Let us note first that b') implies b) by virtue of (21.13.10); conversely, if b') is satisfied and if $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and one has $\operatorname{codim}(Y, X) \geqslant 2$; if we note that $U = X - Y$, $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective (21.13.2), hence condition a) of (21.13.15) amounts to the condition a) of (21.13.14).

a'') For every 1-codimensional cycle Z on X which is principal at the points of S of X, Z is locally principal.

b'') The canonical homomorphism $\operatorname{Pic}(X) \to \operatorname{Pic}(U)$ is surjective.

Let us first prove that a) implies c), since if the 1-codimensional cycle Z on X has its support in one of the sets $X - U_\lambda$, $Z|U_\lambda = 0$, hence $Z|U_\lambda = \operatorname{cyc}(D_\lambda)$, where D_λ is a divisor on U_λ; since X is normal, $Z|U_\lambda$ is locally principal by (21.6.10), (i), and the hypothesis implies that Z is locally principal on X.

Inversely c) implies a), since if Z is a 1-codimensional cycle on X such that $Z|U_\lambda$ is locally principal, the image of $Z|U$ in $\mathfrak{Cl}(U)$ is therefore the image of a unique element $\zeta_0 \in \operatorname{Pic}(U)$ (21.6.11); it follows from (21.6.11) that the image of Z in $\mathfrak{Cl}(X)$ is the image of a unique element $\zeta \in \operatorname{Pic}(X)$, and it is clear then that ζ_0 is the image of ζ. □

Corollary (21.13.15). — *Let* X *be a locally Noetherian normal prescheme,* S *a subset of* X*. The following conditions are equivalent:*

label=a) *Every* 1*-codimensional cycle on* X *which is principal at the points of* S *is locally principal.* **IV-4** | 323 lbbel=b) *For all* $x \in \mathrm{X}$ *which is not a generisation of a point of* S *(in other words, such that* $\{x\} \cap \mathrm{S} = \varnothing$*) and such that* $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$*, the local ring* $\mathscr{O}_{\mathrm{X},x}$ *is parafactorial.*

Remark (21.13.16). — Under the general conditions of (21.13.14), suppose moreover $\varinjlim \mathrm{Pic}(\mathrm{U}_\lambda) = 0$. Then condition *a*) of (21.13.14) is also equivalent to the following:

c) *Every* 1*-codimensional cycle on* X *whose support is contained in one of the sets* $\mathrm{X} - \mathrm{U}_\lambda$ *is locally principal.*

We can restrict to the case where X is irreducible and all the U_λ are non-empty. It is clear that *a*) implies *c*), since if the 1-codimensional cycle Z on X has its support in $\mathrm{X} - \mathrm{U}_\lambda$, $\mathrm{Z}|\mathrm{U}_\lambda = 0$, hence $\mathrm{Z}|\mathrm{U}_\lambda$ is locally principal. Inversely *c*) implies *a*): let Z be a 1-codimensional cycle on X such that $\mathrm{Z}|\mathrm{U}_\lambda$ is locally principal; since X is normal, $\mathrm{Z}|\mathrm{U}_\lambda = \mathrm{cyc}(\mathrm{D}_\lambda)$, where D_λ is a divisor on U_λ. If $\mathrm{D}_\mu = \mathrm{div}(f_\mu)$, f_μ is a regular rational function on U_μ; the hypothesis implies (by virtue of (21.3.4)) that there is a set $\mathrm{U}_\mu \subset \mathrm{U}_\lambda$ such that $\mathrm{D}_\mu|\mathrm{U}_\mu = \mathrm{D}_\lambda|\mathrm{U}_\mu$ is equivalent to 0. If $\mathrm{D}_\mu = \mathrm{div}(f_\mu)$, f_μ may be considered as a regular rational function on X, regular on U_μ; hence $\mathrm{Z}' = \mathrm{cyc}(\mathrm{div}(f_\mu))$, we see therefore that $\mathrm{Z}'' = \mathrm{Z} - \mathrm{Z}'$ has its support in $\mathrm{X} - \mathrm{U}_\mu$; by hypothesis, Z'' is locally principal, whence the conclusion.

The condition $\varinjlim \mathrm{Pic}(\mathrm{U}_\lambda) = 0$ will in particular be satisfied if (U_λ) is the family of open neighbourhoods of a point $z \in \mathrm{X}$, since then U_λ is isomorphic to $\mathscr{O}_{\mathrm{U}_\lambda}$; there exists $\mathrm{U}_\mu \subset \mathrm{U}_\lambda$ such that $\mathscr{L}_\lambda|\mathrm{U}_\mu \cong \mathscr{O}_{\mathrm{U}_\mu}$. We see therefore that, in (21.13.15), if $\mathrm{S} = \{z\}$, conditions *a*) and *b*) are equivalent also to the following:

c) *Every* 1*-codimensional cycle on* X *whose support does not contain* z *is locally principal.*

If moreover $\mathrm{Pic}(\mathrm{X}) = 0$, condition *a*) of (21.13.15) is also equivalent to the following:

c) Every 1-codimensional cycle on X whose support is contained in one of the sets $\mathrm{X} - \mathrm{U}_\lambda$ is locally principal.

21.14. The Ramanujam-Samuel theorem

Theorem (21.14.1). — (Ramanujam-Samuel) — *Let* A *be a local Noetherian ring with maximal ideal* $\mathfrak{m}$, **IV-4** | 323
such that its completion $\widehat{\mathrm{A}}$ *is integral and integrally closed. Let* B *be a local Noetherian ring and* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism making* B *a smooth A-algebra (for the preadic topologies* (**0**, 19.3.1)*) and such that the residue field of* B *is finite over that of* A*. Then every* 1*-codimensional cycle on* $\mathrm{Spec}(\mathrm{B})$ *which is principal at the point* $\mathfrak{p} = \mathfrak{m}\mathrm{B}$*, is a principal* 1*-codimensional cycle.*

Proof. If $k = \mathrm{A}/\mathfrak{m}$ is the residue field, $\mathrm{B}/\mathfrak{m}\mathrm{B}$ is a formally smooth k-algebra (for the preadic topolo- **IV-4** | 324
gies), and in particular integral, in other words $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ is indeed a prime ideal of B, which justifies the statement. It evidently all amounts to proving that every prime ideal $\mathfrak{q}$ of B not contained in $\mathfrak{m}\mathrm{B}$ is *principal*.

Let $\widehat{\mathrm{A}}$, $\widehat{\mathrm{B}}$ be the completions of A and B respectively, so that the maximal ideal of $\widehat{\mathrm{A}}$ is $\mathfrak{m}\widehat{\mathrm{A}}$; we know (**0**, 19.3.6) that $\widehat{\mathrm{B}}$ is a formally smooth $\widehat{\mathrm{A}}$-algebra for the adic topologies. Let k' be the residue field of $\widehat{\mathrm{B}}$, a finite extension of k; there is a local homomorphism $\widehat{\mathrm{A}} \to \mathrm{C}$, where C is a local Noetherian ring which is an $\widehat{\mathrm{A}}$-module *finite* (and *flat*) and $\mathrm{C}/\mathfrak{m}\mathrm{C}$ isomorphic to k' ($\mathbf{0}_{\mathrm{III}}$, 10.3.1); we conclude then from (7.5.1), (7.5.3) and (6.5.4), (ii) that C is *integral* and *integrally closed*. Furthermore, $\mathrm{D} = \widehat{\mathrm{B}} \otimes_{\mathbf{Z}} \mathrm{C}$ is a semi-local complete ring, a direct sum of complete local rings whose one, D_0, has for residue field k' (since $k' \otimes_k k'$ is isomorphic to k'). Since D is formally smooth over C for the preadic topologies; by $\mathrm{D}_0\,/\mathfrak{m}\mathrm{D}_0$ is a k'-algebra formally smooth of residue field k', which implies that it is a k'-isomorphic algebra of power series $k'[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ (**0**, 19.6.4); we conclude, by (**0**, 19.7.5), that D_0 is an algebra of formal power series $\mathrm{C}[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$, and hence *integral and integrally closed* (Bourbaki, *Alg. comm.*, chap. V, §1, n° 4, prop. 14). Since the morphisms $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\widehat{\mathrm{B}})$ and $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\mathrm{C})$ are faithfully flat, we deduce that B and $\widehat{\mathrm{B}}$ are also *integral and integrally closed*. We will thus prove the **IV-4** | 325
theorem in the particular case where A is *complete* and $\mathrm{B} = \mathrm{A}[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$. Let us first show that we can reduce to the case where $n = 1$. Indeed, we proceed by induction on n, and let $f \in \mathfrak{q}$ be an element not belonging to $\mathfrak{p} = \mathfrak{m}\mathrm{B}$; it is a formal power series whose reduced series $\overline{f}_0$ is nonzero; hence, by virtue of the preparation theorem (Bourbaki, *Alg. comm.*, chap. VII, §3, n° 8, prop. 5), for every $f \in \mathfrak{q}$, there exists $g \in \mathrm{B}$ and a polynomial $r \in \mathrm{A}[\mathrm{T}]$ such that $f = gf_0 + r$ and one has thus $r \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$; on the other hand (*loc. cit.*, prop. 6) there exists a distinguished non-constant polynomial $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$ and an invertible element $u \in \mathrm{B}$ such that $f_0 = u\mathrm{F}_0$, hence also $\mathrm{F}_0 \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$; since B is

flat over A[T] ($\mathbf{0}_{\mathrm{I}}$, 7.3.3), it follows from Bourbaki, *Alg. comm.*, chap. VII, §1, nº 10, prop. 15, that $\mathfrak{q}_1$ is a prime ideal of height 1 in A[T]. But on the other hand, we have necessarily $\mathfrak{q}_1 \cap \mathrm{A} = 0$; otherwise, $\mathfrak{q}_1 \cap \mathrm{A}$ would necessarily be of height $\geqslant 1$, and it would follow from (5.5.3) that $\mathfrak{q}_1 = (\mathfrak{q}_1 \cap \mathrm{A})\mathrm{A}[\mathrm{T}]$. But then, since $\mathfrak{q}_1 \subset \mathfrak{m}\mathrm{A}[\mathrm{T}]$, we would have $\mathfrak{q}_1 \subset \mathfrak{m}\mathrm{A}[\mathrm{T}]$, contradicting the hypothesis on $\mathfrak{q}$. If K is the field of fractions of A, $\mathfrak{q}_1\mathrm{K}[\mathrm{T}]$ is therefore a prime ideal distinct from 0 and of K[T], hence of the form $h \cdot \mathrm{K}[\mathrm{T}]$, where $h(\mathrm{T}) = \mathrm{T}^m + a_1\mathrm{T}^{m-1} + \ldots + a_m$ with $m \geqslant 1$ and the $a_i \in \mathrm{K}$, h being irreducible in K[T]. But we have seen above that there exists in $\mathfrak{q}_1$ a distinguished non-constant $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$. t is a root of F_0 in an extension of K and hence divides F_0 *in* K[T]; but since h and F_0 are unitary, the coefficients a_i of h are integers over A (Bourbaki, *Alg. comm.*, chap. V, §1, nº 3, prop. 11), hence belong to A since A is integrally closed.

The statement (21.14.1) is equivalent to the following: □

Corollary (21.14.2). — *Under the hypotheses of (21.14.1) on* A, B *and* ρ, *for every prime ideal* $\mathfrak{q}$ *of* B *not contained in* $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ *and such that* $\dim(\mathrm{B}_{\mathfrak{q}}) \geqslant 2$, *the ring* $\mathrm{B}_{\mathfrak{q}}$ *is parafactorial. In particular, if* $\dim(\mathrm{B} \otimes_{\mathrm{A}} k) > 0$ *(i.e.* $\dim \mathrm{B} > \dim \mathrm{A}$*)* $((\mathbf{0}, 19.7.1)$ *and* $(\mathbf{0}, 6.1.1)))$, *the ring* B *is parafactorial.*

Proof. Indeed, we saw in the proof of (21.14.1) that the conditions of the statement imply that B is integral and integrally closed; and the equivalence of the statements (21.14.1) and (21.14.2) follows then from (21.13.15) applied to $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ and $\mathrm{S} = \{\mathfrak{p}\}$. □

Proposition (21.14.3). — *Let* S *be a normal prescheme,* $f : \mathrm{X} \to \mathrm{S}$ *a smooth morphism.*

label=(i) If S *is locally Noetherian (hence also* X*), then, for all* $x \in \mathrm{X}$ *such that* $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ *and such that* x *is not a maximal point of its fibre* $f^{-1}(f(x))$, *the ring* $\mathscr{O}_{\mathrm{X},x}$ *is parafactorial. Every* **IV-4 | 326** 1*-codimensional cycle* Z *on* X *such that* Supp(Z) *contains no irreducible component of any fibre* X_s, $f^{-1}(s) = \mathrm{X}_s$, *is locally principal.*

lbbel=(ii) Let Y *be a closed subset of* X *containing no irreducible component of any fibre* X_s, $\mathrm{Y}_\eta = \mathrm{Y} \cap \mathrm{X}_\eta$ *is of codimension* $\geqslant 2$ *in* X_η. *Then the pair* (X, Y) *is parafactorial.*

Proof. We note that the conditions of (ii) are satisfied if Y is a closed subset of X such that, for all **IV-4 | 327** $s \in \mathrm{S}$, $\mathrm{Y}_s = \mathrm{Y} \cap \mathrm{X}_s$ is of codimension $\geqslant 2$ in X_s.

To prove (21.14.3), let us note first that under the hypotheses of (i), if we set $\mathrm{Y} = \overline{\{x\}}$, it suffices to verify the conditions of (ii); and we have $\mathrm{Y} \cap \mathrm{X}_s$ is of codimension $\geqslant 2$ in X_s for every s by the hypotheses on x.

For the proof of (i), we first show that $\mathscr{O}_{\mathrm{X},x}$ is parafactorial; since X is normal (17.5.7), hence A is integral and integrally closed, and since $\mathscr{O}_{\mathrm{X},x}$ is a formally smooth $\mathscr{O}_{\mathrm{S},s}$-algebra for the preadic topologies (17.5.3); since Y_s contains no irreducible component of X_s, and since Y_s is of codimension $\geqslant 2$ in X_s, the prime ideal of $\mathscr{O}_{\mathrm{X},x}$ corresponding to the point x is not contained in $\mathfrak{m}_s\mathscr{O}_{\mathrm{X},x}$ since Y contains no irreducible component of X_s. Furthermore, $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant \dim(\mathscr{O}_{\mathrm{S},s}) + 2 \geqslant 2$, so we can apply (21.14.2).

For the proof of (ii), since the conditions and the conclusion are local on S and on X by virtue of (21.13.6), (i), we can restrict to the case where S and X are affine, f is therefore of finite presentation. Since the fibres of f are regular, the hypothesis on Y implies that, for every point $x \in \mathrm{Y}$, $\mathrm{prof}(\mathscr{O}_{\mathrm{X}_{f(s)},x}) = \dim(\mathscr{O}_{\mathrm{X}_{f(s)},x}) \geqslant 1$; it follows from (19.9.8) that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ is *injective*. Furthermore, replacing Y by a larger closed subset following the method of the proof of (19.9.8), and using (21.13.6), (ii), we can suppose that Y is defined by an ideal of *finite type* of the Picard scheme of X, hence the open $\mathrm{U} = \mathrm{X} - \mathrm{Y}$ is *quasi-compact and quasi-separated* and the closed set Y is *constructible*.

To prove (ii), it suffices, for every $\mathscr{O}_{\mathrm{U}}$-Module invertible $\mathscr{L}$ and every point $x \in \mathrm{Y}$, to establish the existence of a neighbourhood open V of x in X such that: 1º the canonical homomorphism $\mathscr{O}_{\mathrm{V}} \to (j_{\mathrm{V}})_*(\mathscr{O}_{\mathrm{U}\cap\mathrm{V}})$ is surjective; 2º there exists an $\mathscr{O}_{\mathrm{V}}$-Module invertible $\mathscr{L}'_{\mathrm{V}}$ such that $\mathscr{L}'_{\mathrm{V}}|\mathrm{U} \cap \mathrm{V} = \mathscr{L}|(\mathrm{U} \cap \mathrm{V})$ (and (21.13.3)).

Set $s = f(x)$; we will see that we can restrict to the case where $\mathrm{S} = \mathrm{S}_0 = \mathrm{Spec}(\mathscr{O}_{\mathrm{S},s})$. Let (S_ν) be a fundamental system of affine open neighbourhoods of s in S, and set $\mathrm{X}_\nu = f^{-1}(\mathrm{S}_\nu)$, $\mathrm{Y}_\nu = \mathrm{Y} \cap \mathrm{X}_\nu$, $\mathrm{U}_\nu = \mathrm{X}_\nu - \mathrm{Y}_\nu$, $j_\nu : \mathrm{U}_\nu \to \mathrm{X}_\nu$ being the canonical injection; $\mathrm{X}_0 = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}_0$ is projective limit of the X_ν and, denoting by $\mathrm{Y}_0 = \mathrm{Y} \cap \mathrm{X}_0$ and $\mathrm{U}_0 = \mathrm{X}_0 - \mathrm{Y}_0$, $j_0 : \mathrm{U}_0 \to \mathrm{X}_0$ the canonical injection, $p_\nu : \mathrm{X}_0 \to \mathrm{X}_\nu$ being the affine morphism, one has $j_*(\mathscr{O}_{\mathrm{U}}) = p_\nu^*((j_\nu)_*(\mathscr{O}_{\mathrm{U}_\nu}))$ (**II**, 1.5.2), and the canonical homomorphism $\rho : \mathscr{O}_X \to j_*(\mathscr{O}_U)$ is none other that $p_\nu^*(\rho_\nu)$, where ρ_ν is the canonical homomorphism. Since by hypothesis ρ_0 is surjective, there exists a sufficiently large ν and $\mathscr{L}_\nu$ such that $\mathscr{L}_0 = p_\nu^*(\mathscr{L}_\nu)$ is invertible. The proof finishes like that of the proposition, when A is a *local* ring: since X is normal and f smooth, S is normal (17.5.7), hence A is integral and integrally closed (17.5.7); since the local rings $\mathscr{O}_{S,s}$ are excellent integrally closed rings (**0**, 7.8.3, (ii)), (iii) and (vi)), A is an inductive limit of its sub-**Z**-algebras *of finite type*; by virtue of (1.8.4.2), there exists an index λ and an A_λ-algebra of finite type B_λ such that $B = B_\lambda \otimes_{A_\lambda} A$. By virtue of (1.8.4.2), there exists an index λ and an A_λ-isomorphism near. Set $S_\lambda = \operatorname{Spec}(A_\lambda)$, $X_\lambda = \operatorname{Spec}(B_\lambda)$; if $\rho_\lambda : X \to X_\lambda$ is the canonical projection, one has $j_*(\mathscr{O}_U) = \rho_\lambda^*((j_\lambda)_*(\mathscr{O}_{U_\lambda}))$; it all amounts to proving that $U = X - Y$ is retrocompact in X. Since the fibres of f are regular, the hypothesis on Y implies that for every point $x \in Y$, $\mathscr{O}_{X,x}$ is parafactorial by (i), and the result then follows from (21.13.10). □

(21.14.4). *Remarks.* — (i) It is possible that the statements (21.14.1) and (21.14.2) remain valid *without hypothesis* on the residue field of B. The statement with this generality would be equivalent, by virtue of (21.13.15) and (21.13.12.1) to the following: let A be a local Noetherian ring, complete, integral, integrally closed, B an arbitrary Noetherian ring which is a formally smooth A-algebra, such that $\dim(B) > \dim(A)$; then B is parafactorial. **IV-4** | 328

(ii) We will see in chap. VI, employing a technique of "finite descent" applied to (21.14.1) (or to (21.14.2)), that the conclusion of (21.14.1) holds valid by replacing the hypothesis that $\widehat{A}$ is *reduced*. If we replace this last condition by $\dim B \geqslant \dim A + 3$, we can even suppress the hypothesis that $\widehat{A}$ is reduced. Similarly, the conclusion of (21.14.3) remains valid by replacing the hypothesis that X is normal by the hypothesis that X is reduced, provided that $\dim(\mathscr{O}_{X,x}) \geqslant \dim(\mathscr{O}_{S,f(x)}) + 2$.

(iii) In chap. III, 3e partie, we prove that if $f : X \to S$ is a smooth morphism, Y a closed locally constructible subset of X such that, for all $s \in S$, with the notations of (21.14.3) $\operatorname{codim}(Y_s, X_s) \geqslant 3$, then the pair (X, Y) is parafactorial. This conclusion is no longer valid when we only suppose $\operatorname{codim}(Y_s, X_s) \geqslant 2$ for all $s \in S$ and when we do not suppose X reduced. For example, let k be a field, $A = k[T]/(T^2)$, $S = \operatorname{Spec}(A)$ (T_1, T_2 indeterminates), so that $X = \mathbb{V}_S^3$, Y being the "zero section"; if s is the unique closed point of S, on a $X_s = \mathbb{V}_k^2$ and Y_s is the "zero section" of X_s. To see that the pair (X, Y) is not parafactorial, it suffices to show that the ring $B = A[T_1, T_2]_{\mathfrak{m}}$, where $\mathfrak{m}$ is the ideal $(T_1) + (T_2)$ which defines Y, is not parafactorial (21.13.10). Let $B_0 = B_{\mathrm{red}}$, and let U and U_0 be the complementary open subsets of the closed point in $\operatorname{Spec}(B)$ and $\operatorname{Spec}(B_0)$; reasoning as in (21.13.9), we have the exact sequence, extension of (21.13.9.1)

$$\Gamma(U, \mathscr{O}_U)^* \longrightarrow \Gamma(U_0, \mathscr{O}_{U_0})^* \longrightarrow H^1(U_0, \mathscr{O}_{U_0}) \longrightarrow \operatorname{Pic}(U) \longrightarrow \operatorname{Pic}(U_0).$$

Now, one has $\Gamma(U, \mathscr{O}_U) = B$, $\Gamma(U_0, \mathscr{O}_{U_0}) = B_0$ since $\operatorname{prof}(B) = \operatorname{prof}(B_0) = 2$ (5.10.5). On the other hand $\operatorname{Pic}(U_0) = 0$ since B_0 is factorial, and $H^1(U_0, \mathscr{O}_{U_0}) \neq 0$ [**?**, 3, Exemple III-1]; since the homomorphism $B^* \to B_0^*$ is surjective, we conclude that $\operatorname{Pic}(U) \neq 0$, hence that B is not parafactorial.

IV-4 | 329

(21.14.4bis). (iv) The result of (21.14.3) was first proved by C. Seshadri [**?**] in the particular case where S is an algebraic normal scheme over an algebraically closed field k and $X = S \times_k T$, where T is a k-algebraic prescheme smooth over k. The proof of Seshadri [**?**, pp. 188–189] is of global nature and uses the theory of Picard schemes. It gives moreover (*loc. cit.*) results such as the following (for which one does not currently possess a proof by local methods). Let S, T be two preschemes locally of finite type over a field k, Z a 1-codimensional cycle on X (considered as S-prescheme); suppose the following conditions are satisfied:

label=1° S and T are geometrically normal over k (6.7.6);

lbbel=2° For every maximal point η of S, the 1-codimensional cycle Z_η on the fibre X_η, having even multiplicity that Z at every point of $X^{(1)} \cap X_\eta$, is locally principal (in other words, is the image of a divisor of X_η, since X_η is normal);

lcbel=3° For all $s \in S$, Z is principal at the maximal points of the fibre X_s.

Then Z is locally principal. In other words, X being normal (6.14.1), every 1-codimensional cycle Z on X such that for every maximal $x \in X$ which is not maximal in its fibre X_s and which does not belong to any of the "generic fibres" X_η (which implies $\dim(\mathscr{O}_{X,x}) \geqslant 2$ by virtue of (6.1.1)), the local ring $\mathscr{O}_{X,x}$ is parafactorial, by virtue of (21.12.15) [1].

21.15. Relative divisors

(21.15.1). Let S be a prescheme, $f : X \to S$ a *flat* morphism and locally of finite presentation. We defined in (20.6.1) the sheaf of rings $\mathscr{M}_{X/S}$ of the *germs of meromorphic functions on* X *relative to* S, sub-sheaf of $\mathscr{M}_X$; it is clear that the canonical injection $\mathscr{O}_X \hookrightarrow \mathscr{M}_X$ (20.1.4.1) maps $\mathscr{O}_X$ into a sub-sheaf of $\mathscr{M}_{X/S}$, with which we identify it. Let $\mathscr{M}^*_{X/S}$ be the sheaf (in multiplicative groups) of germs of invertible sections of $\mathscr{M}^*_X$ which is thus a sub-sheaf of $\mathscr{M}^*_X$ and contains $\mathscr{O}^*_X$ as sub-sheaf.

Definition (21.15.2). — We call *sheaf of divisors on* X *relative to* S, or *sheaf of divisors on* X *transversal to* f, and denote $\mathscr{D}iv_{X/S}$ the quotient sheaf (of commutative groups) $\mathscr{M}^*_{X/S}/\mathscr{O}^*_X$; the sections of this sheaf over X are called *divisors relative to* S, or *divisors on* X *transversal to* f; they form a commutative group denoted $\mathrm{Div}(X/S)$.

It is clear that $\mathscr{D}iv_{X/S}$ identifies canonically with a sub-sheaf of $\mathscr{D}iv_X$, and hence $\mathrm{Div}(X/S)$ with a subgroup of $\mathrm{Div}(X)$, which we denote also *additively*.

For every open U of X, we have $\mathscr{M}^*_{X/S}|U = \mathscr{M}^*_{U/S}$, hence $\mathscr{D}iv_{X/S}|U = \mathscr{D}iv_{U/S}$, and the sheaf $\mathscr{D}iv_{X/S}$ is therefore equal to the presheaf $U \mapsto \mathrm{Div}(U/S)$. **IV-4** | 330

Since $\mathscr{M}^*_{X/S}$ is a sub-sheaf of $\mathscr{M}^*_X$, the definitions, notations and formulas relative to divisors of sections of $\mathscr{M}_X$ over X (21.1.3) apply without change to sections of $\mathscr{M}_{X/S}$ over X.

(21.15.3). The structure of sheaf of ordered groups on $\mathscr{D}iv_X$ (21.1.6) induces on the sub-sheaf $\mathscr{D}iv_{X/S}$ a structure of sheaf of ordered groups, for which the sheaf of monoids of germs of positive sections is $\mathscr{D}iv^+_{X/S} = \mathscr{D}iv^+_X \cap \mathscr{D}iv_{X/S}$, which we denote $\mathscr{D}iv^+_{X/S}$. We have $\Gamma(X, \mathscr{D}iv^+_{X/S}) = \mathrm{Div}^+(X) \cap \mathrm{Div}(X/S)$; we denote this sub-monoid of $\mathrm{Div}(X/S)$ by $\mathrm{Div}^+(X/S)$, and it consists of elements $\geqslant 0$ with respect to the ordered group structure on $\mathrm{Div}(X/S)$. It follows from (21.5.1) that $\mathscr{D}iv^+_{X/S}$ is the image in $\mathscr{D}iv_{X/S}$ of the sub-sheaf of monoids

$$\mathscr{O}^{\mathrm{reg}}_{X/S} = \mathscr{O}_X \cap \mathscr{M}^*_{X/S} \tag{21.15.3.1}$$

For every open U of X, $\Gamma(U, \mathscr{O}^{\mathrm{reg}}_{X/S})$ is the set of sections t of $\mathscr{O}_X$ over U such that t is regular and that $1/t$ belongs to $\Gamma(U, \mathscr{M}_{X/S})$, which means, with the notations of (20.6.1), that $t \in \mathrm{T}_{X/S}(U)$, so that the sheaf $\mathscr{O}^{\mathrm{reg}}_{X/S}$ is none other than the sheaf denoted $\mathscr{T}_{X/S}$ in (20.6.1). We can therefore write

$$\mathscr{D}iv^+_{X/S} = \mathscr{O}^{\mathrm{reg}}_{X/S}/\mathscr{O}^*_X \tag{21.15.3.2}$$

up to canonical isomorphism.

(21.15.3bis). Let $D \in \mathrm{Div}^+(X/S)$ and consider the closed sub-prescheme $Y(D)$ defined by the ideal $\mathscr{J}_X(D)$ of $\mathscr{O}_X$ (21.6.5); by virtue of what precedes, for all $x \in Y(D)$, there is an open neighbourhood U of x in X and a section $t \in \mathrm{T}_{X/S}(U)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U \cdot t$; since $x \in Y(D)$, the image $t_{f(x)}$ of t in $\Gamma(U \cap X_{f(x)}, \mathscr{O}_{X_{f(x)}})$ belongs to the maximal ideal of $\mathscr{O}_{X_{f(x)},x}$, and furthermore, by definition, for all $s \in S$, the image t_s of t in $\Gamma(U \cap X_s, \mathscr{O}_{X_s})$ is a regular section. We deduce therefore from (11.3.8) and (19.2.4) that the canonical injection $Y(D) \to X$ is *transversely regular relative to* S *and of codimension* 1. The converse being immediate, we see that one can *identify* canonically the positive divisors on X relative to S with the closed sub-preschemes Y of X such that the canonical injection $Y \to X$ is a transversely regular immersion relative to S and of codimension 1. We will ordinarily make this identification.

Proposition (21.15.4). — *Let* D *be a divisor on* X, $\mathscr{J}_X(D)$ *the ideal fractionnaire invertible corresponding* (21.2.8) *and* (21.2.9). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that, for all* $x \in X$, $(\mathscr{J}_X(D) \cap \mathscr{M}^*_{X/S})_x \neq 0$ *(or equivalently, that* $\mathscr{J}_X(D) \subset \mathscr{M}_{X/S}$ *and* $\mathscr{J}_X(D)^{-1} \subset \mathscr{M}_{X/S}$*).*

Proof. Indeed, to say that $D \in \mathrm{Div}(X/S)$ means that for all $x \in X$, there exists a neighbourhood open U of x and a section $f \in \Gamma(U, \mathscr{M}^*_{X/S})$ such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}^*_X)$; since $\mathscr{J}_X(D)|U$ is the fractional ideal $\mathscr{O}_U \cdot f$, the proposition follows immediately. □

Proposition (21.15.5). — *Let* D *be a divisor on* X, $\mathscr{O}_X(D)$ *the fractional ideal invertible and* s_D *the meromorphic regular section of* $\mathscr{O}_X(D)$ *defined canonically by* D (21.2.8) *and* (21.2.9). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that* $s_D \in \Gamma(X, \mathscr{M}_{X/S}(\mathscr{O}_X(D)))$. **IV-4** | 331

Proof. Indeed, if U is an open of X such that $\mathrm{D}|\mathrm{U} = \mathrm{div}_{\mathrm{U}}(f)$, where $f \in \Gamma(\mathrm{U}, \mathscr{M}_{\mathrm{X}}^{*})$, to say that $s_{\mathrm{D}} \in \Gamma(\mathrm{U}, \mathscr{M}_{\mathrm{X}/\mathrm{S}}(\mathscr{O}_{\mathrm{X}}(\mathrm{D})))$ means, by virtue of the definitions (20.6.2), that $f^{-1} \in \Gamma(\mathrm{U}, \mathscr{M}_{\mathrm{X}/\mathrm{S}})$, whence the proposition. □

The interpretation of the divisors on X by means of the classes $\mathrm{cl}(\mathscr{L}, s)$ (21.2.11) allows us therefore to interpret the elements of $\mathrm{Div}(\mathrm{X}/\mathrm{S})$ as the pairs $(\mathscr{L}, s)$ (up to isomorphism) such that $\mathscr{L}$ is an $\mathscr{O}_{\mathrm{X}}$-Module invertible and that s is a meromorphic section of $\mathscr{L}$ regular relative to S (20.6.5), (iii)).

Proposition (21.15.6). — *Let* D *be a divisor on* X *relative to* S, *and suppose that for all* $x \in \mathrm{X}$ *such that* $\mathrm{prof}(\mathscr{O}_{\mathrm{X}_{f(s)},x}) = 1$, *we have* $\mathrm{D}_x \geqslant 0$ *(resp.* $\mathrm{D}_x = 0$*). Then* $\mathrm{D} \geqslant 0$ *(resp.* $\mathrm{D} = 0$*).*

Proof. As in (21.1.8), we can restrict to the case where $\mathrm{D} = \mathrm{div}(\varphi)$, where φ is a meromorphic regular function relative to S, and it all amounts to seeing that if $\mathrm{D}_x \geqslant 0$ everywhere then φ is everywhere defined in X. But this hypothesis means that, if $\mathrm{T} = \mathrm{X} - \mathrm{dom}(\varphi)$, we have $\mathrm{prof}(\mathscr{O}_{\mathrm{X}_{f(s)},x}) \geqslant 2$ for all $x \in \mathrm{T}$, and it suffices to conclude by applying (20.6.6). □

(21.15.7). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : \mathrm{X}' \to \mathrm{X}$ be an S-morphism. If the S-morphism g is *flat*, we know (21.4.5) that the inverse image by g of every divisor on X is defined; if furthermore $\mathrm{D} \in \mathrm{Div}(\mathrm{X}/\mathrm{S})$, it follows from the definition (21.15.2) and from (20.6.8) that $g^{*}(\mathrm{D}) \in \mathrm{Div}(\mathrm{X}'/\mathrm{S})$.

(21.15.8). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : \mathrm{X}' \to \mathrm{X}$ be an S-morphism *finite* and *flat*. We note that g is also necessarily of finite presentation (1.4.3), (1.4.6) and (1.6.3), hence $g_{*}(\mathscr{O}_{\mathrm{X}'})$ is a $\mathscr{O}_{\mathrm{X}}$-Module *locally free* (2.1.12), in other words g is a locally free morphism (18.2.7); for all $s \in \mathrm{S}$, the corresponding morphism $g_s : \mathrm{X}'_s \to \mathrm{X}_s$ is also finite and locally free. We deduce then from (21.5.2) and (20.6.1) that for every open U of X and every section $t' \in \Gamma(g^{-1}(\mathrm{U}), \mathscr{T}_{\mathrm{X}'/\mathrm{S}})$, the norm $\mathrm{N}_{\mathrm{X}'/\mathrm{X}}(t')$ belongs to $\Gamma(\mathrm{U}, \mathscr{T}_{\mathrm{X}/\mathrm{S}})$; the reasoning of (21.5.3) proves then that for every $\mathscr{O}_{\mathrm{X}'}$-Module invertible $\mathscr{L}'$ and every meromorphic section u' of $\mathscr{L}'$ above S, the norm $\mathscr{L} = \mathrm{N}_{\mathrm{X}'/\mathrm{X}}(\mathscr{L}')$ is a meromorphic section of X, regular relative to S. The interpretation of the direct image of divisors relatives to S given in (21.15.5) and the definition of a direct image of divisors (21.5.5) prove then that *for every divisor* $\mathrm{D}' \in \mathrm{Div}(\mathrm{X}'/\mathrm{S})$, we have $g_{*}(\mathrm{D}') \in \mathrm{Div}(\mathrm{X}/\mathrm{S})$.

(21.15.9). Consider finally an arbitrary morphism $\mathrm{S}' \to \mathrm{S}$, and (under the hypotheses of (21.15.1)) set **IV-4** | 332
$\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{S}'$, which is flat and locally of finite presentation over S′; if $p : \mathrm{X}' \to \mathrm{X}$ is the canonical projection, we have seen (20.6.9) that there is a canonical homomorphism $p^{*}(\mathscr{M}_{\mathrm{X}/\mathrm{S}}) \to \mathscr{M}_{\mathrm{X}'/\mathrm{S}'}$, which transforms every section of $\mathscr{M}_{\mathrm{X}/\mathrm{S}}$ over U, *regular relative to* S, into a section of $\mathscr{M}_{\mathrm{X}'/\mathrm{S}'}$ regular relative to S′ (20.6.5), (iii); we conclude then from the definition (21.15.2) and the right exactness of the functor p^{*}, that the preceding homomorphism defines by passage to quotients a canonical homomorphism

$$\mathrm{Div}(\mathrm{X}/\mathrm{S}) \longrightarrow \mathrm{Div}(\mathrm{X}'/\mathrm{S}') \tag{21.15.9.1}$$

which evidently transforms elements of $\mathrm{Div}^{+}(\mathrm{X}/\mathrm{S})$ into elements of $\mathrm{Div}^{+}(\mathrm{X}'/\mathrm{S}')$. We also set $\mathrm{Div}(\mathrm{X}'/\mathrm{S}') = \mathrm{Div}_{\mathrm{X}/\mathrm{S}}(\mathrm{S}')$ (resp. $\mathrm{Div}^{+}(\mathrm{X}'/\mathrm{S}') = \mathrm{Div}^{+}_{\mathrm{X}/\mathrm{S}}(\mathrm{S}')$), and we see immediately that we have thus defined two contravariant functors

$$\mathrm{Div}_{\mathrm{X}/\mathrm{S}} : \mathbf{Sch}_{/\mathrm{S}} \longrightarrow \mathbf{Ab}, \qquad \mathrm{Div}^{+}_{\mathrm{X}/\mathrm{S}} : \mathbf{Sch}_{/\mathrm{S}} \longrightarrow \mathbf{Ens}$$

from the category of S-preschemes into that of commutative groups (resp. sets). We will see further (chap. VI) the important cases where the functor $\mathrm{Div}^{+}_{\mathrm{X}/\mathrm{S}}$ is *representable* ($\mathbf{0}_{\mathrm{III}}$, 8.1.8).

For every divisor $\mathrm{D} \in \mathrm{Div}(\mathrm{X}/\mathrm{S})$, the image of D by the homomorphism (21.15.9.1) is none other than the *inverse image* $p^{*}(\mathrm{D})$ (in the sense of (21.4.2)): the existence of this inverse image and the preceding assertions are immediate consequences of (20.6.5), (iii) and (20.6.9).

The elements of $\mathrm{Div}(\mathrm{X}'/\mathrm{S}')$ are often called, by abuse of language, "*families of divisors on* X *relative to* S, *parametrised by the S-prescheme* S′"; this terminology is used especially when it concerns positive divisors.

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